The long-range Ising CFT: research notes

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Combined public edition

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Preface

These seven research notes accompany the study of the long-range Ising CFT. The public repository accompanying the paper (`https://github.com/Canonical111/LRI_public`) supplies the code, data, and replays in the seven folders named below; its root `README.md` and each folder's `results_map.md` give the entry points. The published folder notes remain authoritative. Each paragraph below is an excerpt from that part's own abstract or summary, with its citations left to that part. Numbering and bibliographies remain local to each part.

GFF. We determine the thermal scaling dimension of the long-range Ising model in the expansion parameter $\epsilon = 2s - d$. The analytic result through $C_5(d)$ is expressed in an explicitly defined master basis, with its graph weights, subtraction prescription, regulator expansions and finite special-function representations derived below.

SRI. We collect the expansions of the thermal scaling dimension $\Delta_{\mathcal{E}}$ of the critical long-range Ising model about the point where it joins the short-range Ising universality class, in $d = 1, 2, 3$.

HT_Pert. This note documents every perturbative Hamiltonian-truncation (HT) number quoted in the paper *The long-range Ising CFT: perturbative and nonperturbative approaches* and in its Supplemental Material, and how to reproduce each from the folder `HT_Pert/` of the public repository.

HT. The long-range Ising (LRI) fixed point in d dimensions is reached from the generalized free field (GFF) ϕ of dimension Δ_σ by the relevant ϕ^4 deformation. In radial quantization the GFF becomes a set of free oscillators.

NRG. We compute the scaling dimension $\Delta_{\mathcal{E}}$ of the energy operator of the one-dimensional long-range Ising model, with couplings decaying as $1/r^{1+s}$, at thirteen values of the spin dimension $\Delta_\sigma = (1-s)/2$ between 0.249 and 0.001.

MC. We determine, by cluster Monte Carlo simulation, the scaling dimension $\Delta_{\mathcal{E}}$ of the leading $\mathbb{Z}_2$-even scalar of the ferromagnetic long-range Ising model with couplings $J(r) \propto r^{-(d+s)}$, as a function of the exactly known spin dimension $\Delta_\sigma = (d-s)/2$, in $d = 1, 2, 3$.

Resummation. We reconstruct the thermal scaling dimension $\Delta_{\mathcal{E}}$ of the critical long-range Ising model across its whole interacting window in $d = 1, 2, 3$, by Borel resummation of the generalized-free-field (GFF) ϵ expansion through eighth order combined with the expansion about the short-range end.

AI agents and reproducibility

AI systems have recently contributed results in amplitudes [1, 2], QCD [3], integrable chains [4] and mathematics [5]. To our knowledge, language-model agents had not computed new critical exponents or CFT data; reinforcement learning has recovered known ones [6, 7], and neural networks trained on crossing symmetry reconstruct conformal correlators and partition functions [8, 9]. This part describes how the work of this study was organized and how its results were checked.

Architecture of the collaboration. The work was organized in layers (Fig. 1). The author set the direction: the problem, the five descriptions and key choices of method, among them computing exact asymptotic forms to guide the numerical fits and proposing the transcendental bases in which to seek exact coefficients. The author also set the rules for the clusters, approved plans and production runs, and decided between options the agents presented; the logs contain no derivation or production code by the author. OpenAI's GPT-6 astra drafted the initial high-performance codes for the clusters and computed the coefficients near the short-range end. Anthropic's Claude Code agents, in host sessions that exchanged messages and spawned subagents, reviewed and extended these codes, ran the campaigns, derived and proved closed forms, wrote the fit protocols and the research notes, and drafted the article. OpenAI's Codex, running GPT-6 astra in sessions separate from those that did the work, reviewed plans, codes and results.

How the results were checked. We did not rely on any single computation. Four layers of checks were used, and each result section of the article states which applied.

1. *Review before production.* Production plans and codes were reviewed by Codex, in most cases until approved; where the author stopped a review early or a last fix was not inspected, the research notes record it.

2. *Rules fixed in advance.* Reporting rules, such as error bars of four sampling standard errors, were fixed before the results where possible; where a rule was fixed afterwards and confirmed by the author, the research notes that follow say so. For the MC fits this held only in part: in $d = 1$ and $d = 3$ some fit choices were made or amended after data had been seen, among them the amendment under which the $d = 3$ value at $s = 1.6$ is accepted. Part VI records when each rule was fixed relative to the data.

3. *Independent recomputation.* Key results were recomputed by separately written codes or methods: the eight-loop coefficients in $d = 2, 3$, the short-range coefficients $C_5^{\mathrm{SR}}(1)$ to $C_8^{\mathrm{SR}}(1)$ and $C_2^{\mathrm{SR}}(2)$, the Hamiltonian-truncation solvers and the Wilson chain of NRG. Results reviews by Codex covered most of the final coefficients.

4. *Universality.* The five descriptions compute the same $\Delta_{\mathcal{E}}$ with unrelated errors (Sec. 3.5 of the article).

These checks caught errors. The author's own earlier calculation of the four-loop coefficient gave $C_4(1) = 45.3896$; when the Claude Code agents and GPT-6 astra checked it, both found 45.42488010356, the value in Table 1 of the article. An audit requested by the author found that an earlier version of our MC report presented a fit-model spread as a standard deviation. The checks do not cover an error shared by all codes of one description; only the comparison across descriptions does.

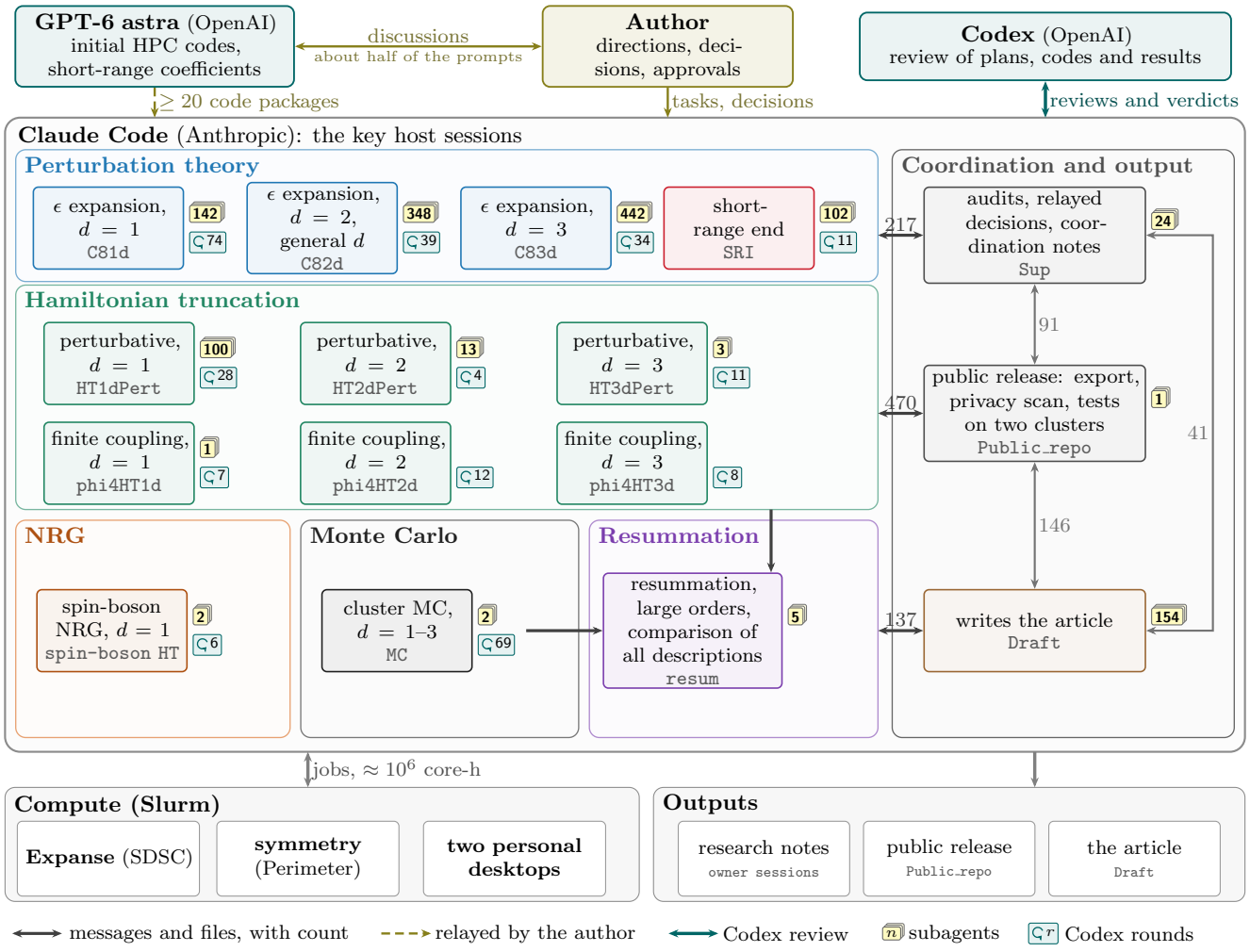


Figure 1: Collaboration of the AI agents. GPT-6 astra drafted the initial codes in discussions with the author (olive; dashed arrows: work relayed by the author). Inside the Claude Code frame, sessions that each own one topic produce the results, grouped and colored by method; Sup coordinates them, Public_repo assembles the public release, resum combines the numbers of all descriptions, and Draft writes the article. Badges count the subagents each session spawned (yellow, 1381 in all) and the rounds of review by Codex (teal, about 310 in all); arrows count the messages between sessions. The bottom row shows where the jobs ran and what the work produced. Of the 35 Claude Code sessions, not counting subagents, the key ones are shown.

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Part I

Thermal dimensions in the long-range Ising model

Analytic coefficients through fifth order and numerical sixth- to eighth-order coefficients

Research calculation record

Condensed research note | 16 October 2026

Abstract

We determine the thermal scaling dimension of the long-range Ising model in the expansion parameter $\epsilon = 2s - d$. The analytic result through $C_5(d)$ is expressed in an explicitly defined master basis, with its graph weights, subtraction prescription, regulator expansions and finite special-function representations derived below. At $d = 1$, the result uses only fixed Gamma/polygamma values and ordinary generalized hypergeometric values; the proof of the $\mathcal{K}$ reduction, the logarithmic-wheel reduction and the five-value evaluation of $P_5(1)$ are included in full. At $d = 2$ the primitive periods factor in the same scalar values, and the nonprimitive masters $\mathcal{K}_2$, $\mathcal{L}_2$ and Z_2 are evaluated in closed form, so that $C_5(2)$ is an explicit expression in those values, $\Gamma(\frac{1}{4})$, π, $\log 2$, Catalan's constant and $\zeta(3)$. At $d = 3$ the remaining rank-two structure is identified without claiming an unavailable scalar summation. For generic $0 < d < 4$ the masters $\mathcal{L}_d$ and Z_d are reduced to at most three samples of finitely many hypergeometric functions with at most two summation indices at unit argument and their first and second derivatives in the rim exponent, using a proof that the top Fourier modes of products of balanced kernels vanish; $\mathcal{K}_d$ is reduced, under stated analytic inputs, to one Laurent coefficient of such a sample, and, by a second route, to a polygamma part plus one two-fold Mellin–Barnes integral. A finite representation of a reduced analytic expression in nine special-function values gives $C_6(1) = 14331.68703275805$; exact sums over all six-loop graphs at increasing quadrature degree, which do not use that expression, give $14331.6870328(5)$, in agreement to 1.0×10^{-8}. Numerical finite-forest calculations give $C_6(2) = 362.366 \pm 0.004$, $C_6(3) = 33.5754 \pm 0.0005$, $C_7(1) = -322550 \pm 41$, $C_7(2) = -3876.1 \pm 2.1$, $C_7(3) = -220.35 \pm 0.13$, $C_8(1) = (8.060 \pm 0.017) \times 10^6$, $C_8(2) = 45990 \pm 33$ and $C_8(3) = 1602.6 \pm 3.9$. The integration kernels, regulator-dependent fixed-point extraction, local subtractions, sampling designs, covariance propagation and convergence checks supporting these estimates are retained. Numerical allowances are empirical, not rigorous intervals or bounds on truncating the epsilon expansion. Part V records a second observable built from the same graphs and master functions, the bulk one-point coefficient a_{ϕ^2} of the auxiliary bulk field through ϵ^6 at general d, from a calculation supplied by the author whose checks are internal to this framework. This note records one common graph/renormalization calculation; it is not an independent field-theoretic certification of its higher-loop results.

Part I

Conclusions, conventions and renormalization

1 Scope and current coefficients

The model is the long-range Ising model: spins $\sigma_i = \pm 1$ on the sites of $\mathbb{Z}^d$ with ferromagnetic energy

$$\mathcal{H}_{\rm lat} = -\sum_{i<j} J(|i-j|)\,\sigma_i\sigma_j, \qquad J(r) \sim r^{-(d+s)},$$

in the critical long-range regime above $s = d/2$. Its continuum description is a Gaussian field with covariance proportional to $|p|^{-s}$ in momentum space and to $|x|^{-(d-s)}$ in position space, normalized to coefficient one, with a quartic interaction of engineering dimension $2s - d = \epsilon$. The observable is the scaling dimension of the thermal operator $\mathcal{E} = [\phi^2]$:

$$\epsilon = 2s - d, \qquad \Delta_\phi = \frac{d-\epsilon}{4}, \qquad \Delta_{\mathcal{E}}(d,\epsilon) = \sum_{n\geq 0} C_n(d)\epsilon^n. \tag{1}$$

The general-dimensional long-range regime is fixed $0 < d < 4$ and sufficiently small positive ϵ; noninteger dimensions and intermediate meromorphic integrals have the continuation prescriptions stated in the relevant sections. This is a local expansion near $s = d/2$, not an unresummed approximation over the entire interacting regime.

Order	$d = 1$	$d = 2$	$d = 3$
C_0	$1/2$	1	$3/2$
C_1	$-1/6$	$-1/6$	$-1/6$
C_2	0.657131263981	0.308065413582	0.181221785406
C_3	-3.719856322902	-0.874850012653	-0.353581520433
C_4	45.42488010356	5.048868431	1.228063545
C_5	-731.13947662454	-38.833645896	-5.850351421
C_6	14331.68703275805	362.366 ± 0.004	33.5754 ± 0.0005
C_7	-322550 ± 41	-3876.1 ± 2.1	-220.35 ± 0.13
C_8	$(8.060 \pm 0.017)10^6$	45990 ± 33	1602.6 ± 3.9

The first six rows use the special-function evaluations, not the noisier lower-order checks produced during high-loop integration; their closed forms are collected in Section 1.1. The entry for $C_6(1)$ is the finite special-function representation of Section 1.2. It evaluates a reduced analytic expression that is taken as an input and is not derived in this note (Section 1.3). The independent deterministic sum over all graphs at quadrature degree 22 (Section 22) gives 14331.6870328(5). It has no sampling error; its uncertainty is the empirical convergence-and-rounding allowance of Section 22.7, and the representation lies 1.0×10^{-8} below its unrounded value 14331.68703276838. For the remaining entries of the last three rows, the allowances above and the sampling standard errors below are *different* quantities:

Coefficient	Underlying estimator	Sampling standard error
$C_6(2)$	362.36647176375726	0.0007997307
$C_6(3)$	33.57542664131543	0.0001185495235
$C_7(1)$	−322549.8888129889	10.08791295
$C_7(2)$	−3876.1260927949925	0.5244496691
$C_7(3)$	−220.35448047392794	0.0318188006
$C_8(1)$	8059766.723129955	4209.576151
$C_8(2)$	45990.00263614446	8.216820958
$C_8(3)$	1602.6052229663408	0.9516489064

Each allowance is the output of a stated rule applied to the sampling standard error and, where they exist, to measured convergence steps. The table gives the rule for each entry and when it was fixed: before the production run, during it but before its result was known, or after the result. "Four SE" means four sampling standard errors; a half-split estimate, called a half-stream estimate in Sections 28 and 31, uses only the first or only the second half of the replicates of every random stream.

Entry	Allowance	Rule	Fixed
$C_6(1)$, graph sum	$\pm 4.9 \cdot 10^{-7}$	the last change with quadrature degree, plus the difference of the two cutoffs, a floating-point allowance and the lower-loop convergence (Section 22.7)	before production
$C_6(2)$	±0.004	four SE, rounded up to a multiple of 0.001 (Section 26.7)	before the result was known; rounding step refined from 0.01 afterwards
$C_6(3)$	±0.0005	four SE, rounded up to a multiple of 0.0001 (Section 26.8)	before production; rounding step refined from 0.001 afterwards
$C_7(1)$	±41	the largest of four SE, twice the last refinement step plus four of its SE, twice the degree six to seven check step plus four of its SE, and the half-split displacement, rounded up to two significant figures (Section 27.8)	before production
$C_7(2)$	±2.1	the larger of four SE and the largest half-split deviation, rounded up to a multiple of 0.1 (Section 28)	after the result (the rule of an earlier, lower-precision $C_7(2)$ calculation)
$C_7(3)$	±0.13	four SE, rounded up to a multiple of 0.01 (Section 29)	before the result was known
$C_8(1)$	±17000	as for $C_7(1)$, with the degree eight to nine check step, and with twice the last degree step of the exact loop-five input as a fifth candidate (Section 30.1)	before production; batch count raised afterwards
$C_8(2)$	±33	as for $C_7(2)$, rounded up to a multiple of 1 (Section 31)	after the result
$C_8(3)$	±3.9	four SE, rounded up to a multiple of 0.1 (Section 32)	after the result

For $C_7(1)$ the four candidates are 40.35, 39.50, 22.49 and 12.58, so the allowance is four sampling standard errors. The radial quadrature of $C_7(1)$ stops at degree six, and the rule adds no model of the truncation beyond it. Adding the measured degree six to seven step and a geometric tail at the last step ratio gives the post hoc estimate $C_7(1) \approx -322524 \pm 38$ (753); its central value lies inside the quoted interval. That tail model agrees with the lower-order checks at fifth and sixth

order but is not validated at seventh order, where the step ratio is larger. The result is therefore $C_7(1) = -322550 \pm 41$, and the truncation-aware value is a check (Section 27.9). For $C_8(1)$ the rule and the design were fixed before production, but the number of production batches was not. Under the stopping rule set first, nine batches give 8061480 ± 5300 with allowance ± 22000; after interim results of five to eight batches had been seen, all fourteen planned batches were completed (Section 30.3). At degree eight the radial quadrature is converged well inside the allowance. For $C_8(2)$ and $C_8(3)$ four sampling standard errors exceed the distances of both half-split estimates from the full estimate. The coverage of the allowance on $C_8(3)$ is conditional on a finite variance, for which Section 32.5 gives empirical evidence, not a bound.

Two eighth-order results are checked by a separately developed code. At $d = 2$ it shares the method and the per-graph input files of loops three to seven and gives $C_8(2) = 46006.1 \pm 13.4$ (sampling standard error), $+1.03$ combined standard errors from the estimator (Section 31.7). At $d = 3$ it shares the method and the seventh-order integrator and gives $C_8(3) = 1600.98 \pm 3.97$, -0.40 combined standard errors from the estimator (Section 32.7). The additional estimator digits are not certified physical digits. Numerical errors do not cover shared mistakes in the graph/renormalization framework, and are not bounds on the omitted orders in (1).

Part V (Section 34) adds a second observable built from the same graphs and master functions: the bulk one-point coefficient a_{ϕ^2} of the composite $:\Phi^2:$ of the auxiliary bulk field, through ϵ^6 at fixed $0 < d < 4$, which is the horizontal coordinate of the bootstrap bounds of [8]. It comes from a calculation supplied by the author with its own code. Its checks are internal to the master framework of this note, together with comparisons against this note's own inputs (Section 34.12); Section 34.1 states what it establishes.

1.1 Closed forms through fifth order

The entries C_0 to C_5 of the table are values of exact expressions, not statistical estimates. This subsection states the expressions together with every definition they need; each formula below repeats one derived later in the note, and symbols with more than one meaning in the note are listed in Section 1.4. At $d = 1$ every step is proved in this note (Part II). At $d = 2$ the closed form of $C_5(2)$ is proved in this note. At $d = 3$ the inputs of $C_5(3)$, namely Q_3, $P_4(3)$, $P_5(3)$, $\mathcal{L}_3$, Z_3 and $\mathcal{K}_3$, are evaluated numerically. The values of Q_3, $P_4(3)$ and $P_5(3)$ from their fixed formulas agree to 10^{-74} or better between evaluations at 90 and 100 working digits. Independent evaluations confirm Q_3 and $P_4(3)$ to about 10^{-79} and $2 \cdot 10^{-86}$ (a 100-digit spectral evaluation), $P_5(3)$ to about $2 \cdot 10^{-53}$ (a 60-digit integration of the Gamma kernels), and $\mathcal{L}_3$, Z_3 and $\mathcal{K}_3$ to $1.2 \cdot 10^{-93}$ or better. Propagated into $C_5(3)$ these differences are below 10^{-54}, well below one unit in the last of the 50 digits printed for it. These agreements are not interval bounds. The spectral formula for $\mathcal{K}_d$ below is proved at $d = 2$ in Section 18.9; at other d its derivation uses input (M) of Section 21.1 and the extraction of $[\tau^2]$ under the trace, which is proved only at $d = 2$ (Section 21.12). The value of $\mathcal{K}_3$ does not depend on (M): the two-fold Barnes representation (588), derived without it, reproduces it to $4 \cdot 10^{-100}$. Section 1.3 lists what is proved, what is conditional and what is numerical.

Any dimension. With $q = d/4$, the polygamma functions $\psi_m(z) = d^{m+1} \log \Gamma(z)/dz^{m+1}$ and the combinations

$$A = \psi_0(2q) + \psi_0(1) - 2\psi_0(q), \quad R = \psi_1(2q) - \psi_1(1),$$
$$U = \psi_2(q), \qquad\qquad V = \psi_2(1) + \psi_2(2q), \quad X = \psi_3(2q) - \psi_3(1).$$

the coefficients are

$$\boxed{\begin{aligned} C_0(d) &= \frac{d}{2}, & C_1(d) &= -\frac{1}{6}, \quad C_2(d) = \frac{A}{9},\\ C_3(d) &= \frac{A^2+R-8Q}{108},\\ 648C_4(d) &= 16P_4 - A^3 - 8AQ - 3AR + \frac{53}{2}U - 3V. \end{aligned}}$$

$$\boxed{\begin{aligned} 46656C_5(d) = &-1536(P_5+\mathcal{K}_d) + 384AP_4 - 576\mathcal{L}_d - 1728Q^2\\ &+ 720A^2Q + 1008QR + 7152Z_d\\ &- 15A^4 - 90A^2R + 414AU - 108AV - 117R^2 - 75X. \end{aligned}}$$

Here Q_d, $P_4(d)$ and $P_5(d)$, written Q, P_4 and P_5 in the boxed formulas, are the primitive (zig-zag) periods with three, four and five loops, normalized by h_d^{L-1} with $h_d = \Gamma(d/2)/\pi^{d/2}$, and $\mathcal{K}_d$, $\mathcal{L}_d$ and Z_d are nonprimitive master data. All of them are spectral integrals. With

$$\begin{aligned} d_0(d) &= 1, & d_\ell(d) &= \frac{(2\ell+d-2)\Gamma(\ell+d-2)}{\Gamma(d-1)\Gamma(\ell+1)} \quad (\ell \geq 1),\\ a_\ell &= \frac{\ell}{2} + \frac{q}{2}, & b_\ell &= a_\ell + q,\\ t_\ell(\nu) &= \Gamma(2q)\frac{\Gamma(a_\ell + \mathrm{i}\nu)\Gamma(a_\ell - \mathrm{i}\nu)}{\Gamma(b_\ell + \mathrm{i}\nu)\Gamma(b_\ell - \mathrm{i}\nu)}. \end{aligned}$$

and the trace

$$\langle f \rangle_d = \frac{1}{\pi}\sum_{\ell=0}^{\infty} d_\ell(d) \int_0^\infty f(\ell,\nu)\, d\nu.$$

the four-loop and three-loop periods are

$$3Q_d = \langle t_\ell^3 \rangle_d, \qquad P_4(d) = \langle t_\ell^4 \rangle_d.$$

and, with

$$\begin{aligned} w_1 &= -\psi_0(q) + \frac{1}{2}\Re\{\psi_0(a_\ell + \mathrm{i}\nu) + \psi_0(b_\ell + \mathrm{i}\nu)\},\\ w_2 &= \frac{1}{8}\Re\{\psi_1(a_\ell + \mathrm{i}\nu) - \psi_1(b_\ell + \mathrm{i}\nu)\}. \end{aligned}$$

the two wheel master data are

$$\boxed{\mathcal{L}_d = \langle 4t_\ell^4 w_1 \rangle_d, \qquad Z_d = \left\langle t_\ell^3 \left(w_1^2 + \frac{w_2}{3} \right) \right\rangle_d.}$$

For $\mathcal{K}_d$ put $c_* = \psi_0(1) + \psi_0(2q)$ and, at $a = a_\ell$ and $b = b_\ell$,

$$\begin{aligned} p &= \Re\{\psi_0(a + \mathrm{i}\nu) + \psi_0(b + \mathrm{i}\nu)\}, & h &= \Im\{\psi_0(a + \mathrm{i}\nu) - \psi_0(b + \mathrm{i}\nu)\},\\ j &= \Re\{\psi_1(a + \mathrm{i}\nu) - \psi_1(b + \mathrm{i}\nu)\}, & k &= \Im\{\psi_1(a + \mathrm{i}\nu) + \psi_1(b + \mathrm{i}\nu)\},\\ r &= \Re\{\psi_2(a + \mathrm{i}\nu) + \psi_2(b + \mathrm{i}\nu)\}, & E_2 &= -\frac{h^2}{2} - \frac{5j}{8}. \end{aligned}$$

Then

$$\boxed{\mathcal{K}_d = -\frac{AU}{4} + \frac{1}{4}\left\langle t_\ell^2 \left[(c_* - p)E_2 - \frac{hk}{2} - \frac{5r}{32} \right] \right\rangle_d.}$$

The master $\mathcal{K}_d$ is defined as the finite second-order coefficient of a subtracted two-loop master graph, and the spectral formula above is its evaluation. The two-fold Mellin–Barnes integral (588), derived in position space without that formula (Section 21.10), agrees to about 10^{-100} with the

proved values at $d = 1$ (76) and $d = 2$ (271) and with the residue series at $d = 3$ (Section 21.11). The trace forms converge directly for $\frac{2}{3} < d < 4$ (Q_d, Z_d), for $d > \frac{1}{2}$ (P_4, $\mathcal{L}_d$) and for $0 < d < 4$ ($\mathcal{K}_d$); elsewhere they are continued analytically.

The five-loop primitive needs the two-vector zig-zag spectrum. With $p = d/2 - 1$ (not the p of the formula for $\mathcal{K}_d$),

$$\tau_n^{(d)}(\nu) = (-1)^n \pi^{d/2} \frac{\Gamma(n/2 + d/8 + \mathrm{i}\nu)\Gamma(n/2 + d/8 - \mathrm{i}\nu)}{\Gamma(n/2 + 3d/8 + \mathrm{i}\nu)\Gamma(n/2 + 3d/8 - \mathrm{i}\nu)},$$
$$\rho_n^{(d)}(\nu) = \frac{\Gamma(p)\Gamma(n+2p)(n+p)}{2\pi^{3d/2+1}\Gamma(2p)n!} \left[(n+p)^2 + 4\nu^2\right] \frac{\Gamma(p + 2\mathrm{i}\nu)\Gamma(p - 2\mathrm{i}\nu)}{\Gamma(2\mathrm{i}\nu)\Gamma(-2\mathrm{i}\nu)}.$$

the periods with $L = 3, 4, 5$ loops are, for $2 < d < 4$,

$$\boxed{P_L(d) = h_d^{L-1} \sum_{n=0}^{\infty} \int_0^{\infty} \rho_n^{(d)}(\nu) \, [\tau_n^{(d)}(\nu)]^{L+2} d\nu, \qquad L = 3, 4, 5.}$$

and, for $0 < d < 2$, where the first pair of Gamma poles crosses the real contour,

$$P_L(d) = h_d^{L-1} \Bigg\{ \sum_{n=0}^{\infty} \int_0^{\infty} \rho_n^{(d)}(\nu) [\tau_n^{(d)}(\nu)]^{L+2} d\nu + \sum_{n=1}^{\infty} \frac{\Gamma(n+2p)(n+p)n(n+2p)}{2\pi^{3d/2} n! \Gamma(-p)} [\tau_n^{(d)}(-\mathrm{i}p/2)]^{L+2} \Bigg\}.$$

At removable singular values such as $d = 1, 2$ the Gamma ratios are taken in their joint analytic limit. The masters can be reduced further, to a few two-variable hypergeometric samples and their low-order parameter jets. For generic d this is a finite representation, not a reduction to scalar values.

One dimension. At $d = 1$ every coefficient is a fixed value. With $q = \frac{1}{4}$, $\bar{q} = \frac{3}{4}$ and

$$\boxed{\begin{array}{ll} \mathsf{A} = \psi_0(2q) + \psi_0(1) - 2\psi_0(q), & \mathsf{D} = \psi_0(\bar{q}) - \psi_0(q), \\ \mathsf{T} = \psi_1(q), & \overline{\mathsf{T}} = \psi_1(\bar{q}), \\ \mathsf{U} = \psi_2(q), & \overline{\mathsf{U}} = \psi_2(\bar{q}). \end{array}}$$

whose values are, with Catalan's constant $\mathsf{G} = \sum_{n \geq 0} (-1)^n/(2n+1)^2$,

$$\begin{array}{cc} \mathsf{A} = A, & \mathsf{D} = \pi, \\ \mathsf{T} + \overline{\mathsf{T}} = 2\pi^2, & \mathsf{T} - \overline{\mathsf{T}} = 16\mathsf{G}, \\ \mathsf{U} + \overline{\mathsf{U}} = -112\zeta(3), & \overline{\mathsf{U}} - \mathsf{U} = 4\pi^3, \\ H = \dfrac{5\mathsf{T} - \overline{\mathsf{T}}}{6}, & \log 2 = \dfrac{\mathsf{A} - \mathsf{D}}{4}. \end{array}$$

and with $Q = \Gamma(\frac{1}{4})^4/\pi$, the coefficients are

$$\boxed{\begin{array}{ll} C_0 = \dfrac{1}{2}, & C_1 = -\dfrac{1}{6}, \quad C_2 = \dfrac{\mathsf{A}}{9}, \\ C_3 = \dfrac{\mathsf{A}^2 + (\mathsf{T} + \overline{\mathsf{T}})/6 - 8Q}{108}. & \end{array}}$$

$$\boxed{648 C_4 = 16 P_4 - \mathsf{A}^3 - \frac{\mathsf{A}}{2}(\mathsf{T} + \overline{\mathsf{T}}) - 8\mathsf{A}Q + \frac{365\mathsf{U} - 6\overline{\mathsf{U}}}{14}.}$$

$$\boxed{\begin{aligned}46656C_5 = &-1536P_5 + 96(7\mathsf{A}-\mathsf{D})P_4 + 384\mathsf{D}B_- - 1728Q^2\\ &+16Q(45\mathsf{A}^2+383\mathsf{T}-64\overline{\mathsf{T}}) - 768\mathsf{T}(3\mathsf{T}-\overline{\mathsf{T}})\\ &-15\mathsf{A}^4 - 15\mathsf{A}^2(\mathsf{T}+\overline{\mathsf{T}})\\ &-\frac{6\mathsf{A}}{7}(431\mathsf{U}+18\overline{\mathsf{U}}) - \frac{83}{4}(\mathsf{T}+\overline{\mathsf{T}})^2.\end{aligned}}$$

The periods are ordinary hypergeometric values:

$$P_4 = 13B_- - 3B_+, \qquad P_5 = 16\pi^{3/2}\big[-35X_0 + (22\sqrt{2}-13)X_+ + (22\sqrt{2}+13)X_-\big],$$

with

$$\begin{aligned}B_- &= \Gamma(\tfrac14)^4 F_-, \quad B_+ = \frac{64\pi^4}{\Gamma(\frac14)^4}F_+,\\ X_0 &= \mathcal{Y}_2, \qquad X_+ = c_8^7\mathcal{Y}_3 - s_8^7\mathcal{Y}_5, \quad X_- = c_8^7\mathcal{Y}_1 + s_8^7\mathcal{Y}_7.\\ F_- &= {}_4F_3\left(\begin{matrix}\frac14,\frac14,\frac12,\frac12\\ \frac34,\frac34,1\end{matrix};1\right), \quad F_+ = {}_4F_3\left(\begin{matrix}\frac34,\frac34,\frac12,\frac12\\ \frac54,\frac54,1\end{matrix};1\right),\\ c_8 &= \cos(\pi/8), \qquad s_8 = \sin(\pi/8),\\ \mathcal{Y}_r &= \frac{r}{8}\left[\frac{\Gamma((r+2)/8)}{\Gamma((r+6)/8)}\right]^7 {}_9F_8\left(\begin{matrix}1,1+\frac{r}{8},((r+2)/8)^{\times 7}\\ \frac{r}{8},((r+6)/8)^{\times 7}\end{matrix};-1\right),\\ &\quad r\in\{1,2,3,5,7\}.\end{aligned}$$

Here $(a)^{\times n}$ stands for n copies of the parameter a, and ${}_pF_q$ is the series of Section 1.4. At $r=2$ one upper and one lower parameter 1 cancel, so $\mathcal{Y}_2$ is an ${}_8F_7$ value. No elementary evaluation or minimality of these values is asserted.

Two dimensions. With $a = \log 2$,

$$\boxed{Q_2 = \frac16\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4 = \frac{\Gamma(1/4)^8}{24\pi^4}.}$$

$$\boxed{\begin{aligned}C_0(2) &= 1, \qquad C_1(2) = -\frac16, \quad C_2(2) = \frac{4a}{9},\\ C_3(2) &= \frac{4a^2}{27} - \frac{\Gamma(1/4)^8}{324\pi^4},\\ 648C_4(2) &= 16P_4(2) - 64a^3 - 32aQ_2 - 359\zeta(3),\\ 46656C_5(2) &= -1536\{P_5(2)+\mathcal{K}_2\} + 1536aP_4(2) - 576\mathcal{L}_2 - 1728Q_2^2\\ &\quad + (11520a^2 + 28608\mathsf{G})Q_2 - 3840a^4 - 21456a\zeta(3).\end{aligned}}$$

The two nonprimitive masters are $\mathcal{L}_2 = -2\log 2\, P_4(2)$, proved in Section 18.7 for the series definition of $\mathcal{L}_2$, and $\mathcal{K}_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2$, proved in Section 18.8 for its master-path definition. The formula for $C_5(2)$ below also uses $Z_2 = 4\mathsf{G}Q_2$ (265). The primitive periods reuse the one-dimensional values above (proved in Section 18.5):

$$\boxed{P_4(2) = \frac{B_-^2 - B_+^2}{4\pi^3}.}$$

$$\boxed{\begin{aligned}P_5(2) = \frac{8}{\pi}\big[&25X_0^2 - 28\sqrt{2}\,X_0(X_+ + X_-)\\ &+ 18(X_+^2 + X_-^2) + 28X_+X_-\big].\end{aligned}}$$

Hence

$$\boxed{\begin{aligned}972\,C_5(2) = &-32P_5(2) + 56\log 2\,P_4(2) - 36Q_2^2 + \big(240\log^2 2 + 596\mathsf{G}\big)Q_2\\ &- 1024\mathsf{G}^2 + 449\,\zeta(3)\log 2 - 80\log^4 2.\end{aligned}}$$

so $C_5(2)$ needs only $\Gamma(\frac{1}{4})$, π, $\log 2$, G, $\zeta(3)$, $F_\pm$ and the five values $\mathcal{Y}_r$.

Three dimensions. At $q = \frac{3}{4}$ the polygamma combinations give $A = 2 + 4\log 2 - \pi$ and $R = \frac{\pi^2}{3} - 4$, so $C_2(3) = (2 + 4\log 2 - \pi)/9$. The higher coefficients follow from the formulas for any dimension with the $d = 3$ values of the masters, which reduce to fixed two-variable hypergeometric functions. No reduction of them to scalar ordinary ${}_pF_q$ values is known or asserted. Evaluated this way, $C_5(3) = -5.850351420751327229217944230497744300110997346378 3\ldots$ to the 50 digits shown.

1.2 A finite special-function representation of $C_6(1)$

The value of $C_6(1)$ in the table is a finite formula in nine special-function values. It comes from an analytic calculation in the package `C6_research_note/`. That calculation takes a reduced analytic expression for the six-loop coefficient as its input and converts each of its sectors into one value at unit argument; Appendix B gives the conversion. With $A = \pi + 4\log 2$, $Q = \Gamma(\frac{1}{4})^4/\pi$ and Catalan's constant G,

$$\begin{aligned}2187\,C_6(1) = E &- \frac{476}{3}\pi^3(32\mathsf{G} + 9\pi^2) - 2176\big[(32\mathsf{G} + 9\pi^2)L_3 + 128\,C_{212}\big]\\ &- 4\pi^2\rho_4\,\mathcal{G}_4 + 192\pi\sqrt{290}\,\mathcal{G}_6 + 16\pi^3\rho_8\,\mathcal{G}_8 - 393\pi\sqrt{89}\,\mathcal{H}_W\\ &+ 24\pi^{3/2}(\rho_7\,\mathcal{G}_7 + 196\pi\sqrt{185}\,\mathcal{H}_7) + 7680\pi^3\,\mathcal{H}_*,\end{aligned} \tag{2}$$

where E is the polynomial

$$\begin{aligned}E = &\frac{Q}{8}\Big[63A^3 + 120A\mathsf{G} + 79A\pi^2 + 2013\pi^3 + 54204\zeta(3)\Big] + Q^2(29\pi - 249A) + \frac{21}{640}A^5 + \frac{7}{64}A^3\pi^2\\ &+ \frac{1149}{64}A^2\pi^3 + \frac{7863}{16}A^2\zeta(3) + 192A\mathsf{G}(16\mathsf{G} + 3\pi^2) + \frac{14853}{640}A\pi^4 - \frac{544}{3}\mathsf{G}\pi^3\\ &+ \frac{37631}{128}\pi^5 + \frac{722279}{48}\pi^2\zeta(3) - \frac{805959}{8}\zeta(5).\end{aligned}$$

The nine values are the following.

- $L_3 = \mathrm{Ls}_3(\frac{\pi}{2})$ and $C_{212} = \mathrm{Cl}_{2,1,2}(\frac{\pi}{2})$, a log-sine integral and a multiple Clausen value:

$$\mathrm{Ls}_3(\sigma) = -\int_0^\sigma \log^2\!\big|2\sin\tfrac{\theta}{2}\big|\,d\theta, \qquad \mathrm{Cl}_{2,1,2}(\sigma) = \sum_{n>m>k\geq 1}\frac{\cos(n\sigma)}{n^2mk^2}.$$

- Four Meijer G values at unit argument, in the convention of [22]:

$$\mathcal{G}_4 = G^{2,3}_{5,5}\left(1\,\middle|\,\begin{matrix}(\frac{1}{2})^{\times 4},\alpha_4\\ 0^{\times 4},\alpha_4\end{matrix}\right), \qquad \mathcal{G}_6 = G^{4,5}_{7,7}\left(1\,\middle|\,\begin{matrix}(\frac{1}{2})^{\times 6},\alpha_6\\ 0^{\times 6},\alpha_6\end{matrix}\right),$$

$$\mathcal{G}_8 = G^{3,8}_{11,11}\left(1\,\middle|\,\begin{matrix}-\frac{1}{4},(\frac{1}{2})^{\times 8},\alpha_{8,+},\alpha_{8,-}\\ 0^{\times 8},\frac{3}{4},\alpha_{8,+},\alpha_{8,-}\end{matrix}\right),$$

$$\mathcal{G}_7 = G^{3,8}_{10,10}\left(1\,\middle|\,\begin{matrix}-\frac{1}{4},(\frac{1}{2})^{\times 7},\eta_{7,+},\eta_{7,-}\\ 0^{\times 7},\frac{3}{4},\eta_{7,+},\eta_{7,-}\end{matrix}\right),$$

with the parameters

$$C_d = 96Q - 42A^2 - 18A\pi - 1227\pi^2, \qquad C_s = 60Q - \tfrac{105}{4}A^2 - 9A\pi - \tfrac{2469}{4}\pi^2,$$
$$\rho_4 = \sqrt{C_d^2 + C_s^2}, \qquad \alpha_4 = \frac{1}{\pi}\arctan\frac{C_s}{C_d}, \qquad \alpha_6 = \frac{1}{\pi}\arctan\frac{11}{13},$$
$$\rho_8 = \sqrt{2177203 - 1028322\sqrt{2}},$$
$$\alpha_{8,\pm} = \frac{1}{2\pi}\Big[\arctan\frac{556}{1233 - 417\sqrt{2}} \pm \mathrm{i}\,\mathrm{arcosh}\,\frac{1233 - 139\sqrt{2}}{\rho_8}\Big],$$
$$q_{7,2} = 259A - 49\pi, \qquad q_{7,1} = 481A - 539\pi, \qquad q_{7,0} = 407A - 1127\pi,$$
$$d_7 = \sqrt{q_{7,1}^2 - 4q_{7,2}q_{7,0}}, \qquad \rho_7 = \sqrt{(q_{7,2} - q_{7,0})^2 + q_{7,1}^2}, \qquad \eta_{7,\pm} = \frac{1}{\pi}\arctan\frac{q_{7,1} \pm d_7}{2q_{7,2}},$$

with principal values throughout.

- Three bivariate H values at unit arguments, $\mathcal{H}_W$, $\mathcal{H}_7$ and $\mathcal{H}_*$, that is, two-fold Mellin–Barnes integrals. The first is

$$\mathcal{H}_W = \frac{1}{(2\pi\mathrm{i})^2}\int_{\frac14-\mathrm{i}\infty}^{\frac14+\mathrm{i}\infty}\int_{\frac18-\mathrm{i}\infty}^{\frac18+\mathrm{i}\infty}\frac{N(s)\,\Gamma(t)\Gamma(t+\frac12)\Gamma(\frac12-t)\Gamma(1-t)}{(\frac12-s-t)^2}\,dt\,ds,$$
$$N(s) = \frac{\Gamma(s)^2\Gamma(\frac12-s)^3}{\Gamma(1-s)^2\Gamma(\frac12+s)\Gamma(s+\alpha_W)\Gamma(1-s-\alpha_W)},$$

with $\alpha_W = \pi^{-1}\arctan\frac58$. The other two are

$$\mathcal{H}_7 = \frac{1}{(2\pi\mathrm{i})^2}\int_{\frac14-\mathrm{i}\infty}^{\frac14+\mathrm{i}\infty}\int_{\frac{1}{16}-\mathrm{i}\infty}^{\frac{1}{16}+\mathrm{i}\infty}\frac{\mathcal{A}_7(s)\cot(2\pi t)}{(\frac12-t)(\frac12-s-t)}\,dt\,ds,$$
$$\mathcal{A}_7(s) = \frac{\Gamma(s)^3\Gamma(\frac12-s)^7\Gamma(\frac54-s)}{\Gamma(1-s)^3\Gamma(-s)\Gamma(\frac14-s)\prod_{b\in\{\beta_7,\bar\beta_7\}}\Gamma(s+b)\Gamma(1-s-b)},$$
$$\mathcal{H}_* = \frac{1}{(2\pi\mathrm{i})^2}\int_{\frac14-\mathrm{i}\infty}^{\frac14+\mathrm{i}\infty}\int_{\frac14-\mathrm{i}\infty}^{\frac14+\mathrm{i}\infty}\frac{\Psi_+(s)\Psi_-(t)}{\Gamma(1-s-t)}\,dt\,ds,$$
$$\Psi_\sigma(s) = \frac{\Gamma(s)^3\Gamma(\frac12-s)^3}{\Gamma(\frac12+s)\prod_{b\in\{\beta_\sigma,\bar\beta_\sigma\}}\Gamma(s+b)\Gamma(1-s-b)},$$

where $\beta_7 = \pi^{-1}\arctan\big((13+\mathrm{i}\sqrt{139})/14\big)$ and, for $\sigma = \pm1$, $\beta_\sigma = \pi^{-1}\arctan\big(B_\sigma/2 + \mathrm{i}\sqrt{C_\sigma - B_\sigma^2/4}\big)$ with $B_\sigma = \frac{21}{20} + \sigma e_*$, $C_\sigma = \frac{\lambda_*}{10} + \sigma e_*\chi_*$, $e_* = \sqrt{40\lambda_* - 279}/20$, $\chi_* = 2(21\lambda_* - 150)/(40\lambda_* - 279)$, and λ_* is the root in $(15, 16)$ of $4\lambda^3 - 112\lambda^2 + 909\lambda - 2250 = 0$. The bar denotes the complex conjugate; all inverse tangents are principal (complex for β_7 and β_σ).

The formula evaluates to

$$C_6(1) = 14331.687032758054320445463255246217118317 32\ldots$$

The three bivariate values carry the three sectors of the reduced expression for which no evaluation in standard constants is known: the wheel sum W_0, the seventh-sector sum M_7 (608) and the pairing P_a of primitive hypergeometric functions (609) (Section 22.9). The count of nine describes how the terms are packaged, not the number of independent constants.

The reduced expression is an input: it is derived neither in this note nor in the package. Two of its Laurent coefficients were missing from its first version and were added by a correction; they contribute +659.279 to $C_6(1)$. The deterministic sum over all six-loop graphs at quadrature degree

22 does not use it and gives 14331.6870328(5). The representation lies 1.03×10^{-8} below its unrounded value, 14331.68703276838, which is 0.021 of its allowance; the comparison tests the reduced expression at the level of 3×10^{-11} relative. This confirms the reduced expression and its correction numerically; it does not prove them. The package encloses an earlier form of the reduced expression, (958), in an interval of width below 1.2×10^{-145} (Section B.8.5). That form reaches (604) only through exact identities of the package that this note does not reproduce, among them two inherited normalizations of the wheel periods; the later forms, reassembled from the same intervals with M_7 enclosed through the tail bound (956), have the same width. The enclosure therefore holds for the reduced expression conditionally on those identities, and it encloses the reduced expression, not $C_6(1)$. An independent review (`LRI_C6_closed_form_review/`) recomputed the input and the nine values with separate codes and found no error.

1.3 Status of the analytic results

The table separates what this note proves from what it takes as an input or confirms only numerically. "Proved here" means that the note contains the complete argument; "conditional" names the unproved input; "numerical" means agreement between independent evaluations, without an interval bound unless one is stated. The numerical coefficients $C_6(d)$ for $d = 2, 3$, $C_7(d)$ and $C_8(d)$ are sampling estimates with the allowances of Section 1 and are not listed.

Statement	Where	Status
One dimension		
Graph weights, subtraction and RG assembly of C_0–C_5	Secs. 2, 3, 13; App. A	proved here; every four- and five-loop residue is listed exactly
$\mathcal{K}$ (76)	Sec. 8.3	proved here
Wheel reduction of $\mathcal{L}$	Sec. 9.2	proved here
$P_5(1)$ in five fixed values (29)	Sec. 10.7	proved here
Graph integration of $C_5(1)$	Sec. 14	numerical: agrees with the closed form to $6 \cdot 10^{-13}$ with the span cutoff and $3 \cdot 10^{-12}$ with the maxgap cutoff, at degree 24; not a bound
$C_6(1) = 14331.68703275805\ldots$ (2)	Secs. 1.2, 22.9; App. B	conditional: the reduced expression (604) is taken as an input and derived neither here nor in `C6_research_note/`; its conversion to the nine values is derived in App. B, from three inherited elliptic-moment evaluations (911)–(913) and published identities; the interval of Section B.8.5 encloses an earlier form of the reduced expression, which reaches (604) only through package identities not reproduced here, among them two inherited wheel normalizations; confirmed numerically by the graph sum to $3 \cdot 10^{-11}$ relative
$C_6(1) = 14331.6870328(5)$, graph sum	Sec. 22	numerical: exact sum over all graphs at quadrature degree 22, with an empirical convergence allowance
Two dimensions		
$Q_2 = \Gamma(\frac{1}{4})^8/(24\pi^4)$	(192), (267)	proved here (complex Selberg integral, gamma reflection)
$Z_2 = 4\mathsf{G}Q_2$ (265)	Sec. 16.7	proved here: a complex Selberg integral for the two-screening integral (264), with the wheel jets (244)

Statement	Where	Status
$P_4(2) = (B_-^2 - B_+^2)/(4\pi^3)$ (333)	Sec. 18.5	proved here: finite-Fourier two-mode theorem and the elliptic moments of Section 9; the first derivation of b cites the mixed-moment integral and the Stieltjes identity of [25]; the second (Section 18.5.2) uses neither
$P_5(2)$, quadratic in X_0, $X_\pm$ (334)	Sec. 18.5	proved here: square-lattice factorization (342), the one-dimensional three-mode function (144) and exact residue algebra
$\mathcal{L}_2 = -2\log 2\, P_4(2)$ (270), for the series (256)	Sec. 18.7	proved here: elliptic lemmas of Section 9 and one contour integral, with the Stieltjes identity of [25] cited. Three further proofs avoid that identity: rim–spoke reflection (Section 18.7.6), Parseval pairings on the unit circle (Section 18.7.8) and the Fourier representation of the two modes (Section 18.7.9); the last two use the half-strip evaluation of the elliptic moments (Section 18.7.7)
$\mathcal{K}_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2$ (271), for the master-path definition (238)	Secs. 18.8, 18.9.8	proved here: position-space proof (rotation lemma, sum of two squares, Laurent extraction) with the one-dimensional lemmas of Section 8.3; a second position-space proof of (377), by Riemann's bilinear relation (Section 18.9.8)
$\mathcal{K}_2$ also equals the series (260) and the spectral form (248)	Secs. 18.9.6, 18.9.7	proved here: regularization in the rim exponent, residues on the square lattice and periodization, with the one-dimensional lemmas of Section 8; for the series also the per-spin residue identity, proved at $d = 2$ with explicit arc bounds
Mixed-trace step (249), for $0 < \tau < \frac{1}{2}$	Sec. 18.9.5	proved here at $d = 2$, in the full-Mellin-line reading: the closed-form trace (390) with (377); at other d it is input (M) of Section 21.1
Closed form of $C_5(2)$ (272)	Sec. 17	proved here: exact algebra from the rows above and (223)
General dimension		
C_0–$C_5(d)$ (222), (223)	Sec. 16.3	proved by exact computation: exact symbolic residues of the 26 four-loop and 129 five-loop classes (certificate and exact audit in `lri_general_special_functions_update/`)
Spectral and series forms of Q_d, P_4, $\mathcal{L}_d$, Z_d (243), (246), (255)–(257)	Secs. 16.4.1, 16.6	proved here
Spectral and series forms of $\mathcal{K}_d$ (248), (260)	Secs. 16.4.2, 21.12	at $d = 2$ proved here (Section 18.9); at other d conditional on input (M) and on extracting $[\tau^2]$ under the trace
Finite-Fourier theorem of the primitive sector and finite evaluation rules for Q_d, $P_4(d)$, $P_5(d)$	Sec. 18	proved here for $\frac{1}{2} < q < 1$, then by continuation

Statement	Where	Status
Double residue series of $P_5(d)$ (281)	Sec. 17.4; checks in 17.7	proved here for $\frac{1}{2} < q < 1$; the extension by uniform convergence to $\frac{2}{5} < d < 4$ uses convergence orders whose elementary estimates are stated but not written out; checked in double precision at $d = 2, \frac{13}{5}, 3$
Finite Horn representation of $\mathcal{L}_d$, Z_d, $\mathcal{K}_d$ (298)	Sec. 17	proved here for $\frac{2}{3} < d < 2$, then by continuation of the complete combination
Finite Fourier theorem at a general rim exponent, top-mode theorem and its product form, residue functional	Secs. 20.1.1, 20.1.2, 20.1.3; status in 20.10	proved here, with complete proofs in Sec. 20.1. Reviewed independently in the form before the merge (`LRI_C5_general_d/review/VERIFICATION.md`): the finite Fourier theorem, the top-mode theorem in its wheel-only form, the residue functional and the block decomposition; the product form separately (`review/lemmaG-proof`). Not reviewed: the explicit constants of (465), (466), (467) (Sec. 21.6) and the contour of the residue-functional proof, changed at the merge
Sample formulas for $\mathcal{L}_d$, Z_d and the reduced $C_5(d)$	Secs. 20.2–20.7; checks in 20.9	sample formulas proved (Sections 20.3 and 20.4); reduced $C_5(d)$ derived (exact algebra, rederived symbolically in a separate review), conditional on (M), and on (H) on the jet circle, unless $\mathcal{K}_d$ comes from (588); confirmed numerically to about 10^{-48}
Constant (560) of the two-wheel sample and bilateral identity (561)	Sec. 21.2; status in 21.12	proved here from the case $r = 2$ of the top-mode theorem, the residue functional and the two-wheel convolution; not independently reviewed; constancy and value confirmed numerically
Growth theorem for the R family of $\mathcal{K}_d$	Secs. 21.4, 21.6; status in 21.12	proved here; parts (a) (sufficiency), (b) and (c) are the case $r = 3$ of the proof of the top-mode theorem, which was reviewed independently (Section 20.10); the sharpness of $\mathbb{H}_q$ and the explicit constants not independently reviewed
Sample formulas for $\mathcal{K}_d$	Secs. 21.7, 21.8; status in 21.12	given (M) and (J), proved on $\mathbb{H}_q$ under the conditions of Section 21.12; the residue functional and the three-node rule not independently reviewed; at $d = 2$, (M) is proved for real $0 < \tau < \frac{1}{2}$ (Section 18.9.5; complex τ is not addressed there) and (J) in Section 18.8.4; on the jet circle also conditional on (H), at every d
Barnes representation of $\mathcal{K}_d$ (588)	Sec. 21.10	derived here without (M), (J), (H); the statement re-derived by separate AI sessions; the printed derivation not reviewed
Three dimensions		
Determinant and rank-two identification of Q_3, $P_4(3)$, $P_5(3)$, and the complex-node variant (438)	Secs. 19.4, 19.5, 19.6	proved here

Statement	Where	Status
Specialization of $\mathcal{L}_d$ and Z_d to $d = 3$	Sec. 20.6; conditions in 20.10	derived here: exact algebra on two identities proved in Section 20.6 (the half-shift proof at a general rim exponent, with Dixon's and Dougall's sums); the identities re-derived by separate AI sessions, the printed derivation not reviewed; confirmed numerically to 10^{-51}
Values of Q_3, $P_4(3)$, $P_5(3)$ and $C_5(3)$	Sec. 19.10	numerical: independent evaluations agree to about 10^{-79} (Q_3), $2 \cdot 10^{-86}$ ($P_4(3)$) and $2 \cdot 10^{-53}$ ($P_5(3)$), which changes $C_5(3)$ by less than 10^{-54}; not interval bounds. $C_5(3)$ does not depend on (M), through (588)
No ${}_3F_2(1)$ or ${}_4F_3(1)$ relation for $\mathfrak{H}_3(0)$ of Q_3 (405) in a bounded search; no relation for $\mathsf{r}_3(3)$, $\mathcal{K}_3$	Secs. 19.10, 20.8, 21.9	numerical only: bounded catalogue and PSLQ searches in stated bases, not independence proofs; no search is stated for $P_4(3)$, $P_5(3)$
Bulk one-point coefficient (Part V)		
$a^{(2)}_{\phi^2}$, $a^{(3)}_{\phi^2}$	Sec. 34.6.1	derived there; agree with [8]
$a^{(4)}_{\phi^2}$–$a^{(6)}_{\phi^2}$ and the Gamma-factor form (820)	Secs. 34.3, 34.6	calculation supplied by the author (16 October 2026); checks internal to the master framework: the five- and six-vertex graphs are reconstructed from master jets and checked with other regulator directions, not integrated one by one; the crossed kite rests on the package's Mellin–Barnes representation
b_5 and c_5 separately (834), (835)	Secs. 34.5, 34.12	$3c_5 - b_5$ equals (229) exactly; the split agrees numerically with graph sums at $d = 1$ and with the seventh- and eighth-order productions at $d = 2, 3$
Map to $C_j(d)$ (829)	Sec. 34.4	exact algebra between two calculations with shared inputs; validates neither
Exact $a^{(n)}_{\phi^2}(1)$, $a^{(n)}_{\phi^2}(2)$ and the closed $d = 2$ map (862)	Secs. 34.9, 34.8	exact algebra on the proved $d = 1$, $d = 2$ masters, conditional on H_4 of the supplied calculation
Comparison with the numerical bootstrap	Sec. 34.11	exploratory: readings of published figures, no bootstrap problem solved; the spread of prescriptions is not an error bar

1.4 Notation and the logical route

In the one-dimensional coordinate prescription the covariance is $\langle\phi(x)\phi(0)\rangle_0 = |x|^{-(1-\epsilon)/2}$ and the interaction is $g\phi^4/4!$. The graph weights and MS recursion are fixed first. The one-dimensional master reductions then give the full analytic coefficient; the residue certificate of Appendix A lists the residue of every four- and five-loop class. The general-dimensional sections replace the one-dimensional integrals by normalized Schwinger and spectral integrals, derive the master jets and provide their finite-function representations; Sections 20 and 21 shorten those of the three nonprimitive masters to a few two-variable hypergeometric samples and their low-order parameter jets. The numerical sections retain a separate, regulator-dependent finite-scheme formulation and its complete error accounting. Part V uses the same graphs, subtractions and masters for a second observable, the bulk one-point coefficient. Appendix A lists the residue certificate, and Appendix B derives the special-function representation of $C_6(1)$ quoted in Section 1.2.

Symbol	Meaning
$C_n(d)$	Physical coefficient of ϵ^n in $\Delta_{\mathcal{E}}$.
b_L, c_L	L-loop beta and mass-anomalous-dimension coefficients; lowercase c_L is not C_L.
z	Analytic regulator, set equal to physical ϵ only after the required Laurent or Taylor data have been retained.
d, q	Spatial dimension and $q = d/4$ in the general-dimensional sections. In the five-index one-dimensional master $\mathcal{J}(a,b,c,d,e)$, d is only an edge exponent.
$\widehat{b}_L(z), \widehat{c}_L(z)$	Finite-scheme RG functions, whose z dependence cannot be dropped before fixed-point extraction.
$P_3(d), Q_d$	Primitive normalization $P_3(d) = 3Q_d$. At $d = 1$, $Q = \Gamma(1/4)^4/\pi$.
$\mathcal{K}_d, \mathcal{L}_d, Z_d$	Explicit nonprimitive master data; $\mathcal{K}_d$ is not an elliptic integral. Reduced at generic d in Sections 20 and 21.
$K(m), \mathbf{K}(k)$	Complete elliptic integrals in parameter and modulus conventions: $\mathbf{K}(k) = K(k^2)$. A complementary argument is not a derivative.
$\psi_m(z)$	$d^{m+1} \log \Gamma(z)/dz^{m+1}$; ψ_0 is digamma.
G	Catalan's constant; distinct from graph symbols and matrix-argument hypergeometric functions.
a_{ϕ^2}, $a^{(n)}_{\phi^2}(d)$, $a^{(n)}$	The bulk one-point coefficient of $:\Phi^2:$ and its coefficient of ϵ^n (Part V); not the squared defect OPE coefficient $\lambda^2_{\phi\phi\mathcal{E}}$.
(M)	Input of Section 21.1: the mixed trace $J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_d$ as an identity of meromorphic functions of τ, with the trace (554) read on the full Mellin line and continued from its strip of convergence; proved at $d = 2$ for real $0 < \tau < \frac{1}{2}$ (Section 18.9.5).
(J)	Input of Section 21.1: the jet $J_t - J_t^\Gamma = \mathcal{K}_d t^2 + O(t^3)$ on the master path (238), whose orders t^0, t^1 are used explicitly; at $d = 2$ its low orders are rederived in Section 18.8.4.
(H)	Continuation hypothesis of Section 21.8: the continued three-node samples (581), (585) are jointly holomorphic in (t, τ) on a domain that contains the disk bounded by the jet circle and meets $\mathbb{H}_q$, off a closed discrete set.
$\mathbb{H}_q$	The half-plane $\operatorname{Re}\tau > \tau_*(q) = \max(0, 2q-1, \frac{1}{2}-2q)$ (571), on which the lattice sum of the R family converges absolutely (Section 21.4).
jet circle	The circle $\lvert\tau\rvert = r_\tau$ around the pole $\tau = 0$ on which the τ-jet of (586) is taken as a Cauchy integral; it does not lie in $\mathbb{H}_q$ (Section 21.8).
$[\tau^2]$ under the trace	The interchange of the Taylor coefficient $[\tau^2]$ at $\tau = 0$ with the trace, $[\tau^2]\langle t_\ell^2(R_\tau - S_\tau)\rangle_d = \langle t_\ell^2 E_2\rangle_d$, used to derive (248) from (238) (Sections 16.4.2, 21.3); justified only at $d = 2$ (Section 18.9.2).
Part III families	
b, $b_{\max}$, b_{spec}	The rim exponent of a balanced wheel with spoke exponent $d - b$ (242), complex in general; the critical wheel has $b = d/2$. Sections 20 and 21 write $\sigma = b/2$ and $g = (d-b)/2$ (g_b in Section 21). The spectral integral (460) represents W_r for $0 < \operatorname{Re} b < b_{\text{spec}} = d - \max(d, 1)/r$, and the finite Fourier theorem of Section 20.1.1 holds for $0 < \operatorname{Re} b < b_{\max}(r, d)$ (463), with $b_{\max} \le b_{\text{spec}}$. Not b_L, b_ℓ or the mode b of (340).

Symbol	Meaning
$W_r(d;b)$, $t_{\ell,b}$, N_b, Ξ_r	The r-wheel $W_r(d;b) = \langle t_{\ell,b}^r \rangle_d$ (242), with $t_{\ell,b}(\nu) = N_b\, g_\ell(\mathrm{i}\nu;b)$ for real b and the normalization N_b of (460); at $b = d/2$, $t_{\ell,b} = t_\ell$ and $N_{d/2} = \Gamma(2q)$. $\Xi_r(t)$ (304) and $\Xi_r(t;b)$ (462) are the periodized samples, and Ξ_2 (Sections 20.9, 21.2) is the case $r = 2$; Ξ_{R} (570) belongs to the R family, and Ξ_Γ (Section 24.1) is unrelated. The rim sample $\mathfrak{t}_3(b)$, first used in Section 19, is defined in (495).
$T_{\ell n}$, $T_{mn}(x)$	$T_{\ell n}$ (251), with $S_{\ell n}$, $a_{\ell n}$, $b_{\ell n}$ there (not S_τ, a_ℓ, b_ℓ), and $T_{\ell n}(b)$ at a rim exponent b (490). $T_{mn}(x) = c_m(x)c_n(x)$ (345) has the first index $m = \ell + n$, so that $T_{mn}(\frac{1}{2}) = T_{\ell n}$ at $d = 2$; $T_\ell(z)$, with one index, is a product kernel of Section 20.1.2.
J_t, $J_{2\tau}$, $C(\tau)$, R_τ, S_τ	The master path $J_t = J_d(\frac{d}{2}, \frac{d}{2}, \frac{d}{2}, d-t, \frac{d}{2})$ of the jet (237), its Gamma part J_t^Γ and $\mathcal{K}_d = [t^2](J_t - J_t^\Gamma)$ (238). With $t = 2\tau$, $C(\tau) = \Gamma(2q)\Gamma(\tau)/\Gamma(2q-\tau)$ and the kernels $R_\tau(s)$, $S_\tau(s)$ (249) form the mixed trace of input (M); the R and S families of $\mathcal{K}_d$ are those of R_τ and S_τ. Used in Sections 16.4.2, 18.8, 18.9 and 21. $C(\tau)$ is not a coefficient C_j, R_τ is not R, and S_τ is not $S_{\ell n}$.
μ_1, μ_2, η_m, θ_m, λ_m, γ_C	Polygamma combinations at the arguments $a_{\ell n}$ of (251): λ_m (252) and η_m, θ_m, μ_1, μ_2 (253), used in the series of Section 16.5. The λ_j of (274) and the η_r, θ_r, μ_1, μ_2 of (275) are the same combinations, with $m = \ell + n$; Sections 18.7.1, 20.5 and 20.4.4 continue them in the rim exponent. $\gamma_C(a) = \Gamma(a)/\Gamma(1-a)$ (Sections 15, 16.7, 18.7.6, 20.6).
$\mathcal{N}_d$, $\mathcal{N}_x = N_x$	$\mathcal{N}_d = \mathcal{K}_d + \frac{3}{8}\mathcal{L}_d - \frac{149}{32}Z_d$ (403), the only combination in which the nonprimitive masters enter $C_5(d)$. $N_x = \Gamma(1-x)/\Gamma(x)$ (345) is written $\mathcal{N}_x$ in Sections 20.1.3 and 20.6: one object with two names, unrelated to $\mathcal{N}_d$; it equals N_b at $d = 2$, $x = \frac{b}{2}$.
Symbols with more than one meaning; each is defined where it is used	
h	$h_d = \Gamma(d/2)/\pi^{d/2}$, and an $\Im\psi_0$ combination in the formula for $\mathcal{K}_d$ (both Section 1.1); the propagator exponent $(1-\epsilon)/2$ or $(1-z)/2$ (Sections 2, 3); the Gamma ratios $h_b(z)$ (Section 20.6) and $h_x(\sigma)$ (Section 18.7.9); further local meanings in Sections 17.4, 18.9.7, 19, 20.1, 22.9, 23, 25, 28 and 32.7.
p	an $\Re\psi_0$ combination in the formula for $\mathcal{K}_d$, and $p = d/2 - 1$ in the five-loop spectral formula (both Section 1.1); $p = D/2 - 1$ in Section 10; local regulators, momenta, edge powers, exponent sums, pole orders and auxiliary polynomials elsewhere, for example in Sections 18.7.6, 18.9.4, 19.5, 20.1.4 and 28.
B, B, $\mathcal{B}$, $\mathfrak{B}$	Euler's $B(u,v)$; $\mathsf{B}(u,v) = \int_{\mathbb{R}} \lvert x\rvert^{-u}\lvert 1-x\rvert^{-v}\,dx$ (Section 7); $\mathcal{B}(g)$ and $B(g)$, the same reduced beta function (Sections 2, 22, 23.1); $\mathcal{B}_L(z)$, the mass-insertion amplitude (Section 3); $B_\pm$, hypergeometric values (Section 1.1), not B_σ, $\sigma = \pm 1$, of Section 1.2; B_d (230); the residue weights $B_{r,k}(\sigma)$ (Section 20.1.3); the Bernoulli polynomials $B_n(x)$ (Sections 19.10, 21.8); $\mathcal{B}(x)$, a mode amplitude (349), and $\mathcal{B}_\tau$ (Section 18.9.5); local B_k, $B_i(n)$ (Sections 18.9.2, 18.9.7); $\mathfrak{B}_{\mathrm{kn}}$ (Section 23.1), unrelated to $\mathfrak{B}_c$ and the transforms $\mathfrak{B}_{1/4}$, $\mathfrak{B}_{1/8}$ of Appendix B.

Symbol	Meaning
H, H_7, $\mathcal{H}_7$, $\mathfrak{H}_7$	$H = 2\pi^2/3 + 8\mathsf{G}$ (Sections 1.1, 4); H_7, a ${}_7F_6$ value (Section 9); $\mathcal{H}_7$, a bivariate H function, with $\mathcal{H}_W$, $\mathcal{H}_*$ (Section 1.2); $\mathfrak{H}_7$, a two-variable Horn function (Section 19), with the Horn series $\mathfrak{H}[\cdots]$ (290) and the values $\mathfrak{H}_+$, $\mathfrak{H}_c$ (Section 20.4.3); (H), the continuation hypothesis (first block), and the half-plane $\mathbb{H}_q$; $H(a;v)$ (274); $\mathcal{H}_x$ (348); H_6^{χ} and $H_6(d)$, bare channels, not ${}_6F_5$ values (Sections 22, 23.1); the harmonic numbers H_n (Section 22.9, Appendix B); the Hardy space H^2 (Section 18.7.8); $H(\tau)$ (Section 18.9.7); H_{R}, H_{M}, H_{L} (Section 21.7); further local meanings in Sections 11, 14, 23 and 27.
G_4, $\mathcal{G}_4$, $\mathcal{G}_L$, G_n, G	$G_4 = \Gamma(\frac{1}{4})^4$ (Section 9); $\mathcal{G}_4$, $\mathcal{G}_6$, $\mathcal{G}_7$, $\mathcal{G}_8$, Meijer G values (Section 1.2); $\mathcal{G}_L(z)$, a Gamma prefactor (Section 25); $\mathcal{G}_x(s)$, a Gamma product (Sections 18.7.6, 18.7.9), and $\mathcal{G}_\alpha(n)$ (584); $G_n(z)$, coupling counterterms (Section 3); G_0, G_1, G_2 (236); $\mathsf{G} = \mathsf{G}$, Catalan's constant; $\mathfrak{G}(\tau)$, with $\mathfrak{F}(\tau)$ (75), not the graph set $\mathfrak{G}_L$ (Section 3); local $G_{\ell k}$, $G(\beta)$, $G_\ell(s)$ (Sections 17.4, 18.7.7, 18.9.3).
Q, Q_d, $\mathcal{Q}_3$	$Q = \Gamma(\frac{1}{4})^4/\pi = Q_1$; Q_d, the three-loop period, written Q in the general-dimensional formulas of Section 1.1; $Q = Q_3$ in Section 24; $\mathcal{Q}_3 = [\tau^3]\mathcal{Q}(\tau)^2$ (Section 18.8).
R, U, T, $\mathcal{T}$	polygamma combinations (R, U) and values (T, U) in Section 1.1; integrals or functions in Sections 5, 7 and 9, those of Section 9 reused in Section 18.7; U is also the allowance of Section 22.7; $\mathcal{R}$ has different meanings in Sections 3 and 24, and $\mathcal{R}_{\ell n}$, $\mathcal{R}_+$, $\mathcal{R}_{\mathrm{m}}$ are lattice sums (Sections 16.5, 20.3); the R family of $\mathcal{K}_d$ is that of R_τ (Part III families); R, the number of replicates (Sections 26.11, 31.7); U_G, U_0, U_F, Symanzik polynomials, and R, a parameter sum (Sections 23, 25, 28); $T_\tau(s)$, $T_{\mathsf{h}}(\eta)$ (Sections 18.9.3, 18.9.4) and T_{R}, T_{L} (Sections 21.4, 21.6); $\mathcal{T}_d$, the corrected tetrahedron (Section 15), against $\mathcal{T} = \sum_n a_n^4 h_n(8h_n^2 + \delta_n)$ in the wheel identities of Section B.4.
E	$\mathcal{E} = [\phi^2]$; E_2 in the formula for $\mathcal{K}_d$ (Section 1.1), unrelated to the polynomials E_m of Section 18; E, a polynomial in constants (Section 1.2); $E(a,b)$, the Euler integral of K (Sections 8, 18.9.4); $E_{\mathfrak{k}}$, $E^\sharp$ (Section 18.7.6); E_0, E_0^*, E_*^* (Sections 19, 19.5); E, the number of edges of a graph (Section 26); the line sets E_B, $E(W)$ (Sections 28, 32.7).
Z	$\mathcal{Z}_g$, Z_r and the counterterm coefficients $Z_n(z)$ (Section 3); Z_d, the wheel master, with $Z_2 = 4\mathsf{G}Q_2$ (265).
D, D	$\mathsf{D} = \pi$ (Section 1.1); D, the dimension in Section 10; $D = \pi\cot\pi q$ (459), with S, C there; D_4, D_5 and in general D_L, the combinations $3c_L - b_L$ (Sections 2, 15), with D_5^{prim} (Section 16.3). Not $3c_L - b_L$: D_3, D_5, polylogarithm values (Sections 22.9, B.7), and D_4, D_6, wheel sums (Section B.4); D_2, the two-loop kernel of Section 23.8; $\widetilde{D}_j$ (274); D_s, D_r (Sections 16.3, 16.4.1); the sine products $D_q(t)$, $D_b(t)$, D_T, D_{R} (Sections 18, 20.1, 20.1.2, 21.4); $\mathfrak{D}_L$ (Section 20.2); $D(\ell)$ (Section 18.9.7); local $\mathcal{D}(\tau)$, $\mathcal{D}_2$, $\mathcal{D}_3$ (Section 18.9.2), D_{F} (Section 18.7.9) and $\mathcal{D}_4(y)$ (Section B.8.2).
$\mathcal{K}$, K	$\mathcal{K}$ and $\mathcal{K}_d$, the nonprimitive master; PP and, in Sections 23–24, $\mathcal{K}$, the pole-part projector; K_G, the forest kernels of Sections 25–28; $K(m)$, $\mathbf{K}(k)$, elliptic integrals; $\boldsymbol{K}_{\leq 4}$, the loop one-to-four vector of Section 27; $K_\ell(s)$, the five-loop primitive kernel (303); K_5, the complete graph (Sections 16.3, 28.6); K_n, a Pochhammer ratio (Section 18.7.6) and the sums K_n (Section B.1); $\mathfrak{K}$, a compact set (Section 21.6).

Symbol	Meaning
$G_{L,d}$	the Gamma prefactor of Sections 25 and 26; its two normalizations differ by a constant, so the log-jets $a_{L,j}$ and ℓ_j agree for $j \geq 1$.
r_j, r_k	r_j, series coefficients (Section 16.5); r_k, s_k, Laurent coefficients (555), with $\mathsf{r}_{-1}, \dots, \mathsf{r}_2$ in (564) and $\mathsf{r}_3 = [\tau^3]\langle t_\ell^2 R_\tau \rangle_d$ (Section 21); $r_3 = 2Q_2$, the primitive three-loop residue (Section 23); r_1, r_2 (841), the coefficients of the ratio $b_0^{\Phi,-}/b_0^{\Phi,+}$ of bulk-to-defect coefficients (Section 34.6.1), not the r_1, r_2 of (564).
w, $\mathfrak{w}$	w_1, w_2, spectral weights (Sections 1.1, 20.4.4); the graph weights $w_4 = w_A$ and $w_2 = w_B$ (Sections 3, 22); $w_H = 3w_B - w_A$; $w(k) = \log \frac{k}{1-k^2}$ (Sections 9, 18.7.7); $w_{\mathcal{K}}$, the weight of (248) at $d = 2$ (Section 18.9.1); $w_j(\tau)$, the node weights (580); $\mathfrak{w}_{r,h}$, spin-tail weights (Section 19.10), and $\mathfrak{w}_{r,j}(b)$, node weights (484).
z, z_1, z_2	z, the analytic regulator; z_1, z_2, deformation coordinates (Sections 7, 16.3) or Mellin–Barnes variables (Section 21.10); z is also a local complex variable in proofs where the regulator does not enter or is called τ (Sections 18.7.6, 18.7.8, 18.9.2, 20.1), and $z_\ell = \frac{\ell}{2} + \mathrm{i}\nu$ (Section 20.1.4).
D_G, ϖ	local meanings in Sections 23 and 25 (D_G); Mellin–Barnes weights (Section 21.10), hypergeometric values (Section 22.9) and Binet's function $\varpi(t)$ (Section 18.7.6) (ϖ).
MOM	the finite zero-momentum scheme of Section 25.
a, $\mathfrak{a}$	$a = \log 2$ and $a_\ell = \frac{\ell}{2} + \frac{q}{2}$ (Section 1.1), with $a_\ell = \frac{\ell+\sigma}{2}$ at a rim exponent (460); the finite Laurent coefficients $a_{L,k}$ (Sections 3, 15); the series terms $a_n(t)$ (Section 8); the log-jets $a_{L,j}$ (Sections 25, 26); a_n in Section 22.9; $a^{(n)}$ and the amplitudes $\mathsf{a}_n(\epsilon)$ (Part V); $a = 8N_{\mathrm{e}}/\pi$, a mode at $d = 2$ (340); $a = a_\ell$ at fixed spin (Section 18.9.1); $a = \frac{1}{2} + \frac{z}{4}$ (Sections 23, 28, 31.7); the prefactors a_r^0, a_r^+, a_r^c (Sections 18.4, 20.4.3); a_G (Sections 23, 28.6). $\mathfrak{a}_3$, $\mathfrak{b}_3$, $\mathfrak{a}_4$, $\mathfrak{b}_4$, mode amplitudes (Section 19), continued in b by (543); $\mathfrak{a}(x)$, $\mathfrak{b}(x)$ and $\mathfrak{s}(x)$ (Section 18.7.6) are not these or the Horn values $\mathfrak{s}_0$, $\mathfrak{s}_{1/2}$ (Sections 20.2, 20.3).
A, $\mathcal{A}$	A and A_d (176), so $A_2 = 4\log 2$ and $A_3 = 2 + 4\log 2 - \pi$; $A = \mathsf{A} = A_1$ in Sections 1.1 and 1.2; the Horn blocks A_r^0, A_r^+ (321) (Sections 20.1.3, 20.4.3); local A_k, B_k and $A_i(m)$ (Sections 18.9.2, 18.9.7), not A_d. $\mathcal{A}_L(z)$, $\mathcal{A}_L(d, z)$, the bare amplitudes (Sections 3, 15), and $\mathcal{A}_L^\chi$ (Section 22); $\mathcal{A}_{\mathrm{s}}(t)$ (Section 10); $\mathcal{A}_{\ell k}$ (Sections 18, 17.4); $\mathcal{A}(x)$, a mode amplitude (349); $\mathcal{A}_\tau$ (Section 18.9.5); $\mathcal{A}(h, z)$ (691) and $\mathcal{A}(x, z)$ (Section 27); $\mathcal{A}_7(s)$ (Section 1.2).
b, c	b_L, c_L (first block), and the rim exponent b (Part III families); $b = 8M_{\mathrm{e}}/\pi$, a mode at $d = 2$ (340); $b = b_\ell$ at fixed spin (Section 18.9.1); $b = \frac{1}{2} + \frac{3z}{4}$ (Section 31.7); $b_0^{\Phi,\pm}$, bulk-to-defect coefficients (Section 34.6.1); c_j (274), so the c_5, c_6 of (280) are not loop coefficients; $c_n = (\frac{1}{2})_n/n!$ (Sections 18.5.2, 18.7, 18.9.7); $c_k(x)$ (345); c_* (247); $c_\tau = \cos \pi\tau$ (Sections 18.8, 18.9); the Fourier coefficients $c_{r,k}$ (Sections 18, 20.1.1) and $c_k(\tau)$ (Section 21.4); generic constants c, c_0 (Sections 18.9.2, 21.6).

Symbol	Meaning
C, S	$C_n(d)$ (first block); $C(\tau)$ (Part III families); $S = \sin \pi q$, $C = \cos \pi q$ (459) (Sections 20.2–20.7); the Horn blocks $C_r(t)$ (321), for example $C_3(t)$, not $C_3(d)$ (Sections 20.1.3, 20.4.3); (C), the condition (469); generic constants C, C_0, C_r, C_q, C_K, $C_{\mathfrak{R}}$ in bounds (Sections 18.7.6, 18.9.2, 20.1, 20.4.1, 21.6); C_d, C_s, C_σ, C_{212} of Section 1.2; the zonal polynomials $C_\kappa^{(2)}$ (Sections 19.6, 19.10); C (Part V); $S_{\ell n}$ (251), the periodized sums $S_r(t)$, S_{R} (Sections 18, 20.1, 21.4) and the sectors S_{pol}, S_6, S_8 (Section 22.9).
F	$F_4(v)$, $F_2(v)$, the bare amplitudes (Section 3); $F_1(d,z)$, $F_2(d,z)$ of (199) (Sections 15, 28, 31.7); the polygamma polynomials F_3, F_4, $F_{3,1}$, $F_{4,1}$ (254) (Sections 16.5, 20.3, 20.4.4); $F(\tau)$ and its coefficients F_k (555), so that $F_3 = \mathsf{r}_3 - \mathsf{s}_3$ in Section 21.3 is not the polynomial F_3; $F(s)$ (263); $F_{r,j}$ (Section 17); $F_\pm$, ${}_4F_3$ values (Section 1.1); local F_{L}, F_{R} (Section 18.7.6) and $F(\beta)$, $G(\beta)$, $L(\beta)$ (Section 18.7.7).
g	the coupling (Section 2); the Barnes kernels $g_\ell(s)$, $g_\ell(s;b)$ (Sections 16–21) and $\mathsf{g}_\ell(s)$ (Section 18.7.6); $g = (d-b)/2$, the spoke half-exponent (Section 20), written g_b in Section 21; the free exponents g_i of f_{g_i} (Section 20.1.2); $g(k) = \mathbf{K}(k')$ (Sections 9, 18.7.7); $g(m,n)$ (Section 18.9.7); g_{alt} (Section 34.5).
I	$I_m(z)$ (Section 3); the Euler moments $I_1, \ldots, I_6$ (Sections 8, 18.8, 18.9.4, 18.9.8); I_1, I_3, I_5, I_7 (Section 10); the pairings I_{jk} (Section 22.9, Appendix B); identity matrices I_2, I_L (Sections 19.6, 19.9, 23); I, the number of model terms (Section 21.8); local I_ℓ, I_{L}, I_{R} and intervals I, I_{F} (Sections 20.1.2, 18.7.6, 18.7, 18.7.9).
J, J	$\mathcal{J}_m(z)$ (Section 3) and the master $\mathcal{J}(a,b,c,d,e)$ (first block); the jet J_d (237) and the master path J_t, $J_{2\tau}$ (Part III families); $J_0(q)$ (Section 15); J_1, J_3, J_5 (Section 10); $J_r(a;v)$ (274); the bare channels J_5^χ, $J_5(d)$ (Sections 22, 23.1); the Jacobians J, J_L, $J_{f,L}$ (Sections 26.13, 28.7); $J(\lvert i-j\rvert)$, the lattice coupling (Section 1); $\mathsf{J}_0(x)$, $\mathsf{J}_1(x)$, Barnes integrals (Section 18.7.6), not J_d of Section 34.6.1 at $d = 0, 1$.
P	the periods $P_3(d) = 3Q_d$, $P_4(d)$, $P_5(d)$ (first block); P_W, a wheel period (Sections 12, 16.4.1); the lattice weights $P_\pm$, P_{m} (Section 20.3) and P_{R}, P_{M}, P_{L} (Sections 21.6, 21.7); P_{10}, the retained ten primitive six-loop kernels (Sections 26.7, 26.12–26.14); P_a, a pairing (609); local P_f, P_{h} (Section 18.9.4).
q	$q = d/4$ (first block); the nome $q = e^{-\pi \mathbf{K}(k')/\mathbf{K}(k)}$ (Section 18.7.9); the coefficients q_n of $\mathsf{q}(z)$ (Section 18.7.8) and the sequence q_n of Section B.4; the parameters $q_{7,j}$ of Section 1.2.
s, ς	the long-range exponent s (Section 1); the Mellin variable s (Sections 16.4.2, 21.1); $s_b = \sin \frac{\pi b}{2}$, $\varsigma_b = \cos \frac{\pi b}{2}$ (Section 20); $s_\tau = \sin \pi\tau$ (Sections 18.8, 18.9); ς_f, sign vectors (Section 18.8); s_{K}, a parametric excess (Section 19.9).
s_G	$s_G(x) = \frac{1}{4}\sum_e \log x_e - \frac{L}{2}\log S(x)$, the regulator logarithm (Section 26); $s_G = \sum_i e_i(e_i - 1)$, the leg count of a graph (Section 28).
t	the path parameter of J_t (238), not $t = d - e$ of (233); the periodization variable of $S_r(t)$, $\Xi_r(t)$ (Sections 18, 20, 21.4); the wheel eigenvalues t_ℓ, $t_{\ell,b}$ (Section 1.1, Part III families); a direction parameter in Section 34.6.1.

Symbol	Meaning
u, v, σ	u_j, the coefficients of g_* (Section 2); u, v, σ with τ (Section 8) or ϱ (Sections 18.8.4, 18.9.1), Taylor data of $\mathfrak{F}$, $\mathfrak{G}$; $u(b)$, $v(b)$, the weights of (496), with u_0, v_0 (Section 20.4.5); local u, v (Section 20.6); u_h (314); $v_m(x)$ (Section 18.7.9). $\sigma_i = \pm 1$, the spins (Section 1), and $\sigma = \pm 1$ (Section 1.2); $\sigma = b/2$ (Sections 20, 21), a free offset in Section 20.1.2; $\sigma = \operatorname{Im} t$ (Section 19.5); $\widehat{\sigma}$, standard errors (Part IV); further local meanings in Sections 7, 10, 15, 16.3, 16.7 and 23.
V, X, $\mathcal{X}$, X	V, X, polygamma constants (221); X_0, $X_\pm$, the one-dimensional values of Section 1.1, against X_0, X_h at $d = 3$ (421), with $X_0 = \Xi_4(0; \frac{3}{2})$ (Section 20.6); $\mathsf{X}_r(t)$ (Section 19); $\mathcal{X}_d(\tau)$ (555), with $\mathcal{X}_2(\tau)$ (Section 18.9.1); the samples $\mathcal{X}_0$, $\mathcal{X}_{1/2}$, $\mathcal{X}_T$ (Sections 20.2, 20.3), where the subscript labels a node, not a dimension; the model factors $X(m)$ (Section 21.8); V, V_0, elliptic moments (Sections 9, 18.7.7); $V_r(t)$, $V_r^{(0)}(t)$ (Sections 19, 19.5); the variances V_C, V_{pf}, V_Δ, $V_{f\Delta}$ (Section 28) and $\widehat{V}_G$, $\widehat{V}_{\mathrm{bench}}$ (Section 26.11); further local meanings in Sections 11, 20.1, 23 and 34.2.
x_0, m_0	x_0, a vertex fixed at the origin (Sections 3, 9, 10, 34.6.1); x_0, a Gamma ratio (583); m_0, an integer cutoff (Section 21.6), and m_0 of Section B.2.2; $\mathfrak{m}_0$, a moment (Section 18.7.9).
C_d, C_s, C_σ, d_7, $q_{7,j}$, ρ_4, ρ_7, ρ_8; W_0, M_7, P_a	parameters of the nine values of Section 1.2, used again in Section 22.9 and Appendix B: C_d, C_s, C_σ, B_σ, d_7, $q_{7,j}$, ρ_4, ρ_7, ρ_8, $\eta_{7,\pm}$ and λ_* are not $C_n(d)$, $B_\pm$, d_ℓ, q, $\rho_n^{(d)}$, η_m or λ_m. Section 1.2 uses the wheel sum W_0, the seventh-sector sum M_7 and the pairing P_a before their definitions (608), (609); W_0 is not a wheel W_r.
β	$\beta_{\mathrm{RG}}(g)$ (Section 2); α, β, γ, constants of the five-loop primitive (Sections 10.7, 18.5); $\beta_\ell = a_\ell + g$ (460); the free shifts β_i of (469) (Sections 20.1.2, 21.5); β_k, β_n, digamma sequences (Sections 18.7, 18.7.8); β_f (Section 18.8); β_7, β_σ (Section 1.2); the bubbles β_{ij} (Section 28).
γ	$\gamma_r(g)$ (Section 2); α, β, γ, constants of the five-loop primitive (Section 10.7); Euler's constant γ_E (Section 8), written γ in Section 18.7.9; γ_C (Part III families); the Fourier coefficients γ_m (Sections 20.1.1, 20.1.2, 21.6); γ, a divergent subgraph (Sections 28, 32.7).
η	η_m, polygamma combinations (Part III families); $\eta_r = 1/(8\pi r)$, the imaginary offset of the nodes (312) (Sections 20.1.3, 20.1.4); η and η_{R} (571), decay exponents of the growth bounds (Sections 18, 20.1, 20.1.1, 21.4); $\eta_{7,\pm}$, parameters of $\mathcal{G}_7$ (Section 1.2); η, the remainder of (728); a local real variable in Sections 18.7.7, 18.7.9, 18.8, 18.9.4, 18.9.8, 21.2 and 24.1.
θ	θ_m and the θ_r of (275), polygamma combinations (Part III families); $\theta = \theta_0 = 2q$ or $\theta_1 = 2q - 1$, the index of the measure block M_θ (Section 17.4); $\theta = -\frac{1}{4}$, a parameter of a complex Selberg integral (Section 15); $\theta_i(z)$, Stirling remainders (Section 21.6); θ_{cv}, a control coefficient (Section 26.14); elsewhere an angle.
ϑ	the auxiliary coordinate of the edge exponents α_e (Section 34.6.5); an angle, with $f_\vartheta = \mathbf{K}(\sin\vartheta)$ and $g_\vartheta = \mathbf{K}(\cos\vartheta)$ (Section 18.7.8), and a local angle in Section 18.8; $\vartheta_\tau(x)$, a Gamma ratio (Section 18.9.2); $\vartheta = 2\psi_1(q) - \psi_1(2q)$ (Section 21.10).
κ	$\kappa(d, z)$ (174); $\kappa_{\theta,j}$, κ_j^0 (280); $\kappa(S)$, a nominal endpoint power (Section 26.11); local exponents, partitions and constants in Sections 18.8, 18.7.6, 18.7.7, 19.6, 20.1.2, 20.2, 20.5 and 22.9. The exponent $\varkappa$ (Sections 20.1.2, 21.6) is a different letter.

Symbol	Meaning
λ	λ_m and λ_j, the same polygamma combinations, continued as $\lambda_j(x)$ (Part III families); λ_j, local to the wheel-tail paragraph of Section 19.10; λ_k, the Stirling coefficients (583); λ_e, the completion exponents (Sections 7, 16.3); the bare couplings λ_0 (Section 15), λ_B and λ (Sections 34.2, 34.5); λ_*, a cubic root (Sections 1.2, B.2.5); $\lambda^2_{\phi\phi\mathcal{E}}$, a defect OPE coefficient (Part V); scale factors in Part IV.
τ	the regulator of the master path, $t = 2\tau$, and of R_τ, S_τ, $C(\tau)$ (Part III families); $\tau_n(\nu)$, $\tau_n^{(d)}(\nu)$, the zig-zag spectra (Sections 1.1, 10, 15); τ, a Taylor coefficient of $\mathfrak{G}$ (Section 8), written ϱ in Section 18.9.1; $\tau_*(q)$ (571); local parameters in Sections 15, 16.7 and 18.7.6, and the sums τ_n of Section B.4.
ϕ, φ, Φ	the field ϕ (Sections 1.4, 34.2) and the bulk field Φ (Part V); $\phi(x) = \Gamma(\frac{1}{4}+x)/\Gamma(\frac{3}{4}+x)$ (Section 18.9.2); the angle ϕ (Section 18.7.9); $\phi(R, R')$ (Section 18.8); $\varphi_x(s)$ (345), $\varphi_\tau(y)$ (Section 18.9.8), $\varphi(z)$, $\bar\varphi(z)$ (501) and φ_m (Section 20.6); Φ_2 (Section 17.2), $\Phi_t(m)$, $\widehat\Phi$ (Sections 21.4–21.8), Φ_-, Φ_0, Φ_+ (Section 22.9) and $\Phi_{\le 4}$ (Section 32.7); local meanings in Sections 5, 18.7 and 20.1.4.
ψ, Ψ	ψ_m, the polygamma functions (first block), written $\psi = \psi_0$ in Sections 28, 34.6.1 and B.5; in the proof of the twofold-convolution lemma of Section 18.7.9, $\psi = \theta - \frac{\phi}{2}$ and $\psi_0 = \pi - \frac{\phi}{2}$ are angles; $\Psi_\sigma(s)$ (Section 1.2), $\Psi(\sigma)$ (Section 18.7.9), $\Psi_{r,m}(v; b)$ (Section 20.6) and $\Psi_t(n)$, $\widehat\Psi$ (Sections 21.4–21.8).
ω, Ω	$\omega(\rho)$, the constant of the Gamma-ratio bound (465) (Sections 20.1, 21.6), against ω_j, the sine shifts of the top-mode proof (Section 20.1.2); $\omega = w(\kappa)+\log 2$ (Section 18.7.7); ω_τ (Section 18.8); $\omega = \pm 1$ (Section 19); ω_b, ω_c (Appendix A); $\Omega_r(t)$, $\Omega(t; x)$, $\Omega_r(t; x)$, Ω_{R}, periodized wheels (Sections 18.5, 18.7, 20.6, 21.9); the simplex measure $d\Omega$ (Part IV).
sans-serif letters	local to the section that defines them, except A, D, T, U (Section 1.1), B and $\mathsf{G} = \mathsf{G}$. Part V uses H_j, e_j, f_j, l_j, S_n, w_n, L_n, J_d, Y, C, $\mathsf{K}_\times$, $\mathsf{K}_\parallel$, $\mathsf{c}_\times$, a_n, r_1, r_2, $\mathsf{F}_{\mathrm{sch}}$, $\mathsf{f}_k^{\mathrm{sch}}$, to keep them apart from h_d, $B_\pm$, ℓ_j, W_r, $\mathcal{L}_d$, $\mathcal{J}$, H and C_n. Part III uses its own in Sections 17–21, for example c_j (345), X_r, r_k, s_k, P_τ, F_i, E, N_q and the sequences x_m, y_j, z_n (486); Section 28 uses P_2, P_3, p_2, p_3, and Appendix B the matrices M, N. Clashes: the strip T (Section 18.7.7) and the transpose T have the glyph of $\mathsf{T} = \psi_1(q)$; the arrays $\mathsf{A}_{M,\alpha}$, $\mathsf{B}_{M,\alpha}$ (298) and $\mathsf{U}_\pm$, $\mathsf{V}_\pm$ (Section 18.4) are not A, B, U; J_0, J_1 (Section 18.7.6) are not J_d; a_j (Section 18.7.6) is not a_n; r_1, r_2 are not r_k; and P_x, Q_x (Section 18.7.9) are functions, not the pole sets P_T, P_τ (Sections 20.1.2, 21.4).

For reference,

$$K(m) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - m\sin^2\theta}}, \quad B(u,v) = \frac{\Gamma(u)\Gamma(v)}{\Gamma(u+v)}, \quad (a)_n = \frac{\Gamma(a+n)}{\Gamma(a)}.$$

Our ordinary hypergeometric convention is

$${}_pF_q\left(\begin{matrix}a_1,\dots,a_p\\ b_1,\dots,b_q\end{matrix}; x\right) = \sum_{n=0}^{\infty} \frac{\prod_i (a_i)_n}{\prod_j (b_j)_n} \frac{x^n}{n!}.$$

For $p = q + 1$, positive real parametric excess ensures absolute convergence at $x = 1$ in the nonsingular cases used here [22]. Repeated parameters may not be cancelled along a deformation

unless they agree as functions of that deformation, not merely at its evaluation point. Contours, endpoint subtractions and continuation rules are part of the master definitions, not optional numerical conventions. In particular, $1/4 \mapsto d/4$ alone does not generalize a one-dimensional formula: $3d/4$ and $1 - d/4$ coincide only at $d = 1$.

Evidence and reproducibility. The derivations and definitions below specify the retained calculation. Published lower-order results and general mathematical identities are cited separately from this project's higher-loop claims. Algebraic or master-level agreement does not independently validate every graph. The public repository of this work contains the code, graph arrays, numerical inputs and records behind every quoted number; Section 33 gives the package map, in which package paths are relative to the folder `GFF/code/` of that repository, and the provenance of every merged block. These files supply reproducibility data, not unstated definitions in the formulas.

2 Renormalization-group derivation

Normalize the free covariance and perturbations by

$$\langle \phi(x)\phi(0)\rangle_0 = |x|^{-h}, \qquad h = \frac{1-\epsilon}{2}, \qquad S_{\rm int} = \int dx \left(\frac{g_0}{4!}\phi^4 + \frac{r_0}{2}\phi^2\right). \tag{3}$$

After subtraction of the critical local mass power divergence, the long-range kinetic term has no wave-function counterterm. This is the field-theory framework of Ref. [7]. In the mass-anomalous-dimension convention used here,

$$\beta_{\rm RG}(g) = -\epsilon g + \sum_{L\geq 1} b_L g^{L+1}, \qquad \gamma_r(g) = \sum_{L\geq 1} c_L g^L, \tag{4}$$

$$\beta_{\rm RG}(g_*) = 0, \qquad \Delta_{\mathcal{E}} = 1 - \nu^{-1} = \frac{1-\epsilon}{2} + \gamma_r(g_*). \tag{5}$$

Here b_4, c_4 are *loop coefficients*, whereas C_4, C_5 are coefficients in the physical ϵ expansion. None is Euler's beta function $B(u,v) = \Gamma(u)\Gamma(v)/\Gamma(u+v)$.

2.1 Lagrange inversion

Put

$$\mathcal{B}(g) = b_1 + b_2 g + b_3 g^2 + b_4 g^3 + b_5 g^4 + \cdots, \qquad \epsilon = g_* \mathcal{B}(g_*). \tag{6}$$

Then, for $n \geq 1$,

$$\boxed{C_n = -\tfrac{1}{2}\delta_{n1} + \frac{1}{n}[g^{n-1}]\frac{\gamma_r'(g)}{\mathcal{B}(g)^n}.} \tag{7}$$

This follows directly by applying Lagrange–Bürmann inversion to $\gamma_r(g_*(\epsilon))$. Equivalently, for $g_* = \sum_j u_j \epsilon^j$,

$$\begin{aligned} &u_1 = b_1^{-1}, && u_2 = -b_2 b_1^{-3}, \\ &u_3 = 2b_2^2 b_1^{-5} - b_3 b_1^{-4}, \\ &u_4 = -5b_2^3 b_1^{-7} + 5b_2 b_3 b_1^{-6} - b_4 b_1^{-5}, \\ &u_5 = 14 b_2^4 b_1^{-9} - 21 b_2^2 b_3 b_1^{-8} + (6b_2 b_4 + 3b_3^2) b_1^{-7} - b_5 b_1^{-6}. \end{aligned} \tag{8}$$

The coefficients through three loops in this normalization are

$$\begin{aligned} &b_1 = 3, && c_1 = 1, && b_2 = -6A, && c_2 = -A, \\ &b_3 = 6Q + \frac{87}{4}A^2 - \frac{3}{4}\pi^2 - 4\pi, && c_3 = \frac{7}{2}A^2 - \frac{1}{6}\pi^2 - \frac{4}{3}\pi. \end{aligned} \tag{9}$$

Consequently,

$$\boxed{C_4 = \frac{60A^3 - 8Ab_3 + 18Ac_3 + 3c_4 - b_4}{243},} \tag{10}$$

and

$$\boxed{\begin{aligned} C_5 = \frac{1}{729}\big[&336A^4 - 60A^2b_3 + 108A^2c_3 + b_3^2 - 3b_3c_3 \\ &- 10Ab_4 + 24Ac_4 + 3c_5 - b_5\big]. \end{aligned}} \tag{11}$$

At fifth order the new combination is $D_5 = 3c_5 - b_5$. The four-loop combination $D_4 = 3c_4 - b_4$, sufficient for C_4, is not sufficient by itself for C_5. Extending a four-loop fixed-point series without five-loop input would miss $D_5/729$.

3 The complete graph and subtraction prescription

This section fixes the full four- and five-loop calculation, including nonprimitive graphs. Write z for an auxiliary regulator, equal to ϵ on the physical regulator path, and set $h = (1 - z)/2$.

3.1 Ordered graphs and exact weights

For loop order L, let $\mathfrak{G}_L$ be the set of symmetric edge-multiplicity matrices on $0, \dots, L$ satisfying

$$\begin{aligned} &m_{ii} = 0, \quad m_{ij} \in \{0, 1, 2, 3\}, \quad \sum_{i<j} m_{ij} = 2L, \quad d_i := \sum_j m_{ij} \in \{2, 3, 4\}, \\ &\sum_{\substack{i \in S \\ j \notin S}} m_{ij} \geq 2 \quad \text{for every nonempty proper vertex subset } S. \end{aligned} \tag{12}$$

The cut condition expresses connectedness and one-particle irreducibility. Put $e_i = 4 - d_i$, so $\sum_i e_i = 4$, and use the rational weights

$$\boxed{w_4(m) = \frac{24}{\prod_i e_i! \prod_{i<j} m_{ij}!}, \qquad w_2(m) = \frac{w_4(m)}{12} \sum_i e_i(e_i - 1).} \tag{13}$$

The mass insertion can be represented by choosing two external half-edges at the same vertex of a quartic graph, which yields the second weight. All ordered vertex labels are retained. The perturbative vertex factorial cancels the sum over orderings; there is no additional factor $(L + 1)!$ in the following ordered integrals. The enumerated counts for $L = 1, \dots, 5$ are

$$|\mathfrak{G}_L| = 1,\ 6,\ 85,\ 1890,\ 59145. \tag{14}$$

These are not counts of isomorphism classes.

A zero-dimensional benchmark of the weights. The weights (13) can be checked without enumerating graphs. In zero dimensions put

$$M_j(g) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^j \exp\Big[-\frac{x^2}{2} - \frac{g}{24}(x^4 - 6x^2 + 3)\Big]\, dx;$$

the normal-ordered interaction removes tadpoles. With r coupled to $x^2/2$, the Legendre transformation gives the amputated one-particle-irreducible four-point function and the mass insertion,

$$\Gamma^{(4)} = 3\frac{M_0^2}{M_2^2} - \frac{M_4 M_0^3}{M_2^4}, \qquad \partial_r \Gamma^{(2)}\big|_{r=0} = \frac{M_0 M_4 - M_2^2}{2M_2^2}, \tag{15}$$

whose Gaussian moments give the exact series

$$\Gamma^{(4)} = g - \tfrac{3}{2}g^2 + \tfrac{15}{4}g^3 - \tfrac{105}{8}g^4 + \tfrac{893}{16}g^5 - \tfrac{8795}{32}g^6 + \tfrac{97987}{64}g^7 - \tfrac{10951025}{1152}g^8 + \tfrac{49985915}{768}g^9 - \cdots,$$
$$\partial_r\Gamma^{(2)} = 1 - \tfrac{1}{2}g + \tfrac{3}{4}g^2 - \tfrac{49}{24}g^3 + \tfrac{343}{48}g^4 - \tfrac{2875}{96}g^5 + \tfrac{83711}{576}g^6 - \tfrac{2763929}{3456}g^7 + \tfrac{33994099}{6912}g^8 - \cdots.$$

At every loop order the weights obey

$$\sum_{m\in\mathfrak{G}_L} w_4(m) = (-1)^L(L+1)!\,[g^{L+1}]\Gamma^{(4)}, \qquad \sum_{m\in\mathfrak{G}_L} w_2(m) = (-1)^L(L+1)!\,[g^L]\,\partial_r\Gamma^{(2)}. \tag{16}$$

A class G with automorphism group $\mathrm{Aut}(G)$ has $(L+1)!/|\,\mathrm{Aut}(G)|$ vertex labelings, so the class weights (754) sum to $(-1)^L[g^{L+1}]\Gamma^{(4)}$ and $(-1)^L[g^L]\partial_r\Gamma^{(2)}$. For $L = 1,\dots,5$ the ordered sums are 3, $\frac{45}{2}$, 315, $\frac{13395}{2}$, $\frac{395775}{2}$ and 1, $\frac{9}{2}$, 49, $\frac{1715}{2}$, $\frac{43125}{2}$; the five- and six-loop sums of the enumeration in the table after (597) are those of (16). A second enumerator, which builds the adjacency matrices row by row and tests connectedness and the cut condition of (12) by a bridge search, rather than building the graphs edge by edge and testing every cut, reproduced every ordered-label graph and every rational weight (13); its record is not part of the repository. The class inventories of the eighth-order calculation for $L = 1,\dots,8$ (Section 30.2), whose loops one to seven reproduce those of the seventh-order calculation, satisfy the class form exactly (`LRI_C8_HPC/gen/zero_dim_check.py`, with its output in `LRI_C8_HPC/inputs8/verification/logs/c8gen_fin_54591916.out`). The zero-dimensional integral is a combinatorial test of the weights, not an approximation to a critical exponent.

With $x_0 = 0, x_L = 1$, define

$$\begin{aligned} \mathcal{J}_m(z) &= \int_{0<x_1<\cdots<x_{L-1}<1} \prod_{i<j}(x_j - x_i)^{-hm_{ij}}\, d^{L-1}x, \\ I_m(z) &= \frac{\mathcal{J}_m(z)}{Lz}, \qquad \mathcal{A}_L(z) = \sum_m w_4(m) I_m(z), \qquad \mathcal{B}_L(z) = \sum_m w_2(m) I_m(z). \end{aligned} \tag{17}$$

The factor $1/(Lz)$ is the overall scale integral with the total vertex span cut off at one. Translation has been removed. The mass power divergence is treated by analytic continuation and critical mass subtraction, rather than by discarding the associated nontrivial graphs.

3.2 Minimal subtraction

The bare amplitudes are

$$F_4(v) = v + \sum_{L\ge1}(-1)^L\mathcal{A}_L(z)v^{L+1}, \qquad F_2(v) = 1 + \sum_{L\ge1}(-1)^L\mathcal{B}_L(z)v^L, \tag{18}$$

with $\mathcal{A}_1 = 3/z$, $\mathcal{B}_1 = 1/z$. Write

$$g_0 = \mu^z\mathcal{Z}_g(g,z), \quad r_0 = \mu^s r Z_r(g,z), \quad \mathcal{Z}_g = g + \sum_{n\ge2} G_n(z)g^n, \quad Z_r = 1 + \sum_{n\ge1} Z_n(z)g^n. \tag{19}$$

The coefficients G_n, Z_n contain only pole terms. If PP selects negative Laurent powers of z, finiteness of $F_4(\mathcal{Z}_g)$ and $Z_r F_2(\mathcal{Z}_g)$ gives

$$G_n = -\,\mathrm{PP}[g^n]F_4(\mathcal{Z}_{g,<n}), \qquad Z_n = -\,\mathrm{PP}[g^n]\big[Z_{r,<n}F_2(\mathcal{Z}_g)\big]. \tag{20}$$

The RG coefficients follow from the simple poles:

$$\boxed{b_L = L[z^{-1}]G_{L+1}, \qquad c_L = L[z^{-1}]Z_L.} \tag{21}$$

Equivalently, the pole equations are

$$z\mathcal{Z}_g + \beta_{\mathrm{RG}}\partial_g\mathcal{Z}_g = 0, \qquad \beta_{\mathrm{RG}}\partial_g Z_r + \gamma_r Z_r = 0. \tag{22}$$

They imply the useful recursions

$$G_n = \frac{\sum_{\ell=1}^{n-1} b_\ell (n-\ell) G_{n-\ell}}{(n-1)z}, \qquad Z_n = \frac{\sum_{\ell=1}^{n} [(n-\ell)b_\ell + c_\ell] Z_{n-\ell}}{nz}, \tag{23}$$

where $G_1 = Z_0 = 1$.

3.3 Primitive versus nonprimitive: the precise distinction

A primitive diagram has an overall ultraviolet divergence and no divergent proper subdiagram. A nonprimitive diagram contains subdivergences and must be evaluated together with its counterterms. For example, a two-edge bubble has the local integral

$$\int_0^\ell r^{-1+z}\, dr = \frac{1}{z} + \log \ell + O(z).$$

If it occurs inside a larger graph, its pole subtraction multiplies the quotient graph. In particular,

$$\left(-\frac{p}{z}\right)\left(\frac{q_{-1}}{z} + q_0 + O(z)\right) = -\frac{pq_{-1}}{z^2} - \frac{pq_0}{z} + O(1),$$

so the larger graph's new simple pole depends on the lower-loop *finite part* q_0. Primitive periods alone do not determine it. The forest formula implements these products systematically [21].

The phrase "remaining nonprimitive" used during the calculation also included the lower-loop fixed-point terms. Precisely,

$$C_5 = C_5^{\text{primitive}} + \frac{\mathcal{R} - 10Ab_4 + 24Ac_4 + 3c_5^{\text{nonprim}} - b_5^{\text{nonprim}}}{729},$$

where $\mathcal{R} = 336A^4 - 60A^2 b_3 + 108A^2 c_3 + b_3^2 - 3b_3 c_3$. The later separation into primitive, wheel-plus-assembly, and rest sectors is stated in Section 13. Those pieces are bookkeeping within the chosen normalization; the universal object is their sum.

3.4 Laurent orders needed for the fifth coefficient

Let $a_{L,k} = [z^k]\mathcal{A}_L$ and $d_{L,k} = [z^k]\mathcal{B}_L$. Expanding (20) gives

$$\begin{aligned}
b_4 &= 36Aa_{2,0} - 216a_{2,1} + 48a_{3,0} - 4a_{4,-1}, \\
c_4 &= 26Ad_{2,0} - 140d_{2,1} + 40d_{3,0} - 4d_{4,-1}, \qquad (24) \\
b_5 &= 5a_{5,-1} - 75a_{4,0} + 450a_{3,1} - 60Aa_{3,0} - 1350a_{2,2} + 585Aa_{2,1} - 5b_3 a_{2,0}, \\
c_5 &= 5d_{5,-1} - 65d_{4,0} + 325d_{3,1} - \frac{95}{2}Ad_{3,0} - \frac{2275}{3}d_{2,2} \\
&\quad + \frac{725}{2}Ad_{2,1} - \frac{10b_3 + 5c_3}{3}d_{2,0}. \qquad (25)
\end{aligned}$$

The lower-loop data must therefore be retained through

Loop order	Required Laurent powers in $\mathcal{A}_L, \mathcal{B}_L$
2	$z^{-2}, z^{-1}, 1, z, z^2$
3	$z^{-3}, z^{-2}, z^{-1}, 1, z$
4	$z^{-4}, z^{-3}, z^{-2}, z^{-1}, 1$
5	$z^{-5}, z^{-4}, z^{-3}, z^{-2}, z^{-1}$

The finite four-loop terms $a_{4,0}, d_{4,0}$ are new inputs relative to the computation of C_4. Simplifying C_4 alone cannot determine them.

What the higher-pole identities cannot detect. The pole equations (22) fix the higher poles of G_n and Z_n from the lower-order simple poles. They do not fix a new simple pole. For example, the shift

$$b_5 \longmapsto b_5 + \delta b_5, \qquad \mathcal{Z}_g \longmapsto \mathcal{Z}_g + \frac{\delta b_5}{5z}\, g^6$$

preserves $z\mathcal{Z}_g + \beta_{\mathrm{RG}}\partial_g \mathcal{Z}_g = 0$ through g^6: the three new terms are $\frac{1}{5}\delta b_5 g^6$, $-\frac{6}{5}\delta b_5 g^6$ and $\delta b_5 g^6$. It changes the physical coefficient by $\delta C_5 = -\delta b_5/729$. More generally $\delta C_5 = (3\delta c_5 - \delta b_5)/729$ when the lower loops are held fixed. Agreement of the higher poles with their renormalization-group values, used as a check in Sections 14 and 22.8 and at seventh and eighth order, therefore tests the lower-order inputs and the pipeline but cannot detect missing new simple-pole information. This caveat was written by OpenAI GPT-6 astra in the research archive behind the sixth-order packages (package `LRI_C6_D2_HPC/reference/`).

Part II

One-dimensional analytic coefficients through fifth order

4 Final one-dimensional formulas

We use

$$\epsilon = 2s - 1, \qquad \Delta_\phi = \frac{1-\epsilon}{4}, \qquad \Delta_{\mathcal{E}} = \frac{1}{2} - \frac{\epsilon}{6} + C_2\epsilon^2 + C_3\epsilon^3 + C_4\epsilon^4 + C_5\epsilon^5 + O(\epsilon^6).$$

Here $\mathcal{E} = [\phi^2]$ is the thermal operator. The covariance is $\langle\phi(x)\phi(0)\rangle_0 = |x|^{-(1-\epsilon)/2}$, and the interaction is $g\phi^4/4!$. Set

$$A = \pi + 4\log 2, \qquad Q = \frac{\Gamma(1/4)^4}{\pi}, \qquad H = \frac{2\pi^2}{3} + 8\mathsf{G}, \qquad \mathsf{G} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}. \tag{26}$$

Thus G is Catalan's constant. It is not a graph amplitude.

The complete reduction obtained from the graph calculation is

$$\boxed{\begin{aligned} 46656C_5 = {} & -1536(P_5 + \mathcal{K}) + 384AP_4 - 576\mathcal{L} - 1728Q^2 \\ & + 48Q(15A^2 + 149H + 7\pi^2) - 15A^4 - 30A^2\pi^2 \\ & - 828A\pi^3 - 21456A\zeta(3) - 83\pi^4. \end{aligned}} \tag{27}$$

The four constants $P_4, P_5, \mathcal{L}, \mathcal{K}$ are evaluated in Sections 8–10; their graph-normalized definitions and complete reductions are included. The preferred fixed-parameter presentation of P_5 is given next.

4.1 The five fixed hypergeometric values for P_5

For $r \in \{1, 2, 3, 5, 7\}$ define

$$\boxed{\begin{aligned} \mathcal{Y}_r = {} & \frac{r}{8}\left[\frac{\Gamma((r+2)/8)}{\Gamma((r+6)/8)}\right]^7 \\ & \times {}_9F_8\left(\begin{matrix} 1,\ 1+r/8,\ ((r+2)/8)^{\times 7} \\ r/8,\ ((r+6)/8)^{\times 7} \end{matrix}; -1\right). \end{aligned}} \tag{28}$$

Put $c_8 = \cos(\pi/8)$ and $s_8 = \sin(\pi/8)$. Then

$$\boxed{\begin{aligned} P_5 = 16\pi^{3/2}\Big\{ & -35\mathcal{Y}_2 \\ & + (22\sqrt{2}-13)(c_8^7\mathcal{Y}_3 - s_8^7\mathcal{Y}_5) \\ & + (22\sqrt{2}+13)(c_8^7\mathcal{Y}_1 + s_8^7\mathcal{Y}_7)\Big\}. \end{aligned}} \tag{29}$$

Every parameter is fixed and positive; all arguments are the ordinary -1. Each series has excess $3/2$ and is absolutely convergent. For $r = 2$ one matching numerator/denominator parameter 1 cancels, so

$$\mathcal{Y}_2 = \frac{\pi^{7/2}}{4}\,{}_8F_7\left(\begin{matrix}\frac{5}{4}, (\frac{1}{2})^{\times 7} \\ \frac{1}{4}, 1^{\times 6}\end{matrix}; -1\right). \tag{30}$$

Thus (29) uses four ${}_9F_8$ values and one ${}_8F_7$ value. It has no parameter derivative, Taylor-coefficient extraction, coalescence limit, extra summation, multivariate function, or unstated analytic-continuation sheet. Section 10.7 proves the identity using a regularized bilateral sum with exactly three Fourier harmonics. Inserting it in the polygamma form of C_5 makes the complete result an ordinary fixed-parameter hypergeometric expression. No elementary evaluation or minimality of this five-value basis is asserted.

4.2 The reflected-quarter polygamma basis

The following is the derivative-free form of every coefficient $C_0, \ldots, C_5$. The evaluation of $\mathcal{K}$ used here is proved in Section 8.3, and the wheel reduction in Section 9.2.

4.2.1 A basis adapted to $1/4$ and $3/4$

Put $q = 1/4$ and $\bar{q} = 1 - q = 3/4$. With the convention $\psi_m(z) = d^{m+1}\log\Gamma(z)/dz^{m+1}$, define

$$\boxed{\begin{aligned} &\mathsf{A} = \psi_0(2q) + \psi_0(1) - 2\psi_0(q), && \mathsf{D} = \psi_0(\bar{q}) - \psi_0(q), \\ &\mathsf{T} = \psi_1(q), && \overline{\mathsf{T}} = \psi_1(\bar{q}), \\ &\mathsf{U} = \psi_2(q), && \overline{\mathsf{U}} = \psi_2(\bar{q}). \end{aligned}} \tag{31}$$

An overline here means evaluation at the reflected argument, not complex conjugation. There is no differentiation with respect to q in the coefficient formulas. At $q = 1/4$, the exact dictionary is

$$\begin{aligned} \mathsf{A} &= A, & \mathsf{D} &= \pi, \\ \mathsf{T} + \overline{\mathsf{T}} &= 2\pi^2, & \mathsf{T} - \overline{\mathsf{T}} &= 16\mathsf{G}, \\ \mathsf{U} + \overline{\mathsf{U}} &= -112\zeta(3), & \overline{\mathsf{U}} - \mathsf{U} &= 4\pi^3, \\ H &= \frac{5\mathsf{T} - \overline{\mathsf{T}}}{6}, & \log 2 &= \frac{\mathsf{A} - \mathsf{D}}{4}. \end{aligned} \tag{32}$$

These follow from the polygamma reflection and multiplication identities [32, 33]. An entirely quarter-argument version of A, apart from the reference value at 1, is

$$\mathsf{A} = \frac{4\psi_0(1) - 5\psi_0(q) + \psi_0(\bar{q})}{3} \qquad (q = 1/4).$$

We retain the gamma normalization Q and the ordinary periods P_4, P_5. For clarity, the wheel normalizations are

$$B_- = \Gamma(q)^4 F_-, \qquad B_+ = \frac{64\pi^4}{\Gamma(q)^4}F_+, \qquad P_4 = 13B_- - 3B_+.$$

If desired as a uniform notation, $Q = \Gamma(q)^4/\mathsf{D}$ and $B_+ = 64\mathsf{D}^4F_+/\Gamma(q)^4$ at $q = 1/4$. This is not an evaluation of Q or the periods in polygamma values alone.

4.2.2 The two master reductions

The proved polygamma evaluation and the digamma-prefactor wheel identity are

$$\boxed{\begin{aligned}\mathcal{K} &= \frac{1}{2}\left[\mathsf{T}(3\mathsf{T}-\overline{\mathsf{T}})+\mathsf{AU}\right],\\ \mathcal{L} &= -\frac{3\mathsf{A}-\mathsf{D}}{6}P_4-\frac{2\mathsf{D}}{3}B_-.\end{aligned}} \tag{33}$$

Equivalently,

$$\mathcal{L} = -\frac{13\mathsf{A}-3\mathsf{D}}{2}B_- + \frac{3\mathsf{A}-\mathsf{D}}{2}B_+.$$

The first identity eliminates the entire six-${}_3F_2$ parameter derivative in the definition of $\mathcal{K}$. The second removes the parameter derivative in the definition of the logarithmic wheel but does not eliminate the two wheel periods $B_\pm$. Both are equalities to the original master definitions, not merely equalities among proposed right-hand sides.

4.2.3 C_0 through C_5

Writing $\Delta_{\mathcal{E}} = \sum_{n=0}^{5} C_n\epsilon^n + O(\epsilon^6)$, the additional coefficient formulas are

$$\boxed{\begin{aligned}&C_0 = \frac{1}{2}, && C_1 = -\frac{1}{6}, \quad C_2 = \frac{\mathsf{A}}{9},\\ &C_3 = \frac{\mathsf{A}^2+(\mathsf{T}+\overline{\mathsf{T}})/6-8Q}{108}.\end{aligned}} \tag{34}$$

$$\boxed{\begin{aligned}648C_4 = 16P_4 - \mathsf{A}^3 - \frac{\mathsf{A}}{2}(\mathsf{T}+\overline{\mathsf{T}}) - 8\mathsf{A}Q\\ + \frac{365\mathsf{U}-6\overline{\mathsf{U}}}{14}.\end{aligned}} \tag{35}$$

$$\boxed{\begin{aligned}46656C_5 = &-1536P_5 + 96(7\mathsf{A}-\mathsf{D})P_4 + 384\mathsf{D}B_- - 1728Q^2\\ &+16Q(45\mathsf{A}^2+383\mathsf{T}-64\overline{\mathsf{T}}) - 768\mathsf{T}(3\mathsf{T}-\overline{\mathsf{T}})\\ &-15\mathsf{A}^4 - 15\mathsf{A}^2(\mathsf{T}+\overline{\mathsf{T}})\\ &-\frac{6\mathsf{A}}{7}(431\mathsf{U}+18\overline{\mathsf{U}}) - \frac{83}{4}(\mathsf{T}+\overline{\mathsf{T}})^2.\end{aligned}} \tag{36}$$

Section 8.3 proves the master identity used in this substitution. All explicit logarithmic, zeta, Catalan, and power-of-π coefficients in these expressions are written using fixed-argument ψ_0, ψ_1, ψ_2. The non-polygamma input consists of the gamma normalization Q, the two wheel periods, and P_5; their original definitions and normalizations have not changed.

For checking (36), the wheel term is exactly

$$384AP_4 - 576\mathcal{L} = 96(7\mathsf{A}-\mathsf{D})P_4 + 384\mathsf{D}B_-.$$

An exact symbolic check (`lri_psi_complete_update/verification/symbolic_checks.json`) compares each C_n with the formulas of Section 4, in particular (27), after substituting (32) and (33). In particular, no arithmetic inference is based on fitting decimal values of the periods.

5 The full-line Selberg formula used in the expansions

For

$$M(u,v,w)=\iint_{\mathbb{R}^2}|xy|^{-u}|(1-x)(1-y)|^{-v}|x-y|^{-w}\,dx\,dy,$$

let

$$U=S_2(1-u,1-v,-w/2),\qquad c=\cos(\pi w/2), \tag{37}$$

$$\theta_u=\pi(u+w/2),\quad \theta_v=\pi(v+w/2), \tag{38}$$

$$T=\pi(u+v+w),\qquad \Phi=\pi(u+v+w/2), \tag{39}$$

where

$$S_2(a,b,c)=\frac{\Gamma(a)\Gamma(b)\Gamma(a+c)\Gamma(b+c)\Gamma(1+2c)}{\Gamma(a+b+c)\Gamma(a+b+2c)\Gamma(1+c)}.$$

The real-line chamber sum is

$$\begin{aligned}M=U\Bigg[1-\frac{2c(\sin\theta_u+\sin\theta_v)}{\sin T}\\ +\frac{\sin\theta_v\sin(\pi v)+\sin\theta_u\sin(\pi u)+2c\sin\theta_u\sin\theta_v}{\sin T\sin\Phi}\Bigg].\end{aligned} \tag{40}$$

Apparent removable singularities are treated in the whole expression. The elementary Selberg evaluation is standard [13]; the real-line chamber weighting follows by splitting each integration variable into $(-\infty,0),(0,1),(1,\infty)$ and mapping the resulting chambers to the Selberg square; the trigonometric factors are fixed by analytic continuation of those chamber integrals. The formula is stated in full so that no separate chamber table is an input. This formula supplies every symmetric two-variable expansion used here.

6 Root and subtraction conventions

For an L-loop graph the rooted bare integral has the overall factor $2/(Lz)$, with the two root positions fixed at zero and one and their separation integrated with a unit infrared cutoff. Roots must not both lie in a proper superficially divergent collision set. In the code, the root-safety test includes power-divergent and non-one-particle-irreducible collision sets as well as logarithmic ones. This avoids spurious infrared poles in a rooted integral. The counterterm forest itself still contains the proper logarithmic one-particle-irreducible subgraphs; induced contractions that create tadpoles vanish in the stated prescription.

In a quotient graph, the roots are inherited from the parent graph. Choosing arbitrary new quotient roots before subtracting can change finite terms and is not a legitimate shortcut. Counterterms of proper subgraphs can be computed with their own safe roots, since their local pole coefficients are independent of that infrared choice. These distinctions were enforced when producing the residue certificate and the numerical comparisons.

7 Why the remaining diagrams need only three second-order expansions

The nonprimitive graph calculation requires only the following three second-order master expansions.

A vertex connected to two other vertices can be integrated using

$$\mathsf{B}(u,v)=\int_{\mathbb{R}}|x|^{-u}|1-x|^{-v}\,dx=\sqrt{\pi}\,\frac{\Gamma((1-u)/2)\Gamma((1-v)/2)\Gamma((u+v-1)/2)}{\Gamma(u/2)\Gamma(v/2)\Gamma((2-u-v)/2)}. \tag{41}$$

It replaces its two edges by one edge of power $u+v-1$. A block with two internal vertices and two boundary vertices reduces to

$$\mathcal{J}(a,b,c,d,e)=\iint_{\mathbb{R}^2}\frac{dx\,dy}{|x|^a|1-x|^b|y|^c|1-y|^d|x-y|^e}. \tag{42}$$

A generic chamber decomposition of (42) is a finite sum of ordinary ${}_3F_2(1)$ functions. The crucial further simplification is that the full five-loop coefficient does *not* require its arbitrary-parameter form: only its second-order regulator expansions at three critical index patterns occur.

Complete the five active indices to the ten edges on vertices $(0,1,x,y,\infty)$, ordered lexicographically:

$$\begin{aligned}\boldsymbol{\lambda}=(&3-a-b-c-d-e,\ a,\ c,\ b+d+e-1,\ b,\ d,\\ &a+c+e-1,\ e,\ 2-a-b-e,\ 2-c-d-e).\end{aligned} \tag{43}$$

Every completed vertex has incident power sum two. Real-line conformal changes of variables permit permutations of the five completed vertices, followed by fixing three positions to $0,1,\infty$. This gives an exact S_5 action on the five active indices. Zero-edge representatives factor into (41).

After this action and the zero-edge factorizations, the critical representatives needed are

$$T=(1/2,1/2,1/2,1/2,1/2),\quad A_0=(1/2,1/2,1/2,1/2,1),\quad B_0=(1/2,1/2,1/2,1,1/2). \tag{44}$$

The exceptional higher-connectivity graphs are reduced to these same patterns by the regulator-corrected moves in Section 11.

At five loops a nontrivial two-internal-vertex block is multiplied by at most two further simple-pole factors and the overall scale pole. An analytic remainder of total regulator degree at least three cannot then contribute to the simple pole. Consequently, the three second-order expansions below suffice. This order counting also covers the finite lower-loop terms required by the five-loop forests.

7.1 Regular tetrahedron pattern: a gamma-valued Hessian

At T, $\mathcal{J}=3Q$ and all first index derivatives vanish. On the balanced completed variations, the unique permutation-invariant quadratic form is proportional to $\sum_e(\delta\lambda_e)^2$. Evaluation along a symmetric Selberg path fixes it:

$$\boxed{\mathcal{J}=3Q+\frac{QH}{4}\sum_{e=1}^{10}(\delta\lambda_e)^2+O(3).} \tag{45}$$

Here $O(3)$ denotes analytic terms of total regulator degree at least three.

For an explicit check, put

$$M(u,v,w)=\mathcal{J}(u,v,u,v,w).$$

The full-line Selberg evaluation gives

$$M(1/2,1/2,1/2)=3Q,\qquad \nabla M=0,\qquad \frac{\nabla^2 M}{3Q}=H\begin{pmatrix}2&1&1\\1&2&1\\1&1&1\end{pmatrix}. \tag{46}$$

The appearance of Catalan's constant comes from $\psi_1(1/4)=\pi^2+8\mathsf{G}$ and $\psi_1(3/4)=\pi^2-8\mathsf{G}$. The Selberg formula used to establish these identities is recalled in Section 5 [13].

7.2 Pattern A_0: no new regular quadratic constant

Write

$$a = u + p, \quad c = u - p, \quad b = v + q, \quad d = v - q, \quad e = w, \qquad \sigma_0 = 2 - 2u - w, \quad \sigma_1 = 2 - 2v - w.$$

Subtracting the endpoint pole differences from the symmetric integral gives

$$\boxed{\mathcal{J} = M(u, v, w) + R(w)\left(\frac{p^2}{\sigma_0} + \frac{q^2}{\sigma_1}\right) + O(3),} \tag{47}$$

where, through the needed order,

$$R(w) = \pi^3 + 28\zeta(3) - \frac{A}{2}[\pi^3 + 28\zeta(3)](w - 1) + O((w - 1)^2). \tag{48}$$

In (47), the remainder has total scaling degree at least three when all local regulators are scaled together; higher terms from the endpoint pole residues are included in this notation. The diagonal collision residue is independent of p, q and cancels in the difference. The two endpoint residues yield $R(w) = \partial_x^2 \mathsf{B}(x, w)|_{x=(2-w)/2}$.

After the full endpoint pole parts have been removed, symmetry permits only two additional second-order constants, multiplying $p^2 + q^2$ and pq. Both vanish. To determine them, set $u = v = 1/2$, $w = 1 - t$, and use the two unique-star paths

$$p = t,\ q = 0: \quad \mathcal{J} = \mathsf{B}(1/2 + t, 1/2)\mathsf{B}(1 - t, 1 - t), \tag{49}$$

$$p = q = t/2: \quad \mathcal{J} = \mathsf{B}((1 + t)/2, (1 + t)/2)\mathsf{B}(1 - t, 1 - t). \tag{50}$$

Expansion of these exact gamma products through t^2 fixes both regular quadratic constants to zero. The symbolic verification code is not part of the public repository.

7.3 Pattern B_0: exactly one new mixed derivative

Use local coordinates

$$p_1 = 1 - d, \qquad p_2 = 2 - b - d - e, \tag{51}$$

$$z_1 = 3 - a - b - c - d - e, \qquad z_2 = 2 - c - d - e. \tag{52}$$

Equivalently,

$$d = 1 - p_1, \quad b = 1 + p_1 - e - p_2, \quad c = 1 + p_1 - e - z_2, \quad a = e - p_1 + p_2 + z_2 - z_1.$$

The base point has $p_1 = p_2 = z_1 = z_2 = 0$, $e = 1/2$. The only local pole hyperplanes are $p_1 = 0$ and $p_2 = 0$. A convenient complete pole part is

$$\mathcal{P} = \frac{2}{p_1}\mathsf{B}(e + p_2 + z_2 - z_1, 1 - p_2) + \frac{2}{p_2}\mathsf{B}(e, 1 + p_1 - e) - \frac{4}{p_1 p_2}. \tag{53}$$

After subtracting it, the function is jointly analytic in all five local coordinates.

On $z_2 = 0$, the y integration is a unique three-point star. On $z_1 = 0$, the completed graph has a zero edge. These exact restrictions are

$$U_2(z_1) = \mathsf{B}(1 + p_1 - e, 1 - p_1)\mathsf{B}(e + p_2 - z_1, 1 - p_2), \tag{54}$$

$$U_1(z_2) = \mathsf{B}(e - p_1 + p_2 + z_2, 1 + p_1 - e - z_2)\mathsf{B}(1 + p_1 - e - p_2, 1 - p_1), \tag{55}$$

$$U_0 = U_2(0) = U_1(0). \tag{56}$$

Analytic interpolation between the two restrictions leaves only one second-order mixed term. Explicitly,

$$\boxed{\begin{aligned}\mathcal{J} &= U_2(z_1) + U_1(z_2) - U_0 + z_1 z_2 \mathcal{K} \\ &\quad + \frac{2}{p_1}[\mathsf{B}(e + p_2 + z_2 - z_1, 1 - p_2) - \mathsf{B}(e + p_2 - z_1, 1 - p_2) \\ &\qquad - \mathsf{B}(e + p_2 + z_2, 1 - p_2) + \mathsf{B}(e + p_2, 1 - p_2)] + O(3).\end{aligned}} \tag{57}$$

The subtracted analytic difference vanishes on both $z_1 = 0$ and $z_2 = 0$, hence is divisible by $z_1 z_2$. Its coefficient at the base point is precisely the one constant $\mathcal{K}$. Any dependence of that coefficient on another local index raises the total degree to at least three and is not needed.

This explains why the many apparent ordinary-hypergeometric parameter derivatives in the remaining graph classes collapse to a single new constant rather than a large independent basis.

8 Evaluation of $\mathcal{K}$

8.1 From the original master to six Euler integrals

In (57), choose

$$p_1 = p_2 = z_1 = z_2 = t, \qquad e = 1/2.$$

Then

$$J_t = \mathcal{J}(1/2, 1/2, 1/2, 1 - t, 1/2)$$

and the gamma interpolation becomes

$$J_t^{\Gamma} = \mathsf{B}(1/2, 1 - t)^2 + \frac{2}{t}\,[\mathsf{B}(1/2, 1 - t) - \mathsf{B}(1/2 + 2t, 1 - t)]. \tag{58}$$

Thus

$$J_t - J_t^{\Gamma} = t^2 \mathcal{K} + O(t^3). \tag{59}$$

Integrating x in (42) gives the full-line elliptic star $F(y)$. Fold its three real-line intervals onto $(0, 1)$, obtaining

$$\begin{aligned}J_t = 4 \int_0^1 [K(x) + K(1 - x)]\big[x^{-1/2}(1 - x)^{t-1} \\ + x^{-t}(1 - x)^{t-1} + x^{-1/2}(1 - x)^{-t}\big]\, dx.\end{aligned} \tag{60}$$

This integral is initially convergent for small positive t; its regulated poles are handled explicitly by multiplication by t^2.

For the beta-weighted elliptic moment,

$$E(a, b) = \int_0^1 x^{a-1}(1 - x)^{b-1} K(x)\, dx, \tag{61}$$

Euler integration gives

$$E(a, b) = \frac{\pi}{2} B(a, b)\, {}_3F_2\left(\begin{matrix}1/2, 1/2, a \\ 1, a + b\end{matrix}; 1\right). \tag{62}$$

A Thomae transformation gives the alternative form

$$E(a, b) = \frac{\pi}{2} \frac{\Gamma(b)^2}{\Gamma(b + 1/2)^2}\, {}_3F_2\left(\begin{matrix}b, b, 1 - a \\ b + 1/2, b + 1/2\end{matrix}; 1\right). \tag{63}$$

These follow from the standard beta transform and unity-argument hypergeometric identities [22]. Apply the appropriate form to each of the six terms in (60). Multiplication by t^2 gives exactly (68). Since

$$V_r(t) = t\mathsf{B}(1/2 + rt, 1 - t),$$

(59) proves the regular fourth-derivative formula (70).

For an additional check, the gamma interpolation alone has

$$[t^2]J_t^{\Gamma} = \frac{A^4}{6} - \frac{A^2\pi^2}{6} - \frac{A\pi^3}{3} - 12A\zeta(3) - \frac{\pi^4}{180}. \tag{64}$$

The endpoint-subtracted logarithmic coefficient of (60), minus this expression, is a second numerical route to $\mathcal{K}$.

8.2 An analytic representation of the regulated master

All derivatives in this section act on the displayed parameters; the argument of every hypergeometric function remains one. Define

$$f_1(t) = {}_3F_2\left(\begin{matrix}t, t, 1/2\\ 1/2 + t, 1/2 + t\end{matrix}; 1\right), \qquad f_2(t) = {}_3F_2\left(\begin{matrix}t, t, t\\ 1/2 + t, 1/2 + t\end{matrix}; 1\right), \tag{65}$$

$$f_3(t) = {}_3F_2\left(\begin{matrix}1/2, 1/2, t\\ 1, 1/2 + t\end{matrix}; 1\right), \qquad f_4(t) = {}_3F_2\left(\begin{matrix}1/2, 1/2, t\\ 1, 1\end{matrix}; 1\right), \tag{66}$$

$$f_5(t) = {}_3F_2\left(\begin{matrix}1/2, 1/2, 1/2\\ 1, 3/2 - t\end{matrix}; 1\right), \qquad f_6(t) = {}_3F_2\left(\begin{matrix}1/2, 1/2, 1 - t\\ 1, 3/2 - t\end{matrix}; 1\right). \tag{67}$$

Put

$$\begin{aligned}\widehat{J}(t) = 2\pi\Big\{ &\frac{\Gamma(1+t)^2}{\Gamma(1/2+t)^2}[f_1(t) + f_2(t)] \\ &+ \frac{t\sqrt{\pi}\,\Gamma(1+t)}{\Gamma(1/2+t)} f_3(t) + t\Gamma(1+t)\Gamma(1-t) f_4(t) \\ &+ \frac{t^2\sqrt{\pi}\,\Gamma(1-t)}{\Gamma(3/2-t)}[f_5(t) + f_6(t)]\Big\},\end{aligned} \tag{68}$$

and, for $r = 0, 2$,

$$V_r(t) = 2\sqrt{\pi}\,\frac{\Gamma(1/4 - rt/2)\Gamma(1 + t/2)\Gamma(1/4 + (r-1)t/2)}{\Gamma(1/4 + rt/2)\Gamma(1/2 - t/2)\Gamma(1/4 + (1-r)t/2)}. \tag{69}$$

The new constant has the regular parameter-derivative evaluation

$$\boxed{\mathcal{K} = \frac{1}{24}\,\frac{d^4}{dt^4}\left[\widehat{J}(t) - V_0(t)^2 - 2\big(V_0(t) - V_2(t)\big)\right]\Big|_{t=0}}. \tag{70}$$

The bracket starts at order t^4. Every gamma factor is regular near zero, and the six hypergeometric series have parametric excess either $1/2$ or $1 - t$. Their defining series and the required derivatives therefore converge near the evaluation point [22]. No singular defining series has to be assigned a value at $t = 0$ in (70).

Two numerical evaluations that do not use the result of the proof below give

$$\mathcal{K} = 39.331569128259039057503187151591786152304\ldots: \tag{71}$$

a 32-node Cauchy evaluation of the fourth derivative in (70) at 45 working digits, and the endpoint-subtracted logarithmic integral obtained from (60), which agree to about 4.1×10^{-33} (a 16-node

evaluation at 40 digits differs by 3.7×10^{-17}). The hypergeometric evaluation uses gamma-ratio asymptotics and Hurwitz-zeta tail summation, the integral explicitly subtracted endpoint expansions. At 110 working digits the endpoint integral agrees with the proved value (76) to 6×10^{-109} (`lri_psi_complete_update/verification/numerical_checks.json`). Equation (70) is an intermediate analytic representation; the following proof reduces this exact master to fixed polygamma values.

Conventions and special values. We use the derivative-order convention

$$\psi_m(z) = \frac{d^{m+1}}{dz^{m+1}} \log \Gamma(z), \qquad m = 0, 1, 2, \dots. \tag{72}$$

Thus $\psi_0 = \psi = \Gamma'/\Gamma$ is the digamma function, $\psi_1 = \psi'$ is trigamma, and $\psi_2 = \psi''$ is tetragamma. These are ordinary one-variable polygamma functions, not the multivariate functions sometimes denoted by uppercase Ψ_1, Ψ_2. In particular, a square such as $\psi_1(1/4)^2$ means an ordinary product, not $\psi_2(1/4)$. The standard series and rational-argument identities [32, 33] give

$$\begin{aligned} \psi_0(1) &= -\gamma_E, & \psi_0(1/2) &= -\gamma_E - 2\log 2, \\ \psi_0(1/4) &= -\gamma_E - \frac{\pi}{2} - 3\log 2, & \psi_0(3/4) &= -\gamma_E + \frac{\pi}{2} - 3\log 2, \\ \psi_1(1/4) &= \pi^2 + 8\mathsf{G}, & \psi_1(3/4) &= \pi^2 - 8\mathsf{G}, \\ \psi_2(1/4) &= -2\pi^3 - 56\zeta(3), & \psi_2(3/4) &= 2\pi^3 - 56\zeta(3). \end{aligned} \tag{73}$$

For example, for $m \geq 1$,

$$\psi_m(z) = (-1)^{m+1} m! \sum_{n=0}^{\infty} (n+z)^{-m-1}.$$

Splitting the positive odd integers into the residue classes 1 and 3 modulo four yields the displayed quarter-argument values. At order two, $\sum(4n+1)^{-3} + \sum(4n+3)^{-3} = 7\zeta(3)/8$ and their difference is $\pi^3/32$; the latter also follows from the reflection formula in [32]. The project constants therefore obey the exact dictionary

$$\boxed{\begin{aligned} A &= \psi_0(1/2) + \psi_0(1) - 2\psi_0(1/4), \\ H &= \psi_1(1/4) - \frac{\pi^2}{3}. \end{aligned}} \tag{74}$$

8.3 The compact evaluation and its proof

The proof starts from the original master path (59) and does not assume a numerical integer relation, the wheel reduction, or an unproved hypergeometric summation. The main step is a bilinear identity for a pair of ordinary ${}_3F_2(1)$ functions. The periodicity argument below includes the growth estimate needed to rule out a nonconstant periodic factor.

8.4 Statement and notation

Write $\ell = \log 2$, and define the parameter functions

$$\begin{aligned} \mathfrak{F}(t) &= {}_3F_2\left(\begin{matrix} 1/2, 1/2, t \\ 1, 1 \end{matrix}; 1\right), \\ \mathfrak{G}(t) &= {}_3F_2\left(\begin{matrix} t, t, t \\ 1/2+t, 1/2+t \end{matrix}; 1\right). \end{aligned} \tag{75}$$

These are f_4 and f_2, respectively, in the six-function formula. They are unrelated to the fixed wheel values $F_\pm$ or Catalan's constant G. We will prove

$$\boxed{\begin{aligned}\mathcal{K} &= 128\mathsf{G}^2 + 24\pi^2\mathsf{G} - 4\pi^3\ell - 28(\pi+4\ell)\zeta(3)\\ &= 2\psi_1(1/4)^2 - \pi^2\psi_1(1/4) + \frac{A}{2}\psi_2(1/4).\end{aligned}} \tag{76}$$

The two equalities have different roles: the first evaluates the original master, and the second is the rational-argument polygamma dictionary.

8.5 Reducing the six elliptic moments to two

Keep the *parameter* convention $K(x)$, and put

$$E(a,b) = \int_0^1 x^{a-1}(1-x)^{b-1}K(x)\,dx.$$

For real $0 < t < 1$, define

$$\begin{aligned} I_1 &= E(1/2,t), & I_2 &= E(t,1/2), & I_3 &= E(1-t,t),\\ I_4 &= E(t,1-t), & I_5 &= E(1/2,1-t), & I_6 &= E(1-t,1/2).\end{aligned}$$

Then (60) says $J_t = 4\sum_{j=1}^6 I_j$. Set $s_t = \sin(\pi t)$ and $c_t = \cos(\pi t)$. The four exact moment relations are

$$\boxed{\begin{aligned} I_1 &= c_tI_3 + s_tI_4, & I_5 &= s_tI_3 - c_tI_4,\\ I_2 &= s_tI_3, & I_6 &= s_tI_4.\end{aligned}} \tag{77}$$

Here is a contour proof, including the branch signs. Integrate $z^{-1/2}(1-z)^{t-1}K(z)$ on an upper-half-plane contour, using principal powers continued from $0 < z < 1$. The boundary values for $x > 1$ and $x < 0$ are [34, 23]

$$K(x+\mathrm{i}0) = x^{-1/2}\{K(1/x) + \mathrm{i}K(1-1/x)\}, \qquad K(x) = (1-x)^{-1/2}K\left(\frac{x}{x-1}\right).$$

The three real intervals contribute

$$-\mathrm{i}I_5 + I_1 - e^{-\mathrm{i}\pi t}(I_3 + \mathrm{i}I_4) = 0.$$

Real and imaginary parts give the first two relations. The analogous contour for $z^{-1/2}(1-z)^{t-1}K(1-z)$ gives

$$-I_5 - \mathrm{i}I_6 + I_2 - e^{-\mathrm{i}\pi t}I_4 = 0,$$

and hence the last two. At $z = 0$ the worst singularity is $z^{-1/2}\log z$; at $z = 1$ it is $(1-z)^{t-1}\log(1-z)$. The small arcs therefore vanish. The large arc is $O(R^{t-1}\log R)$ and vanishes for $t < 1$. These estimates justify the contours throughout the indicated interval; the resulting identities continue meromorphically in t.

Euler integration and the Thomae transformation already used in the main text give [22]

$$I_3 = \frac{\pi}{2}\frac{\Gamma(t)^2}{\Gamma(1/2+t)^2}\mathfrak{G}(t), \qquad I_4 = \frac{\pi^2}{2s_t}\mathfrak{F}(t).$$

Consequently,

$$\boxed{\begin{aligned} J_t = 2\pi(1 + c_t + 2s_t)&\frac{\Gamma(t)^2}{\Gamma(1/2+t)^2}\mathfrak{G}(t)\\ &+ 2\pi^2\left(2 + \tan\frac{\pi t}{2}\right)\mathfrak{F}(t).\end{aligned}} \tag{78}$$

This is an exact meromorphic reduction of the six-function master, not a truncation in t.

8.6 A contiguous relation with an explicit telescoping certificate

Write

$$a_n(t) = \frac{(1/2)_n^2 (t)_n}{(n!)^3}, \qquad \mathfrak{F}(t) = \sum_{n=0}^{\infty} a_n(t).$$

Direct rational algebra proves the termwise identity

$$\begin{aligned} t^2 a_n(t+1) - (2t^2 - 2t + 3/4) a_n(t) + (t-1)^2 a_n(t-1) \\ = \frac{(n+1)^3 a_{n+1}(t)}{n+t} - \frac{n^3 a_n(t)}{n+t-1}. \end{aligned} \tag{79}$$

For generic $\Re t < 0$ the right-hand side telescopes to zero: $a_n(t) = O(n^{\Re t - 2})$, so the upper boundary is $O(n^{\Re t})$, and the lower boundary vanishes. Apparent exceptional parameter values are handled by analytic continuation. Thus

$$\boxed{t^2 \mathfrak{F}(t+1) - (2t^2 - 2t + 3/4)\mathfrak{F}(t) + (t-1)^2 \mathfrak{F}(t-1) = 0.} \tag{80}$$

The defining series is holomorphic for $\Re t < 1$. This recurrence extends $\mathfrak{F}$ meromorphically to the plane, with possible poles only at positive integers.

8.7 The discrete Wronskian is constant, not merely periodic

The function $\mathfrak{F}(1-t)$ solves the same recurrence. Its discrete Wronskian with $\mathfrak{F}(t)$ is

$$W(t) = t^2 \{\mathfrak{F}(t+1)\mathfrak{F}(1-t) - \mathfrak{F}(t)\mathfrak{F}(-t)\}. \tag{81}$$

Equation (80) gives $W(t) = W(t-1)$. To determine this periodic function, first use the Thomae identity

$$\mathfrak{F}(1-t) = \frac{\Gamma(t)}{\Gamma(1-t)\Gamma(1/2+t)^2} \mathfrak{G}(t). \tag{82}$$

Since $\mathfrak{G}$ is analytic near zero and $\mathfrak{G}(0) = 1$, we have $\mathfrak{F}(1-t) = 1/(\pi t) + O(1)$. The apparent singularity of W at zero is removable and

$$W(0) = -\frac{1}{\pi^2}.$$

All possible poles of W are integers. Periodicity therefore makes each one removable, and W is entire.

A growth bound is essential here. The double Euler representation is

$$\mathfrak{F}(t) = \frac{1}{\pi^2} \int_0^1 \int_0^1 \frac{(1-uv)^{-t}}{\sqrt{uv(1-u)(1-v)}}\, du\, dv, \qquad \Re t < 1. \tag{83}$$

It follows by expanding $(1-uv)^{-t}$ and applying two beta integrals; absolute convergence at $u = v = 1$ holds precisely for $\Re t < 1$. In particular, $|\mathfrak{F}(x + \mathrm{i}y)| \le \mathfrak{F}(x)$. For $-1/2 \le \Re t \le 1/2$, the four functions $\mathfrak{F}(t), \mathfrak{F}(-t), \mathfrak{F}(t-1), \mathfrak{F}(-t-1)$ are uniformly bounded by $\mathfrak{F}(1/2)$, independently of $\Im t$. Solving (80) at t and $-t$ expresses $\mathfrak{F}(1+t)$ and $\mathfrak{F}(1-t)$ in terms of those bounded functions. Their rational coefficients are bounded for $|\Im t| \ge 1$. Hence

$$W(t) = O(1 + |\Im t|^2) \quad \text{uniformly in a full vertical strip of width one.}$$

An entire one-periodic function with this polynomial vertical growth is constant. Explicitly, its nth Fourier coefficient can be integrated on any horizontal translate of a period; its modulus is

bounded by a polynomial in $|y|$ times $e^{2\pi n y}$. Sending $y \to -\infty$ for $n > 0$ and $y \to +\infty$ for $n < 0$ kills every nonzero mode. We conclude that $W(t) \equiv -1/\pi^2$, or

$$\boxed{\mathfrak{F}(t)\mathfrak{F}(-t) - \mathfrak{F}(1+t)\mathfrak{F}(1-t) = \frac{1}{\pi^2 t^2}.} \tag{84}$$

This argument supplies both the normalization and the reason that a nonconstant periodic multiplier is impossible.

Apply (82) at t and $-t$, and use gamma reflection. The equivalent identity, analytic near zero, is

$$\boxed{\cos^2(\pi t)\,\mathfrak{G}(t)\mathfrak{G}(-t) + \pi^2 t^2\,\mathfrak{F}(t)\mathfrak{F}(-t) = 1.} \tag{85}$$

No numerical input has been used in this proof.

8.8 Only one combination of higher Taylor coefficients is needed

Write ordinary Taylor coefficients, without hidden factorials, as

$$\mathfrak{F}(t) = 1 + ut + vt^2 + O(t^3), \qquad \mathfrak{G}(t) = 1 + \sigma t^3 + \tau t^4 + O(t^5).$$

The absence of the first two terms in $\mathfrak{G} - 1$ follows immediately from its three upper parameters t, t, t. Extracting $[t^4]$ in (85) gives

$$2\tau + \frac{\pi^4}{3} + \pi^2(2v - u^2) = 0,$$

hence the exact elimination rule

$$\boxed{\tau + \pi^2 v = \frac{\pi^2 u^2}{2} - \frac{\pi^4}{6}.} \tag{86}$$

Neither v nor τ has to be evaluated individually. The two lower coefficients are elementary inverse-binomial sums:

$$\begin{aligned} u &= \sum_{n\ge1} \frac{\binom{2n}{n}^2}{16^n n} = 4\ell - \frac{8\mathsf{G}}{\pi}, \\ \sigma &= \sum_{n\ge1} \frac{16^n}{n^3\binom{2n}{n}^2} = 8\pi\mathsf{G} - 14\zeta(3). \end{aligned} \tag{87}$$

For completeness, both evaluations follow without an external summation assumption. Differentiate (83) at zero, set $u = \sin^2\theta$, $v = \sin^2\phi$ in that integral, and use

$$\int_0^{\pi/2} \log(1 - a\sin^2\phi)\,d\phi = \pi \log\frac{1+\sqrt{1-a}}{2}.$$

This elementary identity follows by differentiating in a and matching at $a = 0$. It gives

$$u = -\frac{16}{\pi}\int_0^{\pi/4} \log\cos x\,dx = 4\ell - \frac{8\mathsf{G}}{\pi}.$$

For σ, expand

$$(\arcsin x)^2 = \frac{1}{2}\sum_{n\ge1} \frac{4^n x^{2n}}{n^2\binom{2n}{n}}$$

and integrate with $x = \sin\theta$, using $\int_0^{\pi/2} \sin^{2n-1}\theta\,d\theta = 4^n/(2n\binom{2n}{n})$. The result is

$$\begin{aligned} \sigma &= 4\int_0^{\pi/2} \frac{\theta^2}{\sin\theta}\,d\theta = -8\int_0^{\pi/2} \theta\log\tan(\theta/2)\,d\theta \\ &= 8\pi\sum_{k\ge0}\frac{(-1)^k}{(2k+1)^2} - 16\sum_{k\ge0}\frac{1}{(2k+1)^3} = 8\pi\mathsf{G} - 14\zeta(3). \end{aligned}$$

In the last step one may use an Abel factor in the cosine series $\log\tan(\theta/2) = -2\sum_{k\ge0}\cos((2k+1)\theta)/(2k+1)$ and then take its limit. The endpoint logarithms are integrable, and integration yields absolutely convergent sums.

8.9 Extraction of the original master coefficient

The exact gamma interpolation in the main text gives

$$\mathcal{K} = [t^2]J_t - g_2, \qquad g_2 = \frac{A^4}{6} - \frac{A^2\pi^2}{6} - \frac{A\pi^3}{3} - 12A\zeta(3) - \frac{\pi^4}{180}.$$

Here $[t^2]J_t$ is a Laurent coefficient. No ordinary derivative at a pole is being taken. To expand (78), use

$$\frac{\pi\Gamma(1+t)^2}{\Gamma(1/2+t)^2} = \exp\left(4\ell t - \frac{\pi^2}{3}t^2 + 4\zeta(3)t^3 - \frac{7\pi^4}{90}t^4 + O(t^5)\right).$$

Direct coefficient extraction now yields

$$\begin{aligned}\mathcal{K} = {} & 4A\sigma + 4\tau + \pi^3 u + 4\pi^2 v - 32\ell^2\pi^2 - 8\ell\pi^3 \\ & + 112\ell\zeta(3) + \frac{2\pi^4}{3} + 28\pi\zeta(3).\end{aligned} \tag{88}$$

Substituting (86) first reduces this to

$$\mathcal{K} = 4A\sigma + 2\pi^2u^2 + \pi^3u - 32\ell^2\pi^2 - 8\ell\pi^3 + 28A\zeta(3).$$

The elementary values (87) then give exactly

$$\mathcal{K} = 128\mathsf{G}^2 + 24\pi^2\mathsf{G} - 4\pi^3\ell - 28A\zeta(3).$$

Finally (73) proves (76) and (33). Thus the compact evaluation is an analytic consequence of the original master definition. This completes the proof.

Scope of the proof. The proof evaluates the one-dimensional master constant defined by (70). It proves a local coefficient identity by means of a global parameter identity; it does not claim that the whole deformed master J_t is a gamma quotient, and it does not establish a corresponding identity in other dimensions.

Numerical checks of the identities. The proof uses no numerical input. Its identities were also tested numerically, separately from the proof. These tests are consistency checks, not premises of the proof, and none of them is an interval bound. Besides the 110-digit endpoint comparison quoted after (70), the record `lri_psi_complete_update/verification/numerical_checks.json` contains the following tests. A direct evaluation of $\mathfrak{F}$ and $\mathfrak{G}$ at $t = 0.2$ and 50 working digits satisfies the bilinear identity (85) to 2.6728×10^{-51}. At 75 working digits, with two settings of the asymptotic tail summation, the six-function master $\widehat{J}(t)/t^2$ of (68) and the two-function reduction (78) agree to better than 2×10^{-68} at $t = 0.2$, 0.37 and $0.12 + 0.07\mathrm{i}$, and (85) holds to better than 4×10^{-72} at the two real points. At the complex point the 75-digit bilinear residual is only 3×10^{-59}, and 6×10^{-48} with the longer tail, because the gamma-ratio tail cancels there. Rerun at 130 working digits, the residual has modulus 1.39×10^{-102} (`lri_psi_complete_update/verification/complex_bilinear_130.json`); the less accurate 75-digit values are kept in the record. At 110 digits the integral forms of u and σ given after (87) reproduce their closed forms to 8×10^{-112} and 7×10^{-111}. Finally, the coefficients $C_0,\dots,C_5$ of Section 4 and their reflected-quarter forms agree in exact symbolic algebra (Section 4.2) and numerically at 70 working digits, with largest difference 9.3×10^{-69}, in C_5 (`lri_psi_complete_update/verification/coefficient_values.json`). The numerical comparison reuses the 240-digit values of $F_\pm$ (Section 9) from an earlier record, stored beside it as `wheel_inputs.json`, and recomputes P_5 from its gamma moment. None of these checks bears on the offset of the earlier full-graph control, which Section 14 explains.

9 The wheel periods P_4 and $\mathcal{L}$

9.1 The four-loop wheel: two ${}_4F_3(1)$ values

Let $G_4 = \Gamma(1/4)^4$, and define

$$F_- = {}_4F_3\left(\begin{matrix}1/4,1/4,1/2,1/2\\3/4,3/4,1\end{matrix};1\right), \qquad F_+ = {}_4F_3\left(\begin{matrix}3/4,3/4,1/2,1/2\\5/4,5/4,1\end{matrix};1\right).$$

Then

$$\boxed{P_4 = 13G_4F_- - \frac{192\pi^4}{G_4}F_+.} \tag{89}$$

It evaluates to 2229.52206107094196429452519974 85 .... It follows that the two-${}_4F_3$ expression of the fourth coefficient is

$$\boxed{648C_4 = 208G_4F_- - \frac{3072\pi^4}{G_4}F_+ - A^3 - A\pi^2 - 53\pi^3 - 1436\zeta(3) - 8AQ.} \tag{90}$$

For comparison with the earlier elliptic formulas, define

$$H_4 = {}_4F_3\left(\begin{matrix}1/2,1/2,1/2,1/2\\1,1,1\end{matrix};1\right), \quad H_7 = {}_7F_6\left(\begin{matrix}5/4,(1/2)^{\times 6}\\1/4,1,1,1,1,1\end{matrix};1\right).$$

The two elimination identities are

$$\begin{aligned} H_4 &= \frac{G_4}{4\pi^3}F_- - \frac{16\pi}{G_4}F_+, \\ H_7 &= \frac{G_4}{2\pi^4}F_- + \frac{32}{G_4}F_+. \end{aligned} \tag{91}$$

Indeed, write $\mathbf{K}(k) = K(k^2)$, $k' = \sqrt{1-k^2}$ and $N_\nu = \int_0^1 k^\nu \mathbf{K}(k)\mathbf{K}(k')\,dk$. Euler integration gives

$$N_\nu = \frac{\pi^2}{8}\frac{\Gamma(a)^2}{\Gamma(a+1/2)^2}{}_4F_3\left(\begin{matrix}1/2,1/2,a,a\\1,a+1/2,a+1/2\end{matrix};1\right), \qquad a = \frac{\nu+1}{2}. \tag{92}$$

The Landen substitution $k = (1-t)/(1+t)$, together with the mixed-to-square moment identity, gives $N_{-1/2} - N_{1/2} = 2N_0$ and $N_{-1/2} + N_{1/2} = 4\int_0^1 \mathbf{K}(k)^2\,dk$. Using $N_0 = \pi^3 H_4/8$ and $\int_0^1 \mathbf{K}(k)^2\,dk = \pi^4 H_7/32$ proves the stated identities. These elliptic-moment evaluations are discussed in [25].

9.1.1 From the wheel graph to the elliptic period

For the wheel edges $02, 03, 04, 12, 13, 14, 24, 34$, fix $x_0 = 0, x_1 = 1$. The two remaining outer-vertex integrations factor, giving

$$P_4 = \int_{\mathbb{R}} \frac{F(x)^2}{\sqrt{|x(1-x)|}}\,dx, \qquad F(x) = \int_{\mathbb{R}} \frac{du}{\sqrt{|u(1-u)(u-x)|}}.$$

Splitting at $u = 0, x, 1$ gives $F(x) = 4[K(x) + K(1-x)]$ for $0 < x < 1$. The transformations

$$F(x) = x^{-1/2}F(1/x) \quad (x > 1), \qquad F(x) = (1-x)^{-1/2}F\left(\frac{x}{x-1}\right) \quad (x < 0)$$

fold the tails. Hence

$$P_4 = 16\int_0^1 \frac{1+\sqrt{x}+\sqrt{1-x}}{\sqrt{x(1-x)}}[K(x) + K(1-x)]^2\,dx.$$

The ordered tetrahedron satisfies

$$\int_{0<x<y<1} \frac{dx\,dy}{\sqrt{xy(1-x)(1-y)(y-x)}} = \frac{1}{2}S_2\left(\frac{1}{2},\frac{1}{2},-\frac{1}{4}\right) = \frac{\Gamma(1/4)^4}{4\pi};$$

its full-line period is $3Q$. The three-loop calculation of [7], with the tetrahedron value corrected in [12], uses this normalization; the integral above is a Selberg integral [13].

9.1.2 The moment identities and their contour proof

Let $M = \int_0^1 \mathbf{K}(k)^2\,dk$, $N = N_0$, and $J = \int_0^1 \mathbf{K}(k)\mathbf{K}(k')/k'\,dk$ with $k' = \sqrt{1-k^2}$. Euler integration of the complementary moment gives

$$\int_0^1 k^r \mathbf{K}(k')\,dk = \frac{\pi}{4}\frac{\Gamma((r+1)/2)^2}{\Gamma((r+2)/2)^2}, \quad r > -1.$$

Expanding the other $\mathbf{K}$ proves the formula for N_ν above. The square-moment evaluation is $M = \pi^4 H_7/32$ [25]. To see the mixed-to-square identity $J = 2M$, integrate $\mathbf{K}(z)^2$ around the upper semicircle from 1 to -1. Cauchy's theorem gives $-2M$ on the arc. The boundary-value Landen relation

$$e^{i\theta}\mathbf{K}(e^{2i\theta}) = \tfrac{1}{2}[\mathbf{K}(\cos\theta) + i\mathbf{K}(\sin\theta)], \quad 0 < \theta < \pi/2,$$

with the limit taken from inside the disk, makes the real part of that arc integral equal to $-J$; the imaginary part is odd under $\theta \mapsto \pi/2 - \theta$. Integrable logarithmic endpoint singularities are handled by an interior limit. Thus $J = 2M$.

The real Landen substitution $k = (1-t)/(1+t)$ preserves the mixed measure:

$$\mathbf{K}(k)\mathbf{K}(k')|dk| = \mathbf{K}(t)\mathbf{K}(t')dt.$$

The sum and difference of the half-integer weights transform to $2/t'$ and $2t/t'$, respectively. Therefore

$$N_{-1/2} + N_{1/2} = 2J = 4M, \qquad N_{-1/2} - N_{1/2} = 2N_0.$$

Since

$$N_{-1/2} = \frac{G_4}{16}F_-, \quad N_{1/2} = \frac{4\pi^4}{G_4}F_+, \quad N_0 = \frac{\pi^3}{8}H_4,$$

these prove both elimination identities. Folding the wheel integral gives $P_4 = 16(20M + 16N)$, and consequently the stated two-${}_4F_3$ result.

9.1.3 Four-loop counterterm combination

The residue certificate, including the wheel $b_4^{\text{wheel}} = -6P_4$, gives

$$3c_4 - b_4 = 6P_4 + 45AQ + \frac{405A^3 - 27A\pi^2 - 159\pi^3}{8} - 8\pi A - \frac{1077}{2}\zeta(3).$$

Inserting this in the fourth-order RG formula eliminates the intermediate πA terms and yields the compact C_4. This is an analytic reduction within the graph calculation, not an additional external benchmark.

$$B_- = G_4 F_-, \qquad B_+ = \frac{64\pi^4}{G_4}F_+, \qquad P_4 = 13B_- - 3B_+. \tag{93}$$

The derivative-free formula is

$$\boxed{\begin{aligned}\mathcal{L} &= -(5\pi + 26\log 2)B_- + (\pi + 6\log 2)B_+ \\ &= -\frac{13A - 3\pi}{2}B_- + \frac{3A - \pi}{2}B_+.\end{aligned}} \tag{94}$$

$$\boxed{\begin{aligned}\mathcal{L} = -\frac{2}{3}\Big\{&[\psi_0(1) - \psi_0(1/4)]P_4 \\ &+ [\psi_0(3/4) - \psi_0(1/4)]B_-\Big\}.\end{aligned}} \tag{95}$$

The two digamma differences are $\pi/2 + 3\log 2$ and π, respectively. Numerically, $\mathcal{L} = -5805.301629548802380141179877001 38\ldots$.

9.2 The logarithmic wheel: full contour proof

We give the contour proof of the logarithmic-wheel reduction. It proves (94) and (95), independently of the evaluation of $\mathcal{K}$. The symbols M, N, U, V, and R below denote local elliptic moments. They are distinct from the $V_r(t)$ of Section 8.

9.3 Conventions and the folded integral

In this section put $\ell = \log 2$ and use $B_\pm$ from (93). The elliptic integral uses the *modulus* convention:

$$\mathbf{K}(k) = \frac{\pi}{2}{}_2F_1(1/2, 1/2; 1; k^2), \qquad k' = \sqrt{1-k^2}, \qquad f(k) = \mathbf{K}(k), \quad g(k) = \mathbf{K}(k').$$

The prime in k' denotes the complementary modulus, not differentiation. In the parameter convention, $K(x) = \mathbf{K}(\sqrt{x})$; the full-line definition below uses it. Define

$$\begin{aligned} M &= \int_0^1 f(k)^2\,dk, & N &= \int_0^1 f(k)g(k)\,dk, \\ U_0 &= \int_0^1 \frac{f(k)^2}{k'}\,dk, & V_0 &= \int_0^1 \frac{f(k)g(k)}{k'}\,dk, \\ U &= \int_0^1 \frac{f(k)^2\log(kk')}{k'}\,dk, & V &= \int_0^1 \frac{f(k)g(k)\log(kk')}{k'}\,dk, \\ w(k) &= \log\frac{k}{1-k^2}, & R &= \int_0^1 f(k)^2 w(k)\,dk. \end{aligned}$$

All these integrals converge.

The wheel master is defined on the full line by

$$\mathcal{L} = \int_{\mathbb{R}} \frac{F(x)^2\log|x(1-x)|}{\sqrt{|x(1-x)|}}\,dx, \qquad F(x) = \int_{\mathbb{R}} \frac{du}{\sqrt{|u(1-u)(u-x)|}}.$$

For $0 < x < 1$, $F(x) = 4[K(x) + K(1-x)]$. Folding the two tails gives

$$\mathcal{L} = 16\int_0^1 [K(x)+K(1-x)]^2\Big[\frac{\log x + \log(1-x)}{\sqrt{x(1-x)}} + \frac{\log(1-x) - 2\log x}{\sqrt{1-x}} + \frac{\log x - 2\log(1-x)}{\sqrt{x}}\Big]\,dx.$$

Substitution $x = k^2$ and reflection of the second tail term give the useful identity

$$\frac{\mathcal{L}}{128} = U + V + \int_0^1 (f+g)^2 w\,dk. \tag{96}$$

9.4 Real Landen identities

Let $T(k) = (1-k)/(1+k)$. Standard Landen transformations [31, 34] give

$$f(T(k)) = \frac{1+k}{2}g(k), \qquad g(T(k)) = (1+k)f(k).$$

Consequently,

$$f(T)g(T)|T'| = fg, \qquad g(T)^2|T'| = 2f^2, \qquad f(T)^2|T'| = \tfrac{1}{2}g^2. \tag{97}$$

In particular, $\int_0^1 g^2\,dk = 2M$. Since

$$w(T(k)) = -w(k) - 2\ell,$$

we have

$$\int_0^1 fgw\,dk = -\ell N, \qquad \int_0^1 g^2 w\,dk = -2R - 4\ell M. \tag{98}$$

Combining these with (96) yields

$$\frac{\mathcal{L}}{128} = U + V - R - 4\ell M - 2\ell N. \tag{99}$$

It remains to evaluate just three logarithmic combinations U, V, R.

9.5 Evaluation of U by the Stieltjes transform

Use the standard identity (Wan, Theorem 1, equation (15)) [25]:

$$f(t)^2 = \frac{4}{\pi}\int_0^1 \frac{kf(k)g(k)}{1-t^2k^2}\,dk, \qquad 0 \le t < 1. \tag{100}$$

Integration with respect to dt gives

$$M = \frac{4}{\pi}\int_0^1 fg\ \operatorname{artanh} k\,dk.$$

Applying T and $\operatorname{artanh} T(k) = -\frac{1}{2}\log k$ therefore gives

$$\int_0^1 fg \log k\,dk = -\frac{\pi}{2}M. \tag{101}$$

Also $\log(1+T(k)) = \ell - \log(1+k)$, so

$$\int_0^1 fg \log(1+k)\,dk = \frac{\ell}{2}N. \tag{102}$$

Integrating (100) against $dt/\sqrt{1-t^2}$ proves $U_0 = 2N$, using the elementary integral $\int_0^{\pi/2}(1-k^2\sin^2\theta)^{-1}\,d\theta = \pi/(2k')$ and then complementing k.

For the logarithmic version, the tangent substitution gives

$$\int_0^{\pi/2} \frac{\log(\sin\theta\cos\theta)}{1-k^2\sin^2\theta}\,d\theta = -\frac{\pi}{2k'}\log\frac{(1+k')^2}{k'}.$$

Insert this into (100) and complement the outer variable:

$$\begin{aligned} U &= -2\int_0^1 \frac{kfg}{k'}\log\frac{(1+k')^2}{k'}\,dk \\ &= -2\int_0^1 fg\,[2\log(1+k) - \log k]\,dk. \end{aligned}$$

Equations (101)–(102) now prove

$$\boxed{U = -\pi M - 2\ell N.} \tag{103}$$

The integral interchanges are justified by positivity for the nonlogarithmic identities and by Tonelli applied to the negative logarithm for the last one; the resulting moments are finite.

9.6 Evaluation of R by a first-quadrant contour

Continue $\mathbf{K}(z)$ from its series at zero and put

$$W(z) = \operatorname{Log} z - \operatorname{Log}(1 - z^2),$$

with the logarithms analytic in the first quadrant and their indicated upper boundary values on the real axis. Integrate $\mathbf{K}(z)^2 W(z)$ around the first quadrant. The large arc vanishes because the integrand is $O(\log^3 |z|/|z|^2)$. The small arcs at 0 and 1 also vanish; their singularities are only integrable powers of logarithms. Thus

$$\int_0^\infty \mathbf{K}(x + i0)^2 W(x + i0)\, dx = i \int_0^\infty \mathbf{K}(iy)^2 W(iy)\, dy. \tag{104}$$

The reciprocal and imaginary-modulus connection formulas [31, 34] are

$$\mathbf{K}(x + i0) = \frac{1}{x}[f(1/x) + ig(1/x)] \quad (x > 1), \qquad \mathbf{K}(iy) = \frac{1}{\sqrt{1 + y^2}} f\left(\frac{y}{\sqrt{1 + y^2}}\right).$$

Furthermore $W(x+i0) = w(1/x)+i\pi$ for $x > 1$ and $\operatorname{Im} W(iy) = \pi/2$. The real part of the left-hand side of (104) is therefore

$$2R - \int_0^1 g^2 w\, dk - 2\pi N = 4R + 4\ell M - 2\pi N.$$

On the right-hand side, set $y = k/k'$; the real part is

$$-\frac{\pi}{2} \int_0^\infty \mathbf{K}(iy)^2\, dy = -\frac{\pi}{2} U_0 = -\pi N.$$

Equating these gives

$$\boxed{R = \frac{\pi}{4} N - \ell M.} \tag{105}$$

In particular, the extra πN term is a branch contribution; discarding the imaginary part of the reciprocal-modulus continuation would lose it.

9.7 Evaluation of V by an upper-semicircle contour

Integrate the same $\mathbf{K}(z)^2 W(z)$ over the upper half of the unit disk. The integral over the diameter from -1 to 1 is

$$2R + i\pi M.$$

On the semicircle, put $z = e^{2i\theta}$, $0 < \theta < \pi/2$. The boundary-value Landen identity, with continuation from the interior of the disk, reads

$$e^{i\theta} \mathbf{K}(e^{2i\theta}) = \frac{1}{2}[\mathbf{K}(\cos\theta) + i\mathbf{K}(\sin\theta)],$$

This follows by applying $\mathbf{K}(z) = (1+z)^{-1}\mathbf{K}(2\sqrt{z}/(1+z))$: as z approaches $e^{2i\theta}$ from inside the disk, its transformed modulus approaches $\sec\theta$ from the upper side. The reciprocal-modulus formula then fixes the displayed plus sign. Also,

$$W(e^{2i\theta}) = -\log(4 \sin\theta \cos\theta) + \frac{i\pi}{2}.$$

Write $f_\theta = \mathbf{K}(\sin\theta)$, $g_\theta = \mathbf{K}(\cos\theta)$. Then

$$\mathbf{K}(z)^2\, dz = \left[-f_\theta g_\theta + \frac{i}{2}(g_\theta^2 - f_\theta^2)\right] d\theta.$$

Terms containing $g_\theta^2 - f_\theta^2$ times a symmetric factor integrate to zero. The complete arc integral is therefore

$$V + 2\ell V_0 - \frac{i\pi}{2} V_0.$$

Cauchy's theorem implies, from its imaginary and real parts respectively,

$$V_0 = 2M, \qquad V = -2R - 4\ell M.$$

Together with (105), this proves

$$\boxed{V = -\frac{\pi}{2} N - 2\ell M.} \tag{106}$$

The endpoint limits are valid because the logarithmic singularities are integrable and the small indentations contribute zero.

9.8 Cancellation and reduction to the two ${}_4F_3$ values

Substituting (103), (105), and (106) into (99) gives

$$\boxed{\mathcal{L} = -(128\pi + 640\ell)M - (96\pi + 512\ell)N.} \tag{107}$$

To eliminate M, N, introduce

$$N_\nu = \int_0^1 k^\nu f(k) g(k)\, dk, \qquad \nu > -1.$$

Expanding f and using the beta moment of g yields

$$N_\nu = \frac{\pi^2}{8} \frac{\Gamma((\nu+1)/2)^2}{\Gamma((\nu+2)/2)^2} {}_4F_3\left(\begin{matrix} 1/2, 1/2, (\nu+1)/2, (\nu+1)/2 \\ 1, (\nu+2)/2, (\nu+2)/2 \end{matrix} ; 1 \right).$$

Hence $N_{-1/2} = B_-/16$ and $N_{1/2} = B_+/16$. Applying T to their sum and difference gives

$$N_{-1/2} + N_{1/2} = 2V_0 = 4M, \qquad N_{-1/2} - N_{1/2} = 2N.$$

Thus

$$M = \frac{B_- + B_+}{64}, \qquad N = \frac{B_- - B_+}{32}.$$

Substitution into (107) proves (94). The nonlogarithmic folded integral gives $P_4 = 320M + 256N = 13B_- - 3B_+$, which proves (95). No parameter-differentiation identity is required.

10 The five-loop primitive period

10.1 Primitive graph and normalization

A zig-zag decompletion has edges

$$02, 03, 04, 05, 13, 14, 15, 24, 25, 35.$$

Fixing $x_0 = 0, x_5 = 1$ and performing two independent star integrations gives

$$P_5 = \int_{\mathbb{R}^2} \frac{F(u)F(v)}{\sqrt{|v(1-u)(u-v)|}}\, du\, dv, \quad F(x) = \int_{\mathbb{R}} \frac{dt}{\sqrt{|t(1-t)(t-x)|}}. \tag{108}$$

This integral is convergent and has no subdivergence subtractions. The zig-zag class has automorphism order 2 and contributes $24P_5$ to b_5. The other two primitive classes share the product period $(3Q)^2 = 9Q^2$ and have automorphism orders 16 and 12. Their combined contribution is

$$\left(\frac{48}{16} + \frac{48}{12}\right) 9Q^2 = 63Q^2.$$

All primitive mass weights vanish because the four external legs are on distinct vertices. This yields $C_5^{\text{primitive}} = -(63Q^2 + 24P_5)/729$ in the present normalization.

10.2 Spectral representation and continuation to one dimension

The starting point is the conformal-triangle spectral representation of zig-zag graphs in [24], in particular its eigenvalues, spectral measure and two-point reduction, equations (5.23), (5.27) and (6.2). In the present loop convention its eigenvalue power is $k = L+2$, where L is the contribution's loop order in the quartic beta function.

Continuing the spectral formula to $D = 1$ requires more than substituting $D = 1$ into the density on the real spectral axis. A pole crossing contributes a discrete sum. Define

$$\tau_n(\nu) = (-1)^n\sqrt{\pi}\left|\frac{\Gamma(n/2+1/8+i\nu)}{\Gamma(n/2+3/8+i\nu)}\right|^2, \qquad n = 0, 1, \tag{109}$$

$$\sigma_j = (-1)^j\sqrt{\pi}\,\frac{\Gamma(j/2-1/8)\Gamma(j/2+3/8)}{\Gamma(j/2+1/8)\Gamma(j/2+5/8)}, \qquad j = 1, 2, \ldots. \tag{110}$$

For the zig-zag family, the one-dimensional period is

$$\boxed{P_L = \frac{1}{\pi^2}\int_0^\infty \nu\tanh(2\pi\nu)\left[\tau_0(\nu)^k + \tau_1(\nu)^k\right]\,d\nu + \frac{1}{2\pi^2}\sum_{j=1}^{\infty}(j-\tfrac{1}{2})\sigma_j^k, \qquad k = L+2.} \tag{111}$$

The first term alone is not the period.

10.2.1 Origin of the discrete sum

Here is the contour-continuation detail needed to fix its sign and normalization. Set $p = D/2-1$ and specialize the propagator parameter in [24] to $\beta = D/4$. The combined density is

$$\rho_n^{(D)}(\nu) = \frac{\Gamma(p)\Gamma(n+2p)(n+p)}{2\pi^{3D/2+1}\Gamma(2p)n!}\left[(n+p)^2+4\nu^2\right] \times \frac{\Gamma(p+2i\nu)\Gamma(p-2i\nu)}{\Gamma(2i\nu)\Gamma(-2i\nu)}. \tag{112}$$

The corresponding eigenvalue is

$$\tau_n^{(D)}(\nu) = (-1)^n\pi^{D/2}\frac{\Gamma(n/2+D/8+i\nu)\Gamma(n/2+D/8-i\nu)}{\Gamma(n/2+3D/8+i\nu)\Gamma(n/2+3D/8-i\nu)}.$$

For $D > 2$ the spectral formula uses the real ν contour. As D passes below 2, the poles $\nu = \pm ip/2$ exchange sides. Retaining the analytically continued contour gives a correction

$$-2\pi i\,\mathrm{Res}_{\nu=-ip/2}\,\rho_n^{(D)}(\nu)\tau_n^{(D)}(\nu)^k.$$

Before taking $D \to 1$, its coefficient is

$$\frac{\Gamma(n+2p)(n+p)n(n+2p)}{2\pi^{3D/2}n!\Gamma(-p)}\,\tau_n^{(D)}(-ip/2)^k. \tag{113}$$

The $n = 0$ crossing cancels identically. For $n \geq 1$, the limit of the prefactor in (113) is $(n-1/2)/(2\pi^2)$, while the eigenvalue becomes (110). This gives the second term of (111).

On the real axis, only $n = 0, 1$ remain in the pointwise $D \to 1$ limit, each with density $\nu\tanh(2\pi\nu)/\pi^2$. The fact that higher-spin densities vanish on this axis does not remove their crossed-pole residues. Losing these residues would already spoil the known three-loop tetrahedron normalization.

The continuation and residue reduction here specialize the spectral construction of [24]; they are not an independently published evaluation of this Ising coefficient.

10.3 Closing the contour and summing the gamma poles

Define

$$R_\alpha(x) = \frac{\Gamma(\alpha+x)\Gamma(\alpha-x)}{\Gamma(\alpha+1/4+x)\Gamma(\alpha+1/4-x)}. \tag{114}$$

With $x = i\nu$, closing the continuous spectral integral to the right expresses it as

$$P_L^{\text{cont}} = \pi^{k/2-1} \sum_{\Re x>0} \operatorname{Res}_x \left\{ x \tan(2\pi x) \left[R_{1/8}(x)^k + (-1)^k R_{5/8}(x)^k \right] \right\}. \tag{115}$$

One can use expanding rectangular contours avoiding the real poles. Gamma reflection and Stirling estimates give vanishing outer-contour contributions for $k > 4$.

There are two pole families. The tangent poles are $x_j = (2j-1)/4$, $j \geq 1$, where

$$R_{5/8}(x_j) = -R_{1/8}(x_j), \qquad \sqrt{\pi} R_{1/8}(x_j) = \sigma_j.$$

Their residues in (115) give exactly

$$-\frac{1}{2\pi^2} \sum_{j=1}^{\infty} (j - \tfrac{1}{2}) \sigma_j^k.$$

They cancel the discrete term of (111). The complete period is therefore given just by the gamma poles at $x = \alpha + n$, $n \geq 0$, for $\alpha = 1/8, 5/8$.

10.4 A Barnes integral with no unwanted tangent poles

Set

$$s = 2x - \frac{1}{4}, \qquad g(s) = \frac{\Gamma(s+\frac{1}{2})}{\Gamma(s+1)}, \qquad C(s) = \csc(\pi s), \qquad D(s) = \cot(\pi s).$$

Reflection and duplication imply

$$R_{1/8}(x) = g(s)\left[1 - \cot(\pi s/2)\right], \tag{116}$$

$$R_{5/8}(x) = g(s)\left[1 + \tan(\pi s/2)\right]. \tag{117}$$

The two gamma-pole families have become $s = 0, 1, 2, \ldots$. Define

$$S(s) = \tan(\pi s + \pi/4)\{[1 - \cot(\pi s/2)]^7 - [1 + \tan(\pi s/2)]^7\}. \tag{118}$$

Its unwanted tangent poles are at $s = 1/4 + j$. Their residues equal those of $16\sqrt{2}\,\csc[\pi(s-1/4)]$. Consequently the kernel

$$\mathcal{R}(s) = S(s) - 16\sqrt{2}\,\csc[\pi(s-1/4)]$$

has only integer poles on the right of $\Re s = -1/4$ and retains the desired residues. A rational trigonometric simplification gives

$$\boxed{\begin{aligned}\mathcal{R}(s) = {} & -128C + 160C^3 + 192C^5 - 128C^7 \\ & - 16DC - 400DC^3 + 192DC^5.\end{aligned}} \tag{119}$$

The SymPy check `lri_fixed_hypergeometric_update/public/kernel_identity.py` verifies this exactly after replacing $\tan(\pi s/2)$ by an indeterminate.

Closing a vertical contour to the right is clockwise. Also, $x\,dx = (s+1/4)\,ds/4$. Therefore

$$P_5 = -\frac{\pi^{5/2}}{4} \frac{1}{2\pi \mathrm{i}} \int_{-1/4-\mathrm{i}\infty}^{-1/4+\mathrm{i}\infty} (s+\tfrac{1}{4}) g(s)^7 \mathcal{R}(s)\,ds. \tag{120}$$

No poles lie on this line. On rectangles avoiding the real poles, gamma-ratio asymptotics and the trigonometric factors justify the vanishing of the closing contributions. The remaining integer-pole sum converges absolutely.

10.5 The reflection that removes four derivative orders

For odd j, write

$$I_j = \frac{1}{2\pi\mathrm{i}} \int_{-1/4-\mathrm{i}\infty}^{-1/4+\mathrm{i}\infty} (s+\tfrac{1}{4}) g(s)^7 C(s)^j \, ds, \tag{121}$$

$$J_j = \frac{1}{2\pi\mathrm{i}} \int_{-1/4-\mathrm{i}\infty}^{-1/4+\mathrm{i}\infty} (s+\tfrac{1}{4}) g(s)^7 D(s) C(s)^j \, ds. \tag{122}$$

The substitution $s \mapsto -1/2 - s$ preserves the vertical line. Gamma reflection and elementary trigonometry give

$$\boxed{g(-\tfrac{1}{2}-s) = -D(s)g(s), \qquad C(-\tfrac{1}{2}-s) = -\frac{C(s)}{D(s)}, \qquad D(-\tfrac{1}{2}-s) = \frac{1}{D(s)}.} \tag{123}$$

Together with $D^2 = C^2 - 1$, these imply

$$\boxed{I_7 = 0, \qquad I_5 = I_3, \qquad J_5 = 0, \qquad J_3 = J_1.} \tag{124}$$

For example, $I_7 = -I_7$. For I_3 the reflected integrand gives $I_3 = -I_7 + 2I_5 - I_3$, whence $I_3 = I_5$. Similarly, $J_5 = -J_5$ and $J_1 = -J_5 + 2J_3 - J_1$.

Substitution into (119)–(120) yields

$$\boxed{P_5 = \frac{\pi^{5/2}}{4}(128 I_1 - 352 I_3 + 416 J_1).} \tag{125}$$

The pole order is now at most three rather than seven. This is the precise mechanism behind the reduction from sixth to second parameter derivatives.

10.5.1 Summing the three remaining pole families

Define the analytic function

$$\mathcal{A}_{\mathrm{s}}(t) = \sum_{n=0}^{\infty} (-1)^n (n + \tfrac{1}{4} + t) \left[\frac{\Gamma(n+\frac{1}{2}+t)}{\Gamma(n+1+t)}\right]^7. \tag{126}$$

Termwise parameter derivatives through the needed order are absolutely convergent near zero. Using $\csc[\pi(n+t)] = (-1)^n \csc(\pi t)$ and $\cot[\pi(n+t)] = \cot(\pi t)$, closing to the right gives

$$\boxed{\begin{aligned} I_1 &= -\frac{\mathcal{A}_{\mathrm{s}}(0)}{\pi}, \\ I_3 &= -\frac{\mathcal{A}_{\mathrm{s}}''(0) + \pi^2 \mathcal{A}_{\mathrm{s}}(0)}{2\pi^3}, \\ J_1 &= -\frac{\mathcal{A}_{\mathrm{s}}'(0)}{\pi^2}. \end{aligned}} \tag{127}$$

Here the minus signs are the clockwise-contour signs. For the third formula, $\csc(\pi t)\cot(\pi t) = (\pi t)^{-2} - 1/6 + O(t^2)$ has no simple-pole term. Substitution into (125) gives

$$P_5 = \frac{4}{\sqrt{\pi}} \left[11\mathcal{A}_{\mathrm{s}}''(0) - 26\pi \mathcal{A}_{\mathrm{s}}'(0) + 3\pi^2 \mathcal{A}_{\mathrm{s}}(0)\right]. \tag{128}$$

This intermediate identity follows by substituting the three residues into (125). The finite-periodization proof below eliminates all three Taylor operations from the final evaluation.

10.6 Derivation of the positive gamma integral

On the same vertical line, put $s = -1/4 + \mathrm{i}y$ and $G(y) = |\Gamma(1/4 + \mathrm{i}y)|^2$. Gamma reflection gives

$$g(s) = \frac{G(y)}{\sqrt{2}\,\pi}\big[\cosh(\pi y) - \mathrm{i}\sinh(\pi y)\big]. \tag{129}$$

Writing $z = \cosh(\pi y) - \mathrm{i}\sinh(\pi y)$, one has $C(s) = -\sqrt{2}/z$ and $D(s) = -\overline{z}/z$. Taking conjugate halves of the contour integral yields

$$I_1 = \frac{1}{8\pi^8}\int_0^\infty y G(y)^7\big[\sinh^3(2\pi y) - 3\sinh(2\pi y)\big]\,dy, \tag{130}$$

$$I_3 = -\frac{1}{2\pi^8}\int_0^\infty y G(y)^7\sinh(2\pi y)\,dy, \tag{131}$$

$$J_1 = \frac{1}{4\pi^8}\int_0^\infty y G(y)^7\sinh(2\pi y)\cosh(2\pi y)\,dy. \tag{132}$$

Inserting these into (125) and using $4\sinh^3 v = \sinh(3v) - 3\sinh v$ and $2\sinh v\cosh v = \sinh(2v)$ proves (133). This derivation fixes the coefficients $1, 13, 29$ and the factor $\pi^{-11/2}$ analytically; they were not obtained by fitting a numerical period.

$$\boxed{P_5 = \frac{1}{\pi^{11/2}}\int_0^\infty y|\Gamma(\tfrac{1}{4} + \mathrm{i}y)|^{14}[\sinh(6\pi y) + 13\sinh(4\pi y) + 29\sinh(2\pi y)]\,dy.} \tag{133}$$

The integrand is $O(y^2)$ at zero and $O(y^{-5/2}e^{-\pi y})$ at infinity by the Gamma-ratio asymptotics [27]; it is positive on the integration interval. This provides an independent convergent numerical representation of the same period.

10.7 Finite periodization and the five fixed values

10.7.1 The unilateral and bilateral sums

Use the meromorphic function

$$f(s) = \left(s + \frac{1}{4}\right)\left[\frac{\Gamma(s + \frac{1}{2})}{\Gamma(s+1)}\right]^7, \qquad \mathcal{A}_{\mathrm{s}}(t) = \sum_{n=0}^{\infty}(-1)^n f(n+t). \tag{134}$$

This is the same unilateral sum as (126). By taking out the $n = 0$ factors and using $(t + \frac{1}{4} + n)/(t + \frac{1}{4}) = (t + \frac{5}{4})_n/(t + \frac{1}{4})_n$,

$$\begin{aligned}\mathcal{A}_{\mathrm{s}}(t) = {} & \left(t + \frac{1}{4}\right)\left[\frac{\Gamma(t + \frac{1}{2})}{\Gamma(t+1)}\right]^7 \\ & \times {}_9F_8\left(\begin{matrix}1, t + \frac{5}{4}, (t + \frac{1}{2})^{\times 7} \\ t + \frac{1}{4}, (t+1)^{\times 7}\end{matrix}; -1\right).\end{aligned} \tag{135}$$

Consequently

$$\mathcal{Y}_r = \mathcal{A}_{\mathrm{s}}\left(\frac{r-2}{8}\right), \qquad r = 1, 2, 3, 5, 7. \tag{136}$$

The parameter sum difference in (135) is $3/2$; gamma-ratio asymptotics give absolute and locally uniform convergence away from poles. In particular all five evaluation points in (136) are nonsingular and have positive hypergeometric parameters [22, 27].

Now extend the index in (134) to all integers:

$$\Sigma(t) = \sum_{n\in\mathbb{Z}}(-1)^n f(n+t), \qquad \Xi(t) = \cos^7(\pi t)\,\Sigma(t). \tag{137}$$

These expressions are initially defined away from the half integers. Gamma reflection gives the two useful identities

$$\frac{\Gamma(-s)}{\Gamma(\frac{1}{2}-s)} = -\cot(\pi s)\frac{\Gamma(s+\frac{1}{2})}{\Gamma(s+1)}, \qquad f(-\tfrac{1}{2}-s) = \cot^7(\pi s) f(s). \tag{138}$$

The minus sign from the linear factor in f cancels the seventh power of the minus sign in the first identity. These are direct consequences of the standard gamma reflection formula [26]. Writing a negative summation index as $n = -m-1$ and using (138) shows, including the alternating sign, that

$$\boxed{\begin{aligned}\Sigma(t) &= \mathcal{A}_{\mathrm{s}}(t) - \tan^7(\pi t)\mathcal{A}_{\mathrm{s}}(\tfrac{1}{2}-t),\\ \Xi(t) &= \cos^7(\pi t)\mathcal{A}_{\mathrm{s}}(t) - \sin^7(\pi t)\mathcal{A}_{\mathrm{s}}(\tfrac{1}{2}-t).\end{aligned}} \tag{139}$$

Thus evaluating Ξ requires only two ordinary unilateral hypergeometric values, with no new special-function convention.

10.7.2 Why only three harmonics survive

We give the analytic growth argument explicitly, because periodicity alone would not imply a finite Fourier polynomial. For a compact set of t avoiding half integers, Stirling's ratio estimate and (138) imply

$$f(n+t) = O(|n|^{-5/2}) \qquad (n \to \pm\infty),$$

locally uniformly. Indeed the positive-index estimate follows from $\Gamma(s+1/2)/\Gamma(s+1) \sim s^{-1/2}$; reflection reduces the negative-index estimate to the same right-half-plane bound. Hence Σ is meromorphic. Its only possible poles are at half integers and have order at most seven. Multiplication by $\cos^7(\pi t)$ removes them. The same estimates after multiplication are locally uniform at a half integer, so the removal applies to the entire series, not just to individual summands. Therefore Ξ is entire.

Reindexing the absolutely convergent bilateral sum gives

$$\Sigma(t+1) = -\Sigma(t), \qquad \Xi(t+1) = \Xi(t).$$

For $0 \le x \le 1$ and $|y| \ge 1$, the reflection factors are uniformly bounded. The right-half-plane gamma estimate therefore also gives

$$|\Sigma(x+\mathrm{i}y)| \le C\sum_{n\in\mathbb{Z}}(1+|n+\mathrm{i}y|)^{-5/2} = O(|y|^{-3/2}). \tag{140}$$

It follows that

$$|\Xi(x+\mathrm{i}y)| = O\Big(e^{7\pi|y|}|y|^{-3/2}\Big). \tag{141}$$

Let $c_k = \int_0^1 \Xi(x)e^{-2\pi\mathrm{i}kx}\,dx$. Entirety and periodicity allow the contour to be shifted vertically. For $k \ge 4$, shift down to $\operatorname{Im} t = -Y$ and use (141) to obtain

$$|c_k| \le CY^{-3/2}e^{(7-2k)\pi Y} \longrightarrow 0.$$

Shifting upward gives $c_k = 0$ for $k \le -4$. Thus only the seven Fourier modes $-3,\dots,3$ are possible.

There is one more symmetry. Reflection and reindexing imply

$$\Sigma(-\tfrac{1}{2}-t) = \cot^7(\pi t)\Sigma(t), \qquad \Xi(\tfrac{1}{2}-t) = -\Xi(t).$$

In the second equation periodicity has been used after multiplying by $\cos^7(\pi t)$. If $\Xi(t) = \sum_{k=-3}^{3} c_k e^{2\pi\mathrm{i}kt}$, this gives $c_0 = 0$ and $c_{-k} = -(-1)^k c_k$. Consequently

$$\boxed{\Xi(t) = \alpha\cos(2\pi t) + \beta\sin(4\pi t) + \gamma\cos(6\pi t).} \tag{142}$$

The coefficients are real because (139) is real at nonsingular real t. This establishes the three-dimensional space analytically; the coefficients have not been fitted by assuming an unproved trigonometric ansatz.

10.7.3 Exact finite sampling and evaluation of the period

Set

$$\begin{aligned} X_0 &= \Xi(0) = \mathcal{Y}_2, \\ X_+ &= \Xi(\tfrac{1}{8}) = c_8^7\mathcal{Y}_3 - s_8^7\mathcal{Y}_5, \\ X_- &= \Xi(-\tfrac{1}{8}) = c_8^7\mathcal{Y}_1 + s_8^7\mathcal{Y}_7. \end{aligned} \tag{143}$$

Equation (142) gives exactly

$$\begin{aligned} \alpha &= \frac{X_0}{2} + \frac{X_+ + X_-}{2\sqrt{2}}, \\ \beta &= \frac{X_+ - X_-}{2}, \\ \gamma &= \frac{X_0}{2} - \frac{X_+ + X_-}{2\sqrt{2}}. \end{aligned} \tag{144}$$

These are fixed, finite evaluation points; no spacing tends to zero and there is no interpolation remainder.

For completeness, the link to the previously derived Barnes integrals can be made without a numerical differentiation step. The first equation in (139) gives $\Sigma(t) = \mathcal{A}_{\rm s}(t) + O(t^7)$ at zero, since $\mathcal{A}_{\rm s}(1/2 - t)$ is analytic there. Hence

$$\begin{aligned} \mathcal{A}_{\rm s}(0) &= \alpha + \gamma, \\ \mathcal{A}'_{\rm s}(0) &= 4\pi\beta, \\ \mathcal{A}''_{\rm s}(0) &= \pi^2(3\alpha - 29\gamma). \end{aligned} \tag{145}$$

Substitution of (145) into (128) gives

$$P_{\bar{5}} = 16\pi^{3/2}(9\alpha - 26\beta - 79\gamma).$$

Using (144) therefore yields

$$P_5 = 16\pi^{3/2}[-35X_0 + (22\sqrt{2} - 13)X_+ + (22\sqrt{2} + 13)X_-], \tag{146}$$

which proves (29). Differentiation is used only in the derivation of (128), not in the final fixed-value formula.

10.7.4 All five parameter arrays

For clarity, the repeating parameters and the shifted pair in the five functions are all displayed below. Each row also has the numerator parameter 1, as in (28).

r	shifted upper	seven equal uppers	shifted lower	seven equal lowers
1	9/8	3/8	1/8	7/8
2	5/4	1/2	1/4	1
3	11/8	5/8	3/8	9/8
5	13/8	7/8	5/8	11/8
7	15/8	9/8	7/8	13/8

The positive denominator parameters rule out singular defining series. The only cancellation used is the valid, fixed-parameter one at $r = 2$. The eighths arise from the choice of the exact sampling points $t = \pm 1/8$, not from a change of physical dimension or from a new regulator. There is no claim that the five individual values are independent or that their appearance is minimal.

The eighths in the parameters arise from the fixed sample nodes $t = 0, \pm 1/8$. They are not new regulators or a change in physical dimension. At 125 working digits the independently evaluated Gamma integral and the five-value expression differed by 3.63×10^{-121}; this is a numerical consistency check rather than an interval enclosure. The period begins

$$P_5 = 28599.29317656210353664647702142205749032124603126 4 \\ 16921356927935177776442703824905147018458419833882\ldots.$$

11 Restoring finite regulator corrections

The preceding expansions are inserted in the forest calculation, not merely in bare uniqueness integrals. The general forest framework is standard [21]; the concrete deformations and order-by-order corrections below are part of this calculation.

Let the original edge powers be $m_e(1-z)/2$ and introduce

$$a_e(z,d) = m_e(1-z)/2 + d\, v_e.$$

The chosen v_e have zero sum over the entire graph and over every proper logarithmically divergent subgraph. Therefore d changes shapes of the integrals but not their logarithmic degrees. At fixed d, write

$$C_G(z,d) = \sum_{k\geq 1} \frac{c_{G,k}(d)}{z^k}, \qquad r_G(d) = c_{G,1}(d). \tag{147}$$

The higher-pole identity is

$$L_G c_{G,k}(d) = \sum_{\gamma \subsetneq G} L_\gamma r_\gamma(d) c_{G/\gamma,k-1}(d), \qquad k \geq 2, \tag{148}$$

where the sum is over connected logarithmic one-particle-irreducible proper subgraphs. In particular, the simple residue belongs to the subgraph, not to the quotient.

Define the higher-pole part

$$H_G(z,d) = \frac{1}{L_G} \sum_{\gamma \subsetneq G} \frac{L_\gamma r_\gamma(d)}{z}\, C_{G/\gamma}(z,d). \tag{149}$$

Let $\mathfrak{F}'$ denote nonempty sets of disjoint proper logarithmic subgraphs; nested subgraphs are already included inside each C_γ. The physical residue is obtained from an integrable diagonal $d = z$ by

$$\boxed{\begin{aligned} r_G(0) = {} & -\operatorname{Res}_{z=0}\left[I_G(z,z) + \sum_{F\in\mathfrak{F}'} \Big(\prod_{\gamma\in F} C_\gamma(z,z) \Big) I_{G/F}(z,z) \right] \\ & -\operatorname{Res}_{z=0} H_G(z,z). \end{aligned}} \tag{150}$$

Crucially, $C_\gamma(z,z)$ means first form the fixed-d counterterm and then set $d = z$. It is not the counterterm obtained by setting $d = z$ before subtraction.

To prove (150), expand the fixed-d forest-subtracted amplitude. Its poles are $-C_G(z,d)$. On the diagonal, the coefficient of z^{-1} receives $-r_G(0)$ and also Taylor coefficients of the higher poles. Equation (148) identifies these additional terms as $-\operatorname{Res} H_G(z,z)$, giving the stated formula. This retains precisely the finite regulator corrections that were missing from bare uniqueness reductions.

11.1 Seven regulator-corrected graph classes

For six vertices use edge-word order

$$(01, 02, 03, 04, 05, 12, 13, 14, 15, 23, 24, 25, 34, 35, 45).$$

Class numbers are zero-based positions of lexicographically minimal edge words in the sorted 129-class list. The shifts used are:

Class	Nonzero v_e	Fixed roots
52	$v_{13} = -2,\ v_{23} = 2$	$0, 4$
56	$v_{14} = -2,\ v_{24} = 2$	$0, 4$
60	$v_{14} = -2,\ v_{45} = 2$	$0, 2$
79	$v_{23} = 2,\ v_{15} = -2$	$0, 4$
80	$v_{12} = -2,\ v_{13} = 2$	$0, 4$
91	$v_{14} = 2,\ v_{15} = -2,\ v_{25} = 2,\ v_{34} = -2$	$0, 4$
127	$v_{05} = -4,\ v_{14} = 2,\ v_{23} = 2$	$1, 2$

The exceptional four-loop class with word `0011101211` uses $v_{24} = 2, v_{34} = -2$, with roots $0, 1$. Its shift is antisymmetric under a graph automorphism, so the first derivative of its simple residue vanishes. Its higher-pole derivatives are still retained through (148).

Ordinary degree-three unique stars reduce six of the five-loop classes. Class 79 uses a two-internal-vertex conformal three-point block. On its diagonal both internal vertices have incident power sum two, and conformal covariance fixes the external dependence. With $h = (1 - z)/2$, its bare integral becomes

$$I^U_{79}(z) = \frac{2}{5z} M(h, h, h + 2z)\mathsf{B}(h, 1 - 3z)\mathsf{B}(h, 1 - 4z). \tag{151}$$

It is gamma-valued. The regulator-restoration terms in (150) use the tetrahedron Hessian (46). They give

$$r_{79} = 3Q\left[-\frac{13A^2}{20} + \frac{\pi^2}{12} + \frac{37H}{60}\right] = Q\left[-\frac{39A^2}{20} + \frac{89\pi^2}{60} + \frac{74\mathsf{G}}{5}\right]. \tag{152}$$

Dropping those restoration terms would not give this residue.

The complete resulting seven residues are recorded compactly below. Define

$$X = -\frac{25A^4}{12} + \frac{5A^2\pi^2}{6} + \frac{52A\pi^3}{15} + \frac{512A\zeta(3)}{5} - \frac{11\pi^4}{900}, \tag{153}$$

$$Y = Q\left[-\frac{39A^2}{20} + \frac{89\pi^2}{60} + \frac{74\mathsf{G}}{5}\right]. \tag{154}$$

Then

$$r_{52} = Q\left[-\frac{3A^2}{2} + \frac{41\pi^2}{30} + \frac{76\mathsf{G}}{5}\right], \tag{155}$$

$$r_{56} = X - \frac{16}{5}\mathcal{K}, \tag{156}$$

$$r_{60} = -\frac{47A^4}{240} - \frac{11A^2\pi^2}{120} + \frac{19A\pi^3}{60} + \frac{43A\zeta(3)}{5} + \frac{3\pi^4}{400}, \tag{157}$$

$$r_{79} = Y, \tag{158}$$

$$r_{80} = X + \frac{4}{5}\mathcal{K}, \tag{159}$$

$$r_{91} = -\frac{79A^4}{80} - \frac{43A^2\pi^2}{360} + \frac{17A\pi^3}{12} + \frac{117A\zeta(3)}{5} - \frac{623\pi^4}{3600}, \tag{160}$$

$$r_{127} = Y. \tag{161}$$

These are unweighted counterterm residues. Their graph weights, words, and the residue expressions for all other required classes are in the residue certificate of Appendix A, and in machine-readable form in `lri_numerical_resolution/verification/inputs/residue_certificate.json`.

12 The six nonprimitive wheel-insertion topologies

A primitive graph has no ultraviolet-divergent proper subgraph. A nonprimitive graph requires subtraction of such subgraphs in addition to its overall pole. The six classes in this section contain a one-loop bubble or a four-loop wheel subgraph. They were evaluated separately from the remaining master-integral classes and must be included exactly once.

Using the edge-word order of Section 11, define $d_G = -5(3w_2 - w_4)/|\operatorname{Aut} G|$. The unweighted counterterm residues are

Class	Edge word	$\lvert\operatorname{Aut} G\rvert$	d_G	r_G
51	000110111111110	4	15/2	$-2AP_4/5 - 3\mathcal{L}/10$
78	000111101110111	2	15	$-2AP_4/5 - \mathcal{L}/10$
99	000120111111100	2	30	$-2AP_4/5 - \mathcal{L}/10$
113	000121101110110	1	60	$-2AP_4/5 + \mathcal{L}/10$
124	001110111112000	8	15/2	$-2AP_4/5 - \mathcal{L}/2$
128	001111011201100	4	15	$-2AP_4/5 - 3\mathcal{L}/10$

The table includes the subdivergence subtractions; these are not bare integrals. Weighted summation gives

$$(3c_5 - b_5)_{\text{wheel insertions}} = -54AP_4 - 9\mathcal{L}. \tag{162}$$

Since $b_4^{\text{wheel}} = -6P_4$ and $c_4^{\text{wheel}} = 0$, the lower-loop assembly supplies $60AP_4$. Thus

$$C_5^{\text{wheel+assembly}} = \frac{(-54AP_4 - 9\mathcal{L}) + 60AP_4}{729} = \frac{2AP_4 - 3\mathcal{L}}{243}. \tag{163}$$

The finite star expansion and the regulated wheel derivatives reduce the six residues to P_4 and $\mathcal{L}$. The test programs are not part of the public repository; the detailed derivation is given below. This sector is not the entire nonprimitive contribution.

12.1 Derivation of the residue table

12.1.1 First regulator derivative of the full-line star

Introduce

$$F_h(x) = \int_{\mathbb{R}} |u(1-u)(u-x)|^{-h}\, du.$$

For $0 < x < 1$, split the real line at $0, x, 1$ to obtain

$$\begin{aligned} F_h(x) = B(1-h,1-h)\Big[&x^{1-2h}{}_2F_1(h,1-h;2-2h;x) \\ &+ (1-x)^{1-2h}{}_2F_1(h,1-h;2-2h;1-x)\Big] \\ + B(1-h,3h-1)\Big[&{}_2F_1(h,3h-1;2h;x) + {}_2F_1(h,3h-1;2h;1-x)\Big]. \end{aligned} \tag{164}$$

At $h = 1/2$, the lower-parameter derivative terms cancel in the sum. Euler's transformation [23] gives

$$\partial_a {}_2F_1(a,b;1;x)\big|_{a=b=1/2} = -\frac{1}{2}\log(1-x)\,{}_2F_1(1/2,1/2;1;x).$$

Using this in (164), the beta-function derivative terms cancel as well, leaving

$$\boxed{\partial_h F_h(x)|_{h=1/2} = -\log|x(1-x)|F(x).} \tag{165}$$

The tails follow from $F_h(x) = x^{1-3h}F_h(1/x)$ for $x > 1$ and the corresponding negative-tail transformation.

In the opposite-outer-root representation of the wheel, put

$$P_W(z) = \int_{\mathbb{R}} |x(1-x)|^{-h} F_h(x)^2\,dx, \qquad h = (1-z)/2.$$

Equation (165) implies

$$\boxed{P_W'(0) = \frac{3}{2}\mathcal{L}.} \tag{166}$$

12.1.2 Edge derivatives and root changes

In this rooted wheel, the fixed roots are opposite outer vertices. The remaining variables are the center x and the two other outer vertices. Let D_eP denote the derivative with respect to an independent edge power at the point where all eight powers are $1/2$.

The completed wheel is the octahedral graph. For conformally balanced index variations, averaging over its edge-transitive automorphism group makes the first variation vanish. In the opposite-root representation, the allowed deformations obey zero total edge shift and zero sum of spoke shifts. It follows that the index gradient is constant separately on the four rim edges and the four spokes. The two root-to-center spokes differentiate the explicit factor $|x(1-x)|^{-1/2}$, so each contributes $-\mathcal{L}/2$. Equation (165) for one outer integrated vertex then fixes the rim derivative. Thus

$$\boxed{D_{\text{spoke}}P = -\frac{1}{2}\mathcal{L}, \qquad D_{\text{rim}}P = -\frac{1}{4}\mathcal{L}.} \tag{167}$$

Changing the fixed outer pair from opposite to adjacent gives, by the scale Jacobian,

$$P_W^{\text{adj}\,\prime}(0) = P_W^{\text{opp}\,\prime}(0) + 4D_{\text{rim}}P = \frac{1}{2}\mathcal{L}.$$

Only these first-derivative statements are needed; they are not identities for arbitrary higher index variations.

12.1.3 Bubble insertions on a rim or spoke

The full-line convolution coefficient is

$$\mathsf{B}(u,v) = \sqrt{\pi}\,\frac{\Gamma((1-u)/2)\Gamma((1-v)/2)\Gamma((u+v-1)/2)}{\Gamma(u/2)\Gamma(v/2)\Gamma((2-u-v)/2)}.$$

In particular, $\mathsf{B}(h,2h) = 2/z + A + O(z)$. For classes 99 and 113 a degree-three bubble endpoint can be integrated exactly. The rooted bare integral is

$$I_G(z) = \frac{2}{5z}\mathsf{B}(h,2h)P_e(z), \qquad P_e'(0) = P_W'(0) - D_eP.$$

The bubble counterterm contributes $-P_W(z)/z^2$ to the subtraction bracket. Its simple-pole extraction yields

$$r_G = -\frac{2}{5}AP_4 + \frac{1}{5}P_W'(0) + \frac{4}{5}D_eP.$$

The modified quotient edge is a spoke for class 99 and a rim for class 113. Equations (166)–(167) give their table entries.

12.1.4 Splitting the degree-four center

For classes 124 and 128 one needs the finite part of a star with one doubled edge. The local distributional expansion is

$$\begin{aligned}&\int_{\mathbb{R}} |u-y|^{-h}|u-z_1|^{-h}|u-x|^{-1+z}\,du\\&\quad= |x-y|^{-h}|x-z_1|^{-h}\left[\frac{2}{z}+A+2\log\frac{|x-y|\,|x-z_1|}{|y-z_1|}\right]+O(z).\end{aligned}\tag{168}$$

One derivation shifts the two single-edge powers from h to $h+z$, applies the unique star–triangle integral, and restores the finite logarithmic terms from the local bubble residue. Those logarithms cannot be dropped.

After subtraction,

$$r_G=-\frac{2}{5}AP_4+\frac{1}{5}P'_W(0)-\frac{4}{5}\mathcal{L}_{\triangle},$$

where $\mathcal{L}_{\triangle}$ inserts the logarithm in (168) into the wheel. For opposite neighbors, $\mathcal{L}_{\triangle}=\mathcal{L}$. For adjacent neighbors, (167) gives $\mathcal{L}_{\triangle}=3\mathcal{L}/4$. This proves classes 124 and 128.

12.1.5 A wheel inserted into the outer one-loop graph

Classes 51 and 78 have only the wheel as a proper ultraviolet-divergent subgraph. Invert about the degree-two root and fix the other root at one. The additional powers produced by inversion differ from their zero-regulator values by $O(z)$; their logarithms vanish at the only proper divergent collision, so they do not affect the constant term of the angular integral. One obtains

$$J_G(z)=\mathsf{B}(h+2z,1-4z)P_W^{\text{root pair}}(z)+O(z),\quad \mathsf{B}(h+2z,1-4z)=\frac{1}{2z}+A+O(z).$$

The wheel counterterm times the bubble quotient contributes only a double pole. Hence

$$r_G=-\frac{2}{5}AP_4-\frac{1}{5}P_W^{\text{root pair}\,\prime}(0).$$

The pair is opposite for class 51 and adjacent for class 78, proving the remaining entries.

13 Assembly of the full coefficient

Write

$$\beta(g)=-\epsilon g+\sum_{L\ge1}b_Lg^{L+1},\qquad \gamma_r(g)=\sum_{L\ge1}c_Lg^L.$$

The long-range kinetic term has no wave-function counterterm in this setup [7]. At the nonzero fixed point,

$$\Delta_{\mathcal{E}}=(1-\epsilon)/2+\gamma_r(g_*),\qquad \beta(g_*)=0.$$

With $b_1=3, c_1=1, b_2=-6A, c_2=-A$ and

$$b_3=6Q+\frac{87}{4}A^2-\frac{3}{4}\pi^2-4\pi,\qquad c_3=\frac{7}{2}A^2-\frac{1}{6}\pi^2-\frac{4}{3}\pi,$$

Lagrange inversion gives

$$\begin{aligned}729C_5=&\,336A^4-60A^2b_3+108A^2c_3+b_3^2-3b_3c_3\\&-10Ab_4+24Ac_4+3c_5-b_5.\end{aligned}\tag{169}$$

The primitive and wheel-insertion sectors are

$$C_5^{\text{primitive}} = -\frac{63Q^2 + 24P_5}{729}, \qquad C_5^{\text{wheel+assembly}} = \frac{2AP_4 - 3\mathcal{L}}{243}. \tag{170}$$

The result for all other subtracted terms, including their lower-loop assembly terms, is

$$\boxed{\begin{aligned} C_5^{\text{rest}} = &-\frac{5A^4}{15552} + \frac{5A^2Q}{324} - \frac{5A^2\pi^2}{7776} - \frac{23A\pi^3}{1296} - \frac{149A\zeta(3)}{324} \\ &+ \frac{298\mathsf{G}Q}{243} - \frac{8\mathcal{K}}{243} + \frac{4Q^2}{81} + \frac{319Q\pi^2}{2916} - \frac{83\pi^4}{46656}. \end{aligned}} \tag{171}$$

The coefficient of the only new master constant is $-8/243$. Adding the three sectors and using $H = 2\pi^2/3 + 8\mathsf{G}$ proves (27) by exact algebra.

The enumeration and weighting are fixed by the following graph construction. With $e_i = 4 - \sum_j m_{ij}$,

$$w_4 = \frac{24}{\prod_i e_i! \prod_{i<j} m_{ij}!}, \qquad w_2 = \frac{w_4}{12} \sum_i e_i(e_i - 1),$$

and an unlabeled L-loop class contributes

$$b_L = L(-1)^L \frac{w_4}{|\operatorname{Aut} G|} r_G, \qquad c_L = L(-1)^L \frac{w_2}{|\operatorname{Aut} G|} r_G.$$

The 129 five-loop classes are exhausted by 92 explicitly evaluated remaining classes, six wheel-insertion classes, three primitive classes, and 28 classes whose $3w_2 - w_4$ vanishes. The 26 non-wheel four-loop classes are also included. These disjoint sets cover the full enumeration; no unknown residue is hidden in (171).

14 Independent full-graph numerical control

The following control checks the assembled coefficient independently of special-function evaluation. Only the precision-preserving formulation and its converged comparison are needed here. The code and records are in the package `lri_numerical_resolution/` (Section 33).

The graph control uses the two coordinate cutoffs and meromorphic Chebyshev quadrature defined in Section 22, specialized to five loops: the cutoff "maxgap" requires the largest gap between consecutive ordered vertices to be below one, and the cutoff "span" requires the total vertex span to be below one. Below, N_{lower} is the quadrature degree of the lower-loop inputs. In particular its integrand is (600) with $L = 5$, and its radial moments are (601). The bare amplitudes are renormalized by (20) before applying (11). Thus the special-function period values are not inputs to this independent integration.

The combined bare simple pole is

$$H_{5,-1} = [z^{-1}](3\mathcal{B}_5 - \mathcal{A}_5).$$

Its exact sensitivity, with all lower-loop inputs held fixed, is

$$\delta C_5 = \frac{5}{729}\, \delta H_{5,-1}.$$

Indeed the new fifth-order contribution is $(3c_5 - b_5)/729$, while (25) gives $\delta(3c_5 - b_5) = 5\delta H_{5,-1}$. The program therefore forms the exact integer channel weights before integration; this removes 9,540 zero-weight ordered graphs from the 59,145-graph inventory, rather than subtracting two noisy channel sums only at the end. Quadrature weights, smooth-kernel evaluation, within-group

accumulation and final Laurent assembly are all precision-controlled. A same-degree comparison with the full two-channel calculation changes C_5 by less than 1.7×10^{-12}. The combined channel has 335 rather than 342 sector exponent groups, and exhaustive exact enumeration shows that the groups with exponent $a = -2$ disappear from it: its most negative exponent is $-\frac{3}{2}$. At degree 18, replacing only the binary64 five-loop kernel by the extended one, with the same lower-loop inputs, changes C_5 by -7.957×10^{-9} (span) and -7.965×10^{-9} (maxgap), most of the offset of the earlier version described after the table below. For an independent pole check, the renormalization-group equations predict the higher poles $H_{5,-p}$, $p \geq 2$, from the lower loops; the largest residuals are 1.96×10^{-11} (span) and 1.74×10^{-11} (maxgap). They are diagnostics only and do not certify the simple pole, which carries the new information.

With lower-loop degree $N_{\text{lower}} = 32$, the complete results are

Degree	Cutoff	Graph value of C_5	Graph value minus closed form
22	span	-731.1394766245375925	$+2.2334 \times 10^{-12}$
22	maxgap	-731.1394766245106192	$+2.9207 \times 10^{-11}$
24	span	-731.1394766245391892	$+6.3667 \times 10^{-13}$
24	maxgap	-731.1394766245368057	$+3.0201 \times 10^{-12}$

The comparison target is the special-function evaluation $-731.13947662453982585386912658252 7\ldots$. These are observed residuals, not rigorous bounds on either representation. The cutoff difference alone is not an accuracy certificate: a shared regular contribution can have a common error that cancels in that difference. The complete-channel and precision/degree comparisons are therefore separate checks. An earlier version of this control, whose quadrature kernel used binary64 arithmetic, gave -731.1394766152787 with the span cutoff, 9.26×10^{-9} above the closed form. The precision-preserving formulation removes that offset; the analytic coefficient was not changed. The signed error ledger of the earlier value is in `lri_numerical_resolution/verification/records/resolution_audit.json` (field `error_budgets`).

14.1 Further checks of the analytic coefficient

The following checks of the one-dimensional result were run with the analytic calculation and do not use the full graph integration above.

- *Exact algebra.* The two regular quadratic coefficients at pattern A_0 vanish; the fourth-order weighted sum reproduces the analytic $3c_4 - b_4$, including all gamma and zeta terms; every five-loop class is accounted for exactly once in the nonzero, primitive, wheel or zero-weight sets; and the rationally weighted residue table sums to (171), whose addition to the primitive and wheel sectors gives (27). No coefficient is inferred from a numerical integer-relation fit (`lri_numerical_resolution/verification/records/symbolic_audit.json`).

- *Seven subtracted graphs.* With extended precision in the quadrature weights, integrands and accumulation, seven classes were integrated anew, with the analytic residues used only as comparison targets. At interpolation degree 18 the numerical minus analytic residues are -1.02×10^{-9} (class 52), -4.95×10^{-9} (56), $+3.31 \times 10^{-10}$ (60), -2.25×10^{-9} (79), -5.19×10^{-10} (80), -5.82×10^{-11} (91) and $+2.27 \times 10^{-10}$ (127); at degree 22 those of classes 56 and 79 fall to -1.02×10^{-11} and -1.92×10^{-11}. The lower-loop inputs of these comparisons were evaluated as bare span-cutoff integrals. The checks cover gamma-valued and $\mathcal{K}$-dependent residues; their record is not part of the repository.

- *Fresh evaluation of the masters.* Evaluating all four masters without stored period values, at 65 working digits with 64 Cauchy nodes for the parameter derivatives, reproduces $C_5 = -731.13947662453982585386912658252723066 25\ldots$; a separate 28-digit run agrees to its

requested precision (record not part of the repository; the replay of Section 33 recomputes C_5 from the closed forms).

- *Regenerated certificate.* In the code delivery that preceded this note, the symbolic task regenerated all 26 non-wheel four-loop and 92 retained five-loop residues of Appendix A from the graph-class data, the master expansions and the forest rules, without reading a saved residue; all 118 comparisons were exact. That record is not part of the repository. Regenerating the table is stronger than adding the saved rows, but it uses the same master identities and reduction code.

These tests, like the graph integration, do not establish the physical five-loop coefficient independently of the shared graph and renormalization framework.

14.2 External lattice comparison: informative but inconclusive

The Monte Carlo data of Uzelac, Glumac and Aničić concern the same classical long-range chain, with $J(r) \propto r^{-(1+s)}$ [11]. Their Table II gives $y_t = \nu^{-1}$, which here is $1 - \Delta_{\mathcal{E}}$, at three couplings. An earlier comparison resummed the series of y_t through ϵ^5 in six Borel–Padé–Leroy forms: the approximants [3/2] and [2/3] of the Borel–Leroy transform $\sum_k c_k t^k/\Gamma(k+b+1)$, with $b = 0, 1, 2$, followed by the Laplace integral $\int_0^\infty e^{-t} t^b [L/M](\epsilon t)\, dt$. None of the six approximants has a positive real pole at these couplings. No Monte Carlo value was fitted.

s	ϵ	y_t of [11]	Resummation range
0.6	0.2	0.50 ± 0.02	0.51738–0.51774
0.7	0.4	0.51 ± 0.02	0.51518–0.51886
0.8	0.6	0.47 ± 0.02	0.50021–0.51231

The third row shows a tension, and it must not be omitted from the interpretation: at $s = 0.8$ all six resummed values lie above the Monte Carlo interval, the lowest by 0.010. The resummation spread is not a confidence interval. Changing C_5 by 20% in either direction still gives pole-free resummed values inside the Monte Carlo interval nearest to the expansion point, at $s = 0.6$, so these data do not isolate the fifth coefficient. With C_5 lowered by 20%, however, the three [3/2] approximants acquire positive real Borel poles at all three couplings, and only the three [2/3] values remain; with C_5 raised by 20%, all six stay pole-free. The closed-form $C_0, \ldots, C_5$ of Section 4 reproduce the three ranges to the digits shown: `lri_numerical_resolution/public/lattice_comparison.py` recomputes the ranges, the absence of positive Borel poles and the 20% statement. No simulation was run for this note.

Part III

Arbitrary-dimensional analytic coefficients and specializations

15 General-dimensional normalization and lower-order derivation

15.1 Dimension, regulator and normalization

In this section, d always denotes the spatial dimension, not the dummy exponent called d in some earlier one-dimensional subintegrals. Take

$$0 < d < 4, \qquad \epsilon = 2s - d, \qquad \zeta = \frac{s}{2} = \frac{d+\epsilon}{4}, \qquad \Delta_\phi = \frac{d-\epsilon}{4}. \tag{172}$$

The physical long-range kinetic term has $0 < \zeta < 1$; at fixed $d < 4$ this holds for sufficiently small positive ϵ. Noninteger d is defined by analytic continuation of the dimensionally parameterized integrals. Analytically continuing individual periods to $d = 4$ is useful as a check, but it is not the same as replacing this expansion by the short-range Wilson–Fisher expansion: the absence of a wave-function counterterm, used below, is a long-range statement [7, 12].

After tuning the relevant local mass term to the critical surface,

$$\Delta_{\mathcal{E}}(d,\epsilon) = d - \nu^{-1} = \frac{d-\epsilon}{2} + \gamma_r(g_*), \qquad \beta(g) = -\epsilon g + \sum_{\ell\geq 1} b_\ell(d) g^{\ell+1}, \qquad \gamma_r(g) = \sum_{\ell\geq 1} c_\ell(d) g^\ell. \tag{173}$$

We use a minimal-subtraction convention with $b_1 = 3$, $c_1 = 1$ and an exactly pole-only one-loop bubble. Write z for the analytic regulator and introduce the infrared covariance $(p^2+\mu^2)^{-(d+z)/4}$. The bare coupling normalization is

$$\lambda_0 = \mu^z \kappa(d,z) \mathcal{Z}_g(g,z), \qquad \kappa(d,z) = (4\pi)^{d/2} \frac{\Gamma((d+z)/2)}{\Gamma(1+z/2)}. \tag{174}$$

The parameter μ is an infrared regulator, not the coefficient of the physical ϕ^2 perturbation. A direct momentum integration gives

$$\kappa(d,z) \int \frac{d^d p}{(2\pi)^d} (p^2+1)^{-(d+z)/2} = \frac{2}{z}. \tag{175}$$

Hence the one-loop four-point and mass amplitudes are $3/z$ and $1/z$. The extra z-dependent normalization in (174) must be kept before extracting Laurent coefficients. Finite parts computed in this massive-Schwinger prescription must not be mixed with finite parts from the one-dimensional ordered-span prescription of Section 3.

15.2 The evaluated coefficients through the third order

Let

$$\begin{aligned} q &= d/4, & A_d &= \psi_0(2q) + \psi_0(1) - 2\psi_0(q), \\ R_d &= \psi_1(2q) - \psi_1(1), & M_d &= \frac{\Gamma(-q)\Gamma(2q)^2}{\Gamma(3q)} = -\frac{\Gamma(1-q)\Gamma(2q)^2}{q\Gamma(3q)}, \end{aligned} \tag{176}$$

and let $\mathcal{T}_d$ be the corrected tetrahedron beta-function coefficient α_{I_4} of [12]. Its independent integral definition is given in (188); put

$$Q_d = \frac{\mathcal{T}_d}{6}. \tag{177}$$

In particular $Q_1 = Q$, but Q_d is not defined by replacing the argument of the Gamma functions in the one-dimensional value Q. The coefficient formulas are

$$\boxed{\begin{aligned} &C_0(d) = \frac{d}{2}, && C_1(d) = -\frac{1}{6}, \quad C_2(d) = \frac{A_d}{9}, \\ &C_3(d) = \frac{A_d^2 + R_d - 8Q_d}{108} = \frac{A_d^2 + R_d}{108} - \frac{\mathcal{T}_d}{81}. \end{aligned}} \tag{178}$$

Here and below the polygamma convention is $\psi_m(z) = d^{m+1}\log\Gamma(z)/dz^{m+1}$. The corresponding three-loop MS coefficients in (174) are

$$\boxed{\begin{aligned} &b_1 = 3, && c_1 = 1, && b_2 = -6A_d, \quad c_2 = -A_d, \\ &b_3 = 6Q_d + \frac{87}{4}A_d^2 - \frac{9}{4}R_d + M_d, && c_3 = \frac{7}{2}A_d^2 - \frac{1}{2}R_d + \frac{1}{3}M_d. \end{aligned}} \tag{179}$$

The M_d term cancels from the physical $C_3(d)$, but it cannot be dropped from the RG functions used at fourth and fifth order.

At $d = 1$,

$$A_1 = A, \qquad R_1 = \frac{\pi^2}{3}, \qquad M_1 = -4\pi, \qquad Q_1 = \frac{\Gamma(1/4)^4}{\pi}. \tag{180}$$

This recovers (9) and the one-dimensional C_3 of Section 4. Equation (176) also displays a concrete obstruction to blind argument substitution: $3q$ and $1-q$ cancel in M_d only at $d = 1$. In addition, the reflected identity for R_d is not obtained by keeping $\psi_1(q) + \psi_1(1-q) = 2\pi^2$ away from $q = 1/4$.

15.2.1 Derivation from the published three-loop calculation

The three-loop beta functions of [7], Eqs. (2.13) and (2.14) there, are sums over graphs with coefficients α_X: α_D for the one-loop graph, α_S for the two-loop graph, and α_U, α_T and $\alpha_{I_1}, \ldots, \alpha_{I_4}$ for the seven three-loop graphs, each the normalized graph amplitude of Eq. (2.12) there. The tetrahedron coefficient α_{I_4} given in [7] was corrected in [12]; the corrected value is required here. For clarity, let $t_1 = \psi_1(1)$, $t_q = \psi_1(q)$, $t_2 = \psi_1(2q)$ and $J = J_0(q)$. The coefficients quoted in [12], specialized to one scalar, are

$$\begin{aligned} \alpha_U = \alpha_{I_2} &= -t_1 - t_q + 2t_2 + J, \\ \alpha_T &= \tfrac{1}{2}A_d^2 + \tfrac{1}{2}t_1 + t_q - \tfrac{3}{2}t_2 - J, \\ \alpha_{I_1} &= \tfrac{3}{2}A_d^2 + \tfrac{1}{2}(t_1 - t_2), \qquad \alpha_{I_3} = M_d/3, \qquad \alpha_{I_4} = 6Q_d. \end{aligned} \tag{181}$$

An explicit definition, useful for reproducing the intermediate cancellation, is

$$\begin{aligned} J_0(q) = \frac{1}{\Gamma(q)^2}\sum_{n=1}^{\infty} &\frac{\Gamma(n+2q)\Gamma(n+q)^2}{n\,n!\,\Gamma(2q+2n)} \\ &\times \left[2\psi_0(n+1) - \psi_0(n) - 2\psi_0(n+q) - \psi_0(n+2q) + 2\psi_0(2q+2n)\right]. \end{aligned} \tag{182}$$

After dividing the zero-momentum one-loop normalization out of the coupling, write $\alpha_S/\alpha_D^2 = -A_d + s_1 z + O(z^2)$. The expansion in [12] gives

$$s_1 = \frac{1}{4}A_d^2 + \frac{3}{4}t_1 + t_q - \frac{7}{4}t_2 - J. \tag{183}$$

The two-loop bare amplitudes are then

$$\mathcal{A}_2 = \frac{9}{z^2} + \frac{3A_d}{z} - 3s_1 + O(z), \qquad \mathcal{B}_2 = \frac{2}{z^2} + \frac{A_d}{2z} - \frac{s_1}{2} + O(z). \tag{184}$$

Consequently, at $z = 0$, the zero-momentum and MS variables obey

$$g_{\mathrm{ZM}} = g_{\mathrm{MS}} - 3s_1 g_{\mathrm{MS}}^3 + O(g^4), \qquad r_{\mathrm{ZM}} = r_{\mathrm{MS}}(1 - \tfrac{1}{2}s_1 g_{\mathrm{MS}}^2 + O(g^3)). \tag{185}$$

These finite transformations imply

$$\begin{aligned} b_3^{\mathrm{MS}} &= 3\alpha_T + 12\alpha_U + 12\alpha_{I_1} + 3\alpha_{I_3} + \mathcal{T}_d + 9s_1, \\ c_3^{\mathrm{MS}} &= \alpha_T + 2\alpha_{I_1} + \alpha_U + \alpha_{I_3}. \end{aligned} \tag{186}$$

Substitution yields (179): all J and t_q terms cancel. Alternatively, the physical coefficient can be checked directly before choosing the MS scheme:

$$\begin{aligned} C_3 &= -\frac{1}{81}\left[-12A_d^2 + 9s_1 + 9\alpha_U + 6\alpha_{I_1} + \mathcal{T}_d\right] \\ &= \frac{A_d^2 + R_d}{108} - \frac{\mathcal{T}_d}{81}. \end{aligned} \tag{187}$$

The cancellation is symbolic; it is not an integer-relation fit.

15.3 A specified tetrahedron at every dimension

The equal-index specialization of the corrected kite representation in [12], eqs. (3.8)–(3.11), is

$$\boxed{\begin{aligned} \mathcal{T}_d = \frac{2\Gamma(2q)^2}{\Gamma(q)^4} \int_{-q/3-\mathrm{i}\infty}^{-q/3+\mathrm{i}\infty} \frac{du}{2\pi\mathrm{i}} \int_{-q/3-\mathrm{i}\infty}^{-q/3+\mathrm{i}\infty} \frac{dv}{2\pi\mathrm{i}} \\ \times \frac{\Gamma(-u)^2\Gamma(-v)^2\Gamma(q+u)\Gamma(q+v)\Gamma(q+u+v)^2\Gamma(-u-v)}{\Gamma(q-u)\Gamma(q-v)\Gamma(2q+u+v)}. \end{aligned}} \tag{188}$$

Both contours run upwards. Every numerator Gamma argument has positive real part on these contours for $q > 0$, so no pole separation is left implicit. If $u = -q/3 + \mathrm{i}x$, $v = -q/3 + \mathrm{i}y$, the exponential part of Stirling's bound is $\exp[-\pi(|x| + |y| + |x+y|)]$; the double integral is absolutely convergent. There are no hidden regulator derivatives in this representation. It is a twofold Mellin–Barnes integral, not a claim of an ordinary single-variable hypergeometric evaluation for generic d.

For an independent geometric normalization, let $|e| = 1$ and

$$\begin{aligned} P_3^{\mathrm{raw}}(d) &= \int_{\mathbb{R}^d} d^dx \int_{\mathbb{R}^d} d^dy \, \left[x^2(x-e)^2y^2(y-e)^2(x-y)^2\right]^{-q}, \\ h_d &= \frac{\Gamma(d/2)}{\pi^{d/2}}, \\ \mathcal{T}_d &= 2h_d^2 P_3^{\mathrm{raw}}(d), \qquad Q_d = \frac{h_d^2}{3} P_3^{\mathrm{raw}}(d). \end{aligned} \tag{189}$$

The spatial integral defines the same analytic family; the MB formula provides a direct meaning when d is not an integer.

15.3.1 An exact two-dimensional Gamma evaluation

For $d = 2$ identify $\mathbb{R}^2$ with $\mathbb{C}$. Then

$$P_3^{\mathrm{raw}}(2) = \int_{\mathbb{C}^2} \frac{d^2z\, d^2w}{|z|\,|1-z|\,|w|\,|1-w|\,|z-w|}. \tag{190}$$

Write $\gamma_C(a) = \Gamma(a)/\Gamma(1-a)$ for real equal holomorphic and antiholomorphic parameters. The complex Selberg identity [6], with $n = 2$, $\sigma = \tau = 1/2$, $\theta = -1/4$, gives

$$\begin{aligned} P_3^{\mathrm{raw}}(2) &= 2\pi^2 \prod_{j=1}^{2} \frac{\gamma_C(1/2 + (j-1)\theta)^2 \gamma_C(j\theta)}{\gamma_C(1+j\theta)\gamma_C(\theta)} \\ &= \frac{\pi^2}{2}\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4. \end{aligned} \tag{191}$$

The convergence inequalities hold strictly: $\theta > -1/2$, $\sigma+\theta = \tau+\theta = 1/4 > 0$, $\sigma+\tau+\theta = 3/4 < 1$, and $\sigma+\tau+2\theta = 1/2 < 1$. Thus this is a convergent integral evaluation, not a formal continuation of a divergent Selberg integral. Using $h_2 = 1/\pi$ yields

$$\boxed{\begin{aligned} \mathcal{T}_2 &= \left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4, \qquad Q_2 = \frac{1}{6}\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4, \\ C_2(2) &= \frac{4\log 2}{9}, \qquad C_3(2) = \frac{4\log^2 2}{27} - \frac{1}{81}\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4. \end{aligned}} \tag{192}$$

The fact that quarter-arguments reappear at $d = 2$ here is a consequence of the complex Selberg reduction, not a justification for replacing arguments in the $d = 1$ formulas.

15.4 Dimension-independent graph data and the d-dimensional kernel

Use precisely the finite matrix set in (12): $L+1$ vertices, $2L$ internal edges, $m_{ij} \in \{0,1,2,3\}$, internal degrees between two and four, and every nontrivial cut containing at least two edges. The weights w_4, w_2 are those in (13). In a full d-dimensional integral there is *no ordering of vertices on a line*. Therefore the labeled sum now includes the vertex factorial:

$$\boxed{\begin{aligned} \mathcal{A}_L(d,z) &= \frac{1}{(L+1)!}\sum_{m\in\mathfrak{G}_L} w_4(m)\mathcal{I}_m(d,z), \\ \mathcal{B}_L(d,z) &= \frac{1}{(L+1)!}\sum_{m\in\mathfrak{G}_L} w_2(m)\mathcal{I}_m(d,z). \end{aligned}} \tag{193}$$

Equivalently, sum one representative of each vertex-isomorphism class with weights $w_4/|\operatorname{Aut}_v(m)|$, $w_2/|\operatorname{Aut}_v(m)|$. Parallel-edge factorials are already present in w_4, w_2; Aut_v counts vertex permutations, not a second set of edge permutations.

Let the $2L$ individual internal lines, including parallel lines, be labeled $e = 1, \dots, 2L$. The first Symanzik polynomial and simplex measure are

$$U_m(x) = \sum_{\text{spanning trees } T}\prod_{e\notin T} x_e, \qquad d\Omega = \delta\!\left(1 - \sum_{e=1}^{2L} x_e\right)\prod_{e=1}^{2L} dx_e. \tag{194}$$

Put $a = (d+z)/4$ and $r(d,z) = \Gamma(2a)/\Gamma(1+z/2)$. The complete normalized graph kernel is

$$\boxed{\mathcal{I}_m(d,z) = \frac{r(d,z)^L\Gamma(Lz/2)}{\Gamma(a)^{2L}}\int_{x_e\ge 0} d\Omega\,\frac{\prod_{e=1}^{2L} x_e^{a-1}}{U_m(x)^{d/2}}.} \tag{195}$$

For a primitive graph it suffices to set $z = 0$ in the simplex factor when computing the residue. For nonprimitive graphs one must first meromorphically continue the integral and apply the subtraction recursion; the simplex integral at $z = 0$ need not converge.

To derive (195), apply a Schwinger parameter to every $(p^2+1)^{-a}$, perform the L Gaussian momentum integrals, and set the Schwinger parameters equal to λx_e. The powers of the common scale are

$$\lambda^{2La-1-dL/2} = \lambda^{Lz/2-1}, \qquad \int_0^\infty e^{-\lambda}\lambda^{Lz/2-1}d\lambda = \Gamma(Lz/2).$$

The factors $(4\pi)^{-Ld/2}$ cancel against (174). For the one-loop bubble $U = x_1 + x_2 = 1$ on the simplex, (195) gives exactly $2/z$, confirming both the factor of two and the vertex-factorial convention.

Power subgraphs and continuation. For fixed $0 < d < 4$ a proper unmarked two-point subgraph has a local power divergence, not an independent wave-function counterterm. We use the analytic critical-mass prescription of [7], as in the one-dimensional calculation: its local mass power divergence is removed by critical tuning and its analytic continuation is retained. In particular, graphs with triple lines or vacuum-attached pieces are not silently dropped. An ordinary domain from which to continue the raw Schwinger integrals is real $z > d/3$: for an induced subgraph with v vertices and e edges, the ultraviolet condition is $ez + d(2v - 2 - e) > 0$, and the worst tadpole-free two-vertex case is $e = 3$. Continuing to $z = 0$ implements the specified analytic regularization; subsequent logarithmic subdivergences are removed by MS. No cutoff on individual divergent simplex faces is to be substituted for this prescription.

L	Labeled matrices	Vertex classes	Primitive classes	Nonzero labeled $3w_2 - w_4$
1	1	1	1	0
2	6	2	0	3
3	85	8	1	55
4	1890	27	1	1470
5	59145	129	3	49605

These counts are for the displayed matrix set, not for a minimal nonprimitive master basis. Some classes can disappear or combine only after analytic continuation and subtraction. The inventory file `lri_general_special_functions_update/verification/graph_inventory.json`, written by `code/graph_inventory.py` of that package, prints all 167 classes through five loops, their exact rational weights, expanded edge lists and every monomial of U_m. The polynomials were checked independently against the matrix-tree determinant at exact integer edge parameters (record `verification/inventory_checks.json`). The identity

$$\frac{1}{(L+1)!}\sum_m w_i(m)\mathcal{I}_m = \sum_{[m]} \frac{w_i(m)}{|\operatorname{Aut}_v(m)|}\mathcal{I}_m \tag{196}$$

follows from orbit–stabilizer and was also checked in the enumeration.

15.5 An all-regulator two-loop master in ordinary hypergeometric functions

There are only two two-loop vertex classes. The one-vertex-reducible class has amplitude $(2/z)^2$; denote the other by $S_d(z)$. Their exact weights give

$$\mathcal{A}_2 = \frac{3}{z^2} + 3S_d(z), \qquad \mathcal{B}_2 = \frac{1}{z^2} + \frac{1}{2}S_d(z). \tag{197}$$

For the same $a = (d+z)/4$, put $b = (d+3z)/4$ and define

$$\begin{aligned} F_1(d,z) &= {}_3F_2\left(\begin{matrix} d/2, z/2, a \\ 1-z/2, a+1/2 \end{matrix}; \frac{1}{4}\right), \\ F_2(d,z) &= {}_3F_2\left(\begin{matrix} z, 2a, b \\ 1+z/2, b+1/2 \end{matrix}; \frac{1}{4}\right), \\ W(d,z) &= \frac{r(d,z)^2\Gamma(1+z)\Gamma(1-z/2)\Gamma(b)^2}{\Gamma(a)^2\Gamma(d/2)\Gamma(2b)}. \end{aligned} \tag{198}$$

Then the full meromorphic two-loop amplitude is

$$\boxed{S_d(z) = \frac{4F_1(d,z) - 2W(d,z)F_2(d,z)}{z^2}.} \tag{199}$$

The two defining ${}_3F_2$ series have argument $1/4$, strictly inside the convergence disk. Near $z=0$ their parameters and the Gamma factors in W are regular. Thus (199) determines all of the two-loop Laurent data needed at fourth and fifth order, not just the residue. The pole is explicit and the numerator is analytic.

15.5.1 Derivation and a residue check

Number the edges so that the Symanzik polynomial is $U=(x_1+x_2)(x_3+x_4)+x_3x_4$. Set $x_1+x_2=u$, $x_3=(1-u)v$, $x_4=(1-u)(1-v)$. The beta integral over the split of x_1,x_2 and then over u reduces the simplex to

$$\begin{aligned} S_d(z) =& \frac{r(d,z)^2\Gamma(z)}{\Gamma(a)^4}B(a,a)B(2a,z/2) \\ &\times \int_0^1 [v(1-v)]^{a-1}\,{}_2F_1\!\left(\frac{d}{2},\frac{z}{2};\frac{d}{2}+z;1-v+v^2\right)dv. \end{aligned} \tag{200}$$

Apply the Gauss connection formula at $1-v(1-v)$ [23]. Its two pieces have powers $[v(1-v)]^0$ and $[v(1-v)]^{z/2}$, respectively. For either piece, termwise beta integration uses

$$\frac{B(c+n,c+n)}{B(c,c)}=\frac{(c)_n^2}{(2c)_{2n}}=\frac{(c)_n}{4^n(c+1/2)_n}.$$

This gives exactly F_1,F_2. The first prefactor simplifies to $4/z^2$; the second to $-2W/z^2$. The derivation starts where the beta integrals converge and then continues meromorphically.

At $z=0$, $F_1=F_2=W=1$. The $n\ge 1$ terms of the defining series show $\partial_z F_2(d,0)=2\partial_z F_1(d,0)$, while $\partial_z \log W(d,0)=-A_d/2$. It follows directly that

$$4F_1-2WF_2=2+A_dz-s_1z^2+O(z^3),\qquad S_d(z)=\frac{2}{z^2}+\frac{A_d}{z}-s_1+O(z). \tag{201}$$

Thus the known two-loop pole is recovered without evaluating the intermediate hypergeometric derivatives separately. Derivatives appear here only in the proof of a Taylor identity; the all-regulator formula (199) is an ordinary fixed-argument evaluation.

15.6 Complete higher-order MS assembly

For all the following quantities use the same prescription (174)–(195). Define

$$a_{L,k}(d)=[z^k]\mathcal{A}_L(d,z),\qquad e_{L,k}(d)=[z^k]\mathcal{B}_L(d,z). \tag{202}$$

The letter e here is a mass-amplitude Laurent coefficient, not an external half-edge count e_i. The following finite recipe makes the general-dimensional C_4,C_5 representation unambiguous.

Set

$$\begin{aligned} F_4(v)&=v+\sum_{L=1}^{5}(-1)^L\mathcal{A}_Lv^{L+1}, & F_2(v)&=1+\sum_{L=1}^{5}(-1)^L\mathcal{B}_Lv^L,\\ \mathcal{Z}_g&=g+\sum_{n=2}^{6}G_n(z)g^n, & Z_r&=1+\sum_{n=1}^{5}Z_n(z)g^n. \end{aligned} \tag{203}$$

All G_n,Z_n are pure pole parts. Recursively impose

$$\begin{aligned} G_n&=-\operatorname{PP}[g^n]F_4(\mathcal{Z}_{g,<n}), & Z_n&=-\operatorname{PP}[g^n]\{Z_{r,<n}F_2(\mathcal{Z}_g)\},\\ b_L&=L[z^{-1}]G_{L+1}, & c_L&=L[z^{-1}]Z_L. \end{aligned} \tag{204}$$

Here PP retains negative powers of z. This is also an explicit forest subtraction: multiplication by lower counterterms is carried out before selecting a new simple pole. Keeping only primitive graphs would not implement (204).

Write $A = A_d$, $R = R_d$, $M = M_d$, $Q = Q_d$ temporarily in the formulas below. Expanding the recursion gives

$$\begin{aligned} b_4 &= 36Aa_{2,0} - 216a_{2,1} + 48a_{3,0} - 4a_{4,-1}, \\ c_4 &= 26Ae_{2,0} - 140e_{2,1} + 40e_{3,0} - 4e_{4,-1}, \end{aligned} \tag{205}$$

$$\begin{aligned} b_5 &= 5a_{5,-1} - 75a_{4,0} + 450a_{3,1} - 60Aa_{3,0} - 1350a_{2,2} + 585Aa_{2,1} - 5b_3a_{2,0}, \\ c_5 &= 5e_{5,-1} - 65e_{4,0} + 325e_{3,1} - \frac{95}{2}Ae_{3,0} - \frac{2275}{3}e_{2,2} \\ &\quad + \frac{725}{2}Ae_{2,1} - \frac{10b_3 + 5c_3}{3}e_{2,0}. \end{aligned} \tag{206}$$

Every $a_{L,k}, e_{L,k}$ in these formulas is the coefficient of the explicit finite integral sum (193), with U defined in (194); there are no unspecified diagram labels. The code inventory is a convenience for enumerating this finite definition, not a replacement for it.

For completeness, the required orders are

Loop order	Highest Laurent power	Required analytic input
2	z^2	Full formula (199)
3	z^1	Master expansions through the stated order
4	z^0	Master finite parts and the primitive period
5	z^{-1}	Subtracted simple poles and primitive periods

The combinations of these finite data required by the physical coefficients are reduced in Section 16. They cannot be discarded merely because the primitive residues are known.

15.6.1 The coefficient formulas through fifth order

Lagrange inversion applied to $\epsilon = g_*(3 + b_2g_* + b_3g_*^2 + b_4g_*^3 + b_5g_*^4)$ gives

$$C_n(d) = -\frac{1}{2}\delta_{n1} + \frac{1}{n}[g^{n-1}]\frac{\gamma_r'(g)}{(3 + b_2g + b_3g^2 + b_4g^3 + b_5g^4)^n}. \tag{207}$$

Define $D_4(d) = 3c_4(d) - b_4(d)$ and $D_5(d) = 3c_5(d) - b_5(d)$. Substituting the evaluated (179) gives the collected general-d formulas

$$\boxed{243C_4(d) = D_4 - 51A^3 + 9AR - 48AQ - 2AM,} \tag{208}$$

$$\boxed{\begin{aligned} 729C_5(d) = D_5 + 10AD_4 - 6Ac_4 - \frac{5541}{16}A^4 - \frac{51}{4}A^2M - 162A^2Q + \frac{315}{8}A^2R \\ + 6MQ - \frac{3}{4}MR + 36Q^2 - 18QR + \frac{27}{16}R^2. \end{aligned}} \tag{209}$$

Equations (193), (195), and (205)–(206) are part of these formulas: D_4, c_4, D_5 are completely specified Laurent functionals, not asserted new elementary constants. Nevertheless, leaving them as integrals is a weaker endpoint than the finite hypergeometric reduction obtained at $d = 1$. In particular, a known value of D_4 would not by itself determine c_4 or the finite four-loop amplitudes needed for C_5.

An equivalent form, identical to (10) and (11) and convenient for checking signs, is

$$243C_4 = 60A^3 - 8Ab_3 + 18Ac_3 + 3c_4 - b_4,$$
$$729C_5 = 336A^4 - 60A^2b_3 + 108A^2c_3 + b_3^2 - 3b_3c_3 - 10Ab_4 + 24Ac_4 + 3c_5 - b_5. \tag{210}$$

All terms in (208)–(209) have been checked by exact symbolic algebra, including independent extraction of the simple-pole coefficients in (205)–(206) from the RG pole equations. Substituting the established one-dimensional RG data into the universal assembly reproduces the retained $C_4(1), C_5(1)$ formulas. That last check is an algebraic specialization, not a fresh numerical evaluation of the new Schwinger integrals at $d = 1$.

15.6.2 The combined five-loop channel survives in arbitrary d

The exact integrand weight used in the numerical resolution of the one-dimensional problem was combinatorial:

$$w_H = 3w_2 - w_4 = \frac{w_4}{4}\left[\sum_i e_i(e_i - 1) - 4\right]. \tag{211}$$

It therefore cancels the same 9540 of the 59145 labeled five-loop matrices in every dimension. With lower-loop data held fixed,

$$\boxed{\delta C_5(d) = \frac{5}{729}\,\delta[z^{-1}]\{3\mathcal{B}_5(d,z) - \mathcal{A}_5(d,z)\}.} \tag{212}$$

This does not remove the $a_{4,0}, e_{4,0}$ terms. It identifies the five-loop channel that should be integrated directly, before a large numerical subtraction, when performing a general-d evaluation.

15.7 General-dimensional primitive periods

The spectral construction extends the primitive periods, but not by keeping the one-dimensional five hypergeometric parameter lists. Let $P_L^{\mathrm{raw}}(d)$ be the conformal zig-zag decompletion with edge exponents $q = d/4$, two fixed points at distance one and the remaining $L - 1$ vertices integrated over $\mathbb{R}^d$. Normalize

$$P_L(d) = h_d^{L-1} P_L^{\mathrm{raw}}(d), \qquad h_d = \Gamma(d/2)/\pi^{d/2}. \tag{213}$$

Thus $P_3(d) = 3Q_d$, $P_4(1) = P_4$, and $P_5(1) = P_5$. The raw coordinate integrals at integer dimension and their continued spectral or parametric representations fix the graph definition.

Set $p = d/2 - 1$ and $k = L + 2$. The eigenvalue and measure obtained from [24], eqs. (5.23), (5.27), and (6.2), are

$$\tau_n^{(d)}(\nu) = (-1)^n \pi^{d/2} \frac{\Gamma(n/2 + d/8 + \mathrm{i}\nu)\Gamma(n/2 + d/8 - \mathrm{i}\nu)}{\Gamma(n/2 + 3d/8 + \mathrm{i}\nu)\Gamma(n/2 + 3d/8 - \mathrm{i}\nu)}, \tag{214}$$

$$\rho_n^{(d)}(\nu) = \frac{\Gamma(p)\Gamma(n+2p)(n+p)}{2\pi^{3d/2+1}\Gamma(2p)n!}\left[(n+p)^2 + 4\nu^2\right]\frac{\Gamma(p + 2\mathrm{i}\nu)\Gamma(p - 2\mathrm{i}\nu)}{\Gamma(2\mathrm{i}\nu)\Gamma(-2\mathrm{i}\nu)}. \tag{215}$$

For $2 < d < 4$ the ordinary real-axis representation is

$$\boxed{P_L(d) = h_d^{L-1} \sum_{n=0}^{\infty} \int_0^{\infty} \rho_n^{(d)}(\nu)\,[\tau_n^{(d)}(\nu)]^{L+2} d\nu, \qquad L = 3, 4, 5.} \tag{216}$$

The angular quantum number n runs over an infinite tower. Its presence is one reason the finite $d = 1$ hypergeometric formula cannot be lifted by changing quarter-arguments alone.

For $0 < d < 2$, $-1 < p < 0$ and the first pair of Gamma poles crosses the real spectral contour. Continuing the contour as in the earlier one-dimensional derivation adds

$$P_L(d) = h_d^{L-1}\Bigg\{ \sum_{n=0}^{\infty} \int_0^{\infty} \rho_n^{(d)}(\nu)[\tau_n^{(d)}(\nu)]^{L+2} d\nu + \sum_{n=1}^{\infty} \frac{\Gamma(n+2p)(n+p)n(n+2p)}{2\pi^{3d/2} n!\Gamma(-p)} [\tau_n^{(d)}(-ip/2)]^{L+2} \Bigg\}. \tag{217}$$

The $n = 0$ crossing vanishes identically. At removable singular parameter values such as $d = 1, 2$, take the joint analytic limit of the Gamma ratios, not the separate factors. There are no additional measure-pole crossings inside $0 < d < 2$; the eigenvalue Gamma arguments for the $n \geq 1$ residues stay away from their pole sets. The residue coefficient is $-2\pi i \operatorname{Res}_{\nu=-ip/2}(\rho\tau^{L+2})$, fixing its sign. At $d = 1$, formula (217) reduces to the continuous-plus-discrete formula (111) of Section 10. The numerical code `lri_general_special_functions_update/code/primitive_spectral.py` (record `verification/primitive_spectral_checks.json`) implements only (216) with $d > 2$; it does not silently substitute $d < 2$ into that real-axis code.

As a normalization check, at $d = 4$ one has $\tau_n = (-1)^n 4\pi^2/[(n+1)^2 + 4\nu^2]$. An elementary beta integration then gives the graph-period limits

$$P_3(4) = 6\zeta(3), \qquad P_4(4) = 20\zeta(5), \qquad P_5(4) = \frac{441}{8}\zeta(7). \tag{218}$$

Equivalently, $\mathcal{T}_4 = 12\zeta(3)$. These are checks of the period normalization, not assertions that the complete long-range RG functions have a regular short-range limit; M_d, for example, has a pole at $d = 4$.

15.7.1 Primitive weights and the product periods

The primitive classes at $L = 3, 4, 5$ have vertex automorphism orders 24, 8, and $(12, 16, 2)$ respectively. The four external legs are on distinct vertices, so their mass weights vanish. At five loops, the first two completed graphs have a three-vertex cut with two components of two vertices each. Fix the cut vertices at $0, e, \infty$; then the two remaining integrations decouple into two tetrahedron integrals. Since every completed vertex has four propagators of exponent $d/4$, the conformal normalization gives $P_3(d)^2 = 9Q_d^2$, independently of d. The third class is the zig-zag $P_5(d)$. Consequently the primitive pieces are

$$\boxed{\begin{array}{ll} b_3^{\mathrm{prim}}(d) = 6Q_d, & b_4^{\mathrm{prim}}(d) = -6P_4(d), \\ b_5^{\mathrm{prim}}(d) = 63Q_d^2 + 24P_5(d), & c_L^{\mathrm{prim}}(d) = 0 \quad (L = 3, 4, 5). \end{array}} \tag{219}$$

Their direct new-loop contributions to C_4, C_5 are therefore $2P_4(d)/81$ and $-[63Q_d^2 + 24P_5(d)]/729$. The lower-loop RG assembly and nonprimitive graphs remain essential. The inventory file `grap h_inventory.json` supplies explicit completed edge lists and the cut vertices for the product factorizations, rather than identifying them only by topology names.

15.8 Numerical checks of the lower-order and primitive data

These checks were run by OpenAI GPT-6 astra in the general-dimensional calculation (package `lri_general_special_functions_update/`, scripts `tetrahedron.py`, `two_loop_master.py`, `primitive_spectral.py`, `symbolic_checks.py` and `inventory_checks.py` in `code/`; records of the same names in `verification/`). They are numerical consistency residuals, not interval enclosures.

Tetrahedron. A double-precision tensor trapezoidal quadrature of (188) used the contours $\Re u = \Re v = -q/3$, spacings 0.015 and 0.01, and imaginary cutoffs 8 and 10. The finer run differs from the exact values at $d = 1, 2, 4$ ((180), (192) and $\mathcal{T}_4 = 12\zeta(3)$) by about 3.65×10^{-13}, 6.43×10^{-14} and 5.63×10^{-15}. The resulting values, with $C_2(d)$ and $C_3(d)$ from (178), are (record `C0_to_C3_values.csv`)

d	$\mathcal{T}_d$	$C_2(d)$	$C_3(d)$
1	330.008919265	0.657131264	−3.719856323
1.5	142.350694846	0.427213506	−1.612227399
2	76.628287192	0.308065414	−0.874850013
2.5	46.354193318	0.233416481	−0.535556041
3	30.102615387	0.181221785	−0.353581520
3.5	20.493170714	0.142032326	−0.246028075

The rows at $d = 1, 2, 3$ agree with the table of Section 1. At $d = 1.5$ and 2.5 the record values of $\mathcal{T}_d$ (the full double-precision values in `C0_to_C3_values.csv`) agree with $6Q_d$ from the independent spectral code of `LRI_C5_general_d/references/reference_values.json` to 1.0×10^{-13} and 2.2×10^{-13} absolute (comparison made for this note). The rounded numerical values published with the corrected tetrahedron [12] are useful benchmarks, but their last printed digits are not used as inputs.

Two-loop master. At 75 working digits the two-${}_3F_2$ formula (199) was compared with the Euler integral (200) at $(d, z) = (1, 0.4)$, $(2, 0.3)$ and $(3, 0.35)$. The differences were 1.38×10^{-44}, below 2.2×10^{-58} and below 4.5×10^{-75}. Cauchy extraction of the analytic numerator of (199), with radii 0.025 and 0.02 and 72 and 80 nodes, gives the Laurent data through z^2 at $d = 1, 2, 3$; changing the settings changes the extracted coefficients by less than 2.0×10^{-69}. At $d = 1$ the finite parts in this massive-Schwinger prescription are $[z^0]\mathcal{A}_2 \simeq -55.190665$ and $[z^0]\mathcal{B}_2 \simeq -9.198444$. They differ from the ordered-span finite parts of Section 3, as they must; the two sets are not interchangeable (Section 15).

Primitive spectral formula. Gauss–Legendre quadrature of (216) after a map of the spectral half-line, with 192 and 320 nodes and Euler acceleration of the alternating angular sums, gives at $d = 3$ $P_3(3) \simeq 15.05130769366$, equal to $\mathcal{T}_3/2 = 3Q_3$ of the separate Mellin–Barnes calculation, and $P_5(3) \simeq 240.34031776$. The $P_4(3)$ run is an unextrapolated positive partial sum: 62.0583263456, 62.0583277437, 62.0583279184 and 62.0583279403 at 128, 256, 512 and 1024 angular modes, with no bound claimed for the omitted tail. All three agree with the high-precision values of Section 19.10 to the accuracy of the quadrature. The same code at $d = 4$ reproduces the four-dimensional periods $P_3 = 6\zeta(3)$ and $P_5 = \frac{441}{8}\zeta(7)$ within 4.3×10^{-13} and 3.4×10^{-12}. None of these is a determination of the complete $C_4(3)$ or $C_5(3)$.

Exact algebra and graph data. The symbolic check reconstructs the MS three-loop coefficients (179) from the zero-momentum ones, shows the cancellation of J_0, expands the Lagrange inversion through fifth order, derives (205)–(206) independently from the pole equations, and recovers the one-dimensional coefficient formulas by substitution. The inventory check verifies all 167 Kirchhoff polynomials by exact matrix-tree arithmetic, the orbit factors, the primitive mass weights, the two product factorizations of (219) and the 9540 zero combined weights of (211). No general-dimensional nonprimitive four- or five-loop integral is evaluated numerically in these checks.

16 General-dimensional master reduction and collected coefficients

16.1 Conventions and the collected answer

Here d is the physical spatial dimension, not the exponent-shift parameter d of Section 11. Set

$$q = \frac{d}{4}, \qquad \epsilon = 2s - d, \qquad \Delta_\phi = \frac{d-\epsilon}{4}, \qquad \Delta_{\mathcal{E}} = \sum_{j=0}^{5} C_j(d)\epsilon^j + O(\epsilon^6). \tag{220}$$

The physical perturbative range is fixed $0 < d < 4$ and sufficiently small positive ϵ. The endpoint $d = 4$ is not substituted into unrenormalized power-subgraph formulas. Define

$$\begin{aligned} A &= \psi_0(2q) + \psi_0(1) - 2\psi_0(q), & R &= \psi_1(2q) - \psi_1(1), \\ U &= \psi_2(q), & V &= \psi_2(1) + \psi_2(2q), \quad X = \psi_3(2q) - \psi_3(1). \end{aligned} \tag{221}$$

Here and below $Q = Q_d$, $P_4 = P_4(d)$ and $P_5 = P_5(d)$ have the normalization of (213): $P_3 = 3Q$. The three additional dimension-dependent functions are denoted by $\mathcal{K}_d, \mathcal{L}_d, Z_d$. Explicit series definitions are (256), (257) and (260). They are not free parameters. Sections 20 and 21 give shorter finite-sample evaluations of all three, and (553) is (223) written in them.

The complete collected coefficients are

$$\boxed{\begin{aligned} C_0(d) &= \frac{d}{2}, & C_1(d) &= -\frac{1}{6}, \quad C_2(d) = \frac{A}{9}, \\ C_3(d) &= \frac{A^2 + R - 8Q}{108}, \\ 648C_4(d) &= 16P_4 - A^3 - 8AQ - 3AR + \frac{53}{2}U - 3V. \end{aligned}} \tag{222}$$

$$\boxed{\begin{aligned} 46656C_5(d) = &-1536(P_5 + \mathcal{K}_d) + 384AP_4 - 576\mathcal{L}_d - 1728Q^2 \\ &+ 720A^2Q + 1008QR + 7152Z_d \\ &- 15A^4 - 90A^2R + 414AU - 108AV - 117R^2 - 75X. \end{aligned}} \tag{223}$$

In particular, the extra Gamma factor associated with power-divergent mass subgraphs cancels from the explicit coefficients of both physical answers. It must nevertheless be retained until the graph and fixed-point assembly is complete.

Exact one-dimensional specialization. At $d = 1$, the dictionary is

$$\begin{aligned} R &= \frac{\pi^2}{3}, & U &= -2\pi^3 - 56\zeta(3), \quad V = -16\zeta(3), \\ X &= \frac{14\pi^4}{15}, \quad Z_1 = QH, & H &= \frac{2\pi^2}{3} + 8\mathsf{G}, \\ \mathcal{K}_1 &= \mathcal{K}, & \mathcal{L}_1 &= \mathcal{L}. \end{aligned} \tag{224}$$

Substitution into (222)–(223) reproduces the one-dimensional coefficients of Section 4 (for C_5, (27)) exactly. In particular, the $7152Z_d$ term becomes $48 \cdot 149QH$. Neither $Z_d = Q_dH_d$ with a guessed H_d, nor the one-dimensional polygamma formula for $\mathcal{K}$, is assumed away from $d = 1$.

16.2 The evaluated RG combinations

For comparison with the assembly formulas (208)–(210), define

$$\begin{aligned} M &= \frac{\Gamma(-q)\Gamma(2q)^2}{\Gamma(3q)} = -\frac{\Gamma(1-q)\Gamma(2q)^2}{q\Gamma(3q)}, \\ N &= \psi_0(-q) + \psi_0(3q) - \psi_0(1) - \psi_0(2q), \qquad D_j = 3c_j - b_j. \end{aligned} \tag{225}$$

For $0 < q < 1$, the last Gamma representation has only positive arguments. The recurrence $\psi_0(-q) = \psi_0(1-q) + 1/q$ likewise avoids a negative argument. The exact fourth-order results are

$$b_4 = -\frac{393}{4}A^3 - \frac{11}{2}AM - 54AQ + \frac{117}{4}AR - \frac{3}{2}MN - 6P_4 - \frac{165}{8}U + \frac{9}{4}V, \tag{226}$$

$$c_4 = -\frac{127}{8}A^3 - \frac{7}{6}AM - 3AQ + \frac{51}{8}AR - \frac{1}{2}MN - \frac{57}{16}U + \frac{3}{8}V, \tag{227}$$

$$D_4 = \frac{405}{8}A^3 + 2AM + 45AQ - \frac{81}{8}AR + 6P_4 + \frac{159}{16}U - \frac{9}{8}V. \tag{228}$$

Separate the fifth-order nonprimitive nonwheel part from its wheel and primitive contributions:

$$\begin{aligned} D_5^{\mathrm{rem}} = &-\frac{16347}{64}A^4 - \frac{57}{4}A^2M - \frac{1179}{4}A^2Q + \frac{3159}{32}A^2R - 3AMN \\ &- \frac{3657}{32}AU + \frac{189}{16}AV - 24\mathcal{K}_d - 6MQ + \frac{3}{4}MR + \frac{135}{4}QR \\ &- \frac{225}{64}R^2 - \frac{75}{64}X + \frac{447}{4}Z_d, \\ D_5 = D_5^{\mathrm{rem}} &- 54AP_4 - 9\mathcal{L}_d - 63Q^2 - 24P_5. \end{aligned} \tag{229}$$

These expressions determine the three combinations D_4, c_4 and D_5 left open in (208)–(209). Inserting them into (210), together with (179), gives (222)–(223); the M, MN and MR terms cancel. The script `lri_general_special_functions_update/code/general_special/exact_audit.py` checks this substitution in exact arithmetic. It computes b_3 and c_3 from the three-loop graph rows instead of taking (179) as input.

16.3 Why the graph reduction extends to general dimension

Normalize every coordinate integration with $h_d = \Gamma(d/2)/\pi^{d/2}$. The two-point convolution coefficient is

$$B_d(a,b) = \Gamma(d/2)\frac{\Gamma((d-a)/2)\Gamma((d-b)/2)\Gamma((a+b-d)/2)}{\Gamma(a/2)\Gamma(b/2)\Gamma((2d-a-b)/2)}. \tag{230}$$

It multiplies $|x-y|^{-(a+b-d)}$. It is symmetric under permutations of $a, b, 2d-a-b$. The star–triangle relation applies when its three absolute-distance exponents sum to $2d$. These are dimension-dependent Gamma identities, not replacements of special one-dimensional constants [24].

A reduced two-vertex master is

$$J_d(a,b,c,f,e) = h_d^2 \int_{\mathbb{R}^d} d^dx \int_{\mathbb{R}^d} d^dy \, \frac{1}{|x|^a|x-\hat{e}|^b|y|^c|y-\hat{e}|^f|x-y|^e}, \qquad |\hat{e}| = 1. \tag{231}$$

Integrals outside their ordinary convergence domain are defined by analytic continuation and the critical-mass prescription of [7] used in Section 15. Its conformal completion is a balanced K_5 with ordered edges $(01, 02, 03, 04, 12, 13, 14, 23, 24, 34)$ and exponents

$$\begin{aligned} \boldsymbol{\lambda} = (&3d-a-b-c-f-e,\ a,\ c,\ b+f+e-d,\ b,\ f, \\ &a+c+e-d,\ e,\ 2d-a-b-e,\ 2d-c-f-e). \end{aligned} \tag{232}$$

Each completed vertex has incident exponent sum $2d$; permutation of its five vertices gives the exact completion covariance. The needed Taylor data reduce to three critical patterns.

16.3.1 Pattern A: the Gamma-determined jet

For the critical point $(d/2, d/2, d/2, d/2, d)$ set

$$\begin{aligned} &t = d-e, \qquad p_0 = 2d-a-c-e, \quad p_1 = 2d-b-f-e, \\ &u = (a-c)/2, \quad v = (b-f)/2, \qquad \sigma = p_0 + p_1. \end{aligned} \tag{233}$$

An explicit pole subtraction is

$$\begin{aligned}\mathcal{P}_A = \frac{2}{p_0}B_d\left(\frac{d+t}{2}-u,d-t\right)+\frac{2}{p_1}B_d\left(\frac{d+t}{2}-v,d-t\right)\\ +\frac{2}{t}B_d(d-p_0,d-p_1)-\frac{4}{t}\left(\frac{1}{p_0}+\frac{1}{p_1}\right).\end{aligned} \tag{234}$$

The analytic remainder through total degree two is

$$\boxed{J_d-\mathcal{P}_A=\sigma\left(\frac{AR}{2}+\frac{U}{8}\right)+t\sigma\left(\frac{A^2R}{8}+\frac{AU}{16}-\frac{R^2}{8}\right)-\frac{AV}{8}\sigma^2+O(3).} \tag{235}$$

This is fixed without evaluating a new master. On each of $a+b+e=2d$, $c+f+e=2d$, and $a+b+c+f+e=3d$, a completion edge vanishes and star–triangle reduction gives a product of two copies of (230). After (234), the remainder is analytic. An analytic difference vanishing on these three distinct hyperplanes is divisible by their product. It consequently has no term of total degree at most two. Expanding the Gamma products therefore determines (235) uniquely. The exact interpolation in `lri_general_special_functions_update/code/general_special/derive_A.py` uses rational directions only; it is finite polynomial linear algebra, not numerical fitting.

16.3.2 Pattern B: one genuinely independent quadratic coefficient

At $(d/2,d/2,d/2,d,d/2)$ put the following local coordinates; p_1 here is not the p_1 of (233), which equals p_2 here: $p_1=d-f$, $p_2=2d-b-f-e$, $z_1=3d-a-b-c-f-e$, $z_2=2d-c-f-e$. Define

$$\begin{aligned}G_2&=B_d(d+p_1-e,d-p_1)B_d(e+p_2-z_1,d-p_2),\\ G_1&=B_d(e-p_1+p_2+z_2,d+p_1-e-z_2)B_d(d+p_1-e-p_2,d-p_1),\\ G_0&=B_d(d+p_1-e,d-p_1)B_d(e+p_2,d-p_2),\\ \Delta_B&=B_d(e+p_2+z_2-z_1,d-p_2)-B_d(e+p_2-z_1,d-p_2)\\ &\quad -B_d(e+p_2+z_2,d-p_2)+B_d(e+p_2,d-p_2).\end{aligned} \tag{236}$$

The needed jet is

$$J_d=G_2+G_1-G_0+\frac{2}{p_1}\Delta_B+\mathcal{K}_d z_1z_2+O(3). \tag{237}$$

The Gamma comparison agrees on $z_1=0$ and $z_2=0$. Completion symmetry and analyticity make the degree-two difference proportional to z_1z_2. Its normalization is fixed by the one-parameter path below. Its parameter t is not the $t=d-e$ of (233):

$$\begin{aligned}J_t&=J_d(d/2,d/2,d/2,d-t,d/2),\\ J_t^{\Gamma}&=B_d(d/2,d-t)^2+\frac{2}{t}\left[B_d(d/2,d-t)-B_d(d/2+2t,d-t)\right],\\ \mathcal{K}_d&=[t^2](J_t-J_t^{\Gamma}).\end{aligned} \tag{238}$$

This coefficient extraction is a *proof and normalization definition*; the final evaluation (260) has neither t nor a derivative. At $d=1$ it is exactly the one-dimensional master $\mathcal{K}$ of (57), not an unrelated constant with a matching numerical value.

16.3.3 Pattern T: tetrahedron symmetry

At $a=b=c=f=e=d/2$, all ten completion edges equal $d/2$. Balanced variations satisfy $\sum_{e\ni v}\delta\lambda_e=0$ at every vertex. Here e labels the ten completion edges; it is not the exponent e

of (231). There is no invariant linear form, and every invariant quadratic form on this balanced subspace is proportional to $\sum_e(\delta\lambda_e)^2$. Indeed the coefficients of a general permutation-invariant quadratic form depend only on whether two edges coincide, meet, or are disjoint; the balance equations reduce the latter two sums to the first. Thus

$$\boxed{J_d = 3Q_d + \frac{Z_d}{4}\sum_{e=1}^{10}(\delta\lambda_e)^2 + O(3).} \tag{239}$$

This argument applies in every dimension; the coefficient Z_d itself is not fixed by symmetry. A central-edge-only change by t has squared completion variation $6t^2$. A balanced three-wheel path has $18t^2$, which fixes the factors of three and nine used below.

16.3.4 Forest sum and six wheel rows

The graph set, exact Wick weights and subtraction prescription are those already specified in (193)–(195). Equivalently, a rooted logarithmic L-loop coordinate graph has common radial factor $2/(Lz)$ and edge exponents $m_{ij}(d-z)/2$. Its proper disjoint forests are subtracted recursively; nested forests occur inside the subgraph counterterms. The local counterterm is

$$Z_G = -\,\mathrm{PP}\left[I_G + \sum_{\varnothing\neq\mathcal{F}}\left(\prod_{\gamma\in\mathcal{F}} Z_\gamma\right) I_{G/\mathcal{F}}\right].$$

The shift method of Section 11, used on exceptional graphs, preserves the sum of edge deformations in every divergent subgraph. Only convolution, star–triangle, completion covariance and the jets above enter. Consequently its combinatorics is dimension independent. Critical base exponents scale with d; regulator slopes do *not* scale with d.

The residue certificate for these classes consists of the files `rows_4_0_27.json` and `rows_5_0_129.json` in `lri_general_special_functions_update/code/general_special/`. They list all 26 nonprimitive four-loop classes and the 92 nonzero five-loop classes that are neither wheels nor primitive. At five loops six further classes are wheels, three are primitive, and 28 have zero combined weight. The sum is 129 classes. For each class the files give the canonical word, the rational weights and the exact symbolic residue. They give (226)–(229); no period value is used to obtain these polynomial coefficients.

For completeness the six wheel residues are $-2AP_4/5+(\alpha D_s+\beta D_r)/5$, where $D_s = \frac{1}{4}\partial_a P_W$ and $D_r = \frac{1}{4}\partial_b P_W$ are the spoke and rim derivatives at $a = b = d/2$ of the rooted four-wheel P_W of Section 16.4.1. The class index is the zero-based position in the five-loop list of `lri_general_special_functions_update/code/general_special/C5_wheel_nonprimitive/reduced_classes.json`. The data are:

Class index	combined weight	α	β
51	15/2	2	2
78	15	2	−2
99	30	2	−2
113	60	−2	2
124	15/2	6	−2
128	15	6	−6

Their sum is $-54AP_4 + 24D_s - 12D_r = -54AP_4 - 9\mathcal{L}_d$. The completion/product argument behind (219) gives $D_5^{\mathrm{prim}} = -63Q^2 - 24P_5$.

16.4 A common spectral kernel for all nonprimitive master data

Let

$$\begin{aligned} d_0(d) &= 1, & d_\ell(d) &= \frac{(2\ell+d-2)\Gamma(\ell+d-2)}{\Gamma(d-1)\Gamma(\ell+1)} \quad (\ell \geq 1), \\ a_\ell &= \frac{\ell}{2}+\frac{q}{2}, & b_\ell &= a_\ell + q, \\ t_\ell(\nu) &= \Gamma(2q)\frac{\Gamma(a_\ell+\mathrm{i}\nu)\Gamma(a_\ell-\mathrm{i}\nu)}{\Gamma(b_\ell+\mathrm{i}\nu)\Gamma(b_\ell-\mathrm{i}\nu)}. \end{aligned} \tag{240}$$

Here a_ℓ, b_ℓ are spin-dependent Gamma arguments, not the RG coefficients b_j of (173). At $d=1$, $(d_0, d_1, d_2, \ldots) = (1, 1, 0, \ldots)$; at $d=2$ the multiplicities are $(1, 2, 2, \ldots)$. At noninteger $d<1$ they are continued Gamma ratios, not positive integer multiplicities. For notational economy write

$$\langle f \rangle_d = \frac{1}{\pi}\sum_{\ell=0}^{\infty} d_\ell(d) \int_0^\infty f(\ell,\nu)\, d\nu. \tag{241}$$

This is the *one-vector wheel kernel*, not the two-vector zig-zag measure in (215). The primitive $P_5(d)$ keeps the latter representation and its stated continuation prescription.

For a balanced wheel with rim exponent b and spoke exponent $d-b$, the commuting H_α construction of [24] gives eigenvalue

$$t_{\ell,b}(\nu) = \Gamma(d/2)\frac{\Gamma((d-b)/2)}{\Gamma(b/2)}\left|\frac{\Gamma(\ell/2+b/4+\mathrm{i}\nu)}{\Gamma(\ell/2+d/2-b/4+\mathrm{i}\nu)}\right|^2, \qquad W_r(d;b) = \langle t_{\ell,b}^r \rangle_d. \tag{242}$$

One can also obtain it directly by applying the normalized convolution operator $h_d|x-y|^{-b}|y|^{b-d}$ to homogeneous harmonic functions. The spherical-harmonic addition theorem gives the factor $d_\ell/(2\pi)$ on the full Mellin line, hence d_ℓ/π on the positive half-line. In particular

$$3Q_d = \langle t_\ell^3 \rangle_d, \qquad P_4(d) = \langle t_\ell^4 \rangle_d. \tag{243}$$

At the graph-period check point $d=4$, $t_\ell = (\nu^2 + (\ell+1)^2/4)^{-1}$ and $d_\ell = (\ell+1)^2$, giving $W_3 = 6\zeta(3)$ and $W_4 = 20\zeta(5)$.

16.4.1 The logarithmic wheel and the tetrahedron Hessian

Let a rooted four-wheel have four spokes of exponent a and four rims of exponent b, and let $P_W(a,b)$ be its period, normalized as in (242), so that $P_W(d-b,b) = W_4(d;b)$. At $a=b=d/2$ its completion is the octahedron. The completed exponents are four a's, four b's, two $-2d+3a+2b$'s, two $2d-a-2b$'s, and the three absent-edge exponents $2d-4a$, $4d-4a-4b$, 0. Their vertex sums equal $2d$. The octahedral symmetry has two edge orbits: 12 present and three absent. The total variation is zero, while the sum over the present edges is $8\,\delta a + 4\,\delta b$. Its invariant first derivative therefore obeys $\partial_a P_W = 2\partial_b P_W$, or $D_s = 2D_r$ when each derivative is divided by four. It follows that

$$\mathcal{L}_d = \frac{4}{3}(D_r - 2D_s) = -4D_r = \partial_b W_4(d;b)|_{b=d/2}, \qquad Z_d = \frac{1}{9}\partial_b^2 W_3(d;b)\Big|_{b=d/2}. \tag{244}$$

These derivatives are used only in the proof. For direct evaluation put

$$\begin{aligned} w_1 &= -\psi_0(q) + \frac{1}{2}\Re\{\psi_0(a_\ell+\mathrm{i}\nu) + \psi_0(b_\ell+\mathrm{i}\nu)\}, \\ w_2 &= \frac{1}{8}\Re\{\psi_1(a_\ell+\mathrm{i}\nu) - \psi_1(b_\ell+\mathrm{i}\nu)\}. \end{aligned} \tag{245}$$

Then

$$\boxed{\mathcal{L}_d = \langle 4t_\ell^4 w_1 \rangle_d, \qquad Z_d = \left\langle t_\ell^3 \left(w_1^2 + \frac{w_2}{3} \right) \right\rangle_d.} \tag{246}$$

The first derivative $\langle 3t_\ell^3 w_1 \rangle_d$ vanishes identically; it is a useful normalization and tail-control check.

16.4.2 A convergent single-kernel formula for the subtracted master

Set $c_* = \psi_0(1) + \psi_0(2q)$ and, at $a = a_\ell, b = b_\ell$, define the local abbreviations below. Here h is not h_d, j and k are not indices, and r is not the wheel order:

$$\begin{aligned} p &= \Re\{\psi_0(a+\mathrm{i}\nu) + \psi_0(b+\mathrm{i}\nu)\}, \quad & h &= \Im\{\psi_0(a+\mathrm{i}\nu) - \psi_0(b+\mathrm{i}\nu)\}, \\ j &= \Re\{\psi_1(a+\mathrm{i}\nu) - \psi_1(b+\mathrm{i}\nu)\}, \quad & k &= \Im\{\psi_1(a+\mathrm{i}\nu) + \psi_1(b+\mathrm{i}\nu)\}, \\ r &= \Re\{\psi_2(a+\mathrm{i}\nu) + \psi_2(b+\mathrm{i}\nu)\}, \quad & E_2 &= -\frac{h^2}{2} - \frac{5j}{8}. \end{aligned} \tag{247}$$

The complete subtraction collapses to

$$\boxed{\mathcal{K}_d = -\frac{AU}{4} + \frac{1}{4}\left\langle t_\ell^2 \left[(c_* - p)E_2 - \frac{hk}{2} - \frac{5r}{32}\right]\right\rangle_d .} \tag{248}$$

Unlike the unsubtracted J_t of (238), this integral has no ultraviolet pole. Its large-spin summand is $O(\ell^{-3}\log\ell)$.

Here is a derivation that fixes its finite part. Write $t = 2\tau$ in (238) and $s = \mathrm{i}\nu$. Here and below s is the Mellin variable, not the long-range exponent s of (220). The mixed trace of two critical wheel operators and one operator of rim exponent $d - 2\tau$ is $J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_d$, where (the letters are local: $C(\tau)$ is not a coefficient C_j, R_τ is not R of (221), and S_τ is not $S_{\ell n}$ of (251))

$$\begin{aligned} C(\tau) &= \frac{\Gamma(2q)\Gamma(\tau)}{\Gamma(2q-\tau)}, \\ R_\tau(s) &= \frac{\Gamma(a+s)\Gamma(b-\tau-s)}{\Gamma(a+\tau+s)\Gamma(b-s)}, \\ S_\tau(s) &= \left[\frac{\Gamma(a-\tau/4+s)\Gamma(a-\tau/4-s)\Gamma(b+s)\Gamma(b-s)}{\Gamma(b+\tau/4+s)\Gamma(b+\tau/4-s)\Gamma(a+s)\Gamma(a-s)}\right]^2 . \end{aligned} \tag{249}$$

At $d = 2$ this mixed-trace step is proved, for $0 < \tau < \frac{1}{2}$ and in its full-Mellin-line reading, in Section 18.9.5. Its reading for complex τ and its status at other d are stated as input (M) in Section 21.1. The two-wheel convolution is elementary: $W_2(d;b') = B_d(d-b', 2b')$. Taking $b' = d/2 - \tau$ gives the exact identity

$$J_{2\tau} = C(\tau)^2\left[\frac{\Gamma(q-\tau/2)}{\Gamma(q+\tau/2)}\right]^4 + C(\tau)\langle t_\ell^2(R_\tau - S_\tau)\rangle_d. \tag{250}$$

On the Mellin line, the real difference starts at order τ^2:

$$\Re(R_\tau - S_\tau) = E_2\tau^2 + \left[-pE_2 - \frac{hk}{2} - \frac{5r}{32}\right]\tau^3 + O(\tau^4).$$

Using $C(\tau) = \tau^{-1} + c_* + O(\tau)$ and $t = 2\tau$ yields the integral in (248). The coefficient of t^2 in the explicit Gamma term of (250) is

$$\frac{A^4}{6} - \frac{A^2R}{2} + \frac{R^2}{8} + \frac{A(2V-U)}{12} - \frac{X}{48},$$

whereas that of J_t^Γ is

$$\frac{A^4}{6} - \frac{A^2R}{2} + \frac{R^2}{8} + \frac{A(V+U)}{6} - \frac{X}{48}.$$

Their difference is exactly $-AU/4$. This derives (248) from the master definition (238), rather than from a numerical relation search, given the mixed-trace step and the extraction of $[\tau^2]$ under the trace. Both are proved at $d = 2$ (Sections 18.9.2 and 18.9.5). At other d the first is input (M) of Section 21.1, and the second is an unproved interchange of $[\tau^2]$ with the trace (Section 21.12); both are confirmed numerically in Section 21.11.

16.5 Explicit polygamma–hypergeometric series: no integrals or parameter derivatives

All quantities needed for the nonprimitive terms can now be written using positive-argument Gamma and polygamma functions and integer sums. For $\ell, n \geq 0$ define

$$\begin{aligned} T_{\ell n} &= \frac{(q)_{\ell+n}(1-q)_n}{(2q)_{\ell+n}\, n!}, & S_{\ell n} &= \frac{(2q)_{\ell+n}\, n!}{(3q)_{\ell+n}(1+q)_n}, \\ a_{\ell n} &= \ell + n + q, & b_{\ell n} &= \ell + n + 2q, \\ \chi_m(n;q) &= (-1)^m \psi_m(n+1-q) + \psi_m(q) - (-1)^m \psi_m(1-q). \end{aligned} \tag{251}$$

In terms of (240), $a_{\ell n} = 2a_\ell + n$ and $b_{\ell n} = a_\ell + b_\ell + n$; they are not a_ℓ, b_ℓ. Reflection gives $\chi_m(n;q) = \psi_m(q-n)$, but the positive-argument expression is preferable for evaluating it. Thus the natural pair $q = d/4$ and $1-q$ remains explicit; no trigonometric replacement is required.

For $m = 1, 2, 3, 4$ set (these λ_m are not the completion exponents λ_e of (232))

$$\begin{aligned} \lambda_m = \psi_{m-1}(a_{\ell n}) - \psi_{m-1}(b_{\ell n}) - (-1)^m \chi_{m-1}(n;q) \\ + (1 + (-1)^m)\psi_{m-1}(1) - \psi_{m-1}(n+1). \end{aligned} \tag{252}$$

The following are explicit polygamma combinations, not differentiation operators:

$$\begin{aligned} \eta_m &= \frac{1}{2}\{\psi_m(a_{\ell n}) + (-1)^m \chi_m(n;q)\}, & m &= 1, 2, 3, \\ \theta_m &= \frac{1}{4}\{\psi_{m+1}(a_{\ell n}) - (-1)^m \chi_{m+1}(n;q)\}, & m &= 1, 2, \\ \mu_1 &= \frac{1}{2}\{\psi_0(a_{\ell n}) - \psi_0(q) + \psi_0(n+1-q) - \psi_0(1-q)\}, \\ \mu_2 &= \frac{1}{4}\{\psi_1(a_{\ell n}) - \psi_1(q) + \psi_1(n+1-q) - \psi_1(1-q)\}. \end{aligned} \tag{253}$$

All subscripts ℓ, n on the right are suppressed in the following polynomials:

$$\begin{aligned} F_3 &= \frac{9}{2}\lambda_1^2 + \frac{3}{2}\lambda_2, \\ F_{3,1} &= 9\lambda_1\eta_1 + \frac{3}{2}\eta_2, \\ F_{3,2} &= 9\eta_1^2 + 9\lambda_1\theta_1 + \frac{3}{2}\theta_2, \\ F_4 &= \frac{32}{3}\lambda_1^3 + 8\lambda_1\lambda_2 + \frac{2}{3}\lambda_3, \\ F_{4,1} &= 32\lambda_1^2\eta_1 + 8\eta_1\lambda_2 + 8\lambda_1\eta_2 + \frac{2}{3}\eta_3. \end{aligned} \tag{254}$$

The subscripts $1, 2$ in these labels name the displayed polynomials; they are not a request to differentiate a special function.

The primitive wheel periods and the two corresponding nonprimitive master data are

$$\boxed{\begin{aligned} Q_d &= \frac{1}{3}\sum_{\ell=0}^{\infty} d_\ell(d) \sum_{n=0}^{\infty} T_{\ell n}^3 F_3, \\ P_4(d) &= -\sum_{\ell=0}^{\infty} d_\ell(d) \sum_{n=0}^{\infty} T_{\ell n}^4 F_4. \end{aligned}} \tag{255}$$

$$\boxed{\mathcal{L}_d = -\sum_{\ell=0}^{\infty} d_\ell(d) \sum_{n=0}^{\infty} T_{\ell n}^4 (4\mu_1 F_4 + F_{4,1}).} \tag{256}$$

$$\boxed{Z_d = \frac{1}{9}\sum_{\ell=0}^{\infty} d_\ell(d) \sum_{n=0}^{\infty} T_{\ell n}^3 \left[(9\mu_1^2 + 3\mu_2)F_3 + 6\mu_1 F_{3,1} + F_{3,2}\right].} \tag{257}$$

These are ordinary nonnegative-integer series at fixed dimension. They contain only Pochhammer symbols and $\psi_0, \ldots, \psi_3$ at the explicit arguments above.

For $\mathcal{K}_d$ define, for $j = 0, 1, 2, 3$,

$$r_j = \frac{1}{2}\lambda_{j+1} + \psi_j(b_{\ell n}) + \psi_j(n+1) - (1-(-1)^j)\psi_j(1), \qquad c_* = \psi_0(1) + \psi_0(2q), \tag{258}$$

and the polynomial

$$\begin{aligned}\mathcal{R}_{\ell n} = {} & \frac{r_0 - c_*}{2}(\lambda_1^3 + 3\lambda_1\lambda_2 + \lambda_3) + \frac{3r_1}{4}(\lambda_1^2 + \lambda_2) - \frac{\lambda_1 r_2 + r_3}{4} \\ & - \frac{\lambda_1^4}{12} + \frac{\lambda_1^2\lambda_2}{8} + \frac{7\lambda_1\lambda_3}{24} + \frac{\lambda_2^2}{8} + \frac{\lambda_4}{12}.\end{aligned} \tag{259}$$

Then the explicit result is

$$\boxed{\mathcal{K}_d = -\frac{AU}{4} + \sum_{\ell=0}^{\infty} d_\ell(d) \left\{\sum_{n=0}^{\infty}\left[\frac{M^2 S_{\ell n}^2}{8} - \frac{T_{\ell n}^2 \mathcal{R}_{\ell n}}{4}\right]\right\}.} \tag{260}$$

The square brackets must be formed before summing: they are the two pole families of the same convergent integral. The inner n sum is performed before the outer spin sum. Separating the two families or rearranging this as an unrestricted double sum is not justified; for example, absolute double summability fails in part of the physical range even though the stated iterated sum converges.

Domains and continuation. For $2/3 < d < 4$, the Q_d, Z_d series and their Barnes representations converge directly; the corresponding lower threshold for P_4 and $\mathcal{L}_d$ is $d > 1/2$. The formulas cover $d = 1, 2, 3$ without any continuation. The subtracted $\mathcal{K}_d$ spin integral and its ordered residue sum converge for $0 < d < 4$. Numerically, at $d = 0.8$ the literal bracket of (260) loses its digits to cancellation far out in the radial tail, so Section 21.11 evaluates the canceled form (289) there instead. For smaller positive dimensions in the other cases, use analytic continuation of the displayed special functions from their common convergence strip, equivalently the convergent normalized coordinate/Schwinger periods. Do not evaluate individual divergent spin integrals and assign them finite numbers. The analytically continued multiplicities at $d < 1$ provide cancellations that need not survive taking absolute values term by term.

Why these are hypergeometric-series data, rather than new graph names. The coefficient $d_\ell T_{\ell n}^r$ is a two-variable Horn-type hypergeometric term: its ratios under $\ell \mapsto \ell + 1$ and $n \mapsto n + 1$ are rational. Away from removable exceptional dimensions,

$$d_\ell(d) = \frac{(4q-2)_\ell (2q)_\ell}{(2q-1)_\ell \, \ell!}.$$

Together with $(q)_{\ell+n}^r/(2q)_{\ell+n}^r$ and $(1-q)_n^r/(n!)^r$, this prints the full parameter array of the underlying Kampé de Fériet-type series. The extra polygamma polynomials are explicit resonant weights; the weighted series is not falsely labeled as an unweighted ordinary ${}_pF_q$. Most importantly, (251)–(260) are executable definitions in standard Gamma and polygamma functions, with no unsupplied master value.

16.6 Residue proof of the series

Consider the even Barnes moment for one fixed spin,

$$I_r(a,g)=\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\left[\frac{\Gamma(a+s)\Gamma(a-s)}{\Gamma(a+g+s)\Gamma(a+g-s)}\right]^r. \tag{261}$$

For the wheel at $b=d/2$, $a=a_\ell$ and $g=q$. Closing an upward contour to the right is clockwise, so the integral is the negative sum of residues at $s=a+n$. Put $s=a+n+u$. Then

$$\frac{\Gamma(a+s)\Gamma(a-s)}{\Gamma(a+q+s)\Gamma(a+q-s)}=-\frac{T_{\ell n}}{\Gamma(2q)\,u}\exp\left(\sum_{m\ge1}\frac{\lambda_m}{m!}u^m\right). \tag{262}$$

This follows from Gamma reflection and

$$\Gamma(-n-u)=\frac{(-1)^{n+1}\Gamma(1-u)}{n!\,u\prod_{j=1}^{n}(1+u/j)}.$$

The regular logarithmic derivative of this last factor at order m is $(1+(-1)^m)\psi_{m-1}(1)-\psi_{m-1}(n+1)$, which is precisely the last line of (252). The second and third complete exponential Bell polynomials give F_3 and F_4. Their signs are opposite because the pole orders are three and four. This proves (255).

For a general balanced rim exponent put $x=b/2$, $g=d/2-x$. The normalized residue prefactor is

$$-\frac{(x)_{\ell+n}(1-g)_n}{(d/2)_{\ell+n}\,n!}.$$

Its logarithmic first and second b derivatives at $b=d/2$ are μ_1,μ_2. The corresponding first and second derivatives of λ_m are the already displayed η_m,θ_m. Differentiating the convergent residue identities in the proof and then substituting these *explicit* formulas yields (256)–(257). No differentiation remains in those evaluation formulas.

For the subtracted master there are two right-hand pole families. Write

$$g(s)=\frac{\Gamma(a+s)\Gamma(a-s)}{\Gamma(b+s)\Gamma(b-s)},\quad \ell(s)=\frac{g'(s)}{g(s)},\quad p(s)=\frac{1}{2}\{\psi_0(a+s)+\psi_0(a-s)+\psi_0(b+s)+\psi_0(b-s)\}.$$

Here $\ell(s)$ is the logarithmic derivative of $g(s)$, not the spin ℓ. The holomorphic continuation of the weight in (248) is

$$F(s)=(c_*-p)\left(\frac{\ell(s)^2}{8}-\frac{5\ell'(s)}{16}\right)+\frac{\ell(s)p'(s)}{4}-\frac{5p''(s)}{32}. \tag{263}$$

Primes in this proof act on the elementary Gamma/polygamma kernel, not on an unevaluated master. At $s=a+n+u$,

$$\ell(s)=-u^{-1}+\sum_{m\ge1}\lambda_m\frac{u^{m-1}}{(m-1)!},\qquad p(s)=\frac{1}{2u}+\sum_{j\ge0}r_j\frac{u^j}{j!}.$$

Multiplying these finite jets through the required order gives the negative residue $-T_{\ell n}^2\mathcal{R}_{\ell n}/4$ of $\Gamma(2q)^2g(s)^2F(s)/4$. At the other family, $s=b+n+u$, $g(s)^2$ has a double zero but $F(s)=-1/(2u^3)+O(u^{-2})$. The resulting negative residue is $M^2S_{\ell n}^2/8$. This proves both terms and the sign in (260). The local residue polynomial was also checked independently by Cauchy circles around both families at integer and noninteger dimensions.

16.7 An additional exact two-dimensional master evaluation

The dimension-two balanced three-wheel is a complex Selberg integral. For b close to one take, in the complex Selberg identity [6], the parameters $n = 2$, $\sigma = b/2$, $\tau = 1 - b/2$ and $\theta = -b/4$. These names are local to this paragraph; they are not the σ of (233), the τ of (249) or the θ_m of (253). With $\gamma_C(x) = \Gamma(x)/\Gamma(1-x)$, its normalized value simplifies to

$$\boxed{W_3(2;b) = \frac{1}{2}\frac{\gamma_C(b/4)^3}{\gamma_C(3b/4)}.} \tag{264}$$

All convergence inequalities hold in a neighborhood of $b = 1$: $0 < b < 4/3$ suffices for the displayed two-screening integral. At $b = 1$ its logarithmic derivative vanishes, and its logarithmic second derivative is

$$\frac{3}{4}\{\psi_1(1/4) - \psi_1(3/4)\} = 12\mathsf{G}.$$

Combining this with (244) gives the exact check

$$\boxed{Z_2 = 4\mathsf{G}\,Q_2, \qquad Q_2 = \frac{1}{6}\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4.} \tag{265}$$

This is a derivation from a convergent integral identity, not an integer-relation fit to the numerical value of Z_2.

A second derivation of (264) and of (265), from Dixon's theorem and the two-node formula of Section 20.2 without the complex Selberg integral, is given in Section 20.6, (548)–(550); it holds for $0 < \operatorname{Re} b < \frac{4}{3}$.

16.8 Special-function coefficient checks

The one-dimensional specialization recovers the analytic formulas of Part II. The general master calculations give, at the precision relevant here,

d	$C_4(d)$	$C_5(d)$
2	5.048868431	−38.833645896
3	1.228063545	−5.850351421

These are master/assembly evaluations, not a second full-graph calculation in arbitrary dimension. The two-dimensional exact result $Z_2 = 4\mathsf{G}Q_2$ supplies an independent master check. The closed forms (270), (271) make the $d = 2$ entry of C_5 exact through (272).

Master values at $d = 1, 2, 3$. The six masters of (223) take the following values, correctly rounded to 30 significant digits:

Master	d	Value	Source
Q_d	1	55.001486544162982619920766 2385	(180)
	2	12.771381198646957657749763 6845	(267)
	3	5.017102564553855369590854845 69	(405)
$P_4(d)$	1	2229.52206107094196429452519975	(93)
	2	250.48731558186728553185175688 8	(333)
	3	62.058327943406672466463388584 1	(422)
$P_5(d)$	1	28599.29317656210353664647702 14	(29)
	2	1548.08264023312061513243907395	(334)
	3	240.34031775908354981636654638 8	(425)
$\mathcal{K}_d$	1	39.33156912825903905750318715 16	(76)
	2	3.51810913859025224164909017031	(271)
	3	1.14859464952717933655664631081	(260), record below
$\mathcal{L}_d$	1	−5805.30162954880238014117987700	(94)
	2	−347.24915312320113105847678307 5	(270)
	3	−62.71058496838766934729617343 35	records below
Z_d	1	764.93103019958857704016177876 9	(224)
	2	46.79258307232969665673882587 80	(265)
	3	8.82092489306009248553686732817	records below

At $d = 1$ and $d = 2$ every entry is a closed form of this note, evaluated at 60 digits; the entries agree with all values of these masters printed elsewhere in the note. At $d = 3$, Q_3, $P_4(3)$ and $P_5(3)$ are the fixed rational formulas of Section 19, evaluated at 90 digits with 160 summed terms and a Hurwitz-zeta tail expansion of order 64 (`LRI_C5_d3/inputs/general_d3/period_values.json`); Section 19.10 compares them with a less converged 55-digit run and with independent integrations of the Gamma kernels. $\mathcal{K}_3$ is the 430-digit ball-arithmetic evaluation of (260) (`LRI_C5_d3/wf_K3/K3_value.json`); at $d = 3$ that series rests on input (M) of Section 21.1 and on the extraction of $[\tau^2]$ under the trace (Section 21.12). The two-fold Barnes integral (588), which uses neither, reproduces it to $4 \cdot 10^{-100}$ (Section 21.11). $\mathcal{L}_3$ and Z_3 come from the $d = 3$ formulas of Section 20.6, which rest on the half-shift identities of Section 20.6 (records `LRI_C5_d3/wf_L3/L3_final_values.json`, about 158 digits, and `LRI_C5_d3/wf_Z3/Z3_values.json`, about 250 digits). An independent 100-digit evaluation of the spectral traces of Section 16.4, with the derivatives in b of (244) and the τ-expansion of (250) taken on Cauchy circles (`LRI_C5_d3/run_C_combined.json`), agrees with $\mathcal{L}_3$, Z_3 and $\mathcal{K}_3$ to 1.2×10^{-93}, 2.9×10^{-98} and 1.5×10^{-96}, and with $3Q_3$ and $P_4(3)$ to 3.5×10^{-79} and 2.1×10^{-86}. These agreements are numerical, not interval bounds. Values at other dimensions (and Z_d to 36 digits) are in the validation tables of Sections 20.3.1 and 20.9; the first, lower-precision evaluations are described in the paragraph on controls below.

An independent route to $P_5(2)$. The spectral formula (216) has a regular limit $d \to 2$. Rename its angular index n (of (214) and (215)) to ℓ. At $d = 2$ one has $p = 0$ and $h_2 = 1/\pi$, the joint limit of the Gamma ratios in (215) gives the multiplicities $d_\ell(2) = 1, 2, 2, \ldots$, and $\tau_\ell^{(2)}(\nu) = (-1)^\ell \pi\, t_\ell(\nu)$, with $t_\ell(\nu)$ of (240) at $d = 2$: there $\Gamma(2q) = 1$ and $t_\ell(\nu) = |\Gamma(\ell/2 + \frac{1}{4} + \mathrm{i}\nu)/\Gamma(\ell/2 + \frac{3}{4} + \mathrm{i}\nu)|^2$. For $L = 5$,

$$P_5(2) = \frac{1}{2\pi} \sum_{\ell=0}^{\infty} (-1)^\ell d_\ell(2) \int_0^\infty (\ell^2 + 4\nu^2)\, t_\ell(\nu)^7 \, d\nu. \tag{266}$$

OpenAI GPT-6 astra evaluated (266) in double precision, in the package `lri_general_special_functions_update/` (script `code/general_special/spectral_evaluate.py`, record `verification/general_special/spectral_master_values.json`, entry `zigzag_d2`). With 192, 384 and 768 quadrature nodes it agrees with the table to 6×10^{-11}, and the same limit for $L = 3$ reproduces $P_3(2) = 3Q_2$. This route uses neither the square factorization of Section 18.5 nor the finite sampling of Section 18.

Controls of the master evaluations. These controls belong to the same general-dimensional calculation (folder `code/general_special/` of `lri_general_special_functions_update/`, records in `verification/general_special/`). The first values of Q_d, $P_4(d)$, $\mathcal{L}_d$, Z_d and $\mathcal{K}_d$ came from the common spectral kernel of Section 16.4 (`spectral_evaluate.py`, record `spectral_master_values.json`). That evaluator is independent of the two-contour kite representation, which was used separately to check $\mathcal{K}_d$ after extracting its pinch residue (`kappa_mb.py`, record `kappa_mb_checks.json`). The common-kernel runs vary the spin cutoff (128, 256, 512), the number of quadrature nodes (384, 512, 768) and the number of terms of an angular-tail expansion. That expansion follows from centered Gamma ratios and even inverse powers of $\ell + (d-2)/2$; its fitted coefficients accelerate the integration and are not an ansatz for a master identity. At $d = 2$ and $d = 3$ the two-contour and common-kernel values of $\mathcal{K}_d$ agree to about 10^{-11} or better. The slower two-contour run at $d = 1$ is not assigned that accuracy; there the one-dimensional closed form is the comparison target. Local Cauchy residues test both families of (260) (`local_residue_checks.json`), and raw finite sums of the series converge towards the spectral values (`ordered_series_checks.json`); raw rectangular truncations are not assigned an error bound, and the iterated convention of (260) is kept. Exact SymPy calculations (`exact_audit.py`) check the three-loop benchmark, both fixed-point assemblies, the cancellation of M and N, both one-dimensional specializations, the tetrahedron quadratic normalization and the local $\mathcal{K}$ residue polynomial. The digits of these runs are superseded by the high-precision values of Section 16.8 (see also Sections 17.2, 19.10, 20.9 and 21.11); they remain an independent lower-precision route. They are master-level checks, not a second general-dimensional full-graph calculation, and not interval enclosures.

The one-dimensional polygamma form of $\mathcal{K}$ does not lift to general d. The combination $2\psi_1(q)^2 - 3R\psi_1(q) + A\psi_2(q)/2$, with A, R of (221), equals $\mathcal{K}$ at $d = 1$ but gives $25.3748796\ldots$ at $d = 2$, whereas $\mathcal{K}_2 = 3.5181091\ldots$ (record `kappa_mb_checks.json`).

17 Finite multivariable representations for the masters

This section writes the masters of (223) in standard function classes. For $\mathcal{L}_d$, Z_d and $\mathcal{K}_d$ it reaches finite combinations of unweighted multivariable Horn series at unit arguments (Section 17.6 and the finite arrays after it). This is a function-class statement, not a reduction to a few ordinary ${}_pF_q$ values: the combinations are long, outside a common convergence strip only the complete combination may be continued in d, and the weighted double series of Section 16.5 remain preferable for numerical work. For the primitive $P_5(d)$ the section gives a derivative-free double residue series (Section 17.4); the shorter finite sampling of the primitives is Section 18, and at $d = 2$ all masters have closed forms (Section 17.2). Apart from those closed forms, the constructions of this section were written by OpenAI GPT-6 astra (package `lri_general_closed_update/`).

17.1 The two-dimensional normalization and its immediate consequences

At $d = 2$ the tetrahedron normalization (177), with $\mathcal{T}_2$ from (192), gives

$$\boxed{Q_2 = \frac{1}{6}\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4 = \frac{\Gamma(1/4)^8}{24\pi^4}.} \tag{267}$$

Indeed, reflection gives $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$, so the fourth power of the ratio is $\Gamma(1/4)^8/(4\pi^4)$ [26]. The factor $1/6$ is that of $Q_d = \mathcal{T}_d/6$ in (177); it must not be dropped. Numerically,

$$Q_2 = 12.771381198646957657749763684524008422037284985934 7\ldots, \qquad Z_2 = \frac{\mathsf{G}\,\Gamma(1/4)^8}{6\pi^4}. \tag{268}$$

The second identity is $Z_2 = 4\mathsf{G}Q_2$ of (265), written in the same Gamma normalization.

For $a = \log 2$, the dimension-two specialization of all coefficients is

$$\begin{aligned}
C_0(2) &= 1, & C_1(2) &= -\frac{1}{6}, \quad C_2(2) = \frac{4a}{9},\\
C_3(2) &= \frac{4a^2}{27} - \frac{\Gamma(1/4)^8}{324\pi^4},\\
648C_4(2) &= 16P_4(2) - 64a^3 - 32aQ_2 - 359\zeta(3),\\
46656C_5(2) &= -1536\{P_5(2) + \mathcal{K}_2\} + 1536aP_4(2) - 576\mathcal{L}_2 - 1728Q_2^2\\
&\quad + (11520a^2 + 28608\mathsf{G})Q_2 - 3840a^4 - 21456a\zeta(3).
\end{aligned} \tag{269}$$

The remaining dimension-two masters in this display are the fixed hypergeometric functions specified below, not new undetermined constants; the two nonprimitive ones, $\mathcal{K}_2$ and $\mathcal{L}_2$, are evaluated in closed form in Section 17.2. None of these identities is inferred from numerical values.

17.2 Closed forms of the two-dimensional nonprimitive masters

The two masters of (269) that are neither primitive periods nor Gamma values have closed forms in constants that are already present:

$$\boxed{\mathcal{L}_2 = -2\log 2\, P_4(2) = -\frac{A}{2}\, P_4(2) = -\frac{\log 2}{2\pi^3}(B_-^2 - B_+^2),} \tag{270}$$

$$\boxed{\mathcal{K}_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2 = \frac{1}{2}\Big[\frac{\{\psi_1(\frac{1}{4}) - \psi_1(\frac{3}{4})\}^2}{4} + AU\Big].} \tag{271}$$

Here $A = 4\log 2$ and $U = \psi_2(\frac{1}{2}) = -14\zeta(3)$ are the values of (221) at $d = 2$, $B_\pm$ are the retained one-dimensional values of (330), and $\psi_1(\frac{1}{4}) - \psi_1(\frac{3}{4}) = 16\mathsf{G}$ by (73). The second form of (271) reads $\mathcal{K}_2 = \frac{1}{2}[\Phi_2 + AU]$ with $\Phi_2 = 64\mathsf{G}^2$. The one-dimensional result (76) has the same shape, $\mathcal{K} = \frac{1}{2}[\Phi_1 + AU]$ with $\Phi_1 = \psi_1(\frac{1}{4})\{3\psi_1(\frac{1}{4}) - \psi_1(\frac{3}{4})\}$ and the $d = 1$ values of A and U.

Status. Both identities are proved; no numerical identification enters either proof.

- (270) is proved in Section 18.7 for $\mathcal{L}_2$ defined by the series (256) at $d = 2$. The same proof establishes the two single-sum relations (356), which were first found numerically. The wheel-derivative reading $\partial_b W_4(2; b)|_{b=1} = -2\log 2\, W_4(2; 1)$ of (244), where $W_4(2; 1) = P_4(2)$ by (243), is the form in which $\mathcal{L}_d$ enters (229). It follows because the function $\mathcal{W}(x)$ of Section 18.7 equals $W_4(2; 2x)$ for $\frac{2}{5} \le x \le \frac{3}{5}$. This is the per-spin residue evaluation of (261) at rim exponent $2x$, written out at $d = 2$ as (360) in Section 18.7.6; it is also the case $d = 2$ of (479) with (539). Since the two functions agree on an interval, so do their derivatives. The wheel-derivative reading has also been proved directly, by a rim–spoke reflection identity (Section 18.7.6).
- (271) is proved in Section 18.8 for $\mathcal{K}_2$ defined by the master path (238). The proof starts from the position-space integral and uses neither (248) nor (260). The series (260) and the spectral form (248) are also proved to equal the right side of (271), in Section 18.9, which also proves the mixed-trace step at $d = 2$ and gives a second position-space proof of (377).

Three further proofs were found independently. Two of them prove (270) through the rim derivative (347) and the residue functional (350) of Section 18.7, but obtain the derivatives of the two modes differently: one evaluates the single sums of (352) by Hardy-space Parseval pairings on the unit

circle (Section 18.7.8), the other differentiates a Fourier (Cauchy beta integral) representation of the two-mode coefficients (Section 18.7.9). The logarithmic elliptic identities they need are proved in Section 18.7.7 by half-strip contours and the real Landen transformation, without Wan's Stieltjes identity. The third proves that the right side of (248) at $d = 2$ equals (271). It regularizes (250) in the rim exponent, evaluates the regulated trace by residues on the square lattice and periodizes the resulting single sums; it is written out in Sections 18.9.2–18.9.6. Its statements for (260) and (238) rest on the per-spin residue identity and on the mixed-trace step, which Sections 18.9.7 and 18.9.5 prove at $d = 2$.

Numerical confirmation. These checks are not proof steps. A 70-digit Euler–Maclaurin evaluation of the square-lattice factorization of (256), and of (260) through the limit $\theta \to 0$ of its factorized single sums at $z = e^{\mathrm{i}\theta}$, gives the residuals 4.5×10^{-69} for (270) and 2.0×10^{-70} for (271). The factorized single sums of (260) diverge individually, so the unrestricted double sum is not used; the limit $\theta \to 0$ of the lattice sum with weight $e^{\mathrm{i}(m-n)\theta}$ preserves the spin order and equals the ordered spin sum (part (c) of the proposition on the series and the strip limit, Section 18.9.7). An independent evaluation from the spectral kernel, with $\mathcal{L}_2 = \partial_b W_4(2;b)|_{b=1}$ as in (244) and $\mathcal{K}_2$ from the spectral form (248), gives 1.7×10^{-79} and 3.1×10^{-62}. To fifty digits,

$$\begin{aligned} P_4(2) &= 250.48731558186728553185175688791140716835571092358\ldots, \\ \mathcal{L}_2 &= -347.24915312320113105847678307522316920816760199487\ldots, \\ \mathcal{K}_2 &= 3.51810913859025224164909017030773952262267280947 10\ldots, \\ P_5(2) &= 1548.08264023312061513243907394578934038811023232 29\ldots. \end{aligned}$$

The five-loop coefficient. Inserting (270) and (271) into (269) and dividing by 48 gives (exact algebra)

$$\boxed{\begin{aligned} 972\,C_5(2) = {} & -32P_5(2) + 56\log 2\,P_4(2) - 36Q_2^2 + (240\log^2 2 + 596\mathsf{G})Q_2 \\ & - 1024\mathsf{G}^2 + 449\,\zeta(3)\log 2 - 80\log^4 2. \end{aligned}} \tag{272}$$

In the general-dimensional form (223) the two substitutions are $384AP_4 - 576\mathcal{L}_2 = 672AP_4$ and $-1536\mathcal{K}_2 = -768(\Phi_2 + AU)$ at $d = 2$. Thus $C_5(2)$ requires only $\Gamma(\frac{1}{4})$, π, $\log 2$, G, $\zeta(3)$, the two ${}_4F_3(1)$ values $F_\pm$ of (331) and the five fixed values (332) that enter (334). All of them already occur in the one-dimensional coefficients, and no multivariable value of Section 17.6 is needed at $d = 2$. Numerically (272) gives $C_5(2) = -38.8336458962253605529369 8\ldots$, in agreement with the table of Section 1.

Remaining gaps. There is no gap in the proofs of (270) and (271) as stated. The following related points are not settled in this note.

- Wan's Stieltjes identity (100), which enters the proof of Section 18.7 through $U_0 = 2N_{\mathrm{e}}$ and (101), is cited rather than reproved there, exactly as in Section 9.2. The proofs of Sections 18.7.6, 18.7.8 and 18.7.9 do not use it. The half-strip contours of Section 18.7.7 prove $U_0 = 2N_{\mathrm{e}}$ and reprove (103), (106) and (105) without it, and Section 18.5.2 reproves (340), hence (333), without it; among the proofs of (270) only the first still uses it, through (101).
- The derivation of (248) in Section 16.4 uses the mixed-trace representation $J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_d$ stated before (250). At $d = 2$ it is proved for $0 < \tau < \frac{1}{2}$, in its full-Mellin-line (real-part) reading only, in Section 18.9.5; its reading for complex τ and its status at other d are input (M) of Section 21.1. The closed forms here do not use it.
- The four-exponent generalization of the rim–spoke reflection ratio of Section 18.7.6 has been checked numerically but not proved; it is not used.

17.3 A common residue dictionary

For nonnegative integers ℓ, n put $m = \ell + n$, $N = \ell + 2n$, and

$$\begin{aligned} T_{\ell n} &= \frac{(q)_{\ell+n}(1-q)_n}{(2q)_{\ell+n} n!}, \\ d_0(4q) &= 1, \qquad d_{j+1}(4q) = 4q\frac{(2q+1)_j(4q-1)_j}{(2q)_j(2)_j}, \quad j \geq 0. \end{aligned} \tag{273}$$

The second line is a nonsingular way to implement angular multiplicities. At $d = 1$ the upper parameter $4q - 1 = 0$ terminates the positive-spin series at $j = 0$, leaving just $\ell = 0, 1$. At $d = 2$ every positive-spin multiplicity is two. At noninteger $d < 1$ this is an analytic multiplicity, not a positive integer count of modes.

Define four useful classes of explicit scalar sequences:

$$\begin{aligned} \Delta_r(a, b; v) &= \psi_r(v + a) - \psi_r(v + b), \qquad r \geq 0, \\ J_r(a; v) &= \psi_r(v + a), \qquad r \geq 1, \\ H(a; v) &= \psi_0(v + a) - \psi_0(a), \\ c_j &= (-1)^{j+1}\psi_{j-1}(q) - \psi_{j-1}(1 - q) + (1 + (-1)^j)\psi_{j-1}(1), \\ \widetilde{D}_j &= \Delta_{j-1}(q, 2q; m) + \Delta_{j-1}(1 - q, 1; n), \qquad \lambda_j = c_j + \widetilde{D}_j. \end{aligned} \tag{274}$$

Here $j \leq 6$ suffices. All appearances of ψ_r with a summation index will be eliminated by finite Horn insertions below. Only c_j and the other fixed-argument prefactors retain polygamma notation.

For the lower wheel functions, define, with the indicated finite ranges,

$$\begin{aligned} \eta_r &= \frac{(-1)^r\psi_r(q) - \psi_r(1-q)}{2} + \frac{J_r(q; m) + J_r(1 - q; n)}{2}, && 1 \leq r \leq 3, \\ \theta_r &= -\frac{(-1)^r\psi_{r+1}(q) + \psi_{r+1}(1-q)}{4} + \frac{J_{r+1}(q; m) + J_{r+1}(1 - q; n)}{4}, && 1 \leq r \leq 2, \\ \mu_1 &= \frac{H(q; m) + H(1 - q; n)}{2}, \\ \mu_2 &= \frac{J_1(q; m) + J_1(1 - q; n) - \psi_1(q) - \psi_1(1 - q)}{4}. \end{aligned} \tag{275}$$

The ordinary finite polynomials needed are

$$\begin{aligned} F_3 &= \frac{9}{2}\lambda_1^2 + \frac{3}{2}\lambda_2, & F_{3,1} &= 9\lambda_1\eta_1 + \frac{3}{2}\eta_2, \\ F_{3,2} &= 9\eta_1^2 + 9\lambda_1\theta_1 + \frac{3}{2}\theta_2, \\ F_4 &= \frac{32}{3}\lambda_1^3 + 8\lambda_1\lambda_2 + \frac{2}{3}\lambda_3, \\ F_{4,1} &= 32\lambda_1^2\eta_1 + 8\eta_1\lambda_2 + 8\lambda_1\eta_2 + \frac{2}{3}\eta_3. \end{aligned} \tag{276}$$

Thus the already proved wheel definitions can be written as

$$\boxed{\begin{aligned} Q_d &= \frac{1}{3}\sum_{\ell,n\geq 0} d_\ell(d) T_{\ell n}^3 F_3, \\ P_4(d) &= -\sum_{\ell,n\geq 0} d_\ell(d) T_{\ell n}^4 F_4, \\ \mathcal{L}_d &= -\sum_{\ell,n\geq 0} d_\ell(d) T_{\ell n}^4 (4\mu_1 F_4 + F_{4,1}), \\ Z_d &= \frac{1}{9}\sum_{\ell,n\geq 0} d_\ell(d) T_{\ell n}^3 \{(9\mu_1^2 + 3\mu_2)F_3 + 6\mu_1 F_{3,1} + F_{3,2}\}. \end{aligned}} \tag{277}$$

The subscripts on $F_{r,j}$ are labels of printed polynomials, not unperformed derivatives. Equations (273)–(277) are also the exact finite inputs for the unweighted Horn construction.

17.4 A double residue series for $P_5(d)$

This subsection completes the residue dictionary above for the remaining primitive (implementation `code/general_closed/primitive_series.py` of `lri_general_closed_update/`). Like (255) for Q_d and $P_4(d)$, the result is an explicit double series, with no Barnes integral and no parameter derivative. At $d = 1, 2, 3$ the coefficients use the fixed-value formulas (29), (334) and (425); for other d the stated evaluator is (319). The series is an independent representation and a check of them. The proof below uses objects defined in Section 18, which follows: the Barnes form (303) with the kernel (300), the measure block (324), the array $\mathcal{A}_{\ell k}$ and the values $G_{\ell k}$ of (309), the two-channel identity (325) and the polynomials E_m of (316).

In this subsection θ is $\theta_0 = 2q$ or $\theta_1 = 2q - 1$, and M_θ is the measure block of (324); this θ is not a wheel sequence θ_r of (275). At the shifted arguments $q + N$, $N = 0, 1, 2, \ldots$, the relation $\Gamma(x - N) = (-1)^N \Gamma(x)/(1 - x)_N$ gives

$$M_\theta(q + N) = \frac{\Gamma(\theta + q + N)\Gamma(\theta - q - N)}{\Gamma(q + N)\Gamma(-q - N)} = M_\theta(q)\, \frac{(\theta + q)_N (1 + q)_N}{(q)_N (1 + q - \theta)_N}. \tag{278}$$

In the second channel below only $\ell \geq 1$ contributes, so $N \geq 1$ there, and one uses

$$M_\theta(q + N) = M_\theta(q + 1)\, \frac{(\theta + q + 1)_{N-1} (2 + q)_{N-1}}{(q + 1)_{N-1} (2 + q - \theta)_{N-1}}, \qquad N \geq 1, \tag{279}$$

rather than the prefactor $M_\theta(q)$ of (278). For $\theta = 2q - 1$ that prefactor contains $\Gamma(3q - 1)$, which is singular at $q = \frac{1}{3}$; there, for $N \geq 1$, it multiplies the vanishing $(3q - 1)_N$. The value $M_{2q-1}(q)$ is itself singular at $q = \frac{1}{3}$, so the $\ell = 0$ terms of the second channel are omitted rather than weighted by zero. In the block notation of (296), $\ell = j + 1$ and $N - 1 = j + 2n$. The first channel uses both $\ell = 0$ and $\ell \geq 1$.

With c_j, $\widetilde{D}_j$ and Δ_r of (274) and $N = \ell + 2n$ of (273), put, for $1 \leq j \leq 6$,

$$\begin{aligned} \kappa_{\theta,j} &= \kappa_j^0 + 7\widetilde{D}_j + 2^j \{\Delta_{j-1}(\theta + q, q; N) + \Delta_{j-1}(1 + q, 1 + q - \theta; N)\}, \\ \kappa_j^0 &= \begin{cases} (7 - 2^{j+1})\, c_j, & j \text{ odd}, \\ 7c_j, & j \text{ even}. \end{cases} \end{aligned} \tag{280}$$

Thus $(\kappa_1^0, \ldots, \kappa_6^0) = (3c_1, 7c_2, -9c_3, 7c_4, -57c_5, 7c_6)$, the same in both channels. This follows from reflection and recurrence in the proof below, not from a fit. The $\kappa_{\theta,j}$ are unrelated to the normalization $\kappa(d, z)$ of (174). Let E_6 be the polynomial of (316), displayed in (317); $E_6(x_1, \ldots, x_6)$ is the coefficient of u^6 in $\exp(\sum_{j=1}^6 x_j u^j/j!)$. Then

$$\boxed{\begin{aligned} P_5(d) = \frac{1}{2\Gamma(2q)^2} \sum_{\ell,n \geq 0} (-1)^\ell d_\ell(d)\, T_{\ell n}^7 \Big[& M_{2q}(q + N)\, E_6(\kappa_{2q,1}, \ldots, \kappa_{2q,6}) \\ & + \ell(\ell + 4q - 2)\, M_{2q-1}(q + N)\, E_6(\kappa_{2q-1,1}, \ldots, \kappa_{2q-1,6}) \Big]. \end{aligned}} \tag{281}$$

Every ingredient is explicit and free of derivatives. The double series converges absolutely for $\frac{2}{5} < d < 4$; smaller positive dimensions are reached by analytic continuation of the same master.

Proof. Take $\frac{1}{2} < q < 1$ and start from the Barnes form (303), with $p = 2q - 1$ and a_ℓ, b_ℓ, $g_\ell(s)$ of (300); in this strip the contour separates the pole families. The line points upwards and is closed to the right, so each spin integral is minus the sum of its right residues. There are two families of

right poles: $s = a_\ell + n$, $n \geq 0$, of order seven, from $g_\ell(s)^7$, and the measure poles $s = (p+k)/2$, $k \geq 0$, from $\Gamma(p-2s)$. The second family must not simply be discarded term by term.

Measure poles. Since $\operatorname{Res}_{s=(p+k)/2} \Gamma(p-2s) = (-1)^{k+1}/(2\,k!)$ and $\Gamma(p+k)\Gamma(-p-k) = (-1)^{k+1}\pi/[(p+k)\sin\pi p]$, minus the residue at $s = (p+k)/2$, times the spin weight $(-1)^\ell d_\ell$, is $-\Gamma(2p+1)\sin(\pi p)\,\mathcal{A}_{\ell k}/(4\pi)$ before the overall factor $\Gamma(2q)^5/2$, with $\mathcal{A}_{\ell k}$ and $G_{\ell k} = g_\ell((p+k)/2)$ of (309). The multiplicity d_k in $\mathcal{A}_{\ell k}$ arises as $\Gamma(2p+k)(p+k)/k! = \Gamma(2p+1)d_k/2$, by (240) with $d-2 = 2p$. The factor is finite in the initial strip and independent of ℓ, k. Gamma reflection gives $G_{k\ell} = (-1)^{\ell-k} G_{\ell k}$ for integer ℓ, k (Section 18), and the power seven is odd, so

$$\mathcal{A}_{k\ell} = -\mathcal{A}_{\ell k}, \qquad \mathcal{A}_{\ell\ell} = 0, \qquad \sum_{\ell,k\geq 0} \mathcal{A}_{\ell k} = 0. \tag{282}$$

The last statement uses absolute convergence. For $\frac{1}{2} < q < 1$ and all $\ell, k \geq 0$, with $h = |\ell - k|$,

$$|\mathcal{A}_{\ell k}| \leq \text{const} \cdot (1+\ell+k)^{q-3}\, h\, (1+h)^{-7q}. \tag{283}$$

Indeed, $d_\ell(d) = O((1+\ell)^{4q-2})$ by (240) and $4q-2 > 0$, so $d_\ell d_k \leq \text{const} \cdot (1+\ell+k)^{8q-4}$; next, $|(\ell+p)^2 - (k+p)^2| = h(\ell+k+2p) \leq 2h(1+\ell+k)$; finally, Gamma-ratio estimates [27] bound the first ratio in $G_{\ell k}$ by a constant times $(1+\ell+k)^{-q}$ and, for $\ell \geq k$, the second by a constant times $(1+h)^{-q}$, and $|G_{\ell k}| = |G_{k\ell}|$ covers $\ell < k$. The right side of (283) is summable over $\ell, k \geq 0$: at fixed $\ell + k$ each value of h occurs at most twice, so the factor $h(1+h)^{-7q}$ summed over these pairs is at most $2\sum_{h\geq 0} h(1+h)^{-7q}$, finite since $7q > 2$, and $(1+\ell+k)^{q-3}$ is summable over $\ell + k$ since $q < 2$. These estimates justify the exchange $\ell \leftrightarrow k$, so the measure poles cancel by an exact involution; no divergent sum is discarded. (The periodized kernel (304) needs the finer statement that canceling pairs lie in the same congruence class; it is shown after (309).)

The two channels. Since $\Gamma(p+1\pm 2s) = (p \pm 2s)\Gamma(p \pm 2s)$,

$$[(\ell+p)^2 - 4s^2]\Gamma(p+2s)\Gamma(p-2s) = \Gamma(p+1+2s)\Gamma(p+1-2s) + \ell(\ell+2p)\Gamma(p+2s)\Gamma(p-2s).$$

After division by $\Gamma(2s)\Gamma(-2s)$ this is (325) at $V = 2s$, because $p+1 = 2q$ and $2p = 4q-2$. At $s = a_\ell + n + u$ one has $V = q + N + 2u$, so the two channels are $M_{2q}(q+N+2u)$ and $\ell(\ell+4q-2)M_{2q-1}(q+N+2u)$.

Local expansion. At $s = a_\ell + n + u$, (262) gives $g_\ell(s) = -T_{\ell n}\exp(\sum_{j\geq 1}\lambda_j u^j/j!)/(\Gamma(2q)u)$, with $\lambda_j = c_j + \widetilde{D}_j$ of (274); these extend the λ_m of (252) to $j \leq 6$. For either channel,

$$\log\frac{M_\theta(q+N+2u)}{M_\theta(q+N)} = \sum_{j\geq 1}\frac{u^j}{j!}\Big\{2^j\left[\psi_{j-1}(\theta+q+N) - \psi_{j-1}(q+N)\right] + (-2)^j\left[\psi_{j-1}(\theta-q-N) - \psi_{j-1}(-q-N)\right]\Big\}.$$

Write $\mathrm{ct}_m(x) = \frac{d^m}{dx^m}\pi\cot(\pi x)$, so that reflection reads $\psi_m(x) = (-1)^m\psi_m(1-x) - \mathrm{ct}_m(x)$ [26]. Applied to the two negative arguments, the polygamma part of the second bracket becomes $2^j\Delta_{j-1}(1+q, 1+q-\theta; N)$, and, since ct_m has period one, its cotangent part is $(-2)^j[\mathrm{ct}_{j-1}(-q) - \mathrm{ct}_{j-1}(\theta-q)]$. In both channels $\theta - q \equiv q \pmod 1$, and $\mathrm{ct}_m(-x) = (-1)^{m+1}\mathrm{ct}_m(x)$, so this equals $(-2)^j[(-1)^j - 1]\,\mathrm{ct}_{j-1}(q)$. It vanishes for even j. For odd j it is $2^{j+1}\mathrm{ct}_{j-1}(q) = -2^{j+1}c_j$, because then $c_j = \psi_{j-1}(q) - \psi_{j-1}(1-q) = -\mathrm{ct}_{j-1}(q)$. Adding $7\lambda_j$ from $g_\ell(s)^7$ gives exactly $\kappa_{\theta,j}$ of (280). Hence $g_\ell(s)^7 M_\theta(q+N+2u)$ equals $-(T_{\ell n}/\Gamma(2q))^7\, M_\theta(q+N)\, u^{-7}\exp(\sum_{j\geq 1}\kappa_{\theta,j}u^j/j!)$, and minus its residue at $u = 0$ is $(T_{\ell n}/\Gamma(2q))^7 M_\theta(q+N)E_6(\kappa_{\theta,1},\ldots,\kappa_{\theta,6})$, the sign coming from the seventh power. Summing over n and ℓ with the weights $(-1)^\ell d_\ell$ and multiplying by $\Gamma(2q)^5/2$ gives (281) for $\frac{1}{2} < q < 1$. The local expansion is a device of the proof; the final polynomial needs no coefficient extraction.

Convergence and continuation. For fixed ℓ the terms of (281) are $O(n^{-10q})$; for fixed n each channel is $O(\ell^{q-2})$; in the joint large-index region they are $O((\ell+n)^{-6q-2})$. These orders follow from

$d_\ell(d) = O(\ell^{4q-2})$, $T_{\ell n} = O((1+\ell+n)^{-q}(1+n)^{-q})$, $M_\theta(q+N) = O(N^{2\theta})$ and the boundedness of $\kappa_{\theta,j}$; the elementary estimates are not written out. Hence the series converges absolutely for $\frac{1}{10} < q < 1$, locally uniformly away from the removable parameter singularities treated by (279). Uniform convergence extends the identity from $\frac{1}{2} < q < 1$ to this range, and smaller positive dimensions follow by analytic continuation of the same master; no finite subtraction is added. □

The numerical checks of (281) are collected with the other checks of this section in Section 17.7.

17.5 Making the $\mathcal{K}_d$ cancellation explicit

The paired convention of (260) is essential. We expose its common non-summable part before passing to separate Horn terms. Put

$$\begin{aligned} x &= \pi\cot(\pi q), \qquad P = \pi^2, \\ R_\infty &= \frac{(P+x^2)^2}{2} = \frac{\pi^4}{2\sin^4(\pi q)}, \\ J_j &= J_j(2q;m) + J_j(1;n), \quad 1 \le j \le 3, \\ H &= H(2q;m) + H(1;n). \end{aligned} \tag{284}$$

Here P is just π^2 and is not a graph period. The first four fixed constants in (274) simplify to

$$(c_1,c_2,c_3,c_4) = \left(-x, -x^2 - \frac{2}{3}P, -2x^3 - 2Px, -6x^4 - 8Px^2 - \frac{28}{15}P^2\right). \tag{285}$$

In particular $c_1^3 + 3c_1c_2 + c_3 = 0$. This identity removes an otherwise standalone logarithmic large-index term.

The full remaining polynomial is

$$\begin{aligned} \mathcal{P} = {} & \frac{\widetilde{D}_1^4}{6} - \frac{2x\widetilde{D}_1^3}{3} + \frac{5}{4}\widetilde{D}_1^2\widetilde{D}_2 + \frac{3}{4}\widetilde{D}_1^2 J_1 - \left(\frac{13P}{12} + \frac{x^2}{4}\right)\widetilde{D}_1^2 \\ & - \frac{5x}{2}\widetilde{D}_1\widetilde{D}_2 + \frac{5}{12}\widetilde{D}_1\widetilde{D}_3 - \frac{3x}{2}\widetilde{D}_1 J_1 - \frac{1}{4}\widetilde{D}_1 J_2 + \left(\frac{4}{3}Px + x^3\right)\widetilde{D}_1 \\ & + \frac{1}{2}\widetilde{D}_2^2 + \frac{3}{4}\widetilde{D}_2 J_1 + \left(-\frac{11P}{12} + \frac{x^2}{4}\right)\widetilde{D}_2 - \frac{5x}{12}\widetilde{D}_3 - \frac{1}{24}\widetilde{D}_4 \\ & + H\left(\frac{1}{2}\widetilde{D}_1^3 - \frac{3}{2}x\widetilde{D}_1^2 + \frac{3}{2}\widetilde{D}_1\widetilde{D}_2 - P\widetilde{D}_1 - \frac{3}{2}x\widetilde{D}_2 + \frac{1}{2}\widetilde{D}_3\right) \\ & - \frac{P}{2}J_1 + \frac{x}{4}J_2 - \frac{1}{4}J_3. \end{aligned} \tag{286}$$

Every monomial contains at least one $\widetilde{D}_j$ or J_j; the possible H factor does not occur alone. Direct exact algebra in the residue polynomial gives $\mathcal{R}_{\ell n} = R_\infty + \mathcal{P}$.

Gauss summation [4] gives a second useful identity:

$$\begin{aligned} r(c) &= \frac{\Gamma(c)^2}{\Gamma(c-q)\Gamma(c+q)} = {}_2F_1\left(\begin{matrix} q, -q \\ c \end{matrix}; 1\right), \\ \delta(c) &= r(c) - 1 = -\frac{q^2}{c}\,{}_3F_2\left(\begin{matrix} 1, 1-q, 1+q \\ 2, c+1 \end{matrix}; 1\right). \end{aligned} \tag{287}$$

The second line follows by isolating the zeroth term and shifting the remaining index. Both series converge for the positive c used here. Using $S_{\ell n}$ and M of (260), gamma reflection shows

$$\frac{M^2 S_{\ell n}^2}{T_{\ell n}^2} = 2R_\infty\, r(m+2q)^2 r(n+1)^2. \tag{288}$$

Therefore the exact canceled form is

$$\boxed{\mathcal{K}_d = -\frac{AU}{4} + \frac{1}{4}\sum_{\ell,n\ge 0} d_\ell(d) T_{\ell n}^2 \Big\{ R_\infty\big[(1+\delta(m+2q))^2(1+\delta(n+1))^2 - 1\big] - \mathcal{P}\Big\}.} \tag{289}$$

The bracket involving δ has exactly eight nonconstant monomials, with coefficients $\binom{2}{i}\binom{2}{j}$, $0 \le i,j \le 2$, $(i,j) \ne (0,0)$. Together with (286), this is a finite polynomial input to Horn functions; the two original divergent family sums have not been assigned independent values.

For $0 < q < 1/2$, the monomial sums obtained this way are absolutely convergent. In the joint large-index region the uncancelled base scales as $O((\ell+n)^{-2})$, and every printed monomial supplies at least one additional inverse power, up to harmless logarithms. At fixed n, the remaining angular bound is $O(\ell^{2q-2}\log\ell)$, which explains the upper endpoint $q = 1/2$. At fixed ℓ, the radial bound is summable for positive q. The endpoint and larger dimensions must be treated by continuation of the *complete finite combination*, not by separately summing divergent monomial series at argument one. The paired iterated series remains a direct alternative there.

17.6 The standard hypergeometric function and the finite parameter rules

For finite lists of parameter–vector pairs define the ordinary multivariable Horn-type generalized hypergeometric series

$$\mathfrak{H}\left[\begin{matrix}(a_\alpha, \boldsymbol{u}_\alpha)_\alpha \\ (b_\beta, \boldsymbol{v}_\beta)_\beta\end{matrix}; \boldsymbol{z}\right] = \sum_{\boldsymbol{n}\in\mathbb{N}_0^r} \frac{\prod_\alpha (a_\alpha)_{\boldsymbol{u}_\alpha\cdot\boldsymbol{n}}}{\prod_\beta (b_\beta)_{\boldsymbol{v}_\beta\cdot\boldsymbol{n}}} \prod_{i=1}^r \frac{z_i^{n_i}}{n_i!}. \tag{290}$$

All vectors used below have nonnegative integer entries. Repeated pairs are allowed. Thus each coefficient ratio on increasing a single index is an explicit rational function of the indices. This is the standard Horn/Gamma-ratio class, not a new LRI-specific function whose evaluation has been left unspecified. It includes generalized Lauricella-type arrays; the $\ell + 2n$ vectors in the primitive make the more general Horn notation convenient [3, 5].

A finite set of identities is sufficient to print all its parameters. For $a, b > 0$, nonnegative integer v and the indicated orders,

$$\begin{aligned}
J_r(a;v) &= (-1)^{r+1} r! \sum_{k\ge 0} \frac{1}{(v+a+k)^{r+1}}, && r \ge 1,\\
\Delta_r(a,b;v) &= (-1)^{r+1} r!(b-a) \sum_{j=0}^{r} \sum_{k\ge 0} \frac{1}{(v+a+k)^{r+1-j}(v+b+k)^{j+1}}, && r \ge 0,\\
H(a;v) &= v \sum_{k\ge 0} \frac{1}{(a+k)(a+v+k)},\\
\delta(a+v) &= -q^2 \sum_{k\ge 0} \frac{(1-q)_k (1+q)_k}{(2)_k (a+v)_{k+1}}.
\end{aligned} \tag{291}$$

The first three follow from polygamma series and a finite partial-fraction identity [32]; the last is (287). For a product, assign a *different* index k to every factor. The choices $0 \le j \le r$ are finite, so expanding any of the printed master polynomials produces finitely many unweighted series. There is no harmonic-number or polygamma factor left in their coefficients.

Every reciprocal affine power is encoded by one pair of parameter rows:

$$(a + \boldsymbol{v}\cdot\boldsymbol{n})^{-s} = a^{-s} \frac{(a)^s_{\boldsymbol{v}\cdot\boldsymbol{n}}}{(a+1)^s_{\boldsymbol{v}\cdot\boldsymbol{n}}}. \tag{292}$$

For the last identity in (291), use

$$\frac{1}{(a+v)_{k+1}} = \frac{1}{a}\frac{(a)_v}{(a+1)_{v+k}}. \tag{293}$$

An otherwise absent $k!$ is supplied by an upper pair $(1, \boldsymbol{e}_k)$ in (290), which cancels its standard denominator. This fixes all factorial conventions explicitly.

The polynomial factor v in a harmonic insertion, and the primitive's $\ell(\ell+4q-2)$, also require no Euler derivative. Expand any resulting power of an index into falling factorials:

$$n^s = \sum_{j=0}^{s} \left\{ {s \atop j} \right\} n^{\underline{j}}, \qquad (m+j)^{\underline{j}} = j!\frac{(j+1)_m}{(1)_m}. \tag{294}$$

Shift $n = m + j$ in each term. Each old parameter pair $(a, \boldsymbol{v})$ is shifted to $(a + v_n j, \boldsymbol{v})$, with a prefactor $(a)_{v_n j}$ in the numerator or denominator as appropriate. Multiply by $j! z_n^j$ and add the upper/lower pair $(j+1, \boldsymbol{e}_n)/(1, \boldsymbol{e}_n)$. The braces in (294) are Stirling numbers of the second kind, so this operation is a finite algebraic rule.

17.6.1 Explicit starting arrays

For any of the powers $r = 2, 3, 4, 7$, split the original double sum into $\ell = 0$ and $\ell = j + 1$. Before inserting weights, the zero-spin block has scalar prefactor one and rows

$$\begin{matrix} (1;1),\ (q;1)^{\times r},\ (1-q;1)^{\times r} \\ (2q;1)^{\times r},\ (1;1)^{\times r} \end{matrix}, \qquad z_n = 1. \tag{295}$$

The positive-spin block has scalar prefactor $4q/2^r$ and rows

$$\begin{matrix} (1;1,0),(1;0,1),(2q+1;1,0),(4q-1;1,0),(q+1;1,1)^{\times r},(1-q;0,1)^{\times r} \\ (2q;1,0),(2;1,0),(2q+1;1,1)^{\times r},(1;0,1)^{\times r} \end{matrix}, \qquad (z_j, z_n) = (1,1). \tag{296}$$

Here semicolons separate a scalar parameter from its index vector; these are precisely the pairs in (290).

17.6.2 An explicit parameter-array example

For illustration, it is enough to print one unambiguous insertion instead of thousands of nearly identical arrays. The zero-spin contribution of $T_{0n}^3 \Delta_0(q, 2q; n)$ alone is

$$-\frac{1}{2q}\mathfrak{H}\left[\begin{matrix} (1;1,0),(1;0,1),(q;1,0)^{\times 3},(1-q;1,0)^{\times 3},(q;1,1),(2q;1,1) \\ (2q;1,0)^{\times 3},(1;1,0)^{\times 3},(q+1;1,1),(2q+1;1,1) \end{matrix};(1,1)\right]. \tag{297}$$

Its coefficient is obtained from $\Delta_0(q, 2q; n) = -q\sum_k [(n+q+k)(n+2q+k)]^{-1}$. There is no remaining derivative or finite-index harmonic sum in (297). All other entries follow the same printed rules. The two-variable example is deliberately elementary; products insert additional variables, up to the bounds in the following table.

17.7 Finite arrays, scope of continuation, and what is now closed

The scalar polynomial inputs for the array construction are precisely (276), (277), (286)–(289). They have finite degree. Applying (291)–(294) therefore gives a finite list of conventional parameter arrays for every master:

$$M(q) = M_{\text{scalar}}(q) + \sum_{\alpha=1}^{N_M} p_{M,\alpha}(q)\, \mathfrak{H}[\mathsf{A}_{M,\alpha}(q); \mathsf{B}_{M,\alpha}(q); \boldsymbol{z}_{M,\alpha}], \tag{298}$$

where M is one of $\mathcal{L}, Z, \mathcal{K}$. The primitive masters use the smaller finite-sample representation of Section 18; their residue series (the Q_d and $P_4(d)$ lines of (277), and (281)) are not converted into arrays here, and Section 18.6 compares the counts such a conversion gives. Only $M = \mathcal{K}$ has a separate scalar term, $-AU/4$. The coefficients $p_{M,\alpha}$ use rational functions of q, Gamma values and *fixed-argument* polygamma values through ψ_3. All parameter rows are explicit affine functions of q with rational coefficients, and the nonprimitive argument vectors contain only 1. No unspecified generic-index kite function or Feynman integral is included in (298).

The supplied direct compiler cancels identical parameter pairs and merges literally identical arrays but does not attempt contiguous relations or permutations of all auxiliary axes. Its nonminimal counts are

Master	Number of fixed Horn values	Maximum variables per value
$\mathcal{L}_d$	244	6
Z_d	318	6
$\mathcal{K}_d$	386	6

These counts quantify an important tradeoff: a finite unweighted function-class expression has been reached, but the mechanical expansion is *not shorter* than the weighted double series. It gives finite function-class closure, not a short scalar-hypergeometric evaluation. The counts assert neither minimality nor independence. A much shorter representation is obtained by a different route in Sections 20 and 21: periodization in the rim exponent and the top-mode theorem (473) reduce $\mathcal{L}_d$ and Z_d to the values and first and second rim-exponent derivatives of eight functions with at most two summation indices (the inventory after (553)), and, under the inputs stated in Section 21.1, $\mathcal{K}_d$ to a third-order τ-jet of three-node samples of three two-variable Horn functions (586). These forms keep parameter derivatives, so they lie outside the unweighted class of (298).

The array files, their scalar-parameter dictionary, and the compiler are included in `lri_general_closed_update/` under `code/general_closed/` and `verification/general_closed/horn_arrays/`. Each JSON row records the scalar prefactor, the full upper and lower parameter–vector lists with multiplicities, and the argument vector. The equations above specify a finite way of generating those same lists without needing the program; the files are a compact machine-readable certificate, not a replacement for a missing mathematical definition.

Continuation is part of the formula. There is a nonempty common absolute-convergence strip

$$\frac{1}{6} < q < \frac{1}{2}, \qquad \text{equivalently } \frac{2}{3} < d < 2, \tag{299}$$

for this finite construction. This follows from the wheel bounds and the canceled $\mathcal{K}$ bounds. The tail insertions have fixed sign or are finite signed sums of positive denominators there, and their absolute sums have the same bounds as the weights they replace. This justifies each finite expansion and the added summation indices in that strip.

For other $0 < d < 4$, use the unique dimensional analytic continuation of *the complete finite combination* from (299). At a removable exceptional value, evaluate that continued combination, not a separate pole part of each summand. In particular some individual $\mathcal{K}$ Horn series diverge at unity for $d \geq 2$, even though their continued combination is the finite physical master. This is not an extra free regulator, nor does it introduce an adjustable finite part; it is the continuation already fixed by the original subtracted master. It does mean that the result must not be advertised as separately convergent unity-argument series for every physical dimension. Direct paired summation or the original independent master integral is often numerically preferable at these points.

Thus (222)–(223), together with the finite parameter rules, give all coefficients through $C_5(d)$ in a specified standard multivariable hypergeometric class. Sections 20 and 21 shorten this considerably, at the price of low-order parameter derivatives; (553) is $C_5(d)$ in that form. A finite reduction to

ordinary ${}_pF_q$ values for generic d remains a different, stronger question. It is not asserted by these results, and the searches of Section 20.8, which found no such reduction of the remaining constants, are numerical and rule nothing out. At $d = 2$ such a reduction holds for the whole coefficient: Section 17.2 expresses $C_5(2)$ through ${}_4F_3(1)$, ${}_9F_8(-1)$ and ${}_8F_7(-1)$ values and fixed constants.

Checks of the finite multivariable construction. These checks belong to the package `lri_general_closed_update/` (scripts `check_new_identities.py`, `check_primitive_sums.py`, `check_horn_compiler.py` and `check_insertion_prefactors.py` in `code/general_closed/`; records `identity_checks.json`, `primitive_sum_checks.json`, `horn_compiler_checks.json` and `insertion_prefactor_checks.json` in `verification/general_closed/`). They are consistency checks, not interval bounds. The Gamma-ratio and eighth-power forms of Q_2 in (267), evaluated separately at 60 working digits, differ by about -4.98×10^{-60}. The same run checks the canceled polynomial (286), the normalization (288) and the shifted Gauss identity (287) at $q = 1/4, 3/10, 1/2, 3/4$ and several nonzero indices, with residuals below 2.0×10^{-58}. The Horn compiler is checked by 1038 exact rational comparisons of normalized coefficients at $q = 3/10$, over 519 sampled rows and including inserted tail indices, and by 12 falling-factorial shift identities. A further 336 exact rational checks cover the scalar prefactor of every type of tail insertion, and exact algebra verifies the polynomial E_6 and both dimension-two substitutions in (269). These checks verify the parameter encoding; they do not sum every continued multivariable endpoint function numerically.

For the double residue series (281) of Section 17.4: at 60 digits, Cauchy circles around poles of both channels reproduce the local residues, with largest normalized residual below 8.8×10^{-55}; the antisymmetry (282) was checked for unequal spins at $q = \frac{3}{5}$ and $\frac{3}{4}$; and the constants κ_j^0 agree with the unreduced reflection expressions. Double-precision sums of (281), with 112 spins, an Euler transform of the alternating spin sum and radial cutoffs 128 to 1024, were compared with spectral quadrature of (216) at 224 and 352 nodes:

d	residue-sum value of $P_5(d)$	residue sum minus 352-node control
2	1548.0826402331213	-4.43×10^{-11}
13/5	473.7956970211414	-6.37×10^{-12}
3	240.3403177590616	-2.77×10^{-11}

The displayed digits are outputs, not certified digits. Against the high-precision values of Section 16.8 the residue sums differ by $+6.8 \times 10^{-13}$ at $d = 2$ and -2.2×10^{-11} at $d = 3$. At $d = 1$ the raw radial series converges much more slowly: against a 60-digit evaluation of the Gamma integral its residual falls from 1.05×10^{-5} at cutoff 128 to 5.86×10^{-8} at cutoff 1024. No high-precision agreement is claimed there; the five-value formula (29) remains the evaluator at $d = 1$.

The rigorous parts of the construction are the algebraic normalization, the local residue identities, the involutive cancellation of the measure poles of $P_5(d)$ (Section 17.4) and the finite Horn conversion in its stated convergence domain, followed by the prescribed continuation; the numerical residuals corroborate these steps but are not interval bounds. The physical identification of the coefficients rests on the graph reduction and forest assembly of Sections 15 and 16; no independent general-dimensional five-loop graph calculation is claimed.

18 Finite periodization of the primitive sector

18.1 The three Barnes kernels

Put $q = d/4$, $p = 2q - 1$ and

$$\begin{aligned} a_\ell &= (\ell + q)/2, & b_\ell &= a_\ell + q, \\ g_\ell(s) &= \frac{\Gamma(a_\ell + s)\Gamma(a_\ell - s)}{\Gamma(b_\ell + s)\Gamma(b_\ell - s)}, & D_q(t) &= \sin(\pi t)\sin\big(\pi(t+q)\big). \end{aligned} \tag{300}$$

Use $d_\ell = d_\ell(4q)$ from (240). A version with no spurious $0/0$ at $d = 1, 2$ is

$$d_0 = 1, \qquad d_{k+1} = 4q\,\frac{(2q+1)_k(4q-1)_k}{(2q)_k(2)_k}, \qquad k \geq 0. \tag{301}$$

The normalized wheel periods are

$$W_r = \Gamma(2q)^r \sum_{\ell \geq 0} d_\ell \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\, g_\ell(s)^r, \qquad W_3 = 3Q_d, \quad W_4 = P_4(d). \tag{302}$$

For the five-loop primitive define

$$\begin{aligned} K_\ell(s) &= \big[(\ell+p)^2 - 4s^2\big]\frac{\Gamma(p+2s)\Gamma(p-2s)}{\Gamma(2s)\Gamma(-2s)} g_\ell(s)^7, \\ P_5(d) &= \frac{\Gamma(2q)^5}{2} \sum_{\ell \geq 0} (-1)^\ell d_\ell \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\, K_\ell(s). \end{aligned} \tag{303}$$

Initially take $1/2 < q < 1$ so that the contours separate the pole families without crossed-pole terms. The result below is then continued with the same prescription as the original primitive master. The kernels and normalization are the wheel and conformal-triangle kernels already specified in this note, based on the operator framework of [24]. No new renormalization convention is introduced.

For the proof only, periodize the kernels:

$$\begin{aligned} S_r(t) &= \sum_{\ell \geq 0} d_\ell \sum_{n \in \mathbb{Z}} g_\ell(a_\ell + n + t)^r, & r &= 3, 4, \\ S_7(t) &= \sum_{\ell \geq 0} (-1)^\ell d_\ell \sum_{n \in \mathbb{Z}} K_\ell(a_\ell + n + t), & & \\ \Xi_r(t) &= D_q(t)^r S_r(t), & r &= 3, 4, 7. \end{aligned} \tag{304}$$

The infinite sums in (304) are proof devices, not additional operations in the finite evaluation rule below.

18.2 Why finitely many Fourier modes suffice

18.2.1 Wheel poles, reflection and growth

For $r = 3, 4$ the only pole classes of S_r modulo one are $t = 0$ and $t = -q$, each of order at most r. Gamma reflection gives, for $n \geq 0$,

$$g_\ell(a_\ell + n + t) = -\frac{\sin \pi(q - t)}{\sin \pi t}\,\frac{\Gamma(\ell + n + q + t)\Gamma(n + 1 - q + t)}{\Gamma(\ell + n + 2q + t)\Gamma(n + 1 + t)}. \tag{305}$$

Reflection, recurrence and the Gamma-ratio asymptotics used here are standard identities [26, 27]; the periodization argument is given explicitly rather than inferred from those identities alone.

On compact sets avoiding the pole classes, split the lattice into $n \geq 0$, $n \leq -\ell - 1$, and $-\ell \leq n \leq -1$. In the middle part put $n = -j - 1$, $\ell = j + k + 1$; then

$$g_\ell(a_\ell + n + t) = \frac{\Gamma(k + q + t)\Gamma(j + 1 - t)}{\Gamma(k + 2q + t)\Gamma(j + q + 1 - t)}. \tag{306}$$

These bounds give locally uniform convergence, including after the possible pole factors are cleared. Consequently Ξ_r is entire. Evenness of $g_\ell(s)$ and the integer reindexing $n \mapsto -\ell - n$ give

$$\Xi_r(t+1) = \Xi_r(t), \qquad \Xi_r(-q-t) = \Xi_r(t). \tag{307}$$

For $t = x + \mathrm{i}y$, $0 \le x \le 1$, $|y| \ge 1$, the right-hand sum is bounded by a constant times

$$\sum_{\ell,n\ge 0} (1+\ell)^{4q-2}(1+|\ell+n+\mathrm{i}y|)^{-rq}(1+|n+\mathrm{i}y|)^{-rq}.$$

The middle expression (306) has the analogous bound with weight $(1+j+k)^{4q-2}$. Integral comparison, initially for $1/2 < q < 1$, gives

$$|S_r(x+\mathrm{i}y)| = O(|y|^{-2q(r-2)}), \qquad |\Xi_r(x+\mathrm{i}y)| = O(e^{2\pi r|y|}|y|^{-2q(r-2)}). \tag{308}$$

For example the endpoint $\ell \gg n, |y|$ has power $\ell^{q(4-r)-2}$, which is summable in this strip. This also controls the endpoints of the scaled integral, not just simultaneous large indices.

18.2.2 Five-loop measure poles cancel class by class

The extra measure poles of K_ℓ occur at $2s = \pm(p+k)$, $k \ge 0$. Their *weighted* right residues have the form

$$\begin{aligned} \mathcal{A}_{\ell k} &= (-1)^\ell d_\ell d_k\big[(\ell+p)^2 - (k+p)^2\big]G^7_{\ell k}, \\ G_{\ell k} &= \frac{\Gamma((\ell+k+3q-1)/2)}{\Gamma((\ell+k+5q-1)/2)}\frac{\Gamma((\ell-k+1-q)/2)}{\Gamma((\ell-k+1+q)/2)}. \end{aligned} \tag{309}$$

Gamma reflection [26], applied to the second ratio, gives $G_{k\ell} = (-1)^{\ell-k}G_{\ell k}$ and hence $\mathcal{A}_{k\ell} = -\mathcal{A}_{\ell k}$; the diagonal residue is zero. For the present purpose cancellation of the *total* measure-residue sum is not by itself sufficient. One must check the congruence classes. A right measure pole is at

$$t = \frac{q-1+k-\ell}{2} \pmod 1.$$

Exchanging k, ℓ changes this by an integer, so the canceling residues are in the same class. A left measure pole is at $t = (1-3q-k-\ell)/2 \pmod 1$, also unchanged by the exchange. Thus the extra poles cancel locally in the spin-summed periodization, not only after the final contour integration. Absolute convergence in the initial strip, which the bound (283) provides, permits the pairing. Coincident pole parameters are handled by continuation of the complete expression, not by keeping a separate singular residue.

After this cancellation only the two order-seven Gamma-pole classes remain; Ξ_7 is entire and satisfies (307). The measure contributes at most a factor of order $(1+\ell+n+|y|)^{4q}$ to the bound behind (308), so

$$|S_7(x+\mathrm{i}y)| = O(|y|^{-6q}), \qquad |\Xi_7(x+\mathrm{i}y)| = O(e^{14\pi|y|}|y|^{-6q}). \tag{310}$$

The endpoint $\ell \gg n, |y|$ now has power ℓ^{q-2}, again summable for $q < 1$.

18.2.3 Finite-dimensionality theorem

For an entire one-periodic function the Fourier coefficient is unchanged when its integration contour is translated vertically. Apply this to (308) or (310). For mode $k \ge r$, translation down by $\mathrm{i}Y$ bounds its absolute value by $Ce^{2\pi(r-k)Y}Y^{-\eta}$ with $\eta > 0$; it therefore vanishes, including the

boundary mode $k = r$. Upward translation deals with $k \le -r$. Reflection (307) pairs the remaining modes. Thus, exactly,

$$\boxed{\Xi_r(t) = \sum_{k=0}^{r-1} c_{r,k}(q)\cos\bigl(2\pi k(t+q/2)\bigr), \qquad r = 3, 4, 7.} \tag{311}$$

For real q the coefficients are real, since the meromorphic kernel is real on nonsingular real arguments. This is a proved finite Fourier space, not a fitted trigonometric ansatz. For the wheels the top coefficient also vanishes: $c_{3,2}(q) = c_{4,3}(q) = 0$ for $\frac{2}{r} < d < 4$, by (473), proved in Section 20.1.2 for a general rim exponent. The sums in (318) may therefore stop at $k = r - 2$ for $r = 3, 4$; the rules below remain exact as written. No such statement is made for $r = 7$, whose kernel is not a product of balanced factors.

18.3 A finite evaluation rule, with no derivatives

A stable fixed sampling grid is

$$\begin{aligned} &N_r = 2r - 1, \qquad \eta_r = \frac{1}{8\pi r}, \quad t_j = -\frac{q}{2} + \frac{j}{N_r} + \mathrm{i}\eta_r, \quad j = 0, \dots, r-1, \\ &Y_j = \Re\Xi_r(t_j). \end{aligned} \tag{312}$$

The nonzero imaginary part avoids every real pole class. It is a fixed number, not a regulator to be sent to zero. Cosine orthogonality gives

$$\boxed{\begin{aligned} c_{r,0} &= \frac{1}{N_r}\left(Y_0 + 2\sum_{j=1}^{r-1} Y_j\right), \\ c_{r,k} &= \frac{2}{N_r\cosh(k/(4r))}\left[Y_0 + 2\sum_{j=1}^{r-1} Y_j\cos\frac{2\pi kj}{N_r}\right], \quad 1 \le k < r. \end{aligned}} \tag{313}$$

Only finite sums appear here. For complex q, use all $2r - 1$ complex samples and the ordinary discrete Fourier transform, rather than using complex conjugation to reconstruct half of them.

The residue functional can also be printed without differentiating anything. For $1 \le h < r$ set

$$\begin{aligned} L_h(q) &= (-1)^{h+1}\psi_{h-1}(1-q) - \psi_{h-1}(q), \\ u_h(q) &= r\left[(1 + (-1)^h)\psi_{h-1}(1) - L_h(q)\right]. \end{aligned} \tag{314}$$

These particular polygamma combinations are elementary reflection values. If $c = \cot(\pi q)$, an implementation requiring no higher polygamma calls may use

$$\begin{aligned} &\frac{u_1}{r\pi} = -c, && \frac{u_2}{r\pi^2} = c^2 + \frac{4}{3}, && \frac{u_3}{r\pi^3} = -2c(1+c^2), \\ &\frac{u_4}{r\pi^4} = \frac{32}{15} + 8c^2 + 6c^4, && \frac{u_5}{r\pi^5} = -8c(1+c^2)(2+3c^2), \\ &\frac{u_6}{r\pi^6} = \frac{1024}{63} + 136c^2 + 240c^4 + 120c^6. \end{aligned} \tag{315}$$

Only $u_1, \dots, u_{r-1}$ are used. These follow from reflection and the integer polygamma values, and introduce no additional period constants.

Define finite polynomials E_m, only for $0 \le m \le 6$, by

$$E_0 = 1, \qquad E_m(x_1, \dots, x_m) = \frac{1}{m}\sum_{h=1}^{m} \frac{x_h}{(h-1)!} E_{m-h}(x_1, \dots, x_{m-h}). \tag{316}$$

This is a finite polynomial definition, not a parameter derivative. For orientation,

$$\begin{aligned} E_2 &= (x_1^2 + x_2)/2, & E_3 &= (x_1^3 + 3x_1x_2 + x_3)/6, \\ 720E_6 &= x_1^6 + 15x_1^4x_2 + 20x_1^3x_3 + 45x_1^2x_2^2 + 15x_1^2x_4 \\ &\quad + 60x_1x_2x_3 + 6x_1x_5 + 15x_2^3 + 15x_2x_4 + 10x_3^2 + x_6. \end{aligned} \tag{317}$$

Write

$$\begin{aligned} B_{r,k}(q) &= \frac{1}{2}\big[e^{\mathrm{i}\pi kq}E_{r-1}(u_1 + 2\pi\mathrm{i}k, u_2, \dots, u_{r-1}) \\ &\qquad + e^{-\mathrm{i}\pi kq}E_{r-1}(u_1 - 2\pi\mathrm{i}k, u_2, \dots, u_{r-1})\big], \\ \mathcal{E}_r(q; \boldsymbol{Y}) &= -\frac{\sum_{k=0}^{r-1} B_{r,k}(q)c_{r,k}}{(\pi \sin \pi q)^r}. \end{aligned} \tag{318}$$

Then the finite primitive formulas are

$$\boxed{Q_d = \frac{\Gamma(2q)^3}{3}\mathcal{E}_3, \qquad P_4(d) = \Gamma(2q)^4\mathcal{E}_4, \qquad P_5(d) = \frac{\Gamma(2q)^5}{2}\mathcal{E}_7.} \tag{319}$$

To check the sign and factors, close each Barnes contour to the right. Its orientation is clockwise, so its value is minus the sum of right residues. That sum is $\mathrm{Res}_{t=0} S_r(t)$. In $D_q(t)^{-r}$ the leading coefficient is $(\pi \sin \pi q)^{-r}t^{-r}$; the logarithmic coefficients of the analytic factor are exactly (314). Expanding a cosine then gives (318). Residues are used in this proof only; the final formula (319) contains finite arithmetic and fixed special-function values.

18.4 Explicit one- and two-variable functions for every sample

For finite upper/lower lists of parameter–integer-vector pairs use

$$\mathfrak{H}[\mathsf{U}; \mathsf{V}; \boldsymbol{z}] = \sum_{\boldsymbol{n} \ge 0} \frac{\prod_{(a,v)\in\mathsf{U}} (a)_{v\cdot\boldsymbol{n}}}{\prod_{(b,w)\in\mathsf{V}} (b)_{w\cdot\boldsymbol{n}}} \prod_i \frac{z_i^{n_i}}{n_i!}. \tag{320}$$

This is the unweighted Horn convention already used in the note. Signed integer-vector entries are permitted; $(a)_{-m}$ means $\Gamma(a-m)/\Gamma(a)$, not a truncated product. Consecutive coefficient ratios are rational in the indices. There are no index-dependent polygammas in the following arrays. The function definition contains its usual series, just as the definition of ${}_pF_q$ does; the master formula has no additional outer infinite sum. This distinction is not a claim that a two-variable function has become a one-variable function.

For two variables put $\boldsymbol{e} = (1,0)$, $\boldsymbol{f} = (0,1)$, $\boldsymbol{c} = \boldsymbol{e} + \boldsymbol{f}$. Repeated rows are denoted by a superscript $\times r$. Define

$$\begin{aligned} A_r^0(t) &= a_r^0(t)\, {}_{2r+1}F_{2r}\left(\begin{matrix} 1, (q+t)^{\times r}, (1-q+t)^{\times r} \\ (2q+t)^{\times r}, (1+t)^{\times r} \end{matrix}; 1\right), \\ a_r^0(t) &= \left[\frac{\Gamma(q+t)\Gamma(-t)}{\Gamma(2q+t)\Gamma(q-t)}\right]^r, \\ A_r^+(t) &= a_r^+(t)\mathfrak{H}[\mathsf{U}_+; \mathsf{V}_+; (1,1)], \\ a_r^+(t) &= 4q\left[\frac{\Gamma(q+1+t)\Gamma(-t)}{\Gamma(2q+1+t)\Gamma(q-t)}\right]^r, \\ C_r(t) &= a_r^c(t)\mathfrak{H}[\mathsf{U}_c; \mathsf{V}_c; (1,1)], \\ a_r^c(t) &= 4q\left[\frac{\Gamma(q+t)\Gamma(1-t)}{\Gamma(2q+t)\Gamma(1+q-t)}\right]^r. \end{aligned} \tag{321}$$

The complete two-variable arrays are

$$\begin{aligned}
\mathsf{U}_+ &= \{(q+1+t,\boldsymbol{c})^{\times r},(1,\boldsymbol{e}),(2q+1,\boldsymbol{e}),(4q-1,\boldsymbol{e}),(1,\boldsymbol{f}),(1-q+t,\boldsymbol{f})^{\times r}\},\\
\mathsf{V}_+ &= \{(2q+1+t,\boldsymbol{c})^{\times r},(2q,\boldsymbol{e}),(2,\boldsymbol{e}),(1+t,\boldsymbol{f})^{\times r}\},\\
\mathsf{U}_c &= \{(2q+1,\boldsymbol{c}),(4q-1,\boldsymbol{c}),(1,\boldsymbol{e}),(1-t,\boldsymbol{e})^{\times r},(1,\boldsymbol{f}),(q+t,\boldsymbol{f})^{\times r}\},\\
\mathsf{V}_c &= \{(2q,\boldsymbol{c}),(2,\boldsymbol{c}),(1+q-t,\boldsymbol{e})^{\times r},(2q+t,\boldsymbol{f})^{\times r}\}.
\end{aligned} \tag{322}$$

For $r=3,4$ every sample is therefore

$$\boxed{\Xi_r(t)=D_q(t)^r\big[A_r^0(t)+A_r^+(t)+A_r^0(1-q-t)+A_r^+(1-q-t)+C_r(t)\big].} \tag{323}$$

The right tail gives the first two functions, evenness maps the left tail to $1-q-t$, and (306) gives the last function. Equation (301) then proves every Pochhammer row in (322). This is also a direct coefficient-by-coefficient proof of the sample formula.

18.4.1 The eight primitive sample blocks

The $r=7$ arrays are obtained from the same printed blocks by finite row insertions, as follows. Put

$$v=q+2t,\quad \theta_0=2q,\quad \theta_1=2q-1,\qquad M_\theta(v)=\frac{\Gamma(\theta+v)\Gamma(\theta-v)}{\Gamma(v)\Gamma(-v)}. \tag{324}$$

The identity underlying the two channels is

$$\big[(\ell+p)^2-V^2\big]\frac{\Gamma(p+V)\Gamma(p-V)}{\Gamma(V)\Gamma(-V)}=M_{2q}(V)+\ell(\ell+4q-2)M_{2q-1}(V). \tag{325}$$

For clarity, a "block" below includes its scalar prefactor, both lists and its argument vector.

Define a right insertion $\mathcal{R}_{\theta,w,h}$ on any block by multiplying its prefactor by $M_\theta(w)$ and adjoining the rows

$$\text{upper: } (\theta+w,h),(1+w,h),\qquad \text{lower: } (w,h),(1+w-\theta,h). \tag{326}$$

It multiplies its coefficient by $M_\theta(w+h\cdot\boldsymbol{n})$. This follows by reflecting the negative Gamma argument and using that $h\cdot\boldsymbol{n}$ is an integer. A middle insertion $\mathcal{C}_{\theta,w,h}$ multiplies the prefactor by $M_\theta(w)$ and adjoins instead

$$\text{upper: } (\theta+w,h),(\theta-w,-h),\qquad \text{lower: } (w,h),(-w,-h). \tag{327}$$

This is the same identity before reflecting the index difference.

Start with A_7^0,A_7^+,C_7. Let A^- be A_7^+ with a minus sign in its prefactor and arguments $(-1,1)$; let C^- be C_7 with a minus sign and arguments $(-1,-1)$. Let $\widehat{A}^-$ be obtained from A^- by multiplying the prefactor by $4q-1$, replacing its upper row $(4q-1,\boldsymbol{e})$ by $(4q,\boldsymbol{e})$ and its lower row $(2,\boldsymbol{e})$ by $(1,\boldsymbol{e})$. Define $\widehat{C}^-$ with the same replacements along $\boldsymbol{c}$. These replacements supply the factor $\ell(\ell+4q-2)$ in the second channel of (325).

There are three right blocks and two middle blocks:

$$\begin{aligned}
Z(t)&=\mathcal{R}_{\theta_0,v,2}A_7^0(t),\\
P_0(t)&=\mathcal{R}_{\theta_0,v+1,\boldsymbol{e}+2\boldsymbol{f}}A^-(t),\quad P_1(t)=\mathcal{R}_{\theta_1,v+1,\boldsymbol{e}+2\boldsymbol{f}}\widehat{A}^-(t),\\
C_0(t)&=\mathcal{C}_{\theta_0,v-1,\boldsymbol{f}-\boldsymbol{e}}C^-(t),\quad C_1(t)=\mathcal{C}_{\theta_1,v-1,\boldsymbol{f}-\boldsymbol{e}}\widehat{C}^-(t).
\end{aligned} \tag{328}$$

The single-variable vector in the first line is the integer 2. Duplication of those step-two Pochhammers turns $Z(t)$ into an ordinary ${}_{19}F_{18}(1)$. All other blocks have only two indices. The complete fixed sample is

$$\boxed{\begin{aligned}\Xi_7(t)=D_q(t)^7\big[&Z(t)+P_0(t)+P_1(t)\\&+Z(1-q-t)+P_0(1-q-t)+P_1(1-q-t)+C_0(t)+C_1(t)\big].\end{aligned}} \tag{329}$$

Equations (312)–(329) specify every parameter, argument and prefactor using finite lists; the JSON templates in `lri_general_periodization_update/verification/general_periodization/arrays/` are machine-readable copies, not missing definitions.

Convergence and continuation. The proofs above start at $1/2 < q < 1$. The right and middle Gamma-ratio estimates give direct absolute convergence of the displayed blocks for $q > 1/(2r)$ in a wheel and $q > 1/10$ in the primitive, always with $q < 1$. The endpoint in a positive-spin right block imposes the additional condition $q(4-r) < 1$ for a wheel and $q < 1$ for the primitive; these are automatic in the stated ranges. Thus the new Q, P_4, P_5 formulas are direct endpoint-series formulas for $d > 2/3, 1/2, 2/5$, respectively. For smaller positive dimensions use their continued complete combinations. Exceptional real parameter coincidences are dealt with by the same continuation; the off-real sampling points themselves are nonsingular. In particular $d = 1, 2, 3$ are valid specializations, though the retained five ordinary values remain a better presentation for $P_5(1)$.

18.5 A stronger result at two dimensions: reuse the one-dimensional values

In this subsection B_-, B_+ and X_0, X_+, X_- refer to the *retained one-dimensional* values, not to new functions of d:

$$\begin{aligned} B_- &= \Gamma(\tfrac14)^4 F_-, \quad B_+ = \frac{64\pi^4}{\Gamma(\frac14)^4} F_+, \\ X_0 &= \mathcal{Y}_2, \qquad\qquad X_+ = c_8^7 \mathcal{Y}_3 - s_8^7 \mathcal{Y}_5, \quad X_- = c_8^7 \mathcal{Y}_1 + s_8^7 \mathcal{Y}_7. \end{aligned} \tag{330}$$

For a self-contained dimension-two parameter dictionary, these are

$$\begin{aligned} F_- &= {}_4F_3\left(\begin{matrix}\frac14,\frac14,\frac12,\frac12\\ \frac34,\frac34,1\end{matrix};1\right), \quad F_+ = {}_4F_3\left(\begin{matrix}\frac34,\frac34,\frac12,\frac12\\ \frac54,\frac54,1\end{matrix};1\right), \\ c_8 &= \cos(\pi/8), \qquad\qquad s_8 = \sin(\pi/8), \end{aligned} \tag{331}$$

$$\begin{aligned} \mathcal{Y}_r &= \frac{r}{8}\left[\frac{\Gamma((r+2)/8)}{\Gamma((r+6)/8)}\right]^7 {}_9F_8\left(\begin{matrix}1, 1+\frac{r}{8}, ((r+2)/8)^{\times 7}\\ \frac{r}{8}, ((r+6)/8)^{\times 7}\end{matrix};-1\right), \\ & r \in \{1,2,3,5,7\}. \end{aligned} \tag{332}$$

At $r = 2$ one upper and lower parameter 1 cancel, giving an ordinary ${}_8F_7$; all parameter lists are fixed and positive rational. The new identities are

$$\boxed{P_4(2) = \frac{B_-^2 - B_+^2}{4\pi^3}.} \tag{333}$$

$$\boxed{\begin{aligned} P_5(2) = \frac{8}{\pi}\big[&25X_0^2 - 28\sqrt{2}\, X_0(X_+ + X_-) \\ &+ 18(X_+^2 + X_-^2) + 28 X_+ X_-\big]. \end{aligned}} \tag{334}$$

Thus no new hypergeometric value is needed for either two-dimensional primitive. The four-loop result uses the same two ${}_4F_3(1)$ values; the five-loop result uses exactly the preferred four ${}_9F_8(-1)$ and one ${}_8F_7(-1)$ with positive rational parameters. The formulas are quadratic rather than linear. By themselves they do not evaluate the two-dimensional nonprimitive sector; that evaluation is completed in Section 17.2, with the proofs in Sections 18.7 and 18.8.

18.5.1 Factorization of the wheel lattice

At $q = 1/2$ the multiplicities are $d_0 = 1$, $d_{\ell>0} = 2$. In a right half-lattice put $k = \ell + n$, $j = n$. Gamma reflection gives

$$g_\ell(a_\ell + n + t) = -\cot(\pi t)\, \frac{\Gamma(k + \frac12 + t)}{\Gamma(k + 1 + t)} \frac{\Gamma(j + \frac12 + t)}{\Gamma(j + 1 + t)}.$$

The diagonal has multiplicity one and the two off-diagonal half-lattices have multiplicity two. They therefore form a full square lattice. Define the meromorphic bilateral series and its regularization

$$\begin{aligned} B_r(t) &= \sum_{n\in\mathbb{Z}} \left[\frac{\Gamma(n+\frac12+t)}{\Gamma(n+1+t)}\right]^r, \\ \Omega_r(t) &= \cos^r(\pi t) B_r(t). \end{aligned} \tag{335}$$

The index sums are absolute for $r > 2$. Direct reflection or the square lattice just described gives

$$\Xi_r(t) = (-1)^r \Omega_r(t)^2 \qquad (d = 2). \tag{336}$$

For $r = 4$, the growth and reflection argument behind (311) now yields only two modes:

$$\Omega_4(t) = a + b\cos\bigl(2\pi(t+\tfrac14)\bigr). \tag{337}$$

There is no fitting in this statement: Ω_4 is entire, one-periodic, even about $-1/4$, and has growth $O(e^{4\pi|\Im t|}|\Im t|^{-1})$, which excludes the boundary second mode.

Let $M_{\rm e} = \int_0^1 \mathbf{K}(k)^2\,dk$ and $N_{\rm e} = \int_0^1 \mathbf{K}(k)\mathbf{K}(k')\,dk$, with the modulus convention of Section 9.2. The established elliptic identities give

$$M_{\rm e} = \frac{B_- + B_+}{64}, \qquad N_{\rm e} = \frac{B_- - B_+}{32}. \tag{338}$$

Here is a short analytic determination of both coefficients in (337). At $t = 0$ the reflected half of Ω_4 vanishes to fourth order. Hence

$$a = \pi^2\, {}_4F_3\left(\begin{matrix}\frac12,\frac12,\frac12,\frac12\\ 1,1,1\end{matrix};1\right) = \frac{8N_{\rm e}}{\pi}.$$

For $N(\nu) = \int_0^1 k^\nu \mathbf{K}(k)\mathbf{K}(k')\,dk$, the mixed moment beta integral of [25] gives

$$N(\nu) = \frac{\pi^2}{8}\sum_{n\ge0} \frac{(\frac12)_n^2}{(n!)^2}\frac{\Gamma(n+(\nu+1)/2)^2}{\Gamma(n+(\nu+2)/2)^2}. \tag{339}$$

Differentiation in this convergent *proof* representation gives $\Omega_4'(0) = 32N'(0)/\pi$. The already proved Stieltjes–Landen identity (101) says $N'(0) = -\pi M_{\rm e}/2$. As $\Omega_4'(0) = -2\pi b$, it follows that

$$a = \frac{8N_{\rm e}}{\pi}, \qquad b = \frac{8M_{\rm e}}{\pi}. \tag{340}$$

No derivative remains in these coefficients. Substitute $\Xi_4 = (a + b\cos(2\pi(t+1/4)))^2$ into (318). At $q = 1/2$ the only contributing coefficient is $c_{4,1} = 2ab$, and $P_4(2) = 8ab/\pi = 512M_{\rm e}N_{\rm e}/\pi^3$. Equation (333) follows immediately.

18.5.2 A second determination of b, without the Stieltjes identity

The value $a = 8N_{\rm e}/\pi$ above uses only Euler integration. The value of b above rests on the mixed-moment series (339) and on (101), whose proof uses the Stieltjes identity (100) of [25]. The following derivation uses neither. It was written by the Claude Code session that wrote the third and fourth proofs of (270) in Section 18.7 (package `LRI_C5_d2_simplify/`, folder `proof/`).

Put $c_n = (\frac12)_n/n!$, so that $\Gamma(n+\frac12)/\Gamma(n+1) = \sqrt{\pi}\,c_n$. At $t = 0$ the reflected half of Ω_4 vanishes to fourth order and $\cos^4\pi t$ has zero derivative, so

$$-2\pi b = \Omega_4'(0) = 4\pi^2 \sum_{n\ge0} c_n^4\bigl[\psi_0(n+\tfrac12) - \psi_0(n+1)\bigr].$$

Item (ii) of the pairing proposition (368) below evaluates this single sum by Parseval's identity on the unit circle. It uses only the generating function (366) (first line) and the circle values (367), and gives $b = 4V_0/\pi$, with $V_0 = \int_0^1 \mathbf{K}(k)\mathbf{K}(k')\,dk/k'$ as defined before (96). The imaginary part of the upper-semicircle contour that proves (106) gives $V_0 = 2M_\mathrm{e}$; that contour uses the boundary-value Landen relation and not (100). Hence $b = 8M_\mathrm{e}/\pi$, which is (340). The Fourier representation of Section 18.7.9 gives the same value independently, $b = v_1(\frac{1}{2})/(4\pi) = 4V_0/\pi$. For a the same two routes give $a = 4U_0/\pi$, and $U_0 = 2N_\mathrm{e}$ is proved without (100) in Section 18.7.7. Since (338) uses only $V_0 = 2M_\mathrm{e}$ and the real Landen substitution (the reduction to the two ${}_4F_3$ values after (107)), (333) is proved without the Stieltjes identity and without the mixed-moment series. As a check, not a proof step, the 70-digit single sums give $8ab/\pi - (B_-^2 - B_+^2)/(4\pi^3) = -1.1 \times 10^{-67}$ (`LRI_C5_d2_simplify/proof/synth/synth_checks.log`).

18.5.3 The five-loop square and the retained five values

Put

$$\begin{aligned} G(s) &= (2s+\tfrac{1}{2})\left[\frac{\Gamma(s+\frac{1}{2})}{\Gamma(s+1)}\right]^7, \\ \Omega_\mathrm{a}(t) &= \cos^7(\pi t)\sum_{n\in\mathbb{Z}}(-1)^n G(n+t). \end{aligned} \tag{341}$$

At $q = 1/2$ the measure in (303) is $\ell^2 - 4s^2$. On the square lattice it becomes $-(2k+\frac{1}{2}+2t)(2j+\frac{1}{2}+2t)$ and $(-1)^\ell = (-1)^{k-j}$. Consequently

$$\Xi_7(t) = \Omega_\mathrm{a}(t)^2 \qquad (d = 2). \tag{342}$$

The sign is plus: the negative quadratic measure and the odd seventh power of the Gamma-reflection sign cancel.

Most usefully, $G(s) = 2f(s)$ for the *one-dimensional* function in (134). Thus

$$\Omega_\mathrm{a}(t) = 2\Xi^{(1)}(t) = 2\big[\alpha\cos 2\pi t + \beta\sin 4\pi t + \gamma\cos 6\pi t\big], \tag{343}$$

where α, β, γ are exactly those in (144). Applying the finite residue polynomial gives

$$P_5(2) = \frac{8}{\pi}\left(\alpha^2 - 14\alpha\gamma + 8\beta^2 + 113\gamma^2\right). \tag{344}$$

One convenient exact algebra check is that at $q = 1/2$

$$(B_{7,0}, B_{7,1}, \ldots, B_{7,6}) = \pi^6(20, 0, 28, 0, 84, 0, 924).$$

Expanding the square in (343) and using the normalization $P_5 = \mathcal{E}_7/2$ gives (344); inserting (144) gives (334). This proves the quadratic factorization without a numerical integer-relation search.

18.6 Numerical checks of the finite periodization

These checks were run by OpenAI GPT-6 astra in the calculation that introduced the periodization (package `lri_general_periodization_update/`; scripts in `code/general_periodization/`, records `wheel_periodization.json`, `horn_template_checks.json`, `p5_periodization.json`, `d2_factorization.json`, `d2_control.json`, `d2_inherited_periods.json` and `exact_algebra.json` in `verification/general_periodization/`). They are numerical consistency checks, not interval bounds.

Function counts. The finite evaluation (319) with the arrays of Section 18.4 needs far fewer special-function values than the unweighted Horn arrays obtained by applying the rules of Section 17.6 to the wheel series (277) and to the derivative-free double-residue series (281) of $P_5(d)$:

	Horn values	variables	periodization values, at most	variables
Q_d	20	4	15	2
$P_4(d)$	56	5	20	2
$P_5(d)$	15060	8	56	2

For $P_5(d)$ the Horn count is 6024 for the first channel of (281) and 9036 for the second. The sampling counts are the five blocks of (323) at each of the r nodes of (312) ($r = 3, 4$) and the eight blocks of (329) at the seven nodes for $r = 7$. The counts include the one-variable subblocks, and taking the real part of a complex value counts as one evaluation. They are upper bounds and assert neither minimality nor independence. The gain is the elimination of the auxiliary indices that encoded residue polygammas, not a proof that the multivariable functions reduce to ordinary ones.

Fixed-spin controls. At 35 working digits, fixed-spin wheel samples were evaluated as ordinary hypergeometric functions and compared with real-axis Barnes integrals. The cases $(q, r, \ell) = (0.6, 3, 0)$, $(0.7, 3, 2)$, $(0.6, 4, 1)$ and $(0.25, 4, 0)$ differ by less than 3×10^{-30}, and further off-grid samples check the finite Fourier polynomial (311). These test building blocks of the theorem, not a rederived graph sum.

Parameter arrays and finite transforms. At 65 working digits, 44 coefficients of the Horn blocks, including the signed-index middle blocks, were compared with the Gamma kernels at $q = 1/4, 1/2, 0.63, 3/4$; the largest normalized residual is 3.81×10^{-65}. The cosine transforms (313) were checked on every basis mode for $r = 3, 4, 7$, with residuals below 4.75×10^{-66}. The exact derivation of the arrays is (305)–(306) and the finite row rules; these are coefficient checks.

Noninteger dimensions. The spin-summed finite-sampling expression for $P_5(d)$ was compared with spectral quadrature of (216), with 640 nodes and 160 spin terms; the sampling computation used 128 spin terms, Euler acceleration and doubled radial cutoffs. The difference is -1.39×10^{-9} at $d = 2.4$ and $+7.38 \times 10^{-8}$ at $d = 3$, and the off-grid periodization residuals are below 2.3×10^{-11}. These runs use double-precision Gamma kernels inside wider accumulations, so their residuals are not promoted to high precision because the finite weights were computed at 50 digits. The five-loop residue functional is sensitive to cancellation. Imaginary-axis Chebyshev nodes far from the real axis are particularly ill conditioned; the grid (312) controls this amplification much better, but it is not an error certificate.

Two dimensions. The square factorizations of Section 18.5 were evaluated first at 55 digits, using Gamma-ratio series with a Hurwitz-zeta asymptotic tail and a finite Euler transform for the alternating kernel. A 70-digit rerun changed the direct cutoff from 96 to 144, the tail order from 44 to 58 and the Euler settings from $(20, 260)$ to $(32, 340)$, and used the grid (312). The Q_2 value agrees with $\Gamma(1/4)^8/(24\pi^4)$ to about 67 decimal places, and $P_4(2)$ and $P_5(2)$ are the values quoted in Section 17.2. The retained values $F_\pm$ and the five $\mathcal{Y}_r$ were then recomputed with literal ordinary hypergeometric calls at 60 working digits. The formulas (333) and (334) differ from the 70-digit square-factorization values by -5.97×10^{-59} and $+1.59 \times 10^{-58}$. The quadratic change of basis from (344) to (334) was checked by exact symbolic algebra. No stored period value was an input to either route.

What the finite Fourier theorem does not give. The theorem (311) concerns the Barnes-shift variable t. It does not imply finite dependence on an independent rim exponent b, so it cannot replace the first and second b derivatives that define $\mathcal{L}_d$ and Z_d in (244) by a finite sampling grid in b; that would be an unjustified interpolation. Sections 20 and 21 treat the rim exponent by a separate periodization (Section 20.1) and keep low-order parameter derivatives.

18.7 Proof of the closed form of $\mathcal{L}_2$

We prove (270) for the series (256). The rim exponent of the wheel lattice of Section 18.5 is deformed away from one, the two-mode statement (337) is proved for every nearby rim exponent, and the two derivative sums that appear are evaluated with the elliptic lemmas of Section 9.2 and one further contour integral. This is the one-dimensional contour proof of Section 9.2, transferred to the square lattice. Throughout, $d = 2$, $q = \frac{1}{2}$, $c_k = (\frac{1}{2})_k/k!$ (not the constants c_j of (274)), and

$$\alpha_k = \psi_0(k+\tfrac{1}{2}) - \psi_0(k+1), \qquad \beta_k = \psi_0(k+\tfrac{1}{2}) - \psi_0(\tfrac{1}{2}),$$

that is, $\alpha_k = \Delta_0(\frac{1}{2}, 1; k)$ and $\beta_k = H(\frac{1}{2}; k)$ in the notation of (274). The letters a, b denote the coefficients of (337). The functions f, g, the weight w and the moments U_0, R are those of Section 9.2, whose M, N are $M_{\mathrm{e}}, N_{\mathrm{e}}$ of (338).

The first proof (Sections 18.7.1–18.7.5) and the second (Section 18.7.6) were written by a Claude Code session with prover and reviewer subagents (package `LRI_C5_d2_proof/`, folders `L2_singlesums/` and `L2_structural/`). Another Claude Code session, with prover, verifier and synthesizer subagents, wrote the half-strip evaluation of Section 18.7.7, the third and fourth proofs (Sections 18.7.8 and 18.7.9) and the remark on the integer-relation form of the collapse inside the first proof (package `LRI_C5_d2_simplify/`, folder `proof/`; the third proof is in its subfolder `L2-elliptic/`, the half-strip evaluation and the fourth proof in `L2-fourier/`). That session had first found (270) and the relations (356) numerically, in the top level of the same package.

18.7.1 The series as a rim derivative

Let $I = [\frac{2}{5}, \frac{3}{5}]$. For $x \in I$ and $m, n, k \geq 0$ put

$$\varphi_x(s) = \frac{\Gamma(s+x)}{\Gamma(s+1)}, \qquad N_x = \frac{\Gamma(1-x)}{\Gamma(x)}, \qquad c_k(x) = \frac{(x)_k}{k!}, \qquad T_{mn}(x) = c_m(x)c_n(x),$$
$$\lambda_j(x) = \Delta_{j-1}(x, 1; m) + \Delta_{j-1}(x, 1; n) + \mathsf{c}_j(x), \qquad \sum_{j\geq 1} \mathsf{c}_j(x)\frac{u^j}{j!} = \log\frac{\pi u}{\sin \pi u} + \log\frac{\sin \pi(x+u)}{\sin \pi x}, \tag{345}$$

so that $\mathsf{c}_1 = \pi \cot \pi x$, $\mathsf{c}_2 = \frac{\pi^2}{3} - \frac{\pi^2}{\sin^2 \pi x}$ and $\mathsf{c}_3 = \frac{2\pi^3 \cos \pi x}{\sin^3 \pi x}$; at $x = \frac{1}{2}$ these are the constants c_j of (274). Let $F_4(x)$ be the polynomial F_4 of (254) evaluated at these $\lambda_j(x)$.

One summand as a residue. Gamma reflection [26] gives $N_x \sin(\pi x)\varphi_x(m)\varphi_x(n)/\pi = T_{mn}(x)$, and $\partial_u^j \log \varphi_x(k+u)|_{u=0} = \Delta_{j-1}(x, 1; k)$. Hence, as Laurent expansions at $u = 0$,

$$N_x \frac{\sin \pi(x+u)}{\sin \pi u}\, \varphi_x(m+u)\varphi_x(n+u) = \frac{T_{mn}(x)}{u} \exp\Big(\sum_{j\geq 1} \lambda_j(x)\frac{u^j}{j!}\Big). \tag{346}$$

Since F_4 is the complete Bell coefficient $[u^3] \exp(4\sum_j \lambda_j u^j/j!)$, the residue at $u = 0$ of the fourth power of the left side of (346) is $T_{mn}^4 F_4$. Equation (346) is the two-dimensional form of (262) for rim exponent $2x$.

Identification at $x=\frac12$. Put $m=\ell+n$. Then $T_{mn}(\frac12)=T_{\ell n}$ of (251), and the $\lambda_j(\frac12)$, $j\le 3$, are the λ_j of (252), since $\mathsf{c}_1(\frac12)=\mathsf{c}_3(\frac12)=0$ and $\mathsf{c}_2(\frac12)=-\frac{2\pi^2}{3}=2[\psi_1(1)-\psi_1(\frac12)]$. Moreover $\frac12\partial_x\log T_{mn}|_{x=1/2}=\mu_1$ and $\frac12\partial_x\lambda_j|_{x=1/2}=\eta_j$ of (253), because $\partial_x\Delta_{j-1}(x,1;k)=\psi_j(k+x)$ and $\frac12\mathsf{c}_j'(\frac12)=(-\frac{\pi^2}{2},0,-\pi^4)=\frac12((-1)^j-1)\psi_j(\frac12)$. As $F_{4,1}=\sum_j(\partial F_4/\partial\lambda_j)\,\eta_j$, the summand of (256) is

$$T_{\ell n}^4(4\mu_1F_4+F_{4,1})=\tfrac12\,\partial_x\big[T_{mn}(x)^4F_4(x)\big]_{x=1/2}.$$

These identifications are exact algebra in independent polygamma symbols (machine checked in `LRI_C5_d2_proof/L2_singlesums/verify_symbolic.py`, §§1–3 and 5).

The square lattice. For $x\in I$ and $k\ge0$, Stirling's formula [27] gives $0<c_k(x)\le C(k+1)^{x-1}$. The quantities $\Delta_{j-1}(x,1;k)$, $\psi_j(k+x)$ ($j\ge1$), c_j and c_j' are bounded, and $|\psi_0(k+x)-\psi_0(x)|\le C\log(k+2)$. Hence $T_{mn}^4F_4$ and its x-derivative are bounded by $Cc_m(x)^4c_n(x)^4\log(m+2)\log(n+2)$, which is summable over $\mathbb{N}_0^2$ uniformly on I because $4(x-1)\le-\frac85$. The summands are symmetric in m,n, and at $d=2$ the multiplicities of (240) are $d_0=1$ and $d_{\ell\ge1}=2$. Therefore $\sum_\ell d_\ell\sum_n y_{\ell+n,n}=\sum_{m,n\ge0}y_{mn}$ for every symmetric absolutely summable array (y_{mn}). Define

$$\mathcal{W}(x)=-\sum_{m,n\ge0}T_{mn}(x)^4F_4(x),\qquad x\in I.$$

By (255), $\mathcal{W}(\frac12)=P_4(2)$. By (256) and termwise differentiation (Weierstrass M-test),

$$\mathcal{L}_2=\tfrac12\,\mathcal{W}'(\tfrac12).\tag{347}$$

The function $\mathcal{W}(x)$ equals the wheel $W_4(2;2x)$ of (242) (see the Status paragraph of Section 17.2); the proof below does not use this.

Residue form. Put $\mathcal{H}_x(u)=\sum_{k\ge0}\varphi_x(k+u)^4$. On the circle $|u|=\frac{1}{10}$ we have $|\varphi_x(k+u)|\le C(k+1)^{x-1}$ uniformly, so the double series of fourth powers of the left side of (346) converges uniformly there, to $N_x^4\sin^4\pi(x+u)\,\mathcal{H}_x(u)^2/\sin^4\pi u$. Integrating term by term over the circle,

$$\mathcal{W}(x)=-N_x^4\,\mathrm{Res}_{u=0}\,\frac{\sin^4\pi(x+u)\,\mathcal{H}_x(u)^2}{\sin^4\pi u}.\tag{348}$$

18.7.2 Two Fourier modes for every rim exponent

For $x\in I$ define

$$\Omega(t;x)=\sin^4\pi(t+x)\,\mathcal{H}_x(t)+\sin^4(\pi t)\sum_{j\ge0}\Big[\frac{\Gamma(j+1-t)}{\Gamma(j+2-x-t)}\Big]^4.$$

At $x=\frac12$ this is $\Omega_4(t)$ of (335). We claim

$$N_x^2\,\Omega(t;x)=\mathcal{A}(x)+\mathcal{B}(x)\cos2\pi(t+\tfrac{x}{2}),\qquad \mathcal{A}(\tfrac12)=a,\quad \mathcal{B}(\tfrac12)=b,\tag{349}$$

with $\mathcal{A},\mathcal{B}$ real analytic on the interior of I.

Gamma reflection gives, for $j\ge0$ and $m\in\mathbb{Z}$,

$$\sin^4(\pi t)\Big[\frac{\Gamma(j+1-t)}{\Gamma(j+2-x-t)}\Big]^4=\sin^4\pi(t+x)\,\varphi_x(t-j-1)^4,$$
$$\sin^4(\pi t)\,\varphi_x(-m-x-t)^4=\sin^4\pi(t+x)\,\varphi_x(m+t)^4.$$

By the first identity, $\Omega(t;x)=\sin^4\pi(t+x)\sum_{n\in\mathbb{Z}}\varphi_x(n+t)^4$ away from the poles, which is one-periodic. By the second, with $n\mapsto-n$, $\Omega(-x-t;x)=\Omega(t;x)$. The two series in the definition

converge locally uniformly away from $t \in -x - \mathbb{N}_0$ and $t \in 1 + \mathbb{N}_0$, respectively. Their poles there have order at most four and are canceled by the zeros of $\sin^4 \pi(t+x)$ and $\sin^4 \pi t$, so Ω is entire. For $\Re z \geq 0$, $|z| \geq 1$ and $x \in I$, Stirling's formula with its error bound in the closed right half-plane [27] gives, uniformly in x,

$$\Big|\frac{\Gamma(z+x)}{\Gamma(z+1)}\Big| \leq C|z|^{x-1}, \qquad \Big|\frac{\Gamma(z)}{\Gamma(z+1-x)}\Big| \leq C|z|^{x-1}.$$

With $z = n + t$ and $z = j + 1 - t$, for $t = \sigma + \mathrm{i}y$, $0 \leq \sigma \leq 1$, $|y| \geq 1$, this gives $|\Omega(t;x)| \leq Ce^{4\pi|y|} \sum_{n\geq 0} (n^2+y^2)^{2x-2} \leq C' e^{4\pi|y|} |y|^{4x-3}$, and $4x - 3 < 0$. The argument of (311) now applies: shifting the contour of the k-th Fourier coefficient to $\Im t = \mp Y$ removes every mode with $|k| \geq 2$, including $|k| = 2$ because of the decaying factor $|y|^{4x-3}$. The reflection $\Omega(-x-t) = \Omega(t)$ gives $\omega_1 = e^{2\pi \mathrm{i} x} \omega_{-1}$ for the two remaining nonconstant coefficients, which yields the form (349), and $\mathcal{A}, \mathcal{B}$ are real because Ω is real on the real axis. Finally $\mathcal{A}(x) = N_x^2 \int_0^1 \Omega(\sigma + \mathrm{i}; x)\, d\sigma$ and $\mathcal{B}(x) = 2e^{\mathrm{i}\pi x} N_x^2 \int_0^1 \Omega(\sigma+\mathrm{i};x) e^{2\pi\mathrm{i}(\sigma+\mathrm{i})} d\sigma$. On $\Im t = 1$ both series converge locally uniformly for complex x near I, so $\mathcal{A}$ and $\mathcal{B}$ are real analytic. At $x = \frac{1}{2}$, $N_{1/2} = 1$ and (349) is (337).

18.7.3 The residue functional and an exact reduction

By the definition of Ω, $\Omega(u;x) - \sin^4 \pi(x+u) \mathcal{H}_x(u)$ equals $\sin^4(\pi u)$ times a function analytic for $|u| < 1$, and $\sin \pi(x+u) \neq 0$ near $u = 0$. In (348) the factor $N_x^2 \sin^4 \pi(x+u) \mathcal{H}_x(u)$ may therefore be replaced by (349), since the difference contributes only terms regular at $u = 0$. Hence

$$\begin{gathered} \mathcal{W}(x) = -\big[\rho_0 \mathcal{A}^2 + 2\rho_1 \mathcal{A}\mathcal{B} + \rho_2 \mathcal{B}^2\big], \qquad \rho_p(x) = \mathrm{Res}_{u=0} \frac{\cos^p(2\pi u + \pi x)}{\sin^4 \pi u \, \sin^4 \pi(u+x)}, \\ \rho_0 = -\frac{4(\cos 2\pi x + 4)\cos \pi x}{\pi \sin^7 \pi x}, \qquad \rho_1 = \frac{4(4\sin^2 \pi x - 5)}{\pi \sin^7 \pi x}, \qquad \rho_2 = \frac{4(\sin^2 \pi x - 5)\cos \pi x}{\pi \sin^7 \pi x}. \end{gathered} \tag{350}$$

At $x = \frac{1}{2}$ one has $\rho_0 = \rho_2 = 0$, $\rho_1 = -\frac{4}{\pi}$, $\rho_0' = 12$, $\rho_1' = 0$ and $\rho_2' = 16$. So $\mathcal{W}(\frac{1}{2}) = 8ab/\pi$, the value of $P_4(2)$ found after (340), and by (347)

$$\mathcal{L}_2 = -6a^2 - 8b^2 + \frac{4}{\pi}\big[\mathcal{A}'(\tfrac{1}{2})\, b + a\, \mathcal{B}'(\tfrac{1}{2})\big].$$

At $t = 0$ the second series in Ω carries the factor $\sin^4 \pi t$. Evaluating (349) and its t-derivative at $t = 0$ gives, for $x \in I$,

$$\begin{gathered} N_x^2 \sin^4(\pi x)\, \mathcal{H}_x(0) = \mathcal{A} + \mathcal{B} \cos \pi x, \\ N_x^2 \big[4\pi \sin^3 \pi x \cos \pi x\, \mathcal{H}_x(0) + \sin^4 \pi x\, \partial_u \mathcal{H}_x(0)\big] = -2\pi \mathcal{B} \sin \pi x. \end{gathered}$$

The series $\mathcal{H}$ is jointly analytic in (u,x) near $(0, \frac{1}{2})$ and may be differentiated termwise. At $x = \frac{1}{2}$ we have $N_{1/2} = 1$, $(N_x^2)' = -4\psi_0(\frac{1}{2})$, $\varphi_{1/2}(k) = \sqrt{\pi}\, c_k$ and

$$\begin{aligned} \mathcal{H} &= \pi^2 \sum_k c_k^4, & \partial_u \mathcal{H} &= 4\pi^2 \sum_k c_k^4 \alpha_k, \\ \partial_x \mathcal{H} &= 4\pi^2 \sum_k c_k^4 \psi_0(k+\tfrac{1}{2}), & \partial_x \partial_u \mathcal{H} &= 4\pi^2 \sum_k c_k^4 \big[4\psi_0(k+\tfrac{1}{2})\alpha_k + \psi_1(k+\tfrac{1}{2})\big]. \end{aligned}$$

The two relations and their x-derivatives at $\frac{1}{2}$ then give $a = \pi^2 \sum_k c_k^4$ and $b = -2\pi \sum_k c_k^4 \alpha_k$; the terms in $\psi_0(\frac{1}{2})$ cancel, and

$$\begin{aligned} \mathcal{A}'(\tfrac{1}{2}) &= \pi b + 4\pi^2 \mathcal{S}_1, & \mathcal{B}'(\tfrac{1}{2}) &= 2\pi a - 2\pi \mathcal{S}_2, \\ \mathcal{S}_1 &= \sum_{k\geq 0} c_k^4 \beta_k, & \mathcal{S}_2 &= \sum_{k \geq 0} c_k^4 \big[4\alpha_k \beta_k + \psi_1(k+\tfrac{1}{2})\big]. \end{aligned} \tag{351}$$

All four sums converge absolutely, since $c_k^4 \sim (\pi k)^{-2}$. Consequently

$$\boxed{\mathcal{L}_2 = 2a^2 - 4b^2 + 16\pi b\,\mathcal{S}_1 - 8a\,\mathcal{S}_2, \qquad P_4(2) = \frac{8ab}{\pi}.} \tag{352}$$

This is an exact identity; it uses no relation among the single sums.

Remark (the integer-relation form of the collapse). The integer-relation search that first found (270) returned, in the present normalization, $\mathcal{L}_2 = 3a^2 - 2b^2 + 8\pi b\,\mathcal{S}_1 - 12a\,\mathcal{S}_2$. It differs from (352) by $a^2 + 2b^2 - 8\pi b\,\mathcal{S}_1 - 4a\,\mathcal{S}_2$, which vanishes only by the single-sum relations (356) below. The four numbers a^2, b^2, $\pi b\,\mathcal{S}_1$ and $a\,\mathcal{S}_2$ therefore satisfy a $\mathbb{Z}$-linear relation, and the searched form is one member of a one-parameter family of representations. Only (352) follows from the structure without those relations, and it is the form used here; both give the same value. The algebra is checked exactly by `LRI_C5_d2_simplify/proof/synth/synth_checks.py`.

18.7.4 Generating functions and elliptic moments

For $0 < k < 1$,

$$\begin{aligned}
\sum_{n\ge 0} c_n^2 \alpha_n k^{2n} &= -g(k) - \frac{2}{\pi} f(k) \log k, \\
\sum_{n\ge 0} c_n^2 \beta_n k^{2n} &= -\frac{2}{\pi} f(k) \log k', \\
\sum_{n\ge 0} c_n^2 \bigl[4\alpha_n\beta_n + 2\psi_1(n+\tfrac12)\bigr] k^{2n} &= 4g(k)\log k' + 2\pi f(k) + \frac{8}{\pi} f(k) \log k \, \log k'.
\end{aligned} \tag{353}$$

To prove these, put $z = k^2$, $\Phi(x,z) = {}_2F_1(x,x;1;z)$ and

$$\Lambda_x(z) = \sum_{n\ge 0} \frac{(x)_n^2}{n!^2} \bigl[2\psi_0(n+1) - 2\psi_0(n+x) - \log z\bigr] z^n = \frac{\Gamma(x)^2}{\Gamma(2x)}\, {}_2F_1(x,x;2x;1-z),$$

the logarithmic case $m = 0$ of the connection formula 15.8.10 of [23]. At $x = \frac12$ it gives $\Lambda_{1/2}(z) = \pi\, {}_2F_1(\frac12,\frac12;1;1-z) = 2g(k)$, which is the first line, since $\sum_n c_n^2 k^{2n} = \frac{2}{\pi} f(k)$. Euler's transformation $\Phi(x,z) = (1-z)^{1-2x}\Phi(1-x,z)$ (15.8.1 of [23]), differentiated at $x = \frac12$, gives $\partial_x \Phi(\frac12, z) = -\log(1-z)\,\Phi(\frac12,z)$. Termwise, $\partial_x \Phi(\frac12,z) = 2\sum_n c_n^2 \beta_n z^n$, which is the second line. The connection formula 15.8.4 of [23] for the parameters $(x,x;1)$, with its Gamma coefficients simplified by reflection, reads

$$\Lambda_x(z) = (1-z)^{1-2x}\Lambda_{1-x}(z) + 2\pi \cot(\pi x)\,\Phi(x,z), \qquad 0 < x < 1,\ x \ne \tfrac12.$$

For fixed $|z| < 1$ the series converge locally uniformly in x, so both sides are analytic on $0 < x < 1$ and the identity also holds at $x = \frac12$. Its x-derivative there is $\partial_x \Lambda|_{1/2} = -\log(1-z)\,\Lambda_{1/2}(z) - \pi^2 \Phi(\frac12, z)$. Termwise, $\partial_x \Lambda|_{1/2} = -\sum_n c_n^2 [4\alpha_n\beta_n + 2\psi_1(n+\frac12)] z^n - 2\log z \sum_n c_n^2 \beta_n z^n$, and with the second line this is the third.

Write $\mathrm{FG}[\phi] = \int_0^1 f(k) g(k) \phi(k)\, dk$ and $\mathrm{GG}[\phi] = \int_0^1 g(k)^2 \phi(k)\, dk$. Section 9.2 gives the following values: $\mathrm{GG}[1] = 2M_\mathrm{e}$ by (97); $\mathrm{FG}[w] = -\log 2\, N_\mathrm{e}$ and $\mathrm{GG}[w] = -2R - 4\log 2\, M_\mathrm{e}$ by (98); $\mathrm{FG}[\log k] = -\frac{\pi}{2} M_\mathrm{e}$ by (101); $R = \frac{\pi}{4} N_\mathrm{e} - \log 2\, M_\mathrm{e}$ by (105); and $U_0 = 2N_\mathrm{e}$. One further input is needed:

$$\mathrm{GG}[\log k] = \int_0^1 \mathbf{K}(k')^2 \log k \, dk = -\pi N_\mathrm{e}. \tag{354}$$

It is the real part of the first-quadrant identity (104) with $W(z)$ replaced by $\mathrm{Log}\, z$. The arc estimates are unchanged: the integrand is $O(\log^3 |z|/|z|^2)$ at infinity and has integrable logarithmic singularities at 0 and 1. After $x = 1/k$ on $(1,\infty)$ the boundary value $\mathrm{Log}(x + \mathrm{i}0) = -\log k$

multiplies $(f+\mathrm{i}g)^2$, so the two contributions $\pm\int_0^1 f^2\log k\,dk$ cancel and the real part of the left side is $\mathrm{GG}[\log k]$. On the imaginary axis $\mathrm{Log}(\mathrm{i}y)=\log y+\frac{\mathrm{i}\pi}{2}$, so after $y=k/k'$ the real part of the right side is $-\frac{\pi}{2}U_0=-\pi N_\mathrm{e}$. Since $\log k'=\frac12(\log k-w)$, it follows that

$$\mathrm{FG}[\log k']=-\frac{\pi}{4}M_\mathrm{e}+\frac{\log 2}{2}N_\mathrm{e},\qquad \mathrm{GG}[\log k']=\log 2\,M_\mathrm{e}-\frac{\pi}{4}N_\mathrm{e}. \tag{355}$$

18.7.5 The single sums and the conclusion

The complementary moment of Section 9, $\int_0^1 k^r g(k)\,dk=\frac{\pi}{4}\Gamma(\frac{r+1}{2})^2/\Gamma(\frac{r+2}{2})^2$ for $r>-1$, and its r-derivative at $r=2n$ (dominated convergence) give

$$c_n^2=\frac{4}{\pi^2}\int_0^1 k^{2n}g(k)\,dk,\qquad c_n^2\alpha_n=\frac{4}{\pi^2}\int_0^1 k^{2n}\log k\;g(k)\,dk.$$

In each single sum write one factor c_n^2, or $c_n^2\alpha_n$, in this form, and sum the remaining factor with (353). For $\mathcal{S}_2$ use

$$c_n^4\big[4\alpha_n\beta_n+\psi_1(n+\tfrac12)\big]=\tfrac12\{c_n^2\big[4\alpha_n\beta_n+2\psi_1(n+\tfrac12)\big]\cdot c_n^2+2c_n^2\alpha_n\cdot 2c_n^2\beta_n\}.$$

The interchanges of sum and integral are justified by Tonelli's theorem, since $c_n^2>0$, $\alpha_n<0$ and $\beta_n\ge 0$ have fixed signs, and, for the first product above, by Fubini's theorem, since that product is absolutely summable. The two contributions $\pm\frac{16}{\pi^3}\int_0^1 fg\log k\log k'\,dk$ cancel, and

$$\sum_n c_n^4=\frac{8N_\mathrm{e}}{\pi^3},\qquad \sum_n c_n^4\alpha_n=-\frac{4M_\mathrm{e}}{\pi^2},\qquad \mathcal{S}_1=-\frac{8}{\pi^3}\,\mathrm{FG}[\log k'],\qquad \mathcal{S}_2=\frac{8}{\pi^2}\,\mathrm{GG}[\log k']+\frac{4}{\pi}N_\mathrm{e}.$$

Hence $a=8N_\mathrm{e}/\pi$ and $b=8M_\mathrm{e}/\pi$, in agreement with (340), and (355) gives the two single-sum relations

$$\boxed{\mathcal{S}_1=\frac{\pi b-2a\log 2}{4\pi^2},\qquad \mathcal{S}_2=\frac{a}{4}+\frac{b\log 2}{\pi}.} \tag{356}$$

With $\hat a=a/\pi^2=\sum_k c_k^4$ and $\hat b=b/\pi^2=-\frac{2}{\pi}\sum_k c_k^4\alpha_k$ they read $\mathcal{S}_1=\frac14(\pi\hat b-2\hat a\log 2)$ and $\mathcal{S}_2=\frac{\pi^2}{4}\hat a+\pi\hat b\log 2$, the two relations first found numerically. Inserting (356) into (352), the a^2 and b^2 terms cancel:

$$\mathcal{L}_2=2a^2-4b^2+\frac{4b}{\pi}(\pi b-2a\log 2)-8a\Big(\frac{a}{4}+\frac{b\log 2}{\pi}\Big)=-\frac{16\log 2}{\pi}\,ab=-2\log 2\,P_4(2).$$

With $P_4(2)=(B_-^2-B_+^2)/(4\pi^3)$ from (333), this proves (270). Equivalently, (351) and (356) say $\mathcal{A}'(\frac12)=2\pi b-2\log 2\,a$ and $\mathcal{B}'(\frac12)=\frac{3\pi}{2}a-2\log 2\,b$, that is, $(4^x\mathcal{A})'=2\pi\,4^x\mathcal{B}$ and $(4^x\mathcal{B})'=\frac{3\pi}{2}\,4^x\mathcal{A}$ at $x=\frac12$.

Checks. These are not proof steps. The closed forms of $\sum c_k^4$, $\sum c_k^4\alpha_k$, $\mathcal{S}_1$ and $\mathcal{S}_2$ agree with 70-digit Euler–Maclaurin values of the single sums to 3×10^{-70}. With $\mathcal{A}'$ and $\mathcal{B}'$ obtained from the termwise x-derivative of $\Omega(t;x)$ at generic t, (351) and the chain $\frac12\mathcal{W}'(\frac12)=\mathcal{L}_2$ hold to 10^{-82}. The generating functions (353) and the moments (354), (355) hold to about 35 digits.

18.7.6 A second proof by rim–spoke reflection

An independent proof works directly with the wheel derivative of (244). This subsection states the result and its consequences and then proves them in full. For $\frac14<x<\frac34$ it shows that

$$W_4(2;2x)=\frac{2^{4-4x}\Gamma(1-x)^4\Gamma(x)^6}{\pi^{11/2}\Gamma(2x-\frac12)}\big[\mathfrak{a}(x)\mathfrak{b}(1-x)+\mathfrak{b}(x)\mathfrak{a}(1-x)\big],$$

where $\mathfrak{a}(x)+\mathfrak{b}(x)\cos 2\pi\tau$ is the two-mode form of the entire function

$$\pi^4\sum_{k\in\mathbb{Z}}\bigl[\Gamma(1-\tfrac{x}{2}+k+\tau)\,\Gamma(1-\tfrac{x}{2}-k-\tau)\bigr]^{-4},$$

with $\mathfrak{a}(\frac12)=a$ and $\mathfrak{b}(\frac12)=b$. The inputs are the square-lattice factorization of the wheel for every rim exponent, the growth argument of (311), and a Barnes integral of well-poised type. That integral is proved in its terminating case from the Pfaff–Saalschütz sum (16.4.3 of [22]), Whipple's transformations (16.4.14 and 16.4.15 of [22]) and Barnes' first lemma (DLMF 5.13.3), and is extended by Carlson's theorem. The bracket is invariant under $x\to 1-x$, which exchanges rim and spoke exponents. Hence, with $\gamma_C(z)=\Gamma(z)/\Gamma(1-z)$ as in (264),

$$\frac{W_4(2;b)}{W_4(2;2-b)}=2^{4-4b}\,\frac{\gamma_C(b/2)^2}{\gamma_C(b-\frac12)},\qquad \frac12<b<\frac32,$$

and at the fixed point $b=1$ only the Gamma prefactor contributes to the logarithmic derivative. This gives $\partial_b\log W_4(2;b)|_{b=1}=\frac12(-4\log 2-4\psi_0(\frac12)+6\psi_0(\frac12)-2\psi_0(\frac12))=-2\log 2$. As a check, the factorized form agrees with a direct spectral evaluation of $W_4(2;2x)$ at $x=\frac25,\frac12,\frac35$ to 2×10^{-70}.

The machine checks of the proof below are `LRI_C5_d2_proof/L2_structural/checks_all.wl` (log `checks_all.log`); the 70-digit spectral check just quoted is recorded in `LRI_C5_d2_proof/review_L2_structural_numerics_r0/`.

Notation. Throughout this subsubsection $d=2$, the rim exponent is $b=2x$, and $N_x=\Gamma(1-x)/\Gamma(x)=\gamma_C(1-x)$ and $\varphi_x(s)=\Gamma(s+x)/\Gamma(s+1)$ are as in (345). Since $\Gamma(d/2)=1$, the eigenvalue (242) is $t_{\ell,2x}(\nu)=N_x\,\mathfrak{g}_\ell(\mathrm{i}\nu)$ with

$$\mathfrak{g}_\ell(s)=\frac{\Gamma(\frac{\ell}{2}+\frac{x}{2}+s)\,\Gamma(\frac{\ell}{2}+\frac{x}{2}-s)}{\Gamma(\frac{\ell}{2}+1-\frac{x}{2}+s)\,\Gamma(\frac{\ell}{2}+1-\frac{x}{2}-s)},$$

the kernel $g_\ell(s;2x)$ of (460) at $d=2$, and $W_4(2;2x)=\langle t_{\ell,2x}^4\rangle_2$ with $d_0=1$, $d_{\ell\ge1}=2$. For $0<x<\frac34$ let

$$\Omega(t;x)=\pi^4\sum_{k\in\mathbb{Z}}\bigl[\Gamma(1+k+t)\,\Gamma(1-x-k-t)\bigr]^{-4}=\sin^4\pi(t+x)\sum_{k\in\mathbb{Z}}\varphi_x(k+t)^4,$$

where the second form follows termwise from Gamma reflection [26]. For $x\in I$ this is the function $\Omega(t;x)$ of (349), and $\Omega(t;\frac12)=\Omega_4(t)$ of (335). The function displayed above in the variable τ is $\Omega(\tau-\frac{x}{2};x)$, and its two modes are written $\mathfrak{a}(x)$, $\mathfrak{b}(x)$. The statement to be proved is

$$W_4(2;2x)=\mathfrak{p}(x)\bigl[\mathfrak{a}(x)\mathfrak{b}(1-x)+\mathfrak{b}(x)\mathfrak{a}(1-x)\bigr],\qquad \mathfrak{p}(x)=\frac{2^{4-4x}\Gamma(1-x)^4\Gamma(x)^6}{\pi^{11/2}\Gamma(2x-\frac12)},\qquad \tfrac14<x<\tfrac34. \tag{357}$$

At $x=\frac12$, $\mathfrak{p}(\frac12)=4/\pi$ and (357) is $P_4(2)=8ab/\pi$.

Lemma (two modes of a reciprocal-Gamma kernel). *Let $0<\kappa_1,\dots,\kappa_4<1$ with $\kappa=\kappa_1+\cdots+\kappa_4>\frac52$, and let $\mathfrak{k}(s)=\prod_{j=1}^4[\Gamma(\kappa_j+s)\Gamma(\kappa_j-s)]^{-1}$, an entire even function. Then*

(i) $|\mathfrak{k}(s)|\le C\,e^{4\pi|\Im s|}(1+|s|)^{4-2\kappa}$ *for all $s\in\mathbb{C}$;*

(ii) $E_{\mathfrak{k}}(\tau)=\sum_{k\in\mathbb{Z}}\mathfrak{k}(k+\tau)$ *converges absolutely and locally uniformly, and $E_{\mathfrak{k}}$ is entire, even and one-periodic;*

(iii) $E_{\mathfrak{k}}(\tau)=\int_{\mathbb{R}}\mathfrak{k}(s)\,ds+2\cos(2\pi\tau)\int_{\mathbb{R}}\mathfrak{k}(s)\cos(2\pi s)\,ds$.

Proof. (i) By reflection, $[\Gamma(\kappa_j+s)\Gamma(\kappa_j-s)]^{-1} = \pi^{-1}\sin\pi(\kappa_j-s)\,\Gamma(s+1-\kappa_j)/\Gamma(s+\kappa_j)$. For $\Re s \ge 0$ the Gamma ratio has no poles, because $1-\kappa_j>0$, and it is $O((1+|s|)^{1-2\kappa_j})$ by the Gamma-ratio asymptotics [27] for $|s|\ge 1$ and by continuity on the compact rest; moreover $|\sin\pi(\kappa_j-s)| \le e^{\pi|\Im s|}$. For $\Re s \le 0$ use $\mathfrak{k}(-s)=\mathfrak{k}(s)$. (ii) On a compact set K, (i) gives $|\mathfrak{k}(k+\tau)| \le C_K(1+|k|)^{4-2\kappa}$, and $2\kappa-4>1$; the Weierstrass test applies. Evenness and periodicity follow by re-indexing. (iii) For $\tau=u+\mathrm{i}y$ with $0\le u\le 1$, (i) and comparison with $\int_{\mathbb{R}}(1+|v+\mathrm{i}y|)^{4-2\kappa}\,dv$ give $|E_{\mathfrak{k}}(u+\mathrm{i}y)| \le C' e^{4\pi|y|}(1+|y|)^{5-2\kappa}$, with $5-2\kappa<0$. For an entire one-periodic function the coefficient $e_m=\int_0^1 E_{\mathfrak{k}}(u+\mathrm{i}y)e^{-2\pi\mathrm{i}m(u+\mathrm{i}y)}\,du$ does not depend on y, so $|e_m| \le C' e^{2\pi m y+4\pi|y|}(1+|y|)^{5-2\kappa}$. Letting $y\to-\infty$ for $m\ge 2$ and $y\to+\infty$ for $m\le -2$ gives $e_m=0$; for $|m|=2$ this uses $(1+|y|)^{5-2\kappa}\to 0$. Evenness gives $e_{-1}=e_1$. Finally $\int_{\mathbb{R}}|\mathfrak{k}|<\infty$, so by Fubini $e_0=\sum_k\int_0^1\mathfrak{k}(k+u)\,du=\int_{\mathbb{R}}\mathfrak{k}$ and $e_1=\int_{\mathbb{R}}\mathfrak{k}(s)e^{-2\pi\mathrm{i}s}\,ds=\int_{\mathbb{R}}\mathfrak{k}(s)\cos(2\pi s)\,ds$. □

This is the growth argument of (311) applied to a reciprocal-Gamma kernel. Two cases are used.

(a) $\kappa_j=1-\frac{x}{2}$ for all j, with $0<x<\frac34$, so that $\kappa=4-2x>\frac52$. Then $\pi^4 E_{\mathfrak{k}}(\tau)=\Omega(\tau-\frac{x}{2};x)$, so that for $0<x<\frac34$

$$\Omega(t;x)=\mathfrak{a}(x)+\mathfrak{b}(x)\cos 2\pi(t+\tfrac{x}{2}),\qquad \mathfrak{a}(x)=\pi^4\int_{\mathbb{R}}\frac{ds}{\mathcal{G}_x(s)^4},\qquad \mathfrak{b}(x)=2\pi^4\int_{\mathbb{R}}\frac{\cos(2\pi s)\,ds}{\mathcal{G}_x(s)^4}, \tag{358}$$

with $\mathcal{G}_x(s)=\Gamma(1-\frac{x}{2}+s)\Gamma(1-\frac{x}{2}-s)$. On I, (349) gives $\mathfrak{a}=\mathcal{A}/N_x^2$ and $\mathfrak{b}=\mathcal{B}/N_x^2$, so $\mathfrak{a}(\frac12)=a$ and $\mathfrak{b}(\frac12)=b$. Evaluating at $t=-\frac{x}{2}$ and $t=\frac12-\frac{x}{2}$, the two combinations $\mathfrak{a}\pm\mathfrak{b}$ are well-poised bilateral series: $\mathfrak{a}(x)+\mathfrak{b}(x)=\sin^4\frac{\pi x}{2}\sum_{k\in\mathbb{Z}}[\Gamma(k+\frac{x}{2})/\Gamma(k+1-\frac{x}{2})]^4$ and $\mathfrak{a}(x)-\mathfrak{b}(x)=\cos^4\frac{\pi x}{2}\sum_{k\in\mathbb{Z}}[\Gamma(k+\frac12+\frac{x}{2})/\Gamma(k+\frac32-\frac{x}{2})]^4$.

(b) $\kappa_j=\frac12+p_j$ with $0<p_j<\frac12$ and $p=p_1+\cdots+p_4>\frac12$, so that $\kappa=2+p>\frac52$. Then

$$E^\sharp(\tau)=\sum_{k\in\mathbb{Z}}\prod_{j=1}^4\big[\Gamma(\tfrac12+p_j+k+\tau)\,\Gamma(\tfrac12+p_j-k-\tau)\big]^{-1}$$

is a constant plus a multiple of $\cos 2\pi\tau$. If $p_j=\frac{x}{2}$ for all j, then $\pi^4E^\sharp(\tau)=\Omega(\tau-\frac{1-x}{2};1-x)$, because $1-\frac{1-x}{2}=\frac12+\frac{x}{2}$.

Proposition (the lattice step). *For $0<x<\frac34$ put $\mathcal{F}_x(s)=[\Gamma(\frac{x}{2}+s)\Gamma(\frac{x}{2}-s)]^4$ and*

$$\mathsf{J}_0(x)=\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,\mathcal{F}_x(s),\qquad \mathsf{J}_1(x)=\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,\cos(2\pi s)\,\mathcal{F}_x(s),$$

which converge absolutely because $|\Gamma(\frac{x}{2}+\mathrm{i}y)|^2\sim 2\pi|y|^{x-1}e^{-\pi|y|}$. Then

$$\begin{aligned} W_4(2;2x) &= -N_x^4\,\mathrm{Res}_{u=0}\frac{\Omega(u;x)^2}{\sin^4\pi u\,\sin^4\pi(u+x)} = \frac{N_x^4}{\pi^4}\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,\Omega(s-\tfrac{x}{2};x)\,\mathcal{F}_x(s)\\ &= \frac{N_x^4}{\pi^4}\big[\mathfrak{a}(x)\mathsf{J}_0(x)+\mathfrak{b}(x)\mathsf{J}_1(x)\big]. \end{aligned} \tag{359}$$

In particular, with $\mathcal{W}$ of (348),

$$W_4(2;2x)=\mathcal{W}(x)\qquad (x\in I). \tag{360}$$

Proof. (i) *Spectral form.* Since $t_{\ell,2x}(\nu)$ is even in ν, (241) gives

$$W_4(2;2x)=N_x^4\sum_{\ell\ge0}d_\ell\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,\mathsf{g}_\ell(s)^4.$$

The double sum and integral converge absolutely for $x<\frac34$: at fixed ℓ, $t_{\ell,2x}(\nu)^4=O(\nu^{8x-8})$, and $\int_0^\infty t_{\ell,2x}^4\,d\nu=O(\ell^{8x-7})$.

(ii) *Lattice identity.* For $\ell \geq 0$ and $n \in \mathbb{Z}$, Gamma reflection applied to $\Gamma(-n-u)$ and to $1/\Gamma(1-x-n-u)$ gives

$$\mathsf{g}_\ell(\tfrac{\ell}{2}+\tfrac{x}{2}+n+u) = -\frac{\sin\pi(x+u)}{\sin\pi u}\,\varphi_x(\ell+n+u)\,\varphi_x(n+u),$$

so $-N_x\,\mathsf{g}_\ell(\frac{\ell}{2}+\frac{x}{2}+n+u)$ is the left side of (346) with $m=\ell+n$. With $(k,j)=(\ell+n,n)$ the weights $d_0=1$, $d_{\ell\geq 1}=2$ turn $\sum_{\ell\geq 0} d_\ell \sum_{n\in\mathbb{Z}}$ into the sum over all $(k,j)\in\mathbb{Z}^2$, because the summand $\varphi_x(k+u)^4\varphi_x(j+u)^4$ is symmetric in k,j. Hence

$$\mathsf{S}(u) = \sum_{\ell\geq 0} d_\ell \sum_{n\in\mathbb{Z}} \mathsf{g}_\ell(\tfrac{\ell}{2}+\tfrac{x}{2}+n+u)^4 = \frac{\sin^4\pi(x+u)}{\sin^4\pi u}\Big[\sum_{k\in\mathbb{Z}}\varphi_x(k+u)^4\Big]^2 = \frac{\Omega(u;x)^2}{\sin^4\pi u\,\sin^4\pi(u+x)}.$$

On a circle $|u|=r$ with $0<r<\min(x,1-x)$ one has $|\varphi_x(k+u)|\leq C(1+|k|)^{x-1}$ uniformly, and $4x-4<-1$, so the double series converges absolutely and uniformly there. This is (539) at $d=2$.

(iii) *Residues of the two-dimensional integral.* For each ℓ the poles of g_ℓ in $\Re s>0$ are at $s=\frac{\ell}{2}+\frac{x}{2}+n$, $n\geq 0$, from $\Gamma(\frac{\ell}{2}+\frac{x}{2}-s)$; those of $\Gamma(\frac{\ell}{2}+\frac{x}{2}+s)$ lie in $\Re s<0$. By reflection,

$$\mathsf{g}_\ell(s) = \frac{\Gamma(s+\frac{\ell}{2}+\frac{x}{2})}{\Gamma(s+\frac{\ell}{2}+1-\frac{x}{2})}\cdot\frac{\Gamma(s-\frac{\ell}{2}+\frac{x}{2})}{\Gamma(s-\frac{\ell}{2}+1-\frac{x}{2})}\cdot\frac{\sin\pi(\frac{x}{2}+s-\frac{\ell}{2})}{\sin\pi(\frac{x}{2}-s+\frac{\ell}{2})}.$$

On the arcs $|s-\frac{\ell}{2}-\frac{x}{2}|=N+\frac{1}{2}$ in $\Re s\geq 0$, which pass midway between consecutive poles, the sine ratio is bounded uniformly in N, and for large N both Gamma ratios are $O(|s|^{x-1})$. So $\mathsf{g}_\ell^4=O(|s|^{8x-8})$ there, and since the arcs have length $O(N)$ and $8x-7<0$ their contributions vanish. Closing the contour to the right, which is clockwise, $\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\mathsf{g}_\ell^4 = -\sum_{n\geq 0}\operatorname{Res}_{u=0}\mathsf{g}_\ell(\frac{\ell}{2}+\frac{x}{2}+n+u)^4$. For $n<0$ the function $\mathsf{g}_\ell(\frac{\ell}{2}+\frac{x}{2}+n+u)$ is regular at $u=0$: there $\Gamma(-n-u)$ is regular, and so is $\Gamma(\ell+n+x+u)$, since $x\notin\mathbb{Z}$. Summing over ℓ with the weights d_ℓ, and using the uniform convergence of (ii) on $|u|=r$ to take the sums inside one contour integral, gives $W_4(2;2x)=-N_x^4\operatorname{Res}_{u=0}\mathsf{S}(u)$, the first equality in (359).

For $k\leq -1$, $1/\Gamma(k+1+u)$ has a simple zero at $u=0$, so $\varphi_x(k+u)^4=O(u^4)$, and all terms of S that contain such a factor are regular at $u=0$. Hence $\operatorname{Res}_{u=0}\mathsf{S}(u)=\operatorname{Res}_{u=0}[\sin^4\pi(x+u)\,\mathcal{H}_x(u)^2/\sin^4\pi u]$ with $\mathcal{H}_x$ of (348), and for $x\in I$ that equation gives (360). This is the per-spin residue evaluation of (261) at rim exponent $2x$; it is also the case $d=2$ of (479) with (539).

(iv) *Reflection.* For $k\in\mathbb{Z}$, $1/\Gamma(k+1+u)=(-1)^{k+1}\sin(\pi u)\Gamma(-k-u)/\pi$, so $\varphi_x(k+u)^4=\pi^{-4}\sin^4(\pi u)\,\mathcal{F}_x(\frac{x}{2}+k+u)$. With $\mathcal{P}_x(u)=\sum_{k\in\mathbb{Z}}\mathcal{F}_x(\frac{x}{2}+k+u)$ this gives $\Omega(u;x)=\pi^{-4}\sin^4\pi(x+u)\sin^4(\pi u)\,\mathcal{P}_x(u)$, hence

$$\frac{\Omega(u;x)^2}{\sin^4\pi u\,\sin^4\pi(u+x)} = \pi^{-4}\,\Omega(u;x)\,\mathcal{P}_x(u).$$

(v) *Residues of the one-dimensional integral.* The function $\Omega(s-\frac{x}{2};x)=\mathfrak{a}+\mathfrak{b}\cos 2\pi s$ is entire, with $|\Omega(s-\frac{x}{2};x)|\leq(|\mathfrak{a}|+|\mathfrak{b}|)e^{2\pi|\Im s|}$. Writing $\Gamma(\frac{x}{2}-s)=\pi/[\sin\pi(\frac{x}{2}-s)\Gamma(1-\frac{x}{2}+s)]$ gives $|\mathcal{F}_x(s)|\leq C|s|^{4x-4}e^{-4\pi|\Im s|}$ on the arcs $|s-\frac{x}{2}|=N+\frac{1}{2}$ in $\Re s\geq 0$. So the integrand is $O(|s|^{4x-4})$ there, and the arc contributions vanish because $4x-3<0$. The poles of $\mathcal{F}_x$ in $\Re s>0$ are at $s=\frac{x}{2}+n$, $n\geq 0$, and therefore

$$\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,\Omega(s-\tfrac{x}{2};x)\mathcal{F}_x(s) = -\sum_{n\geq 0}\operatorname{Res}_{u=0}\big[\Omega(n+u;x)\,\mathcal{F}_x(\tfrac{x}{2}+n+u)\big] = -\operatorname{Res}_{u=0}\big[\Omega(u;x)\,\mathcal{P}_x(u)\big],$$

by the periodicity of Ω and the uniform convergence of $\mathcal{P}_x$ near $u=0$; the terms with $n<0$ are regular at $u=0$. With (iv) this is the second equality of (359), and (358) gives the third. □

Lemma (a well-poised Barnes integral). *Let* $\Re \mathsf{a}_j > 0$ *(*$j = 1,2,3$*),* $\mathsf{a} = \mathsf{a}_1 + \mathsf{a}_2 + \mathsf{a}_3$ *and* $\Re(\mathsf{a} + \tfrac12 - \mathsf{b}) > 0$*. Then*

$$\begin{aligned}\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}} \frac{\prod_{j=1}^{3}\Gamma(\mathsf{a}_j+s)\Gamma(\mathsf{a}_j-s)}{\Gamma(\mathsf{b}+s)\Gamma(\mathsf{b}-s)} \\ = \frac{2\sqrt{\pi}\,2^{2\mathsf{b}-2\mathsf{a}}\prod_{i<j}\Gamma(\mathsf{a}_i+\mathsf{a}_j)\prod_i\Gamma(2\mathsf{a}_i)}{\Gamma(\mathsf{a}+\frac12-\mathsf{b})\prod_i\Gamma(\mathsf{a}_i+\mathsf{b})}\,{}_4F_3\!\left(\begin{matrix}2\mathsf{b}-1,\ \mathsf{b}-\mathsf{a}_1,\ \mathsf{b}-\mathsf{a}_2,\ \mathsf{b}-\mathsf{a}_3\\ \mathsf{b}+\mathsf{a}_1,\ \mathsf{b}+\mathsf{a}_2,\ \mathsf{b}+\mathsf{a}_3\end{matrix};1\right).\end{aligned} \tag{361}$$

The ${}_4F_3$ *is well-poised and converges, since its parameter excess is* $1 + 2\mathsf{a} - 2\mathsf{b}$*.*

At $\mathsf{b} = \mathsf{a}_3$ this is Barnes' first lemma (DLMF 5.13.3). At $\mathsf{b} = \frac12$ it is the de Branges–Wilson integral (DLMF 5.13.5) with one parameter equal to 0, since $\Gamma(s)\Gamma(-s)/[\Gamma(2s)\Gamma(-2s)] = 4\cos\pi s = 4\pi/[\Gamma(\frac12+s)\Gamma(\frac12-s)]$. Write I_{L} and I_{R} for the two sides of (361).

Proof. (a) *Terminating case* $\mathsf{a}_3 = \mathsf{b} + n$, $n \in \mathbb{N}_0$. Take $\mathsf{a}_1, \mathsf{a}_2 > 0$ and $\mathsf{b} > \frac12$ real and generic ($\mathsf{b} - \mathsf{a}_1, \mathsf{b} - \mathsf{a}_2 \notin \mathbb{Z}$). Then $\Gamma(\mathsf{a}_3 \pm s)/\Gamma(\mathsf{b} \pm s) = (\mathsf{b}+s)_n(\mathsf{b}-s)_n$. The Pfaff–Saalschütz sum (16.4.3 of [22]), with its parameters taken as $\mathsf{a}_2 + s$, $\mathsf{a}_2 - s$, $\mathsf{a}_2 + \mathsf{b}$ and $1 + \mathsf{a}_2 - \mathsf{b} - n$, gives

$$(\mathsf{b}+s)_n(\mathsf{b}-s)_n = (\mathsf{a}_2+\mathsf{b})_n(\mathsf{b}-\mathsf{a}_2)_n \sum_{m=0}^{n} \frac{(-n)_m(\mathsf{a}_2+s)_m(\mathsf{a}_2-s)_m}{m!\,(\mathsf{a}_2+\mathsf{b})_m(1+\mathsf{a}_2-\mathsf{b}-n)_m}.$$

Since $(\mathsf{a}_2 \pm s)_m\Gamma(\mathsf{a}_2 \pm s) = \Gamma(\mathsf{a}_2 + m \pm s)$, Barnes' first lemma evaluates each term: $\int \frac{ds}{2\pi\mathrm{i}}\Gamma(\mathsf{a}_1 + s)\Gamma(\mathsf{a}_1 - s)\Gamma(\mathsf{a}_2 + m + s)\Gamma(\mathsf{a}_2 + m - s) = \Gamma(2\mathsf{a}_1)\Gamma(\mathsf{a}_1 + \mathsf{a}_2 + m)^2\Gamma(2\mathsf{a}_2 + 2m)/\Gamma(2\mathsf{a}_1 + 2\mathsf{a}_2 + 2m)$. With $(2c)_{2m} = 4^m(c)_m(c+\frac12)_m$ this gives

$$\begin{aligned} I_{\mathrm{L}} &= (\mathsf{a}_2+\mathsf{b})_n(\mathsf{b}-\mathsf{a}_2)_n\,\frac{\Gamma(2\mathsf{a}_1)\Gamma(2\mathsf{a}_2)\Gamma(\mathsf{a}_1+\mathsf{a}_2)^2}{\Gamma(2\mathsf{a}_1+2\mathsf{a}_2)}\,F_{\mathrm{L}},\\ F_{\mathrm{L}} &= {}_4F_3\!\left(\begin{matrix}-n,\ \mathsf{a}_1+\mathsf{a}_2,\ \mathsf{a}_2,\ \mathsf{a}_2+\frac12\\ \mathsf{a}_2+\mathsf{b},\ 1+\mathsf{a}_2-\mathsf{b}-n,\ \mathsf{a}_1+\mathsf{a}_2+\frac12\end{matrix};1\right).\end{aligned}$$

On the right, $F_{\mathrm{R}} = {}_4F_3(2\mathsf{b}-1, \mathsf{b}-\mathsf{a}_1, \mathsf{b}-\mathsf{a}_2, -n; \mathsf{b}+\mathsf{a}_1, \mathsf{b}+\mathsf{a}_2, 2\mathsf{b}+n; 1)$ is the terminating very-well-poised ${}_7F_6$ with first parameter $2\mathsf{b}-1$ and further parameters $\mathsf{b}-\mathsf{a}_1$, $\mathsf{b}-\mathsf{a}_2$, $\mathsf{b}-\frac12$, b, $-n$: the pair $\mathsf{b}-\frac12$, b cancels the very-well-poised factor and itself. Whipple's transformation (16.4.15 of [22], with $f = -n$, $(b,c) \to (\mathsf{b}-\mathsf{a}_1, \mathsf{b}-\mathsf{a}_2)$ and $(d,e) \to (\mathsf{b}-\frac12, \mathsf{b})$) gives

$$F_{\mathrm{R}} = \frac{(2\mathsf{b})_n(\frac12)_n}{(\mathsf{b}+\frac12)_n(\mathsf{b})_n}\,{}_4F_3\!\left(\begin{matrix}\mathsf{a}_1+\mathsf{a}_2,\ \mathsf{b}-\frac12,\ \mathsf{b},\ -n\\ \mathsf{b}+\mathsf{a}_1,\ \mathsf{b}+\mathsf{a}_2,\ \frac12-n\end{matrix};1\right),$$

and Whipple's transformation of a balanced ${}_4F_3$ (16.4.14 of [22], with $(a;b,c;d,e,f) \to (\mathsf{a}_1+\mathsf{a}_2; \mathsf{b}-\frac12, \mathsf{b}; \mathsf{b}+\mathsf{a}_2, \mathsf{b}+\mathsf{a}_1, \frac12-n)$) turns the last ${}_4F_3$ into $\frac{(\mathsf{b}-\mathsf{a}_2)_n(\frac12-n-\mathsf{a}_1-\mathsf{a}_2)_n}{(\mathsf{b}+\mathsf{a}_1)_n(\frac12-n)_n}F_{\mathrm{L}}$. Hence $F_{\mathrm{R}} = K_n F_{\mathrm{L}}$ with

$$K_n = \frac{(2\mathsf{b})_n(\frac12)_n(\mathsf{b}-\mathsf{a}_2)_n(\frac12-n-\mathsf{a}_1-\mathsf{a}_2)_n}{(\mathsf{b}+\frac12)_n(\mathsf{b})_n(\mathsf{b}+\mathsf{a}_1)_n(\frac12-n)_n}.$$

The prefactor of I_{R} times K_n equals the prefactor of I_{L}. Indeed, with $\mathsf{a}_3 = \mathsf{b}+n$: $\Gamma(\mathsf{a}_1+\mathsf{b}+n) = \Gamma(\mathsf{a}_1+\mathsf{b})(\mathsf{b}+\mathsf{a}_1)_n$, $\Gamma(\mathsf{a}_2+\mathsf{b}+n) = \Gamma(\mathsf{a}_2+\mathsf{b})(\mathsf{a}_2+\mathsf{b})_n$, $\Gamma(2\mathsf{b}+2n)(2\mathsf{b})_n/\Gamma(2\mathsf{b}+n) = (2\mathsf{b})_{2n} = 4^n(\mathsf{b})_n(\mathsf{b}+\frac12)_n$, and $(\frac12)_n(\frac12-n-c)_n/(\frac12-n)_n = (\frac12+c)_n$ for $c = \mathsf{a}_1+\mathsf{a}_2$, which cancels against $\Gamma(\mathsf{a}+\frac12-\mathsf{b}) = \Gamma(c+\frac12)(c+\frac12)_n$. What remains is $2\sqrt{\pi}\,2^{-2c}\Gamma(2c)/[\Gamma(c)\Gamma(c+\frac12)] = 1$, by duplication [26]. So $I_{\mathrm{L}} = I_{\mathrm{R}}$ when $\mathsf{a}_3 = \mathsf{b}+n$.

(b) *Carlson's theorem.* Keep $\mathsf{a}_1, \mathsf{a}_2 > 0$ and a generic $\mathsf{b} \in (\frac12, \frac12 + \mathsf{a}_1 + \mathsf{a}_2)$, put $\mathsf{a}_3 = \mathsf{b} + z$ with $\Re z \ge 0$, and let $f(z) = (I_{\mathrm{L}} - I_{\mathrm{R}})/\Gamma(\mathsf{b}+z)^2$. Then f is analytic for $\Re z > -\varepsilon$, and $f(n) = 0$ for $n \in \mathbb{N}_0$ by (a). It is bounded on $\Re z \ge 0$. First, by the Gamma bound below with $w = \mathsf{b} + z$,

$$\Big|\frac{I_{\mathrm{L}}}{\Gamma(\mathsf{b}+z)^2}\Big| \le \frac{e^{2/3}}{2\pi}\int_{\mathbb{R}} \frac{|\Gamma(\mathsf{a}_1+\mathrm{i}y)|^2|\Gamma(\mathsf{a}_2+\mathrm{i}y)|^2}{|\Gamma(\mathsf{b}+\mathrm{i}y)|^2}\,dy < \infty.$$

Second, by duplication, $I_{\rm R}/\Gamma(\mathsf{b}+z)^2$ is a constant times $\frac{\Gamma(\mathsf{a}_1+\mathsf{b}+z)\Gamma(\mathsf{a}_2+\mathsf{b}+z)\Gamma(\mathsf{b}+\frac12+z)}{\Gamma(\mathsf{a}_1+\mathsf{a}_2+\frac12+z)\Gamma(2\mathsf{b}+z)\Gamma(\mathsf{b}+z)}\,F_{\rm R}(z)$, where $F_{\rm R}(z)$ is the ${}_4F_3$ of (361) at $\mathsf{a}_3=\mathsf{b}+z$. The Gamma ratio has equal total weights in numerator and denominator and no poles for $\Re z\ge 0$, so it is bounded there [27]. For $\Re z\ge 0$ and $k\ge 0$, $|k-z|\le|k+z|\le|k+2\mathsf{b}+z|$, so $|(-z)_n/(2\mathsf{b}+z)_n|\le 1$ and $|F_{\rm R}(z)|\le\sum_n|\frac{(2\mathsf{b}-1)_n(\mathsf{b}-\mathsf{a}_1)_n(\mathsf{b}-\mathsf{a}_2)_n}{n!\,(\mathsf{b}+\mathsf{a}_1)_n(\mathsf{b}+\mathsf{a}_2)_n}|<\infty$, the terms being $O(n^{2\mathsf{b}-2-2\mathsf{a}_1-2\mathsf{a}_2})$ with exponent below -1. A function that is analytic and bounded on $\Re z\ge 0$ and vanishes at every nonnegative integer is identically zero (Carlson's theorem [36]). So $f\equiv 0$.

(c) *Continuation.* Both sides of (361) are analytic on the convex domain $\{\Re\mathsf{a}_j>0,\ \Re(\mathsf{a}+\frac12-\mathsf{b})>0\}\subset\mathbb{C}^4$. On the left, the imaginary axis separates the poles of $\Gamma(\mathsf{a}_j+s)$ from those of $\Gamma(\mathsf{a}_j-s)$ and the integrand decays like $e^{-2\pi|\Im s|}$, locally uniformly. On the right, $1/[\Gamma(\mathsf{b}+\mathsf{a}_j)(\mathsf{b}+\mathsf{a}_j)_n]=1/\Gamma(\mathsf{b}+\mathsf{a}_j+n)$ is entire and the series converges locally uniformly. By (b) the two sides agree for real $(\mathsf{a}_1,\mathsf{a}_2,\mathsf{b})$ in an open subset of $\mathbb{R}^3$ and $\Re\mathsf{a}_3\ge\mathsf{b}$. Continuation in a_3, and then in $(\mathsf{a}_1,\mathsf{a}_2,\mathsf{b})$ (an analytic function that vanishes on an open subset of $\mathbb{R}^3$ vanishes identically), gives equality on the whole domain. □

Lemma (a uniform Gamma bound). *For $\Re w\ge\frac12$ and real y, $|\Gamma(w+\mathrm{i}y)\Gamma(w-\mathrm{i}y)|\le e^{2/3}|\Gamma(w)|^2$.*

Proof. Binet's first formula [37] gives, for $\Re\zeta>0$, $\log\Gamma(\zeta)=(\zeta-\frac12)\log\zeta-\zeta+\frac12\log 2\pi+\int_0^\infty\varpi(t)e^{-t\zeta}\,\frac{dt}{t}$ with $\varpi(t)=\frac12-\frac1t+\frac{1}{e^t-1}=\frac12[\coth\frac t2-\frac2t]$. The expansion $\coth v=\frac1v+2v\sum_{n\ge1}(v^2+n^2\pi^2)^{-1}$ (DLMF 4.36.3) gives $0<\coth v-\frac1v<\frac v3$ for $v>0$, hence $0<\varpi(t)/t<\frac{1}{12}$, and the remainder integral has modulus at most $1/(12\Re\zeta)\le\frac16$. In $\log|\Gamma(w+\mathrm{i}y)\Gamma(w-\mathrm{i}y)/\Gamma(w)^2|$ the terms $-\zeta$ and $\frac12\log 2\pi$ cancel. The main terms give $\mathfrak{g}(y)+\mathfrak{g}(-y)-2\mathfrak{g}(0)$ with $\mathfrak{g}(t)=\Re[(w+\mathrm{i}t-\frac12)\log(w+\mathrm{i}t)]$, and, with $\zeta=w+\mathrm{i}t$, $\mathfrak{g}''(t)=-\Re[\zeta^{-1}+\frac12\zeta^{-2}]\le(\frac12-\Re w)/|\zeta|^2\le 0$. So $\mathfrak{g}$ is concave and $\mathfrak{g}(y)+\mathfrak{g}(-y)-2\mathfrak{g}(0)\le 0$. The four remainders contribute at most $\frac23$. □

Proposition (Barnes–reciprocal duality). *Let $0<p_j<\frac12$ $(j=1,\dots,4)$, $p=p_1+\cdots+p_4>\frac12$, and*

$$\mathfrak{l}(p_1,\dots,p_4)=\frac{2^{2-2p}\prod_{i<j}\Gamma(p_i+p_j)\prod_i\Gamma(2p_i)}{\pi^{9/2}\,\Gamma(p-\frac12)}.$$

Then, with $E^\sharp$ of case (b) above, for all $\tau\in\mathbb{C}$

$$\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}[\cos 2\pi s+\cos 2\pi\tau]\prod_{j=1}^4\Gamma(p_j+s)\Gamma(p_j-s)=4\pi^7\,\mathfrak{l}(p_1,\dots,p_4)\,E^\sharp(\tau).\tag{362}$$

Proof. The left side is a constant plus a multiple of $\cos 2\pi\tau$, and so is the right side, by case (b). Both sides are symmetric in the p_j. It therefore suffices to prove equality at $\tau_4=\frac12+p_4$ when $p_3\ne p_4$: by symmetry it then holds at $\tau_3=\frac12+p_3$ as well, and $\cos 2\pi\tau_j=-\cos 2\pi p_j$ takes different values at $j=3,4$, because $\cos 2\pi(\cdot)$ is injective on $(0,\frac12)$; a constant plus a multiple of $\cos 2\pi\tau$ is fixed by its values at two different values of $\cos 2\pi\tau$. The case $p_3=p_4$ follows by continuity of both sides in the p_j (dominated convergence on the left, uniform convergence of $E^\sharp$ on the right).

Left side at τ_4. By reflection and $\cos 2\pi s-\cos 2\pi p_4=2\sin\pi(p_4+s)\sin\pi(p_4-s)$, $(\cos 2\pi s-\cos 2\pi p_4)\Gamma(p_4+s)\Gamma(p_4-s)=2\pi^2/[\Gamma(1-p_4+s)\Gamma(1-p_4-s)]$. So the left side is $2\pi^2$ times the left side of (361) with $(\mathsf{a}_1,\mathsf{a}_2,\mathsf{a}_3,\mathsf{b})=(p_1,p_2,p_3,1-p_4)$; the condition of that lemma is $p>\frac12$.

Right side at τ_4. For $k\ge0$ the factor $j=4$ contains $1/\Gamma(-k)=0$. For $k=-1-n$ with $n\ge0$ the summand is $\prod_j[\Gamma(p_j+p_4-n)\Gamma(1+p_j-p_4+n)]^{-1}$, and $1/\Gamma(z-n)=(-1)^n(1-z)_n/\Gamma(z)$ gives

$$E^\sharp(\tau_4)=\prod_{j=1}^4\frac{1}{\Gamma(p_j+p_4)\Gamma(1+p_j-p_4)}\;{}_4F_3\!\left(\begin{matrix}1-2p_4,\ 1-p_1-p_4,\ 1-p_2-p_4,\ 1-p_3-p_4\\ 1+p_1-p_4,\ 1+p_2-p_4,\ 1+p_3-p_4\end{matrix};1\right),$$

the ${}_4F_3$ of (361) at $\mathfrak{b} = 1 - p_4$. The prefactors agree: after cancelling $\prod_{j\le 3}\Gamma(p_j + p_4)$ and $\Gamma(2p_4)$, $4\pi^7\mathfrak{l}/\prod_j[\Gamma(p_j + p_4)\Gamma(1 + p_j - p_4)]$ reduces to $4\pi^{5/2}2^{2-2p}\prod_{i<j\le 3}\Gamma(p_i + p_j)\prod_{i\le 3}\Gamma(2p_i)/[\Gamma(p - \frac{1}{2})\prod_{j\le 3}\Gamma(1 + p_j - p_4)]$, which is $2\pi^2$ times the prefactor of (361) at $\mathfrak{b} = 1 - p_4$. □

Corollary (the symmetric case). *For* $\frac{1}{4} < x < \frac{3}{4}$ *and* $\mathfrak{l}_x = \mathfrak{l}(\frac{x}{2}, \frac{x}{2}, \frac{x}{2}, \frac{x}{2})$, *that is* $\mathfrak{l}_x = 2^{2-4x}\Gamma(x)^{10}/(\pi^{9/2}\Gamma(2x - \frac{1}{2}))$,

$$\mathsf{J}_0(x) = 4\pi^3\,\mathfrak{l}_x\,\mathfrak{b}(1 - x), \qquad \mathsf{J}_1(x) = 4\pi^3\,\mathfrak{l}_x\,\mathfrak{a}(1 - x).$$

Proof. Take all $p_j = \frac{x}{2}$ in (362), so that $p = 2x > \frac{1}{2}$. The left side is $\mathsf{J}_1(x) + \mathsf{J}_0(x)\cos 2\pi\tau$. By case (b) and (358) at $1 - x$, which lies in $(0, \frac{3}{4})$, the right side is $4\pi^7\mathfrak{l}_x\pi^{-4}\Omega(\tau - \frac{1-x}{2}; 1 - x) = 4\pi^3\mathfrak{l}_x[\mathfrak{a}(1 - x) + \mathfrak{b}(1 - x)\cos 2\pi\tau]$. Compare the two modes. □

Proof of (357). By (359) and the corollary, $W_4(2; 2x) = N_x^4\pi^{-4}\,4\pi^3\mathfrak{l}_x[\mathfrak{a}(x)\mathfrak{b}(1-x)+\mathfrak{b}(x)\mathfrak{a}(1-x)]$, and $\frac{4}{\pi}N_x^4\mathfrak{l}_x = \mathfrak{p}(x)$. □

Corollary (rim–spoke reflection). *For* $\frac{1}{2} < b < \frac{3}{2}$,

$$\frac{W_4(2; b)}{W_4(2; 2 - b)} = \frac{\mathfrak{p}(b/2)}{\mathfrak{p}(1 - b/2)} = 2^{4-4b}\,\frac{\gamma_C(b/2)^2}{\gamma_C(b - \frac{1}{2})}.$$

Proof. The bracket $\mathfrak{s}(x) = \mathfrak{a}(x)\mathfrak{b}(1-x)+\mathfrak{b}(x)\mathfrak{a}(1-x)$ is invariant under $x \to 1-x$, and $W_4(2; 2x) > 0$ because its spectral integrand is positive, so $\mathfrak{s} \neq 0$. Hence $W_4(2; 2x)/W_4(2; 2-2x) = \mathfrak{p}(x)/\mathfrak{p}(1-x) = 2^{4-8x}\Gamma(x)^2\Gamma(\frac{3}{2} - 2x)/[\Gamma(1 - x)^2\Gamma(2x - \frac{1}{2})] = 2^{4-8x}\gamma_C(x)^2/\gamma_C(2x - \frac{1}{2})$; put $b = 2x$. □

Corollary (the logarithmic wheel). $\partial_b W_4(2; b)|_{b=1} = -2\log 2\,P_4(2)$, *and* $\mathcal{L}_2$ *of* (256) *equals* $-2\log 2\,P_4(2)$.

Proof. The kernel $\mathcal{G}_x(s)^{-4}$ of (358) is entire in (x, s). Part (i) of the two-mode lemma holds verbatim for complex κ_j with $0 < \Re\kappa_j < 1$; for $\kappa_j = 1 - \frac{x'}{2}$ it gives $|\mathcal{G}_{x'}(s)|^{-4} \le C(1 + |s|)^{4\Re x' - 4}$ for real s, locally uniformly in x'. Cauchy's estimate on a circle $|x' - x| = r$ then gives $|\partial_x\mathcal{G}_x(s)^{-4}| \le C(1 + |s|)^{4(x+r)-4}$, which is integrable when $x + r < \frac{3}{4}$. By dominated convergence $\mathfrak{a}, \mathfrak{b} \in C^1(0, \frac{3}{4})$, so $\mathfrak{s} \in C^1(\frac{1}{4}, \frac{3}{4})$, and $\mathfrak{s}(1 - x) = \mathfrak{s}(x)$ gives $\mathfrak{s}'(\frac{1}{2}) = 0$. Therefore

$$\frac{d}{dx}W_4(2; 2x)\Big|_{x=\frac{1}{2}} = \mathfrak{p}'(\tfrac{1}{2})\,\mathfrak{s}(\tfrac{1}{2}) = (\log\mathfrak{p})'(\tfrac{1}{2})\,W_4(2; 1),$$
$$(\log\mathfrak{p})'(x) = -4\log 2 - 4\psi_0(1 - x) + 6\psi_0(x) - 2\psi_0(2x - \tfrac{1}{2}),$$

and $(\log\mathfrak{p})'(\frac{1}{2}) = -4\log 2$, the three $\psi_0(\frac{1}{2})$ terms cancelling. With $b = 2x$, $\partial_b W_4(2; b)|_{b=1} = -2\log 2\,W_4(2; 1)$, and $W_4(2; 1) = P_4(2)$ by (243). This is the wheel-derivative reading (244) of $\mathcal{L}_2$. For the series reading, (360) and (347) give $\mathcal{L}_2 = \frac{1}{2}\mathcal{W}'(\frac{1}{2}) = \partial_b W_4(2; b)|_{b=1}$. □

Remarks. (1) The reflection $b \to 2 - b$ exchanges the rim and spoke exponents of the two-dimensional wheel. It is not a graph symmetry, since planar duality maps the wheel to itself with the same exponents. It is a hypergeometric identity, (361) and (362), between a Barnes integral of Gamma products and a periodization of reciprocal Gamma products. $\mathcal{L}_2$ is the derivative at the fixed point $b = 1$, where only the Gamma prefactor $\mathfrak{p}$ contributes. (2) This route gives only the combination of single sums that $\mathcal{L}_2$ needs, not the two relations (356) separately. (3) The duality (362) is proved for four different parameters, but the lattice step with four different rim exponents was not written out; the resulting four-exponent ratio is the numerical statement listed under the remaining gaps of Section 17.2. (4) Checks, not proof steps: `checks_all.log` records the residue

functional (350) in exact algebra; (357) at $x = \frac{1}{2}, \frac{2}{5}, \frac{1}{3}, \frac{3}{10}, 0.27, 0.6, 0.7$ to all 35 working digits; $(\log \mathfrak{p})'(\frac{1}{2}) = -4\log 2$ and the reflection ratio exactly; (362) at complex τ and (361) at complex parameters to about 20 digits; and the identities $F_{\mathrm{R}} = K_n F_{\mathrm{L}}$ and the prefactor ratio of step (a) exactly for $n = 0, \dots, 6$. The script `extra_checks.wl` of the same folder tests the Gamma bound on random samples.

18.7.7 The elliptic moments by half-strip contours

The two further proofs below use the elliptic moments U_0, V_0, U, V, R defined before (96). This subsubsection evaluates all of them by one method, which does not use the Stieltjes identity (100). The notation is the one introduced with these moments: the modulus convention for $\mathbf{K}$, $f(k) = \mathbf{K}(k)$, $g(k) = \mathbf{K}(k')$, the weight $w(k) = \log \frac{k}{1-k^2}$, and the moments M, N, which are $M_{\mathrm{e}}, N_{\mathrm{e}}$ of (338); $T(k) = \frac{1-k}{1+k}$ is the Landen map of (97).

Proposition (elliptic moments).

$$\begin{aligned} U_0 = 2N_{\mathrm{e}}, \qquad V_0 = 2M_{\mathrm{e}}, \qquad R = \frac{\pi}{4} N_{\mathrm{e}} - \log 2\, M_{\mathrm{e}}, \\ U + \log 2\, U_0 = -\pi M_{\mathrm{e}}, \qquad V + \log 2\, V_0 = -\frac{\pi}{2} N_{\mathrm{e}}. \end{aligned} \tag{363}$$

Equivalently (103), (106), (105), $U_0 = 2N_{\mathrm{e}}$ *and* $V_0 = 2M_{\mathrm{e}}$ *hold, and they are proved here without* (100).

Proof. Angle form. With $k = \sin\beta$ one has $dk/k' = d\beta$ and $kk' = \frac{1}{2}\sin 2\beta$. So, with $F(\beta) = \mathbf{K}(\sin\beta)$, $G(\beta) = \mathbf{K}(\cos\beta)$ and $L(\beta) = \log\sin 2\beta$,

$$\begin{aligned} U_0 = \int_0^{\pi/2} F^2\, d\beta, \qquad V_0 = \int_0^{\pi/2} FG\, d\beta, \\ U + \log 2\, U_0 = \int_0^{\pi/2} F^2 L\, d\beta, \quad V + \log 2\, V_0 = \int_0^{\pi/2} FGL\, d\beta. \end{aligned}$$

(1) *Holomorphy.* On the strip $\mathsf{T} = \{0 < \Re\beta < \frac{\pi}{2}\}$ the functions F, G and L (principal branch) are holomorphic. For $\beta = \alpha + \mathrm{i}\eta$ with $0 < \alpha < \frac{\pi}{2}$, $\sin\beta = \sin\alpha\cosh\eta + \mathrm{i}\cos\alpha\sinh\eta$ has positive real part and is real only for $\eta = 0$, where it lies in $(0,1)$; hence $\sin^2\beta$ is real only there and avoids the cut $[1,\infty)$ of $\mathbf{K}$ as a function of k^2. The same holds for $\cos\beta = \cos\alpha\cosh\eta - \mathrm{i}\sin\alpha\sinh\eta$. Next, $\sin 2\beta = \sin 2\alpha\cosh 2\eta + \mathrm{i}\cos 2\alpha\sinh 2\eta$ is real only if $\eta = 0$ or $\alpha = \frac{\pi}{4}$, and then it is positive.

(2) *Boundary values.* For $y > 0$ put $\kappa = \tanh y$ and $\kappa' = \operatorname{sech} y$. The imaginary-modulus transformation $\mathbf{K}(\mathrm{i}k/k') = k'\mathbf{K}(k)$ and the reciprocal-modulus transformation $\mathbf{K}(X + \mathrm{i}0) = X^{-1}[\mathbf{K}(X^{-1}) + \mathrm{i}\mathbf{K}(\sqrt{1-X^{-2}})]$ for $X > 1$ [34] give, as limits from inside T,

$$\begin{aligned} F(\mathrm{i}y) &= \kappa' f(\kappa), & F(\tfrac{\pi}{2} + \mathrm{i}y) &= \kappa'[g(\kappa) + \mathrm{i}f(\kappa)], & L(\mathrm{i}y) &= w(\kappa) + \log 2 + \tfrac{\mathrm{i}\pi}{2}, \\ G(\mathrm{i}y) &= \kappa'[g(\kappa) - \mathrm{i}f(\kappa)], & G(\tfrac{\pi}{2} + \mathrm{i}y) &= \kappa' f(\kappa), & L(\tfrac{\pi}{2} + \mathrm{i}y) &= w(\kappa) + \log 2 - \tfrac{\mathrm{i}\pi}{2}. \end{aligned} \tag{364}$$

Indeed $\sin(\epsilon + \mathrm{i}y) \to \mathrm{i}\sinh y$ with $\mathrm{i}\sinh y = \mathrm{i}\kappa/\kappa'$; $\sin(\frac{\pi}{2} - \epsilon + \mathrm{i}y) \to \cosh y + \mathrm{i}0$; $\cos(\epsilon + \mathrm{i}y) \to \cosh y - \mathrm{i}0$; $\cos(\frac{\pi}{2} - \epsilon + \mathrm{i}y) \to -\mathrm{i}\sinh y$, and $\mathbf{K}$ is even; $\sin 2(\epsilon + \mathrm{i}y) \to \mathrm{i}\sinh 2y$ and $\sin 2(\frac{\pi}{2} - \epsilon + \mathrm{i}y) \to -\mathrm{i}\sinh 2y$, both from the right half-plane; finally $\log\sinh 2y = \log(2\kappa/\kappa'^2) = w(\kappa) + \log 2$.

(3) *Bounds.* Let $\Theta \in \{F^2, FG, F^2L, FGL\}$. With the boundary values included, $|\Theta(\beta)| \le C(1 + |\log|\beta||^3 + |\log|\beta - \frac{\pi}{2}||^3)$ for $0 \le \Im\beta \le 1$, because $\mathbf{K}(k) = O(1 + |\log(1-k^2)|)$ uniformly as $k^2 \to 1$ in $\mathbb{C} \setminus [1,\infty)$ (from the expansion in powers of k'^2, DLMF 19.12.1) and $L(\beta) = O(1 + |\log|\beta|| + |\log|\beta - \frac{\pi}{2}||)$. For $\Im\beta \ge 1$, $|\Theta(\beta)| \le C(1 + \Im\beta)^3 e^{-2\Im\beta}$, because $|\sin\beta|, |\cos\beta| \ge \sinh\Im\beta$, $\mathbf{K}(k) = O(|k|^{-1}\log|k|)$ as $|k| \to \infty$ [34], and $L = O(1 + \Im\beta)$. These two uniform estimates, which include the one-sided boundary values of (2), are standard and are stated here without further detail.

(4) *Half-strip identity.* Cauchy's theorem on the rectangle $[\epsilon, \frac{\pi}{2}-\epsilon]\times[0,Y]$, followed by $\epsilon\to 0$ (dominated convergence with the bounds of (3) and locally uniform convergence to the boundary values) and $Y\to\infty$ (the top edge is $O(Y^3e^{-2Y})$), gives

$$\int_0^{\pi/2}\Theta(\beta)\,d\beta=-\mathrm{i}\int_0^{\infty}\bigl[\Theta(\tfrac{\pi}{2}+\mathrm{i}y)-\Theta(\mathrm{i}y)\bigr]\,dy. \tag{365}$$

(5) *Evaluation.* Write f,g for $f(\kappa),g(\kappa)$ and $\omega=w(\kappa)+\log 2$; all integrals below are over $\kappa\in(0,1)$ and converge, since the integrands have at most $\log^3$ endpoint singularities. With $dy=d\kappa/\kappa'^2$ and (364), (365) gives

$$U_0=2\int fg-\mathrm{i}\int(g^2-2f^2),\qquad V_0=2\int f^2,$$
$$U+\log 2\,U_0=\int\bigl[2fg\,\omega-\tfrac{\pi}{2}g^2\bigr]-\mathrm{i}\int\bigl[(g^2-2f^2)\omega+\pi fg\bigr],\qquad V+\log 2\,V_0=\int\bigl[2f^2\omega-\pi fg\bigr].$$

The left sides are real, so the imaginary parts vanish: $U_0=2N_{\mathrm{e}}$, $\int g^2=2M_{\mathrm{e}}$, $V_0=2M_{\mathrm{e}}$, $\int(g^2-2f^2)\omega=-\pi N_{\mathrm{e}}$, and $U+\log 2\,U_0=2\int fg\,\omega-\pi M_{\mathrm{e}}$, $V+\log 2\,V_0=2\int f^2\omega-\pi N_{\mathrm{e}}$.

(6) *Landen.* By (97), $f(T)g(T)|T'|=fg$ and $g(T)^2|T'|=2f^2$, and $w(T(\kappa))=-w(\kappa)-2\log 2$ gives $\omega(T(\kappa))=-\omega(\kappa)$. Substituting $\kappa\to T(\kappa)$ gives $\int fg\,\omega=-\int fg\,\omega=0$ and $\int g^2\omega=-2\int f^2\omega$. With (5), $-4\int f^2\omega=-\pi N_{\mathrm{e}}$, that is $\int f^2\omega=\frac{\pi}{4}N_{\mathrm{e}}$. Therefore $U+\log 2\,U_0=-\pi M_{\mathrm{e}}$, $V+\log 2\,V_0=\frac{\pi}{2}N_{\mathrm{e}}-\pi N_{\mathrm{e}}=-\frac{\pi}{2}N_{\mathrm{e}}$ and $R=\int f^2w=\int f^2\omega-\log 2\,M_{\mathrm{e}}=\frac{\pi}{4}N_{\mathrm{e}}-\log 2\,M_{\mathrm{e}}$. With $U_0=2N_{\mathrm{e}}$ and $V_0=2M_{\mathrm{e}}$ these are $U=-\pi M_{\mathrm{e}}-2\log 2\,N_{\mathrm{e}}$, $V=-\frac{\pi}{2}N_{\mathrm{e}}-2\log 2\,M_{\mathrm{e}}$ and $R=\frac{\pi}{4}N_{\mathrm{e}}-\log 2\,M_{\mathrm{e}}$, that is (103), (106) and (105). □

The half-strip contours thus reprove the three logarithmic evaluations of the one-dimensional wheel by a single method. Conversely, (363) follows from (103), (106), (105) and $U_0=2N_{\mathrm{e}}$, $V_0=2M_{\mathrm{e}}$ as proved in Section 9.2; that second proof uses (100). As a check, not a proof step, a 60-digit run (symmetry, Slurm job 567960; `LRI_C5_d2_simplify/proof/L2-fourier/check_strip_dps60.log`) confirms the boundary values (364) to 4×10^{-62}, the four strip identities and the Landen steps to 5×10^{-63}, and the identities for $U+\log 2\,U_0$ and $V+\log 2\,V_0$ to 2×10^{-63} and 2×10^{-65}.

18.7.8 A third proof: Parseval pairings on the unit circle

This proof keeps the rim derivative (347), the residue functional (350), the jets (351) and the reduction (352) of Section 18.7. It evaluates the four single sums $a=\pi^2\sum_k c_k^4$, $b=-2\pi\sum_k c_k^4\alpha_k$, $\mathcal{S}_1$ and $\mathcal{S}_2$ differently: each is a Parseval pairing of two power series on the unit circle, and the circle values turn the pairings into the elliptic moments U_0, V_0, U, V defined before (96). With (363) no step uses (100). Here c_n, α_n, β_n are as in Section 18.7,

$$y_1(z)={}_2F_1(\tfrac12,\tfrac12;1;z)=\sum_{n\ge0}c_n^2z^n,\qquad y_3(z)={}_2F_1(\tfrac12,\tfrac12;1;1-z),$$

and, as in the upper-semicircle contour that proves (106), $f_\vartheta=\mathbf{K}(\sin\vartheta)$ and $g_\vartheta=\mathbf{K}(\cos\vartheta)=f_{\pi/2-\vartheta}$ for $0<\vartheta<\frac{\pi}{2}$.

Generating functions on the slit disk. For $|z|<1$, $z\notin(-1,0]$, with principal logarithms,

$$\sum_{n\ge0}c_n^2\alpha_nz^n=-\frac{\pi}{2}\,y_3(z)-\frac12\log z\;y_1(z),$$
$$\sum_{n\ge0}c_n^2\beta_nz^n=-\frac12\log(1-z)\,y_1(z), \tag{366}$$
$$\sum_{n\ge0}c_n^2\bigl[4\alpha_n\beta_n+2\psi_1(n+\tfrac12)\bigr]z^n=\pi\log(1-z)\,y_3(z)+\pi^2y_1(z)+\log z\,\log(1-z)\,y_1(z).$$

At $z = k^2$ with $0 < k < 1$, where $y_1 = \frac{2}{\pi} f(k)$ and $y_3 = \frac{2}{\pi} g(k)$, these are the three lines of (353). The left sides are power series with polynomially bounded coefficients, hence analytic for $|z| < 1$. The right sides are analytic on the slit disk, since y_1 is analytic off $[1, \infty)$, y_3 off $(-\infty, 0]$ and $\log z$ off $(-\infty, 0]$. The slit disk is connected and the two sides agree on $(0, 1)$, so they agree on the whole slit disk.

Lemma (values on the circle). *For $0 < \vartheta < \frac{\pi}{4}$,*

$$y_1(e^{4\mathrm{i}\vartheta}) = \frac{1}{\pi}\, e^{-\mathrm{i}\vartheta}[g_\vartheta + \mathrm{i} f_\vartheta], \qquad y_3(e^{-4\mathrm{i}\vartheta}) = \frac{2}{\pi}\, e^{\mathrm{i}\vartheta} f_\vartheta. \tag{367}$$

Proof. The function y_1 is analytic on $\mathbb{C} \setminus [1, \infty)$, and $y_1(e^{4\mathrm{i}\vartheta}) = \frac{2}{\pi}\mathbf{K}(e^{2\mathrm{i}\vartheta})$. The boundary-value Landen relation $e^{\mathrm{i}\vartheta}\mathbf{K}(e^{2\mathrm{i}\vartheta}) = \frac{1}{2}[\mathbf{K}(\cos\vartheta) + \mathrm{i}\mathbf{K}(\sin\vartheta)]$, proved with the upper-semicircle contour behind (106), gives the first identity. (The coefficients 1 and i of the reciprocal-modulus value used there can be confirmed at the endpoint. For $s > 1$, $y_1(s + \mathrm{i}0)$ solves the hypergeometric equation on $(1, \infty)$, whose solutions are spanned by $s^{-1/2} y_1(1/s)$ and $s^{-1/2} y_1(1 - 1/s)$. As $s \downarrow 1$ the first behaves like $\pi^{-1} \log \frac{16}{s-1}$ and the second tends to 1, while $y_1(z) = \pi^{-1}[4 \log 2 - \log(1 - z)] + o(1)$ as $z \to 1$ [23] gives $y_1(s + \mathrm{i}0) = \pi^{-1}[\log \frac{16}{s-1} + \mathrm{i}\pi] + o(1)$. So the coefficients are 1 and i.) For the second identity, both sides of ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 1 - w^2) = \frac{2}{1+w}\, {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; (\frac{1-w}{1+w})^2)$ are analytic for $\Re w > 0$, where $w^2 \notin (-\infty, 0]$ and $|\frac{1-w}{1+w}| < 1$, and they agree on $(0, 1)$, where this is the real Landen identity $f(T(k)) = \frac{1+k}{2} g(k)$ behind (97). Take $w = e^{-2\mathrm{i}\vartheta}$. Then $1 - w^2 = 1 - e^{-4\mathrm{i}\vartheta}$, $(1 - w)/(1 + w) = \mathrm{i} \tan\vartheta$ and $2/(1 + w) = e^{\mathrm{i}\vartheta}/\cos\vartheta$, and Pfaff's transformation [23] gives ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; -\tan^2\vartheta) = \cos\vartheta\, {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; \sin^2\vartheta) = \frac{2}{\pi}\cos\vartheta\, f_\vartheta$. □

Proposition (pairings).

$$a = \frac{4U_0}{\pi}, \qquad b = \frac{4V_0}{\pi}, \qquad \mathcal{S}_1 = -\frac{2}{\pi^3}(U + 2\log 2\, U_0), \qquad \mathcal{S}_2 = \frac{a}{2} + \frac{4}{\pi^2}(V + 2\log 2\, V_0). \tag{368}$$

Proof. If $\mathsf{p}(z) = \sum_n p_n z^n$ and $\mathsf{q}(z) = \sum_n q_n z^n$ have real square-summable coefficients, they lie in the Hardy space H^2, have radial limits almost everywhere, and $\sum_n p_n q_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} \mathsf{p}(e^{\mathrm{i}\theta})\, \mathsf{q}(e^{-\mathrm{i}\theta})\, d\theta$. All coefficient sequences used below are square summable: $c_n^2 = O(1/n)$, $c_n^2 \beta_n = O(\log n/n)$, $c_n^2 \alpha_n = O(n^{-2})$ and $c_n^2[4\alpha_n\beta_n + 2\psi_1(n + \frac{1}{2})] = O(\log n/n^2)$. For $\theta \in (-\pi, \pi)\setminus\{0\}$ the radius to $e^{\mathrm{i}\theta}$ lies in the slit disk, and the right sides of (366) are continuous up to $e^{\mathrm{i}\theta}$, so the radial limits are the values of the closed forms. On the circle $z = e^{\mathrm{i}\theta}$ one has $\log \bar{z} = -\mathrm{i}\theta$ and $\log(1-z) + \log(1-\bar{z}) = 2\log|2\sin\frac{\theta}{2}|$; $\Im \log(1 - \bar{z})$ is odd in θ; $|y_1(z)|^2$ is even; and $y_1(e^{-\mathrm{i}\theta}) y_3(e^{\mathrm{i}\theta}) = \overline{y_1(e^{\mathrm{i}\theta}) y_3(e^{-\mathrm{i}\theta})}$, since y_1, y_3 have real Taylor coefficients. All integrands below have only integrable logarithmic singularities. For $0 < \theta < \pi$ and $\theta = 4\vartheta$, (367) gives $|y_1(e^{\mathrm{i}\theta})|^2 = \pi^{-2}(f_\vartheta^2 + g_\vartheta^2)$ and $y_1(e^{\mathrm{i}\theta}) y_3(e^{-\mathrm{i}\theta}) = 2\pi^{-2}(f_\vartheta g_\vartheta + \mathrm{i} f_\vartheta^2)$. The fold $\vartheta \mapsto \frac{\pi}{2} - \vartheta$ exchanges f and g and fixes $\sin 2\vartheta$; so for any weight $\chi(\sin 2\vartheta)$ it turns $\int_0^{\pi/4}(f^2 + g^2)\chi\, d\vartheta$ into $\int_0^{\pi/2} f^2 \chi\, d\vartheta$ and $\int_0^{\pi/4} f g\, \chi\, d\vartheta$ into $\frac{1}{2}\int_0^{\pi/2} f g\, \chi\, d\vartheta$. Finally, by the angle form in the proof of (363) (with $\beta = \vartheta$), $\int_0^{\pi/2} f_\vartheta^2 \log(2\sin 2\vartheta)\, d\vartheta = U + 2\log 2\, U_0$ and $\int_0^{\pi/2} f_\vartheta g_\vartheta \log(2\sin 2\vartheta)\, d\vartheta = V + 2\log 2\, V_0$.

(i) $a/\pi^2 = \sum c_n^2 \cdot c_n^2 = \frac{1}{2\pi}\int |y_1|^2\, d\theta = \frac{4}{\pi^3}\int_0^{\pi/4}(f_\vartheta^2 + g_\vartheta^2)\, d\vartheta = \frac{4U_0}{\pi^3}$.

(ii) $-b/(2\pi) = \sum c_n^2 \cdot c_n^2 \alpha_n$ is the pairing of y_1 with the first line of (366). The integrand is $-\frac{\pi}{2} y_1(z) y_3(\bar{z}) + \frac{\mathrm{i}\theta}{2}|y_1(z)|^2$, and the second term is odd. So $b = \pi\Re\int_0^{\pi} y_1(e^{\mathrm{i}\theta}) y_3(e^{-\mathrm{i}\theta})\, d\theta = \frac{8}{\pi}\int_0^{\pi/4} f_\vartheta g_\vartheta\, d\vartheta = \frac{4V_0}{\pi}$.

(iii) $\mathcal{S}_1 = \sum c_n^2 \cdot c_n^2 \beta_n = -\frac{1}{4\pi}\int \log(1 - \bar{z})\, |y_1(z)|^2\, d\theta$, by the second line of (366). Dropping the odd part and setting $\theta = 4\vartheta$,

$$\mathcal{S}_1 = -\frac{2}{\pi^3}\int_0^{\pi/4} \log(2\sin 2\vartheta)(f_\vartheta^2 + g_\vartheta^2)\, d\vartheta = -\frac{2}{\pi^3}\int_0^{\pi/2} \log(2\sin 2\vartheta) f_\vartheta^2\, d\vartheta = -\frac{2}{\pi^3}(U + 2\log 2\, U_0).$$

(iv) Since $c_n^4[4\alpha_n\beta_n+\psi_1(n+\frac12)] = (2c_n^2\beta_n)(c_n^2\alpha_n) + c_n^2\cdot\frac12 c_n^2[4\alpha_n\beta_n+2\psi_1(n+\frac12)]$, $\mathcal{S}_2$ is the sum of two pairings. By (366) the combined integrand is

$$\pi\log|2\sin\tfrac{\theta}{2}|\;y_1(z)y_3(\bar z)+\frac{\pi^2}{2}|y_1(z)|^2-\mathrm{i}\theta\log|2\sin\tfrac{\theta}{2}|\;|y_1(z)|^2.$$

The last term is odd. The middle one contributes $\frac{\pi^2}{2}\cdot\frac{a}{\pi^2}=\frac{a}{2}$. The first one contributes $\frac{2}{\pi^2}\int_0^\pi \log(2\sin\frac{\theta}{2}) f_{\theta/4}g_{\theta/4}\,d\theta = \frac{4}{\pi^2}\int_0^{\pi/2}\log(2\sin 2\vartheta) f_\vartheta g_\vartheta\,d\vartheta = \frac{4}{\pi^2}(V+2\log 2\,V_0)$. □

Conclusion. Inserting (363) into (368) gives $a=8N_{\mathrm e}/\pi$ and $b=8M_{\mathrm e}/\pi$, which reproves (340), and

$$\mathcal{S}_1=\frac{2M_{\mathrm e}}{\pi^2}-\frac{4\log 2\,N_{\mathrm e}}{\pi^3}=\frac{\pi b-2a\log 2}{4\pi^2},\qquad \mathcal{S}_2=\frac{2N_{\mathrm e}}{\pi}+\frac{8\log 2\,M_{\mathrm e}}{\pi^2}=\frac{a}{4}+\frac{b\log 2}{\pi},$$

which are (356). With (352) this gives $\mathcal{L}_2=-2\log 2\,P_4(2)$ exactly as at the end of Section 18.7, and with (333) it proves (270). The inputs are Euler's and Pfaff's transformations, the connection formulas of [23] behind (353), the Landen transformations, Parseval's identity in H^2, the boundary-value Landen relation and the half-strip contours of Section 18.7.7; the Stieltjes identity is not used.

Remarks. (1) The pairings (368) use only the circle values. Given $U_0=2N_{\mathrm e}$ and $V_0=2M_{\mathrm e}$, the first relation of (356) is therefore equivalent to $U+\log 2\,U_0=-\frac{\pi}{2}V_0$, that is to (103), and the second to $V+\log 2\,V_0=-\frac{\pi}{4}U_0$, that is to (106). The two single-sum relations are the unit-circle images of the logarithmic evaluations of the one-dimensional wheel. (2) Two relations among second-order logarithmic moments were found numerically while this route was developed; they are not used. With $\langle\chi\rangle=\int_0^1 f(k)g(k)\chi(k)\,dk$, $2\langle\log^2 k\rangle-\langle\log k\log(1-k)\rangle-3\langle\log k\log(1+k)\rangle=\frac{\pi^2}{2}N_{\mathrm e}+\pi\log 2\,M_{\mathrm e}$, which is equivalent to the second relation of (356) through a second-order moment representation that has only been sketched, and $32\langle\log^2(1+k)\rangle=22\langle\log^2 k\rangle-7\pi^2N_{\mathrm e}+8\log^2 2\,N_{\mathrm e}$. Both come from an integer-relation search at 40 digits (`LRI_C5_d2_simplify/proof/L2-elliptic/probe_moments.py`) and are numerical statements only. (3) Checks, not proof steps: the scripts `check_parseval.py`, `collapse_sympy.py`, `check_collapse.py` and `check_rimderivative.py` of the same folder test the generating functions, the circle values and the pairings at 30 digits, the rim derivative (347) with the 70-digit single sums, and the collapse algebra exactly (no logs of these runs are shipped); the synthesis check `LRI_C5_d2_simplify/proof/synth/synth_checks.log` gives, with the 70-digit single sums, residuals 9×10^{-72} and 1.1×10^{-69} for the two relations (356), 7×10^{-68} for (352), and 1.7×10^{-67} for $\mathcal{L}_2+16\log 2\,ab/\pi$.

18.7.9 A fourth proof: the Fourier representation of the two modes

This proof keeps the rim derivative (347) and the residue functional (350), but obtains the modes $\mathcal{A}$, $\mathcal{B}$ and their x-derivatives at $x=\frac12$ without single sums and without growth estimates. Cauchy's beta integral writes the Fourier coefficients of $\Omega(t;x)$ as convolutions of $(2\cos\frac{\theta}{2})^{-x}$ on $(-\pi,\pi)$, which have compact support, and at $x=\frac12$ these convolutions are complete elliptic integrals. Throughout, $I_{\mathrm F}=(\frac38,\frac58)$, which contains I, and $D_{\mathrm F}=\{x\in\mathbb{C}:|x-\frac12|<\frac18\}$. For real $x\in I_{\mathrm F}$ put

$$h_x(\sigma)=\frac{1}{\Gamma(1-\frac{x}{2}+\sigma)\,\Gamma(1-\frac{x}{2}-\sigma)},\qquad \mathsf{P}_x(\theta)=(2\cos\tfrac{\theta}{2})^{-x}\,\mathbf{1}_{(-\pi,\pi)}(\theta),$$
$$\mathsf{Q}_x=\mathsf{P}_x*\mathsf{P}_x,\qquad v_m(x)=(\mathsf{Q}_x*\mathsf{Q}_x)(2\pi m).$$

Thus $h_x=\mathcal{G}_x^{-1}$ in the notation of (358). By reflection, $h_x(\sigma)=\sin\pi(1-\frac{x}{2}-\sigma)\,\Gamma(\frac{x}{2}+\sigma)/[\pi\Gamma(1-\frac{x}{2}+\sigma)]$, so h_x is real, even and continuous on $\mathbb{R}$ with $|h_x(\sigma)|\le C(1+|\sigma|)^{x-1}$; hence $h_x^2\in L^2(\mathbb{R})$ and $h_x^4\in L^1(\mathbb{R})$. The terms of $\Omega(t;x)$ satisfy $\pi^4[\Gamma(1+s)\Gamma(1-x-s)]^{-4}=\pi^4h_x(\sigma)^4$ at $s=\sigma-\frac{x}{2}$.

Lemma (Cauchy's beta integral). *For real $c > -1$ and real σ,*

$$\int_{-\pi}^{\pi} (2\cos\tfrac{\theta}{2})^c\, e^{\mathrm{i}\sigma\theta}\, d\theta = \frac{2\pi\,\Gamma(c+1)}{\Gamma(1+\frac{c}{2}+\sigma)\,\Gamma(1+\frac{c}{2}-\sigma)}.$$

Proof. Put $s = \sigma - \frac{c}{2}$ and $w = e^{\mathrm{i}\theta}$. For $|\theta| < \pi$, $1 + w = 2\cos\frac{\theta}{2}\, e^{\mathrm{i}\theta/2}$, so with principal powers $(2\cos\frac{\theta}{2})^c e^{\mathrm{i}\sigma\theta} = (1+w)^c w^s$, and the integral equals $-\mathrm{i}\oint_{|w|=1}(1+w)^c w^{s-1}\,dw$. Let $\Re s > 0$. The integrand is holomorphic in the disk slit along $(-1, 0]$ and integrable at $w = 0$. For $0 < \rho < 1$, collapsing the circle $|w| = \rho$ onto the two sides of the slit $[-\rho, 0]$, where $w^{s-1} = r^{s-1}e^{\pm\mathrm{i}\pi(s-1)}$ at $w = -r$, gives $\oint_{|w|=\rho}(1+w)^c w^{s-1}\,dw = 2\mathrm{i}\sin(\pi s)\int_0^\rho (1-r)^c r^{s-1}\,dr$. As $\rho \to 1$ the right side tends to $2\mathrm{i}\sin(\pi s)\,\mathrm{B}(s, c+1)$; the left side tends to the integral over the unit circle by dominated convergence, with the majorant $|1+\rho e^{\mathrm{i}\theta}|^c \le (4\rho)^{c/2}|\cos\frac{\theta}{2}|^c$ for $-1 < c < 0$ (since $|1+\rho e^{\mathrm{i}\theta}|^2 - 4\rho\cos^2\frac{\theta}{2} = (1-\rho)^2$) and $|1+\rho e^{\mathrm{i}\theta}|^c \le 2^c$ for $c \ge 0$. Hence the integral is $2\sin(\pi s)\Gamma(s)\Gamma(c+1)/\Gamma(s+c+1) = 2\pi\Gamma(c+1)/[\Gamma(1-s)\Gamma(1+c+s)]$, which is the claim. Both sides are entire in σ, so the restriction $\Re s > 0$ is removed by continuation. □

With $c = -x$ the lemma reads $h_x(\sigma) = (2\pi\Gamma(1-x))^{-1}\int_{\mathbb{R}} \mathsf{P}_x(\theta) e^{\mathrm{i}\sigma\theta}\,d\theta$.

Lemma (the twofold convolution). *For $0 < |\phi| < 2\pi$ put $k = \cos\frac{\phi}{4}$ and $k' = |\sin\frac{\phi}{4}|$. For real $x \in I_{\mathrm{F}}$,*

$$\mathsf{Q}_x(\phi) = 2\int_0^{\pi/2} \frac{(2k\cos\eta)^{1-2x}}{\sqrt{1-k^2\sin^2\eta}}\,d\eta = (2k)^{1-2x}\,\mathrm{B}(\tfrac12, 1-x)\,{}_2F_1(\tfrac12, \tfrac12; \tfrac32 - x; k^2). \tag{369}$$

Both expressions define holomorphic functions of $x \in D_{\mathrm{F}}$ that agree there, and, uniformly for $x \in D_{\mathrm{F}}$,

$$|\mathsf{Q}_x(\phi)| \le C\,|\sin\tfrac{\phi}{2}|^{-1/4}, \qquad |\partial_x \mathsf{Q}_x(\phi)| \le C\,|\sin\tfrac{\phi}{2}|^{-1/3}. \tag{370}$$

At $x = \frac12$,

$$\mathsf{Q}_{1/2}(\phi) = 2\mathbf{K}(k), \qquad \partial_x \mathsf{Q}_x(\phi)\big|_{x=1/2} = \pi\mathbf{K}(k') - 2\mathbf{K}(k)\log(4kk'). \tag{371}$$

Proof. By evenness take $0 < \phi < 2\pi$. The integrand of $\mathsf{P}_x * \mathsf{P}_x$ is supported on $\theta \in (\phi - \pi, \pi)$, and $4\cos\frac{\theta}{2}\cos\frac{\phi-\theta}{2} = 2[\cos\psi - \cos\psi_0]$ with $\psi = \theta - \frac{\phi}{2}$ and $\psi_0 = \pi - \frac{\phi}{2}$. So $\mathsf{Q}_x(\phi) = \int_{-\psi_0}^{\psi_0}[2(\cos\psi - \cos\psi_0)]^{-x}\,d\psi$. The substitution $\sin\frac{\psi}{2} = k\sin\eta$ with $k = \sin\frac{\psi_0}{2} = \cos\frac{\phi}{4}$ gives $2(\cos\psi - \cos\psi_0) = 4k^2\cos^2\eta$ and $d\psi = 2k\cos\eta\,d\eta/\sqrt{1-k^2\sin^2\eta}$, which is the η-integral. Expanding $(1-k^2\sin^2\eta)^{-1/2}$ and integrating termwise with $\int_0^{\pi/2}\sin^{2n}\eta\cos^{1-2x}\eta\,d\eta = \Gamma(n+\frac12)\Gamma(1-x)/[2\Gamma(n+\frac32 - x)]$ gives the ${}_2F_1$. For fixed ϕ both expressions are holomorphic in $x \in D_{\mathrm{F}}$ and agree for real x, hence on D_{F}.

Bounds: for $x \in D_{\mathrm{F}}$, $\xi = \Re(1-2x) \in (-\frac14, \frac14)$, and since $k\cos\eta \le 1$, $|(2k\cos\eta)^{1-2x}| = (2k\cos\eta)^\xi \le 2(k\cos\eta)^{-1/4}$ and $|\partial_x(2k\cos\eta)^{1-2x}| \le 4|\log(2k\cos\eta)|(k\cos\eta)^{-1/4} \le C(k\cos\eta)^{-1/3}$. For $0 < s' < 1$, $\int_0^{\pi/2}(\cos\eta)^{-s'}(1-k^2\sin^2\eta)^{-1/2}\,d\eta \le C_{s'}k'^{-s'}$: on $[0, \frac{\pi}{4}]$ the integrand is bounded, and on $[\frac{\pi}{4}, \frac{\pi}{2}]$ one uses $(1-k^2\sin^2\eta)^{1/2} \ge \max(\cos\eta, k'/\sqrt{2})$ and the variable $\cos\eta$. Since $kk' = \frac12|\sin\frac{\phi}{2}|$, (370) follows, and differentiation under the η-integral is justified by the same majorants.

At $x = \frac12$ the η-integral is $2\mathbf{K}(k)$, and $\partial_x\mathsf{Q}_x|_{x=1/2} = -4\log(2k)\mathbf{K}(k) - 4\int_0^{\pi/2}\log\cos\eta\,(1 - k^2\sin^2\eta)^{-1/2}\,d\eta$. The next lemma gives $-4\log(2k)\mathbf{K}(k) - 2\mathbf{K}(k)\log\frac{k'}{k} + \pi\mathbf{K}(k')$, which is (371). □

Lemma (a logarithmic moment). *For $0 < k < 1$,*

$$\int_0^{\pi/2} \frac{\log\cos\eta}{\sqrt{1-k^2\sin^2\eta}}\,d\eta = \frac12\mathbf{K}(k)\log\frac{k'}{k} - \frac{\pi}{4}\mathbf{K}(k').$$

Proof. With $\eta = \mathrm{am}(u, k)$ the left side is $\int_0^{\mathbf{K}}\log\mathrm{cn}\,u\;du$, $\mathbf{K} = \mathbf{K}(k)$. The theta-quotient form of the Jacobian functions [35] gives $\mathrm{cn}\,u = \sqrt{k'/k}\,\theta_2(\zeta, q)/\theta_4(\zeta, q)$ with $\zeta = \pi u/(2\mathbf{K})$ and $q = e^{-\pi\mathbf{K}(k')/\mathbf{K}(k)}$,

and the product formulas [35] give

$$\log\frac{\theta_2(\zeta,q)}{\theta_4(\zeta,q)} = \log(2q^{1/4}) + \log\cos\zeta + \sum_{n\ge1}\bigl[\log(1+2q^{2n}\cos 2\zeta + q^{4n}) - \log(1-2q^{2n-1}\cos 2\zeta + q^{4n-2})\bigr].$$

For $0 \le r < 1$, $\log(1 \pm 2r\cos 2\zeta + r^2) = 2\Re\log(1 \pm re^{2\mathrm{i}\zeta})$ has mean zero over $[0,\pi]$, hence over $[0,\frac{\pi}{2}]$ by the symmetry $\zeta \mapsto \pi - \zeta$, and the series converges uniformly. The mean of $\log\cos\zeta$ over $[0,\frac{\pi}{2}]$ is $-\log 2$. So the mean of $\log\mathrm{cn}$ over $[0,\mathbf{K}]$ is $\frac12\log\frac{k'}{k} + \frac14\log q$, and $\int_0^{\mathbf{K}}\log\mathrm{cn}\,u\;du = \frac{\mathbf{K}}{2}\log\frac{k'}{k} - \frac{\pi}{4}\mathbf{K}(k')$. □

Proposition (two modes from compact Fourier support). *For real $x \in I_{\mathrm{F}}$ and all $t \in \mathbb{C}$, $\Omega(t;x) = \mathfrak{a}(x) + \mathfrak{b}(x)\cos 2\pi(t+\frac{x}{2})$, with*

$$\mathfrak{a}(x) = \pi^4\int_{\mathbb{R}} h_x^4\,d\sigma = \frac{\pi\,v_0(x)}{8\,\Gamma(1-x)^4}, \qquad \mathfrak{b}(x) = 2\pi^4\int_{\mathbb{R}} h_x^4\cos(2\pi\sigma)\,d\sigma = \frac{\pi\,v_1(x)}{4\,\Gamma(1-x)^4}. \tag{372}$$

Consequently, on I, $\mathcal{A}(x) = \sin^2(\pi x)\,v_0(x)/(8\pi)$ and $\mathcal{B}(x) = \sin^2(\pi x)\,v_1(x)/(4\pi)$.

Proof. Let $\omega_m = \int_0^1 \Omega(t;x)e^{-2\pi\mathrm{i}mt}\,dt$. The terms of Ω are integrable on the real line, so unfolding (Fubini) and $s = \sigma - \frac{x}{2}$ give $\omega_m = \pi^4 e^{\mathrm{i}\pi mx}\int_{\mathbb{R}} h_x(\sigma)^4 e^{-2\pi\mathrm{i}m\sigma}\,d\sigma$. By the Cauchy beta integral with $c = -x$ and Fubini ($\mathsf{P}_x \in L^1$), $h_x(\sigma)^2 = (2\pi\Gamma(1-x))^{-2}\int_{\mathbb{R}}\mathsf{Q}_x(\phi)e^{\mathrm{i}\sigma\phi}\,d\phi$. Since $\mathsf{Q}_x \in L^1\cap L^2$ by (370), Plancherel's theorem gives, with $\widehat{\chi}(\theta) = \int\chi(\sigma)e^{-\mathrm{i}\theta\sigma}\,d\sigma$, the identity $\widehat{h_x^2} = \mathsf{Q}_x/(2\pi\Gamma(1-x)^2)$ in L^2. Parseval's identity for the L^2 functions h_x^2 and $h_x^2 e^{\mathrm{i}\xi\sigma}$ then gives

$$\int_{\mathbb{R}} h_x^4 e^{-\mathrm{i}\xi\sigma}\,d\sigma = \frac{1}{2\pi}\int_{\mathbb{R}}\widehat{h_x^2}(\theta)\,\overline{\widehat{h_x^2}(\theta-\xi)}\,d\theta = \frac{(\mathsf{Q}_x * \mathsf{Q}_x)(\xi)}{8\pi^3\Gamma(1-x)^4}.$$

Thus $\omega_m = \pi e^{\mathrm{i}\pi mx}v_m(x)/(8\Gamma(1-x)^4)$. Now Q_x is even, nonnegative and supported in $[-2\pi,2\pi]$, so $\mathsf{Q}_x * \mathsf{Q}_x$ is even and supported in $[-4\pi,4\pi]$, and at $\xi = \pm4\pi$ the convolution integral runs over a single point. Hence $v_{-1} = v_1$ and $v_m = 0$ for $|m| \ge 2$. A continuous one-periodic function whose Fourier coefficients vanish for $|m| \ge 2$ equals the corresponding trigonometric polynomial (Fejér), so for real t, $\Omega(t;x) = \frac{\pi}{8\Gamma(1-x)^4}[v_0(x) + 2v_1(x)\cos(2\pi t + \pi x)]$, and by analyticity for all t. The integral forms follow from ω_0 and ω_1. Finally $N_x^2/\Gamma(1-x)^4 = [\Gamma(x)\Gamma(1-x)]^{-2} = \sin^2(\pi x)/\pi^2$ and (349) give the expressions for $\mathcal{A}$ and $\mathcal{B}$. □

For $x \in I$ this is a second proof of (349): the modes with $|m| \ge 2$ are removed by the compact support of $\mathsf{Q}_x * \mathsf{Q}_x$, without growth estimates. The same argument should work for every real $x < \frac34$, where Q_x behaves like $|\phi|^{1-2x}$ near 0 and is still square integrable; this extension is a sketch, is not proved here and is not used (the general case is (358)).

Lemma (the convolutions at $x = \frac12$). *v_0 and v_1 extend holomorphically to D_{F}, with $v_m'(x) = \int_{\mathbb{R}}\partial_x[\mathsf{Q}_x(\phi)\mathsf{Q}_x(2\pi m - \phi)]\,d\phi$. At $x = \frac12$,*

$$v_0 = 32\,U_0, \qquad v_1 = 16\,V_0, \qquad v_0' = 32\bigl[\pi V_0 - 2U - 4\log 2\,U_0\bigr], \qquad v_1' = 16\bigl[\pi U_0 - 2V - 4\log 2\,V_0\bigr]. \tag{373}$$

Proof. Use the holomorphic extension (369). By (370), $|\mathsf{Q}_x(\phi)\mathsf{Q}_x(2\pi m - \phi)|$ and its x-derivative are bounded, uniformly for $x \in D_{\mathrm{F}}$, by $C|\sin\frac{\phi}{2}|^{-7/12}$ on the support ($|\phi| < 2\pi$ for $m = 0$, $0 < \phi < 2\pi$ for $m = 1$), which is integrable. Holomorphy and differentiation under the integral follow. By evenness, $v_0' = 4\int_0^{2\pi}(\partial_x\mathsf{Q}_x)\mathsf{Q}_x\,d\phi$ and $v_1' = 2\int_0^{2\pi}(\partial_x\mathsf{Q}_x)(\phi)\,\mathsf{Q}_x(2\pi-\phi)\,d\phi$. Substitute $\phi = 2\pi - 4\beta$ with $\beta \in (0,\frac{\pi}{2})$: then $\cos\frac{\phi}{4} = \sin\beta$, $\cos\frac{2\pi-\phi}{4} = \cos\beta$ and $4kk' = 2\sin 2\beta$, so (371) gives $\mathsf{Q}_{1/2}(\phi) = 2F(\beta)$, $\mathsf{Q}_{1/2}(2\pi-\phi) = 2G(\beta)$ and $\partial_x\mathsf{Q}_x(\phi)|_{1/2} = \pi G(\beta) - 2F(\beta)[\log 2 + \log\sin 2\beta]$, with F, G as in the proof of (363). With $d\phi = 4\,d\beta$, the angle forms given there and $\int_0^{\pi/2}G^2\,d\beta = U_0$, this gives (373). □

Conclusion. By (372), since $\sin^2\pi x$ has zero derivative at $x=\frac12$, and by (373),

$$a=\frac{v_0(\frac12)}{8\pi}=\frac{4U_0}{\pi},\qquad b=\frac{v_1(\frac12)}{4\pi}=\frac{4V_0}{\pi},\qquad \mathcal{A}'(\tfrac12)=\frac{v_0'(\frac12)}{8\pi},\qquad \mathcal{B}'(\tfrac12)=\frac{v_1'(\frac12)}{4\pi}.$$

With (363), $a=8N_{\mathrm e}/\pi$, $b=8M_{\mathrm e}/\pi$, $-2U-4\log 2\,U_0=2\pi M_{\mathrm e}-4\log 2\,N_{\mathrm e}$ and $-2V-4\log 2\,V_0=\pi N_{\mathrm e}-4\log 2\,M_{\mathrm e}$, so that

$$\mathcal{A}'(\tfrac12)=16M_{\mathrm e}-\frac{16\log 2}{\pi}N_{\mathrm e}=2\pi b-2\log 2\,a,\qquad \mathcal{B}'(\tfrac12)=12N_{\mathrm e}-\frac{16\log 2}{\pi}M_{\mathrm e}=\frac{3\pi}{2}a-2\log 2\,b. \tag{374}$$

Inserted into the expression for $\mathcal{L}_2$ displayed after (350),

$$\mathcal{L}_2=-6a^2-8b^2+\frac{4}{\pi}\Big[(2\pi b-2\log 2\,a)\,b+a\Big(\frac{3\pi}{2}a-2\log 2\,b\Big)\Big]=-\frac{16\log 2}{\pi}\,ab=-2\log 2\,P_4(2).$$

The rotation part $(2\pi b,\frac{3\pi}{2}a)$ of (374) cancels $-6a^2-8b^2$ exactly, and the diagonal part $-2\log 2$ leaves $-2\log 2$ times the coefficient $8/\pi$ of ab in $P_4(2)=8ab/\pi$. With (351), (374) is equivalent to (356), which this route therefore proves once more; (374) are the relations quoted after (356) and in (540). The inputs are Cauchy's beta integral, Plancherel's theorem, Stirling's formula, the theta-function form of cn and the half-strip contours of Section 18.7.7; the Stieltjes identity is not used.

Remark (a real-line form). Put $\mathfrak{m}_j=\int_{\mathbb R}h_{1/2}(\sigma)^4\cos(2\pi j\sigma)\,d\sigma$, so that $\mathfrak{m}_0=a/\pi^4$ and $\mathfrak{m}_1=b/(2\pi^4)$, and $\Psi(\sigma)=\psi_0(\frac34+\sigma)+\psi_0(\frac34-\sigma)$. Then

$$\int_{\mathbb R}h_{1/2}^4\,\Psi\,d\sigma=2\pi\mathfrak{m}_1-(5\log 2+2\gamma)\mathfrak{m}_0,\qquad \int_{\mathbb R}h_{1/2}^4\,\Psi\cos(2\pi\sigma)\,d\sigma=\frac{3\pi}{8}\mathfrak{m}_0-(5\log 2+2\gamma)\mathfrak{m}_1.$$

To see this, differentiate the integral forms (372) under the integral. One has $\partial_x h_x^4=2h_x^4\Psi_x$ with $\Psi_x(\sigma)=\psi_0(1-\frac{x}{2}+\sigma)+\psi_0(1-\frac{x}{2}-\sigma)$, which is entire in σ because the poles of $\psi_0(1-\frac{x}{2}-\sigma)$ sit at the fourth-order zeros of h_x^4; the reflected form of h_x gives the majorant $|\partial_x h_x^4|\le C\log(2+|\sigma|)(1+|\sigma|)^{4x-4}$, uniformly for x near $\frac12$. So $\mathfrak{a}'(\frac12)=2\pi^4\int h_{1/2}^4\Psi$ and $\mathfrak{b}'(\frac12)=4\pi^4\int h_{1/2}^4\Psi\cos 2\pi\sigma$. Since $\mathfrak{a}=\mathcal{A}/N_x^2$, $\mathfrak{b}=\mathcal{B}/N_x^2$ and $(N_x^{-2})'(\frac12)=4\psi_0(\frac12)$, (374) gives $\mathfrak{a}'(\frac12)=2\pi b-(10\log 2+4\gamma)a$ and $\mathfrak{b}'(\frac12)=\frac{3\pi}{2}a-(10\log 2+4\gamma)b$, which are the displayed identities. By duplication and reflection, $\Psi(\sigma)=\psi_0(\frac12+2\sigma)+\psi_0(\frac12-2\sigma)-2\log 2+\pi\sec 2\pi\sigma$, where the combination is regular; so the first identity also reads $\int_{\mathbb R}h_{1/2}^4[\psi_0(\frac12+2\sigma)+\psi_0(\frac12-2\sigma)+\pi\sec 2\pi\sigma]\,d\sigma=2\pi\mathfrak{m}_1-(3\log 2+2\gamma)\mathfrak{m}_0$. This form is not used.

Checks. These are not proof steps. A 60-digit run (symmetry, Slurm job 567960; logs `LRI_C5_d2_simplify/proof/L2-fourier/check_reduction_dps60.log` and `check_strip_dps60.log`, script `check_reduction.py`) gives: the Cauchy beta integral to 1.4×10^{-42}; $\mathsf{Q}_{1/2}=2\mathbf{K}$ and (371) to 6×10^{-65}; $a=4U_0/\pi$ and $b=4V_0/\pi$ to 4×10^{-65}; the derivatives from (373) against (351) with the 70-digit single sums to 6×10^{-63}; (374) to 5×10^{-63} and 3×10^{-64}; and the collapse with these elliptic values of a, b, $\mathcal{A}'$, $\mathcal{B}'$ against the 70-digit $\mathcal{L}_2$ to 7×10^{-62}. The direct quadrature of Q_x at $x=0.6$ is limited by its endpoint singularity (2×10^{-26}). The symbolic collapse is checked exactly by `collapse.py` (log `collapse.log`).

18.8 Proof of the closed form of $\mathcal{K}_2$

We prove (271) for $\mathcal{K}_2=[t^2](J_t-J_t^{\Gamma})$ of (238) at $d=2$. Write $t=2\tau$. Since $h_2=1/\pi$, the path master is the kite (671) at $\boldsymbol{p}=(1,1,1,2-2\tau,1)$:

$$J_{2\tau}=\frac{1}{\pi^2}\int_{\mathbb C^2}\frac{d^2x\,d^2y}{|x|\,|1-x|\,|y|\,|1-y|^{2-2\tau}\,|x-y|},\qquad 0<\tau<1.$$

We use the parameter convention $K(m)$ of Section 1.4, the moments

$$E(a,b)=\int_0^1 x^{a-1}(1-x)^{b-1}K(x)\,dx,$$

the six moments $I_1,\dots,I_6$ listed before (77), evaluated at $t=\tau$, and the functions $\mathfrak{F},\mathfrak{G}$ of (75). Put $s_\tau=\sin\pi\tau$ and $c_\tau=\cos\pi\tau$. At $d=1$ the master is a linear functional of the six moments, (60); at $d=2$ it is a quadratic form in them. This proof uses none of the periodization machinery of this Section; it is placed next to the proof for $\mathcal{L}_2$ for convenience. It was written by the Claude Code session that wrote the first proof of Section 18.7 (package `LRI_C5_d2_proof/`, folder `K2_selberg/`); the closed form (271) itself was first found numerically by the other session named there.

18.8.1 Real cells and elliptic moments

For $\xi=(\xi_1,\xi_2)\in\mathbb{R}^2$ let

$$(L_1,\dots,L_5)=(\xi_1,\,1-\xi_1,\,\xi_2,\,1-\xi_2,\,\xi_2-\xi_1),\qquad(\kappa_1,\dots,\kappa_5)=(-\tfrac12,-\tfrac12,-\tfrac12,\tau-1,-\tfrac12),$$

and $\Upsilon(\xi)=\prod_f|L_f(\xi)|^{\kappa_f}$. Then $\pi^2J_{2\tau}=\int_{\mathbb{C}^2}\prod_f|L_f(x,y)|^{2\kappa_f}d^2x\,d^2y$, where $L_f(x,y)$ is the same affine form at complex arguments. The five lines $L_f=0$ cut $\mathbb{R}^2$ into twelve open cells R: for ξ_2 in one of $(-\infty,0)$, $(0,1)$, $(1,\infty)$, the points $0,1,\xi_2$ cut the ξ_1-line into four intervals. On each cell the sign vector $\varsigma(R)=(\operatorname{sgn}L_f)_f$ is constant, and distinct cells have distinct sign vectors. For $0<\tau<1$,

$$\Upsilon_R:=\int_R\Upsilon(\xi)\,d\xi=2I_{j(R)},\tag{375}$$

with $j(R)$ given by

cell	j	cell	j	cell	j
$\xi_1<\xi_2<0$	6	$0<\xi_1<\xi_2<1$	1	$\xi_2<0,\ 1<\xi_1$	5
$\xi_2<\xi_1<0$	5	$0<\xi_2<\xi_1<1$	2	$0<\xi_2<1<\xi_1$	1
$\xi_1<0<\xi_2<1$	2	$0<\xi_1<1<\xi_2$	3	$1<\xi_1<\xi_2$	4
$\xi_1<0,\ 1<\xi_2$	4	$\xi_2<0<\xi_1<1$	6	$1<\xi_2<\xi_1$	3

Indeed $\Upsilon\ge0$, so by Tonelli we may integrate over ξ_1 first. For $e_1<e_2<e_3$ and $P(\xi_1)=|(\xi_1-e_1)(\xi_1-e_2)(\xi_1-e_3)|^{-1/2}$, the substitutions $\xi_1=e_3-(e_3-e_1)/\sin^2\theta$, $e_1+(e_2-e_1)\sin^2\theta$, $e_3-(e_3-e_2)\sin^2\theta$ and $e_1+(e_3-e_1)/\sin^2\theta$ on the four intervals give

$$\int_{-\infty}^{e_1}P=\int_{e_2}^{e_3}P=2rK(m),\qquad\int_{e_1}^{e_2}P=\int_{e_3}^{\infty}P=2rK(1-m),$$
$$m=\frac{e_3-e_2}{e_3-e_1},\qquad r=(e_3-e_1)^{-1/2}.$$

With $\{e_i\}=\{0,1,\xi_2\}$, the outer weight $|\xi_2|^{-1/2}|1-\xi_2|^{\tau-1}r$ becomes the Euler weight $v^{-1/2}(1-v)^{\tau-1}$, $v^{-1/2}(1-v)^{-\tau}$ or $v^{-\tau}(1-v)^{\tau-1}$ after $v=\xi_2$ on $(0,1)$, $v=-\xi_2/(1-\xi_2)$ on $(-\infty,0)$ or $v=1/\xi_2$ on $(1,\infty)$, respectively. The arguments of K become v or $1-v$, and the table follows. All twelve integrals are finite for $0<\tau<1$.

18.8.2 The rotation lemma

For each form write $|L_f(x,y)|^2=\alpha_f+\beta_f^2$, where $x=x_1+\mathrm{i}x_2$, $y=y_1+\mathrm{i}y_2$, $\alpha_f=L_f(x_1,y_1)^2$, and β_f is the linear part of L_f evaluated at (x_2,y_2); this holds because the L_f have real coefficients. Put

$\xi=(x_1-x_2,y_1-y_2)$ and $\eta=(x_1+x_2,y_1+y_2)$. Then $\alpha_f-\beta_f^2=L_f(\xi)L_f(\eta)$ and $dx_1\,dx_2\,dy_1\,dy_2=\frac14\,d\xi\,d\eta$. We claim

$$\pi^2 J_{2\tau}=-\frac14\sum_{R,R'}e^{\mathrm{i}\pi\phi(R,R')}\,\Upsilon_R\Upsilon_{R'},\qquad \phi(R,R')=\sum_{f:\ \varsigma_f(R)\neq\varsigma_f(R')}\kappa_f, \tag{376}$$

the sum running over all 144 ordered pairs of cells.

(a) *Convergence.* Since $\alpha+\beta^2\geq|\alpha-\beta^2|$ and all $\kappa_f<0$, the integrand $\prod_f(\alpha_f+\beta_f^2)^{\kappa_f}$ is at most $\Upsilon(\xi)\Upsilon(\eta)$, whose integral is $\frac14(\int_{\mathbb{R}^2}\Upsilon)^2=4(\sum_j I_j)^2<\infty$ by (375). In particular $J_{2\tau}$ converges for $0<\tau<1$.

(b) *An analytic family.* For $\Re z>0$ put $\Psi(z)=z^2\int_{\mathbb{R}^4}\prod_f(\alpha_f+z^2\beta_f^2)^{\kappa_f}\,dx_1\,dx_2\,dy_1\,dy_2$, with principal powers; off a null set $\alpha_f+z^2\beta_f^2\notin(-\infty,0]$. If $|\arg z|\leq\theta_0<\frac{\pi}{2}$ and $r_1\leq|z|\leq r_2$, the inequality $|\alpha+Be^{\mathrm{i}\vartheta}|\geq(\alpha+B)\cos\frac{\vartheta}{2}$ for $\alpha,B\geq0$ gives $|\alpha_f+z^2\beta_f^2|\geq\cos\theta_0\,\min(1,r_1^2)\,(\alpha_f+\beta_f^2)$. The integrand is therefore dominated by a constant multiple of the integrable function of (a), and Ψ is holomorphic on $\Re z>0$ by Morera's theorem, Fubini's theorem and dominated convergence. For real $z>0$ the substitution $(x_2,y_2)\to(x_2,y_2)/z$ gives $\Psi(z)=\pi^2J_{2\tau}$, so Ψ is constant by the identity theorem.

(c) *The boundary value.* Let $z=e^{\mathrm{i}\theta}$ with $\theta\uparrow\frac{\pi}{2}$. Since $|\alpha_f+e^{2\mathrm{i}\theta}\beta_f^2|\geq|\alpha_f-\beta_f^2|$, the integrand is dominated by $\Upsilon(\xi)\Upsilon(\eta)$ for every θ. Since $\Im(\alpha_f+e^{2\mathrm{i}\theta}\beta_f^2)>0$ when $\beta_f\neq0$, almost everywhere $(\alpha_f+e^{2\mathrm{i}\theta}\beta_f^2)^{\kappa_f}\to|L_f(\xi)L_f(\eta)|^{\kappa_f}e^{\mathrm{i}\pi\kappa_f[L_f(\xi)L_f(\eta)<0]}$. Dominated convergence and $z^2\to-1$ give $\pi^2J_{2\tau}=-\frac14\iint\Upsilon(\xi)\Upsilon(\eta)\exp(\mathrm{i}\pi\sum_f\kappa_f[\varsigma_f(\xi)\neq\varsigma_f(\eta)])\,d\xi\,d\eta$. The phase is constant on each product of two cells, which proves (376).

This is the Kawai–Lewellen–Tye (KLT) factorization [43] behind (674), proved here directly at the point needed.

18.8.3 The master path as a sum of two squares

With $\omega_\tau=e^{\mathrm{i}\pi\tau}$ the phase in (376) is $(-\mathrm{i})^n(-\omega_\tau)^m$, where n counts the forms with exponent $-\frac12$ whose signs differ on R and R', and $m=1$ if the sign of $1-\xi_2$ differs and $m=0$ otherwise. Collecting the 144 terms with the table of (375), a finite exact computation (machine checked in `LRI_C5_d2_proof/K2_selberg/region_sum_matrix.py`), gives, with $\omega=\omega_\tau$,

$$\begin{aligned}\pi^2J_{2\tau}=4\big[&\mathrm{i}I_1I_2-\mathrm{i}\omega I_1I_4-\mathrm{i}I_1I_5+I_1I_6-\mathrm{i}\omega I_2I_3-\omega I_2I_4+I_2I_5\\&+\mathrm{i}I_3I_4+\mathrm{i}\omega I_3I_5-\omega I_3I_6+\mathrm{i}\omega I_4I_6+\mathrm{i}I_5I_6\big].\end{aligned}$$

Now insert the moment relations (77) at $t=\tau\in(0,1)$ and $\omega_\tau=c_\tau+\mathrm{i}s_\tau$. (The second contour in the proof of (77) also needs, on $x<0$, the boundary value $K(1-x-\mathrm{i}0)=(1-x)^{-1/2}[K(\frac{1}{1-x})-\mathrm{i}K(\frac{-x}{1-x})]$. It follows from the displayed formula for $K(x+\mathrm{i}0)$, $x>1$, by Schwarz reflection, and that proof uses it without displaying it.) All terms except $s_\tau^2(I_3^2+I_4^2)$ cancel; for instance, the coefficient of I_3I_4 becomes $2\mathrm{i}-2\omega_\tau(s_\tau+\mathrm{i}c_\tau)=0$. Euler's integral gives $I_3=\pi^2\mathfrak{F}(1-\tau)/(2s_\tau)$ and $I_4=\pi^2\mathfrak{F}(\tau)/(2s_\tau)$, because $B(\tau,1-\tau)=\pi/s_\tau$. Therefore, for $0<\tau<1$,

$$\boxed{J_{2\tau}=\pi^2\big[\mathfrak{F}(\tau)^2+\mathfrak{F}(1-\tau)^2\big]=\frac{4}{\pi^2}s_\tau^2(I_3^2+I_4^2)=\frac{4}{\pi^2}\big[s_\tau I_1I_4+I_2I_5\big].} \tag{377}$$

The last form is (674) at this point: with $a=b=A'=c=-\frac12$ and $B'=\tau-1$, the four factors $\mathcal{O}(a,b,A',B',c)$, $\mathcal{M}(a,b,A',B',c)$, $\mathcal{O}(A',B',a,b,c)$ and $\mathcal{M}(A',B',a,b,c)$ take the values $2I_1$, $2I_4$, $2s_\tau I_3=2I_2$ and $2I_5$. That identity is thereby proved here; it is not an input. Equation (377) is the two-dimensional partner of (78), with no cross term. It is symmetric under $t\to2-t$, as completion covariance requires. At $\tau=\frac12$ Dixon's sum [22] gives $\mathfrak{F}(\frac12)=\Gamma(\frac14)^4/(4\pi^3)$, so $J_1=\Gamma(\frac14)^8/(8\pi^4)=3Q_2$, in agreement with (267).

18.8.4 Laurent extraction

Write $\mathfrak{F}(\tau) = 1 + u\tau + v\tau^2 + O(\tau^3)$ and $\mathfrak{G}(\tau) = 1 + \sigma\tau^3 + \varrho\tau^4 + O(\tau^5)$, where ϱ is the coefficient called τ in (86) and $\mathfrak{G} - 1 = O(\tau^3)$ because every term $n \geq 1$ of $\mathfrak{G}$ carries $(\tau)_n^3$. By (82), $\pi\mathfrak{F}(1-\tau) = \tau^{-1}\mathcal{Q}(\tau)\mathfrak{G}(\tau)$ for $0 < \tau < 1$, with $\mathcal{Q}(\tau) = \pi\Gamma(1+\tau)/[\Gamma(1-\tau)\Gamma(\frac{1}{2}+\tau)^2]$. The series of $\mathfrak{F}$ converges locally uniformly on $\Re\tau < 1$, its terms being $O(n^{\Re\tau-2})$; $\mathfrak{G}$ is holomorphic on $|\tau| < \frac{1}{2}$; and $\mathcal{Q}$ is holomorphic and zero-free there, with $\mathcal{Q}(0) = 1$. Hence

$$J_{2\tau} = \pi^2\mathfrak{F}(\tau)^2 + \tau^{-2}\mathcal{Q}(\tau)^2\mathfrak{G}(\tau)^2$$

is meromorphic on $|\tau| < \frac{1}{2}$, with a double pole at $\tau = 0$ only. By the identity theorem it is the continuation in which (238) takes its Laurent coefficient. At $d = 2$, (230) gives

$$J^{\Gamma}_{2\tau} = \mathcal{V}(\tau)^2 + \frac{\mathcal{V}(\tau) - \widetilde{\mathcal{V}}(\tau)}{\tau}, \qquad \mathcal{V} = \frac{\Gamma(\tau)\Gamma(\frac{1}{2}-\tau)}{\Gamma(1-\tau)\Gamma(\frac{1}{2}+\tau)}, \quad \widetilde{\mathcal{V}} = \frac{\Gamma(\tau)\Gamma(\frac{1}{2}+\tau)\Gamma(\frac{1}{2}-2\tau)}{\Gamma(1-\tau)\Gamma(\frac{1}{2}-\tau)\Gamma(\frac{1}{2}+2\tau)},$$

which are $B_2(1, 2-2\tau)$ and $B_2(1+4\tau, 2-2\tau)$. The series $\log\Gamma(1+\tau) = -\gamma_E\tau + \sum_{k\geq 2}(-1)^k\zeta(k)\tau^k/k$ and $\log\Gamma(\frac{1}{2}+\tau) = \frac{1}{2}\log\pi + \sum_{k\geq 1}\psi_{k-1}(\frac{1}{2})\tau^k/k!$, with $\psi_0(\frac{1}{2}) = -\gamma_E - 2\log 2$, $\psi_1(\frac{1}{2}) = \frac{\pi^2}{2}$, $\psi_2(\frac{1}{2}) = -14\zeta(3)$ and $\psi_3(\frac{1}{2}) = \pi^4$ [32], give

$$\log\mathcal{Q}(\tau)^2 = 8\log 2\,\tau - \pi^2\tau^2 + 8\zeta(3)\tau^3 - \frac{\pi^4}{6}\tau^4 + O(\tau^5),$$
$$J^{\Gamma}_{2\tau} = \frac{e^{8\tau\log 2}}{\tau^2} - 20\zeta(3)\tau - 48\zeta(3)\log 2\,\tau^2 + O(\tau^3).$$

The last coefficient is four times the coefficient of t^2 in J^{Γ}_t stated after (250), evaluated at $q = \frac{1}{2}$. Write $\mathcal{Q}_k = [\tau^k]\mathcal{Q}(\tau)^2$; in particular $\mathcal{Q}_3 = \frac{256}{3}\log^3 2 - 8\pi^2\log 2 + 8\zeta(3)$ and $\mathcal{Q}_4 = \frac{512}{3}\log^4 2 - 32\pi^2\log^2 2 + 64\zeta(3)\log 2 + \frac{\pi^4}{3}$. The Laurent coefficients are

order	$J_{2\tau}$	$J^{\Gamma}_{2\tau}$
τ^{-2}	1	1
τ^{-1}	$8\log 2$	$8\log 2$
τ^0	$\pi^2 + \mathcal{Q}_2 = 32\log^2 2$	$32\log^2 2$
τ^1	$2\pi^2 u + 2\sigma + \mathcal{Q}_3$	$\frac{256}{3}\log^3 2 - 20\zeta(3)$
τ^2	$\pi^2(u^2 + 2v) + 2\varrho + 16\sigma\log 2 + \mathcal{Q}_4$	$\frac{512}{3}\log^4 2 - 48\zeta(3)\log 2$

The orders $\tau^{-2}, \tau^{-1}, \tau^0$ cancel identically. At order τ the difference is $2[\sigma + \pi^2 u - 4\pi^2\log 2 + 14\zeta(3)]$, which vanishes by (87); this is the two-dimensional instance of $J_t - J^{\Gamma}_t = O(t^2)$ in (237), here derived rather than assumed. At order τ^2 the coefficients v and ϱ enter only through $2(\varrho + \pi^2 v) = \pi^2 u^2 - \frac{\pi^4}{3}$, which is (86). Since $t = 2\tau$, $[t^2] = \frac{1}{4}[\tau^2]$, and

$$\mathcal{K}_2 = \frac{\pi^2}{2}u^2 + 4\sigma\log 2 - 8\pi^2\log^2 2 + 28\zeta(3)\log 2.$$

With $u = 4\log 2 - 8\mathsf{G}/\pi$ and $\sigma = 8\pi\mathsf{G} - 14\zeta(3)$ from (87), this is $\mathcal{K}_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2$. Eliminating σ with the order-τ relation instead gives $\mathcal{K}_2 = \frac{\pi^2}{2}(u - A)^2 + \frac{1}{2}AU$ with $\pi(A - u) = 8\mathsf{G}$, the second form of (271). All coefficient algebra here is exact and machine checked (`LRI_C5_d2_proof/K2_selberg/laurent_K2_sympy.py` and `LRI_C5_d2_proof/K2_selberg/laurent_K2.wl`).

The one-dimensional inputs (82), (86) and (87) concern $\mathfrak{F}$ and $\mathfrak{G}$ only and are proved in Section 8.3. In that proof (82) is also used at $-t$, with $\mathfrak{F}(1+t)$ continued by (80). This is legitimate: both sides of (82) are meromorphic in a punctured disk around $t = 0$ and agree for $0 < t < \frac{1}{2}$, so they agree there identically.

18.8.5 Further proofs and checks

Section 18.9 gives a second position-space proof of (377), by Riemann's bilinear relation (Section 18.9.8). It also proves at $d = 2$ that the spectral form (248) and the series (260) equal (271), and it proves the mixed-trace step for $0 < \tau < \frac{1}{2}$; its items (i)–(v) say where each statement is proved. Together with Section 18.8.4, these results show that at $d = 2$ the definition (238), the spectral form and the series agree; the agreement of the definition with the series does not use the mixed-trace step.

Checks. These are not proof steps. The closed form (377) agrees with a two-dimensional quadrature of the y-integral, with the inner star (398) inserted, to between 38 and 68 digits at seven values of τ. Direct nested quadratures of the twelve cell integrals agree with $2I_j$ to between 18 and 24 digits, and (250), evaluated with an independent spectral code, reproduces (377) to 54 digits at $\tau = 0.1$ and $\tau = 0.23$. At $\tau = \frac{3}{10}$ the twelve-term form above, with the moments computed by direct quadrature, agrees with $\pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2] = 47.790785413226103671\ldots$ to 37 digits.

18.9 Spectral and series forms of $\mathcal{K}_2$, the mixed-trace step, and a second position-space proof

Section 18.8 proves (271) for the master-path definition (238). This subsection proves, at $d = 2$, the statements that connect that definition with the spectral form (248) and the series (260).

(i) For $0 < \tau < \frac{1}{2}$ the two regulated traces of (250) are evaluated in closed form: $C(\tau)\langle t_\ell^2 R_\tau\rangle_2 = \pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2]$ and $\langle t_\ell^2 S_\tau\rangle_2 = C(\tau)\mathcal{X}_2(\tau)^2$ (Section 18.9.5).

(ii) With (377) this proves the mixed-trace representation $J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_2$ stated before (250), at $d = 2$, for $0 < \tau < \frac{1}{2}$ and in the full-Mellin-line reading (Section 18.9.5).

(iii) The coefficient $[\tau^2]$ may be taken under the trace, and the right side of (248) at $d = 2$ equals $32\mathsf{G}^2 - 28\zeta(3)\log 2$; as a by-product, $\langle t_\ell^2 E_2\rangle_2 = -21\zeta(3)$ (Sections 18.9.2 and 18.9.6).

(iv) Each inner n sum of (260) at $d = 2$ equals the corresponding spin term of (248), and the spin sum converges absolutely. Hence (260) equals (271), and the strip limit used for the 70-digit evaluation of Section 17.2 computes exactly this ordered sum (Section 18.9.7).

(v) A second proof of (377) from the position-space integral evaluates the inner star by Riemann's bilinear relation and the outer integral by a Stokes pairing on the slit plane (Section 18.9.8).

All five statements are proved here. Besides standard theorems (Stirling's formula, Gamma reflection, Gauss's and Thomae's unity-argument identities, the residue theorem, Liouville's theorem, Green's theorem, dominated convergence and the identity theorem), the proofs use only one-dimensional lemmas of Section 8: the Euler integral for $E(a, b)$, the moment relations (77), the Thomae relation (82), and the Taylor data (87) and (86). These concern the elliptic moments and the two functions $\mathfrak{F}, \mathfrak{G}$ only, and do not depend on the dimension. No numerical identification enters. The results are specific to $d = 2$: at other dimensions the mixed-trace step remains input (M) of Section 21.1, and the interchange of $[\tau^2]$ with the trace is not proved.

The material comes from two packages, both written by Claude Code sessions:

- package `LRI_C5_d2_simplify/` (folders `proof/` and `proof/K2-parseval/`), by the session that wrote the third and fourth proofs of (270): the regularized spectral route, Sections 18.9.2–18.9.6, apart from the parts listed in the next item;

- package `LRI_C5_d2_proof/`, by the session that wrote the first proof of Section 18.7 and the proof of Section 18.8: in folder `K2_gauss/`, the sharper estimate at the end of Section 18.9.2, the second evaluation of the single sums (Section 18.9.4), a second derivation of the Laurent extraction of Section 18.9.6 and the per-spin identity of Section 18.9.7 with its arc bounds; in folders `K2_d1template/` and `K2_d1template_fix1/`, the revised contour bounds used in Section 18.9.3, the third evaluation of the single sums, the mixed-trace corollary of Section 18.9.5 and the Riemann bilinear proof of Section 18.9.8.

Both packages contain the absolute convergence of the series and the strip limit of Section 18.9.7.

18.9.1 Conventions at $d = 2$

Throughout, $d = 2$ and $q = \frac{1}{2}$. Then $a_\ell = \frac{\ell}{2} + \frac{1}{4}$, $b_\ell = \frac{\ell}{2} + \frac{3}{4}$, $d_0 = 1$, $d_\ell = 2$ for $\ell \geq 1$, and $\Gamma(2q) = 1$, so $t_\ell(\nu) = g_\ell(\mathrm{i}\nu)$ with the kernel g_ℓ of (300). At fixed spin we write $a = a_\ell$, $b = b_\ell$; then $b - a = \frac{1}{2}$ and $a + b = \ell + 1$. The quantities of (249) are

$$C(\tau) = \frac{\Gamma(\tau)}{\Gamma(1-\tau)} = \frac{1}{\tau} + c_* + O(\tau), \qquad c_* = 2\psi_0(1), \qquad \mathcal{X}_2(\tau) = \Big[\frac{\Gamma(\frac{1}{2} - \frac{\tau}{2})}{\Gamma(\frac{1}{2} + \frac{\tau}{2})}\Big]^2,$$

where $\mathcal{X}_2$ is $\mathcal{X}_d$ of (555) at $d = 2$. The constants of (221) are $A = 4\log 2$ and $U = \psi_2(\frac{1}{2}) = -14\zeta(3)$, so $-AU/4 = 14\zeta(3)\log 2$. We write

$$w_{\mathcal{K}} = (c_* - p)E_2 - \frac{hk}{2} - \frac{5r}{32}$$

for the weight in (248), with p, h, j, k, r, E_2 of (247). Since h, k are odd in ν and p, j, r are even, E_2 and $w_{\mathcal{K}}$ are even in ν.

The kernel $R_\tau(s)$ is not even in s. As in (554), every trace in this subsection is read on the full Mellin line,

$$\langle f \rangle_2 = \frac{1}{2\pi} \sum_{\ell \geq 0} d_\ell \int_{\mathbb{R}} f(\ell, \nu)\, d\nu.$$

For f even in ν this is (241). For real τ, $R_\tau(-\mathrm{i}\nu) = \overline{R_\tau(\mathrm{i}\nu)}$, so $\langle t_\ell^2 R_\tau \rangle_2$ is $\frac{1}{\pi}\sum_\ell d_\ell \int_0^\infty \Re\{t_\ell^2 R_\tau\}\, d\nu$, the real-part reading used in (250). We put $s_\tau = \sin \pi\tau$ and $c_\tau = \cos \pi\tau$, as in Section 18.8, and $\xi = \frac{\ell}{2} + \mathrm{i}\nu$, $\bar{\xi} = \frac{\ell}{2} - \mathrm{i}\nu$ (not the real points ξ of Section 18.8), so that $\Re\xi = \ell/2 \geq 0$, $a \pm \mathrm{i}\nu = \frac{1}{4} + \xi$ or $\frac{1}{4} + \bar{\xi}$, and $b \pm \mathrm{i}\nu = \frac{3}{4} + \xi$ or $\frac{3}{4} + \bar{\xi}$. The functions $\mathfrak{F}, \mathfrak{G}$ are those of (75), with the Taylor data of Section 18.8.4:

$$\begin{aligned} &\mathfrak{F}(\tau) = 1 + u\tau + v\tau^2 + O(\tau^3), \qquad \mathfrak{G}(\tau) = 1 + \sigma\tau^3 + \varrho\tau^4 + O(\tau^5), \\ &u = 4\log 2 - \frac{8\mathsf{G}}{\pi}, \qquad \sigma = 8\pi\mathsf{G} - 14\zeta(3), \qquad \varrho + \pi^2 v = \frac{\pi^2 u^2}{2} - \frac{\pi^4}{6}. \end{aligned} \tag{378}$$

Here ϱ is the coefficient called τ in (86); the letter τ is the regulator throughout this subsection.

18.9.2 Analytic regularization in the rim exponent

Lemma (uniform Gamma ratios). *Let $\Lambda_0 \subset \mathbb{C}$ be compact and $c_0 > 0$. There is C_0 such that for all $\alpha, \beta \in \Lambda_0$ and all z with $\Re z \geq 0$ and $\Re(z + \alpha) \geq c_0$,*

$$\Big|\frac{\Gamma(z+\alpha)}{\Gamma(z+\beta)} - (1+z)^{\alpha-\beta}\Big| \leq \frac{C_0\,|(1+z)^{\alpha-\beta}|}{1+|z|}, \qquad \Big|\frac{\Gamma(z+\alpha)}{\Gamma(z+\beta)}\Big| \leq C_0(1+|z|)^{\Re(\alpha-\beta)}.$$

The same holds for z in a closed sector $|\arg z| \leq \pi - \delta$, $|z| \geq R$, after enlarging C_0 and R.

Proof. For $|z|$ large, $z + \alpha$ and $z + \beta$ lie in a sector $|\arg w| \leq \pi - \delta'$, where $\log\Gamma(w) = (w - \frac{1}{2})\log w - w + \frac{1}{2}\log 2\pi + O(|w|^{-1})$ holds uniformly [27]. Expanding gives $\log\Gamma(z+\alpha) - \log\Gamma(z+\beta) = (\alpha - \beta)\log(1+z) + O(|z|^{-1})$ uniformly in $\alpha, \beta \in \Lambda_0$. On the remaining compact set of z both sides are continuous, because $1/\Gamma$ is entire and $\Re(z + \alpha) \geq c_0$. □

Lemma (majorant). *Fix $0 < r < \frac{1}{2}$. There is C_r such that for all $\ell \geq 0$, $\nu \in \mathbb{R}$ and complex $|\tau| \leq r$,*

$$\left|t_\ell(\nu)^2 \left[R_\tau(\mathrm{i}\nu) - S_\tau(\mathrm{i}\nu)\right]\right| \leq C_r(1+|\xi|)^{-3+2r}. \tag{379}$$

Hence $\mathcal{D}(\tau) = \langle t_\ell^2(R_\tau - S_\tau)\rangle_2$ converges absolutely and uniformly on $|\tau| \leq r$, and $\mathcal{D}$ is holomorphic in $|\tau| < \frac{1}{2}$. The function $\mathcal{D}$ is $F(\tau)$ of (555) at $d = 2$.

Proof. With $\phi(x) = \Gamma(\frac{1}{4}+x)/\Gamma(\frac{3}{4}+x)$ one has $t_\ell = \phi(\xi)\phi(\bar{\xi})$,

$$R_\tau(\mathrm{i}\nu) = \frac{\Gamma(\frac{1}{4}+\xi)}{\Gamma(\frac{1}{4}+\tau+\xi)} \cdot \frac{\Gamma(\frac{3}{4}-\tau+\bar{\xi})}{\Gamma(\frac{3}{4}+\bar{\xi})}, \qquad S_\tau(\mathrm{i}\nu) = \vartheta_\tau(\xi)\vartheta_\tau(\bar{\xi}),$$
$$\vartheta_\tau(x) = \Big[\frac{\Gamma(\frac{1}{4}-\frac{\tau}{4}+x)\,\Gamma(\frac{3}{4}+x)}{\Gamma(\frac{1}{4}+x)\,\Gamma(\frac{3}{4}+\frac{\tau}{4}+x)}\Big]^2.$$

Every Gamma function in a numerator has an argument with real part at least $\frac{1}{4} - \frac{r}{4} > 0$. The lemma on uniform Gamma ratios gives $|t_\ell|^2 \leq C(1+|\xi|)^{-2}$ and

$$R_\tau(\mathrm{i}\nu),\ S_\tau(\mathrm{i}\nu) = (1+\xi)^{-\tau}(1+\bar{\xi})^{-\tau}\Big(1 + O\Big(\frac{1}{1+|\xi|}\Big)\Big)$$

uniformly in $|\tau| \leq r$. Since $\arg(1+\bar{\xi}) = -\arg(1+\xi)$, the product $(1+\xi)^{-\tau}(1+\bar{\xi})^{-\tau}$ equals $|1+\xi|^{-2\tau}$, whose modulus is at most $(1+|\xi|)^{2r}$. Hence

$$R_\tau(\mathrm{i}\nu) - S_\tau(\mathrm{i}\nu) = |1+\xi|^{-2\tau}\, O\big((1+|\xi|)^{-1}\big) \qquad \text{uniformly in } |\tau| \leq r, \tag{380}$$

and (379) follows. The majorant is summable: $\int_{\mathbb{R}}(1+|\frac{\ell}{2}+\mathrm{i}\nu|)^{-3+2r}\,d\nu \leq C(1+\ell)^{-2+2r}$ and $\sum_\ell (1+\ell)^{-2+2r} < \infty$. The integrands are holomorphic in τ, so dominated convergence and Morera's theorem give holomorphy. □

Corollary (regularized form). *(i) $\mathcal{D}(\tau) = \mathcal{D}_2\tau^2 + \mathcal{D}_3\tau^3 + O(\tau^4)$, with $\mathcal{D}_2 = \langle t_\ell^2 E_2\rangle_2$ and $\mathcal{D}_3 = \langle t_\ell^2[-pE_2 - \frac{hk}{2} - \frac{5r}{32}]\rangle_2$. (ii) $C(\tau)\mathcal{D}(\tau)$ is holomorphic in $|\tau| < \frac{1}{2}$, and $[\tau^2]\{C(\tau)\mathcal{D}(\tau)\} = \langle t_\ell^2 w_{\mathcal{K}}\rangle_2$. Hence the right side of* (248) *at $d = 2$ is*

$$-\frac{AU}{4} + \frac{1}{4}\langle t_\ell^2 w_{\mathcal{K}}\rangle_2 = 14\zeta(3)\log 2 + \frac{1}{4}[\tau^2]\{C(\tau)\mathcal{D}(\tau)\}. \tag{381}$$

(iii) For real $0 < \tau < \frac{1}{2}$, $\mathcal{D}(\tau) = \langle t_\ell^2 R_\tau\rangle_2 - \langle t_\ell^2 S_\tau\rangle_2$, and both traces converge absolutely.

Proof. (i) By Cauchy's formula on $|\tau| = r$ and Fubini's theorem, justified by (379), $[\tau^n]\mathcal{D} = \langle t_\ell^2\,[\tau^n](R_\tau - S_\tau)\rangle_2$. Taylor expansion of the Gamma functions gives

$$\log R_\tau(s) = \sum_{k\geq 1} \frac{(-\tau)^k\psi_{k-1}(b-s) - \tau^k\psi_{k-1}(a+s)}{k!},$$
$$\log S_\tau(s) = 2\sum_{k\geq 1} \frac{(-\frac{\tau}{4})^k\{\psi_{k-1}(a+s)+\psi_{k-1}(a-s)\} - (\frac{\tau}{4})^k\{\psi_{k-1}(b+s)+\psi_{k-1}(b-s)\}}{k!}.$$

Expanding the exponentials to third order and symmetrizing under $s \to -s$ gives the finite polynomial identity

$$\tfrac{1}{2}\big[(R_\tau - S_\tau)(s) + (R_\tau - S_\tau)(-s)\big] = E_2\tau^2 + \Big[-pE_2 - \frac{hk}{2} - \frac{5r}{32}\Big]\tau^3 + O(\tau^4),$$

with p, h, j, k, r of (247) written in the polygamma values at $a \pm s$, $b \pm s$. This is the expansion displayed after (250). It is checked in exact computer algebra, at $q = \frac{1}{2}$ in `LRI_C5_d2_simplify/proof/K2-parseval/symbolic_checks.py` (part A) and for arbitrary real a, b in `LRI_C5_d2_proof/K2_gauss/check_expansion.wl`. The parts odd in ν integrate to zero in the full-line trace. (ii)

$C(\tau)=\tau^{-1}\Gamma(1+\tau)/\Gamma(1-\tau)$ has a simple pole with residue 1, so $[\tau^2](C\mathcal{D})=\mathcal{D}_3+c_*\mathcal{D}_2=\langle t_\ell^2 w_{\mathcal{K}}\rangle_2$. (iii) For real $\tau\in(0,\frac12)$ the lemma on uniform Gamma ratios gives $|t_\ell^2 R_\tau|, |t_\ell^2 S_\tau|\le C(1+|\xi|)^{-2-2\tau}$, which is summable. □

Part (i) supplies, at $d=2$, the interchange of $[\tau^2]$ with the trace that the derivation of (248) after (250) uses.

A sharper estimate for the even part. The bound (379) does not use the cancellation in the even part $D^{\mathrm{e}}_\tau(\nu)=\frac12[R_\tau(\mathrm{i}\nu)+R_\tau(-\mathrm{i}\nu)]-S_\tau(\mathrm{i}\nu)$. Fix $\rho_0=\frac{1}{32}$ and write $z=a+\mathrm{i}\nu$. There is C such that, for all $\ell\ge0$, $\nu\in\mathbb{R}$ and $|\tau|\le\rho_0$,

$$|t_\ell(\nu)|\le C(1+|z|)^{-1},\qquad |D^{\mathrm{e}}_\tau(\nu)|\le C|\tau|^2(1+|z|)^{-2+2\rho_0}. \tag{382}$$

Proof. The first bound is $|\Gamma(w)/\Gamma(w+\frac12)|\le C|w|^{-1/2}$ for $\Re w\ge\frac14$ [27], with compactness. For the second, all arguments $w\in\{a\pm s, b\pm s\}$ have $\Re w\ge\frac14>\rho_0$, so the series $\log\Gamma(w+\alpha\tau)-\log\Gamma(w)=\sum_{k\ge1}\psi_{k-1}(w)(\alpha\tau)^k/k!$ with $|\alpha|\le1$ converge. The polygamma functions obey $|\psi_{k-1}(w)|\le(k-1)!\sum_{n\ge0}|w+n|^{-k}\le(k-1)!\,4^kC_0(1+|w|)^{-2}$ for $k\ge3$, and $\psi_1(w)-\psi_1(w+\frac12)=O(|w|^{-2})$, $\psi_{k-1}(w)=O(|w|^{1-k})$, $\psi_0(w)=\log w+O(|w|^{-1})$. Put $A_k=\psi_{k-1}(a+s)+\psi_{k-1}(a-s)$, $B_k=\psi_{k-1}(b+s)+\psi_{k-1}(b-s)$, $\Delta_{\mathrm{e}}=\frac12[\log R_\tau(s)+\log R_\tau(-s)]-\log S_\tau(s)$ and $\varepsilon_{\mathrm{o}}=\frac12[\log R_\tau(s)-\log R_\tau(-s)]$. The two expansions above give

$$[\tau^k]\Delta_{\mathrm{e}}=\frac{1}{k!}\Big[-(\tfrac12+2(-4)^{-k})A_k+(\tfrac{(-1)^k}{2}+2\cdot4^{-k})B_k\Big].$$

The coefficient vanishes for $k=1$; it is $-\frac{5}{16}(A_2-B_2)=O(|z|^{-2})$ for $k=2$ and $-\frac{15}{192}(A_3+B_3)=O(|z|^{-2})$ for $k=3$; and it is $O((8^k/k)|z|^{-2})$ for $k\ge4$. Hence $|\Delta_{\mathrm{e}}|\le C|\tau|^2|z|^{-2}$. Similarly $|\varepsilon_{\mathrm{o}}|\le C|\tau||z|^{-1}$; its τ^1 coefficient is $\frac12[(\psi_0(b+s)-\psi_0(a+s))-(\psi_0(b-s)-\psi_0(a-s))]$. Finally $|\log S_\tau|\le2|\tau|\log(2+|z|)+C|\tau|$. Since $D^{\mathrm{e}}_\tau=S_\tau(e^{\Delta_{\mathrm{e}}}\cosh\varepsilon_{\mathrm{o}}-1)$, it is bounded by $C(2+|z|)^{2\rho_0}(|\Delta_{\mathrm{e}}|+|\varepsilon_{\mathrm{o}}|^2)$, which proves (382) for $|z|\ge z_0$. For $|z|\le z_0$ (finitely many spins and bounded ν) compactness and $D^{\mathrm{e}}_\tau=O(\tau^2)$ suffice. □

Consequently $C(\tau)D^{\mathrm{e}}_\tau$, whose pole at $\tau=0$ is cancelled, is holomorphic on $|\tau|\le\rho_0$, and by the maximum principle $|t_\ell^2C(\tau)D^{\mathrm{e}}_\tau|\le C(1+|z|)^{-4+2\rho_0}$ there. Since $[\tau^2]\{C(\tau)D^{\mathrm{e}}_\tau\}=w_{\mathcal{K}}$ (by the expansion in the proof of part (i) of the corollary and $C(\tau)=\tau^{-1}+c_*+O(\tau)$), Cauchy's estimate gives $|t_\ell^2 w_{\mathcal{K}}|\le C(1+|z|)^{-4+2\rho_0}$, so the spin term of (248) at $d=2$ is $O(\ell^{-3+2\rho_0})$, consistent with the decay $O(\ell^{-3}\log\ell)$ stated after (248).

18.9.3 Residue families on the square lattice

For $0<\tau<\frac12$ define the lattice functions (the sans-serif letters are local to this subsection)

$$\begin{aligned}\mathsf{h}(x)&=\frac{\Gamma(x+\frac12)^3}{\Gamma(x+1)^2\,\Gamma(x+\frac12+\tau)},&\quad \mathsf{k}(x)&=\frac{\Gamma(x+1-\tau)^3}{\Gamma(x+\frac32-\tau)^2\,\Gamma(x+1)},\\ \mathsf{m}(x)&=\Big[\frac{\Gamma(x+\frac{1-\tau}{2})}{\Gamma(x+1)}\Big]^2,&\quad \mathsf{n}(x)&=\frac{\Gamma(x+\frac12)^2\,\Gamma(x+1-\tau)}{\Gamma(x+1)^3}.\end{aligned} \tag{383}$$

For $x\ge0$, Stirling's formula gives $\mathsf{h},\mathsf{k},\mathsf{m},\mathsf{n}=O((1+x)^{-1-\tau})$. Since h'/h and m'/m are combinations of digamma differences, which are $O(1/(1+x))$, also $\mathsf{h}',\mathsf{m}'=O((1+x)^{-2-\tau})$. Note the shift relation $\mathsf{h}(x+\frac12-\tau)=\mathsf{k}(x)$. The single sums

$$\begin{aligned}\Sigma_{\mathsf{h}}&=\sum_{m\ge0}\mathsf{h}(m),&\quad \Sigma'_{\mathsf{h}}&=\sum_{m\ge0}\mathsf{h}'(m),&\quad \Sigma_{\mathsf{k}}&=\sum_{m\ge0}\mathsf{k}(m),\\ \Sigma_{\mathsf{m}}&=\sum_{m\ge0}\mathsf{m}(m),&\quad \Sigma'_{\mathsf{m}}&=\sum_{m\ge0}\mathsf{m}'(m),&\quad \Sigma_{\mathsf{n}}&=\sum_{k\ge0}\mathsf{n}(k)\end{aligned}$$

therefore converge absolutely for $\tau > 0$. At fixed spin the two integrands are

$$\Theta^{\mathrm{R}}_\ell(s) = g_\ell(s)^2 R_\tau(s) = \frac{\Gamma(a+s)^3\,\Gamma(a-s)^2\,\Gamma(b-\tau-s)}{\Gamma(b+s)^2\,\Gamma(b-s)^3\,\Gamma(a+\tau+s)}, \qquad \Theta^{\mathrm{S}}_\ell(s) = g_\ell(s)^2 S_\tau(s) = g_\ell(s;1-\tau)^2,$$

where $g_\ell(s;1-\tau)$ is the balanced kernel of rim exponent $1-\tau$ of (557), that is $\Gamma(a'+s)\Gamma(a'-s)/[\Gamma(b'+s)\Gamma(b'-s)]$ with $a' = a - \frac{\tau}{4}$ and $b' = b + \frac{\tau}{4}$. Thus $\langle t_\ell^2 R_\tau\rangle_2 = \sum_\ell d_\ell \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}} \Theta^{\mathrm{R}}_\ell(s)$, and similarly for S_τ.

Lemma (per-spin residues). *Let $\ell \geq 0$, $0 < \tau < \frac{1}{2}$ and $m = \ell + n$. Then, with absolutely convergent series,*

$$\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\,\Theta^{\mathrm{R}}_\ell(s) = \sum_{n\geq 0}\big[\rho_{\mathsf{h}}(m,n) + \rho_{\mathsf{k}}(m,n)\big], \qquad \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\,\Theta^{\mathrm{S}}_\ell(s) = \sum_{n\geq 0}\rho_{\mathsf{m}}(m,n), \tag{384}$$

$$\rho_{\mathsf{h}}(m,n) = -\frac{1}{\pi^2 c_\tau}\Big[\mathsf{h}'(m)\mathsf{h}(n) + \mathsf{h}(m)\mathsf{h}'(n) + \pi\tan(\pi\tau)\,\mathsf{h}(m)\mathsf{h}(n)\Big], \qquad \rho_{\mathsf{k}}(m,n) = \frac{s_\tau^3}{\pi c_\tau^2}\,\mathsf{k}(m)\mathsf{k}(n),$$

$$\rho_{\mathsf{m}}(m,n) = -\frac{1}{\pi^2}\Big[\cos^2(\tfrac{\pi\tau}{2})\big(\mathsf{m}'(m)\mathsf{m}(n) + \mathsf{m}(m)\mathsf{m}'(n)\big) + \pi s_\tau\,\mathsf{m}(m)\mathsf{m}(n)\Big].$$

Proof. Poles. In $\Re s > 0$ the poles of Θ^{R}_ℓ are the double poles $s = a+n$, from $\Gamma(a-s)^2$, and the simple poles $s = b-\tau+n$, from $\Gamma(b-\tau-s)$, $n \geq 0$; the other factors are finite and nonzero there because $\tau \notin \frac{1}{2}+\mathbb{Z}$ and $\tau \notin \mathbb{Z}$. The two families are disjoint because $b-\tau-a = \frac{1}{2}-\tau \notin \mathbb{Z}$, and none lies on $\Re s = 0$, since $a > 0$ and $b-\tau \geq \frac{3}{4}-\tau > 0$. The factors with left poles are holomorphic on $\Re s \geq 0$.

Arcs. Gamma reflection of the three factors with right poles, together with $\sin\pi(b-s) = \cos\pi(a-s)$, gives the identity $\Theta^{\mathrm{R}}_\ell = G_\ell\,T_\tau$, with

$$G_\ell(s) = \Big[\frac{\Gamma(a+s)\Gamma(1-b+s)}{\Gamma(b+s)\Gamma(1-a+s)}\Big]^2 \frac{\Gamma(a+s)\Gamma(1-b+s)}{\Gamma(a+\tau+s)\Gamma(1-b+\tau+s)},$$
$$T_\tau(s) = \cot^2\pi(a-s)\,\frac{\cos\pi(a-s)}{\sin\pi(b-\tau-s)}.$$

For fixed ℓ, G_ℓ is holomorphic on $\{\Re s \geq 0,\ |s| \geq b\}$; there every Gamma argument stays in a sector $|\arg| \leq \frac{3}{4}\pi$, and Stirling's formula gives $|G_\ell(s)| \leq C_\ell |s|^{-2-2\tau}$. Inside $|s| < b$ the factors have poles and zeros that cancel pairwise in Θ^{R}_ℓ: the triple poles of $\Gamma(1-b+s)^3$ at $s = b-1-k$ against the triple zeros of $\cot^2\pi(a-s)\cos\pi(a-s)$; the double poles of $\cot^2\pi(a-s)$ at $s = a-k$, $k \geq 1$, against the double zeros of $1/\Gamma(1-a+s)^2$; and the poles of $1/\sin\pi(b-\tau-s)$ at $s = b-\tau-k$, $k \geq 1$, against the zeros of $1/\Gamma(1-b+\tau+s)$. The residue theorem is therefore applied to Θ^{R}_ℓ itself. The factor T_τ is 1-periodic and bounded on $|\Im s| \geq 1$, and it is bounded on the circles $|s| = \rho_N := a + N + \frac{3}{4} - \frac{\tau}{2}$, which stay at distance at least $\frac{1}{4} - O(\rho_N^{-1})$ from both pole families. So $|\Theta^{\mathrm{R}}_\ell| \leq C\rho_N^{-2-2\tau}$ on the right half-circles, and the arcs, together with the parts of the Mellin line beyond ρ_N (where $|\Theta^{\mathrm{R}}_\ell(\mathrm{i}\nu)| \leq C(1+|\nu|)^{-2-2\tau}$), contribute $O(\rho_N^{-1-2\tau})$. The right half-discs are traversed clockwise, so

$$\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\,\Theta^{\mathrm{R}}_\ell = -\sum_{n\geq 0}\Big[\mathrm{Res}_{s=a+n} + \mathrm{Res}_{s=b-\tau+n}\Big]\Theta^{\mathrm{R}}_\ell,$$

the residue series converging absolutely by the bounds below. (The number of poles of each family inside a given arc is N or $N+1$; this does not matter, since the series converges absolutely.)

Residues. At $s = a+n+u$ one has $a+s = m+\frac{1}{2}+u$, $b+s = m+1+u$ and $a+\tau+s = m+\frac{1}{2}+\tau+u$, while reflection gives

$$\frac{\Gamma(a-s)^2\,\Gamma(b-\tau-s)}{\Gamma(b-s)^3} = \frac{\cos^3\pi u}{\sin^2\pi u\,\cos\pi(\tau+u)}\,\mathsf{h}(n+u),$$

the signs $(-1)^{2n}$, $(-1)^n$ and $(-1)^{3n}$ of the three reflected factors cancelling. Hence $\Theta^{\mathrm{R}}_\ell = (\pi u)^{-2}\Psi_\tau(u)\,\mathsf{h}(m+u)\mathsf{h}(n+u)$ with $\Psi_\tau(u) = (\pi u)^2\cos^3\pi u/[\sin^2\pi u\cos\pi(\tau+u)] = c_\tau^{-1}(1+\pi u\tan\pi\tau + O(u^2))$, and the residue is $\pi^{-2}[c_\tau^{-1}(\mathsf{h}'(m)\mathsf{h}(n)+\mathsf{h}(m)\mathsf{h}'(n)) + \pi s_\tau c_\tau^{-2}\mathsf{h}(m)\mathsf{h}(n)] = -\rho_{\mathsf{h}}(m,n)$. At $s = b-\tau+n+u$ only $\Gamma(b-\tau-s) = \Gamma(-n-u)$ is singular, with residue $-(-1)^n/n!$ in s. At $u=0$ the remaining factors are $\Gamma(a+s)^3/[\Gamma(b+s)^2\Gamma(a+\tau+s)] = \mathsf{k}(m)$ and

$$\frac{\Gamma(a-s)^2}{\Gamma(b-s)^3} = \frac{\Gamma(\tau-\frac12-n)^2}{\Gamma(\tau-n)^3} = (-1)^n\frac{s_\tau^3}{\pi c_\tau^2}\,\frac{\Gamma(n+1-\tau)^3}{\Gamma(n+\frac32-\tau)^2},$$

so the residue is $-(s_\tau^3/\pi c_\tau^2)\mathsf{k}(m)\mathsf{k}(n) = -\rho_{\mathsf{k}}(m,n)$.

The S family. The function $\Theta^{\mathrm{S}}_\ell = g_\ell(s;1-\tau)^2$ has in $\Re s>0$ only the double poles $s = a'+n$. Put $x_\tau = \frac{1-\tau}{2}$; then $2a' = \ell + x_\tau$, $a'+b' = \ell+1$ and $b'-a' = 1-x_\tau$. At $s = a'+n+u$, reflection gives

$$g_\ell(s;1-\tau) = -\frac{\sin\pi(x_\tau+u)}{\sin\pi u}\,\frac{\Gamma(m+x_\tau+u)\,\Gamma(n+x_\tau+u)}{\Gamma(m+1+u)\,\Gamma(n+1+u)},$$
$$\Theta^{\mathrm{S}}_\ell = \frac{\sin^2\pi(x_\tau+u)}{\sin^2\pi u}\,\mathsf{m}(m+u)\mathsf{m}(n+u).$$

Since $\sin^2\pi(x_\tau+u)/\sin^2\pi u = (\pi u)^{-2}[\sin^2\pi x_\tau + \pi u\sin 2\pi x_\tau + O(u^2)]$, $\sin\pi x_\tau = \cos\frac{\pi\tau}{2}$ and $\sin 2\pi x_\tau = s_\tau$, the residue is $\pi^{-2}[\pi s_\tau\mathsf{m}(m)\mathsf{m}(n) + \cos^2(\frac{\pi\tau}{2})(\mathsf{m}'(m)\mathsf{m}(n)+\mathsf{m}(m)\mathsf{m}'(n))] = -\rho_{\mathsf{m}}(m,n)$. On the circles $|s| = a'+N+\frac12$, at distance $\frac12$ from the poles, reflection and Stirling's formula give $|g_\ell(s;1-\tau)| \le C|s|^{-1-\tau}$, so the arcs vanish and the second identity of (384) follows as the first. Absolute convergence follows from $|\rho_{\mathsf{h}}|, |\rho_{\mathsf{k}}|, |\rho_{\mathsf{m}}| \le C(1+m)^{-1-\tau}(1+n)^{-1-\tau}$. □

Proposition (factorized traces). *For $0<\tau<\frac12$,*

$$\begin{aligned}\langle t_\ell^2 R_\tau\rangle_2 &= -\frac{1}{\pi^2 c_\tau}\Big[2\Sigma_{\mathsf{h}}\Sigma'_{\mathsf{h}} + \pi\tan(\pi\tau)\,\Sigma^2_{\mathsf{h}}\Big] + \frac{s_\tau^3}{\pi c_\tau^2}\,\Sigma^2_{\mathsf{k}},\\ \langle t_\ell^2 S_\tau\rangle_2 &= -\frac{1}{\pi^2}\Big[2\cos^2(\tfrac{\pi\tau}{2})\,\Sigma_{\mathsf{m}}\Sigma'_{\mathsf{m}} + \pi s_\tau\,\Sigma^2_{\mathsf{m}}\Big].\end{aligned} \tag{385}$$

Proof. If ρ is symmetric on $\mathbb{Z}^2_{\ge0}$ and absolutely summable, then $\sum_{\ell\ge0} d_\ell\sum_{n\ge0}\rho(\ell+n,n) = \sum_{m,n\ge0}\rho(m,n)$: the spin ℓ labels the diagonal $m-n=\ell$, $d_0=1$ counts the main diagonal, and $d_\ell = 2$ counts the two diagonals $m-n=\pm\ell$, which carry equal sums. Apply this to $\rho_{\mathsf{h}}+\rho_{\mathsf{k}}$ and to ρ_{m}, which are symmetric and absolutely summable for $\tau>0$, and factor the absolutely convergent double sums. □

This is the spectral form, at $\tau>0$, of the square-lattice factorization (336). With $\hat{\mathsf{h}}(z) = \sum_m \mathsf{h}(m)z^m$ and similarly for the other sequences, the Fourier series $\sum_{m,n}(\rho_{\mathsf{h}}+\rho_{\mathsf{k}})(m,n)e^{\mathrm{i}(m-n)\theta}$ is the same bilinear combination of $\hat{\mathsf{h}}, \hat{\mathsf{h}}', \hat{\mathsf{k}}$ and their conjugates at $z = e^{\mathrm{i}\theta}$. For $\tau>0$ all coefficient sequences are absolutely summable, these functions are continuous on the closed disk, and the strip limit $\theta\to0$ is evaluation at $z=1$. At $\tau=0$ the same limit needs the logarithmic singular parts of the single sums (Section 18.9.7).

18.9.4 The single sums in closed form

Lemma (two-mode periodization). *For $0<\tau<\frac12$,*

$$\Sigma'_{\mathsf{h}} = -\pi s_\tau\,\Sigma_{\mathsf{n}}, \qquad \Sigma_{\mathsf{h}} = c_\tau\,\Sigma_{\mathsf{n}} + s_\tau^2\,\Sigma_{\mathsf{k}}. \tag{386}$$

Proof. Bilateral sum. For $k\ge0$, reflection gives identically in η

$$\mathsf{h}(-k-1+\eta) = T_{\mathsf{h}}(\eta)\,\mathsf{n}(k+\tfrac12-\eta), \qquad T_{\mathsf{h}}(\eta) = \frac{\sin^2(\pi\eta)\,\cos\pi(\eta+\tau)}{\cos^3(\pi\eta)},$$

since the three factors $\Gamma(-k-\frac12+\eta)^3$, $\Gamma(-k+\eta)^{-2}$ and $\Gamma(-k-\frac12+\tau+\eta)^{-1}$ produce the signs $(-1)^{3k+3}$, $(-1)^{2k}$ and $(-1)^{k+1}$, which cancel. The series $\sum_{m\ge0}\mathsf{h}(m+\eta)$ and $\sum_{k\ge0}\mathsf{n}(k+\frac12-\eta)$ converge locally uniformly and are holomorphic for $\Re\eta>-\frac12$ and $\Re\eta<1$ respectively, because $\mathsf{h},\mathsf{n}=O(|x|^{-1-\tau})$ and the poles of n lie at $-\frac12-\mathbb{Z}_{\ge0}$ and $-1+\tau-\mathbb{Z}_{\ge0}$. Hence the bilateral sum $\Sigma^{\mathbb{Z}}_{\mathsf{h}}(\eta)=\sum_{m\in\mathbb{Z}}\mathsf{h}(m+\eta)$ satisfies, on the strip $-\frac12<\Re\eta<1$,

$$\Omega_{\mathsf{h}}(\eta):=\cos^3(\pi\eta)\,\Sigma^{\mathbb{Z}}_{\mathsf{h}}(\eta)=\cos^3(\pi\eta)\sum_{m\ge0}\mathsf{h}(m+\eta)+\sin^2(\pi\eta)\cos\pi(\eta+\tau)\sum_{k\ge0}\mathsf{n}(k+\tfrac12-\eta),$$

and the right side is holomorphic there. *Entire and antiperiodic.* $\Sigma^{\mathbb{Z}}_{\mathsf{h}}$ is 1-periodic and $\cos^3(\pi\eta)$ changes sign under $\eta\mapsto\eta+1$. Since the strip has width $\frac32>1$, Ω_{h} extends to an entire function with $\Omega_{\mathsf{h}}(\eta+1)=-\Omega_{\mathsf{h}}(\eta)$. *Growth.* For $0\le\Re\eta\le1$ and $|\Im\eta|=|y|\ge1$, the lemma on uniform Gamma ratios in the sector $|\arg|\le\pi-\delta$ for $|y|\ge R$, and compactness for $1\le|y|\le R$ (there are no poles off the real axis), give $|\sum_{m\ge0}\mathsf{h}(m+\eta)|,|\sum_{k\ge0}\mathsf{n}(k+\frac12-\eta)|\le C\sum_{k\ge0}|k+\mathrm{i}y|^{-1-\tau}\le C'|y|^{-\tau}$; for the term with real part in $[-\frac12,0)$ the argument stays in such a sector because $|y|\ge1$. Hence $|\Omega_{\mathsf{h}}(\eta)|\le Ce^{3\pi|y|}|y|^{-\tau}$. *Two modes.* An entire function of period 2 that changes sign under $\eta\mapsto\eta+1$ has an expansion $\sum_{k\text{ odd}}\hat\omega_k e^{\mathrm{i}\pi k\eta}$, with $\hat\omega_k=\frac12\int_0^2\Omega_{\mathsf{h}}(x+\mathrm{i}y)e^{-\mathrm{i}\pi k(x+\mathrm{i}y)}\,dx$ for every y. Thus $|\hat\omega_k|\le Ce^{3\pi|y|}|y|^{-\tau}e^{\pi ky}$. Letting $y\to-\infty$ for $k\ge3$ and $y\to+\infty$ for $k\le-3$ gives $\hat\omega_k=0$ for $|k|\ge3$; for $|k|=3$ this uses the factor $|y|^{-\tau}$, that is $\tau>0$. Therefore $\Omega_{\mathsf{h}}(\eta)=\alpha_0\cos\pi\eta+\beta_0\sin\pi\eta$. *Three evaluations.* At $\eta=0$ (value and derivative; the second term of Ω_{h} has a double zero there, and termwise differentiation is legitimate), $\alpha_0=\Sigma_{\mathsf{h}}$ and $\pi\beta_0=\Sigma'_{\mathsf{h}}$. At $\eta=\frac12$, $\beta_0=\Omega_{\mathsf{h}}(\frac12)=\cos(\frac{\pi}{2}+\pi\tau)\Sigma_{\mathsf{n}}=-s_\tau\Sigma_{\mathsf{n}}$. At $\eta=\frac12-\tau$ the second term vanishes, $\cos^3(\frac{\pi}{2}-\pi\tau)=s_\tau^3$, and $\sum_m\mathsf{h}(m+\frac12-\tau)=\Sigma_{\mathsf{k}}$ by the shift relation, so $\alpha_0 s_\tau+\beta_0c_\tau=s_\tau^3\Sigma_{\mathsf{k}}$. These three relations give (386). □

Lemma (one-mode periodization). *For $0<\tau<\frac12$ and $x_\tau=\frac{1-\tau}{2}$,*

$$\Sigma_{\mathsf{m}}=\Gamma(\tau)\,\mathcal{X}_2(\tau),\qquad\Sigma'_{\mathsf{m}}=-2\pi\tan(\tfrac{\pi\tau}{2})\,\Sigma_{\mathsf{m}},\qquad\sum_{m\in\mathbb{Z}}\mathsf{m}(m+\eta)=\Sigma_{\mathsf{m}}\frac{\sin^2\pi x_\tau}{\sin^2\pi(x_\tau+\eta)}.\tag{387}$$

Proof. Gauss's summation [4] gives $\Sigma_{\mathsf{m}}=\Gamma(x_\tau)^2{}_2F_1(x_\tau,x_\tau;1;1)=\Gamma(x_\tau)^2\Gamma(1-2x_\tau)/\Gamma(1-x_\tau)^2=\Gamma(\tau)\mathcal{X}_2(\tau)$, since $1-2x_\tau=\tau>0$. Reflection gives $\mathsf{m}(-k-1+\eta)=\frac{\sin^2\pi\eta}{\sin^2\pi(x_\tau+\eta)}\tilde{\mathsf{m}}(k-\eta)$ with $\tilde{\mathsf{m}}(x)=[\Gamma(x+1)/\Gamma(x+2-x_\tau)]^2$. Hence

$$\sin^2\pi(x_\tau+\eta)\sum_{m\in\mathbb{Z}}\mathsf{m}(m+\eta)=\sin^2\pi(x_\tau+\eta)\sum_{m\ge0}\mathsf{m}(m+\eta)+\sin^2(\pi\eta)\sum_{k\ge0}\tilde{\mathsf{m}}(k-\eta)$$

is holomorphic on $-x_\tau<\Re\eta<1$, a strip of width larger than 1; it is 1-periodic, hence entire, and it is $O(e^{2\pi|y|}|y|^{-\tau})$. Its Fourier modes $e^{2\pi\mathrm{i}k\eta}$ with $k\ne0$ vanish by the argument of the previous proof (for $|k|=1$ through the factor $|y|^{-\tau}$), so it is a constant c_{m}. The reflected part has a double zero at $\eta=0$, so $c_{\mathsf{m}}=\sin^2(\pi x_\tau)\Sigma_{\mathsf{m}}$, which is the last identity, and $\Sigma'_{\mathsf{m}}/\Sigma_{\mathsf{m}}=\frac{d}{d\eta}\log\sin^{-2}\pi(x_\tau+\eta)\big|_0=-2\pi\cot\pi x_\tau=-2\pi\tan\frac{\pi\tau}{2}$. □

The last identity in (387) is an instance of Dougall's bilateral ${}_2H_2$ summation; the proof above does not assume it.

Lemma (Euler and Thomae). *For $0<\tau<\frac12$,*

$$\Sigma_{\mathsf{n}}=\pi\Gamma(1-\tau)\,\mathfrak{F}(1-\tau)=\frac{\pi\Gamma(\tau)}{\Gamma(\frac12+\tau)^2}\,\mathfrak{G}(\tau),\qquad\Sigma_{\mathsf{k}}=\frac{\Gamma(1-\tau)^3}{\Gamma(\frac32-\tau)^2}\,\mathfrak{G}(1-\tau)=\Gamma(\tau)\Gamma(1-\tau)^2\,\mathfrak{F}(\tau).\tag{388}$$

Proof. The first form of each is the definition of the series: $\Sigma_{\mathsf{n}}=\sum_k\Gamma(k+\frac12)^2\Gamma(k+1-\tau)/k!^3$ and $\Sigma_{\mathsf{k}}=\sum_k\Gamma(k+1-\tau)^3/[\Gamma(k+\frac32-\tau)^2k!]$. The second forms follow from (82) at $t=\tau$ and at

$t = 1-\tau$; the latter reads $\mathfrak{G}(1-\tau) = \Gamma(\frac{3}{2}-\tau)^2\Gamma(\tau)\Gamma(1-\tau)^{-1}\mathfrak{F}(\tau)$, and its ${}_3F_2(1)$ has parametric excess $\tau > 0$. □

In particular $s_\tau\Sigma_{\mathsf{k}} = \pi\Gamma(1-\tau)\mathfrak{F}(\tau)$, since $\Gamma(\tau)\Gamma(1-\tau) = \pi/s_\tau$.

Second evaluation: a bilateral Laurent lemma. The derivative sums can also be obtained without the growth argument for Ω_{h}. Let $f \in \{\mathsf{h}, \mathsf{m}\}$, with $\beta_f = \frac{1}{2}$ and pole order $p_f = 3$ for h, and $\beta_f = x_\tau$, $p_f = 2$ for m.

(a) For each $\eta_0 > 0$ there is C_{η_0} with $|f(K)| \le C_{\eta_0}(1+|K|)^{-1-\tau}$ whenever $\operatorname{dist}(K, -\beta_f - \mathbb{Z}_{\ge 0}) \ge \eta_0$. For $\Re K \ge 0$ this is Stirling's formula. For $\Re K < 0$ use the reflection identities

$$\mathsf{h}(K) = \frac{\sin^2(\pi K)\,\sin\pi(K+\frac{1}{2}+\tau)}{\sin^3\pi(K+\frac{1}{2})}\,\frac{\Gamma(-K)^2\,\Gamma(\frac{1}{2}-\tau-K)}{\Gamma(\frac{1}{2}-K)^3},$$
$$\mathsf{m}(K) = \frac{\sin^2(\pi K)}{\sin^2\pi(K+x_\tau)}\,\frac{\Gamma(-K)^2}{\Gamma(1-x_\tau-K)^2}\,:$$

the trigonometric factors are 1-periodic, bounded away from the poles, and have finite limits as $\Im K \to \pm\infty$, and the Gamma ratios are $O(|K|^{-1-\tau})$ by Stirling's formula in $\Re(-K) > 0$.

(b) $P_f(u) = \sum_{k\in\mathbb{Z}} f(k+u)$ is meromorphic and 1-periodic, with poles only in $-\beta_f + \mathbb{Z}$, of order at most p_f; its Laurent coefficients a_{-j} at $u = -\beta_f$ are the sums of the termwise Laurent coefficients (by (a), the convergence is absolute and locally uniform off the poles and uniform on small circles around $-\beta_f$). Moreover $\sup_{0\le\Re u\le 1}|P_f(u)| \le C\sum_k |k+u|^{-1-\tau} \le C'|\Im u|^{-\tau} \to 0$ as $|\Im u| \to \infty$.

(c) $P_f(u) = a_{-2}\,\pi^2/\sin^2\pi(u+\beta_f) + a_{-3}\,\pi^3\cos\pi(u+\beta_f)/\sin^3\pi(u+\beta_f)$, with $a_{-3} = 0$ for m. Indeed, since $\pi^2/\sin^2\pi\epsilon = \epsilon^{-2}+O(1)$, $\pi^3\cos\pi\epsilon/\sin^3\pi\epsilon = \epsilon^{-3}+O(\epsilon)$ and $\pi\cot\pi\epsilon = \epsilon^{-1}+O(\epsilon)$, the function $P_f - a_{-2}\pi^2\sin^{-2}\pi(u+\beta_f) - a_{-3}\pi^3\cos\pi(u+\beta_f)\sin^{-3}\pi(u+\beta_f) - a_{-1}\pi\cot\pi(u+\beta_f)$ is entire, 1-periodic and bounded, hence constant by Liouville's theorem. Its limits at $\pm\mathrm{i}\infty$ are $\pm\mathrm{i}\pi a_{-1}$, so $a_{-1} = 0$ and the constant vanishes.

(d) $\sum_{k\ge 0} f'(k) = P_f'(0)$: for $k \ge 1$, $f(u-k)$ has a double zero at $u = 0$, from $1/\Gamma(K+1)^2$, while the other factors are finite at $K = -k$ because $\frac{1}{2}-k$, $\frac{1}{2}+\tau-k$ and $x_\tau - k$ are not poles; termwise differentiation of the two halves of P_f is justified by local uniform convergence.

For m, only $\mathsf{m}(-j-x_\tau+\epsilon) \simeq \epsilon^{-2}/[j!^2\Gamma(1-x_\tau-j)^2]$ contributes at $u = -x_\tau$, and $1/\Gamma(1-x_\tau-j) = (-1)^j\sin(\pi x_\tau)\Gamma(x_\tau+j)/\pi$, so $a_{-2} = \pi^{-2}\sin^2(\pi x_\tau)\Sigma_{\mathsf{m}}$. Then (c) is the last identity of (387), and (d) gives $\Sigma'_{\mathsf{m}} = -2\pi\cot(\pi x_\tau)\Sigma_{\mathsf{m}}$. For h, only the terms $\mathsf{h}(-\frac{1}{2}-j+\epsilon)$ have a pole at $u = -\frac{1}{2}$. With $\Gamma(-j+\epsilon)^3 = (-1)^j\epsilon^{-3}/j!^3 + \ldots$, $1/\Gamma(\frac{1}{2}-j)^2 = \Gamma(\frac{1}{2}+j)^2/\pi^2$ and $1/\Gamma(\tau-j) = (-1)^j s_\tau\Gamma(1+j-\tau)/\pi$,

$$a_{-3} = \frac{s_\tau}{\pi^3}\sum_{j\ge 0}\frac{\Gamma(\frac{1}{2}+j)^2\,\Gamma(1+j-\tau)}{j!^3} = \frac{s_\tau}{\pi^3}\,\Sigma_{\mathsf{n}},$$

which converges since $\tau > 0$. With $\beta_f = \frac{1}{2}$, (c) reads $P_{\mathsf{h}}(u) = a_{-2}\pi^2\sec^2\pi u - a_{-3}\pi^3\tan(\pi u)\sec^2(\pi u)$, so (d) gives $\Sigma'_{\mathsf{h}} = P'_{\mathsf{h}}(0) = -\pi^4 a_{-3} = -\pi s_\tau\Sigma_{\mathsf{n}}$, the first identity of (386). (As a by-product, $P_{\mathsf{h}}(0) = a_{-2}\pi^2 = \Sigma_{\mathsf{h}}$, because $\mathsf{h}(-k) = 0$ for $k \ge 1$.) The undifferentiated sum follows from the Euler integral of Section 8: with the moments I_j listed before (77), at $t = \tau$, termwise integration with $c_n^2 B(n+\frac{1}{2},\tau) = \Gamma(\tau)\mathsf{h}(n)/\pi$, where $c_n = (\frac{1}{2})_n/n!$, gives

$$\Sigma_{\mathsf{h}} = \frac{\pi^{3/2}}{\Gamma(\frac{1}{2}+\tau)}\,{}_3F_2\left(\begin{matrix}\frac{1}{2},\frac{1}{2},\frac{1}{2}\\ 1,\frac{1}{2}+\tau\end{matrix};1\right) = \frac{2I_1}{\Gamma(\tau)} = \frac{\pi c_\tau\Gamma(\tau)}{\Gamma(\frac{1}{2}+\tau)^2}\,\mathfrak{G}(\tau) + \frac{\pi^2}{\Gamma(\tau)}\,\mathfrak{F}(\tau), \tag{389}$$

the last step by $I_1 = c_\tau I_3 + s_\tau I_4$ of (77) and the values of I_3, I_4 stated after it. By (388) the two terms are $c_\tau\Sigma_{\mathsf{n}}$ and $s_\tau^2\Sigma_{\mathsf{k}}$, which reproves the second identity of (386).

Third evaluation: the logarithmic expansion of K. The derivative sum is also a single elliptic moment. For $0<x<1$ the logarithmic connection formula [23] gives

$$K(1-x)=\frac{1}{2}\sum_{n\ge 0}c_n^2x^n\,[2\psi_0(n+1)-2\psi_0(n+\tfrac12)-\log x],$$

with positive terms. Since $\int_0^1 x^{n-1/2}(1-x)^{\tau-1}\log x\,dx=B(n+\frac12,\tau)[\psi_0(n+\frac12)-\psi_0(n+\frac12+\tau)]$ and $\mathsf{h}'(n)/\mathsf{h}(n)=3\psi_0(n+\frac12)-2\psi_0(n+1)-\psi_0(n+\frac12+\tau)$, Tonelli's theorem gives

$$I_2=E(\tau,\tfrac12)=\int_0^1 x^{-1/2}(1-x)^{\tau-1}K(1-x)\,dx=-\frac{\Gamma(\tau)}{2\pi}\,\Sigma_{\mathsf{h}}',$$

so $\Sigma_{\mathsf{h}}'=-2\pi I_2/\Gamma(\tau)$. By (77), $I_2=s_\tau I_3=\frac{\pi^2}{2}\mathfrak{F}(1-\tau)$ and $I_1=c_\tau I_3+s_\tau I_4=\frac{\pi^2}{2}[\mathfrak{F}(\tau)+\cot(\pi\tau)\mathfrak{F}(1-\tau)]$. Hence

$$\Sigma_{\mathsf{h}}=\frac{\pi^2}{\Gamma(\tau)}\,[\mathfrak{F}(\tau)+\cot(\pi\tau)\,\mathfrak{F}(1-\tau)],\qquad \Sigma_{\mathsf{h}}'=-\frac{\pi^3}{\Gamma(\tau)}\,\mathfrak{F}(1-\tau),\qquad \Sigma_{\mathsf{k}}=\Gamma(\tau)\Gamma(1-\tau)^2\,\mathfrak{F}(\tau),$$

in agreement with (386) and (388). This route uses the elliptic moments and the moment relations in place of periodization.

18.9.5 The regulated traces and the mixed-trace step

Theorem (regulated traces). *For $0<\tau<\frac12$ both traces converge absolutely, and*

$$\boxed{C(\tau)\langle t_\ell^2R_\tau\rangle_2=\pi^2\,[\mathfrak{F}(\tau)^2+\mathfrak{F}(1-\tau)^2],\qquad \langle t_\ell^2S_\tau\rangle_2=C(\tau)\,\mathcal{X}_2(\tau)^2.}\tag{390}$$

Proof. Insert (386) into (385). Since $-2\Sigma_{\mathsf{h}}\Sigma_{\mathsf{h}}'=2\pi s_\tau\Sigma_{\mathsf{n}}\Sigma_{\mathsf{h}}$ and $\Sigma_{\mathsf{h}}=c_\tau\Sigma_{\mathsf{n}}+s_\tau^2\Sigma_{\mathsf{k}}$,

$$-\frac{1}{\pi^2c_\tau}\Big[2\Sigma_{\mathsf{h}}\Sigma_{\mathsf{h}}'+\pi\frac{s_\tau}{c_\tau}\Sigma_{\mathsf{h}}^2\Big]=\frac{s_\tau}{\pi c_\tau^2}\,\Sigma_{\mathsf{h}}(2c_\tau\Sigma_{\mathsf{n}}-\Sigma_{\mathsf{h}})=\frac{s_\tau}{\pi c_\tau^2}(c_\tau^2\Sigma_{\mathsf{n}}^2-s_\tau^4\Sigma_{\mathsf{k}}^2).$$

Adding the k family, the factors c_τ^{-2} cancel:

$$\langle t_\ell^2R_\tau\rangle_2=\frac{s_\tau}{\pi}\,[\Sigma_{\mathsf{n}}^2+s_\tau^2\Sigma_{\mathsf{k}}^2]=\pi s_\tau\Gamma(1-\tau)^2\,[\mathfrak{F}(1-\tau)^2+\mathfrak{F}(\tau)^2],$$

by (388). Multiplication by $C(\tau)$, with $C(\tau)\,\pi s_\tau\Gamma(1-\tau)^2=\pi s_\tau\Gamma(\tau)\Gamma(1-\tau)=\pi^2$, gives the first identity. For the second, (387) in (385) gives

$$\langle t_\ell^2S_\tau\rangle_2=\frac{\Sigma_{\mathsf{m}}^2}{\pi}\Big[4\cos^2(\tfrac{\pi\tau}{2})\tan(\tfrac{\pi\tau}{2})-s_\tau\Big]=\frac{s_\tau\Gamma(\tau)^2\mathcal{X}_2(\tau)^2}{\pi}=C(\tau)\,\mathcal{X}_2(\tau)^2.\qquad\square$$

The second identity is (556) at $d=2$, which Section 21.2 derives at every dimension from the two-wheel convolution $W_2(d;b')=B_d(d-b',2b')$; here it is obtained from Gauss's sum and the periodization (Dougall) identity of (387). The second and third evaluations of the single sums give the same result: with $\mathcal{A}_\tau=\Gamma(\tau)\mathfrak{G}(\tau)/\Gamma(\frac12+\tau)^2$ and $\mathcal{B}_\tau=\pi\mathfrak{F}(\tau)/\Gamma(\tau)$ one has $\Sigma_{\mathsf{h}}=\pi(c_\tau\mathcal{A}_\tau+\mathcal{B}_\tau)$, $\Sigma_{\mathsf{h}}'=-\pi^2s_\tau\mathcal{A}_\tau$ and $\Sigma_{\mathsf{k}}=\pi\mathcal{B}_\tau/s_\tau^2$, and (385) becomes $\langle t_\ell^2R_\tau\rangle_2=\pi s_\tau\mathcal{A}_\tau^2+\pi\mathcal{B}_\tau^2/s_\tau$, that is

$$C(\tau)\langle t_\ell^2R_\tau\rangle_2=\pi^2\mathfrak{F}(\tau)^2+s_\tau^2\frac{\Gamma(\tau)^4}{\Gamma(\frac12+\tau)^4}\,\mathfrak{G}(\tau)^2,$$

which is (390) by (82). The assembly with the third evaluation is checked in exact algebra in `LRI_C5_d2_proof/K2_d1template_fix1/exact_algebra_v2.wl` (part S2), and the periodization route numerically (Section 18.9.9). At $d=1$ the regulated master is linear in $\mathfrak{F}$ and $\mathfrak{G}$ (78); at $d=2$ it is a sum of two squares, which is where G^2 comes from: $u=4\log2-8\mathsf{G}/\pi$ enters the

extraction below through $2\pi^2u^2$. The two-mode lemma is the second-power analogue of (337); here the decay $|y|^{-\tau}$ supplied by the regulator is what removes the third harmonics. The periodization route uses neither elliptic integrals nor the moment relations, only residues, Gauss's sum, Thomae's relation and periodization, whereas the position-space proofs of (377) relate the same single sums to the elliptic moments. At $\tau > 0$ the only single sums are $\Sigma_{\mathsf{h}}, \Sigma'_{\mathsf{h}}, \Sigma_{\mathsf{k}}, \Sigma_{\mathsf{m}}, \Sigma'_{\mathsf{m}}$ and Σ_{n}, all absolutely convergent; the divergent single sums of the factorization at $\tau = 0$, and their finite parts, are never evaluated.

Corollary (the mixed-trace step at $d = 2$). *For $0 < \tau < \frac{1}{2}$,*

$$J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_2, \tag{391}$$

with the trace read on the full Mellin line, equivalently as the real part of the half-line integral (241)*. This is input (M) of Section 21.1 at $d = 2$ on this interval.*

Proof. By (377), $J_{2\tau} = \pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2]$ for $0 < \tau < 1$, and by (390) the right side of (391) has the same value for $0 < \tau < \frac{1}{2}$. □

The proofs of (377) in Section 18.8 and in Section 18.9.8 use neither the trace nor (M), and the proof of (390) does not use (M); so (391) is proved, not assumed. With the second identity of (390), which is the two-wheel identity at $d = 2$, it gives (250) at $d = 2$ for $0 < \tau < \frac{1}{2}$, and with part (i) of the corollary of Section 18.9.2 the derivation of (248) from (238) given after (250) is complete at $d = 2$. As a normalization check, at $\tau = \frac{1}{2}$ one has $R_{1/2} = t_\ell$ and $C(\frac{1}{2}) = 1$, so the right side of (391) is $\langle t_\ell^3\rangle_2 = 3Q_2$ by (243), while (377) gives $J_1 = 3Q_2$ by Dixon's sum. The point $\tau = \frac{1}{2}$ lies outside the interval of the corollary, because the two pole families of Θ_ℓ^{R} merge there, so this is a consistency check only. The reading of (M) for complex τ, and (M) at other dimensions, are not addressed here.

18.9.6 The spectral form at $d = 2$

Theorem. *The right side of* (248) *at $d = 2$ equals* $32\mathsf{G}^2 - 28\zeta(3)\log 2$.

Proof. By part (iii) of the corollary of Section 18.9.2 and by (390), for $0 < \tau < \frac{1}{2}$,

$$C(\tau)\mathcal{D}(\tau) = \pi^2\mathfrak{F}(\tau)^2 + \pi^2\mathfrak{F}(1-\tau)^2 - C(\tau)^2\mathcal{X}_2(\tau)^2. \tag{392}$$

As in Section 18.8.4, (82) gives $\pi\mathfrak{F}(1-\tau) = \tau^{-1}\mathcal{Q}(\tau)\mathfrak{G}(\tau)$ with $\mathcal{Q}(\tau) = \pi\Gamma(1+\tau)/[\Gamma(1-\tau)\Gamma(\frac{1}{2}+\tau)^2]$, holomorphic and zero-free on $|\tau| < \frac{1}{2}$ with $\mathcal{Q}(0) = 1$; and $\tau C(\tau)\mathcal{X}_2(\tau)$ is holomorphic there with value 1 at 0. So the right side of (392) is meromorphic on $|\tau| < \frac{1}{2}$. The left side is holomorphic there by the corollary of Section 18.9.2. Two functions meromorphic on a punctured disk that agree on $(0, \frac{1}{2})$ agree identically, so their Laurent coefficients at 0 coincide. From $\log\Gamma(c+x) - \log\Gamma(c) = \sum_k \psi_{k-1}(c)x^k/k!$, $\psi_0(\frac{1}{2}) - \psi_0(1) = -2\log 2$, $\psi_k(1) = (-1)^{k+1}k!\,\zeta(k+1)$ and $\psi_k(\frac{1}{2}) = (2^{k+1}-1)\psi_k(1)$ [32],

$$\log\mathcal{Q}(\tau)^2 = 8\log 2\,\tau - \pi^2\tau^2 + 8\zeta(3)\tau^3 - \frac{\pi^4}{6}\tau^4 + O(\tau^5),$$
$$\log\big[\tau C(\tau)\mathcal{X}_2(\tau)\big]^2 = 8\log 2\,\tau + \zeta(3)\tau^3 + O(\tau^5).$$

The first is the series of Section 18.8.4. Hence $C\mathcal{D} = \pi^2\mathfrak{F}(\tau)^2 + \tau^{-2}\{\mathcal{Q}(\tau)^2\mathfrak{G}(\tau)^2 - [\tau C(\tau)\mathcal{X}_2(\tau)]^2\}$, and with (378) and the coefficients $\mathcal{Q}_k = [\tau^k]\mathcal{Q}(\tau)^2$ of Section 18.8.4,

$$\begin{aligned}
[\tau^{-2}]\,C\mathcal{D} &= 1 - 1 = 0, \qquad [\tau^{-1}]\,C\mathcal{D} = 8\log 2 - 8\log 2 = 0,\\
[\tau^0]\,C\mathcal{D} &= \pi^2 + \mathcal{Q}_2 - 32\log^2 2 = 0,\\
[\tau^1]\,C\mathcal{D} &= 2\pi^2 u + 2\sigma + 7\zeta(3) - 8\pi^2\log 2,\\
[\tau^2]\,C\mathcal{D} &= 2(\varrho + \pi^2 v) + \pi^2u^2 + \frac{\pi^4}{3} + 56\zeta(3)\log 2\\
&\qquad - 32\pi^2\log^2 2 + 16\sigma\log 2.
\end{aligned} \tag{393}$$

In the last line the τ^2 coefficient of $\pi^2\mathfrak{F}^2$ is $\pi^2(u^2+2v)$, that of $\tau^{-2}\mathcal{Q}^2\mathfrak{G}^2$ is $\mathcal{Q}_4+2\sigma\mathcal{Q}_1+2\varrho$ with $\mathcal{Q}_1=8\log 2$ and $\mathcal{Q}_4$ of Section 18.8.4, and that of $\tau^{-2}[\tau C\mathcal{X}_2]^2$ is $\frac{512}{3}\log^4 2+8\zeta(3)\log 2$; in the line before, $\mathcal{Q}_3$ of Section 18.8.4 and $[\tau^3][\tau C\mathcal{X}_2]^2=\frac{256}{3}\log^3 2+\zeta(3)$ enter. The vanishing of the first three coefficients is the consistency required by $\mathcal{D}=O(\tau^2)$. The cancellation rule in (378) removes v, ϱ and π^4: $[\tau^2]\,C\mathcal{D}=2\pi^2u^2-32\pi^2\log^2 2+16\sigma\log 2+56\zeta(3)\log 2$. With u and σ from (378), $2\pi^2u^2=32\pi^2\log^2 2-128\pi\mathsf{G}\log 2+128\mathsf{G}^2$ and $16\sigma\log 2=128\pi\mathsf{G}\log 2-224\zeta(3)\log 2$, so $[\tau^2]\,C\mathcal{D}=128\mathsf{G}^2-168\zeta(3)\log 2$. Finally (381) gives

$$14\zeta(3)\log 2+\tfrac14\big(128\mathsf{G}^2-168\zeta(3)\log 2\big)=32\mathsf{G}^2-28\zeta(3)\log 2. \qquad \square$$

Corollary (a spectral moment).

$$\langle t_\ell^2E_2\rangle_2=\mathcal{D}_2=[\tau^1]\,C\mathcal{D}=2\pi^2u+2\sigma+7\zeta(3)-8\pi^2\log 2=-21\zeta(3). \tag{394}$$

This is (567) at $d=2$, $\frac32\psi_2(\frac12)=-21\zeta(3)$, proved here without (M) or (J), and with the interchange of $[\tau^2]$ and the trace justified by the corollary of Section 18.9.2.

Relation to the definition. With (391), $J_{2\tau}-J^\Gamma_{2\tau}=C(\tau)\mathcal{D}(\tau)-[J^\Gamma_{2\tau}-C(\tau)^2\mathcal{X}_2(\tau)^2]$, and by (563) at $d=2$ the bracket is $\frac32U\tau+AU\tau^2+O(\tau^3)$. So the order-$\tau$ coefficient of $J_{2\tau}-J^\Gamma_{2\tau}$ is $[\tau^1]C\mathcal{D}-\frac32U=-21\zeta(3)+21\zeta(3)=0$, which is the vanishing found in Section 18.8.4, and $\frac14[\tau^2](J_{2\tau}-J^\Gamma_{2\tau})=\frac14[\tau^2]C\mathcal{D}-\frac{AU}{4}$ is the right side of (381). At $d=2$ the definition (238), the spectral form (248) and the regulated trace therefore give the same expression identically, not only the same number. The exact algebra of both routes is checked in `LRI_C5_d2_simplify/proof/K2-parseval/symbolic_checks.py` (part B) and `LRI_C5_d2_simplify/proof/synth/synth_checks_K2.py`.

18.9.7 The series at $d=2$

At $q=\frac12$ the data of (251) and (260) are $M^2=16$, $T_{\ell n}=c_mc_n$ and $S_{\ell n}=e_me_n$, with $m=\ell+n$, $c_k=(\frac12)_k/k!=\Gamma(k+\frac12)/[\sqrt{\pi}\,\Gamma(k+1)]$ and $e_k=k!/(\frac32)_k$. Write

$$D(\ell)=\sum_{n\ge0}\Big[\frac{M^2S_{\ell n}^2}{8}-\frac{T_{\ell n}^2\mathcal{R}_{\ell n}}{4}\Big],$$

so that (260) at $d=2$ reads $-\frac{AU}{4}+\sum_\ell d_\ell D(\ell)$.

Lemma (local forms). *Let $u\to0$. At $s=a+n+u$,*

$$g_\ell(s)=-\pi\cot(\pi u)\,c(m+u)\,c(n+u),\qquad c(x)=\frac{\Gamma(x+\frac12)}{\sqrt{\pi}\,\Gamma(x+1)},$$

with $\psi_0(a+s)=\psi_0(m+\frac12+u)$, $\psi_0(b+s)=\psi_0(m+1+u)$, $\psi_0(a-s)=\psi_0(n+1+u)+\pi\cot\pi u$ and $\psi_0(b-s)=\psi_0(n+\frac12+u)-\pi\tan\pi u$. At $s=b+n+u$,

$$g_\ell(s)=\tan(\pi u)\,\frac{\Gamma(m+1+u)\,\Gamma(n+1+u)}{\Gamma(m+\frac32+u)\,\Gamma(n+\frac32+u)},$$

with $\psi_0(a+s)=\psi_0(m+1+u)$, $\psi_0(b+s)=\psi_0(m+\frac32+u)$, $\psi_0(a-s)=\psi_0(n+\frac32+u)-\pi\tan\pi u$ and $\psi_0(b-s)=\psi_0(n+1+u)+\pi\cot\pi u$.

Proof. Use $\Gamma(z)\Gamma(1-z)=\pi/\sin\pi z$ and $\psi_0(1-z)-\psi_0(z)=\pi\cot\pi z$ [26, 32]; for example $\Gamma(-n-u)/\Gamma(\frac12-n-u)=[\sin\pi(\frac12-n-u)/\sin\pi(-n-u)]\,\Gamma(n+\frac12+u)/\Gamma(n+1+u)=-\cot(\pi u)\,\Gamma(n+\frac12+u)/\Gamma(n+1+u)$. $\square$

The products $c(m)c(n) = T_{\ell n}$ and $\Gamma(k+1)/\Gamma(k+\frac{3}{2}) = 2e_k/\sqrt{\pi}$ connect these local forms with (251). Inserting them into the holomorphic weight $F(s)$ of (263) and expanding in u gives the $q = \frac{1}{2}$ case of the residues computed in Section 16.6:

$$\mathrm{Res}_{s=a+n}\,\frac{g_\ell^2 F}{4} = \frac{T_{\ell n}^2 \mathcal{R}_{\ell n}}{4}, \qquad \mathrm{Res}_{s=b+n}\,\frac{g_\ell^2 F}{4} = -2e_m^2 e_n^2 = -\frac{M^2 S_{\ell n}^2}{8}. \tag{395}$$

Both are identities between Laurent coefficients, polynomial in the independent symbols $\psi_j(m+\frac{1}{2}), \psi_j(m+1), \psi_j(n+\frac{1}{2}), \psi_j(n+1)$ (or their shifts by $\frac{1}{2}$) with exact π, $\psi_j(1)$ and $\psi_j(\frac{1}{2})$; they are verified exactly in `LRI_C5_d2_proof/K2_gauss/check_residue_series.wl`.

Proposition (the per-spin identity). *For every $\ell \geq 0$ the inner series $D(\ell)$ converges, and*

$$D(\ell) = \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\,\frac{g_\ell(s)^2 F(s)}{4} = \frac{1}{4\pi}\int_0^\infty t_\ell(\nu)^2\, w_{\mathcal{K}}\, d\nu. \tag{396}$$

Proof. On $s = \mathrm{i}\nu$ one has $g_\ell'/g_\ell = \psi_0(a+s) - \psi_0(a-s) - \psi_0(b+s) + \psi_0(b-s) = 2\mathrm{i}h$, its derivative is $2j$, and $p(\mathrm{i}\nu) = p$, $p'(\mathrm{i}\nu) = \mathrm{i}k$, $p''(\mathrm{i}\nu) = r$ for the function $p(s)$ of Section 16.6. Hence $F(\mathrm{i}\nu) = (c_* - p)(-\frac{h^2}{2} - \frac{5j}{8}) - \frac{hk}{2} - \frac{5r}{32} = w_{\mathcal{K}}$, which is real and even in ν; so is $g_\ell(\mathrm{i}\nu) = t_\ell(\nu)$. This gives the second equality. For the first, close the contour to the right along $|s| = \rho_N := a + N + \frac{3}{4}$; these arcs stay at distance at least $\frac{1}{4}$ from both pole families $a + \mathbb{Z}_{\geq 0}$ and $b + \mathbb{Z}_{\geq 0}$. On the arcs $|g_\ell(s)| \leq C|s|^{-1}$: indeed $\Gamma(a+s)/\Gamma(b+s) = O(|s|^{-1/2})$ by Stirling's formula for $\Re s \geq 0$, and

$$\frac{\Gamma(a-s)}{\Gamma(b-s)} = \frac{\sin\pi(b-s)}{\sin\pi(a-s)} \cdot \frac{\Gamma(1-b+s)}{\Gamma(1-a+s)},$$

where the trigonometric ratio is bounded at distance at least $\frac{1}{4}$ from $a + \mathbb{Z}$ and has finite limits as $\Im s \to \pm\infty$, and the Gamma ratio is $O(|s|^{-1/2})$. In the same way $\psi_0(a-s) = \psi_0(1-a+s) - \pi\cot\pi(a-s)$, and the analogous formula for $\psi_0(b-s)$, give $|F| \leq C\log^2|s|$ on the arcs. The arc integrals are therefore $O(\rho_N^{-1}\log^2\rho_N) \to 0$. At fixed ℓ each residue family converges absolutely, with terms $O(n^{-2}\log^4 n)$, and the residues are summed in increasing n. Closing upward to the right is clockwise, so (395) gives $\int \frac{ds}{2\pi\mathrm{i}}\frac{g_\ell^2 F}{4} = -\sum_n[\mathrm{Res}_{a+n} + \mathrm{Res}_{b+n}] = D(\ell)$. □

This is the per-spin identity behind (260), at $d = 2$, with the contour estimates that Section 16.6 does not write out.

Proposition (the series and the strip limit). *(a) For each $r \in (0, \frac{1}{2})$ there is C_r' with $|t_\ell(\nu)^2 w_{\mathcal{K}}(\ell,\nu)| \leq C_r'(1+|\xi|)^{-3+2r}$; in particular $\langle |t_\ell^2 w_{\mathcal{K}}|\rangle_2 < \infty$. (b) $\sum_\ell d_\ell |D(\ell)| < \infty$, and (260) at $d = 2$ equals $-\frac{AU}{4} + \frac{1}{4}\langle t_\ell^2 w_{\mathcal{K}}\rangle_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2$. (c) Let $g(m,n)$ be the symmetric square-lattice summand with $g(\ell+n,n)$ the summand of $D(\ell)$, written as a finite sum $\sum_i \alpha_i A_i(m)B_i(n)$ of products of real sequences of bounded variation that are $O(\log^3(m+2)/(m+1))$. Put $\hat{A}_i(z) = \sum_m A_i(m)z^m$. Then $h(\theta) = \sum_i \alpha_i \hat{A}_i(e^{\mathrm{i}\theta})\overline{\hat{B}_i(e^{\mathrm{i}\theta})}$ is continuous on $(0, 2\pi)$, its Fourier coefficients are $D(|\ell|)$, and $\lim_{\theta\to 0} h(\theta) = \sum_{\ell\geq 0} d_\ell D(\ell)$.* Such a factorization, with 27 distinct single sequences, is the exact symbolic one behind the 70-digit evaluation (`LRI_C5_d2_simplify/factor_K2.py`, output `K2_factorized.json`); each sequence is c_k^2 or e_k^2 times polygamma differences with at most one factor $\psi_0(k+1)$.

Proof. (a) Let $H(\tau) = \frac{1}{2}C(\tau)[(R_\tau - S_\tau)(\mathrm{i}\nu) + (R_\tau - S_\tau)(-\mathrm{i}\nu)]$. It is holomorphic in $|\tau| < \frac{1}{2}$, since the pole of C is cancelled by the double zero of the even part, and $[\tau^2]H = w_{\mathcal{K}}$ by the corollary of Section 18.9.2. By (380), $|H| \leq C(1+|\xi|)^{-1+2r}$ on $|\tau| = r$, and Cauchy's estimate gives $|w_{\mathcal{K}}| \leq r^{-2}C(1+|\xi|)^{-1+2r}$; multiply by $t_\ell^2 \leq C(1+|\xi|)^{-2}$. (b) By (396), $|D(\ell)| \leq \frac{1}{8\pi}\int_{\mathbb{R}} |t_\ell^2 w_{\mathcal{K}}|\, d\nu$, so (a) gives absolute convergence of the spin sum, and the value follows from the theorem of Section 18.9.6. (c) By Dirichlet's test (bounded variation and $A_i(m) \to 0$) the series $\hat{A}_i(e^{\mathrm{i}\theta})$ converge locally uniformly on $(0, 2\pi)$, so h is continuous there. Since $A_i, B_i \in \ell^2$, the product $\hat{A}_i\overline{\hat{B}_i}$

is integrable on the circle, and its ℓth Fourier coefficient is the correlation $\sum_n A_i(n+\ell)B_i(n)$, with $A_i(k)=0$ for $k<0$. Summing over i, the coefficient is $\sum_n g(n+\ell,n)=D(\ell)$ for $\ell\ge 0$, and $D(|\ell|)$ for $\ell<0$ by the symmetry of g. By (b) these coefficients are absolutely summable, so the Fourier series converges uniformly to a continuous function equal to h almost everywhere, hence everywhere on $(0,2\pi)$. Letting $\theta\to 0$ gives $\sum_{\ell\in\mathbb{Z}} D(|\ell|)=\sum_{\ell\ge 0} d_\ell D(\ell)$. □

At $d=2$ the inner series of (260) therefore sum in an absolutely convergent spin order, although the unrestricted double series is not absolutely convergent: its square-lattice single sums diverge like $\sum_k \log^p k/k$. Part (c) shows that the strip limit $\theta\to 0$ of the lattice sum with weight $e^{\mathrm{i}(m-n)\theta}$, used for the 70-digit evaluation of Section 17.2 (`LRI_C5_d2_simplify/eval_K2.py`), computes exactly this ordered sum. The sharper bound (382) gives $D(\ell)=O(\ell^{-3+2\rho_0})$.

Remark (why the single sums at $\tau=0$ are avoided). This is a numerical observation and is not used. At $\tau=0$ the $\mathcal{R}$ part of the bilinear form $g(m,n)$ has rank 9, and the $S_{\ell n}$ part adds one more. In the variables $\psi_{j-1}(k+\frac12)-\psi_{j-1}(\frac12)$ and $\psi_{j-1}(k+1)-\psi_{j-1}(1)$, the span of all parameter derivatives of ${}_2F_1(\frac12+\alpha,\frac12+\beta;1+\gamma;z)$ through total order four has dimension 22, and the row space of the form has three directions outside it, at weights 2, 3 and 4 (`LRI_C5_d2_proof/K2_gauss/rowspace.py`). An integer-relation search on the 70-digit single sums (`LRI_C5_d2_proof/K2_gauss/pslq_singles.py`) indicates that individual sums contain constants absent from the final answer, for example

$$\sum_k c_k^2[\psi_2(k+\tfrac12)-\psi_2(k+1)] \overset{\text{num}}{=} 3\pi^3+4\pi\log^2 2-64\mathsf{G}\log 2-128\,\Im\,\mathrm{Li}_3(\tfrac{1+\mathrm{i}}{2}),$$

while the weight-five sums are not identified in a small basis. The regulator $\tau>0$ makes every single sum convergent, and the only non-Gamma functions left are $\mathfrak{F}$ and $\mathfrak{G}$, whose unknown coefficients v,ϱ cancel by (86).

18.9.8 A second position-space proof: Riemann's bilinear relation

This subsubsection proves (377) from the integral

$$J_{2\tau}=\frac{1}{\pi^2}\int_{\mathbb{C}^2}\frac{d^2x\,d^2y}{|x|\,|1-x|\,|y|\,|1-y|^{2-2\tau}\,|x-y|},\qquad 0<\tau<1,$$

of Section 18.8, without the cell decomposition and rotation of that section. Here d^2x is Lebesgue measure on $\mathbb{C}\cong\mathbb{R}^2$, so $dx\wedge d\bar{x}=-2\mathrm{i}\,d^2x$; $K(m)$ is the parameter convention, holomorphic on $\mathbb{C}\setminus[1,\infty)$; $I_1,\dots,I_6$ are the moments listed before (77), at $t=\tau$; $\mathbb{C}_{\text{cut}}=\mathbb{C}\setminus((-\infty,0]\cup[1,\infty))$; all powers are principal; and the letters ω,Ω_y,Π_j below are local (they are not the ω_τ of Section 18.8 or the Ω_r of (335)).

Lemma (properties of K). (i) For $m\in\mathbb{C}\setminus[1,\infty)$, $K(m)=\frac12\int_0^1 ds/\sqrt{s(1-s)(1-ms)}$, by $s=\sin^2\theta$ in the definition of Section 1.4. (ii) For $m\in\mathbb{C}\setminus[1,\infty)$, $K(m)=(1-m)^{-1/2}K(\frac{m}{m-1})$ [23]; in particular $K(x)=(1-x)^{-1/2}K(\frac{x}{x-1})$ for $x<0$, with $\frac{x}{x-1}\in(0,1)$, and $K(1-x)=x^{-1/2}K(1-\frac1x)$ for $x>1$. (iii) For $x>1$, $K(x\pm\mathrm{i}0)=x^{-1/2}[K(\frac1x)\pm\mathrm{i}K(1-\frac1x)]$. (iv) For $|1-m|<1$, $m\notin[1,\infty)$, $K(m)=\frac12\sum_n c_n^2(1-m)^n[2\psi_0(n+1)-2\psi_0(n+\frac12)-\log(1-m)]$ [23]. (v) There is C such that, for all $m\in\mathbb{C}\setminus[1,\infty)$ and for the boundary values (iii), $|K(m)|\le C(1+|\log|1-m||)$ for $|m|\le 3$ and $|K(m)|\le C|m|^{-1/2}(1+\log|m|)$ for $|m|\ge 3$.

Proof. (iii) In (i) put $m=x\pm\mathrm{i}\eta$ with $\eta\downarrow 0$. For $s>1/x$, $1-ms\to(1-xs)\mp\mathrm{i}0$ with $1-xs<0$, so $(1-ms)^{-1/2}\to\pm\mathrm{i}(xs-1)^{-1/2}$. Since $|1-ms|\ge|1-xs|$ and $(s(1-s)|1-xs|)^{-1/2}\in L^1(0,1)$, dominated convergence applies. On $(0,1/x)$ substitute $s=\sigma/x$, and on $(1/x,1)$ substitute $s=1-(1-\frac1x)\sigma$; the two pieces become $x^{-1/2}K(1/x)$ and $\pm\mathrm{i}x^{-1/2}K(1-1/x)$. The restriction of

K to the upper (lower) half-plane continues analytically across $(1,\infty)$, because the hypergeometric equation has analytic coefficients there, so K extends continuously from each side with these values (the + case is the formula used in the proof of (77)). (v) If $|1-m|\le\frac12$, use (iv) and $|\log(1-m)|\le|\log|1-m||+\pi$. If $|m|\le 3$ and $|1-m|\ge\frac12$, use (i) and $|1-ms|^2=(1-s\Re m)^2+s^2(\Im m)^2$: for $\Re m\le\frac34$, $|1-ms|\ge\frac14$; for $\Re m\ge\frac54$, $|1-ms|\ge|1-s\Re m|$ and $\int_0^1(s(1-s)|1-xs|)^{-1/2}ds$ is bounded for $x\in[\frac54,3]$; for $\frac34<\Re m<\frac54$, $(\Im m)^2\ge\frac{3}{16}$, so $|1-ms|\ge\frac38$ for $s\le\frac12$ and $|1-ms|\ge\frac{\sqrt3}{8}$ for $s\ge\frac12$. If $|m|\ge 3$, (ii) gives $K(m)=(1-m)^{-1/2}K(\frac{m}{m-1})$ with $|1-\frac{m}{m-1}|=|m-1|^{-1}\le\frac12$, which reduces to the first case. The boundary values obey the same bounds by continuity. □

Green's theorem on slit domains. Below, Green's theorem $\int_V d\alpha=\oint_{\partial V}\alpha$, or Cauchy's theorem, is applied to a domain $V=\{r<|z-e|$ for the marked points e, $|z|<R\}\setminus\Sigma$, where Σ is a finite union of segments (slits) and $\alpha=\mathsf{g}\,d\bar z$ or $\mathsf{g}\,dz$. In every case g is real-analytic and bounded on V and extends continuously to each open slit from each side, and $d\alpha$ is integrable. Delete the η-neighborhood of Σ; on the closure of what remains α is C^1, so Green's theorem applies. As $\eta\to0$, the area integral converges by dominated convergence; the integrals over the two curves parallel to a slit converge to the integrals of the one-sided boundary values, by uniform convergence on compact parts of the open slits and boundedness; and the remaining boundary pieces have length $O(\eta)$. So the slit edges enter with the positive orientation of V: each edge is traversed with V on its left.

Reduction to the inner star. The integrand is positive, so by Tonelli's theorem

$$J_{2\tau}=\frac{1}{\pi^2}\int_{\mathbb{C}}d^2y\,|y|^{-1}|1-y|^{2\tau-2}\mathcal{I}(y),\qquad \mathcal{I}(y)=\int_{\mathbb{C}}\frac{d^2x}{|x|\,|1-x|\,|x-y|}\in(0,\infty], \tag{397}$$

with both sides in $[0,\infty]$. For $y\notin\{0,1\}$, $\mathcal{I}(y)<\infty$: the three factors are locally integrable in $\mathbb{R}^2$, and the integrand is $O(|x|^{-3})$ at infinity.

Lemma (the inner star is a Riemann bilinear relation). *For $y\in\mathbb{C}_{\mathrm{cut}}$,*

$$\boxed{\mathcal{I}(y)=8\,\Re[K(y)\,\overline{K(1-y)}].} \tag{398}$$

Proof. Fix $y\in\mathbb{C}_{\mathrm{cut}}$ and put $\Pi_y(x)=x(x-1)(x-y)$.

(a) The cut. Let $\Gamma_y=[0,y]\cup[y,1]\cup[1,+\infty)$, oriented from 0 to ∞. Since $y\notin(-\infty,0]\cup[1,\infty)$, the segment $[0,y]$ does not meet $[1,\infty)$; $[y,1]$ meets $[1,\infty)$ only at 1, and $[0,y]$ meets $[y,1]$ only at y (for $y\in(0,1)$ the three pieces are collinear). So Γ_y is a simple arc from 0 to ∞, and $\mathbb{C}\setminus\Gamma_y$ is simply connected. Π_y has no zero there, so there is a holomorphic ω on $\mathbb{C}\setminus\Gamma_y$ with $\omega^2=1/\Pi_y$, and $|\omega|^2=1/(|x||1-x||x-y|)$.

(b) Edge values and jumps. Near any point of an open piece of Γ_y, Π_y has two holomorphic square roots on a small disk, and ω equals $\pm1/$(that root) on each side. Hence ω has continuous boundary values ω_L (left of Γ_y) and ω_R (right of Γ_y) on each open piece, and the ratio $\omega_R/\omega_L\in\{\pm1\}$ is a constant $\varepsilon_1,\varepsilon_2,\varepsilon_3$ on $(0,y)$, $(y,1)$, $(1,\infty)$. A small circle around 0 crosses Γ_y once, on $(0,y)$; since Π_y has a simple zero inside, $\Pi_y^{-1/2}$ changes sign along the circle, so $\varepsilon_1=-1$. A small circle around y crosses $(0,y)$ and $(y,1)$ once each, so $\varepsilon_1\varepsilon_2=-1$ and $\varepsilon_2=+1$. A small circle around 1 crosses $(y,1)$ and $(1,\infty)$, so $\varepsilon_3=-1$. The direction of crossing does not matter, because $\varepsilon_k^{-1}=\varepsilon_k$.

(c) Primitive. $|\omega|\le C|x-e|^{-1/2}$ near $e\in\{0,y,1\}$, and $|\omega|\le C|x|^{-3/2}$ for large $|x|$. Let Ω_y be the primitive of ω on $\mathbb{C}\setminus\Gamma_y$ with $\Omega_y(x)\to0$ as $x\to0$; the limit exists by the Cauchy criterion, since radial and circular paths near 0 give $|\Omega_y(x')-\Omega_y(x'')|\le C(|x'|^{1/2}+|x''|^{1/2})$. Ω_y is bounded: it is continuous up to the edges from each side, has limits at $0,y,1$ within each adjacent sector, and for $|x|\ge R_0:=2+|y|$ it differs from its value on the circle $|x|=R_0$ by the integral along a

radial segment, which does not meet Γ_y (for $\arg x \neq 0$) and is $O(R_0^{-1/2})$. The edge values satisfy $d\Omega_{y,L} = \omega_L\,dx$ and $d\Omega_{y,R} = \omega_R\,dx$ along Γ_y. At the corners y and 1 the left edges of the two adjacent pieces border the same sector, and so do the right edges; $|\omega|$ is integrable there, so the edge values are continuous through the corners. Put

$$\Pi_1 = \int_0^y \omega_L\,dx, \qquad \Pi_2 = \int_y^1 \omega_L\,dx, \qquad \Pi_3 = \int_1^\infty \omega_L\,dx \qquad (\text{along } \Gamma_y).$$

By (b) the edge values are: on $(0,y)$, $\Omega_{y,L} = \int_0^x \omega_L$ and $\Omega_{y,R} = -\Omega_{y,L}$; at y, $\pm\Pi_1$; on $(y,1)$, $\Omega_{y,L} = \Pi_1 + \int_y^x \omega_L$ and $\Omega_{y,R} = -\Pi_1 + \int_y^x \omega_L$; at 1, $\Pi_1+\Pi_2$ and $-\Pi_1+\Pi_2$; on $(1,\infty)$, $\Omega_{y,L} = \Pi_1+\Pi_2+\int_1^x \omega_L$ and $\Omega_{y,R} = -\Pi_1 + \Pi_2 - \int_1^x \omega_L$. Thus $\Omega_{y,L} + \Omega_{y,R} = 0$ on $(0,y)$, $\Omega_{y,L} - \Omega_{y,R} = 2\Pi_1$ on $(y,1)$, and $\Omega_{y,L} + \Omega_{y,R} = 2\Pi_2$ on $(1,\infty)$.

(d) Stokes. On $\mathbb{C}\setminus\Gamma_y$, $d(\Omega_y\bar\omega\,d\bar x) = \omega\bar\omega\,dx\wedge d\bar x = -2\mathrm{i}|\omega|^2 d^2x$. Apply Green's theorem on slit domains to $\{x\notin\Gamma_y:\ |x|<R,\ |x-e|>r,\ e=0,y,1\}$. The positively oriented boundary runs along the left edge of Γ_y in its direction and along the right edge against it, so

$$-2\mathrm{i}\int |\omega|^2 d^2x = \int_{\Gamma_y} (\Omega_{y,L}\bar\omega_L - \Omega_{y,R}\bar\omega_R) d\bar x + (\text{circle terms}).$$

The circle terms vanish in the limit: on $|x|=r$, $|\Omega_y|\le Cr^{1/2}$ and $|\omega|\le Cr^{-1/2}$, so the term is $O(r)$; on $|x-y|=r$ and $|x-1|=r$, Ω_y is bounded and $|\omega|\le Cr^{-1/2}$, so the term is $O(r^{1/2})$; on $|x|=R$, Ω_y is bounded and $|\omega|\le CR^{-3/2}$, so the term is $O(R^{-1/2})$. The edge integrands are integrable. Letting $r\to 0$ and $R\to\infty$ (monotone convergence on the left) gives $-2\mathrm{i}\,\mathcal{I}(y) = \int_{\Gamma_y}(\Omega_{y,L}\bar\omega_L - \Omega_{y,R}\bar\omega_R)\,d\bar x$. By (b) and (c) the integrand is 0 on $(0,y)$, $2\Pi_1\bar\omega_L$ on $(y,1)$ and $2\Pi_2\bar\omega_L$ on $(1,\infty)$. Hence $-2\mathrm{i}\,\mathcal{I}(y) = 2\Pi_1\bar\Pi_2 + 2\Pi_2\bar\Pi_3$.

(e) Cauchy. The same exhaustion applied to $\omega\,dx$ gives $\int_{\Gamma_y}(\omega_L-\omega_R)\,dx = 0$, that is $2\Pi_1+0+2\Pi_3=0$, so $\Pi_3 = -\Pi_1$. Therefore $-2\mathrm{i}\,\mathcal{I}(y) = 2(\Pi_1\bar\Pi_2 - \bar\Pi_1\Pi_2) = 4\mathrm{i}\,\Im(\Pi_1\bar\Pi_2)$, that is $\mathcal{I}(y) = 2\,\Im(\bar\Pi_1\Pi_2)$. This is Riemann's bilinear relation for the holomorphic differential dx/w on the elliptic curve $w^2 = \Pi_y(x)$, whose periods are $2\Pi_1$ and $2\Pi_2$; the factor $\frac12$ in $\mathcal{I} = \frac12\Im(\overline{2\Pi_1}\,2\Pi_2)$ is the inverse of the degree 2 of the covering $(x,w)\mapsto x$.

(f) The periods. On $[0,y]$ put $x = ys$. Then $\Pi_y(ys) = y^2 s(1-s)(1-ys)$, and $s(1-s)(1-ys)\notin(-\infty,0]$ for $s\in(0,1)$ because $ys\notin[1,\infty)$. Hence $\omega_L(ys) = \eta_1/(y\sqrt{s(1-s)(1-ys)})$ with a constant $\eta_1\in\{\pm1\}$, and by (i) of the lemma on K, $\Pi_1 = 2\eta_1 K(y)$. On $[y,1]$ put $x = 1-(1-y)\sigma'$. Then $\Pi_y = -(1-y)^2\sigma'(1-\sigma')(1-(1-y)\sigma')$, with $\sigma'(1-\sigma')(1-(1-y)\sigma')\notin(-\infty,0]$, so $\omega_L = \eta_2/(\mathrm{i}(1-y)\sqrt{\sigma'(1-\sigma')(1-(1-y)\sigma')})$ and, with $dx = -(1-y)d\sigma'$, $\Pi_2 = -2\mathrm{i}\eta_2 K(1-y)$. So (e) gives $\mathcal{I}(y) = 8\epsilon_y\Re[K(y)\overline{K(1-y)}]$ with $\epsilon_y = -\eta_1\eta_2\in\{\pm1\}$.

(g) The sign. $\mathcal{I}(y)>0$, so $\Re[K(y)\overline{K(1-y)}]\neq 0$ on $\mathbb{C}_{\mathrm{cut}}$. This function is continuous on the connected set $\mathbb{C}_{\mathrm{cut}}$ and equals $K(\frac12)^2>0$ at $y=\frac12$, so it is positive everywhere and $\epsilon_y\equiv 1$. □

For real $y\in(0,1)$ the sign can also be read off directly: continuation along the upper edge gives $\omega_L = |\Pi_y|^{-1/2}$ on $(0,y)$, $\mathrm{i}|\Pi_y|^{-1/2}$ on $(y,1)$ and $-|\Pi_y|^{-1/2}$ on $(1,\infty)$, hence $\Pi_1 = 2K(y)$, $\Pi_2 = 2\mathrm{i}K(1-y)$ and $\Pi_3 = -2K(y)$ (using $\int_1^\infty dx/\sqrt{\Pi_y} = 2K(y)$ via $x=1/s$), which confirms $\Pi_3=-\Pi_1$ and $\mathcal{I}(y) = 8K(y)K(1-y)$. The algebra of (d)–(f) is checked exactly in `LRI_C5_d2_proof/K2_d1template_fix1/exact_algebra_v2.wl` (part S0a).

Edge values of the outer integrand. Let $\varphi_\tau(y) = y^{-1/2}(1-y)^{\tau-1}$, $\mathsf{u} = \varphi_\tau K(y)$ and $\mathsf{v} = \varphi_\tau K(1-y)$. These are holomorphic on $\mathbb{C}_{\mathrm{cut}}$, and $|\varphi_\tau|^2 = |y|^{-1}|1-y|^{2\tau-2}$, so by (398)

$$|y|^{-1}|1-y|^{2\tau-2}\,\mathcal{I}(y) = 4(\mathsf{u}\bar{\mathsf{v}} + \bar{\mathsf{u}}\mathsf{v}) = 8\Re(\mathsf{u}\bar{\mathsf{v}}), \qquad y\in\mathbb{C}_{\mathrm{cut}}. \tag{399}$$

Define the positive functions

$$f_-(x) = |x|^{-1/2}(1-x)^{\tau-1}K(x), \quad g_-(x) = |x|^{-1/2}(1-x)^{\tau-3/2}K(\tfrac{1}{1-x}) \quad (x<0),$$
$$f_+(x) = x^{-1}(x-1)^{\tau-1}K(\tfrac1x), \qquad g_+(x) = x^{-1}(x-1)^{\tau-1}K(1-\tfrac1x) \qquad (x>1).$$

The principal powers have the one-sided values $(x \pm \mathrm{i}0)^{-1/2} = \mp \mathrm{i}|x|^{-1/2}$ for $x < 0$ and $(1-(x \pm \mathrm{i}0))^{\tau-1} = -(x-1)^{\tau-1}e^{\mp \mathrm{i}\pi\tau}$ for $x > 1$. With (ii) and (iii) of the lemma on K the edge values are

$$\begin{aligned} x<0: &\quad \mathsf{u}(x \pm \mathrm{i}0) = \mp \mathrm{i} f_-(x), \qquad \mathsf{v}(x \pm \mathrm{i}0) = -f_-(x) \mp \mathrm{i} g_-(x), \\ x>1: &\quad \mathsf{u}(x \pm \mathrm{i}0) = -e^{\mp \mathrm{i}\pi\tau}\big(f_+(x) \pm \mathrm{i} g_+(x)\big), \qquad \mathsf{v}(x \pm \mathrm{i}0) = -e^{\mp \mathrm{i}\pi\tau} g_+(x). \end{aligned} \tag{400}$$

For the v entry at $x < 0$: $1 - y = (1-x) \mp \mathrm{i}0$ with $1 - x > 1$, so (iii) gives $K(1-y) = (1-x)^{-1/2}[K(\frac{1}{1-x}) \mp \mathrm{i}K(\frac{x}{x-1})]$, and $K(\frac{x}{x-1}) = (1-x)^{1/2}K(x)$ by (ii). For $x > 1$, $K(1-x) = x^{-1/2}K(1-\frac{1}{x})$ by (ii), so v jumps only through φ_τ. The substitutions $w = \frac{x}{x-1}$, $w = \frac{1}{1-x}$, $w = \frac{1}{x}$ and $w = \frac{1}{x}$ give (exact check S0f)

$$\int_{-\infty}^{0} f_- = I_5, \quad \int_{-\infty}^{0} g_- = I_6, \quad \int_1^{\infty} f_+ = I_3, \quad \int_1^{\infty} g_+ = I_4, \quad \int_0^1 \mathsf{u} = I_1, \quad \int_0^1 \mathsf{v} = I_2. \tag{401}$$

By (v) of the lemma on K: near 0, $|\mathsf{u}| \le C|y|^{-1/2}$ and $|\mathsf{v}| \le C|y|^{-1/2}(1 + |\log|y||)$; near 1, $|\mathsf{u}| \le C|1-y|^{\tau-1}(1 + |\log|1-y||)$ and $|\mathsf{v}| \le C|1-y|^{\tau-1}$; for $|y| \ge 4$, $|\mathsf{u}|, |\mathsf{v}| \le C|y|^{\tau-2}(1 + \log|y|)$; and on compact subsets of $\mathbb{C} \setminus \{0,1\}$ both are bounded, uniformly up to the edges. Hence $\mathsf{u}\bar{\mathsf{v}} \in L^1(\mathbb{C})$ for $0 < \tau < 1$, and $J_{2\tau} < \infty$ by (397) and (399). The moment relations (77) at $t = \tau$ are the real and imaginary parts of Cauchy's theorem for u and v on the upper half-plane, $\int_{\mathbb{R}} \mathsf{u}(x + \mathrm{i}0)\,dx = \int_{\mathbb{R}} \mathsf{v}(x + \mathrm{i}0)\,dx = 0$, read with (400) and (401); this is the contour proof given after (77).

Lemma (the outer integral). *For $0 < \tau < 1$,*

$$\int_{\mathbb{C}} |y|^{-1}|1-y|^{2\tau-2}\,\mathcal{I}(y)\,d^2y = 4\big[I_5^2 + 2s_\tau I_1 I_4 - I_4^2\big]. \tag{402}$$

Proof. (a) Primitive. $\mathbb{C}_{\mathrm{cut}}$ is star-shaped with respect to every point of $(0,1)$: a segment from $c \in (0,1)$ to $y \in \mathbb{C}_{\mathrm{cut}} \setminus \mathbb{R}$ meets $\mathbb{R}$ only at c. Put $\mathsf{u}_1(y) = \int_{[0,y]} \mathsf{u}(z)\,dz$. The integral converges since $|\mathsf{u}| \le C|z|^{-1/2}$ near 0, and it differs from $\int_{[1/2,y]} \mathsf{u}$ by a constant (Cauchy's theorem for triangles), so u_1 is a holomorphic primitive of u on $\mathbb{C}_{\mathrm{cut}}$. Along $[0,y]$, $|1 - \rho e^{\mathrm{i}\theta}| \ge |1-\rho|$; with the bounds above, $|\mathsf{u}(\rho e^{\mathrm{i}\theta})| \le C\rho^{-1/2}|1-\rho|^{\tau-1}(1 + |\log|1-\rho||)$ for $\rho \le 3$ and $\le C\rho^{\tau-2}(1+\log\rho)$ for $\rho \ge 3$, so u_1 is bounded on $\mathbb{C}_{\mathrm{cut}}$. Moreover $|\mathsf{u}_1(y)| \le C|y|^{1/2}$ near 0, and $\mathsf{u}_1(y) = I_1 + \int_{[1,y]} \mathsf{u}$ (Cauchy's theorem for the triangle $0, 1, y$), so $\mathsf{u}_1(y) \to I_1$ as $y \to 1$.

(b) Edge values of u_1. With $\hat{f}_-(x) = \int_x^0 f_-$, $\hat{f}_+(x) = \int_1^x f_+$ and $\hat{g}_+(x) = \int_1^x g_+$,

$$\mathsf{u}_1(x \pm \mathrm{i}0) = \pm \mathrm{i}\hat{f}_-(x) \quad (x<0), \qquad \mathsf{u}_1(x \pm \mathrm{i}0) = I_1 - e^{\mp \mathrm{i}\pi\tau}\big(\hat{f}_+(x) \pm \mathrm{i}\hat{g}_+(x)\big) \quad (x>1).$$

(c) Stokes. On $\mathbb{C}_{\mathrm{cut}}$, $d(\mathsf{u}_1\bar{\mathsf{v}}\,d\bar{y}) = \mathsf{u}\bar{\mathsf{v}}\,dy \wedge d\bar{y} = -2\mathrm{i}\,\mathsf{u}\bar{\mathsf{v}}\,d^2y$. Apply Green's theorem on slit domains to $\{y \in \mathbb{C}_{\mathrm{cut}} : r < |y| < R,\ |y-1| > r\}$. The upper edges of both slits are traversed in the $+x$ direction and the lower edges in the $-x$ direction. With $[\mathsf{g}]_-^+ = \mathsf{g}(x + \mathrm{i}0) - \mathsf{g}(x - \mathrm{i}0)$,

$$-2\mathrm{i}\int \mathsf{u}\bar{\mathsf{v}}\,d^2y = \int_{-R}^{-r} [\mathsf{u}_1\bar{\mathsf{v}}]_-^+\,dx + \int_{1+r}^{R} [\mathsf{u}_1\bar{\mathsf{v}}]_-^+\,dx + (\text{circle terms}).$$

(d) Circle terms. On $|y| = r$: $|\mathsf{u}_1| \le Cr^{1/2}$ and $|\mathsf{v}| \le Cr^{-1/2}(1 + |\log r|)$, so the term is $O(r(1 + |\log r|))$. On $|y-1| = r$: $|\mathsf{u}_1| \le C$ and $|\mathsf{v}| \le Cr^{\tau-1}$, so the term is $O(r^\tau)$. On $|y| = R$: $|\mathsf{u}_1| \le C$ and $|\mathsf{v}| \le CR^{\tau-2}(1+\log R)$, so the term is $O(R^{\tau-1}(1+\log R))$. All three vanish in the limit, because $0 < \tau < 1$.

(e) Jumps. From (400) and (b),

$$\begin{aligned} x<0: &\quad [\mathsf{u}_1\bar{\mathsf{v}}]_-^+ = (\mathrm{i}\hat{f}_-)(-f_- + \mathrm{i}g_-) - (-\mathrm{i}\hat{f}_-)(-f_- - \mathrm{i}g_-) = -2\mathrm{i}\,\hat{f}_- f_-, \\ x>1: &\quad [\mathsf{u}_1\bar{\mathsf{v}}]_-^+ = -2\mathrm{i}s_\tau I_1\, g_+ + 2\mathrm{i}\,\hat{g}_+ g_+. \end{aligned}$$

(f) Limit. Since $\mathsf{u}\bar{\mathsf{v}} \in L^1$, the area integral converges. On $(-\infty, 0)$, $\hat{f}_-' = -f_-$, $\hat{f}_-(0) = 0$ and $\hat{f}_-(-\infty) = I_5$, so $\int_{-\infty}^0 (-2\mathrm{i}\hat{f}_- f_-)\,dx = -\mathrm{i}I_5^2$. On $(1, \infty)$, $\hat{g}_+' = g_+$, $\hat{g}_+(1) = 0$ and $\hat{g}_+(\infty) = I_4$, so $\int_1^\infty (-2\mathrm{i}s_\tau I_1 g_+ + 2\mathrm{i}\hat{g}_+ g_+)\,dx = -2\mathrm{i}s_\tau I_1 I_4 + \mathrm{i}I_4^2$. Hence

$$\int_{\mathbb{C}_{\mathrm{cut}}} \mathsf{u}\bar{\mathsf{v}}\, d^2y = \frac{\mathrm{i}}{2}(-\mathrm{i}I_5^2 - 2\mathrm{i}s_\tau I_1 I_4 + \mathrm{i}I_4^2) = \frac{1}{2}(I_5^2 + 2s_\tau I_1 I_4 - I_4^2).$$

(g) By (399), the left side of (402) is $8\Re \int \mathsf{u}\bar{\mathsf{v}}\, d^2y = 4(I_5^2 + 2s_\tau I_1 I_4 - I_4^2)$; the complement of $\mathbb{C}_{\mathrm{cut}}$ has measure zero. □

Second evaluation. Take instead the primitive $\mathsf{v}_1(y) = \int_{[1,y]} \mathsf{v}$ of v, based at 1. On $x < 0$ it has edge values $\mathsf{v}_1(x \pm \mathrm{i}0) = -I_2 + \hat{f}_-(x) \pm \mathrm{i}\hat{g}_-(x)$, with $\hat{g}_-(x) = \int_x^0 g_-$, and the same steps give $\int \mathsf{v}\bar{\mathsf{u}}\, d^2y = \frac{1}{2}(I_4^2 + 2I_2 I_5 - I_5^2)$ (exact check S0c). Since $\int \mathsf{v}\bar{\mathsf{u}} = \overline{\int \mathsf{u}\bar{\mathsf{v}}}$ and both values are real, the two evaluations must agree; given (77) they do, the difference of the two brackets being $2I_4^2(1 - s_\tau^2 - c_\tau^2) = 0$. Their average, $\int \mathsf{u}\bar{\mathsf{v}}\, d^2y = \frac{1}{2}(s_\tau I_1 I_4 + I_2 I_5)$, is an identity of the bracket polynomials that needs no relation among the I_j; it is the bilinear form of (674) at this point, the last form of (377).

Assembly. By (397), (402), (77) and the values $I_3 = \pi^2 \mathfrak{F}(1-\tau)/(2s_\tau)$, $I_4 = \pi^2 \mathfrak{F}(\tau)/(2s_\tau)$ of Section 18.8,

$$J_{2\tau} = \frac{4}{\pi^2}(I_5^2 + 2s_\tau I_1 I_4 - I_4^2) = \frac{4}{\pi^2} s_\tau^2 (I_3^2 + I_4^2) = \pi^2 \left[\mathfrak{F}(1-\tau)^2 + \mathfrak{F}(\tau)^2\right] < \infty,$$

which is (377); the other forms there follow from $I_2 = s_\tau I_3$, $I_6 = s_\tau I_4$ and the average above. All of this algebra is checked exactly in `LRI_C5_d2_proof/K2_d1template_fix1/exact_algebra_v2.wl` (parts S0a–S0f). This proof uses neither the cell table (375), nor the rotation lemma (376), nor (674).

Remarks. (1) *Power counting.* Convergence already follows from the bounds above; for completeness, the integrand of $J_{2\tau}$ in $\mathbb{R}^4$ has the following strict margins: $x \to 0$, $x \to 1$, $x \to y$ and $y \to 0$, margin 1; $y \to 1$, margin 2τ; $x, y \to 0$ jointly, $4 - 3 = 1$; $x, y \to 1$ jointly, $4 - (4 - 2\tau) = 2\tau$; $x \to \infty$ alone, $2 - 3 = -1$; $y \to \infty$ alone, $2 - (4 - 2\tau) = 2\tau - 2$; $x, y \to \infty$ jointly, $4 - (6 - 2\tau) = 2\tau - 2$; $x, y \to \infty$ with $x - y$ bounded, total $2\tau - 3$, with $|x-y|^{-1}$ integrable. Mixed configurations factorize into these cases. For $0 < \tau < 1$ every margin has the required sign, so the point $(1, 1, 1, 2 - 2\tau, 1)$ lies in the interior of the convergence domain in all five exponents. (2) *Relation to $d = 1$.* At $d = 1$, integrating x over $\mathbb{R}$ gives the full-line elliptic star, which is linear in K, and folding gives the linear functional (60). At $d = 2$, integrating x over $\mathbb{C}$ gives the bilinear star (398), and the y integration becomes the bilinear Stokes pairing (402). The six Euler moments, Euler integration and the moment relations are common to both dimensions. The higher Taylor coefficients enter the extraction of Section 18.8.4 as $2(\varrho + \pi^2 v)$, against $4(\varrho + \pi^2 v)$ in (88) at $d = 1$.

18.9.9 Status and checks

Statement	Status	Inputs
Majorant (379), regularized form (381), $[\tau^2]$ under the trace; sharper bound (382)	proved	Stirling [27]; the expansion after (250), checked exactly
Per-spin residues (384), factorized traces (385)	proved	reflection, Stirling
Single sums (386), (387), (388), (389) (three evaluations)	proved	Gauss [4], (82), (77), Euler integral, [23]
Regulated traces (390)	proved	the rows above
Mixed-trace step (391) at $d = 2$, $0 < \tau < \frac{1}{2}$, full-line reading	proved	(390), (377)
Right side of (248) at $d = 2$ equals $32\mathsf{G}^2 - 28\zeta(3)\log 2$; $\langle t_\ell^2 E_2\rangle_2 = -21\zeta(3)$	proved	(87), (86)
Per-spin identity (396); (260) at $d = 2$ equals the same; strip limit	proved	(395) (exact algebra), the factorization of `factor_K2.py` for part (c)
Inner star (398), outer integral (402), second proof of (377)	proved	(77), Euler integral, [23]
(M) for complex τ; (M) and $[\tau^2]$ under the trace at $d \neq 2$	not addressed	input (M) of Section 21.1

Checks. None of the steps above uses numerics; the following are checks only, each with its record in the package named.

- *Periodization and traces* (`check_periodization.py` in `LRI_C5_d2_simplify/proof/K2-parseval/`): at $\tau = 0.13$, 0.3 and 0.41 both identities (386), the first two of (387), the Euler–Thomae step (388) and the first identity of (390) hold to 10^{-34} or better; for example $\langle t_\ell^2 R_{0.3}\rangle_2 = 20.73663823644757695029316\ldots$ from both sides.
- *Residues* (`check_perspin.py` in the same folder): (384) at $\tau = 0.3$, $\ell = 0, 3$, Barnes integral against residue sum, to 6×10^{-19}; (396) at $\ell = 0, 1$ to 10^{-21}; the factorized summand of `factor_K2.py` against (259) at four lattice points to 4×10^{-20}. The residues of (384) by Cauchy circles agree with the formulas to 6×10^{-54}, and the identity $\Theta_\ell^{\mathrm{R}} = G_\ell T_\tau$ holds at a test point to 10^{-78} (`LRI_C5_d2_proof/review_K2_d1template_numerics_r0/logs/a01_9_20.568115.log`).
- *Single sums*: the third evaluation, and (385) computed from direct sums, agree with the closed forms and with $\pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2]$ to 3.3×10^{-67} at $\tau = \frac{1}{7}, \frac{3}{10}, \frac{9}{20}$ (`LRI_C5_d2_proof/review_K2_d1template_numerics_r0/logs/a4b_*.log`). In the second evaluation, Gauss's sum, the Thomae step and the undifferentiated single sums hold to about 10^{-50}, the derivative identities to 10^{-42} (limited by finite differences), and the formula of part (c) for P_{h} at six points (τ, u) to about 24 digits (`LRI_C5_d2_proof/K2_gauss/logs/check_singles.log`, `check_dLambda.log`, `check_bilateral.log`).
- *Algebra*: the expansion after (250), both Laurent extractions and (395) are exact (`K2-parseval/symbolic_checks.out`, `synth/synth_checks_K2.log`, `K2_gauss/logs/check_expansion.log`, `check_extraction.log`, `check_residue_series.log`); the Taylor inputs (378) hold to 10^{-23} (`check_taylor_inputs.out`).
- *Spectral moment*: a direct spectral evaluation of $\langle t_\ell^2 E_2\rangle_2$ on symmetry (spins $\ell < 4096$, Richardson extrapolation in ℓ; job 567967, `K2-parseval/e2check-567967.out`) gives -25.243194966351479993, equal to $-21\zeta(3)$ within 7.3×10^{-24}. A direct spectral integration of $C(\tau)\langle t_\ell^2(R_\tau - S_\tau)\rangle_2$ at $\tau = \frac{1}{10}, \frac{1}{5}$ agrees with the right side of (392) to about 14 digits (`K2_gauss/logs/check_endtoend.log`).
- *Extraction without the cancellation rule*: a 128-point Cauchy integral on $|\tau| = \frac{1}{8}$, at 90 digits, of the true ${}_3F_2$ functions in (392), which does not use (86), gives $-\frac{AU}{4} + \frac{1}{4}[\tau^2]C\mathcal{D}$ equal to

$32\mathsf{G}^2 - 28\zeta(3)\log 2$ to 88 digits, $[\tau^1]C\mathcal{D} = -21\zeta(3)$ to 20 digits, and agreement with the 70-digit series value to 2×10^{-65}, the truncation of that value (`K2_gauss/logs/check_K2_cauchy.log`).

- *Position space*: the edge values (400) hold to 7×10^{-30}, and the Stokes expressions and moment relations to 2×10^{-28}, at $\tau = 0.3, 0.45, 0.6$ (`LRI_C5_d2_proof/K2_d1template_fix1/check_boundary_values_mp.out`); (398) agrees with an independent Feynman-parameter representation to 56–58 digits at fourteen points (`check_lemma1_star.out`); a fresh composite run at $\tau = \frac{37}{100}$, inserting (398) into (397) and integrating over y, gives $J_{0.74} = 41.788097966990073475823724927016212 7\ldots$, equal to $\pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2]$ to 38 digits (`composite_tau037.out`).
- *The tetrahedron point*: at $\tau = \frac{1}{2}$, (377) gives $J_1 = 2\pi^2\mathfrak{F}(\frac{1}{2})^2 = 3Q_2$ (Dixon's sum, checked to 578 digits), and its second derivative, $\frac{\pi^2}{2}[\mathfrak{F}'(\frac{1}{2})^2 + \mathfrak{F}(\frac{1}{2})\mathfrak{F}''(\frac{1}{2})]$, equals $Z_2 = 4\mathsf{G}Q_2$ of (265) to 77 digits (`review_K2_d1template_numerics_r0/logs/c6b.568116.log`). By (239), a change of the single exponent f by δ has squared completion variation $4\delta^2$, so this is a check of (377) against the complex Selberg evaluation of Z_2.

18.10 Scope of the primitive reduction

The finite sample formulas use at most two-variable hypergeometric functions for generic dimension; at $d = 2$ the factorizations of Section 18.5 use the same scalar values as at $d = 1$. For generic dimension the nonprimitive functions are the explicit finite multivariable combinations of Section 17; Sections 20 and 21 extend the periodization of this section to a general rim exponent and to the family R_τ of (249), and reduce them to two-variable samples and their low-order parameter jets. At $d = 2$ all three are known in closed form: $Z_2 = 4\mathsf{G}Q_2$ by (265), and $\mathcal{L}_2$, $\mathcal{K}_2$ by (270) and (271) (Sections 18.7 and 18.8). Their physical occurrence can be collected as

$$\mathcal{N}_d = \mathcal{K}_d + \frac{3}{8}\mathcal{L}_d - \frac{149}{32}Z_d. \tag{403}$$

This is a change of basis, not an additional evaluation. In this combination (223) reads

$$\boxed{\begin{aligned} 46656C_5(d) = {} & -1536(P_5(d) + \mathcal{N}_d) + 384AP_4(d) - 1728Q_d^2 + 720A^2Q_d + 1008Q_dR \\ & - 15A^4 - 90A^2R + 414AU - 108AV - 117R^2 - 75X, \end{aligned}} \tag{404}$$

since $1536 \cdot \frac{3}{8} = 576$ and $1536 \cdot \frac{149}{32} = 7152$. Thus $\mathcal{K}_d$, $\mathcal{L}_d$ and Z_d enter $C_5(d)$ only through $\mathcal{N}_d$, which appears together with $P_5(d)$. This is a remark on the structure of (223); Sections 20 and 21 reduce the three constituents separately. The finite Fourier spectra used in Sections 20 and 21 belong to the one-parameter families through which the masters are defined, the rim exponent in (244) and τ in (249); for $\mathcal{K}_d$ this route also rests on the inputs of Section 21.1.

19 Three dimensions: half-shift determinants and rank-two functions

The half-shift reduction and the rank-two identification of this section, with their checks, were written by OpenAI GPT-6 astra, in the packages `lri_d3_reduction_update/` and `lri_d3_matrix_update/` (Section 33). The integer dimension $d = 3$ admits a further reduction of the primitive periods Q_3, $P_4(3)$ and $P_5(3)$, but not the square factorization of $d = 2$ (Section 18.5). Its multiplicity $d_\ell = 2\ell + 1$ produces an antisymmetric determinant on two lattices offset by one half (Section 19.4). This eliminates one Fourier-parity sector of the samples (311) in favor of ordinary hypergeometric values. The complementary sector is not fixed by that identity. The formulas below

evaluate it by a small, explicitly specified set of two-variable Horn values, which are identified, from Section 19.6 on, with real rank-two matrix-argument hypergeometric functions. The formulas contain no parameter derivatives, Barnes integrals or external infinite sums. They do not prove that the remaining two-variable functions are finite combinations of ordinary ${}_pF_q$ values.

The tetrahedron takes the form

$$\boxed{Q_3 = 4\pi^2 \mathfrak{H}_3(0) - \frac{\Gamma(\frac{1}{4})^{10}}{48\pi^{7/2}}.} \tag{405}$$

Here Q_3 means $Q_{d=3}$, not the one-dimensional three-loop coefficient. The fixed *two-variable* function $\mathfrak{H}_3(0)$ is fully specified in (414) below. In particular (405) is not a Gamma-only evaluation of the tetrahedron.

For the real formulas with positive rational parameters proved below, the numbers of distinct function evaluations are bounded as in the following table. For comparison, its second column counts the values of functions of at most two variables in the generic-dimensional samples (323) and (329) at the nodes (312): five blocks at each of the r nodes for $r = 3, 4$, and eight blocks at each of the seven nodes for $r = 7$, one-variable blocks included and the real part of a complex value counted once.

Period	Generic-d samples, at most	Two-variable values	Ordinary values	Total
Q_3	15	1	0	1
$P_4(3)$	20	2	3	5
$P_5(3)$	56	5	18	23

Gamma factors and finite arithmetic are not counted. The counts are upper bounds, not independence or minimality statements. The fixed complex nodes of Section 19.5 reduce the two-variable counts further, to one for $P_4(3)$ and two for $P_5(3)$, at the price of fixed complex parameters.

19.1 Ordinary hypergeometric moments

Write $h(x) = \Gamma(x + \frac{1}{8})/\Gamma(x + \frac{7}{8})$. For $\omega = \pm 1$ and nonnegative integer m define the ordinary function

$$\mathfrak{U}^{(\omega)}_{r,m}(u) = u^m h(u)^r \, {}_{r+m+1}F_{r+m}\left(\begin{matrix} 1, (u+\frac{1}{8})^{\times r}, (u+1)^{\times m} \\ (u+\frac{7}{8})^{\times r}, u^{\times m} \end{matrix}; \omega\right). \tag{406}$$

Repeated parameters with multiplicity zero are absent. Identical fixed upper and lower parameters may be canceled. Its series is $\sum_{n\geq 0} \omega^n (n+u)^m h(n+u)^r$. The required pairs are $(r, m) = (3, 0), (3, 1), (4, 0), (4, 1), (7, 1), (7, 3)$; all satisfy $3r/4 - m > 1$, so the displayed ordinary series converge absolutely at both arguments $\omega = 1$ and -1 [22].

Use the following finite combination, also entirely in ordinary functions:

$$\boxed{\begin{aligned} \Psi^{(\omega)}_{r,m}(u) = {} & \sin^r(\pi(u+\tfrac{1}{8}))\mathfrak{U}^{(\omega)}_{r,m}(u) \\ & + \omega(-1)^m \sin^r(\pi(\tfrac{1}{8}-u))\mathfrak{U}^{(\omega)}_{r,m}(1-u). \end{aligned}} \tag{407}$$

In all real formulas below $0 < u < 1$; all upper and lower parameters of the uncancelled ordinary functions are positive rational numbers. At $u = 1/8$ the second term vanishes exactly, so its function need not be computed. The symbol ω is a hypergeometric argument, not the physical expansion parameter ϵ.

The finite-mode identities needed here are

$$\boxed{\begin{aligned}&\Psi^{(+)}_{3,0}(u)=\mathfrak{a}_3\cos\pi u, && \Psi^{(+)}_{3,1}(u)=\mathfrak{b}_3\sin\pi u,\\ &\Psi^{(+)}_{4,0}(u)=\mathfrak{a}_4+\mathfrak{c}_4\cos 2\pi u, && \Psi^{(+)}_{4,1}(u)=\mathfrak{b}_4\sin 2\pi u,\\ &\Psi^{(-)}_{7,m}(u)=\sum_{j=1}^{3}h_{m,j}\sin(2\pi ju), && m=1,3.\end{aligned}} \tag{408}$$

The coefficient $\mathfrak{c}_4$ will cancel from every answer. The useful amplitudes have the fixed-value definitions

$$\begin{aligned}\mathfrak{a}_3&=\frac{\Psi^{(+)}_{3,0}(1/8)}{\cos(\pi/8)}, \quad \mathfrak{b}_3=\frac{\Psi^{(+)}_{3,1}(1/8)}{\sin(\pi/8)},\\ \mathfrak{a}_4&=\Psi^{(+)}_{4,0}(1/4), \quad \mathfrak{b}_4=\frac{\Psi^{(+)}_{4,1}(1/8)}{\sin(\pi/4)}.\end{aligned} \tag{409}$$

For $m=1,3$ let $Y_{m,-}=\Psi^{(-)}_{7,m}(1/8)$, $Y_{m,0}=\Psi^{(-)}_{7,m}(1/4)$ and $Y_{m,+}=\Psi^{(-)}_{7,m}(3/8)$. The three coefficients $h_{m,j}$, unrelated to the function $h(x)$ above, are explicitly

$$\begin{aligned}h_{m,1}&=\frac{Y_{m,0}}{2}+\frac{Y_{m,-}+Y_{m,+}}{2\sqrt{2}},\\ h_{m,2}&=\frac{Y_{m,-}-Y_{m,+}}{2},\\ h_{m,3}&=-\frac{Y_{m,0}}{2}+\frac{Y_{m,-}+Y_{m,+}}{2\sqrt{2}}.\end{aligned} \tag{410}$$

This is a finite sine transform on a proved three-mode space, not numerical interpolation of an arbitrary function.

19.1.1 Two Gamma evaluations within the ordinary sector

Dixon's sum and the Rogers–Dougall sum [22, eqs. 16.4.4 and 16.4.9] give

$$\boxed{\mathfrak{a}_3\mathfrak{b}_3=\frac{\Gamma(\frac14)^{10}}{16\sqrt{2}\,\pi^4}, \qquad \mathfrak{b}_4=\frac{\Gamma(\frac14)^6}{16\pi^{5/2}}.} \tag{411}$$

For clarity the particular sums used in this step are

$$\begin{aligned}{}_3F_2\left(\begin{matrix}(\frac14)^{\times 3}\\ 1,1\end{matrix};1\right)&=\frac{\Gamma(\frac98)\Gamma(\frac58)}{\Gamma(\frac54)\Gamma(\frac78)^2\Gamma(\frac34)},\\ {}_4F_3\left(\begin{matrix}\frac98,(\frac14)^{\times 3}\\ \frac18,1,1\end{matrix};1\right)&=\frac{\Gamma(\frac58)\Gamma(\frac18)}{\Gamma(\frac54)\Gamma(\frac34)\Gamma(\frac38)^2},\\ {}_5F_4\left(\begin{matrix}\frac98,(\frac14)^{\times 4}\\ \frac18,1,1,1\end{matrix};1\right)&=\frac{\sqrt{\pi}}{\Gamma(\frac54)\Gamma(\frac34)^3}.\end{aligned} \tag{412}$$

For the second identity insert a canceled pair 5/8 into the Rogers–Dougall ${}_5F_4$, with parameters $(a,b,c,d)=(1/4,1/4,1/4,5/8)$ in the notation of [22, eq. 16.4.9]; this d is not the dimension. For the third take $(a,b,c,d)=(1/4,1/4,1/4,1/4)$. Their convergence conditions hold. Substitution into (409), followed by reflection and duplication [26], proves (411). Thus the Gamma powers in (405) are analytic evaluations, not PSLQ fits.

19.2 The remaining two-variable functions and complete sample formulas

Use the Horn convention (320). Put $\boldsymbol{e}=(1,0)$, $\boldsymbol{f}=(0,1)$, $\boldsymbol{c}=(1,1)$ and $\boldsymbol{v}=(1,2)$, and set

$$\begin{aligned}\mathsf{U}_r(t)&=\{(1,\boldsymbol{e}),(1,\boldsymbol{f}),(\tfrac{3}{2},\boldsymbol{e}),(\tfrac{3}{4}+t,\boldsymbol{c})^{\times r},(\tfrac{1}{4}+t,\boldsymbol{f})^{\times r}\},\\ \mathsf{V}_r(t)&=\{(\tfrac{1}{2},\boldsymbol{e}),(\tfrac{3}{2}+t,\boldsymbol{c})^{\times r},(1+t,\boldsymbol{f})^{\times r}\}.\end{aligned}\tag{413}$$

For $r=3,4$ define the fixed two-variable function

$$\boxed{\mathfrak{H}_r(t)=\mathfrak{H}[\mathsf{U}_r(t);\mathsf{V}_r(t);(1,1)].}\tag{414}$$

For $r=7$ adjoin three upper/lower row pairs:

$$\begin{aligned}x_t&=t+\tfrac{5}{8},\qquad & y_t&=t+\tfrac{1}{8},\quad v_t=2t+\tfrac{3}{4},\\ \mathfrak{H}_7(t)&=\mathfrak{H}\Big[\mathsf{U}_7(t)\cup\{(1+x_t,\boldsymbol{c}),(1+y_t,\boldsymbol{f}),(1+v_t,\boldsymbol{v})\};\\ &\qquad \mathsf{V}_7(t)\cup\{(x_t,\boldsymbol{c}),(y_t,\boldsymbol{f}),(v_t,\boldsymbol{v})\};(-1,1)\Big].\end{aligned}\tag{415}$$

Every row is printed here; there is no unevaluated coefficient-extraction instruction. The two upper rows with parameter 1 cancel the two factorials of the Horn convention. The ratio $(3/2)_\ell/(1/2)_\ell=2\ell+1$ supplies the three-dimensional angular multiplicity. The three additional pairs supply x, y and $x+y$. All real nodes used below have positive rational rows.

For compact scalar prefactors write

$$\begin{aligned}G_r(t)&=\left[\frac{\Gamma(\frac{3}{4}+t)\Gamma(\frac{1}{4}+t)}{\Gamma(\frac{3}{2}+t)\Gamma(1+t)}\right]^r,\\ V_r(t)&=G_r(t)\mathfrak{H}_r(t),\qquad & r&=3,4,\\ V_7(t)&=4x_ty_tv_tG_7(t)\mathfrak{H}_7(t).\end{aligned}\tag{416}$$

The symbol $V_r(t)$ here is local to this section; it is not the polygamma constant V in the general-dimensional coefficient formula.

Let $\mathsf{X}_r(t)=\Xi_r(t)|_{q=3/4}$, with Ξ_r as in (304), and define

$$\begin{aligned}t^\dagger&=\tfrac{1}{4}-t,\qquad & f(t)&=-\tfrac{1}{2}\cos 2\pi t,\\ D(t)&=\sin\pi t\sin\pi(t+\tfrac{3}{4}),\quad & T(t)&=\tan\pi(\tfrac{3}{4}+2t),\quad z_t=\tfrac{7}{8}-t.\end{aligned}\tag{417}$$

A sample in a wheel is now just

$$\boxed{\begin{aligned}\mathsf{X}_r(t)&=f(t)^rV_r(t)+f(t^\dagger)^rV_r(t^\dagger)\\ &\qquad+2D(t)^r\big[\mathfrak{U}^{(+)}_{r,1}(x_t)\mathfrak{U}^{(+)}_{r,0}(z_t)+\mathfrak{U}^{(+)}_{r,0}(x_t)\mathfrak{U}^{(+)}_{r,1}(z_t)\big],\qquad r=3,4.\end{aligned}}\tag{418}$$

The corresponding five-loop sample is

$$\boxed{\begin{aligned}\mathsf{X}_7(t)&=T(t)\Big\{f(t)^7V_7(t)-f(t^\dagger)^7V_7(t^\dagger)\\ &\qquad+8D(t)^7\big[\mathfrak{U}^{(-)}_{7,3}(x_t)\mathfrak{U}^{(-)}_{7,1}(z_t)-\mathfrak{U}^{(-)}_{7,1}(x_t)\mathfrak{U}^{(-)}_{7,3}(z_t)\big]\Big\}.\end{aligned}}\tag{419}$$

Only fixed ordinary and two-variable functions appear. In particular the middle triangular part of the (ℓ,n) lattice of (304) factorizes into ordinary products. At $t=0$ the left and middle pieces vanish, giving the especially simple nonsingular formulas

$$\boxed{\begin{aligned}\mathsf{X}_3(0)&=-2\sqrt{2}\,\pi^{3/2}\mathfrak{H}_3(0),\\ \mathsf{X}_4(0)&=4\pi^2\mathfrak{H}_4(0),\\ \mathsf{X}_7(0)&=\frac{15\sqrt{2}}{8}\pi^{7/2}\mathfrak{H}_7(0).\end{aligned}}\tag{420}$$

These are values of already regularized functions, not singular parameter limits to be taken by the reader.

19.3 Primitive formulas with only real rational parameters

Let $c_8 = \cos(\pi/8)$ and $s_8 = \sin(\pi/8)$, as in (331). For the four-loop answer use

$$\begin{aligned} X_0 &= 4\pi^2 \mathfrak{H}_4(0), \\ X_h &= \frac{1}{32}\left[\frac{\Gamma(\frac{7}{8})\Gamma(\frac{3}{8})}{\Gamma(\frac{13}{8})\Gamma(\frac{9}{8})}\right]^4 \mathfrak{H}_4(\tfrac{1}{8}) + 4 s_8^8 \mathfrak{U}_{4,0}^{(+)}(\tfrac{3}{4})\mathfrak{U}_{4,1}^{(+)}(\tfrac{3}{4}), \\ \mathfrak{a}_4 &= c_8^4 \mathfrak{U}_{4,0}^{(+)}(\tfrac{1}{4}) + s_8^4 \mathfrak{U}_{4,0}^{(+)}(\tfrac{3}{4}), \qquad \mathfrak{b}_4 = \frac{\Gamma(\frac{1}{4})^6}{16\pi^{5/2}}. \end{aligned} \tag{421}$$

Here $X_0 = \mathsf{X}_4(0)$ and $X_h = \mathsf{X}_4(1/8)$; they are not the one-dimensional X_0, X_+, X_- of Section 18. The answer is

$$\boxed{P_4(3) = 2\pi X_0 - 10\pi X_h + (14\sqrt{2} - 20)\pi \mathfrak{a}_4 \mathfrak{b}_4.} \tag{422}$$

This uses two $\mathfrak{H}_4$ values, two ordinary ${}_5F_4(1)$ values, and one ordinary ${}_6F_5(1)$ value, plus Gamma factors. No claim that the two $\mathfrak{H}_4$ values have become ordinary functions is intended.

For $P_5(3)$ form the two ordinary-product constants

$$\begin{aligned} \alpha &= h_{3,2}h_{1,1} - h_{3,1}h_{1,2}, \\ \beta &= h_{3,2}h_{1,3} - h_{3,3}h_{1,2}, \\ \theta(t) &= 2\pi(t + \tfrac{3}{8}), \qquad w(t) = \cos\theta(t), \\ \mathcal{O}(t) &= 16 w(t)(1 - w(t)^2)\left[-\alpha - 3\beta + 4\beta w(t)^2\right]. \end{aligned} \tag{423}$$

All $h_{m,j}$ were given by the finite ordinary evaluations (410). Use three regularized samples, with $u_j = \cos(4\pi(t_j + 3/8))$ the variable of (435) below:

$$\begin{aligned} t_0 &= 0, & t_+ &= \tfrac{1}{16}, & t_- &= -\tfrac{1}{16}, \\ u_0 &= 0, & u_+ &= \tfrac{1}{\sqrt{2}}, & u_- &= -\tfrac{1}{\sqrt{2}}, \\ E_j &= \frac{\mathsf{X}_7(t_j) - \mathcal{O}(t_j)}{1 - u_j}, & & j = 0, +, -. \end{aligned} \tag{424}$$

Equations (419) and (420) specify each sample without derivatives. The result is

$$\boxed{\begin{aligned} P_5(3) = 4\sqrt{2}\,\pi^{3/2}\left[-35E_0 + \left(22 + \frac{13}{\sqrt{2}}\right)E_+ + \left(22 - \frac{13}{\sqrt{2}}\right)E_-\right] \\ - 448\pi^{3/2}(2\beta - \alpha). \end{aligned}} \tag{425}$$

The only two-variable values are

$$\mathfrak{H}_7(0), \quad \mathfrak{H}_7(\tfrac{1}{16}), \quad \mathfrak{H}_7(-\tfrac{1}{16}), \quad \mathfrak{H}_7(\tfrac{3}{16}), \quad \mathfrak{H}_7(\tfrac{5}{16}). \tag{426}$$

All their uncancelled parameters are positive rational, and their arguments are $(-1, 1)$. The ten ordinary values for $h_{1,j}, h_{3,j}$ use $u = 1/8, 1/4, 3/4, 3/8, 5/8$ and $m = 1, 3$. Eight further ordinary values in the two middle determinants use $u = 9/16, 11/16, 13/16, 15/16$ and $m = 1, 3$. In total $P_5(3)$ therefore uses eighteen ordinary values. The excess of every one of these ordinary series is $17/4 - m > 0$.

19.4 Proof of the dimension-three determinant

19.4.1 The half-offset lattice, not a dimension-two square

At $q = 3/4$ set $k = \ell + n$, $j = n$ in the bilateral sums (304), with g_ℓ and a_ℓ as in (300), and put

$$x = k + t + 5/8, \qquad y = j + t + 1/8, \qquad u = t + 1/8.$$

Gamma reflection gives, as a meromorphic identity,

$$g_\ell(a_\ell+n+t)=-\frac{\sin\pi(3/4-t)}{\sin\pi t}\,h(x)h(y),\qquad d_\ell=2\ell+1=2(x-y). \tag{427}$$

Here $k\ge j$, with both indices otherwise bilateral. The two sequences are $h(k+u+1/2)$ and $h(j+u)$; unlike $d=2$, they are not the same sequence on the same integer lattice.

The five-loop Gamma measure becomes elementary:

$$\boxed{\frac{\Gamma(\frac12+v)\Gamma(\frac12-v)}{\Gamma(v)\Gamma(-v)}=-v\tan\pi v.} \tag{428}$$

This follows directly from the two reflection formulas in [26]. At a sampled pole-family shift $v=2s=x+y$,

$$(\ell+\tfrac12)^2-v^2=-4xy,\qquad(-1)^\ell=(-1)^{k-j}.$$

Multiplying by d_ℓ gives the polynomial $8xy(x^2-y^2)$ times the common tangent $T(t)$ and the alternating sign. Splitting the bilateral lattice into its two infinite tails and its middle part now proves (418)–(419). For example the middle part has $\ell=j+k+1$, $j,k\ge0$, and $x=k+t+5/8$, $y=j+7/8-t$. Its weight is $2(x+y)$ in a wheel and the rank-two polynomial $8(x^3y-xy^3)T(t)$ in the primitive. These separate into the displayed ordinary products.

19.4.2 Completing only the antisymmetric half

A shift $t\mapsto t+1/2$ interchanges the two lattices. In a wheel the factor $x-y$ changes sign. In the primitive the extra alternating sign also changes sign. After reindexing the two complementary half-planes, the full-square sums factor into determinants. The result is

$$\begin{aligned}
&\mathsf{X}_r(t)-(-1)^r\mathsf{X}_r(t+\tfrac12)\\
&\quad=2(-1)^r\big[\Psi^{(+)}_{r,1}(u+\tfrac12)\Psi^{(+)}_{r,0}(u)-\Psi^{(+)}_{r,0}(u+\tfrac12)\Psi^{(+)}_{r,1}(u)\big],\quad r=3,4,\\
&\mathsf{X}_7(t)-\mathsf{X}_7(t+\tfrac12)\\
&\quad=-8T(t)\big[\Psi^{(-)}_{7,3}(u+\tfrac12)\Psi^{(-)}_{7,1}(u)-\Psi^{(-)}_{7,1}(u+\tfrac12)\Psi^{(-)}_{7,3}(u)\big].
\end{aligned} \tag{429}$$

The factors are obtained from $\sin\pi(t+1/4)\sin\pi(t+3/4)=\cos(2\pi t)/2$. All moments involved obey $3r/4-m>1$, so the product sums used in the reindexing are absolutely convergent. The formulas extend across removable trigonometric singularities by equality of the complete meromorphic expressions, not by discarding a pole family.

For the proof of the finite modes, reflection gives

$$h(-x)=\frac{\sin\pi(x+1/8)}{\sin\pi(1/8-x)}h(x).$$

It turns (407) into the regularized bilateral moment $\sin^r\pi(u+1/8)\sum_{n\in\mathbb{Z}}\omega^n(n+u)^mh(n+u)^r$. This proof representation has all poles canceled, and satisfies

$$\begin{aligned}
\Psi(u+1)&=(-1)^r\omega\Psi(u),&\qquad\Psi(-u)&=(-1)^m\Psi(u),\\
|\Psi(x+\mathrm{i}y)|&=O\Big(e^{r\pi|y|}|y|^{m+1-3r/4}\Big)&&(|y|\to\infty).
\end{aligned} \tag{430}$$

The growth estimate follows from centered Gamma-ratio asymptotics and absolute integral comparison [27]. Periodicity or antiperiodicity supplies integer or half-integer Fourier frequencies. Vertical contour translation excludes their boundary frequencies when $m+1-3r/4<0$. Reflection then leaves exactly the modes in (408). This proves finite-dimensionality, rather than assuming it from a few samples.

Substituting those modes in (429) yields

$$\boxed{\begin{aligned}&\mathsf{X}_3(t)+\mathsf{X}_3(t+\tfrac12)=-2\mathfrak{a}_3\mathfrak{b}_3,\\&\mathsf{X}_4(t)-\mathsf{X}_4(t+\tfrac12)=4\mathfrak{a}_4\mathfrak{b}_4\cos\theta(t),\\&\mathsf{X}_7(t)-\mathsf{X}_7(t+\tfrac12)=2\mathcal{O}(t).\end{aligned}} \tag{431}$$

In the cosine convention (311), the first two lines prove $c_{3,2}=0$, $c_{3,0}=-\mathfrak{a}_3\mathfrak{b}_3$, $c_{4,3}=0$, and $c_{4,1}=2\mathfrak{a}_4\mathfrak{b}_4$. The last line fixes

$$c_{7,1}=-4\alpha-4\beta,\qquad c_{7,3}=4\alpha+8\beta,\qquad c_{7,5}=-4\beta. \tag{432}$$

Of these, $c_{3,2}=0$ and $c_{4,3}=0$ hold in every dimension, by the top-mode theorem (473) of Section 20.1.2; the identifications $c_{3,0}=-\mathfrak{a}_3\mathfrak{b}_3$ and $c_{4,1}=2\mathfrak{a}_4\mathfrak{b}_4$ and the coefficients (432) are the additional restrictions special to three dimensions. The identifications of $c_{3,0}$ and $c_{4,1}$ also hold at a general rim exponent b, by the same proof (Section 20.6); the $d=3$ forms of $\mathcal{L}_d$ and Z_d in Section 20.6 use them.

19.4.3 Finite residue algebra and the remaining even polynomial

For a self-contained normalization check, the weights $B_{r,k}/\pi^{r-1}$ in (318) at $q=3/4$ are

$$\begin{aligned}&r=3:\quad(8,-6\sqrt2,12),\\&r=4:\quad(32,-24\sqrt2,40,-40\sqrt2),\\&r=7:\quad(2624,-1904\sqrt2,2912,-2352\sqrt2,4032,-3696\sqrt2,7392).\end{aligned} \tag{433}$$

Together with (319) at $q=3/4$, that is $Q_3=\Gamma(3/2)^3\mathcal{E}_3/3$, $P_4(3)=\Gamma(3/2)^4\mathcal{E}_4$, and $P_5(3)=\Gamma(3/2)^5\mathcal{E}_7/2$, these weights give (405) and (422) immediately. The clockwise right closure supplies the minus sign in $\mathcal{E}_r$.

For $r=7$ there are also exact zeros

$$\mathsf{X}_7(\tfrac18)=\mathsf{X}_7(\tfrac58)=0. \tag{434}$$

Indeed $v=\ell+3/4+2n+2t$ is an integer at either point, and the measure in (428) vanishes. No Gamma pole of g_ℓ occurs there. The termwise vanishing is legitimate in the absolutely convergent regularized sum. Hence the even part is exactly

$$\mathsf{X}_7(t)-\mathcal{O}(t)=(1-u)(p_0+p_1u+p_2u^2),\qquad u=\cos(4\pi(t+3/8)). \tag{435}$$

The even-mode contribution to the $r=7$ residue weight is $-32(9p_0+13p_1+22p_2)$; the odd contribution is $1792\sqrt2(2\beta-\alpha)$. For every quadratic polynomial p,

$$9p_0+13p_1+22p_2=-35p(0)+\left(22+\frac{13}{\sqrt2}\right)p(1/\sqrt2)+\left(22-\frac{13}{\sqrt2}\right)p(-1/\sqrt2). \tag{436}$$

This is verified by comparing three coefficients. It proves (425) without a derivative, infinitesimal step, or numerical interpolation error.

Crucially, (431) does not determine the complementary parity sector. It leaves one coefficient for $r=3$, two for $r=4$, and three for the even $r=7$ polynomial. Equations (414) and (415) evaluate that remaining information. This explains why the dimension-two square proof of Section 18.5 cannot simply be repeated with the replacement $d_\ell=2\ell+1$. It does not prove that some different identity could never reduce the residual Horn values further.

19.5 A variant with fixed complex nodes

The real rational formulas above are the preferred presentation when parameter simplicity matters. Two further exact finite-node identities give smaller two-variable counts. Define

$$\begin{aligned} t_4^* &= -\tfrac{3}{8} + \frac{\mathrm{i}\log 2}{4\pi}, \qquad & t_7^* &= -\tfrac{3}{8} + \frac{\mathrm{i}}{4\pi}\operatorname{arcosh}\frac{22}{13}, \\ E_0^* &= \mathsf{X}_7(0) - \mathcal{O}(0), & E_*^* &= \mathsf{X}_7(t_7^*) - \mathcal{O}(t_7^*), \end{aligned} \tag{437}$$

with $\mathcal{O}$ as in (423); E_0^* is the E_0 of (424). Then

$$\boxed{\begin{aligned} P_4(3) &= 8\pi\left[3\sqrt{2}\,\mathfrak{a}_4\mathfrak{b}_4 - \mathsf{X}_4(t_4^*)\right], \\ P_5(3) &= \frac{2\sqrt{2}\,\pi^{3/2}}{11}\left[29E_0^* - \frac{2197}{9}E_*^*\right] - 448\pi^{3/2}(2\beta-\alpha). \end{aligned}} \tag{438}$$

Both nodes lie on the line $\operatorname{Re} t = -\frac{3}{8}$. There every $\cos(2\pi k(t+\frac{3}{8}))$ is real, so the samples (311), whose coefficients are real, are real.

Proof. Since Ξ_r is entire, (311) and (435) are identities of entire functions whose right sides are trigonometric polynomials, so they hold at complex t. By (311) at $q = \frac{3}{4}$, with $c_{4,3} = 0$ and $c_{4,1} = 2\mathfrak{a}_4\mathfrak{b}_4$ (Section 19.4), $\mathsf{X}_4(t) = c_{4,0} + 2\mathfrak{a}_4\mathfrak{b}_4\cos\theta(t) + c_{4,2}\cos 2\theta(t)$ with $\theta(t)$ as in (423). The weights (433), inserted in (318) and (319), give $P_4(3) = -\pi\big(8c_{4,0} - 6\sqrt{2}\,c_{4,1} + 10c_{4,2}\big)$. At t_4^* one has $\cos\theta = \cosh(\frac{1}{2}\log 2) = \frac{3}{2\sqrt{2}}$ and $\cos 2\theta = \cos\big(4\pi(t_4^*+\frac{3}{8})\big) = \frac{5}{4}$, which gives the first line. For the second, (435) gives $E_0^* = p(0)$ and, since $u = \cos\big(4\pi(t_7^*+\frac{3}{8})\big) = \frac{22}{13}$ and $1 - \frac{22}{13} = -\frac{9}{13}$, $E_*^* = -\frac{9}{13}\,p(\frac{22}{13})$. The rule

$$9p_0 + 13p_1 + 22p_2 = \frac{29}{22}\,p(0) + \frac{169}{22}\,p(\tfrac{22}{13}),$$

exact for every quadratic p (compare the coefficients of 1, u and u^2), replaces (436) in the proof of (425); $169 \cdot 13 = 2197$. □

A sample on this line needs only one Horn value. By (413) and (416), with $x = x_t + k$, $y = y_t + n$ and $k = \ell + n$ as in (427), $V_r(t) = \sum_{k\ge n\ge 0} 2(x-y)h(x)^r h(y)^r$ for $r = 3, 4$ and $V_7(t) = 8\sum_{k\ge n\ge 0}(-1)^{k-n}xy(x^2-y^2)h(x)^7h(y)^7$. A shift $t \mapsto t+1$ moves x and y by one and so removes the boundary row $n = 0$; the sum of that row is denoted $V_r^{(0)}(t)$:

$$\begin{aligned} V_r(t+1) &= V_r(t) - V_r^{(0)}(t), \\ V_r^{(0)}(t) &= 2h(y_t)^r\left[\mathfrak{U}_{r,1}^{(+)}(x_t) - y_t\,\mathfrak{U}_{r,0}^{(+)}(x_t)\right], \qquad r = 3, 4, \\ V_7^{(0)}(t) &= 8h(y_t)^7\left[y_t\,\mathfrak{U}_{7,3}^{(-)}(x_t) - y_t^3\,\mathfrak{U}_{7,1}^{(-)}(x_t)\right]. \end{aligned} \tag{439}$$

Let $t = -\frac{3}{8} + \mathrm{i}\sigma$ with $\sigma > 0$. Then $t^\dagger = 1 + \bar{t}$, and since V_r is real on the real axis, $V_r(t^\dagger) = \overline{V_r(t) - V_r^{(0)}(t)}$. The other ordinary factors reduce by $\mathfrak{U}_{r,m}^{(\omega)}(u+1) = \omega\big[\mathfrak{U}_{r,m}^{(\omega)}(u) - u^m h(u)^r\big]$. This proves the counts of one $\mathfrak{H}_4$ value for $P_4(3)$ and two $\mathfrak{H}_7$ values for $P_5(3)$, with no second complex Horn value hidden in the samples. The nodes are fixed constants, not regulators. They do introduce fixed complex parameters, some with negative real part (at t_4^* the row $\frac{1}{4} + t$ of (413) has real part $-\frac{1}{8}$); there are no denominator poles at the stated nodes. No minimality claim is made for this variant. Its numerical agreement with (422) and (425) is given in Section 19.10.

19.6 The real rank-two hypergeometric normalization

This subsection and the next two identify the residual functions $\mathfrak{H}_r$ with standard real matrix-argument hypergeometric functions of rank two; Section 19.9 then records a matrix transformation

that maps the tetrahedron to itself and so does not reduce it. The identification is exact and coefficientwise, not a numerical fit. It does not reduce them to ordinary one-variable ${}_pF_q$ values: the ordinary-hypergeometric endpoint reached for the primitive periods at $d = 2$ (Section 18.5) is not reached at $d = 3$. What it adds is a classification: the residual objects are low-order functions on a rank-two symmetric cone, not unclassified Horn arrays. The rational-node formulas above remain the preferred explicit specification of the periods.

For a partition $\kappa = (m, n)$ with $m \geq n \geq 0$ (here m is a part of κ, not the moment order m of (406)), write

$$[a]_\kappa = (a)_m(a - \tfrac{1}{2})_n, \qquad |\kappa| = m + n. \tag{440}$$

From here to the end of this section the Jack parameter is $\alpha = 2$, the matrix size is two, and the physical dimension is $d = 3$. This α is unrelated to the constant α of (423). These three numbers refer to different notions. Let $C_\kappa^{(2)}$ denote the C-normalized real zonal polynomial, so that $\sum_{|\kappa|=k} C_\kappa^{(2)}(X) = (\operatorname{tr} X)^k$. We use the standard definition [38, 39]

$${}_pF_q^{(\alpha=2)}\left(\begin{matrix}\boldsymbol{a}\\ \boldsymbol{b}\end{matrix}; zI_2\right) = \sum_{m\geq n\geq 0} \frac{\prod_i [a_i]_{(m,n)}}{\prod_j [b_j]_{(m,n)}} \frac{C^{(2)}_{(m,n)}(I_2)}{(m+n)!}\, z^{m+n}. \tag{441}$$

This definition fixes the conventional special function; it is not a claim that writing a summation as a function symbol has removed its rank-two structure.

The identity that matches the angular multiplicity is

$$\boxed{\frac{C^{(2)}_{(m,n)}(I_2)}{(m+n)!} = \frac{2(m-n)+1}{(\frac{3}{2})_m n!} = \frac{2(m-n)+1}{[\frac{3}{2}]_{(m,n)}}.} \tag{442}$$

Here is an explicit hook-product verification, including normalization. If $a(s)$ and $l(s)$ are the arm and leg lengths of a box of κ, the Jack hook product is

$$\begin{aligned} j_{(m,n)}(2) &= \prod_{s\in(m,n)} (2a(s)+l(s)+1)(2a(s)+l(s)+2) \\ &= 2^{2(m+n)} \frac{(\frac{3}{2})_m\, m!\, n!\, (\frac{1}{2})_n}{2(m-n)+1}. \end{aligned} \tag{443}$$

The specialization formula $C_\kappa^{(2)}(I_2)/|\kappa|! = 2^{2|\kappa|}[1]_\kappa / j_\kappa(2)$ then gives (442). In particular the factor $2\ell + 1$ with $\ell = m - n$ is exactly the rank-two real-zonal weight after canceling one $[3/2]_\kappa$. This is why a real matrix-argument function appears.

19.7 A standard ${}_3F_2$ of matrix argument for the tetrahedron

Retain the residual functions $\mathfrak{H}_r(t)$ of (414) and (415) and put

$$a = t + \frac{3}{4}, \qquad b = t + \frac{3}{2}. \tag{444}$$

For $r = 3, 4$ their exact identification is

$$\boxed{\mathfrak{H}_r(t) = {}_{r+1}F_r^{(\alpha=2)}\left(\begin{matrix}a^{\times r},\ \frac{3}{2}\\ b^{\times r}\end{matrix}; I_2\right).} \tag{445}$$

To verify it, use $m = \ell + n$ in the explicit Horn arrays. The summand of either side is

$$(2\ell+1)\left[\frac{(a)_{\ell+n}(a-\frac{1}{2})_n}{(b)_{\ell+n}(b-\frac{1}{2})_n}\right]^r. \tag{446}$$

On the matrix side this follows from canceling $[3/2]_{(m,n)}$ against (442). Thus no summation theorem or exchange of a conditionally convergent sum is being assumed. The endpoint sums here are absolutely convergent: along $m \to \infty$ with n fixed the majorant is $m^{1-3r/4}$, and the remaining n factors are summable for $r = 3, 4$.

At $t = 0$ one upper parameter $3/2$ cancels a lower parameter. The tetrahedron formula (405) therefore becomes

$$\boxed{Q_3 = 4\pi^2 {}_3F_2^{(\alpha=2)}\left(\begin{matrix}\frac{3}{4},\frac{3}{4},\frac{3}{4}\\ \frac{3}{2},\frac{3}{2}\end{matrix}; I_2\right) - \frac{\Gamma(\frac{1}{4})^{10}}{48\pi^{7/2}}.} \tag{447}$$

The same value is the rim sample $\mathfrak{t}_3(b)$ of (495) at $d = 3$ and $b = \frac{3}{2}$: there $\sigma = g = \frac{3}{4}$, and the summand $d_\ell T_{\ell n}(\frac{3}{2})^3$ of (490) is (446) with $r = 3$, $t = 0$. Its second b-derivative enters Z_3 (Section 20.6). Both remaining four-loop values likewise have short conventional arrays:

$$\boxed{\begin{aligned}\mathfrak{H}_4(0) &= {}_4F_3^{(\alpha=2)}\left(\begin{matrix}(\frac{3}{4})^{\times 4}\\ (\frac{3}{2})^{\times 3}\end{matrix}; I_2\right),\\ \mathfrak{H}_4(\tfrac{1}{8}) &= {}_5F_4^{(\alpha=2)}\left(\begin{matrix}(\frac{7}{8})^{\times 4},\frac{3}{2}\\ (\frac{13}{8})^{\times 4}\end{matrix}; I_2\right).\end{aligned}} \tag{448}$$

Substitution in (422) gives an equivalent matrix-argument presentation of $P_4(3)$ without changing its ordinary moment factors. Neither of these matrix functions is to be evaluated by an ordinary scalar `hyper` call.

A local coefficient check makes the distinction unambiguous. With the same upper and lower parameters as in (447),

$$\begin{aligned}{}_3F_2^{(\alpha=2)}(\tfrac{3}{4},\tfrac{3}{4},\tfrac{3}{4};\tfrac{3}{2},\tfrac{3}{2};zI_2) &= 1 + \frac{3}{8}z + \frac{2769}{12800}z^2 + \frac{15113}{102400}z^3 + O(z^4),\\ {}_3F_2(\tfrac{3}{4},\tfrac{3}{4},\tfrac{3}{4};\tfrac{3}{2},\tfrac{3}{2};z) &= 1 + \frac{3}{16}z + \frac{1029}{12800}z^2 + \frac{9317}{204800}z^3 + O(z^4).\end{aligned} \tag{449}$$

The first coefficients already differ. This checks the normalization; it does not preclude an identity with some *other* scalar hypergeometric functions or parameters.

19.8 Five loops: an exact three-term standard-function dictionary

The polynomial insertion in $\mathfrak{H}_7$ can also be expressed using ordinary parameter shifts of standard rank-two functions, rather than a bespoke extra Horn row. Set (this v is local; it is not v_t of (415))

$$v = t + \frac{5}{8}, \qquad \delta_v = v(v - \tfrac{1}{2}). \tag{450}$$

Define three standard matrix-argument functions, all at $-I_2$:

$$\begin{aligned}\mathcal{J}_A(t) &= {}_9F_8^{(\alpha=2)}\left(\begin{matrix}a^{\times 7},\frac{3}{2},v+2\\ b^{\times 7},v\end{matrix}; -I_2\right),\\ \mathcal{J}_B(t) &= {}_{10}F_9^{(\alpha=2)}\left(\begin{matrix}a^{\times 7},\frac{3}{2},v+1,v+1\\ b^{\times 7},v,v\end{matrix}; -I_2\right),\\ \mathcal{J}_C(t) &= {}_9F_8^{(\alpha=2)}\left(\begin{matrix}a^{\times 7},\frac{3}{2},v+1\\ b^{\times 7},v\end{matrix}; -I_2\right).\end{aligned} \tag{451}$$

Then the exact dictionary is

$$\boxed{\mathfrak{H}_7(t) = \frac{\delta_{v+1}\mathcal{J}_A(t) - \delta_v\mathcal{J}_B(t) - \mathcal{J}_C(t)}{2v - \frac{1}{2}}.} \tag{452}$$

The five arguments required by (425), listed in (426), are $t = 0, \pm 1/16, 3/16, 5/16$. They all have $v > 1/2$, and all generalized Pochhammer factors in these arrays have positive real starting arguments. At $t = 0$, canceling one common $3/2$ changes the three orders to ${}_8F_7$, ${}_9F_8$ and ${}_8F_7$, still *of matrix argument*. No coalescence limit or parameter differentiation is used.

Here is the coefficientwise proof, which also fixes the sign of the matrix argument. For $\kappa = (m, n)$ put $x = m + v$, $y = n + v - 1/2$. Direct Pochhammer cancellation gives

$$\begin{aligned} \frac{[v+1]_\kappa}{[v]_\kappa} &= \frac{xy}{\delta_v}, \\ \frac{[v+2]_\kappa}{[v]_\kappa} &= \frac{xy(x+1)(y+1)}{\delta_v \delta_{v+1}}. \end{aligned} \tag{453}$$

The only polynomial identity needed is

$$xy(x+y) = xy(x+1)(y+1) - (xy)^2 - xy. \tag{454}$$

Consequently the right-hand side of (452) multiplies the common coefficient (446), with $r = 7$, by $xy(x+y)/(\delta_v(2v-1/2))$, exactly the three polynomial ratios in the Horn array (415) of $\mathfrak{H}_7$. Finally $(-1)^\ell = (-1)^{m-n} = (-1)^{m+n}$, so the matrix argument is $-I_2$, *not* $\mathrm{diag}(-1, 1)$. Each of the three endpoint series separately converges absolutely. For example, the largest polynomial insertion has degree two in each of m, n; its absolute summand is bounded by a constant times $(m+1)^{-9/4}(n+1)^{-13/4}$ on $m \geq n$, which is summable.

This dictionary standardizes the function class, but increases the number of displayed functions per Horn block from one to three. It is not advertised as a more economical numerical formula. In particular, it does not eliminate the complementary ordered-lattice information isolated by the half-shift determinant.

19.9 A checked transformation that does not finish the tetrahedron

The matrix Kummer transformation [39, Eq. (35.8.5)] is a useful diagnostic. For $s_{\mathrm{K}} = b_1 + b_2 - a_1 - a_2 - a_3$ it writes the ${}_3F_2^{(\alpha=2)}$ at I_2 with the parameter array $(a_1, a_2, a_3; b_1, b_2)$ as the one with the array

$$(a_1, a_2, a_3; b_1, b_2) \longmapsto (b_1 - a_1, b_1 - a_2, a_3; b_1, s_{\mathrm{K}} + a_3), \tag{455}$$

times the prefactor $\Gamma_2(b_2)\Gamma_2(s_{\mathrm{K}})/[\Gamma_2(b_2 - a_3)\Gamma_2(s_{\mathrm{K}} + a_3)]$. Here $\Gamma_2(a) = \sqrt{\pi}\,\Gamma(a)\Gamma(a - \frac{1}{2})$ is the multivariate Gamma function of order two, and the conditions $\mathrm{Re}\, b_2, \mathrm{Re}\, s_{\mathrm{K}} > \frac{1}{2}$ hold below. For the array of (447), $s_{\mathrm{K}} = \frac{3}{4}$: the transformed array is identical, and the prefactor equals one. Parameter permutations add nothing, because the three upper parameters and the two lower parameters are separately equal. Transformations generated by this map and these permutations therefore expose no canceling parameter and no terminating series. This is a statement about this explicit route, not a proof that a different reduction to ordinary ${}_pF_q$ values cannot exist. The script `code/rank_two_checks.py` of `lri_d3_matrix_update/` checks the fixed point exactly.

The determinantal formulas for *Hermitian* matrix-argument functions have Jack parameter $\alpha = 1$, whereas (442) fixes $\alpha = 2$ here. Transplanting a Hermitian formula would change the coefficients. The distinction between real and Hermitian matrix-argument transformations is made explicitly in [40]; it explains why the familiar determinant machinery is not itself the missing scalar identity.

19.10 Three-dimensional values and limits of the reduction

The fixed rational formulas give

$$\begin{aligned} Q_3 &= 5.017102564553855369590854845690 18\ldots, \\ P_4(3) &= 62.058327943406672466463388584096 7\ldots, \\ P_5(3) &= 240.340317759083549816366546388004\ldots. \end{aligned}$$

The digits shown are correctly rounded; ... marks omitted digits. Fresh integrations of the Gamma kernels on the real axis, which do not use these formulas, agree with these values to about 10^{-51}, 10^{-52} and 10^{-53}, respectively. For Q_3 and $P_4(3)$ there is a stronger independent check: the 100-digit spectral evaluation of Section 16.8 agrees with $3Q_3$ and $P_4(3)$ to 3.5×10^{-79} and 2.1×10^{-86}. For $P_5(3)$ the kernel integration is the only independent check. The formulas themselves, evaluated at 55 and at 90 working digits with changed truncation and tail order, differ by 2.0×10^{-53}, 2.9×10^{-53} and 1.6×10^{-49}; the 55-digit run is the less accurate one, since the 90-digit values agree with a further 100-digit evaluation to better than 10^{-74}. These are numerical residuals, not interval bounds. The rank-two identity is a coefficientwise identity with Jack parameter $\alpha=2$: neither its superscript nor its matrix argument may be dropped. No complete reduction of these remaining values, or of $\mathcal{N}_3$ ((403) at $d=3$), to scalar ordinary ${}_pF_q$ values is asserted.

How the values were obtained and checked. The formula side sums the ordinary and two-variable series of the formulas above to a truncation point and adds an asymptotic tail built from the Stirling expansion of the Gamma ratios and Hurwitz zeta values. The runs at 55 and 90 working digits used truncation 80 with tail order 36 and truncation 160 with tail order 64 (`code/general_d3/verify_d3.py`, record `verification/general_d3/period_values.json` of `lri_d3_reduction_update/`). The 100-digit run used truncation 220 and tail order 80 (`code/recompute_d3_periods.py`, record `verification/fresh_primitive_values.json` of `lri_d3_matrix_update/`); it repeats the formula side and is not an independent integral. The tails are numerical accelerations, not interval bounds and not fits.

The independent check (`code/general_d3/check_spectral.py`, record `verification/general_d3/fresh_spectral.json`) integrates the real-axis Gamma kernels afresh, without reading stored period values, at 45 and 60 working digits with different quadrature breakpoints and spin cutoffs. Its wheel spin tail comes from analytically integrated centered-Stirling coefficients, not from a fit to the answer. With the symbols λ_j, e_j and $\mathfrak{w}_{r,h}$ local to this paragraph, put $e_0=1$ and

$$\begin{aligned}\lambda_j&=-\frac{rB_{2j+1}(\frac18)}{j(2j+1)},\qquad e_j=\frac1j\sum_{k=1}^{j}k\lambda_k e_{j-k},\\ \mathfrak{w}_{r,h}&=2\Gamma(\tfrac32)^r\,\Gamma(\tfrac{3r}{2}+2h-1)\sum_{j=0}^{h}\frac{e_je_{h-j}}{\Gamma(\frac{3r}{4}+2j)\,\Gamma(\frac{3r}{4}+2h-2j)},\end{aligned}\tag{456}$$

where $B_n(x)$ are the Bernoulli polynomials. In $\langle t_\ell^r\rangle_3$ of Section 1.1, which is $3Q_3$ for $r=3$ and $P_4(3)$ for $r=4$, the spins $\ell<\ell_{\mathrm c}$ are integrated directly, and the sum over the spins $\ell\ge\ell_{\mathrm c}$ is approximated by the truncated asymptotic series $\sum_{h=0}^{h_{\max}}\mathfrak{w}_{r,h}\,\zeta(\frac{3r}{2}-2+2h,\ell_{\mathrm c}+\frac12)$, with the Hurwitz zeta function; the 60-digit run used $\ell_{\mathrm c}=72$ and $h_{\max}=32$. This is a numerical acceleration, not an extra operation in (405) or (422). The five-loop kernel uses an alternating spin sum, accelerated separately by an Euler transformation (232 spins in the 60-digit run). In that run the formula values minus the kernel values are -1.47×10^{-51} for Q_3, -1.53×10^{-52} for $P_4(3)$ and -2.41×10^{-53} for $P_5(3)$. These residuals are the differences of `period_values.json` (90 digits) and `fresh_spectral.json` (60 digits), also listed in the delivered record `verification/general_d3/comparison_summary.json`.

At 75 working digits, evaluations at the complex points $t=0.063+0.041\mathrm{i}$ and $t=-0.082+0.027\mathrm{i}$ confirm the half-shift determinants (429), with largest residual 2.42×10^{-67}, and evaluations at two complex points u confirm the finite modes (408) to 5×10^{-70} (`verification/general_d3/identities.json`). In the same run the complex-node formulas (438) differ from (422) and (425) by 7.0×10^{-71} and 2.7×10^{-65}. Twenty-five exact symbolic checks of the finite residue algebra, the sample rules and the polynomial identities, and 288 comparisons of rational Horn coefficients, pass (`code/general_d3/check_exact.py`, record `verification/general_d3/exact_algebra.json`). Six literal ordinary-hypergeometric evaluations at 40 working digits agree with the tail evaluator to 2.06×10^{-38} (`verification/general_d3/literal_hypergeometric.json`). All these residuals

are observed values, not bounds. They are master-level checks, not a new three-dimensional full-graph calculation or a rigorous enclosure.

Checks of the rank-two identification. The script `code/rank_two_checks.py` of `lri_d3_matrix_update/` (record `verification/rank_two_checks.json`) builds the Jack hook products independently and compares them with (442) on all 91 partitions $\kappa = (m, n)$ with $m \le 12$. On the same partitions it compares the coefficients of (445) for $r = 3, 4$ at $t = 0, \frac{1}{8}, \frac{2}{7}$, and those of (452) at the five nodes (426), with the Horn coefficients. With the polynomial identity (454), the Kummer fixed point of Section 19.9 and the normalization $\sum_{|\kappa|=k} C_\kappa^{(2)}(I_2)/k! = 2^k/k!$ for $k \le 12$, these are 1107 exact checks, all passing. Direct expansions of the matrix functions and of the Horn series agree through total degree 90 at $z = \frac{1}{4}$ (for $\mathfrak{H}_3(0)$ and $\mathfrak{H}_4(\frac{1}{8})$) and $z = -\frac{1}{4}$ (for $\mathfrak{H}_7(\pm\frac{1}{16})$), with z as in (441), at 90 working digits. These are coefficient and interior-series checks, not interval bounds on the endpoint values.

A bounded search for a scalar form of the tetrahedron. Consider the 6479 convergent ordinary values $F = {}_3F_2(\boldsymbol{a}; \boldsymbol{b}; 1)$ and $F = {}_4F_3(\boldsymbol{a}; \boldsymbol{b}; 1)$ whose upper parameters lie in $\{\frac{1}{8}, \frac{2}{8}, \dots, 1\}$ and lower parameters in $\{\frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{3}{2}\}$, with positive excess and no upper parameter equal to a lower one. For each of them PSLQ was run on the vector $(\mathfrak{H}_3(0), 1, \Gamma(\frac{1}{4})^{10}/\pi^{11/2}, F)$ at 85 working digits, with tolerance 10^{-57}, coefficients up to 10^6 and at most 800 iterations. No relation with a nonzero coefficient of $\mathfrak{H}_3(0)$ was returned (`code/scalar_catalog_search.py` of `lri_d3_matrix_update/`, record `verification/scalar_catalog_search.json`). This finite search is not an independence result and does not rule out other parameter arrays, arguments, prefactors or nonlinear identities. It concerns the tetrahedron only.

The open primitive problem. The smallest open primitive question is to evaluate the single rank-two ${}_3F_2^{(\alpha=2)}$ of (447) by finitely many ordinary scalar hypergeometric values. The four- and five-loop complements need the related rank-two functions (448) and (452). No such scalarization is proved here. Reducing these primitive functions would still leave $\mathcal{N}_3$ for $C_5(3)$.

Coefficients through fourth order. At $d = 3$, that is $q = \frac{3}{4}$, the constants (221) are

$$\begin{aligned} &A = 2 + 4\log 2 - \pi, \quad R = \frac{\pi^2}{3} - 4, \qquad U = 2\pi^3 - 56\zeta(3), \\ &V = 16 - 16\zeta(3), \qquad X = \frac{14\pi^4}{15} - 96, \end{aligned} \tag{457}$$

by reflection, recurrence and the integer and half-integer polygamma values [32, 33]. Here V and X are these constants, not $V_r(t)$ of (416), $\mathsf{X}_r(t)$ or X_0, X_h of (421). Then (222) reads

$$\boxed{\begin{aligned} C_0(3) = \tfrac{3}{2}, \qquad C_1(3) = -\tfrac{1}{6}, \qquad C_2(3) = \tfrac{A}{9}, \\ 108\, C_3(3) = A^2 + \tfrac{\pi^2}{3} - 4 - 8Q_3, \\ 648\, C_4(3) = 16P_4(3) - A^3 - 8AQ_3 - \pi^2 A + 12A + 53\pi^3 - 1436\zeta(3) - 48, \end{aligned}} \tag{458}$$

with Q_3 from (405) and $P_4(3)$ from (422). This gives $C_4(3) = 1.228063545327623079\ldots$; it is a reevaluation of the coefficient formula, not a new four-loop graph integration.

With these values and those of the $d = 3$ nonprimitive masters $\mathcal{L}_3$, Z_3 and $\mathcal{K}_3$ (Section 16.8), (223) gives

$$C_5(3) = -5.850351420751327229217944230497744300110997346378 3\ldots$$

to the 50 digits shown (`LRI_C5_d3/c5_value.json`). The formula-side values of Q_3, $P_4(3)$ and $P_5(3)$ are stable to 10^{-74} or better, and the independent checks above and in Section 16.8 confirm

Q_3, $P_4(3)$, $P_5(3)$ and the nonprimitive masters to about 10^{-79}, $2\cdot10^{-86}$, $2\cdot10^{-53}$ and $1.2\cdot10^{-93}$ or better. Propagated through (223), these differences change $C_5(3)$ by less than 10^{-54}: the largest part, about 8×10^{-55}, comes from $P_5(3)$, since $\partial C_5/\partial P_5 = -1536/46656 \approx -0.033$. $\mathcal{K}_3$ is the 430-digit value of (260); the two-fold Barnes integral (588), which uses none of the inputs of Section 21.1, reproduces it to $4\cdot10^{-100}$ (Section 21.11), and the generic-dimensional route of Sections 20 and 21 reproduces $C_5(3)$ to $5\cdot10^{-45}$ (Section 20.9).

20 Generic-dimensional reduction of $\mathcal{L}_d$ and Z_d

The nonprimitive masters $\mathcal{L}_d$ and Z_d of (223) are rim jets of the balanced wheels, $\mathcal{L}_d = \partial_b W_4(d;b)|_{b=d/2}$ and $Z_d = \frac{1}{9}\partial_b^2 W_3(d;b)|_{b=d/2}$ (244). Section 18 periodizes the wheels only at $b=d/2$. This section extends the finite periodization to a general rim exponent b (Section 20.1) and proves that the top Fourier modes of every product of balanced kernels satisfying the convergence condition (C) of (469) vanish in every dimension, in particular for every balanced wheel (Section 20.1.2). With these two statements it gives, for $\mathcal{L}_d$ and Z_d, in place of the six-variable Horn representation of (298) (244 and 318 values), samples at three nodes, real for real b: five two-variable Horn functions at unit argument for $\mathcal{L}_d$, and two ${}_7F_6(1)$ values and three two-variable Horn values for Z_d, together with their first and second b-derivatives (Sections 20.2–20.7; the unweighted, derivative-free form (298) remains valid). $\mathcal{K}_d$ is treated in Section 21; $P_5(d)$ is not treated here. Section 20.2 states the sample formulas of the two wheels, and Sections 20.3 and 20.4 prove them for the four-wheel and the three-wheel and write them out in Horn values. The package `LRI_C5_general_d/` holds the code and records: `foundation/` for Section 20.1, `L/` for Section 20.3, `Z/` for Section 20.4 and `specialize/` for the rest; the review records are in `review/`.

Provenance. This section, with the code and records in `foundation/`, `L/`, `Z/` and `specialize/`, was written by a Claude Code session. Its complete proofs, with the detailed checks, were added on 16 October 2026 by the same session, from that session's separate notes, which are not part of the repository. A second Claude Code session, which wrote Section 21, supplied two inputs: the explicit constants of (465), (466) and (467), taken from its growth proof (Section 21.6; package `LRI_C5_general_d/K_general/`), and the spectral reference values `LRI_C5_general_d/references/reference_values.json`, computed with its code `LRI_C5_general_d/refvals/hp_general_d.py`. The $d=3$ identities of Section 20.6 state their own provenance. Section 20.10 says which parts the reviews covered.

Status. The general-b statements of Section 20.1, the lattice identities (539) and the sample formulas of Section 20.2, with the two-node formula (496), have written proofs here (Sections 20.1, 20.3, 20.4 and 20.6). So do the $d=3$ identities (544) and (545) of Section 20.6, whose printed derivation has not been reviewed. The formulas (492)–(553) are *derived*: exact algebra on top of these theorems, confirmed numerically (Section 20.9). The reviews and checks named in this section and in Section 21 were made by separate AI sessions with their own codes; there was no human referee. Statements that a further reduction does not exist are *numerical only*. Section 20.10 lists the status of every statement, the conditional inputs and what each review covered. The evaluation of $\mathcal{K}_d$ quoted in Section 20.7 is that of Section 21, which has its own status list (Section 21.12) and was not part of the review `LRI_C5_general_d/review/VERIFICATION.md`.

Notation. In this section $q=d/4$. Except in Section 20.1.2, where σ is a free offset and g in f_g a free exponent, $\sigma=b/2$ and $g=(d-b)/2=2q-\sigma$ is the spoke half-exponent; the Barnes kernel g_ℓ carries its spin index. Further $s_b=\sin\frac{\pi b}{2}$, $\varsigma_b=\cos\frac{\pi b}{2}$, and in Sections 20.2–20.7, at $b=d/2$,

$$S=\sin\pi q,\qquad C=\cos\pi q,\qquad D=\pi\cot\pi q=\psi_0(1-q)-\psi_0(q). \tag{459}$$

In Section 20.1 and in the bounds of the proofs of Sections 20.3 and 20.4.1, C (also C_K) denotes a generic constant; (C) is the condition (469), and $C_r(t)$ is a block of Section 20.1.3. This D is not the dimension D of Section 10 and not the functions $D_b(t)$, $D_T(t)$ of (462), (470). A, R, U, V, X are the polygamma values (221); at $q=\frac14$, $A=\mathsf{A}$ and $U=\mathsf{U}$ of (31), and $D=\mathsf{D}=\pi$. The calligraphic $\mathcal{A},\mathcal{B}$ are the mode amplitudes of (349), not polygamma values. Likewise the samples $\mathcal{X}_T,\mathcal{X}_0,\mathcal{X}_{1/2}$ of (489) are unrelated to X, the R family of $\mathcal{K}_d$ is unrelated to R, and the simplex variable V of Section 20.1.2 is unrelated to V. A prime denotes ∂_b. Functions of b such as $\mathfrak{t}_3(b)$ of (495) also depend on d; where d is fixed it is written first, as in $\mathfrak{t}_3(1;b)$. Some letters have meanings here that are local to this section: g is the spoke half-exponent, not the coupling; $\sigma=b/2$ is a rim offset, not a spin; s_b is a sine, not the exponent s; S and C are the trigonometric values (459), not C_n, $C(\tau)$ or the condition (C); and $\omega(\rho)$ of (465) is a constant, unrelated to the numbers ω_j of Step 1 of the proof of (471).

20.1 Finite periodization at a general rim exponent

Kernel. For $0<d<4$ and a rim exponent b (complex in general) put

$$N_b=\frac{\Gamma(2q)\Gamma(g)}{\Gamma(\sigma)},\qquad a_\ell=\frac{\ell+\sigma}{2},\qquad \beta_\ell=a_\ell+g,\qquad g_\ell(s;b)=\frac{\Gamma(a_\ell+s)\Gamma(a_\ell-s)}{\Gamma(\beta_\ell+s)\Gamma(\beta_\ell-s)},$$
$$W_r(d;b)=\langle t_{\ell,b}^{\,r}\rangle_d=N_b^{\,r}\sum_{\ell\ge0}d_\ell\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,g_\ell(s;b)^r. \tag{460}$$

The eigenvalue of (242) is $t_{\ell,b}(\nu)=N_b\,g_\ell(\mathrm{i}\nu;b)$ for real b; for complex b, W_r and $\langle t_{\ell,b}^{\,r}\rangle_d$ denote the analytic continuation (460), with $|\cdot|^2$ replaced by $\Gamma(a_\ell+\mathrm{i}\nu)\Gamma(a_\ell-\mathrm{i}\nu)/(\Gamma(\beta_\ell+\mathrm{i}\nu)\Gamma(\beta_\ell-\mathrm{i}\nu))$. Here $2a_\ell=\ell+\sigma$ and $a_\ell+\beta_\ell=\ell+2q$. At $b=d/2$ one has $\sigma=g=q$, $N_{d/2}=\Gamma(2q)$, β_ℓ is the b_ℓ of (300), and (460) is (302). The multiplicities (301) obey $|d_\ell|\le C(1+\ell)^{d-2}$ and

$$\sum_{\ell\ge0}d_\ell z^\ell=(1+z)(1-z)^{1-d},\qquad |z|<1. \tag{461}$$

For $\operatorname{Re}b>0$ the imaginary axis separates the two pole families of every kernel, and the integral in (460) converges absolutely for $2r\operatorname{Re}g>\max(d,1)$. So (460) represents W_r on $0<\operatorname{Re}b<b_{\mathrm{spec}}=d-\max(d,1)/r$.

On the contour $|g_\ell(\mathrm{i}\nu;b)|$ is comparable to $(1+\ell+|\nu|)^{-2\operatorname{Re}g}$. The ν-integral therefore converges absolutely for $2r\operatorname{Re}g>1$ and the spin sum for $2r\operatorname{Re}g>d$; together these are the condition just stated. The integral also converges at some $\operatorname{Re}b\le0$, but there it no longer represents W_r. The function $W_r(d;b)$ is singular at $b=0$, where the $\ell=0$ contour is pinched (a pole of order $r-1$), and at $b=b_{\mathrm{spec}}$, where the spin or the ν integration diverges. For b near $\frac{d}{2}$ the disc $|b-\frac{d}{2}|<\min(\frac{d}{2},\,b_{\mathrm{spec}}-\frac{d}{2})$ lies in this strip of analyticity. The bound $|d_\ell|\le C(1+\ell)^{d-2}$ holds for every $0<d<4$ and is proved below, (466); for $d<1$ the d_ℓ with $\ell\ge2$ are negative.

Periodization and strip. Define

$$S_r(t;b)=\sum_{\ell\ge0}d_\ell\sum_{n\in\mathbb{Z}}g_\ell(a_\ell+n+t;b)^r,\qquad D_b(t)=\sin(\pi t)\sin\bigl(\pi(t+\sigma)\bigr),\qquad \Xi_r(t;b)=D_b(t)^r\,S_r(t;b). \tag{462}$$

At $b=d/2$ these are S_r, D_q, Ξ_r of (304); the subscript of D_q there is $\sigma=q$, so D_q is D_b at $b=2q$, not at $b=q$. Put $\alpha=r\operatorname{Re}g$ and

$$b_{\max}(r,d)=\min\Bigl\{d-\frac{d}{r},\;d-\frac{2(d-1)}{r},\;d-\frac{1}{r}\Bigr\}, \tag{463}$$

that is $d-\frac1r$ for $d\le1$, $\frac{(r-1)d}{r}$ for $1\le d\le2$ and $d-\frac{2(d-1)}{r}$ for $d\ge2$. The condition $\operatorname{Re}b<b_{\max}$ is the conjunction of (i) $2\alpha>d$, (ii) $\alpha>d-1$ and (iii) $2\alpha>1$. One has $b_{\max}\le b_{\mathrm{spec}}$, with equality for $d\le2$, and $b=d/2$ lies in $0<b<b_{\max}(r,d)$ iff $\frac{2}{r}<d<4$: $d>\frac12$ for W_4 and $d>\frac23$ for W_3.

Lattice split and estimates. Let $f(z)=\Gamma(z)/\Gamma(z+g)$. For $\ell\ge 0$, $n\in\mathbb{Z}$, $j,k\ge 0$ and t off the classes $t\equiv 0$, $t\equiv -\sigma \pmod 1$, Gamma reflection [26] gives

$$\begin{aligned} g_\ell(a_\ell+n+t) &= f(\ell+\sigma+n+t)\,f(-n-t)\\ &= -\frac{\sin\pi(g-t)}{\sin\pi t}\,\frac{\Gamma(\ell+n+\sigma+t)\,\Gamma(n+1-g+t)}{\Gamma(\ell+n+2q+t)\,\Gamma(n+1+t)},\\ g_{j+k+1}(a_{j+k+1}-j-1+t) &= \frac{\Gamma(k+\sigma+t)\,\Gamma(j+1-t)}{\Gamma(k+2q+t)\,\Gamma(j+1+g-t)},\\ g_\ell(a_\ell+n+t) &= g_\ell\big(a_\ell+(-\ell-n-1)+(1-\sigma-t)\big). \end{aligned} \tag{464}$$

At $b=d/2$ the first two are (305) and (306). The lattice splits into the right part $n\ge 0$, the middle part $-\ell\le n\le -1$ and the left part $n\le -\ell-1$; by the last identity the left part is the right part at the mirror point $1-\sigma-t$. The Gamma-ratio bound $|\Gamma(z+u)/\Gamma(z+v)|\le C|z|^{\mathrm{Re}(u-v)}$ [27] gives, on compact sets avoiding the pole classes, $|g_\ell(a_\ell+n+t)|^r\le C(1+\ell+n)^{-\alpha}(1+n)^{-\alpha}$ in the right part and $C(1+k)^{-\alpha}(1+j)^{-\alpha}$ in the middle part. For $t=x+\mathrm{i}y$ with x bounded and $|y|\ge y_0>|\operatorname{Im}\sigma|+1$ the same holds with $1+\ell+n, 1+n, 1+k, 1+j$ replaced by $\ell+n+|y|,\dots,j+|y|$. The constants are uniform for b in compact subsets of $\operatorname{Re} b>0$. Finally $\sum_{\ell,n\ge 0}(1+\ell)^{d-2}(1+\ell+n)^{-\alpha}(1+n)^{-\alpha}<\infty$ iff $\alpha>\max(\frac{d}{2},d-1,\frac12)$, that is iff $\operatorname{Re} b<b_{\max}$; under this condition the version with Y in place of the 1's is $O(Y^{\max(d,1)-2\alpha}(1+\log Y))$, and the same holds for the middle sum. The paragraph *Proofs of the estimates* below proves them in full, with explicit constants in (465) and (467).

Proofs of the estimates. *The identities* (464). For $s=a_\ell+n+t$ one has $a_\ell+s=\ell+\sigma+n+t$, $a_\ell-s=-n-t$, $\beta_\ell+s=\ell+\sigma+n+t+g$ and $\beta_\ell-s=g-n-t$, which is the factorized form. Gamma reflection [26] gives, for every integer n,

$$\frac{\Gamma(-n-t)}{\Gamma(g-n-t)}=-\frac{\sin\pi(g-t)}{\sin\pi t}\,\frac{\Gamma(n+1-g+t)}{\Gamma(n+1+t)},$$

and $\ell+\sigma+n+t+g=\ell+2q+n+t$; this is the second line. The substitution $\ell=j+k+1$, $n=-j-1$ in the factorized form gives the third line. Finally g_ℓ is even in s, and $-(a_\ell+n+t)=a_\ell+(-\ell-n-1)+(1-\sigma-t)$ because $2a_\ell=\ell+\sigma$.

The Gamma-ratio bound. Stirling's formula with its error bound [27] states that for $|\arg w|<\pi$, with principal branches, $\operatorname{Ln}\Gamma(w)=(w-\frac12)\operatorname{Ln} w-w+\frac12\ln 2\pi+\epsilon_\Gamma(w)$ with $|\epsilon_\Gamma(w)|\le \sec^2(\frac12\arg w)/(12|w|)$. From it: let $\rho\ge 0$ and $\omega(\rho)=\rho+\frac32\rho^2+\frac{11}{3}\rho^3+\frac43$. If $\operatorname{Re} z\ge 0$, $|z|\ge\max(1,2\rho)$ and $|u|,|v|\le\rho$, then $\Gamma(z+u)$ and $\Gamma(z+v)$ are finite and nonzero, and

$$\begin{aligned} \frac{\Gamma(z+u)}{\Gamma(z+v)} &= z^{u-v}e^{\theta},\qquad |\theta|\le\frac{\omega(\rho)}{|z|},\\ e^{-\omega(\rho)/|z|}e^{-\frac{\pi}{2}|\operatorname{Im}(u-v)|}|z|^{\mathrm{Re}(u-v)} &\le \Big|\frac{\Gamma(z+u)}{\Gamma(z+v)}\Big| \le e^{\omega(\rho)/|z|}e^{\frac{\pi}{2}|\operatorname{Im}(u-v)|}|z|^{\mathrm{Re}(u-v)}. \end{aligned} \tag{465}$$

Proof. Put $\xi=u/z$, so $|\xi|\le\frac12$, and $w=z+u=z(1+\xi)$. Then $|w|\ge|z|/2\ge\frac12$ and $|\arg w|\le|\arg z|+|\arg(1+\xi)|\le\frac{\pi}{2}+\frac{\pi}{6}$, so w is neither a pole nor a zero of Γ, and $|\epsilon_\Gamma(w)|\le\sec^2(\pi/3)/(12|w|)\le 2/(3|z|)$. The imaginary part of $\operatorname{Ln} z+\operatorname{Ln}(1+\xi)$ lies in $[-\frac{2\pi}{3},\frac{2\pi}{3}]$, so $\operatorname{Ln} w=\operatorname{Ln} z+\operatorname{Ln}(1+\xi)$. Write $\operatorname{Ln}(1+\xi)=\xi-\xi^2/2+\xi_3$ with $|\xi_3|\le|\xi|^3/(3(1-|\xi|))\le\frac23|\xi|^3$. Multiplying out with $z\xi=u$,

$$(z+u-\tfrac12)\operatorname{Ln}(1+\xi)=u+h(z,u),\qquad h(z,u)=\frac{u^2}{2z}-\frac{u}{2z}-\frac{u^3}{2z^2}+\frac{u^2}{4z^2}+(z+u-\tfrac12)\xi_3.$$

With $|u|\le\rho\le|z|/2$, $|z|\ge 1$ and $|z+u-\frac12|\le 2|z|$, each term is bounded, and $|h(z,u)|\le(\frac{\rho}{2}+\frac34\rho^2+\frac{11}{6}\rho^3)/|z|$. Hence $\operatorname{Ln}\Gamma(z+u)=(z+u-\frac12)\operatorname{Ln} z-z+\frac12\ln 2\pi+h(z,u)+\epsilon_\Gamma(z+u)$. Subtracting the same identity for v gives $\operatorname{Ln}\Gamma(z+u)-\operatorname{Ln}\Gamma(z+v)=(u-v)\operatorname{Ln} z+\theta$ with $|\theta|\le 2(\frac{\rho}{2}+\frac34\rho^2+\frac{11}{6}\rho^3)/|z|+\frac{4}{3|z|}=\omega(\rho)/|z|$. Exponentiating gives the first line of (465). The

second follows from $|z^{u-v}| = |z|^{\mathrm{Re}(u-v)} e^{-\mathrm{Im}(u-v)\arg z}$ and $|\arg z| \le \frac{\pi}{2}$. □ The bound is a version of the large-z asymptotics of a Gamma ratio [27] that is uniform in u, v and in the closed right half-plane, with an explicit constant. It is used below for arguments $m + c + t$ with an integer $m \ge 0$ and a bounded constant c: one puts $z = m + t + X$, where $X \ge 0$ bounds $|\mathrm{Re}\, t|$, so that $\mathrm{Re}\, z \ge m$, and absorbs $c - X$ into u or v; arguments $m + c - t$ are treated in the same way.

The multiplicities. For $q \ne \frac{1}{4}$, (301) gives $d_{k+1} = \frac{2}{\Gamma(4q-1)}(2q+k)\frac{\Gamma(k+4q-1)}{\Gamma(k+2)}$. By (465) with $z = k \ge 6$ and $\rho = 3$, $\Gamma(k+4q-1)/\Gamma(k+2) = k^{4q-3}e^{\theta}$ with $|\theta| \le \omega(3)/k$, uniformly in $q \in (0,1)$. The finitely many $k < 6$ are continuous in q, and so is $2/\Gamma(4q-1)$, also at $q = \frac{1}{4}$, where it vanishes; near $q = \frac{1}{4}$ one uses the form (301), with $(4q-1)_k = (4q-1)(4q)_{k-1}$, and at $q = \frac{1}{4}$ the multiplicities are $(1, 1, 0, 0, \dots)$. Hence

$$|d_\ell| \le C_q\,(1+\ell)^{d-2}, \qquad d_\ell\,(1+\ell)^{2-d} \to \frac{2}{\Gamma(d-1)} \quad (\ell \to \infty), \tag{466}$$

with $C_q = \sup_\ell |d_\ell|(1+\ell)^{2-d}$ finite and bounded on compact subsets of $0 < q < 1$, across $d = 1$ included (the limit is 0 at $d = 1$).

The right and middle bounds. Let $\mathrm{Re}\, b > 0$ and let K be a compact set off the classes $t \equiv 0$ and $t \equiv -\sigma$. In the second line of (464), (465) bounds $\Gamma(\ell+n+\sigma+t)/\Gamma(\ell+n+2q+t)$ by $C(1+\ell+n)^{-\mathrm{Re}\, g}$ and $\Gamma(n+1-g+t)/\Gamma(n+1+t)$ by $C(1+n)^{-\mathrm{Re}\, g}$, since $\sigma - 2q = -g$, for all indices with $\ell + n$, respectively n, above a bound fixed by K and b. In the third line it bounds $\Gamma(k+\sigma+t)/\Gamma(k+2q+t)$ and $\Gamma(j+1-t)/\Gamma(j+1+g-t)$ by $C(1+k)^{-\mathrm{Re}\, g}$ and $C(1+j)^{-\mathrm{Re}\, g}$. The prefactor $\sin\pi(g-t)/\sin\pi t$ is bounded on K. Each ratio depends on a single combination of the summation indices ($\ell + n$, n, k or j). A ratio with a small index therefore occurs for infinitely many lattice points but takes only finitely many values; these are continuous on K and bounded there one by one. Raising to the power r gives the bounds stated above, $(1+\ell+n)^{-\alpha}(1+n)^{-\alpha}$ and $(1+k)^{-\alpha}(1+j)^{-\alpha}$ with $\alpha = r\,\mathrm{Re}\, g$. Now let $t = x + \mathrm{i}y$ with x in a fixed bounded interval and $|y| \ge y_0 > |\mathrm{Im}\,\sigma| + 1$. The numerator arguments $\ell + n + \sigma + t$, $n + 1 - g + t$ and $k + \sigma + t$ have imaginary parts of modulus at least $|y| - |\mathrm{Im}\,\sigma| > 1$ (note $\mathrm{Im}\, g = -\mathrm{Im}\,\sigma$), and $j + 1 - t$ has $|\mathrm{Im}| = |y| > 1$; so no Gamma function in the numerators meets a pole, and the prefactor is bounded. With z chosen as above, $|z|$ is comparable to the index plus $|y|$. For $|y|$ or the index above $2\rho + 1$, where ρ bounds the shifts, (465) applies directly; the remaining factor values, with small index and bounded $|y| \ge y_0$, are finitely many continuous functions on a compact set that meets no pole. This gives the bounds with $1+\ell+n$, $1+n$, $1+k$, $1+j$ replaced by $\ell+n+|y|$, $n+|y|$, $k+|y|$, $j+|y|$. For complex b this is why y_0 must exceed $|\mathrm{Im}\,\sigma|+1$: the class $-\sigma+\mathbb{Z}$ lies on $\mathrm{Im}\, t = -\mathrm{Im}\,\sigma$. All quantities are continuous in b, so the constants are uniform for b in a compact set B of $\mathrm{Re}\, b > 0$, provided K avoids the pole classes of every $b \in B$, $y_0 > 1 + \max_B |\mathrm{Im}\,\sigma|$, and α is replaced by $r \min_B \mathrm{Re}\, g$.

The lattice sums. Let $0 < d < 4$ and $\alpha > \max(\frac{d}{2}, d-1, \frac{1}{2})$, and put $\bar\alpha = 2\alpha - \max(d-1, 0)$, so that $\bar\alpha > 1$ and $1 - \bar\alpha = \max(d,1) - 2\alpha$. Then for $Y \ge 1$

$$\begin{aligned} \Sigma^{\mathrm{r}}(Y) &= \sum_{\ell,n\ge 0} (1+\ell)^{d-2}(\ell+n+Y)^{-\alpha}(n+Y)^{-\alpha} \le C^{\mathrm{r}}\, Y^{\max(d,1)-2\alpha}(1+\log Y), \\ \Sigma^{\mathrm{m}}(Y) &= \sum_{j,k\ge 0} (1+j+k)^{d-2}(k+Y)^{-\alpha}(j+Y)^{-\alpha} \le C^{\mathrm{m}}\, Y^{d-2\alpha}, \end{aligned} \tag{467}$$

where, with $C_1 = 1 + 2^{\alpha-d+1}/(\alpha-d+1)$,

$$C^{\mathrm{r}} = C_1\Big[1 + \frac{1}{\bar\alpha - 1} + \frac{1}{(\bar\alpha-1)^2}\Big], \qquad C^{\mathrm{m}} = \begin{cases} 2^{d-1}C_1\big(1+\frac{1}{\alpha-1}\big), & d \ge 2, \\ \Big[\frac{2}{d} + \frac{2^{\alpha-d/2}}{\alpha-d/2}\Big]^2, & d < 2. \end{cases}$$

The two expressions for C^{m} are each continuous on their closed ranges, and C_1 and C^{r} are continuous on the open set of (d, α) defined by the condition; so all the constants are bounded on its compact

subsets, across $d = 1$ and $d = 2$ included. Since $Y^{d-2\alpha} \le Y^{\max(d,1)-2\alpha}(1+\log Y)$ for $Y \ge 1$, the middle sum obeys the bound of the right sum as well. The left part of a lattice sum, with k in place of n, has the form of the right one. *Proof.* (i) For $M \ge 1$ and an exponent $e \in (-2, 2)$ with $\alpha > e+1$ put $\Sigma_e(M) = \sum_{\ell\ge 0}(1+\ell)^e(\ell+M)^{-\alpha}$ and $u = 1+\ell$. Since $\ell + M \ge \max(u, M)$, $\Sigma_e(M) \le M^{-\alpha}\sum_{1\le u\le M} u^e + \sum_{u>M} u^{e-\alpha}$. Here $\sum_{1\le u\le M} u^e \le M^{\max(e+1,0)}(1+\log M)$: for $e > 0$ each term is at most M^e; for $-1 \le e \le 0$, $u^e \le M^{e+1}/u$; for $e < -1$, $u^e \le 1/u$; and $\sum_{u\le M} 1/u \le 1 + \log M$. Since $u^{e-\alpha}$ decreases, $\sum_{u>M} u^{e-\alpha} \le \int_{\lfloor M\rfloor}^{\infty} v^{e-\alpha}dv \le 2^{\alpha-e-1}M^{e+1-\alpha}/(\alpha-e-1)$, using $\lfloor M\rfloor \ge M/2$. With $e = d-2$ this gives $\Sigma_{d-2}(M) \le C_1 M^{\max(d-1,0)-\alpha}(1+\log M)$, and for $d \ge 2$ the same without the logarithm. (ii) The right sum is $\Sigma^{\mathrm{r}}(Y) = \sum_{n\ge 0}(n+Y)^{-\alpha}\Sigma_{d-2}(n+Y) \le C_1\sum_{n\ge 0}(n+Y)^{-\bar\alpha}(1+\log(n+Y))$. The function $v \mapsto v^{-\bar\alpha}(1+\log v)$ decreases on $[1,\infty)$ because $\bar\alpha \ge 1$, so the sum is at most

$$Y^{-\bar\alpha}(1+\log Y) + \int_Y^{\infty} v^{-\bar\alpha}(1+\log v)\,dv = Y^{-\bar\alpha}(1+\log Y) + Y^{1-\bar\alpha}\Big[\frac{1+\log Y}{\bar\alpha-1} + \frac{1}{(\bar\alpha-1)^2}\Big],$$

which is at most $[1 + \frac{1}{\bar\alpha-1} + \frac{1}{(\bar\alpha-1)^2}]\,Y^{1-\bar\alpha}(1+\log Y)$; times C_1 this is the bound with C^{r}. (iii) The middle sum for $d \ge 2$: by $(1+j+k)^{d-2} \le 2^{d-2}[(1+j)^{d-2}+(1+k)^{d-2}]$, $\Sigma^{\mathrm{m}}(Y) \le 2^{d-1}\Sigma_{d-2}(Y)\sum_k(k+Y)^{-\alpha}$, with $\Sigma_{d-2}(Y) \le C_1 Y^{d-1-\alpha}$ and $\sum_k(k+Y)^{-\alpha} \le (1+\frac{1}{\alpha-1})Y^{1-\alpha}$, because $\alpha > d-1 \ge 1$. (iv) The middle sum for $d < 2$: here $1+j+k \ge \sqrt{(1+j)(1+k)}$, so $(1+j+k)^{d-2} \le [(1+j)(1+k)]^{(d-2)/2}$ and $\Sigma^{\mathrm{m}}(Y) \le \Sigma_{(d-2)/2}(Y)^2$. Since $(d-2)/2 \in (-1,0)$, $\sum_{1\le u\le Y} u^{(d-2)/2} \le Y^{d/2}/(d/2)$, and the tail needs $\alpha > d/2$; this gives the stated C^{m}. □

The condition is also necessary. At $Y = 1$, (467) gives $\sum_{\ell,n\ge 0}(1+\ell)^{d-2}(1+\ell+n)^{-\alpha}(1+n)^{-\alpha} < \infty$ for $\alpha > \max(\frac{d}{2}, d-1, \frac{1}{2})$; the converse also holds. Split the sum into $n \le \ell$ and $n > \ell$. For $n \le \ell$, $1+\ell+n$ is comparable to $1+\ell$, and $\sum_{n\le\ell}(1+n)^{-\alpha}$ to 1, $1+\log(1+\ell)$ or $(1+\ell)^{1-\alpha}$ for $\alpha > 1$, $\alpha = 1$, $\alpha < 1$; this part converges iff $\alpha > d-1$ (when $\alpha > 1$), $d < 2$ (when $\alpha = 1$) or $2\alpha > d$ (when $\alpha < 1$). For $n > \ell$, $1+\ell+n$ is comparable to $1+n$, and $\sum_{\ell<n}(1+\ell)^{d-2}$ to n^{d-1}, $1+\log n$ or 1 for $d > 1$, $d = 1$, $d < 1$; this part converges iff $2\alpha > \max(d, 1)$. The conjunction of the two conditions is $\alpha > \max(\frac{d}{2}, d-1, \frac{1}{2})$: the second gives the first and last entries, and if it holds, the first part gives $\alpha > d-1$ in each of its three cases (for $\alpha < 1$ the condition $2\alpha > d$ forces $d < 2$, so $\alpha > \frac{d}{2} > d-1$).

The entries of $b_{\max}$. With $\alpha = r\,\mathrm{Re}\,g = r(d - \mathrm{Re}\,b)/2$ the condition reads $\mathrm{Re}\,b < b_{\max}(r,d)$. Its three entries come from three regions of the lattice: (i) $2\alpha > d$ from the region where ℓ and n are both large, (ii) $\alpha > d-1$ from the end $\ell \gg n$, and (iii) $2\alpha > 1$ from the end $n \gg \ell$. For $r = 4$, (463) is $d - \frac{1}{4}$, $\frac{3}{4}d$, $\frac{d+1}{2}$ on the ranges $d \le 1$, $1 \le d \le 2$, $d \ge 2$; for $r = 3$ it is $d - \frac{1}{3}$, $\frac{2}{3}d$, $\frac{d+2}{3}$. Some values, with the spectral bound of (460):

d	0.8	1	1.25	1.5	2	2.5	2.75	3
$b_{\max}(3,d)$	0.4667	0.6667	0.8333	1	1.3333	1.5	1.5833	1.6667
$b_{\mathrm{spec}}(3,d)$	0.4667	0.6667	0.8333	1	1.3333	1.6667	1.8333	2
$b_{\max}(4,d)$	0.55	0.75	0.9375	1.125	1.5	1.75	1.875	2
$b_{\mathrm{spec}}(4,d)$	0.55	0.75	0.9375	1.125	1.5	1.875	2.0625	2.25

The two bounds differ only for $d > 2$ (paragraph *Beyond* $b_{\max}$ of Section 20.1.4). At $d = 2$ and $r = 4$, $b_{\max} = \frac{3}{2}$ is $x = \frac{b}{2} < \frac{3}{4}$ for the one-variable sum $\Omega(t;x)$ of (349), a range that contains the interval used in Section 18.7; at $d = 3$, $b_{\max}(4,3) = 2$ is the range "every $b < 2$" of the $d = 3$ paragraph of Section 20.6. For $r = 3$, $b_{\max}(3,2) = \frac{4}{3}$ is the range of (264), and $b_{\max}(3,3) = \frac{5}{3}$. The exponent η of part (c) of Section 20.1.1 is $2\alpha - \max(d,1) > 0$ throughout the strip, by (i) and (iii), so the exclusion of the boundary mode needs no extra condition.

20.1.1 Finite Fourier space

Let $r \ge 2$, $0 < d < 4$ and $0 < \mathrm{Re}\,b < b_{\max}(r,d)$, $\sigma \notin \mathbb{Z}$. Then:

(a) $S_r(t;b)$ converges absolutely and locally uniformly off the classes $t \equiv 0$ and $t \equiv -\sigma$; it is one-periodic and meromorphic, with poles of order at most r in these two classes only.

(b) $\Xi_r(t;b)$ is entire, $\Xi_r(t+1;b) = \Xi_r(t;b)$ and $\Xi_r(-\sigma - t;b) = \Xi_r(t;b)$.

(c) For $0 \le x \le 1$ and $|y| \ge 2 + |\operatorname{Im} b|$, $|\Xi_r(x + \mathrm{i}y;b)| \le C\, e^{2\pi r|y|}|y|^{-\eta}(1 + \log|y|)$ with $\eta = r(d - \operatorname{Re} b) - \max(d,1) > 0$ (a decay exponent, unrelated to the weights η_m of (253)).

(d) There are coefficients $c_{r,k}(q,b)$, analytic in b on the strip and real for real b, with

$$\Xi_r(t;b) = \sum_{k=0}^{r-1} c_{r,k}(q,b)\cos\bigl(2\pi k(t + \tfrac{b}{4})\bigr). \tag{468}$$

$\Xi_r(t;b)$ is jointly analytic in (t,b), so every sample at a fixed node is analytic in b.

Proof. (a) By the right and middle bounds and (467), the right and middle parts converge absolutely and uniformly on compact sets off the pole classes, and so does the left part, which is the right part at the mirror point. Each term is meromorphic in t. By (464) poles come only from $\Gamma(-n-t)$ ($t \in \mathbb{Z}$) and $\Gamma(\ell + \sigma + n + t)$ ($t \in -\sigma + \mathbb{Z}$), each to order at most r. If t_0 is such a point, $(t - t_0)^r$ times each term is analytic in a small disc around t_0; the series of these functions converges uniformly on the boundary circle, hence, by the maximum principle, in the disc, so $(t - t_0)^r S_r$ is analytic near t_0. Periodicity follows from $n \mapsto n + 1$ in an absolutely convergent sum. (b) D_b^r vanishes to order r on both classes, so Ξ_r is entire, and $D_b(t+1) = D_b(t)$. Evenness of g_ℓ gives $g_\ell(a_\ell + n - \sigma - t) = g_\ell(a_\ell + (-\ell - n) + t)$, and $n \mapsto -\ell - n$ is a bijection of $\mathbb{Z}$, so $S_r(-\sigma - t) = S_r(t)$; also $D_b(-\sigma - t) = \sin\pi(\sigma + t)\sin\pi t = D_b(t)$. (c) $|D_b(x + \mathrm{i}y)| \le Ce^{2\pi|y|}$. For $|y| \ge 2 + |\operatorname{Im} b| = 2 + 2|\operatorname{Im}\sigma|$ the large-$|y|$ bounds apply at t and, for the left part, at the mirror point $1 - \sigma - t$, whose imaginary part has modulus at least $|y| - |\operatorname{Im}\sigma| \ge \max(|y|/2,\, 2 + |\operatorname{Im}\sigma|)$. With (467) at $Y = |y|$ (or $|y|/2$), each of the three parts of S_r is bounded by $C|y|^{\max(d,1) - 2\alpha}(1 + \log|y|)$, and $\eta = 2\alpha - \max(d,1)$. (d) As for (311), the coefficient $\gamma_m = \int_0^1 \Xi_r(x + \mathrm{i}y)e^{-2\pi\mathrm{i}m(x + \mathrm{i}y)}dx$ does not depend on y (Cauchy's theorem on a rectangle, the vertical sides cancelling by periodicity). For $m \ge r$ take $y = -Y$ with $Y \ge 2 + |\operatorname{Im} b|$; by (c), $|\gamma_m| \le Ce^{2\pi(r-m)Y}Y^{-\eta}(1 + \log Y) \to 0$, also for $m = r$ since $\eta > 0$. For $m \le -r$ take $y = +Y$. So $\Xi_r = \sum_{|m| \le r-1} \gamma_m e^{2\pi\mathrm{i}mt}$. The reflection in (b), $\Xi_r(-\sigma - t) = \sum_m \gamma_{-m}e^{2\pi\mathrm{i}m\sigma}e^{2\pi\mathrm{i}mt}$, gives $\gamma_m = \gamma_{-m}e^{2\pi\mathrm{i}m\sigma}$. With $c_{r,0} = \gamma_0$ and $c_{r,k} = 2\gamma_k e^{-\mathrm{i}\pi k\sigma} = 2\gamma_{-k}e^{\mathrm{i}\pi k\sigma}$ one has $\gamma_k e^{2\pi\mathrm{i}kt} + \gamma_{-k}e^{-2\pi\mathrm{i}kt} = c_{r,k}\cos(2\pi k(t + \frac{b}{4}))$, which is (468). For the analyticity in b let B be a compact subset of the strip and $y_B = 2 + \max_B |\operatorname{Im} b|$. On the segment $t = x + \mathrm{i}y_B$, $0 \le x \le 1$, the series for S_r converges uniformly in $(x,b) \in [0,1] \times B$, by the large-$|y|$ bounds with constants uniform on B and (467) with $\alpha = r\min_B \operatorname{Re} g > \max(\frac{d}{2}, d-1, \frac{1}{2})$. Its terms are analytic in b there, since the segment stays at distance at least one from both pole classes, and $D_b(t)^r$ is entire in (t,b). Hence $\Xi_r(x + \mathrm{i}y_B;b)$ is analytic in b, and so is $\gamma_m(b) = \int_0^1 \Xi_r(x + \mathrm{i}y_B;b)e^{-2\pi\mathrm{i}m(x + \mathrm{i}y_B)}dx$ on the interior of B. The same argument, with the Cauchy formula in t on small circles around the pole classes, shows that $\Xi_r(t;b)$ is jointly analytic in (t,b). For real b, Ξ_r is real on the real axis, so $\gamma_{-m} = \overline{\gamma_m}$ and the $c_{r,k}$ are real. □ (The point $\sigma \in \mathbb{Z}$ lies in the strip only at $b = 2$ when $r \ge 3$ and $d > (2r-2)/(r-2)$, never for $r = 3$ or for $r = 4$, $d \le 3$; it is treated in the paragraph *Integer* σ below. The masters use only b near $\frac{d}{2}$, where $0 < \sigma < 1$.)

Integer σ. In the strip, $\sigma \in \mathbb{Z}$ can occur only at the real point $b = 2$, because $b_{\max} \le d - \frac{d}{r} < d < 4$. The point $b = 2$ lies in the strip iff $b_{\max}(r,d) > 2$, that is iff $r \ge 3$ and $d > (2r-2)/(r-2)$. This never happens for $r = 3$; it happens for $d > 3$ when $r = 4$, for $d > \frac{8}{3}$ when $r = 5$, and for $d > (2r-2)/(r-2)$, which decreases to 2, for larger r. At $b = 2$ the two pole classes merge. Ξ_r is still entire: the poles of S_r in $\mathbb{Z}$ have order at most $2r$, and $D_2(t)^r = (-1)^r \sin^{2r}\pi t$. The analyticity argument in (d) does not use $\sigma \notin \mathbb{Z}$, so the $c_{r,k}$ are analytic across $b = 2$, and the finite Fourier statement holds there as well. The top-mode statement (473) holds there by continuity, and also directly, since (471) allows every $\sigma \in \mathbb{C}$. Only the weights of (479) and of (484), which carry

$(\sin\pi\sigma)^{-r}$, are singular at $b=2$. W_r is analytic there, so these poles cancel in the sum (paragraph *Radius of the sample jets* of Section 20.1.4). None of the cases used in this note ($r=3,4$ with $d\le 3$) contains $b=2$ in its strip.

20.1.2 The top modes vanish

The top-mode statement holds for every product of balanced kernels that satisfies the convergence condition (C) below, with different spoke exponents and independent shifts of the left and right Gamma arguments; the wheels are the case of equal factors. Write $f_g(w)=\Gamma(w)/\Gamma(w+g)$, so that $f=f_g$ in (464). In this subsection g_i,α_i,β_i ($i=1,\dots,r$) are numbers attached to the i-th factor: the kernel $g_\ell(s;b)$ is always written with its arguments, and α_i,β_i are unrelated to $\alpha=r\,\mathrm{Re}\,g$ and to β_ℓ in (460). Let $0<d<4$, $r\ge 2$ and $\sigma\in\mathbb{C}$, and let $g_i,\alpha_i,\beta_i\in\mathbb{C}$ satisfy

$$\text{(B)}\quad \sigma+\alpha_i+\beta_i+g_i=\frac{d}{2}\ \ (i=1,\dots,r),\qquad \text{(C)}\quad \varkappa:=\sum_{i=1}^{r}\mathrm{Re}\,g_i>\max\Big(\frac{d}{2},\,d-1,\,\frac{1}{2}\Big). \tag{469}$$

Put

$$\begin{aligned} T_\ell(z)&=\prod_{i=1}^{r} f_{g_i}(\ell+\sigma+\alpha_i+z)\,f_{g_i}(\beta_i-z),\qquad S_T(t)=\sum_{\ell\ge 0} d_\ell\sum_{n\in\mathbb{Z}} T_\ell(n+t),\\ D_T(t)&=\prod_{i=1}^{r}\sin\pi(t+\sigma+\alpha_i)\,\sin\pi(t-\beta_i),\qquad \Xi_T=D_T S_T. \end{aligned} \tag{470}$$

Here $T_\ell(z)$, with one index, is unrelated to $T_{\ell n}(b)$ of (490). Then S_T converges absolutely and locally uniformly off the $2r$ pole classes $-\sigma-\alpha_i+\mathbb{Z}$ and $\beta_i+\mathbb{Z}$; Ξ_T is entire and one-periodic, $\Xi_T=\sum_{|m|\le r-1}\gamma_m e^{2\pi i m t}$, with γ_m analytic in (σ,α_i,g_i) on the convex set defined by (C) (β_i fixed by (B)); and

$$\boxed{\gamma_{1-r}=\gamma_{r-1}=0,\qquad \Xi_T(t)=\sum_{|m|\le r-2}\gamma_m\,e^{2\pi i m t},} \tag{471}$$

equivalently $S_T(t)=O(e^{-4\pi|\mathrm{Im}\,t|})$ as $|\mathrm{Im}\,t|\to\infty$. Individual $\mathrm{Re}\,g_i$ may be zero or negative; only their sum is constrained. It is proved below, and Section 21.5 applies it to the R family of $\mathcal{K}_d$.

Mixed wheels and wheels. A product $\prod_i g_\ell(s+\delta_i;b_i)$ of kernels (460) with rim exponents b_i and shifts δ_i, sampled on the lattice $s=\frac{\ell+\sigma}{2}+n+t$ with any offset σ, is $T_\ell(n+t)$ with

$$g_i=\frac{d-b_i}{2},\qquad \alpha_i=\frac{1}{2}\Big(\frac{b_i}{2}-\sigma\Big)+\delta_i,\qquad \beta_i=\frac{1}{2}\Big(\frac{b_i}{2}-\sigma\Big)-\delta_i, \tag{472}$$

since $g_\ell(s;b)=f_{(d-b)/2}(\frac{\ell}{2}+\frac{b}{4}+s)\,f_{(d-b)/2}(\frac{\ell}{2}+\frac{b}{4}-s)$. Then (B) holds automatically, (C) says that the mean rim exponent satisfies $\frac{1}{r}\sum_i\mathrm{Re}\,b_i<b_{\max}(r,d)$, and the shifts are free. For the wheels, $b_i=b$, $\delta_i=0$ and $\sigma=b/2$, one has $\alpha_i=\beta_i=0$, $g_i=g$, $T_\ell(n+t)=g_\ell(a_\ell+n+t;b)^r$ and $D_T=D_b^r$, so $\Xi_T=\Xi_r$, $\varkappa=\alpha$, and (C) is $\mathrm{Re}\,b<b_{\max}$. With $c_{r,r-1}=2\gamma_{r-1}e^{-i\pi(r-1)\sigma}$ this gives, under the hypotheses of Section 20.1.1,

$$\boxed{c_{r,r-1}(q,b)=0,\qquad \Xi_r(t;b)=\sum_{k=0}^{r-2}c_{r,k}(q,b)\cos\big(2\pi k(t+\tfrac{b}{4})\big),} \tag{473}$$

and equivalently $S_r(t;b)=O(e^{-4\pi|\mathrm{Im}\,t|})$. This wheel statement is the one used in the rest of this section; Section 21 uses the product case $r=3$ and the case $r=2$ of (471).

The R family of $\mathcal{K}_d$. Section 21.5 applies (471) to the R family of $\mathcal{K}_d$, a product of three balanced kernels that is not a wheel. On the τ-circle used for $\mathcal{K}_d$ that application rests in addition on the continuation hypothesis of Section 21.8; the separate lattice-sum code `LRI_C5_general_d/review/lemmaG-numeric` of Section 20.9 confirms it at two points of that kind (top modes below $2\cdot10^{-47}$ at 54 digits).

Proof of (471). Its steps are as follows; the paragraph *Details of the top-mode proof* after the proof gives in full the estimates and computations that the steps state in short form. Let $Y_0 = 1+\max_i\max(|\operatorname{Im}(\sigma+\alpha_i)|,|\operatorname{Im}\beta_i|)$, so that all pole classes lie in $|\operatorname{Im}t|\le Y_0-1$.

Step 0 (estimates and finite modes). Split the lattice into the right part $n\ge0$, the middle part $\ell=j+k+1$, $n=-j-1$, and the left part $n=-\ell-1-k$ ($j,k\ge0$); there is no mirror map in general. Gamma reflection writes the factors with large negative argument as $f_g(\beta-n-t)=\frac{\sin\pi(\beta+g-t)}{\sin\pi(\beta-t)}\frac{\Gamma(n+1-\beta-g+t)}{\Gamma(n+1-\beta+t)}$ in the right part and $f_g(\sigma+\alpha-1-k+t)=\frac{\sin\pi(\sigma+\alpha+g+t)}{\sin\pi(\sigma+\alpha+t)}\frac{\Gamma(k+2-\sigma-\alpha-g-t)}{\Gamma(k+2-\sigma-\alpha-t)}$ in the left part. The Gamma-ratio bound applied to every factor gives a power $-\operatorname{Re}g_i$ of a base common to all factors, so the exponents add up to $-\varkappa$, and the lattice sums of Section 20.1 apply with $\alpha=\varkappa$. Step 0 of the paragraph *Details of the top-mode proof* carries this out: S_T converges absolutely and locally uniformly off the pole classes; for $0\le x\le1$ and $|y|\ge\max(Y_0,2)$, $|S_T(x+\mathrm{i}y)|\le C|y|^{\max(d,1)-2\varkappa}(1+\log|y|)$; Ξ_T is entire and one-periodic, its modes $|m|\ge r$ vanish, and the γ_m are analytic in the parameters.

Step 1 (two half-planes). On $\operatorname{Im}t>Y_0$, S_T is analytic, one-periodic and tends to zero, so $S_T=\sum_{m\ge1}s_m^+w^m$ with $w=e^{2\pi\mathrm{i}t}$; on $\operatorname{Im}t<-Y_0$, $S_T=\sum_{m\ge1}s_m^-w^{-m}$. Let ω_j run over the $2r$ numbers $\sigma+\alpha_i$ and $-\beta_i$. Since $\sin\pi(t+\omega)=\frac{\mathrm{i}}{2}e^{-\mathrm{i}\pi(t+\omega)}(1-e^{2\pi\mathrm{i}\omega}w)=-\frac{\mathrm{i}}{2}e^{\mathrm{i}\pi(t+\omega)}(1-e^{-2\pi\mathrm{i}\omega}w^{-1})$, one has $D_T=\kappa_+w^{-r}\prod_j(1-e^{2\pi\mathrm{i}\omega_j}w)=\kappa_-w^r\prod_j(1-e^{-2\pi\mathrm{i}\omega_j}w^{-1})$ with $\kappa_\pm=(-\frac14)^re^{\mp\mathrm{i}\pi\sum_j\omega_j}$, and comparing coefficients gives

$$\gamma_{1-r}=\kappa_+s_1^+,\qquad \gamma_{r-1}=\kappa_-s_1^-.$$

For the wheels the reflection in Section 20.1.1(b) relates the two. In general there is no such symmetry at complex parameters, and the two coefficients are treated separately.

Step 2 (unfolding). For $Y>Y_0$ the double series converges absolutely on the segment $x+\mathrm{i}Y$, $0\le x\le1$ (this uses (C)), so

$$s_1^+=\int_0^1 S_T(x+\mathrm{i}Y)\,e^{-2\pi\mathrm{i}(x+\mathrm{i}Y)}\,dx=\sum_{\ell\ge0}d_\ell I_\ell,\qquad I_\ell=\int_{\mathbb{R}+\mathrm{i}Y}T_\ell(z)\,e^{-2\pi\mathrm{i}z}\,dz,\qquad \sum_\ell|d_\ell||I_\ell|<\infty.$$

Step 3 (Laplace form and I_ℓ). Assume first $\operatorname{Re}g_i>0$ for all i (removed in Step 7). Rotating the ray of Euler's integral $\Gamma(g)f_g(w)=\int_0^\infty e^{-wu}(1-e^{-u})^{g-1}du$, with $u=\mp\mathrm{i}v$ through the first or the fourth quadrant, gives, first for $\operatorname{Re}w>0$ and then by analytic continuation,

$$\begin{aligned} f_g(w)&=\frac{-\mathrm{i}}{\Gamma(g)}\int_0^\infty e^{\mathrm{i}wv}(1-e^{\mathrm{i}v})^{g-1}dv \qquad(\operatorname{Im}w>0),\\ f_g(w)&=\frac{\mathrm{i}}{\Gamma(g)}\int_0^\infty e^{-\mathrm{i}wv}(1-e^{-\mathrm{i}v})^{g-1}dv \quad(\operatorname{Im}w<0),\end{aligned}\tag{474}$$

with principal powers. The singularities at $v\in2\pi\mathbb{N}$ are integrable, since $|(1-e^{\pm\mathrm{i}v})^{g-1}|\le e^{\pi|\operatorname{Im}g|/2}|2\sin\frac{v}{2}|^{\operatorname{Re}g-1}$. On $\operatorname{Im}z=Y$ every left argument $\ell+\sigma+\alpha_i+z$ has imaginary part at least 1 and every right argument β_i-z at most -1. Let $a_\pm$ be the convolutions on $[0,\infty)$ of $\mathsf{h}_i^+(v)=e^{\mathrm{i}\alpha_iv}(1-e^{\mathrm{i}v})^{g_i-1}$ and of $\mathsf{h}_i^-(v)=e^{-\mathrm{i}\beta_iv}(1-e^{-\mathrm{i}v})^{g_i-1}$, that is $a_\pm(V)=\int_{\Delta_V}\prod_i\mathsf{h}_i^\pm(v_i)\,d\mu$ with $\Delta_V=\{v_i\ge0,\sum_iv_i=V\}$. By Fubini, $\prod_i f_{g_i}(\ell+\sigma+\alpha_i+z)=c_+\int_0^\infty e^{\mathrm{i}(\ell+\sigma+z)V}a_+(V)\,dV$ and $\prod_i f_{g_i}(\beta_i-z)=c_-\int_0^\infty e^{\mathrm{i}zV}a_-(V)\,dV$, where $c_\pm=\prod_i(\mp\mathrm{i})/\Gamma(g_i)$ and $c_+c_-=\kappa_\Gamma:=\prod_i\Gamma(g_i)^{-2}$.

Endpoint behavior. Fix $0<\delta<\frac{\pi}{2}$, and let $\lambda(V)=1+|\log|2\pi-V||$ if $\varkappa=1$ and $\lambda=1$ otherwise (this λ is local to Step 3 and unrelated to the weights λ_m of Section 20.5). Then

$$|a_\pm(V)|\le\begin{cases}C\,V^{\varkappa-1}, & 0<V\le\pi,\\ C\,|2\pi-V|^{\min(\varkappa-1,0)}\lambda(V), & \pi\le V\le2\pi+\delta.\end{cases}\tag{475}$$

The paragraph *Details of the top-mode proof* proves (475) from $|\mathsf{h}_i^{\pm}(v)| \le C|2\sin\frac{v}{2}|^{\operatorname{Re} g_i - 1}$ and Dirichlet's integral. For $\varkappa > 1$ the bound at 2π is attained in general, and at $\varkappa = 1$ it is at most logarithmic. Hence, near $V = 0$,

$$a_+(V)\, a_-(2\pi - V) = O(V^{\,\varkappa - 1 + \min(\varkappa - 1, 0)}) \qquad (\text{times at most } 1 + |\log V| \text{ at } \varkappa = 1), \tag{476}$$

and symmetrically near $V = 2\pi$. The exponent exceeds -1 iff $2\varkappa > 1$.

Along $z = x + \mathrm{i}Y$ the two products are $c_\pm$ times the Fourier transforms $\widehat{a_\pm^Y}(x) = \int e^{\mathrm{i}xV} a_\pm^Y(V)\, dV$ of $a_+^Y(V) = e^{\mathrm{i}(\ell+\sigma)V - YV} a_+(V)$ and $a_-^Y(V) = e^{-YV} a_-(V)$, extended by zero to $V \le 0$; both lie in $L^1(\mathbb{R})$. So $T_\ell(z) = \kappa_\Gamma \widehat{a_+^Y * a_-^Y}(x)$. Since $|f_{g_i}(w)| \le C(1 + |w|)^{-\operatorname{Re} g_i}$ on horizontal lines at distance at least one from the real axis, $\widehat{a_+^Y * a_-^Y} = O(|x|^{-2\varkappa})$ is integrable by $2\varkappa > 1$. On $[0, 2\pi + \delta]$ the functions $a_\pm^Y$ are square integrable by (475) and $2\varkappa > 1$, so $a_+^Y * a_-^Y$ is continuous at $V = 2\pi$. Fourier inversion at this point gives

$$I_\ell = 2\pi\kappa_\Gamma \int_0^{2\pi} e^{\mathrm{i}(\ell+\sigma)V} a_+(V)\, a_-(2\pi - V)\, dV = 2\pi\kappa_\Gamma \int_{\widetilde{\Delta}} e^{\mathrm{i}(\ell+\sigma)V} \prod_{i=1}^{r} \mathsf{h}_i^+(v_i)\, \mathsf{h}_i^-(v_i')\, d\mathrm{vol}, \tag{477}$$

where $\widetilde{\Delta} = \{(v, v') \in [0,\infty)^{2r} : V + V' = 2\pi\}$, $V = \sum_i v_i$, $V' = \sum_i v_i'$, and $d\mathrm{vol} = dv_1 \cdots dv_r\, dv_1' \cdots dv_{r-1}'$ is its Lebesgue measure, invariant under permutations of the $2r$ coordinates.

Step 4 (spin sum). By Step 2 and Abel's theorem, $s_1^+ = \lim_{\rho\uparrow 1} \sum_\ell d_\ell \rho^\ell I_\ell$. For $\rho < 1$ the sum may be taken inside (477), since $a_+(V)a_-(2\pi - V) \in L^1(0, 2\pi)$ by (476), and it produces (461) at $z = \rho e^{\mathrm{i}V}$. For $\frac{1}{2} \le \rho < 1$, $|1 - \rho e^{\mathrm{i}V}| \ge \sqrt{2}|\sin\frac{V}{2}|$, so the integrand is dominated by $C|\sin\frac{V}{2}|^{-\max(d-1,0)}|a_+(V)a_-(2\pi - V)|$. This function is bounded on compact subsets of $(0, 2\pi)$ and is $O(V^{\,\varkappa - 1 + \min(\varkappa-1,0) - \max(d-1,0)})$ at both ends (with $2\pi - V$ at the upper end and at most a logarithm at $\varkappa = 1$). The bound is integrable iff $\varkappa + \min(\varkappa - 1, 0) > \max(d - 1, 0)$, which is exactly (C): for $\varkappa \ge 1$ it reads $\varkappa > d - 1$, and for $\varkappa < 1$ it reads $2\varkappa > \max(d, 1)$, which forces $d < 2$ and hence $\varkappa > \frac{d}{2} > d - 1$. For the wheels this is the full hypothesis $b < b_{\max}$, not only $b < b_{\mathrm{spec}}$ (Step 2 uses it as well). Dominated convergence gives

$$s_1^+ = 2\pi\kappa_\Gamma \int_{\widetilde{\Delta}} \Psi\, d\mathrm{vol}, \qquad \Psi = e^{\mathrm{i}\sigma V}(1 + e^{\mathrm{i}V})(1 - e^{\mathrm{i}V})^{1-d} \prod_{i=1}^{r} \mathsf{h}_i^+(v_i)\, \mathsf{h}_i^-(v_i').$$

Step 5 (balance and exchange). For $0 < v < 2\pi$, $1 - e^{\pm\mathrm{i}v} = 2\sin\frac{v}{2}\, e^{\pm\mathrm{i}(v-\pi)/2}$, so that $(1 - e^{\mathrm{i}v})^{g-1} = e^{\mathrm{i}(g-1)(v-\pi)}(1 - e^{-\mathrm{i}v})^{g-1}$ for principal powers, and by (B)

$$\mathsf{h}_i^+(v) = e^{-\mathrm{i}\pi(g_i - 1)}\, e^{\mathrm{i}(\frac{d}{2} - 1 - \sigma)v}\, \mathsf{h}_i^-(v) \qquad (0 < v < 2\pi), \tag{478}$$

with the same rate $\frac{d}{2} - 1 - \sigma$ for every factor. In the interior of $\widetilde{\Delta}$ all coordinates lie in $(0, 2\pi)$. With $1 + e^{\mathrm{i}V} = 2\cos\frac{V}{2}\, e^{\mathrm{i}V/2}$ and $(1 - e^{\mathrm{i}V})^{1-d} = (2\sin\frac{V}{2})^{1-d} e^{\mathrm{i}(1-d)(V-\pi)/2}$ the rates in V cancel, and

$$\Psi = e^{\mathrm{i}\pi(d-1)/2 - \mathrm{i}\pi\sum_i (g_i - 1)}\, 2\cos\frac{V}{2}\Big(2\sin\frac{V}{2}\Big)^{1-d} \prod_{i=1}^{r} \mathsf{h}_i^-(v_i)\, \mathsf{h}_i^-(v_i').$$

The exchange $\iota(v, v') = (v', v)$ maps $\widetilde{\Delta}$ onto itself, preserves $d\mathrm{vol}$ and sends V to $2\pi - V$. Every factor of Ψ is ι-invariant except $\cos\frac{V}{2}$, which changes sign. So $\Psi \circ \iota = -\Psi$, $s_1^+ = 0$ and $\gamma_{1-r} = 0$. Integrated over the fibers of V, this is the odd one-variable integral $s_1^+ = 2\pi\kappa_\Gamma e^{\mathrm{i}\pi(d-1)/2 - \mathrm{i}\pi\sum_i (g_i-1)} \int_0^{2\pi} a_-(V)a_-(2\pi - V)(2\sin\frac{V}{2})^{1-d} 2\cos\frac{V}{2}\, dV$. For the wheels, $a_-(V) = e^{-\mathrm{i}(g-1)(V - r\pi)/2}\mathsf{p}(V)$ with $\mathsf{p}(V) = \int_{\Delta_V} \prod_i (2\sin\frac{v_i}{2})^{g-1} d\mu$, positive for real b, and the prefactor becomes $2\pi e^{\mathrm{i}\pi(\sigma + 1/2)}/\Gamma(g)^{2r}$: the integrand is real up to a constant phase. Equivalently, writing $\Psi = 2\cos\frac{V}{2}(2\sin\frac{V}{2})^{1-d}\, \Pi\, e^{\mathrm{i}\Theta}$ with $\Pi = \prod_i (2\sin\frac{v_i}{2} \cdot 2\sin\frac{v_i'}{2})^{g_i - 1}$, the phase obeys $\Theta - \Theta \circ \iota =$

$\sum_i(\sigma+\alpha_i+\beta_i+g_i-\frac{d}{2})(v_i-v_i')$ for all (v,v'). So per-factor balance, not total balance, makes it ι-invariant.

Step 6 (the other top mode). Since d and the d_ℓ are real, $\overline{\Xi_T(\bar t;p)}=\Xi_T(t;\bar p)$ for the parameter point $p=(\sigma,\alpha_i,g_i)$, and $\bar p$ again satisfies (B) and (C). Hence $\gamma_{r-1}(p)=\overline{\gamma_{1-r}(\bar p)}=0$ by Steps 1–5 at $\bar p$. The direct computation on $\operatorname{Im} z=-Y$ gives the same: the two forms of (474) exchange their roles, every exponential has i replaced by $-$i, and the integrand is again odd under ι.

Step 7 (continuation). $\operatorname{Re} g_i>0$ was used only in Steps 3–5. The γ_m are analytic on the convex, hence connected, set (C), and $\gamma_{\pm(r-1)}$ vanish on its nonempty open part where all $\operatorname{Re} g_i>0$; so they vanish on all of it. □

Details of the top-mode proof. The steps above state some estimates and computations in short form; this paragraph gives them in full, and with it the proof of (471) is complete. Throughout, β_i is given by (B). The set (C) is an open half-space of $\mathbb{C}^{2r+1}\cong\mathbb{R}^{4r+2}$, hence convex and connected. For a compact set B in it, let Y_B be the maximum of Y_0 and $\varkappa_B$ the minimum of $\varkappa$ over B; then $\varkappa_B>\max(\frac{d}{2},d-1,\frac{1}{2})$. Let P_T be the union of the $2r$ classes $-\sigma-\alpha_i+\mathbb{Z}$ and $\beta_i+\mathbb{Z}$.

Step 0. In the right part the Gamma-ratio bound (465) is applied to the left factors $f_{g_i}(\ell+\sigma+\alpha_i+n+t)$ as they stand (base $1+\ell+n$) and to the right factors in the first reflected form (base $1+n$); in the middle part to all factors as they stand, $f_{g_i}(k+\sigma+\alpha_i+t)$ and $f_{g_i}(j+1+\beta_i-t)$ (bases $1+k$ and $1+j$); in the left part to the left factors in the second reflected form (base $1+k$) and to the right factors as they stand, $f_{g_i}(\ell+k+1+\beta_i-t)$ (base $1+\ell+k$). Factor i contributes the power $-\operatorname{Re} g_i$ of a base comparable, uniformly in i, to one of these, so the exponents add up to $-\varkappa$ whatever the signs of the individual $\operatorname{Re} g_i$. On a compact set K off P_T the sine ratios of the reflected forms are bounded, and, as in the paragraph *Proofs of the estimates*, the finitely many factor values with small arguments are bounded on K one by one, although each occurs for infinitely many lattice points. This gives $|T_\ell(n+t)|\le C_K(1+\ell+n)^{-\varkappa}(1+n)^{-\varkappa}$, $C_K(1+k)^{-\varkappa}(1+j)^{-\varkappa}$ and $C_K(1+k)^{-\varkappa}(1+\ell+k)^{-\varkappa}$ in the three parts. For $|y|\ge Y_0$: by (B), $\operatorname{Im}(\beta_i+g_i)=-\operatorname{Im}(\sigma+\alpha_i)$ and $\operatorname{Im}(\sigma+\alpha_i+g_i)=-\operatorname{Im}\beta_i$. So every Gamma argument in a numerator, namely $\ell+n+\sigma+\alpha_i+t$, $n+1-\beta_i-g_i+t$, $k+\sigma+\alpha_i+t$, $j+1+\beta_i-t$, $k+2-\sigma-\alpha_i-g_i-t$ and $\ell+k+1+\beta_i-t$, and every sine argument in a denominator, has imaginary part of modulus at least $|y|-Y_0+1\ge1$. No numerator Gamma function then meets a pole, the sine ratios are bounded, and the bounds hold with each base $1+\cdots$ replaced by $\cdots+|y|$. The constants are uniform for parameters in B, with $\varkappa_B$ in place of $\varkappa$ and $|y|\ge Y_B$, provided K avoids P_T for every parameter point of B. For the wheels the right and middle bounds are those of the paragraph *Proofs of the estimates*. With $|d_\ell|\le C(1+\ell)^{d-2}$ and (467) at $\alpha=\varkappa$ (the left sum has the form of the right one, with k in place of n) the three parts converge absolutely and uniformly on compact subsets off P_T. Each term is meromorphic in t; its poles come from $\Gamma(\ell+\sigma+\alpha_i+n+t)$ and $\Gamma(\beta_i-n-t)$, are simple, and lie on the classes $-\sigma-\alpha_i+\mathbb{Z}$ and $\beta_i+\mathbb{Z}$. At a point t_0 of P_T every term therefore has a pole of order at most the number of the $2r$ classes that contain t_0; multiplied by $(t-t_0)$ to that power, the series converges uniformly on a small circle around t_0, so the product with S_T is analytic near t_0, as in part (a) above. Periodicity follows from $n\mapsto n+1$. Each factor of D_T has simple zeros exactly on one class, so D_T vanishes at t_0 to the same order, and Ξ_T is entire and one-periodic. For $0\le x\le1$ and $|y|\ge\max(Y_0,2)$, the large-$|y|$ bounds and (467) give $|S_T(x+\mathrm{i}y)|\le C|y|^{\max(d,1)-2\varkappa}(1+\log|y|)$, and $|D_T(x+\mathrm{i}y)|\le Ce^{2\pi r|y|}$. Taking $y=-Y$ for $m\ge r$ and $y=Y$ for $m\le-r$ in the Fourier coefficient gives $|\gamma_m|\le Ce^{2\pi(r-|m|)Y}Y^{\max(d,1)-2\varkappa}(1+\log Y)\to0$, because $2\varkappa>\max(d,1)$ by (C). For the analyticity, on the segment $t=x+\mathrm{i}Y_B$, $0\le x\le1$, which stays at distance at least one from P_T for every parameter point of B, the series of S_T converges uniformly in x and in the parameters, its terms are analytic in the parameters, and D_T is entire in t and the parameters. Hence $\gamma_m=\int_0^1\Xi_T(x+\mathrm{i}Y_B)e^{-2\pi\mathrm{i}m(x+\mathrm{i}Y_B)}dx$ is analytic on the interior of B.

Step 1. The coefficients of the expansion in the upper half-plane with $m\le0$ vanish: the coefficient integral $\int_0^1 S_T(x+\mathrm{i}y)e^{-2\pi\mathrm{i}m(x+\mathrm{i}y)}dx$ does not depend on $y>Y_0$, its exponential is bounded by one

for $m \le 0$, and $S_T \to 0$ uniformly as $y \to \infty$. The relation $\gamma_{1-r} = \kappa_+ s_1^+$ compares the coefficients of the lowest power w^{1-r} in $\Xi_T = D_T S_T$ with the expansion of Step 0; the lower half-plane is the same.

Step 3: the Laplace forms (474). For $\operatorname{Re} w > 0$, $\Gamma(g) f_g(w) = B(w,g) = \int_0^\infty e^{-wu}(1-e^{-u})^{g-1}du$ (put $x = e^{-u}$ in Euler's integral). Substitute $u = -\mathrm{i}v$, with v on the positive imaginary axis, and rotate the ray to $[0,\infty)$ through the first quadrant. If also $\operatorname{Im} w > 0$, then $|e^{\mathrm{i}wv}| = e^{-\operatorname{Im}(wv)}$ decays exponentially in the closed quadrant, since $\arg w + \arg v \in (0,\pi)$ there. Moreover $1 - e^{\mathrm{i}v}$ has nonnegative real part in the quadrant and vanishes only at $v \in 2\pi\mathbb{N}_0$, where the integrand is $O(|v - 2\pi k|^{\operatorname{Re} g - 1})$, which is integrable. These points lie on the boundary of the quadrant: $v = 0$ at its corner and $v = 2\pi k$, $k \ge 1$, on its real side. A small quarter circle of radius ϵ at 0 and small semicircles at $2\pi k$, $k \ge 1$, contribute $O(\epsilon^{\operatorname{Re} g})$ each and nothing in the limit. This gives the first formula for $\operatorname{Re} w > 0$, $\operatorname{Im} w > 0$. The substitution $u = \mathrm{i}v$ and the fourth quadrant give the second formula for $\operatorname{Re} w > 0$, $\operatorname{Im} w < 0$. Both right sides are analytic on the respective half-planes, because $|(1-e^{\pm\mathrm{i}v})^{g-1}| \le e^{\pi|\operatorname{Im} g|/2}|2\sin\frac{v}{2}|^{\operatorname{Re} g - 1}$ is 2π-periodic and locally integrable; so (474) holds on the whole half-planes. For real g the second formula is the complex conjugate of the first; for complex g it is the separate rotation just described. On $\operatorname{Im} z = Y > Y_0$ one has $|e^{\mathrm{i}(\ell+\sigma+\alpha_i+z)v}| = e^{-(Y+\operatorname{Im}(\sigma+\alpha_i))v} \le e^{-v}$ and $|e^{-\mathrm{i}(\beta_i - z)v}| = e^{-(Y-\operatorname{Im}\beta_i)v} \le e^{-v}$, so the two products of Step 3 are absolutely convergent integrals and Fubini applies.

Step 3: the endpoint bound (475). On $[0,3\pi]$ the exponentials in $\mathsf{h}_i^\pm$ and the factors $e^{-\operatorname{Im} g_i \arg(1-e^{\pm\mathrm{i}v})}$ are bounded, so $|\mathsf{h}_i^\pm(v)| \le C|2\sin\frac{v}{2}|^{\operatorname{Re} g_i - 1}$ there. The modulus $|2\sin\frac{v}{2}|$ lies between cv and v on $(0,\pi+\delta]$ and between $c|2\pi - v|$ and $|2\pi - v|$ on $[\frac{\pi}{2}, 3\pi)$. Hence, whatever the sign of $\operatorname{Re} g_i - 1$, $|\mathsf{h}_i^\pm(v)| \le C v^{\operatorname{Re} g_i - 1}$ on the first interval and $|\mathsf{h}_i^\pm(v)| \le C|2\pi - v|^{\operatorname{Re} g_i - 1}$ on the second. For $V \le \pi$ every coordinate of Δ_V is at most π, and Dirichlet's integral gives $|a_\pm(V)| \le C\int_{\Delta_V}\prod_i v_i^{\operatorname{Re} g_i - 1}d\mu = CV^{\varkappa-1}\prod_i \Gamma(\operatorname{Re} g_i)/\Gamma(\varkappa)$. For $\pi \le V \le 2\pi + \delta$ at most one coordinate of a point of Δ_V exceeds $V/2$. Where none does, all coordinates are below $\pi + \delta$, and the same Dirichlet bound gives $O(1)$. Where $v_j > V/2$, write $s = V - v_j < V/2$ and $\epsilon_V = 2\pi - V \in [-\delta, \pi]$, so that $2\pi - v_j = \epsilon_V + s$. Bounding factor j by $C|\epsilon_V + s|^{\operatorname{Re} g_j - 1}$ and integrating the other $r-1$ factors over their simplex of size s (Dirichlet again, with exponent $\varkappa - \operatorname{Re} g_j > 0$) leaves

$$C\int_0^{V/2} |\epsilon_V + s|^{\operatorname{Re} g_j - 1} s^{\varkappa - \operatorname{Re} g_j - 1} ds.$$

For $\operatorname{Re} g_j \ge 1$ this is bounded. For $\operatorname{Re} g_j < 1$, the part $s \le 2|\epsilon_V|$ is $|\epsilon_V|^{\varkappa-1}$ times the convergent integral $\int_0^2 |1 \pm u|^{\operatorname{Re} g_j - 1} u^{\varkappa - \operatorname{Re} g_j - 1} du$ (substitute $s = |\epsilon_V|u$), and on $s > 2|\epsilon_V|$ one has $|\epsilon_V + s| \ge s/2$, which leaves $C\int_{2|\epsilon_V|}^{2\pi} s^{\varkappa-2} ds$. Both are $O(|\epsilon_V|^{\min(\varkappa-1,0)}\lambda)$. Summing over j proves (475). At $\varkappa = 1$ the bound is at most logarithmic: when $\operatorname{Im}\sum_i g_i \ne 0$, $a_\pm(2\pi - \epsilon_V)$ stays bounded and oscillates like $\epsilon_V^{\mathrm{i}\operatorname{Im}\sum_i g_i}$ (checked numerically in the review `LRI_C5_general_d/review/lemmaG-proof` for $r = 2$, $g = (0.45 + 0.3\mathrm{i},\, 0.55)$), and the logarithm can occur only when $\operatorname{Im}\sum_i g_i = 0$. Only the upper bound is used. For the wheels at real b, by dominated convergence, $a_-(2\pi^-) = e^{-\mathrm{i}(g-1)(2-r)\pi/2}\mathsf{p}(2\pi)$ is finite and nonzero, so for $\varkappa > 1$ the exponent in (476) is $\varkappa - 1$, not $2\varkappa - 2$.

Step 3: the formula (477). The functions $a_\pm^Y$ lie in $L^1(\mathbb{R})$: $|a_+^Y(V)|$ is at most the convolution of the functions $Ce^{-(Y+\operatorname{Im}(\sigma+\alpha_i))v}|2\sin\frac{v}{2}|^{\operatorname{Re} g_i - 1}$, each integrable on $[0,\infty)$, and likewise for a_-^Y. The bound $|f_{g_i}(w)| \le C(1+|w|)^{-\operatorname{Re} g_i}$ on horizontal lines at distance at least one from the real axis follows from Gamma reflection and (465). For $U < 2\pi + \delta$ the convolution $(a_+^Y * a_-^Y)(U) = \int_0^U a_+^Y(V)a_-^Y(U-V)dV$ involves $a_\pm^Y$ only on $[0, 2\pi+\delta]$, where they are square integrable by (475) and $2\varkappa > 1$, and a convolution of two square-integrable functions is continuous. Since the convolution and its Fourier transform are both integrable, Fourier inversion holds at this point of continuity, $\int_{\mathbb{R}} \widehat{a_+^Y * a_-^Y}(x)e^{-2\pi\mathrm{i}x}dx = 2\pi(a_+^Y * a_-^Y)(2\pi)$. Together with $e^{-2\pi\mathrm{i}z} = e^{2\pi Y}e^{-2\pi\mathrm{i}x}$, and the factor $e^{-2\pi Y}$ in $(a_+^Y * a_-^Y)(2\pi)$, this is the first form of (477). The second form is Fubini; the absolute integrand obeys the same bounds. The measure $d\mathrm{vol}$ is $(2r)^{-1/2}$ times the surface measure of the hyperplane $V + V' = 2\pi$, hence invariant under every permutation of the $2r$ coordinates.

Step 4. For $\frac{1}{2} \le \rho < 1$, $|1-\rho e^{\mathrm{i}V}|^2 = (1-\rho)^2 + 4\rho\sin^2\frac{V}{2} \ge 2\sin^2\frac{V}{2}$, so $|(1+\rho e^{\mathrm{i}V})(1-\rho e^{\mathrm{i}V})^{1-d}| \le C|\sin\frac{V}{2}|^{-\max(d-1,0)}$. Dominated convergence gives first the one-variable form $s_1^+ = 2\pi\kappa_\Gamma \int_0^{2\pi} e^{\mathrm{i}\sigma V}(1+e^{\mathrm{i}V})(1-e^{\mathrm{i}V})^{1-d} a_+(V) a_-(2\pi-V)\, dV$, and the form on $\widetilde{\Delta}$ is Fubini, since after integration over the fibers of V the function $|\Psi|$ is dominated by the same integrable function. For $d>2$ the entry $\varkappa > d-1$ of (C) is stronger than the spectral condition $2\varkappa > d$, and by (476) the argument does not extend past $b_{\max}$.

Step 6. Conjugating the expansion of Step 1 at $\bar p$ gives the expansion in the lower half-plane at p, so $s_m^-(p) = \overline{s_m^+(\bar p)}$ and $\gamma_{r-1}(p) = \kappa_- s_1^-(p) = \kappa_- \overline{s_1^+(\bar p)} = 0$; this is equivalent to the relation $\gamma_m(\bar p) = \overline{\gamma_{-m}(p)}$ used above. At real parameters it is the reality $\gamma_{-m} = \overline{\gamma_m}$; at complex parameters it relates two different parameter points and is not a symmetry of a single Ξ_T.

Step 7. The part of (C) where all $\operatorname{Re} g_i > 0$ is open and nonempty; it contains the points with all $\operatorname{Re} g_i = \varkappa/r$.

The equivalent form. On $\operatorname{Im} t > Y_0$, $D_T = \kappa_+ w^{-r}(1+O(w))$ with $\kappa_+ \ne 0$, and $\Xi_T = \sum_{|m|\le r-2}\gamma_m w^m = O(e^{2\pi(r-2)\operatorname{Im} t})$; so $S_T = \Xi_T/D_T = O(e^{-4\pi\operatorname{Im} t})$ as $\operatorname{Im} t \to +\infty$. Conversely, this decay forces $s_1^+ = 0$: let $Y \to \infty$ in its coefficient integral. The lower half-plane is the same with κ_-. For the wheels, with $c_{r,r-1} = 2\gamma_{r-1}e^{-\mathrm{i}\pi(r-1)\sigma}$, this is the equivalent form stated after (473).

The wheel case and the earlier proof. For the wheels this proof is the earlier, wheel-only proof that the review `LRI_C5_general_d/review/VERIFICATION.md` checked (proof review in `LRI_C5_general_d/review/foundation-rigor`; Section 20.10), with Steps 3–4 as corrected after that review, up to three changes: the parameters are complex throughout (the earlier version reduced to real b by analyticity in b), the second top mode comes from Step 6 instead of the reflection of part (b) of Section 20.1.1, and Step 5 is written through (478).

Two ingredients enter: the balance of every single factor (Step 5) and the factor $1+z$ in (461), which is odd about $V=\pi$. Both are essential: controls that break either one have nonzero top modes. The numerical tests of the theorem and of its proof, with these controls, are collected in Section 20.9 (item *Top-mode theorem and its product form*).

Relation to the periodization at $b=d/2$ and to the half-shift reduction. At $b=d/2$ and $\frac{2}{r} < d < 4$, (473) sharpens (311) for the wheels: $c_{3,2}(q)=0$ and $c_{4,3}(q)=0$ in every such dimension, and the sum in (318) stops at $k=r-2$. It says nothing about Ξ_7: the kernel K_ℓ of (303) carries the extra factor $[(\ell+p)^2-4s^2]\Gamma(p+2s)\Gamma(p-2s)/(\Gamma(2s)\Gamma(-2s))$ and the multiplicities $(-1)^\ell d_\ell$, so it is not of the form (470). In Section 19 the relations $c_{3,2}=0$ and $c_{4,3}=0$ come out of the half-shift determinant (431); they are the $d=3$ cases of (473), which holds for every d and every b in the strip. What is special to three dimensions is the identification of the surviving modes with products of ordinary moment amplitudes, $c_{3,0} = -\mathfrak{a}_3\mathfrak{b}_3$ and $c_{4,1} = 2\mathfrak{a}_4\mathfrak{b}_4$, and the odd $r=7$ coefficients (432). At $d=1$ and $d=2$, (473) also follows from the one-variable lattices (539) below (at $d=2$, $b=1$ and $r=4$ it is (337)); at generic d only the proof of Section 20.1.2 gives it.

20.1.3 Residue functional, explicit weights and blocks

For $0 < \operatorname{Re} b < b_{\max}(r,d)$ and $\sigma \notin \mathbb{Z}$,

$$W_r(d;b) = -N_b^r \operatorname{Res}_{t=0} \frac{\Xi_r(t;b)}{D_b(t)^r} = -\Big[\frac{N_b}{\pi\sin\pi\sigma}\Big]^r \sum_{k=0}^{r-2} B_{r,k}(\sigma)\, c_{r,k}(q,b), \tag{479}$$

where $B_{r,k}$ is (318) with the logarithmic coefficients (314), both with q replaced by σ. The proof follows the one after (319); it is given in full in the paragraph *The residue functional in full* below.

The weights depend on b only, not on q. For $r = 3, 4$,

$$\boxed{\begin{aligned} W_3(d;b) &= -\frac{2N_b^3}{\pi s_b^5}\Big[(1+2\varsigma_b^2)\,c_{3,0} + 3\varsigma_b\,c_{3,1}\Big],\\ W_4(d;b) &= \frac{4N_b^4}{\pi s_b^7}\Big[\varsigma_b(3+2\varsigma_b^2)\,c_{4,0} + (1+4\varsigma_b^2)\,c_{4,1} + 5\varsigma_b\,c_{4,2}\Big]. \end{aligned}} \tag{480}$$

The weights of the omitted top modes, $B_{3,2} = 6\pi^2/s_b^2$ and $B_{4,3} = -20\pi^3/s_b^3$, are nonzero, so the formulas below depend on (473).

The residue functional in full. *Proof of* (479). Let $0 < \operatorname{Re} b < b_{\max}(r,d)$ and $\sigma \notin \mathbb{Z}$; then $b_{\max} \le b_{\mathrm{spec}}$, so (460) represents W_r. For each spin close the Barnes contour to the right by the arcs of the circles $|s - a_\ell| = n_0 + \frac12$, $n_0 \in \mathbb{N}$, that lie in $\operatorname{Re} s \ge 0$. The part of the disc $|s-a_\ell| < n_0 + \frac12$ in $\operatorname{Re} s > 0$ contains the right poles $a_\ell + n$, $0 \le n \le n_0$, and no left pole, since the left poles $-a_\ell - n$ have negative real part. The arcs stay at distance at least $\frac12$ from $a_\ell + \mathbb{Z}$, because $|s - a_\ell - m| \ge |n_0 + \frac12 - |m|| \ge \frac12$ for every integer m, so the sine ratio produced by reflecting $\Gamma(a_\ell - s)/\Gamma(\beta_\ell - s)$ is bounded on them, and $g_\ell(s)^r = O(|s|^{-2r\operatorname{Re} g})$ there; since $2r\operatorname{Re} g > 1$, the arcs contribute nothing in the limit. (The package's own proof closes the contour on the circles $|s| = a_\ell + N + \frac12$, whose radius is real only for real b. The circles centred at a_ℓ keep the distance $\frac12$ from $a_\ell + \mathbb{Z}$ for every complex b. This change was made when the proof was merged into this note, on 16 October 2026, and has not been reviewed.) The closure is clockwise, so the ℓ-th integral is $-\sum_{n\ge0}\operatorname{Res}_{s=a_\ell+n} g_\ell(s)^r$, and $\operatorname{Res}_{s=a_\ell+n} g_\ell(s)^r = \operatorname{Res}_{t=0} g_\ell(a_\ell+n+t)^r$. In S_r the point $t=0$ is a pole only of the terms with $n \ge 0$: for $n<0$, $\Gamma(-n-t)$ is regular at $t=0$, and $t = 0 \equiv -\sigma$ would need $\sigma \in \mathbb{Z}$. By the uniform convergence on a small circle $|t| = \epsilon$ (paragraph *Proofs of the estimates*), $\sum_\ell d_\ell \sum_{n\ge0}\operatorname{Res}_{s=a_\ell+n} g_\ell^r = \operatorname{Res}_{t=0} S_r = \operatorname{Res}_{t=0} \Xi_r/D_b^r$, which is the first equality of (479). Next,

$$\log\frac{(\pi t\sin\pi\sigma)^r}{D_b(t)^r} = r\log\frac{\pi t}{\sin\pi t} - r\log\frac{\sin\pi(t+\sigma)}{\sin\pi\sigma} = \sum_{h\ge1} u_h\frac{t^h}{h!},$$

with u_h the coefficients (314) at $q\to\sigma$, $u_h(\sigma) = r[(1+(-1)^h)\psi_{h-1}(1) - (-1)^{h+1}\psi_{h-1}(1-\sigma) + \psi_{h-1}(\sigma)]$. Indeed $\log(\pi t/\sin\pi t) = \sum_{m\ge1}\zeta(2m)t^{2m}/m$, with $2(h-1)!\,\zeta(h) = (1+(-1)^h)\psi_{h-1}(1)$ for even h, and $\partial_\sigma^h \log\sin\pi\sigma = -\psi_{h-1}(\sigma) - (-1)^h\psi_{h-1}(1-\sigma)$ [32]. With the polynomials E_m of (316), which are the Taylor coefficients of $\exp\sum_h x_h t^h/h!$, therefore $\operatorname{Res}_{t=0} e^{2\pi i m t} D_b(t)^{-r} = (\pi\sin\pi\sigma)^{-r} E_{r-1}(u_1 + 2\pi i m, u_2, \dots, u_{r-1})$. Inserting $\Xi_r = \sum_{|m|\le r-2}\gamma_m e^{2\pi i m t}$, which is (473), gives the exponential form

$$W_r(d;b) = -\Big[\frac{N_b}{\pi\sin\pi\sigma}\Big]^r \sum_{|m|\le r-2}\gamma_m\,E_{r-1}(u_1 + 2\pi i m, u_2, \dots, u_{r-1}), \tag{481}$$

and $\gamma_{\pm k} = \frac12 c_{r,k} e^{\pm i\pi k\sigma}$ $(k \ge 1)$, $\gamma_0 = c_{r,0}$ turn it into (479) with $B_{r,k}(\sigma) = \frac12[e^{i\pi k\sigma}E_{r-1}(u_1 + 2\pi i k, u_2, \dots) + e^{-i\pi k\sigma}E_{r-1}(u_1 - 2\pi i k, u_2, \dots)]$, which is (318) at $q\to\sigma$. The functional depends on b only, through σ; the dimension enters only through N_b and the $c_{r,k}(q,b)$. By (315) at $q\to\sigma$ the u_h are elementary, $u_1/(r\pi) = -\varsigma_b/s_b$, $u_2/(r\pi^2) = \varsigma_b^2/s_b^2 + \frac43$ and $u_3/(r\pi^3) = -2(\varsigma_b/s_b)(1+\varsigma_b^2/s_b^2)$, and for the b-derivatives of the weights $\partial_\sigma u_h = u_{h+1} - r(1-(-1)^h)\psi_h(1)$. For $r = 3, 4$,

$$\begin{gathered} B_{3,0} = \frac{2\pi^2(1+2\varsigma_b^2)}{s_b^2}, \qquad B_{3,1} = \frac{6\pi^2\varsigma_b}{s_b^2}, \qquad B_{3,2} = \frac{6\pi^2}{s_b^2},\\ B_{4,0} = -\frac{4\pi^3\varsigma_b(3+2\varsigma_b^2)}{s_b^3}, \qquad B_{4,1} = -\frac{4\pi^3(1+4\varsigma_b^2)}{s_b^3}, \qquad B_{4,2} = -\frac{20\pi^3\varsigma_b}{s_b^3}, \qquad B_{4,3} = -\frac{20\pi^3}{s_b^3}, \end{gathered} \tag{482}$$

and (480) follows. □ At $d = 2$ and $r = 4$, with $x = \frac{b}{2}$, N_b equals $\mathcal{N}_x$ of Section 20.6, and inserting $c_{4,0}, c_{4,1}, c_{4,2}$ of the $d = 2$ paragraph there into (480) reproduces the weights ρ_0, ρ_1, ρ_2 of (350). At $b = 1$ ($\varsigma_b = 0$, $s_b = 1$, $N_1 = 1$) it reduces to $P_4(2) = 4c_{4,1}/\pi = 8\mathcal{A}(\frac12)\mathcal{B}(\frac12)/\pi$.

Each sample is a sum of five blocks, (483) below. The blocks are (321) with the arrays (322), where the rim parameter q is replaced by σ in $q+t$, $q+1+t$ and in the mirror point, and the spoke parameter q by g in $1-q+t$, $1+q-t$ and $\Gamma(q-t)$. Every parameter is affine in (q,b), and all five blocks converge absolutely exactly for $\operatorname{Re} b < b_{\max}$ (paragraph *Proof of the block decomposition* below). The sections below do not use generic nodes. They sample at the pole node and at the reflection centers, where the blocks collapse to fewer functions.

Proof of the block decomposition. Write A_r^0, A_r^+, C_r for the blocks (321) with the substitutions just stated; at $b=\frac{d}{2}$ they are the blocks of (323). The claim is

$$\Xi_r(t;b) = D_b(t)^r\big[A_r^0(t)+A_r^+(t)+A_r^0(1-\sigma-t)+A_r^+(1-\sigma-t)+C_r(t)\big] \qquad (t\notin\mathbb{Z}\cup(-\sigma+\mathbb{Z})). \tag{483}$$

In the right part with $\ell=0$, the factorized form in (464) and

$$f(\sigma+n+t)=\frac{\Gamma(\sigma+t)}{\Gamma(2q+t)}\frac{(\sigma+t)_n}{(2q+t)_n}, \qquad f(-n-t)=\frac{\Gamma(-t)}{\Gamma(g-t)}\frac{(1-g+t)_n}{(1+t)_n},$$

which use $\sigma+g=2q$, give the series of $A_r^0(t)$ term by term; the upper parameter 1 of the ${}_{2r+1}F_{2r}$ cancels $n!$. For $\ell=j+1$, $f(j+1+\sigma+n+t)=[\Gamma(1+\sigma+t)/\Gamma(1+2q+t)]\,(1+\sigma+t)_{j+n}/(1+2q+t)_{j+n}$, and the multiplicity $d_{j+1}=4q\,(2q+1)_j(4q-1)_j/((2q)_j(2)_j)$ of (301) supplies the prefactor $4q$ and the rows $(2q+1,\boldsymbol{e}),(4q-1,\boldsymbol{e});(2q,\boldsymbol{e}),(2,\boldsymbol{e})$; the rows $(1,\boldsymbol{e})$ and $(1,\boldsymbol{f})$ cancel the factorials of (320). This gives $A_r^+(t)$. In the middle part, the third line of (464),

$$\frac{\Gamma(k+\sigma+t)}{\Gamma(k+2q+t)}\,\frac{\Gamma(j+1-t)}{\Gamma(j+1+g-t)}=\frac{\Gamma(\sigma+t)\Gamma(1-t)}{\Gamma(2q+t)\Gamma(1+g-t)}\,\frac{(\sigma+t)_k(1-t)_j}{(2q+t)_k(1+g-t)_j},$$

with the multiplicity $d_{j+k+1}=4q\,(2q+1)_{j+k}(4q-1)_{j+k}/((2q)_{j+k}(2)_{j+k})$, gives $C_r(t)$, with the multiplicity rows along $\boldsymbol{c}$. By the last line of (464) the left part is the right part at $1-\sigma-t$. All the series converge absolutely by (467), so the rearrangement is justified, and multiplying by $D_b(t)^r$ gives (483). Each block also converges directly under part of the condition: the ${}_{2r+1}F_{2r}(1)$ has parameter excess $r(2q+t)+r(1+t)-1-r(\sigma+t)-r(1-g+t)=2rg-1$, so it needs condition (iii); A_r^+ needs (i)–(iii), and C_r needs (i) and (ii). Together this is exactly $\operatorname{Re} b<b_{\max}$. At integer d the multiplicity rows terminate: $d_{j+1}/(4q)$ is δ_{j0} at $d=1$, 1 at $d=2$ and $1+\frac{2}{3}j$ at $d=3$.

Removable points of individual blocks. As functions of t the five blocks are sub-sums of S_r, so each is analytic off $\mathbb{Z}\cup(-\sigma+\mathbb{Z})$. In the factorized form, a Gamma prefactor times a Horn value, an individual Horn value has poles where one of its lower parameters is a non-positive integer, cancelled by a zero of the same order of its prefactor. This happens for $A_r^0(t)$ at $t\in-2q-\mathbb{N}_0$, for $A_r^0(1-\sigma-t)$ at $t\in1+g+\mathbb{N}_0$, for $A_r^+(t)$ at $t\in-1-2q-\mathbb{N}_0$, for $A_r^+(1-\sigma-t)$ at $t\in2+g+\mathbb{N}_0$, and for $C_r(t)$ at $t\in-2q-\mathbb{N}_0$ and $t\in1+g+\mathbb{N}_0$. At these points (483) holds by continuity. For real b the points are real, and the nodes of (484) below, which have $\operatorname{Im} t_j=\eta_r>0$, never meet them.

20.1.4 Fixed nodes, jets, evaluation of the samples, and the continuation beyond $b_{\max}$

The formulas of this subsection are exact algebra on Sections 20.1.1–20.1.3; the continuation beyond $b_{\max}$ is derived, not proved (Section 20.10). Sections 20.2–20.7 do not use this subsection. It describes the form implemented by the general-b code `LRI_C5_general_d/foundation/wheel_general_b.py` (function `W(r,d,b,dps)`), which Section 20.9 tests.

Nodes that do not move with b. Let N be either $N_r=2r-1$ of (312) or the reduced value $N_r'=2r-3$ permitted by (473), and $M=(N-1)/2$. Put

$$t_j=-\frac{q}{2}+\frac{j}{N}+\mathrm{i}\eta_r, \qquad j=0,\dots,N-1,$$

with $\eta_r = 1/(8\pi r)$ of (312); for $N = N_r$ the first r of them are the nodes of (312). Since Ξ_r is a trigonometric polynomial with exponential modes $|m| \le M$, the discrete Fourier transform of the N complex samples recovers all modes without aliasing, $\gamma_m = N^{-1}\sum_j \Xi_r(t_j;b)e^{-2\pi i m t_j}$. Inserted into (481) this gives, for $0 < \operatorname{Re} b < b_{\max}(r,d)$ and $\sigma \notin \mathbb{Z}$,

$$\boxed{\begin{aligned} W_r(d;b) &= N_b^r \sum_{j=0}^{N-1} \mathfrak{w}_{r,j}(b)\, \Xi_r(t_j;b), \\ \mathfrak{w}_{r,j}(b) &= -\frac{1}{N(\pi \sin \pi\sigma)^r} \sum_{|m|\le M} E_{r-1}(u_1 + 2\pi i m, u_2, \dots, u_{r-1})\, e^{-2\pi i m t_j}. \end{aligned}} \tag{484}$$

The weights are elementary: by (315) at $q \to \sigma$ they are polynomials in π, $\cot\frac{\pi b}{2}$ and $1/\sin\frac{\pi b}{2}$, times $e^{-2\pi i m t_j}$. The nodes do not depend on b, and every factor is analytic in b on the strip (the samples by part (d) of Section 20.1.1); so the product rule and (244) give, with $\partial_b \log N_b = -\frac12[\psi_0(g) + \psi_0(\sigma)]$,

$$\begin{aligned} \mathcal{L}_d &= \sum_j \Big[\partial_b(N_b^4 \mathfrak{w}_{4,j})\, \Xi_4(t_j;b) + N_b^4 \mathfrak{w}_{4,j}\, \partial_b \Xi_4(t_j;b)\Big]_{b=d/2}, \\ Z_d &= \frac19 \sum_j \Big[\partial_b^2(N_b^3 \mathfrak{w}_{3,j})\, \Xi_3 + 2\partial_b(N_b^3 \mathfrak{w}_{3,j})\, \partial_b \Xi_3 + N_b^3 \mathfrak{w}_{3,j}\, \partial_b^2 \Xi_3\Big]_{b=d/2}. \end{aligned} \tag{485}$$

Here $\partial_b^k \Xi_r(t_j;b) = \sum_{p\le k} \binom{k}{p} \partial_b^{k-p}[D_b(t_j)^r]\, \partial_b^p[\cdots]$, with $\partial_b D_b(t)^r = \frac{r\pi}{2}\cot\pi(t+\sigma)\, D_b(t)^r$, and the bracket is the sum of the five blocks of (483). The b-derivatives of the blocks at fixed t_j are first and second derivatives of the ${}_{2r+1}F_{2r}(1)$ values and of the two-variable Horn values with respect to their parameters, which are all affine in b with slopes $\pm\frac12$, together with polygamma values from the Gamma prefactors; in the mirrored blocks the argument $1-\sigma-t_j$ moves with b, which is again a parameter derivative. In this form $\mathcal{L}_d$ and Z_d are finite combinations, with elementary coefficients, of one- and two-variable hypergeometric values at unit argument and their first and second parameter derivatives. With the reduced nodes W_3 needs 3 complex samples and W_4 needs 5. The b-dependent symmetric nodes $-\frac{\sigma}{2} + \frac{j}{N_r} + i\eta_r$ with the real cosine transform (313) are equally valid, and need 2 and 3 real samples; their b-derivatives also contain t-derivatives of the samples, which is why the fixed nodes are used for the jets. The code implements the symmetric nodes as a cross-check (option `nodes='note'`).

Radius of the sample jets. $W_r(d;b)$ is analytic up to b_{spec}, but the individual samples $\Xi_r(t_j;b)$ and the coefficients $c_{r,k}(q,b)$ with $k \le r-2$ have a simple pole at $b = b_{\max}$ when $b_{\max} < b_{\text{spec}}$, that is for $d > 2$ (nonzero residue in every case computed; paragraph *Beyond* $b_{\max}$ below). Moreover, at a fixed node the individual blocks have poles in b where $\sigma + t_j \in \mathbb{Z}$ (at complex b, since $\operatorname{Im} t_j = \eta_r \ne 0$); they are removable in $\Xi_r(t_j;b)$ because of the zero of $D_b(t_j)^r$. The derivatives in (485) at $b = \frac{d}{2}$ are unaffected. When they are computed from Taylor series or Cauchy circles of the samples, however, the radius must stay below $b_{\max} - \frac{d}{2}$; for $d \ge 2$ this is $\frac12$ for $r = 4$ and $\frac{4-d}{6}$ for $r = 3$. A second restriction concerns the weights: $\mathfrak{w}_{r,j}(b)$ carries $(\sin\pi\sigma)^{-r}$ and has a pole of order r at $b = 2$. The disc $|b - \frac{d}{2}| < b_{\max} - \frac{d}{2}$ contains $b = 2$ exactly when $b_{\max} > 2$ (paragraph *Integer* σ of Section 20.1.1): for $r = 4$ when $d > 3$, and for $r = 5$ when $d > \frac83$. There the samples and W_r are analytic, and the poles of the weights cancel only in the sum (484). Cauchy circles of the weighted products $N_b^r \mathfrak{w}_{r,j} \Xi_r(t_j;b)$, or of the weights alone, must then stay inside $|b - \frac{d}{2}| < |2 - \frac{d}{2}|$. The jets (485) at $b = \frac{d}{2} \ne 2$ are unaffected, and the cases used in this note ($r = 3, 4$ with $d \le 3$) do not meet $b = 2$.

Evaluation of the Horn values. The module `LRI_C5_general_d/foundation/hypsums.py` evaluates the sums of the five blocks as an exact finite part plus an asymptotic tail, and `wheel_general_b.py` assembles (483) and (484). Every sequence that occurs is a Gamma-ratio sequence y_n, a constant times a product of integer powers of Pochhammer ratios $(e)_n/(e')_n$. Its large-n expansion

$\mathsf{y}_n \sim \sum_k \upsilon_k n^{-p_\mathsf{y}-k}$ follows from the Bernoulli-polynomial expansion of $\log\Gamma(n+e)-\log\Gamma(n+e')$ [27]. *Single sums:* $\sum_{n\ge0}\mathsf{y}_n = \sum_{n<N_\mathrm{H}}\mathsf{y}_n + \sum_k \upsilon_k\,\zeta(p_\mathsf{y}+k, N_\mathrm{H})$, with Hurwitz zeta values [28] computed by Euler–Maclaurin summation in which every term carries the factor $N_\mathrm{H}^{-p_\mathsf{y}-k}$, so that they have relative precision for complex exponents. (The routine `mpmath.zeta(s,a)` has an absolute error floor of about 10^{-P} at P working digits for complex s and is not used.) The same formula, with the Hurwitz values continued in the exponent, is the analytic continuation (the zeta-regularized value) when the series diverges. *Two-variable sums:* A_r^+ and C_r have the form $\sum_{m\ge0}\mathsf{x}_m\mathsf{q}_m$ with a Cauchy convolution $\mathsf{q}_m = \sum_{n=0}^m \mathsf{y}_{m-n}\mathsf{z}_n$. For A_r^+, $\mathsf{x}_m = [(1+\sigma+t)_m/(1+2q+t)_m]^r$, $\mathsf{y}_j = d_{j+1}/(4q)$ and $\mathsf{z}_n = [(1-g+t)_n/(1+t)_n]^r$; for C_r, $\mathsf{x}_m = d_{m+1}/(4q)$, $\mathsf{y}_j = [(1-t)_j/(1+g-t)_j]^r$ and $\mathsf{z}_k = [(\sigma+t)_k/(2q+t)_k]^r$. The code computes q_m exactly for $m < M_\mathrm{H}$ and uses its large-m expansion for $m \ge M_\mathrm{H}$. If $\mathsf{y}_j \sim \sum_a \upsilon_a j^{-p_\mathsf{y}-a}$ and $\mathsf{z}_n \sim \sum_c \upsilon'_c n^{-p_\mathsf{z}-c}$, the generating function of y at $x = e^{-w} \to 1$ has the singular part $\sum_a \upsilon_a\Gamma(1-p_\mathsf{y}-a)w^{p_\mathsf{y}+a-1}$ and the regular part $\sum_k \mathsf{m}_k^\mathsf{y}(-w)^k/k!$, where m_k^y is the zeta-regularized moment "$\sum_j j^k\mathsf{y}_j$" (Mellin asymptotics of harmonic sums [29]; transfer at the singularity $x=1$ [30]), and likewise for z. Multiplying and transferring back gives three families,

$$\begin{aligned}
\mathsf{q}_m &\sim \sum_i \big[\mathfrak{e}_i^{(1)} m^{-p_\mathsf{y}-i} + \mathfrak{e}_i^{(2)} m^{-p_\mathsf{z}-i} + \mathfrak{e}_i^{(3)} m^{1-p_\mathsf{y}-p_\mathsf{z}-i}\big],\\
\mathfrak{e}_i^{(1)} &= \sum_{a+k=i} \upsilon_a \frac{(p_\mathsf{y}+a)_k}{k!}\mathsf{m}_k^\mathsf{z}, \qquad \mathfrak{e}_i^{(2)} = \sum_{c+k=i} \upsilon'_c \frac{(p_\mathsf{z}+c)_k}{k!}\mathsf{m}_k^\mathsf{y},\\
\mathfrak{e}_i^{(3)} &= \sum_{a+c=i} \upsilon_a\upsilon'_c \frac{\Gamma(1-p_\mathsf{y}-a)\Gamma(1-p_\mathsf{z}-c)}{\Gamma(2-p_\mathsf{y}-p_\mathsf{z}-a-c)}.
\end{aligned} \tag{486}$$

The tail $\sum_{m\ge M_\mathrm{H}}\mathsf{x}_m\mathsf{q}_m$ is then a finite combination of three batches of Hurwitz values. For the multiplicity sequence at non-integer d the regularized moments are known in closed form. By (461), $\sum_j (d_{j+1}/4q)e^{-jw} = [(1+e^{-w})(1-e^{-w})^{1-d}-1]e^w/(4q)$, whose first term has only non-integer powers of w; hence

$$\mathsf{m}_k^\mathsf{y} = \frac{(-1)^{k+1}}{4q} \qquad (\mathsf{y}_j = d_{j+1}/(4q),\ d\notin\mathbb{Z}), \tag{487}$$

which the numerical moments reproduce. The other moments are computed with a head $N_\mathrm{H} \le M_\mathrm{H}/2$, so that the cancellation in $\sum_{n<N_\mathrm{H}} n^k\mathsf{z}_n$ is suppressed by M_H^{-k} in the tail. *Exceptional parameters:* for non-integer d the coefficients in (486) have removable poles when $r(d-b)/2 \in \mathbb{Z}$ (the $\log m$ resonance of two families). At such points `W` returns the mean over 8 points of a circle of radius about $10^{-(P+10)/8}$ around b, which is exact for analytic W_r up to the eighth power of the radius, with extra working digits. At $d = 1, 2, 3$ the multiplicities are polynomial in j (or δ_{j0}); then only convergent moments enter, the coefficient of the third family is $\upsilon_a\upsilon'_c(-p_\mathsf{y}-a)!/(1-p_\mathsf{z}-c)_{1-p_\mathsf{y}-a}$, the middle block is a finite sum of products of single moments, and no resonance occurs inside the strip. *Parameters and cost:* the defaults are $M_\mathrm{H} = \max(80, 1.6P')$, $K_\mathrm{H} = \max(30, 0.8P'+10)$ asymptotic terms and $N_\mathrm{H} = M_\mathrm{H}/2$, with $P' = P+15$ guard digits at P requested digits. One W_4 at 50 digits takes about 45 s on one Intel core of the symmetry cluster, and all 846 evaluations of the checks of the general-b code in Section 20.9 took 6 minutes on one 64-core AMD node. The accuracy of this method is established by convergence in $(M_\mathrm{H}, K_\mathrm{H}, N_\mathrm{H})$ and by the independent checks, not by interval bounds. *Independent spectral code:* `LRI_C5_general_d/foundation/spectral.py` evaluates (460) on the Mellin line, by tanh–sinh quadrature for the spins $\ell < 40$ and, for $\ell \ge 40$, by the Stirling expansion in $z_\ell = \frac{\ell}{2} + \mathrm{i}\nu$ with $\frac{1}{2\pi}\int_\mathbb{R} z_\ell^{-e}\bar z_\ell^{-e'}d\nu = \ell^{1-e-e'}\Gamma(e+e'-1)/(\Gamma(e)\Gamma(e'))$ and a Hurwitz-zeta spin tail. It shares no code with the finite-sample formula except the Hurwitz routine.

Beyond b_max. For $d > 2$ one has $b_\mathrm{max} < b_\mathrm{spec}$; the ℓ-endpoint condition (ii) binds. The continuation meant here is the continuation in b of each block value at a fixed node t. Each block of (483) is a finite part, plus Hurwitz zeta values with exponents affine in b, plus convergent remainders, and the evaluation above computes exactly this continuation (zeta regularization of the tails). So the samples $\Xi_r(t_j; b)$ and the $c_{r,k}$ continue meromorphically in b. On $b_\mathrm{max} \le b < b_\mathrm{spec}$

their only pole is at $b_{\max}$: the next candidates $b_{\max}+2k/r$ exceed $b_{\rm spec}$, since $b_{\rm spec}-b_{\max}=(d-2)/r<2/r$. Both sides of (484) are meromorphic and agree on the strip, so the identity persists on $b_{\max}\le b<b_{\rm spec}$ with continued block values. The pole at $b_{\max}$ was derived in the review (`LRI_C5_general_d/review/foundation-rigor`); the derivation is an endpoint asymptotic sketch, not a written proof. The divergence at $b_{\max}$ comes from the ends $\ell\to\infty$ of the lattice at fixed n and at fixed $\ell+n$. There $d_\ell\sim 2\ell^{d-2}/\Gamma(d-1)$ by (466), $f(\ell+c)^r\sim\ell^{-rg}$, and $\sum_\ell \ell^{d-2-rg}$ has the pole $1/(rg-d+1)=-2/(r(b-b_{\max}))$. Collecting these ends gives

$$\begin{aligned}\operatorname{Res}_{b=b_{\max}}\Xi_r(t;b)&=-\frac{4}{r\,\Gamma(d-1)}\,D_b(t)^r\big[\Phi_{\max}(-t)+\Phi_{\max}(\sigma+t)\big],\\ \Phi_{\max}(u)&=\sum_{m\in\mathbb{Z}}\Big[\frac{\Gamma(m+u)}{\Gamma(m+u+g)}\Big]^r,\end{aligned}\tag{488}$$

at $\sigma=b_{\max}/2$ and $g=(d-b_{\max})/2$, where $rg=d-1>1$. The residue has three properties. (1) Its top mode is proportional to $\cos(\pi(rg+2-d)/2)$, which vanishes at $b_{\max}$; this follows from Steps 1, 3 and 5 of the proof of (471) applied to $\Phi_{\max}$, with $a_\pm(2\pi)$ finite because $rg>1$, and it is consistent with (473) holding up to the boundary. (2) Its lower modes are nonzero in every case computed, so the continued $c_{r,k}$ with $k\le r-2$ have a simple pole. (3) The functional annihilates it: $\Phi_{\max}(\sigma+t)$ is regular at $t=0$, and $\operatorname{Res}_{t=0}\Phi_{\max}(-t)=-\sum_{n\ge0}\operatorname{Res}_{z=-n}f(z)^r=0$, because f^r has poles only in $-\mathbb{N}_0$ and is $O(|z|^{-rg})$ on the circles $|z|=n_0+\frac12$, with $rg>1$. Hence W_r has no pole at $b_{\max}$, as the spectral representation requires. The same review checked this numerically at 20 digits for $(d,r)=(2.5,4)$, $(2.75,3)$, $(3.5,4)$ and $(3.5,3)$, with its scripts `bmax_residue.py`, `bmax_normalization.py` and `bmax_compare.py` (symmetry job 569252 for the first two): the top mode of (488) is $5\cdot10^{-21}$ to $2\cdot10^{-18}$ of the largest mode, and the lower modes are of order one; the functional applied to (488) vanishes to $2\cdot10^{-21}$–10^{-18} relative; at $d=2.5$, $r=4$ the predicted residues of $c_{4,0},\dots,c_{4,3}$, $(-9.109258,\,-11.819680,\,-2.719024,\,0)$, agree with the estimate $\frac{\delta}{2}[c_{4,k}(b_{\max}+\delta)-c_{4,k}(b_{\max}-\delta)]$ from the continued code to $O(\delta^2)$, at most $4\cdot10^{-5}$ at $\delta=10^{-3}$ and $4\cdot10^{-7}$ at $\delta=10^{-4}$. This also confirms that the zeta-regularized evaluation is the analytic continuation. The continued formula agrees with the spectral values at $(d,b)=(2.5,1.8)$ and $(2.5,1.76)$ for $r=4$ and $(2.5,1.55)$ for $r=3$ to 10^{-14} at 15 digits, and at 50 digits as in Section 20.9. For $d<1$ condition (iii) binds and coincides with the spectral bound. Nothing downstream uses the continuation: every use of these formulas in this note is at $b=\frac{d}{2}$, inside the strip.

20.2 Samples, modes and the two masters

By (473) and (480), W_4 needs three modes and W_3 two. In the variable $\tilde t=t+\frac{b}{4}$ the trigonometric polynomial is $\sum_k c_{r,k}\cos 2\pi k\tilde t$, even about $\tilde t=0$ and $\tilde t=\frac12$. We sample it at the pole node $t=0$ ($\tilde t=\frac{b}{4}$) and at these reflection centers.

Four-wheel. Three real samples fix the three modes:

$$\mathcal{X}_T(b)=N_b^4\Xi_4(0;b),\qquad \mathcal{X}_0(b)=N_b^4\Xi_4(-\tfrac{b}{4};b),\qquad \mathcal{X}_{1/2}(b)=N_b^4\Xi_4(\tfrac12-\tfrac{b}{4};b).\tag{489}$$

With $x=b/4$, $y=d/2-b/4$ (in the definitions of $\mathfrak{h}_u,\mathfrak{s}_u$ only; elsewhere in this section $t=x+\mathrm{i}y$, and x is the argument of $\Omega_r(t;x)$ in Section 20.6) they are five two-variable Horn functions at argument $(1,1)$: $\mathcal{X}_T=(\pi s_b)^4\mathfrak{h}_x$, $\mathcal{X}_0=\kappa_0^4(\mathfrak{s}_0+2\xi_0^4\mathfrak{h}_1)$, $\mathcal{X}_{1/2}=\kappa_h^4(\mathfrak{s}_{1/2}+2\xi_h^4\mathfrak{h}_{1/2})$, where

$$\mathfrak{h}_u=\sum_{\ell,n\ge0}d_\ell\Big[\frac{(u+x)_{\ell+n}}{(u+y)_{\ell+n}}\Big]^4\Big[\frac{(u+1-y)_n}{(u+1-x)_n}\Big]^4,\qquad \mathfrak{s}_u=\sum_{j,k\ge0}d_{j+k+2u}\Big[\frac{(u+x)_j(u+x)_k}{(u+y)_j(u+y)_k}\Big]^4,$$

$\kappa_0=N_b\sin^2(\pi x)\Gamma(x)^2/\Gamma(y)^2$, $\kappa_h=N_b\cos^2(\pi x)\Gamma(\frac12+x)^2/\Gamma(\frac12+y)^2$, $\xi_0=\frac{x(1-y)}{y(1-x)}$, $\xi_h=\frac{1-2y}{1-2x}$. The index u of $\mathfrak{h}_u$ and $\mathfrak{s}_u$ is unrelated to the weight $u(b)$ of (496). In particular $\mathfrak{h}_x(d;b)=\sum d_\ell T_{\ell n}(b)^4$

with

$$T_{\ell n}(b) = \frac{(\sigma)_{\ell+n}(1-g)_n}{(2q)_{\ell+n}\, n!}, \tag{490}$$

which at $b = d/2$ is the $T_{\ell n}$ of (251).

Section 20.3 proves these sample formulas for every r, (510)–(512).

By (473) the samples are $N_b^{-4}\mathcal{X}_0 = c_{4,0} + c_{4,1} + c_{4,2}$, $N_b^{-4}\mathcal{X}_{1/2} = c_{4,0} - c_{4,1} + c_{4,2}$ and $N_b^{-4}\mathcal{X}_T = c_{4,0} + \varsigma_b c_{4,1} + c_{4,2}\cos\pi b$. This system has determinant $4\sin^2\pi\sigma \neq 0$. Solving it and inserting into (480) gives, for every b in the strip,

$$W_4(d;b) = \frac{2}{\pi s_b^7}\Big[(1+\varsigma_b)^2(1+2\varsigma_b)\mathcal{X}_0 - (1-\varsigma_b)^2(1-2\varsigma_b)\mathcal{X}_{1/2} - 2\varsigma_b\mathcal{X}_T\Big]. \tag{491}$$

At $b = d/2$, and differentiating once (the product rule done once; all samples at $b = d/2$),

$$P_4(d) = \frac{2}{\pi S^7}\Big[(1+C)^2(1+2C)\mathcal{X}_0 - (1-C)^2(1-2C)\mathcal{X}_{1/2} - 2C\mathcal{X}_T\Big], \tag{492}$$

$$\begin{aligned}\mathcal{L}_d = &-\frac{1}{S^8}\Big[(1+C)^2(4+9C+8C^2)\mathcal{X}_0 + (1-C)^2(4-9C+8C^2)\mathcal{X}_{1/2} - 2(1+6C^2)\mathcal{X}_T\Big]\\ &+\frac{2}{\pi S^7}\,\mathfrak{D}_L,\end{aligned} \tag{493}$$

$$\mathfrak{D}_L = (1+C)^2(1+2C)\,\mathcal{X}_0' - (1-C)^2(1-2C)\,\mathcal{X}_{1/2}' - 2C\,\mathcal{X}_T'. \tag{494}$$

Written out in the five Horn values and their first b-derivatives, (493) is (519), derived in Section 20.3. The derivative data enter only through the single frozen-weight combination $\mathfrak{D}_L$, the b-derivative of the bracket of (492) with its coefficients held at $b = d/2$. The first line of (493) is the derivative of the coefficients of (491).

Three-wheel. Two real samples suffice (proofs in Section 20.4.2): the pole node $t = 0$ and the self-mirror node $t_b = \frac{1}{2} - \frac{b}{4}$, where $1 - \sigma - t_b = t_b$ and $\tilde{t} = \frac{1}{2}$. At $t = 0$ only the leading coefficients of the triple poles survive, and at t_b the kernel is sampled on the midpoint lattice, because $a_\ell + n + t_b = n + \frac{\ell+1}{2}$. With

$$\mathfrak{t}_3(b) = \sum_{\ell,n\geq 0} d_\ell T_{\ell n}(b)^3, \qquad \widehat{M}_3(b) = \sum_{\ell\geq 0} d_\ell \sum_{s\in\frac{\ell+1}{2}+\mathbb{Z}} [N_b g_\ell(s;b)]^3, \tag{495}$$

one has $N_b^3\Xi_3(0;b) = -(\pi s_b)^3\mathfrak{t}_3$ and $N_b^3\Xi_3(t_b;b) = (\frac{1+\varsigma_b}{2})^3\widehat{M}_3$. Note that $\widehat{M}_3$ is the bare lattice sum: the factor $D_b(t_b)^3 = (1+\varsigma_b)^3/8$ carried by the sample is written separately. The two samples are $c_{3,0} + \varsigma_b c_{3,1}$ and $c_{3,0} - c_{3,1}$, and (480) gives

$$W_3(d;b) = u(b)\mathfrak{t}_3(b) + v(b)\widehat{M}_3(b), \qquad u(b) = \frac{2\pi^2(1+2\varsigma_b)}{s_b^2}, \qquad v(b) = \frac{\varsigma_b(1+\varsigma_b)^2}{2\pi s_b^3}. \tag{496}$$

$\mathfrak{t}_3$ is a ${}_7F_6(1)$ plus one two-variable Horn value, $\widehat{M}_3$ a ${}_7F_6(1)$ plus two. In the block decomposition of Section 20.1.3, $\widehat{M}_3$ is the sum of a right-part sum (the ${}_7F_6(1)$ and one Horn value) and a middle-part sum (one Horn value); the count of Section 20.7 treats these two as separate functions. In terms of the normalized zero mode $\hat{c}_0 = N_b^3 c_{3,0} = \frac{\varsigma_b(1+\varsigma_b)^2}{8}\widehat{M}_3 - \frac{\pi^3 s_b^3}{1+\varsigma_b}\mathfrak{t}_3$, (496) reads $W_3 = \frac{6\pi^2}{s_b^2}\mathfrak{t}_3 + \frac{4}{\pi s_b^3}\hat{c}_0$. Then, at $b = d/2$,

$$3Q_d = \frac{6\pi^2}{S^2}\mathfrak{t}_3 + \frac{4}{\pi S^3}\hat{c}_0, \tag{497}$$

$$Z_d = \frac{2\pi^2}{3S^2}\mathfrak{t}_3'' - \frac{\pi^2\cos(\pi d/2)}{6S^2}Q_d + \frac{1}{9}\Big[\frac{\pi(1+C^2)}{S^5}\hat{c}_0 - \frac{4C}{S^4}\hat{c}_0' + \frac{4}{\pi S^3}\hat{c}_0''\Big]. \tag{498}$$

Here $\partial_b W_3(d;d/2)=0$ has been used to remove $\mathfrak{t}_3'$. Its proof is Pattern T of Section 16.3, (239), together with (244): on the balanced K_5 subspace there is no linear form invariant under the permutation group S_5 (not the lattice sum S_r of (462) at $r=5$), and the balanced three-wheel path is a balanced variation. ((246) records the vanishing only as a check.) The formula (498), and the Z_d part of (553), depend on this input; the full jet formula $9Z_d=(u\mathfrak{t}_3+v\widehat{M}_3)''$ and (497) do not. At $d=1,2$ the input also follows from the closed forms of Section 20.6. The second derivatives enter only through $\frac{1}{9}(u\mathfrak{t}_3''+v\widehat{M}_3'')$ with $u=\frac{2\pi^2(1+2C)}{S^2}$, $v=\frac{C(1+C)^2}{2\pi S^3}$, and the first derivatives only through $\hat{c}_0'$.

Analyticity and termwise derivatives. The five Horn functions, $\mathfrak{t}_3$ and $\widehat{M}_3$ are analytic on the strip, and their b-derivatives are the termwise derivatives. The bounds of Section 20.1 hold uniformly for b in compact subsets of the strip, with α above the threshold $\max(\frac{d}{2},d-1,\frac{1}{2})$ by a fixed margin; the polygamma weights produced by differentiation grow only logarithmically and cost part of that margin (for the five Horn functions, the proof of (518) in Section 20.3; for the three-wheel, Section 20.4.1).

20.3 Proofs of the four-wheel samples and the explicit formula for $\mathcal{L}_d$

The code `LRI_C5_general_d/L/L_general.py` evaluates the formulas of this subsection. It proves the sample formulas stated after (489) for every $r\ge 2$, writes (491) and (493) out in the five Horn values and their first b-derivatives, and collects the checks of that form (Section 20.3.1).

Setting and status. Throughout, $r\ge 2$, $0<d<4$, $0<\operatorname{Re} b<b_{\max}(r,d)$ of (463) and $\sigma\notin\mathbb{Z}$, the hypotheses of Section 20.1.1. As in Section 20.2,

$$x=\frac{b}{4},\qquad y=\frac{d}{2}-\frac{b}{4},\qquad \sigma=2x,\qquad g=y-x,\qquad x+y=2q,\qquad \tilde{t}=t+x, \tag{499}$$

and in this subsection x,y always have this meaning. For $d\neq 2$ the multiplicities (240) have the very-well-poised form

$$d_\ell=\frac{(4q-2)_\ell\,(2q)_\ell}{(2q-1)_\ell\,\ell!}, \tag{500}$$

since $\Gamma(d-1)=(d-2)\Gamma(d-2)$, $(2\ell+d-2)/(d-2)=(2q)_\ell/(2q-1)_\ell$ and $\Gamma(\ell+d-2)/(\Gamma(d-2)\,\ell!)=(4q-2)_\ell/\ell!$; here $(2q-1)_\ell\neq 0$ because $2q-1\in(-1,1)$ and $2q\neq 1$. The form (301) has no $0/0$ at $d=2$, where $d_\ell=2-\delta_{\ell 0}$. Status: the lattice form (504)–(505), the samples (510)–(512) and the derivative series (518) are proved here, from the estimates of Section 20.1 and the finite Fourier theorem of Section 20.1.1. The formulas (514), (515), (517) and (519) are derived: exact algebra on these statements, on the top-mode theorem (473) and on the residue functional (480), checked symbolically. All numbers in Section 20.3.1 are numerical. A separate review with its own codes (`LRI_C5_general_d/review/L-rigor/`) rederived every step of the package's own derivation by hand and symbolically and found no wrong formula (for the steps added in this note, see Section 20.10); its corrections (the continuity step at $t=0$ in the proof of (510), the restriction of the three-dimensional moment statement to $\mathfrak{s}_0,\mathfrak{s}_{1/2}$, and the list of digamma arguments) are incorporated below.

Lattice sums in the variable $\tilde{t}$. Let

$$\varphi(z)=\frac{\Gamma(z+x)}{\Gamma(z+y)},\qquad \bar{\varphi}(z)=\frac{\Gamma(z+1-y)}{\Gamma(z+1-x)}, \tag{501}$$

$$\begin{aligned}\mathcal{R}_+(u)&=\sum_{\ell,n\ge 0} d_\ell\big[\varphi(\ell+n+u)\,\bar{\varphi}(n+u)\big]^r,\\ \mathcal{R}_{\mathrm{m}}(\tilde{t})=\sum_{j,k\ge 0} d_{j+k+1}\big[\varphi(k+\tilde{t})\,\varphi(j+1-\tilde{t})\big]^r,&\qquad \mathcal{R}_{\mathrm{s}}(\tilde{t})=\sum_{j,k\ge 0} d_{j+k}\big[\varphi(k+\tilde{t})\,\varphi(j-\tilde{t})\big]^r,\end{aligned} \tag{502}$$

$$P_+(\tilde t)=\big[\sin\pi(\tilde t+x)\sin\pi(\tilde t-y)\big]^r,\qquad P_-(\tilde t)=\big[\sin\pi(\tilde t-x)\sin\pi(\tilde t+y)\big]^r,$$
$$P_{\rm m}(\tilde t)=\big[\sin\pi(\tilde t-x)\sin\pi(\tilde t+x)\big]^r=D_b(t)^r. \tag{503}$$

For $t=\tilde t-x$ off the pole classes $t\equiv 0$ and $t\equiv-\sigma$,

$$\Xi_r(t;b)=P_+(\tilde t)\,\mathcal R_+(\tilde t)+P_-(\tilde t)\,\mathcal R_+(1-\tilde t)+P_{\rm m}(\tilde t)\,\mathcal R_{\rm m}(\tilde t) \tag{504}$$
$$=P_+(\tilde t)\,\mathcal R_+(1+\tilde t)+P_-(\tilde t)\,\mathcal R_+(1-\tilde t)+P_{\rm m}(\tilde t)\,\mathcal R_{\rm s}(\tilde t), \tag{505}$$

with absolutely convergent series. The second form is manifestly even in $\tilde t$: $P_+(-\tilde t)=P_-(\tilde t)$, $P_{\rm m}$ is even, and $\mathcal R_{\rm s}$ is even (exchange j and k).

Proof. Split the lattice of (462) into the right part $n\ge 0$, the middle part $\ell=j+k+1$, $n=-j-1$ ($j,k\ge 0$) and the left part $n\le-\ell-1$, as in Section 20.1. By the estimates there and Section 20.1.1(a) the three parts converge absolutely off the pole classes, so they may be summed separately and rearranged.

Right part. By the first line of (464), $g_\ell(a_\ell{+}n{+}t)=f(\ell{+}\sigma{+}n{+}t)\,f(-n{-}t)$ with $f(z)=\Gamma(z)/\Gamma(z{+}g)$. Since $\sigma+t=\tilde t+x$ and $x+g=y$, $f(\ell+\sigma+n+t)=\varphi(\ell+n+\tilde t)$. Gamma reflection in the form $\Gamma(z-n)\Gamma(1-z+n)=(-1)^n\pi/\sin\pi z$ [26], used for $z=x-\tilde t$ and $z=y-\tilde t$, gives

$$f(-n-t)=\frac{\Gamma(x-\tilde t-n)}{\Gamma(y-\tilde t-n)}=\frac{\sin\pi(y-\tilde t)}{\sin\pi(x-\tilde t)}\,\bar\varphi(n+\tilde t).$$

Since $D_b(t)=\sin\pi(\tilde t-x)\sin\pi(\tilde t+x)$, the prefactor is $D_b(t)\sin\pi(y-\tilde t)/\sin\pi(x-\tilde t)=\sin\pi(\tilde t+x)\sin\pi(\tilde t-y)$, and the right part is $P_+(\tilde t)\mathcal R_+(\tilde t)$.

Left part. By the last line of (464), $n\mapsto n'=-\ell-n-1\ge 0$ maps the left part bijectively onto the right part at the mirror point $t'=1-\sigma-t$, that is at $\tilde t'=1-\tilde t$. Since $D_b(1-\sigma-t)=\sin\pi(\sigma+t)\sin\pi t=D_b(t)$, the left part is $P_+(1-\tilde t)\mathcal R_+(1-\tilde t)$, and $P_+(1-\tilde t)=P_-(\tilde t)$.

Middle part. In the second line of (464), $k{+}\sigma{+}t=k{+}\tilde t{+}x$, $k{+}2q{+}t=k{+}\tilde t{+}y$, $j{+}1{-}t=j{+}1{-}\tilde t{+}x$ and $j+1+g-t=j+1-\tilde t+y$, so the summand is $[\varphi(k+\tilde t)\varphi(j+1-\tilde t)]^r$ with multiplicity d_{j+k+1}, and the prefactor is $D_b(t)^r=P_{\rm m}(\tilde t)$. This proves (504).

Second form. $\mathcal R_+(\tilde t)-\mathcal R_+(1{+}\tilde t)=\sum_\ell d_\ell[\varphi(\ell{+}\tilde t)\,\bar\varphi(\tilde t)]^r$ is the column $n=0$. From $\sin\pi(\tilde t-y)\,\Gamma(1-y+\tilde t)=-\pi/\Gamma(y-\tilde t)$ and $1/\Gamma(1-x+\tilde t)=\sin\pi(x-\tilde t)\,\Gamma(x-\tilde t)/\pi$,

$$\sin\pi(\tilde t+x)\sin\pi(\tilde t-y)\,\bar\varphi(\tilde t)=\sin\pi(\tilde t+x)\sin\pi(\tilde t-x)\,\varphi(-\tilde t), \tag{506}$$

so $P_+(\tilde t)[\mathcal R_+(\tilde t)-\mathcal R_+(1+\tilde t)]=P_{\rm m}(\tilde t)\sum_\ell d_\ell[\varphi(\ell+\tilde t)\varphi(-\tilde t)]^r$. This is the row $j=0$ of $\mathcal R_{\rm s}(\tilde t)$, and the rows $j\ge 1$ of $\mathcal R_{\rm s}(\tilde t)$ are $\mathcal R_{\rm m}(\tilde t)$ after $j\to j+1$. All rearrangements are of absolutely convergent series. □

The three samples. Normalize the sums by their first terms: for $u\in\{x,\frac12,1\}$ and $u\in\{0,\frac12\}$ respectively,

$$\mathfrak h_u^{(r)}=\sum_{\ell,n\ge0}d_\ell\Big[\frac{(u+x)_{\ell+n}}{(u+y)_{\ell+n}}\Big]^r\Big[\frac{(u+1-y)_n}{(u+1-x)_n}\Big]^r,\qquad \mathfrak s_u^{(r)}=\sum_{j,k\ge0}d_{j+k+2u}\Big[\frac{(u+x)_j\,(u+x)_k}{(u+y)_j\,(u+y)_k}\Big]^r, \tag{507}$$

so that $\mathcal R_+(u)=[\varphi(u)\bar\varphi(u)]^r\mathfrak h_u^{(r)}$, $\mathcal R_{\rm s}(0)=\varphi(0)^{2r}\mathfrak s_0^{(r)}$ and $\mathcal R_{\rm m}(\frac12)=\varphi(\frac12)^{2r}\mathfrak s_{1/2}^{(r)}$. For $r=4$ these are the $\mathfrak h_u$, $\mathfrak s_u$ of Section 20.2, and the superscript is dropped; for $r=3$, $\mathfrak h_x^{(3)}$ is $\mathfrak t_3$ of (495). At $u=x$ one has $u+x=\sigma$, $u+y=2q$, $u+1-y=1-g$ and $u+1-x=1$, so $\mathfrak h_x^{(r)}=\sum_{\ell,n}d_\ell T_{\ell n}(b)^r$ with $T_{\ell n}(b)$ of (490). In the notation (320), with $\boldsymbol e=(1,0)$, $\boldsymbol f=(0,1)$, $\boldsymbol c=\boldsymbol e+\boldsymbol f$, the first summation

index ℓ (or j), the second n (or k), and argument $(1,1)$,

$$\begin{aligned}
\mathfrak{h}_u^{(r)} &= \mathfrak{H}\big[\{(u+x,\boldsymbol{c})^{\times r},(4q-2,\boldsymbol{e}),(2q,\boldsymbol{e}),(1,\boldsymbol{f}),(u+1-y,\boldsymbol{f})^{\times r}\};\\
&\qquad \{(u+y,\boldsymbol{c})^{\times r},(2q-1,\boldsymbol{e}),(u+1-x,\boldsymbol{f})^{\times r}\};(1,1)\big],\\
\mathfrak{s}_0^{(r)} &= \mathfrak{H}\big[\{(4q-2,\boldsymbol{c}),(2q,\boldsymbol{c}),(1,\boldsymbol{e}),(1,\boldsymbol{f}),(x,\boldsymbol{e})^{\times r},(x,\boldsymbol{f})^{\times r}\};\\
&\qquad \{(2q-1,\boldsymbol{c}),(1,\boldsymbol{c}),(y,\boldsymbol{e})^{\times r},(y,\boldsymbol{f})^{\times r}\};(1,1)\big],\\
\mathfrak{s}_{1/2}^{(r)} &= 4q\,\mathfrak{H}\big[\{(2q+1,\boldsymbol{c}),(4q-1,\boldsymbol{c}),(1,\boldsymbol{e}),(1,\boldsymbol{f}),(\tfrac12+x,\boldsymbol{e})^{\times r},(\tfrac12+x,\boldsymbol{f})^{\times r}\};\\
&\qquad \{(2q,\boldsymbol{c}),(2,\boldsymbol{c}),(\tfrac12+y,\boldsymbol{e})^{\times r},(\tfrac12+y,\boldsymbol{f})^{\times r}\};(1,1)\big].
\end{aligned} \tag{508}$$

The first two use (500) and hold for $d\neq 2$; the third uses (301) and holds for every d. The upper pairs $(1,\boldsymbol{e})$, $(1,\boldsymbol{f})$ cancel the factorials of (320) where the summand has none. At $b=d/2$ ($x=\frac{q}{2}$, $y=\frac{3q}{2}$) the parameters are

$$\begin{aligned}
\mathfrak{h}_x &= \sum_{\ell,n} d_\ell \Big[\tfrac{(q)_{\ell+n}}{(2q)_{\ell+n}}\Big]^4\Big[\tfrac{(1-q)_n}{n!}\Big]^4 = \sum_{\ell,n} d_\ell T_{\ell n}^4, \quad \mathfrak{h}_1 = \sum_{\ell,n} d_\ell \Big[\tfrac{(1+\frac{q}{2})_{\ell+n}}{(1+\frac{3q}{2})_{\ell+n}}\Big]^4\Big[\tfrac{(2-\frac{3q}{2})_n}{(2-\frac{q}{2})_n}\Big]^4,\\
\mathfrak{h}_{1/2} &= \sum_{\ell,n} d_\ell \Big[\tfrac{(\frac{1+q}{2})_{\ell+n}}{(\frac{1+3q}{2})_{\ell+n}}\Big]^4\Big[\tfrac{(\frac{3-3q}{2})_n}{(\frac{3-q}{2})_n}\Big]^4, \qquad \mathfrak{s}_0 = \sum_{j,k} d_{j+k}\Big[\tfrac{(\frac{q}{2})_j(\frac{q}{2})_k}{(\frac{3q}{2})_j(\frac{3q}{2})_k}\Big]^4,\\
\mathfrak{s}_{1/2} &= \sum_{j,k} d_{j+k+1}\Big[\tfrac{(\frac{1+q}{2})_j(\frac{1+q}{2})_k}{(\frac{1+3q}{2})_j(\frac{1+3q}{2})_k}\Big]^4,
\end{aligned} \tag{509}$$

with $T_{\ell n}$ of (251). Thus $\mathfrak{h}_x$ is the unweighted version of the residue series (255). With $\kappa_0,\kappa_h,\xi_0,\xi_h$ of Section 20.2, that is $\kappa_0=N_b\sin^2(\pi x)\varphi(0)^2$, $\kappa_h=N_b\cos^2(\pi x)\varphi(\frac12)^2$, $\xi_0=x(1-y)/(y(1-x))$ and $\xi_h=(1-2y)/(1-2x)$, the samples are

$$N_b^r\,\Xi_r(0;b) = (-1)^r(\pi s_b)^r\,\mathfrak{h}_x^{(r)}, \tag{510}$$

$$N_b^r\,\Xi_r(-\tfrac{b}{4};b) = (-1)^r\kappa_0^r\big[\mathfrak{s}_0^{(r)}+2\xi_0^r\mathfrak{h}_1^{(r)}\big], \tag{511}$$

$$N_b^r\,\Xi_r(\tfrac12-\tfrac{b}{4};b) = \kappa_h^r\big[\mathfrak{s}_{1/2}^{(r)}+2\xi_h^r\mathfrak{h}_{1/2}^{(r)}\big]. \tag{512}$$

For $r=4$ these are the expressions for $\mathcal{X}_T,\mathcal{X}_0,\mathcal{X}_{1/2}$ given after (489). For $r=3$, (510) is $N_b^3\Xi_3(0;b)=-(\pi s_b)^3\mathfrak{t}_3$, which Section 20.4.2 also proves directly, (521), and comparison of (512) with the midpoint sample (522) gives $(\frac{1+\varsigma_b}{2})^3\widehat{M}_3=\kappa_h^3[\mathfrak{s}_{1/2}^{(3)}+2\xi_h^3\mathfrak{h}_{1/2}^{(3)}]$: the right and left parts, equal at $\tilde t=\frac12$, give $2\kappa_h^3\xi_h^3\mathfrak{h}_{1/2}^{(3)}$ (whose row $\ell=0$ is a ${}_7F_6(1)$), and the middle part gives $\kappa_h^3\mathfrak{s}_{1/2}^{(3)}$.

Proof. (i) The node $t=0$, $\tilde t=x$. It lies in the pole class $t\equiv 0$, so (504) is first extended to it. Assume $g\neq 1$. Written in t, the terms of $\mathcal{R}_+(\tilde t)$ are $[\Gamma(\ell+n+\sigma+t)\Gamma(n+1-g+t)/(\Gamma(\ell+n+2q+t)\Gamma(n+1+t))]^r$, the Gamma ratios of the first line of (464) without their prefactor $-\sin\pi(g-t)/\sin\pi t$, which has moved into P_+. The terms of $\mathcal{R}_+(1-\tilde t)$ are the same ratios at $1-\sigma-t$, $[\Gamma(\ell+n+1-t)\Gamma(n+2-2q-t)/(\Gamma(\ell+n+1+2q-\sigma-t)\Gamma(n+2-\sigma-t))]^r$, and the terms of $\mathcal{R}_{\mathrm m}(\tilde t)$ are the ratios of the second line of (464). All their poles lie in $-\sigma-\mathbb{N}_0$, $g-1-\mathbb{N}_0$, $\mathbb{N}=\{1,2,\dots\}$ and $2-2q+\mathbb{N}_0$. None of these sets contains 0, since $\operatorname{Re}\sigma>0$, $q<1$, $g\neq1$ and $\operatorname{Re}g<2$. Choose a closed disc $|t|\le\delta$ that avoids these sets and the class $-\sigma+\mathbb{Z}$. On it the shifts range over a compact set on which no Gamma argument meets a pole, so the Gamma-ratio bound of Section 20.1 holds uniformly, and the terms, multiplicities included, are bounded by $C(1+\ell)^{d-2}(1+\ell+n)^{-\alpha}(1+n)^{-\alpha}$ (the first two sums) and $C(1+j+k)^{d-2}(1+k)^{-\alpha}(1+j)^{-\alpha}$ (the middle sum), $\alpha=r\operatorname{Re}g$. By the lattice-sum criterion of Section 20.1, which holds since $\operatorname{Re}b<b_{\max}$, the three series converge uniformly on the disc. Hence both sides of (504) are analytic on the disc (Ξ_r is entire by Section 20.1.1(b), the weights are entire); they agree on the punctured disc, so they agree at $t=0$. At $\tilde t=x$ the weights P_- and $P_{\mathrm m}$ vanish, so only the right sum survives, with $P_+(x)=[\sin\pi\sigma\,\sin\pi(x-y)]^r=(-1)^r[\sin\pi\sigma\sin\pi g]^r$. With $\varphi(x)=\Gamma(\sigma)/\Gamma(2q)$, $\bar\varphi(x)=\Gamma(1-g)$ and $\Gamma(g)\Gamma(1-g)\sin\pi g=\pi$,

$$N_b\sin\pi\sigma\sin\pi g\;\varphi(x)\bar\varphi(x)=\pi\sin\pi\sigma=\pi s_b,$$

which together with $\mathcal{R}_+(x) = [\varphi(x)\bar\varphi(x)]^r \mathfrak{h}_x^{(r)}$ gives (510). At the isolated point $g = 1$, that is $b = d-2$ (in the strip only for $d > 2$), $\bar\varphi(x)$ has a pole. There (510) follows by continuity in b, since both sides are analytic in b: the left side by Section 20.1.1(d), the right side because the terms of $\mathfrak{h}_x^{(r)} = \sum d_\ell T_{\ell n}(b)^r$ are polynomials in b and obey the same majorant uniformly near $b = d-2$. The masters use only b near $\frac{d}{2}$, where $g = q < 1$.

(ii) The node $\tilde t = 0$, $t = -\frac{b}{4}$. It is off the pole classes, since $t \equiv 0$ or $t \equiv -\sigma$ would need $\frac{b}{4} \in \mathbb{Z}$. In (505), $P_+(0) = P_-(0) = (-1)^r[\sin\pi x \sin\pi y]^r$ and $P_{\rm m}(0) = (-1)^r \sin^{2r}\pi x$, so $\Xi_r = (-1)^r\{2[\sin\pi x \sin\pi y\, \varphi(1)\bar\varphi(1)]^r \mathfrak{h}_1^{(r)} + [\sin^2(\pi x)\varphi(0)^2]^r \mathfrak{s}_0^{(r)}\}$.

(iii) The node $\tilde t = \frac12$, $t = \frac12 - \frac{b}{4}$. It is off the pole classes, since either class would need $b \equiv 2 \pmod 4$, that is $\sigma \in \mathbb{Z}$. In (504), $P_+(\frac12) = P_-(\frac12) = [\cos\pi x \cos\pi y]^r$ and $P_{\rm m}(\frac12) = \cos^{2r}\pi x$, so $\Xi_r = 2[\cos\pi x \cos\pi y\, \varphi(\frac12)\bar\varphi(\frac12)]^r \mathfrak{h}_{1/2}^{(r)} + [\cos^2(\pi x)\varphi(\frac12)^2]^r \mathfrak{s}_{1/2}^{(r)}$.

Finally, $\varphi(1) = \frac{x}{y}\varphi(0)$ and $\bar\varphi(1) = \frac{1-y}{1-x}\Gamma(1-y)/\Gamma(1-x)$ with $\Gamma(1-y)/\Gamma(1-x) = (\sin\pi x/\sin\pi y)\varphi(0)$ by reflection, and $\bar\varphi(\frac12) = \Gamma(\frac32 - y)/\Gamma(\frac32 - x) = \frac{1-2y}{1-2x}(\cos\pi x/\cos\pi y)\varphi(\frac12)$ by $\Gamma(\frac12 + z)\Gamma(\frac12 - z) = \pi/\cos\pi z$. Hence

$$\sin\pi x \sin\pi y\ \varphi(1)\bar\varphi(1) = \xi_0 \sin^2(\pi x)\,\varphi(0)^2, \qquad \cos\pi x \cos\pi y\ \varphi(\tfrac12)\bar\varphi(\tfrac12) = \xi_h \cos^2(\pi x)\,\varphi(\tfrac12)^2, \tag{513}$$

and multiplying (ii) and (iii) by N_b^r gives (511) and (512). □

The three-node formula in Horn values. By (473) the three samples are, in units of N_b^4, $c_{4,0} + c_{4,1} + c_{4,2}$ at $\tilde t = 0$, $c_{4,0} - c_{4,1} + c_{4,2}$ at $\tilde t = \frac12$ and $c_{4,0} + \varsigma_b c_{4,1} + \cos(\pi b)\, c_{4,2}$ at $t = 0$. The determinant is $4\sin^2\pi\sigma = 4s_b^2 \neq 0$, and

$$N_b^4 c_{4,1} = \frac{\mathcal{X}_0 - \mathcal{X}_{1/2}}{2}, \qquad N_b^4 c_{4,0} = \frac{\mathcal{X}_0 + \mathcal{X}_{1/2}}{2} - N_b^4 c_{4,2},$$
$$N_b^4 c_{4,2} = \frac{1}{2s_b^2}\Big[\frac{\mathcal{X}_0 + \mathcal{X}_{1/2}}{2} + \varsigma_b N_b^4 c_{4,1} - \mathcal{X}_T\Big].$$

Inserting these into (480) gives (491), and with (510)–(512) at $r = 4$, for every b in the strip,

$$\boxed{\begin{aligned} W_4(d;b) = &-\frac{4\pi^3\varsigma_b}{s_b^3}\,\mathfrak{h}_x + \frac{2(1+\varsigma_b)^2(1+2\varsigma_b)\kappa_0^4}{\pi s_b^7}\big[\mathfrak{s}_0 + 2\xi_0^4\mathfrak{h}_1\big] \\ &- \frac{2(1-\varsigma_b)^2(1-2\varsigma_b)\kappa_h^4}{\pi s_b^7}\big[\mathfrak{s}_{1/2} + 2\xi_h^4\mathfrak{h}_{1/2}\big]. \end{aligned}} \tag{514}$$

For $r = 3$ and $0 < \operatorname{Re} b < b_{\max}(3,d)$ the solve of the two samples at $t = 0$ and $\tilde t = \frac12$ is the proof of (496) in Section 20.4.2; it gives $W_3(d;b) = -\frac{2N_b^3}{\pi s_b^5}[(1+2\varsigma_b)\Xi_3(0;b) - 2\varsigma_b(1-\varsigma_b)\Xi_3(\frac12 - \frac{b}{4};b)]$, and with (510) and (512) at $r = 3$ this is

$$W_3(d;b) = \frac{2\pi^2(1+2\varsigma_b)}{s_b^2}\,\mathfrak{h}_x^{(3)} + \frac{4\varsigma_b(1-\varsigma_b)\kappa_h^3}{\pi s_b^5}\big[\mathfrak{s}_{1/2}^{(3)} + 2\xi_h^3\mathfrak{h}_{1/2}^{(3)}\big], \tag{515}$$

which is (496) with its midpoint sample written in Horn values. Neither formula holds without the top-mode theorem: a mode $c_{4,3}$ would enter W_4 with the nonzero weight $B_{4,3} = -20\pi^3/s_b^3$ of Section 20.1.3 and would not be fixed by three samples. The agreement of (514) with independent values at $b \neq d/2$ (Section 20.3.1) is therefore also a numerical test of (473). The exact solve and the insertion into (480) are checked symbolically by `LRI_C5_general_d/L/sym_functional.py`, which also rederives (480) from (318) with q replaced by σ.

The four-wheel period. At $b=d/2$ one has $x=\frac{q}{2}$, $y=\frac{3q}{2}$, $\sigma=g=q$, $N_{d/2}=\Gamma(2q)$, $s_b=S$, $\varsigma_b=C$, so $\kappa_0=\Gamma(2q)\sin^2(\frac{\pi q}{2})\,[\Gamma(\frac{q}{2})/\Gamma(\frac{3q}{2})]^2$, $\kappa_h=\Gamma(2q)\cos^2(\frac{\pi q}{2})\,[\Gamma(\frac{1+q}{2})/\Gamma(\frac{1+3q}{2})]^2$, $\xi_0=\frac{2-3q}{3(2-q)}$ and $\xi_h=\frac{1-3q}{1-q}$. Put, for b near $d/2$,

$$\begin{aligned} \mathsf{k}_0(b)&=\frac{2(1+\varsigma_b)^2\kappa_0^4}{\pi s_b^7}, \qquad \mathsf{k}_h(b)=\frac{2(1-\varsigma_b)^2\kappa_h^4}{\pi s_b^7},\\ \mathfrak{S}_0&=\mathfrak{s}_0+2\xi_0^4\mathfrak{h}_1=\frac{\mathcal{X}_0}{\kappa_0^4}, \qquad \mathfrak{S}_h=\mathfrak{s}_{1/2}+2\xi_h^4\mathfrak{h}_{1/2}=\frac{\mathcal{X}_{1/2}}{\kappa_h^4}, \end{aligned} \tag{516}$$

so that (514) reads $W_4=-\frac{4\pi^3\varsigma_b}{s_b^3}\mathfrak{h}_x+(1+2\varsigma_b)\mathsf{k}_0(b)\mathfrak{S}_0+(2\varsigma_b-1)\mathsf{k}_h(b)\mathfrak{S}_h$. Write $\mathsf{k}_0,\mathsf{k}_h$ for the values at $b=d/2$; with $1+C=2\cos^2\frac{\pi q}{2}$, $1-C=2\sin^2\frac{\pi q}{2}$ and $S=2\sin\frac{\pi q}{2}\cos\frac{\pi q}{2}$,

$$\mathsf{k}_0=\frac{\Gamma(2q)^4}{16\pi}\Big[\frac{\Gamma(\frac{q}{2})}{\Gamma(\frac{3q}{2})}\Big]^8\frac{\tan\frac{\pi q}{2}}{\cos^2\frac{\pi q}{2}}, \qquad \mathsf{k}_h=\frac{\Gamma(2q)^4}{16\pi}\Big[\frac{\Gamma(\frac{1+q}{2})}{\Gamma(\frac{1+3q}{2})}\Big]^8\frac{\cot\frac{\pi q}{2}}{\sin^2\frac{\pi q}{2}}.$$

Then, for $\frac12<d<4$,

$$\boxed{P_4(d)=-\frac{4\pi^3C}{S^3}\,\mathfrak{h}_x+(1+2C)\,\mathsf{k}_0\,\mathfrak{S}_0+(2C-1)\,\mathsf{k}_h\,\mathfrak{S}_h,} \tag{517}$$

which is (492) with the samples written in Horn values. The symbols $\mathsf{k}_0,\mathsf{k}_h,\mathfrak{S}_0,\mathfrak{S}_h$ and $\mathsf{v}_0,\mathsf{v}_h$ below are local to this subsection.

Derivative series. A prime denotes the total derivative d/db at $b=d/2$; in $\mathfrak{h}_x$ the argument $u=x$ moves with b, while $u=\frac12,1$ in $\mathfrak{h}_u$ and $u=0,\frac12$ in $\mathfrak{s}_u$ are fixed. Let $H(a;m)=\psi_0(a+m)-\psi_0(a)$ as in (274), and write $\mathfrak{h}_u[\ell,n]$ and $\mathfrak{s}_u[j,k]$ for the summands of (509). For $\frac12<d<4$ the series

$$\begin{aligned} \mathfrak{h}_x'&=2\sum_{\ell,n\ge0}d_\ell T_{\ell n}^4\big[H(q;\ell+n)+H(1-q;n)\big]=4\sum_{\ell,n\ge0}d_\ell T_{\ell n}^4\mu_1,\\ \mathfrak{h}_u'&=\sum_{\ell,n\ge0}\mathfrak{h}_u[\ell,n]\big[H(u+\tfrac{q}{2};\ell+n)+H(u+\tfrac{3q}{2};\ell+n)\\ &\qquad\qquad+H(u+1-\tfrac{3q}{2};n)+H(u+1-\tfrac{q}{2};n)\big],\\ \mathfrak{s}_u'&=2\sum_{j,k\ge0}\mathfrak{s}_u[j,k]\big[H(u+\tfrac{q}{2};j)+H(u+\tfrac{3q}{2};j)\big], \end{aligned} \tag{518}$$

with $u=\frac12,1$ in the second line, $u=0,\frac12$ in the third and μ_1 of (253), converge absolutely and are the b-derivatives at $b=d/2$. For general r the prefactors $2,1,2$ become $\frac{r}{2},\frac{r}{4},\frac{r}{2}$.

Proof. $x'=\frac14$, $y'=-\frac14$ and $\partial_a\log(a)_m=H(a;m)$. In $\mathfrak{h}_u$ with u fixed, the parameters $u+x$ and $u+1-y$ stand in numerators and move with slope $\frac14$, while $u+y$ and $u+1-x$ stand in denominators and move with slope $-\frac14$, so all four contributions carry the sign $+$, each with weight $\frac{r}{4}$. In $\mathfrak{h}_x$ the numerator parameters $u+x=\sigma$ and $u+1-y=1-g$ move with slope $\frac12$, and $u+y=2q$, $u+1-x=1$ are fixed; this gives weight $\frac{r}{2}$, and at $b=d/2$ the bracket is $2\mu_1$. In $\mathfrak{s}_u$ the contributions are $\frac{r}{4}[H(u+x;j)+H(u+y;j)+H(u+x;k)+H(u+y;k)]$, and the summand is symmetric in $j\leftrightarrow k$, so the k-terms sum to the j-terms. For convergence, let B be a closed disc around $d/2$ inside the strip $0<\operatorname{Re}b<b_{\max}(4,d)$ on which $\sigma\notin\mathbb{Z}$; it exists because $d/2$ lies in the strip for $\frac12<d<4$. On B the summands of the five functions are analytic in b (no denominator Pochhammer symbol vanishes, since $\operatorname{Re}y>0$ and $\operatorname{Re}(1-x)>0$ there), and by the Gamma-ratio bound, whose constants are uniform for parameters in compact sets, they are bounded by $C(1+\ell)^{d-2}(1+\ell+n)^{-\alpha_B}(1+n)^{-\alpha_B}$ for $\mathfrak{h}_u$ and $C(1+j+k)^{d-2}(1+j)^{-\alpha_B}(1+k)^{-\alpha_B}$ for $\mathfrak{s}_u$, with $\alpha_B=\min_B 4\operatorname{Re}g>\max(\frac{d}{2},d-1,\frac12)$. By the lattice-sum criterion of Section 20.1 the five series converge uniformly on B. By Weierstrass' theorem their sums are analytic inside B and

may be differentiated termwise, and Cauchy's estimate on a circle around $d/2$ inside B bounds the differentiated terms by a constant times the same majorant, so the derivative series converge absolutely. The weights $H(a;m)$ grow like $\log(1+m)$, which is the logarithmic loss mentioned in Section 20.2. $\square$

The rim jet in Horn values. For $\frac12 < d < 4$,

$$\boxed{\begin{aligned}\mathcal{L}_d = {}& \frac{2\pi^4(1+2C^2)}{S^4}\,\mathfrak{h}_x - \frac{4\pi^3 C}{S^3}\,\mathfrak{h}'_x \\ & + \mathsf{k}_0\Big\{\big[(1+2C)\mathsf{v}_0 - \pi S\big]\mathfrak{S}_0 + (1+2C)\big[\mathfrak{s}'_0 + 2\xi_0^4\mathfrak{h}'_1 + 2(\xi_0^4)'\mathfrak{h}_1\big]\Big\} \\ & + \mathsf{k}_h\Big\{\big[(2C-1)\mathsf{v}_h - \pi S\big]\mathfrak{S}_h + (2C-1)\big[\mathfrak{s}'_{1/2} + 2\xi_h^4\mathfrak{h}'_{1/2} + 2(\xi_h^4)'\mathfrak{h}_{1/2}\big]\Big\},\end{aligned}} \tag{519}$$

with the five derivatives (518),

$$\begin{aligned}\mathsf{v}_0 &= \partial_b \log \mathsf{k}_0(b)\big|_{b=d/2} = -4\psi_0(q) + 2\psi_0(\tfrac{q}{2}) + 2\psi_0(\tfrac{3q}{2}) + \pi\cot\tfrac{\pi q}{2} - \tfrac{3\pi}{2}\cot\pi q,\\ \mathsf{v}_h &= \partial_b \log \mathsf{k}_h(b)\big|_{b=d/2} = -4\psi_0(q) + 2\psi_0(\tfrac{1+q}{2}) + 2\psi_0(\tfrac{1+3q}{2}) - \pi\tan\tfrac{\pi q}{2} - \tfrac{3\pi}{2}\cot\pi q,\\ (\xi_0^4)' &= 2\xi_0^4\Big[\frac{4}{3q} + \frac{1}{2-q}\Big] + \frac{2\xi_0^3}{3(2-q)}, \qquad (\xi_h^4)' = \frac{4(1-2q)\,\xi_h^3}{(1-q)^2}.\end{aligned}$$

Every coefficient is finite for $\frac12 < d < 4$. Written with the samples, (519) is (493): both are the b-derivative at $d/2$ of (491) $=$ (514), and $\mathcal{X}'_0 = (\kappa_0^4)'\mathfrak{S}_0 + \kappa_0^4\mathfrak{S}'_0$, $\mathcal{X}'_{1/2} = (\kappa_h^4)'\mathfrak{S}_h + \kappa_h^4\mathfrak{S}'_h$, $\mathcal{X}'_T = (\pi^4 s_b^4)'\mathfrak{h}_x + \pi^4 S^4\mathfrak{h}'_x$.

Proof. Near $b = d/2$ one has $0 < \sigma < 1$, so (514) holds on a neighborhood of $d/2$. Every factor is analytic there, the five Horn functions by the proof of (518), so the product rule applies to $W_4 = -\frac{4\pi^3\varsigma_b}{s_b^3}\mathfrak{h}_x + (1+2\varsigma_b)\mathsf{k}_0(b)\mathfrak{S}_0 + (2\varsigma_b-1)\mathsf{k}_h(b)\mathfrak{S}_h$. With $\partial_b\varsigma_b = -\frac{\pi}{2}s_b$ and $\partial_b s_b = \frac{\pi}{2}\varsigma_b$ one finds $\partial_b(-4\pi^3\varsigma_b/s_b^3) = 2\pi^4(1+2\varsigma_b^2)/s_b^4$ and $\partial_b(1+2\varsigma_b) = \partial_b(2\varsigma_b-1) = -\pi s_b$, which give the first line of (519) and the terms $-\pi S\,\mathsf{k}_0\mathfrak{S}_0$, $-\pi S\,\mathsf{k}_h\mathfrak{S}_h$. Next, with $\partial_b \log N_b = -\frac12[\psi_0(g) + \psi_0(\sigma)]$, $\partial_b \log\varphi(0)^2 = \frac12[\psi_0(x) + \psi_0(y)]$, $\partial_b \log\varphi(\frac12)^2 = \frac12[\psi_0(\frac12+x) + \psi_0(\frac12+y)]$, $\partial_b \log\sin^2\pi x = \frac{\pi}{2}\cot\pi x$ and $\partial_b\log\cos^2\pi x = -\frac{\pi}{2}\tan\pi x$,

$$\begin{aligned}\partial_b\log\mathsf{k}_0(b) &= -\frac{\pi s_b}{1+\varsigma_b} - \frac{7\pi\varsigma_b}{2s_b} - 2\psi_0(g) - 2\psi_0(\sigma) + 2\psi_0(x) + 2\psi_0(y) + 2\pi\cot\pi x,\\ \partial_b\log\mathsf{k}_h(b) &= \frac{\pi s_b}{1-\varsigma_b} - \frac{7\pi\varsigma_b}{2s_b} - 2\psi_0(g) - 2\psi_0(\sigma) + 2\psi_0(\tfrac12+x) + 2\psi_0(\tfrac12+y) - 2\pi\tan\pi x.\end{aligned}$$

At $b = d/2$, $\pi S/(1+C) = \pi\tan\frac{\pi q}{2}$, $\pi S/(1-C) = \pi\cot\frac{\pi q}{2}$ and $7\pi C/(2S) = \frac{7\pi}{2}\cot\pi q$; with $\pi\cot\frac{\pi q}{2} - \pi\tan\frac{\pi q}{2} = 2\pi\cot\pi q$ this gives the printed v_0 and v_h, and the terms $(1+2C)\mathsf{v}_0\mathsf{k}_0\mathfrak{S}_0$ and $(2C-1)\mathsf{v}_h\mathsf{k}_h\mathfrak{S}_h$. The remaining terms are $(1+2C)\mathsf{k}_0\mathfrak{S}'_0$ and $(2C-1)\mathsf{k}_h\mathfrak{S}'_h$, with $\mathfrak{S}'_0 = \mathfrak{s}'_0 + 2\xi_0^4\mathfrak{h}'_1 + 2(\xi_0^4)'\mathfrak{h}_1$ and likewise for $\mathfrak{S}'_h$. Finally $(\xi_0^4)' = 4\xi_0^4\partial_b\log\xi_0$ with $\partial_b\log\xi_0 = \frac14[\frac{1}{x} + \frac{1}{y} + \frac{1}{1-x} + \frac{1}{1-y}] = \frac12[\frac{4}{3q} + \frac{1}{2-3q} + \frac{1}{2-q}]$ at $b = d/2$, and $(\xi_h^4)' = 4\xi_h^4\partial_b\log\xi_h$ with $\partial_b\log\xi_h = \frac12[\frac{1}{1-2x} + \frac{1}{1-2y}] = \frac12[\frac{1}{1-q} + \frac{1}{1-3q}]$; the poles at $q = \frac23$ and $q = \frac13$ cancel against $\xi_0 \propto 2-3q$ and $\xi_h \propto 1-3q$, which gives the printed forms. The coefficients are finite for $0 < q < 1$: $S \neq 0$, the trigonometric functions of $\frac{\pi q}{2}$ and $\cot\pi q$ are finite, the digamma arguments are positive, and $2-q, 1-q \neq 0$. $\square$

The closed coefficients were compared with direct symbolic differentiation of (514) at eleven values of d, including $\frac43$, 2 and $\frac83$, to 10^{-40} (`LRI_C5_general_d/L/check_closed_coeffs.py`), and the review of `LRI_C5_general_d/review/L-rigor/` repeated this with its own coefficient functions (Section 20.9).

Moving and fixed nodes. The nodes $\tilde t = 0, \frac12$ are $t = -\frac{b}{4}$ and $t = \frac12 - \frac{b}{4}$, which move with b. Since $\Xi_4(t;b) = \sum_k c_{4,k} \cos 2\pi k\tilde t$ is even about both points, $\partial_t \Xi_4 = 0$ there, and the total b-derivative of these samples equals their b-derivative at fixed t. In the fixed-t form the right half-lattice is differentiated only in its numerator parameters ($\sigma + t$ and $1 - g + t$ move with slope $\frac12$, while $2q + t$ and $1 + t$ are fixed) and the left half-lattice only in its denominator parameters; their average is the symmetric derivative of (518), in which every parameter moves with slope $\pm\frac14$. The node $t = 0$ is fixed by construction. No t-derivative of any sample enters (519).

Relation to the residue series. The series (256) is a residue at $t = 0$: it needs the Laurent data of the right half-lattice at $t = 0$ up to third order, through the polynomials F_4 and $F_{4,1}$ of (254) (with λ_3 and η_3). Formula (519) keeps only the zeroth-order data at $t = 0$, namely $\mathfrak{h}_x = \sum d_\ell T_{\ell n}^4$ and $\mathfrak{h}'_x = 4\sum d_\ell T_{\ell n}^4 \mu_1$, and replaces the higher Laurent coefficients by the values at the two reflection centers. Section 20.5 is the intermediate form with the pole node alone, one Laurent order below (256).

Special dimensions. At $b = d/2$ one has $y = 3x$, and the triple-angle formulas give $\sin \pi y = (1{+}2C)\sin \pi x$ and $\cos \pi y = (2C{-}1)\cos \pi x$, so $P_\pm(0) = (1{+}2C)^r P_{\rm m}(0)$ and $P_\pm(\frac12) = (2C{-}1)^r P_{\rm m}(\frac12)$. In (517) and (519) the sample at $\tilde t = 0$ carries the factor $1 + 2C$, and $\mathfrak{h}_1$ carries in addition ξ_0^4 (or ξ_0^3 in $(\xi_0^4)'$), with $\xi_0 \propto 2 - 3q$; the sample at $\tilde t = \frac12$ carries $2C - 1$, and $\mathfrak{h}_{1/2}$ carries in addition ξ_h^4 or ξ_h^3, with $\xi_h \propto 1 - 3q$. Hence at $d = \frac83$ ($q = \frac23$, $C = -\frac12$, $y = 1$) $\mathfrak{h}_1$ drops out of both P_4 and $\mathcal{L}_d$: its coefficient in $W_4(d;b)$ has a zero of order five at $b = d/2$, a simple zero of $1 + 2\varsigma_b$ times the fourth power of the simple zero of ξ_0. There $\mathfrak{s}_0$ enters $\mathcal{L}_d$ only through its value, with coefficient $-\pi S\mathsf{k}_0$. At $d = \frac43$ ($q = \frac13$, $C = \frac12$, $y = \frac12$) the same holds for $\mathfrak{h}_{1/2}$ and $\mathfrak{s}_{1/2}$. At $d = 2$ ($C = 0$) the coefficient of $\mathfrak{h}'_x$ vanishes, and $\mathfrak{h}_x$ drops out of $P_4(2)$. Thus $P_4(\frac43)$ and $P_4(\frac83)$ need three Horn values, $\mathcal{L}_{4/3}$ and $\mathcal{L}_{8/3}$ four values and three derivatives, and $\mathcal{L}_2$ five values and four derivatives. In the code the vanishing coefficients come out zero to working precision.

The five functions at $d = 1, 2, 3$. At $d = 1$, $d_\ell = \delta_{\ell 0} + \delta_{\ell 1}$, so each $\mathfrak{h}_u$ is a single sum, a ${}_9F_8(1)$ (an ${}_8F_7(1)$ for $u = x$) plus the analogous single sum of the row $\ell = 1$; $\mathfrak{s}_0 = 1 + 2(x/y)^4$, which is $\frac{83}{81}$ at $b = \frac12$ with $\mathfrak{s}'_0 = \frac{64}{243}$, and $\mathfrak{s}_{1/2} = 1$. The samples at $\tilde t = 0$ and $\tilde t = \frac12$ coincide, since $c_{4,1} \equiv 0$ there (Section 20.6). At $d = 2$, $x + y = 1$, so $\bar\varphi = \varphi$, and $d_\ell = 2 - \delta_{\ell 0}$. Put $\chi_m = \varphi(m + u)^r$ and $\mathsf{m}(u) = \sum_{n\ge0}[(u + x)_n/(u + y)_n]^r$, so that $\sum_{m\ge0} \chi_m = \varphi(u)^r \mathsf{m}(u)$. In $\mathcal{R}_+(u)$ set $m = \ell + n$; the weight d_{m-n} is 2 for $m > n$ and 1 for $m = n$, so $\mathcal{R}_+(u) = \sum_{m\ge n\ge0} d_{m-n}\chi_m\chi_n = (\sum_m \chi_m)^2$, the symmetrization of the proof of (539). Likewise $d_{j+k+1} = 2$ and $d_{j+k} = 2 - \delta_{j0}\delta_{k0}$ give $\mathcal{R}_{\rm m}(\frac12) = 2\varphi(\frac12)^{2r}\mathsf{m}(\frac12)^2$ and $\mathcal{R}_{\rm s}(0) = 2[\varphi(0)^r\mathsf{m}(0)]^2 - \varphi(0)^{2r}$, and $\mathsf{m}(0) = 1 + (x/y)^r\mathsf{m}(1)$. Hence

$$\mathfrak{h}_u^{(r)} = \mathsf{m}(u)^2, \qquad \mathfrak{s}_{1/2}^{(r)} = 2\,\mathfrak{h}_{1/2}^{(r)}, \qquad \mathfrak{s}_0^{(r)} = 2\big[1 + (x/y)^r\mathsf{m}(1)\big]^2 - 1 \qquad (d = 2),$$

with $\mathsf{m}(x) = {}_rF_{r-1}(\sigma,\dots,\sigma;1,\dots,1;1)$; for real b, $\mathsf{m}(u) > 0$ is the positive square root of $\mathfrak{h}_u^{(r)}$. The $d = 2$ rows of the test data in Section 20.3.1 satisfy the last two relations to the printed precision. At $d = 3$, $d_\ell = 2\ell + 1$ is linear, so $\mathfrak{s}_0$ and $\mathfrak{s}_{1/2}$ are products of one-variable moments: with $\mathsf{n}_i(u) = \sum_{j\ge0} j^i[(u + x)_j/(u + y)_j]^4$, $\mathfrak{s}_0 = 4\mathsf{n}_0(0)\mathsf{n}_1(0) + \mathsf{n}_0(0)^2$ and $\mathfrak{s}_{1/2} = 4\mathsf{n}_0(\frac12)\mathsf{n}_1(\frac12) + 3\mathsf{n}_0(\frac12)^2$. This does not hold for $\mathfrak{h}_x, \mathfrak{h}_{1/2}, \mathfrak{h}_1$: with $m = \ell + n$ they are nested sums over $m \ge n \ge 0$ with weight $2(m - n) + 1$, genuine two-variable values (rank two). The reduction of $\mathcal{L}_3$ in Section 20.6 works instead with the half-shift structure of the full samples. Its Dougall closed form of the amplitude $\mathfrak{b}_4$ rests on the linear multiplicity $2\ell + 1$, which turns the half-shifted lattice sum into a very-well-poised bilateral series. For non-integer d the multiplicity (500) is not polynomial; no Gamma-product amplitude was found there, and none was searched for systematically. The closed forms of $\mathcal{L}_1$, $\mathcal{L}_2$ and the $d = 3$ reduction are obtained in Section 20.6 from the mode form, not from (519); (519) reproduces all three values numerically (Section 20.3.1).

Function count. At generic $\frac{1}{2} < d < 4$, (517) needs the five two-variable Horn values $\mathfrak{h}_x$, $\mathfrak{h}_{1/2}$, $\mathfrak{h}_1$, $\mathfrak{s}_0$, $\mathfrak{s}_{1/2}$ at argument $(1,1)$, and (519) the same five values and their five first parameter derivatives (518). The coefficients need trigonometric functions of πq and $\frac{\pi q}{2}$; $\Gamma(2q)$, $\Gamma(\frac{q}{2})$, $\Gamma(\frac{3q}{2})$, $\Gamma(\frac{1+q}{2})$, $\Gamma(\frac{1+3q}{2})$; ψ_0 at q, $\frac{q}{2}$, $\frac{3q}{2}$, $\frac{1+q}{2}$, $\frac{1+3q}{2}$ (by the duplication formula, equivalently $\psi_0(q), \psi_0(3q), \psi_0(\frac{q}{2}), \psi_0(\frac{3q}{2})$ and $\log 2$); and rational functions of q. In the unweighted Horn class (320) the derivatives cost 19 three-variable values: each weight $H(a;m) = m\sum_{k\ge 0}[(a+k)(a+m+k)]^{-1}$ of (291) adds one summation index, encoded by (292), and the factor m costs one array for $m = n$ or $m = j$ and two for $m = \ell + n$ (294). This gives 3 arrays for $\mathfrak{h}'_x$, 6 each for $\mathfrak{h}'_{1/2}$ and $\mathfrak{h}'_1$, and 2 each for $\mathfrak{s}'_0$ and $\mathfrak{s}'_{1/2}$. So $\mathcal{L}_d$ needs 5 two-variable and 19 three-variable unweighted Horn values, in place of the 244 values with up to six variables of (298). In the split form used by the code (next paragraph) the five functions are four one-variable values, an ${}_8F_7(1)$ in $\mathfrak{h}_x$, two ${}_9F_8(1)$ in $\mathfrak{h}_{1/2}, \mathfrak{h}_1$ and a ${}_7F_6(1)$ in $\mathfrak{s}_0$, and five two-variable values, each with its first derivative. For $P_4(d)$, the sampling (323) at the nodes (312) uses 20 complex block values (A_4^0, A_4^+ at eight complex points and C_4 at four; 15 with (473)); (517) uses five real Horn values. The five derivative values are the new constants of the generic-dimensional problem. Whether they are independent is not decided here; at $d = 1, 2$ they reduce, and at $d = 3$ they reduce, with the identities of Section 20.6, to the two constants $\mathfrak{d}_H, \mathfrak{d}_a$ (Section 20.6). Contiguous relations do not connect the five functions, since their parameters differ by non-integer shifts (x and $\frac{1}{2}$; Section 20.8).

Evaluation. `LRI_C5_general_d/L/L_general.py` evaluates (514), and (515), for any complex b in the strip, in a split form. Let $\hat{d}_m = d_{m+1}/(4q) = (2q+1)_m(4q-1)_m/((2q)_m(2)_m)$ be the ratio of (301). The row $\ell = 0$ of $\mathfrak{h}_u$ is a one-variable sum, and the rows $\ell = j+1 \ge 1$ are $4q[(u+x)/(u+y)]^r \sum_{j,n\ge 0} \hat{d}_j \mathsf{x}_{j+n}\mathsf{z}_n$ with $\mathsf{x}_m = [(u+1+x)_m/(u+1+y)_m]^r$ and $\mathsf{z}_n = [(u+1-y)_n/(u+1-x)_n]^r$, sequences of the kind x, z of (486). Further $\mathfrak{s}_{1/2} = 4q\sum_{j,k}\hat{d}_{j+k}a_j a_k$ with $a_j = [(\frac{1}{2}+x)_j/(\frac{1}{2}+y)_j]^r$, and $\mathfrak{s}_0 = 1 + 4q(x/y)^r[\sum_k \hat{d}_k\tilde{a}_k + \sum_{j,k}\hat{d}_{j+k}\tilde{a}_j a_k]$ with $a_j = [(x)_j/(y)_j]^r$ and $\tilde{a}_k = [(1+x)_k/(1+y)_k]^r$. Each sum is evaluated as an exact finite part plus an asymptotic tail by the method of the paragraph *Evaluation of the Horn values* of Section 20.1.4, with the routines of `LRI_C5_general_d/foundation/hypsums.py` and the default truncation parameters and guard digits stated there; at $d = 1, 2, 3$ the multiplicities are used as exact polynomials. The b-derivatives are exact: `LRI_C5_general_d/L/jet.py` implements first-order jets (a value and its derivative, combined by the product rule), and `LRI_C5_general_d/L/hypsums_jet.py` is the same algorithm with every operation jet-valued, including the Hurwitz zeta derivative in its first argument and the Bernoulli-polynomial tail coefficients. Evaluating at the jet based at $b = \frac{d}{2}$ returns the value and the derivative of the evaluated expression, that is the series (518) with their tails (now with $\log m$ terms) accelerated in the same way. No finite difference and no Cauchy circle in b enters $\mathcal{L}_d$. One pair $(P_4(d), \mathcal{L}_d)$ at 50 digits takes about 15 s on one AMD core of the symmetry cluster (5 s at 30 digits on a laptop); the earlier route of `LRI_C5_general_d/foundation/`, a 96-point Cauchy circle in b of the finite-sample W_4, took about 45 s per point at 50 digits. The function `L_explicit` evaluates (519) as printed, with the closed-form coefficients of `coeffs_closed` times the five values and the five derivative series; it agrees with the jet derivative of the whole of (514) to working precision, and `LRI_C5_general_d/L/sym_L.py` does the same with coefficients obtained by symbolic differentiation. Usage: `python3 L_general.py d [dps] [b]`, with d given exactly (for example `4/3`) and $b = d/2$ by default; the script prints W_4, W_3 and their b-derivatives and, at $b = d/2$, the value of (519). Resonant parameters, $r(d-b)/2 \in \mathbb{Z}$ at non-integer d, are refused; at $b = d/2$ they occur for $r = 3$ at $d = \frac{4}{3}, \frac{8}{3}$ and never for $r = 4$.

20.3.1 Checks of the five-Horn formula

All runs were made on the symmetry cluster. The records are in `LRI_C5_general_d/L/results/` (`summary.json`, `val50.json`, `val70.json`, `cfd40.json` and `ref_d*.json`), produced by `run_validation.py`, `refcheck_r4.py`, `check_func_derivs.py`, `analyze.py` and `make_tables.py`

in `LRI_C5_general_d/L/`. References: at $d = 1$ the proved closed forms $P_4 = 13B_- - 3B_+$ (Section 4) and (33); at $d = 2$ (333) and $\mathcal{L}_2 = -2\log 2\, P_4(2)$ (270); at $d = 3$ the 160-digit values of `LRI_C5_d3`; at $d = 0.8, 1.25, 1.5, 2.5, 2.75$ the 41- to 50-digit values of the independent spectral code in `LRI_C5_general_d/references/reference_values.json`; at $d = 0.6, \frac{4}{3}, \frac{8}{3}, 3.5$ the same spectral code, run at 32 digits for $r = 4$ only (`LRI_C5_general_d/L/refcheck_r4.py`; at $d = 0.6$ the point $b = d/2$ lies outside the $r = 3$ strip and the $r = 3$ spectral sum does not converge). The spectral code shares no code with the five-Horn formula.

Values of $\mathcal{L}_d$. $\mathcal{L}_d$ from (519) with exact jet derivatives, at the working precision of the run (50 digits; 70 digits at $d = 1, 1.5, 2, 2.5, 3$). "dev." is the relative deviation from the reference, "conv." the relative change of $\mathcal{L}_d$ when all truncation parameters are enlarged by the factor 1.4.

d	$\mathcal{L}_d$ (five-Horn formula)	reference	dev. P_4	dev. $\mathcal{L}_d$	conv. $\mathcal{L}_d$
0.6	−45086.56690064459550422	spectral, own run (32 d.)	$3.6 \cdot 10^{-32}$	$7.3 \cdot 10^{-33}$	$1.9 \cdot 10^{-44}$
0.8	−14228.69045883150419322995007	spectral reference (47 d.)	$1.0 \cdot 10^{-45}$	$2.7 \cdot 10^{-45}$	$3.2 \cdot 10^{-45}$
1	−5805.301629548802380141	closed form (proved)	$1.0 \cdot 10^{-80}$	$3.4 \cdot 10^{-81}$	$< 10^{-80}$
1.25	−2361.293891754120963656812701	spectral reference (49 d.)	$3.3 \cdot 10^{-45}$	$8.7 \cdot 10^{-46}$	$1.4 \cdot 10^{-47}$
4/3	−1818.786953676526007105	spectral, own run (32 d.)	$3.7 \cdot 10^{-24}$	$5.5 \cdot 10^{-24}$	$2.8 \cdot 10^{-49}$
1.5	−1127.932175171310391303895184	spectral reference (50 d.)	$9.7 \cdot 10^{-47}$	$3.6 \cdot 10^{-45}$	$1.2 \cdot 10^{-58}$
2	−347.2491531232011310585	$-2\log 2\, P_4(2)$ (proved)	$4.7 \cdot 10^{-81}$	$8.9 \cdot 10^{-81}$	$< 10^{-80}$
2.5	−136.796077287838337518533388	spectral reference (49 d.)	$3.8 \cdot 10^{-45}$	$2.5 \cdot 10^{-45}$	$2.4 \cdot 10^{-58}$
8/3	−104.035830813482931953	spectral, own run (32 d.)	$2.6 \cdot 10^{-26}$	$3.5 \cdot 10^{-26}$	$2.2 \cdot 10^{-47}$
2.75	−91.22798596406115896228950228	spectral reference (50 d.)	$5.3 \cdot 10^{-46}$	$1.9 \cdot 10^{-46}$	$2.5 \cdot 10^{-47}$
3	−62.7105849683876693473	`LRI_C5_d3` (160 d.)	$6.0 \cdot 10^{-81}$	$1.7 \cdot 10^{-81}$	$< 10^{-80}$
3.5	−31.81370266461412493531	spectral, own run (32 d.)	$2.8 \cdot 10^{-27}$	$3.5 \cdot 10^{-27}$	$9.5 \cdot 10^{-47}$

Away from the center. Relative deviation of (514), (515) and the two-node formula (496) from the 56-digit values of the independent spectral code, from the exact $W_3(2; b)$ of (264), and, at $d = 2$, of the ratio $W_4(2; 1.05)/W_4(2; 0.95)$ from the rim–spoke reflection $W_4(2; b)/W_4(2; 2-b) = 2^{4-4b}\gamma_C(b/2)^2/\gamma_C(b-\frac{1}{2})$, which follows from the formula for $W_4(2; 2x)$ in Section 18.7.6. The columns (514) and (515) come from `LRI_C5_general_d/L/L_general.py`, at 70 digits for $d = 1.5, 2, 2.5$ and at 50 digits otherwise. The column (496) comes from `LRI_C5_general_d/Z/offdiag_check.py` (record `LRI_C5_general_d/Z/results/offdiag60.json`), at 50 digits against the spectral values and at 60 digits against the finite-sample formula (484) with five fixed nodes ("fixed nodes"), which shares the Horn summation code and is therefore not independent; that code also gives the value column. The two W_3 columns differ only at $d = 1.5$ and 2.5, where the working precisions differ.

d	b	$\mathrm{Re}\,W_3(d;b)$ (24 digits)	reference	dev. (514)	dev. (515)	dev. (496)
0.8	0.37	266.761267027041907531165	spectral	$3.5\cdot10^{-51}$	$2.9\cdot10^{-49}$	$2.9\cdot10^{-49}$
0.8	0.43	268.642244930670511767657	spectral	$3.5\cdot10^{-50}$	$1.5\cdot10^{-48}$	$1.5\cdot10^{-48}$
1.25	0.575	107.4291471054848990911 8	spectral	$5.8\cdot10^{-53}$	$2.7\cdot10^{-50}$	$2.7\cdot10^{-50}$
1.25	0.675	108.364244882288165222595	spectral	$6.8\cdot10^{-53}$	$1.1\cdot10^{-49}$	$1.1\cdot10^{-49}$
1.5	0.7	72.7303680788392574835365	spectral	$2.0\cdot10^{-54}$	$1.5\cdot10^{-52}$	$1.5\cdot10^{-50}$
1.5	0.8	73.0949788822424244641612	spectral	$2.3\cdot10^{-53}$	$8.1\cdot10^{-52}$	$1.2\cdot10^{-50}$
2.5	1.2	23.3776070284027512618263	spectral	$8.1\cdot10^{-57}$	$2.4\cdot10^{-54}$	$2.8\cdot10^{-51}$
2.5	1.3	23.403771830554642453335	spectral	$7.9\cdot10^{-56}$	$9.9\cdot10^{-54}$	$3.2\cdot10^{-51}$
2.75	1.325	18.6817874548469398300141	spectral	$4.8\cdot10^{-55}$	$1.6\cdot10^{-51}$	$1.6\cdot10^{-51}$
2.75	1.425	18.6977204798280561759503	spectral	$2.3\cdot10^{-52}$	$7.5\cdot10^{-51}$	$7.5\cdot10^{-51}$
2	0.95		exact (264)	(ratio, below)	$5.8\cdot10^{-81}$	
2	1.05		exact (264)	(ratio, below)	$5.6\cdot10^{-81}$	
2	1.05/0.95		reflection	$9.8\cdot10^{-81}$ (ratio)		
1	0.37	216.134210851145764012993	fixed nodes			$3.4\cdot10^{-62}$
1	$0.3+0.1\,\mathrm{i}$	223.841209588900262170043	fixed nodes			$4.1\cdot10^{-62}$
2	0.8	45.6639529602144152055429	fixed nodes			$4.5\cdot10^{-63}$
2	$1.2+0.05\,\mathrm{i}$	49.6299926202374115388838	fixed nodes			$1.9\cdot10^{-62}$
3	1.3	16.4598664890623652308798	fixed nodes			$5.6\cdot10^{-62}$
3	1.6	15.5022646572795884994581	fixed nodes			$1.2\cdot10^{-62}$
3	$1.45+0.05\,\mathrm{i}$	15.0586428466284212402012	fixed nodes			$3.5\cdot10^{-62}$
1.5	$0.9+0.05\,\mathrm{i}$	86.1127570666953915845079	fixed nodes			$4.6\cdot10^{-56}$
2.5	1.4	25.6947810198617030782933	fixed nodes			$1.8\cdot10^{-55}$
3.5	1.7	10.2964255108971126897525	fixed nodes			$6.3\cdot10^{-55}$
0.8	0.42	263.52442188748393591694	fixed nodes			$3.5\cdot10^{-55}$

Test data. The five Horn values and their total first b-derivatives at $b=d/2$ ($u=x$ moving in $\mathfrak{h}_x$), 32 of the at least 55 digits computed in the 70-digit runs. At $d=1$, $\mathfrak{s}_0=\frac{83}{81}$, $\mathfrak{s}_0'=\frac{64}{243}$ and $\mathfrak{s}_{1/2}=1$ exactly.

d	function	value at $b=d/2$	∂_b at $b=d/2$
1	$\mathfrak{h}_x$	1.1186363871641870683496192575256	1.2852461541784784274144166577 49
	$\mathfrak{h}_1$	2.5987822357262795715169656086966	7.9424211032565657723245715768249
	$\mathfrak{h}_{1/2}$	1.6764843387459078434810926719847	3.9229772806249080806880221308364
	$\mathfrak{s}_0$	1.0246913580246913580246913580247	0.2633744855967078189300411522633 7
	$\mathfrak{s}_{1/2}$	1.0	0.0
1.5	$\mathfrak{h}_x$	1.1773925910607038758559762936 85	1.2933702591309377564756247705028
	$\mathfrak{h}_1$	2.7285923368887621906692408263826	7.0118518320221630889168328700506
	$\mathfrak{h}_{1/2}$	1.6390896218805694790051124408507	2.9123315039974903478917414402412
	$\mathfrak{s}_0$	1.0548507244990532977247501057129	0.4452440264452722619797249734899 7
	$\mathfrak{s}_{1/2}$	2.2461861367840794529465463415088	3.3311337225326803025747929195969
2	$\mathfrak{h}_x$	1.2513473666877450268884376715415	1.4377231145268788593624919295969
	$\mathfrak{h}_1$	2.8105997380603038789311930383846	6.5768099722236687733793558671552
	$\mathfrak{h}_{1/2}$	1.6274006603148409180018539998063	2.4813324518913317498491423929508
	$\mathfrak{s}_0$	1.0836461096219775566294370388355	0.5495504249104147258401786138706
	$\mathfrak{s}_{1/2}$	3.2548013206296818360037079996126	4.9626649037826634996982847859016
2.5	$\mathfrak{h}_x$	1.3376359713522616550547394154586	1.6441674626487503441319240236304
	$\mathfrak{h}_1$	2.8658700646971037650444991547941	6.3128962181069309575437047 13237
	$\mathfrak{h}_{1/2}$	1.6288526582621043253519113017864	2.2658474252647133894689489695958
	$\mathfrak{s}_0$	1.1103501379648233369119496859969	0.6126086566837367097941991100351 6
	$\mathfrak{s}_{1/2}$	4.1368224751542518108391408828496	5.8497720890012283607483552162793
3	$\mathfrak{h}_x$	1.4335280971237659036402062676254	1.8841954627648666804462998053832
	$\mathfrak{h}_1$	2.9047690244187603057569160926227	6.1245414235188740303623918912455
	$\mathfrak{h}_{1/2}$	1.6357377373331371059669409314044	2.1424411510929036014226183493683
	$\mathfrak{s}_0$	1.1349627686583730036642106378911	0.6502812349965942458213304530189 7
	$\mathfrak{s}_{1/2}$	4.9458028842831626652217506484 88	6.3455500640456084878526885322906

Per-function derivative check. For each of the five functions the jet value and derivative at $b=d/2$ were compared, at 40 digits, with the mean and the first Fourier coefficient of its plain (non-jet) values on a 48-point circle of radius $0.05\min(\frac{d}{2}, b_{\max}-\frac{d}{2})$ around $d/2$ (`LRI_C5_general_d/L/check_func_derivs.py`, `LRI_C5_general_d/L/results/cfd40.json`). This checks the jet implementation for every function separately; it is not an independent evaluation. Each cell gives

the relative deviation of the value / of the derivative, the latter relative to $\max(|f'|,1)$ (at $d=1$, $\mathfrak{s}_{1/2}=1$ has $f'=0$).

d	$\mathfrak{h}_x$	$\mathfrak{h}_1$	$\mathfrak{h}_{1/2}$	$\mathfrak{s}_0$	$\mathfrak{s}_{1/2}$
1	$1{\cdot}10^{-45}$ / $3{\cdot}10^{-45}$	$6{\cdot}10^{-46}$ / $4{\cdot}10^{-46}$	$2{\cdot}10^{-45}$ / $1{\cdot}10^{-44}$	$3{\cdot}10^{-45}$ / $2{\cdot}10^{-44}$	$< 10^{-60}$ / $7{\cdot}10^{-56}$
1.5	$4{\cdot}10^{-45}$ / $2{\cdot}10^{-44}$	$4{\cdot}10^{-45}$ / $1{\cdot}10^{-44}$	$3{\cdot}10^{-45}$ / $2{\cdot}10^{-44}$	$4{\cdot}10^{-45}$ / $2{\cdot}10^{-44}$	$5{\cdot}10^{-44}$ / $3{\cdot}10^{-43}$
2	$2{\cdot}10^{-45}$ / $9{\cdot}10^{-45}$	$6{\cdot}10^{-46}$ / $2{\cdot}10^{-45}$	$1{\cdot}10^{-45}$ / $5{\cdot}10^{-45}$	$4{\cdot}10^{-46}$ / $1{\cdot}10^{-44}$	$1{\cdot}10^{-45}$ / $1{\cdot}10^{-45}$
2.5	$2{\cdot}10^{-45}$ / $2{\cdot}10^{-45}$	$1{\cdot}10^{-45}$ / $2{\cdot}10^{-45}$	$8{\cdot}10^{-46}$ / $9{\cdot}10^{-45}$	$2{\cdot}10^{-48}$ / $7{\cdot}10^{-45}$	$3{\cdot}10^{-44}$ / $3{\cdot}10^{-43}$
3	$1{\cdot}10^{-45}$ / $8{\cdot}10^{-45}$	$2{\cdot}10^{-45}$ / $9{\cdot}10^{-46}$	$1{\cdot}10^{-45}$ / $5{\cdot}10^{-45}$	$2{\cdot}10^{-45}$ / $8{\cdot}10^{-45}$	$3{\cdot}10^{-46}$ / $4{\cdot}10^{-45}$

Summary. At $d=1,2,3$, $\mathcal{L}_d$ and $P_4(d)$ agree with the proved or 160-digit values to 10^{-60} or better at 50 digits and to 10^{-80} or better at 70 digits. At the five non-integer dimensions with 41- to 50-digit spectral references the agreement is 10^{-45} to 10^{-47}, that is the accuracy of the references (the internal convergence is 10^{-45} to 10^{-49} at 50 digits and about 10^{-58} at 70 digits); at $d=0.6,\frac{4}{3},\frac{8}{3},3.5$ it is 10^{-24} to 10^{-33}, limited by the 32-digit spectral runs. Away from the center, $W_4(d;b)$ and $W_3(d;b)$ at the ten points $b=d/2\pm0.05$ (±0.03 at $d=0.8$) agree with the 56-digit spectral values to $1.5\cdot10^{-48}$ or better (W_4: $3.5\cdot10^{-50}$ or better), and at $d=2$ with the exact $W_3(2;b)$ and the rim–spoke reflection to 10^{-80}. The $r=3$ by-product $3Q_d$ agrees with the references at the eight dimensions where they exist to 10^{-45} to 10^{-81}, and $\partial_b W_3(d;b)|_{b=d/2}$, which vanishes by (239) with (244), comes out 10^{-47} to 10^{-86} relative to W_3; this tests the jets of the $r=3$ functions. Each jet derivative agrees with the 48-point Cauchy-circle derivative to $3\cdot10^{-43}$ or better at 40 digits (previous table). The accelerated evaluation is an asymptotic-series method; its accuracy rests on the convergence in the truncation parameters and on the independent comparisons above, not on interval bounds.

20.4 The three-wheel at a general rim exponent: proofs

The code of this subsection is in `LRI_C5_general_d/Z/`. A separate review with its own codes rederived the derivations of the package's own text (`LRI_C5_general_d/review/Z-rigor/`; for the steps added in this note, see Section 20.10).

This subsection proves the three-wheel statements of Section 20.2: the two samples, the two-node formula (496) with its zero-mode form, (497) and (498). It also gives the five functions behind $\mathfrak{t}_3$ and $\widehat{M}_3$, the jets of the weights u,v, series forms of the jets and the first-derivative identity. Throughout $r=3$, $0<d<4$, and b lies in the strip

$$0<\operatorname{Re}b<b_{\max}(3,d)=\min\Big\{\frac{2d}{3},\ \frac{d+2}{3},\ d-\frac{1}{3}\Big\}<2$$

of (463), which contains $b=\frac{d}{2}$ if and only if $d>\frac{2}{3}$. On this strip $0<\operatorname{Re}\sigma<1$, so $\sigma\notin\mathbb{Z}$. The inputs are the finite Fourier theorem of Section 20.1.1, the top-mode theorem (473), the residue functional (480), the reflected form (464) and the block decomposition of Section 20.1.3. By (473) the three-wheel sample has two modes,

$$\Xi_3(t;b)=c_{3,0}(q,b)+c_{3,1}(q,b)\cos2\pi\Big(t+\frac{b}{4}\Big). \tag{520}$$

20.4.1 Analyticity in the rim exponent

Statement. $\mathfrak{t}_3$ and $\widehat{M}_3$ of (495), the five functions of (526) below and the weighted series of (529) are analytic in b on the strip. Every b-derivative is the termwise derivative, and the differentiated series again converge locally uniformly.

Proof. Let K be a compact subset of the strip and $\alpha_K=\min_{b\in K}3\operatorname{Re}g$. Since $\operatorname{Re}b<b_{\max}(3,d)$ on K, the conditions (i)–(iii) stated after (463) hold at every point of K with $\alpha=3\operatorname{Re}g$, and

by compactness $\alpha_K > \max(\frac{d}{2}, d-1, \frac{1}{2})$. The estimates of Section 20.1 (paragraph "Lattice split and estimates") come from the Gamma-ratio bound $|\Gamma(z+u)/\Gamma(z+v)| \le C|z|^{\mathrm{Re}(u-v)}$, which is uniform for u, v in compact sets. Here it is applied to Gamma ratios whose parameters σ, g, $2q$ are affine in b, and only its part for t in a compact set is used, not the large-$|\operatorname{Im} t|$ part. Write $T_{\ell n} = [(\sigma)_{\ell+n}/(2q)_{\ell+n}]\,[(1-g)_n/n!]$ as in (490). It is a product of two polynomials in b, bounded on K by $C_K(1+\ell+n)^{-\operatorname{Re} g}$ and $C_K(1+n)^{-\operatorname{Re} g}$. With $|d_\ell| \le C(1+\ell)^{d-2}$ there is a constant C_K with

$$|d_\ell\, T_{\ell n}(b)^3| \le C_K(1+\ell)^{d-2}(1+\ell+n)^{-\alpha_K}(1+n)^{-\alpha_K} \qquad (\ell, n \ge 0,\ b \in K).$$

The terms of $\widehat{M}_3$ are $N_b^3 g_\ell(a_\ell + n + t_b; b)^3$ with $t_b = \frac{1-\sigma}{2}$, since $a_\ell + n + t_b = n + \frac{\ell+1}{2}$. For $b \in K$ the node t_b stays at a positive distance from both pole classes $t \equiv 0$ and $t \equiv -\sigma \pmod 1$, because $\operatorname{Re} t_b$ and $\operatorname{Re}(t_b + \sigma) = \operatorname{Re} \frac{1+\sigma}{2}$ lie in compact subintervals of $(0, \frac{1}{2})$ and $(\frac{1}{2}, 1)$. So the right part ($n \ge 0$, by the first line of (464)) and the middle part (by its second line) obey the two bounds of Section 20.1 with a constant independent of $b \in K$; the finitely many terms with small arguments are continuous in b on K. The left part equals the right part, because $1 - \sigma - t_b = t_b$ and the last identity of (464) holds. These majorants do not depend on $b \in K$, and they are summable by the criterion at the end of the same paragraph, since $\alpha_K > \max(\frac{d}{2}, d-1, \frac{1}{2})$. Every term is analytic in b on the strip: $T_{\ell n}$ is a polynomial in b, and at the fixed lattice points $s = n + \frac{\ell+1}{2}$ the numerator Gamma functions of $N_b g_\ell(s; b)$ have the arguments $\ell + n + \frac{1+\sigma}{2}$, $\frac{\sigma-1}{2} - n$ and g, none of which is a pole for $0 < \operatorname{Re}\sigma < 1$ and $\operatorname{Re} g > 0$. By the Weierstrass M-test and Weierstrass's theorem on locally uniform limits of analytic functions, $\mathfrak{t}_3$ and $\widehat{M}_3$ are analytic on the strip, every b-derivative is the termwise derivative, and the differentiated series again converge locally uniformly. The same holds for the subseries of (526). For the weighted series of (529) it can also be seen directly: the weights grow at most logarithmically in the indices, and replacing α_K by a slightly smaller exponent, still above $\max(\frac{d}{2}, d-1, \frac{1}{2})$, gives summable majorants that are uniform on K. □

20.4.2 The two samples and the two-node formula

Pole node. For b in the strip,

$$\Xi_3(0; b) = -\Big(\frac{\pi s_b}{N_b}\Big)^3 \mathfrak{t}_3(b). \tag{521}$$

This is the case $r = 3$ of (510), which Section 20.3 proves for every r by extending the lattice form (504) to $t = 0$; the proof below takes the limit term by term instead.

Proof. By Section 20.1.1(a) and its proof, $t^3 S_3(t; b) = \sum_\ell d_\ell \sum_n t^3 g_\ell(a_\ell + n + t; b)^3$ converges uniformly near $t = 0$, and only the terms with $n \ge 0$ are singular at $t = 0$: the poles of the middle part in the class $t \equiv 0$ lie at $t \ge 1$, by the second line of (464), the same holds for the left part by the mirror identity, and the class $t \equiv -\sigma$ does not contain 0 since $\sigma \notin \mathbb{Z}$. Since $D_b(t)/t \to \pi s_b$, the value $\Xi_3(0) = \lim_{t\to 0} D_b(t)^3 S_3(t)$ is the sum over $\ell \ge 0$, $n \ge 0$ of $d_\ell \lim_{t\to 0}[D_b(t) g_\ell(a_\ell + n + t)]^3$. By the first line of (464),

$$\begin{aligned} D_b(t)\, g_\ell(a_\ell + n + t) \ &\to\ -\, s_b \sin(\pi g)\, \frac{\Gamma(\ell + n + \sigma)\Gamma(n + 1 - g)}{\Gamma(\ell + n + 2q)\, n!} \\ &=\ -\, s_b \sin(\pi g)\Gamma(1-g)\frac{\Gamma(\sigma)}{\Gamma(2q)}\, T_{\ell n} = -\frac{\pi s_b}{N_b}\, T_{\ell n}, \end{aligned}$$

using $\sin(\pi g)\Gamma(1-g) = \pi/\Gamma(g)$. Cubing and summing gives (521). □

The function $T_{\ell n}$ is the normalized residue prefactor at a general rim exponent: $N_b g_\ell(a_\ell + n + u; b) = -T_{\ell n}(b)/u + O(1)$ for $n \ge 0$ (as in Section 20.5), so $\mathfrak{t}_3 = -\sum_{\ell,n} d_\ell \lim_{u\to 0} u^3 [N_b g_\ell(a_\ell + n + u; b)]^3$ collects only the *leading* coefficients of the triple poles. The residues of the cube need the full Laurent data, which is the origin of the polygamma polynomial F_3 of (255).

Self-mirror node. Let $t_b = \frac{1-\sigma}{2} = \frac{1}{2} - \frac{b}{4}$. For b in the strip, $1 - \sigma - t_b = t_b$, $\cos 2\pi(t_b + \frac{b}{4}) = -1$, $D_b(t_b) = \cos^2 \frac{\pi\sigma}{2} = \frac{1+\varsigma_b}{2}$, and

$$\Xi_3(t_b; b) = \Big(\frac{1+\varsigma_b}{2}\Big)^3 \frac{\widehat{M}_3(b)}{N_b^3}, \qquad \widehat{M}_3(b) = N_b^3\big[2A_3^0(t_b) + 2A_3^+(t_b) + C_3(t_b)\big], \tag{522}$$

with the blocks A_3^0, A_3^+, C_3 of Section 20.1.3, written out in (524) below. In the Horn values of Section 20.3 the same sample is (512) at $r = 3$.

Proof. The first three statements are direct: $D_b(t_b) = \sin\frac{\pi(1-\sigma)}{2} \sin\frac{\pi(1+\sigma)}{2} = \cos^2\frac{\pi\sigma}{2}$ and $2\cos^2\frac{\pi\sigma}{2} = 1 + \cos\pi\sigma = 1 + \varsigma_b$. Since $0 < \operatorname{Re}\sigma < 1$, t_b lies in neither pole class, so $\Xi_3(t_b) = D_b(t_b)^3 S_3(t_b)$ with an absolutely convergent $S_3(t_b)$. Moreover $a_\ell + n + t_b = n + \frac{\ell+1}{2}$; as n runs over $\mathbb{Z}$ this runs over the midpoint lattice $\frac{\ell+1}{2} + \mathbb{Z}$ of (495), hence $N_b^3 S_3(t_b; b) = \widehat{M}_3(b)$. Finally the block decomposition of Section 20.1.3 at a point equal to its own mirror point $1 - \sigma - t$ gives the block form. □

The midpoint lattice is invariant under $s \mapsto -s$, as is $g_\ell(s; b)$.

Proof of (496). By (520), $y_0 := \Xi_3(0; b) = c_{3,0} + \varsigma_b c_{3,1}$ and $y_b := \Xi_3(t_b; b) = c_{3,0} - c_{3,1}$. On the strip $1 + \varsigma_b \neq 0$ ($\varsigma_b = -1$ only for $b \in 2 + 4\mathbb{Z}$), so $c_{3,0} = (y_0 + \varsigma_b y_b)/(1 + \varsigma_b)$ and $c_{3,1} = (y_0 - y_b)/(1 + \varsigma_b)$. Then

$$(1 + 2\varsigma_b^2)c_{3,0} + 3\varsigma_b c_{3,1} = \frac{(1+\varsigma_b)(1+2\varsigma_b)\, y_0 - 2\varsigma_b s_b^2\, y_b}{1+\varsigma_b},$$

and the first line of (480) gives

$$W_3(d; b) = -\frac{2N_b^3(1+2\varsigma_b)}{\pi s_b^5}\, y_0 + \frac{4N_b^3\varsigma_b}{\pi s_b^3(1+\varsigma_b)}\, y_b.$$

Inserting (521) and (522) gives (496) for every b in the strip and every $0 < d < 4$. □ (With the midpoint sample in Horn values the same formula is (515).)

Zero-mode form. With the normalized zero mode $\hat{c}_0(b) = N_b^3 c_{3,0}(q, b) = N_b^3 \int_0^1 \Xi_3(t; b)\, dt$,

$$W_3(d; b) = \frac{6\pi^2}{s_b^2}\mathfrak{t}_3(b) + \frac{4}{\pi s_b^3}\hat{c}_0(b), \qquad \hat{c}_0(b) = \frac{\varsigma_b(1+\varsigma_b)^2}{8}\widehat{M}_3(b) - \frac{\pi^3 s_b^3}{1+\varsigma_b}\mathfrak{t}_3(b). \tag{523}$$

Proof. $\hat{c}_0 = N_b^3(y_0 + \varsigma_b y_b)/(1 + \varsigma_b)$ with (521) and (522) gives the second relation. Since $y_0 = c_{3,0} + \varsigma_b c_{3,1}$, one has $(1 + 2\varsigma_b^2)c_{3,0} + 3\varsigma_b c_{3,1} = 3y_0 - 2s_b^2 c_{3,0}$ identically (no division by ς_b occurs, so $\varsigma_b = 0$ is allowed), and the first line of (480) with (521) gives the first relation. □ This is the form quoted after (496); at $b = \frac{d}{2}$, where $W_3(d; \frac{d}{2}) = 3Q_d$ by (302), it is (497).

At $b = \frac{d}{2}$, (496) is a finite formula for the primitive, $3Q_d = u(\frac{d}{2})\mathfrak{t}_3(\frac{d}{2}) + v(\frac{d}{2})\widehat{M}_3(\frac{d}{2})$, with five real hypergeometric values at the two real nodes 0 and $\frac{1-q}{2}$, whereas (319) uses three complex samples with five blocks each. The extra input is the top-mode theorem (473). For complex b the node t_b is complex; both nodes are real for real b.

20.4.3 The five functions

At $r = 3$ and a general rim exponent the blocks of Section 20.1.3 are

$$\begin{aligned}
A_3^0(t) &= a_3^0(t)\, {}_7F_6\left(\begin{matrix}1, (\sigma+t)^{\times 3}, (1-g+t)^{\times 3} \\ (2q+t)^{\times 3}, (1+t)^{\times 3}\end{matrix}; 1\right), & a_3^0(t) &= \Big[\frac{\Gamma(\sigma+t)\Gamma(-t)}{\Gamma(2q+t)\Gamma(g-t)}\Big]^3, \\
A_3^+(t) &= a_3^+(t)\, \mathfrak{H}_+(t), & a_3^+(t) &= 4q\Big[\frac{\Gamma(1+\sigma+t)\Gamma(-t)}{\Gamma(1+2q+t)\Gamma(g-t)}\Big]^3, \\
C_3(t) &= a_3^c(t)\, \mathfrak{H}_c(t), & a_3^c(t) &= 4q\Big[\frac{\Gamma(\sigma+t)\Gamma(1-t)}{\Gamma(2q+t)\Gamma(1+g-t)}\Big]^3,
\end{aligned} \tag{524}$$

with the two Horn values

$$\mathfrak{H}_+(t)=\sum_{j,n\ge0}\frac{d_{j+1}}{4q}\Big[\frac{(1+\sigma+t)_{j+n}}{(1+2q+t)_{j+n}}\Big]^3\Big[\frac{(1-g+t)_n}{(1+t)_n}\Big]^3,$$
$$\mathfrak{H}_c(t)=\sum_{j,k\ge0}\frac{d_{j+k+1}}{4q}\Big[\frac{(1-t)_j}{(1+g-t)_j}\Big]^3\Big[\frac{(\sigma+t)_k}{(2q+t)_k}\Big]^3. \tag{525}$$

These are (321) and $\mathfrak{H}[\mathsf{U}_+;\mathsf{V}_+;(1,1)]$, $\mathfrak{H}[\mathsf{U}_c;\mathsf{V}_c;(1,1)]$ of (322) at $r=3$ with the rim parameter $q\to\sigma$ and the spoke parameter $q\to g$, as stated in Section 20.1.3. Here $d_{j+1}/(4q)=(2q+1)_j(4q-1)_j/[(2q)_j(2)_j]$ by (301); it equals δ_{j0}, 1 and $1+\frac{2}{3}j$ at $d=1,2,3$.

Split $\mathfrak{t}_3=\mathfrak{t}_{3,0}+\mathfrak{t}_{3,+}$ into the spin $\ell=0$ and $\ell\ge1$, and $\widehat{M}_3=\widehat{M}_{3,0}+\widehat{M}_{3,+}+\widehat{M}_{3,c}$ by (522):

$$\mathfrak{t}_{3,0}(b)={}_7F_6\left(\begin{matrix}1,\sigma,\sigma,\sigma,1-g,1-g,1-g\\2q,2q,2q,1,1,1\end{matrix};1\right),\qquad \mathfrak{t}_{3,+}(b)=4q\Big(\frac{\sigma}{2q}\Big)^3\mathfrak{H}_+(0),$$
$$\widehat{M}_{3,0}(b)=2N_b^3A_3^0(t_b),\qquad \widehat{M}_{3,+}(b)=2N_b^3A_3^+(t_b),\quad \widehat{M}_{3,c}(b)=N_b^3C_3(t_b). \tag{526}$$

The spin-zero row is the ${}_7F_6(1)$ because $T_{0n}=(\sigma)_n(1-g)_n/[(2q)_n\,n!]$, the upper parameter 1 cancelling one $n!$; $\mathfrak{t}_{3,+}$ follows from $T_{j+1,n}=\frac{\sigma}{2q}\frac{(1+\sigma)_{j+n}(1-g)_n}{(1+2q)_{j+n}\,n!}$ and $d_{j+1}=4q\cdot d_{j+1}/(4q)$. (The subscript 0 of $\mathfrak{t}_{3,0}$ and $\widehat{M}_{3,0}$ names the spin-zero block; these are not Fourier coefficients $c_{3,0}$.) At the self-mirror node, $\sigma+t_b=1-t_b=\frac{1+\sigma}{2}$ and $2q+t_b=1+g-t_b=2q+\frac{1-\sigma}{2}$, so the two sequences in $\mathfrak{H}_c(t_b)$ coincide and

$$\mathfrak{H}_c(t_b)=\sum_{j,k\ge0}\frac{d_{j+k+1}}{4q}\,\mathfrak{q}_j\mathfrak{q}_k,\qquad \mathfrak{q}_k=\Big[\frac{(\frac{1+\sigma}{2})_k}{(2q+\frac{1-\sigma}{2})_k}\Big]^3.$$

The remaining parameters at t_b are $1-g+t_b=\frac{3+\sigma}{2}-2q$ and $1+t_b=\frac{3-\sigma}{2}$. Every parameter is affine in b, so the b-derivatives of the five functions are directional parameter derivatives (table below). The prefactors a_3^0,a_3^+,a_3^c at t_b and N_b contribute only polygamma values at fixed arguments. So $\mathfrak{t}_3$ is one ${}_7F_6(1)$ and one two-variable Horn value, and $\widehat{M}_3$ is one ${}_7F_6(1)$ and two two-variable Horn values, as stated in Section 20.2.

function	upper parameters (slope in b)	lower parameters (slope in b)	at $b=d/2$
$\mathfrak{t}_{3,0}$: ${}_7F_6$ at $t=0$	$\sigma^{\times3}$ $(\frac12)$, $(1-g)^{\times3}$ $(\frac12)$	$(2q)^{\times3}$, $1^{\times3}$ (0)	$q,\ 1-q;\ 2q,\ 1$
$\mathfrak{H}_+(0)$	$1+\sigma$ $(\frac12)$, $1-g$ $(\frac12)$	$1+2q$, 1 (0)	$1+q,\ 1-q;\ 1+2q,\ 1$
$A_3^0(t_b)$: ${}_7F_6$	$\frac{1+\sigma}{2}$ $(\frac14)$, $\frac{3+\sigma}{2}-2q$ $(\frac14)$	$2q+\frac{1-\sigma}{2}$ $(-\frac14)$, $\frac{3-\sigma}{2}$ $(-\frac14)$	$\frac{1+q}{2}$, $\frac{3-3q}{2}$; $\frac{1+3q}{2}$, $\frac{3-q}{2}$
$\mathfrak{H}_+(t_b)$	$\frac{3+\sigma}{2}$ $(\frac14)$, $\frac{3+\sigma}{2}-2q$ $(\frac14)$	$1+2q+\frac{1-\sigma}{2}$ $(-\frac14)$, $\frac{3-\sigma}{2}$ $(-\frac14)$	$\frac{3+q}{2}$, $\frac{3-3q}{2}$; $\frac{3+3q}{2}$, $\frac{3-q}{2}$
$\mathfrak{H}_c(t_b)$	$\frac{1+\sigma}{2}$ $(\frac14)$, twice	$2q+\frac{1-\sigma}{2}$ $(-\frac14)$, twice	$\frac{1+q}{2}$; $\frac{1+3q}{2}$

The table lists the parameters of the five hypergeometric functions of (526) (each parameter repeated three times, $r=3$) and their slopes in b. The multiplicity factor $d_{j+1}/(4q)$ (or $d_{j+k+1}/(4q)$) of the Horn values and the prefactor $4q$ do not depend on b.

20.4.4 Rim jets and their series

Primes denote ∂_b; by Section 20.4.1 every function above is analytic in b on the strip. For $\frac23<d<4$, at $b=\frac{d}{2}$,

$$9Z_d=u''\mathfrak{t}_3+2u'\mathfrak{t}_3'+u\mathfrak{t}_3''+v''\widehat{M}_3+2v'\widehat{M}_3'+v\widehat{M}_3'', \tag{527}$$

with $\mathfrak{t}_3^{(k)}=\mathfrak{t}_{3,0}^{(k)}+\mathfrak{t}_{3,+}^{(k)}$ and $\widehat{M}_3^{(k)}=\widehat{M}_{3,0}^{(k)}+\widehat{M}_{3,+}^{(k)}+\widehat{M}_{3,c}^{(k)}$ the jets of the five functions (526), and, with S,C of (459),

$$u=\frac{2\pi^2(1+2C)}{S^2},\quad u'=-\frac{2\pi^3(1+C+C^2)}{S^3},\quad u''=\frac{\pi^4(1+5C+2C^2+C^3)}{S^4},$$
$$v=\frac{C(1+C)^2}{2\pi S^3},\qquad v'=-\frac{(1+C)^2(1+2C)}{4S^4},\quad v''=\frac{\pi(1+C)^3(2+C)}{4S^5}. \tag{528}$$

Proof. By (244), $9Z_d = \partial_b^2 W_3(d;b)$ at $b = \frac{d}{2}$, which lies in the strip for $\frac{2}{3} < d < 4$. Differentiate (496) twice, termwise by Section 20.4.1, using $\partial_b s_b = \frac{\pi}{2}\varsigma_b$ and $\partial_b \varsigma_b = -\frac{\pi}{2} s_b$; for example $u' = -2\pi^3[s_b^2 + \varsigma_b(1+2\varsigma_b)]/s_b^3$ and $s_b^2 + \varsigma_b + 2\varsigma_b^2 = 1 + \varsigma_b + \varsigma_b^2$. □ Formula (527) is exact differentiation and does not use the input $\partial_b W_3(d;\frac{d}{2}) = 0$.

Series form of the jets. On the strip,

$$\begin{aligned} \mathfrak{t}_3' &= 3\sum_{\ell,n\ge 0} d_\ell T_{\ell n}^3 \mu_1, & \mathfrak{t}_3'' &= \sum_{\ell,n\ge 0} d_\ell T_{\ell n}^3 (9\mu_1^2 + 3\mu_2), \\ \widehat{M}_3' &= 3\sum_{\ell\ge 0} d_\ell \sum_s [N_b g_\ell(s;b)]^3 w_1, & \widehat{M}_3'' &= \sum_{\ell\ge 0} d_\ell \sum_s [N_b g_\ell(s;b)]^3 (9w_1^2 + 3w_2), \end{aligned} \tag{529}$$

with $s \in \frac{\ell+1}{2} + \mathbb{Z}$ and the explicit weights (arguments $(\ell,n;b)$ of μ_m and $(\ell,s;b)$ of w_m suppressed)

$$\begin{aligned} \mu_1 &= \tfrac{1}{2}\big[\psi_0(\sigma+\ell+n) - \psi_0(\sigma) + \psi_0(1-g+n) - \psi_0(1-g)\big], \\ \mu_2 &= \tfrac{1}{4}\big[\psi_1(\sigma+\ell+n) - \psi_1(\sigma) + \psi_1(1-g+n) - \psi_1(1-g)\big], \\ w_1 &= -\tfrac{1}{2}\big[\psi_0(g) + \psi_0(\sigma)\big] + \tfrac{1}{4}\textstyle\sum_\pm \big[\psi_0(a_\ell \pm s) + \psi_0(\beta_\ell \pm s)\big], \\ w_2 &= \tfrac{1}{4}\big[\psi_1(g) - \psi_1(\sigma)\big] + \tfrac{1}{16}\textstyle\sum_\pm \big[\psi_1(a_\ell \pm s) - \psi_1(\beta_\ell \pm s)\big]. \end{aligned} \tag{530}$$

At $b = \frac{d}{2}$, μ_1, μ_2 are those of (253), and w_1, w_2 are the spectral weights of (245) continued to $s = \mathrm{i}\nu$. Consequently

$$9Z_d = \sum_{\ell,n\ge 0} d_\ell T_{\ell n}^3 \big[(9\mu_1^2{+}3\mu_2)\,u{+}6\mu_1\,u'{+}u''\big] + \sum_{\ell\ge 0} d_\ell \sum_{s\in\frac{\ell+1}{2}+\mathbb{Z}} [N_b g_\ell(s;b)]^3 \big[(9w_1^2{+}3w_2)\,v{+}6w_1\,v'{+}v''\big], \tag{531}$$

at $b = \frac{d}{2}$, with $u, \dots, v''$ of (528).

Proof. $\mu_1 = \partial_b \log T_{\ell n}$ and $\mu_2 = \partial_b \mu_1$, since $\partial_b \sigma = \frac{1}{2} = \partial_b(1-g)$. At fixed s, $w_1 = \partial_b \log[N_b g_\ell(s;b)]$ and $w_2 = \partial_b w_1$, since $\partial_b a_\ell = \frac{1}{4} = -\partial_b \beta_\ell$ and $\partial_b \log N_b = -\frac{1}{2}[\psi_0(g) + \psi_0(\sigma)]$. The midpoint lattice does not move with b, so (529) are the termwise derivatives of (495), which Section 20.4.1 justifies; (531) is (527) with these series. The weights were also checked term by term against numerical differentiation, to 10^{-30} (`LRI_C5_general_d/Z/check_weights.py`). □

Comparison with (257). The series (257) is $9Z_d = \sum d_\ell T_{\ell n}^3[(9\mu_1^2 + 3\mu_2)F_3 + 6\mu_1 F_{3,1} + F_{3,2}]$, with the index-dependent triple-pole weight F_3 of (254) and its rim derivatives $F_{3,1}, F_{3,2}$. In (531) the same structure appears with $F_3, F_{3,1}, F_{3,2}$ replaced by the index-independent elementary numbers u, u', u'', at the price of the midpoint-lattice term. In that term $\frac{1}{9}\widehat{M}_3''$ is the midpoint-lattice sum of the spectral integrand $t_\ell^3(w_1^2 + w_2/3)$ of (246), whose integral is Z_d itself.

20.4.5 The first-derivative identity and the simplified formula for Z_d

First rim derivative. $\partial_b W_3(d;\frac{d}{2}) = 0$ is equivalent to $u'\mathfrak{t}_3 + u\mathfrak{t}_3' + v'\widehat{M}_3 + v\widehat{M}_3' = 0$, that is, at $b = \frac{d}{2}$,

$$8\pi^3 S^2(1+2C)\,\mathfrak{t}_3' + 2SC(1+C)^2\,\widehat{M}_3' = 8\pi^4 S(1+C+C^2)\,\mathfrak{t}_3 + \pi(1+C)^2(1+2C)\,\widehat{M}_3, \tag{532}$$

and, in the zero-mode form (523), to

$$3\pi^3 S^2\,\mathfrak{t}_3' + 2S\,\hat{c}_0' = 3\pi C\,(\pi^3 S\,\mathfrak{t}_3 + \hat{c}_0). \tag{533}$$

Proof. The first is $4\pi S^4\,\partial_b W_3$ with (528). For the second write $W_3 = u_0\mathfrak{t}_3 + v_0\hat{c}_0$ with $u_0 = 6\pi^2/s_b^2$ and $v_0 = 4/(\pi s_b^3)$, by (523); at $b = \frac{d}{2}$, $u_0' = -6\pi^3 C/S^3$ and $v_0' = -6C/S^4$, and (533) is $\frac{\pi S^4}{2}\,\partial_b W_3 = 0$. □

Given the input $\partial_b W_3(d;\frac{d}{2}) = 0$, which is proved by Pattern T (239) with (244) (Section 20.2), the first derivative of the residue sum is not independent:

$$\mathfrak{t}_3' = \frac{\pi C}{S}\,\mathfrak{t}_3 + \frac{C}{\pi^2 S^2}\,\hat{c}_0 - \frac{2}{3\pi^3 S}\,\hat{c}_0' \qquad \big(b = \tfrac{d}{2}\big) \tag{534}$$

for every d. In the basis $(\mathfrak{t}_3, \widehat{M}_3)$, (532) eliminates $\mathfrak{t}_3'$ for $d \neq \frac{8}{3}$ (where $1+2C=0$), or $\widehat{M}_3'$ for $d \neq 2$ (where $C=0$). At $d=1$, (533) reads $\mathfrak{t}_3'(1;\frac{1}{2}) = \pi\,\mathfrak{t}_3(1;\frac{1}{2})$, which Dixon's closed form (546) confirms. At $d=3$, with the Gamma form of $\hat{c}_0(3;b)$ (Section 20.6), it reads $\mathfrak{t}_3'(3;\frac{3}{2}) = -\pi\,\mathfrak{t}_3(3;\frac{3}{2}) + \Gamma(\frac{1}{4})^{10}/(64\pi^{9/2})$, an identity also found in the $d=3$ computation of `LRI_C5_d3/wf_Z3/`.

Proof of (498). Assume $\partial_b W_3(d;\frac{d}{2}) = 0$. At $b=\frac{d}{2}$, with $\hat{c}_0^{(k)} = \partial_b^k \hat{c}_0|_{b=d/2}$,

$$Z_d = \frac{2\pi^2}{3S^2}\,\mathfrak{t}_3'' - \frac{\pi^4\cos(\pi d/2)}{3S^4}\,\mathfrak{t}_3 + \frac{1}{9}\Big[\frac{3\pi}{S^3}\,\hat{c}_0 - \frac{4C}{S^4}\,\hat{c}_0' + \frac{4}{\pi S^3}\,\hat{c}_0''\Big], \tag{535}$$

and, with (497), equivalently (498). Indeed, at $b=\frac{d}{2}$,

$$u_0' = -\frac{6\pi^3 C}{S^3}, \qquad u_0'' = \frac{3\pi^4(1+2C^2)}{S^4}, \qquad v_0' = -\frac{6C}{S^4}, \qquad v_0'' = \frac{3\pi(1+3C^2)}{S^5}.$$

Insert (534) into $9Z_d = W_3'' = u_0''\mathfrak{t}_3 + 2u_0'\mathfrak{t}_3' + u_0\mathfrak{t}_3'' + v_0''\hat{c}_0 + 2v_0'\hat{c}_0' + v_0\hat{c}_0''$. The coefficient of $\mathfrak{t}_3$ becomes $u_0'' + 2u_0'\pi C/S = 3\pi^4(1-2C^2)/S^4 = -3\pi^4\cos(\pi d/2)/S^4$; that of $\hat{c}_0$ becomes $v_0'' + 2u_0'C/(\pi^2 S^2) = 3\pi(1-C^2)/S^5 = 3\pi/S^3$; that of $\hat{c}_0'$ becomes $2v_0' - 4u_0'/(3\pi^3 S) = -4C/S^4$. This is (535). Replacing $\mathfrak{t}_3$ by $\frac{S^2}{6\pi^2}(3Q_d - \frac{4}{\pi S^3}\hat{c}_0)$ from (497) turns the $\mathfrak{t}_3$ term into $-\frac{\pi^2\cos(\pi d/2)}{6S^2}Q_d + \frac{2\pi\cos(\pi d/2)}{9S^5}\hat{c}_0$, and $3S^2 + 2\cos 2\pi q = 1 + C^2$ gives (498). □

The same computation shows that the full jet formula (527) and (535) differ exactly by $-\frac{2\pi C}{9S}\,\partial_b W_3(d;\frac{d}{2})$ (the review `LRI_C5_general_d/review/Z-rigor/` confirmed this symbolically). So (498) depends on the input only through this term, which is absent at $d=2$ where $C=0$. In (535) the second derivatives enter only through $\frac{2\pi^2}{3S^2}\mathfrak{t}_3'' + \frac{4}{9\pi S^3}\hat{c}_0''$, whose second-derivative part is $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$, exactly as in (527). The data needed in (535) are $\mathfrak{t}_3$, $\mathfrak{t}_3''$ and the jets of the zero mode, which is computed from the two nodes by (523); so the complete set of inputs remains $\{\mathfrak{t}_3^{(k)}, \widehat{M}_3^{(k)}\}_{k\le 2}$, that is, at most 15 special-function numbers (two ${}_7F_6(1)$ values and three two-variable Horn values, each with its first and second b-derivatives), subject to the two exact relations $W_3(d;\frac{d}{2}) = 3Q_d$ and (532).

The count. The 15 numbers enter only through the six combinations $\mathfrak{t}_3^{(k)}, \widehat{M}_3^{(k)}$, $k\le 2$, and the second derivatives only through the single combination $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$. After the two exact relations, Z_d is an elementary multiple of Q_d plus an elementary combination of three further numbers, for instance $\mathfrak{t}_3$ (or $\hat{c}_0$), $\hat{c}_0'$ and $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$ at $b=\frac{d}{2}$; this is the count of Section 20.7. The choice works at every d, because the coefficients $6\pi^2/S^2$ and $4/(\pi S^3)$ of $\mathfrak{t}_3$ and $\hat{c}_0$ in (497), and $3\pi^3 S^2$ and $2S$ of $\mathfrak{t}_3'$ and $\hat{c}_0'$ in (533), never vanish for $0<d<4$. At $d=1$ ($\hat{c}_0 \equiv 0$) and at $d=2$ ($\widehat{M}_3 \equiv 0$) only the second-derivative combination remains, and Section 20.6 evaluates it in closed form. All functions used here belong to the classes of (321) and (322), with the rim parameter σ and the spoke parameter g separated.

20.5 The pole node alone: residue series of one lower Laurent order

At $t=0$ the left and middle half-lattices carry $[\sin\pi t]^r$, so $\Xi_r^{(k)}(0;b)$ for $k \le r-1$ comes from the right half-lattice alone. With $r-1$ modes, $k \le r-2$ suffices. Let $\lambda_m(\ell,n;b)$ be (252) with the rim and spoke separated,

$$\lambda_m = \psi_{m-1}(\ell+n+\sigma) - \psi_{m-1}(\ell+n+2q) - (-1)^m\psi_{m-1}(g-n) + (1+(-1)^m)\psi_{m-1}(1) - \psi_{m-1}(n+1),$$

so that $N_b g_\ell(a_\ell + n + u; b) = -\frac{T_{\ell n}(b)}{u}\exp\sum_m \lambda_m \frac{u^m}{m!}$ as in (262). Then, with $\kappa_b = \pi\cot\frac{\pi b}{2}$,

$$\boxed{W_3(d;b) = \sum_{\ell,n\ge 0} d_\ell T_{\ell n}^3\big[2\pi^2 - 6\kappa_b\lambda_1\big], \qquad W_4(d;b) = \sum_{\ell,n\ge 0} d_\ell T_{\ell n}^4\Big[4\kappa_b(4\lambda_1^2 + \lambda_2 - \tfrac{\pi^2}{3}) - 8\pi^2\lambda_1\Big].} \tag{536}$$

At $b = d/2$, $\kappa_b = D$ and the λ_m are those of (252):

$$3Q_d = \sum d_\ell T_{\ell n}^3 (2\pi^2 - 6D\lambda_1), \qquad P_4(d) = \sum d_\ell T_{\ell n}^4 \Big[4D(4\lambda_1^2 + \lambda_2 - \tfrac{\pi^2}{3}) - 8\pi^2\lambda_1\Big], \tag{537}$$

to be compared with F_3 (quadratic, λ_2) and F_4 (cubic, λ_3) of (255). Differentiating (536) in b (termwise, as for (256); $\partial_b\lambda_m = \eta_m$, $\partial_b\eta_m = \theta_m$, $\partial_b \log T_{\ell n} = \mu_1$, $\partial_b\mu_1 = \mu_2$ of (253); $\kappa_b' = -\frac{\pi^2}{2S^2}$, $\kappa_b'' = \frac{\pi^3 C}{2S^3}$):

$$\begin{aligned}
&\mathcal{L}_d = \sum d_\ell T_{\ell n}^4 [4\mu_1 \widehat{F}_4 + \widehat{F}_4'], && 9Z_d = \sum d_\ell T_{\ell n}^3 [(9\mu_1^2 + 3\mu_2)\widehat{F}_3 + 6\mu_1 \widehat{F}_3' + \widehat{F}_3''],\\
&\widehat{F}_4 = 4D(4\lambda_1^2 + \lambda_2 - \tfrac{\pi^2}{3}) - 8\pi^2\lambda_1, && \widehat{F}_4' = -\tfrac{2\pi^2}{S^2}(4\lambda_1^2 + \lambda_2 - \tfrac{\pi^2}{3}) + 4D(8\lambda_1\eta_1 + \eta_2) - 8\pi^2\eta_1,\\
&\widehat{F}_3 = 2\pi^2 - 6D\lambda_1, && \widehat{F}_3' = \tfrac{3\pi^2}{S^2}\lambda_1 - 6D\eta_1, \qquad \widehat{F}_3'' = -\tfrac{3\pi^3 C}{S^3}\lambda_1 + \tfrac{6\pi^2}{S^2}\eta_1 - 6D\theta_1.
\end{aligned} \tag{538}$$

Here $\widehat{F}_3$, $\widehat{F}_4$ are the weights of (537), the analogues of F_3, F_4 of (254), and primes are their b-derivatives. These replace (256) and (257): the polygamma weights drop by one order (λ_3, η_3 no longer occur in $\mathcal{L}_d$, and $\lambda_2, \eta_2, \theta_2$ no longer occur in Z_d). In this form $P_4, \mathcal{L}_d$ need one Horn family and its parameter derivatives of order ≤ 2 (shift) and ≤ 1 (rim); Q_d, Z_d need one family with orders ≤ 1 and ≤ 2. Status: derived (exact algebra, `LRI_C5_general_d/specialize/sym_taylor.py`); (536) is confirmed numerically, and (538) through b-derivatives of (536).

20.6 Specialization to $d = 1, 2, 3$

The one-variable function behind $d = 1$ and $d = 2$. Let $\Omega_r(t; x) = \sin^r \pi(t + x) \sum_{n\in\mathbb{Z}} [\Gamma(n + t + x)/\Gamma(n + t + 1)]^r$. At $x = \frac{1}{2}$ this is the $\Omega_r(t)$ of (335), and $\Omega_4(t; x)$ is the $\Omega(t; x)$ of (349). For every b in the strip,

$$d = 1: \ \Xi_r(t; b) = (-1)^r 2^{-rb}\, \Omega_r(2t; b), \qquad d = 2: \ \Xi_r(t; b) = (-1)^r\, \Omega_r(t; \tfrac{b}{2})^2. \tag{539}$$

Proof at $d = 1$. Here $d_0 = d_1 = 1$, $d_\ell = 0$ for $\ell \ge 2$, $2q = \frac{1}{2}$ and $1 - g = \frac{1}{2} + \sigma$. In the reflected form (464), valid for every $n \in \mathbb{Z}$, Legendre's duplication formula $\Gamma(w)\Gamma(w + \frac{1}{2}) = 2^{1-2w}\sqrt{\pi}\,\Gamma(2w)$ [26] turns the $\ell = 0$ and $\ell = 1$ Gamma ratios at $z = n + t$ into $2^{1-2\sigma}\Gamma(2z + b)/\Gamma(2z + 1)$ and $2^{1-2\sigma}\Gamma(2z + 1 + b)/\Gamma(2z + 2)$. These are the terms $m = 2n$ and $m = 2n + 1$ of the bilateral sum of $\Omega_r(2t; b)$, so the two rows merge into one lattice. The prefactor is $D_b(t) \sin \pi(g - t)/\sin \pi t = \sin \pi(t + \sigma) \cos \pi(t + \sigma) = \frac{1}{2} \sin \pi(2t + b)$. The sign of (464) gives $(-1)^r$, and the powers of two combine to $2^{-r}2^{r(1-2\sigma)} = 2^{-rb}$.

Proof at $d = 2$. Here $d_0 = 1$, $d_\ell = 2$ for $\ell \ge 1$ (by (461)), $2q = 1$ and $g = 1 - \sigma$. With $\phi(z) = \Gamma(z + \sigma)/\Gamma(z + 1)$ (local to this proof; not the φ of (501)) the factorized form in (464) and Gamma reflection give $f(\ell + \sigma + n + t) = \phi(\ell + n + t)$ and $f(-n - t) = -[\sin \pi(\sigma + t)/\sin \pi t]\, \phi(n + t)$, so $D_b(t)\, g_\ell(a_\ell + n + t) = -\sin^2 \pi(t + \sigma)\, \phi(n + t)\phi(\ell + n + t)$. With $\chi_m = \phi(m + t)^r$ the multiplicities turn $\sum_{n\in\mathbb{Z}} \sum_{\ell\ge 0} d_\ell \chi_n \chi_{n+\ell}$ into the symmetrization $(\sum_{m\in\mathbb{Z}} \chi_m)^2$. All sums converge absolutely, since $|\chi_m| \asymp |m|^{-r \operatorname{Re} g}$ and $r \operatorname{Re} g > 1$ is $\operatorname{Re} b < b_{\max}(r, 2)$. This is the second identity. Both identities are first obtained off the pole classes and then hold everywhere, since both sides are entire. □

Since $\Omega_r(t + 1; x) = (-1)^r \Omega_r(t; x)$ and Ω_r grows like $e^{r\pi|\operatorname{Im} t|}$ times a negative power, the frequencies of Ω_r lie in $\mathbb{Z} + \frac{r}{2}$ and have modulus at most $\frac{r}{2} - 1$. So at $d = 1, 2$ (539) gives (473) directly, and at $d = 1$ more: only the modes $k \equiv r \pmod 2$ occur, so $c_{4,1} \equiv 0$ and $c_{3,0} \equiv 0$.

Two amplitudes and their first-derivative relations. By (349), $\mathcal{N}_x^2 \Omega_4(t; x) = \mathcal{A}(x) + \mathcal{B}(x) \cos 2\pi(t + \frac{x}{2})$ with $\mathcal{N}_x = \Gamma(1 - x)/\Gamma(x)$, the N_x of (345), and Section 18.7 proves, at the single point $x = \frac{1}{2}$,

$$\mathcal{A}' = 2\pi\mathcal{B} - 2\log 2\, \mathcal{A}, \qquad \mathcal{B}' = \tfrac{3\pi}{2}\mathcal{A} - 2\log 2\, \mathcal{B}, \qquad \mathcal{A}(\tfrac{1}{2}) = \frac{8N_{\mathrm{e}}}{\pi}, \quad \mathcal{B}(\tfrac{1}{2}) = \frac{8M_{\mathrm{e}}}{\pi}, \tag{540}$$

with $M_{\rm e}=(B_-+B_+)/64$, $N_{\rm e}=(B_--B_+)/32$ of (338) and x-derivatives. These are first-derivative relations at $x=\frac12$ (from (351) and the elliptic and Landen identities (356)), not a differential system in x; equivalently $(4^x\mathcal{A})'=2\pi\,4^x\mathcal{B}$ and $(4^x\mathcal{B})'=\frac{3\pi}{2}\,4^x\mathcal{A}$ there. They are what makes the rim derivative collapse at $d=1$ and $d=2$: both dimensions sample the *same* Ω_4 at the *same* point $x=\frac12$ ($b=\frac12$ at $d=1$, $b/2=\frac12$ at $d=2$).

$d=1$**.** By (539) with $x=b$: $c_{4,0}=2^{-4b}\mathcal{A}(x)/\mathcal{N}_x^2$, $c_{4,1}\equiv0$, $c_{4,2}=2^{-4b}\mathcal{B}(x)/\mathcal{N}_x^2$, so $\mathcal{X}_0\equiv\mathcal{X}_{1/2}$ and only two samples are independent. At $b=\frac12$ ($N_b=\sqrt{\pi}$, $\varsigma_b=s_b=2^{-1/2}$, $\partial_b\log N_b=-\psi_0(\frac14)$), (540) gives

$$\begin{aligned}P_4(1)&=32\pi\mathcal{A}+40\pi\mathcal{B},\\ \mathcal{L}_1&=-2(\pi+\log2)\,P_4(1)+52\pi^2\mathcal{A}+64\pi^2\mathcal{B}=-(96\pi+512\log2)N_{\rm e}-(128\pi+640\log2)M_{\rm e},\end{aligned}\tag{541}$$

which is (107), that is (33): $\mathcal{L}_1=-\frac{3\mathsf{A}-\mathsf{D}}{6}P_4-\frac{2\mathsf{D}}{3}B_-$ with $\mathsf{A}=\pi+4\log2$, $\mathsf{D}=\pi$ and $B_-=8\pi(c_{4,0}+2c_{4,2})$. The frozen derivative is elementary in the two amplitudes, $\mathfrak{D}_L=\sqrt2\,(5\mathcal{X}_0'-\mathcal{X}_T')=\frac{\sqrt2\pi^2}{4}\big[(\frac{31\pi}{2}-8\log2)\mathcal{A}+(19\pi-10\log2)\mathcal{B}\big]$, and the single Horn derivative is $\mathfrak{h}_x'=4\sum_k c_k(\frac12)^4[\psi_0(k+\frac12)-\psi_0(\frac12)]=4\mathcal{S}_1$ of (351), with $c_k(x)$ of (345) and $\mathfrak{h}_x(1;b)={}_4F_3(b,b,b,b;1,1,1;1)$.

$d=2$**.** By (539) with $x=b/2$: $c_{4,0}=(\mathcal{A}^2+\frac12\mathcal{B}^2)/\mathcal{N}_x^4$, $c_{4,1}=2\mathcal{A}\mathcal{B}/\mathcal{N}_x^4$, $c_{4,2}=\mathcal{B}^2/(2\mathcal{N}_x^4)$, that is $\mathcal{X}_T=\mathcal{A}^2$, $\mathcal{X}_0=(\mathcal{A}+\mathcal{B})^2$, $\mathcal{X}_{1/2}=(\mathcal{A}-\mathcal{B})^2$ at $b=1$. Here $C=0$, so $D=0$ and ς_b passes through zero: the weight of $\mathcal{X}_T'$ in (494) vanishes, and $\partial_b\log(N_b/\mathcal{N}_x)=0$. Then

$$\mathcal{L}_2=-(6\mathcal{A}^2+8\mathcal{B}^2)+\frac{2}{\pi}(\mathcal{X}_0'-\mathcal{X}_{1/2}')=-(6\mathcal{A}^2+8\mathcal{B}^2)+\frac{4}{\pi}\big[\mathcal{A}'\mathcal{B}+\mathcal{A}\mathcal{B}'\big]=-\frac{16\log2}{\pi}\mathcal{A}\mathcal{B}=-2\log2\,P_4(2),$$

with x-derivatives on $\mathcal{A},\mathcal{B}$ (so $\partial_b=\frac12\partial_x$), which is (350) and (270). The motion of the trigonometric weights ($\partial_b\varsigma_b=-\frac{\pi}{2}$) produces $-(6\mathcal{A}^2+8\mathcal{B}^2)$, which the rotation terms $2\pi\mathcal{B}$, $\frac{3\pi}{2}\mathcal{A}$ of (540) cancel exactly; only the normalization 4^x survives. At $d=1$ the same relations leave $-\frac{2\mathsf{D}}{3}B_-$. If A and D in (33) are replaced by their general-d values A of (221) and D of (459), the B_- term vanishes at $d=2$ and the formula becomes exactly $\mathcal{L}_2=-\frac{A}{2}P_4(2)$ with $A=4\log2$, which is (270). At $d=3$ no analogue of B_- with elementary coefficients in the samples was found (an integer-relation search by lattice reduction at 155 digits, repeated at 125 digits, in `LRI_C5_d3`, found no relation of $\mathcal{L}_3$ with the $P_4(3)$ blocks; a numerical statement). In (537) the same $D=0$ leaves $3Q_2=2\pi^2\sum d_\ell T_{\ell n}^3$ (one unweighted Horn value, the complex Selberg value of (264)) and $P_4(2)=-8\pi^2\sum d_\ell T_{\ell n}^4\lambda_1$.

The half-shift identities at $d=3$ for a general rim exponent. The $d=3$ forms of $\mathcal{L}_d$ and Z_d below need two identities at rim exponents $b\neq\frac32$: $c_{4,1}(b)=2\mathfrak{a}_4(b)\mathfrak{b}_4(b)$, and a Gamma form of $c_{3,0}(b)$. Both follow from the half-shift proof of Section 19.4, repeated at a general b. The derivation is this project's own work: a Claude Code session and its subagents made it in the package `LRI_C5_d3/` (2 October 2026), where it was recorded as a sketch that follows that proof step by step, and three separate AI sessions per statement tried to refute the statements, re-deriving steps by hand or with codes of their own, without finding an error. The derivation is written out here; as printed it has not been reviewed. At $d=3$ one has $q=\frac34$, $\sigma=\frac{b}{2}$ and $g=\frac32-\sigma=\frac{3-b}{2}$. In this paragraph x, y, u and v are local, as in Section 19.4, and

$$h_b(z)=\frac{\Gamma(z+\frac{1-g}{2})}{\Gamma(z+\frac{1+g}{2})},\qquad u=t+\frac{1-g}{2}.$$

At $b=\frac32$, $\frac{1-g}{2}=\frac18$, h_b is the function h of (406) and $u=t+\frac18$.

Lattice. Put $k=\ell+n$, $j=n$, $x=k+u+\frac12$ and $y=j+u$. The first line of (464) gives

$$g_\ell(a_\ell+n+t;b)=-\frac{\sin\pi(g-t)}{\sin\pi t}\,h_b(x)\,h_b(y),\qquad d_\ell=2\ell+1=2(x-y),\tag{542}$$

which is (427) at $b=\frac32$. Since $g=\frac32-\sigma$, the prefactor of Ξ_r in (462) combines as $D_b(t)\cdot(-\sin\pi(g-t)/\sin\pi t)=\frac12\sin2\pi(t+\sigma)=-\sin\pi(u+\frac{1-g}{2})\sin\pi(u+\frac12+\frac{1-g}{2})$, so that

$$\Xi_r(t;b)=(-1)^r\Big[\sin\pi\big(u+\tfrac{1-g}{2}\big)\sin\pi\big(u+\tfrac12+\tfrac{1-g}{2}\big)\Big]^r\sum_{k\ge j}2(x-y)\,h_b(x)^r\,h_b(y)^r,$$

summed over all integers $k\ge j$. At $b=\frac32$ the bracket is $\sin\pi(t+\frac14)\sin\pi(t+\frac34)=\frac12\cos2\pi t$, the factor used in Section 19.4.

Half shift. For $m=0,1$ define the regularized bilateral moments

$$\Psi_{r,m}(v;b)=\sin^r\pi\big(v+\tfrac{1-g}{2}\big)\sum_{n\in\mathbb{Z}}(n+v)^m\,h_b(n+v)^r.$$

By the reflection $h_b(-z)=h_b(z)\sin\pi(z+\frac{1-g}{2})/\sin\pi(\frac{1-g}{2}-z)$ they are the functions (407) with $\omega=+1$ and $\frac18,\frac78$ replaced by $\frac{1-g}{2},\frac{1+g}{2}$; at $b=\frac32$ they are $\Psi^{(+)}_{r,m}$. Their series converge absolutely for $r\,\mathrm{Re}\,g>m+1$, since $|h_b(z)|$ is comparable to $|z|^{-\mathrm{Re}\,g}$. The shift $t\mapsto t+\frac12$ moves u by $\frac12$, exchanges the lattices $\mathbb{Z}+u+\frac12$ of x and $\mathbb{Z}+u$ of y, and changes the sign of the trigonometric bracket. If $r\,\mathrm{Re}\,g>2$, all sums involved converge absolutely, and the reindexing of Section 19.4 gives (429) at general b:

$$\Xi_r(t;b)-(-1)^r\,\Xi_r(t+\tfrac12;b)=2(-1)^r\big[\Psi_{r,1}(u+\tfrac12;b)\Psi_{r,0}(u;b)-\Psi_{r,0}(u+\tfrac12;b)\Psi_{r,1}(u;b)\big].$$

Finite modes. As in (430), $\Psi_{r,m}(v+1;b)=(-1)^r\Psi_{r,m}(v;b)$, $\Psi_{r,m}(-v;b)=(-1)^m\Psi_{r,m}(v;b)$, and $|\Psi_{r,m}(v;b)|=O(e^{r\pi|\mathrm{Im}\,v|}|\mathrm{Im}\,v|^{m+1-r\,\mathrm{Re}\,g})$ as $|\mathrm{Im}\,v|\to\infty$, now with the exponent $r\,\mathrm{Re}\,g$ in place of $\frac{3r}{4}$. Contour translation excludes the boundary frequencies when $m+1<r\,\mathrm{Re}\,g$. For $m\le1$ this is $\mathrm{Re}\,g>\frac12$, that is $\mathrm{Re}\,b<2=b_{\max}(4,3)$, for $r=4$, and $\mathrm{Re}\,g>\frac23$, that is $\mathrm{Re}\,b<\frac53=b_{\max}(3,3)$, for $r=3$, by (463); these are also the conditions of the half shift. Reflection then leaves the modes of (408),

$$\Psi_{4,0}=\mathfrak{a}_4(b)+\mathfrak{c}_4(b)\cos2\pi v,\quad \Psi_{4,1}=\mathfrak{b}_4(b)\sin2\pi v,\quad \Psi_{3,0}=\mathfrak{a}_3(b)\cos\pi v,\quad \Psi_{3,1}=\mathfrak{b}_3(b)\sin\pi v,\tag{543}$$

which define the amplitudes at general b; at $b=\frac32$ they are those of (408).

Fourier coefficients. By (468), $\Xi_r(t;b)=\sum_k c_{r,k}(\frac34,b)\cos2\pi k(u+\frac14)$, since $t+\frac b4=u+\frac14$. Inserting (543) in the half-shift identity gives $\Xi_4(t;b)-\Xi_4(t+\frac12;b)=-4\mathfrak{a}_4\mathfrak{b}_4\sin2\pi u$ and $\Xi_3(t;b)+\Xi_3(t+\frac12;b)=-2\mathfrak{a}_3\mathfrak{b}_3$, while the left sides are $-2c_{4,1}\sin2\pi u+2c_{4,3}\sin6\pi u$ and $2c_{3,0}-2c_{3,2}\cos4\pi u$. Hence, for $0<\mathrm{Re}\,b<2$ and $0<\mathrm{Re}\,b<\frac53$ respectively,

$$c_{4,1}(\tfrac34,b)=2\mathfrak{a}_4(b)\mathfrak{b}_4(b),\qquad c_{3,0}(\tfrac34,b)=-\mathfrak{a}_3(b)\mathfrak{b}_3(b),\tag{544}$$

and $c_{4,3}=c_{3,2}=0$, as (473) also gives. At $b=\frac32$ this is (431).

Gamma evaluations. Let $1<b<\frac53$ for $r=3$ and $1<b<2$ for $r=4$, so that $\frac12<g<1$. At the node $v=\frac{1-g}{2}$ the second term of (407) carries the factor $\sin^r\pi(\frac{1-g}{2}-v)=0$, and $h_b(n+\frac{1-g}{2})=\Gamma(n+1-g)/\Gamma(n+1)$. With $\mathfrak{U}_{r,m}=\sum_{n\ge0}(n+\frac{1-g}{2})^m h_b(n+\frac{1-g}{2})^r$, which is $\mathfrak{U}^{(+)}_{r,m}(\frac{1-g}{2})$ of (406) at general b,

$$\mathfrak{a}_3(b)=\frac{\sin^3(\pi g)}{\sin(\pi g/2)}\,\mathfrak{U}_{3,0},\qquad \mathfrak{b}_3(b)=\frac{\sin^3(\pi g)}{\cos(\pi g/2)}\,\mathfrak{U}_{3,1},\qquad \mathfrak{b}_4(b)=\sin^3(\pi g)\,\mathfrak{U}_{4,1}.$$

Dixon's sum [22, eq. 16.4.4] with $a=b=c=1-g$, and the Rogers–Dougall sum [22, eq. 16.4.9] with $(a,b,c,d)=(1-g,1-g,1-g,1-\frac g2)$, after inserting the canceled pair $1-\frac g2$, and with $(a,b,c,d)=(1-g,1-g,1-g,1-g)$ (in the notation of [22]; these b,c,d are not the rim exponent or the dimension) give

$$\mathfrak{U}_{3,0}=\frac{\Gamma(1-g)^3\,\Gamma(\frac{3-g}{2})\,\Gamma(\frac{3g-1}{2})}{\Gamma(2-g)\,\Gamma(\frac{1+g}{2})^2\,\Gamma(g)},\qquad \mathfrak{U}_{3,1}=\frac{1-g}{2}\,\frac{\Gamma(1-g)^3\,\Gamma(\frac{2-g}{2})\,\Gamma(\frac{3g-2}{2})}{\Gamma(2-g)\,\Gamma(g)\,\Gamma(\frac g2)^2},$$

$$\mathfrak{U}_{4,1}=\frac{1-g}{2}\,\frac{\Gamma(1-g)^4\,\Gamma(2g-1)}{\Gamma(2-g)\,\Gamma(g)^3}.$$

Their convergence conditions hold: $g > \frac{1}{3}$ for Dixon's sum, $g > \frac{2}{3}$ and $g > \frac{1}{2}$ for the two Rogers–Dougall sums. Duplication and reflection [26] then give

$$\mathfrak{a}_3(b)\mathfrak{b}_3(b) = \frac{\pi^5\,\Gamma(3g-2)}{\Gamma(g)^9}, \qquad \mathfrak{b}_4(b) = \frac{\pi^3\,\Gamma(2g-1)}{2\,\Gamma(g)^6}. \tag{545}$$

Both sides are analytic in b on the strips of (544), where the poles of $\Gamma(3g-2)$ and $\Gamma(2g-1)$ ($\mathrm{Re}\,b \geq \frac{5}{3}$ and $\mathrm{Re}\,b \geq 2$) do not occur, so (545) holds there by analytic continuation. At $b = \frac{3}{2}$ it is (411). Consequently $\hat{c}_0(3;b) = N_b^3 c_{3,0} = -N_b^3\pi^5\Gamma(3g-2)/\Gamma(g)^9$, and, with ν_b of the next paragraph, $\nu_b^{1/2}\mathfrak{b}_4 = \pi^3\Gamma(2g-1)/[2\Gamma(\sigma)^2\Gamma(g)^4]$, whose logarithmic b-derivative at $b = \frac{3}{2}$ is $\psi_0(\frac{3}{4}) - \psi_0(\frac{1}{2}) = \frac{\pi}{2} - \log 2$.

Checks (records in `LRI_C5_d3/`; numerical, not interval bounds). A code of one of the verifiers (`wf_vL3num/`, records `modes_b*.json`), which shares no summation routine with the evaluation code `wf_L3/gen3.py`, fits $\Psi_{4,0}$, $\Psi_{4,1}$ and Ξ_4 at 90 digits at $b = 0.4$, 0.8, 1.2, 1.5, 1.6, 1.8 and 1.9: the modes absent from (543), and $c_{4,3}$, are below $3\cdot10^{-69}$, $c_{4,1} - 2\mathfrak{a}_4\mathfrak{b}_4$ is below $2\cdot10^{-68}$, and at the six points other than $b = 0.4$ the function $W_4(3;b)$ assembled from the fitted coefficients agrees with an independent spectral code to $5.1\cdot10^{-50}$ or better. The product $\mathfrak{a}_3\mathfrak{b}_3$ of (545) holds at $b = 1.1$, 1.25, 1.4, 1.5, 1.6 and $1.3 + 0.2\mathrm{i}$ to $2\cdot10^{-60}$ at 60 digits, and, with a second, literal hypergeometric evaluator of the moments (not run at $b = 1.6$), to $6\cdot10^{-29}$ at 30 digits (`wf_Z3_gamma_verify/check_amp.py`, logs `check_dps30.log`, `check_dps60.log`). The checks of the separate review at $b = 0.8$, outside the range of the Gamma evaluation, and at other points are quoted after (551), and the $d = 3$ dictionary below is checked against the 160-digit values in Section 20.9.

$d = 3$. By (544), $c_{4,1}(b) = 2\mathfrak{a}_4(b)\mathfrak{b}_4(b)$ for $0 < b < 2$, with the moment amplitudes of (543), and (545) gives $\mathfrak{b}_4(b)$ in Gamma form, so that $\partial_b[\nu_b^{1/2}\mathfrak{b}_4] = (\frac{\pi}{2} - \log 2)\mathfrak{b}_4$ at $b = \frac{3}{2}$, where $\nu_b = [\Gamma(\frac{3}{2} - \frac{b}{2})/\Gamma(\frac{b}{2})]^4 = N_b^4/\Gamma(\frac{3}{2})^4$ equals one there. Let $X_0 = \Xi_4(0;\frac{3}{2})$; this is the X_0 of (421), not the sample $\mathcal{X}_0$ of (489). The dictionary to (489) is

$$\mathcal{X}_T = N_b^4 X_0, \qquad \mathcal{X}_0 - \mathcal{X}_{1/2} = 4N_b^4\mathfrak{a}_4\mathfrak{b}_4, \qquad \mathfrak{d}_H = \partial_b\big[\nu_b(4c_{4,0}+5c_{4,2})\big] = \frac{16}{\pi^2}\partial_b\big[N_b^4(4c_{4,0}+5c_{4,2})\big],$$
$$\mathfrak{d}_a = \partial_b\big[\nu_b^{1/2}\mathfrak{a}_4\big] = \frac{8}{\pi^2}\frac{\partial_b(N_b^4 c_{4,1})}{\mathfrak{b}_4} - \Big(\frac{\pi}{2} - \log 2\Big)\mathfrak{a}_4,$$

where the first equalities for $\mathfrak{d}_H$ and $\mathfrak{d}_a$ are their definitions. Inserting the dictionary into (493) gives $\mathcal{L}_3 = 4\pi P_4(3) - 2\pi^2 X_0 + 6\sqrt{2}\pi(\pi - 2\log 2)\mathfrak{a}_4\mathfrak{b}_4 - 2\pi\mathfrak{d}_H + 12\sqrt{2}\pi\mathfrak{b}_4\mathfrak{d}_a$. The derivative constants occur only in the combination $-2\pi\mathfrak{d}_H + 12\sqrt{2}\pi\mathfrak{b}_4\mathfrak{d}_a$. It has the same derivative content as $\widehat{P}'$, the b-derivative at $b = \frac{3}{2}$ of $\widehat{P}(b) = -2\pi\nu_b(4c_{4,0}+5c_{4,2}) + 6\sqrt{2}\pi\nu_b c_{4,1}$, in terms of which the 160-digit evaluation (`LRI_C5_d3/wf_L3/`) writes $\mathcal{L}_3 = 4\pi P_4(3) - 2\pi^2 X_0 + \widehat{P}'$. The two differ by value terms:

$$\frac{2}{\pi S^7}\mathfrak{D}_L = \widehat{P}' - 8\sqrt{2}\,N_b^4 c_{4,1} + 32\,N_b^4 c_{4,2} \qquad (b = \tfrac{3}{2}),$$

numerically -1332.48 against $\widehat{P}' = 274.55$; the identity holds to 10^{-57} with the 160-digit $d = 3$ values stored in `LRI_C5_d3/wf_L3/L3_final_values.json`. So $\mathfrak{D}_L$ is not literally $\widehat{P}'$. The split into a Horn-sector and an ordinary-sector constant follows the rank-two structure. The collapse fails at $d = 3$ because the lattice no longer reduces to a one-variable bilateral sum: the multiplicity $2\ell + 1$ gives rank two, the Dougall factor $\mathfrak{b}_4$ closes but $\mathfrak{a}_4'$ and the Horn-sector derivative do not (lattice reduction at 155 digits, `LRI_C5_d3`; numerical), and $D = -\pi \neq 0$.

Z_d at $d = 1, 2, 3$. In (498) the inputs are $\mathfrak{t}_3(b)$ and $\hat{c}_0(b)$. At $d = 1$, (539) makes Ξ_3 odd under $t \to t + \frac{1}{2}$, so $\hat{c}_0 \equiv 0$; with $\cos\frac{\pi}{2} = 0$ and $\mathfrak{t}_3(1;b) = {}_3F_2(b,b,b;1,1;1)$, which is Dixon's sum [22], this gives $Z_1 = \frac{4\pi^2}{3}\mathfrak{t}_3''(\frac{1}{2}) = QH$ of (224). At $d = 2$ the same symmetrization as in the proof

of (539) gives $\widehat{M}_3 \equiv 0$ (the midpoint sum is the square of a bilateral sum that is odd under $m \to -1-m$) and $\mathfrak{t}_3(2;b) = [{}_3F_2(\sigma,\sigma,\sigma;1,1;1)]^2$, Dixon's sum squared, because $T_{\ell n} = \chi_{\ell+n}\chi_n$ with $\chi_m = (\sigma)_m/m!$. Hence $\hat{c}_0 = -\pi^3 s_b^3 \mathfrak{t}_3/(1+\varsigma_b)$ and $u\mathfrak{t}_3 = \frac{1}{2}\gamma_C(b/4)^3/\gamma_C(3b/4)$, the complex Selberg value (264), here obtained from Dixon's theorem; $Z_2 = 4\mathsf{G}Q_2$ follows from it alone (proofs in the next paragraph, (548)–(550)). At $d=3$, $\hat{c}_0(b) = -N_b^3\pi^5\Gamma(3g-2)/\Gamma(g)^9$, by (544) and (545). Then $\hat{c}_0, \hat{c}_0', \hat{c}_0''$ are elementary, the zero-mode part of (498) is $\sqrt{2}\,(4\pi + 32\mathsf{G}/\pi)\,\hat{c}_0(\frac{3}{2})$, and $Z_3 = \frac{4\pi^2}{3}\mathfrak{t}_3''(\frac{3}{2}) - (4\mathsf{G} + \frac{\pi^2}{2})\frac{\Gamma(1/4)^{10}}{16\pi^{7/2}}$. The collapse stops at $\mathfrak{t}_3''$, because $\mathfrak{t}_3(3;b)$ is a genuine two-variable Horn function.

Proofs of the three-wheel statements at $d = 1, 2, 3$. $d=1$. For $0 < b < \frac{2}{3} = b_{\max}(3,1)$,

$$\mathfrak{t}_3(1;b) = {}_3F_2\left(\begin{matrix}b,b,b\\1,1\end{matrix};1\right) = \frac{\Gamma(1+\frac{b}{2})\,\Gamma(1-\frac{3b}{2})}{\Gamma(1+b)\,\Gamma(1-\frac{b}{2})^2\,\Gamma(1-b)}, \qquad W_3(1;b) = \frac{6\pi^2}{s_b^2}\,\mathfrak{t}_3(1;b). \tag{546}$$

Proof. At $d=1$ only the modes $k \equiv r \pmod 2$ occur (after (539)), so $c_{3,0} \equiv 0$, $\hat{c}_0 \equiv 0$, and (523) gives $W_3 = 6\pi^2\mathfrak{t}_3/s_b^2$. At $d=1$, $d_0 = d_1 = 1$ and $d_\ell = 0$ for $\ell \geq 2$ (the factor $(4q-1)_j$ in (301) vanishes for $j \geq 1$), $2q = \frac{1}{2}$ and $1-g = \frac{1}{2} + \sigma$. Legendre's duplication $(x)_n(x+\frac{1}{2})_n = 4^{-n}(2x)_{2n}$ [26] gives $T_{0n} = (b)_{2n}/(2n)!$ and $T_{1n} = (b)_{2n+1}/(2n+1)!$. Hence $\mathfrak{t}_3(1;b) = \sum_{m\geq 0}[(b)_m/m!]^3$, which is Dixon's well-poised ${}_3F_2$ with $a=b=c$ [22]; it converges for $\operatorname{Re} b < \frac{2}{3} = b_{\max}(3,1)$, and Dixon's theorem gives the Gamma product. □ Equivalently, the vanishing zero mode is the relation $\widehat{M}_3(1;b) = 8\pi^3 s_b^3\,\mathfrak{t}_3(1;b)/[\varsigma_b(1+\varsigma_b)^3]$ between the two lattices.

Z_1 from Dixon's theorem. Since $\hat{c}_0 \equiv 0$ for all b, $W_3(1;b) = 6\pi^2\mathfrak{t}_3(1;b)/s_b^2$ on $0 < b < \frac{2}{3}$. At $b = \frac{1}{2}$ one has $s_b = \varsigma_b = 2^{-1/2}$, and the logarithmic derivatives of the Gamma product (546), with the recurrences $\psi_m(z+1) = \psi_m(z) + (-1)^m m!\, z^{-m-1}$ and $\psi_1(\frac{1}{2}) = \frac{\pi^2}{2}$, are

$$(\log \mathfrak{t}_3)'(\tfrac{1}{2}) = \psi_0(\tfrac{3}{4}) - \psi_0(\tfrac{1}{4}) = \pi, \qquad (\log \mathfrak{t}_3)''(\tfrac{1}{2}) = \tfrac{5}{2}\psi_1(\tfrac{1}{4}) - \tfrac{1}{2}\psi_1(\tfrac{3}{4}) - \pi^2 = \pi^2 + 24\mathsf{G},$$

using $\psi_1(\frac{1}{4}) = \pi^2 + 8\mathsf{G}$ and $\psi_1(\frac{3}{4}) = \pi^2 - 8\mathsf{G}$. Further $\mathfrak{t}_3(1;\frac{1}{2}) = \Gamma(\frac{5}{4})\Gamma(\frac{1}{4})/[\Gamma(\frac{3}{2})\Gamma(\frac{3}{4})^2\Gamma(\frac{1}{2})] = \Gamma(\frac{1}{4})^4/(4\pi^3)$ by (546) and $\Gamma(\frac{3}{4}) = \sqrt{2}\,\pi/\Gamma(\frac{1}{4})$, so that (497) gives $Q_1 = 4\pi^2\mathfrak{t}_3(1;\frac{1}{2}) = \Gamma(\frac{1}{4})^4/\pi$. The first derivative of $W_3(1;b)$ at $b = \frac{1}{2}$ is $12\pi^2\mathfrak{t}_3[(\log\mathfrak{t}_3)' - \pi] = 0$, so the input $\partial_b W_3 = 0$ holds here by Dixon's theorem; the second derivative is $24\pi^4\mathfrak{t}_3 - 24\pi^3\mathfrak{t}_3' + 12\pi^2\mathfrak{t}_3'' = 12\pi^2\mathfrak{t}_3''$. Hence, with $\mathfrak{t}_3''/\mathfrak{t}_3 = (\log\mathfrak{t}_3)'' + [(\log\mathfrak{t}_3)']^2 = 2\pi^2 + 24\mathsf{G}$,

$$Z_1 = \frac{4\pi^2}{3}\,\mathfrak{t}_3''(1;\tfrac{1}{2}) = Q_1\Big(\frac{2\pi^2}{3} + 8\mathsf{G}\Big), \tag{547}$$

the value $Q_1 H$ of (224), here obtained from Dixon's theorem without Pattern T.

$d=2$. For $0 < \operatorname{Re} b < \frac{4}{3} = b_{\max}(3,2)$, $\widehat{M}_3(2;b) = 0$, and $\mathfrak{t}_3(2;b)$ is the square of Dixon's sum at $\sigma = \frac{b}{2}$:

$$\mathfrak{t}_3(2;b) = \Big[{}_3F_2\left(\begin{matrix}\sigma,\sigma,\sigma\\1,1\end{matrix};1\right)\Big]^2 = \Big[\frac{\Gamma(1+\frac{b}{4})\,\Gamma(1-\frac{3b}{4})}{\Gamma(1+\frac{b}{2})\,\Gamma(1-\frac{b}{4})^2\,\Gamma(1-\frac{b}{2})}\Big]^2 = \frac{s_b^2}{4\pi^2(1+2\varsigma_b)}\,\frac{\gamma_C(b/4)^3}{\gamma_C(3b/4)}, \tag{548}$$

with $\gamma_C(x) = \Gamma(x)/\Gamma(1-x)$ as in (264). Hence $c_{3,0} = c_{3,1}$ and

$$W_3(2;b) = u(b)\,\mathfrak{t}_3(2;b) = \frac{1}{2}\,\frac{\gamma_C(b/4)^3}{\gamma_C(3b/4)}, \tag{549}$$

which is the complex Selberg value (264). Note also $\mathfrak{t}_3(2;b) = \mathfrak{t}_3(1;\frac{b}{2})^2$.

Proof. At $d=2$: $g = 1-\sigma$, $d_0 = 1$, $d_\ell = 2$ for $\ell \geq 1$. Put $\varrho = \frac{1+\sigma}{2}$ and $\varphi_m = \Gamma(m+\varrho)/\Gamma(m+2-\varrho)$, $m \in \mathbb{Z}$ (both local to this proof). For $s = n + \frac{\ell+1}{2}$ one has $a_\ell + s = \ell + n + \varrho$, $a_\ell - s = \varrho - 1 - n$ and

$\beta_\ell \pm s = a_\ell \pm s + 2 - 2\varrho$. Gamma reflection [26] gives $\Gamma(\varrho-1-n)/\Gamma(1-\varrho-n) = -\varphi_n$, because $\sin\pi(1-\varrho-n)/\sin\pi(\varrho-1-n) = -1$. So $g_\ell(n+\frac{\ell+1}{2}) = -\varphi_{\ell+n}\varphi_n$ and

$$\frac{\widehat{M}_3}{N_b^3} = -\sum_{\ell\geq 0} d_\ell \sum_{n\in\mathbb{Z}} \varphi_{\ell+n}^3\varphi_n^3 = -\sum_{m\geq n} (2-\delta_{mn})\,\varphi_m^3\varphi_n^3 = -\Big(\sum_{m\in\mathbb{Z}} \varphi_m^3\Big)^2.$$

The same reflection gives $\varphi_{-1-m} = -\varphi_m$, so the bilateral sum vanishes. All sums converge absolutely: $|\varphi_m| \asymp |m|^{\operatorname{Re}\sigma-1}$, and $3(1-\operatorname{Re}\sigma) > 1$ is $\operatorname{Re} b < \frac{4}{3}$. Then $y_b = c_{3,0} - c_{3,1} = 0$ by (522).

At the pole node the same rearrangement applies. At $d=2$ one has $2q=1$ and $1-g=\sigma$, so $T_{\ell n} = \chi_{\ell+n}\chi_n$ with $\chi_m = (\sigma)_m/m!$, and $\mathfrak{t}_3 = \sum_{m\geq n\geq 0}(2-\delta_{mn})\chi_m^3\chi_n^3 = (\sum_{m\geq 0}\chi_m^3)^2$. The inner sum is Dixon's ${}_3F_2(\sigma,\sigma,\sigma;1,1;1)$, as at $d=1$ with $b\to\sigma$ [22]; it converges absolutely for $\operatorname{Re}(2-3\sigma) > 0$, again $\operatorname{Re} b < \frac{4}{3}$. This is the first equality of (548), and Dixon's theorem gives the second. For the third put $x = \frac{b}{4}$. Gamma reflection and $\Gamma(1+x) = x\Gamma(x)$ give

$$\Big[\frac{\Gamma(1+x)\Gamma(1-3x)}{\Gamma(1+2x)\Gamma(1-x)^2\Gamma(1-2x)}\Big]^2 \frac{\gamma_C(3x)}{\gamma_C(x)^3} = \frac{x^2\,\Gamma(3x)\Gamma(1-3x)}{[\Gamma(1+2x)\Gamma(1-2x)]^2\,\Gamma(x)\Gamma(1-x)} = \frac{\sin\pi x\,\sin^2 2\pi x}{4\pi^2\sin 3\pi x},$$

and $\sin 3\pi x = \sin\pi x\,(1+2\cos 2\pi x)$ turns the right side into $s_b^2/(4\pi^2(1+2\varsigma_b))$. Finally (496) with $\widehat{M}_3 = 0$ gives $W_3(2;b) = u(b)\mathfrak{t}_3(2;b)$, and $u(b)\,s_b^2/(4\pi^2(1+2\varsigma_b)) = \frac{1}{2}$. □

This is the square structure of the two-dimensional lattice (539) at the two nodes: at the self-mirror node the square vanishes, and at the pole node it is the square of Dixon's sum. It derives the complex Selberg value (264) from (496) and Dixon's theorem, without the complex Selberg integral. Its agreement with (264), which Section 16.7 obtains from that integral, is an exact check of (496) and of the theorems it rests on (Sections 20.1.1, 20.1.2, 20.1.3) at $d=2$ on the range $0 < b < \frac{4}{3}$ where both derivations hold.

Z_2 from Dixon's theorem. Since $\widehat{M}_3 \equiv 0$, (527) gives $9Z_2 = (u\mathfrak{t}_3)''(1)$ with $u\mathfrak{t}_3$ of (549). Its logarithmic derivative vanishes at $b=1$ and its logarithmic second derivative there is $\frac{3}{4}[\psi_1(\frac{1}{4}) - \psi_1(\frac{3}{4})] = 12\mathsf{G}$, as in Section 16.7, so $(u\mathfrak{t}_3)''(1) = 12\mathsf{G}\,(u\mathfrak{t}_3)(1)$; with $3Q_2 = u(1)\mathfrak{t}_3(2;1) = 2\pi^2\,\mathfrak{t}_3(1;\frac{1}{2})^2 = \Gamma(\frac{1}{4})^8/(8\pi^4)$ this gives

$$Z_2 = 4\mathsf{G}\,Q_2, \tag{550}$$

which is (265). Here it follows from Dixon's theorem alone, without the complex Selberg integral and without the input $\partial_b W_3 = 0$ (which holds here because the logarithmic derivative vanishes).

$d=3$. At $d=3$, $\Gamma(2q) = \Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$ and $N_b^3 = \frac{\pi^{3/2}}{8}\Gamma(g)^3/\Gamma(\sigma)^3$. With $c_{3,0}(b) = -\mathfrak{a}_3(b)\mathfrak{b}_3(b) = -\pi^5\Gamma(3g-2)/\Gamma(g)^9$, the Gamma form (544), (545), one has $\frac{4}{\pi s_b^3}\hat{c}_0 = -\frac{\pi^{11/2}\Gamma(3g-2)}{2s_b^3\Gamma(\sigma)^3\Gamma(g)^6}$ with $3g-2 = \frac{5-3b}{2}$, and (523) becomes

$$W_3(3;b) = \frac{6\pi^2\,\mathfrak{t}_3(3;b)}{\sin^2(\pi b/2)} - \frac{\pi^{11/2}\,\Gamma(\frac{5-3b}{2})}{2\sin^3(\pi b/2)\,\Gamma(\frac{3-b}{2})^6\,\Gamma(\frac{b}{2})^3}, \tag{551}$$

the general-b formula of the $d=3$ computation (`LRI_C5_d3/wf_Z3/`). By Section 20.6 the Gamma form holds on the whole strip $0 < \operatorname{Re} b < \frac{5}{3} = b_{\max}(3,3)$. A run of the separate review with the block code of `LRI_C5_general_d/Z/` confirms it numerically at $b = 0.8$, below the range $1 < b < \frac{5}{3}$ of the original Gamma evaluation in `LRI_C5_d3/`, at $b = 1.2$, 1.35, 1.6 and at the complex point $b = 1.45 + 0.05\mathrm{i}$ (`LRI_C5_general_d/review/Z-rigor/d3_gamma_offcenter.log`); an own code of that review confirms it at $b = 1.2$ and $b = 1.45 + 0.05\mathrm{i}$ (Section 20.9).

Z_3. Here $S = \frac{1}{\sqrt{2}}$, $C = -\frac{1}{\sqrt{2}}$ and $\cos\frac{3\pi}{2} = 0$. With the Gamma form, $\log(-\hat{c}_0) = \text{const} + \log\Gamma(3g-2) - 3\log\Gamma(\sigma) - 6\log\Gamma(g)$, so at $b = \frac{3}{2}$ ($g = \sigma = \frac{3}{4}$)

$$(\log(-\hat{c}_0))' = \tfrac{3}{2}[\psi_0(\tfrac{3}{4}) - \psi_0(\tfrac{1}{4})] = \tfrac{3\pi}{2}, \qquad (\log(-\hat{c}_0))'' = \tfrac{9}{4}[\psi_1(\tfrac{1}{4}) - \psi_1(\tfrac{3}{4})] = 36\mathsf{G},$$

and $\hat c_0(\frac32)=-\Gamma(\frac14)^{10}/(2^{15/2}\pi^{5/2})$. In (535) the coefficients are $3\pi/S^3=6\sqrt2\pi$, $-4C/S^4=8\sqrt2$ and $4/(\pi S^3)=8\sqrt2/\pi$, so the zero-mode part is $\frac{\sqrt2}{9}[6\pi+12\pi+8(\frac{9\pi}{4}+\frac{36\mathsf G}{\pi})]\hat c_0=\sqrt2\,(4\pi+\frac{32\mathsf G}{\pi})\,\hat c_0(\frac32)$ and

$$Z_3=\frac{4\pi^2}{3}\,\mathfrak{t}_3''(3;\tfrac32)+\sqrt2\Big(4\pi+\frac{32\mathsf G}{\pi}\Big)\hat c_0(\tfrac32)=\frac{4\pi^2}{3}\,\mathfrak{t}_3''(3;\tfrac32)-\Big(4\mathsf G+\frac{\pi^2}{2}\Big)\frac{\Gamma(1/4)^{10}}{16\pi^{7/2}},\tag{552}$$

the formula of the paragraph above. It uses the Gamma form (545) and the input $\partial_b W_3=0$ (through (535)). At $d=1,2$ the whole function $\mathfrak{t}_3(d;b)$ is a Gamma product (Dixon's sum and its square); at $d=3$, $\mathfrak{t}_3''(3;\frac32)$ remains as the one new constant.

	$\mathcal{L}_d$ (four-wheel)	Z_d (three-wheel)
$d=1$	duplication to $\Omega_4(\cdot;\frac12)$; relations (540); B_- term $\propto D$ remains	$\hat c_0\equiv0$; $\mathfrak{t}_3$ Dixon
$d=2$	square of $\Omega_4(\cdot;\frac12)$; relations (540); $D=0$ leaves only $-2\log2$	$\widehat{M}_3\equiv0$; $\mathfrak{t}_3$ Dixon squared (= Selberg)
$d=3$	rank two; only $\mathfrak{b}_4$ (Dougall) closes; $\mathfrak{D}_L$ not reduced	$\hat c_0$ Gamma; $\mathfrak{t}_3''$ not reduced
generic	$\mathfrak{D}_L$ not reduced	$\hat c_0$, $\hat c_0'$, $\frac19(u\mathfrak{t}_3''+v\widehat{M}_3'')$ not reduced

20.7 The five-loop coefficient

Inserting (492), (493), (497), (498) into (223) (exact algebra; A,R,U,V,X of (221)) gives

$$\begin{aligned}46656\,C_5(d)=\;&-1536\big[P_5(d)+\mathcal{K}_d\big]-1728Q_d^2+\Big[720A^2+1008R-\frac{1192\pi^2\cos(\pi d/2)}{S^2}\Big]Q_d\\&+k_0\mathcal{X}_0+k_{1/2}\mathcal{X}_{1/2}+k_T\mathcal{X}_T-\frac{1152}{\pi S^7}\mathfrak{D}_L+\frac{4768\pi^2}{S^2}\mathfrak{t}_3''\\&+\frac{2384}{3}\Big[\frac{\pi(1+C^2)}{S^5}\hat c_0-\frac{4C}{S^4}\hat c_0'+\frac{4}{\pi S^3}\hat c_0''\Big]\\&-15A^4-90A^2R+414AU-108AV-117R^2-75X,\end{aligned}\tag{553}$$

$$\begin{aligned}k_0&=(1+C)^2\Big[\frac{768A(1+2C)}{\pi S^7}+\frac{576(4+9C+8C^2)}{S^8}\Big],\\k_{1/2}&=(1-C)^2\Big[-\frac{768A(1-2C)}{\pi S^7}+\frac{576(4-9C+8C^2)}{S^8}\Big],\\k_T&=-\frac{1536AC}{\pi S^7}-\frac{1152(1+6C^2)}{S^8},\qquad Q_d=\frac{2\pi^2}{S^2}\mathfrak{t}_3+\frac{4}{3\pi S^3}\hat c_0.\end{aligned}$$

The term $384AP_4-576\mathcal{L}_d$ of (223) is $\sum_i k_i\mathcal{X}_i-\frac{1152}{\pi S^7}\mathfrak{D}_L$, and $7152Z_d$ contributes the $\mathfrak{t}_3''$, $\hat c_0$ and $\cos(\pi d/2)Q_d$ terms. The masters $P_5(d)$ and $\mathcal{K}_d$ enter (553) only through $-1536[P_5(d)+\mathcal{K}_d]$, as in (223). $P_5(d)$ is (319). $\mathcal{K}_d$ is defined by (238). It is evaluated by (260), whose derivation uses the mixed-trace step (input (M) of Section 21.1) and the extraction of $[\tau^2]$ under the trace, by the Mellin–Barnes form (588), which uses none of the inputs of Section 21.1, or by (565); any of these may be substituted in this term without changing the rest of (553). With (565),

$$-1536\,\mathcal{K}_d=384AU-576c_*U-384(\mathsf r_3-\mathsf s_3),$$

where $c_*=\psi_0(1)+\psi_0(2q)$ is that of (247) and $\mathsf r_3,\mathsf s_3$ are the Laurent coefficients of (555), with $\mathsf r_3=[\tau^3]\langle t_\ell^2R_\tau\rangle_d$ (sans-serif; not the r_3 of (259) or of (637)). This is exact given (M) and (J); (J) is the path form of (237), on which (223) itself rests. When $\mathsf r_3$ is taken from three-node samples on the jet circle, as in (586), hypothesis (H) of Section 21.8 is needed as well. On the range $\frac23<d<4$ of (553) the caveat for $d\le\frac47$ of Section 21.1 does not arise.

master	before	with (553)
$P_4(d)$	(323): 20 complex block values	3 samples: 5 two-variable Horn values $\mathfrak{h}_x, \mathfrak{h}_1, \mathfrak{h}_{1/2}, \mathfrak{s}_0, \mathfrak{s}_{1/2}$
$\mathcal{L}_d$	(298): 244 values, ≤ 6 variables	the same 5 functions and their first b-derivatives, in the one combination $\mathfrak{D}_L$
Q_d	(319): 3 complex samples	$\mathfrak{t}_3$, $\widehat{M}_3$: 2 one-variable ${}_7F_6(1)$ and 3 two-variable Horn values
Z_d	(298): 318 values, ≤ 6 variables	the 5 functions of the Q_d row with first and second b-derivatives, through $\hat{c}_0'$ and $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$
$\mathcal{K}_d$	(298): 386 values, ≤ 6 variables	Section 21: a polygamma part plus r_3, a third-order τ-jet of three-node samples of three two-variable Horn functions (586), under the inputs of Section 21.1; or one two-fold Barnes integral (588)
$P_5(d)$	(329)	not treated in this section

Apart from P_5 and $\mathcal{K}_d$, $C_5(d)$ needs eight hypergeometric functions with at most two summation indices at unit argument: five for $r = 4$ ($\mathfrak{h}_x, \mathfrak{h}_1, \mathfrak{h}_{1/2}, \mathfrak{s}_0, \mathfrak{s}_{1/2}$) and three for $r = 3$ ($\mathfrak{t}_3$ and the right-part and middle-part sums of $\widehat{M}_3$, Section 20.2). Split into one-variable rows and two-variable remainders, they give six one-variable rows, namely ${}_8F_7$, two ${}_9F_8$ and ${}_7F_6$ for $r = 4$ and two ${}_7F_6$ for $r = 3$, plus eight two-variable Horn values. $C_5(d)$ also needs the first b-derivatives of all eight functions, the second b-derivatives of the three functions for $r = 3$, and elementary numbers (π; $\sin, \cos$ of $\pi q, \pi q/2$; Γ, ψ_0, ψ_1 at $q/2, q, 3q/2, 2q, (1 \pm q)/2, (1 + 3q)/2, (3 \pm q)/2$). After the exact relations they enter as eight numbers per dimension: $\mathcal{X}_T, \mathcal{X}_0, \mathcal{X}_{1/2}, \mathfrak{D}_L$ and $\mathfrak{t}_3, \hat{c}_0, \hat{c}_0', \frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$. For Q_d and Z_d together these are four numbers. Since Q_d is one combination of $\mathfrak{t}_3$ and $\hat{c}_0$, (497), Z_d needs three numbers beyond the value of Q_d: one of $\mathfrak{t}_3, \hat{c}_0$, then $\hat{c}_0'$ and $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$.

20.8 Further reductions tried

- *Frozen weights* (exact): one derivative constant for $\mathcal{L}_d$, one first- and one second-derivative constant for Z_d (above).
- *Pole node only* (exact): (536)–(538), one Horn family per wheel, one Laurent order below (255)–(257). It trades functions for parameter derivatives, so in the unweighted Horn class it is not shorter than (553).
- *Node choices.* $t = 0$ is the only node with a single block; the reflection centers need two; a general node three. A real two-node form of W_4 using $t = 0$ exists only for $|\cos\frac{\pi b}{2}| \le \frac{1}{2}$ (second node $\cos 2\pi\tilde{t}_2 = 3\varsigma_b/(1 + 2\varsigma_b^2)$, a general point moving with b), so it does not shorten $\mathcal{L}_d$.
- *Summation theorems for the zero-spin rows.* The row of $\mathfrak{h}_x$ (and of $\mathfrak{t}_3$) is a well-poised ${}_{2r}F_{2r-1}(1)$ with $a = \sigma$, pairs $(\sigma, 1)$, $(1 - g, 2q)$, not very-well-poised; the rows at the reflection centers are not well-poised for generic b. Dixon and Dougall do not apply for $r \ge 3$; at $d = 1$ only the sum of the $\ell = 0, 1$ rows is summable. Contiguous relations do not connect the functions (shifts $x, \frac{1}{2}, 1$ are non-integer apart). Negative.
- *Gamma form of* $\hat{c}_0$ (true at $d = 1, 2, 3$): PSLQ at 80 digits at $d = \frac{4}{5}, \frac{8}{5}, \frac{12}{5}, \frac{16}{5}$ of $\log|\hat{c}_0|$ against log of $\pi, 2, 5, \frac{1+\sqrt{5}}{2}, \Gamma(\frac{1}{5}), \Gamma(\frac{2}{5})$ (all Gamma and trigonometric values at multiples of $\frac{1}{10}$): no relation with coefficients below 10^{11}. Negative, numerical.
- *Elementary closure of* $\mathfrak{D}_L$ (true at $d = 1, 2$): PSLQ at $d = \frac{4}{3}, \frac{8}{3}$ of $\mathcal{L}_d$ against $\{\mathcal{X}_T, \mathcal{X}_0, \mathcal{X}_{1/2}\} \times \{1, \sqrt{3}, \frac{\log 2}{\pi}, \frac{\log 3}{\pi}, \frac{\sqrt{3}\log 2}{\pi}, \frac{\sqrt{3}\log 3}{\pi}\}$ (95 digits): no relation stable under a change of precision; none with coefficients below about 10^4. Negative, numerical.

- *Polygamma closure of Z_d/Q_d and $\mathfrak{t}_3''/\mathfrak{t}_3$* (true at $d = 1, 2$, (547), (550)). A bounded PSLQ probe at 70 digits, on the 90-digit values of the two-node item of Section 20.9, at $d = 1.25, 1.5, 2.5$, against the 15 constants 1, π^2, $\psi_1(q)$, $\psi_1(2q)$, $\psi_1(\frac{q}{2})$, $\psi_1(1-\frac{q}{2})$, $\psi_1(\frac{3q}{2})$, $\psi_1(3q)$, $\psi_0(q)^2$, $\psi_0(q)\psi_0(2q)$, $\psi_0(2q)^2$, $\pi\psi_0(q)$, $\pi\psi_0(2q)$, $\pi^2/\sin^2\pi q$, $\pi\cot\pi q$ (coefficient bound 10^4): at $d = 1.5$ only a relation internal to the basis (target coefficient zero), at $d = 1.25$ and 2.5 only relations with coefficients of size 10^3–10^4, the size expected of spurious relations at this precision (`LRI_C5_general_d/Z/pslq_quick.py`, `LRI_C5_general_d/Z/results/pslq_quick.log`). Negative, numerical; not an independence proof. No closed form of $\hat{c}_0$ is known at generic d: it vanishes at $d = 1$, equals $-\pi^3 s_b^3 \mathfrak{t}_3/(1+\varsigma_b)$ with $\mathfrak{t}_3$ the square of Dixon's sum at $d = 2$, and is a Gamma product in b at $d = 3$. The $d = 2$ and $d = 3$ Gamma forms have different shapes, so a uniform Gamma product in (d, b) is not expected.

20.9 Checks

Computations on the symmetry computing cluster of the Perimeter Institute (`LRI_C5_general_d/specialize/results/`) and the separate review (`LRI_C5_general_d/review/`). The package replay, `make -f public.mk replay`, compares with the records only the numbers listed in `GFF/results/closed_forms_general/reduction_d/expected.json`. These do not include the numbers of Sections 20.1.4, 20.3 and 20.4, nor those of the items below on the general-b periodization, on the review probes and the product form in more detail, and on the two-node formula; those numbers are read from the records named there, and no replay target compares them with the text.

- (539) against the five-Horn samples: $d = 1$, $b = \frac{1}{2}, 0.45$ and $d = 2$, $b = 1, 1.1$, $r = 3, 4$: 10^{-44}–10^{-46}. Relations (540) and $\mathcal{A}, \mathcal{B}$ against $B_\pm$: $5 \cdot 10^{-46}$. $P_4(1), \mathcal{L}_1, P_4(2), \mathcal{L}_2$ from the modes: $\leq 2 \cdot 10^{-45}$.

- $d = 3$ dictionary against the 160-digit values of `LRI_C5_d3/wf_L3`: $c_{4,k}$, X_0, $\mathfrak{a}_4$, $\mathfrak{b}_4$, $\mathfrak{d}_H$, $\mathfrak{d}_a$, $P_4(3)$, $\mathcal{L}_3$ all to $1.2 \cdot 10^{-51}$ (50 digits); $\mathcal{L}_3$ formula $5 \cdot 10^{-64}$.

- (536) against the multi-node values ($d = 1.5, 2.5$; $b = d/2$, $d/2 + 0.04$; $r = 3, 4$): $5 \cdot 10^{-21}$–$3 \cdot 10^{-23}$ (Cauchy circle in the shift).

- (492), (493), (497), (498) against the five-Horn and two-node codes at $d = 1$, 2, 3, 0.8, 1.25, 1.5, 2.5, 2.75 (50 digits): $\leq 2.1 \cdot 10^{-48}$; the block $384AP_4 - 576\mathcal{L} - 1728Q^2 + (720A^2 + 1008R)Q + 7152Z$ and its reduced form (553): $\leq 1.2 \cdot 10^{-48}$; against closed forms ($d = 1, 2$) $\leq 4 \cdot 10^{-50}$, against the independent spectral references $4 \cdot 10^{-50}$ (50-digit reference values, 60-digit samples; `LRI_C5_general_d/review/joint/numerics-end-to-end`), at $d = 3$ $6 \cdot 10^{-34}$ (limited by the 33-digit Q_3 used).

- The explicit five-Horn forms (514) and (519) against the proved values at $d = 1, 2$, the 160-digit values at $d = 3$, the independent spectral references at nine non-integer dimensions and off-center spectral values at ten points, with per-function checks of the jet derivatives: Section 20.3.1.

- End to end, with $P_5(d)$ from (29), (334) and (425) (values in Section 16.8) and $\mathcal{K}_d$ from (565) (r_3 from the jets of Section 21.11):

$$C_5(1) = -731.139476624539825853869126582527231,$$
$$C_5(2) = -38.833645896225360552936982467532923 2,$$
$$C_5(3) = -5.850351420751327229217944230497744300111.$$

  They agree with $C_5(1)$ of (36), printed to 33 decimals in Section 14, to $3 \cdot 10^{-34}$ (all printed digits), with (272) to $2 \cdot 10^{-50}$ and with `LRI_C5_d3` to $5 \cdot 10^{-45}$ (`LRI_C5_general_d/review/joint/numerics-end-to-end`).

- *Top-mode theorem and its product form* (Section 20.1.2). The balance of every single factor and the factor $1+z$ in (461) are both essential. An unbalanced wheel with the same pole classes ($d=1$, $r=4$, $\sigma=0.2$, spoke half-exponent 0.25 instead of 0.3) has $|\gamma_{\pm 3}|$ equal to 1.7% of the largest mode, against 10^{-41} for the balanced kernel at 40 digits; an independent 50-digit code (`LRI_C5_general_d/review/topmode-numeric`) finds unbalanced top modes of $6.7\cdot 10^{-4}$ to $4.6\cdot 10^{-2}$, linear in the imbalance, and $1.9\cdot 10^{-3}$ when the factor $1+z$ is dropped. For balanced wheels the same code finds $|c_{r,r-1}|/\max_k |c_{r,k}|$ between $2.4\cdot 10^{-51}$ and $1.6\cdot 10^{-49}$ at 22 configurations with $r=2,\dots,6$, $0.8\le d\le 3.2$ and complex b, against surviving modes of $1.6\cdot 10^{-3}$ to 1. For products of different factors, a separate 48-digit lattice-sum code (`LRI_C5_general_d/review/lemmaG-numeric`; reruns at 54 digits on other compute nodes) finds top modes between $2.8\cdot 10^{-48}$ and $4.7\cdot 10^{-44}$ of the largest mode, at the level of the vanishing boundary modes $\gamma_{\pm r}$, at 12 mixed wheels ($r=3,4$; $d=1.3$, 2.4, 3.1; unequal b_i, real and complex shifts and offsets, one with $\operatorname{Re} g_1<0$), $6.5\cdot 10^{-47}$–$5.2\cdot 10^{-45}$ at four R kernels with real τ and $3.0\cdot 10^{-46}$ at one wheel; its reruns give at most $9.3\cdot 10^{-54}$. The two top modes vanish separately, also where $|\gamma_{r-2}|$ and $|\gamma_{2-r}|$ differ by up to 79% of the largest mode, and they stay below $2\cdot 10^{-47}$ (54 digits) at four analytically continued points on the boundary of (C) or outside it, two of them in the R family. A per-factor imbalance δ_{bal} at fixed total balance switches them on linearly, with relative top modes 0.047 down to 0.0013 for $\delta_{\mathrm{bal}}=0.15$ down to 0.0015. The per-spin code of Section 21.11 (`LRI_C5_general_d/K_general/note_fragment/topmode/perspin_topmode.py`; its modes agree with those of the 48-digit lattice-sum code to 10^{-15}) finds top modes of at most $6.5\cdot 10^{-16}$ of the largest mode, at the level of its extrapolation, for five R kernels (real and complex τ) and two random mixed kernels ($r=3,4$), against $3\cdot 10^{-3}$ to $6\cdot 10^{-2}$ for controls that keep the total balance but break (B), use multiplicities of another dimension, or shift one right argument. The short check `LRI_C5_general_d/foundation/check_topG.py` (30 digits, about 20 s on a laptop) finds 10^{-24} to $9\cdot 10^{-16}$, falling like the truncation error, at four mixed points with $r=3,4$ and $d=0.6$, 1.7, 3.3, three of them with one $\operatorname{Re} g_i<0$ (Step 7), against $9\cdot 10^{-10}$ to $2\cdot 10^{-5}$ for controls with right shifts ± 0.1 at fixed total balance; it also confirms (478) and (472) to 10^{-29}. No further mode vanishes in general: at the points tested, $c_{5,3}$ and $c_{6,4}$ are $1.6\cdot 10^{-3}$ to 0.16 of the largest mode. The proof review `LRI_C5_general_d/review/lemmaG-proof` tested Steps 1–6 numerically in unbalanced cases, and the endpoint exponents of (475) (item on the product form below).

- Independent review, own codes: top modes as quoted in the item above; the printed formulas of Sections 20.2–20.6 and (553) rederived symbolically with difference 0 ((538) by hand); off-center W_3, W_4 from independent samples (exact big-integer convolutions) to 44–53 digits; P_4 and $\mathcal{L}_d$ against spectral references to 31–39 and 31–38 digits; the coefficient functions of the five-Horn formula for $\mathcal{L}_d$ at 11 values of d to $1.2\cdot 10^{-38}$; Z_1, Z_2, Z_3 to 30.7, 35.2, 29.5 digits; the reduced C_5 block against the reference-master block to 30.4–35.6 digits at 8 values of d.

- *The general-b periodization* (Section 20.1). The checks use the code `wheel_general_b.py` with `hypsums.py` of `LRI_C5_general_d/foundation/`, driven by `run_validation.py` on the symmetry cluster (job 568940 on one 64-core AMD node, 50 requested digits and 15 guard digits; supplementary job 568959); records `foundation/results/val50.json` and `valx.json`, with the comparison `val50_analysis.txt` and `val50_summary.json` written by `analyze_val.py`. References: the closed forms of Section 1.1 at $d=1,2$ (for P_4, Q, $\mathcal{L}$ and Z), the archived $d=3$ values (≥ 90 digits), and at non-integer d the independent spectral values of `LRI_C5_general_d/references/reference_values.json` (41–50 digits at $b=\frac{d}{2}$ and 56 digits off it when these runs were made; the file was later rewritten with 50 significant digits).

  - At $b=\frac{d}{2}$, with the fixed nodes of (484) ($N=N_r$, complex discrete Fourier transform) and with the b-dependent nodes and the real cosine transform (313) (which at $b=\frac{d}{2}$ is

literally the rule of Section 18), $W_4 = P_4(d)$ and $W_3 = 3Q_d$ agree with the references to $2.2 \cdot 10^{-66}$–$1.5 \cdot 10^{-63}$ at $d = 1, 2, 3$ (the working precision) and to $9.7 \cdot 10^{-47}$–$3.8 \cdot 10^{-45}$ at $d = 0.8, 1.25, 1.5, 2.5, 2.75$, which is the accuracy of the references there (42–50 digits).

- At $b \neq \frac{d}{2}$ ($b = \frac{d}{2} \pm 0.05$, at $d = 0.8$ $b = \frac{d}{2} \pm 0.03$; $r = 3, 4$; the five non-integer d, 20 points), the fixed-node values agree with the spectral references to $3.1 \cdot 10^{-55}$–$3.0 \cdot 10^{-49}$, and the b-dependent nodes (not run at $d = 0.8$) to $3.3 \cdot 10^{-54}$–$7.7 \cdot 10^{-50}$. A second spectral code, `foundation/spectral.py`, which shares only the Hurwitz routine, agrees at six of these points ($d = 1.5$, $b = 0.7, 0.8$, $r = 3, 4$; $d = 2.5$, $b = 1.2$, $r = 3, 4$) to $8.2 \cdot 10^{-54}$–$1.9 \cdot 10^{-50}$. At $d = 2$ the exact $W_3(2; b)$ of (264) is reproduced at $b = 0.8$ and 1.2 to $1.3 \cdot 10^{-65}$ and $7.0 \cdot 10^{-65}$.
- Structure: $|c_{r,r-1}| / \max_k |c_{r,k}|$ lies between $4 \cdot 10^{-66}$ and $9 \cdot 10^{-49}$ at every point; at an extra point $t_* = 0.2371 + 0.0917\mathrm{i}$ the trigonometric polynomial reproduces $\Xi_r(t_*; b)$ to at most $1.0 \cdot 10^{-62}$ at $d = 1, 2, 3$ and $1.2 \cdot 10^{-49}$ at non-integer d; at $d = 1$ also $c_{4,1} = c_{4,3} = 0$ and $c_{3,0} = c_{3,2} = 0$, and at $d = 2$ $c_{3,0} = c_{3,1}$, to working precision, as (539) requires. The reduced nodes $N = N_r'$ agree with $N = N_r$ to $6.4 \cdot 10^{-65}$ ($d = 2$, $b = 1$, $r = 4$), $4.1 \cdot 10^{-63}$ ($d = 3$, $b = 1.5$, $r = 3$) and $1.2 \cdot 10^{-53}$–$2.5 \cdot 10^{-50}$ at four non-integer points.
- The b-dependence: Cauchy integrals of the finite-sample $W_r(d; b)$ over 96 points of the circle $|b - \frac{d}{2}| = \rho$ (phases $2\pi(k + \frac{1}{2})/96$, conjugate symmetry used), with ρ one third of the distance to the nearest singularity of W_r or to $b_{\max}$, so that the truncation error is below about 3^{-96}, give $\mathcal{L}_d = \partial_b W_4$ and $Z_d = \frac{1}{9}\partial_b^2 W_3$ at $b = \frac{d}{2}$ to $3.0 \cdot 10^{-63}$–$1.0 \cdot 10^{-46}$ relative at $d = 1, 2, 3$ and to $1.2 \cdot 10^{-46}$–$3.5 \cdot 10^{-45}$ at $d = 0.8, 1.25, 1.5, 2.5, 2.75$ (references of 41–50 digits). The first derivative $\partial_b W_3(d; \frac{d}{2})$, which vanishes by (239) with (244), comes out between 10^{-61} and 10^{-44}. The sentence after the table that follows this list compares these values with the other evaluations.
- The jet formula (485) evaluated literally at 30 digits (`foundation/sample_jets.py`, job 568967; the per-block values and b-derivatives at the nodes are stored in `foundation/results/sample_jets.json`), with the sample derivatives at the reduced fixed nodes taken from Cauchy circles of 48 points of radius one sixth of the distance to the nearest singularity of the samples: $\mathcal{L}_d$ and Z_d at $d = 1, 1.5, 2, 2.5, 3$ to $2.4 \cdot 10^{-31}$–$2.8 \cdot 10^{-30}$, and $W_r(d; \frac{d}{2})$ to $5.3 \cdot 10^{-32}$–$2.7 \cdot 10^{-30}$.
- Beyond $b_{\max}$ (Section 20.1.4): the continued Horn values against `spectral.py` at $(d, b, r) = (2.5, 1.8, 4)$, $(2.5, 1.55, 3)$, $(2.75, 1.95, 4)$ and $(2.75, 1.65, 3)$ to $4.7 \cdot 10^{-45}$, $4.9 \cdot 10^{-48}$, $2.1 \cdot 10^{-41}$ and $1.7 \cdot 10^{-49}$, limited by the cancellation near the pole of the $c_{r,k}$ at $b_{\max}$. At the resonant points $(d, b, r) = (\frac{4}{3}, \frac{2}{3}, 3)$ and $(2.5, 1.5, 4)$, where $r(d - b)/2 \in \mathbb{Z}$ and the circle mean is used, to $3.5 \cdot 10^{-61}$ and $1.5 \cdot 10^{-61}$.

d	$P_4(d) = W_4(d; \frac{d}{2})$	$3Q_d = W_3(d; \frac{d}{2})$
0.8	`4403.5659705473683681`	`259.83164202536385403`
1.25	`1120.1530930090638933`	`104.22818434903994884`
1.5	`632.7128141805458393`	`71.175347423099135692`
2.5	`118.19407871190995074`	`23.177096659039552764`
2.75	`84.737926139821720537`	`18.545946016773332171`

The table gives the finite-sample $P_4(d)$ and $3Q_d$ (fixed nodes), within $3.8 \cdot 10^{-45}$ of the spectral references. The values of $\mathcal{L}_d$ and Z_d from the b-circles at these five dimensions (`foundation/results/val50_summary.json`) are within $3.5 \cdot 10^{-45}$ of the references and agree in every printed digit with the table of $\mathcal{L}_d$ in Section 20.3.1 and the table of Z_d in the two-node item below.

- *Caution on the structural checks.* The periodicity, reflection and off-grid checks test the structure of Section 20.1.1, not the accuracy of the samples. They cannot see a sample

error that is itself a reflection-symmetric trigonometric polynomial of degree at most r; a t-independent error in a tail amplitude, multiplied by $D_b(t)^r$ and the Gamma prefactors, is of this kind. Only the top and boundary modes and the comparisons with independent references test accuracy. Errors of this kind occurred in the reviewer's own code and were caught only by the top and boundary modes and the references.

- *The review probes of Sections 20.1.1–20.1.3, in more detail.* `LRI_C5_general_d/review/topmode-numeric` (own lattice sums by two-dimensional Euler–Maclaurin summation with tanh–sinh quadrature, 50 digits, symmetry jobs 569215 and 569260–569313): besides the top modes quoted above, the boundary modes are $4 \cdot 10^{-51}$ to $2.1 \cdot 10^{-49}$; at $r = 2$, $d = 1.3$, $b = 0.6 + 0.1\mathrm{i}$, where $\alpha = 0.7$ barely exceeds the bound 0.65, Ξ_2 is constant to $2.7 \cdot 10^{-51}$; the weights $B_{r,k}$ of (479) agree with a direct series residue to $1.7 \cdot 10^{-51}$–$7.8 \cdot 10^{-49}$; W_r agrees with the spectral references to 47–50 digits; the stored $c_{r,k}$ of `foundation/results/val50.json` agree to $1.4 \cdot 10^{-50}$–$4.2 \cdot 10^{-50}$; the unbalanced controls quoted above stand against a floor of 10^{-30}. `LRI_C5_general_d/review/foundation-rigor` (double-precision brute-force lattice sums, `brute_xi.py`, 17–25 nodes, symmetry jobs 569242 and 569252): the five blocks of (483) agree with the brute-force sums to $2.5 \cdot 10^{-16}$–$1.4 \cdot 10^{-14}$, and the top modes are $1.9 \cdot 10^{-15}$ to $2.7 \cdot 10^{-14}$ in the fast cases ($d = 3.5$; $d = 0.9$, with negative multiplicities; complex b); in the slow cases the residuals halve when the cutoffs are doubled, so they are truncation error. These checks also cover $\sigma > 1$ at $d = 3.9$.

- *The checks of the product form, in more detail.* The proof check `LRI_C5_general_d/review/lemmaG-proof` (record `review.json`; the test scripts `t1_laplace.py` to `t5b_r4.py` saved no logs) tested: the Laplace forms (474) for complex g, also at $\mathrm{Re}\, w < 0$, to 10^{-20}–10^{-26}; the relation $\gamma_{1-r} = \kappa_+ s_1^+$ with s_1^+ from Step 4 and all its constants, against lattice sums in unbalanced cases, where the top mode is nonzero, so that the derivation and not only the final cancellation is tested: $7 \cdot 10^{-11}$ ($r = 2$, real), $2 \cdot 10^{-8}$ ($r = 2$, complex; limited by the lattice truncation), $2 \cdot 10^{-13}$ ($r = 2$, total balance only), 10^{-12} ($r = 3$, $g = (1, 2, 3)$, complex) and $3 \cdot 10^{-12}$ ($r = 4$, $g = (1, 1, 2, 2)$, $d = 2.9$, total balance only), and 10^{-22}–$2 \cdot 10^{-27}$ in balanced cases; the lower half-plane of Step 6, by the direct formula and by the conjugation, to $3 \cdot 10^{-12}$ at a complex unbalanced point with $r = 2$, where $|\gamma_1| \neq |\gamma_{-1}|$; and the endpoint exponents of (475), with fitted slopes -0.300 at $\varkappa = 0.7$ and -0.600 at $\varkappa = 0.4$ on both sides of 2π, bounded at $\varkappa = 1.2$, and bounded and oscillating at $\varkappa = 1$ with complex g_i. As a check that the mechanism does not prove too much, its formal extension to $r = 1$ at $d = 2$ predicts $\sum_{m \in \mathbb{Z}} \Gamma(m + a)/\Gamma(m + a + g) = 0$ for $\mathrm{Re}\, g > 1$, a known corollary of Dougall's bilateral sum. The 48-digit code `LRI_C5_general_d/review/lemmaG-numeric` (record `probe_report.json`) shares nothing with the general-b code, with `check_topG.py` or with the $\mathcal{K}_d$ code. It samples Ξ_T at $2r + 1$ points on a horizontal line and Fourier transforms; the boundary modes $\gamma_{\pm r}$, which vanish by Step 0, set the noise floor. It splits S_T into its right, middle and left parts, each a sum over a quadrant, evaluated by exact head sums, exact Abel–Plana tails and the corner double integral (tanh–sinh rules matched to the endpoint exponents, Gauss–Legendre panels); Gamma ratios come from its own asymptotic series, and `mpmath.zeta` is not used. Main runs are at 48 digits; every case inside (C) and every continued point was rerun at 54 digits with finer settings and a disjoint set of nodes (symmetry jobs 569336–569347 and 569461). Validation: closed-form quadrant sums to $1.1 \cdot 10^{-52}$; main against fine runs to $5 \cdot 10^{-44}$; the 50-digit modes of `topmode-numeric` at one wheel reproduced to all 40 printed digits, and the modes of the per-spin code of Section 21.11 at three of its points to its accuracy of 10^{-15}. Controls: a per-factor imbalance at fixed total balance ($\beta_1 \to \beta_1 + \delta_{\mathrm{bal}}$, $\beta_2 \to \beta_2 - \delta_{\mathrm{bal}}$) at a real mixed wheel with $r = 3$, $d = 1.3$ gives relative top modes 0.047, 0.030, 0.0116, 0.0041, 0.00128 for $\delta_{\mathrm{bal}} = 0.15$, 0.05, 0.015, 0.005, 0.0015, with $\gamma_{-2}/\delta_{\mathrm{bal}}$ converging to about $1.72 + 1.30\mathrm{i}$ with an $O(\delta_{\mathrm{bal}})$ correction, so the onset is linear, while $\gamma_{\pm 3}$ stay at 1.4–$1.6 \cdot 10^{-44}$; at the complex version of that point and complex δ_{bal} the two top modes are nonzero and unrelated (0.025 and 0.015 at $\delta_{\mathrm{bal}} = 0.05 + 0.03\mathrm{i}$, 0.0032

and 0.0018 at $\delta_{\mathrm{bal}} = 0.005 + 0.003\mathrm{i}$); an R kernel with right shifts ± 0.15, a control of the per-spin code, gives 0.0135 and 0.0029, as that code does; outside (C), at two mixed wheels with $2\varkappa - d = -0.06$ and (B) exact, the plain lattice sum diverges (largest $|\gamma_m|$ about 10^{102} at truncation depth 26 and 10^{134} at depth 36), while the analytically continued sum (the leading power of the divergent corner integral subtracted and restored analytically) stays a trigonometric polynomial of degree $r-1$ with vanishing top modes ($6.8\cdot 10^{-42}$–$3.7\cdot 10^{-41}$ at 48 digits, at most $6.1\cdot 10^{-50}$ at 54). The continuation reproduces a synthetic closed form outside its convergence region to $7.6\cdot 10^{-17}$, and continued and plain evaluations agree to $8\cdot 10^{-44}$ at a point inside (C). So (C) acts as a convergence hypothesis, as Step 7 suggests. One complex case was also evaluated on the lines $\operatorname{Im} t = 0.2$ and -0.25, with modes agreeing to $1.2\cdot 10^{-42}$. All error bars of this probe are empirical (convergence ladders, independent reruns, must-vanish modes), with no interval bounds. Endpoint exponents near 0.1 converge slowly in the quadrature step; the cases avoid that regime or were cross-checked. The continued points test only failures of $2\varkappa > d$ by less than one; the τ-circle of Section 21.8 at $d = 2.5$, where also $\varkappa > d-1$ fails, was not tested.

- *Two-node formula for the three-wheel* (Section 20.4). Code in `LRI_C5_general_d/Z/`, records in `LRI_C5_general_d/Z/results/`. `z_blocks.py` evaluates the five functions of (526) at any complex b with the accelerated Horn sums of `LRI_C5_general_d/foundation/hypsums.py` and `wheel_general_b.py` (exact finite part, Stirling tails, Euler–Maclaurin Hurwitz zeta values with relative precision, three-family convolution asymptotics, exact regularized multiplicity moments) and assembles (527), its reduced forms and (535). The main engine `z_jets.py` reruns the same algorithms with every parameter in the ring $\mathbb{C}[\varepsilon]/(\varepsilon^3)$, $b = \frac{d}{2} + \varepsilon$, so the b-derivatives of the five functions are exact derivatives of the accelerated evaluation, with no numerical differentiation; at jet order 0 it reproduces the block values of the general-b code to working precision. `z_cauchy.py` computes the same jets from Cauchy circles in b; it is a cross-check, and it is used at the resonant dimensions $\frac{3d}{4} \in \mathbb{Z}$ ($d = \frac{4}{3}, \frac{8}{3}$), where the accelerated sums have removable singularities at $b = \frac{d}{2}$. `Z_general.py` is the entry point (Z_d to 25 digits in about 5 s on one laptop core); the full 90-digit set of the table below took 12 s on one 64-core node of the symmetry cluster.

  References: at $d = 1, 2$ the closed forms (224) and (265); at $d = 3$ the 250-digit value of `LRI_C5_d3/wf_Z3/`; at $d = 0.8$, 1.25, 1.5, 2.5, 2.75 the independent spectral values of `LRI_C5_general_d/references/reference_values.json` (spectral code `LRI_C5_general_d/refvals/hp_general_d.py`); at $d = \frac{4}{3}$, 1.75, $\frac{8}{3}$, 3.5, 3.75 new runs of the same spectral code, copied unchanged to `LRI_C5_general_d/Z/spectral_ref/` (45 digits, 64-point b-circle). The spectral code shares no code with the finite-sample evaluation. At $d = 0.7$, just above the threshold $d = \frac{2}{3}$, the spectral run did not converge: its own $\partial_b W_3$ came out as -83, and its b-circle passes within 0.013 of the spectral convergence bound $b_{\mathrm{spec}} = d - \frac{1}{3} = 0.367$ of Section 20.1. There only a consistency check is available: a b-circle of the general-b finite-sample formula with five fixed complex nodes and all five blocks per node (`LRI_C5_general_d/Z/fd_circle_check.py`), which shares the Horn summation code and is therefore not independent.

d	Z_d (two-node formula, 36 digits shown)	reference	rel. dev.	$\lvert\partial_b W_3\rvert$	engine, digits
1	764.931030199588577040161778769488397	(224)	$4.0{\cdot}10^{-91}$	$1.1{\cdot}10^{-97}$	jets, 90
2	46.792583072329696656738825877981 0842	(265)	$3.9{\cdot}10^{-91}$	$3.5{\cdot}10^{-98}$	jets, 90
3	8.8209248930600924855368673281740 8087	$d=3$, 250 digits	$4.4{\cdot}10^{-91}$	$5.3{\cdot}10^{-98}$	jets, 90
0.8	1870.912407752717377849941062290 72511	spectral, 46 digits	$2.0{\cdot}10^{-45}$	$4.7{\cdot}10^{-70}$	jets, 90
1.25	312.281876942721713720205721011622893	spectral, 41 digits	$1.9{\cdot}10^{-46}$	$1.7{\cdot}10^{-73}$	jets, 90
1.5	149.908309291859708760442421585801485	spectral, 44 digits	$2.9{\cdot}10^{-45}$	$4.3{\cdot}10^{-73}$	jets, 90
2.5	18.787028196469216775295577525903 24	spectral, 44 digits	$1.2{\cdot}10^{-45}$	$2.1{\cdot}10^{-74}$	jets, 90
2.75	12.672126145166654840113185919395 7078	spectral, 44 digits	$1.2{\cdot}10^{-46}$	$1.8{\cdot}10^{-74}$	jets, 90
4/3	240.897293321182080101413442492954567	new spectral run	$1.0{\cdot}10^{-39}$	$4.3{\cdot}10^{-44}$	Cauchy, 60
1.75	80.431085634404387403291158005793 2935	new spectral run	$1.4{\cdot}10^{-40}$	$1.7{\cdot}10^{-73}$	jets, 90
8/3	14.394625919159569839228806546254 5566	new spectral run	$1.0{\cdot}10^{-39}$	$4.3{\cdot}10^{-43}$	Cauchy, 60
3.5	4.604077324137569309169079543257723 11	new spectral run	$1.4{\cdot}10^{-40}$	$1.4{\cdot}10^{-74}$	jets, 90
3.75	3.426911330730984835562506781385969 49	new spectral run	$1.4{\cdot}10^{-40}$	$1.5{\cdot}10^{-74}$	jets, 90
0.7	3193.828118855016790427553349977467 35	fixed-node circle*	$7.9{\cdot}10^{-40}$	$7.8{\cdot}10^{-70}$	jets, 90

The table gives Z_d from (527) against the references (records `LRI_C5_general_d/Z/results/Z_values.json`, `zj90.json`, `zc60x.json`, `compare_refs.log`). "rel. dev." is relative to the reference. The spectral references are good to the number of digits stated, or to about 10^{-39} for the new runs. The deviations of the five rows marked "spectral" were computed with the reference file as it stood before it was rebuilt with 50 printed digits (`compare_refs.log` predates the rebuild); the stated accuracies of the references did not change. $|\partial_b W_3|$ is the first rim derivative at $b=\frac{d}{2}$ computed from the same jets; it must vanish. *At $d=0.7$ the reference is the fixed-node b-circle described above, a consistency check only.

The two-node identity (496) itself was tested away from $b=\frac{d}{2}$, including complex b (`LRI_C5_general_d/Z/offdiag_check.py`, record `LRI_C5_general_d/Z/results/offdiag60.json`), against the 56-digit spectral values of `LRI_C5_general_d/references/reference_values.json` (evaluated at 50 digits) and against the five-fixed-node finite-sample formula (60 digits; not independent, as above). The values and deviations are in the last column of the table in the paragraph *Away from the center* of Section 20.3.1, next to those of the five-Horn formulas.

Block-level checks with the 90-digit exact jets (value, first and second b-derivative at $b=\frac{d}{2}$; `LRI_C5_general_d/Z/check_blocks.py`, `check_formulas.py`, record `results/check_formulas.log`): at $d=1$, $\mathfrak{t}_3$ against (546) to $1.0\cdot10^{-91}$, $1.5\cdot10^{-91}$, $7.3\cdot10^{-91}$, and $\hat{c}_0$ (which must vanish) at most $4.1\cdot10^{-90}$, $5.9\cdot10^{-89}$, $1.7\cdot10^{-87}$ in absolute value; at $d=2$, $\mathfrak{t}_3$ against (548) to $1.3\cdot10^{-92}$, $7.0\cdot10^{-91}$, $8.5\cdot10^{-91}$, and $\widehat{M}_3$ (which must vanish) at most $7.0\cdot10^{-105}$, $9.0\cdot10^{-98}$, $8.0\cdot10^{-97}$; at $d=3$, $\mathfrak{t}_3$ against the values of $\mathfrak{t}_3(3;b)$ and its first two derivatives at $b=\frac{3}{2}$ from the independent $d=3$ code (`LRI_C5_d3/wf_Z3/`) to $1.3\cdot10^{-90}$, $1.1\cdot10^{-90}$, $1.1\cdot10^{-90}$, and $\hat{c}_0$ against $-N_b^3\pi^5\Gamma(3g-2)/\Gamma(g)^9$ to $1.2\cdot10^{-90}$, $1.8\cdot10^{-90}$, $1.4\cdot10^{-90}$; at all d, the jets (528) against numerical differentiation to $5.2\cdot10^{-110}$. Also, at all twelve dimensions of the 90-digit runs, (532), (533), (534), (535) and (498) hold to about 10^{-73} and (497) to $3\cdot10^{-87}$; the $d=1$ relation between the two lattices holds to $4.6\cdot10^{-91}$, and the $d=3$ identity for $\mathfrak{t}_3'(3;\frac{3}{2})$ to $4.5\cdot10^{-91}$.

Summary. (i) The two-node identity holds away from $b=\frac{d}{2}$, also for complex b, to 10^{-48}–10^{-51} against the spectral values and to 10^{-55}–10^{-63} against the fixed-node formula. (ii) Z_1, Z_2, Z_3 agree with their references to $4\cdot10^{-91}$ with the exact-jet engine at 90 digits, and to about 10^{-60} at 60 digits. (iii) At the five dimensions with spectral references the agreement is 10^{-45}–10^{-46}, the stated precision of those references; at the five dimensions of the new spectral runs it is 10^{-39}–10^{-40}, the precision of those runs. (iv) The first rim derivative computed from the same jets vanishes to 10^{-70}–10^{-98} (exact jets, 90 digits) and to $4\cdot10^{-43}$ (Cauchy engine). (v) The exact-jet values at 60 and 90 working digits agree to 10^{-53}–10^{-61} at 12 dimensions, and the Cauchy-circle engine (64 points; its error is the aliasing of the circle) agrees with the 90-digit jets to 10^{-42}–10^{-55} at 10 dimensions. (vi) The block-level identities above hold to about 10^{-90}. The Cauchy circles in b use the same blocks and are a consistency check, not an independent evaluation. The separate review (`LRI_C5_general_d/review/Z-rigor/`, own codes) found in addition: the direct midpoint-lattice sum at $d=1$ against $8\pi^3 s_b^3\mathfrak{t}_3/[\varsigma_b(1+\varsigma_b)^3]$

to 10^{-37}–10^{-42} at $b = 0.2, 0.3, 0.5$; $\mathfrak{t}_3(2;b)$ against (548) to about 10^{-30} at five points including complex b, and the reflection identities behind $\widehat{M}_3(2;b) = 0$ to about 10^{-30}; at $d = 3$ an own evaluation of $\mathfrak{t}_3$ and $\widehat{M}_3$ reproducing the Gamma form of $\hat{c}_0$ to $3 \cdot 10^{-30}$ ($b = 1.2$) and $3 \cdot 10^{-28}$ ($b = 1.45 + 0.05\mathrm{i}$). That review did not check the resonant dimensions $d = \frac{4}{3}, \frac{8}{3}$, where the only reference is the spectral code. All these numbers are numerical evidence from asymptotic-series evaluations, not interval bounds; together with the top-mode tests above they also support (473) at $r = 3$, and at $d = 2$ the identity (549) is an exact check of it.

20.10 Status, limits and open points

- *Proved, with written proofs:* the finite Fourier theorem (Section 20.1.1), the top-mode theorem (473) and its product form (471), the residue functional (479) and the block decomposition; the identities (539); the sample formulas of Section 20.2 (Section 20.3, (504)–(512), with the derivative series (518); for the three-wheel, (521) and (522) of Section 20.4.2) and the analyticity of the samples in b. Checks: a separate review with its own codes (`LRI_C5_general_d/review/VERIFICATION.md`, proof review in `LRI_C5_general_d/review/foundation-rigor`) rederived by hand the proofs of Section 20.1 in the form they had before they were merged into this note: the reflected forms and the estimates with unspecified constants, the finite Fourier theorem, the top-mode proof in its earlier wheel-only form, the residue functional, the block decomposition and the fixed-node formula. It also rederived the identities (539); the proof of the $d = 2$ identity printed here was written after it. It did not cover the explicit constants of (465), (466) and (467), which come from Section 21.6 and have had no independent review, nor the contour in the proof of (479), which was changed at the merge (paragraph *The residue functional in full* of Section 20.1.3), nor the derivation of the $d = 3$ identities in Section 20.6. Within the review a second reader checked parts of the top-mode proof, and a 50-digit code of its own (`LRI_C5_general_d/review/topmode-numeric`) tested the statements. The review found no printed formula wrong and one wrong intermediate estimate (the endpoint exponent in Steps 3–4 of the top-mode proof); Steps 3–4 as printed here use exactly the hypothesis $b < b_{\max}$. A second separate check of the proof in product form, which contains the wheel case, is in `LRI_C5_general_d/review/lemmaG-proof`; it also tested the endpoint exponents of (475) numerically. Beyond the wheel case, the product form needs the product estimates with $\varkappa = \sum_j \operatorname{Re} g_i$, the endpoint bound in product form, the kernel identity (478) with the exchange on Δ, the conjugation step for the second top mode and the continuation in the exponents. The check in `LRI_C5_general_d/review/lemmaG-proof` rederived every step by hand and tested Steps 1–6 numerically in unbalanced cases; it found no fatal or major issue and seven minor wording and documentation items, which are applied here. A separate 48-digit lattice-sum code (`LRI_C5_general_d/review/lemmaG-numeric`; Section 20.9) confirms (471) numerically. The check in `LRI_C5_general_d/review/lemmaG-proof` covered the statements (471) and (473), including the R-family parameters used in Section 21.5 and the equivalence of (C) with $\tau \in \mathbb{H}_q$ there; Section 21.5 uses these statements. An earlier separate proof of the case $r = 3$, written for the $\mathcal{K}_d$ calculation, is superseded by the proof printed here; where the two differ, that review found the earlier threshold $|\operatorname{Im} t| \geq 2$ insufficient for complex parameters, which is why the proof here uses Y_0. In the case $r = 3$, Step 0 of the proof gives parts (a) (sufficiency), (b) and (c) of the growth theorem of Section 21.4; the sharpness of $\mathbb{H}_q$ and the explicit constants are proved in Section 21.6, which that review did not cover. The use of (471) for the R family of $\mathcal{K}_d$ at continued points also needs the continuation of the regularized samples in τ (hypothesis (H) of Section 21.8). The reviews `LRI_C5_general_d/review/L-rigor/` and `LRI_C5_general_d/review/Z-rigor/` read the derivations of Sections 20.3 and 20.4 in the package's own text. Details that this note writes out where that text only states them, for instance the majorants in the proof of (518) and the regularity of the middle and left parts at $t = 0$ in the proof of (521), were added at the merge and have not been reviewed.

- *Derived*, meaning exact algebra on the above: (491), (492)–(498), (536)–(538), the specializations of Section 20.6 at $d = 1, 2, 3$ (at $d = 3$ with the identities (544) and (545) of Section 20.6), and (553). All were rederived symbolically by the review of `LRI_C5_general_d/review/VERIFICATION.md`, the $d = 3$ specializations with the two identities as inputs, and are confirmed numerically (Section 20.9); (538) only through b-derivatives of (536), not by literal summation. The explicit forms (514), (515), (517) and (519) of Section 20.3 are derived in the same sense; their coefficients were also compared with direct symbolic differentiation (Section 20.3).

- *Fixed-node form and continuation.* The fixed-node formula (484) and the jets (485) are exact algebra on (479), confirmed numerically (Section 20.9); Sections 20.2–20.7 do not use them. The continuation of the samples beyond $b_{\max}$ (Section 20.1.4) is derived as the continuation of each block value at a fixed node; the residue (488) is an endpoint asymptotic derivation made in the review `LRI_C5_general_d/review/foundation-rigor`, not a written proof, and it is confirmed numerically and not used downstream.

- *Three-wheel, Section 20.4.* Proved here, given the theorems of Sections 20.1.1–20.1.3: the analyticity statement of Section 20.4.1, the two samples (521), (522), the two-node formula (496) for every b in the strip and every $0 < d < 4$, and its zero-mode form (523). Derived (exact differentiation and algebra, rederived symbolically by the review `LRI_C5_general_d/review/Z-rigor/`): (527), (529), (531), the equivalence of $\partial_b W_3(d; \frac{d}{2}) = 0$ with (532) and with (533), (534) and (535); of these only (534) and (535), hence (498), use $\partial_b W_3(d; \frac{d}{2}) = 0$. Proved here at $d = 1, 2$, from Dixon's theorem and without Pattern T: (546), (547), $\widehat{M}_3(2; b) = 0$, (548), (549) and (550). At $d = 3$, (551) and (552) follow with the Gamma form of $\hat{c}_0(3; b)$, proved in Section 20.6, whose printed derivation has not been reviewed. Open: whether $\mathfrak{t}_3''$, $\widehat{M}_3'$, $\widehat{M}_3''$ (equivalently the zero-mode jets) reduce further at generic d.

- *Conditional inputs.* (498) and the Z_d part of (553) use $\partial_b W_3(d; d/2) = 0$. This input is proved, by Pattern T (239) with (244); it is listed here only to record that (498) depends on it, while (497) and the full jet formula $9Z_d = (u\mathfrak{t}_3 + v\widehat{M}_3)''$ do not. The $d = 3$ specializations of Z_d and $\mathcal{L}_d$ use the Gamma form of $\hat{c}_0$ and the identity $c_{4,1}(b) = 2\mathfrak{a}_4(b)\mathfrak{b}_4(b)$. Both are proved in Section 20.6, which repeats the half-shift proof of Section 19.4 at a general rim exponent; separate AI sessions re-derived both statements, and the printed derivation has not been reviewed. They are confirmed numerically: the Gamma form also at $b = 0.8$ and complex b, and the $d = 3$ dictionary to $1.2 \cdot 10^{-51}$ (Section 20.9). If $\mathcal{K}_d$ is taken from (565), (553) is conditional on (M), and on (H) when r_3 is sampled on the jet circle (Section 21.12); with (588) it is not.

- *Numerical:* all values of Section 20.9, the evaluation of the Horn values and their derivatives (asymptotic series, no interval bounds), and every statement that a further reduction does not exist (Section 20.8; PSLQ bounds in stated bases, not independence proofs).

- *Ranges.* $\frac{1}{2} < d < 4$ for $P_4, \mathcal{L}_d$ and $\frac{2}{3} < d < 4$ for Q_d, Z_d, where $b = d/2$ lies in the strip; the continuation below these thresholds was not tested. At $d = \frac{4}{3}, \frac{8}{3}$ the accelerated evaluation of the Z_d jets has removable resonances and uses Cauchy circles in b. The continuation of the samples beyond $b_{\max}$ for $d > 2$ (a simple pole at $b_{\max}$ whose residue the functional annihilates; Section 20.1.4) is not used anywhere. For $C_5(d)$ the range is $\frac{2}{3} < d < 4$.

- *Outside this section:* $\mathcal{K}_d$ (Section 21, status in Section 21.12) and $P_5(d)$. Since $P_5(d)$ was evaluated to high precision only at $d = 1, 2, 3$ (in double precision also at $d = 2.4$ and $\frac{13}{5}$, Sections 18.6 and 17.7), $C_5(d)$ itself was evaluated only at $d = 1, 2, 3$ ($C_5(1) = -731.139476624\ldots$, $C_5(2) = -38.8336458962\ldots$, $C_5(3) = -5.85035142075\ldots$; full values in Section 20.9); the reduced block of (553) without $P_5, \mathcal{K}_d$ was evaluated at eight dimensions.

- *Open:* whether $\mathfrak{D}_L$, $\hat{c}_0$, $\hat{c}_0'$ and $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$ reduce further at generic d, as they do at $d = 1, 2$ and partly at $d = 3$.

21 The nonprimitive master $\mathcal{K}_d$ at generic dimension

This section evaluates the nonprimitive master $\mathcal{K}_d$ of (223) for generic $0 < d < 4$ (for $d \le \frac{4}{7}$ conditional on the separating-contour construction, Section 21.1). For $\mathcal{K}_d$ it gives, in place of the 386 Horn values of (298) with up to six variables each, a low-order parameter jet of two-variable functions, obtained by periodizing the family R_τ of (249) as in Sections 18 and 20.1. There are three steps.

1. The S family of (249) is elementary in every dimension (Section 21.2).
2. Given the inputs (M) and (J) of Section 21.1, the Gamma subtraction of (238) fixes the R family through order τ^2. The only non-elementary number in $\mathcal{K}_d$ is then the Laurent coefficient $\mathsf{r}_3 = [\tau^3]\langle t_\ell^2 R_\tau\rangle_d$ (Section 21.3).
3. $\langle t_\ell^2 R_\tau\rangle_d$ is a three-node sample of an entire periodic function with the Fourier modes $|k| \le 1$, on the half-plane $\mathbb{H}_q$ of (571) with $\operatorname{Re}\tau < \frac{3}{2}q$, $\tau \ne q$, and, by the continuation hypothesis (H) of Section 21.8, on the circle that gives the jet. Each sample consists of three two-variable Horn functions at argument $(1,1)$ (Sections 21.4–21.8).

By a second route, which uses none of these inputs, $\mathcal{K}_d$ is a polygamma part plus one two-fold Barnes integral (Section 21.10). The specializations to $d = 1, 2, 3$ reproduce (76), (271) and the 430-digit (260) value of $\mathcal{K}_3$ (Section 21.9). Section 21.11 collects the numerical checks, and Section 21.12 states what is proved, what is conditional and what is numerical. Codes and logs are in the package folder `LRI_C5_general_d/K_general/`. The material of this section is this project's own work, done in Claude Code sessions in that folder; the proof of the growth theorem in Section 21.6 was written there as a separate proof file and is merged here in full.

21.1 Conventions and two inputs from the master path

Put $q = d/4$. We use A, R, U, V, X of (221), $c_* = \psi_0(1) + \psi_0(2q)$ of (247), $C(\tau)$, R_τ, S_τ of (249), and d_ℓ, a_ℓ, b_ℓ of (240). For a rim exponent b (not to be confused with b_ℓ), $g_\ell(s;b)$ and N_b are those of (460); we write $a_\ell(b) = \ell/2 + b/4$ and $g_b = (d-b)/2$ for the a_ℓ and g there (so $a_\ell(b) + g_b$ is its β_ℓ), and keep a_ℓ, b_ℓ for their values at $b = d/2$, as in (240). For real b, $t_{\ell,b}(\nu) = N_b\, g_\ell(\mathrm{i}\nu; b)$ in (242); for complex b, $t_{\ell,b}$ and $\langle t_{\ell,b}^r\rangle_d$ denote the analytic continuation (460), in which $|\cdot|^2$ of (242) is replaced by $\Gamma(a+\mathrm{i}\nu)\Gamma(a-\mathrm{i}\nu)/(\Gamma(\beta+\mathrm{i}\nu)\Gamma(\beta-\mathrm{i}\nu))$ with $a = a_\ell(b)$, $\beta = a_\ell(b) + g_b$. Further, $g_\ell(s; d/2) = g_\ell(s)$ is the kernel of (300). We write $t_\ell(s) = \Gamma(2q) g_\ell(s)$, so that $t_\ell(\mathrm{i}\nu)$ is (240). The kernel R_τ is not even in s, so every trace is read on the full Mellin line,

$$\langle t_\ell^2 R_\tau\rangle_d = \Gamma(2q)^2 \sum_{\ell\ge 0} d_\ell \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\, g_\ell(s)^2 R_\tau(s). \tag{554}$$

For real τ the integrand of (554) takes complex-conjugate values at $s = \pm\mathrm{i}\nu$, so this is the real-part reading of (241) used in (248) and in (391) (Section 18.9.5); for complex τ it is the continuation of that trace, in the sense fixed in (M) below. For even kernels such as $t_\ell^2 S_\tau$ it coincides with (241). Put

$$\mathcal{X}_d(\tau) = \Big[\frac{\Gamma(q-\tau/2)}{\Gamma(q+\tau/2)}\Big]^2, \qquad F(\tau) = \langle t_\ell^2 (R_\tau - S_\tau)\rangle_d = \sum_k F_k \tau^k,$$
$$\mathsf{r}_k = [\tau^k]\langle t_\ell^2 R_\tau\rangle_d, \qquad \mathsf{s}_k = [\tau^k]\langle t_\ell^2 S_\tau\rangle_d. \tag{555}$$

The Laurent coefficients r_k are not the polygamma combinations r_j of (258). In this section $C(\tau)$ is that of (249) and is written with its argument; a bare C, also with a subscript such as C_ℓ, is a generic constant; $C_5(d)$ is the coefficient of (223). The constants S, C, D of (459) are not used, and $\mathcal{X}_d(\tau)$ is unrelated to the samples $\mathcal{X}_0, \mathcal{X}_{1/2}, \mathcal{X}_T$ of Section 20.2.

Two statements of Section 16 about the master path J_t of (238) enter below.

(M) *Mixed trace.* $J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_d$, as stated before (250), as an identity of meromorphic functions of τ. The expression (554) converges absolutely, with the Mellin line separating the left and right pole families of every spin, only for $\max(0, \frac{1}{2} - 2q) < \operatorname{Re}\tau < \frac{3}{2}q$. Outside this strip, and in particular near $\tau = 0$, $\langle t_\ell^2 R_\tau\rangle_d$ denotes its meromorphic continuation, which by (M) is $J_{2\tau}/C(\tau)$; the r_k of (555) are its Laurent coefficients at $\tau = 0$. The strip is nonempty iff $d > \frac{4}{7}$. For $d \le \frac{4}{7}$ there is no strip to continue from: the trace must then be defined with contours that separate the pole families for finitely many spins, which is only sketched (Section 21.7), and the statements below that involve $\langle t_\ell^2 R_\tau\rangle_d$ or r_k are conditional on that construction. (M) is proved at $d = 2$ for $0 < \tau < \frac{1}{2}$ ((391) in Section 18.9.5). At other d it is confirmed numerically (Section 21.11).

(J) *Jet.* $J_t - J_t^\Gamma = \mathcal{K}_d t^2 + O(t^3)$ on the path of (238). This is (237) restricted to that path, and at $d = 1$ it is (59). Section 16 derives (237) from completion symmetry and analyticity; it is listed here as an input because its lower orders t^0, t^1 on the path are used explicitly, in (564) and (589). At $d = 2$ its low orders are rederived in Section 18.8.4, and at non-integer d it is confirmed numerically (Section 21.11).

21.2 The S family is elementary

Closed form of the S family. Let $0 < d < 4$ and $\max(0, \frac{1}{2} - 2q) < \operatorname{Re}\tau < 2q$; this strip is nonempty for $d > \frac{1}{2}$. Then

$$\boxed{\langle t_\ell^2 S_\tau\rangle_d = C(\tau)\,\mathcal{X}_d(\tau)^2 = \frac{\Gamma(2q)\,\Gamma(\tau)\,\Gamma(q-\tau/2)^4}{\Gamma(2q-\tau)\,\Gamma(q+\tau/2)^4},} \tag{556}$$

and everywhere else $\langle t_\ell^2 S_\tau\rangle_d$ denotes the meromorphic continuation of the right side in τ and d. Equivalently $C(\tau)\langle t_\ell^2 S_\tau\rangle_d = C(\tau)^2\mathcal{X}_d(\tau)^2$, the first term of (250); (556) makes that term an explicit statement about the S family.

Proof. Put $b' = d/2 - \tau$. Then $a_\ell(b') = a_\ell - \tau/4$, $a_\ell(b') + g_{b'} = b_\ell + \tau/4$ and $N_{b'} = \Gamma(2q)\Gamma(q + \tau/2)/\Gamma(q - \tau/2)$, so (249) gives the kernel identity

$$t_\ell(s)^2 S_\tau(s) = \Gamma(2q)^2 g_\ell(s;b')^2 = \mathcal{X}_d(\tau)\,\big[N_{b'}\,g_\ell(s;b')\big]^2. \tag{557}$$

Hence $\langle t_\ell^2 S_\tau\rangle_d = \mathcal{X}_d(\tau)\langle t_{\ell,b'}^2\rangle_d = \mathcal{X}_d(\tau)W_2(d;b')$, the balanced two-wheel of (242). A two-wheel is a single convolution, so $W_2(d;b') = B_d(d-b', 2b')$ by (230); this is the identity used before (250), and spectrally it is Parseval's identity for the dilatation-invariant operator with eigenvalue $t_{\ell,b'}$. For complex b' it extends because both sides are holomorphic in τ on the strip, the trace by locally uniform absolute convergence. With $(d-b', 2b') = (2q+\tau, 4q-2\tau)$, (230) gives $B_d = C(\tau)\mathcal{X}_d(\tau)$. Finally, $|g_\ell(\mathrm{i}\nu;b')| \asymp (1+\ell+|\nu|)^{-2\operatorname{Re} g_{b'}}$ with $g_{b'} = q + \tau/2$, so the ν-integral and the spin sum converge absolutely iff $4\operatorname{Re} g_{b'} > \max(d, 1)$; together with $\operatorname{Re} b' > 0$, which keeps the $\ell = 0$ contour between its two pole families, this is the stated strip. □

Laurent form. Expanding $\log\Gamma$ gives $\log[\tau\langle t_\ell^2 S_\tau\rangle_d] = \sum_{h\ge1} x_h\tau^h/h!$ with

$$\begin{gathered} x_h = \psi_{h-1}(1) - (-1)^h\psi_{h-1}(2q) - [h \text{ odd}]\,2^{3-h}\psi_{h-1}(q), \\ (x_1, x_2, x_3, x_4) = (c_* - 4\psi_0(q),\ -R,\ V-U,\ -X). \end{gathered} \tag{558}$$

Thus $\mathsf{s}_{m-1} = [u^m]\exp\sum_h x_h u^h/h!$, which is the polynomial $E_m(x_1,\dots,x_m)$ of (316):

$$\begin{gathered} \mathsf{s}_{-1} = 1, \qquad \mathsf{s}_0 = x_1, \qquad \mathsf{s}_1 = \tfrac{1}{2}(x_1^2 + x_2), \qquad \mathsf{s}_2 = \tfrac{1}{6}(x_1^3 + 3x_1x_2 + x_3), \\ \mathsf{s}_3 = \tfrac{1}{24}(x_1^4 + 6x_1^2x_2 + 4x_1x_3 + 3x_2^2 + x_4). \end{gathered} \tag{559}$$

Remarks. (i) At $d=2$, (556) is the value of $\langle t_\ell^2 S_\tau\rangle_2$ obtained in (390) (Section 18.9.5) from Gauss's sum and the bilateral (Dougall) identity of (387) on the square lattice. At generic d no Dixon, Dougall or Thomae evaluation is needed. (ii) By (557) the spin-summed periodization of $t_\ell^2 S_\tau$ is a balanced two-wheel of rim exponent $d/2-\tau$. For $\tau\in\mathbb{H}_q$ of (571) its Ξ is constant in t, by the case $r=2$ of (471) (here r counts the factors of the product; it is unrelated to the Laurent coefficients r_k) with $\alpha_i=\beta_i=-\tau/4$ and $g_i=q+\tau/2$, that is (472) with $b_i=d/2-\tau$; (556) does not use this. (iii) For a rim exponent b with $0<\operatorname{Re} b<b_{\max}(2,d)$ put $\sigma=b/2$, as in Section 20, so that $g_b=2q-\sigma$. By the case $r=2$ of (473), $\Xi_2(t;b)=c_{2,0}(q,b)$ does not depend on t. The residue functional (479) at $r=2$, where $B_{2,0}(\sigma)=-2\pi\cot\pi\sigma$, together with $W_2(d;b)=B_d(d-b,2b)$ from (230) and N_b of (460), gives the constant:

$$\Xi_2(t;b)=c_{2,0}(q,b)=\frac{\pi\sin^3(\pi\sigma)\,W_2(d;b)}{2\cos(\pi\sigma)\,N_b^2}=\frac{\pi\sin^3(\pi\sigma)\,\Gamma(\sigma)^4\,\Gamma(2q-2\sigma)}{2\cos(\pi\sigma)\,\Gamma(2q)\,\Gamma(g_b)^4\,\Gamma(2\sigma)}. \tag{560}$$

At $b=d/2-\tau$, that is $\sigma=q-\frac{\tau}{2}$, with $\tau\in\mathbb{H}_q$ and $\operatorname{Re}\tau<2q$, this reads $\Gamma(2q)^2\Xi_2=\pi\sin^3(\pi\sigma)\langle t_\ell^2 S_\tau\rangle_d/(2\cos\pi\sigma)$, by (556). With the Gamma-reflected form of $g_\ell(a_\ell+n+t)$ in (464), $D_b^2 S_2=c_{2,0}$ becomes a bilateral identity, for $t\notin\mathbb{Z}\cup(-\sigma+\mathbb{Z})\cup(g_b+\mathbb{Z})$:

$$\sum_{\ell\ge0}d_\ell\sum_{n\in\mathbb{Z}}\Big[\frac{\Gamma(\ell+n+\sigma+t)}{\Gamma(\ell+n+2q+t)}\Big]^2\Big[\frac{\Gamma(n+1-g_b+t)}{\Gamma(n+1+t)}\Big]^2=\frac{c_{2,0}(q,b)}{\sin^2\pi(t+\sigma)\,\sin^2\pi(g_b-t)}. \tag{561}$$

At $d=2$ the left side is the square of a single bilateral sum, and (561) is the square of Dougall's ${}_2H_2$ sum, an instance of which is the last identity of (387) (with $x_\tau=\sigma$ and $\eta=t$); at generic d it is its analogue for the multiplicities d_ℓ. The package record `LRI_C5_general_d/K_general/S_part/onemode.jsonl` (code `onemode_check.py` there) confirms the constancy of Ξ_2 at $b=d/2-\tau$ and the value (560), at four nodes each, to $4\cdot10^{-17}$ at seven points $0.8\le d\le3.5$ and to $2\cdot10^{-39}$ at two points at $d=1$.

21.3 Low orders of the R family and the single new number

At $t=2\tau$, (230) gives $B_d(2q,4q-2\tau)=C(\tau)\Gamma(q-\tau)/\Gamma(q+\tau)$ and $B_d(2q+4\tau,4q-2\tau)=C(\tau)\Gamma(q-2\tau)\Gamma(q+\tau)/[\Gamma(q+2\tau)\Gamma(q-\tau)]$, so the Gamma interpolation of (238) is

$$J^{\Gamma}_{2\tau}=C(\tau)^2\frac{\Gamma(q-\tau)^2}{\Gamma(q+\tau)^2}+\frac{C(\tau)}{\tau}\Big[\frac{\Gamma(q-\tau)}{\Gamma(q+\tau)}-\frac{\Gamma(q-2\tau)\Gamma(q+\tau)}{\Gamma(q+2\tau)\Gamma(q-\tau)}\Big]. \tag{562}$$

Expanding $\log\Gamma$ gives the exact Laurent series

$$\Delta(\tau):=J^{\Gamma}_{2\tau}-C(\tau)^2\mathcal{X}_d(\tau)^2=\frac{3}{2}U\tau+AU\tau^2+O(\tau^3). \tag{563}$$

The orders $\tau^{-2},\tau^{-1},\tau^0$ cancel identically, and $[\tau^2]\Delta=AU$ is -4 times the difference $-AU/4$ of the t^2 coefficients stated after (250). A symbolic check for general q is `LRI_C5_general_d/K_general/note_fragment/fragment_checks/check_delta_symbolic.wls`.

Low orders. Assume (M) and (J). Then $F(\tau)=\frac{3}{2}U\tau^2+F_3\tau^3+O(\tau^4)$; that is,

$$\mathsf{r}_{-1}=1,\qquad \mathsf{r}_0=\mathsf{s}_0,\qquad \mathsf{r}_1=\mathsf{s}_1,\qquad \mathsf{r}_2=\mathsf{s}_2+\tfrac{3}{2}U, \tag{564}$$

and

$$\boxed{\mathcal{K}_d=\frac{U}{8}\big[\psi_0(1)+\psi_0(2q)+4\psi_0(q)\big]+\frac{\mathsf{r}_3-\mathsf{s}_3}{4},} \tag{565}$$

with s_3 the polygamma polynomial (559). Equivalently,

$$\mathcal{K}_d=-\frac{AU}{4}+\frac{1}{4}[\tau^2]\Big\{C(\tau)\langle t_\ell^2 R_\tau\rangle_d-C(\tau)^2\mathcal{X}_d(\tau)^2\Big\}. \tag{566}$$

Equation (566) uses only (M), (563) and the definition (238), since $[\tau^2](J_{2\tau} - J^{\Gamma}_{2\tau}) = 4\mathcal{K}_d$ for the Laurent coefficients whatever the lower orders are; here (J) enters only (564) and (565).

Proof. By (M) and (556), $J_{2\tau} = C(\tau)^2\mathcal{X}_d(\tau)^2 + C(\tau)F(\tau)$. By (J) at $t = 2\tau$, $J_{2\tau} - J^{\Gamma}_{2\tau} = 4\mathcal{K}_d\tau^2 + O(\tau^3)$. Subtracting and using (563),

$$C(\tau)F(\tau) = \tfrac{3}{2}U\tau + (AU + 4\mathcal{K}_d)\tau^2 + O(\tau^3).$$

This is (566). Since $C(\tau) = \tau^{-1} + c_* + O(\tau)$, it follows that $F(\tau) = \frac{3}{2}U\tau^2 + [AU + 4\mathcal{K}_d - \frac{3}{2}c_*U]\tau^3 + O(\tau^4)$, which is (564). With $F_3 = \mathsf{r}_3 - \mathsf{s}_3$ and $A = c_* - 2\psi_0(q)$ this gives (565). □

The spectral moment. On the Mellin line $\mathrm{Re}(R_\tau - S_\tau) = E_2\tau^2 + O(\tau^3)$, with the weight $E_2 = -h^2/2 - 5j/8$ of (247), as displayed after (250) (h, j the weights of (247), not the index h of (558) or the node index j of (580); this E_2 is not the polynomial E_2 of (316), which gives s_1 in (559)). If $[\tau^2]$ may be taken under the trace (the step used in the derivation of (248), justified only at $d = 2$, by the corollary of Section 18.9.2), then $F_2 = \langle t_\ell^2 E_2\rangle_d$, and (564), which assumes (M) and (J), gives

$$\boxed{\langle t_\ell^2 E_2\rangle_d = \frac{3}{2}\,\psi_2\Big(\frac{d}{4}\Big).} \tag{567}$$

At $d = 2$ this is the value $-21\zeta(3)$ of (394); by (73) it is $-3\pi^3 - 84\zeta(3)$ at $d = 1$ and $3\pi^3 - 84\zeta(3)$ at $d = 3$.

Consequence. Equation (565) puts all non-elementary content of $\mathcal{K}_d$ into the single number r_3. The following sections evaluate $\langle t_\ell^2 R_\tau\rangle_d$, and hence r_3, by finite sampling.

21.4 The R family: lattice form, growth and finite Fourier space

Lattice form. Let $f^{\mathrm{R}}_\ell(s) = \Gamma(2q)^2 g_\ell(s)^2 R_\tau(s)$ be the summand of (554). In Sections 21.4–21.9, t is the periodization variable, not the path parameter of J_t. Direct substitution gives, for $\ell \geq 0$ and $n \in \mathbb{Z}$,

$$\begin{gathered} f^{\mathrm{R}}_\ell(a_\ell + n + t) = \Gamma(2q)^2\,\Phi_t(\ell + n)\,\Psi_t(n), \\ \Phi_t(m) = \frac{\Gamma(m+q+t)^3}{\Gamma(m+2q+t)^2\Gamma(m+q+\tau+t)}, \qquad \Psi_t(n) = \frac{\Gamma(-n-t)^2\Gamma(q-\tau-n-t)}{\Gamma(q-n-t)^3}. \end{gathered} \tag{568}$$

Reflection gives $\Psi_t(n) = T_{\mathrm{R}}(t)\Phi_{t+1-2q}(n)$ with $T_{\mathrm{R}}(t) = \sin^3\pi(q-t)/[\sin^2(\pi t)\sin\pi(q-\tau-t)]$. The summand is therefore one Gamma ratio $\widetilde{\Phi}(w) = \Gamma(w+q)^3/[\Gamma(w+2q)^2\Gamma(w+q+\tau)]$ on the two lattices $w = m+t$ and $w = n+t+1-d/2$, with the Toeplitz weight d_{m-n}, where $\sum_\ell d_\ell z^\ell = (1+z)(1-z)^{1-d}$. A half-shift as in (429) maps this structure to itself only if $2(1-d/2) \in \mathbb{Z}$, that is at $d = 1, 2, 3$; at generic d the half-shift factorization of (429) is not available.

Product structure. In the notation (460) the kernel is a product of three balanced factors:

$$t_\ell(s)^2 R_\tau(s) = \Gamma(2q)^2\, g_\ell(s;\tfrac{d}{2})^2\, g_\ell\Big(s + \frac{\tau - q}{2};\, d - 2\tau\Big). \tag{569}$$

Indeed $a_\ell(d-2\tau) \pm (s + \frac{\tau-q}{2})$ equal $a_\ell + s$ and $b_\ell - \tau - s$, and $a_\ell(d-2\tau) + \tau \pm (s + \frac{\tau-q}{2})$ equal $a_\ell + \tau + s$ and $b_\ell - s$. Each factor has rim plus spoke exponent d, but the third has rim exponent $d - 2\tau$ and its own shift. Equation (569) is the kernel form of the mixed trace of two critical wheel operators and one operator of rim exponent $d - 2\tau$ (Section 16.4), and $C(\tau) = N_{d-2\tau}$.

Periodization. Define

$$S_{\mathrm{R}}(t;\tau)=\sum_{\ell\ge 0} d_\ell \sum_{n\in\mathbb{Z}} f_\ell^{\mathrm{R}}(a_\ell+n+t), \qquad \Xi_{\mathrm{R}}=D_{\mathrm{R}}S_{\mathrm{R}},$$
$$D_{\mathrm{R}}(t)=\sin^2(\pi t)\sin^3\bigl(\pi(t+q)\bigr)\sin\bigl(\pi(t-q+\tau)\bigr). \tag{570}$$

The pole classes are $\mathsf{P}_\tau=\mathbb{Z}\cup(-q+\mathbb{Z})\cup(q-\tau+\mathbb{Z})$, of orders at most $2,3,1$, matching the zeros of D_{R}. Put

$$\tau_*(q)=\max\bigl(0,\,2q-1,\,\tfrac12-2q\bigr), \qquad \mathbb{H}_q=\{\tau\in\mathbb{C}:\operatorname{Re}\tau>\tau_*(q)\},$$
$$\eta_{\mathrm{R}}=2\operatorname{Re}\tau-\max(0,1-4q). \tag{571}$$

On $\mathbb{H}_q$ one has $\eta_{\mathrm{R}}>0$.

Growth and finite modes. Let $0<d<4$.

(a) For $t\notin\mathsf{P}_\tau\cup(q+\mathbb{Z})$ the double series S_{R} converges absolutely iff $\tau\in\mathbb{H}_q$. For $\tau\in\mathbb{H}_q$ it converges locally uniformly off P_τ, and S_{R} is meromorphic and one-periodic in t, with poles only in P_τ of orders at most those of the zeros of D_{R}.

(b) For every compact set of pairs (q,τ) with $\tau\in\mathbb{H}_q$ there are constants Y and C with $|S_{\mathrm{R}}(x+\mathrm{i}y)|\le C|y|^{-\eta_{\mathrm{R}}}(1+\log|y|)$ for all real x and $|y|\ge Y$. The right part ($n\ge 0$), the middle part ($-\ell\le n\le -1$) and the left part ($n\le -\ell-1$) obey this bound separately for $0\le x\le 1$.

(c) Ξ_{R} is entire and one-periodic in t, jointly holomorphic on $\mathbb{C}\times\mathbb{H}_q$, and $\Xi_{\mathrm{R}}(t;\tau)=\sum_{|k|\le 2}c_k(\tau)e^{2\pi\mathrm{i}kt}$ with c_k holomorphic on $\mathbb{H}_q$.

Proof. Section 21.6 gives the complete proof, with explicit constants. Parts (a) (sufficiency), (b) and (c) are also the case $r=3$ of Step 0 of the proof of (471), with the parameters of Section 21.5, for which $\varkappa=2q+\operatorname{Re}\tau$ in (469) and $\max(d,1)-2\varkappa=-\eta_{\mathrm{R}}$. In outline: Gamma reflection writes each summand of the right, middle and left parts as a trigonometric factor times two Gamma ratios, each of total exponent $-(2q+\tau)$; a Stirling bound that is uniform in the closed right half-plane and two lattice-sum estimates give (b) and the convergence in (a); matching lower bounds give the divergence outside $\mathbb{H}_q$; and once D_{R} has cleared the poles, (b) removes the modes $|k|\ge 3$. □

The exponent η_{R} plays the role of $2q(r-2)$ in (308). The estimate holds on the whole half-plane $\mathbb{H}_q$; only the residue functional of Section 21.7 needs $\operatorname{Re}\tau<\frac32 q$. Periodicity and analyticity improve (b) to $S_{\mathrm{R}}=O(e^{-2\pi|\operatorname{Im}t|})$ on $\mathbb{H}_q$ (Section 21.6), while each part decays only like a power.

21.5 The top modes vanish

The kernel (569) is a product of three balanced kernels with different rim exponents and shifts, so the product form (471) of the top-mode theorem of Section 20.1.2 applies to it; its wheel case (473) alone does not.

The R family has the modes $|k|\le 1$. For $0<d<4$ and $\tau\in\mathbb{H}_q$, the coefficients $c_{\pm2}$ in part (c) of the growth theorem of Section 21.4 vanish:

$$\boxed{\Xi_{\mathrm{R}}(t;\tau)=c_0(\tau)+c_1(\tau)e^{2\pi\mathrm{i}t}+c_{-1}(\tau)e^{-2\pi\mathrm{i}t},} \tag{572}$$

equivalently $S_{\mathrm{R}}=O(e^{-4\pi|\operatorname{Im}t|})$.

Proof. By (569), $f_\ell^{\mathrm{R}}(a_\ell+n+t)=\Gamma(2q)^2T_\ell(n+t)$ with T_ℓ of (470) for $r=3$, $\sigma=q$, $(g_1,g_2,g_3)=(q,q,\tau)$, $\alpha_i=0$ and $(\beta_1,\beta_2,\beta_3)=(0,0,q-\tau)$; this is (472) with $b=(\frac{d}{2},\frac{d}{2},d-2\tau)$ and $\delta=(0,0,\frac{\tau-q}{2})$.

Then D_T, S_T and Ξ_T of (470) are D_{R}, $\Gamma(2q)^{-2}S_{\mathrm{R}}$ and $\Gamma(2q)^{-2}\Xi_{\mathrm{R}}$, so $\gamma_m = \Gamma(2q)^{-2}c_m$. Condition (B) of (469) holds factor by factor, and (C) reads $2q + \operatorname{Re}\tau > \max(2q, 4q-1, \frac{1}{2})$, that is $\tau \in \mathbb{H}_q$. □

A three-wheel has the same exponential modes; there reflection pairs $k = \pm 1$ into the single cosine mode $c_{3,1}$, while R_τ is not even and $c_{\pm 1}$ are independent. For complex τ no reflection relates c_2 and c_{-2}; Step 6 of the proof of (471) treats them separately.

Remarks. (i) Read as a wheel with the single pair $(\sigma, g_3) = (q, \tau)$, the third factor is off balance by $\tau - q$; only the per-factor balance (B) holds, which is why the product form is needed; with its own shift $\beta_3 = q - \tau$ the factor is balanced and (478) applies. (ii) Per-factor balance is needed: total balance does not suffice, and the multiplicities must be those of the dimension d in (B), as the following two controls show numerically. At $d = 1.7$, $\tau = 0.55 + 0.3\mathrm{i}$, a control kernel with right shifts $\beta = (0.15, -0.15, q - \tau)$, which keeps the total balance, has $|c_{-2}|/\max_k |c_k| = 1.4 \cdot 10^{-2}$, and the R kernel with the multiplicities of $d' = 1.9$, for which the phase Θ of Step 5 of the proof of (471) obeys $\Theta - \Theta \circ \iota = (d - d')(V - \pi)$, has $6 \cdot 10^{-2}$; for the balanced kernel the ratio is below 10^{-15}, within the spread of the spin-sum extrapolation. (iii) (471) also contains the S family ($r = 2$, rim exponent $d/2 - \tau$, by (557)). The consequences of (473) for Section 19 are stated in Section 20.1.2, paragraph "Relation to the periodization at $b = d/2$ and to the half-shift reduction".

21.6 Proof of the growth theorem

This subsection proves part (b) of the growth theorem of Section 21.4 with explicit constants, derives (a) and (c) from Step 0 of the proof of (471), and proves the converse in (a), that is the sharpness of $\mathbb{H}_q$, and the exponential decay of S_{R}. It uses three estimates proved in Section 20.1: the Gamma-ratio bound (465), the multiplicity bound (466) and the lattice sums (467). Throughout, $0 < q < 1$, $d = 4q$, $t = x + \mathrm{i}y$ and $\varkappa = 2q + \operatorname{Re}\tau$; log, arg $\in (-\pi, \pi]$ and complex powers are principal; and, besides the generic constants C of Section 21.1, a bare lowercase c is a positive generic constant. In terms of $\varkappa$, $\tau \in \mathbb{H}_q$ means

$$\varkappa > 2q \quad (\text{that is, } \operatorname{Re}\tau > 0), \qquad \varkappa > 4q - 1, \qquad 2\varkappa > 1, \tag{573}$$

which is the condition of (467) with $\alpha = \varkappa$; its three entries come from the region where both lattice indices are large, from the end $\ell \to \infty$ and from the end $n \to \infty$ (paragraph *The entries of* $b_{\max}$ of Section 20.1). So $\tau_* = \frac{1}{2} - 2q$ for $q \le \frac{1}{4}$, $\tau_* = 0$ for $\frac{1}{4} \le q \le \frac{1}{2}$ and $\tau_* = 2q - 1$ for $q \ge \frac{1}{2}$, and $\eta_{\mathrm{R}} = 2\varkappa - \max(d, 1)$.

Reflected form of the three parts. Put

$$\begin{aligned} \mathsf{F}_1(z) &= \frac{\Gamma(z+q)^3}{\Gamma(z+2q)^2\,\Gamma(z+q+\tau)}, & \mathsf{F}_2(z) &= \frac{\Gamma(z+1-q)^3}{\Gamma(z+1)^2\,\Gamma(z+1-q+\tau)}, \\ \mathsf{F}_3(z) &= \frac{\Gamma(z)^2\,\Gamma(z+q-\tau)}{\Gamma(z+q)^3}, & \mathsf{F}_4(z) &= \frac{\Gamma(z+1-2q)^2\,\Gamma(z+1-q-\tau)}{\Gamma(z+1-q)^3}, \end{aligned} \tag{574}$$

so that $\mathsf{F}_1 = \widetilde{\Phi}$ and $\Phi_t(m) = \mathsf{F}_1(m + t)$ in (568) (the F_i are not the Laurent coefficients F_k of (555)). The reflection formula $\Gamma(w)\Gamma(1-w) = \pi/\sin \pi w$ [26] and $\sin \pi(c - n) = (-1)^n \sin \pi c$ give, for integers $n, j, p \ge 0$,

$$\Psi_t(n) = T_{\mathrm{R}}(t)\,\mathsf{F}_2(n + t), \qquad \Psi_t(-j-1) = \mathsf{F}_3(j + 1 - t), \qquad \Phi_t(-1-p) = T_{\mathrm{L}}(t)\,\mathsf{F}_4(p + 1 - t),$$

with T_{R} of Section 21.4 and $T_{\mathrm{L}}(t) = \sin^2 \pi(2q - 1 + t)\, \sin \pi(q + \tau - 1 + t)/\sin^3 \pi(q - 1 + t)$; the reflection signs combine to $(-1)^{4n} = (-1)^{6p} = 1$. Hence the right part ($n \ge 0$), the middle part

($n=-j-1$, $\ell=j+k+1$) and the left part ($n=-\ell-1-p$) of $S_{\mathrm R}$ are

$$\begin{aligned} S^{\rightarrow}(t) &= \Gamma(2q)^2\,T_{\mathrm R}(t)\sum_{\ell,n\ge0} d_\ell\,\mathsf F_1(\ell+n+t)\,\mathsf F_2(n+t),\\ S^{|}(t) &= \Gamma(2q)^2\sum_{j,k\ge0} d_{j+k+1}\,\mathsf F_1(k+t)\,\mathsf F_3(j+1-t),\\ S^{\leftarrow}(t) &= \Gamma(2q)^2\,T_{\mathrm L}(t)\sum_{\ell,p\ge0} d_\ell\,\mathsf F_4(p+1-t)\,\mathsf F_3(\ell+p+1-t), \end{aligned} \tag{575}$$

that is $\Gamma(2q)^2P_{\mathrm R}H_{\mathrm R}$, $\Gamma(2q)^2P_{\mathrm M}H_{\mathrm M}$ and $\Gamma(2q)^2P_{\mathrm L}H_{\mathrm L}$ of (581), and $S_{\mathrm R}=S^{\rightarrow}+S^{|}+S^{\leftarrow}$ wherever the double series converges absolutely. Since $R_\tau(-s)\neq R_\tau(s)$, the left part is not the right part at a reflected point, unlike the wheels of Section 20.1, and it is estimated separately. For $0\le x\le1$ every argument z in (575) is $m+t$ or $m+1-t$ with an integer $m\ge0$ (one of $\ell+n$, n, k, j, p, $\ell+p$), so $\operatorname{Re}z\ge m$ and $|z|\ge\max(m,|y|)$.

Gamma ratios and multiplicities. Let $\rho\ge2+|\tau|$. Every parameter in (574) then has modulus at most ρ, and each $\mathsf F_i$ is a product of three ratios of (465),

$$\begin{aligned} \mathsf F_1 &= \Big[\frac{\Gamma(z+q)}{\Gamma(z+2q)}\Big]^2\frac{\Gamma(z+q)}{\Gamma(z+q+\tau)}, \quad \mathsf F_2 = \Big[\frac{\Gamma(z+1-q)}{\Gamma(z+1)}\Big]^2\frac{\Gamma(z+1-q)}{\Gamma(z+1-q+\tau)},\\ \mathsf F_3 &= \Big[\frac{\Gamma(z)}{\Gamma(z+q)}\Big]^2\frac{\Gamma(z+q-\tau)}{\Gamma(z+q)}, \quad \mathsf F_4 = \Big[\frac{\Gamma(z+1-2q)}{\Gamma(z+1-q)}\Big]^2\frac{\Gamma(z+1-q-\tau)}{\Gamma(z+1-q)}, \end{aligned}$$

whose exponents $u-v$ add up to $-2q-\tau$. Since $z^az^b=z^{a+b}$ for principal powers, (465) gives, for $\operatorname{Re}z\ge0$, $|z|\ge2\rho$ and $i=1,\dots,4$,

$$\begin{gathered} \mathsf F_i(z)=z^{-2q-\tau}e^{\theta_i(z)}, \qquad |\theta_i(z)|\le\frac{3\omega(\rho)}{|z|},\\ e^{-3\omega(\rho)/|z|-\frac{\pi}{2}|\operatorname{Im}\tau|}\,|z|^{-\varkappa}\le|\mathsf F_i(z)|\le e^{3\omega(\rho)/|z|+\frac{\pi}{2}|\operatorname{Im}\tau|}\,|z|^{-\varkappa}. \end{gathered} \tag{576}$$

By (466), $|d_\ell|\le C_q(1+\ell)^{d-2}$ with C_q bounded for q in compact subsets of $(0,1)$, across $d=1$, and $d_\ell(1+\ell)^{2-d}\to2/\Gamma(d-1)$; the limit vanishes only at $q=\frac14$, so for $q\neq\frac14$ also $|d_\ell|\ge c\,\ell^{d-2}$ for large ℓ. By (301), with $(4q-1)_k=(4q-1)(4q)_{k-1}$, all $d_\ell\ge0$ for $q\ge\frac14$, while for $q<\frac14$ ($d<1$) $d_\ell<0$ for every $\ell\ge2$.

Trigonometric factors. *Lemma.* Let $c_{\mathrm T}=8/(1-e^{-2\pi})^3\approx8.045$. If $|y|\ge1$ and $|y+\operatorname{Im}\tau|\ge1$, then $|T_{\mathrm R}(t)|,|T_{\mathrm L}(t)|\le c_{\mathrm T}e^{\pi|\operatorname{Im}\tau|}$. For all t, $|D_{\mathrm R}(t)|\le e^{6\pi|y|+\pi|\operatorname{Im}\tau|}$, and if $|y|\ge1+|\operatorname{Im}\tau|$, then $|D_{\mathrm R}(t)|\ge\big(\frac{1-e^{-2\pi}}{2}\big)^6e^{6\pi|y|-\pi|\operatorname{Im}\tau|}$.

Proof. From $|\sin\pi w|^2=\sin^2(\pi\operatorname{Re}w)+\sinh^2(\pi\operatorname{Im}w)$ one has $\sinh(\pi|\operatorname{Im}w|)\le|\sin\pi w|\le e^{\pi|\operatorname{Im}w|}$, and $\sinh(\pi b)\ge\frac12(1-e^{-2\pi})e^{\pi b}$ for $b\ge1$. The sines of $T_{\mathrm R}$ have imaginary parts $-y$ (numerator, three times) and y, y, $-(y+\operatorname{Im}\tau)$ (denominator); those of $T_{\mathrm L}$ have y, y, $y+\operatorname{Im}\tau$ (numerator) and y three times (denominator); those of $D_{\mathrm R}$ have y five times and $y+\operatorname{Im}\tau$ once. Insert these bounds and use $|y|-|\operatorname{Im}\tau|\le|y+\operatorname{Im}\tau|\le|y|+|\operatorname{Im}\tau|$. □

Proof of (b), with constants. With $\alpha=\varkappa$, the sums $\Sigma^{\mathrm r}(Y)$ and $\Sigma^{\mathrm m}(Y)$ of (467) obey $\Sigma^{\mathrm r}(Y)\le C^{\mathrm r}Y^{-\eta_{\mathrm R}}(1+\log Y)$ and $\Sigma^{\mathrm m}(Y)\le C^{\mathrm m}Y^{-2\operatorname{Re}\tau}$ for $Y\ge1$, since $\max(d,1)-2\varkappa=-\eta_{\mathrm R}$ and $d-2\varkappa=-2\operatorname{Re}\tau$. Let $\mathfrak K$ be a compact set of pairs (q,τ) with $0<q<1$ and $\tau\in\mathbb H_q$, and take maxima over $\mathfrak K$. Put $\rho_{\mathfrak K}=2+\max|\tau|$ and $Y_{\mathfrak K}=2\rho_{\mathfrak K}$, which exceeds $1+\max|\operatorname{Im}\tau|$; let $c_{\mathfrak K}=2^{\varkappa}\exp\big(3\omega(\rho_{\mathfrak K})/Y_{\mathfrak K}+\frac{\pi}{2}|\operatorname{Im}\tau|\big)$, a function of (q,τ), and

$$C_{\mathfrak K}=\max\,\Gamma(2q)^2\,C_q\,c_{\mathfrak K}^2\,\big[c_{\mathrm T}e^{\pi|\operatorname{Im}\tau|}C^{\mathrm r}+2^{\max(d-2,0)}C^{\mathrm m}\big],$$

with C^{r}, C^{m} at $\alpha=\varkappa$; it is finite because C_q, C^{r} and C^{m} are bounded on compact sets. *For $(q,\tau)\in\mathfrak{K}$ and $|y|\ge Y_{\mathfrak{K}}$ the three series* (575) *converge absolutely for $0\le x\le 1$, $|S^{\rightarrow}|$ and $|S^{\leftarrow}|$ are at most $C_{\mathfrak{K}}|y|^{-\eta_{\mathrm{R}}}(1+\log|y|)$, $|S^{|}|$ is at most $C_{\mathfrak{K}}|y|^{-2\operatorname{Re}\tau}$, and for all real x*

$$\begin{aligned}|S_{\mathrm{R}}(x+\mathrm{i}y)|&\le 3C_{\mathfrak{K}}\,|y|^{-\eta_{\mathrm{R}}}(1+\log|y|),\\ |\Xi_{\mathrm{R}}(x+\mathrm{i}y)|&\le 3C_{\mathfrak{K}}\,e^{\pi|\operatorname{Im}\tau|}e^{6\pi|y|}\,|y|^{-\eta_{\mathrm{R}}}(1+\log|y|).\end{aligned}\tag{577}$$

This is (b) with $Y=Y_{\mathfrak{K}}$ and $C=3C_{\mathfrak{K}}$, the bound on the parts holding for $0\le x\le 1$.

Proof. Fix $(q,\tau)\in\mathfrak{K}$, $0\le x\le 1$ and $|y|\ge Y_{\mathfrak{K}}$. Every argument z in (575) has $\operatorname{Re}z\ge 0$ and $|z|\ge\max(m,|y|)\ge\frac12(m+|y|)$, in particular $|z|\ge Y_{\mathfrak{K}}=2\rho_{\mathfrak{K}}$, so (576) with $\rho=\rho_{\mathfrak{K}}$ gives $|\mathsf{F}_i(z)|\le c_{\mathfrak{K}}(m+|y|)^{-\varkappa}$. The lemma on trigonometric factors applies, since $|y|\ge 1+|\operatorname{Im}\tau|$. With $|d_\ell|\le C_q(1+\ell)^{d-2}$ and $|d_{j+k+1}|\le C_q(2+j+k)^{d-2}\le 2^{\max(d-2,0)}C_q(1+j+k)^{d-2}$, the series (575) are dominated termwise:

$$|S^{\rightarrow}|,\ |S^{\leftarrow}|\le\Gamma(2q)^2c_{\mathrm{T}}e^{\pi|\operatorname{Im}\tau|}C_q\,c_{\mathfrak{K}}^2\,\Sigma^{\mathrm{r}}(|y|),\qquad |S^{|}|\le\Gamma(2q)^2\,2^{\max(d-2,0)}C_q\,c_{\mathfrak{K}}^2\,\Sigma^{\mathrm{m}}(|y|);$$

for the left part the dominating sum $\sum_{\ell,p\ge0}(1+\ell)^{d-2}(\ell+p+|y|)^{-\varkappa}(p+|y|)^{-\varkappa}$ is again $\Sigma^{\mathrm{r}}(|y|)$. This gives the bounds on the parts. Their absolute convergence also justifies $S_{\mathrm{R}}=S^{\rightarrow}+S^{|}+S^{\leftarrow}$ for $0\le x\le 1$; for other x one uses $S_{\mathrm{R}}(t+1)=S_{\mathrm{R}}(t)$, the shift $n\mapsto n+1$ of the absolutely convergent double series. Finally $|\Xi_{\mathrm{R}}|\le|D_{\mathrm{R}}||S_{\mathrm{R}}|$ and the bound on D_{R}. □

The constant is explicit but crude, because $\omega(\rho)$ grows like $\frac{11}{3}\rho^3$. For $|y|\ge Y\ge Y_{\mathfrak{K}}$ one may replace $Y_{\mathfrak{K}}$ by Y in $c_{\mathfrak{K}}$; as $Y\to\infty$ the constant decreases to the same expression with $c_{\mathfrak{K}}$ replaced by $2^{\varkappa}e^{\pi|\operatorname{Im}\tau|/2}$ (Section 21.11, item Growth constants).

Proof of (a) and (c). Sufficiency in (a), and (c), are the case $r=3$ of Step 0 of the proof of (471), carried out in full in the paragraph *Details of the top-mode proof* of Section 20.1.2, with the parameters of Section 21.5: $\sigma=q$, $(g_1,g_2,g_3)=(q,q,\tau)$, $\alpha_i=0$ and $(\beta_1,\beta_2,\beta_3)=(0,0,q-\tau)$. Then $f_\ell^{\mathrm{R}}(a_\ell+n+t)=\Gamma(2q)^2T_\ell(n+t)$, $D_T=D_{\mathrm{R}}$, $\varkappa=2q+\operatorname{Re}\tau$, and condition (C) of (469) is $\tau\in\mathbb{H}_q$. The six pole classes of Step 0 are $-q+\mathbb{Z}$ three times, $\mathbb{Z}$ twice and $q-\tau+\mathbb{Z}$ once, so $\mathsf{P}_T=\mathsf{P}_\tau$: through $\Gamma(\ell+n+q+t)^3$, $\Gamma(-n-t)^2$ and $\Gamma(q-\tau-n-t)$ a term of S_{R} has poles of orders at most 3 on $-q+\mathbb{Z}$, 2 on $\mathbb{Z}$ and 1 on $q-\tau+\mathbb{Z}$, added where classes meet, which are the zero orders of D_{R}. Step 0 gives absolute and locally uniform convergence off P_τ; uniform convergence of $(t-t_0)^o$ times the series on a small circle around each $t_0\in\mathsf{P}_\tau$, with o the order of the zero of D_{R} at t_0, hence the pole orders in (a); periodicity by $n\mapsto n+1$; an entire one-periodic Ξ_{R}; the vanishing of the modes $|k|\ge3$, the boundary modes included because $\eta_{\mathrm{R}}=2\varkappa-\max(d,1)>0$; and coefficients $c_k=\Gamma(2q)^2\gamma_k$ that are holomorphic in the parameters, hence in $\tau\in\mathbb{H}_q$. As a trigonometric polynomial in t with holomorphic coefficients, Ξ_{R} is jointly holomorphic on $\mathbb{C}\times\mathbb{H}_q$. The converse in (a) is proved in the paragraph *Sharpness of* $\mathbb{H}_q$ below. □

Exponential decay of the total. For $y>Y_{\mathfrak{K}}$ there are no poles, since P_τ lies on the lines $\operatorname{Im}t=0$ and $\operatorname{Im}t=-\operatorname{Im}\tau$; so $S_{\mathrm{R}}(t)=\widehat{S}(e^{2\pi\mathrm{i}t})$ with $\widehat{S}$ holomorphic on $0<|w|<e^{-2\pi Y_{\mathfrak{K}}}$. By (577), $\widehat{S}(w)\to0$ as $w\to0$, so $\widehat{S}$ extends holomorphically with $\widehat{S}(0)=0$ (Riemann's theorem on removable singularities). The maximum principle for $\widehat{S}(w)/w$ on $|w|\le e^{-2\pi(Y_{\mathfrak{K}}+1)}$ then gives $|S_{\mathrm{R}}(x+\mathrm{i}y)|\le e^{2\pi(Y_{\mathfrak{K}}+1)}M_{\mathfrak{K}}\,e^{-2\pi y}$ for $y\ge Y_{\mathfrak{K}}+1$, where $M_{\mathfrak{K}}=3C_{\mathfrak{K}}(Y_{\mathfrak{K}}+1)^{-\eta_{\mathrm{R}}}(1+\log(Y_{\mathfrak{K}}+1))$ bounds $|S_{\mathrm{R}}|$ on $\operatorname{Im}t=Y_{\mathfrak{K}}+1$; for $y\le-Y_{\mathfrak{K}}-1$ one uses $e^{-2\pi\mathrm{i}t}$ in place of w. So $S_{\mathrm{R}}=O(e^{-2\pi|\operatorname{Im}t|})$ and $\Xi_{\mathrm{R}}=O(e^{4\pi|\operatorname{Im}t|})$, uniformly in x and on $\mathfrak{K}$, while numerically the parts decay only like powers (Section 21.11, item Growth): the exponential decay of the total is a cancellation between the parts. Equation (572) improves this to $S_{\mathrm{R}}=O(e^{-4\pi|\operatorname{Im}t|})$.

Sharpness of $\mathbb{H}_q$. *Let $0<q<1$ and $\tau\notin\mathbb{H}_q$. For every $t\notin\mathsf{P}_\tau\cup(q+\mathbb{Z})$ the double series $S_{\mathrm{R}}(t)$ does not converge absolutely.* So the lattice sum, and with it any estimate of the type (b), exists exactly on $\mathbb{H}_q$, up to the discrete set of exceptional t; this is the converse in (a).

Proof. Fix t, and put $\rho = 2+|\tau|$ and $m_0 = \lceil 2|\operatorname{Re} t| + 2 + 2\rho \rceil$. For an integer $m \geq m_0$ the arguments $z = m + t$ and $z = m + 1 - t$ of (575) have $\operatorname{Re} z \geq m - |\operatorname{Re} t| \geq \max(2\rho, \frac{1}{2}(1+m))$, so (576) gives $c\,|z|^{-\varkappa} \leq |\mathsf{F}_i(z)| \leq C|z|^{-\varkappa}$, and $\mathsf{F}_i(z) \neq 0$ there; $T_{\mathrm{R}}(t) \neq 0$ because $t \notin q + \mathbb{Z}$; and for $q \neq \frac{1}{4}$, $|d_\ell| \geq c\,\ell^{d-2}$ for large ℓ. One of the conditions (573) fails.

- $2\varkappa \leq 1$. The terms of the right part with $\ell = 0$, $d_0 \Gamma(2q)^2 T_{\mathrm{R}}(t) \mathsf{F}_1(n+t)\mathsf{F}_2(n+t)$, have modulus at least $c\,n^{-2\varkappa}$ for large n and are not summable. This case contains $q = \frac{1}{4}$, where $\tau_* = 0 = \frac{1}{2} - 2q$.
- $2\varkappa > 1$ and $\varkappa \leq 4q - 1$, so $q > \frac{3}{8}$. Fix $j = j_0 \geq m_0$ in the middle part. Its terms $d_{j_0+k+1}\Gamma(2q)^2 \mathsf{F}_1(k+t)\mathsf{F}_3(j_0+1-t)$, with $\mathsf{F}_3(j_0+1-t) \neq 0$ fixed, have modulus at least $c\,k^{4q-2-\varkappa}$ for large k, and $4q - 2 - \varkappa \geq -1$.
- $2\varkappa > 1$, $\varkappa > 4q - 1$ and $\varkappa \leq 2q$, so $q > \frac{1}{4}$ and $\operatorname{Re}\tau \leq 0$. In the middle part the block $N \leq j, k < 2N$, for large N, has N^2 terms, each of modulus at least $c\,N^{4q-2}N^{-2\varkappa}$, because $j + k + 1$ lies between N and $4N$ and $|k+t|, |j+1-t| \leq 3N$. The block contributes at least $c\,N^{-2\operatorname{Re}\tau} \geq c$, and the sum over the blocks $N = 2^i$ diverges. □

Is the exponent η_{R} attained? For $q \geq \frac{1}{4}$, where all $d_\ell \geq 0$ and $\eta_{\mathrm{R}} = 2\operatorname{Re}\tau$, the bound on the right and left parts is attained numerically: $\max_x |S^{\rightarrow}||y|^{\eta_{\mathrm{R}}}$ tends to a constant at $d = 1.25$, 1.5, 2.5 and 3 (records `n5_case0.log` and `n5_case2.log` to `n5_case5.log` in `LRI_C5_general_d/K_general/note_fragment/growth/checks/`). Numerically it is not attained at $d = 0.8$ ($q < \frac{1}{4}$), the only such point tested: there the right and left parts decay like $|y|^{-2\operatorname{Re}\tau}$, faster than $|y|^{-\eta_{\mathrm{R}}}$ (`n5_case1.log`). The absolute values taken in the proof lose the cancellation $\sum_\ell d_\ell = \lim_{z\to 1}(1+z)(1-z)^{1-4q} = 0$ of the negative multiplicities.

21.7 Residue functional, three-node rule and the Horn samples

Three-node rule. Let $\tau \in \mathbb{H}_q$ with $\operatorname{Re}\tau < \frac{3}{2}q$ and $\tau \neq q$. Then

$$\langle t_\ell^2 R_\tau \rangle_d = -\operatorname{Res}_{t=0} \frac{\Xi_{\mathrm{R}}}{D_{\mathrm{R}}} - \operatorname{Res}_{t=q-\tau} \frac{\Xi_{\mathrm{R}}}{D_{\mathrm{R}}} = \sum_{|k|\leq 1} \rho_k c_k, \tag{578}$$

$$\rho_k = -\frac{2\pi \mathrm{i} k - \pi\big[3\cot \pi q + \cot \pi(\tau - q)\big]}{\pi^2 \sin^3(\pi q)\sin \pi(\tau - q)} - \frac{e^{2\pi \mathrm{i} k(q-\tau)}}{\pi \sin^2 \pi(q-\tau)\, \sin^3 \pi(2q-\tau)}, \tag{579}$$

and for every real t_0 with $t_0 + \frac{j}{3} \notin \mathsf{P}_\tau$,

$$\boxed{\langle t_\ell^2 R_\tau \rangle_d = \sum_{j=0}^{2} w_j(\tau)\, \Xi_{\mathrm{R}}\big(t_0 + \tfrac{j}{3}; \tau\big), \qquad w_j(\tau) = \frac{1}{3} \sum_{|k|\leq 1} \rho_k(\tau)\, e^{-2\pi \mathrm{i} k(t_0 + j/3)}.} \tag{580}$$

Proof. The left poles of f_ℓ^{R} are $s = -a_\ell - n$, and the right poles are $s = a_\ell + n$ and $s = b_\ell - \tau + n$, $n \geq 0$, that is $t = 0$ and $t = q - \tau$ on the lattice $s = a_\ell + n + t$. The imaginary axis separates the two families because $a_\ell \geq \frac{q}{2} > 0$ and $\operatorname{Re}(b_\ell - \tau) \geq \frac{3}{2}q - \operatorname{Re}\tau > 0$. The two right families are disjoint because $q - \tau \notin \mathbb{Z}$, which on the strip excludes only $\tau = q$. Moreover $2q - \tau \notin \mathbb{Z}$ holds automatically: $2q - \tau = m \in \mathbb{Z}$ would give $\operatorname{Re}\tau = 2q - m$, which is either $\geq 2q > \frac{3}{2}q$ or $\leq 2q - 1 \leq \tau_*(q)$.

The trace converges absolutely. Put $\varkappa = 2q + \operatorname{Re}\tau$ (the $\varkappa$ of (469) for this kernel). On the axis, the Gamma-ratio bound (465), which gives $|\Gamma(z+u)/\Gamma(z+v)| \leq C|z|^{\operatorname{Re}(u-v)}$ for $\operatorname{Re} z \geq 0$ and $|z|$ large, applied with $z = a_\ell \pm \mathrm{i}\nu$, gives $|f_\ell^{\mathrm{R}}(\mathrm{i}\nu)| \leq C(a_\ell + |\nu|)^{-2\varkappa}$ for $a_\ell + |\nu|$ large. So each ν-integral converges ($2\varkappa > 1$), and with $|d_\ell| \leq C(1+\ell)^{d-2}$, $\sum_\ell |d_\ell| \int |f_\ell^{\mathrm{R}}(\mathrm{i}\nu)|\, d\nu \leq C \sum_{\ell \geq 1} \ell^{4q-2}\ell^{1-2\varkappa} < \infty$, since $4q - 1 - 2\varkappa = -1 - 2\operatorname{Re}\tau$ and $\operatorname{Re}\tau > 0$.

Close each Barnes contour of (554) to the right, on arcs $|s| = R_N \to \infty$, $\operatorname{Re} s \geq 0$, with radii chosen so that the arcs keep a distance at least $\frac{1}{8}$ from both right lattices. On the arcs, reflection of $\Gamma(a_\ell - s)^2\Gamma(b_\ell - \tau - s)/\Gamma(b_\ell - s)^3$ [26] gives a ratio of three sines over three sines, bounded away from the lattices and for large $|\operatorname{Im} s|$, times two Gamma ratios of the type above. Hence $|f_\ell^{\mathrm{R}}(s)| \leq C_\ell |s|^{-4q-2\operatorname{Re}\tau}$ with $4q + 2\operatorname{Re}\tau > 1$, so the arcs do not contribute, and the integral is minus the sum of the right residues. The terms of S_{R} with $n < 0$ are holomorphic at $t = 0$ and $t = q-\tau$, since a pole there would need $q \in \mathbb{Z}$, $q-\tau \in \mathbb{Z}$ or $2q-\tau \in \mathbb{Z}$. Uniform convergence on small circles (Section 21.6) therefore identifies the sums of residues with the residues of $S_{\mathrm{R}} = \Xi_{\mathrm{R}}/D_{\mathrm{R}}$, and (572) gives $\sum_{|k|\leq 1} \rho_k c_k$. The three-point discrete Fourier transform is exact for modes $|k| \leq 1$. □

Part (c) of the growth theorem alone gives a five-node rule, with the nodes $t_0 + j/5$ and the modes $|k| \leq 2$. By (572), three nodes give the same $\mathcal{K}_d$ at three fifths of the sampling cost (Section 21.11). On the jet circle at $d = 2$, ρ_k has a triple pole at $\tau = 0$, from $\sin^3 \pi(2q - \tau)$, which cancels in the sum; there $\tau = 0$ is also a spurious pole of individual pieces of the continued samples (Section 21.8), and the exceptional set E of hypothesis (H), stated in Section 21.8, still meets the closed disk $|\tau| \leq r_\tau$ only in $\{0\}$.

Separating contours (sketch). The strip of (578), $\tau \in \mathbb{H}_q$ with $\operatorname{Re}\tau < \frac{3}{2}q$, is nonempty iff $d > \frac{4}{7}$. For $\operatorname{Re}\tau \geq \frac{3}{2}q$ the imaginary axis no longer separates the pole families for the finitely many spins with $\operatorname{Re}(b_\ell - \tau) \leq 0$. For $d \leq \frac{4}{7}$ this happens at every $\tau \in \mathbb{H}_q$, because then $\tau_*(q) = \frac{1}{2} - 2q \geq \frac{3}{2}q$. For those spins one replaces the axis by a contour that separates the left pole family from the two right ones. The proof above then gives (578), for $\tau \notin q + \mathbb{Z}$, for the resulting separating-contour trace. That trace is jointly meromorphic in (q, τ). Its poles lie at $\tau \in 2q + \mathbb{N}_0$, where the left family $s = -a_\ell - n$ and the right family $s = b_\ell - \tau + n'$ pinch the contour ($\tau = \ell + 2q + n + n'$). There the factor $1/\sin^3 \pi(2q - \tau)$ of ρ_k in (579) is singular as well, and $\tau = 2q$, which has $\operatorname{Re}\tau \geq \frac{3}{2}q$, lies in $\mathbb{H}_q$ for $q > \frac{1}{8}$. The trace agrees with the axis trace wherever the latter is defined, so it is the meromorphic continuation of the trace in τ and, for $d \leq \frac{4}{7}$, also in q. This is a sketch: the construction is not carried out in full in this note (Section 21.12).

The samples as Horn functions. Put

$$\widehat{\Phi}(m) = \frac{(q+t)_m^3}{(2q+t)_m^2 (q+\tau+t)_m}, \qquad \widehat{\Psi}(n) = \frac{(1-q+t)_n^3}{(1+t)_n^2 (1-q+\tau+t)_n},$$
$$\widehat{\Psi}_{\mathrm{M}}(m) = \frac{(1-t)_m^2 (1+q-\tau-t)_m}{(1+q-t)_m^3}, \quad \widehat{\Phi}_{\mathrm{L}}(p) = \frac{(2-2q-t)_p^2 (2-q-\tau-t)_p}{(2-q-t)_p^3}.$$

For $\tau \in \mathbb{H}_q$ and $t \notin \mathsf{P}_\tau$,

$$\boxed{\Xi_{\mathrm{R}}(t;\tau) = D_{\mathrm{R}}(t)\,\Gamma(2q)^2 \big[P_{\mathrm{R}} H_{\mathrm{R}} + P_{\mathrm{M}} H_{\mathrm{M}} + P_{\mathrm{L}} H_{\mathrm{L}}\big],} \tag{581}$$

with $P_{\mathrm{R}} = \Phi_t(0)\Psi_t(0)$, $P_{\mathrm{M}} = \Phi_t(0)\Psi_t(-1)$, $P_{\mathrm{L}} = \Phi_t(-1)\Psi_t(-1)$ from (568), and

$$H_{\mathrm{R}} = \sum_{\ell,n\geq 0} d_\ell \widehat{\Phi}(\ell+n)\widehat{\Psi}(n), \qquad H_{\mathrm{M}} = \sum_{j,k\geq 0} d_{j+k+1} \widehat{\Psi}_{\mathrm{M}}(j)\widehat{\Phi}(k),$$
$$H_{\mathrm{L}} = \sum_{\ell,p\geq 0} d_\ell \widehat{\Psi}_{\mathrm{M}}(\ell+p)\widehat{\Phi}_{\mathrm{L}}(p).$$

In the convention of (320), with $\boldsymbol{e}=(1,0)$, $\boldsymbol{f}=(0,1)$ and $\boldsymbol{c}=\boldsymbol{e}+\boldsymbol{f}$, these are $H_{\mathrm{R}}=\mathfrak{H}[\mathsf{U}_{\mathrm{R}};\mathsf{V}_{\mathrm{R}};(1,1)]$, $H_{\mathrm{M}}=4q\,\mathfrak{H}[\mathsf{U}_{\mathrm{M}};\mathsf{V}_{\mathrm{M}};(1,1)]$ and $H_{\mathrm{L}}=\mathfrak{H}[\mathsf{U}_{\mathrm{L}};\mathsf{V}_{\mathrm{L}};(1,1)]$, where

$$\begin{aligned}
\mathsf{U}_{\mathrm{R}} &= \{(4q-2,\boldsymbol{e}),(2q,\boldsymbol{e}),(q+t,\boldsymbol{c})^{\times 3},(1,\boldsymbol{f}),(1-q+t,\boldsymbol{f})^{\times 3}\},\\
\mathsf{V}_{\mathrm{R}} &= \{(2q-1,\boldsymbol{e}),(2q+t,\boldsymbol{c})^{\times 2},(q+\tau+t,\boldsymbol{c}),(1+t,\boldsymbol{f})^{\times 2},(1-q+\tau+t,\boldsymbol{f})\},\\
\mathsf{U}_{\mathrm{M}} &= \{(4q-1,\boldsymbol{c}),(2q+1,\boldsymbol{c}),(1,\boldsymbol{e}),(1-t,\boldsymbol{e})^{\times 2},(1+q-\tau-t,\boldsymbol{e}),\\
&\qquad (1,\boldsymbol{f}),(q+t,\boldsymbol{f})^{\times 3}\},\\
\mathsf{V}_{\mathrm{M}} &= \{(2q,\boldsymbol{c}),(2,\boldsymbol{c}),(1+q-t,\boldsymbol{e})^{\times 3},(2q+t,\boldsymbol{f})^{\times 2},(q+\tau+t,\boldsymbol{f})\},\\
\mathsf{U}_{\mathrm{L}} &= \{(4q-2,\boldsymbol{e}),(2q,\boldsymbol{e}),(1-t,\boldsymbol{c})^{\times 2},(1+q-\tau-t,\boldsymbol{c}),\\
&\qquad (1,\boldsymbol{f}),(2-2q-t,\boldsymbol{f})^{\times 2},(2-q-\tau-t,\boldsymbol{f})\},\\
\mathsf{V}_{\mathrm{L}} &= \{(2q-1,\boldsymbol{e}),(1+q-t,\boldsymbol{c})^{\times 3},(2-q-t,\boldsymbol{f})^{\times 3}\}.
\end{aligned} \tag{582}$$

Proof. On the right part ($n\ge 0$) the Pochhammer form of (568) gives $\Phi_t(\ell+n)\Psi_t(n)=P_{\mathrm{R}}\widehat{\Phi}(\ell+n)\widehat{\Psi}(n)$; the reflection signs combine to $(-1)^{6n}=1$. On the middle part put $n=-j-1$ and $\ell=j+k+1$, and on the left part $n=-\ell-1-p$. The multiplicities are $d_\ell=(4q-2)_\ell(2q)_\ell/[(2q-1)_\ell\,\ell!]$ (Section 16.5) for the $\boldsymbol{e}$ rows and (301) for the $\boldsymbol{c}$ rows of H_{M}, and the rows $(1,\boldsymbol{e})$, $(1,\boldsymbol{f})$ cancel the factorials of (320). Absolute convergence on $\mathbb{H}_q$ (Section 21.4) justifies the regrouping. □

The subscripts $\mathrm{R},\mathrm{M},\mathrm{L}$ of $H,P,\mathsf{U},\mathsf{V}$ and of $T_{\mathrm{R}},T_{\mathrm{L}}$ name the right, middle and left parts of the lattice; on S,D,Ξ,η,Ω the subscript R names the R family.

All parameters are affine in (q,t,τ). The series H_{R} and H_{L} converge absolutely on $\mathbb{H}_q$, and H_{M} on $\operatorname{Re}\tau>\max(0,2q-1)$. At $d=2$ the rows $(4q-2,\boldsymbol{e})/(2q-1,\boldsymbol{e})$ are $0/0$; there one splits off $\ell=0$ as in (321). At $d=3$ and $\tau=q$, where $R_q=g_\ell$, the rows of H_{R} reduce to those of $\mathfrak{H}_3(t)$ in (413).

21.8 Continuation in τ and the jet

The jet of (565) is taken on a circle $|\tau|=r_\tau$ around the pole $\tau=0$ of $\langle t_\ell^2 R_\tau\rangle_d$. Since $\operatorname{Re}\tau>0$ is one of the conditions defining $\mathbb{H}_q$, no such circle lies in $\mathbb{H}_q$; for $d=0.8$, 2.5, 2.75 and 3 the whole circle used for the values of Section 21.11 ($r_\tau=0.04$ or 0.08) lies outside it. The samples are continued as follows.

Each of $\widehat{\Phi},\widehat{\Psi},\widehat{\Psi}_{\mathrm{M}},\widehat{\Phi}_{\mathrm{L}}$ behaves as a constant times m^{a-1} with $a=1-2q-\tau$. With $\mathsf{b}_\alpha(m)=\Gamma(m+\alpha)/\Gamma(m+1)$, subtract an I-term model, $X(m)=\sum_{i<I}x_i\mathsf{b}_{a-i}(m)+\delta X(m)$ with $\delta X=O(m^{a-1-I})$. Here X, Y stand for any of the four factors above and x_i, y_i for their model coefficients; they are not the constant X of (221) or the x_h of (558). The coefficients x_i follow from the Bernoulli-polynomial expansion of $\log X(m)$: each is the Gamma ratio x_0 below times a polynomial in Bernoulli polynomials of the parameters. Each factor has the form $X(m)=\prod_{i=1}^{3}(u_i)_m/(v_i)_m$, with upper and lower parameters u_i,v_i read off above (local names) and $\sum_i(u_i-v_i)=a-1$. As $m\to\infty$ [27],

$$\begin{aligned}
X(m) &\sim x_0\,m^{a-1}\exp\sum_{k\ge 1}\lambda_k m^{-k}, \qquad x_0=\prod_{i=1}^{3}\frac{\Gamma(v_i)}{\Gamma(u_i)},\\
\lambda_k &= \frac{(-1)^{k+1}}{k(k+1)}\sum_{i=1}^{3}[B_{k+1}(u_i)-B_{k+1}(v_i)],
\end{aligned} \tag{583}$$

with the Bernoulli polynomials B_{k+1} (not the B_d of (230) or the weights $B_{r,k}$ of (479)). The basis function b_{a-i} has the same expansion with the single pair $(u,v)=(a-i,1)$ and the constant 1. Matching the coefficients of m^{a-1-j}, $j<I$, gives $x_1,\dots,x_{I-1}$ by a triangular solve with unit diagonal, starting from x_0. The model sums are closed Gamma forms, by Gauss's sum [4] after

$(2q)_\ell/(2q-1)_\ell = 1+\ell/(2q-1)$, and by Chu–Vandermonde:

$$\begin{aligned}
\mathcal{G}_\alpha(n) &:= \sum_{\ell\ge 0} d_\ell\, \mathsf{b}_\alpha(\ell+n) = \frac{\Gamma(n+\alpha)\Gamma(2-4q-\alpha)\,(2n+2-4q+\alpha)}{\Gamma(n+3-4q)\Gamma(1-\alpha)},\\
\mathcal{S}(\alpha,\beta) &:= \sum_{n\ge 0} \mathcal{G}_\alpha(n)\,\mathsf{b}_\beta(n) = \frac{\Gamma(\alpha)\Gamma(\beta)\Gamma(2-4q-\alpha-\beta)\,(2-4q+\alpha-\beta)}{\Gamma(1-\alpha)\Gamma(3-4q-\beta)},\\
\widetilde{\mathcal{G}}_\alpha(j) &:= \sum_{k\ge 0} d_{j+k+1}\mathsf{b}_\alpha(k) = \frac{d_{j+1}\Gamma(\alpha)\Gamma(j+2)\Gamma(2-4q-\alpha)}{\Gamma(3-4q)\Gamma(j+2-\alpha)}\Big[2-4q-\alpha+\frac{(4q-1+j)\alpha}{2q+j}\Big],\\
\mathcal{V}(\alpha,\beta) &:= \sum_{j,k\ge 0} d_{j+k+1}\mathsf{b}_\beta(j)\,\mathsf{b}_\alpha(k) = \frac{2(1-2q)\Gamma(\alpha)\Gamma(\beta)\Gamma(2-4q-\alpha-\beta)\,(4q-\alpha-\beta)}{\Gamma(3-4q)\Gamma(2-\alpha-\beta)},
\end{aligned} \tag{584}$$

valid where the sums converge and by continuation elsewhere. For the Toeplitz blocks, H_{R} with $(X,Y)=(\widehat{\Phi},\widehat{\Psi})$ and H_{L} with $(X,Y)=(\widehat{\Psi}_{\mathrm{M}},\widehat{\Phi}_{\mathrm{L}})$, and for the middle block H_{M} with $(X,Y)=(\widehat{\Phi},\widehat{\Psi}_{\mathrm{M}})$, this gives

$$\begin{aligned}
\sum_{\ell,n\ge 0} d_\ell X(\ell+n)Y(n) &= \sum_{\ell,n\ge 0} d_\ell\,\delta X(\ell+n)Y(n) + \sum_{n\ge 0}\Big[\sum_i x_i\mathcal{G}_{a-i}(n)\Big]\delta Y(n)\\
&\quad + \sum_{i,i'} x_i y_{i'}\mathcal{S}(a-i,a-i'),\\
\sum_{j,k\ge 0} d_{j+k+1}Y(j)X(k) &= \sum_{j,k\ge 0} d_{j+k+1}\,\delta Y(j)\,\delta X(k) + \sum_{j\ge 0}\delta Y(j)\sum_i x_i\widetilde{\mathcal{G}}_{a-i}(j)\\
&\quad + \sum_{k\ge 0}\delta X(k)\sum_i y_i\widetilde{\mathcal{G}}_{a-i}(k) + \sum_{i,i'} x_i y_{i'}\mathcal{V}(a-i,a-i').
\end{aligned} \tag{585}$$

For $I\ge 1$ the two-variable remainders converge absolutely in a neighborhood of $\tau=0$ for every $0<q<1$, and so do the one-variable remainders. For $d\neq 2$ the only singularity at $\tau=0$ is $\Gamma(2\tau)$, in $\mathcal{S}(a,a)$ and $\mathcal{V}(a,a)$; at $d=2$ the two spurious families below also contain $\tau=0$, and these poles of individual pieces cancel in the sum. Spurious poles of individual pieces, at $\tau\in 1-2q+\mathbb{Z}$, at $\tau\in 2q-1-\mathbb{Z}$ and at $\tau=q$, bound the admissible radius of the circle. On $\mathbb{H}_q$ the right sides of (585) equal the lattice sums, by an absolutely convergent rearrangement.

Hypothesis (H) and transfer. (H) states that there are a bounded connected open set $\mathcal{U}$ containing the closed disk $|\tau|\le r_\tau$ and meeting $\mathbb{H}_q$, and a closed discrete set $\mathsf{E}\subset\mathcal{U}$ with $\mathsf{E}\cap\{|\tau|\le r_\tau\}\subseteq\{0\}$, such that the continued samples (581), (585) are jointly holomorphic in (t,τ) for $\tau\in\mathcal{U}\setminus\mathsf{E}$ and $t\notin\mathsf{P}_\tau$, locally uniformly. The construction above gives this, except that the uniform remainder bounds of the inverse-factorial expansions [27] are not written out. The radius r_τ is also taken so small that the disk contains no zero of $\sin\pi(\tau-q)$ or $\sin\pi(2q-\tau)$ other than $\tau=0$, so that the weights w_j of (580) have no other singularity there. Given (H), the Fourier coefficients $c_k(\tau)=\int_0^1 \Xi_{\mathrm{R}}(x+\mathrm{i}y_1;\tau)e^{-2\pi\mathrm{i}k(x+\mathrm{i}y_1)}dx$, with $y_1>1+\sup_{\mathcal{U}}|\operatorname{Im}\tau|$, so that the segment avoids P_τ, are holomorphic on $\mathcal{U}\setminus\mathsf{E}$: the integrand is jointly continuous and holomorphic in τ, so by Fubini's theorem its integral over the boundary of every small triangle in $\mathcal{U}\setminus\mathsf{E}$ vanishes, and Morera's theorem applies. For $\tau\in(\mathcal{U}\cap\mathbb{H}_q)\setminus\mathsf{E}$, $c_k(\tau)$ is the k-th Fourier coefficient of the entire one-periodic function $\Xi_{\mathrm{R}}(\cdot\,;\tau)$, which does not depend on the line $\operatorname{Im}t=y_1$. The set $\mathcal{U}\setminus\mathsf{E}$ is connected, since a closed discrete subset does not disconnect a planar domain, and $(\mathcal{U}\cap\mathbb{H}_q)\setminus\mathsf{E}$ is a nonempty open subset of it. The coefficients with $|k|\ge 2$ vanish on that subset by (572), hence identically by the identity theorem. Both sides of $\Xi_{\mathrm{R}}(t;\tau)=\sum_{|k|\le 1}c_k(\tau)e^{2\pi\mathrm{i}kt}$ are holomorphic on the set of (t,τ) with $\tau\in\mathcal{U}\setminus\mathsf{E}$ and $t\notin\mathsf{P}_\tau$, which is connected because its base $\mathcal{U}\setminus\mathsf{E}$ and its fibers $\mathbb{C}\setminus\mathsf{P}_\tau$ are; they agree for $\tau\in(\mathcal{U}\cap\mathbb{H}_q)\setminus\mathsf{E}$, so they agree on all of it, at the real nodes included. The right side of (580) is then holomorphic on the connected set $(\mathcal{U}\setminus\mathsf{E})\cup\mathbb{H}_q$ off the poles of the

ρ_k, and equals the trace on the residue strip, so it is the continuation of $\langle t_\ell^2 R_\tau\rangle_d$, and

$$\boxed{\begin{aligned}\mathcal{K}_d &= -\frac{AU}{4}+\frac{1}{4}[\tau^2]\Big\{C(\tau)\sum_{j=0}^{2} w_j(\tau)\,\Xi_{\mathrm{R}}\big(t_0+\tfrac{j}{3};\tau\big)-C(\tau)^2\mathcal{X}_d(\tau)^2\Big\},\\ \mathsf{r}_3 &= [\tau^3]\sum_{j=0}^{2} w_j(\tau)\,\Xi_{\mathrm{R}}\big(t_0+\tfrac{j}{3};\tau\big).\end{aligned}} \tag{586}$$

Thus, for $d\neq 2$, r_3 is a finite combination, with elementary coefficients, of the τ-derivatives of order at most three at $\tau=0$ of nine two-variable remainder sums (three blocks at three nodes), of one-variable remainder sums, and of Gamma values. At $d=2$ the triple pole of ρ_k and the spurious poles of individual pieces at $\tau=0$ raise the orders of the derivatives that enter; see the open item on a form adapted to $d=2$ (Section 21.12). In the computations below the jet is taken with a Cauchy circle; explicit polygamma-weighted two-variable series for these derivatives have not been written out.

No part-by-part estimate on the jet circle. The transfer cannot be replaced by a direct estimate of the lattice parts on the circle. Heuristically, for large $|\operatorname{Im} t|$ the right part behaves like a t-dependent factor times $\sum_{n\geq 0}(n+t)^{-1-2\tau}\approx t^{-2\tau}/(2\tau)$: on $\mathbb{H}_q$ it decays like $|\operatorname{Im} t|^{-2\operatorname{Re}\tau}$ (item Growth of Section 21.11, for $q\geq\frac14$), and for $\operatorname{Re}\tau<0$ it grows like $|\operatorname{Im} t|^{2|\operatorname{Re}\tau|}$. The total decays fast: given (H), $|\Xi_{\mathrm{R}}(x+\mathrm{i}y;\tau)|\leq\sum_{|k|\leq 1}|c_k(\tau)|\,e^{2\pi|y|}$, and since $|\sin\pi w|\geq\sinh\pi|\operatorname{Im} w|$ for each of the six sine factors of D_{R}, $S_{\mathrm{R}}=\Xi_{\mathrm{R}}/D_{\mathrm{R}}=O(e^{-4\pi|y|})$ for $|y|\geq 1+|\operatorname{Im}\tau|$. At six points of the jet circles ($d=0.8,1.25,1.5,2.5,3$; five of the six outside $\mathbb{H}_q$) the largest of the continued right, middle and left parts, times $|y|^{2\operatorname{Re}\tau}$, is constant to a few percent for $y=1,2,3$, while $|\Xi_{\mathrm{R}}|e^{-2\pi|y|}$ is constant; at $y=3$ the total is $1.2\cdot10^{15}$ to $1.6\cdot10^{16}$ times smaller than the largest part (records `n6_case0.log` to `n6_case5.log`, with the `.json` files of the same names, in `LRI_C5_general_d/K_general/note_fragment/growth/checks/`). On the circle the decay of the total is thus a cancellation between the parts, which no part-by-part bound can give; this is why (H) is a continuation statement.

Status of (H) and a route to a proof. The construction (585) is expected to satisfy (H). Its kernels (584) depend on (q,τ) only and are meromorphic in τ; their poles, and the points $\tau\in q+\mathbb{Z}$ where pole classes merge, belong to E. By (583) its model coefficients are the Gamma ratios x_0 times polynomials in Bernoulli polynomials of the parameters, hence holomorphic in (t,τ) wherever the x_0 are. Each model term gains one power in the remainder sums (the package uses $I=26$ to 30). Only the locally uniform remainder bound [27] is missing. An independent proof of (H), not carried out, could proceed as follows for the right block, with t in a compact strip. (i) Subtract from each of the two Gamma ratios of the right part, $\widetilde{\Phi}(z)$ at $z=\ell+n+t$ and $\widetilde{\Phi}(z+1-2q)$ at $z=n+t$ (Section 21.4), the single ratio $\mathfrak{m}_c(z)=\Gamma(z+c)/\Gamma(z+c+2q+\tau)$ with a constant c. The Gamma-ratio bound (465) makes the differences $O(|z|^{-\varkappa-1})$, $\varkappa=2q+\operatorname{Re}\tau$, and the remainder double sums then converge absolutely for $\operatorname{Re}\tau>\max(-\frac12,2q-2,-2q)$. (ii) The spin sum of $\mathfrak{m}_c$ is a Beta integral against $\sum_\ell d_\ell x^\ell=(1+x)(1-x)^{1-d}$ of (461), which gives two Beta functions. (iii) The remaining n-sum of $\mathfrak{m}_c$ times these Beta functions is a ${}_3F_2(1)$ with one upper parameter equal to 1 and parameter excess 2τ. A two-term Thomae relation, [22, eq. 16.4.11] applied twice, maps a ${}_3F_2(1)$ with upper parameters $(1,u,u')$, lower parameters (v,v') and excess $s=v+v'-1-u-u'$ (local names) to $\Gamma(v)\Gamma(v')\Gamma(s)/[\Gamma(s+u)\Gamma(s+u')]$ times the ${}_3F_2(1)$ with upper parameters $(v-1,v'-1,s)$ and lower parameters $(s+u,s+u')$, whose excess is 1. With $s=2\tau$ this continues the n-sum as $\Gamma(2\tau)$ times a ${}_3F_2(1)$ that converges for all τ. The left block is a Toeplitz sum of the same type; the middle block needs the two-variable kernels $\widetilde{\mathcal{G}}$ and $\mathcal{V}$ of (584).

21.9 Specializations to $d = 1, 2, 3$

$d = 1$. Here $d_\ell = (1, 1, 0, \dots)$ and the offset is $1 - d/2 = \frac{1}{2}$. The two spins merge into one sum over consecutive pairs $(w, w+\frac{1}{2})$ of the lattice $t+\frac{1}{2}\mathbb{Z}$, which has period $\frac{1}{2}$ in t, while $D_{\mathrm{R}}T_{\mathrm{R}} = -\frac{1}{8}\cos^3(2\pi t)$ is antiperiodic under $t \to t+\frac{1}{2}$. Hence $\Xi_{\mathrm{R}}(t+\frac{1}{2}) = -\Xi_{\mathrm{R}}(t)$: only $c_{\pm 1}$ survive, and $c_0 = c_{\pm 2} = 0$. Numerically, $C(\tau)$ times the sampled $\langle t_\ell^2 R_\tau\rangle_1$ reproduces (78) with the path parameter of J_t set to 2τ. With $\mathcal{K}_1 = \mathcal{K}$ of (76) (see (224)), (565) gives, conditionally on (M) at $d = 1$ (confirmed numerically above), the closed form

$$\begin{aligned}\mathsf{r}_3(1) &= \mathsf{s}_3(1) + 4\mathcal{K}_1 - 2\big(\pi^3 + 28\zeta(3)\big)\big(3\gamma_E + 7\log 2 + \pi\big)\\ &= 767.24520944465678609047981573762 24\dots,\end{aligned}$$

because $\frac{U}{8}[\psi_0(1) + \psi_0(\frac{1}{2}) + 4\psi_0(\frac{1}{4})] = \frac{1}{2}(\pi^3 + 28\zeta(3))(3\gamma_E + 7\log 2 + \pi)$ at $q = \frac{1}{4}$. No further condition enters: (J) at $d = 1$ is (59), on which (76) already rests.

$d = 2$. Here the offset vanishes and $d_\ell = (1, 2, 2, \dots)$. Then

$$\Psi_t(n) = \frac{\cos^3(\pi t)}{\sin^2(\pi t)\cos\pi(t+\tau)}\,\Phi_t(n), \qquad \Xi_{\mathrm{R}}(t) = -\Omega_{\mathrm{R}}(t)^2, \qquad \Omega_{\mathrm{R}}(t) = \cos^3(\pi t)\sum_{m\in\mathbb{Z}}\Phi_t(m),$$

because the lattice sum becomes the square of $\sum_m \Phi_t(m)$. The function Ω_{R} is antiperiodic, and, being entire with $|\Omega_{\mathrm{R}}|^2 = |\Xi_{\mathrm{R}}| = O(e^{2\pi|\operatorname{Im} t|})$ by (572), it has only the two modes $e^{\pm i\pi t}$. This is the square-lattice structure of Sections 18.9.3–18.9.5, where the trace is evaluated in closed form, $C(\tau)\langle t_\ell^2 R_\tau\rangle_2 = \pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2]$, that is (377); the generic samples reproduce it numerically. At $d = 2$ both (M) and (J) are proved, (M) for $0 < \tau < \frac{1}{2}$ in Section 18.9.5 and (J) in Section 18.8.4, so (565) and (271) give the exact value

$$\begin{aligned}\mathsf{r}_3(2) &= \mathsf{s}_3(2) + 128\mathsf{G}^2 - 168\,\zeta(3)\log 2 - 42\gamma_E\zeta(3)\\ &= 35.63714944308140680727039972168179\dots.\end{aligned}$$

$d = 3$. Here the offset is $\frac{1}{2}$ and $d_\ell = 2\ell + 1$. The half-shift argument of (429), repeated for this kernel, gives the following for $\frac{1}{2} < \tau < \frac{9}{8}$; the derivation is not written out here, and the identity is checked numerically at six points to $1.1 \cdot 10^{-24}$ (Section 21.11).

$$\Xi_{\mathrm{R}}(t) + \Xi_{\mathrm{R}}(t+\tfrac{1}{2}) = -\frac{\pi}{16}\cos^3(2\pi t)\,[\mathcal{M}^1_{\mathrm{o}}\mathcal{M}^0_{\mathrm{e}} - \mathcal{M}^0_{\mathrm{o}}\mathcal{M}^1_{\mathrm{e}}], \qquad \mathcal{M}^k_{\mathrm{e,o}} = \sum_{n\in\mathbb{Z}}(n+u)^k\,\widetilde{\Phi}(n+u-\tfrac{5}{8})\big|_{q=3/4},$$

with $u = t + \frac{1}{8}$ (e) or $u = t + \frac{5}{8}$ (o) and $\widetilde{\Phi}$ of Section 21.4, so that $\widetilde{\Phi}(w-\frac{5}{8})|_{q=3/4} = \Gamma(w+\frac{1}{8})^3/[\Gamma(w+\frac{7}{8})^2\Gamma(w+\frac{1}{8}+\tau)]$; at $\tau = q = \frac{3}{4}$ this is the cube of the function $\Gamma(x+\frac{1}{8})/\Gamma(x+\frac{7}{8})$ of (406). By (572) the left side is the constant $2c_0$. So c_0 is an ordinary 2×2 determinant of one-variable bilateral moments, while this identity does not reduce $c_{\pm 1}$; this is the R analogue of (431). No closed form of $\mathcal{K}_3$ is known. The residue series (260), whose derivation away from $d = 2$ uses (M) and the extraction of $[\tau^2]$ under the trace (Section 21.12), summed in ball arithmetic, gives 430 digits (`LRI_C5_d3/wf_K3/K3_value.json`), $\mathcal{K}_3 = 1.14859464952717933655664631081067318763\dots$; the position-space value of Section 21.10, which uses neither, agrees with it to $3.6\cdot 10^{-100}$ (Section 21.11). With this value, (565), conditionally on (M) and (J) at $d = 3$, gives $\mathsf{r}_3(3) = 4.51175935823600578025662089796157 5\dots$. A PSLQ search for $\mathcal{K}_3$ at 420 digits found no integer relation in the bases listed in `LRI_C5_d3/wf_K3/pslq_K3_results_420.json`.

Consistency. At $d = 1, 2, 3$ the five-node version of (586) (nodes $t_0 + j/5$, modes $|k| \le 2$) agrees with (76), (271) and the 430-digit (260) value of $\mathcal{K}_3$ to $5 \cdot 10^{-37}$, $5 \cdot 10^{-48}$ and $1.4 \cdot 10^{-43}$ (relative), and the position-space value of Section 21.10 agrees with them to about 10^{-100} (Section 21.11). The Laurent coefficients $\mathsf{r}_3(1)$ and $\mathsf{r}_3(2)$ of the sampled jets agree with the closed forms above to 10^{-32}.

21.10 Position-space representation

By completion symmetry $J_t = J_d(d/2, d/2, d/2, d-t, d/2) = J_d(d-t, d/2, d/2, d/2, d/2)$. The y-integral is then the one-loop star with indices (q,q,q), which has a two-fold Mellin–Barnes representation, and the x-integral is a bubble. With $\mathsf{N}_q = \Gamma(2q)^2/\Gamma(q)^4$ and $z_{12} = z_1 + z_2$,

$$J_t = \mathsf{N}_q \iint \frac{dz_1\,dz_2}{(2\pi\mathrm{i})^2}\,\Gamma(-z_1)^2\Gamma(-z_2)^2\Gamma(q+z_{12})^2\frac{\Gamma(q+z_2)}{\Gamma(q-z_2)}\,\frac{\Gamma(\frac{t}{2}+z_1)\Gamma(q-\frac{t}{2}-z_{12})}{\Gamma(2q-\frac{t}{2}-z_1)\Gamma(q+\frac{t}{2}+z_{12})}, \tag{587}$$

on $-\frac{t}{2} < \operatorname{Re} z_1 < 0$, $-q < \operatorname{Re} z_2 < 0$, $\operatorname{Re} z_{12} > -q$. Moving $\operatorname{Re} z_1$ to the left of $-\frac{t}{2}$ crosses one simple pole. By Barnes' first lemma its residue is $J_t^{\mathrm{pole}} = \Gamma(2q)\Gamma(\frac{t}{2})^2\Gamma(q-\frac{t}{2})^4/[\Gamma(q)^4\Gamma(2q-t)]$, and the remaining integral is holomorphic at $t=0$. The coefficient $[t^2]$ of $J_t - J_t^{\Gamma}$ is therefore the sum of the polygamma part $\mathcal{K}_d^{\Gamma} := [t^2](J_t^{\mathrm{pole}} - J_t^{\Gamma})$ and the coefficient $\mathcal{I}_d$ of t^2 in the remaining integral:

$$\boxed{\mathcal{K}_d = \mathcal{K}_d^{\Gamma} + \mathcal{I}_d,} \tag{588}$$

$$\mathcal{K}_d^{\Gamma} = \frac{A^2\vartheta}{2} + \frac{A}{2}\big[\psi_2(2q) - \psi_2(q)\big] - \frac{R\vartheta}{4} + \frac{\vartheta^2}{8} + \frac{2\psi_3(q) - 7\psi_3(2q)}{48}, \qquad \vartheta = 2\psi_1(q) - \psi_1(2q),$$

$$\mathcal{I}_d = \mathsf{N}_q \iint \frac{dz_1\,dz_2}{(2\pi\mathrm{i})^2}\,\mathfrak{M}_0\,\frac{\varpi_1^2 + \varpi_2}{2},$$

$$\mathfrak{M}_0 = \frac{\Gamma(-z_1)^2\Gamma(z_1)}{\Gamma(2q-z_1)}\,\frac{\Gamma(-z_2)^2\Gamma(q+z_2)}{\Gamma(q-z_2)}\,\Gamma(q+z_{12})\Gamma(q-z_{12}),$$

$$\varpi_1 = \tfrac{1}{2}\big[\psi_0(z_1) + \psi_0(2q-z_1) - \psi_0(q-z_{12}) - \psi_0(q+z_{12})\big],$$

$$\varpi_2 = \tfrac{1}{4}\big[\psi_1(z_1) - \psi_1(2q-z_1) + \psi_1(q-z_{12}) - \psi_1(q+z_{12})\big],$$

on $-q < \operatorname{Re} z_1, \operatorname{Re} z_2 < 0$, $\operatorname{Re} z_{12} > -q$ (for example $\operatorname{Re} z_1 = \operatorname{Re} z_2 = -q/3$). By (J) the orders t^0 and t^1 of $J_t - J_t^{\Gamma}$ vanish, which gives two Barnes identities:

$$\mathsf{N}_q \iint \frac{dz_1\,dz_2}{(2\pi\mathrm{i})^2}\,\mathfrak{M}_0 = \psi_1(2q) - 2\psi_1(q), \qquad \mathsf{N}_q \iint \frac{dz_1\,dz_2}{(2\pi\mathrm{i})^2}\,\mathfrak{M}_0\,\varpi_1 = -\Big[A\vartheta + \frac{\psi_2(2q)}{2} - \psi_2(q)\Big]. \tag{589}$$

This route uses only the definition (238), which defines $\mathcal{K}_d$ as the coefficient $[t^2]$; it uses none of (M), (J), (H), and not the spectral kernel. Its agreement with (586) at the level of 10^{-41} to 10^{-47} (Section 21.11) is therefore a numerical test of (M), and of (H) on the jet circle, at non-integer d. The completed graph of J_t can be put into this star-and-bubble form in 60 ways (choice of the vertex sent to infinity, the two external vertices, the star vertex and the bubble vertex). A computer enumeration shows that each of the 60 two-fold Mellin–Barnes integrands contains a squared Gamma function, so this framework has no derivative-free residue expansion of J_t. The kite with general indices can also be written as a sum of four Kampé de Fériet series $F^{2:2;2}_{2:1;1}(1,1)$ and two Horn double series. This form is not derived or used in this note; it is recorded, with its convergence conditions, in `LRI_C5_general_d/K_general/position_space/position_space_summary.json`.

21.11 Numerical confirmation

Values. The first table gives $\mathcal{K}_d$ from (588), evaluated to 100 digits by the trapezoid rule in the ball arithmetic of the Arb library [48] on $\operatorname{Re} z_1 = \operatorname{Re} z_2 = -q/3$ (`LRI_C5_general_d/K_general/position_space/K_d_values_mb_D100.json`), and r_3 obtained from it through (565). That column assumes (M) and (J); it agrees with the τ^3 coefficients of the 34-digit Laurent tables of the directly sampled five-node jets to $3\cdot10^{-34}$ (relative). It is printed to 34 digits, so that (565) with the printed r_3 reproduces the printed $\mathcal{K}_d$ to $3\cdot10^{-32}$ (relative; 10^{-33} against the 100-digit values). The digits shown are correctly rounded; … marks omitted digits.

d	$\mathcal{K}_d$, position space (588)	$\mathsf{r}_3 = [\tau^3]\langle t_\ell^2 R_\tau\rangle_d$ by (565)
0.8	92.372770239922771632645391899036…	1894.360867513991151448939858484147…
1	39.331569128259039057503187151598…	767.245209444656786090479815737622 4…
1.25	17.203345729921688327085720106549…	302.137977543021229758469687902 9900…
1.5	9.018297401594820547084743104 6793…	136.568544321226036920660372299 8423…
2	3.518109138590252241649090170 3077…	35.637149443081406807270399721681 79…
2.5	1.840501295428588030730181860 2723…	11.467317760441838785767751689501 30…
2.75	1.429249188660312000099175536 8839…	6.984719816576732523926795215898772…
3	1.148594649527179336556646310 8107…	4.511759358236005780256620897961575…

Routes. The second table gives relative deviations from the position-space values. "Spectral reference" is the τ-circle evaluation of (248) in `LRI_C5_general_d/references/reference_values.json` (50 printed digits, of which that file states 45 to 47 to be correct). "Abel–Plana" is (248) with an Abel–Plana spin tail at 60 digits, equal to the residue series (289) to better than 10^{-53} for $d \le 2.5$ and to $5\cdot10^{-49}$ at $d = 2.75$ (`LRI_C5_general_d/K_general/refcheck/`). The two jets are the first line of (586), that is (566), evaluated with the five nodes $t_0 + j/5$ (modes $|k| \le 2$) and with the three nodes of (580), on a Cauchy circle of 48 points and radius 0.08 (0.04 at $d = 0.8$), at 72 digits. The three-node circles at $d = 1.5$, 2.5 and 3 were computed by the separate check code of an independent AI review of Sections 20 and 21 (`LRI_C5_general_d/review/joint/numerics-end-to-end/k3_jets.json`); through (565) instead of (566) they give $3.8\cdot10^{-45}$, $5.0\cdot10^{-44}$ and $-8.1\cdot10^{-43}$ (`LRI_C5_general_d/K_general/note_fragment/test_build_joint/check_k3_routes.log`). The circles at $d = 0.8$, 2.5, 2.75 and 3 lie entirely outside $\mathbb{H}_q$. The last column has the opposite sign convention: it gives the position-space value minus the reference, relative to the reference, for the references (76), (271) and the 430-digit (260) value of $\mathcal{K}_3$, the last conditional on (M) and on the extraction of $[\tau^2]$ under the trace.

d	spectral reference	Abel–Plana	five-node jet	three-node jet	position space vs (76), (271), (260)
0.8	$-1.9\cdot10^{-49}$	$8.2\cdot10^{-57}$	$-8.0\cdot10^{-47}$	$-8.1\cdot10^{-47}$	–
1	–	–	$-4.8\cdot10^{-37}$	–	$5.2\cdot10^{-101}$
1.25	$1.8\cdot10^{-49}$	$-2.4\cdot10^{-55}$	$-1.0\cdot10^{-41}$	$-1.0\cdot10^{-41}$	–
1.5	$1.9\cdot10^{-49}$	$2.1\cdot10^{-56}$	$-9.6\cdot10^{-46}$	$-1.4\cdot10^{-45}$	–
2	–	–	$5.2\cdot10^{-48}$	–	$-1.5\cdot10^{-100}$
2.5	$5.2\cdot10^{-51}$	$-1.2\cdot10^{-55}$	$5.4\cdot10^{-44}$	$4.1\cdot10^{-44}$	–
2.75	$2.1\cdot10^{-50}$	$-8.3\cdot10^{-56}$	$6.0\cdot10^{-44}$	$-3.5\cdot10^{-44}$	–
3	–	–	$-1.4\cdot10^{-43}$	$-9.5\cdot10^{-43}$	$3.6\cdot10^{-100}$

Further checks. All are numerical, with empirical error control and no interval enclosures.

- *S family.* (556) agrees with the spectral trace at 36 points ($0.8 \le d \le 3.5$, real and complex τ) to $2\cdot10^{-38}$ or better, with an independent real-axis lattice route at 18 points to $4\cdot10^{-37}$ or better, and with the continued spectral trace on τ-circles to $8\cdot10^{-29}$.
- *Low orders.* With the 34-digit Laurent tables of the five-node jets, $|\mathsf{r}_0 - \mathsf{s}_0|$, $|\mathsf{r}_1 - \mathsf{s}_1|$ and $|\mathsf{r}_2 - \mathsf{s}_2 - \frac{3}{2}U|$ are at most $5\cdot10^{-32}$ at all eight d, and (565) reproduces $\mathcal{K}_d$ to 10^{-33}. In the 72-digit jets $|F_2 - \frac{3}{2}U|$ lies between 10^{-47} and $4\cdot10^{-35}$.
- *Inputs (M) and (J).* The mixed trace (M) holds pointwise at three complex τ on the τ-circles of the spectral reference (paragraph Routes) at $d = 0.8, 1.25, 1.5, 2.5, 2.75$ to $4\cdot10^{-32}$. For (J), at $d = 1.25$ the full quadratic jet of $J_d - (G_2 + G_1 - G_0 + \frac{2}{p_1}\Delta_B)$ of (237) at its base point $(d/2, d/2, d/2, d, d/2)$, reconstructed from its restrictions to 18 random lines through that point, has only the $z_1 z_2$ coefficient, equal to $\mathcal{K}_d$ to $2\cdot10^{-12}$; the other 14 coefficients are below $3\cdot10^{-12}$. The orders t^0, t^1, that is (589), hold to 10^{-101} at all eight d.

- *Samples.* (581) with (585) agrees with the spectral trace at eight (d,τ) points, three of them outside $\mathbb{H}_q$, to $1.1\cdot10^{-19}$ or better (relative; at $d=0.8,1.25$ against the spectral trace with an analytic ν-tail). The termwise identities behind (568) and (582), the kernels (584) and ρ_k of (579) hold to $6\cdot10^{-39}$ (`LRI_C5_general_d/K_general/note_fragment/fragment_checks/check_blocks.py`). At $d=1,2$ the generic samples reproduce (78) and (377) at two values of τ each, to 10^{-21} and $2\cdot10^{-19}$ (absolute), and at $d=3$ the determinant form of Section 21.9 holds to $1.1\cdot10^{-24}$ at six points.

- *Growth.* At six points of $\mathbb{H}_q$ ($0.8\le d\le3$, real and complex τ) the three parts obey the bound of Section 21.4, which is attained for $q\ge\frac14$, while the total $S_{\mathrm R}$ is up to 10^{25} times smaller and decays like $e^{-4\pi|\operatorname{Im}t|}$.

- *Growth constants.* These test the estimates used in Section 21.6. Stirling's error bound, the input of (465), holds at 4000 random points with $|\arg w|\le0.97\pi$ and $0.05\le|w|\le10^4$, the largest ratio of $|\epsilon_\Gamma(w)|$ to the bound being 0.99999973; in (465), $|\theta||z|/\omega(\rho)\le0.29$ on random samples for $\rho=0.25,\dots,3$, the corner $|z|=\max(1,2\rho)$, $\arg z\to\pm\frac{\pi}{2}$ included. For $0.02\le q\le0.99$ the constant C_q of (466), evaluated over $\ell\le2\cdot10^4$, lies between 1.04 and 7.9, and $d_\ell(1+\ell)^{2-d}$ approaches $2/\Gamma(d-1)$. At ten pairs (q,α), some close to the thresholds (573), and $Y=2,\dots,512$, the sums $\Sigma^{\mathrm r}(Y)$ and $\Sigma^{\mathrm m}(Y)$ of (467) (FFT convolutions of 2^{22} terms with a fitted tail) stay below their bounds, the largest ratio being 0.81, and the exact limits of $Y^{2\alpha-\max(d,1)}\Sigma^{\mathrm r}(Y)$ (for $d\ne1$) and $Y^{2\alpha-d}\Sigma^{\mathrm m}(Y)$ as $Y\to\infty$ lie below $C^{\mathrm r}$ and $C^{\mathrm m}$. The Horn-block code of (581), used for the item Growth, reproduces the three parts (575), computed as Richardson-extrapolated direct lattice sums in ball arithmetic, at twelve complex t with $|\operatorname{Im}t|\le4$ ($d=0.8,1.5,2.5,3.4$, $\operatorname{Re}\tau=2.37$ or 2.6) to between $2\cdot10^{-17}$ and $5\cdot10^{-9}$ (relative), within the error of the direct sums. At the six points of the item Growth, $C_{\mathfrak K}$ of Section 21.6 for the single point $\mathfrak K=\{(q,\tau)\}$ is $4\cdot10^{31}$ to $4\cdot10^{44}$; its limit for large $|y|$ is $2.6\cdot10^3$ to $2.7\cdot10^4$, which is 200 to 1000 times the observed maximum of $|S^{\rightarrow}|$, $|S^{|}|$, $|S^{\leftarrow}|$ times $|y|^{\eta_{\mathrm R}}/(1+\log|y|)$. Records: `n1_stirling.log`, `n2_mult.log`, `n3_lattice.log`, `n4_crossval.log` and `n8_constants.log` in `LRI_C5_general_d/K_general/note_fragment/growth/checks/`.

- *Top modes.* With the Horn-block code at eleven points, $0.3\le d\le3.9$, including two continued points, $|c_{\pm2}|/\max_k|c_k|$ lies between $6\cdot10^{-33}$ and $3\cdot10^{-27}$ and falls with the truncation error. An independent per-spin code gives ratios below 10^{-15} at five points of $\mathbb{H}_q$, where single spins contribute $O(1)$, each below the spread of its spin-sum extrapolation (at most $8.5\cdot10^{-16}$); three unbalanced controls give between $3\cdot10^{-3}$ and $6\cdot10^{-2}$. At six points of the jet circles the continued samples have $|c_{\pm2}|/\max_k|c_k|\le1.3\cdot10^{-37}$.

- *Three nodes.* At $d=0.8,1.25,2.75$ the three-node $\mathcal K_d$ equals the five-node value to $3\cdot10^{-49}$, $3.5\cdot10^{-47}$ and $9.5\cdot10^{-44}$, at 0.5 to 0.65 of the wall time per circle point. At the two points of the item Top modes nearest to $d=4$, $d=3.7$ and 3.9, the three-node and five-node traces differ by $8\cdot10^{-24}$ and $1.5\cdot10^{-21}$ (relative), more than the top modes, which are about 10^{-27} of the largest mode. The cause is the conditioning of the residue functional, not the rule: at $d=3.9$, $\tau=1.2{+}0.3\mathrm i$, the weights (579) have $|\rho_k|\approx2{\cdot}10^4$, through the factor $\cot(\pi q)/\sin^3(\pi q)$, against a trace of modulus about 9, so (578) cancels about five digits and amplifies the truncation error of the samples, for both rules alike; at $d=3.9$ the difference is $5\cdot10^{-14}$ with the coarser of the two truncations (`modes_d3.7_t0.95+0.15j.json` and `modes_d3.9_t1.2+0.3j.json` in `LRI_C5_general_d/K_general/note_fragment/topmode/results/`).

- *Position space.* Two contours and step sizes agree to $3{\cdot}10^{-55}$ (relative) at $d=0.8,1.25,1.5,2.5$. An independent double-precision evaluation of (588) and (589), written from the formulas as printed here, reproduces $\mathcal K_d$ to $7\cdot10^{-14}$ (relative) and the two identities to 10^{-12} absolute ($3\cdot10^{-15}$ relative) at all eight d (`LRI_C5_general_d/K_general/note_fragment/fragment_checks/check_barnes.py`).

- *A caution.* The literal bracket of (260), summed with the Abel–Plana tails of `LRI_C5_general_d/K_general/refcheck/routeB_residue.py`, fails numerically at $d = 0.8$, the only $d < 1$ tested: it gives 70.3 at 40 digits and 2539.5 at 60 digits in place of $\mathcal{K}_{0.8} = 92.37\ldots$, while the canceled form (289), summed by the same code, works. For $1.25 \le d \le 3$ the two brackets agree to the working precision. The cause is numerical. The two brackets agree term by term, also at non-integer and complex indices, but the literal one is a difference of two nearly equal terms of order n^{-d} in the radial index n, and that code fixes its constants at 20 digits beyond its working precision, so the difference keeps no correct digit beyond $n \approx 10^{80}$ at 60 digits (10^{60} at 40 digits). At $d = 0.8$ the radial tail of that code reaches $n \approx 10^{394}$ ($3.5 \cdot 10^{279}$), where n times the corrupted term is not small; at $d = 1.25$ it stops near $n \approx 10^{93}$ (10^{67}), and n times the error stays below 10^{-70} (10^{-50}), far below the working precision (`check_kbracket_d08.log` and `check_kbracket_d08_huge.log` in `LRI_C5_general_d/K_general/note_fragment/test_build_joint/`). This does not bear on the convergence statement after (260).

21.12 Status

Proved. The following have written proofs, in this section or in Section 20. Here an independent review is a separate AI session that re-derived the statement and tried to refute it, and a numerical probe is a separate code. Unless stated otherwise, a statement has had no independent review.

- The closed form (556) and its Laurent form (559): Gamma algebra, the kernel identity (557), and the two-wheel convolution $W_2(d; b') = B_d(d - b', 2b')$ of (230), also used for (250). Three separate AI sessions re-derived it independently and tried to refute it; none found an error.
- The constant (560) and the bilateral identity (561) (Section 21.2, remark (iii)): consequences of the case $r = 2$ of (473), of (479) and of (230); the constancy and the value at $b = d/2 - \tau$ are confirmed numerically there. No independent review.
- The Laurent series (563): exact Gamma algebra, checked symbolically for general q.
- The lattice and product forms (568) and (569), the sample formula (581) with the arrays (582), and the kernels (584): exact derivations. Separate AI sessions re-derived the lattice and product forms, the sample formula and the arrays and tried to refute them, without finding an error; all of them, the kernels included, are rechecked numerically to $6 \cdot 10^{-39}$ (Section 21.11, item Samples).
- The growth theorem of Section 21.4, proved in Section 21.6: growth bound with explicit constants, finite Fourier space with $|k| \le 2$, and sharpness of $\mathbb{H}_q$. Parts (a) (sufficiency), (b) and (c) are the case $r = 3$ of (471); an independent AI review re-derived every step of that proof, Step 0 included, by hand (Section 20.10). The sharpness of $\mathbb{H}_q$ and the explicit constants of Section 21.6 have had no independent review.
- The theorem (572) (R family, modes $|k| \le 1$) on $\mathbb{H}_q$. It is the case $r = 3$ of the product form (471) of the top-mode theorem, which has a written proof (Section 20.1.2). An independent AI review re-derived every step of that proof by hand and found no fatal or major issue (record `LRI_C5_general_d/review/lemmaG-proof`); it also checked the R-family parameters of Section 21.5 and the equivalence of condition (C) of (469) with $\tau \in \mathbb{H}_q$. A separate 48-digit lattice-sum code finds the top modes of four R kernels below $5.2 \cdot 10^{-45}$ of the largest mode (Section 20.9; `LRI_C5_general_d/review/lemmaG-numeric`).
- The residue functional (578) and the three-node rule (580) for $\tau \in \mathbb{H}_q$, $\operatorname{Re}\tau < \frac{3}{2}q$, $\tau \ne q$.
- The representation (587) and (588): exact derivation in Section 21.10 from (238) (completion symmetry, the two-fold Mellin–Barnes form of the one-loop star, Barnes' first lemma). It

uses none of (M), (J), (H), since (238) defines $\mathcal{K}_d$ as the coefficient $[t^2]$. Separate AI sessions re-derived the statement and tried to refute it, without finding an error; the derivation printed in Section 21.10 has not been reviewed. A separate code, written from the formulas as printed here, re-evaluates them (Section 21.11, item Position space; `LRI_C5_general_d/K_general/note_fragment/fragment_checks/check_barnes.py`).

Conditional.

- (564) and (565) are exact given (M) and (J), and (566) given (M) alone. The note proves (M) only at $d = 2$; at other d it is numerical. Its strongest test is the agreement of (586) with the position-space route, which uses none of (M), (J), (H); on the jet circle this agreement tests (M) and (H) together. (J) is the path form of (237), derived in Section 16, on which (223) itself rests. At $d = 2$ both inputs are proved, (M) for $0 < \tau < \frac{1}{2}$ in Section 18.9.5 and (J) in Section 18.8.4, so there these statements and the value of $\mathsf{r}_3(2)$ in Section 21.9 are proved. Elsewhere, including the values of $\mathsf{r}_3(1)$ and $\mathsf{r}_3(3)$ in Section 21.9 and the r_3 column of the first table of Section 21.11, they are conditional on (M) and (J). At $d = 1$, (J) is (59), on which (76) itself rests, so there $\mathsf{r}_3(1)$ is conditional on (M) alone.
- The two Barnes identities (589) hold given (J).
- (567) needs, in addition, $[\tau^2]$ under the trace. The derivation of (248) uses the same step, which is justified only at $d = 2$ (corollary of Section 18.9.2); there (567) is (394).
- On the jet circle, (572), (580) and (586) need (H), the analyticity in τ of the continued samples; the first line of (586) needs in addition only (M), through (566). None of the circles used in Section 21.11 lies in $\mathbb{H}_q$.
- For $d \le \frac{4}{7}$ the strip of (M) is empty; there the residue functional, and with it the definition of $\langle t_\ell^2 R_\tau \rangle_d$ and of the r_k, is only sketched. For $\mathrm{Re}\,\tau \ge \frac{3}{2}q$ the residue functional is only sketched (Section 21.7, paragraph "Separating contours").
- For $C_5(d)$ these inputs reduce to (M), plus (H) when r_3 is sampled on the jet circle as in (586): (J) adds no condition, since (223) already rests on (237), and with $\mathcal{K}_d$ from (588) none of (M), (J), (H) enters. The range of (553) is $\frac{2}{3} < d < 4$, so the caveat for $d \le \frac{4}{7}$ never affects $C_5(d)$.

Numerical only. All digit agreements of Section 21.11, and the $d = 3$ determinant form of c_0 in Section 21.9. The 100-digit values are not interval enclosures: ball arithmetic bounds the rounding error, while discretization and truncation are controlled empirically. The tested range is $0.8 \le d \le 3$ for $\mathcal{K}_d$ and $0.3 \le d \le 3.9$ for the top modes. A reduction of r_3 to known constants was not attempted at generic d; at $d = 3$ the PSLQ search was negative. The half-shift factorization of (429) is not available at generic d, and there is no derivative-free Mellin–Barnes residue expansion of J_t (Section 21.10).

Open.

- An independent review of the sharpness of $\mathbb{H}_q$ and of the explicit constants of the growth theorem (Section 21.6).
- A proof of (M) at generic d, and a written proof of (H) with uniform remainder bounds.
- Explicit polygamma-weighted two-variable series for r_3 (the τ-derivatives of the remainders in (585)), and a form adapted to $d = 2$, where individual pieces have removable singularities.

- A PSLQ search on r_3 or $\mathcal{K}_d$ with the 100-digit values, keeping $n \log_{10}(\max|c|)$ at least ten digits below the working precision and confirming any relation at a second precision.
- Dimensions $d < 0.8$, and the five-loop primitive kernel, which (471) does not cover.
- A confluent series for J_t on the path of (238) for $d \geq 2$ at small t. The confluent form recorded in `LRI_C5_general_d/K_general/position_space/position_space_summary.json` holds only for $0 < d < 2 + 2\,\mathrm{Re}\,t$; the confluent limit of the formula with four Kampé de Fériet and two Horn series mentioned at the end of Section 21.10, which would cover $d \geq 2$, has not been written.

Part IV

Numerical sixth- to eighth-order coefficients

22 The sixth-order coefficient in one dimension

22.1 Result, convention, and scope

This section extends the one-dimensional calculation by one perturbative order. It does not extend the arbitrary-dimensional formula to sixth order and does not alter any fifth-order identity of the analytic parts of this note. The regulator and normalization remain

$$\epsilon = 2s - 1, \qquad \Delta_{\mathcal{E}} = \frac{1}{2} - \frac{\epsilon}{6} + \sum_{n=2}^{6} C_n(1)\epsilon^n + O(\epsilon^7). \tag{590}$$

The result of the graph-based numerical calculation is

$$\boxed{C_6(1) = 14331.68703277 \pm 0.00000049,} \tag{591}$$

that is, $C_6(1) = 14331.6870328(5)$. The value is the mean of the two infrared-cutoff evaluations at quadrature degree $N_* = 22$. Both are computed by exact summation over every nonzero six-loop graph and every gap ordering, as described in Section 22.5. The quoted uncertainty is the pre-registered empirical allowance of Section 22.7. It adds the last observed change with degree, the residual difference between the two cutoffs, a measured floating-point allowance, and the convergence of the lower-loop inputs. It is not a rigorous interval enclosure or a probability guarantee, and it is not a bound on the omitted orders of the ϵ series.

This is a six-loop graph integration. It is not a Padé prediction, large-order fit, interpolation in dimension, or extrapolation from the five previously known physical coefficients, which do not determine C_6. No Monte Carlo sampling enters the final value. An earlier evaluation combined exhaustive degree-six sums, deterministic hard-sector refinements and Monte Carlo corrections for the higher degrees. It gave 14331.77075 with sampling standard error 0.14344, quoted as 14331.8 ± 0.5. The exact sums show that this estimate was high by 0.084, or 0.58 of its standard error; its sampled level differences are consistent with the exact ones (Section 22.8).

Section 22.9 compares (591) with a finite special-function representation of a reduced analytic expression for $C_6(1)$, derived separately; the two agree to 1.0×10^{-8}. Section 1.2 quotes that representation, 14331.68703275805, as the value of $C_6(1)$, and (591) is its independent numerical confirmation.

22.2 Exactly which new amplitudes are required

Write the beta function and mass anomalous dimension in the minimal-subtraction conventions of Sections 2 and 3:

$$\beta(g) = -\epsilon g + g^2 B(g), \quad B(g) = 3 + b_2 g + \cdots + b_6 g^5, \qquad \gamma_r(g) = g + c_2 g^2 + \cdots + c_6 g^6. \tag{592}$$

Since $\epsilon = g_* B(g_*)$, Lagrange inversion gives, for $n \geq 2$,

$$C_n = \frac{1}{n}[g^{n-1}]\gamma_r'(g) B(g)^{-n}. \tag{593}$$

With $A = \pi + 4\log 2$, $b_2 = -6A$, and $c_2 = -A$, the exact sixth-order assembly is

$$\begin{aligned} 2187 C_6 = {} & 3c_6 - b_6 + 2016A^5 - 448A^3 b_3 + 672A^3 c_3 - 84A^2 b_4 + 168A^2 c_4 \\ & + \frac{49}{3} A b_3^2 - 42A b_3 c_3 - 12A b_5 + 30A c_5 + \frac{7}{3} b_3 b_4 - 4 b_3 c_4 - 3 b_4 c_3. \end{aligned} \tag{594}$$

The script `LRI_C6_1D_HPC/reference/verify_algebra.py` is an independent exact-rational polynomial check of (594). It also checks the sensitivities $\partial C_6/\partial c_6 = 1/729$ and $\partial C_6/\partial b_6 = -1/2187$.

Let $\mathcal{A}_L^\chi(\epsilon)$ and $\mathcal{B}_L^\chi(\epsilon)$ denote the bare coupling and mass amplitudes for the cutoff χ. These are amplitudes, not the scalar A or the function $B(g)$ in (592). Define

$$H_6^\chi = [\epsilon^{-1}](3\mathcal{B}_6^\chi - \mathcal{A}_6^\chi), \qquad J_5^\chi = [\epsilon^0](8\mathcal{B}_5^\chi - 3\mathcal{A}_5^\chi). \tag{595}$$

The only new five- and six-loop inputs to the physical assembly are these two numbers. Specifically,

$$\boxed{C_6 = R_6^\chi + \frac{4}{243} J_5^\chi - \frac{2}{729} H_6^\chi.} \tag{596}$$

Here R_6^χ is obtained from the same Laurent minimal-subtraction recursion, setting the five-loop finite terms and six-loop simple poles to zero, while retaining all previously required poles. Two-loop amplitudes are now needed through ϵ^3, three-loop amplitudes through ϵ^2, and four-loop amplitudes through ϵ^1. Thus R_6^χ contains newly calculated higher Laurent data even though it contains no new six-loop master.

For clarity, the new contributions to the renormalization coefficients are $b_6 = 108 a_{5,0} - 6 a_{6,-1} + \cdots$ and $c_6 = 96 e_{5,0} - 6 e_{6,-1} + \cdots$, where a, e are the coefficients of $\mathcal{A}, \mathcal{B}$. The combination $3c_6 - b_6 = 36 J_5 - 6 H_6 + \cdots$ proves the factors and signs in (596). Unit shifts of J_5 and H_6 were also checked through the full high-precision recursion.

22.3 Exhaustive graph enumeration and exact weights

At L loops the ordered-coordinate representation has $L+1$ vertices and $2L$ internal edges. An integer multiplicity $m_{ij} \in \{0, 1, 2, 3\}$ is assigned to each pair. Tadpoles are omitted, as in the five-loop graph set (12); every vertex has degree between two and four, and every nontrivial cut has at least two crossing edges. The latter requirement enforces connectivity and one-particle irreducibility.

If $e_i = 4 - \deg(i)$, the rational Wick weights (13) are, with $w_A = w_4$ and $w_B = w_2$,

$$w_A(G) = \frac{24}{\prod_i e_i! \prod_{i<j} m_{ij}!}, \qquad w_B(G) = \frac{w_A(G)}{12} \sum_i e_i(e_i - 1), \qquad w_H = 3 w_B - w_A. \tag{597}$$

All weights are stored exactly as integers in units of $1/1728$. The fresh enumeration gives

	Five loops	Six loops
Labeled 1PI graphs	59,145	2,427,005
Zero w_H graphs	9,540	289,590
Nonzero w_H graphs	49,605	2,137,415
$\sum_G w_A$	197,887.5	7,716,476.25
$\sum_G w_B$	21,562.5	732,471.25
$\sum_G w_H$	$-133,200$	$-5,519,062.5$

These are labeled graph counts, not counts of topological isomorphism classes or master integrals. Every surviving six-loop w_H is negative; zero-weight graphs are removed by exact integer arithmetic before integration. The 720 gap orderings reduce to 360 after spatial reversal; this is an exact symmetry reduction, not a sampled subset of orderings.

22.4 Meromorphic quadrature and the two cutoffs

Let $D = L - 1$ and order the L consecutive gaps by magnitude. If R is the largest gap, the smaller ones are $Rt_0, Rt_0t_1, \ldots, Rt_0 \cdots t_{D-1}$, with $0 \le t_j \le 1$. For an edge e, factor its distance as

$$|x_i - x_j| = R \left(\prod_{j<r_e} t_j \right) Q_e(t), \qquad Q_e(t) > 0, \quad b_j = \sum_{e:r_e>j} m_e, \qquad a_j = L - 2 - j - \frac{b_j}{2}. \tag{598}$$

Here r_e is the rank of the largest gap crossed by that edge. Each Q_e is a sum of products of consecutive t_j, including a constant term 1. Define

$$H(t) = 1 + t_0 + t_0t_1 + \cdots + t_0 \cdots t_{D-1}. \tag{599}$$

The two infrared prescriptions are $R < 1$ (maxgap) and $RH(t) < 1$ (span). Integrating the common scale exactly, a weighted graph-sector contributes

$$\frac{w_G}{L\epsilon} \int_{[0,1]^D} \prod_j dt_j \, t_j^{a_j + \epsilon b_j/2} \prod_e Q_e(t)^{-m_e/2} \exp\left\{ \epsilon \left[\frac{1}{2} \sum_e m_e \log Q_e(t) - L \, \mathbf{1}_{\chi=\text{span}} \log H(t) \right] \right\}. \tag{600}$$

The integral is meromorphically continued in ϵ before taking its Laurent coefficients. It is not an ordinary convergent integral at $\epsilon = 0$ in every sector.

Each smooth factor is tensor-interpolated on the degree-N Chebyshev grid. Integration of the interpolation polynomial is then exact as a meromorphic function. For one monomial,

$$\int_0^1 t^{a+\epsilon b/2+k} \, dt = \frac{1}{a + k + 1 + \epsilon b/2}. \tag{601}$$

If $a + k + 1 = 0$, its contribution is the explicit pole $2/(b\epsilon)$; otherwise the usual geometric expansion gives every needed Laurent term. This implements endpoint subtraction without estimating a regulated integral by a small nonzero ϵ.

The quadrature weights are generated with 100-decimal-digit arithmetic and stored both in binary64 and as pairs of binary64 numbers whose sum is accurate to about 10^{-32}. The binary64 tables agree bitwise with an independent 80-digit generation, apart from entries that vanish identically, and they reproduce the exact Laurent coefficients of polynomial test integrands at every degree used. The arithmetic of the sums themselves is described in Section 22.5.

22.5 Exact summation at increasing degree

Let $\boldsymbol{H}^{(N)}$ collect the twelve six-loop Laurent coefficients at quadrature degree N: powers -6 through -1 for maxgap, followed by the same powers for span. Each entry is a finite sum over all

2,137,415 nonzero six-loop graphs, all 360 reversal classes, and all $(N+1)^5$ tensor nodes of the interpolated integrand (600). These sums are evaluated in full for $N = 6, 8, \ldots, 22$; no graph, sector or node is sampled. Two exact reorganizations make this affordable. Graphs with the same bucket vector (b_j) in a given sector share their Laurent weight tensor, and the edge factors $\prod_e Q_e^{-m_e/2}$ and the logarithms in (600) factor into two halves of the non-adjacent edges. The complete sequence through $N = 22$ used about 45 node-hours on 64-core processors.

Each graph-sector term is assigned to one of two strata according to its exponents a_j in (598). Let p be the number of axes with integral $a_j \le -1$, which carry a Laurent pole in (601). The hard stratum is

$$\mathcal{S} = \{p \ge 2\} \ \cup \ \{\text{some } a_j < -1\}, \tag{602}$$

and the light stratum is its complement, in which every axis satisfies $a_j \ge -1$ and at most one axis produces a pole. The strata are summed in different arithmetic.

- The light stratum uses binary64 vector arithmetic on blocks of eight nodes. Within each block the contributions of all weight groups are accumulated with error-free two-sum compensation; the blocks are then reduced in extended precision in a fixed order. Each sector value is therefore independent of the thread count, the division of sectors among jobs, and restarts. Its sensitivity to random one-unit perturbations in the last place of every node datum and weight is at most 7×10^{-10} in C_6 and does not grow with N.
- The hard stratum uses 80-bit extended precision (64-bit significand) throughout, with the weights read from the double-length tables. In binary64 the hard-stratum error grows with the degree because of cancellation within the stratum: the difference between binary64 and extended precision is 5×10^{-9}, 7.6×10^{-8} and 1.5×10^{-7} in C_6 at $N = 6$, 8 and 10.

Plain binary64 accumulation of the sector sums, as in the earlier deterministic degree-six evaluation, shifts C_6 by about 10^{-5} at $N = 6$ and 2×10^{-5} at $N = 8$. The earlier degree-six value differs from the present one by about 10^{-5} for this reason.

Before production, the implementation was required to reproduce the earlier deterministic degree-six sums and the earlier hard-stratum sums at $N = 6, 8, 10$ within those records' own summation errors, to give the complete sums as light plus hard stratum, and to agree between binary64 and extended precision within the measured errors. Identical executables give bitwise identical binary64 results on AMD and Intel processors; extended-precision results agree to 1.7×10^{-16} relative, the difference coming from vendor-specific x87 transcendental instructions.

22.6 Five-loop and lower-loop inputs

The five-loop data in (596) are obtained by the same exact summation over the five-loop graphs, with the 120 gap orderings reduced to 60 by spatial reversal; a graph is removed exactly when its weight in the combination being summed vanishes, $w_H = 3w_B - w_A$ for $\mathcal{H}_5$ (9,540 of the 59,145 graphs) or $8w_B - 3w_A$ for $\mathcal{J}_5$, in extended precision, at degrees $M = 12, 16, \ldots, 40$. The coupling and mass amplitudes are separated through $\mathcal{A}_5 = 8\mathcal{H}_5 - 3\mathcal{J}_5$ and $\mathcal{B}_5 = 3\mathcal{H}_5 - \mathcal{J}_5$, where $\mathcal{H}_5$ and $\mathcal{J}_5$ are the combinations $3\mathcal{B}_5 - \mathcal{A}_5$ and $8\mathcal{B}_5 - 3\mathcal{A}_5$. Their poles enter R_6^χ, and the finite part of $\mathcal{J}_5$ is J_5^χ. From $M = 28$ onward, successive changes of C_6 are at most 4.4×10^{-9}, below the measured five-loop rounding sensitivity of 1.0×10^{-8}. The two-, three- and four-loop amplitudes, through the orders listed after (596), are evaluated at degree 28; the change from degree 22 is -1.54×10^{-9} (maxgap) and $+4.14 \times 10^{-10}$ (span) in C_6. Recomputing them in binary64 instead of extended precision changes C_6 by 6.0×10^{-8}, which implies an extended-precision rounding error of about 3×10^{-11}. The inputs used in the final assembly are

	maxgap	span
R_6^χ, lower degree 28, five-loop poles at $M = 40$	39814.47028245011	81364.48627171610
J_5^χ, $M = 40$	−231603.7671545272	−1836182.1032583138
$K_\chi = R_6^\chi + \frac{4}{243} J_5^\chi$	36002.06259266366	51139.26646499489

so that $C_6^\chi(N) = K_\chi - \frac{2}{729} H_6^\chi(N)$ for each cutoff.

22.7 Degree sequence, stopping rule and uncertainty

N	$C_6^{\mathrm{maxgap}}(N)$	$C_6^{\mathrm{span}}(N)$	maxgap−span	$\rho(N)$
6	14334.549146861	14332.554509427	1.99	5.5×10^{-10}
8	14331.816869671	14331.694925294	0.122	1.9×10^{-10}
10	14331.694513177	14331.686134511	8.4×10^{-3}	7.4×10^{-10}
12	14331.687501161	14331.686917319	5.8×10^{-4}	1.9×10^{-9}
14	14331.687061994	14331.687022076	4.0×10^{-5}	4.1×10^{-9}
16	14331.687034432	14331.687031833	2.6×10^{-6}	2.9×10^{-9}
18	14331.687032845	14331.687032694	1.5×10^{-7}	3.3×10^{-8}
20	14331.687032736	14331.687032727	9.1×10^{-9}	4.5×10^{-9}
22	14331.687032770	14331.687032767	2.5×10^{-9}	8.8×10^{-8}

The convergence is geometric. Each step of two degrees reduces the change by a factor of about 0.06 for maxgap and 0.1 for span, and the two cutoffs, which differ by 2.0 at $N = 6$, approach a common value. The estimator, the uncertainty and the stopping rule were fixed before any degree beyond $N = 8$ was computed. Degrees 6 to 16 were unconditional. From $N = 16$ on, the next degree was computed unless both last changes $\Delta^\chi(N) = C_6^\chi(N) - C_6^\chi(N-2)$ and the cutoff difference were below $2\rho(N)$, where $\rho(N)$ is the floating-point estimate described below. At $N = 22$ the changes are $+3.3 \times 10^{-8}$ and $+4.0 \times 10^{-8}$ and the cutoff difference is 2.5×10^{-9}, all below $2\rho(22) = 1.8 \times 10^{-7}$, so the sequence stopped at $N_* = 22$.

The headline value is $\frac{1}{2}[C_6^{\mathrm{maxgap}}(N_*) + C_6^{\mathrm{span}}(N_*)]$, with the allowance

$$U = \mathrm{ceil}_2\Big(2 \max_\chi |\Delta^\chi(N_*)| + |C_6^{\mathrm{maxgap}} - C_6^{\mathrm{span}}| + 4\rho(N_*) + U_5 + U_{5,\mathrm{round}} + U_{\mathrm{low}}\Big), \tag{603}$$

where ceil_2 rounds up to two significant figures. The terms are 8.0×10^{-8}, 2.5×10^{-9}, 3.5×10^{-7}, 8.7×10^{-9}, 4.1×10^{-8} and 3.1×10^{-9}, so that $U = 4.9 \times 10^{-7}$.

- $\rho(N)$ is a rounding estimate. A fixed subset of 36 sectors is recomputed with every node datum and weight moved by one unit in the last place, in random directions, separately for the two strata. The resulting changes are scaled to all 360 sectors and multiplied by a calibration factor of 1.03. That factor is the largest observed ratio of the true binary64 error, measured against extended precision, to the same perturbation estimate in binary64 at $N = 6, 8, 10$.
- U_5 is twice the change of C_6 between $M = 36$ and $M = 40$. $U_{5,\mathrm{round}}$ is four times the calibrated perturbation estimate for the $M = 40$ five-loop data. U_{low} is twice the lower-degree change quoted above.

The allowance is dominated by the rounding term. Increasing the degree further would not reduce it. The separately measured rounding of the lower-loop amplitudes, about 3×10^{-11}, does not change U.

22.8 Independent checks

The poles ϵ^{-6} through ϵ^{-2} of $3\mathcal{B}_6 - \mathcal{A}_6$ are fixed by lower-order renormalization, so their errors test the whole six-loop pipeline at every degree. For maxgap, the double-pole error is 9.2×10^{-2}, 8.0×10^{-4}, 5.2×10^{-6} and 6.9×10^{-8} at $N = 10, 14, 18, 22$, and the triple-pole error at $N = 22$ is 2.7×10^{-9}; span behaves in the same way. The two cutoffs, whose six-loop integrands and subtraction constants differ, agree to 2.5×10^{-9} at N_*.

The earlier Monte Carlo evaluation sampled the light-stratum differences between degrees. Comparing its level estimates with the exact differences, in units of its standard errors, gives -0.57 and -0.60 for $8-6$, $+0.86$ and $+1.33$ for $10-8$, and $+0.09$ and $+1.26$ for the all-graph $12-10$ diagnostic (maxgap and span). That estimator was therefore consistent within its errors; its final value was 0.58 standard errors above the present one.

The seventh-order calculation of Section 27 produces C_6 as a by-product, in a different representation: coordinate forests, randomized integration, and radial quadrature through degree six. Its degree-six value 14332.43593 ± 0.30500 lies $+2.46$ of its standard errors above (591). Adding its measured degree-six to degree-seven check step gives 14331.98210, at $+0.97$; adding further a geometric tail model gives 14331.62055, at -0.22. The degree-six tension is the radial truncation of that representation, discussed in Section 27; agreement beyond the measured step is conditional on the tail model.

22.9 A finite special-function representation

A separate analytic calculation, recorded in the package `C6_research_note/`, starts from an analytic reduction of the six-loop coefficient to explicitly defined periods and series. It converts that expression into a finite formula in nine special-function values. The reduction itself is derived neither in this note nor in the package. We state it as the input (604) and compare its value with (591). Two of its leading Laurent coefficients, $R_{147,-3} = R_{148,-3} = \frac{2}{5}$ in the package's numbering, were missing from the first version of the input and were added by a correction; their contribution is given below.

In this subsection, for $n \geq 0$,

$$a_n = \binom{2n}{n} 4^{-n}, \qquad h_n = \log 2 + H_n - H_{2n}, \qquad \delta_n = \psi_1(n+\tfrac{1}{2}) - \psi_1(n+1), \qquad y_n = n + \tfrac{1}{4},$$

with $H_n^{(p)} = \sum_{j=1}^{n} j^{-p}$ and $H_n = H_n^{(1)}$; $\delta_n > 0$ is a trigamma difference, not a Kronecker delta. With $A = \pi + 4\log 2$, $Q = \Gamma(\frac{1}{4})^4/\pi$ and Catalan's constant G, the input reads

$$2187\, C_6(1) = E + S_{\mathrm{pol}} + S_{\mathrm{low}} + S_6 + S_8 - 393\pi^3 W_0 + 24\pi^3 M_7 + 96 P_a, \tag{604}$$

with the known polynomial

$$\begin{aligned} E = {} & \frac{Q}{8}\left[63A^3 + 120A\mathsf{G} + 79A\pi^2 + 2013\pi^3 + 54204\zeta(3)\right] + Q^2(29\pi - 249A) + \frac{21}{640}A^5 + \frac{7}{64}A^3\pi^2 \\ & + \frac{1149}{64}A^2\pi^3 + \frac{7863}{16}A^2\zeta(3) + 192A\mathsf{G}(16\mathsf{G} + 3\pi^2) + \frac{14853}{640}A\pi^4 - \frac{544}{3}\mathsf{G}\pi^3 \\ & + \frac{37631}{128}\pi^5 + \frac{722279}{48}\pi^2\zeta(3) - \frac{805959}{8}\zeta(5), \end{aligned} \tag{605}$$

which already contains the restored coefficients, and the polylogarithmic sector

$$S_{\mathrm{pol}} = 4352\big[(32\mathsf{G} + 9\pi^2)D_3 - 64 D_5\big], \qquad D_3 = \operatorname{Im}\operatorname{Li}_{2,1}(\mathrm{i}, 1), \quad D_5 = \operatorname{Re}\operatorname{Li}_{2,1,2}(\mathrm{i}, 1, 1), \tag{606}$$

where $\mathrm{Li}_{s_1,\dots,s_k}(z_1,\dots,z_k)=\sum_{n_1>\cdots>n_k\ge1}\prod_j z_j^{n_j}n_j^{-s_j}$. The three ordinary sectors are

$$\begin{aligned}
S_{\mathrm{low}} &= 4\pi^3 C_d\varpi_4 + 2\pi^4 C_s\varpi_6,\\
C_d &= 96Q - 42A^2 - 18A\pi - 1227\pi^2, \qquad C_s = 60Q - \tfrac{105}{4}A^2 - 9A\pi - \tfrac{2469}{4}\pi^2,\\
S_6 &= 24\Big[\frac{6Q^3}{\pi}(2\Phi_- - 1) - 104\pi^5\Phi_0 + \frac{2^{18}\pi^5}{Q^3}\Phi_+\Big],\\
S_8 &= \pi^6(4932 - 5004\sqrt{2})\varpi_8 + 556\pi^2\big[(17+12\sqrt{2})Y_1 - (17-12\sqrt{2})Y_7\big],
\end{aligned} \tag{607}$$

with the hypergeometric values at unit argument (all parametric excesses are positive)

$$\begin{gathered}
\varpi_4 = {}_4F_3((\tfrac12)^{\times4};1^{\times3};1), \quad \varpi_6 = {}_7F_6(\tfrac54,(\tfrac12)^{\times6};\tfrac14,1^{\times5};1), \quad \varpi_8 = {}_9F_8(\tfrac54,(\tfrac12)^{\times8};\tfrac14,1^{\times7};1),\\
\Phi_- = {}_7F_6(1,(\tfrac14)^{\times6};(\tfrac34)^{\times6};1), \quad \Phi_0 = {}_6F_5((\tfrac12)^{\times6};1^{\times5};1), \quad \Phi_+ = {}_7F_6(1,(\tfrac34)^{\times6};(\tfrac54)^{\times6};1),\\
Y_r = t\Big[\frac{\Gamma(t+\frac14)}{\Gamma(t+\frac34)}\Big]^8\,{}_{10}F_9(1,1+t,(t+\tfrac14)^{\times8};t,(t+\tfrac34)^{\times8};1), \qquad t=\frac{r}{8}.
\end{gathered}$$

The three remaining sectors are sums with no known evaluation in standard constants:

$$\begin{aligned}
W_0 &= \sum_{n\ge0} a_n^4\big[8+5\pi(4n+1)a_n^2\big]\delta_n, \qquad M_7 = \sum_{n\ge0}(-1)^n a_n^7 F_n,\\
F_n &= (98H_{2n} + 74\log2 - 6\pi)T_n - 98\pi\kappa_n - 98(11u_n - 13\pi y_n)\Big(\frac{\pi^2}{3} + 4H_{2n}^{(2)}\Big)\\
&\quad - 8624y_n\big[\zeta(3) - H_{2n}^{(3)}\big],\\
T_n &= 80\pi^2 y_n - 26\pi u_n + 11v_n, \qquad \kappa_n = 28\pi^2 y_n - 10\pi u_n + 3v_n,\\
u_n &= 1 - 14y_n h_n, \qquad v_n = 196y_n h_n^2 - 28h_n + 7y_n(\delta_n - \pi^2),
\end{aligned} \tag{608}$$

and

$$P_a = \pi^4(15I_{22} + 24I_{11} - 4I_{00}), \qquad I_{ij} = \int_0^1 \frac{p_i(x)\,p_j(1-x)}{\pi^{i+j}\sqrt{x(1-x)}}\,dx, \tag{609}$$

with $p_0 = v(x)^2$, $p_1 = \sqrt{Q}\,v(x)\,w(1-x)$, $p_2 = Q\,w(1-x)^2$, $v(x) = {}_2F_1(\frac14,\frac14;1;x)$ and $w(x) = {}_2F_1(\frac14,\frac14;\frac12;x)$. The sum W_0 includes $n=0$, whose term is $(8+5\pi)\pi^2/3$. Its terms decay like $14/(\pi^2 n^4)$; those of M_7 alternate in sign and are $O(n^{-5/2}\log n)$; the pairings have integrable endpoint singularities, so all three converge absolutely.

Special-function form. Appendix B converts every sector of (604) into one value at unit argument, using gamma reflection, sine addition, beta integration and Mellin–Barnes transforms. The result is the nine-value formula (2), whose values and parameters Section 1.2 defines. In particular $D_3 = -\frac12\,\mathrm{Ls}_3(\frac{\pi}{2}) - \frac{7\pi^3}{192}$, $D_5 = \mathrm{Cl}_{2,1,2}(\frac{\pi}{2})$, $W_0 = \sqrt{89}\,\mathcal{H}_W/\pi^2$, $\pi^{3/2}M_7 = \rho_7\,\mathcal{G}_7 + 196\pi\sqrt{185}\,\mathcal{H}_7$ and $P_a = 80\pi^3\mathcal{H}_*$. The count of nine values describes how the terms are packaged, not the number of independent constants: sine addition merges two G values into one with an angle parameter, and α_4 depends on Q, A and π.

Value and comparison with the graph sum. The representation evaluates to

$$C_6^{\mathrm{rep}}(1) = 14331.687032758054320445463255246217118317 32\ldots,$$

with the following sector contributions, in units of C_6:

$E/2187$	-1334.069267305	$S_8/2187$	4446.133424647
$S_{\mathrm{pol}}/2187$	-30.841591973	$-393\pi^3 W_0/2187$	-437.775914461
$S_{\mathrm{low}}/2187$	-944.671764724	$24\pi^3 M_7/2187$	-5217.069054363
$S_6/2187$	3148.152054246	$96P_a/2187$	14701.829146691

It lies 1.03×10^{-8} below the unrounded value 14331.68703276838 of (591), which is 0.021 of the allowance U. The two added Laurent coefficients contribute +659.278983 to C_6; without them (604) would give 13672.408. Because (591) is computed from the graphs directly, without (604), the agreement tests the reduction and its correction numerically at the level of 3×10^{-11} relative. It is not a proof of the reduction.

The package encloses $C_6^{\mathrm{rep}}(1)$ in an outward-rounded interval of width below 1.2×10^{-145}, using explicit bounds on the truncated tails of the series for W_0, M_7 and P_a. That enclosure is rigorous for (604) as stated, conditional on the identities the package uses (Section B.8.5); the tail of M_7 is bounded by (956). An independent review (`LRI_C6_closed_form_review/`) re-evaluated (604) from the definitions above to about 65 digits, recomputed the nine values with separate codes (the two-fold integrals directly, to 9–14 digits), and checked the algebra of the conversions. It found no error. Some conversions rest on further identities. The coefficient series for S_6 and S_8 and the two periodization identities for the wheel are derived in Appendix B, the latter from three elliptic-moment evaluations that the package takes over without derivation, (911)–(913); the Barnes representation of Bailey type [14] and a hypergeometric symmetry are published identities. All were confirmed numerically, the series and periodization identities to better than 10^{-63}. For W_0, M_7 and P_a no reduction to standard constants is known, and a PSLQ search in the package found no relation in the three bases it tested.

22.10 Result in the physical series

Combining the retained analytic lower-order coefficients with the sixth-order value of Section 22.9 gives

$$\begin{aligned}\Delta_{\mathcal{E}} = \frac{1}{2} - \frac{\epsilon}{6} + 0.657131263981\,\epsilon^2 - 3.719856322902\,\epsilon^3 \\ + 45.42488010356\,\epsilon^4 - 731.13947662454\,\epsilon^5 + 14331.68703275805\,\epsilon^6 + O(\epsilon^7).\end{aligned} \tag{610}$$

The graph sum of this section gives the sixth-order coefficient as 14331.6870328(5), in agreement to 1.0×10^{-8}. Neither number is a bound on the omitted $O(\epsilon^7)$ term.

23 Finite Schwinger kernels and local subtraction

Unless stated otherwise, this section fixes $d = 2$, $A = 4\log 2$, $a = (2+z)/4$ and $r(2,z) = 1$. The kernel normalization is exactly (195). For a primitive L-loop graph define

$$a_G = [z^{-1}]\mathcal{I}_G(2,z), \qquad \Pi_G = \frac{L}{2}a_G = \frac{1}{\pi^L}\int d\Omega\,\frac{\prod_e x_e^{-1/2}}{U_G(x)}. \tag{611}$$

The perturbative sign is not included in a_G. At six loops every primitive mass weight vanishes, so

$$C_{6,\mathrm{prim}}(2) = \frac{2}{729}\sum_{G\ \text{primitive}} w_A(G)a_G.$$

The following local one- and two-subgraph constructions define the finite kernels that the previous two-dimensional estimator (Section 26.13) retained as numerical inputs, (741). The 2026 two-dimensional production integrates all six-loop kernels in every replicate (Section 26.7). Their MS conversions are given separately so that their finite-scheme values are not confused with partial MS contributions. The numerical values quoted in this section are archived records of that superseded estimator. They and the programs that produced them are kept in the research archive `LRI_complete_research_bundle.zip` (packages `lri_C6_d2_primitive_update/`, `lri_C6_d2_subtraction_update/` and `lri_C6_d2_two_forest_update/`), which is not part of this repository; no current coefficient depends on them.

23.1 The sixth-order assembly at general dimension

The material of this subsection was written by OpenAI GPT-6 astra in the earlier sixth-order calculations at $d = 2$ and $d = 3$ (package `LRI_C6_D2_HPC/reference/`), with one exception: the value of $\mathfrak{B}_{\rm kn}/2187$ below was computed from the closed forms by a Claude Code session on 16 October 2026, when this subsection was merged into the note; the earlier calculation had evaluated it from numerical master values. The same session rechecked the polynomial (615) and the six-loop totals.

Assembly. The sixth-order formula (594) is the case $n = 6$ of the Lagrange formula (593), an identity between polynomials in the renormalization-group coefficients. In the bubble-normalized MS prescription (174), with

$$\beta(g) = -\epsilon g + g^2 B(g), \quad B(g) = 3 - 6A_d g + b_3 g^2 + \cdots + b_6 g^5, \qquad \gamma_r(g) = g - A_d g^2 + \sum_{j=3}^{6} c_j g^j,$$

it therefore holds at every d with A replaced by A_d of (176); in particular $A_2 = 4\log 2$ and $A_3 = 2 + 4\log 2 - \pi$. All b_j, c_j in one assembly must belong to the same prescription. The sensitivities to c_6 and b_6 quoted after (594) hold at every d as well, and the physical $C_0, \ldots, C_5$ at a fixed dimension cannot supply the independent six-loop combination $3c_6 - b_6$. For the bare amplitudes (193) put

$$J_5(d) = [z^0](8\mathcal{B}_5(d,z) - 3\mathcal{A}_5(d,z)), \qquad H_6(d) = [z^{-1}](3\mathcal{B}_6(d,z) - \mathcal{A}_6(d,z)). \tag{612}$$

The Laurent recursion (204) then gives, exactly as (596) at $d = 1$,

$$\boxed{C_6(d) = R_6(d) + \frac{4}{243} J_5(d) - \frac{2}{729} H_6(d),} \tag{613}$$

where $R_6(d)$ is the result of the same recursion with the five-loop finite parts and the six-loop simple poles set to zero and all other inputs retained. The new bare terms are $b_6 = 108a_{5,0} - 6a_{6,-1} + \cdots$ and $c_6 = 96e_{5,0} - 6e_{6,-1} + \cdots$ in the notation (202), and their combination in (594) is exactly $(36J_5 - 6H_6)/2187$. The two-, three-, four- and five-loop amplitudes are needed through z^3, z^2, z^1 and z^0, and the six-loop ones through z^{-1}, each with all its higher poles; the one-dimensional span and maxgap finite parts belong to a different prescription and are not inputs at $d \neq 1$. These identities were checked by exact rational polynomial arithmetic; the checker is in the research archive `LRI_complete_research_bundle.zip`, which is not part of this repository.

A fifth-order channel that reenters. At fifth order 28 vertex-isomorphism classes have $w_4 = 3w_2$ and drop out of $3\mathcal{B}_5 - \mathcal{A}_5$, whose weight w_H is (211). In J_5 their weight is $8w_2 - 3w_4 = -w_2 \neq 0$, so all 28 reenter at sixth order. Reusing only the fifth-order combined-channel data would omit them. This is consistent with the fifth-order result: these terms cancel there, but not here.

Six-loop inventory. The 2,427,005 labeled six-loop 1PI graphs of the table after (597) fall into 660 vertex-isomorphism classes. With the weights divided by $|\,\mathrm{Aut}\,|$, as in the right side of (196),

$$\sum_G w_A = \frac{97987}{64}, \qquad \sum_G w_B = \frac{83711}{576}, \qquad \sum_G (3w_B - w_A) = -\frac{105125}{96}.$$

Times 7! these are the six-loop totals of the table after (597), and the first two are the zero-dimensional coefficients of (16). Every class has twelve separately indexed internal edges, and its first Symanzik polynomial is stored as a list of complementary spanning-tree masks; the six-loop inventory contains 140,781 such monomials.

MS bookkeeping at $d = 2$. The MS-converted sector values below are quoted together with the five-loop mass terms $-6Ac_5/2187$. The reason is the following exact identity. Use the MS values of b_3, c_3 (179), b_4, c_4 (226)–(227) and $D_5 = 3c_5 - b_5$ (229), and eliminate $b_5 = 3c_5 - D_5$ in (594); since $-12Ab_5 + 30Ac_5 = 12AD_5 - 6Ac_5$, at $d = 2$

$$\boxed{2187\, C_6(2) = \mathfrak{B}_{\rm kn} + 3c_6(2) - b_6(2) - 6A\, c_5(2), \qquad A = 4\log 2.} \tag{614}$$

With the dictionary $R = 0$, $M = -4$, $N = 4 - A$, $U = -14\zeta(3)$, $V = -4\zeta(3)$ and $X = 0$ of (221) and (225) at $q = 1/2$, writing $Q = Q_2$ and $P_j = P_j(2)$,

$$\begin{aligned}\mathfrak{B}_{\rm kn} = &-\frac{14385}{16}A^5 - 1719A^3Q + 186A^3 - \frac{771}{2}A^2P_4 + \frac{164985}{16}A^2\zeta(3) + 174A^2 \\ &- 288A\mathcal{K}_2 - 108A\mathcal{L}_2 - 288AP_5 - 852AQ^2 + 240AQ + 1341AZ_2 - \frac{16}{3}A \\ &- 84P_4Q + 32P_4 + \frac{5511}{2}Q\zeta(3) + 144Q - 718\zeta(3).\end{aligned} \tag{615}$$

Every coefficient follows by exact expansion of $\mathcal{R}(b_3, c_3, b_4, c_4) + 12AD_5$, with $\mathcal{R}$ of (676). With $Z_2 = 4\mathsf{G}Q_2$ (265), the closed forms (270), (271), and $P_4(2)$, $P_5(2)$ of Section 17.2,

$$\frac{\mathfrak{B}_{\rm kn}}{2187} = -1285.866857775584\ldots, \qquad C_6(2) = -1285.866857775584\ldots + \frac{3c_6 - b_6 - 24\log 2\, c_5}{2187}.$$

The fifth-order physical coefficient fixes D_5 but not c_5 separately, so c_5 is a genuine input of (614). This decomposition was used to organize the partial MS sectors below; the current estimate does not use it, since it computes the physical combination in the finite scheme of Section 25.

Why a different representation was needed at $d = 2$. The same 660 six-loop classes occur in every dimension, so the graph count is not what makes $d = 2$ harder. On the line, seven vertices can be ordered and replaced by six positive gaps; removing the common scale leaves five dimensionless gap variables, every distance is a sum of consecutive gaps, and the collision sectors (598) have an efficient implementation. The two-dimensional Schwinger representation has 12 edge parameters and an 11-dimensional projective simplex. (This describes the representation, not a minimal dimension: in coordinate space seven points have 14 real coordinates, of which translation, rotation and scale remove four.) Without a linear ordering and with many more relative configurations, naive tensor quadrature and untuned sampling are much less effective. An earlier attempt at $d = 2, 3$ sampled the unsubtracted Schwinger sector integrals with Taylor endpoint subtraction, both uniformly and with a subset dynamic-programming proposal. The separate large endpoint subtraction terms caused severe variance and loss of precision: the error propagated to $C_6(2)$ from H_6 alone remained of order 10^5, and no value was accepted. More samples do not repair a poor subtraction representation. The common finite kernels, the exact root-scale coefficients and the regulator-aware fixed-point assembly of this section and of Sections 24 and 25 address that obstacle directly. Two dimensions also have advantages: complex-plane factorization, the complex Selberg integral (191) and the bilinear ${}_3F_2$ form (674), which allow many counterterms to be evaluated analytically.

23.2 Exact primitive inventory and a product reduction

Use the complete 660-class six-loop inventory. A graph is declared primitive only if every proper induced vertex subset S with $|S| \geq 2$ satisfies $e(S) < 2(|S| - 1)$. This selects ten classes. All edge labels, parallel-line labels, rational Wick weights and spanning-tree polynomials come from the unchanged input inventory. The class indices in this section are zero-based indices in the project's 660-class six-loop inventory (archived as `data/inventory6.json` in the research archive), not

indices in any external graph catalog; the same indices identify the six-loop kernels in `LRI_C6_D2_HPC/inputs/inventory.json`.

Completing each selected graph to a four-regular graph on eight vertices partitions the ten classes into five isomorphism classes:

Group	Uncompleted class indices	$\sum w_A$	Period
0	$639, 654, 657$	16	$P_3(2)P_4(2)$
1	648	12	Π_1
2	$649, 651, 655$	48	Π_2
3	$653, 658$	12	Π_3
4	659	1/6	Π_4

Each equality within a group follows from conformal completion, not from numerical comparison. In the present two-dimensional normalization each critical coordinate edge has absolute-distance power one, and each completed vertex has incident power four. Inversion of an integration variable therefore cancels its Jacobian, proving invariance under the choice of the deleted vertex. This is the weighted completion argument; the general completion and product identities are reviewed in [42].

For the representative 639, the three-vertex separator is $\{4, 5, 6\}$. Removing it leaves components $\{0, 1\}$ and $\{2, 3, 7\}$. Adding the uniquely required triangles on the separator completes the components to K_5 and the octahedral graph. Fixing the separator at $0, 1, \infty$ makes their integrations independent. Thus their normalized periods multiply, with no additional measure factor:

$$\Pi_0 = P_3(2)P_4(2) = 3Q_2P_4(2). \tag{616}$$

The explicit separator and child-graph isomorphisms are checked in the archived record

`lri_C6_d2_primitive_update/verification/primitive_reduction_checks.json`

of the research archive. The complete primitive contribution can therefore be written

$$\boxed{C_{6,\mathrm{prim}}(2) = \frac{2}{2187}\left[48Q_2P_4(2) + 12\Pi_1 + 48\Pi_2 + 12\Pi_3 + \frac{1}{6}\Pi_4\right].} \tag{617}$$

Only the last four periods require new numerical integration. The first term is evaluated from the retained ordinary-hypergeometric formula for $P_4(2)$ and the exact Gamma value of Q_2:

$$\begin{aligned} Q_2 &= \frac{\Gamma(1/4)^8}{24\pi^4}, \\ P_3(2)P_4(2) &= 9597.206978165420901472979104294093 3\ldots, \\ \frac{96}{2187}Q_2P_4(2) &= 140.425525057747356583052277703434 38\ldots. \end{aligned} \tag{618}$$

The unreduced ten-class integration is retained as an independent check of this reduction; the product value is not fitted to its output.

23.3 Positive tropical sampling and a stable cycle determinant

At $d = 2$, the primitive residue of the normalized Schwinger integral is

$$a_G = \frac{2}{L\pi^L}\int_{x_e\geq 0} d\Omega\, \frac{\prod_{e=1}^{2L} x_e^{-1/2}}{U_G(x)}. \tag{619}$$

There is no endpoint analytic continuation in this primitive integral. We use tropical importance sampling of the positive projective integrand, in the framework of [41]. Here its specialized construction is spelled out to fix the normalization.

In an edge ordering σ, set the largest edge parameter to one and use $x_{\sigma(j)} = \prod_{k=j}^{2L-1} t_k$ for $j < 2L$. If $h(S)$ is the cycle rank of the first $|S|$ edges, write

$$\alpha(S) = \frac{|S| - 2h(S)}{2}, \qquad U_G = \left(\prod_{j=1}^{2L-1} t_j^{h(S_j)} \right) \widetilde{U}_G. \tag{620}$$

All proper-prefix $\alpha(S_j)$ are strictly positive for the selected graphs. The integral over a sector has the form $\int \prod_j t_j^{\alpha(S_j)-1} dt_j / \widetilde{U}_G$. The exact sector-normalization recursion is

$$Z_G(\varnothing) = 1, \qquad Z_G(S) = \frac{\sum_{e \in S} Z_G(S \setminus \{e\})}{\alpha(S)} \quad (S \neq E), \qquad Z_G(E) = \sum_{e \in E} Z_G(E \setminus \{e\}). \tag{621}$$

At each backward step choose the removed edge with probability proportional to $Z_G(S \setminus \{e\})$. Independently sample $t_j = U_j^{1/\alpha(S_j)}$, with U_j uniform on $(0, 1)$. For graph weights w_G, sample graphs proportionally to $w_G Z_G(E)$. If $W = \sum_G w_G Z_G(E)$, each sample of the weighted bare residue is simply

$$Y = \frac{2W}{L\pi^L} \frac{1}{\widetilde{U}_G}. \tag{622}$$

Thus the estimator is positive and bounded; it does not sample a difference of divergent contributions. This positivity does not extend to the omitted nonprimitive subtraction problem.

A direct evaluation of $\widetilde{U}_G$ sums all spanning-tree monomials. For the production runs we instead construct a minimum spanning tree in the sampled edge order. Let f_i be its L chords and let e range over its L tree edges. Define

$$V_{ie} = \begin{cases} \sigma_{ie} \sqrt{x_e / x_{f_i}}, & e \text{ lies in the fundamental cycle of } f_i, \\ 0, & \text{otherwise,} \end{cases} \qquad \sigma_{ie} \in \{+1, -1\}. \tag{623}$$

The matrix-tree identity gives the scaled form

$$\boxed{\widetilde{U}_G = \det(I_L + V V^{\mathsf{T}}).} \tag{624}$$

To see the normalization, the fundamental-cycle matrix has one unit entry on its own chord and zero entries on all other chords. Factoring $\prod_i x_{f_i}$ from its Gram determinant leaves $\det(I + VV^{\mathsf{T}})$. The minimum-spanning-tree property makes that product the leading Hepp monomial and ensures $x_e \le x_{f_i}$ on each fundamental cycle. Hence $|V_{ie}| \le 1$ and every eigenvalue of $I + VV^{\mathsf{T}}$ is at least one. Cholesky evaluation avoids the poor conditioning of an unscaled Laplacian near a sector boundary.

An independent 60-digit check compares the monomial sum with this determinant on 180 configurations, including exponentially small sector variables. The largest relative difference is 4.49×10^{-61}. Production runs also perform pointwise checks before accumulating the batch estimates. The graph input and both C++ implementations are in the research archive (package `lri_C6_d2_primitive_update/`), which is not part of this repository.

23.4 Normalization and the exact set of graphs

Work at $d = 2$, $\epsilon = 2s - 2$. With the regulator z used for Laurent coefficients, $a = \frac{1}{2} + z/4$ and the loop normalization (195) becomes

$$r(2, z) = 1, \qquad I_G(z) = \frac{\Gamma(Lz/2)}{\Gamma(\frac{1}{2} + z/4)^{2L}} \int_{\Delta_{2L-1}} \frac{\prod_e x_e^{a-1}}{U_G(x)} \, d\Omega, \quad U_G = \sum_T \prod_{e \notin T} x_e. \tag{625}$$

Here every internal line is a separate Schwinger variable, including parallel lines. The weights w_A, w_B are the rational vertex-isomorphism weights of the six-loop inventory used throughout this section; set $w_H = 3w_B - w_A$. No additional vertex factorial is introduced in this section.

Let $\mathcal{S}_{26}$ consist of the classes with exactly one connected, proper, one-particle-irreducible logarithmic subgraph γ, and no power-divergent subgraph. Both γ and G/γ are primitive. The class IDs and aggregate weights are as follows; a number j in the second column denotes the unchanged inventory label `L6G`j, not its zero-based array index.

$h = \ell(\gamma)$	Class numbers	Number	$\sum w_H$
1	524, 529, 539, 563, 575, 587, 588, 596, 599, 607, 613, 614, 617, 619, 620, 621, 627	17	$-279/2$
3	642	1	-4
5	638, 639, 641, 647, 648, 651, 653, 657	8	$-93/4$

The total weight is $-667/4$. The archived machine-readable record gives every edge list, the subgraph edge mask, the rational weights, and the exact isomorphism to the primitive subgraph and quotient. A separate checker also reconstructs their spanning-tree polynomials; its 158 exact checks include both aggregate conversion identities below.

23.5 A common-integrand subtraction with no large endpoint difference

Let $h = \ell(\gamma)$, $k = L - h$, and use $\rho = G/\gamma$ for the quotient. Define

$$X_\gamma = \sum_{e\in\gamma} x_e, \quad X_\rho = \sum_{e\notin\gamma} x_e, \quad S = X_\gamma + X_\rho, \quad U_0 = U_\gamma U_\rho, \quad U_\times = U_G - U_0. \tag{626}$$

The edge variables of the quotient are the original edges outside γ. The monomials of U_0 are a subset of those of U_G; therefore $U_\times$ has nonnegative coefficients. This is checked exactly, not by evaluating the polynomials at selected points.

The auxiliary amplitude is

$$\boxed{D_G(z) = I_G(z) - \frac{k}{L} I_\gamma(z) I_\rho(z).} \tag{627}$$

Its common projective kernel is

$$\boxed{\frac{1}{U_G} - \frac{X_\rho}{S\,U_0} = \frac{X_\gamma U_0 - X_\rho U_\times}{S\,U_0\,U_G}.} \tag{628}$$

In simplex gauge $S = 1$; retaining S makes the homogeneity correct in any Hepp gauge. This is not a replacement of the product by a local approximation: its integral is exactly the product term in (627).

Proof of the factor k/L. In the product integral, separate the two sets of variables by $x_\gamma = \lambda u$ and $x_\rho = (1-\lambda)v$, with u, v each on a unit simplex. Their scale factors are $\lambda^{hz/2-1}(1-\lambda)^{kz/2-1}$. Without a scale weight the beta integral exactly combines the two Gamma prefactors into $\Gamma(Lz/2)$. Inserting $X_\rho/S = 1-\lambda$ multiplies the beta integral by

$$\frac{B(hz/2, 1+kz/2)}{B(hz/2, kz/2)} = \frac{k}{L}. \tag{629}$$

The identity holds in the ordinary convergence region and then by meromorphic continuation. Hence the subtraction has the correct normalization at nonzero z, not just at leading pole order.

Endpoint cancellation. When γ shrinks, $U_\times$ has at least one extra power of the subgraph scale compared with U_0. The first numerator term in (628) has this power through X_γ. The artificial singularity when the complementary factor shrinks is removed similarly, using the fact that it is not a proper subdivergence of the original G. Since the two factors are primitive, these are the only zero exponents introduced by the product denominator. The cancellation is applied to the monomial exponents before floating point evaluation. The implementation never forms two nearly equal, individually singular reciprocal polynomials.

More explicitly, order the $2L$ edges, write the ordered Schwinger variables as products of $t_j \in (0,1)$, and let E_j be the first j edges. If

$$h_0(E_j) = \min_{m \in \operatorname{supp} U_0} |m \cap E_j|, \qquad \alpha_j = \frac{j}{2} - h_0(E_j) + \mathbf{1}_{E_j = E_\gamma \text{ or } E_j = E_\rho}, \tag{630}$$

then every proper-sector α_j is positive for these graphs. The indicated powers are stripped from the numerator and t_j is sampled with density $\alpha_j t_j^{\alpha_j - 1}$. Sector probabilities are proportional to the corresponding product of $1/\alpha_j$ and are generated by a subset dynamic program. The normalization for the simple residue is $2/(L\pi^L)$. The remaining rational kernel is bounded on each closed sector after these structural powers have been removed; this distinguishes it from an earlier estimator of the same graphs, which sampled the unsubtracted integrands with importance weights taken from their logarithmic moments and was much noisier.

Define the quantity directly measured by this sampler as

$$Y_{26} = \sum_{G \in \mathcal{S}_{26}} w_H(G)[z^{-1}]D_G(z). \tag{631}$$

Y_{26} is not an MS residue. Its conversion is indispensable.

23.6 Conversion to the same MS coefficient as the six-loop assembly

Write the primitive factors as

$$I_\gamma = \frac{r_\gamma}{z} + f_\gamma + O(z), \qquad I_\rho = \frac{r_\rho}{z} + f_\rho + O(z). \tag{632}$$

For a single proper primitive subdivergence the incomplete MS operation is $R'_G I_G = I_G - (r_\gamma/z) I_\rho$. Taking only the simple pole gives

$$\boxed{[z^{-1}]R'_G I_G = [z^{-1}]D_G - \frac{h}{L} r_\gamma f_\rho + \frac{k}{L} f_\gamma r_\rho.} \tag{633}$$

Higher poles of $R'_G I_G$ are not assumed to vanish. Equation (633) is a Laurent-coefficient identity and does not confuse this auxiliary kinematic subtraction with MS.

There are three primitive five-loop inventory classes. In zero-based array notation they are 123, 125 and 126, with coupling weights $2, 3/2, 12$, respectively. Let f_i denote their individual unweighted finite parts, evaluated in the massive normalization (625), and put

$$a^{\mathrm{p}}_{5,0} = 2f_{123} + \frac{3}{2} f_{125} + 12 f_{126}. \tag{634}$$

The finite parts of conformally equivalent completed graphs are not identified: completion identities fix their residues, not their mass-dependent finite terms. All three terms in (634) were integrated separately.

The one-loop bubble is exactly $I_1 = 2/z$, with zero finite part. For the $h = 3$ class the two primitive factors are isomorphic, so its conversion terms in (633) cancel. Exact aggregation over the other 25 classes gives

$$\boxed{M_{26} := \sum_{G \in \mathcal{S}_{26}} w_H [z^{-1}] R'_G I_G = Y_{26} + \frac{5}{2} a^{\mathrm{p}}_{5,0}.} \tag{635}$$

For example, the three finite-part coefficients on the right are $5, 15/4, 30$, which are exactly $5/2$ times the weights in (634). The direct contribution to the physical assembly is

$$\boxed{\Theta_{26} = \frac{(3c_6 - b_6)_{26}}{2187} = -\frac{2}{729} M_{26}.} \tag{636}$$

The last sign follows from the even-loop sign in the bare amplitudes: $(3c_6 - b_6)_{26} = -6M_{26}$.

That earlier estimator measured the same weighted simple-pole coefficient of the unsubtracted amplitudes,

$$H_{26}^{\mathrm{bare}} = \sum_{G \in \mathcal{S}_{26}} w_H(G)[z^{-1}] I_G(z).$$

For a comparison with it, let r_3, f_3 be the unweighted primitive three-loop residue and finite part. A second exact aggregate identity is

$$\boxed{H_{26}^{\mathrm{bare}} = Y_{26} - \frac{31}{2} a_{5,0}^{\mathrm{p}} - 4 r_3 f_3, \qquad r_3 = 2Q_2.} \tag{637}$$

Comparing Y_{26} directly to H_{26}^{bare} would therefore give a spurious normalization discrepancy.

23.7 The two-subgraph classes and a common finite kernel

There are 40 six-loop classes with exactly two proper logarithmic subgraphs and no power-divergent subgraph: 23 have a nested pair and 17 have a vertex-disjoint pair. Every decomposition used here has two one-loop primitive leaves and one four-loop primitive leaf. At five loops there are 12 analogous classes (seven nested, five disjoint); only five of them carry a nonzero mass weight. Their primitive leaves are two bubbles and the three-loop tetrahedron.

For G let $U = U_G$ and let U_1, U_2 be its leading polynomials when respectively γ_1, γ_2 shrink. For a subgraph of loop number h_i, U_i consists exactly of the monomials of U containing h_i edges from γ_i. The common face is $W = U_{12}$. Set

$$X = U_1 - W, \qquad Y = U_2 - W, \qquad Z = U - U_1 - U_2 + W. \tag{638}$$

Thus $U = W + X + Y + Z$, $U_1 = W + X$, $U_2 = W + Y$. The four polynomials partition the spanning-tree monomials and have nonnegative coefficients. For nested subgraphs put

$$\sigma_1 = \sum_{e \in \gamma_1} x_e, \quad \sigma_2 = \sum_{e \in \gamma_2 \setminus \gamma_1} x_e, \quad \sigma_3 = \sum_{e \notin \gamma_2} x_e, \qquad S = \sigma_1 + \sigma_2 + \sigma_3. \tag{639}$$

For disjoint subgraphs, the three sums run over γ_1, γ_2 and their complement.

The two auxiliary forest kernels, including the common projective scale weight but not the common propagator measure, are

$$K_{\mathrm{n}} = \frac{1}{U} - \frac{\sigma_2 + \sigma_3}{S U_1} - \frac{\sigma_3}{S U_2} + \frac{\sigma_3}{S W}, \tag{640}$$

$$K_{\mathrm{d}} = \frac{1}{U} - \frac{\sigma_2 + \sigma_3}{S U_1} - \frac{\sigma_1 + \sigma_3}{S U_2} + \frac{\sigma_3}{S W}. \tag{641}$$

These are auxiliary subtractions, not MS graph contributions; the conversion follows below. For stable evaluation write either kernel as

$$K_\star = \frac{N_\star}{S\, U\, U_1\, U_2\, W}, \tag{642}$$

where the numerators are the following explicit finite polynomials:

$$\begin{aligned} N_{\mathrm{n}} &= \sigma_1 W(W+X)(W+Y) - \sigma_2 (Y+Z) W(W+Y) \\ &\quad + \sigma_3 \left[XY(2W+X+Y+Z) - ZW^2 \right], \end{aligned} \tag{643}$$

$$\begin{aligned} N_{\mathrm{d}} &= -\sigma_1 W(W+X)(X+Z) - \sigma_2 W(W+Y)(Y+Z) \\ &\quad + \sigma_3 \left[XY(2W+X+Y+Z) - ZW^2 \right]. \end{aligned} \tag{644}$$

Equations (643)–(644) are exact rational identities. The code evaluates these polynomials, not a floating-point subtraction of the four singular fractions.

Why the subtracted integrals converge. In a Hepp sector, choose the largest edge equal to one and write $x_{\pi(j)} = \prod_{k=j}^{2L-1} t_k$. For a proper edge subset s let $h_P(s)$ be the smallest degree of the polynomial P in the edges of s. For a scale sum σ_i this is one if every edge in its group belongs to s, and zero otherwise. Let $h_N(s)$ be the minimum degree among the nonzero products displayed in (643) or (644). No cancellation of leading terms is used to increase this minimum. The residual exponent is

$$\alpha_s = \frac{|s|}{2} + h_N(s) - h_U(s) - h_{U_1}(s) - h_{U_2}(s) - h_W(s). \tag{645}$$

An exact integer check over all proper edge subsets gives $\alpha_s \geq 1/2$ for each of the 40 six-loop and 12 five-loop kernels: there are $40(2^{12}-2) + 12(2^{10}-2) = 176024$ checks. This controls all nested endpoint limits, not just separate limits of individual edges. It establishes a common analytic neighborhood of $z = 0$ for the angular integral after removing the overall $\Gamma(Lz/2)$ pole.

For completeness, the sector weights are computed by

$$D(\varnothing) = 1, \qquad D(s) = \frac{1}{\alpha_s} \sum_{e \in s} D(s \setminus \{e\}), \qquad \alpha_{\mathrm{all}} := 1. \tag{646}$$

A descending edge choice has probability proportional to $D(s \setminus \{e\})$; each continuous coordinate has density $\alpha_s t^{\alpha_s - 1}$. Graphs are sampled proportionally to $|w_G| D(\mathrm{all})$. After dividing numerator and denominator polynomials by their exact leading Hepp monomials, the residual factor has no endpoint pole. The overall residue prefactor is $2/(L\pi^L)$. The numerical estimates below include the graph sign; positivity of the primitive estimator is not asserted for the forest estimator.

23.8 Exact conversion of the auxiliary residues to MS

The conversion needs finite parts, and at this order also the coefficient linear in z of a primitive graph. They cannot be replaced by period residues. Use the unweighted amplitudes

$$I_1 = \frac{2}{z}, \qquad I_p = \frac{r_p}{z} + f_p + a_p z + O(z^2), \qquad p = 3, 4, \tag{647}$$

where

$$r_3 = 2Q_2, \qquad r_4 = P_4(2)/2, \qquad Q_2 = \frac{\Gamma(1/4)^8}{24\pi^4}. \tag{648}$$

For a graph with one logarithmic child γ of loop number h, define

$$D_G(z) = I_G(z) - \frac{L-h}{L} I_\gamma(z) I_{G/\gamma}(z) = \frac{d_G}{z} + f_G + O(z). \tag{649}$$

For the two-loop auxiliary graph this gives

$$D_2 = \frac{A}{z} + f_2 + O(z), \qquad f_2 = -4.836812282568919475092936478902267785676845800388 3\ldots. \tag{650}$$

The value in (650) was checked with the two-${}_3F_2(1/4)$ formula (199) and two Cauchy grids; it is a finite part in the specified massive prescription, not a physical epsilon coefficient.

General nested and disjoint identities. For a nested pair, write $\gamma_1 = G_a$, $\gamma_2 = J$, $J/\gamma_1 = G_b$, $G/\gamma_2 = G_c$, $G/\gamma_1 = T$, with primitive loop numbers h, k, m and $L = h + k + m$. Beta integration of the relative scale variables gives the exact all-z identity

$$D_G = I_G - \frac{k+m}{L} I_a I_T - \frac{m}{L} I_J I_c + \frac{m}{L} I_a I_b I_c. \tag{651}$$

Meanwhile

$$R' I_G = I_G - \frac{r_a}{z} I_T + Z_J I_c, \qquad Z_J = -\,\mathrm{PP}\left(I_J - \frac{r_a}{z} I_b\right). \tag{652}$$

where PP keeps the negative powers of z (the pole part), as in (20). For a disjoint pair with primitive children G_a, G_b and primitive cograph G_c, the analogous identities are

$$D_G = I_G - \frac{L-h}{L} I_a I_{G/G_a} - \frac{L-k}{L} I_b I_{G/G_b} + \frac{m}{L} I_a I_b I_c, \tag{653}$$

$$R' I_G = I_G - \frac{r_a}{z} I_{G/G_a} - \frac{r_b}{z} I_{G/G_b} + \frac{r_a r_b}{z^2} I_c. \tag{654}$$

The factors m/L and $(L-h)/L$ arise from the beta mean of the remaining scale fraction, not from fitted subtraction constants. Substituting (649) into (651)–(654) and extracting z^{-1} proves each graph's conversion. All graph identifications, including the inherited subgraph vertex labels, are checked against the six-loop inventory before the rational weights are collected.

Let $\mathcal{S}_{6,2}$ be the set of the 40 six-loop classes with exactly two proper logarithmic subgraphs, let $Y_{40} = \sum_{G \in \mathcal{S}_{6,2}} w_H(G)[z^{-1}] D_G$, and abbreviate $f_{5,j} = f_{\mathrm{L5G}j}$, $d_{5,j} = d_{\mathrm{L5G}j}$. Define

$$F_5 = \frac{15}{4} f_{5,052} + \frac{15}{2} f_{5,079} + 13 f_{5,100} + 26 f_{5,114} + \frac{13}{4} f_{5,125} + \frac{13}{2} f_{5,129}. \tag{655}$$

The exact aggregate result is

$$\boxed{\begin{aligned} M_{40} &:= \sum_{G \in \mathcal{S}_{6,2}} w_H(G)[z^{-1}] R' I_G \\ &= Y_{40} + F_5 + 9A f_4 - 15 a_4 - 18 f_2 r_4, \\ \Theta_{40} &= -\frac{2}{729} M_{40}. \end{aligned}} \tag{656}$$

Thus the primitive four-loop coefficient a_4, not just its finite part, is essential. The same integrations give the entire one-subdivergence five-loop mass sector:

$$\boxed{c_5^{(1)} = \frac{5}{2} d_{5,052} + 5 d_{5,079} + 3 f_4.} \tag{657}$$

Only the first two of the six fifth-order classes have nonzero mass weights, although all six finite parts enter (655).

A further five-loop mass sector. The mass-carrying two-subgraph classes are exactly L5G053, L5G054, L5G056, L5G077, L5G080, with respective $w_B = (1, 1, 1/2, 1/2, 1)$. All five have nested subdivergences. Put $Y_{5B} = \sum w_B(G)[z^{-1}] D_G$ over this set, and define $f_{4,j}$ using (649) for the indicated fourth-order class. The independent rational collection gives

$$\boxed{c_5^{(2)} = 5\left[Y_{5B} + \frac{2A}{5} f_3 - a_3 - \frac{3}{5} f_2 r_3 - \frac{4}{5} f_{4,013} + \frac{6}{5} f_{4,024}\right].} \tag{658}$$

These classes are disjoint from the one-subgraph and closed Gamma classes. Their contribution to the sixth-order combination is $-6A c_5^{(2)}/2187$, not a sixth-order primitive period.

23.9 Finite-part integrands and numerical results

For any convergent auxiliary kernel D_G, differentiation of the common propagator measure at $z = 0$ gives the logarithmic multiplier

$$\mathcal{J}_L = L \log 2 + \frac{1}{4} \sum_{j=1}^{2L-1} j \log t_j - \frac{L}{2} \log S. \tag{659}$$

Multiplying its residue integrand by (659) gives f_G. For primitive jets through a_p, the code uses the full Taylor coefficients of $z\Gamma(pz/2)/\Gamma(1/2 + z/4)^{2p}$ and the exponential of the logarithmic measure, retaining the exact factorials. No new period identity for the massive finite parts is assumed.

Each of the two main production runs uses three independent streams: 64 000 000 observations for Y_{40}, the six fifth-order auxiliary jets, and the primitive four-loop jets, respectively. Each stream is split into 128 batches. Runs A and B use distinct seeds. Primitive residues are fixed-expectation controls for their finite and linear coefficients; the control slopes are fitted only to excluded one-million-sample pilots. The fourth-order slopes are $(-8.0444695540, 40.8894797679)$ for (f_4, a_4).

Quantity	Run A	Run B
Y_{40}	761.536908 ± 42.870149	790.704780 ± 44.081215
Θ_{40}	220.727969 ± 0.224779	220.412578 ± 0.227748
$c_5^{(1)}$	1651.795516 ± 1.874636	1649.773937 ± 1.603267
$\Theta_{40} - 6Ac_5^{(1)}/2187$	208.163498 ± 0.218856	207.863484 ± 0.223344

The displayed errors are sampling standard errors, not certified uncertainties. Like every partial sum of this section, Θ_{40} is neither an estimate nor a bound for $C_6(2)$; graph-sector contributions are defined only in the stated MS prescription. The main runs agree within one combined standard error for the MS six-loop result. Equal weighting, without fitting to a target value, gives

$$\begin{aligned} Y_{40} &= 776.1208440 \pm 30.7449312, & F_5 &= 6208.4320415 \pm 48.4413942, \\ f_4 &= -957.9657216 \pm 0.0502308, & a_4 &= 4958.8036029 \pm 0.6170590, \\ M_{40} &= -80397.8646408 \pm 58.3185778, & c_5^{(1)} &= 1650.7847264 \pm 1.2333617. \end{aligned} \tag{660}$$

The auxiliary residue Y_{40} and the MS residue M_{40} differ substantially; interchanging them would give an incorrect coefficient. Keeping the within-stream covariance of finite parts and residues yields

$$\boxed{\Theta_{40} - \frac{6A}{2187} c_5^{(1)} = 208.01349115\ldots, \qquad \widehat{\sigma} = 0.15634950\ldots.} \tag{661}$$

Without the primitive controls, the same raw observations give $\Theta_{40} = 220.5459148 \pm 0.1606925$ and the joint combination $207.9871304 \pm 0.1571710$. Controls were not fitted to an assumed value of C_6 or to equality of the two runs.

The additional five-loop calculation uses two independently seeded runs, each with 32 000 000 observations in each of its three streams (Y_{5B}, the two fourth-order auxiliary finite parts, and primitive three-loop jets), again with 128 batches. It gives

$$\begin{aligned} c_5^{(2)} &= 2431.5047824 \pm 1.8267989, \\ -\frac{6A}{2187} c_5^{(2)} &= -18.49537102 \pm 0.01389564. \end{aligned} \tag{662}$$

The two independent estimates of $c_5^{(2)}$ differ by 0.053 combined standard errors. The analysis retains the covariance of f_3 and a_3, as well as that of every residue/finite-part pair from the same stream.

Checks of the two-subgraph kernels. These checks were made by OpenAI GPT-6 astra with the kernels above (research-archive package `lri_C6_d2_two_forest_update/`; the result record is `lri_C6_d2_complete_update/reproduction/lri_C6_d2_two_forest_update/verification/C6_d2_continuation_result.json`). The rational numerator identities (643) and (644) were checked symbolically. An independent scalar implementation compares the normalized floating-point numerator with the original four fractions at 500 digits; over 312 configurations, including a coordinate as small as 10^{-140}, the largest normalized error is 2.49×10^{-16}. This is a pointwise test of the kernel, not an integration error bound.

The third-order nested and disjoint instances of the kernel have the unweighted auxiliary residues

$$[z^{-1}]D_{\mathrm{L3G004}} = A^2, \qquad [z^{-1}]D_{\mathrm{L3G007}} = \frac{A^2 + 4f_2}{3},$$

which follow from the Gamma counterterms of Section 24.1 and the exact two-loop finite part (650) through the conversion of Section 23.8. A sixteen-million-observation control, with the graph weights inserted, deviates by -0.185 and -0.180 standard errors; a separate four-million-observation run by -0.192 and 1.542. For the finite-part extension a sixteen-million-observation two-loop run gives $[z^{-1}]D_2 = 2.77218438 \pm 0.00048568$ and $[z^0]D_2 = -4.83492035 \pm 0.00210898$, at -0.833 and 0.897 standard errors from A and (650). The reference $r_4 = P_4(2)/2$ was evaluated afresh from (333) at 65 digits, and two Cauchy evaluations of f_2 differ by 2.0×10^{-64}; this is consistency of the numerical special-function evaluation, not an interval certificate.

24 Analytic power-sector compensation at $d = 2, 3$

24.1 An exact Gamma-reducible sector

The momentum-space normalization at $d = 2$ has $r(2, z) = 1$. With one nonzero momentum p, a convolution of n parallel propagators with indices α_i reduces to a single propagator of index $\eta = \sum_i \alpha_i - (n - 1)$ and multiplier

$$\mathcal{B}_n(\boldsymbol{\alpha}) = \frac{\Gamma(\eta)}{\Gamma(1-\eta)} \prod_{i=1}^{n} \frac{\Gamma(1-\alpha_i)}{\Gamma(\alpha_i)}, \qquad I_{\parallel}(p) = \mathcal{B}_n(\boldsymbol{\alpha})(p^2)^{-\eta}. \tag{663}$$

The integration convention in this formula is d^2k/π per loop. Successive use of the two-propagator convolution proves the displayed n-propagator version by telescoping Gamma factors. A degree-two vertex with zero momentum injection adds the two adjacent indices. Initially every line has index $1/2 + z/4$; the final line of an accepted L-loop reduction has index $Lz/2$. Thus the bare amplitude is a finite product of Gamma functions at affine half-integer arguments.

This rearrangement is deliberately restricted. All original degree-two vertices must be among the two momentum terminals. Both terminals are inherited under every contraction. An internal index reaching the infrared boundary $\alpha = 1$ at $z = 0$, an identically singular Gamma factor, a tadpole quotient, a scaleless quotient, or an unsolved cograph causes that reduction to be rejected. Power-divergent mass subgraphs are analytically continued in the existing critical-mass prescription; they are not assigned an extra logarithmic vertex counterterm. These restrictions distinguish the accepted rearrangements from an unrestricted infrared rearrangement that would require additional R^* counterterms [21, 2].

As in (652), PP projects a Laurent series onto its negative powers of z. The UV counterterm is computed recursively as

$$Z_G = -\,\mathrm{PP}\left[I_G + \sum_{S \neq \varnothing \text{ proper spinney}} \left(\prod_{\gamma \in S} Z_\gamma \right) I_{G/S} \right]. \tag{664}$$

Spinney components here are vertex-disjoint. Nested subdivergences are already included recursively in Z_γ. Confusing this convention with the edge-compatible scale-derivative forests of the finite kernels of Section 23 changes individual graph counterterms. Formula (664) is the usual MS R-operation; the special graph selection, Gamma reductions and exact arithmetic are the present calculation [21].

If a graph is a one-vertex join of two or more logarithmic blocks, its amplitude factorizes. Induction in (664), or the multiplicativity of the counterterm character for this join, gives

$$Z_{G_1\vee\cdots\vee G_k} = \prod_{j=1}^{k} Z_{G_j}, \qquad k \geq 2. \tag{665}$$

Every factor on the right has strictly negative Laurent degree, so the product has no simple pole. This establishes the simple-pole zero even when the individual finite amplitudes are not evaluated. Only joins whose blocks are all logarithmic are used in this additional zero test.

Exact arithmetic and closed counts. Gamma recurrence, reflection and half-integer polygamma values reduce the Laurent arithmetic to an exact rational polynomial ring in $\ell = \log 2$ and zeta values [26, 32]. Euler's constant cancels explicitly. No integer-relation fit is used. Requiring both the original graph and every necessary cograph/counterterm to close gives

Loop order	1	2	3	4	5	6
Closed Gamma-sector classes	1	2	6	15	47	137

The bare two-terminal graph being reducible by itself is not sufficient: unsupported cograph finite parts are never filled with zero.

Define b_L^Γ, c_L^Γ by precisely this closed class set, with all original rational graph weights. Two useful exact results are

$$c_5^\Gamma = \frac{12692}{3}\ell^4 - 208\ell^2 + \frac{1085}{4}\ell\zeta(3) - \frac{224}{3}\ell - \frac{73}{6}, \tag{666}$$

$$\boxed{\begin{aligned} 2187\,\Xi_\Gamma &:= (3c_6^\Gamma - b_6^\Gamma) - 24\ell c_5^\Gamma \\ &= \frac{5518904}{75}\ell^5 - \frac{216256}{45}\ell^3 + \frac{5763}{20}\ell^2\zeta(3) - 2784\ell^2 + \frac{6368}{15}\ell - 69\zeta(3) + \frac{83397}{640}\zeta(5). \end{aligned}} \tag{667}$$

Numerically,

$$\frac{3c_6^\Gamma - b_6^\Gamma}{2187} = 12.175885035555361667474529\ldots, \qquad \Xi_\Gamma = 4.27467950247038413834713 0\ldots. \tag{668}$$

Neither expression includes the missing graph classes. The full exact counterterm of every accepted graph, not just the weighted totals, is recorded in `sp_exact.json` of `lri_C6_d2_complete_update/reproduction/lri_C6_d2_two_forest_update/analytic/verification/`.

For example, the closed third-order sector gives $b_3^\Gamma = 336\ell^2 - 4$ and $c_3^\Gamma = 52\ell^2 - 4/3$. Restoring the retained wineglass residue $A^2/3$ with its original weights, and the primitive tetrahedron for the beta function, recovers

$$c_3 = \frac{7}{2}A^2 - \frac{4}{3}, \qquad b_3 = \frac{87}{4}A^2 - 4 + 6Q_2. \tag{669}$$

This check takes the wineglass residue and the tetrahedron as inputs; it is not a new independent evaluation of either. The two-loop checks $b_2 = -6A, c_2 = -A$ are entirely within the new Gamma calculation.

Checks of the Gamma sector. These checks were made by OpenAI GPT-6 astra (code `sp_exact.py`, `sp_ms.py` and `check_gamma_coverage.py` in, and records `sp_exact.json` and `sp_ms_90.json` under, `lri_C6_d2_complete_update/reproduction/lri_C6_d2_two_forest_update/analytic/`). The 793 pole coefficients of the exact table were compared with 90-digit Gamma and polygamma arithmetic; the largest absolute difference is below 4.5×10^{-78}. There are 214 accepted terminal-routing instances through six loops. Wherever alternative routings are admitted their complete pole parts agree, and changing $\log p^2$ of the terminal momentum from zero to one leaves the pole parts unchanged; exact arithmetic checks both cancellations before numerical specialization, and the 90-digit residuals are below 3.8×10^{-85} and 2.0×10^{-85}.

24.2 An analytic advance special to two dimensions

Write d^2x/π for the normalized complex-plane measure. Riesz convolution gives

$$\mathcal{B}(u,v) = \frac{\Gamma(1-u/2)\Gamma(1-v/2)\Gamma((u+v)/2-1)}{\Gamma(u/2)\Gamma(v/2)\Gamma(2-(u+v)/2)}. \tag{670}$$

It replaces two distance powers u, v joined at an integrated vertex by the power $u+v-2$. This formula is used meromorphically, with all normalizing factors retained.

For the five-edge two-variable integral define

$$\mathcal{J}(\boldsymbol{p}) = \frac{1}{\pi^2}\int_{\mathbb{C}^2} \frac{d^2x\, d^2y}{|x|^{p_1}|1-x|^{p_2}|y|^{p_3}|1-y|^{p_4}|x-y|^{p_5}}. \tag{671}$$

Put $a = -p_1/2, b = -p_2/2, A' = -p_3/2, B' = -p_4/2, c = -p_5/2$, and

$$\begin{aligned}\mathcal{O}(a,b,A',B',c) &= B(a+1,c+1)B(a+A'+c+2,B'+1)\\ &\quad\times {}_3F_2\left(\begin{matrix}-b, a+1, a+A'+c+2\\ a+c+2, a+A'+B'+c+3\end{matrix};1\right),\end{aligned} \tag{672}$$

$$\begin{aligned}\mathcal{M}(a,b,A',B',c) &= B(a+1,-a-b-c-1)B(B'+1,-A'-B'-c-1)\\ &\quad\times {}_3F_2\left(\begin{matrix}-c, a+1, B'+1\\ -b-c, -A'-c\end{matrix};1\right).\end{aligned} \tag{673}$$

Here A', B' are temporary exponent names, not the RG constant $A = 4\log 2$. A split-contour identity gives

$$\boxed{\begin{aligned}\mathcal{J}(\boldsymbol{p}) = \frac{1}{\pi^2}\big[&\sin(\pi a)\sin(\pi B')\,\mathcal{O}(a,b,A',B',c)\mathcal{M}(a,b,A',B',c)\\ &+\sin(\pi A')\sin(\pi b)\,\mathcal{O}(A',B',a,b,c)\mathcal{M}(A',B',a,b,c)\big].\end{aligned}} \tag{674}$$

Thus the needed nonsymmetric kite is evaluated by four ordinary ${}_3F_2(1)$ values, not by a noisy four-real-dimensional integration. This is a bilinear identity. For complex regulator values the products in (674) are not replaced by products involving complex conjugation.

For clarity, the contour argument starts in a common convergence region. Split the two antiholomorphic contours so that one lies left of zero and the other right of one. The two possible orderings of the remaining holomorphic coordinates give $\mathcal{O}$ and its exchanged version. The discontinuities of the two outer contours give the sine factors in (674). In the mixed domain the changes $x = -u/(1-u)$, $y = 1/(1-v)$ give $\mathcal{M}$ by two Euler beta integrations. The one-variable normalization is fixed by $-\sin(\pi b)B(a+1,b+1)B(-a-b-1,b+1)$; applying the construction twice fixes the positive sign and the π^{-2} in (674). Analytic continuation then proves the meromorphic identity away from its polar divisors. This is the two-integration specialization of the usual KLT contour factorization

[43]; the relation between two-dimensional massless Feynman integrals and bilinear hypergeometric expressions is also established in [45].

The symmetric cases reduce independently to the complex Selberg Gamma product [6]. In particular, setting all five powers to one recovers $\mathcal{J} = 3Q_2$. Three nonsingular symmetric tests, five nontrivial completion transformations at a nonsymmetric point, and a changed-tail hypergeometric calculation verify the normalization separately from the graph recursion. These are consistency checks of the analytic formula, not a fit for its coefficients.

Counterterm closure and infrared conditions. A graph is accepted by the coordinate reduction only when its raw graph, all required subgraphs, and every inherited-root quotient have a declared reduction. A root pair is rejected if it is contained in a proper induced collision set with $e \geq 2(v-1)$. This prevents a spurious infrared singularity being interpreted as a UV counterterm. An unanchored one-attachment massless block is assigned its analytic scaleless value zero only when its superficial degree at $d = 2$ is the nonzero odd number 1 or -1. No logarithmic scaleless factor is assigned zero by this rule. Its logarithmic subtractions are still performed before the graph counterterm is extracted.

Cauchy extraction of the regular function $z^2\mathcal{J}(z)$ uses two settings: 32 nodes at radius 0.02 with 45-digit function evaluation, and 48 nodes at radius 0.016 with 55-digit evaluation. The two runs compare 2699 pole coefficients; their maximum absolute discrepancy is 7.38×10^{-30}. The maximum discrepancy between distinct admissible roots in the higher-precision run is 7.97×10^{-45}. The Gamma-only counterterms of Section 24.1 provide additional independent checks. No zero is inserted because a nonzero result is numerically inconvenient.

The closed graph/subgraph/quotient reductions and the two contractions below cover the nonzero power-sector target. The other logarithmic graph contributions are evaluated by the finite-scheme calculation, without requiring separate numerical values for every individual MS finite part.

24.3 The complete power sector is explicitly accounted for

The only two nonzero-target power graphs not closed by the preceding procedure are L6G106 and L6G117. Both are a sunset insertion on the four-loop primitive L4G027. Put $p = 1 - z/2$. Integrating the chain with powers $p, 3p, p$ gives

$$\mathcal{S}(p) = \frac{\Gamma(1-p/2)^2\Gamma(1-3p/2)\Gamma(5p/2-2)}{\Gamma(p/2)^2\Gamma(3p/2)\Gamma(3-5p/2)}, \qquad \mathcal{S}(1) = -4. \tag{675}$$

Its effective edge power is $5p - 4$. The remaining graph is primitive, so the deformation of this edge does not affect the required simple pole. Each graph has $w_A = 2, w_B = 0$, giving $16P_4(2)$ to b_6. Together they contribute $32P_4(2)$ to b_6 and zero to c_6. This contraction was checked by exact graph isomorphism and not inferred from an expected numerical coefficient.

Let a superscript p denote the sum of graphs containing at least one proper power-divergent subgraph. Define the lower-order polynomial of the constant-coefficient MS assembly, that is, the terms of (594) that contain only A, b_3, c_3, b_4 and c_4, with A at the present dimension (Section 23.1 shows that (594) holds at every dimension with this A):

$$\begin{aligned}\mathcal{R}(b_3, c_3, b_4, c_4) = {} & 2016A^5 - 448A^3b_3 + 672A^3c_3 - 84A^2b_4 + 168A^2c_4 \\ & + \frac{49}{3}Ab_3^2 - 42Ab_3c_3 + \frac{7}{3}b_3b_4 - 4b_3c_4 - 3b_4c_3.\end{aligned} \tag{676}$$

Subtracting the constant-coefficient MS assemblies for the full and power-free graph sets gives the

exact bookkeeping identity

$$\boxed{\begin{aligned} 2187\,\Delta_p &= (3c_6 - b_6)^p - 12Ab_5^p + 30Ac_5^p \\ &\quad + \mathcal{R}(b_3, c_3, b_4, c_4) - \mathcal{R}(b_3 - b_3^p, c_3 - c_3^p, b_4 - b_4^p, c_4 - c_4^p), \\ C_6^{\text{full}}(2) &= C_6^{\text{power-free}}(2) + \Delta_p. \end{aligned}} \tag{677}$$

Only the complete combination $(3c_6 - b_6)^p$, rather than its two individual summands, is needed. Any remaining individual sixth-order power graph with $3w_B = w_A$ has zero weight in this combination and is excluded by an exact rational identity.

The actual MS sums include

$$\begin{aligned} b_3^p &= -4, & c_3^p &= -\frac{4}{3}, \\ b_4^p &= 16A + 24, & c_4^p &= \frac{8}{3}A + 8, \\ b_5^p &= -1540.38932947914290838\ldots, & c_5^p &= -251.621917341000466476\ldots. \end{aligned} \tag{678}$$

The evaluated higher-order part of (677), divided by 2187, is $1.535501191157190785360393154334\ldots$; the lower-order difference is its negative at the computed precision. The two Cauchy settings give $\Delta_p = -8.02 \times 10^{-34}$ and $\Delta_p = 6.91 \times 10^{-47}$, respectively. These residuals are numerical consistency checks, not a proof of an all-orders power-subgraph cancellation theorem. The calculation retains the explicitly evaluated correction (677); it does not omit the power graphs by assumption. Their residual numerical uncertainty is negligible on the scale of the reported $C_6(2)$ uncertainty.

24.4 Power subgraphs: a separate three-dimensional audit

It would not be justified to infer cancellation of power graphs from the two-dimensional result. The correction is instead evaluated in the MS prescription before being added to the finite-scheme power-free result. The coordinate convolution with distance exponents a, b is

$$B_d(a,b) = \Gamma(d/2)\frac{\Gamma((d-a)/2)\Gamma((d-b)/2)\Gamma((a+b-d)/2)}{\Gamma(a/2)\Gamma(b/2)\Gamma((2d-a-b)/2)}. \tag{679}$$

The dimension-three implementation uses affine edge powers $m(3-z)/2$, not the two-dimensional powers with a numerical prefactor changed. It applies the Taylor reductions of the kite master (231) at its critical patterns A, B and T only to their declared orders. Gamma factors, logarithmic counterterms, contractions and safe nonzero-distance roots are retained recursively. A requested coefficient beyond the available master Taylor order is rejected, not assigned zero. Nonlogarithmic one-attachment massless factors are removed only in the specified analytic critical-mass prescription; this test is not used to discard a logarithmic scaleless UV counterterm.

There are 1,5,31 relevant power-containing classes at loops 3,4,5. At six loops all 121 power classes with $3w_B - w_A \neq 0$ are included; the other 52 power classes have exactly zero weight in the sixth-order target. A zero sixth-order target weight does not license dropping a lower-order input. In particular the five-loop graph described next remains necessary.

Two sunset contractions. The graphs L6G106 and L6G117 are sunset insertions on the four-loop primitive. The chain of three effective edges with critical powers $p, 3p, p$, $p = d/2$, has factor

$$M_d = \frac{\Gamma(-q)\Gamma(2q)^2}{\Gamma(3q)}, \qquad M_3 = -\frac{16\pi}{15}. \tag{680}$$

Each graph contributes $-4M_3P_4(3)$ to b_6 and zero to c_6. Only the primitive residue is needed after the contraction; no unknown edge-deformation finite term is inserted. The inherited graph identification and its weights are checked in the audit.

24.5 An exact cancellation avoids two unknown core integrals

The nontrivial extra obstruction in dimension three is a four-loop two-terminal core. Using the terminal labels $2, 3$, its seven internal edges are

$$(01), (02), (03), (03), (12), (12), (13). \tag{681}$$

Every edge has distance power $p = (3 - z)/2$. Equivalently its two-integrated- vertex kite powers are $(p, 2p, 2p, p, p)$. The required expansion has the form

$$J_{\mathrm{core}}(z) = \frac{8}{z^2} + \frac{2A}{z} + u_0 + u_1 z + O(z^2), \qquad A = 2 + 4\log 2 - \pi. \tag{682}$$

The pole coefficients follow from its proper logarithmic subgraphs. The finite coefficients u_0, u_1 are not computed. Their cancellation from the physical target is proved below, rather than assumed.

Let $C(z)$ be the product of the two external chain convolutions and $F(z) = C(z)J_{\mathrm{core}}(z)$. For reference,

$$C(0) = M_3, \qquad C'(0) = \frac{7}{4}M_3(A+N), \qquad N = \psi_0(-q)+\psi_0(3q)-\psi_0(1)-\psi_0(2q) = \frac{62}{15}-\pi-4\log 2. \tag{683}$$

Denote by F_0, F_1 the unknown-dependent regular coefficients. Known pole-dependent terms are retained elsewhere in the audit. Write $\doteq$ for equality only of the dependence on F_0, F_1. If $r_G = [z^{-1}]R'I_G$, then the three affected graphs satisfy

$$\begin{aligned} r_{\mathrm{L5G086}} &\doteq \frac{2}{5}F_0, \\ r_{\mathrm{L6G332}} &\doteq \frac{A}{3}F_0 + \frac{2}{15}F_1, \\ r_{\mathrm{L6G333}} &\doteq \frac{A}{3}F_0 - \frac{2}{15}F_1. \end{aligned} \tag{684}$$

These expressions follow by applying $B_d(d - kz, d/2 - lz) = 2/(kz) + A + O(z)$ to the attached convolutions. For L6G333 the external bubble counterterm contributes $-4F_1/5$; omitting it would spoil the identity.

The five-loop weights are $(w_A, w_B) = (3/4, 1/4)$ and both six-loop weights are $(3/2, 1/4)$. Since $b_L = (-1)^{L+1}Lw_A r_G$ and $c_L = (-1)^{L+1}Lw_B r_G$, the unknown part of the physical sixth-order MS numerator is precisely

$$\boxed{\frac{9}{2}(r_{\mathrm{L6G332}} + r_{\mathrm{L6G333}}) - \frac{15}{2}A\, r_{\mathrm{L5G086}} \doteq 0.} \tag{685}$$

This is an exact rational-polynomial identity in A, F_0, F_1. It eliminates only the two unknown core coefficients, not the other parts of these graphs. Substituting $(u_0, u_1) = (0, 0)$ and $(101, -37)$ in the full graph recursion changes the three individual residues exactly as predicted by (684), to better than 4×10^{-73}. Thus those trial assignments are algebraic checks, not estimates of u_0, u_1. The saved auxiliary individual b_5, b_6 sums obtained with a trial assignment must not be reported as separately evaluated physical MS coefficients.

24.6 Assembled power correction and completeness

The constant-MS lower-order polynomial is the same function $\mathcal{R}$ as in (676); only A changes, to its $d = 3$ value $2 + 4\log 2 - \pi$ of (682). It is repeated here for reference:

$$\begin{aligned} \mathcal{R}(b_3, c_3, b_4, c_4) = {} & 2016A^5 - 448A^3 b_3 + 672A^3 c_3 - 84A^2 b_4 + 168A^2 c_4 \\ & + \frac{49}{3}Ab_3^2 - 42Ab_3c_3 + \frac{7}{3}b_3b_4 - 4b_3c_4 - 3b_4c_3. \end{aligned} \tag{686}$$

A superscript p denotes the complete power-containing graph sum. Subtracting the full and power-free MS assemblies gives

$$\boxed{\begin{aligned} 2187\Delta_p &= (3c_6 - b_6)^p - 12Ab_5^p + 30Ac_5^p \\ &\quad + \mathcal{R}(b_3, c_3, b_4, c_4) - \mathcal{R}(b_3 - b_3^p, c_3 - c_3^p, b_4 - b_4^p, c_4 - c_4^p), \\ C_6^{\text{full}}(3) &= C_6^{\text{pf}}(3) + \Delta_p. \end{aligned}} \tag{687}$$

For the three- and four-loop power pieces the graph and counterterm recursion of this section agrees with

$$\begin{aligned} b_3^p &= M, & c_3^p &= M/3, \\ b_4^p &= -\tfrac{11}{2}AM - \tfrac{3}{2}MN, & c_4^p &= -\tfrac{7}{6}AM - \tfrac{1}{2}MN, \end{aligned} \tag{688}$$

where $M = M_3$. Setting $R = \psi_1(2q) - \psi_1(1)$, $U = \psi_2(q)$, $V = \psi_2(1) + \psi_2(2q)$, $Q = Q_3$, and $P = P_4(3)$, exact assembly algebra gives the lower-order part of (687), $\delta\mathcal{R} = \mathcal{R}(b_3, c_3, b_4, c_4) - \mathcal{R}(b_3 - b_3^p, c_3 - c_3^p, b_4 - b_4^p, c_4 - c_4^p)$ with the complete lower-order coefficients (179), (226) and (227), in closed form; it is $2187\,\Delta_{p,\text{lower}}$ below:

$$\begin{aligned} \delta\mathcal{R} = \frac{M}{24}(&2727A^3 + 603A^2N - 8AM + 72AQ + 63AR \\ &- 216NQ + 27NR - 192P - 318U + 36V). \end{aligned} \tag{689}$$

This polynomial alone does not prove the higher graph sum. The latter was separately computed by the graph and counterterm recursion.

At 90 working digits the two parts of (687), including its factor 1/2187, are

$$\begin{aligned} \Delta_{p,\text{higher}} &= +0.073651697044164869469222945561 1\ldots, \\ \Delta_{p,\text{lower}} &= -0.073651697044164869469222945561 1\ldots. \end{aligned} \tag{690}$$

Their sum is -1.02×10^{-89}. A separate 75-digit run with different core assignments gives -1.94×10^{-75}. A third algebraic stress test replaces the values of the masters Q_3, $P_4(3)$, $\mathcal{K}_3$ and Z_3 by the test numbers 7, 11, 5 and 13, and the core assignments by $(-17, 29)$: the combined residual is 1.52×10^{-65} at 65 working digits. This last test is not a physical evaluation of those masters; it checks that common master-dependent terms cancel consistently in the target.

These are high-precision numerical checks of the *complete* power correction, together with an exact proof of the unknown-core cancellation. They are not an all-orders theorem allowing individual power graphs to be dropped. The tiny computed residual is negligible relative to the sampling error, but it is not an interval-certified error bound. Every nonzero-weight sixth-order power graph and every required fifth-order power input has been covered. Two unrelated non-power four-loop graphs rejected by the coordinate reduction are instead included in the finite-kernel integration and are not a missing power contribution.

25 Finite-scheme sixth-order assembly

25.1 A finite zero-momentum scheme avoids separate MS finite parts

In this subsection only, $\widehat{b}_L(z), \widehat{c}_L(z)$ denote the *epsilon-dependent finite-scheme* RG coefficients of the power-free graph set. They are not the constant MS coefficients. Since the coupling and the mass are defined by bare amplitudes at zero external momentum, this scheme is called finite-MOM (finite momentum subtraction) below. Let

$$\mathcal{A}(h, z) = 1 + \sum_{L=1}^{6} (-1)^L \mathcal{A}_L(z) h^L, \qquad \mathcal{B}(h, z) = 1 + \sum_{L=1}^{6} (-1)^L \mathcal{B}_L(z) h^L, \tag{691}$$

where the graph weights (597) and the bubble normalization are those of Sections 23 and 24. Defining $g = h\mathcal{A}(h,z)$, differentiation at fixed bare parameters gives

$$\beta(g,z) = -z\,h\,\partial_h(h\mathcal{A}(h,z)), \qquad \gamma_r(g,z) = -z\,h\,\partial_h \log \mathcal{B}(h,z). \tag{692}$$

The right sides are reexpanded in g. They have the form

$$\beta = -zg + \sum_{L=1}^{6} \widehat{b}_L(z) g^{L+1}, \qquad \gamma_r = \sum_{L=1}^{6} \widehat{c}_L(z) g^L, \qquad \widehat{b}_1 = 3, \quad \widehat{c}_1 = 1. \tag{693}$$

The fixed-point exponent is unchanged by this finite coupling and operator redefinition: the coupling is reparametrized, and the extra operator-anomalous-dimension term is proportional to β, hence vanishes at the fixed point. This assertion is used only for the same power-free amplitudes in the two prescriptions. The correction for the omitted power-containing amplitudes is separately supplied by (677).

For a compatible forest F of proper logarithmic collision sets, let its blocks be the contracted subgraphs and the remaining root block. Write h_R for the loop number of that root block and I_B for a bare block amplitude. The finite scale-derivative combination is

$$D_G(z) = \sum_F (-1)^{|F|} \frac{h_R}{L} \prod_{B \in F \cup \{G\}} I_B(z). \tag{694}$$

The notation on the product means the forest blocks after their children have been contracted, not a repeated product of unreduced parent graphs. Terms with $h_R = 0$ vanish. Complete rational polynomial comparison with independent inversion of (691) proves

$$\widehat{b}_L(z) = (-1)^{L+1} L z \sum_G w_A(G) D_G(z), \qquad \widehat{c}_L(z) = (-1)^{L+1} L z \sum_G w_B(G) D_G(z). \tag{695}$$

This comparison was rerun at all six loop orders. The sixth-order polynomials contain 532 beta and 383 mass monomials; both differences vanish over the rationals. It is not a numerical agreement at selected values of the bare integrals. The use of common parametric forest kernels is consistent with the general parametric renormalization framework [1].

The independent inversion. The comparison uses the Lagrange–Bürmann inversion of $g = h\mathcal{A}(h,z)$ in (692). With primes denoting ∂_h at fixed z and $\mathcal{A}, \mathcal{B}$ the series (691),

$$\begin{aligned} \widehat{b}_L(z) &= -\frac{z}{L+1}\,[h^L](2h\mathcal{A}' + h^2\mathcal{A}'')\mathcal{A}^{-L-1}, \\ \widehat{c}_L(z) &= -\frac{z}{L}\,[h^{L-1}]\,\frac{d}{dh}\left(\frac{h\mathcal{B}'}{\mathcal{B}}\right)\mathcal{A}^{-L}. \end{aligned} \tag{696}$$

Indeed $[g^n]F(h(g)) = \frac{1}{n}[h^{n-1}]F'(h)\mathcal{A}^{-n}$; for $F = -zh(h\mathcal{A})'$ one has $F' = -z(\mathcal{A} + 3h\mathcal{A}' + h^2\mathcal{A}'')$, and $[h^L](\mathcal{A} + h\mathcal{A}')\mathcal{A}^{-L-1} = 0$ for $L \geq 1$. The same comparison of every monomial was made against the forest polynomials (695). For the six-loop inputs the 487 power-free graphs, with 66,198 forests, combine into 395 nonzero rational kernels; 92 graph kernels vanish identically, and the largest retained kernel has 52 distinct denominators. These scale-derivative coefficients carry explicit regulator dependence; they are not the constant MS coefficients and are never inserted into an MS fixed-point assembly [2]. The inversion formulas (696), the comparison and the counts are those of the earlier two-dimensional calculation by OpenAI GPT-6 astra (package `LRI_C6_D2_HPC/reference/`); the derivation of (696) from $[g^n]F(h(g))$ was written out by a Claude Code session on 16 October 2026, when this paragraph was merged into the note.

25.2 A common, locally subtracted integration kernel

For a forest, let U_F be the product of the Symanzik polynomials of its blocks, embedded in the original edge variables, and let $E_R(F)$ be the root-block edge set. With $S = \sum_e x_e$, define

$$\boxed{K_G(x) = \sum_F (-1)^{|F|} \frac{\sum_{e\in E_R(F)} x_e}{S\, U_F(x)}.} \tag{697}$$

It is homogeneous of degree $-L$. All terms are combined before numerical evaluation. At $d = 2$, the normalization $r(2, z) = \Gamma(1 + z/2)/\Gamma(1 + z/2)$ is identically one, and

$$D_G(z) = \frac{\Gamma(Lz/2)}{\Gamma(1/2 + z/4)^{2L}} \int d\Omega \prod_e x_e^{-1/2+z/4} K_G(x). \tag{698}$$

To verify the factor in (694), integrate first over the relative scales of the blocks. Their Dirichlet parameters are $h_B z/2$, so insertion of the root scale fraction has mean $(h_R z/2)/(Lz/2) = h_R/L$. This proves the exact common-integral identity for positive regulator in its initial convergence region, and then by continuation. No fitted subtraction constant is involved.

The forest cancellations remove the proper logarithmic endpoint singularities. The remaining overall pole is isolated by $\Gamma(Lz/2)$. Consequently the integral in (698) and its required logarithmic moments are finite at $z = 0$. The individual terms of (697) need not be separately integrable there and are never sampled as unrelated singular integrals.

There are 1, 5, 16, 76 and 395 nonzero finite kernels at loop orders 2 through 6. For all 493 kernels, an independent exact-rational check compares the co-tree-mask implementation with a product of Kirchhoff cofactor determinants constructed directly from the forest blocks. Every difference is zero. This validates the common-integral mapping separately from the RG polynomial check.

25.3 Retaining the regulator dependence through sixth order

Write $\widehat{b}_L(z) = \sum_j b_{L,j} z^j$ and $\widehat{c}_L(z) = \sum_j c_{L,j} z^j$. Terms through $j = 6 - L$ are required. In a projective sector with one largest edge fixed at one, put

$$s_L(x) = \frac{1}{4} \sum_e \log x_e - \frac{L}{2} \log S, \qquad \mathcal{G}_L(z) = \frac{\pi^L \Gamma(1 + Lz/2)}{\Gamma(1/2 + z/4)^{2L}}. \tag{699}$$

The normalized Gamma factor has $\mathcal{G}_L(0) = 1$ and

$$\log \mathcal{G}_L(z) = \sum_{j\geq 1} a_{L,j} z^j, \quad a_{L,1} = L \log 2, \quad a_{L,j} = \frac{(-1)^j \zeta(j)}{j} \left[\left(\frac{L}{2}\right)^j - \frac{2L(2^j - 1)}{4^j} \right] \quad (j \geq 2). \tag{700}$$

Only $j \leq 4$ is used. If $t = s_L + a_{L,1}$, the required polynomials are

$$E_0 = 1, \qquad E_1 = t, \quad E_2 = \frac{t^2}{2} + a_{L,2},$$
$$E_3 = \frac{t^3}{6} + a_{L,2} t + a_{L,3}, \qquad E_4 = \frac{t^4}{24} + \frac{a_{L,2} t^2}{2} + a_{L,3} t + a_{L,4} + \frac{a_{L,2}^2}{2}. \tag{701}$$

Integrating $E_j K_G$ with the critical projective measure and multiplying by $2(-1)^{L+1} w_{A,B}/\pi^L$ gives $b_{L,j}, c_{L,j}$. This computes Taylor moments of a finite integral, not a numerical subtraction of Laurent poles.

The two-loop jets are evaluated by the convergent two-${}_3F_2(1/4)$ formula (199) for $S_d(z)$ at $d = 2$, rather than by Monte Carlo. Explicitly,

$$\widehat{b}_2(z) = -6z \left[S_2(z) - \frac{2}{z^2} \right], \qquad \widehat{c}_2(z) = \frac{1}{6} \widehat{b}_2(z). \tag{702}$$

Fresh 65-digit evaluations with radii 0.05 and 0.08 and respectively 64 and 80 Cauchy nodes agree with the saved input through z^4 to better than 6.8×10^{-61}. The five beta coefficients begin

$$\begin{aligned} b_{2,0} &= -16.63553233344, \quad & b_{2,1} &= 29.02087369541, \\ b_{2,2} &= -47.11795797162, \quad & b_{2,3} &= 74.14063002868, \\ b_{2,4} &= -114.64528966290. \end{aligned} \tag{703}$$

Let $\widehat{B}(g,z) = \sum_{L\geq 1} \widehat{b}_L(z) g^{L-1}$. The physical fixed point satisfies $\epsilon = g_* \widehat{B}(g_*, \epsilon)$. Lagrange inversion with z held fixed, followed by $z = \epsilon$, gives

$$\boxed{C_n^{\text{power-free}}(2) = -\frac{1}{2}\delta_{n1} + \sum_{k=1}^{n} \frac{1}{k} [g^{k-1} z^{n-k}] \partial_g \gamma_r(g,z) \widehat{B}(g,z)^{-k}, \qquad n \geq 1.} \tag{704}$$

The constant term is $C_0 = 1$. Only finite polynomials enter this extraction. Independent exact-rational fixed-point substitution and bivariate Lagrange extraction agree in 192 checks, including 24 checks of the complete constant-coefficient MS sixth-order assembly.

The regulator jets cannot be discarded. Substituting only $\widehat{b}_L(0), \widehat{c}_L(0)$ into the constant-coefficient MS assembly (the analogue of (594)) would use a different truncated RG problem. All required jets in (704) are retained, including the two-loop z^4, three-loop z^3, four-loop z^2, and fifth-loop z^1 terms. At sixth order only the combination $3c_{6,0} - b_{6,0}$ is required, as in the MS assembly.

25.4 Importance sampling and retained sector controls

The integration uses Hepp sectors and subset dynamic programming, in the spirit of tropical Feynman quadrature [41]. For each proper nonempty edge subset S, choose a positive proposal exponent $\alpha(S)$. Numerical leading-power diagnostics, performed on the *combined* kernel, guide this choice; they are not presented as rigorous valuations. The normalization is computed without enumerating all $(2L)!$ edge orders:

$$Z(\varnothing) = 1, \qquad Z(S) = \frac{1}{\alpha(S)} \sum_{e \in S} Z(S \setminus \{e\}), \qquad Z(E) = \sum_{e \in E} Z(E \setminus \{e\}). \tag{705}$$

An edge is removed with probability proportional to $Z(S \setminus \{e\})$; the next sector ratio is sampled with density $\alpha(S) t^{\alpha(S)-1}$. The exact reciprocal proposal weight is applied to every measurement. Changing the positive exponents changes the variance, not the target integral. Diagram probabilities are fixed from preceding pilot data with a nonzero uniform floor, and likewise corrected by their reciprocal. A stated sample count is the number of random graph–sector–point draws for an entire loop order, not that number of samples for each individual graph.

Co-tree monomials are evaluated relative to their largest logarithm. A cancellation monitor compares the signed forest value with the sum of absolute term contributions. Evaluation starts in extended precision and is repeated in binary128 when that ratio is below 10^{-9}; a 100-digit fallback is available below 10^{-23} at the second stage. No sample is clipped. The primary run, run C below, invokes binary128 21703 times and never invokes the final fallback. These diagnostics are precision controls, not interval arithmetic.

The 100-digit level of that fallback was later found too low. The following account comes from the three-dimensional sixth-order bundle `lri_C6_d3_complete_update/`, written by OpenAI GPT-6 astra (model stated by the author): a deliberately extreme stress test of the graph L3G007 in that package exposed insufficient relative accuracy, and the fallback was raised to 300 digits before the test was repeated. This is the level used by the later sixth-order runs, including the 2026 productions (Sections 26.7 and 26.8). No production run of the three-dimensional package

called the emergency path, so none used the defective branch, and two same-seed 128,000-sample replays at loops three and six agreed batch for batch before and after the change. None of the runs A–D described next reached that branch: the counters of the final fallback in their records (`lri_C6_d2_complete_update/reproduction/mom_prod_*.json` of the research archive) are zero at every loop order, so their values are unaffected.

Runs A and B, which are unrelated to the two runs of (660), integrate all 395 nonzero sixth-order finite kernels. In run C the 10 primitive, 26 one-subgraph and 40 two-subgraph kernels are replaced by their already measured finite-scheme values, with their uncertainties propagated. The correct replacement is

$$\begin{aligned} K_{76}^{\mathrm{MOM}} &= C_{6,\mathrm{primitive}} - \frac{2}{729}(Y_{26} + Y_{40}) \\ &= 528.2946383532544, \qquad \widehat{\sigma}(K_{76}^{\mathrm{MOM}}) = 0.1373774929551. \end{aligned} \tag{706}$$

It is *not* the sum of the separately MS-converted Θ_{26}, Θ_{40}. This distinction avoids double counting finite-scheme conversion terms. Exact zero weights remove another 28 kernels from the sixth-order physical combination, so the remaining sixth-order integration in C has 291 kernels. Its proposal exponents are 0.85 times those of A and B. A fourth run, D, used the same substitution as C with the proposal exponents of A and B, a different seed and 16,000,000 draws per loop order (A and B used 8,000,000 and C 32,000,000). The measurements behind (706) were made before, and independently of, the loop-order samples and are not used to calibrate an unknown C_6. Runs A–D, combined by inverse covariance with the weights 0.041, 0.039, 0.702 and 0.219, gave the first complete two-dimensional estimate, superseded by the 2026 production (Section 26.7); their records are in the research archive `LRI_complete_research_bundle.zip`, which is not part of this repository, and the input (706) is also in `LRI_C6_D2_HPC/reference/baseline_previous/baseline/lri_C6_d2_complete_update/reproduction/mom_known_sectors.json`.

25.5 The changes required by dimension three

The long-range regulator is varied at fixed dimension, with $q = d/4$ and edge exponent $a = q + z/4$. The general-dimensional renormalization setting follows [7]; the sixth-order numerical work and its evidence are described here and are not part of that three-loop paper.

For a proper logarithmic forest F of a power-free L-loop graph G, let U_F be the product of the first Symanzik polynomials of its contracted blocks, $E_R(F)$ the edges in its root block, and $S = \sum_e x_e$. The correct common kernel in dimension three is

$$\boxed{K_G^{(3)}(x) = \sum_F (-1)^{|F|} \frac{\sum_{e \in E_R(F)} x_e}{S\, U_F(x)^{3/2}}.} \tag{707}$$

The fractional power applies to each complete forest denominator; it cannot be obtained by raising an already summed two-dimensional kernel to 3/2. Each term is evaluated on the same point before the sum is sampled. This preserves local cancellation of logarithmic endpoint singularities. General parametric forest renormalization provides the framework [1, 21]; the retained exact forest-to-RG polynomial identity remains dimension independent.

With $r(d,z) = \Gamma(d/2 + z/2)/\Gamma(1 + z/2)$ and $d = 3$, the corresponding scale-derivative amplitude is

$$D_G(3,z) = \frac{r(3,z)^L \Gamma(Lz/2)}{\Gamma(3/4 + z/4)^{2L}} \int d\Omega \prod_e x_e^{-1/4+z/4}\, K_G^{(3)}(x). \tag{708}$$

Relative block-scale integration has Dirichlet parameters $h_B z/2$. The inserted root fraction therefore has mean h_R/L, proving the same forest coefficient as in the two-dimensional construction. The finite RG coefficients, distinguished by hats from constant MS coefficients, are

$$\widehat{b}_L(z) = (-1)^{L+1} Lz \sum_G w_A(G) D_G(3,z), \qquad \widehat{c}_L(z) = (-1)^{L+1} Lz \sum_G w_B(G) D_G(3,z). \tag{709}$$

The inherited inventory contains 660 sixth-order graph classes. There are 487 power-free classes and 173 classes with a proper power subgraph. After exact forest cancellation, the power-free numerical inputs at $L = 2, 3, 4, 5, 6$ contain respectively $1, 5, 16, 76, 395$ nonzero kernels. Their edge lists and rational weights were checked again against the inventory. This is reuse and checking of the project's graph inventory, not a second independent enumeration.

Dimension-correct analytic prefactors. Unlike at $d = 2$, $r(3, z)$ is not identically one. In a largest-edge projective sector define

$$s_L(x) = \frac{1}{4}\sum_e \log x_e - \frac{L}{2}\log S,$$
$$G_{L,d}(z) = \frac{r(d,z)^L \Gamma(1 + Lz/2)}{\Gamma(q + z/4)^{2L}} \frac{\Gamma(q)^{2L}}{\Gamma(2q)^L}, \qquad q = \frac{3}{4}. \tag{710}$$

Then $G_{L,d}(0) = 1$ and $\log G_{L,d}(z) = \sum_{j\geq 1} a_{L,j} z^j$, where

$$\boxed{a_{L,j} = \frac{L2^{-j}\psi_{j-1}(2q) + [(L/2)^j - L2^{-j}]\psi_{j-1}(1) - 2L4^{-j}\psi_{j-1}(q)}{j!}.} \tag{711}$$

The overall critical normalization is

$$n_L = 2(-1)^{L+1} \frac{\Gamma(2q)^L}{\Gamma(q)^{2L}}. \tag{712}$$

Integrating the coefficients of $\exp(s_L z) G_{L,d}(z)$ against the critical projective kernel and multiplying by $n_L w_{A,B}$ gives the required jets. These coefficients are the finite Bell polynomials (701). Their maximum order is four; no pole-fitting operation is used in the Monte Carlo integration. All 20 Gamma logarithmic jets and five normalizations were freshly checked by independent 65-digit Cauchy evaluation of the analytic Gamma product. The maximum absolute discrepancy is 2.27×10^{-59}.

25.6 Regulator dependence and finite-order extraction

The two-loop input is recomputed from the explicit general-dimensional two-${}_3F_2(1/4)$ formula (199) for $S_d(z)$, using

$$\widehat{b}_2(z) = -6z\left(S_3(z) - \frac{2}{z^2}\right), \qquad \widehat{c}_2(z) = \frac{1}{6}\widehat{b}_2(z). \tag{713}$$

Its beta jets through z^4 are

$$\begin{aligned} b_{2,0} &= -9.7859764118999280, & b_{2,1} &= 13.1771369571188313, \\ b_{2,2} &= -16.3759608428540408, & b_{2,3} &= 19.5740514096297382, \\ b_{2,4} &= -22.7913170046010398. \end{aligned} \tag{714}$$

Two independent 60-digit evaluations use radii $0.05, 0.07$ and respectively 48,64 Cauchy nodes. Their full outputs, rather than these rounded entries, are kept in the package records listed in Section 33.

Let $\widehat{B}(g, z) = \sum_{L\geq 1} \widehat{b}_L(z) g^{L-1}$, with $\widehat{b}_1 = 3, \widehat{c}_1 = 1$, and $\gamma_r(g, z) = \sum_L \widehat{c}_L(z) g^L$. The power-free physical fixed point is defined by $\epsilon = g_* \widehat{B}(g_*, \epsilon)$. Keeping z independent during Lagrange inversion gives

$$\boxed{C_n^{\rm pf}(3) = -\frac{1}{2}\delta_{n1} + \sum_{k=1}^{n} \frac{1}{k}[g^{k-1} z^{n-k}] \partial_g \gamma_r(g, z) \widehat{B}(g, z)^{-k}, \qquad n \geq 1,} \tag{715}$$

with $C_0 = 3/2$. The extraction is from finite polynomials, not from an unevaluated master. The needed orders are z^{6-L} at each loop L. Thus the fifth-loop mass and coupling finite-scheme jets are

included separately, while at six loops only $3c_{6,0} - b_{6,0}$ is needed. Replacing every hatted function by its constant term would be incorrect. An independent rational-polynomial implementation of implicit fixed-point substitution and bivariate Lagrange inversion gives 256 exact checks, including 32 checks of the constant-MS sixth-order assembly.

26 Final numerical design, covariance and precision

Current and superseded estimates. This section serves both dimensions. Only the 2026 productions of Sections 26.7 and 26.8 are current. Each has two campaigns, a and b, with 64 replicates at every loop order, and they give $C_6(2) = 362.366 \pm 0.004$ and $C_6(3) = 33.5754 \pm 0.0005$. The other campaigns described in this Part belong to superseded estimates; they are kept because the productions inherit their kernels, allocations and checks. The superseded sixth-order method-refinement calculation at $d = 2, 3$, called the earlier sixth-order calculation in this section, with its refinement campaigns, previous estimators, method benchmarks (Section 26.11), control experiment (Section 26.14) and method assessment (Section 26.16), was run by OpenAI GPT-6 astra (model stated by the author) and written up by it in a methods note delivered with the package; that note is merged into this section. Its code and records are in `LRI_C6_D2_HPC/reference/`; record paths given below for the earlier calculation are relative to that folder. Three parts were added by a Claude Code session when the methods note was merged, on 16 October 2026: formula (738), rederived from the stored covariance, where it reproduces the reported ratios 0.985 and 0.962; the descriptions of the benchmark variants, of the pilot counts and of the trained maps in Section 26.11, taken from the package code (`code/experiment.cpp`, `code/pilot.py`, `code/train_grid.py`); and the proof of (750).

- Runs A–D of Section 25.4 gave the first complete two-dimensional estimate.
- The previous estimators of Section 26.13 pooled, at each dimension, two baseline campaigns A and B with 64 replicates (Sections 26.2 and 26.4) and two refinement campaigns, also labeled A and B, with 32 replicates (Section 26.12). They gave $C_6(2) = 362.3489 \pm 0.1526$ and $C_6(3) = 33.5682 \pm 0.0242$ (sampling standard errors).

Upper-case letters label superseded campaigns; lower-case a and b label the campaigns of the 2026 productions, here and in Sections 28, 29, 31 and 32.

26.1 Integrated RG Taylor vectors

For a power-free L-loop graph G with $E = 2L$ edges, the retained finite forest kernel is

$$K_G^{(d)}(x) = \sum_F (-1)^{|F|} \frac{\sum_{e \in E_R(F)} x_e}{S(x)\, U_F(x)^{d/2}}, \qquad S(x) = \sum_{e=1}^{E} x_e. \tag{716}$$

The sum is over the compatible forests of (694); U_F is the product of the first Symanzik polynomials of the forest blocks. Terms with identical denominators are combined before floating-point integration. The rational graph weights, normalized edge measure, critical-mass prescription, and signs are those of Section 25. The finite-kernel selection is fixed by exact forest cancellation. There are respectively 5, 16, 76 and 395 nonzero finite kernels at loops 3, 4, 5 and 6. This is a count after exact algebraic cancellation, not the count of all six-loop graph classes.

The dimension-dependent prefactor, including its regulator dependence, is

$$\frac{\Gamma(d/2)^L}{\Gamma(d/4)^{2L}}\, G_{L,d}(z) = \Gamma(1 + Lz/2) \left[\frac{\Gamma(d/2 + z/2)}{\Gamma(1 + z/2)}\right]^L \Gamma(d/4 + z/4)^{-2L}. \tag{717}$$

Here $G_{L,d}$ is the normalized prefactor (710) with $q = d/4$, so $G_{L,d}(0) = 1$; at $d = 2$ it is $\mathcal{G}_L$ of (699). The overall sign and factor are $n_L = 2(-1)^{L+1}\Gamma(d/2)^L/\Gamma(d/4)^{2L}$, as in (712), times the retained coupling or mass graph weight. Put

$$a_{L,j} = [z^j] \log G_{L,d}(z), \qquad s_G(x) = \frac{1}{4}\sum_e \log x_e - \frac{L}{2}\log S(x). \tag{718}$$

At each integration point the Taylor coefficients needed through z^{6-L} are generated by the finite recurrence

$$e_0(x) = 1, \qquad e_n(x) = \frac{1}{n}\sum_{j=1}^{n} j(a_{L,j} + \delta_{j1}s_G(x))e_{n-j}(x). \tag{719}$$

The scalar prefactors and their logarithmic Taylor coefficients were recomputed at 70 decimal working digits and checked against the values stored with the superseded estimator. The separately evaluated two-loop data remain inputs. Four additional independently seeded two-loop sampling checks use both ordering maps in both dimensions and a different, constant-exponent importance density. All five required Taylor coefficients agree with the exact two-loop master (199) within 1.46 estimated standard errors. Their total 8,388,608 samples are controls only and do not enter the final estimate.

For every loop the stored vector contains all required beta-function Taylor coefficients followed by all required mass-anomalous-dimension Taylor coefficients:

$$\boldsymbol{v}_L = (b_{L,0}, \ldots, b_{L,6-L}, c_{L,0}, \ldots, c_{L,6-L}). \tag{720}$$

Their covariance is not discarded. Setting $z = \epsilon$ only after forming these Taylor series, the coefficient map $\mathcal{F}_6$, which returns the coefficient C_6 of $\Delta_{\mathcal{E}}$ at dimension d, is defined by

$$0 = -\epsilon g_* + 3g_*^2 + \sum_{L=2}^{6} \widehat{b}_L(\epsilon) g_*^{L+1} + O(\epsilon^8), \tag{721}$$

$$\Delta_{\mathcal{E}} = \frac{d-\epsilon}{2} + g_* + \sum_{L=2}^{6} \widehat{c}_L(\epsilon) g_*^L + O(\epsilon^7). \tag{722}$$

Using only the values of the finite-scheme RG functions at $z = 0$ would not reproduce this calculation. An independent finite two-variable Lagrange expansion checks the fixed-point iteration used by $\mathcal{F}_6$.

26.2 Graph stratification and fixed pilot allocation

The estimator integrates every graph as a separate stratum, using a fixed sample count n_G within each independent randomized replicate. For a generic component of the integral,

$$\widehat{I}_b = \sum_G w_G \frac{1}{n_G}\sum_{i=0}^{n_G-1} f_G(X_{bGi})/p_G(X_{bGi}). \tag{723}$$

Here p_G is the normalized, unchanged importance density for graph G; there is no inverse random-graph selection probability. This removes random fluctuations in the number of times a graph is visited without changing the integral being estimated.

The counts are powers of two, at least four. A separate pilot and two validation calculations estimate the graphwise variance of the linearized C_6 contribution. These estimates determine the allocation, with a uniform floor; the allocation is frozen before either production campaign. No sample of a campaign is reused as a training pilot for that campaign, and no allocation is selected by agreement with a known physical coefficient. This target-aware allocation is a variance-reduction design, not an identity for C_6 and not a claim of mathematically optimal allocation.

Each graph stratum uses a linearly scrambled Sobol net with a separate random digital shift. Powers-of-two net sizes preserve the intended balance; the first point is retained and the sequence is not thinned. The direction matrices are generated with the documented SciPy Sobol construction, then checked against its public point generator [46]. Every baseline campaign has 64 independent randomized replicates at every loop order, as does each campaign of the 2026 productions; the refinement campaigns of Section 26.12 have 32. The same linear scrambling is shared within a replicate while different graph shifts are independently seeded; covariance is measured from the *total* replicate vector.

Campaign A uses 52-bit digital coordinates and the original sequence of conditional draws for the edge ordering. Campaign B uses 60-bit digital coordinates and the equivalent one-coordinate ordering construction below. The integer net coordinate k is mapped to $(k + 1/2)/2^b$ before taking logarithms. This avoids evaluating an endpoint singularity at zero. Finite digital resolution is itself an approximation; comparing 52 and 60 bits is an empirical control, not a proof of a bound on its bias. The production arithmetic begins at long-double precision in both campaigns; the optional guarded-double experimental build is not used.

26.3 An exact compressed map for the edge ordering

Let S be the set of edges not yet selected and let $D(S)$ be the positive dynamic-programming weight used by the importance sampler. The probability of choosing edge $j \in S$ next is

$$p(j \mid S) = \frac{D(S \setminus \{j\})}{\sum_{k \in S} D(S \setminus \{k\})}. \tag{724}$$

One may use a fresh uniform coordinate for every such draw. Alternatively, start with a single $u \in (0, 1)$, choose the interval $[P_j, P_j + p_j)$ containing u, and replace it by

$$u \longmapsto \frac{u - P_j}{p_j}. \tag{725}$$

Conditional on selecting j, the new variable is again uniform on $(0, 1)$. Induction proves that an entire ordering σ has exactly its original probability $\prod p(j \mid S)$ under this map. The gap coordinates are sampled separately, so the continuous density and the importance weight do not change.

For $E = 2L$, the original map uses $2E - 1$ digital coordinates after graph stratification: E ordering draws and $E - 1$ scale draws. The compressed map uses E coordinates: one ordering coordinate and $E - 1$ scales. At six loops the counts are thus 23 and 12. This reduces the dimension of the sampling map, *not* the physical Schwinger integral. It is not claimed to improve every integrand, and the two campaigns test its empirical behavior rather than assuming an improvement. The probability identity also passed exhaustive exact rational checks for 870 edge permutations in smaller examples.

26.4 Baseline campaigns of the previous estimators

d	Run	$L = 3$	$L = 4$	$L = 5$	$L = 6$	C_6	SE
3	A	4,456,448	9,502,720	32,768,000	67,076,096	33.590772	0.057948
3	B	4,456,448	9,502,720	32,768,000	67,076,096	33.560344	0.053265

The two three-dimensional baseline campaigns use 227,606,528 kernel samples over all four loop orders; pilot and validation samples were additional and excluded. They entered only the previous three-dimensional estimator (740), and the two corresponding two-dimensional campaigns, with 115,969,024 samples each, only the previous two-dimensional estimator. Both are superseded by the productions of Sections 26.7 and 26.8.

Counts in this table are sums over graph strata, not counts per graph. They are the actual rounded power-of-two allocations, not nominal budgets. Both campaigns use the same frozen graph allocation and importance exponents; seeds, net precision and ordering maps differ. The two campaigns are pooled with the fixed weights $1/2, 1/2$ at the level of the RG Taylor vectors. The weights were not optimized after observing the production central values or their empirical variances.

Separate campaigns have independent sampled points.

26.5 Replicate covariance and fixed-point propagation

For campaign t, loop L and $R_{L,t}$ independent randomized replicates, let $v_{L,t,r}$ be the complete vector of coupling, mass and regulator coefficients. Define

$$\begin{aligned} \mu_{L,t} &= \frac{1}{R_{L,t}} \sum_r v_{L,t,r}, \\ \widehat{\Sigma}_{L,t} &= \frac{1}{R_{L,t}(R_{L,t}-1)} \sum_r (v_{L,t,r} - \mu_{L,t})(v_{L,t,r} - \mu_{L,t})^{\mathsf{T}}. \end{aligned} \tag{726}$$

This is the estimated covariance of the mean. The unit of statistical independence is a randomized replicate, not a point within one Sobol net [47]. Independent loops have independent randomizations; within each loop all beta/mass and Taylor-order covariances are retained. The fixed weights, shared-input terms and final Jacobian propagation are given in (740)–(744) below.

No known physical coefficient is used to calibrate the estimator and no outlying replicate is discarded; the quoted central value is the direct pooled estimator. For the previous four-campaign estimators the nonlinear fixed-point map was checked by an independent finite Lagrange expansion, and a delete-one-replicate calculation diagnosed the leading finite-sample bias. These archived checks concern the inherited map and estimator, not the replicate data of the 2026 productions. A replicate bootstrap checks variance propagation. In the previous two-dimensional estimator it drew each shared retained input once per complete bootstrap sample; modeling that input by a normal draw does not prove normality or give an interval certificate. The 2026 productions have no shared retained input.

26.6 The quantity being integrated

For each loop order L, the sampled vector contains the coupling and mass Taylor coefficients through regulator order $6 - L$:

$$\boldsymbol{\mu}_L = (b_{L,0}, \dots, b_{L,6-L}, c_{L,0}, \dots, c_{L,6-L}), \qquad L = 3, 4, 5, 6. \tag{727}$$

The one-loop input is exact, and the two-loop input is the special-function evaluation of Section 25 at each dimension. The physical coefficient is evaluated as a finite polynomial

$$C_6 = \mathcal{F}_6(\boldsymbol{\mu}_3, \boldsymbol{\mu}_4, \boldsymbol{\mu}_5, \boldsymbol{\mu}_6) + \eta. \tag{728}$$

Here $\mathcal{F}_6$ solves the regulator-dependent finite-MOM fixed-point problem before setting $z = \epsilon$; η accounts only for separately retained numerical kernels. Only the previous two-dimensional estimator used such kernels, so $\eta = 0$ in both current estimators. In particular,

$$\frac{\partial C_6}{\partial (b_{6,0}, c_{6,0})} = \frac{(-1, 3)}{2187}. \tag{729}$$

One must not replace the Taylor vectors by their values at $z = 0$ and then apply the constant-coefficient MS assembly. The field-theoretic derivation of these finite vectors and their power-sector compensation is unchanged.

The finite kernel retains the pointwise forest sum (716): every numerator and denominator is evaluated on the same integration point before accumulating the graph estimate. Tropical importance sampling uses the known endpoint geometry of these parametric integrals [41]. More samples cannot in general repair a representation which postpones subtraction of divergent terms until after separate numerical integrations.

26.7 The two-dimensional production of September 2026

At $d = 2$ the current estimate comes from a single new production with two independent campaigns, a and b. It integrates the finite forest kernels (716) of all 5, 16, 76 and 395 nonzero classes at loops 3–6, generates the regulator jets by (719), and applies the fixed-point map (722) with the retained analytic power-sector compensation of Section 24. The code is an optimized, restartable reimplementation of the same calculation. Its kernel keeps the addition order of every polynomial sum and the binary128 and 300-decimal fallbacks at the unchanged thresholds 10^{-9} and 10^{-23}. In its legacy summation mode, replays of all 492 kernels against the original executable agree to a normalized difference below 9×10^{-20}, with identical promotion counts; production uses compensated summation within each task.

Both campaigns have 64 independent randomized replicates at every loop order and use the conditional-draw ordering map with $2E - 1$ digital coordinates. Campaign a uses 52-bit and campaign b 60-bit coordinates, and only campaign a uses, at $L = 6$, the pilot-frozen separable map (737). Every stratum receives the frozen per-replicate count of the refinement allocation (739), raised to at least four, multiplied by 2^{13}; the floor affects only the 28 six-loop kernels whose C_6 projection vanishes exactly. Per replicate this gives 1,107,296,256, 2,164,260,864 and 9,487,515,648 points at $L = 3, 4, 5$, and 38,076,809,216 points for the 385 nonprimitive six-loop kernels. The ten primitive six-loop kernels, which the previous estimator retained as P_{10}, are integrated in every replicate with 5,645,533,184 points per replicate, divided among them by a variance-based (Neyman) allocation frozen from archived paired records of the earlier calculation (`paired_new_a_d2_L6.json`, `paired_new_b_d2_L6.json`, `paired_prod_a_d2_L6.json` and `paired_prod_b_d2_L6.json` in `LRI_C6_D2_HPC/reference/records/`, read by `tools/primitive_allocation.py`). The two campaigns contain 7,229,621,141,504 kernel samples. Group seeds are derived by SHA-256 from a fresh base seed, the campaign, the loop and the role. No production sample was used to choose the allocation, and pilot, smoke and timing samples are excluded. Every required kernel thus has expectation weight one in every replicate, no numerical input is retained, and

$$\boxed{\widehat{C}_6(2) = \mathcal{F}_6\left(\frac{\boldsymbol{\mu}^a + \boldsymbol{\mu}^b}{2}\right), \qquad \widehat{\sigma}_{\text{common}} = 0,} \tag{730}$$

with the covariance (744) at $w_a = w_b = 1/2$.

The production ran on SDSC Expanse as 428 independent 64-thread array elements, 12,535 core-hours in total. Sample identities do not depend on node, thread count or checkpoint boundaries: two blocks of 1024 six-loop tasks run at 64 and at 16 threads gave identical rows, including 152,717 binary128 promotions. The campaigns give

Campaign	$C_6(2)$	Sampling SE
a	362.36672022	0.00097561
b	362.36622331	0.00126747
Combined	362.36647176	0.00079973

They differ by 0.31 combined standard errors. The combined standard error is about half the value 0.0015 projected from the earlier campaigns under Monte Carlo scaling, in which the variance falls as $1/n$ and the standard error as $1/\sqrt{n}$. This single comparison does not determine a convergence

rate; the replicate covariance, not the projection, is reported. The new estimator lies 0.12 previous standard errors from the previous estimator $362.3489138 \pm 0.1526445$.

The reported allowance is four sampling standard errors, 0.0032, rounded up to a multiple of 0.001, which gives $C_6(2) = 362.366 \pm 0.004$. The four-standard-error rule was written down before the production result was available; the rounding step was refined afterwards from the 0.01 first proposed to 0.001, which changes only the rounding of the allowance. An independent two-dimensional calculation through seventh order, in which the mixed local-mass and logarithmic forests are integrated explicitly instead of through the analytic compensation, gives as a by-product $C_6(2) = 362.3727 \pm 0.0086$ (sampling standard error), $+0.73$ combined standard errors from (730). It shares the graph and renormalization framework and therefore does not test that framework. The previous two-dimensional baseline and refinement campaigns, the retained groups (741) and the estimator built from them are superseded; their records are in `LRI_C6_D2_HPC/reference/` (Section 33).

26.8 The three-dimensional production of September 2026

At $d = 3$ the current estimate likewise comes from a single new production with two independent campaigns, a and b. It integrates the finite forest kernels (716) of all 5, 16, 76 and 395 nonzero classes at loops 3–6, the ten primitive six-loop kernels included, generates the regulator jets by (719), and applies the fixed-point map (722) with the retained analytic power-sector compensation of Section 24. The code is the three-dimensional counterpart of the one in Section 26.7, with the binary128 and 300-decimal fallbacks at the same thresholds 10^{-9} and 10^{-23}. In its legacy summation mode it replays the archived three-dimensional refinement streams, 385,744,896 samples on their archived nets and seeds, with fallback counts equal to the archive in every stream, and reproduces the refinement-only estimate 33.56084035 bit for bit; production uses compensated summation within each task. Section 25.4 explains why the last fallback has 300 decimal digits.

Both campaigns have 64 independent randomized replicates at every loop order and use the conditional-draw ordering map with $2E - 1$ digital coordinates and the baseline importance map. Campaign a uses 52-bit and campaign b 60-bit coordinates. Every stratum receives the frozen per-replicate count of the three-dimensional refinement allocation (739), multiplied by 2^{13}. Per replicate this gives 1,241,513,984, 2,499,805,184, 9,743,368,192 and 35,891,576,832 points at $L = 3, 4, 5, 6$, and the two campaigns contain 6,320,161,816,576 kernel samples. Group seeds are derived by SHA-256 from a fresh base seed, the campaign, the loop and the role. No production sample was used to choose the allocation, and the samples of the thread-identity check below are excluded. Every required kernel has expectation weight one in every replicate, no numerical input is retained, and

$$\boxed{\widehat{C}_6(3) = \mathcal{F}_6\left(\frac{\boldsymbol{\mu}^a + \boldsymbol{\mu}^b}{2}\right), \qquad \widehat{\sigma}_{\text{common}} = 0,} \tag{731}$$

with the covariance (744) at $w_a = w_b = 1/2$.

The production ran on SDSC Expanse on 29 September 2026 as 458 independent 64-thread array elements. Sample identities do not depend on node, thread count or checkpoint boundaries: two blocks of 1024 six-loop tasks run at 64 and at 16 threads gave identical rows, including 141,994 binary128 promotions. The campaigns give

Campaign	$C_6(3)$	Sampling SE
a	33.57562307	0.00015062
b	33.57523021	0.00018311
Combined	33.57542664	0.00011855

They differ by 1.66 combined standard errors. The combined standard error is 0.54 of the value

0.00022 projected from the refinement campaigns by plain Monte Carlo scaling, in which the standard error falls as $1/\sqrt{n}$; the replicate covariance, not the projection, is reported. The estimator lies 0.30 standard errors of the superseded estimator (740) from its value 33.5681968 ± 0.0241840, and 0.52 refinement standard errors from the refinement-only estimate.

The reported allowance is four sampling standard errors, 0.00047, rounded up to a multiple of 0.0001, which gives $C_6(3) = 33.5754 \pm 0.0005$. The estimator, the four-standard-error rule and the consistency checks were written down and confirmed before any production result was available; the rounding step was refined afterwards from the 0.001 first registered to 0.0001, which changes only the rounding of the allowance. An independent three-dimensional calculation through seventh order (Section 29), in which the power-containing sector is integrated explicitly instead of through the analytic compensation, gives as a by-product $C_6(3) = 33.57387 \pm 0.00198$ (sampling standard error), -0.79 combined standard errors from the estimator. It shares the graph and renormalization framework and therefore does not test that framework. The previous three-dimensional baseline and refinement campaigns and the estimator (740) built from them are superseded. Their records are in the reference tree that the two- and three-dimensional sixth-order packages share and that is shipped once, in `LRI_C6_D2_HPC/reference/` (`records/`, `baseline_previous/records/` and `verification/method_precision_results.json`); the replay of Section 33 recomputes them from there.

26.9 Result and interpretation of the errors

The current estimates are

$$\boxed{\begin{aligned} C_6(2) &= 362.366 \pm 0.004, \\ C_6(3) &= 33.5754 \pm 0.0005. \end{aligned}} \tag{732}$$

The allowances are four estimated sampling standard errors rounded upward, to a multiple of 0.001 at $d = 2$ and of 0.0001 at $d = 3$. They are empirical numerical allowances, not rigorous enclosures, guaranteed confidence intervals, or truncation errors for the physical ϵ expansion. The underlying estimator digits are recorded for reproducibility, not as independently certified physical digits.

d	Estimator	Sampling standard error
2	362.36647176	0.00079973
3	33.57542664	0.00011855

At $d = 2$ and $d = 3$ the estimates are the 2026 productions of Sections 26.7 and 26.8 alone. Fresh pilot and method-validation samples are excluded. The lower-loop allocations were planned in part from the variances measured in the superseded baseline campaigns (Section 26.4); since the allocation is fixed and the production samples are independently randomized, this does not change the expectation of the estimate. No known C_3, C_4, C_5 is used as a fitted control or imposed as a constraint.

26.10 A faster evaluator with the same integral

Each monomial in a loop-L forest polynomial has exactly L distinct edge factors. Associate a bit mask m to a monomial and let j be one of its set bits. Precompute a directed acyclic product graph obeying

$$p_0 = 1, \qquad p_m = x_j p_{m\setminus\{j\}}. \tag{733}$$

All forest terms share these products. This replaces repeated $\exp(\sum_{e\in m} \log x_e)$ evaluations by one exponential per edge and a shared set of multiplications. For very small coordinates the

same common-monomial scaling as the original implementation is used, with h the largest of the monomial logarithms $\sum_{e\in m}\log x_e$ in the forest sum, as in Section 25.4:

$$p_m = \prod_{e\in m}\exp\left(\log x_e - \frac{h}{L}\right) = \exp\left(\sum_{e\in m}\log x_e - h\right). \tag{734}$$

The common factor is restored outside the forest sum. The direct unscaled path is used only in a range checked against the exponent range of extended precision. The original binary128 and 300-decimal fallbacks for severe cancellation are retained. The numerical kernels are not evaluated at lower precision to obtain the speedup.

An identical-point comparison used 1,095,680 six-loop evaluations for each dimension, with identical seeds, digital matrices, allocations, and importance distributions:

d	Original evaluator (s)	Shared products (s)	Speedup
2	7.88237	2.77982	2.836
3	7.74461	2.69407	2.875

These are elapsed times on one four-CPU machine, not hardware-independent throughput claims. The maximum normalized per-graph replicate difference was below 9.0×10^{-13}. The improvement is in cost per evaluation, not in the variance of an individual sample. Independently constructed Kirchhoff cofactor integrands, built from cofactor determinants rather than from the sampler's list of spanning-tree monomials, supplied 1,038 additional checks over both $d=2$ and $d=3$ and all finite kernels, with reference values at 70 digits at ordinary points and at 650 digits at extreme ones, including endpoint coordinates down to 10^{-130}; the largest normalized discrepancy was 6.88×10^{-18}. Neither comparison is an interval certificate for the full integral.

26.11 Sampling methods tested at matched budgets

The baseline sampler is the one of Section 26.2: independently randomized Sobol nets inside each fixed graph stratum, with power-of-two counts chosen separately for every stratum. The uncertainty is always estimated between independent randomized nets, never from the dependent points within one net [46, 47]. The benchmark target is the six-loop influence (729): with every six-loop finite kernel sampled, let $y_{G,r} = (3c_{6,0} - b_{6,0})_{G,r}/2187$ be the contribution of graph G in replicate r. Its variance is estimated by

$$\widehat{V}_{\text{bench}} = \frac{1}{R}\sum_{G}\widehat{V}_G, \qquad \widehat{V}_G = \frac{1}{R-1}\sum_{r=1}^{R}\left(y_{G,r} - \bar{y}_G\right)^2, \tag{735}$$

the sum of the graph-stratum replicate variances divided by the number R of replicates. It contains no covariance between strata; the production analysis instead keeps the full covariance of the total replicate vectors, (726).

Each pilot comparison used $R=16$ randomizations and 1,095,680 six-loop kernel evaluations. The per-graph counts are powers of two, at least 64, the same at both dimensions, and were fixed in advance from the graph probabilities of an earlier, independent pilot, not from this benchmark (`LRI_C6_D2_HPC/reference/code/pilot.py`). Antithetic sampling used half as many points with two kernel evaluations each. The table gives $\widehat{V}_{\text{bench}}$ relative to the baseline at the same number of kernel evaluations; smaller is better.

Method	$d = 2$	$d = 3$
Randomized Sobol nets (baseline)	1.000	1.000
Independent pseudo-random Monte Carlo	1.899	1.499
Radial coordinates placed first	1.105	1.126
One-coordinate edge-order encoding	1.083	1.055
Endpoint rate 0.6κ	1.667	2.301
Endpoint rate 0.5κ	4.194	6.194
Endpoint rate 1.0κ	4.452	1.797
Radial antithetic pairs	1.584	1.629

The variants are those of `code/experiment.cpp` in the same package. In the baseline layout the E ordering draws and the $E-1$ scale draws of the conditional-draw map alternate in the $2E-1$ net coordinates. "Radial coordinates placed first" puts the $E-1$ scale (radial) coordinates in the first net dimensions and the ordering draws after them. The one-coordinate encoding is the compressed map (725), with E coordinates. The antithetic variant evaluates each point together with its reflection $v \mapsto 1-v$ in the radial coordinates only, keeping the ordering coordinates, and averages the pair. The endpoint rates are defined in the next paragraph. The endpoint-rate and antithetic variants were run in the radial-first layout, so their ratios include the layout effect of the third row; for pseudo-random sampling the layout is immaterial.

The pilot differences are empirical and subject to finite-replicate uncertainty; they are not theorems about the methods. Coordinate reordering, and compression of all order choices into one uniform coordinate, gave no convincing improvement. The compressed map was used only by baseline campaign B (Section 26.2); both refinement campaigns, like both campaigns of the 2026 productions, use the conditional-draw map. Independent pseudo-random sampling was worse than randomized Sobol sampling.

Why a smoother endpoint map was not the winner. Write $\kappa(S)$ for the nominal endpoint power of the sector variable that closes the edge subset S. The proposal exponents $\alpha(S)$ of the baseline sampler, used as in (705), correspond to the nominal endpoint rate $\alpha(S) = 0.8\,\kappa(S)$. The rows "endpoint rate" multiply all of them by 0.75, 0.625 and 1.25, which gives $\alpha = 0.6\kappa$, 0.5κ and 1.0κ; the dynamic-programming normalization is recomputed with the changed exponents and the exact reciprocal proposal weight is applied, so the target integral is unchanged. An illustrative endpoint integral is

$$\int_0^1 t^{\kappa-1} f(t)\,dt = \frac{1}{\alpha}\int_0^1 u^{\kappa/\alpha-1} f(u^{1/\alpha})\,du, \qquad t = u^{1/\alpha}; \tag{736}$$

sampling t with the density $\alpha t^{\alpha-1}$ is this substitution with uniform u. Reducing α makes the endpoint factor $u^{\kappa/\alpha-1}$ smoother, but it simultaneously changes the normalized sampling distribution in every radial coordinate and can move variance into a narrow central region. The variants 0.6κ and 0.5κ lost to the baseline in both dimensions, and so did 1.0κ. Smoothness alone is not a sufficient optimization criterion for these finite, signed, high-dimensional kernels.

A pilot-frozen separable importance map. A separate training stream estimates the conditional second moment in 32 bins for each radial coordinate. Its square root is smoothed, clipped, and blended equally with a uniform density. Inverting this positive piecewise-linear cumulative distribution defines $u = T(v)$. The exact change of measure is

$$\int_{[0,1]^s} f(u)\,du = \int_{[0,1]^s} f(T(v)) \prod_{j=1}^{s} T_j'(v_j)\,dv. \tag{737}$$

All Jacobian factors are included in the kernel. This is a small VEGAS-inspired separable transport, not an implementation or benchmark of the full VEGAS+ algorithm. VEGAS+ additionally

uses adaptive stratification and has documented advantages for some multi-peak and diagonal structures [49].

Fresh validation used 32 randomizations and 4,382,720 evaluations per map. Relative variance per evaluation was

Frozen transport	$d = 2$	$d = 3$
None	1.0000	1.0000
Three trained axes	0.9533	0.9871
All eleven trained axes	0.8985	0.9905

Thus the all-axis map supplied a modest approximately 10% variance reduction at $d = 2$, but no material improvement at $d = 3$. Among the refinement campaigns only campaign A at $d = 2, L = 6$ uses this map; refinement campaign B keeps the baseline map. In the 2026 production only campaign a uses it (Section 26.7). Both three-dimensional campaigns use the baseline map. No map is adapted using its production samples. The mathematical map has endpoints $0, 1$. Delivered decimal maps have endpoint roundoff below 7.8×10^{-16}; a same-point replay with endpoints snapped exactly to $0, 1$ changed the six-loop mean by -2.95×10^{-13}. This is a floating-point consistency check, not a bound on all digital-discretization effects.

The trained maps are built by `code/train_grid.py` of the same package. At six loops each graph has $E - 1 = 11$ radial coordinates. On every axis the training histogram of the second moment is smoothed with the five-bin filter $(1, 2, 3, 2, 1)/9$; its square root is normalized to mean one on that axis and clipped to $[0.15, 4]$, and the density is half this clipped score and half uniform. "Three trained axes" transports, for each graph, the three radial coordinates whose normalized score varies most across the 32 bins and leaves the others uniform; "all eleven" transports every radial coordinate.

Conditional stratification and linear controls. Exact stratification of the first edge-order choice was also tested. The first edge j is chosen with the probability $p(j \mid E)$ of (724); in the stratified variant each choice receives its own randomized net, with a count proportional to $p(j \mid E)$, rounded to a power of two and at least four, and its estimate enters with the weight $p(j \mid E)$. On fresh validation streams with 32 randomizations, its variance times the number of kernel evaluations, relative to the same product for the matched unstratified validation run, was 1.043 at $d = 2$ and 1.062 at $d = 3$. It was not adopted.

Separately, let Σ be the covariance of the estimates $(\widehat{C}_3, \widehat{C}_4, \widehat{C}_5, \widehat{C}_6)$ of the baseline campaigns, and Σ_{low} its 3×3 block for C_3, C_4, C_5. The covariance-optimal linear control by the known C_3, C_4, C_5 would reduce the standard error of $\widehat{C}_6$ by the factor

$$\frac{\widehat{\sigma}_6^{\mathrm{lin}}}{\widehat{\sigma}_6} = \Big(1 - \sum_{k,l=3}^{5} \Sigma_{6k} \big(\Sigma_{\mathrm{low}}^{-1}\big)_{kl} \Sigma_{l6} / \Sigma_{66}\Big)^{1/2}, \tag{738}$$

which is 0.985 at $d = 2$ and 0.962 at $d = 3$. (Formula (738) reproduces both ratios from the covariance stored for the baseline-campaign estimator, field `previous` of `LRI_C6_D2_HPC/reference/verification/method_precision_results.json`.) This is a diagnostic within that particular linear span, not a universal bound, and those controls were neither fitted nor applied. They cannot directly remove six-loop sampling variation, since C_3, C_4, C_5 contain no six-loop input.

26.12 Refinement campaigns and cost-aware allocation

Let a_g/n_g model the variance of the target contribution from graph g, and let c_g denote the measured cost of one sample. Minimizing this model at fixed cost gives

$$\min_{n_g>0} \sum_g \frac{a_g}{n_g}, \qquad \sum_g c_g n_g = T \quad \Longrightarrow \quad n_g = \lambda \sqrt{\frac{a_g}{c_g}}. \tag{739}$$

The derivative of C_6, rather than raw beta-function variance, defines the pilot target. New six-loop allocations use the independent method-validation streams, and lower-loop allocations use the variances measured in the baseline campaigns of Section 26.4; no sample of a refinement campaign sets its own allocation. Each count is rounded to a power of two and frozen before production integration. This is an MC-variance-model allocation used for RQMC, not a claim that the RQMC variance is exactly proportional to $1/n_g$ or that the allocation is globally optimal.

The baseline samples were used twice by the previous estimators: to plan the lower-loop refinement allocations, and as two of the four pooled campaigns of Section 26.13. This does not change the expectation of the pooled jet means. The allocation only sets the variance of the independently randomized refinement samples, and the combination weights, 1/4 for each campaign and, at $d = 2$, 1/2 for the 66 nonprimitive kernels added separately, were fixed before the refinement production (`LRI_C6_D2_HPC/reference/inputs/frozen_production_plan.json`). They are neither inverse-variance weights nor chosen from any result.

Each refinement campaign has 32 independently randomized replicates at each loop order. The per-replicate point counts are

d	$L = 3$	$L = 4$	$L = 5$	$L = 6$
2	135,168	264,192	1,158,144	4,647,936
3	151,552	305,152	1,189,376	4,381,184

The refinement campaigns comprise 385,744,896 kernel evaluations at $d = 3$ and 397,148,160 at $d = 2$. Both are superseded; each row of the table is the frozen allocation that the corresponding 2026 production multiplies by 2^{13} (Sections 26.7 and 26.8). Of the refinement-stage runs, only these two campaigns entered the previous estimators, together with the two baseline campaigns. The 34,144,256 two-dimensional evaluations of the dimension-coupled control experiment of Section 26.14 are counted separately; that experiment entered neither the previous nor the current estimators. Pilot, method-validation, speed-test, and endpoint-replay samples are not included. Campaigns A and B use different seeds and 52-bit and 60-bit digital coordinates, respectively. The source and input hashes were frozen before production and rechecked afterward.

The three-dimensional refinement-campaign determinations, before combination with the baseline campaigns, were

	$C_6(3)$	Sampling SE
Refinement A	33.59417512	0.040029
Refinement B	33.52750475	0.039502
Refinement only	33.56084035	0.028119

At $d = 3$ the campaigns share no retained input, so the standard error of the difference of two estimates is the quadrature sum of their standard errors. Refinement A minus refinement B is +1.19 such standard errors, and the refinement-only value minus the estimator of the two baseline campaigns alone, 33.57555796 ± 0.039354, is -0.30 such standard errors (fields `fresh_campaign_pull` and `fresh_minus_old_pull` of the record `verification/method_precision_results.json` of

the earlier calculation). Pooling all four campaigns in the previous estimator (740) lowered the three-dimensional sampling standard error from 0.039354, for the baseline campaigns alone, to 0.024184, a factor 1.63, which also reflects the larger number of production samples. The 2026 code reproduces the refinement-only value bit for bit by replaying the archived streams (Section 26.8).

At $d = 2$ the corresponding determinations were

	$C_6(2)$	Sampling SE
Refinement A	362.41502790	0.275726
Refinement B	362.46528830	0.273306
Refinement only	362.44015668	0.196310

The refinement campaigns integrated the 66 nonprimitive kernels of the groups (741) but not the ten primitive ones. Each value in this table therefore adds P_{10} once, and each standard error contains $\widehat{\sigma}_P$ once. Because P_{10} is common to the three rows, its variance is removed from campaign-difference tests but not from the uncertainty of each complete value. With it removed, campaigns A and B differ by -0.13 standard errors of the difference, $(\widehat{\sigma}_A^2 + \widehat{\sigma}_B^2 - 2\widehat{\sigma}_P^2)^{1/2}$. The refinement-only value lies $+0.62$ standard errors of the difference, again with the P_{10} variance removed, from the estimator of the baseline campaigns alone, $362.25767208 \pm 0.226352$, which contains P_{10} too (fields `fresh_campaign_pull` and `fresh_minus_old_pull` of `LRI_C6_D2_HPC/reference/verification/method_precision_results.json`). Pooling the four campaigns lowered the sampling standard error of the two-dimensional estimator from 0.226352, for the baseline campaigns alone, to 0.152644, a factor 1.48. This gain also uses more production samples; it is not attributed to the change of sampling algorithm alone. The speedup of the shared-product evaluator (733) lowers the cost per evaluation, a different quantity from the variance per evaluation, and the two are not combined.

26.13 Campaign pooling, retained inputs and covariance

The previous three-dimensional estimator pooled the four campaign jet means, from two baseline and two refinement campaigns, with fixed weights 1/4; every vector is complete:

$$\boxed{\widehat{C}_6(3) = \mathcal{F}_6\left(\frac{\boldsymbol{\mu}^{\mathrm{old,A}} + \boldsymbol{\mu}^{\mathrm{old,B}} + \boldsymbol{\mu}^{\mathrm{new,A}} + \boldsymbol{\mu}^{\mathrm{new,B}}}{4}\right).} \tag{740}$$

There is no separately retained numerical kernel at $d = 3$.

The previous two-dimensional estimator pooled the same four campaign generations, but replaced the 76 six-loop finite kernels of Section 23 by three independently measured groups. In the finite-MOM normalization these were

$$\begin{aligned} P_{10} &= 718.7129769223133, & \widehat{\sigma}_P &= 0.0414043282783, \\ S_{26} &= -188.2890632768604, & \widehat{\sigma}_{26} &= 0.1002179257784, \\ S_{40} &= -2.129275292198507, & \widehat{\sigma}_{40} &= 0.0843482336693. \end{aligned} \tag{741}$$

These are not the separately MS-converted partial sectors Θ_{26}, Θ_{40} of Section 23. Its baseline campaigns omitted all 76 kernels and its refinement campaigns only the ten primitive ones, so it added $P_{10} + \frac{1}{2}S_{66}$, with $S_{66} = S_{26} + S_{40}$, and the common variance

$$\widehat{\sigma}^2_{\mathrm{common}} = \widehat{\sigma}^2_P + \frac{1}{4}(\widehat{\sigma}^2_{26} + \widehat{\sigma}^2_{40}). \tag{742}$$

The 2026 productions (730) and (731) integrate every kernel in every replicate. Both current estimators therefore have $\widehat{\sigma}_{\mathrm{common}} = 0$, and no retained input is averaged away as though independent.

The previous two-dimensional estimator. With the baseline campaigns labeled old and the refinement campaigns new, it was

$$\widehat{C}_6^{\mathrm{prev}}(2) = \mathcal{F}_6\left(\frac{\boldsymbol{\mu}^{\mathrm{old,A}} + \boldsymbol{\mu}^{\mathrm{old,B}} + \boldsymbol{\mu}^{\mathrm{new,A}} + \boldsymbol{\mu}^{\mathrm{new,B}}}{4}\right) + P_{10} + \frac{1}{2}S_{66}. \tag{743}$$

The six-loop vectors of the two baseline campaigns omit all 76 kernels of (741), those of the two refinement campaigns omit only the ten primitive ones, and all lower-loop vectors are complete. Inside the average the 66 nonprimitive kernels therefore have total weight $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ and the ten primitive kernels weight zero, so adding $P_{10} + \frac{1}{2}S_{66}$ gives every required contribution total expectation weight one. The groups P_{10}, S_{26}, S_{40} were measured independently, so the added term has variance $\widehat{\sigma}_P^2 + \frac{1}{4}(\widehat{\sigma}_{26}^2 + \widehat{\sigma}_{40}^2)$, which is (742), and it enters (744) once. The baseline records also contain four-point diagnostic evaluations of the 76 omitted kernels; they are dropped and never enter as measurements. The estimators of the refinement campaigns alone add P_{10} once, with variance $\widehat{\sigma}_P^2$, and that of the baseline campaigns alone adds $P_{10} + S_{66}$. The last sum is the replacement used in run C of Section 25.4: $P_{10} + S_{26} + S_{40} = 528.2946383532544$ and $(\widehat{\sigma}_P^2 + \widehat{\sigma}_{26}^2 + \widehat{\sigma}_{40}^2)^{1/2} = 0.13737749$ agree with K_{76}^{MOM} and its standard error in (706) to all printed digits.

The sampling standard error of (743), propagated with (744), was 0.152644, made of 0.005635, 0.037539, 0.080576 and 0.096763 from loops 3 to 6 and 0.077485 from the common term, combined in quadrature. That of the previous three-dimensional estimator (740) was 0.024184, made of 0.001145, 0.005894, 0.012458 and 0.019839 from loops 3 to 6, with no common term, combined in quadrature (`verification/method_precision_results.json` of the earlier calculation).

For each campaign and loop, let $\widehat{\Sigma}_{L,t}$ be the covariance of its mean estimated from independent randomized replicates. With $J_L = \partial\mathcal{F}_6/\partial\boldsymbol{\mu}_L$, the propagated sampling variance is

$$\widehat{\mathrm{Var}}(\widehat{C}_6) = \sum_{L,t} w_t^2 J_L \widehat{\Sigma}_{L,t} J_L^{\mathsf{T}} + \widehat{\sigma}^2_{\mathrm{common}}. \tag{744}$$

All coupling/mass and Taylor-order covariance is retained. The error budget, in units of C_6, is

d	$L=3$	$L=4$	$L=5$	$L=6$	Common	Total
2	0.000005	0.000113	0.000365	0.000702	0.000000	0.000800
3	0.000001	0.000021	0.000058	0.000101	0.000000	0.000119

Both rows are the 2026 productions, and the common term is zero in both. Contributions in this table combine in quadrature, not by addition of standard errors.

26.14 A dimension-coupled primitive control experiment

This control experiment belongs to the earlier calculation. It concerns only the previous three-dimensional estimator (740), and it entered neither that estimator nor the 2026 production (731).

The Sobol matrices and graph shifts of a three-dimensional campaign can also evaluate the ten primitive six-loop kernels at $d = 2$, on the same randomized net points; each dimension keeps its own importance exponents. This uses common random numbers; it assumes no interpolation in d and no relation between the two physical coefficients. Let $\widehat{P}^{\mathrm{pair}}(2)$ be this paired estimate, normalized as P_{10} in (741), and $\widehat{P}^{\mathrm{ref}}(2)$ an independent estimate of the same quantity with standard error σ_{ref}. For a fixed coefficient θ_{cv},

$$\widehat{C}_6^{\mathrm{cv}}(3) = \widehat{C}_6(3) - \theta_{\mathrm{cv}}(\widehat{P}^{\mathrm{pair}}(2) - \widehat{P}^{\mathrm{ref}}(2)). \tag{745}$$

The difference in parentheses has expectation zero. The paired estimate is correlated with the sampled three-dimensional primitive block; the reference is independent of both and not exact, so

its variance enters once:

$$\operatorname{Var}\widehat{C}_6^{\mathrm{cv}}(3) = \operatorname{Var}\widehat{C}_6(3) + \theta_{\mathrm{cv}}^2(\operatorname{Var}\widehat{P}^{\mathrm{pair}}(2) + \sigma_{\mathrm{ref}}^2) - 2\theta_{\mathrm{cv}}\operatorname{Cov}(\widehat{C}_6(3), \widehat{P}^{\mathrm{pair}}(2)),$$
$$\theta_{\mathrm{cv}}^{\mathrm{opt}} = \frac{\operatorname{Cov}(\widehat{C}_6(3), \widehat{P}^{\mathrm{pair}}(2))}{\operatorname{Var}\widehat{P}^{\mathrm{pair}}(2) + \sigma_{\mathrm{ref}}^2}. \tag{746}$$

The optimum assumes a known population covariance; it does not justify fitting production coefficients without error control.

An independent paired pilot fitted one coefficient for each of the groups P_{10}, S_{26} and S_{40} of (741), the primitive, one-subgraph and two-subgraph blocks: the least-squares slope of the three-dimensional block on its two-dimensional partner over the pilot replicates, that is, the plain ratio of covariance to variance, without the term σ_{ref}^2 of (746). On independent validation streams with 32 randomizations the residual-variance ratios were 0.234, 0.743 and 0.597; these ratios also leave out the reference term. The control doubles the kernel evaluations of a block, so the chosen equal-cost rule kept a block only if twice its ratio was below one (`inputs/cross_primitive_production_plan.json`). Only the primitive block passed, with the frozen values

$$\theta_{\mathrm{cv}} = 0.09955611941934894, \qquad \widehat{P}^{\mathrm{ref}}(2) = P_{10}, \qquad \sigma_{\mathrm{ref}} = \widehat{\sigma}_P. \tag{747}$$

No production sample was used to fit θ_{cv} or to choose the block. Evaluating the primitive block at $d = 2$ on the stored net points of all four campaigns of (740) added 34,144,256 two-dimensional evaluations and no three-dimensional sample, with identical seeds and digital matrices verified (`verification/cross_production_manifest.json`). An independent three-dimensional replay of 894,464 primitive evaluations, the ten primitive graphs in the first two replicates of baseline campaign B and of refinement campaign A, reproduced the stored graph-replicate values with maximum difference zero (`verification/pairing_checks.json`). With the campaign weights 1/4 of (740) the controlled result is

$$\widehat{C}_6^{\mathrm{cv}}(3) = 33.5681256836, \qquad \widehat{\sigma}_{\mathrm{cv}} = 0.0243339897 \tag{748}$$

(`verification/controlled_precision_result.json`), against 33.5681968 ± 0.0241840 for the plain estimator.

The validation variance of the primitive block fell by a factor of about 4.3, but this block carries only about 5.8% of the plain variance. In the production its sampling error fell from 0.00584436 to 0.00325730; the reference adds $\theta_{\mathrm{cv}}\widehat{\sigma}_P = 0.00412205$, giving 0.00525370 in all (`verification/cross_control_budget.json`). In variance, the control lowered the sampling part of the complete coefficient by 9.7×10^{-6} (its six-loop error fell from 0.019839 to 0.019593), while the reference added $(\theta_{\mathrm{cv}}\widehat{\sigma}_P)^2 = 1.70 \times 10^{-5}$. Because the reference term exceeds the gain, the controlled error is slightly larger than the plain 0.02418404; this small difference is not evidence of a general disadvantage of controls. With the reference term included, the optimum of (746) estimated after the production is 0.0579 (same record), a diagnostic that was not used. The experiment shows that reducing the variance of a strongly correlated subintegral need not reduce the uncertainty of the complete coefficient, and that an uncertain auxiliary value cannot be treated as an exact control mean. The plain estimator (740) remained the previous three-dimensional result; the controlled value served only as a consistency check and a method benchmark.

26.15 Final checks and evidence limits

The same uncalibrated samples give the lower-order discrepancy checks

d	C_3 pull	C_4 pull	C_5 pull
2	-0.97	-1.54	-1.07
3	-1.69	-1.09	+0.92

Both rows are the 2026 productions. The reference special-function values are checks only; their numerical values are not refitted or replaced by the Monte Carlo estimates. For the previous estimators, an archived independent finite two-variable Lagrange expansion agreed with the fixed-point iteration to 7×10^{-13} at $d = 2$ and 5×10^{-14} at $d = 3$, and the delete-one-replicate bias estimates were below 3.4×10^{-7} and 6.5×10^{-8} in C_6 units; these check the inherited map $\mathcal{F}_6$, not the 2026 replicate data. Bootstrap resampling of the independent replicate vectors gives sampling errors 0.00080124 at $d = 2$ and 0.00011534 at $d = 3$ (4096 resamples each). The bootstrap is a diagnostic, not an interval certification. The bootstrap means differ from the direct estimates by -1.3×10^{-5} at $d = 2$ and $+2.1 \times 10^{-8}$ at $d = 3$.

For the previous two-dimensional estimator (743), the same uncalibrated samples gave the pulls $+0.09$, $+0.50$ and $+0.60$ for C_3, C_4 and C_5. Its replicate bootstrap, which draws the common inputs once per resample according to (742), gave the sampling error 0.14987068 from 4096 resamples, against the propagated 0.15264446.

For the previous three-dimensional estimator (740) the pulls were $+0.44$, $+0.80$ and -0.21 for C_3, C_4 and C_5, and its replicate bootstrap gave the sampling error 0.02376720 from 4096 resamples, against the propagated 0.02418404 (`verification/method_precision_results.json` and `verification/extra_checks.json` of the earlier calculation).

Verification scope. The final numbers retain the analytic power-sector compensation and the same graph/renormalization framework; narrower sampling errors do not bound shared errors in that framework. The reference C_3–C_5 values were used for comparisons only, not imposed as constraints. The reported results are $C_6(2) = 362.366 \pm 0.004$ and $C_6(3) = 33.5754 \pm 0.0005$; all earlier, less precise full-coefficient estimates and partial sums are superseded.

The method comparison of Sections 26.11 and 26.16 does not bound a shared error in graph generation, renormalization or analytic continuation, and it did not rerun the analytic power-sector compensation. Faster arithmetic and smaller sampling errors are neither an independent field-theoretic validation nor rigorous error enclosures.

26.16 Method selection and methods not adopted

Best supported choice for these integrands. The strongest tested practical combination is local finite forest subtraction (716), tropical importance sampling (705), graph-stratified randomized Sobol nets (Section 26.2), pilot-frozen cost-aware allocation (739), and the shared polynomial products (733) with monitored high-precision fallback. A separately trained transport (737) is only modestly useful at $d = 2$ and unnecessary at $d = 3$ in these tests. The runtime gain and the variance gains are different quantities and are not conflated, and the total precision gain of the refinement campaigns also comes from more production samples (Section 26.12).

Methods not benchmarked here. Rank-one shifted lattice rules with smooth inverse-CDF importance fits are an established alternative for multi-loop integrals, including CPU and CUDA implementations interfaced to pySecDec [50]. That implementation was studied but not benchmarked, so it is not declared better or worse than the present Sobol method. Global tropical sampling across primitive graphs has recently reached much higher primitive loop orders [51]; those positive primitive examples do not by themselves establish its performance for the signed, regulator-dependent renormalized forest combinations of this calculation.

Conditional preintegration. A promising next structural improvement is conditional preintegration: a subset of the coordinates is integrated exactly and only the rest, u_s, is sampled. If Y is

an unbiased single-point estimator, the law of total variance,

$$\mathrm{Var}(Y) = \mathbb{E}\big[\mathrm{Var}(Y \mid u_{\mathrm{s}})\big] + \mathrm{Var}\big(\mathbb{E}[Y \mid u_{\mathrm{s}}]\big), \tag{749}$$

has a nonnegative first term, so

$$\mathrm{Var}\big(\mathbb{E}[Y \mid u_{\mathrm{s}}]\big) \le \mathrm{Var}(Y). \tag{750}$$

This is useful computationally only when the cost of the exact inner integration and any deterministic quadrature error are controlled. No integral was replaced by an unchecked low-order quadrature, and no favorable variance reduction is assumed. The first-edge stratification of Section 26.11, which lost at equal cost, shows why a conditionalization has to be benchmarked at equal cost.

27 Full seventh order in one dimension: coordinate forests and binary-cluster integration

27.1 Result, normalization, and scope

This section continues the numerical calculation to seventh order at $d = 1$, with the unchanged convention

$$\epsilon = 2s - 1, \qquad \Delta_{\mathcal{E}} = \frac{1}{2} - \frac{\epsilon}{6} + \sum_{n=2}^{7} C_n(1)\epsilon^n + O(\epsilon^8). \tag{751}$$

The full graph-based numerical estimate is

$$\boxed{C_7(1) = -322550, \qquad \text{pre-registered empirical allowance: } \pm 41.} \tag{752}$$

The unrounded estimator is -322549.889, with propagated sampling standard error 10.088; a bootstrap over the replicates of every stream gives 10.111. The estimator uses the radial quadrature through degree six. The allowance is the output of a policy fixed before production: the largest of four sampling errors, twice the last refinement plus four of its errors, twice the degree six to seven check step plus four of its errors, and the half-split displacement, rounded up to two significant figures. It is not a rigorous enclosure, a probability guarantee, or an estimate of the error of the truncated physical ϵ series.

The policy value is not a demonstrated total accuracy. It takes the largest of its candidates rather than combining the estimated truncation with the sampling error. The measured degree six to seven step is $+10.40 \pm 0.43$, and a geometric tail beyond degree seven, extrapolated with the last step ratio 0.600, adds $+15.59$. Treating that tail as uncertain by 100% and neglecting the covariance between the estimate and the check level, the truncation-aware estimate is

$$C_7(1) \approx -322524 \pm 38, \tag{753}$$

with the allowance taken as twice the combined error $10.09 \oplus 0.43 \oplus 15.59 = 18.6$. This estimate is post hoc and depends on the tail model, whose ratio at seventh order is larger than at fifth and sixth order (Section 27.9). The intervals of (752) and (753) overlap. Establishing the total error at this level would require radial data at degree eight or an independently validated tail model. The long decimal strings in the machine-readable files specify the estimator, not certified physical digits. In particular, no independent determination of the common graph and renormalization framework is claimed.

Both logarithmically and power-subdivergent diagrams are included. No seventh-order power correction is assigned zero, and no Padé, ratio, large-order, or interpolation estimate is used to fill a missing graph class. The 44 seventh-order primitive classes are integrated directly in a separate campaign, giving -696179.472 ± 0.867. That campaign replaces the primitive contribution

-696147.446 ± 25.368 of the first, smaller seventh-order calculation (23 September 2026, $C_7(1) = -323405 \pm 5862$ sampling standard error; record `GFF/results/C7_1/v1_46batches/archive_original.json`), which it confirms within 1.26 combined standard errors. The primitive number alone is not the physical C_7.

27.2 Complete inventory and exact algebraic coverage

The eight-vertex enumeration was rebuilt from integer edge multiplicities. There are 3986 vertex-isomorphism classes, representing 125894580 labeled 1PI graphs. Of these, 1091 contain an unmarked power-divergent two-point subgraph. The seventh-order $3c_7 - b_7$ target has nonzero weight for 845 of these classes. The primitive sector has 44 classes.

For a graph with vertex automorphism group $\mathrm{Aut}(G)$, internal multiplicities m_{ij}, and external half-edge counts $e_i = 4 - \sum_j m_{ij}$, the class weights are

$$w_A(G) = \frac{24}{|\,\mathrm{Aut}(G)|\prod_i e_i! \prod_{i<j} m_{ij}!}, \qquad w_B(G) = \frac{w_A(G)}{12} \sum_i e_i(e_i - 1). \tag{754}$$

They include the vertex orbit factor and use the full coordinate domain; there is no second vertex-factorial division in the numerical integral. All weights and all forest signs are generated exactly before numerical conversion. The inventory reproduces the previously recorded class and labeled counts.

The weighted coordinate forests were independently compared with formal bare-series inversion, retaining every graph amplitude as an independent symbol and canonically factoring articulation products. The comparison is a coefficientwise rational-polynomial identity, not a test at a few numerical inputs:

Loop order	Coupling monomials	Mass monomials	Nonzero differences
1	1	1	0
2	2	2	0
3	8	6	0
4	29	25	0
5	134	103	0
6	696	525	0
7	4164	2935	0

These counts are taken after articulation products are identified; without that factorization the counts are larger.

An additional graph-by-graph exact reduction gives zero RG polynomials for 691 of the seventh-order classes. Among the 845 power classes with nonzero channel weight, 151 are such exact zeros; the remaining 694 are integrated. For the fifth- and sixth-order power inputs, respectively 24 and 132 nonzero classes remain after the same algebraic zero test. These exclusions are identities, not variance-based pruning. All other nonzero classes have positive production allocation.

The new 44 primitive representatives were matched by completed-graph isomorphism to all 14 periods used by the first calculation. Every aggregated rational weight agrees, including the three lower-period products. The final production integrates these 44 representatives directly; the agreement of that integration with those periods is quoted in Section 27.8.

27.3 One common coordinate-space finite prescription

Computing the power-free and power-containing parts in two different finite prescriptions would require a conversion between them. Here both are computed in one coordinate-space prescription.

An edgewise exponential infrared cutoff is used:

$$C_\mu(x;z)=\frac{2^{z/2}}{\sqrt{\Gamma(1+z)}}|x|^{-(1-z)/2}e^{-\mu|x|}. \tag{755}$$

The one-loop bubble is exactly

$$\int_{\mathbb{R}} C_\mu(x;z)^2\,dx=\frac{2}{z}\mu^{-z}. \tag{756}$$

The edgewise cutoff factorizes on products of subgraphs; replacing it by an independently assigned cutoff on the span of each whole graph would change finite parts and is not done here.

Let the dimensionless bare coupling be x, and write the weighted bare amplitude series as

$$\mathcal{A}(x,z)=1+\sum_{L\ge1}(-1)^L a_L(z)x^L,\qquad \mathcal{B}(x,z)=1+\sum_{L\ge1}(-1)^L e_L(z)x^L. \tag{757}$$

The finite coupling is $g=x\mathcal{A}(x,z)$. The two RG functions follow from differentiating with respect to the common scale:

$$\widehat{\beta}(g,z)+zg=-zx^2\partial_x\mathcal{A}(x,z),\qquad \widehat{\gamma}(g,z)=-zx\partial_x\log\mathcal{B}(x,z),\quad x=x(g,z). \tag{758}$$

Thus the regulator-dependent functions, not just their $z=0$ values, are required. The physical substitution is $z=\epsilon$ after the Taylor jets and finite fixed point have been assembled.

For clarity, the independent algebraic check extracts the coupling and mass coefficients directly from

$$\widehat{b}_L(z)=-\frac{z}{L+1}[x^L]\left(\sum_{j\ge1}j(j+1)(-1)^j a_j(z)x^j\right)\mathcal{A}(x,z)^{-L-1}, \tag{759}$$

$$\widehat{c}_L(z)=-\frac{z}{L}[x^{L-1}]\partial_x\left(\frac{x\partial_x\mathcal{B}(x,z)}{\mathcal{B}(x,z)}\right)\mathcal{A}(x,z)^{-L}. \tag{760}$$

The factor z is carried symbolically before Laurent cancellation. The code does not estimate these expressions by subtracting independently sampled divergent amplitudes.

27.4 Local full-graph forest integrands

For every compatible proper logarithmic forest F, replace its blocks by their quotient graphs and glue the quotients on the original labeled vertices. Let G_F be the resulting edge multigraph. Terms whose quotient is a tadpole are set to zero in the same analytically continued critical mass prescription as the preceding sections. Unmarked power subgraphs are not dropped: their coordinate integrals are continued meromorphically as described below.

For projective coordinates u, put

$$H_F(u)=\sum_{e\in G_F}|u_{e_1}-u_{e_2}|,\qquad R_F(u)=\sum_{e\in\mathrm{root}(F)}|u_{e_1}-u_{e_2}|. \tag{761}$$

The common finite RG integral for an L-loop graph is generated by

$$\mathcal{D}_G(z)=\frac{2^{Lz}\Gamma(1+Lz)}{\Gamma(1+z)^L}\int d\Omega\sum_F(-1)^{|F|}\frac{R_F(u)}{H_F(u)}H_F(u)^{-Lz}\prod_{e\in G_F}|u_{e_1}-u_{e_2}|^{-(1-z)/2}. \tag{762}$$

Then $\widehat{b}_L=(-1)^{L+1}\sum_G w_A(G)\mathcal{D}_G$ and $\widehat{c}_L=(-1)^{L+1}\sum_G w_B(G)\mathcal{D}_G$. The factor R_F/H_F is produced by the scale derivative on the root block, followed by the common scale integral. Explicitly the latter is $\int_0^\infty\lambda^{Lz}\exp(-\lambda H_F)d\lambda$. The exact bare-polynomial check above independently verifies all forest

and root weights. This is a local representation of renormalization in the spirit of scale-dependent forest formulas[1], but the specific kernel and its combinatorial certificate here are part of the present calculation.

All terms in a forest combination are evaluated on the same coordinates. For a fixed forest term the Taylor variable appears through

$$\exp\left\{z\left[\frac{1}{2}\sum_{e}\log|u_{e_1}-u_{e_2}|-L\log H_F(u)\right]\right\}. \tag{763}$$

The common Gamma factor in (762) has logarithmic Taylor coefficients

$$\ell_1 = L\log 2, \qquad \ell_j = \frac{L^j - L}{j!}\psi_{j-1}(1), \quad j \geq 2. \tag{764}$$

These are included through the required order, with all products and Laurent poles retained before the finite coefficients are extracted.

27.5 Binary vertex clusters instead of fully ranked gaps

Order the $L+1$ vertices along the line. A binary tree is obtained by splitting at the largest gap, then repeating inside each of the two subintervals. This partitions the positive-gap domain except for measure-zero ties. The internal nodes are contiguous vertex clusters; there are L of them. Let the root scale be λ, and let $t_C \in [0,1]$ be the ratio between each nonroot node's gap and its parent's gap. Then

$$\delta_C = \lambda \prod_{D\preceq C,\ D\neq\text{root}} t_D, \qquad \left|\frac{\partial(\delta_1,\dots,\delta_L)}{\partial(\lambda,(t_C))}\right| = \lambda^{L-1}\prod_{C\neq\text{root}} t_C^{|C|-2}. \tag{765}$$

The coordinate of every vertex distance is formed as a sum of positive gaps, never by subtracting two nearly equal vertex positions.

Unlike a full ranking of all gaps, this construction does not impose an unnecessary relative ordering between gaps in disjoint clusters. It has $(L+1)!\,\mathrm{Cat}_L$, with $\mathrm{Cat}_L = (2L)!/[L!(L+1)!]$, ordered tree sectors rather than $(L+1)!L!$ ranked-gap sectors, while keeping $L-1$ continuous projective variables. At seven loops there are six continuous cluster ratios. The sector sum is sampled, not exhaustively enumerated.

The tree probabilities are computed without enumerating all trees. For a vertex subset S, let $W_{\{i\}} = 1$ and

$$W_S = \frac{1}{\alpha_S}\sum_{\substack{A\sqcup B=S,\ A,B\neq\varnothing\\ \min S\in A}} 2W_AW_B, \qquad \alpha_{\text{root}} = 1. \tag{766}$$

At a split, choose the unordered pair with probability proportional to W_AW_B and then choose its orientation uniformly. The normalization is $Z = W_{\text{root}}$. A particular ordered tree has probability $1/(Z\prod_{C\neq\text{root}}\alpha_C)$.

For the continuous power-free calculation, the proposal sees every forest term, not only the original graph. With $e_F(C)$ the number of edges internal to C in G_F, choose

$$p_C = |C| - 1 - \frac{1}{2}\max_F e_F(C), \qquad \alpha_C = r\max(\tfrac{1}{2}, p_C), \qquad t_C = u_C^{1/\alpha_C}. \tag{767}$$

Here $r = 0.8$ or 0.7 in the two independent campaigns. The combined tree and radial importance weight is simply

$$Z\prod_{C\neq\text{root}} t_C^{|C|-1-\alpha_C}. \tag{768}$$

Using only the original graph's edge count can miss a more singular forest term and can give infinite sampling variance; this happens for seven-loop forests. The present construction examines the whole finite forest kernel.

The Jacobian is checked by exact rational determinants for 84 random cluster geometries. As an independent normalization check, set $\alpha_C = |C| - 1$ and remove all edge factors. The dynamic program then gives $Z = (L+1)!L$ for $L = 1, \dots, 7$, exactly the full projective volume. Explicit enumeration of the small trees confirms the proposal probabilities with arbitrary rational subset rates.

27.6 Power subgraphs: meromorphic radial integration and paired refinement

In a power-containing graph, a cluster may have negative radial exponent. It is not sampled as an ordinary convergent endpoint integral. For each sampled tree, the regular factor is interpolated on a tensor Chebyshev–Lobatto grid of degree N, and the singular powers are integrated analytically. If $L_i(t) = \sum_k \ell_{ik} t^k$, the one-variable moment is

$$\int_0^1 t^{a+bz} L_i(t)\, dt = \sum_k \frac{\ell_{ik}}{a + k + 1 + bz}, \tag{769}$$

under meromorphic continuation. A zero denominator at $z = 0$ supplies its Laurent pole explicitly; a negative half-integer power is continued by the same rational expression. No endpoint cutoff is fitted or sent to zero numerically. Tensor products of these moments are formed before the finite Taylor coefficient is selected.

For this discrete tree sampling only, the rates use $\alpha_C = r \max(1/2, |p_C|)$. The tree probability is positive and the inverse probability is $Z \prod_C \alpha_C$. It is a probability on a finite set of trees, not a claim that the unregularized radial power integral converges.

Let Y_N denote the complete per-tree Laurent/Taylor vector at degree N. The production estimator uses the exact telescoping organization

$$\mathbb{E}[Y_6] = \mathbb{E}[Y_1] + \mathbb{E}[Y_3 - Y_1] + \mathbb{E}[Y_5 - Y_3] + \mathbb{E}[Y_6 - Y_5]. \tag{770}$$

Within every difference, both resolutions use the same graph, binary tree, and random stream. Different levels use independent streams. This is the multilevel variance-reduction principle[52]; it reduces the noise of the refinement differences without extrapolating from lower physical coefficients. The residual difference between finite degree six and the meromorphic integral is assessed by the successive paired refinements, including the last degree-five/six comparison, not assumed to be zero by definition.

Negative Laurent coefficients of the fully subtracted RG functions vanish. Fixed linear combinations of these exact-zero quantities are used as control variates. The control matrices were fitted to separate coarse pilot samples, with a fixed regularization, and then frozen before production; they were not chosen to match any physical reference. A separate first seventh-order degree-one/three refinement run was used to freeze a cost-aware graph allocation; that training stream is excluded from the final seventh-order refinement mean. Every nonzero remaining stratum keeps positive coverage. The same matrix is applied at every level in (770). Consequently any finite-grid error in a control is also refined by the telescoping calculation. Neither C_5, C_6, nor a conjecture that the seventh-order power correction vanishes is imposed as a constraint.

27.7 Seventh-order assembly and covariance

Write

$$\widehat{\beta}(g, \epsilon) = -\epsilon g + \sum_{L=1}^{7} \widehat{b}_L(\epsilon) g^{L+1}, \qquad \widehat{\gamma}(g, \epsilon) = \sum_{L=1}^{7} \widehat{c}_L(\epsilon) g^L, \tag{771}$$

with $\widehat{b}_1 = 3$, $\widehat{c}_1 = 1$, and retain each jet through ϵ^{7-L}. Solving the finite fixed point is equivalent to

$$C_7 = [\epsilon^7] \sum_{k=1}^{7} \frac{\epsilon^k}{k} [g^{k-1}] \partial_g \widehat{\gamma}(g,\epsilon) \left[3 + \sum_{L=2}^{7} \widehat{b}_L(\epsilon) g^{L-1} \right]^{-k}. \tag{772}$$

In a regulator-independent MS scheme this reduces to $\frac{1}{7}[g^6]\gamma_r'(g)B(g)^{-7}$, but that shorter expression cannot be used on just the constant terms of the present finite-scheme jets.

For fixed one- through four-loop inputs, C_5, C_6, C_7 are affine in all fifth-, sixth-, and seventh-loop jets. Two such high-loop insertions would first contribute beyond order seven. Thus, if v_L is the vector of coupling and mass coefficients (with negative Laurent orders kept for controls),

$$\boldsymbol{C} = \boldsymbol{K}_{\le 4} + \sum_{L=5}^{7} J_L (v_L^{\text{free}} + v_L^{\text{power}}). \tag{773}$$

Here $\boldsymbol{C}$ is the vector of the coefficients C_n, $\boldsymbol{K}_{\le 4}$ its value from the one- through four-loop inputs alone, and J_L the exact matrix of its dependence on v_L. The complete matrices J_L and the lower-loop vectors are in `LRI_C7_HPC/reference/verification/full_affine_assembly.json`, and the frozen control matrices in `LRI_C7_HPC/reference/verification/power_control_training.json`. Thirty-two exact rational tests verify both (772) against an independently iterated fixed point and the affine identity (773).

All coupling/mass and Taylor-order covariances are retained. Each random replicate is projected through the full matrix before computing its covariance. The two power-free campaigns are averaged with equal weights fixed independently of their observed values. The direct primitive campaign is added only once, with variance $(0.8667)^2$. The two independent coarse power campaigns are combined with preassigned sample-count weights. Independent refinement-level covariance matrices are then added. Independent production batches of one frozen design are combined with equal weights, and their replicate covariances are propagated. The finite high-loop assembly itself has no nonlinear Monte Carlo bias, because it is affine.

27.8 Production results and refinement

The final production consists of 399 independent batches of one frozen design, each with its own seeds. The design was fixed before production from a separate pilot: the replicate counts and sample multipliers of every campaign were chosen by minimum-variance allocation at the measured pilot variances and costs, with every multiplier a power of two. Each batch contains the two independent power-free campaigns at five, six and seven loops (128 randomized replicates each); the two independent coarse power campaigns (128 and 64 replicates); the paired refinements $1 \to 3$, $3 \to 5$ and $5 \to 6$ of (770) (128, 128 and 64 replicates); and a paired $6 \to 7$ check level (64 replicates), which is never part of the estimate. The 44 primitive classes are integrated once, in a separate campaign with 2^{23} points per graph and 64 replicates. It gives -696179.472 ± 0.867, which agrees with the period-based value -696147.446 ± 25.368 of the first calculation within 1.26 combined standard errors. The production used about 2.5×10^5 core-hours.

The telescoping resolution sequence for the complete coefficient is

Power-grid degree	Full C_7 estimate	Sampling standard error
1	−331,309.98	6.14
3	−322,614.21	9.21
5	−322,567.22	10.01
6	−322,549.89	10.09

The individual refinement corrections, in absolute units of C_7, are

Paired degrees	Correction	Sampling standard error
$1 \to 3$	+8,695.78	6.86
$3 \to 5$	+46.99	3.93
$5 \to 6$	+17.33	1.21
$6 \to 7$ (check only)	+10.40	0.43

The corrections decrease, but the ratio of successive steps grows, from 0.37 for $5 \to 6$ over $3 \to 5$ to 0.60 for $6 \to 7$ over $5 \to 6$. No limit has been replaced by a fitted extrapolation in (752). The allowance candidates are $4 \times 10.09 = 40.35$, $2 \times 17.33 + 4 \times 1.21 = 39.50$, $2 \times 10.40 + 4 \times 0.43 = 22.49$ and the half-split displacement 12.58, which gives ± 41.

The scatter of the 399 batch estimates about their mean gives $\chi^2 = 348.9$ for 398 degrees of freedom, so the replicate-based errors describe the batch-to-batch variation. A bootstrap over the replicates of every stream gives the standard error 10.11, compared with the covariance value 10.09. Splitting the replicates of every stream into halves gives -322537.30 and -322562.47. The two independent power-free campaigns agree at each loop order to within 0.42 of their combined sampling errors. The largest absolute standardized residual among the 24 monitored spurious Laurent poles is 2.31. With the zero-pole controls the estimate is 71.6 ± 41.0 above the uncontrolled estimator, -322621.5 ± 41.1 (the same convention as at eighth order, Section 30.4).

27.9 Lower-order checks and the radial truncation

The same calculation gives the following uncalibrated lower-order values:

Coefficient	Degree-six value	Sampling error	Reference
$C_5(1)$	-731.174185	0.008355	-731.139477 (exact)
$C_6(1)$	14,332.435930	0.305002	14,331.687033 (deterministic)

The differences are -4.15 and $+2.46$ sampling errors. At this precision the lower orders resolve the truncation of the radial quadrature at degree six. Adding the measured degree six to seven step and a geometric tail with each coefficient's own last step ratio gives

	Degree six	+ step $6 \to 7$	+ geometric tail
$C_5(1)$, difference in errors	-4.15	-1.88	-0.30
$C_6(1)$, difference in errors	$+2.46$	$+0.97$	-0.22
$C_7(1)$, value	-322549.9	-322539.5	-322523.9

The tail ratios are 0.408 for C_5, 0.443 for C_6 and 0.600 for C_7. The lower-order agreement is consistency with the tail model at fifth and sixth order. It does not validate the seventh-order tail, which converges more slowly, and it does not exclude other systematic errors. This is the origin of the truncation-aware estimate (753). Deterministic low-loop assembly also gives $C_3 = -3.71985632290349$ and $C_4 = 45.4248801371166$; their discrepancies from the retained higher-precision values are respectively 1.5×10^{-12} and 3.4×10^{-8}.

The lower-order reference coefficients are used only as checks. They are not fitted into the estimator; in particular, $C_6(1)$ is not an input to the mean in (773). The one- through four-loop inputs are evaluated by deterministic coordinate quadrature. Changing the four-loop degree from 16 to 20 moves the seventh-order assembly by less than 0.001 with the higher-loop inputs held fixed; the preceding degree changes decrease rapidly. This is a convergence observation, not an interval bound.

The tree point evaluator was compared with an independent positive-gap reconstruction at 650 decimal digits in 60 cases, including cluster ratios exponentially smaller than the production minima.

The maximum normalized discrepancy was 1.18×10^{-14}; the extreme-endpoint subset had maximum 3.45×10^{-18}. Continuous integration uses extended precision, binary128 on strongly canceling samples, and a 100-digit fallback for more severe cancellations. The 650-digit comparison tests the values that the evaluator actually returns, including the chosen fallbacks (extended precision in 51 of the 60 cases, binary128 in one and 100 digits in eight), rather than only the uncancelled summands (`LRI_C7_HPC/reference/verification/tree_point_checks.json`). The meromorphic weights are generated at 100 working digits and stored in extended precision. Independent monomial-moment tests of the stored weights include the Laurent-pole and negative-power cases; the largest normalized error in 1152 tests is 1.1×10^{-11} (`LRI_C7_HPC/reference/verification/meromorphic_weight_checks.json`). The production implementation evaluates the radial tensor quadrature in bounded tiles with compensated accumulation. Replayed against binary128 on 112 degree-seven elements, its aggregate projected error in C_7 is 2.8×10^{-12}, compared with 1.9×10^{-11} for the original dense implementation. The worst individual elements, which cancel strongly, reach 4.2×10^{-14} relative. These errors are many orders of magnitude below the sampling error.

The two point streams are randomized differently. In the continuous power-free integration every graph stratum receives a power-of-two number of points of one linearly scrambled Sobol net, and the uncertainty is estimated between independent randomized replicates, not between the correlated points of one net [46]. The export of the direction matrices first reconstructs the sample values of the public SciPy generator and checks them exactly. The power-sector quadrature instead draws its finite tree sectors from independent 64-bit Mersenne Twister streams, seeded per batch and per graph; the paired degrees of (770) evaluate the same sampled trees, which is what makes the paired refinements small in variance. The meromorphic weight files are native extended-precision binaries; on a platform with a different extended-precision format they are regenerated from the shipped code, not read. For the first, smaller calculation, a direct fixed-point assembly of the raw Taylor coefficients, followed by the recorded zero-pole control, agreed with the affine assembly (773) to 1.8×10^{-10}.

28 Full seventh order in two dimensions: mixed local-mass forests

28.1 Result and scope

At $d=2$ the seventh-order coefficient uses the same physical regulator and finite zero-momentum prescription as the sixth-order calculation of Sections 23–26, with $\epsilon=2s-2$,

$$\Delta_{\mathcal{E}}(2,\epsilon)=1-\frac{\epsilon}{2}+\gamma_r(g_*,\epsilon),\qquad \beta(g,z)=-zg+\sum_{L=1}^{7}\widehat{b}_L(z)g^{L+1},\qquad \gamma_r(g,z)=\sum_{L=1}^{7}\widehat{c}_L(z)g^L,\tag{774}$$

$\widehat{b}_1=3$, $\widehat{c}_1=1$ and $\beta(g_*,\epsilon)=0$. Every loop order keeps its Taylor polynomial through z^{7-L} before $z=\epsilon$ is set. Unlike the sixth-order calculation, the power-containing sector is integrated explicitly instead of being taken from an analytic compensation. Write $f_L=(b_L^{\mathrm{pf}},c_L^{\mathrm{pf}})$ and $p_L=(b_L^{\mathrm{p}},c_L^{\mathrm{p}})$ for the power-free and power-containing jets in the same prescription, and $\mathcal{F}$ for the fixed-point series extraction. Then

$$\boxed{C_7(2)=C_7^{\mathrm{pf}}(2)+\Delta_pC_7(2),\qquad \Delta_pC_7=\mathcal{F}_7(f+p)-\mathcal{F}_7(f).}\tag{775}$$

The difference includes lower-loop nonlinear changes; it is not the simple-pole sum of the seven-loop power graphs. The production of Section 28.7 gives

Contribution	Estimate	Sampling standard error
Power-free coefficient $C_7^{\mathrm{pf}}(2)$	−3876.0700590	0.5224502
Power correction $\Delta_pC_7(2)$	−0.0560338	0.0458637
Complete coefficient $C_7(2)$	−3876.1260928	0.5244497

The quoted uncertainty is the empirical reporting allowance

$$E_{\mathrm{emp}} = \max\Big(4\widehat{\sigma},\ \max_j |C_{7,j}^{\mathrm{half}} - C_7|\Big), \quad \text{rounded up,} \tag{776}$$

where $\widehat{\sigma}$ is the sampling standard error and $C_{7,j}^{\mathrm{half}}$, $j = 1, 2$, are the half-stream estimates: the estimator recomputed from the first and from the second half of the replicates of every campaign, loop and sector. The rule is that of an earlier, lower-precision integration of $C_7(2)$ with the same graph set and prescription, called the earlier $C_7(2)$ calculation, or the earlier calculation for short, in this section. The rule was applied to the 2026 production, and its rounding step chosen, after the result was available. Here $4\widehat{\sigma} = 2.098$ exceeds both half-stream deviations, 0.36, and rounding up to a multiple of 0.1 gives

$$\boxed{C_7(2) = -3876.1 \pm 2.1, \qquad \epsilon = 2s - 2.} \tag{777}$$

The earlier calculation rounded the same rule to a multiple of 10 at its precision and reported -3950 ± 310 (estimator -3953.6336215, sampling standard error 76.7310362); the new estimator lies 1.01 of its standard errors from it. The allowance is empirical. It is not a probability guarantee and does not bound an error shared by the graph and renormalization framework; the additional estimator digits are given for reproduction only.

The earlier calculation, its method and its frozen design were written by OpenAI GPT-6 astra (model stated by the author); its code and records are in `LRI_C7_D2_HPC/reference/` (Sections 28.7 and 33). When this method was merged into this section, on 16 October 2026, a Claude Code session wrote out or rechecked four parts: the three-loop check (783)–(785), for which it wrote out the graph weight, the forest cancellation (786), the bound $\mathtt{p}_3/\mathtt{p}_2 \le 1/9$ and the series identity used for the regulator derivative, and checked the quadrature and the Euler and Pfaff steps numerically; the two-loop entries (790)–(791) and the polynomial (792), which it rechecked by a separate Cauchy evaluation and an exact symbolic expansion; the primitive classes of Section 28.6, for which it supplied the argument for $w_B = 0$ and the normalization (793) and compared the 40-digit values with the closed forms; and the variance decomposition (797), which it evaluated with the 2026 record. It also checked that the group configurations of the 2026 production hold the sample counts of Section 28.7.

28.2 Graph weights and complete coverage

An L-loop graph has $L + 1$ vertices and $2L$ individually labeled lines. With edge multiplicities m_{ij} and $e_i = 4 - \deg(i)$,

$$w_A = \frac{24}{|\operatorname{Aut} G| \prod_{i<j} m_{ij}! \prod_i e_i!}, \qquad s_G = \sum_i e_i(e_i - 1), \qquad w_B = \frac{s_G}{12} w_A, \tag{778}$$

so that $3w_B - w_A = (s_G - 4)w_A/4$. This rational weight, not a sampled value, determines the zero-channel exclusions at seven loops; at lower loops the coupling and mass jets are kept separately even when the final-order channel weight vanishes. A proper one-particle-irreducible subgraph with v vertices and e internal lines is logarithmic if $e = 2v-2$ and an unmarked power-divergent two-point subgraph if $e = 2v - 1$. Its loop number is $h = e - v + 1$, so a power subgraph has $h = v$. All proper edge subgraphs, including those not induced by their vertex set, were enumerated independently; their mixed forest maps agree exactly with the production maps for all 1,301 power-containing classes at loops three to seven.

L	All classes	Power-containing	Nonzero mixed kernels	Integrated power classes
3	8	1	1	1
4	27	5	3	3
5	129	31	20	20
6	660	173	107	107
7	3986	1091	709	590

At seven loops 459 classes have $3w_B - w_A = 0$ and 639 further classes have an exactly vanishing mixed forest map. The remaining 2,254 power-free nonprimitive, 44 primitive and 590 power-containing classes are all integrated in every replicate of the production below; there is no omitted nonzero class. At loops three to six the integrated power-free and power-containing classes number 5, 16, 76, 395 and 1, 3, 20, 107.

A graph is power-free if it has no unmarked power-divergent proper two-point subgraph; nonprimitive graphs whose subdivergences are all logarithmic are power-free. The power-free sector partitions as

L	Power-free classes	Nonzero finite kernels	Primitive among them	Zero kernels
2	2	1	0	1
3	7	5	1	2
4	22	16	1	6
5	98	76	3	22
6	487	395	10	92
7	2895	2441	44	454

and at seven loops the 3,986 classes split by sector, kernel and channel weight as

Seven-loop classes	$3w_B - w_A \neq 0$	$3w_B - w_A = 0$	All
Power-free, nonzero kernel	2298	143	2441
Power-free, zero kernel	384	70	454
Power-containing, nonzero kernel	590	119	709
Power-containing, zero kernel	255	127	382
All	3527	459	3986

The first column contains the integrated classes, $2{,}298 = 2{,}254 + 44$ power-free and 590 power-containing, and the $639 = 384 + 255$ classes with a vanishing mixed forest map; the 845 power-containing classes of nonzero channel weight are $590 + 255$. The class-by-class assignment is in `LRI_C7_D2_HPC/reference/verification/full_graph_coverage.json`. The $262 = 143 + 119$ classes with a nonzero kernel and zero channel weight are not integrated here; they first contribute at eighth order (Section 31.2).

The production builders take their subgraphs from induced vertex sets. The independent enumeration of all edge subgraphs also contains the non-induced ones, for example a two-line logarithmic bubble inside a triple-line power insertion, and the exact agreement of the forest maps for all 1,301 power-containing classes shows that the forest terms of such subgraphs cancel and were not dropped by assumption; the three-loop example (786) below shows the cancellation explicitly, and Section 32.7 proves it for every graph of the family. After graph and line labels are matched, all 492 nonzero power-free kernels at loops three to six are identical to the kernels of the sixth-order calculation (Section 25; record `LRI_C7_D2_HPC/reference/LRI_C7_d2_progress/verification/legacy_mapping.json`). A second construction from contracted graph Laplacians and exact rational Kirchhoff determinants, which does not use the monomial selection, reproduces 24 selected seven-loop power-free forest kernels exactly; the analogous construction for the power-containing kernels is the check at 66 configurations quoted below.

28.3 Local mass subtraction and the mixed forest kernel

The Schwinger infrared factor does not perform critical mass tuning: an unmarked self-energy insertion must have its local zero-momentum part subtracted, as for the three-loop graph I_3 in [7, Appendix C.3.4]. In two dimensions its superficial degree at $z = 0$ is one in momenta; the amplitude is even in the external momentum, so the constant Taylor subtraction leaves a remainder of quadratic order. Put

$$U_G(x) = \sum_T \prod_{e \notin T} x_e, \qquad S(x) = \sum_e x_e, \qquad d\Omega = \delta(1 - S(x)) \prod_e dx_e, \tag{779}$$

the sum running over spanning trees. A compatible mixed forest $F = F_{\log} \cup F_{\mathrm{p}}$ consists of nested subgraphs, or edge-disjoint subgraphs meeting in at most one vertex; nested blocks are taken after contracting their children, and U_F is the product of the block Symanzik polynomials. Equivalently, a complement-of-tree monomial m of U_G is kept in U_F if and only if $|m \cap E_\gamma| = h_\gamma$ for every $\gamma \in F$. With the logarithmic-root edge set $E_R^{\log}(F) = E_G \setminus \bigcup_{\gamma \in F_{\log}} E_\gamma$, the finite RG kernel is

$$\boxed{K_G^{\mathrm{full}}(x) = \sum_F (-1)^{|F|} \frac{\sum_{e \in E_R^{\log}(F)} x_e}{S(x)\, U_F(x)}.} \tag{780}$$

Only logarithmic subgraphs are removed from the root numerator. A power subgraph changes the denominator through its local Taylor subtraction but does not remove its interaction order from the root amplitude; excluding its lines from the numerator would implement a different scale derivative and already changes the three-loop answer. To derive (780), apply $1 - T_\gamma^0$, with T_γ^0 setting the external momentum to zero, to every compatible local two-point insertion. Gaussian integration replaces U_G by U_F for the selected power subgraphs, while the vertex interaction order and the common Schwinger mass factor are unchanged. The finite coupling and mass inversion of these critically subtracted amplitudes then gives the same logarithmic root derivative as in the power-free calculation, distributed over the projective block scales as the root fraction $\sum_{e \in E_R^{\log}} x_e / S$. Expanding both operations gives one sign for each selected subgraph. This is a local implementation of parametric forest renormalization [1]. The scale exponents of the blocks are derived in the next paragraph. Identical selected denominators are grouped with integer numerator coefficients and identically zero vectors are removed before any floating-point arithmetic. An independent rational Kirchhoff-determinant implementation agrees exactly at 66 configurations, including endpoint-scaled points.

Scale exponents of the forest blocks. This bookkeeping fixes the numerator of (780). Rescale the lines of a block B of loop number h_B by a common factor λ. The critical line measure $\prod_e x_e^{-1/2+z/4} dx_e$ and the block Symanzik polynomial, of degree h_B, give $\lambda^{|E_B|(1/2+z/4)-h_B}$. A logarithmic block has $|E_B| = 2h_B$ and exponent $h_B z/2$, the Dirichlet parameter of its relative scale in the proof of (698); this produces the root fraction and the sign of each logarithmic forest element. A power block has $|E_B| = 2h_B - 1$ and exponent $-1/2 + (2h_B - 1)z/4$ before its Taylor remainder. Its complement, with $2L - 2h_B + 1$ lines and loop number $L - h_B$, has exponent $+1/2 + (2L - 2h_B + 1)z/4$, the opposite engineering dimension, and the two add to $Lz/2$, the exponent of the whole graph. A mass Taylor operation therefore splits off no logarithmic scale: it carries no further factor h_B/L and introduces no second coupling-inversion root factor. After the logarithmic inversion the root keeps its original lines, including those of the power Taylor factors, and these form the edge set $E_R^{\log}(F)$ of (780). Nested blocks obey the same argument after their children are contracted. In the monomial selection, m is the complement-of-tree mask of the individually labeled lines, so parallel lines are separate variables.

In the normalization $r(2,z)=1$ the power-containing jets are

$$\boxed{\begin{pmatrix}\widehat{b}_L^{\mathrm{p}}(z)\\ \widehat{c}_L^{\mathrm{p}}(z)\end{pmatrix}=\frac{2(-1)^{L+1}\Gamma(1+Lz/2)}{\Gamma(\frac12+z/4)^{2L}}\sum_{G\in\mathrm{p}}\begin{pmatrix}w_A(G)\\ w_B(G)\end{pmatrix}\int d\Omega\prod_e x_e^{-1/2+z/4}K_G^{\mathrm{full}}(x),}\tag{781}$$

with the same prefactor and finite coupling prescription as the power-free data. Before local subtraction a connected power face has radial exponent $e/2-h=-1/2$ at $z=0$; the Taylor remainder raises it to $+1/2$. Logarithmic faces start at zero and their combined subtraction raises the exponent; nested limits are handled by the compatible forest. Regulator derivatives add finite powers of logarithms without removing these margins. The importance maps use the forest denominator valuations and the positive half-integer endpoint floor after subtraction.

Equation (781) is (695) with D_G of (698) and K_G replaced by K_G^{full}, since $Lz\,\Gamma(Lz/2)=2\Gamma(1+Lz/2)$: the prefactor is the logarithmic scale derivative of $\Gamma(Lz/2)$, and at $z=0$ it equals $2(-1)^{L+1}\pi^{-L}$. The power-free jets f_L are given by the same formula with K_G of (697), so no regulator-independent MS coefficient enters the input jets of (775).

Importance rates. Let $\nu_S(P)$ be the smallest degree of a polynomial P in the variables of a proper line subset S, and $N_F=\sum_{e\in E_R^{\log}(F)}x_e$. When the lines of S shrink by a common factor λ, the forest term F of (780), times the line measure at $z=0$, scales as $\lambda^{|S|/2+\nu_S(N_F)-\nu_S(U_F)}$. The builders store the rate

$$\alpha_G(S)=\tfrac12\max\bigl(r_G(S),p_S\bigr),\qquad r_G(S)=\min_F\bigl[\,|S|+2\nu_S(N_F)-2\nu_S(U_F)\bigr],\tag{782}$$

with $p_S=1$ for odd and $p_S=2$ for even $|S|$. The termwise bound $r_G(S)/2$ can be nonpositive where the forest sum cancels. Since the kernel is rational and the measure contributes $|S|/2$, the exponent of the summed kernel lies in $|S|/2+\mathbb{Z}$, and its positivity after subtraction implies the floor $p_S/2$. The floor therefore sharpens a nonpositive termwise bound without changing the integral; Section 31.5 uses the same rule at eighth order and shows that the weight then stays bounded. At seven loops the floor strengthened the rates of 81,632 even subsets in the power-free files (record `LRI_C7_D2_HPC/reference/LRI_C7_d2_progress/verification/parity_rate_update.json`); at loops three to six the power-free rates are the frozen tables of the sixth-order calculation, mapped exactly to the new labels. The campaign factors 0.8 and 0.9 multiply these rates. All such choices change the variance, not the integral, and no interval error is derived from these power-counting arguments.

Routes not used. Subtracting the local part of a self-energy insertion is not the same as discarding the graph, and the absence of wave-function renormalization at this order is not used to delete power amplitudes: in the long-range prescription the local subtraction of a power divergence leaves a finite two-point amplitude [7]. Two other routes to the power sector were tried. A coordinate-space routing audit through the Gamma convolutions, complex Selberg specializations and five-edge kite formulas of Section 24, with all required logarithmic subgraphs and contraction quotients, closed structurally for 740 of the 845 power-containing seven-loop classes of nonzero channel weight and did not close for the other 105. It was an audit of routes, not an evaluation, and no class was assigned zero. An exploratory radial integration by parts did not implement the local subtraction: it recovered a constant term, but its regulator derivatives still produced divergent endpoint integrals. It was rejected, and none of its samples enters a result. The kernel (780) instead performs the local subtraction and the logarithmic forest combinations before any sampling.

28.4 An independent three-loop check

The first power graph G has three external-chain parameters with sum R and internal parameters y_1,y_2,y_3. With $\mathsf{P}_2=y_1y_2+y_1y_3+y_2y_3$, $\mathsf{P}_3=y_1y_2y_3$, $U_G=R\mathsf{P}_2+\mathsf{P}_3$ and $U_0=R\mathsf{P}_2$, its mixed

kernel (780) is exactly

$$K_G^{\mathrm{full}} = \frac{1}{U_G} - \frac{1}{U_0} = -\frac{\mathsf{P}_3}{R\mathsf{P}_2(R\mathsf{P}_2 + \mathsf{P}_3)}, \tag{783}$$

a rational remainder that can be integrated and differentiated at $z = 0$. Let $a = \frac{1}{2} + z/4$, $k = 3a - 1$, and use simplex variables u with $\mathsf{p}_2 = u_1u_2 + u_1u_3 + u_2u_3$ and $\mathsf{p}_3 = u_1u_2u_3$. Integrating the external parameters and the radial variable, and applying a Pfaff transformation, gives

$$\widehat{b}_3^{\mathrm{p}}(z) = -\frac{\Gamma(1 + 3z/2)\,\mathrm{B}(k,k)}{\Gamma(a)^3\Gamma(3a)} \int d\Omega_u\, \mathsf{p}_3^z \mathsf{p}_2^{-3a}\, {}_2F_1\!\left(\frac{3z}{2}, k; 2k; 1 - \frac{\mathsf{p}_3}{\mathsf{p}_2}\right), \tag{784}$$

with $w_A = 1/2$ included. At $z = 0$ the hypergeometric function is one and $\int d\Omega_u\, \mathsf{p}_2^{-3/2} = 2\pi$, so $\widehat{b}_3^{\mathrm{p}}(0) = -4$ and $\widehat{c}_3^{\mathrm{p}} = \widehat{b}_3^{\mathrm{p}}/3$, in agreement with the subtracted I_3 normalization of [7]. The first regulator coefficient is

$$[z]\widehat{b}_3^{\mathrm{p}}(z) = -\frac{2}{\pi} \int d\Omega_u\, \mathsf{p}_2^{-3/2} \left[-\frac{3}{2} + \log \mathsf{p}_3 - \frac{3}{4}\log \mathsf{p}_2 + 3\log \frac{2}{1 + \sqrt{\mathsf{p}_3/\mathsf{p}_2}}\right] = 18.6743207750, \tag{785}$$

from two independently tightened deterministic quadratures differing by 4.6×10^{-10}; this reference uses neither the graph sampler nor its monomial arrays. The production of Section 28.7 gives $\widehat{b}_3^{\mathrm{p}}(0) = -3.99999979 \pm 0.00000042$ and $[z]\widehat{b}_3^{\mathrm{p}} = 18.6743200 \pm 0.0000021$, at $+0.50$ and -0.36 sampling standard errors from these values.

Derivation of (783)–(785)**.** The polynomial $U_G = R\mathsf{P}_2 + \mathsf{P}_3$ fixes the topology of the single three-loop power-containing class. Four vertices lie on a ring. One adjacent pair is joined by the three lines y_1, y_2, y_3, the sunset γ, a power subgraph with $v = 2$, $e = 3$ and $h = 2$. The other three lines are in series, so their parameters enter U_G only through their sum R, and each of their two inner vertices carries two external legs. Hence $|\operatorname{Aut} G| = 2$, the triple line contributes 3! and the legs 2! 2!, so $w_A = 24/48 = 1/2$, and $s_G = 4$ gives $w_B = w_A/3$, which is the statement $\widehat{c}_3^{\mathrm{p}} = \widehat{b}_3^{\mathrm{p}}/3$.

The kernel. The only power subgraph is γ. The logarithmic subgraphs are the three bubbles $\beta_{ij} = \{y_i, y_j\}$ inside γ, which are not induced. Two bubbles share a line and are not compatible, so the compatible forests are $\varnothing$, $\{\gamma\}$, $\{\beta_{ij}\}$ and $\{\beta_{ij}, \gamma\}$. For $\varnothing$ and $\{\gamma\}$ the numerator is S, since a power subgraph removes no line, and the terms are $1/U_G$ and $-1/U_0$ with $U_0 = U_\gamma U_{G/\gamma} = \mathsf{P}_2 R$. For a bubble, $U_{\beta_{ij}} = y_i + y_j$, and G/β_{ij} is the one-vertex join of the loop y_k and the contracted chain, $\{i, j, k\} = \{1, 2, 3\}$. Since γ/β_{ij} is the loop y_k and G/γ the chain,

$$U_{\{\beta_{ij}\}} = (y_i + y_j)\, y_k R = U_{\{\beta_{ij}, \gamma\}}, \tag{786}$$

which are also the monomials of U_G with exactly one of y_i, y_j. Both forests have the numerator $R + y_k$ and they carry opposite signs, so they cancel, and (783) remains.

The integral. With $a = \frac{1}{2} + z/4$ the line measure is $\prod_e x_e^{a-1} dx_e$. Integrating the chain parameters at fixed R gives $R^{3a-1}\Gamma(a)^3/\Gamma(3a)$. Put $R = r$ and $y_i = (1 - r)u_i$ with $\sum_i u_i = 1$. Then $\mathsf{P}_2 = (1 - r)^2 \mathsf{p}_2$, $\mathsf{P}_3 = (1 - r)^3 \mathsf{p}_3$, the Jacobian is $(1 - r)^2$, $\prod_i y_i^{a-1} = (1 - r)^{3a-3} \prod_i u_i^{a-1}$, and $K_G^{\mathrm{full}} = -\mathsf{p}_3/[r(1 - r)\mathsf{p}_2(r\mathsf{p}_2 + (1 - r)\mathsf{p}_3)]$. With $k = 3a - 1$ the r integral is Euler's,

$$-\frac{\mathsf{p}_3}{\mathsf{p}_2} \int_0^1 \frac{r^{k-1}(1 - r)^{k-1}}{r\mathsf{p}_2 + (1 - r)\mathsf{p}_3}\, dr = -\frac{\mathrm{B}(k,k)}{\mathsf{p}_2}\, {}_2F_1\!\left(1, k; 2k; 1 - \frac{\mathsf{p}_2}{\mathsf{p}_3}\right). \tag{787}$$

Its argument is at most -8, because $\mathsf{p}_3/\mathsf{p}_2 = \left(\sum_i u_i^{-1}\right)^{-1} \le 1/9$. The Pfaff transformation ${}_2F_1(1, k; 2k; w) = (1 - w)^{-k}\, {}_2F_1(2k - 1, k; 2k; w/(w - 1))$, with $1 - w = \mathsf{p}_2/\mathsf{p}_3$, $w/(w - 1) = 1 - \mathsf{p}_3/\mathsf{p}_2 \in [8/9, 1)$ and $2k - 1 = 3z/2$, turns the u integrand into $-\mathrm{B}(k,k)\, \mathsf{p}_2^{-1-k} \mathsf{p}_3^k \prod_i u_i^{a-1}\, {}_2F_1(3z/2, k; 2k; 1 - \mathsf{p}_3/\mathsf{p}_2)$. This equals $-\mathrm{B}(k,k)\, \mathsf{p}_3^z \mathsf{p}_2^{-3a}\, {}_2F_1(3z/2, k; 2k; 1 - \mathsf{p}_3/\mathsf{p}_2)$ because $\prod_i u_i = \mathsf{p}_3$ and $4a - 2 = z$. With the prefactor $2\Gamma(1 + 3z/2)/\Gamma(a)^6$ of (781) at $L = 3$, the weight $w_A = 1/2$ and the chain factor $\Gamma(a)^3/\Gamma(3a)$, this is (784).

The two coefficients. At $z = 0$ one has $a = k = \frac{1}{2}$, $\mathrm{B}(\frac{1}{2}, \frac{1}{2}) = \pi$ and $\Gamma(\frac{1}{2})^3\Gamma(\frac{3}{2}) = \pi^2/2$, so the scalar prefactor is $-2/\pi$, and $\int d\Omega_u\, \mathsf{p}_2^{-3/2} = 2\pi$ gives $\widehat{b}_3^{\mathrm{p}}(0) = -4$. The logarithmic derivative of the scalar prefactor at $z = 0$ is $\frac{3}{4}[\psi(\frac{1}{2}) - \psi(\frac{3}{2})] = -\frac{3}{2}$, and $\mathsf{p}_3^z\mathsf{p}_2^{-3a}$ contributes $\log \mathsf{p}_3 - \frac{3}{4}\log \mathsf{p}_2$. Since $(3z/2)_n = O(z)$ for $n \geq 1$, the z dependence of k enters the hypergeometric function only at order z^2, and for $0 \leq \xi < 1$

$$ {}_2F_1\Big(\frac{3z}{2}, \frac{1}{2}; 1; \xi\Big) = 1 + \frac{3z}{2}\sum_{n\geq 1} \frac{(\frac{1}{2})_n}{n!\, n}\, \xi^n + O(z^2) = 1 + 3z \log \frac{2}{1 + \sqrt{1-\xi}} + O(z^2), $$

because $\sum_{n\geq 1}(\frac{1}{2})_n \xi^n/(n!\, n) = \int_0^\xi [(1-t)^{-1/2} - 1]\, dt/t = 2\log[2/(1+\sqrt{1-\xi})]$. At $\xi = 1 - \mathsf{p}_3/\mathsf{p}_2$ the three contributions give (785).

Quadrature. The integrand is symmetric in u. On the sector $u_1 \geq u_2 \geq u_3$ put $u = (1, t, tv)/(1+t+tv)$ with $t, v \in (0, 1)$, and $t = \rho^2$. Then $d\Omega_u\, \mathsf{p}_2^{-3/2} = 2(1 + v + \rho^2 v)^{-3/2} d\rho\, dv$, so the six sectors give the weight $12(1 + v + \rho^2 v)^{-3/2} d\rho\, dv$, free of the algebraic corner singularity, and the logarithms of (785) remain integrable. The adaptive error estimates of the two deterministic quadratures are not interval enclosures. In these variables a separate tanh-sinh quadrature at 20 digits reproduces 2π and 18.6743207750, and (787) and the Pfaff step were confirmed numerically at a sample point.

28.5 Regulator jets and physical assembly

On a projective chart put $h(x) = \frac{1}{4}\sum_e \log x_e - \frac{L}{2}\log S(x)$. The scalar normalization logarithm has Taylor coefficients

$$ a_{L,j} = \frac{(L/2)^j \psi_{j-1}(1) - 2L(1/4)^j \psi_{j-1}(\frac{1}{2})}{j!}, \tag{788} $$

and the integrand jets follow from $e_0 = 1$ and

$$ e_n(x) = \frac{1}{n}\sum_{j=1}^{n} j\big(a_{L,j} + \delta_{j1} h(x)\big) e_{n-j}(x), \qquad n \leq 7 - L, \tag{789} $$

so the maximum orders at loops two to seven are $5, 4, 3, 2, 1, 0$. The two-loop input is the unchanged exact-hypergeometric finite-scheme input and the one-loop input is exact. With all jets present, $g(\epsilon) = \frac{1}{3}[\epsilon - \sum_{L=2}^{7} \widehat{b}_L(\epsilon) g(\epsilon)^L] \bmod \epsilon^8$ is solved by eight iterations from $g = 0$ and substituted in (774). An independent bivariate Lagrange extraction agrees exactly in 24 regulator-dependent rational cases, and 24 further cases agree with the constant-coefficient MS seventh-order polynomial; these test the assembly, not the sampled data. No higher-order coefficient is extrapolated from $C_0, \ldots, C_6$.

By (702) and (199) at $d = 2$, where $r(2, z) = 1$ and $\Gamma(d/2) = 1$,

$$ [z^j]\, \widehat{b}_2(z) = -6\,[z^{j+1}](4F_1(2, z) - 2W(2, z)F_2(2, z)), \qquad \widehat{c}_2 = \tfrac{1}{6}\widehat{b}_2, \qquad j = 0, \ldots, 5. \tag{790} $$

Seventh order needs, beyond (703),

$$ b_{2,5} = 175.39598710648998744915099527723636\ldots. \tag{791} $$

Two Cauchy evaluations at 75 working digits, one with radius 0.05 and 80 nodes and one with radius 0.08 and 96 nodes, agree to 1.75×10^{-68} over the six entries (`LRI_C7_D2_HPC/reference/LRI_C7_d2_progress/verification/two_loop_extension.json`; the stored input is `inputs/two_loop.json` of the same folder). This is a numerical consistency check, not an interval bound. A separate 60-digit Cauchy evaluation, with radii 0.05 and 0.08 and 48 and 64 nodes, reproduces $b_{2,0}, \ldots, b_{2,6}$, including the eighth-order entry of Section 31.2.

For comparison only: in a regulator-independent MS scheme with $b_2 = -6A$, $c_2 = -A$ and $A = A_2 = 4\log 2$ (176), the Lagrange formula (593) at $n = 7$ gives the seventh-order counterpart of (594),

$$\begin{aligned} 6561\, C_7 = {} & 3c_7 - b_7 + 12672A^6 - 3360A^4 b_3 + 4320A^4 c_3 - 672A^3 b_4 + 1152A^3 c_4 \\ & + 192A^2 b_3^2 - 432A^2 b_3 c_3 - 112A^2 b_5 + 240A^2 c_5 + \frac{128}{3} A b_3 b_4 - 64A b_3 c_4 \\ & - 48A b_4 c_3 - 14A b_6 + 36A c_6 - \frac{4}{3} b_3^3 + 4 b_3^2 c_3 + \frac{8}{3} b_3 b_5 - 5 b_3 c_5 \\ & + \frac{4}{3} b_4^2 - 4 b_4 c_4 - 3 b_5 c_3. \end{aligned} \tag{792}$$

It is the polynomial of the 24 constant-coefficient test cases above. An exact symbolic expansion of (593) reproduces all 23 monomials of (792) and, at $n = 6$, (594). The polynomial must not be applied to the constant terms $\widehat{b}_{L,0}, \widehat{c}_{L,0}$ of the finite-scheme jets alone, which define a different truncated problem; the finite-scheme coefficient is extracted from the full jets, and independently by (772), whose form does not depend on d. In both schemes the seven-loop constants enter linearly, with $\partial C_7/\partial(b_7, c_7) = (-1, 3)/6561$, so that $C_7(2) \supset (3\widehat{c}_{7,0} - \widehat{b}_{7,0})/6561$ and the classes with $3w_B = w_A$ drop out at this order.

28.6 Primitive classes: completed periods and three products

A primitive class has $w_B = 0$: a vertex with $e_i \geq 2$ would leave the other L vertices with at least $2L - 2$ lines, a logarithmic subgraph, so each of four vertices carries one external leg and $s_G = 0$. For a primitive graph the forest sum (694) has only its root term, so by (695) and (611), with $a_G = (2/7)\Pi_G$,

$$C_{7,\mathrm{prim}}(2) = \frac{(3c_7 - b_7)_{\mathrm{prim}}}{6561} = -\frac{2}{6561} \sum_{G \text{ primitive}} w_A(G)\, \Pi_G. \tag{793}$$

Completing each of the 44 classes to a four-regular graph on nine vertices partitions them into 14 isomorphism classes, with periods $\Pi_0^{(7)}, \ldots, \Pi_{13}^{(7)}$; the equality of periods within a class is the completion invariance of Section 23. In the zero-based order of the completion dictionary `LRI_C7_D2_HPC/reference/LRI_C7_d2_progress/reference/primitive_completion.json`, the summed weights $\bar{w}_j = \sum w_A$ are

$$(\bar{w}_0, \ldots, \bar{w}_{13}) = \Big(27, \tfrac{27}{2}, 27, 18, 3, 108, 18, \tfrac{27}{4}, 108, 12, 108, 54, 54, 12\Big).$$

Three completed graphs have three-vertex separators: $\Pi_6^{(7)}$ has one, splitting it into two octahedra (four-loop factors); $\Pi_{12}^{(7)}$ has one, splitting it into K_5 and the completed five-loop primitive; and $\Pi_7^{(7)}$ has two, splitting it into three copies of K_5. The factorization behind (616), applied once or twice, gives

$$\Pi_6^{(7)} = P_4(2)^2, \qquad \Pi_7^{(7)} = P_3(2)^3 = 27 Q_2^3, \qquad \Pi_{12}^{(7)} = P_3(2) P_5(2) = 3 Q_2 P_5(2), \tag{794}$$

and therefore

$$\boxed{C_{7,\mathrm{prim}}(2) = -\frac{2}{6561}\Big[18 P_4(2)^2 + \frac{729}{4} Q_2^3 + 162\, Q_2 P_5(2) + \sum_{j \notin \{6,7,12\}} \bar{w}_j \Pi_j^{(7)}\Big].} \tag{795}$$

The remaining sum has eleven periods. The representatives, the weights and the map from the 44 classes to the 14 periods are in the completion dictionary and in `verification/primitive_completion_checks.json` of the same folder. At 40 working digits

$$\begin{aligned} Q_2 &= 12.771381198646957657749763684524008 4\ldots, \\ P_4(2) &= 250.487315581867285531851756887911407\ldots, \\ P_5(2) &= 1548.082640233120615132439073945789 34\ldots, \end{aligned}$$

and the three product terms of (795) contribute $-1436.355740622534051652319212811748 43\ldots$ to $C_7(2)$. These values of Q_2 and $P_4(2)$ agree with (268) and with the fifty-digit $P_4(2)$ of Section 17.2, and solving (272) for $P_5(2)$ with the 36-digit $C_5(2)$ of Section 20.9 reproduces $P_5(2)$ to 34 digits.

The earlier calculation used (795) as follows. Two independently randomized integrations of all 44 classes, each with 16 replicates of 4,096 points per class, measured the three product blocks; all six comparisons with the exact references lie within 1.88 sampling standard errors. The measured blocks were then replaced by the references, which gave the primitive contribution $-8657.4262196 \pm 1.2165092$ (the equal-weight mean of the two integrations, with its sampling standard error). It entered the earlier estimate once, as a retained input. The 2026 production integrates the 44 classes directly in every replicate and does not use the reduction, so (795) remains an exact identity available as a check. The product references have negligible numerical error on the Monte Carlo scale; no interval certification is asserted.

28.7 The 2026 production

The production uses an optimized, restartable reimplementation of the same calculation. Its kernel keeps the reference addition order and the extended-precision first pass, with promotions to binary128 and to the final tier at the unchanged thresholds 10^{-9} and 10^{-23}. The final tier uses 350-decimal GMP arithmetic instead of the original 300-decimal tier; this raises the working precision and was fixed before the production. On the production host both backends reproduce the original executable at all 3,511 graph kernels with zero normalized difference and identical promotion counts, and pass the independent high-precision point checks. Results do not depend on the thread count: 5,776 seven-loop rows run at 64 and at 16 threads were identical, including 31,663 promotions to the GMP tier.

The original executable of the earlier calculation was itself checked against independently ordered high-precision monomial sums: 69 point comparisons for power-free and 174 for power-containing kernels, with maximum errors, normalized by one plus the sum of the absolute forest terms, of 1.64×10^{-32} and 2.0×10^{-32} in binary128 and of 1.85×10^{-302} and 1.3×10^{-302} in the 300-decimal tier. It used compensated accumulation and strict floating-point flags, without fast-math, and shared monomial products in place of repeated exponentials, as in (733), which changes the rounding order but neither the precision nor the expression. These are pointwise arithmetic checks, not bounds on an integral.

Two independent campaigns a and b have 64 randomized replicates for every loop and sector, 52- and 60-bit digital coordinates, and importance rates multiplied by 0.8 and 0.9. Every stratum receives its frozen per-graph allocation from the earlier pilots, at least four points, multiplied by 2^{12}. The 44 primitive classes, which the earlier calculation retained from a separate integration, are integrated with 2^{26} points each per replicate. Counts are powers of two, the base seed is fresh, and a cost model from an independent pilot affects only the placement of tasks. The two campaigns contain 1,094,688,178,176 kernel samples; they ran on the symmetry cluster on 14 AMD and 14 Intel nodes in about 6.5 hours, and every merge and the assembly used the package's own tools.

The frozen design. The per-graph counts were fixed from pilot runs that never entered a result. Each stratum receives a count proportional to the square root of its projected contribution to the variance of C_7 per unit cost, the rule (739) with C_7 in place of C_6, rounded upward to a power of two and frozen before production, separately in the power-free and in the power-containing sector. Per replicate the frozen counts are

Sector	$L=3$	$L=4$	$L=5$	$L=6$	$L=7$	Total
Power-free, nonprimitive	10816	24704	97600	188480	251472	573072
Power-containing	32768	163840	226304	130688	240384	793984

together with 4,096 points for each primitive class (records `LRI_C7_D2_HPC/reference/LRI_C7_d2_progress/verification/frozen_allocation.json` and `LRI_C7_D2_HPC/reference/power_mass/inputs/production_plan.json`). With the factor 2^{12} and the 2^{26} primitive points of the 2026 production, a replicate has $2^{12}(573{,}072 + 793{,}984) + 44 \cdot 2^{26}$ points, and the two campaigns of 64 replicates contain the 1,094,688,178,176 samples quoted above; the group configurations `GFF/data/C7_2/run1-20260930-m4096/groups/*/config.json` hold these counts. The earlier calculation used the frozen counts themselves, with 16 replicates in each of two campaigns per sector, rate multipliers 0.8 and 0.9, and 24,105,472 power-free and primitive and 25,407,488 power-containing kernel evaluations. Its power-free coefficient $-3955.383753 \pm 76.541680$ (sampling standard error) includes the primitive contribution of Section 28.6 once; adding the power correction 1.750131 ± 5.424390 (sampling standard error) gave its estimate. There $4\widehat{\sigma} = 306.9$ exceeded both half-stream deviations, 89.3, and rounding up to a multiple of 10 gave the allowance 310 of Section 28.1.

The campaign means are combined with fixed weights 1/2. With J_L and $J_{f,L}$ the Jacobians of $\mathcal{F}$ at $f + p$ and at f, and $V_{f,L}, V_{p,L}$ the covariances of the campaign-mean input vectors, the complete coefficient and the correction have covariances

$$V_C = \sum_L J_L (V_{f,L} + V_{p,L}) J_L^{\mathsf{T}}, \qquad V_\Delta = \sum_L (J_L - J_{f,L}) V_{f,L} (J_L - J_{f,L})^{\mathsf{T}} + \sum_L J_L V_{p,L} J_L^{\mathsf{T}}. \tag{796}$$

The earlier calculation added a retained primitive variance once; the 2026 production has no shared retained input. The two campaigns and the two half streams give

Estimate	$C_7(2)$	Sampling SE
Campaign a	-3875.3202	0.7266
Campaign b	-3876.9320	0.7565
First half stream	-3876.4894	
Second half stream	-3875.7627	
Combined	-3876.1261	0.5244

The campaigns differ by 1.54 combined standard errors. The error is dominated by the power-free seven-loop graphs:

Sector	$L = 3$	$L = 4$	$L = 5$	$L = 6$	$L = 7$
Power-free	0.0007	0.0047	0.0200	0.0590	0.5187
Power	0.0000	0.0001	0.0013	0.0130	0.0440

Entries are in units of C_7 and combine in quadrature.

28.8 Checks

The same uncalibrated samples give lower-order coefficients, compared with the special-function values of Section 1 and, for C_6, with the two-dimensional production of Section 26.7:

Order	Estimate	Sampling SE	Pull
3	-0.8748513621	1.25×10^{-6}	-1.08
4	5.0488792961	3.01×10^{-5}	$+0.36$
5	-38.8344913344	5.07×10^{-4}	-1.67
6	362.3727456	0.0085778	$+0.73$

The reference values are checks only; no known coefficient is imposed and no power correction is forced to vanish. The measured power corrections at orders four to six are $(5.4 \pm 15.1) \times 10^{-8}$, $(1.10 \pm 1.85) \times 10^{-5}$ and $(3.3 \pm 11.8) \times 10^{-4}$; at order three the difference is at the level of rounding. These are consistent with zero but do not prove a cancellation at seventh order, where the correction is -0.056 ± 0.046. A bootstrap of 4,096 replicate resamples gives standard error 0.52219, against 0.52445 from covariance propagation. The new value lies 1.01 standard errors of the earlier calculation from its estimator, and its power correction lies 0.33 of that calculation's standard errors from its value 1.750131 ± 5.424390 (sampling standard error). Because the power sector is integrated explicitly here, the sixth-order by-product also tests the analytic power-sector compensation used at sixth order. These checks share the graph and renormalization framework and do not validate it. The eighth-order calculation of Section 31 gives $C_7(2) = -3875.55 \pm 0.29$ (sampling standard error) as a by-product, $+0.97$ combined standard errors from (777), and the independent code of Section 31.7 gives -3875.47 ± 0.35, $+1.04$.

Besides (796), the linearization $\delta C_7 = J\,\delta(f+p) = J_f\,\delta f + [(J - J_f)\,\delta f + J\,\delta p]$ gives the power-free variance and the cross-covariance of the power-free coefficient with the correction,

$$V_{\rm pf} = \sum_L J_{f,L} V_{f,L} J_{f,L}^{\mathsf T}, \qquad V_{f\Delta} = \sum_L J_{f,L} V_{f,L} (J_L - J_{f,L})^{\mathsf T}, \qquad V_C = V_{\rm pf} + V_\Delta + V_{f\Delta} + V_{f\Delta}^{\mathsf T}, \tag{797}$$

where the last relation is an exact identity. The record `GFF/results/C7_2/C7_result.json` gives $V_{f\Delta} = -5.09 \times 10^{-6}$ for C_7, a correlation of -2×10^{-4}, and $0.5224502^2 + 0.0458637^2 - 2 \cdot 5.09 \times 10^{-6} = 0.5244497^2$ reproduces the sampling standard error of Section 28.1. In the earlier calculation a retained primitive variance $\sigma^2_{\rm prim}$ entered V_C and $V_{\rm pf}$ once and canceled from V_Δ, so neither the old power-free error nor the common primitive error was averaged down by its two power campaigns.

29 Full seventh order in three dimensions

29.1 Result and scope

At $d = 3$, with $\epsilon = 2s - 3$, the seventh-order coefficient follows the construction of Section 28 with the three-dimensional kernel and normalization of Section 25. With β and γ_r as in (774), $\widehat{b}_1 = 3$ and $\widehat{c}_1 = 1$,

$$\Delta_{\mathcal E}(3, \epsilon) = \frac{3}{2} - \frac{\epsilon}{2} + \gamma_r(g_*, \epsilon), \tag{798}$$

and every loop order keeps its Taylor polynomial through z^{7-L} before $z = \epsilon$ is set. As at $d = 2$, and unlike the three-dimensional sixth-order production of Section 26.8, the power-containing sector is integrated explicitly, and $C_7(3)$ is split as in (775). The production of Section 29.3 gives

Contribution	Estimate	Sampling standard error
Power-free coefficient $C_7^{\rm pf}(3)$	-220.3723574	0.0277073
Power correction $\Delta_p C_7(3)$	0.0178769	0.0156478
Complete coefficient $C_7(3)$	-220.3544805	0.0318188

An earlier three-dimensional seventh-order calculation with the same worker, whose code and records are kept in `LRI_C7_D3_HPC/reference/` (Section 33), reported four sampling standard errors rounded upward, with its two campaigns and its two half streams (the estimates from the first and from the second half of the replicates of every group) as diagnostics. The same rule, rounded up to a multiple of 0.01, was written down before the production result was available and confirmed unchanged afterwards. Here $4\widehat{\sigma} = 0.127$, which gives

$$\boxed{C_7(3) = -220.35 \pm 0.13, \qquad \epsilon = 2s - 3.} \tag{799}$$

Both half-stream estimates deviate from $C_7(3)$ by 0.034, less than $4\widehat{\sigma}$, so the two-dimensional rule (776) gives the same allowance. The earlier $C_7(3)$ calculation reported -228 ± 16 (estimator -227.6181095, sampling standard error 3.8273969); the present estimator lies 1.90 of its standard errors from it. The allowance is empirical. It is not a probability guarantee and does not bound an error shared by the graph and renormalization framework; the additional estimator digits are given for reproduction only.

29.2 Kernel, normalization and coverage

The mixed forest kernel is (780) with every forest denominator U_F replaced by $U_F^{3/2}$, as in (707); the fractional power is applied to each forest term before the terms are summed. The edge measure is $\prod_e x_e^{-1/4+z/4}$ and the normalization is that of (708) and (710)–(712), now with Gamma jets through order $7-L$. The integrand jets follow (789) with $a_{L,j}$ now taken from (711); the logarithmic function $h(x)$ coincides with $s_L(x)$ of (710). In three dimensions an unmarked self-energy insertion has superficial degree 3/2 in momenta at $z=0$. The amplitude is even in the external momentum, so the constant Taylor subtraction again leaves a remainder of quadratic order and the local mass subtraction of Section 28 applies unchanged. Before subtraction a connected power face has radial exponent $3e/4 - 3h/2 = -3/4$ at $z=0$; the Taylor remainder raises it to $+1/4$.

The graph classes, rational weights, zero-channel exclusions and forest maps are those of Section 28. The integrated classes at loops three to seven number $5, 16, 76, 395$ and 2,298 (2,254 nonprimitive and 44 primitive) in the power-free sector and $1, 3, 20, 107, 590$ in the power-containing sector, 3,511 kernels in all. For every one of them the denominator polynomials and weights were checked to be unchanged from the two-dimensional inventory; only the dimension-dependent exponents differ. The two-loop input is (713), now through z^5, with $b_{2,5} = 26.0214190551330710$ in addition to (714); two 65-digit evaluations with radii $0.04, 0.07$ and 64,80 Cauchy nodes agree to 3.3×10^{-57}, and with the sixth-order input to 1.3×10^{-54}. Thirty independent Cauchy evaluations of the Gamma factors agree to 8.3×10^{-54}. The fixed point is solved as in Section 28 and substituted in (798); 64 exact rational cases test the assembly. The seven-loop constants enter only as $C_7(3) \supset (3\widehat{c}_{7,0} - \widehat{b}_{7,0})/6561$.

The single three-loop power graph has the closed form $\widehat{b}_3^{\mathrm{p}}(0) = -16\pi/15 = -3.3510322$, recorded with the earlier calculation, and $\widehat{c}_3^{\mathrm{p}} = \widehat{b}_3^{\mathrm{p}}/3$. The production below gives $\widehat{b}_3^{\mathrm{p}}(0) = -3.3510653 \pm 0.0000262$, -1.26 sampling standard errors from this value.

29.3 The 2026 production

The production worker is byte-identical to that of the earlier calculation. It keeps the extended-precision first pass, with promotions to binary128 and to 350-decimal GMP arithmetic at the unchanged thresholds 10^{-9} and 10^{-23}. Before the campaign, the worker replayed all 168,528 archived production rows on their archived nets and seeds. Of these, 96% reproduced bit for bit and the rest within 1.0×10^{-11} in normalized units, with identical promotion counts. The rebuilt estimator of the earlier calculation was -227.61810947469922, against its recorded value -227.61810947469945. An independent high-precision reimplementation of the sampler and integrand reproduced 13 archived rows across loops three to seven and both sectors. Results do not depend on the thread count: 5,776 seven-loop rows run at 64 and at 16 threads were identical, including 255,396 binary128 and 20,511 GMP promotions. The same rows computed on AMD and on Intel processors differ by at most 2.0×10^{-11} in normalized units, with identical promotion counts; this is about 10^{-7} of the relative statistical error.

Two independent campaigns a and b have 64 randomized replicates for every loop and sector, 52- and 60-bit digital coordinates, and importance rates multiplied by 0.8 and 0.7. Every stratum receives the frozen per-graph allocation of the earlier calculation, at least 64 points, multiplied by 2^{12}. The exception is the 44 seven-loop primitive classes. They are integrated with 2^{22} points

each per replicate, a flat allocation in place of their frozen counts, which changes only the variance. Counts are powers of two and the base seed is fresh. The two campaigns contain 867,784,720,384 kernel samples. They ran on the symmetry cluster on 14 AMD and 14 Intel nodes and completed on 1 October 2026; every merge and the assembly used the package's own tools. The covariances follow (796), and there is no shared retained input. The two campaigns and the two half streams give

Estimate	$C_7(3)$	Sampling SE
Campaign a	-220.3220	0.0431
Campaign b	-220.3870	0.0468
First half stream	-220.3207	
Second half stream	-220.3883	
Combined	-220.3545	0.0318

The campaigns differ by 1.02 combined standard errors. The error is dominated by the seven-loop graphs of both sectors:

Sector	$L=3$	$L=4$	$L=5$	$L=6$	$L=7$
Power-free	0.0000	0.0006	0.0034	0.0119	0.0248
Power	0.0000	0.0002	0.0014	0.0058	0.0145

Entries are in units of C_7 and combine in quadrature.

29.4 Checks

The same uncalibrated samples give lower-order coefficients, compared with the special-function values of Section 1 and, for C_6, with the three-dimensional production of Section 26.8, with both uncertainties combined:

Order	Estimate	Sampling SE	Pull
3	-0.3535806960	3.59×10^{-7}	$+2.30$
4	1.2280727249	8.52×10^{-6}	$+1.08$
5	-5.8503454757	1.52×10^{-4}	$+0.04$
6	33.5738672	0.0019782	-0.79

The reference values are checks only; no known coefficient is imposed and no power correction is forced to vanish. A pull above three in magnitude would have been reported without changing $C_7(3)$; none occurred. The measured power corrections at orders four to six are $(2.0 \pm 2.1) \times 10^{-6}$, $(4.0 \pm 6.4) \times 10^{-5}$ and $(-3.7 \pm 10.2) \times 10^{-4}$; at order three the difference is at the level of rounding. These are consistent with zero but do not prove a cancellation at seventh order, where the correction is 0.018 ± 0.016. A bootstrap of 4,096 replicate resamples gives standard error 0.03204, against 0.03182 from covariance propagation. The power correction lies 1.92 standard errors of the earlier calculation from its value -3.836223 ± 2.007117. Because the power sector is integrated explicitly here, the sixth-order by-product also tests the analytic power-sector compensation used for $C_6(3)$. These checks share the graph and renormalization framework and do not validate it.

30 Full eighth order in one dimension

30.1 Result and scope

The eighth-order coefficient uses the coordinate-forest construction of Section 27, extended by one order:

$$\boxed{C_8(1) = 8059767, \qquad \text{empirical allowance: } \pm 17000,} \tag{800}$$

that is, $C_8(1) = (8.060 \pm 0.017) \times 10^6$. The propagated sampling standard error is 4210; a bootstrap over the replicates of all independent streams gives 4140. The estimator is the mean of fourteen independent production batches of one frozen design (Section 30.3). Their individual values lie between 8041935 and 8089624, each with standard error between 15090 and 16343, and give $\chi^2 = 8.49$ for 13 degrees of freedom about the combined value. The allowance rule was fixed before production and was not changed: the largest of four sampling errors, twice the last refinement plus four of its errors, twice the degree eight to nine check step plus four of its errors, the half-split displacement and twice the last degree step of the exact loop-five input, rounded up to two significant figures. The four-error candidate, $4 \times 4209.58 = 16838.3$, is the largest. The number of production batches was not fixed in advance. Under the stopping rule set first, nine batches give 8061480 ± 5300 with allowance ± 22000; all fourteen planned batches were completed after interim results of five to eight batches had been seen (Section 30.3). As for the lower orders, the allowance is not a rigorous enclosure and does not bound the omitted orders of the ϵ series. The ratio $C_8(1)/C_7(1) = -25.0$ continues the sequence $-12.2, -16.1, -19.6, -22.5$ of the preceding ratios.

30.2 Changes relative to seventh order

The kernel, the binary cluster trees, the meromorphic radial quadrature, the zero-pole controls and the affine assembly (773) are unchanged. Each loop vector carries eight Laurent and Taylor slots per channel instead of seven, and loops up to eight are integrated. Since two insertions of loops five or higher first contribute beyond eighth order, $C_2, \dots, C_8$ are again affine in the loop-five to loop-eight jets. The affine map shows that C_8 depends on loop eight only through the combination $3c_8(0) - b_8(0)$: its two coefficients are in the ratio -3 to rounding, so the classes with $3w_B = w_A$ contribute exactly zero.

The nine-vertex inventory has 26,540 classes. Of these, 7,231 contain a power-divergent subgraph, 248 are primitive and power free, and 3,908 have an identically vanishing forest polynomial. The weighted forests have 2,507,176 coordinate terms. The enumerator and the forest builder are the seven-loop programs generalized to nine vertices with 128-bit packed multiplicity keys. Regenerating loops one to seven reproduces the archived inventories, forest arrays and exact-zero lists byte for byte. The coefficientwise comparison with bare-series inversion holds as a rational polynomial identity through eight loops. The exact class weights satisfy the zero-dimensional sum rules (16) at every loop order through eight. Every power-free class with an articulation vertex is an exact zero, so no class is excluded beyond the exact zeros and the vanishing top-loop channel.

Loops two to five are no longer sampled. Their vectors are deterministic sums over all $(L+1)!\,\mathrm{Cat}_L/2$ reflection-reduced ordered trees, each integrated with the full tensor quadrature: loops two to four at degree 24 for the constant K, and loop five at degrees $6, 8, 10, 12, 14$. The constant reproduces C_2 exactly and C_3 and C_4 to 1.3×10^{-12} and 1.1×10^{-9}. Since C_5 depends only on loops up to five, it is now a deterministic check: at the five degrees it differs from the exact value by -0.94, -0.13, -0.019, -0.0032 and -5.8×10^{-4}. The loop-five contribution to C_8 at degree 14 is -1297392.1, and its last degree step is $+2.8$.

Loops six to eight are sampled. The power-free sector uses the two independent randomized Sobol campaigns of Section 27. The power sector is telescoped as in (770), but to degree eight:

$$\mathbb{E}[Y_8] = \mathbb{E}[Y_1] + \mathbb{E}[Y_3 - Y_1] + \mathbb{E}[Y_5 - Y_3] + \mathbb{E}[Y_6 - Y_5] + \mathbb{E}[Y_7 - Y_6] + \mathbb{E}[Y_8 - Y_7], \tag{801}$$

with a paired degree eight to nine level that is never added to the estimate. With one tree per active eight-loop power graph, one replicate costs about 5, 14, 32 and 67 core-hours at degrees

six, seven, eight and nine. Degree eight is therefore affordable, whereas the seventh-order estimate stopped at degree six, where its truncation dominated.

30.3 Pilot, frozen design and production

An independent pilot of all thirty sampled campaigns fixed the design and trained the zero-pole controls on its coarse power campaigns; it never enters the estimate. It also exposed a sampling hazard. The seventh-order production gives 16372.8 for the non-primitive contribution to the C_7 row. With 64 points per graph and 16 replicates, the two independent seven-loop power-free campaigns of the pilot gave -16793.9 ± 9359.6 and -13230.8 ± 7406.2 (sampling standard errors), that is 33166.7 and 29603.7 below the production value, or 3.5 and 4.0 of their own standard errors. A repetition of the first campaign with 4096 points per graph and 32 replicates gives 16838.1 ± 724.1. That this value agrees to the unit with the four-error candidate 16838.3 of Section 30.1 is a coincidence: the two are unrelated quantities. The randomized estimator is unbiased, but at small point counts its replicate distribution is strongly skewed. The production therefore uses at least 4096 points per graph in every power-free campaign. Per-graph sample counts follow a Neyman allocation from the pilot, and the per-campaign multipliers are split into at most 128 replicates and a power-of-two sample multiplier.

The production consists of fourteen batches with fresh seeds, each of 498 work elements. The first four ran on SDSC Expanse in elements of 16 AMD EPYC 7742 cores and give 8066723 ± 8015 (sampling standard error). Ten further batches of the same design were then planned together, at first under a rule that started no new work after 48 hours and, once the running work had finished, included the batches then complete. That rule gives the first nine batches, 8061480 ± 5300 with allowance ± 22000. After interim results of five to eight batches had been seen, it was decided, on grounds of precision, to complete all fourteen planned batches; from then on every planned batch was to be included once complete, and all were. The design and the allowance rule are unchanged. Seven batches ran on Intel Xeon Gold 6148 nodes of the symmetry cluster and seven on Expanse; each batch ran on one processor type. The means of the two groups of seven differ by 7037 ± 8410. All campaigns are verified against the frozen design, including its counts, multipliers, numerical input hashes and the loop-five input. No seed of the pilot or of the control training is reused. A coverage guard requires every active class of loops six to eight to be integrated with total weight one at every level, including the check level. The calculation used about 9×10^4 physical core-hours (1.4×10^5 as charged, counting hyperthreads on the symmetry cluster), including development, the exact low-loop stage and the pilot.

The kernel was compared on identical tasks with the seventh-order implementation. In power mode the seven common slots agree bitwise at loops five to seven and degrees one to six; the randomized point evaluations agree to 8×10^{-17} relative. A binary128 replay of the contraction gives 2.5×10^{-17} of the row scale at loop eight.

30.4 Composition and lower-order checks

The contributions to C_8 are

Piece	Contribution to C_8	Sampling SE
Loops one to four, K	1531302.35	
Loop five, exact at degree 14	−1297392.1	
Power-free sector, loops six to eight	7879086.8	790.7
Power sector, degree one	316544.2	1690.3
Refinement $1 \to 3$	−364981.8	2887.4
Refinement $3 \to 5$	−4590.2	2346.3
Refinement $5 \to 6$	−26.5	528.4
Refinement $6 \to 7$	−202.2	321.0
Refinement $7 \to 8$	26.3	117.8
Check $8 \to 9$, not added	−113.3	68.1

The error is dominated by the eight-loop refinements $1 \to 3$ and $3 \to 5$ and by the eight-loop coarse level. The last two refinements and the check step are small and statistically consistent with zero. Adding the check step would move the estimate by −113, less than one per cent of the allowance; since it exceeds the last refinement in magnitude, no geometric tail beyond degree nine is fitted. The radial truncation is therefore negligible at the present precision, in contrast to seventh order. With the frozen zero-pole controls the estimate is 11078 above the uncontrolled estimator, 8048689, and its standard error falls from 23730 to 4210. The controls are fixed combinations of 30 Laurent pole coefficients that vanish exactly. Their standardized sample means are at most 2.18 in magnitude; the four above 2 form two pairs of columns that move together, and the squares sum to 39.8, which is only descriptive because the columns are correlated. Projected onto C_8, the control correction, $+11078 \pm 24734$, is consistent with zero, but these residuals alone do not exclude a nonzero pole expectation at the scale of the allowance. Errors of the controls at finite degree are refined with the levels and so enter the refinement terms of the allowance; a failure of the exact vanishing itself would be an error of the framework, which the allowance does not cover. The half-split estimates are 8055451 and 8064082.

The same samples give the lower-order coefficients, uncalibrated:

Coefficient	Estimate	Sampling SE	Reference	Difference
$C_5(1)$	−731.140052	deterministic	−731.139477 (exact)	-5.8×10^{-4}
$C_6(1)$	14332.8222	2.52	14331.68703 (deterministic)	+0.45 SE
$C_7(1)$	−322495.6	74.4	-322549.9 ± 10.1 (Sec. 27)	+0.72 SE

The C_7 comparison combines both sampling errors. Against the truncation-aware seventh-order value (753) the difference is 0.4 combined standard errors. The references are checks only and are not imposed. As at lower orders, these comparisons share the graph and renormalization framework and do not validate it.

31 Full eighth order in two dimensions

31.1 Result and scope

At $d = 2$, with $\epsilon = 2s-2$, the eighth-order coefficient extends the mixed local-mass forest calculation of Section 28 by one order. The renormalization functions of (774) are summed through eight loops,

$$\beta(g,z) = -zg + \sum_{L=1}^{8} \widehat{b}_L(z) g^{L+1}, \qquad \gamma_r(g,z) = \sum_{L=1}^{8} \widehat{c}_L(z) g^L, \tag{802}$$

with $\widehat{b}_1 = 3$, $\widehat{c}_1 = 1$ and $\Delta_{\mathcal{E}}(2,\epsilon) = 1 - \epsilon/2 + \gamma_r(g_*,\epsilon)$, and every loop order keeps its Taylor polynomial through z^{8-L} before $z = \epsilon$ is set. Both sectors are integrated explicitly, and in the notation of (775)

$$C_8(2) = C_8^{\mathrm{pf}}(2) + \Delta_p C_8(2), \qquad \Delta_p C_8 = \mathcal{F}_8(f+p) - \mathcal{F}_8(f). \tag{803}$$

The production of Section 31.4 gives

Contribution	Estimate	Sampling standard error
Power-free coefficient $C_8^{\mathrm{pf}}(2)$	45991.2684796	7.6797436
Power correction $\Delta_p C_8(2)$	−1.2658435	2.9224044
Complete coefficient $C_8(2)$	45990.0026361	8.2168210

The estimate with its sampling standard error is $C_8(2) = 45990.0 \pm 8.2$; the production record quotes this, together with the weight bounds and tail diagnostics of Section 31.5. As for the lower orders, the note quotes instead the reporting allowance of the rule (776) of Section 28, with half streams defined there. Here $4\widehat{\sigma} = 32.87$ exceeds both half-stream deviations, 2.39, and rounding up to a multiple of 1, two significant figures as for $C_7(2)$ and $C_7(3)$, gives

$$\boxed{C_8(2) = 45990 \pm 33, \qquad \epsilon = 2s-2.} \tag{804}$$

The rule was chosen after the result was available. No allowance was registered with the production decision, unlike the four-standard-error rules of $C_6(2)$ and $C_6(3)$, which were written down before their results (Sections 26.7 and 26.8). There is no earlier eighth-order calculation at $d = 2$. A separately developed code, run after this production, gives 46006.1 ± 13.4, $+1.03$ combined standard errors from it (Section 31.7). The exact-path pilot of Section 31.3, with an independent seed, gave 45620 ± 214, 1.73 combined standard errors from the production. The ratio $C_8(2)/C_7(2) = -11.86$ continues the sequence $-5.77, -7.69, -9.33, -10.70$ of the preceding ratios. The allowance is empirical. It is not a probability guarantee and does not bound an error shared by the graph and renormalization framework; the additional estimator digits are given for reproduction only.

31.2 Changes relative to seventh order

The mixed forest kernel (780), the power-sector normalization (781), the graph weights (778), the importance sampler and the default precision policy of Section 28.7 are unchanged. The native integrator is the seven-loop one generalized to sixteen propagators and 31-dimensional digital nets. It keeps an order-seven mode, in which it reproduces the seventh-order production executables bit for bit (Section 31.3).

Regulator jets and low-loop inputs. The integrand jets follow (789) with $n \le 8 - L$, so the maximum orders at loops two to eight are $6, 5, 4, 3, 2, 1, 0$. The Gamma table gains the column

$a_{L,6}$, used only by a two-loop control, and an eight-loop row with normalization $-2/\pi^8$, whose logarithms are not used. The two-loop input gains

$$\widehat{b}_{2,6} = -266.52096796135938173225219261908980094902\ldots, \tag{805}$$

from the same exact-hypergeometric numerator; its first six entries are unchanged. Two Cauchy contours at 75 digits agree to 3.1×10^{-66}, and a direct 130-digit Taylor series confirms the stored value to 7.7×10^{-68}. Every printed Gamma jet is the correctly rounded value of two independent Cauchy contours at 110 digits. The two-loop control graph, integrated by the production code, reproduces the seven entries of $\widehat{b}_2$ within 1.22 sampling standard errors.

Assembly. The physical assembly is a d-general eighth-order module shared with the three-dimensional calculation, used byte for byte and pinned by hash in both. Its floating-point inversion agrees with two exact rational inversions, the bivariate Lagrange formula and the coefficientwise solution of $\beta(g_*, \epsilon) = 0$, to 7.0×10^{-16} of the coefficient scale in 64 random eighth-order cases. With z-independent jets it reproduces the constant-coefficient MS polynomials through eighth order; the eighth-order polynomial, $19683\, C_8 = 3c_8 - b_8 + 42Ac_7 - 16Ab_7 + \cdots$ with 34 monomials after $b_2 = -6A$ and $c_2 = -A$, was derived independently twice and agrees exactly. At order seven the module reproduces the assembly of the seventh-order production (Section 28.7) to 2 ulp. The inversion fixes how the new data enter:

- the top slot of every loop enters additively, $\partial C_8/\partial(\widehat{b}_{L,8-L}, \widehat{c}_{L,8-L}) = (-1, 3)/3^{L+1}$ for $L = 3, \ldots, 8$, without changing a lower C_n. The eight-loop constants therefore enter only as $C_8(2) \supset (3\widehat{c}_{8,0} - \widehat{b}_{8,0})/19683$, through the channel weight $3w_B - w_A$;
- loop seven enters through both channels, $\partial C_8/\partial(\widehat{b}_{7,0}, \widehat{c}_{7,0}) = (-16A_2, 42A_2)/19683$, with $A_2 = -\widehat{b}_{2,0}/6 = 4\log 2$ of (176). This is not proportional to $(-1, 3)$, so the seven-loop classes with $3w_B = w_A$, which Section 28 excludes, now contribute through $-2A_2\widehat{b}_{7,0}/19683$;
- $C_2, \ldots, C_8$ are affine in the jets of loops five to eight taken together, since two insertions of loops five or higher first contribute beyond eighth order; loops two to four enter nonlinearly.

At the production means the three rules are hard gates: the top-slot and loop-seven derivatives agree with their exact values to 6×10^{-16} relative, and the response of C_8 to the jets of loops five to eight is linear to rounding.

The eight-loop sums are complete for $C_8(2)$ but not individually. Since the eight-loop classes with $3w_B = w_A$ are not integrated, the stored $\widehat{b}_{8,0}$ and $\widehat{c}_{8,0}$ omit contributions with $3\,\delta\widehat{c}_{8,0} = \delta\widehat{b}_{8,0}$, which cancel in $3\widehat{c}_{8,0} - \widehat{b}_{8,0}$. They are therefore not the eight-loop coefficients of β and γ_r, and a quantity that depends on them separately, such as the ϵ^8 coefficient of g_*, is not an output of this calculation. At loops three to seven no nonzero class is omitted, so those jets are complete. For the restored seven-loop classes the top-slot combination $3\widehat{c}_{7,1} - \widehat{b}_{7,1}$ vanishes by the weights, but both jets are integrated, as for every other class. The independent code of Section 31.7 also leaves out the eight-loop classes with $3w_B = w_A$, so the caveat applies equally to both codes.

Graph inventory and complete coverage. The nine-vertex inventory has the 26,540 classes of Section 30.2. Its $d = 2$ per-graph input files, called payloads below (line masks, forest terms, weights and importance rates), come from the seven-loop builders of Section 28, generalized to nine vertices with 128-bit class keys. The only dimension-dependent part, the rate table, uses the parity floor of the seven-loop power-free files: every proper edge subset has rate at least $1/2$ if its size is odd and at least 1 if it is even. Regenerating loops three to seven reproduces every production file byte for byte, except the rate lines of the power-free files at loops three to six, whose earlier rate tables are kept unchanged. The same payloads, with the rate line rewritten, serve the three-dimensional calculation.

L	All classes	Power-containing	Nonzero mixed kernels	Integrated classes power-free	power
3	8	1	1	5	1
4	27	5	3	16	3
5	129	31	20	76	20
6	660	173	107	395	107
7	3986	1091	709	2441	709
8	26540	7231	4891	16064	4336

At eight loops 2,254 classes have $3w_B - w_A = 0$ and 3,886 further classes have an exactly vanishing mixed forest map. The remaining 15,816 power-free nonprimitive, 248 primitive and 4,336 power-containing classes are integrated in every replicate. At seven loops all 3,150 classes with a nonvanishing map (2,397 nonprimitive, 44 primitive and 709 power-containing) are integrated, 262 more than in Section 28. In all, 24,173 kernels are integrated, about seven times the 3,511 of the seventh-order calculation, and there is no omitted nonzero class. An eight-loop kernel has up to 4,032 complement-of-tree monomials and 280 forest terms. The exactly vanishing kernels are exactly the classes with an articulation vertex (2,511 power-free and 2,340 power-containing at eight loops), and the exact class weights satisfy the zero-dimensional sum rules (16) at every loop order. An independent edge-subgraph enumeration rederives the mixed forest map of every power-containing class (1,091 at seven and 7,231 at eight loops) without a failure; exact rational Kirchhoff determinants agree with the stored kernels at random rational points for 144 seven- and eight-loop kernels; and an independent count of the compatible families of logarithmic subgraphs (nested, or sharing at most one vertex; the empty family included) gives 11,254,373 over all eight-loop classes, the total recorded with the inventory of the one-dimensional calculation, and agrees with the builders' count for each of the 19,309 power-free classes (`LRI_C8_shared/inventory/inventory8_d2/gates/k_forests.json`).

Power sector and campaigns. The power-containing jets follow (781) through eight loops. Their reporting follows a protocol registered before any eighth-order data. The differences $\Delta_p C_n$ at orders three to seven are null rows, that is, quantities expected to vanish: through sixth order the power sector cancels in the exact coefficients (Section 24), and at seventh order the measured difference is consistent with zero (Section 28). They are flagged when a pull exceeds three in magnitude but never used as gates, and the eighth-order correction is reported and never assumed to vanish. Two independent campaigns a and b again use importance rates multiplied by 0.8 and 0.9 and 52- and 60-bit digital coordinates, with group seeds derived by hashing from a fresh base seed. The campaign means are combined with fixed weights 1/2, and the covariances follow (796) through eight loops; there is no retained input.

31.3 Pilot, design and numerics

An exact-path pilot integrated all 24,173 kernels in both campaigns with 8 replicates of 4,096 points per graph, on three nodes of the symmetry cluster for 227 core-hours. It gave $C_8(2) = 45620 \pm 214$, with campaigns 0.64 standard errors apart and lower-order pulls at most 2.11 (at C_6). It fixed the design and never enters the estimate; its seed is excluded from the production. With its uniform allocation the power-free seven-loop graphs carried 71% of the variance at 2.4% of the cost, and the power-free eight-loop graphs 18% at 71%.

Per-graph counts follow a Neyman allocation, $n_g \propto \sigma_g/\sqrt{c_g}$ with the pilot's per-point standard deviations σ_g and measured costs c_g, on a power-of-two lattice with at least 4,096 points per graph and 128 replicates in every campaign, sector and loop. After the pilot the author chose, from a table of costs at four targets, a predicted standard error of 2.5×10^{-4} of $|C_8| \approx 4.6 \times 10^4$, that is 11.50, which includes a safety factor 1.5 on the raw prediction 7.67. The design, fixed before

the production, gives every power-free eight-loop graph at least 32,768 points per replicate and contains 779,836,456,960 kernel samples in one batch.

Exact path. The default numerics are the precision policy of Section 28.7: a long-double first pass, binary128 below the ratio 10^{-9} and 350-digit GMP arithmetic below 10^{-23}. In order-seven mode the default executables reproduce the seventh-order production executables bit for bit, on both backends, on the two-loop graph and all 3,773 kernels of loops three to seven, including the 262 new seven-loop ones; the order-eight slots common to order seven are identical. Each later kernel speedup (parallel graph loading, and inline correctly rounded binary128 addition, subtraction and multiplication, tested against the compiler library on 2^{32} random operand pairs per operation) had to reproduce the first unoptimized eight-loop build bit for bit on a sample of 1,600 power-free and 400 power-containing eight-loop kernels, on both backends, so the default path is no numerical change. The final executables passed these replays, the 450-digit point checks and the thread-identity tests again on the production cluster.

The faster numerics profile. The production used an opt-in profile, chosen by the author after the pilot and run only by an executable that passed an accuracy gate. Here v is the kernel value at a point and err the same forest sum with every numerator term replaced by its absolute value, so that $|v|/\mathrm{err}$ is the cancellation ratio of Section 25.4. A point that fails the long-double test is first evaluated in double-double arithmetic, with a 106-bit significand, and accepted when $|v| \geq 10^{-20}\,\mathrm{err}$; otherwise, or outside the double-double range, it goes to the unchanged binary128 tier. A point that reaches the multiprecision tier is first evaluated at 77 digits (320 bits) and accepted when $|v| \geq 10^{-63}\,\mathrm{err}$, and otherwise at the unchanged 350 digits. The gate compared the production executable with the exact path on the production hardware in four parts:

(i) on 153,600 seven- and eight-loop points with 450-digit references, every accepted value is within the exact path's bound of 10^{-6} relative, with largest errors 5.2×10^{-12} in double-double and 1.4×10^{-11} in binary128; every 77-digit value equals the 350-digit one after conversion;

(ii) a paired campaign on identical streams, with four replicates of 256 points for every kernel of loops three to eight in both campaigns (193,384 tasks), gives $|\Delta C_8| + 3\,\mathrm{SE}_\Delta = 4.2\times10^{-16}$ against the bound 0.062, one per cent of the standard error at the tightest target considered, with per-task normalized differences at most 5.6×10^{-14}.

(iii) the fractions of points promoted to each tier are unchanged;

(iv) the results are identical at 1, 8, 16 and 64 threads and in eight lanes of eight threads.

The profile runs about 2.5 times faster than the exact path at eight loops. The production design keeps the exact pilot's variances and only re-costs the allocation with a cost model measured under the profile, which (ii) justifies; it costs 0.37 of the exact path at the same target.

31.4 The 2026 production

The batch ran on the symmetry cluster on 6 and 7 October 2026, in lanes of eight threads on AMD EPYC 7543 nodes of one runtime identity: eight nodes for 22.4 to 23.0 hours in a first round and six nodes for 4.4 to 5.6 hours after the 24-hour limit. All 1,129 work indices, 6,192,256 tasks in 24,848 checkpoint blocks, ran once without a failure, in 28.7 hours from submission to the final merge and 13,454 allocated core-hours, 0.93 of the design's estimate. Every merge, the assembly and an independent recomputation of the result record used the package's own tools. With development, the pilot and the tail diagnostics the calculation used about 1.6×10^4 core-hours. The two campaigns and the two half streams give

Estimate	$C_8(2)$	Sampling SE
Campaign a	45993.58	12.15
Campaign b	45986.42	11.06
First half stream	45987.61	
Second half stream	45992.39	
Combined	45990.00	8.22

The campaigns differ by 0.44 combined standard errors. The error is dominated by the power-free eight- and seven-loop graphs:

Sector	$L=3$	$L=4$	$L=5$	$L=6$	$L=7$	$L=8$
Power-free	0.01	0.07	0.22	0.65	3.67	6.71
Power	0.00	0.01	0.11	0.43	1.02	2.70

Entries are in units of C_8 and combine in quadrature. The realized standard error is 1.07 times the raw design prediction, the size expected from per-graph standard deviations estimated with seven degrees of freedom, and well inside the safety factor; per stratum the realized variances at loops seven and eight are 0.98 to 1.32 times the predictions.

31.5 Weight bounds and tail diagnostics

The package's heavy-tail rule, calibrated on synthetic families at $d=3$, computes a fixed-k ($k=16$) Hill estimate [44] $\widehat{\xi}$ of the tail index of the per-point weights of every graph and campaign. At production scale it confirmed 4,832 eight-loop graphs (4,631 power-free and 201 power-containing) as possible infinite-variance cases, with two-campaign mean $\widehat{\xi} > 0.44$ at 32,768 or more points per replicate. Together they carry 27.6% of the variance of $C_8(2)$, and no single graph in one campaign more than 0.08%. Since an infinite variance would invalidate the replicate standard error, the weights were analyzed directly.

In an ordered sector the sampler draws successive scale gaps $z_i = u_i^{1/(\rho_c \alpha_G(S_i))}$, with u_i uniform, S_i the set of lines below the ith gap, $\alpha_G(S)$ the stored importance rate and $\rho_c = 0.8, 0.9$ the campaign multiplier. The weight is

$$t = Z_c\, K_G(x) \prod_i z_i^{|S_i|/2 - \rho_c \alpha_G(S_i)}, \tag{806}$$

with K_G the rational kernel (780) and Z_c a constant. Let $m_G(S) = |S|/2 + \nu_S(K_G)$ be the exact leading exponent of the line measure times K_G, after all forest cancellations, when the lines of a proper subset S are scaled to zero. Near that face $t \sim z^{m_G(S) - \rho_c \alpha_G(S)}$, so the weight is bounded if $m_G(S) \geq \rho_c \alpha_G(S)$ on every face and has a finite second moment if $2m_G(S) > \rho_c \alpha_G(S)$. The builders set $\alpha_G(S) = \max(r_G(S), p_S)/2$, where $r_G(S)/2$ is the smallest exponent of the individual forest terms and p_S is 1 for odd and 2 for even $|S|$. Where $r_G(S) > 0$ a sum cannot have a lower exponent than its terms, so $m_G(S) \geq \alpha_G(S)$. Where $r_G(S) \leq 0$ the forest sum must cancel; since K_G is rational, $m_G(S) \in |S|/2 + \mathbb{Z}$, and $m_G(S) > 0$ already implies $m_G(S) \geq \alpha_G(S)$. Once these cancellations are confirmed, $m_G(S) \geq \alpha_G(S) > \rho_c \alpha_G(S)$ on every face, and the weight is bounded at both rates. Nested limits are covered because every forest denominator U_F is the basis generating polynomial of a matroid, so that a single monomial dominates all nested limits of a sector at once; this rests on the builders' construction and was checked on random orderings.

The cancellations were verified at all 3,959,753 expanded subsets of the 24,173 kernels: every subset with $r_G(S) \leq 0$ of the kernels with the builders' rate tables, and every subset of the 492 power-free kernels at loops three to six, which keep their earlier tables. At each, $\sum_F N_F/U_F$ is expanded in the scaling parameter in exact arithmetic modulo four primes drawn uniformly from $[2^{61}, 2^{62})$, at two random points per prime and graph. A nonzero residue proves a nonvanishing

coefficient, so an error can only overestimate a valuation. With explicit degree and height bounds and the Schwartz–Zippel lemma, the union bound puts the probability of any wrong valuation below 10^{-53}, assuming independent uniform draws. Every stored rate table is reproduced, every required cancellation holds, and the eight trials agree everywhere. Nine archived controls behave as asserted: deleting a forest term or doubling a numerator breaks the cancellation, and rate tables multiplied by 3 or 2.4 give infinite variance with a finite mean at both rates, or at rate b only. Over all seven- and eight-loop kernels the margins are $m_G - \rho_c\alpha_G \geq 0.1$ and 0.05 and $2m_G - \rho_c\alpha_G \geq 0.6$ and 0.55 at rates a and b. Seventeen power-free kernels at loops five and six that keep their earlier rate tables have unbounded weights of finite variance.

The estimated sampling standard error is 8.22. Boundary analysis, supported by this multi-prime randomized cancellation verification of all 24,173 integrated kernels, supports finite second moments at both production rates and bounded weights for all seven- and eight-loop kernels; it is a randomized verification, not a deterministic certificate in characteristic zero. The 4,832 eight-loop tail warnings are consistent with bounded, sharply peaked weights. The confirmed power-free graphs have large canceling forest sums (median 128 terms, against 16 for the cleared graphs), and the fixed-k Hill estimate falls with the number of points in every class, with log-log slopes from -0.34 to -0.17; such finite-sample behavior cannot by itself prove boundedness. Fresh nested sampling supports variance stability in the selected main groups: over 2^{12} to 2^{17} points the slope of $\log(N\,\mathrm{Var})$ is -0.018 $[-0.051, +0.030]$ for 16 confirmed power-free and -0.062 $[-0.110, +0.060]$ for 8 confirmed power-containing graphs, with 95% intervals conditional on the selected panel. In the 380 power-containing eight-loop graphs with a subset that needs two orders of cancellation ("double lift"; 0.058% of the measured variance), a targeted test of the 16 largest contributors (52% of that class's measured variance) shows rare-peak sampling effects, with per-point variance growing over 2^{12} to 2^{19} points in the largest two. Inflating that class's variance by factors 18.9, 45 or 335 gives standard errors 8.26, 8.32 or 8.97; these variance-inflation calculations are sensitivity scenarios, not error bounds. Every graph stays in the estimand.

31.6 Checks

The same uncalibrated samples give lower-order coefficients, compared with the special-function values of Section 1 and, for C_6 and C_7, with the productions of Sections 26.7 and 28, with both uncertainties combined:

Order	Estimate	Sampling SE	Pull
3	-0.8748487258	1.24×10^{-6}	$+1.04$
4	5.0489759126	6.80×10^{-5}	$+1.58$
5	-38.8332791554	9.37×10^{-4}	$+0.39$
6	362.3625083	0.0116244	-0.34
7	-3875.545222	0.289774	$+0.97$

The squared pulls sum to 4.78 for five checks. The reference values are checks only; no known coefficient is imposed and no power correction is forced to vanish. The power null rows at orders four to seven are $(1.37 \pm 4.03) \times 10^{-6}$, $(7.7 \pm 23.5) \times 10^{-5}$, $(2.28 \pm 4.86) \times 10^{-3}$ and $(2.54 \pm 9.00) \times 10^{-2}$, with pulls at most 0.47; at order three the difference vanishes by the weights. These are consistent with zero but do not prove a cancellation at eighth order, where the correction is -1.27 ± 2.92. A bootstrap of 4,096 replicate resamples gives standard error 8.09, against 8.22 from covariance propagation. Because both sectors are integrated explicitly, the sixth-order by-product again tests the analytic power-sector compensation used for $C_6(2)$, and the seventh-order by-product, -3875.55 ± 0.29, is an independent sampling of the seventh-order integrands of Section 28. These checks share the graph and renormalization framework and do not validate it.

31.7 An independent code

A separately developed eighth-order code extends the seventh-order package of Section 28 independently of the package used above. It generates its own nine-vertex classes with 128-bit canonical keys and its own eight-loop payloads, which are different files from those of Section 31.2. It extends the kernel and the regulator jets to eight loops on its own, derives the two-loop and Gamma inputs separately, and solves for the eighth-order fixed point with its own assembly. It integrates the same 24,173 kernels: 16,064 power-free and 4,336 power-containing at eight loops, and 3,150 at seven. It shares with the calculation above the mixed forest kernel (780) and its precision policy, the importance sampler with rates multiplied by 0.8 and 0.9, the digital nets and the seed hashing, the finite renormalization scheme and, byte for byte, the payloads of loops three to seven. It runs only the exact path of Section 31.3 (long double, binary128 and 350-digit GMP), without the double-double and 77-digit tiers.

The independent code, its derivation of the eighth-order extension and the construction checks under the next four headings were written by OpenAI GPT-6 astra (model stated by the author), as were the caveat on the separate eight-loop sums in Section 31.2, the reading of the bootstrap and the reporting rule described below. They were delivered as the package `LRI_C8_D2_HPC_optimized/`, together with an original package that holds the graph payloads, the class generator and its check records (hosted, see `GFF/data/C8_2_check/DATA.md`). A Claude Code session ported the code to the clusters and ran the pilot and the production described afterwards. On 16 October 2026, when the delivered derivation was merged into this subsection, a Claude Code session also compared the shipped inputs of the two packages: the Gamma table of the independent code agrees with that of Section 31.2 to 65 digits, the precision of that table, and its two-loop jets $\widehat{b}_{2,0},\dots,\widehat{b}_{2,6}$ agree with those of the calculation above to 70 digits (largest relative difference 2.9×10^{-70}, at $\widehat{b}_{2,6}$).

Inputs. The Gamma table `inputs/gamma_d2_8.txt` has one row for each $L=2,\dots,8$. A row holds the order $8-L$, the normalization $2(-1)^{L+1}/\pi^L$ and the logarithms $a_{L,0},\dots,a_{L,6}$ of (700), with $a_{L,0}=0$; every row carries all seven logarithms, whether or not its loop uses them.

For the two-loop input, at $d=2$ the parameters of (198) are $a=\frac12+z/4$ and $b=\frac12+3z/4$. The jets are (790) of Section 28, extended to $j=6$: the numerator $z^2S_2(z)=4F_1(2,z)-2W(2,z)F_2(2,z)$ of (199) is analytic at $z=0$, where it equals 2, and the new entry $\widehat{b}_{2,6}$ needs its seventh Taylor coefficient. At $d=2$ the upper parameter $2a=1+z/2$ of F_2 equals its lower parameter $1+z/2$, and the series reduces to

$$F_2(2,z)={}_2F_1\left(\begin{matrix}z,\ b\\ b+\frac12\end{matrix};\frac14\right). \tag{807}$$

The code expands $F_1(2,z)$ and (807), both convergent at the argument $1/4$, in finite Taylor arithmetic in z at 120 decimal digits, where 183 terms of the series suffice, and it expands $\log W(2,z)$ analytically in polygamma values. The result is

$$\widehat{b}_{2,6}=-266.52096796135938173225219261908980094902003629215748\dots.$$

Two evaluations of the unreduced representation on Cauchy contours, of radius 0.05 with 128 nodes and of radius 0.08 with 160 nodes, differ from the Taylor coefficients by at most 5.26×10^{-112}. The first six coefficients are identical as binary64 numbers to the two-loop input of the seventh-order calculation and differ from it by at most 4.06×10^{-70} at high precision. The identity $\widehat{b}_{2,0}=-24\log2$, that is $A_2=4\log2$ in (176), holds exactly and reproduces $C_2(2)=4\log2/9$ of (192). The 100-digit values are in `inputs/two_loop.json`, written by `tools/analytic_inputs.py`; its check record is `verification/analytic_inputs.json` of the original package. These checks concern the special-function input, not the sampled integrals.

Graphs and payloads. The code enumerates the one-particle-irreducible classes with its own canonical keys of 128 bits. At eight loops the keys need up to 69 bits, and 5,068 of them exceed

64. Its class files of loops one to seven are byte for byte those of the seventh-order calculation. At every loop through eight the exact sums of w_A and w_B of (778) over the classes equal the zero-dimensional values, which the code computes separately from normal-ordered Gaussian moments and a Legendre transform. For each of the 4,810 classes of loops three to seven, in both sector inventories (9,620 rows), its forest builder reproduces the weights, the monomial masks, the forest maps and the exact-zero lists of the seventh-order files. The importance-rate arrays are left out of this comparison, because the archived power-free rate arrays were changed after they were generated; the archived rates of loops three to seven are used as supplied, and the rates of the new eight-loop payloads follow the rate formulas of the seventh-order builder. The rates affect only the efficiency of the sampling, since the integral is defined by the complete weighted kernel. At eight loops an edge-subgraph enumeration, which includes the subgraphs not induced by their vertex sets, rederives the mixed forest maps of all 7,231 power-containing classes, 10,451,732 forests in all, without a failure. Thirty-six exact rational evaluations on 18 eight-loop kernels agree with Kirchhoff determinants of the contracted forest blocks.

The census agrees with Section 31.2, with the exclusions taken in the other order. Of the 19,309 power-free and 7,231 power-containing eight-loop classes, 2,511 and 2,340 have an exactly vanishing kernel. Of the remaining 16,798 and 4,891 nonzero kernels, 734 and 555 have $3w_B = w_A$, and 16,064 and 4,336 are integrated (`inputs/coverage.json`). The 4,851 exact zeros and 1,289 channel exclusions are the $2{,}254 + 3{,}886 = 6{,}140$ excluded classes of Section 31.2. At seven loops all 2,441 nonzero power-free kernels, 44 of them primitive, and all 709 nonzero power-containing kernels are integrated, including the 262 classes that the seventh-order production excluded by $3w_B = w_A$ (the split table of Section 28); they are already in the seventh-order reference inventories, although its production payloads omit them. The exact-zero list is computed from the mixed kernel (780) itself. The 3,908 classes with an identically vanishing coordinate-forest polynomial in the one-dimensional calculation (Section 30.2) belong to a different kernel and are not used; the one-dimensional inventory serves only as a check of class counts and topology.

Kernel arithmetic. Besides the 350-digit GMP final tier used in the production, the kernel has an alternate final tier in Boost decimal arithmetic at 300 digits. In order-seven mode, at eight points in each of two streams on all 3,773 kernels of loops three to seven, including the 262 restored ones, all 60,368 point contributions reproduce the arithmetic of the seventh-order kernel exactly on both backends. At 37 points per backend, among them physical eight-loop kernels and cancellation fixtures, the kernel values agree with an independent 450-digit evaluator to 4.55×10^{-304} (decimal) and 2.22×10^{-354} (GMP), in units of the absolute-term scale err of Section 31.3. These are arithmetic comparisons, not interval bounds on the integrals. Restart, chunking, complete coverage, immutable inputs and idempotent merges were tested on native runs. The records are `verification/kernel_checks.json` and `generator/reports/` of the original package.

Assembly. The assembly evaluates the bivariate Lagrange formula (704) through $n = 8$, with $\widehat{B}(g, z) = 3 + \sum_{L=2}^{8} \widehat{b}_L(z) g^{L-1}$ and γ_r of (802). Separately, it solves $g = \frac{1}{3}\big[\epsilon - \sum_{L=2}^{8} \widehat{b}_L(\epsilon) g^L\big] \bmod \epsilon^9$ by nine iterations from $g = 0$ and substitutes the result in $\Delta_{\mathcal{E}} = 1 - \epsilon/2 + \gamma_r(g, \epsilon)$. In 24 exact rational cases with regulator-dependent jets the two constructions agree exactly, and the floating-point assembly used by the production agrees with them to a normalized difference of 2.7×10^{-15}. The same test, `tests/assembly_formula.py` (record `verification/assembly_formula.json` of the original package), confirms $\partial C_8 / \partial \widehat{b}_{8,0} = -1/19683$ and $\partial C_8 / \partial \widehat{c}_{8,0} = 1/6561$, the contribution at eighth order of the seventh-order classes with $3w_B = w_A$, the dependence of C_8 on the new entry $\widehat{b}_{2,6}$, and the affine dependence on the jets of loops five to eight, all as derived in Section 31.2.

Development runs (superseded). Before the port, OpenAI GPT-6 astra ran the delivered code in three complete-coverage runs of its own, each with two campaigns a and b and small point counts. A pilot gave 48626 ± 2633. A refinement with 8 replicates per group and at least 32 points

per eight-loop graph and replicate gave 47068 ± 1854. A second refinement with 16 replicates and at least 128 power-free and 64 power-containing points per eight-loop graph and replicate gave 45512 ± 524, quoted with the code as 45510 ± 520 (sampling standard errors throughout). In the last run the two campaigns differed by 2.26 combined standard errors, the power correction was 153 ± 122, and in 21 of the 25 graphs with the largest variances a single replicate carried at least 80% of the squared deviations in its campaign. This is the small-sample hazard also met at one dimension (Section 30.3). The three values lie $+0.99$, $+0.57$ and -0.94 combined standard errors from the production below. They are superseded by it and enter neither its allocation, which was computed from the pilot below alone, nor its estimate. The records of the first two runs are in the original package; those of the third were not delivered.

Port, pilot and production. The code was adapted to the Expanse cluster without a change to its numerical files. Its optimized kernel reproduced the package's 52 frozen fixtures bit for bit on the production hardware. It used a fresh base seed, checked against the seeds of this section. A pilot with 32 replicates per campaign gave 46178 ± 417 and fixed a Neyman allocation at 28,500 kernel core-hours. The allocation has 64 replicates in every campaign, sector and loop, at least 8,192 points per replicate for every power-free and 4,096 for every power-containing eight-loop graph, and a predicted standard error of 14.97. The production ran on 9 and 10 October 2026 on eleven whole AMD EPYC 7742 nodes for 16.3 to 20.1 hours each. All 342 work indices ran once without a failure, in 207 node-hours (26,526 core-hours), 96.5% of it kernel time.

Estimate	$C_8(2)$	Sampling SE
Campaign a	46012.55	19.70
Campaign b	45999.73	18.23
First half stream	46008.16	
Second half stream	46004.11	
Combined	46006.14	13.42

The campaigns differ by 0.48 combined standard errors, and a bootstrap of 4,096 replicate resamples gives standard error 13.38. The power-free part is 46004.84 ± 12.90 and the explicitly integrated power correction $+1.29 \pm 3.71$. The power-free eight- and seven-loop graphs carry 78% and 14% of the variance, and no single graph more than 0.2%. At orders three to seven the lower-order by-products have pulls -0.01, $+1.75$, $+0.57$, -1.18 and $+1.04$ against the references of Section 31.6; the squared pulls sum to 5.88 for five checks. The power null rows at orders four to six have pulls of at most 0.76.

The bootstrap resamples the 64 replicates of each group. For a mean of R replicates the plug-in bootstrap variance is $(R-1)/R$ times the unbiased estimate (726). To first order the bootstrap standard error is therefore $\sqrt{(R-1)/R}$ times the propagated one: 0.992 here, against the observed $13.38/13.42 = 0.997$, and 0.996 for the 128 replicates of the calculation above, against $8.09/8.22 = 0.984$ (Section 31.6). The finite number of resamples and the nonlinearity of $\mathcal{F}_8$ also enter these ratios. The bootstrap mean differs from the estimator by -0.11, that is -0.008 standard errors (`GFF/results/C8_2_check/prod1/C8_result.json`). Neither a failure of the first-order propagation (796) nor a finite-sample bias of the nonlinear map is therefore visible at this precision. Resampling adds no integration points and cannot reveal regions that no replicate sampled.

Comparison with the calculation above. The combined estimate differs from the estimator of (804) by $+16.1 \pm 15.7$, that is $+1.03$ combined standard errors, well inside the allowance ± 33. The power-free parts differ by 0.90 and the power corrections by 0.54 combined standard errors. The agreement tests the parts that were developed independently: the eight-loop graph generation and payloads, the eighth-order kernel and jet extension, the analytic inputs, the assembly and

the sampling. It does not test what the two codes share. An error in the seventh-order kernel construction, the precision policy, the renormalization scheme or the payloads of loops three to seven would be common to both. The independent estimate is not combined with (804), which is unchanged.

No reporting allowance is assigned to the independent estimate, and its result record carries none. Under the reporting rule that came with the independent code, an allowance and its rule are fixed only after fresh campaigns, half streams, successive refinements, precision-tier replays and the lower-order by-products have been compared, because a sampling standard error alone does not establish final digits; no allowance was fixed for the run above. The estimate therefore enters this note only through its sampling standard error, as a check. The weight bounds of Section 31.5 were established for the rate tables of the payloads of the calculation above. The independent eight-loop payloads carry their own rate tables, built from the seventh-order rate formulas, and the bounds are not claimed for them here.

31.8 Records

The result record, the production, tail and pilot reports and the package are listed in Section 33. The production's plan, design bundle and the merged record of each of its 24 groups, from which the estimator and every number of this section are recomputed, are in `GFF/data/C8_2/`. The full campaign directories of the production `prod1-2.5e-4-ddmp256-20261006-ba68f401e46e` and of the pilot `pilot1-20261006-ba68f401e46e`, with every checkpoint, are archived as listed in `GFF/data/C8_2/DATA.md`. The records of the independent code of Section 31.7 are in `GFF/data/C8_2_check/` and `GFF/results/C8_2_check/`; its full campaign directory is hosted (`GFF/data/C8_2_check/DATA.md`).

32 Full eighth order in three dimensions

32.1 Result and scope

At $d = 3$, with $\epsilon = 2s - 3$, the eighth-order coefficient extends the calculation of Section 29 by one order, as Section 31 extends Section 28. The renormalization functions are summed through eight loops, $\Delta_{\mathcal{E}}(3, \epsilon)$ is (798), and every loop order keeps its Taylor polynomial through z^{8-L} before $z = \epsilon$ is set. Both sectors are integrated explicitly and $C_8(3)$ is split as in (803). The production of Section 32.4 gives

Contribution	Estimate	Sampling standard error
Power-free coefficient $C_8^{\mathrm{pf}}(3)$	1601.9999290	0.7279030
Power correction $\Delta_p C_8(3)$	0.6052940	0.6129303
Complete coefficient $C_8(3)$	1602.6052230	0.9516489

The quoted uncertainty is an allowance, not a standard error. It follows the rule (776), also used for $C_7(3)$ and $C_8(2)$: the larger of four sampling standard errors and the deviations of the two half-stream estimates (computed from the first and from the second half of the replicates of every group), rounded up, here to a multiple of 0.1, which keeps two significant figures as for $C_7(3)$ and $C_8(2)$. The rules for $C_6(3)$ and $C_7(3)$ were written down before their production results. This one was chosen after the result was available and confirmed by the author; no allowance was registered with the production decision. Here $4\widehat{\sigma} = 3.81$ exceeds both half-stream deviations, 0.046, and the rule gives

$$\boxed{C_8(3) = 1602.6 \pm 3.9, \qquad \epsilon = 2s - 3.} \tag{808}$$

There is no earlier eighth-order calculation at $d = 3$. The pilot of Section 32.3, with independent seeds, gave 1619 ± 16, 1.05 combined standard errors from the production. The ratio $C_8(3)/C_7(3) = -7.27$ continues the sequence $-1.95, -3.47, -4.76, -5.74, -6.56$ of the preceding ratios. A separately developed code gives 1600.98 ± 3.97 (sampling standard error), -0.40 combined standard errors from the estimator (Section 32.7). The allowance is empirical. It is not a probability guarantee and does not bound an error shared by the graph and renormalization framework, and its coverage is conditional on the finite variance discussed in Section 32.5. The additional estimator digits are given for reproduction only. The production record quotes the sampling standard error alone, $C_8(3) = 1602.6 \pm 1.0$, together with the tail diagnostics of Section 32.5; the reported uncertainty is the allowance ± 3.9 of (808).

32.2 Changes relative to seventh order

The three-dimensional kernel, edge measure and normalization of Section 29 are unchanged: every forest denominator is raised to $U_F^{3/2}$ before the forest terms are summed, and the Gamma jets (711) now run through order $8 - L$. The graph inventory, the mixed forest maps, the weights, the zero-channel rule and the shared eighth-order assembly are those of Section 31.2, used with the three-dimensional rate lines; the integrated classes at loops three to eight number $5, 16, 76, 395, 2{,}441$ and 16,064 in the power-free sector and $1, 3, 20, 107, 709$ and 4,336 in the power-containing sector, 24,173 kernels in all. The worker is the three-dimensional seven-loop worker generalized to sixteen propagators and 31-dimensional digital nets, with the extended-precision first pass and promotions to binary128 and to 350-digit GMP arithmetic at the unchanged thresholds 10^{-9} and 10^{-23}. In its order-seven mode it reproduces the unchanged seventh-order worker bit for bit on the two-loop graph and all 3,511 kernels of loops three to seven, and it replays the archived seventh-order production. Results do not depend on the thread count and differ between processor types only in the last bits, at most 2.0×10^{-11} in normalized units between AMD and Intel processors; every production job ran on one processor model, AMD EPYC 7543.

Low-loop inputs. The two-loop input gains

$$\widehat{b}_{2,6}(3) = -29.258124480916806558671227221989158045345\ldots, \tag{809}$$

from (713), now through z^6; its first six entries are those of Section 29. Two Cauchy contours at 65 digits agree to 2.7×10^{-56}, and a 95-digit reference contour confirms the stored entries to 1.7×10^{-57}. Every printed Gamma jet, 42 in all, is the correctly rounded value of two independent Cauchy contours at 100 digits. The single three-loop power graph has the closed form $\widehat{b}_3^{\mathrm{p}}(0) = -16\pi/15$; the production gives -3.3509158 ± 0.0000894, $+1.30$ sampling standard errors from it.

Assembly gates. At the production means the top slots enter with $\partial C_8/\partial(\widehat{b}_{L,8-L}, \widehat{c}_{L,8-L}) = (-1, 3)/3^{L+1}$ to 4×10^{-16} relative, so the eight-loop constants enter only as $C_8(3) \supset (3\widehat{c}_{8,0} - \widehat{b}_{8,0})/19683$; loop seven enters through $(-16A_3, 42A_3)/19683$ with $A_3 = -\widehat{b}_{2,0}/6 = 1.6309960686$, to 8×10^{-16}; and the response of C_8 to the jets of loops five to eight is linear to 3×10^{-13}. These three gates are hard; all pass.

32.3 Pilot and design

A staged pilot fixed the design and never enters the estimate; its seeds are excluded from the production. A screen at 4,096 points per replicate (2,732 eight-loop graphs) selected the heavy-tail candidates. A full pilot integrated all 24,173 kernels in both campaigns with 8 replicates of 4,096 points per graph, 16 replicates for the power-containing eight-loop graphs, and gave $C_8(3) = 1619 \pm 16$. A confirmation pilot integrated 710 selected eight-loop graphs at 32,768 points

per replicate, within a cap of a quarter of the full pilot's eight-loop cost. The pilot used 422 core-hours.

Per-graph counts follow the Neyman allocation of Section 31.3 on a power-of-two lattice with at least 4,096 points per graph and 128 replicates in every campaign, sector and loop. For the 710 confirmed graphs the per-point standard deviation is the larger of the full-pilot and confirmation estimates. After the pilot the author chose a target standard error of 1, about 6×10^{-4} of $C_8(3)$, and the cheapest of three power-of-two designs, with a predicted standard error of 0.87 (1.31 with the safety factor 1.5 of Section 31.3) and an estimated cost of 15,492 core-hours, to be topped up if the realized error exceeded 1. Two independent campaigns a and b use importance rates multiplied by 0.8 and 0.7, as in Section 29, and 52- and 60-bit digital coordinates, with group seeds derived from a fresh base seed. The design contains 365,441,318,912 kernel samples.

32.4 The 2026 production

The production ran on the symmetry cluster from 9 to 12 October 2026, in lanes of eight threads on whole AMD EPYC 7543 nodes, at most eight at a time, on ten nodes in all. All 1,941 work indices and 25,714 checkpoint blocks completed. The lanes used 14,970 core-hours, 0.97 of the design's estimate, and the merges 106; with the plan job and the rate probes of Section 32.5 the production used 15,102 core-hours. Every merge, the assembly and an independent re-verification of the result record used the package's own tools. The two campaigns and the two half streams give

Estimate	$C_8(3)$	Sampling SE
Campaign a	1604.684	1.239
Campaign b	1600.527	1.445
First half stream	1602.651	
Second half stream	1602.559	
Combined	1602.605	0.952

The campaigns differ by 2.18 combined standard errors. Linearized with the assembly Jacobian, the difference 4.16 ± 1.90 is spread over loops and sectors, $+1.36$ and $+1.39$ from the power-free and power-containing eight-loop graphs, $+0.90$ from the power-free seven-loop graphs and $+0.19 \pm 0.10$ from the power-free five-loop graphs, with no term above two of its own standard errors. Over the eight-loop graphs the per-graph campaign differences have χ^2 per graph 0.994 (16,064 power-free) and 1.010 (4,336 power-containing), the ten largest carry $+0.19$ and -0.07, and no class of tail index carries the difference. No concentrated discrepancy is therefore seen. This is consistent with a fluctuation, but it does not exclude a small coherent effect of the importance rates spread over many graphs, and it does not by itself establish that the per-graph standard errors are calibrated. The error is dominated by the eight- and seven-loop graphs of both sectors:

Sector	$L=3$	$L=4$	$L=5$	$L=6$	$L=7$	$L=8$
Power-free	0.001	0.010	0.048	0.160	0.362	0.609
Power	0.000	0.003	0.018	0.079	0.220	0.566

Entries are in units of C_8 and combine in quadrature. The realized standard error is 1.09 times the design prediction.

32.5 Tail diagnostics

The fixed-k ($k = 16$) Hill estimate $\widehat{\xi}$ of Section 31.5 listed 10,034 eight-loop graphs with two-campaign mean $\widehat{\xi} > 0.44$ at production length. No graph carries more than 1.65% of the variance of

its group, and none was flagged above 3%. The listed graphs carry 62% of the power-free and 99.5% of the power-containing eight-loop variance, so the finiteness of their variance matters for the quoted error. The exact weight bound of Section 31.5 does not carry over, because the three-dimensional kernel contains $U_F^{3/2}$. Two empirical tests replace it.

First, the importance rates were re-derived on 48 graphs: those with the largest variance shares, representatives of the double-lift classes, two controls and the 16 power-containing eight-loop graphs whose $\widehat{\xi}$ did not fall in the pilot. No asymptotic rate deficit was found. Five apparent deficits were crossovers of the integrand from $t^{1.5}$ to $t^{1.0}$, with a zero of the integrand in between, and the asymptotic exponent 1.0 equals the table rate. This is a probe at one base point, not a certificate.

Second, a true heavy tail keeps $\widehat{\xi}$ constant as the number N of points per replicate grows, while bounded, peaked weights make it fall. For the 441 graphs confirmed in the pilot as possible infinite-variance cases (246 power-free, 195 power-containing), $\widehat{\xi}$ fell from $N = 4{,}096$ to $N = 32{,}768$ as about $N^{-0.23}$ and $N^{-0.18}$. The production, with independent seeds, mostly at intermediate lengths $N = 8{,}192$ and 16,384, reproduces this dependence: the medians of $\widehat{\xi}$ lie at 0.996 to 1.031 of the falling trend interpolated between the pilot lengths, 13 to 38% above the pilot's value at $N = 32{,}768$, which a true heavy tail would keep. The 13 graph-campaigns run beyond the pilot's range, at $N = 65{,}536$, continue to fall, to medians of 0.82 to 0.87 of their $N = 32{,}768$ values, at 0.94 to 1.02 of the extrapolated trend. At matched sample length the campaign shift of $\widehat{\xi}$ in the power-free sector is positive, +0.03 to +0.05, opposite in sign to the shift a rate-limited tail would produce; in the power-containing sector it is inconclusive. The data therefore behave as bounded, peaked weights would. This is empirical evidence, not a bound.

32.6 Checks

The same samples give lower-order coefficients, compared with the special-function values of Section 1 and with the productions of Sections 26.8 and 29, with both uncertainties combined:

Order	Estimate	Sampling SE	Pull
3	-0.3535809354	1.38×10^{-6}	$+0.43$
4	1.2280445577	3.88×10^{-5}	-0.49
5	-5.8499922821	4.86×10^{-4}	$+0.74$
6	33.5763740	0.0050853	$+0.19$
7	-220.3847628	0.0528950	-0.49

The reference values are checks only; no known coefficient is imposed and no power correction is forced to vanish. A pull above three in magnitude would have been reported without changing $C_8(3)$; none occurred. The measured power corrections at orders four to seven are $(8.6 \pm 7.3) \times 10^{-6}$, $(9.2 \pm 13.1) \times 10^{-5}$, $(3.3 \pm 19.5) \times 10^{-4}$ and 0.009 ± 0.024; at order three the difference is at the level of rounding. These are consistent with zero but do not prove a cancellation at eighth order, where the correction is 0.61 ± 0.61. A bootstrap of 4,096 replicate resamples gives standard error 0.940, against 0.952 from covariance propagation. These checks share the graph and renormalization framework and do not validate it.

32.7 An independent code

The eighth-order code of this subsection, `LRI_C8_D3_OPT/`, and its derivation, given below, were written by OpenAI GPT-6 astra (model stated by the author), which was given the seventh-order package of Section 29 and an earlier version of this note; the cluster port and the production were done within this project. The code extends that package independently of the package used above. It regenerates the graph inputs at every loop, restores the seven-loop classes excluded at

seventh order, extends the integrator and the regulator jets to eight loops on its own, derives its own low-loop inputs, and solves for the eighth-order fixed point with its own assembly; it integrates the same 24,173 kernels. It shares with the calculation above the three-dimensional mixed forest kernel and the precision policy of the seventh-order integrator, the importance sampler with rates multiplied by 0.8 and 0.7, the digital nets and the finite renormalization scheme. An error in those shared parts would not be detected.

Graph inputs. The code enumerates its own classes of the graph family of Section 28: connected one-particle-irreducible multigraphs without self-lines, with $L+1$ vertices, $2L$ individually labeled lines and at most four lines at each vertex. Classes are identified by a canonical form from partition refinement, keyed by multiplicity strings, and their counts agree at every loop with those of Section 31.2. An independent nauty computation on two-color incidence graphs (vertices and labeled lines) confirms all 26,540 eight-loop classes and their automorphism orders, and reproduces the archived lower-loop inventories with their exact weights. At eight loops the class weights sum to $\sum_G w_A = 49{,}985{,}915/768$ and $\sum_G w_B = 33{,}994{,}099/6{,}912$, the zero-dimensional four-point and mass-insertion values of (16) in class-weight form.

Its census of exact zeros and channel exclusions is that of Sections 31.2 and 31.7, which also explain why its 4,851 eight-loop zeros differ from the 3,908 of Section 30.2; it integrates $26{,}540 - 4{,}851 - 1{,}289 = 20{,}400$ eight-loop kernels, all exclusions being exact. At seven loops the code restores the 143 power-free and 119 power-containing nonzero classes with $3w_B = w_A$ that Section 29 omits, giving 2,441 and 709 kernels. Its builders reproduce all 3,511 payloads of the seventh-order calculation (loops three to seven), that is, line masks, denominator lists, integer numerators and importance rates, byte for byte after the header line, whose weights are checked against the exact rational values. A coverage record binds the census to the active inventory by its SHA-256 hash, and the assembly refuses to run without it.

Non-induced divergent subgraphs. A subgraph is induced if it contains every line of G between its vertices. The code builds its forests from induced divergent subgraphs only, as do the production builders whose maps the edge-subgraph enumerations of Sections 28 and 31.2 confirm. With parallel lines a graph can also have proper divergent subgraphs that are not induced. The forest terms that contain them cancel exactly, by the following argument. It uses only the graph family and the structure of the forest sum (780) with any fixed power of U_F in its denominators, so it holds in every dimension in which this sum is used, at $d = 2$ and, with $U_F^{3/2}$, at $d = 3$.

Let W be a proper nonempty subset of the vertices of G, $E(W)$ the set of lines with both ends in W and ∂W the set of lines with exactly one end in W. Since G is connected and has no bridge, $|\partial W| \geq 2$, and the degree bound gives

$$2|E(W)| + |\partial W| = \sum_{u \in W} \deg_G(u) \leq 4|W|, \qquad \text{hence} \qquad |E(W)| \leq 2|W| - 1. \tag{810}$$

A proper divergent subgraph does not contain every vertex: on all $L+1$ vertices a logarithmic subgraph would need all $2L$ lines, and a power subgraph one line more than G has. A power subgraph on v vertices has $2v-1$ lines, the maximum allowed by (810), so every proper power subgraph is induced. A proper divergent subgraph γ that is not induced is therefore logarithmic, with $2v-2$ lines on its v vertices, and misses exactly one line ℓ_0 of its induced closure $\bar{\gamma}$, the subgraph of all $2v-1$ lines between the vertices of γ. Adding a line to the one-particle-irreducible γ keeps it one-particle irreducible, so $\bar{\gamma}$ is a proper power subgraph, with loop number $h_{\bar{\gamma}} = v$.

Let m be a complement-of-tree monomial kept by γ, that is $|m \cap E_\gamma| = h_\gamma = v - 1$. The complementary spanning tree then has $v-1$ lines in γ, which form a spanning tree of the vertices of γ, so it cannot also contain ℓ_0, which would close a cycle. Hence $\ell_0 \in m$ and $|m \cap E_{\bar{\gamma}}| = v = h_{\bar{\gamma}}$: the constraint of $\bar{\gamma}$ follows from that of γ, and adding $\bar{\gamma}$ to a forest that contains γ does not change U_F.

Now take a compatible forest F with at least one non-induced logarithmic member, let γ be an inclusion-maximal one, chosen by a fixed rule, and toggle $\bar\gamma$: remove it from F if it is a member and add it otherwise. The result is again a compatible forest. By the definition of compatibility before (780), two members are either nested or edge-disjoint and meeting in at most one vertex. Members inside γ lie inside $\bar\gamma$. Members not nested with γ meet its vertex set, which is that of $\bar\gamma$, in at most one vertex, so they share no line with $\bar\gamma$. Members containing γ are induced, by the maximality of γ if they are logarithmic and by (810) if they are power subgraphs, so they contain $\bar\gamma$. The toggle does not change the set of non-induced logarithmic members, so it selects the same γ again and is an involution without fixed points. It leaves U_F unchanged, by the implied constraint, and the root numerator $\sum_{e\in E_R^{\log}(F)} x_e$, because $\bar\gamma$ is a power member and only logarithmic members remove lines from it; it reverses the sign $(-1)^{|F|}$. The forest terms with a non-induced logarithmic member therefore cancel in pairs, and the sum over forests of induced subgraphs equals the full forest sum. □

By the argument above, the closure of a non-induced divergent subgraph is a proper power subgraph, so power-free graphs have no such subgraph. The code also checks the cancellation by direct enumeration, which therefore needs only the power-containing classes. Its all-edge audit visits every proper subset of labeled lines of all 8,532 power-containing classes of loops three to eight, 492,490,392 subsets, tests one-particle irreducibility, builds the 11,123,924 unreduced mixed forests and reproduces every grouped integer map, the zero maps and the maps with a vanishing top channel included. At eight loops alone it covers the 7,231 power-containing classes, with 40,679 non-induced logarithmic subgraphs and 10,451,732 forests.

Assembly and analytic inputs. The truncated fixed point is obtained by nine iterations of $g \leftarrow \frac{1}{3}[\epsilon - \sum_{L=2}^{8} \widehat{b}_L(\epsilon) g^L] \bmod \epsilon^9$ from $g = 0$, and in 64 exact rational cases, regulator-dependent and constant-coefficient ones included, it agrees exactly with the Lagrange formula (715) carried to eight loops. The response to the jets of loops five to eight has a closed form. Hold the jets of loops one to four fixed and put

$$\Phi_{\le 4}(g,z) = 3g + \sum_{L=2}^{4} \widehat{b}_L(z) g^L, \qquad \gamma_{r,\le 4}(g,z) = g + \sum_{L=2}^{4} \widehat{c}_L(z) g^L, \qquad \Phi_{\le 4}\big(g_{\le 4}(\epsilon),\epsilon\big) = \epsilon, \quad (811)$$

so that $g_{\le 4}(\epsilon) = \epsilon/3 + O(\epsilon^2)$ is the fixed point of the functions (802) truncated after four loops. A change of the jets of loops five to eight changes the exponent by

$$\delta\Delta_{\mathcal{E}}(3,\epsilon) = \sum_{L=5}^{8} g_{\le 4}^L \Big[\delta\widehat{c}_L(\epsilon) - \frac{\partial_g \gamma_{r,\le 4}(g_{\le 4},\epsilon)}{\partial_g \Phi_{\le 4}(g_{\le 4},\epsilon)}\, \delta\widehat{b}_L(\epsilon)\Big] \pmod{\epsilon^9}, \quad (812)$$

with $g_{\le 4} = g_{\le 4}(\epsilon)$. This first variation is exact through order eight, because two insertions of loops five or higher first contribute at order $5 + 5 - 1 = 9$. With $\widehat{b}_L(z) = \sum_j \widehat{b}_{L,j} z^j$ and likewise for $\widehat{c}_L$, the Jacobian of C_8 is therefore, for $L = 5, \dots, 8$ and $0 \le j \le 8 - L$, whatever the jets of loops five to eight,

$$\frac{\partial C_8}{\partial \widehat{c}_{L,j}} = [\epsilon^{8-j}]\, g_{\le 4}^L, \qquad \frac{\partial C_8}{\partial \widehat{b}_{L,j}} = -[\epsilon^{8-j}]\, \frac{\partial_g \gamma_{r,\le 4}(g_{\le 4},\epsilon)}{\partial_g \Phi_{\le 4}(g_{\le 4},\epsilon)}\, g_{\le 4}^L. \quad (813)$$

For $j = 8 - L$ it reduces to the top-slot rule $(-1, 3)/3^{L+1}$ of Section 32.2, so the eight-loop constants enter only as $C_8(3) \supset (3\widehat{c}_{8,0} - \widehat{b}_{8,0})/19683$; for $(L, j) = (7, 0)$, with $A_3 = -\widehat{b}_{2,0}/6$ (Section 32.2) and $\widehat{c}_{2,0} = \widehat{b}_{2,0}/6$, it gives $(-16A_3, 42A_3)/19683$. A seven-loop class with $3w_B = w_A$ has $\delta\widehat{b}_{7,j} = 3\delta\widehat{c}_{7,j}$. Its top slot cancels, but through the mixing with the two-loop input its constant terms give $\delta C_8 = -2A_3\, \delta\widehat{b}_{7,0}/19683 = \widehat{b}_{2,0}\, \delta\widehat{b}_{7,0}/59049$, which is why the code restores these classes.

The code derives its own two-loop and Gamma inputs. At 80 digits, Cauchy contours of radii 0.04 and 0.07 with 80 and 96 nodes agree on all seven two-loop entries to 3.5×10^{-72}; the six entries shared with Section 29 agree with that calculation's input to 4.5×10^{-59}; and a 112-node Cauchy

extraction of the logarithm of the Gamma product checks all 42 Gamma-jet coefficients (loops two to eight, orders one to six) to 1.2×10^{-73}. These are high-precision cross-checks, not interval bounds. By direct comparison of the files `inputs/two_loop.json` and `inputs/gamma_d3_8.txt` of the two packages, the two-loop entries and the Gamma table differ from those of the calculation above by at most 1.7×10^{-57}, the precision of the stored two-loop entries there (Section 32.2), and 4.1×10^{-58}.

Worker. The production used the code's optimized worker, with the 350-digit GMP tier. Its one change to the arithmetic is in the binary128 additions inside the positive denominator polynomials: the 113-bit integer significands are aligned with guard, round and sticky bits and rounded to nearest, ties to even, so each sum is the correctly rounded one; zeros, subnormal and negative operands, exceptional values and possible overflow go to the compiler's addition. Signed forest sums, the addition order, the promotion thresholds and the sampling are unchanged. A test of 12,051,840 additions agrees bit for bit (NaN cases by outcome); on 179 graphs at 40,448 fixed points the optimized and the unoptimized worker give identical coefficient vectors, maximum contributions and promotion counts; 2,432 evaluations at shared points reproduce the jets and promotion counts of the seventh-order worker of Section 29 exactly; and 48 points of 12 eight-loop graphs agree with a 450-digit reference within the tolerances of their precision tiers. Three builds on Expanse gave one binary hash and passed all 14 build and test steps.

Pilot and allocation. The code was delivered with a pilot on its development host: both campaigns with 8 replicates each and 32,768, 8,192, 2,048, 512, 64 and 16 points per graph and replicate at loops three to eight, 21,342,208 evaluations in all. It gave $C_8(3) = 1514.6 \pm 161.7$ (sampling standard error), an exploratory estimate. Its per-graph variances and timings fixed the allocation of the production by the code's Neyman rule: the points per graph are proportional to the square root of the ratio of per-point variance to per-point cost, rounded up to a power of two, at least 4,096, and scaled to the budget. The replay reproduces the allocation file byte for byte. The pilot is not part of the estimate.

Production. The code was adapted to the Expanse cluster without a change to its numerical files. With a fresh base seed, an allocation of about 2,000 core-hours of its development host, 16 replicates per campaign and at least 4,096 points per graph, it ran on 10 and 11 October 2026 on six whole AMD EPYC 7742 nodes, the last 5 of its 1,994 jobs in a separate one-node job, with 3.7×10^{10} evaluations and about 9,800 service units in all. It gives

$$C_8(3) = 1600.98 \pm 3.97 \quad \text{(sampling standard error)}, \tag{814}$$

-0.40 combined standard errors from the estimator. Its campaigns, 1603.53 ± 4.91 and 1598.42 ± 6.24, differ by 0.64 combined standard errors; its lower-order coefficients agree with the references within 1.73 standard errors (pulls -1.41, -1.55, -0.10, $+1.73$ and -0.21 at orders three to seven; the squared pulls sum to 7.40 for these five checks), and its sixth- and seventh-order by-products are 33.599 ± 0.014 and -220.40 ± 0.19. Its power-free part is 1602.61 ± 2.96 and its power correction -1.63 ± 2.64, with covariance 1.9×10^{-4}; they differ from those of Section 32.1 by $+0.20$ and -0.82 combined standard errors. The half streams give 1599.79 and 1602.17, deviations of -1.19 and $+1.19$ from the estimator, and a bootstrap of 4,096 replicate resamples gives standard error 3.78. The eight-loop graphs carry 42.1% (power-free) and 31.5% (power-containing) of the variance, the seven-loop graphs 12.2% and 12.6%, and the six-loop graphs 1.4%. The power null rows at orders four to seven are $(-4.66 \pm 2.22) \times 10^{-5}$, $(-1.91 \pm 4.57) \times 10^{-4}$, $(-1.21 \pm 5.62) \times 10^{-3}$ and 0.072 ± 0.126, with pulls -2.10, -0.42, -0.22 and $+0.57$, whose squares sum to 4.95 for these four null rows.

The quoted error is the covariance of the replicates propagated through the fixed point. The code assigns no allowance, and its result record marks the reportable precision as not established. There

were no tail diagnostics like those of Section 32.5 and no convergence test beyond the campaign and half-stream splits, and neither the delivered science code nor this number had an independent review: the reviews of the port covered only its cluster layer. The independent estimate is not combined with the reported value.

32.8 Records

The production result, the frozen plan and the merged replicate data that reproduce the estimator are in `GFF/results/C8_3/`; the per-graph replicate tables are hosted in `GFF/data/C8_3/`. The verification, the job records and the analyses of Sections 32.4 and 32.5 are in `LRI_C8_D3_HPC/benchmarks/symmetry_2026-10-c8/production/prod1/`. The pilot report is in `pilot/evidence/` and the rate probes in `production/margin_*` of the same `symmetry_2026-10-c8/` folder, and the independent code's production is in `GFF/results/C8_3_check/`. `LRI_C8_D3_HPC/public.mk` recomputes every number of Sections 32.1–32.6 from these records. The numbers of Section 32.7 come from the independent package. `LRI_C8_D3_OPT/public.mk` recomputes those of its production from `GFF/results/C8_3_check/`, except the half streams and the bootstrap error, which need the replicate matrices and are read from its result record. The census, audit, input, worker and pilot checks are the package's delivered evidence (`LRI_C8_D3_OPT/verification/`, `docs/` and `README.md`), its builds are summarized in its `public/REVIEWS.md`, and the differences between the input files follow from the files `inputs/two_loop.json` and `inputs/gamma_d3_8.txt` of the two packages. The complete run tree of the production of Section 32.4 is archived in two copies (Section 33); the native outputs of the independent production are not archived.

33 Reproduction and data dependencies

The code and records behind this note are in the folder `GFF/` of the public repository of this work. Its folder `GFF/code/` holds the packages named in the table below, each at the commit that produced the quoted results; package paths in this note, in the text and in the table, are relative to `GFF/code/`. The records from which the quoted numbers are recomputed are in `GFF/data/`, and the published outputs in `GFF/results/`; these paths are written in full. Every package has a file `public.mk`: `make -f public.mk test` runs its tests, and `make -f public.mk replay` recomputes from that package's shipped records the numbers this note quotes for the results the package produced, and compares them with the quoted digits. Numbers of earlier calculations that a package keeps in `reference/`, checks made when the separate notes were merged into this note, and construction checks whose records are hosted are not all replayed; the rows below say which. Large raw files (checkpoints, graph packs, payload archives) are hosted outside the repository and fetched with `make -f public.mk fetch-data`; the package's `DATA.md` says which target needs them, and its review summary, `REVIEWS.md`, says which reviews it received. The present note is a mathematical condensation, not a production integration campaign. A package's finite parts and graph ordering must be used with that same package's code; files with similar names from different schemes are not interchangeable. Packages of the earlier research archive `LRI_complete_research_bundle.zip` that hold only superseded estimates are not part of the repository. The separate TEX notes that came with several packages are merged into this note and are left out of the repository: their derivations, methods, results and caveats are in the sections cited in the table, and Section 33 says where each block came from. Markdown records of derivation steps that ship inside a package stay there as its records; for example, the delivered stage scripts of `C6_research_note/` read or write such files under `code/stages/` and `inputs/source_snapshots/`. No step of this note cites them; the arguments are written out here, for $C_6(1)$ in Appendix B.

Provenance of the merged material. The table lists the blocks that were merged into this note from separate notes and records on 16 October 2026, with their source and author. None of

these sources names its author: their bylines read "Research calculation record" or "Calculation record", or are empty. The entry "OpenAI GPT-6 astra" records the author's statement that the source was written by that model (for the research bundle and the delivered packages, 13 October 2026; for `C6_research_note/`, 16 October 2026); "author not named" marks a source for which there is no such statement. Claude Code sessions wrote the merged text itself and adapted it to the notation of this note.

Block	Source	Author
Appendix B (derivation of the $C_6(1)$ representation)	`C6_research_note/`: its note and input records	OpenAI GPT-6 astra
Section 27: method, design and checks of the seventh order	`LRI_C7_HPC/reference/`: the two seventh-order notes	OpenAI GPT-6 astra
Sections 1 (lattice model), 1.4 (the caveat on repeated parameters), 3 (zero-dimensional benchmark), 8.3 (checks of $\mathcal{K}$), 9 (decimal value of $\mathcal{L}$), 10 (100-digit value of P_5), 14	the earlier complete note in `LRI_C7_HPC/reference/`	OpenAI GPT-6 astra; the resummation of Section 14.2 recomputed at the merge by a Claude Code session
Sections 34.9, 34.10; the $d = 1$ checks of Section 34.12	`lri_ope_reproducibility/` and [10]	written at the merge by a Claude Code session
Section 16.8, values	the note's closed forms; `LRI_C5_d3/` records	a Claude Code session
Section 16.8, spectral route to $d = 2$	`lri_d3_reduction_update/`: its complete note	OpenAI GPT-6 astra
Section 17.4 (double residue series of $P_5(d)$)	`lri_d3_reduction_update/`: the general-dimensional addition	OpenAI GPT-6 astra; steps the source left implicit written out at the merge by a Claude Code session
(404) and the measure-pole bound	`lri_d3_reduction_update/`: the periodization addition	OpenAI GPT-6 astra
Section 19: opening, count table, Section 19.5, values and dictionary of Section 19.10, catalogue search	`lri_d3_reduction_update/`: the three-dimensional addition	OpenAI GPT-6 astra
Section 19.9, rank-two checks, open problem	`lri_d3_matrix_update/`: its complete note	OpenAI GPT-6 astra
Section 19: the link to $\mathfrak{t}_3(3; \frac{3}{2})$	derived at the merge	a Claude Code session
Section 20.6 (half-shift identities at $d = 3$)	`LRI_C5_d3/`	a Claude Code session (written out at the merge)
Section 21, including the growth proof and the continuation	`LRI_C5_general_d/K_general/`	a Claude Code session
(560)	`LRI_C5_general_d/K_general/`; constant derived at the merge	a Claude Code session
Section 25.4, the 300-digit fallback	`lri_C6_d3_complete_update/`	OpenAI GPT-6 astra
Section 26.14, $d = 3$ rows of the method comparison	`LRI_C6_D2_HPC/reference/`: the numerical-methods note	OpenAI GPT-6 astra; the explanation of the failed control added at the merge by a Claude Code session
Section 32.7	`LRI_C8_D3_OPT/`: its derivation and calculation notes	OpenAI GPT-6 astra; cluster port and production by a Claude Code session
Part V, general d, $d = 3$ and the bootstrap comparison	`lri_ope_reproducibility/`: its note	author not named; supplied by the author on 16 October 2026; the checks of Section 34.12 by Claude Code sessions
Section 17.2: numerical discovery of $\mathcal{L}_2$, $\mathcal{K}_2$ and the single-sum relations	`LRI_C5_d2_simplify/`	a Claude Code session

Block	Source	Author
Sections 18.7 (first and second proofs of $\mathcal{L}_2$) and 18.8 (proof of $\mathcal{K}_2$); in Section 18.9 the sharper estimate, the per-spin identity, the second and third single-sum evaluations, the mixed-trace corollary and the Riemann bilinear proof	`LRI_C5_d2_proof/`	a Claude Code session with prover and reviewer subagents
Sections 18.5.2, 18.7.7–18.7.9; the regularized spectral route of $\mathcal{K}_2$ in Section 18.9	`LRI_C5_d2_simplify/proof/`	a Claude Code session with prover, verifier and synthesizer subagents
Section 20: the generic-dimensional reduction of $\mathcal{L}_d$, Z_d and its proofs	`LRI_C5_general_d/`	a Claude Code session; the explicit constants of (465), (466), (467) come from the growth proof of Section 21.6
Sections 23–25: the blocks merged from the earlier complete note	`LRI_C6_D2_HPC/reference/`: the complete note	OpenAI GPT-6 astra; one closed-form value and one derivation added at the merge by a Claude Code session
Checks merged from the same complete note into Sections 3.4 (what the higher-pole identities cannot detect), 15.8, 16.8 (controls of the master evaluations), 17.7 (checks of the finite multivariable construction) and 18.6	`LRI_C6_D2_HPC/reference/`: the complete note	OpenAI GPT-6 astra
Sections 26.11–26.16: method blocks	`LRI_C6_D2_HPC/reference/`: the methods note	OpenAI GPT-6 astra; the variant descriptions read from the code, the linear-control formula and one proof added at the merge by a Claude Code session
Section 28: merged blocks	`LRI_C7_D2_HPC/reference/`: the method and progress notes	OpenAI GPT-6 astra; the three-loop check, the two-loop entries, the primitive classes and the variance decomposition written out or rechecked at the merge by a Claude Code session
Section 31.7: merged blocks	`LRI_C8_D2_HPC_optimized/`: the calculation note and results fragment	OpenAI GPT-6 astra; the 65- and 70-digit comparison of the two packages' inputs made at the merge by a Claude Code session
Section 34.8, the $d = 2$ rows of Section 34.11 and (863)	written for this note from `lri_ope_reproducibility/`	a Claude Code session; checks in `ope_checks/d2/`

Result or dependency	Package and principal record
One-dimensional analytic C_5	`lri_numerical_resolution/`: graph audit in `verification/records/resolution_audit.json`, with the raw five-loop and lower-loop Laurent records in `verification/records/`, the residue certificate in `verification/inputs/residue_certificate.json` and the kernels that produced the records in `verification/source/`. The packages `lri_psi_complete_update/` and `lri_fixed_hypergeometric_update/` supply the checks of the polygamma evaluation of $\mathcal{K}$ and of the five-value P_5, with the records `verification/numerical_checks.json` and `verification/fixed_hypergeometric_checks.json`. The quoted values are collected in `GFF/results/closed_forms_1/C5_1_quoted.json`, and `make -f public.mk replay` in each package folder recomputes them from the shipped files in under a minute on one core.
General C_0–C_5	`lri_general_special_functions_update/` (with the general-dimensional C_0–C_3, graph-inventory, tetrahedron and two-loop files of Section 15), `lri_general_closed_update/`, `lri_general_periodization_update/`; master evaluators, finite arrays, residue and assembly checks. The quoted values and formulas, with their records, are in `GFF/results/closed_forms_general/C0_C5/expected.json`; `make -f public.mk replay` in each package recomputes them.
Closed forms of $\mathcal{L}_2$, $\mathcal{K}_2$ and $C_5(2)$	`LRI_C5_d2_proof/` (proof routes with their machine checks; the computations behind the quoted checks in `review_*/`; reviews summarized in `REVIEWS.md`) and `LRI_C5_d2_simplify/` (discovery, 70-digit single sums `confirm_dps70.json`, scripts and logs of the three further proofs in `proof/`, whose $\mathcal{L}_2$ parts are written out in Sections 18.7.7–18.7.9; reviews summarized in `REVIEWS.md`); the independent spectral evaluation is `LRI_C5_d3/wf_d2check/d2check_results.json`. Expected values: `GFF/results/closed_forms_2/closed_forms_2.json`; `make -f public.mk replay` in either package recomputes every quoted number.

Result or dependency	Package and principal record
Generic-dimensional reduction of $\mathcal{L}_d$, Z_d and $\mathcal{K}_d$	`LRI_C5_general_d/`: the code and validation records of the general-b periodization (Section 20.1) in `foundation/`; the specialization code and records in `specialize/`; records of the checks in `K_general/note_fragment/growth/checks/`, `K_general/note_fragment/topmode/results/`, `specialize/results/`, `L/results/` (with the code of Section 20.3 in `L/`), `Z/results/` and the end-to-end $C_5(1)$, $C_5(2)$, $C_5(3)$ in `review/joint/numerics-end-to-end/e2e_report.json`; the review records `review/VERIFICATION.md`, `review/lemmaG-proof/` and the check code of `review/` (`lemmaG-numeric/`, `topmode-numeric/`, `joint/numerics-end-to-end/`), with all reviews of both sections, including those of `K_general/note_fragment/`, summarized in `REVIEWS.md` (the review transcripts are not included); independent spectral references `references/reference_values.json`; 100-digit $\mathcal{K}_d$ in `K_general/position_space/K_d_values_mb_D100.json`. `make -f public.mk replay` compares with these records every quoted number of Section 21 (expected values `public/expected_K_d.json`) and the numbers of Section 20 listed in `GFF/results/closed_forms_general/reduction_d/expected.json`, and reruns the fast checks among them. It does not compare the numbers of Sections 20.1.4, 20.3 and 20.4, nor those of the items of Section 20.9 on the general-b periodization, the review probes, the product form in detail and the two-node formula; these were added on 16 October 2026 and are read from the records named there. `make -f public.mk replay-full` recomputes part of those records (`L/results/val50.json`, `L/results/val70.json`, `Z/results/zj*.json`, `Z/results/zc*.json`, `Z/results/compare_refs.log`, `Z/results/check_formulas.log`) and compares them with the shipped records, not with the text; the records in `foundation/results/`, `Z/results/offdiag60.json` and the other files of `L/results/` are recomputed by no target. The $d = 3$ inputs are in `LRI_C5_d3/` (`wf_L3/`, `wf_K3/K3_value.json`), with the quoted values in `GFF/results/closed_forms_3/`.

Result or dependency	Package and principal record
Three-dimensional specialization	`lri_d3_reduction_update/`: `code/general_d3/check_exact.py` (residue algebra, polynomial identities and rational Horn coefficients; `verification/general_d3/exact_algebra.json`, with the eight real-node Horn arrays in `horn_arrays.json` of the same folder), `code/general_d3/verify_d3.py` (formula-side values at 55 and 90 digits, the 75-digit determinant, mode and complex-node checks and the literal hypergeometric checks; `period_values.json`, `identities.json`, `literal_hypergeometric.json`) and `code/general_d3/check_spectral.py` (fresh real-axis Gamma kernels; `fresh_spectral.json`; the residuals are its differences from `period_values.json`, also listed in the delivered record `comparison_summary.json`). `lri_d3_matrix_update/`: `code/rank_two_checks.py` (the 1107 exact checks and the degree-90 interior sums; `verification/rank_two_checks.json`), `code/recompute_d3_periods.py` (the 100-digit values; `verification/fresh_primitive_values.json`) and `code/scalar_catalog_search.py` (the catalogue search of Section 19.10; `verification/scalar_catalog_search.json`, rerun by `make -f public.mk search`). Section 19 quotes these records. The quoted values and their tolerances are in `GFF/results/closed_forms_3/d3_specialization_quoted.json`.
$C_6(1)$	`GFF/results/C6_1/C6_final.json`, with the degree sequence in `assembled.json`, the rounding estimates in `rounding.json`, the stopping decisions in `decisions/`, the pre-registered plan `PLAN.md` and the report `REPORT.md` of the same folder. The results-review checks are `C7_crosscheck.json` and `lower_amplitude_precision/` in that folder. The code is `LRI_C6_1D_HPC/` at git commit `cef6fa7`; the production ran at `8680864`. The merged level records and the engine outputs are in `GFF/data/C6_1/`. The target `replay` of `public.mk` in the code folder regenerates these records and checks every quoted number. The superseded Monte Carlo estimator is not included in the public repository.
Analytic representation of $C_6(1)$	`C6_research_note/`: the reassembly `outputs/reassembly.json`, the interval certificate in `outputs/rerun/complete_refined/` and the task runner `code/reproduce_final.py`, which runs the 22 tasks of the final method (Section B.8). The derivation is Appendix B. The target `replay` of `public.mk` in that folder recomputes the quoted numbers from these records and compares them with `GFF/results/C6_1/analytic/quoted_numbers.json`. The independent review is `LRI_C6_closed_form_review/REVIEW.md`, with its check scripts and their recorded outputs in `checks/` of the same folder.

Result or dependency	Package and principal record
Inherited $d = 2$ finite kernels	Superseded inputs of the previous two-dimensional estimator (the primitive and locally subtracted kernels of Section 23), not additional contributions to add again and not needed to reproduce any current coefficient. The kernels, rates, two-loop input and fixed-point map that the 2026 production uses are byte-identical copies in `LRI_C6_D2_HPC/reference/`, bound by `LRI_C6_D2_HPC/inputs/manifest.json`. The construction records (`lri_C6_d2_primitive_update/`, `lri_C6_d2_subtraction_update/`, `lri_C6_d2_two_forest_update/`) are in the research archive `LRI_complete_research_bundle.zip`, which is not part of this repository.
Power compensation and finite kernels	`lri_C6_d2_complete_update/`: the part of the research-archive package that the two-dimensional power-sector compensation and the audits of Sections 24 and 25 need, namely the saved counterterms, the check scripts and the counterterm chain. `make -f public.mk replay` reruns the checks and compares them with the records in `GFF/results/C6_2/power_compensation/`, among them `GFF/results/C6_2/power_compensation/verification/power_sector_audit.json`; `make -f public.mk regen` regenerates the counterterms from the graph inventory. The three-dimensional compensation is `lri_C6_d3_complete_update/`, with its full-coverage, analytic-extension, power-sector and fixed-point audits; its quoted values are in `GFF/results/C6_3/power_compensation_quoted.json`. The superseded first full sampling run of $C_6(3)$ in that package is not included.
$C_6(2)$	`GFF/results/C6_2/C6_result.json`, with the production log, reporting rule and checks in `GFF/results/C6_2/CAMPAIGN.md`. The frozen plan `manifest.json`, the group configurations and the per-group `merged.json` records are in `GFF/data/C6_2/run-20260928-m8192/`, and the complete run directory with the checkpoints is hosted outside the repository and fetched by `make fetch-data`. The code is `LRI_C6_D2_HPC/` at git commit `6aaf656`; `make -f public.mk replay` there recomputes every number quoted here for $C_6(2)$ from the merged records, and `make -f public.mk remerge` re-merges the checkpoints. The seventh-order cross-check is `GFF/results/C7_2/C7_result.json`. The superseded estimator is `LRI_C6_D2_HPC/reference/verification/method_precision_results.json`.

Result or dependency	Package and principal record
Superseded sixth-order estimators at $d = 2, 3$ and the method benchmarks	`LRI_C6_D2_HPC/reference/`, the earlier method-refinement calculation shared by the two sixth-order packages: the refinement, pilot, validation, speed and endpoint-replay records in `records/` (the baseline campaigns in `baseline_previous/records/`); the per-graph counts, binary Sobol direction matrices, frozen transport tables `grid_d2_top3.txt`, `grid_d2_top11.txt` (and their $d = 3$ counterparts) and the frozen plan `frozen_production_plan.json` in `inputs/`; the source and input hashes in `verification/production_source_hashes.json`. Without new integration, `code/analyze.py` reassembles the previous estimators and their campaign comparisons (`verification/method_precision_results.json`; the rounded reporting convention is stored separately in `verification/reported_values.json`), `code/summarize_methods.py` the benchmarks of Section 26.11 (`verification/method_benchmarks.json`) and `code/extra_checks.py` the finite Lagrange, bootstrap and delete-one checks (`verification/extra_checks.json`). These archived scripts rewrite their output files in `verification/` in place, and on another NumPy build the last bits can differ; `LRI_C6_D2_HPC/tools/reassemble_reference.py` runs the archived `code/analyze.py` for $d = 2$ without writing and compares with the stored result to relative 10^{-12}. The integrator is rebuilt by `code/build.sh` (C++17 with OpenMP, libquadmath and Boost.Multiprecision) and replayed from the shipped direction matrices by `code/replay.py`. `make -f public.mk replay` does not run these archived scripts; the method numbers of Sections 26.11–26.13 that are derived from stored values, such as the linear-control ratios and the refinement pulls, are recomputed by no script.
$C_6(3)$	`GFF/results/C6_3/results_run-20260929-m8192/C6_result.json`, with the frozen plan `manifest.json`, the group configurations and the per-group `merged.json` records in the same folder. The production log, reporting rule and checks are in `GFF/results/C6_3/CAMPAIGN.md`, and the complete run directory is `GFF/data/C6_3/run-20260929-m8192_full.tgz`, hosted outside the repository and fetched by `make fetch-data`. The code is `LRI_C6_D3_HPC/` at git commit `1c6dfb3`; its archived-stream replay is `benchmarks/expanse_2026-09/logs/validation_54510646/archived_stream_replay.json` in that folder, and `make -f public.mk replay` there recomputes every number quoted here for $C_6(3)$ from the merged records. The superseded estimator is `reference/verification/method_precision_results.json` in the reference tree that the package shares with `LRI_C6_D2_HPC/`.

Result or dependency	Package and principal record
Full $C_7(1)$	`GFF/results/C7_1/C7_final.json`, with the frozen design `design_v1.json`, the batch inclusion list `inclusion.txt`, the combined assembly `combined_assembly.json` and the report `REPORT.md` in the same folder; the earlier 46-batch combination is in `v1_46batches/` of that folder. The code is `LRI_C7_HPC/`, exported from git commit `68dc6be`, whose code files equal those of commit `7320afc`, which ran the final combination; the production workers ran commit `3213ce0`, which differs from it only in the verification, assembly, combination and design tools and their tests. Full forest arrays, the retained primitive input and the earlier assembly are in `LRI_C7_HPC/reference/`. The target `replay` of `public.mk` in the code folder recomputes the quoted numbers from these records. The merged replicate records of the 399 batches and of the primitive campaign are hosted data, described in `GFF/data/C7_1/DATA.md`; from them the target `replay-full` reruns the combination.
Full $C_7(2)$	`GFF/results/C7_2/C7_result.json`, with the frozen plan `manifest.json` and the per-group `merged.json` records in `GFF/data/C7_2/run1-20260930-m4096/`; `make -f public.mk replay` in `LRI_C7_D2_HPC/` recomputes the quoted numbers from them for the 2026 production, together with the earlier estimator and power correction (reassembled from `reference/` by `tools/reassemble_reference.py`), the three-loop reference and the coverage tables; the other numbers of the earlier calculation in Section 28 are records in `reference/`, reassembled by `reference/code/reproduce.py`, and are not replayed. The production log and checks are in `GFF/results/C7_2/CAMPAIGN.md`; the code is `LRI_C7_D2_HPC/` at git commit `fb64c47`. The code and records of the earlier calculation (Section 28), with its power-free and primitive campaigns in `reference/LRI_C7_d2_progress/` and its power-sector completion in `reference/power_mass/`, are in `LRI_C7_D2_HPC/reference/`. The checkpoints and task tables of the run are archived, not included.
Full $C_7(3)$	`GFF/results/C7_3/run1-20260930-m4096/C7_result.json`, with the frozen plan `manifest.json`, the group configurations and the per-group `merged.json` records in the same folder. The production log, reporting rule, errata and checks are in `GFF/results/C7_3/CAMPAIGN.md`, and the complete run directory is the hosted `GFF/data/C7_3/run1-20260930-m4096_full.tgz`. The production used `LRI_C7_D3_HPC/`, the export of git commit `4de9412`; the re-verification tool `LRI_C7_D3_HPC/reverify_a6ce987/tools/report.py`, from commit `a6ce987`, reproduces the result. The previous calculation and its construction checks are in `LRI_C7_D3_HPC/reference/`, with its raw records in the hosted `GFF/data/C7_3/reference_records.tar.gz`; the archived-stream replay is in `GFF/results/C7_3/evidence/replay_report.json`, and the reviews are summarized in `GFF/results/C7_3/REVIEWS.md`.

Result or dependency	Package and principal record
Full $C_8(1)$	`GFF/results/C8_1/2026-10_ext1/C8_final_A3/C8_final.json`, the fourteen-batch result, with the report `REPORT.md` one level up, the nine-batch interim in `C8_A1_close/` and the records of both clusters in `records_symmetry/` and `records_expanse/`; the extension rules are `LRI_C8_HPC/docs/EXTENSION_A1.md`, `EXTENSION_A2.md` and `EXTENSION_A3.md`, and the independent reviews are summarized in `LRI_C8_HPC/REVIEWS.md`. The four-batch result `GFF/results/C8_1/2026-10_expanse/C8_final.json` has the report `REPORT.md`, the frozen design `frozen_design/`, the controlled affine map `full_affine8.json`, the batch plans and the combined assembly in the same folder. The merged records of all fourteen batches and of the seven-loop test with 4096 points per graph are in `GFF/data/C8_1/`; `make -f public.mk replay` in `LRI_C8_HPC/` recombines them with the production tools and checks every quoted number. The production code is `LRI_C8_HPC/` at git commit `c0ffa63` (tools of the extension at `15a110e` and `bd13a63`; the public copy is commit `1282e37`, whose code equals `bd13a63`); its nine-vertex inventory is `inputs8/` (the eight-loop coordinate file compressed in `GFF/data/C8_1/inputs8/`), generated by `gen/`, and its kernel regressions against the seventh-order binary are in `validation/`.
Full $C_8(2)$	`GFF/results/C8_2/C8_final.json`, with the tail diagnostics in `GFF/results/C8_2/tailcheck/` and the production's replicate records (plan, design bundle, merged groups) in `GFF/data/C8_2/`. The code is `LRI_C8_D2_HPC/` at git commit `9df5b1c`, whose code equals that of `ba68f40`, the export the production ran; its folder `benchmarks/symmetry_2026-10-c8/` holds the result report (`production/result/README.md`), the tail-check report (`production/tailcheck/README.md`) and the pilot analysis (`pilot/analysis/`), and `make -f public.mk replay` recomputes every number of Sections 31.1–31.6 from the records. Its plan reviews and code inspections are summarized in `LRI_C8_D2_HPC/REVIEWS.md`. The shared nine-vertex inventory is `LRI_C8_shared/inventory/inventory8_d2/`, and the shared eighth-order assembly `LRI_C8_shared/assembly/` is vendored byte for byte in `vendor/c8_assembly/`. The full campaign directories are archived as listed in `GFF/data/C8_2/DATA.md`.

Result or dependency	Package and principal record
Independent check of $C_8(2)$	`GFF/results/C8_2_check/prod1/C8_result.json`, with the frozen plan, the group configurations and merged records in `GFF/data/C8_2_check/prod1/`, the pilot in `GFF/data/C8_2_check/pilot1/` and the analysis in `GFF/results/C8_2_check/prod1_analysis/`. The code is `LRI_C8_D2_HPC_optimized/` at git commit `a791a1f`; `make -f public.mk replay` there recomputes the numbers of the paragraphs on the port, pilot and production and on the comparison in Section 31.7; the construction checks under Inputs, Graphs and payloads, Kernel arithmetic and Assembly, and the development runs, rest on the records of the hosted original package, and the comparison of the two packages' inputs, the bootstrap ratios and the bootstrap mean are not replayed. Its reviews are summarized in `LRI_C8_D2_HPC_optimized/REVIEWS.md`. The full campaign directory and the original package with its graph packs are hosted (`GFF/data/C8_2_check/DATA.md`).
Full $C_8(3)$	`GFF/results/C8_3/C8_result.json`, with the frozen plan `manifest.json`, the group configurations and the per-group `merged.json` records that reproduce the estimator in the same folder; the per-graph replicate tables `graph_replicates.csv` are hosted in `GFF/data/C8_3/`. The production report `README.md`, the tail diagnostics `tail_ntrend/` and the campaign split `ab_split/` are in `LRI_C8_D3_HPC/benchmarks/symmetry_2026-10-c8/production/prod1/`, the rate probes in `production/margin_*`, the priced designs in `production/price/` and the pilot report in `pilot/evidence/` of the same `symmetry_2026-10-c8/` folder. The code is `LRI_C8_D3_HPC/`: the production ran git commit `4fbd1bae`, which is the code shipped, and the evidence is taken from commit `6dd4a75a`; `public.mk` replays every quoted number. The plan, its review log and the results review are not included; they are summarized in `public/REVIEWS.md`. The shared nine-vertex inventory and eighth-order assembly are those of the $C_8(2)$ row. The campaign directories (13 GB) are not included; they are archived in two copies.

Result or dependency	Package and principal record
Independent check of $C_8(3)$	`GFF/results/C8_3_check/runs/prod1/C8_result_54838171.json`, with the frozen plan `manifest.json` and the per-graph statistics `graph_statistics_54838171.json` in the same folder, the frozen allocation `allocations/prod1_2000.json` and the production summary `README.md` in `GFF/results/C8_3_check/`. The code is `LRI_C8_D3_OPT/` at git commit `d5deeb9`; the target `replay` of `public.mk` in that folder recomputes the estimator, its standard error and the quoted checks from the means and covariances stored in the result (`public/replay_check.py`). The port to the clusters, its decisions and its reviews are summarized in `public/PROVENANCE.md` and `public/REVIEWS.md` of the same folder; the logs, the admission ledger and the lane summaries are not included. The graph payloads and the delivered pilot are hosted data, and the native outputs of the campaign are not archived; both are described in `GFF/data/C8_3_check/DATA.md`.
Bulk one-point coefficient (Part V)	`lri_ope_reproducibility/`, supplied by the author on 16 October 2026: the engines in `code/`, the graph inventories and master inputs in `data/`, the coefficient table `results/coefficients.json` and the verification reports in `results/verification/`; the tasks of its runner `run.py` are listed in Section 34.13. `make -f public.mk test` checks the release hashes and runs the quick task in a work folder, `make -f public.mk replay` also reruns the checks of Section 34.12, which are in `ope_checks/` (folders `d1/`, `d2/`, `d3/`, with their recorded outputs), and `make -f public.mk full` reruns the remaining tasks (about ten minutes on one core). The package's own note is merged into Part V and is not part of the repository.

The mathematical definitions and intermediate steps above specify the quantities being computed; raw numerical replicate files are evidence for their estimators, not substitutes for missing master definitions. The file `GFF/REPRODUCE.md` distinguishes the inexpensive replay from saved records from graph regeneration and expensive integration. The numerical allowances throughout this note are empirical and are not upgraded by checksum or source-preservation checks.

Part V

Bulk one-point data

34 The bulk one-point coefficient through sixth order

This part records a second observable built from the graphs and master functions of Parts I–III: the bulk one-point coefficient a_{ϕ^2} of the composite $:\Phi^2:$ of the auxiliary bulk field, through ϵ^6, at fixed $0 < d < 4$. It is the horizontal coordinate of the numerical-bootstrap bounds of [8]. The calculation, its code and its checks were supplied by the author on 16 October 2026 as a reproducibility package; this part contains its derivations, results, conditions and caveats, so that the package is needed only to rerun the computations. The package is `lri_ope_reproducibility/`; package paths

in this part are relative to `GFF/code/`, as in Section 33. Sections 34.1–34.6 hold for general d, and Sections 34.7–34.10 specialize to $d = 1, 2, 3$. Section 34.11 compares with the bootstrap, and Section 34.12 lists what was checked independently for this note.

34.1 Status

The material has four logically different layers.

1. *Definitions and the two lowest orders.* The observable, its normalization and $a^{(2)}_{\phi^2}$, $a^{(3)}_{\phi^2}$ agree with the published values of [8] (Section 34.6.1).

2. *The calculation of* $a^{(4)}_{\phi^2}, a^{(5)}_{\phi^2}, a^{(6)}_{\phi^2}$. It uses the graph set, local subtractions and master functions of this note. Its checks are internal to that framework: the five- and six-vertex graphs are reconstructed from shared master jets and checked with alternative regulator directions, not integrated one by one. The four-vertex finite parts were also evaluated at finite regulator.

3. *The conversion to the thermal coefficients* $C_j(d)$ (Section 34.4). It is exact algebra between two calculations that share their inputs; it does not validate either of them independently.

4. *Finite-coupling estimates* (Section 34.11). The Borel–Padé values and the comparison with published bootstrap figures are exploratory. No bootstrap problem was solved, the bootstrap data are readings of published figures, and the spread between resummation prescriptions is not an error bar.

Residuals of the internal checks, however small, do not bound the omitted orders of the series, the resummation bias, or an error shared by the calculation and its inputs. The coefficients are numbers of a finite-order calculation; $a^{(7)}_{\phi^2}$ and higher are not determined by the thermal coefficients C_6, C_7, C_8 of Part IV.

34.2 The observable

Write the long-range Ising model as the d-dimensional defect of a free scalar Φ in a bulk of dimension $d + 2 - s = 2 + 2\Delta_\phi$ and codimension $2 - s$, the local description of [8, 9]. In the notation of [8] the physical dimension is p, which is our d; the bulk dimension is its d and the codimension its q, whereas here $q = d/4$ throughout. The bulk propagator and the defect interaction are

$$\langle \Phi(X)\Phi(X')\rangle_0 = |X - X'|^{-2\Delta_\phi}, \qquad \phi(x) = \Phi(x, 0), \qquad S_{\rm int} = \frac{\lambda_B}{4!}\int d^d x\, \phi(x)^4, \tag{815}$$

with $\Delta_\phi = (d - \epsilon)/4 = q - \epsilon/4$ and $\epsilon = 2s - d$, $q = d/4$, as in (220). The critical local mass is tuned away, and no wave-function counterterm arises in the long-range regime, so Δ_ϕ is protected. For a bulk point at distance $|y|$ from the defect, define

$$\boxed{a_{\phi^2}(d, \epsilon) = |y|^{2\Delta_\phi} \lim_{X' \to X} \Big[\langle \Phi(X)\Phi(X')\rangle - |X - X'|^{-2\Delta_\phi}\Big], \qquad \langle :\Phi^2: (x, y)\rangle = \frac{a_{\phi^2}(d, \epsilon)}{|y|^{2\Delta_\phi}}.} \tag{816}$$

This is the coincident-limit convention of Appendix B.4.2 of [8] (equation, figure and appendix numbers of [8] refer to its arXiv version 6). For the unit-normalized composite $:\Phi^2: /\sqrt{2}$, divide by $\sqrt{2}$. After tadpole and critical-mass subtraction there is no term of order λ, so

$$a_{\phi^2}(d, \epsilon) = \sum_{n=2}^{6} a^{(n)}_{\phi^2}(d)\, \epsilon^n + O(\epsilon^7). \tag{817}$$

The superscript (n) is an order of ϵ; it is not an index of the Laurent coefficients $a_{L,k}$ of Sections 3 and 15. Below, $a^{(n)}$ abbreviates $a^{(n)}_{\phi^2}(d)$.

Three quantities must be kept apart. The thermal dimension $\Delta_{\mathcal{E}} = \sum_n C_n(d)\epsilon^n$ has the free value $d/2$; the squared defect OPE coefficient $\lambda^2_{\phi\phi\mathcal{E}}$ has the free value 2; and a_{ϕ^2} has the free value 0. Only the last is computed here. For the second, only the one-loop relation is recorded:

$$\lambda^2_{\phi\phi\mathcal{E}} = 2 - 6C_2(d)\,\epsilon + O(\epsilon^2), \tag{818}$$

in agreement with the tree-level and one-loop terms $\sqrt{2}$ and $-A\epsilon/(3\sqrt{2})$ of $\lambda_{\phi\phi\mathcal{E}}$ in Eqs. (3.24) and (3.33) of [8]. It is a separate low-order observable, and the higher-order formulas of this part must not be used for it. For the bootstrap, Eq. (2.16) of [8] allows at least $-0.575408 \le a_{\phi^2} \le 0$ at $d = 3$ and $-\frac{7}{8} \le a_{\phi^2} \le 0$ at $d = 2$. This range contains the generalized-free end, $a_{\phi^2} = 0$, and the short-range end of its Eq. (2.15), where Δ_ϕ is the short-range Ising value, 0.5181489 at $d = 3$ and $\frac{1}{8}$ at $d = 2$; it is compatible with bulk–defect crossing with nonnegative squared defect-channel coefficients.

34.3 The result

With A, R, U, V, X of (221), the masters Q_d, $P_4(d)$, $P_5(d)$, $\mathcal{K}_d$, $\mathcal{L}_d$, Z_d of Section 16, and the reflected polygamma differences

$$\mathsf{e}_j = \psi_j(-q) - \psi_j(q) = \psi_j(1-q) - \psi_j(q) + \frac{j!}{q^{j+1}}, \qquad \mathsf{e}_0 = \pi\cot(\pi q) + \frac{1}{q}, \tag{819}$$

the calculation gives

$$\boxed{a_{\phi^2}(d,\epsilon) = \epsilon^2\,\frac{\Gamma(2q)^3\,\Gamma(-q-\epsilon/4)}{54\,\Gamma(4q)\,\Gamma(q-\epsilon/4)}\Big[1 + \mathsf{H}_1\epsilon + \mathsf{H}_2\epsilon^2 + \mathsf{H}_3\epsilon^3 + \mathsf{H}_4\epsilon^4\Big] + O(\epsilon^7),} \tag{820}$$

with

$$\begin{aligned} \mathsf{H}_1 &= \frac{4}{3}A, \qquad \mathsf{H}_2 = \frac{25}{36}A^2 + \frac{R}{6} - \frac{4}{9}Q_d, \\ \mathsf{H}_3 &= \frac{17}{108}A^3 + \frac{AR}{12} - \frac{4}{9}AQ_d + \frac{4}{27}P_4 + \frac{71}{216}U - \frac{V}{36}, \end{aligned} \tag{821}$$

$$\boxed{\begin{aligned} \mathsf{H}_4 = &-\frac{7A^4}{5184} - \frac{7A^2R}{288} - \frac{7A^2Q_d}{162} + \frac{14AP_4}{81} + \frac{301AU}{864} - \frac{7AV}{144} - \frac{28\mathcal{K}_d}{81} - \frac{7\mathcal{L}_d}{81} - \frac{16P_5}{81} \\ &- \frac{103R^2}{5184} + \frac{7RQ_d}{54} - \frac{14Q_d^2}{81} - \frac{61X}{3888} + \frac{175Z_d}{162}. \end{aligned}} \tag{822}$$

Here $P_4 = P_4(d)$, $P_5 = P_5(d)$, and $6Q_d$ is the corrected tetrahedron $\mathcal{T}_d$ of (177). The leading coefficient is

$$a^{(2)} = \frac{\Gamma(-q)\Gamma(2q)^3}{54\,\Gamma(q)\Gamma(4q)} = -\frac{\Gamma(1-q)\Gamma(2q)^3}{54\,\Gamma(1+q)\Gamma(4q)} < 0. \tag{823}$$

Expanding the Gamma ratio, $\Gamma(-q-\epsilon/4)\Gamma(q)/[\Gamma(q-\epsilon/4)\Gamma(-q)] = \exp\sum_{k\ge1}(-\frac{1}{4})^k\mathsf{e}_{k-1}\epsilon^k/k! = \sum_j \mathsf{f}_j\epsilon^j$, with

$$\begin{aligned} &\mathsf{f}_0 = 1, \quad \mathsf{f}_1 = -\frac{\mathsf{e}_0}{4}, \quad \mathsf{f}_2 = \frac{\mathsf{e}_0^2 + \mathsf{e}_1}{32}, \quad \mathsf{f}_3 = -\frac{\mathsf{e}_0^3 + 3\mathsf{e}_0\mathsf{e}_1 + \mathsf{e}_2}{384}, \\ &\mathsf{f}_4 = \frac{\mathsf{e}_0^4 + 6\mathsf{e}_0^2\mathsf{e}_1 + 3\mathsf{e}_1^2 + 4\mathsf{e}_0\mathsf{e}_2 + \mathsf{e}_3}{6144}, \end{aligned} \tag{824}$$

the ordinary coefficients are the finite convolution

$$\boxed{a^{(n)} = a^{(2)}\sum_{j=0}^{n-2}\mathsf{H}_j\,\mathsf{f}_{n-2-j}, \qquad \mathsf{H}_0 = 1, \quad 2 \le n \le 6.} \tag{825}$$

For example $a^{(3)}/a^{(2)} = \frac{4}{3}A - \frac{\mathsf{e}_0}{4} = \frac{4}{3}A - \frac{\pi}{4}\cot(\pi q) - \frac{1}{d}$. Equations (819)–(825) and the master values suffice to evaluate the coefficients; no graph has to be rebuilt.

The form (820) is a statement about finite orders. Keeping the Gamma ratio unexpanded generates terms beyond ϵ^6; those terms are not predicted. The organization itself is derived: all dependence on the e_j comes from derivatives of the external-leg integral along one direction (Section 34.6.1), and the remaining coefficients H_j contain only A, R, U, V, X and the masters.

34.4 Relation to the thermal coefficients

Eliminating Q_d, P_4 and P_5 with (222)–(223) gives

$$\mathsf{H}_2 = 6C_3 + \frac{23}{36}A^2 + \frac{R}{9}, \qquad \mathsf{H}_3 = 6C_4 + 5AC_3 + \frac{13}{108}A^3 + \frac{7AR}{108} + \frac{U}{12}, \tag{826}$$

$$\mathsf{H}_4 = 6C_5 + 5AC_4 + \frac{5A^2 - R}{6}C_3 + 9C_3^2 - \frac{A^4}{5184} + \frac{7A^2R}{2592} + \frac{235AU}{2592} - \frac{5AV}{432} - \frac{7R^2}{1728} - \frac{47X}{7776} + \frac{13Z_d - \mathcal{L}_d - 12\mathcal{K}_d}{81}. \tag{827}$$

The logarithm of the normalized series is shorter. Define

$$\log \frac{a_{\phi^2}(d,\epsilon)}{a^{(2)}\epsilon^2} = \mathsf{l}_1\epsilon + \mathsf{l}_2\epsilon^2 + \mathsf{l}_3\epsilon^3 + \mathsf{l}_4\epsilon^4 + O(\epsilon^5), \tag{828}$$

a formal Taylor series of a ratio that tends to one (no logarithm of a negative number is taken). Then

$$\boxed{\begin{aligned}
&\mathsf{l}_1 = 12C_2 - \frac{\mathsf{e}_0}{4}, \qquad \mathsf{l}_2 = 6C_3 - \frac{A^2}{4} + \frac{R}{9} + \frac{\mathsf{e}_1}{32},\\
&\mathsf{l}_3 = 6C_4 - 3AC_3 + \frac{19A^3}{324} - \frac{AR}{12} + \frac{U}{12} - \frac{\mathsf{e}_2}{384},\\
&\mathsf{l}_4 = 6C_5 - 3AC_4 + \Big(A^2 - \frac{5}{6}R\Big)C_3 - 9C_3^2 - \frac{11A^4}{576} + \frac{37A^2R}{864} - \frac{53AU}{2592} - \frac{5AV}{432} - \frac{53R^2}{5184}\\
&\qquad - \frac{47X}{7776} + \frac{13Z_d - \mathcal{L}_d - 12\mathcal{K}_d}{81} + \frac{\mathsf{e}_3}{6144},
\end{aligned}} \tag{829}$$

where $A = 9C_2$. Exponentiation gives

$$\begin{aligned}
&\frac{a^{(3)}}{a^{(2)}} = \mathsf{l}_1, \qquad \frac{a^{(4)}}{a^{(2)}} = \mathsf{l}_2 + \frac{\mathsf{l}_1^2}{2}, \qquad \frac{a^{(5)}}{a^{(2)}} = \mathsf{l}_3 + \mathsf{l}_1\mathsf{l}_2 + \frac{\mathsf{l}_1^3}{6},\\
&\frac{a^{(6)}}{a^{(2)}} = \mathsf{l}_4 + \mathsf{l}_1\mathsf{l}_3 + \frac{\mathsf{l}_2^2}{2} + \frac{\mathsf{l}_1^2\mathsf{l}_2}{2} + \frac{\mathsf{l}_1^4}{24}.
\end{aligned} \tag{830}$$

Through $a^{(5)}$ the one-point coefficients are therefore fixed by $C_2, \dots, C_4$, the polygamma values $A = 9C_2$, R, U and $\mathsf{e}_0, \mathsf{e}_1, \mathsf{e}_2$, and the Gamma functions of (823). At sixth order they are not fixed by $C_2, \dots, C_5$: the combination $13Z_d - \mathcal{L}_d - 12\mathcal{K}_d$ is independent of the combination $-1536\mathcal{K}_d - 576\mathcal{L}_d + 7152Z_d$ that enters C_5 in (223), so $a^{(6)}$ needs one more master value at a generic dimension. At $d = 2$ the closed forms of Section 17.2 remove it (Section 34.8). These relations eliminate shared integrals. They do not identify scaling dimensions with OPE data, and they are checked by substitution, not by a new loop calculation.

34.5 Renormalization-group input

The amplitudes of this part are massless position-space integrals, and ϵ itself is their regulator. The coupling is $g = \lambda/h_d = \pi^{d/2}\lambda/\Gamma(d/2)$, with λ the quartic coupling of (815), so that every vertex

integration carries the normalized measure $h_d\, d^d x$ of Section 16.3. It is renormalized by minimal subtraction of the poles in ϵ (coordinate-space MS, with the forest counterterms of Section 34.6.5), with the critical-mass prescription for local power subgraphs, and

$$\beta(g) = -\epsilon g + 3g^2 - 6Ag^3 + b_3 g^4 + b_4 g^5 + b_5 g^6 + \cdots . \tag{831}$$

This is not the massive momentum-space prescription of (173) and (174), which has its own regulator z. The coefficients b_L, c_L of the two prescriptions nevertheless coincide where they were compared: b_3, c_3 and b_4, c_4 of this prescription are those of (179), (226) and (227), with M and N of (225); its $3c_5 - b_5$ is D_5 of (229); and its b_5 agrees numerically with five-loop data of this note at $d = 1, 2, 3$ (Section 34.12, items "Exact algebra" and "Five-loop split"). Finite parts are a different matter: finite parts of a massive-regulator prescription and of the massless position-space prescription must not be interchanged. The package's low-order matching relates a massive-regulator coupling $g_{\rm alt}$ to g through the regulator-dependent factor $\mathsf{F}_{\rm sch}$; with both couplings minimally subtracted,

$$\begin{aligned} \mathsf{F}_{\rm sch}(\epsilon) &= 2^{-\epsilon}\frac{\Gamma(2q+\epsilon/2)}{\Gamma(2q)\Gamma(1+\epsilon/2)}\frac{\Gamma(q-\epsilon/4)^2}{\Gamma(q+\epsilon/4)^2} = 1 + \mathsf{f}_1^{\rm sch}\epsilon + \mathsf{f}_2^{\rm sch}\epsilon^2 + \cdots , \\ g_{\rm alt} &= \mathsf{F}_{\rm sch}\, g - \big[3\mathsf{f}_1^{\rm sch} + 3(\mathsf{f}_2^{\rm sch} + (\mathsf{f}_1^{\rm sch})^2)\epsilon + \cdots\big] g^2 + \big[9(\mathsf{f}_1^{\rm sch})^2 + 6A\,\mathsf{f}_1^{\rm sch} + O(\epsilon)\big] g^3 + \cdots . \end{aligned} \tag{832}$$

The package checks this matching at the fixed point through the order used there (`lri_ope_reproducibility/code/a4/symbolic_check.py`). The higher-order assembly below uses the coordinate-space prescription and its forest counterterms throughout; it imports no unmatched finite amplitude. The fixed point is

$$\begin{aligned} g_* = \frac{\epsilon}{3} + \frac{2A}{9}\epsilon^2 + \Big(\frac{8A^2}{27} - \frac{b_3}{81}\Big)\epsilon^3 + \Big[\frac{40A^3}{81} - \frac{10Ab_3 + b_4}{243}\Big]\epsilon^4 \\ + \frac{672A^4 - 84A^2 b_3 - 12Ab_4 + b_3^2 - b_5}{729}\epsilon^5 + O(\epsilon^6). \end{aligned} \tag{833}$$

Since a_{ϕ^2} starts at g^2, $a^{(6)}$ needs b_5 but not b_6; it needs b_5 itself, not only the thermal combination $D_5 = 3c_5 - b_5$ of (229). With $\widetilde{N} = \psi_1(-q) - \psi_1(3q)$, the five-loop forest sums of the calculation are

$$\begin{aligned} b_5 = {} & \frac{64341A^4}{128} + \frac{2145A^2M}{64} - \frac{17001A^2R}{64} + \frac{3159A^2Q_d}{8} + \frac{393AMN}{32} + 63AP_4 \\ & + \frac{14907AU}{64} - \frac{603AV}{32} + 87\mathcal{K}_d + \frac{51}{4}\mathcal{L}_d + \frac{3}{4}M^2 + \frac{81MN^2}{64} - \frac{9}{8}MR + 6MQ_d \\ & + \frac{81M\widetilde{N}}{64} + 24P_5 + \frac{1839R^2}{128} - \frac{459RQ_d}{8} + 63Q_d^2 + \frac{565X}{128} - \frac{1311Z_d}{8}, \end{aligned} \tag{834}$$

$$\begin{aligned} c_5 = {} & \frac{10549A^4}{128} + \frac{411A^2M}{64} - \frac{3561A^2R}{64} + \frac{267A^2Q_d}{8} + \frac{99AMN}{32} + 3AP_4 \\ & + \frac{2531AU}{64} - \frac{75AV}{32} + 21\mathcal{K}_d + \frac{5}{4}\mathcal{L}_d + \frac{1}{4}M^2 + \frac{27MN^2}{64} - \frac{1}{8}MR \\ & + \frac{27M\widetilde{N}}{64} + \frac{463R^2}{128} - \frac{63RQ_d}{8} + \frac{415X}{384} - \frac{139Z_d}{8}. \end{aligned} \tag{835}$$

Their combination $3c_5 - b_5$ is exactly D_5 of (229); with the Lagrange formula (593), which holds at any d, the coefficients (179)–(227), (834) and (835) give $C_2, \ldots, C_5$ of (222)–(223). The split into b_5 and c_5 is not visible in D_5: zero-weight vertex classes, for example, cancel from D_5 but not from b_5. It was checked separately against this note's own data at $d = 1, 2, 3$ (Section 34.12). The sensitivity is $\partial a^{(6)}/\partial b_5 = -\frac{2}{243}a^{(2)}$, about 2×10^{-4} at $d = 3$.

For $a_{\phi^2} = \sum_{n\ge 2} g^n \mathsf{a}_n(\epsilon)$, sixth order needs the regular Taylor orders

target	a_2	a_3	a_4	a_5	a_6
$a^{(4)}$	ϵ^2	ϵ^1	ϵ^0	–	–
$a^{(5)}$	ϵ^3	ϵ^2	ϵ^1	ϵ^0	–
$a^{(6)}$	ϵ^4	ϵ^3	ϵ^2	ϵ^1	ϵ^0

together with g_* through ϵ^{n-1}. Extending the fixed point alone does not give the next coefficient: each order needs new finite amplitudes.

34.6 Graphs, kernels and finite parts

34.6.1 External legs and the two lowest orders

Put the bulk point at unit distance from the defect. Both bulk legs end on defect vertices x_0, x_1, and every two-point graph reduces to an internal kernel $\propto |x_0 - x_1|^{-2b}$ followed by the external-leg integral

$$\mathsf{J}_d(a,b) = \int d^dx\, d^dy\, \frac{1}{(1+x^2)^a (1+y^2)^a |x-y|^{2b}} = \pi^d \frac{\Gamma(\frac{d}{2}-b)\Gamma(2a+b-d)\Gamma(a+b-\frac{d}{2})^2}{\Gamma(a)^2\Gamma(\frac{d}{2})\Gamma(2a+2b-d)}. \quad (836)$$

The closed form follows from Schwinger parameters (integrate the parameter of $|x-y|$ first, then the two others as a Gamma and a Beta integral) in the region of convergence; at the physical exponents it denotes the critical-mass continuation. A convergent normalization check is $\mathsf{J}_1(1,\frac{1}{4}) = \pi^2$. For an n-vertex graph the $n-2$ internal integrations fix $b = (2n-1)\Delta_\phi - (n-2)\frac{d}{2}$, and with h_d of Section 16

$$\mathsf{L}_n(\epsilon) = h_d^2\, \mathsf{J}_d\Big(\Delta_\phi, (2n-1)\Delta_\phi - \frac{(n-2)d}{2}\Big) = \frac{\Gamma(2q)\Gamma(-q+\frac{2n-1}{4}\epsilon)\Gamma(q-\frac{2n+1}{4}\epsilon)\Gamma(2q-\frac{n}{2}\epsilon)^2}{\Gamma(q-\frac{\epsilon}{4})^2\,\Gamma(4q-n\epsilon)}. \quad (837)$$

All $\mathsf{L}_n(0)$ are equal, to $54a^{(2)}$. Along the direction $(a,b) = (q+t, 3q-t)$, $\mathsf{J}_d = \pi^d\Gamma(2q)\Gamma(-q+t)/[\Gamma(4q)\Gamma(q+t)]$, so the kth derivative of $\log \mathsf{J}_d$ along $\partial_a - \partial_b$ at $(q,3q)$ is exactly e_{k-1} of (819). The Gamma ratio of (820) is, up to a constant, J_d on this line at $t = -\epsilon/4$; this shift is the origin of the factors $(-\frac{1}{4})^k$ in (824). When the assembly is carried out with these outer derivatives kept as independent symbols, they enter only through the combination $\mathsf{H}_4 - \frac{\mathsf{H}_3}{4}\frac{\partial_-\mathsf{J}_d}{\mathsf{J}_d} + \frac{\mathsf{H}_2}{32}\frac{\partial_-^2\mathsf{J}_d}{\mathsf{J}_d} - \frac{\mathsf{H}_1}{384}\frac{\partial_-^3\mathsf{J}_d}{\mathsf{J}_d} + \frac{1}{6144}\frac{\partial_-^4\mathsf{J}_d}{\mathsf{J}_d}$, $\partial_- = \partial_a - \partial_b$ at $(a,b) = (q,3q)$, which is the origin of (820); no all-order statement about the interacting amplitude is assumed.

The first graph is the two-vertex sunset with Wick weight $\frac{1}{6}$. With the normalized bubble

$$\mathsf{Y}(\epsilon) = \Gamma(\tfrac{d}{2})\frac{\Gamma(\frac{\epsilon}{2})^2\Gamma(\frac{d}{2}-\epsilon)}{\Gamma(\frac{d-\epsilon}{2})^2\Gamma(\epsilon)} = \frac{4}{\epsilon} + R\epsilon - \frac{V}{2}\epsilon^2 + \Big(\frac{R^2}{8} + \frac{7X}{48}\Big)\epsilon^3 + O(\epsilon^4), \quad (838)$$

whose constant term vanishes, the renormalized amplitudes of order g^2 and g^3 (at $\mu|y| = 1$) are

$$\mathsf{a}_2 = \frac{\mathsf{L}_2}{6}, \qquad \mathsf{a}_3 = \frac{\mathsf{L}_2}{\epsilon} - \frac{1}{4}\mathsf{Y}\mathsf{L}_3. \quad (839)$$

The pole of a_3 cancels because $\mathsf{L}_2(0) = \mathsf{L}_3(0)$. With $g_* \simeq \epsilon/3$ the sunset gives (823); the sign is that of $\Gamma(-q) < 0$ and involves no positivity statement about a divergent integral. The second arguments of L_2 and L_3 differ by $\epsilon/2$, so the finite part of a_3 is $\frac{1}{2}h_d^2\,\partial_b\mathsf{J}_d$ at $(a,b) = (q,3q)$. Adding the ϵ-dependence of L_2, the finite part of a_3 and the $O(\epsilon^2)$ term of g_* gives $a^{(3)}/a^{(2)} = \frac{4}{3}A + \frac{1}{4}[\psi(q) - \psi(-q)]$. At $d = 2$ and $d = 3$,

$$\begin{aligned} a_{\phi^2}(2,\epsilon) &= -\frac{\epsilon^2}{27} + \Big(\frac{1}{54} - \frac{8\log 4}{81}\Big)\epsilon^3 + O(\epsilon^4), \\ a_{\phi^2}(3,\epsilon) &= -\frac{\pi^{5/2}\epsilon^2}{576\sqrt{2}\,\Gamma(\frac{7}{4})^2} + \frac{4\sqrt{2\pi}}{6075}(13\pi - 28 - 64\log 2)\Gamma(\tfrac{9}{4})^2\epsilon^3 + O(\epsilon^4), \end{aligned} \quad (840)$$

which are the values of Eq. (3.36) of [8] (Section 34.12). The same two coefficients also follow from the exact bulk–defect relation of [8] and the two-loop ratio of its defect-channel coefficients,

without any higher-order one-point input. Let $b_0^{\Phi,-}/b_0^{\Phi,+} = \mathsf{r}_1\epsilon + \mathsf{r}_2\epsilon^2 + \cdots$ be the ratio of the bulk-to-defect coefficients of the two scalar defect modes of Φ, in the notation and normalization of [8]. Its Eqs. (3.25) and (3.34) give

$$\mathsf{r}_1 = \frac{\Gamma(1-q)\Gamma(2q)}{3\sqrt{6}\,\Gamma(1+q)}, \qquad \mathsf{r}_2 = -\frac{\mathsf{r}_1}{12}\big[13\psi(q) + 3\psi(-q) - 8\psi(2q) - 8\psi(1)\big], \tag{841}$$

and expanding the exact relation, its Eq. (2.11), through cubic order gives

$$a^{(2)} = \frac{\Gamma(2q)\Gamma(q)}{\Gamma(4q)\Gamma(-q)}\,\mathsf{r}_1^2, \qquad a^{(3)} = \frac{\Gamma(2q)\Gamma(q)}{\Gamma(4q)\Gamma(-q)}\Big(2\mathsf{r}_1\mathsf{r}_2 + \frac{\mathsf{e}_0}{4}\,\mathsf{r}_1^2\Big). \tag{842}$$

These are (823) and the value of $a^{(3)}/a^{(2)}$ above. The agreement is not independent of all low-order renormalization-group input: the diagrammatic route of this subsection still uses the known two-loop β function, through g_*.

Read literally, two general formulas of [8], its Eqs. (3.35) and (3.47), disagree in places with these values (Section 34.9; the package records the literal evaluations in `lri_ope_reproducibility/results/verification/low_order_checks.json`). These are apparent printed-formula inconsistencies, not conclusions about the authors' numerical code. The conventions used here are fixed by the operational definition (816) and the explicit low-order checks, not by a transcription of those formulas.

34.6.2 Four vertices

A tadpole-free two-point graph with n quartic vertices is a symmetric multiplicity matrix $m_{ij} \in \{0,1,2,3\}$ with zero diagonal, degrees 3 at the two roots, which carry the bulk legs, and 4 elsewhere, and a connected underlying graph. Up to root exchange and permutations of the $n-2$ internal vertices, its Wick weight is

$$w(m) = \frac{|\mathrm{orb}(m)|}{(n-2)!\prod_{i<j} m_{ij}!}, \tag{843}$$

and the sign $(-1)^n$ is kept separately. Reducible graphs are included. Graphs with both bulk legs on one vertex give only local mass and tadpole terms. The numbers of graphs for $n = 2,\dots,6$ are $1, 1, 5, 14, 64$. With edges listed as $(01, 02, 03, 12, 13, 23)$ and squared-distance exponents, the four-vertex graphs are

edge word	weight	internal reduction
$(0,0,3,3,0,1)$	$1/36$	chain $\mathsf{C}(3\Delta_\phi, \Delta_\phi, 3\Delta_\phi)$
$(0,1,2,2,1,1)$	$1/4$	crossed kite $\mathsf{K}_\times$
$(1,0,2,2,0,2)$	$1/8$	chain $\mathsf{C}(2\Delta_\phi, 2\Delta_\phi, 2\Delta_\phi)$
$(1,1,1,1,1,2)$	$1/4$	symmetric kite $\mathsf{K}_\parallel$
$(2,0,1,1,0,3)$	$1/12$	chain $\mathsf{C}(\Delta_\phi, 3\Delta_\phi, \Delta_\phi)$

where the normalized three-link chain is

$$\mathsf{C}(v_1,v_2,v_3) = \Gamma(\tfrac{d}{2})^2 \prod_{i=1}^{3} \frac{\Gamma(\frac{d}{2}-v_i)}{\Gamma(v_i)}\,\frac{\Gamma(v_1+v_2+v_3-d)}{\Gamma(\frac{3d}{2}-v_1-v_2-v_3)}. \tag{844}$$

The two kites are, through the needed order,

$$\mathsf{K}_\times = \frac{8}{\epsilon^2} + \frac{2A}{\epsilon} + 4R + O(\epsilon), \qquad \mathsf{K}_\parallel = \frac{4}{\epsilon^2} + \frac{2A}{\epsilon} + \frac{A^2}{2} + \frac{7}{2}R + O(\epsilon). \tag{845}$$

The symmetric kite is the master (231) at the absolute exponents $(2\Delta_\phi, 2\Delta_\phi, 2\Delta_\phi, 2\Delta_\phi, 4\Delta_\phi)$, that is, Pattern A with the coordinates (233) $t = \epsilon$, $p_0 = p_1 = 2\epsilon$ and $u = v = 0$; its expansion through

ϵ^0 is the Laurent expansion of $\mathcal{P}_A$ in (234), because the remainder (235) is $O(\epsilon)$. The crossed kite is not one of the patterns A, B, T: both of its internal vertices have absolute exponent sum $2d-2\epsilon$. Its finite part comes from contour deformation of a two-fold Mellin–Barnes representation, in which the crossed residues and the single-contour pieces give the displayed terms and the remaining double contour is $O(\epsilon)$. Keeping only the poles would give the wrong $a^{(4)}$. Summing the weights gives the internal kernel

$$\mathsf{S}_4 = \frac{9}{2\epsilon^2} + \frac{A}{\epsilon} + \frac{A^2}{8} + 3R + \frac{M}{9} + \epsilon\Big[\frac{M(A+N)}{18} + AR + \frac{U}{8} - \frac{9}{4}V\Big] + O(\epsilon^2), \tag{846}$$

and the renormalized amplitude

$$\mathsf{a}_4 = \Big(\frac{9}{2\epsilon^2} - \frac{A}{\epsilon}\Big)\mathsf{L}_2 - \frac{9}{4\epsilon}\mathsf{Y}\mathsf{L}_3 + \mathsf{S}_4\mathsf{L}_4. \tag{847}$$

Its finite limit is

$$\mathsf{a}_4(0) = h_d^2\Big[\frac{9}{8}\,\partial_b^2\mathsf{J}_d - A\,\partial_b\mathsf{J}_d + \Big(\frac{A^2}{8} + \frac{3}{4}R + \frac{M}{9}\Big)\mathsf{J}_d\Big]_{(a,b)=(q,3q)}. \tag{848}$$

With a_2 through ϵ^2, a_3 through ϵ and g_* through ϵ^3, the M terms cancel and H_2 of (821) follows. The full finite-ϵ Mellin–Barnes values of both kites and the exact Gamma chains reproduce the assembled $a^{(4)}$ at $d=1,2,3$ by extrapolation $\epsilon\to 0$ to better than 10^{-13}; at $d=2$ the symmetric kite also has the exact complex-Selberg form $2\gamma(a)^2\gamma(a+c)^2\gamma(2c)/[\gamma(2a+c)\gamma(2a+2c)\gamma(c)]$ with $\gamma(z)=\Gamma(z)/\Gamma(1-z)$, $a=1-\Delta_\phi$, $c=-\Delta_\phi$.

34.6.3 Five vertices

The fourteen five-vertex graphs, with edges ordered $(01,02,03,04,12,13,14,23,24,34)$ and roots $0,1$, have the finite internal coefficients below (before the Wick weight). Graphs 1 and 13 are scaleless mass terms. Graph 7 has no finite part but a pole: it is a marginal bubble times the regular tetrahedron, $4Q_d/\epsilon + O(\epsilon)$, and the vanishing balanced first derivative of the tetrahedron (239) is what removes its finite part. At $d=2$ the package also evaluates this graph as an exact three-variable complex-Selberg integral at finite ϵ; with the pole removed, its extrapolation to $\epsilon=0$ is 1.3×10^{-36}, consistent with a zero finite part (task `integrals`, record `lri_ope_reproducibility/results/verification/a5_numeric/verification_results.json`).

graph	edge word	weight	finite internal coefficient
1	0003111300	1/36	0
2	0003120210	1/12	$-M(A+N)$
3	0012102300	1/24	$3M(A+N)$
4	0012111210	1/2	$5A^3/18+65AR/6+43U/36-163V/18$
5	0012120120	1/4	$-7A^3/18+77AR/6+7U/36-247V/18$
6	0012210111	1/4	$-A^3/3+11AR+11U/6-13V/3$
7	0111111111	1/6	0 (simple pole $4Q_d/\epsilon$)
8	1002011310	1/6	$2M(A+N)$
9	1002020220	1/16	$-40V$
10	1002110211	1/4	$2A^3/9+26AR/3+11U/9-166V/9$
11	1011011220	1/8	$8A^3/9+32AR/3+23U/9-160V/9$
12	1011101211	1/2	$3A^3/2+13AR/2+23U/12-17V/2$
13	2001001311	1/24	0
14	2001010221	1/8	$5M(A+N)$

The weighted sum gives

$$\begin{aligned}\mathsf{S}_5 = {}& \frac{18}{\epsilon^3}+\frac{11A}{\epsilon^2}+\frac{b_3/9+3A^2/2+45R/2+4M/3}{\epsilon}\\ &+\frac{7}{8}A^3+M(A+N)+\frac{145}{8}AR+\frac{43}{16}U-\frac{181}{8}V+O(\epsilon),\end{aligned} \tag{849}$$

whose finite term is new integral data; pole cancellation does not fix it. The renormalized amplitude is

$$\mathsf{a}_5=\Big[\frac{18}{\epsilon^3}-\frac{10A}{\epsilon^2}+\frac{b_3}{9\epsilon}\Big]\mathsf{L}_2+\Big[-\frac{27}{2\epsilon^2}+\frac{9A}{4\epsilon}\Big]\mathsf{YL}_3+\frac{12}{\epsilon}\mathsf{S}_4\mathsf{L}_4-\mathsf{S}_5\mathsf{L}_5. \tag{850}$$

The $O(\epsilon)$ term of S_4 reaches the finite part through $12/\epsilon$. After the lower Taylor terms and g_* are inserted, all M and MN terms cancel and H_3 of (821) remains.

34.6.4 Six vertices

The 64 six-vertex graphs give the kernel

$$\begin{aligned}\mathsf{S}_6 = {}& \frac{135}{2\epsilon^4}+\frac{153A}{2\epsilon^3}+\frac{399A^2/8+23M/2+1053R/8+9Q_d}{\epsilon^2}\\ &+\frac{1}{\epsilon}\Big[\frac{365}{16}A^3+\frac{283}{24}AM+\frac{2979}{16}AR+\frac{9}{2}AQ_d+\frac{81}{8}MN+\frac{1}{2}P_4+\frac{985}{32}U-\frac{2733}{16}V\Big]\\ &+\mathsf{w}_6+O(\epsilon).\end{aligned} \tag{851}$$

All four pole coefficients agree with the counterterm constraints, including the parts multiplying derivatives of the external-leg integral; this checks the subtractions and does not determine w_6.

The new data at sixth order are $\mathsf{w}_4 = [\epsilon^2]\mathsf{S}_4$, $\mathsf{w}_5 = [\epsilon]\mathsf{S}_5$, $\mathsf{w}_6 = [\epsilon^0]\mathsf{S}_6$ and b_5:

$$\begin{aligned}\mathsf{w}_4 &= \frac{A^4}{384} + \frac{7A^2M}{288} + \frac{7A^2R}{64} + \frac{7AMN}{144} + \frac{11AU}{192} - \frac{47AV}{96} \\ &\quad + \frac{\mathsf{c}_\times}{4} + \frac{7MN^2}{288} - \frac{M\widetilde{N}}{288} + \frac{103R^2}{128} + \frac{85X}{96}, \qquad (852)\\ \mathsf{w}_5 &= \frac{113A^4}{576} + \frac{55A^2M}{96} + \frac{713A^2R}{96} + \frac{55AMN}{48} + \frac{619AU}{288} - \frac{2071AV}{144} + \frac{9}{2}\mathsf{c}_\times + \frac{55MN^2}{96} + \frac{7MR}{12} \\ &\quad + \frac{7M\widetilde{N}}{96} + \frac{643R^2}{64} + \frac{3}{2}RQ_d + \frac{2179X}{192} - \frac{5Z_d}{12}, \qquad (853)\\ \mathsf{w}_6 &= \frac{1139A^4}{128} + \frac{1697A^2M}{192} + \frac{9057A^2R}{64} + \frac{9}{8}A^2Q_d + \frac{1513AMN}{96} + \frac{2549AU}{64} \\ &\quad - \frac{6545AV}{32} + \frac{189}{4}\mathsf{c}_\times + 5\mathcal{K}_d + \frac{1}{4}\mathcal{L}_d + \frac{11M^2}{108} + \frac{459MN^2}{64} + \frac{89MR}{8} + \frac{123M\widetilde{N}}{64} \\ &\quad + \frac{11345R^2}{128} + \frac{231}{8}RQ_d + \frac{40211X}{384} - \frac{89Z_d}{8}. \qquad (854)\end{aligned}$$

Here $\mathsf{c}_\times$ is the quadratic Taylor coefficient of the crossed-kite pattern along its one undetermined direction. It is kept as a free symbol and cancels:

$$10\cdot\frac{1}{4} - \frac{4}{3}\cdot\frac{9}{2} + \frac{2}{27}\cdot\frac{189}{4} = 0, \qquad (855)$$

because the scalar assembly is

$$\begin{aligned}\mathsf{H}_4 &= \frac{4811A^4}{1296} + \frac{23A^2M}{162} - \frac{839A^2R}{216} + \frac{253A^2Q_d}{81} - \frac{2AMN}{81} + \frac{56AP_4}{81} + \frac{521AU}{324} + \frac{2}{3}AV - \frac{M^2}{729} \\ &\quad - \frac{MR}{18} + \frac{4MQ_d}{81} - \frac{161R^2}{144} - \frac{13RQ_d}{27} + \frac{28Q_d^2}{81} - \frac{35X}{24} + 10\mathsf{w}_4 - \frac{4}{3}\mathsf{w}_5 + \frac{2}{27}\mathsf{w}_6 - \frac{2}{243}b_5. \qquad (856)\end{aligned}$$

Substituting (852)–(854) and (834) removes M, N, $\widetilde{N}$ and $\mathsf{c}_\times$ and gives (822). The masters survive only through $\mathcal{K}_d, \mathcal{L}_d, Z_d, P_5$; no further master enters.

34.6.5 Finite parts and local subtractions

Graphs that are not reduced by convolution (230) and star–triangle are completed conformally as in (232). Their edge exponents are written $\alpha_e = q\,r_e + \epsilon(v_e + \vartheta\,t_e)$ with rational r_e, v_e, t_e; the auxiliary coordinate ϑ along the direction (t_e) separates the logarithmic collision poles. For a UV cluster γ the degree $(|\gamma|-1)\frac{d}{2} - \sum_{e\subset\gamma}\alpha_e$ is ϵ times a linear function of ϑ; IR poles are treated after inversion. At each critical hyperplane the residue factorizes into the normalized anchored subgraph period and its contracted cograph, and the poles are subtracted before the analytic remainder is reconstructed. With m remaining integrations, sixth order needs exponent jets through degree $4-m$. The pole-subtracted remainder is a polynomial of that degree in the deformations; its restrictions to reference hyperplanes with a vanishing completion edge reduce by star–triangle, and exact rational linear algebra checks that these restrictions determine it. The jets used are those of the patterns A, B and T, (235), (237) and (239), together with the wheel derivatives (244); Gamma jets are expanded in abstract polygammas, and no fit in the dimension is made. The one direction not fixed by these restrictions is $\mathsf{c}_\times$.

The local counterterms are the recursive forest formula of Section 16, $Z_G = -\,\mathrm{PP}[I_G + \sum_{\mathcal{F}\neq\varnothing}(\prod_{\gamma\in\mathcal{F}} Z_\gamma) I_{G/\mathcal{F}}]$ over disjoint proper logarithmic subgraphs, with radial factor $2/(L\epsilon)$ for an L-loop vertex graph. With the enumeration's sign convention,

$$b_L = L(-1)^L \sum_G w_4(G)\,[\epsilon^{-1}]Z_G, \qquad c_L = L(-1)^L \sum_G w_2(G)\,[\epsilon^{-1}]Z_G, \qquad (857)$$

over the 129 five-loop vertex classes of Section 16, with w_4 and w_2 the four-point and mass weights. The mass-weighted sum is computed separately; c_5 is not obtained by fitting a thermal coefficient. The reconstruction presumes the master identities and the analytic pole structure of Parts II–III. Alternative rational directions test its consistency inside that framework, not its independence from those inputs.

34.7 Values at $d = 1, 2, 3$

With the closed forms of Section 1.1 at $d = 1, 2$ and the $d = 3$ values of Sections 19.10 and 21.11 (with $\mathcal{L}_3$ and Z_3 from the same evaluation as $C_5(3)$), (825) gives

	$d = 1$	$d = 2$	$d = 3$
$a^{(2)}$	−0.139409946645025	−0.0370370370370370	−0.0254241874995486
$a^{(3)}$	−0.850425350705174	−0.118399443073569	−0.0667823808153469
$a^{(4)}$	1.67987388417139	0.0717109289555818	−0.0279145897482122
$a^{(5)}$	−26.2720127156913	−0.751719824443267	−0.156637777571136
$a^{(6)}$	450.850442331799	6.09946795238583	0.507382663942981
$13Z_d - \mathcal{L}_d - 12\mathcal{K}_d$	15277.4261926043	913.335423400404	163.599472783843

The digits shown are exact roundings: the master values are known to at least 40 digits at all three dimensions. The supplied package evaluates the $d = 3$ nonprimitive masters by double-precision spectral quadrature with analytic spin tails; its $\mathcal{K}_3$, $\mathcal{L}_3$, Z_3 agree with this note's values to about 12 significant digits (largest difference 1.4×10^{-11}, in $\mathcal{L}_3$, a relative 2.2×10^{-13}), its $a^{(6)}(3) = 0.507382663943009$ differs from the table by 2.8×10^{-14}, and its $d = 2$ value before the closed forms of Section 17.2 differed by 2.7×10^{-13}. The table entries are coefficients, not estimates of a_{ϕ^2} at finite ϵ. At $d = 1, 2$ they alternate in sign from $a^{(3)}$ on, and at all three dimensions they grow quickly, so a truncated series at moderate ϵ is not reliable (Section 34.11).

34.8 Two dimensions

At $d = 2$, $q = \frac{1}{2}$ and $A = 4\log 2$, $R = X = 0$, $U = \psi_2(\frac{1}{2}) = -14\zeta(3)$, $V = 2\psi_2(1) = -4\zeta(3)$, $\mathsf{e}_j = j!\,2^{j+1}$. The external-leg ratio is elementary, $\Gamma(-\frac{1}{2} - \frac{\epsilon}{4})/\Gamma(\frac{1}{2} - \frac{\epsilon}{4}) = -2/(1 + \frac{\epsilon}{2})$, so (820) reads

$$a_{\phi^2}(2, \epsilon) = -\frac{\epsilon^2}{27}\,\frac{1 + \mathsf{H}_1\epsilon + \mathsf{H}_2\epsilon^2 + \mathsf{H}_3\epsilon^3 + \mathsf{H}_4\epsilon^4}{1 + \epsilon/2} + O(\epsilon^7). \tag{858}$$

Every master in $\mathsf{H}_1, \ldots, \mathsf{H}_4$ is known in closed form at $d = 2$: Q_2 of (267); $P_4(2)$ and $P_5(2)$ of (333), (334); $\mathcal{L}_2 = -2\log 2\, P_4(2)$ of (270), proved in Section 18.7; $\mathcal{K}_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2$ of (271), proved in Section 18.8 for the master-path definition (238), which is the definition used in (822); and $Z_2 = 4\mathsf{G}Q_2$ of (265), proved in Section 16.7. Given the supplied calculation of $\mathsf{H}_1, \ldots, \mathsf{H}_4$

(Section 34.1), the coefficients are therefore exact. With Q_2, $P_4 = P_4(2)$ and $P_5 = P_5(2)$,

$$\begin{aligned}
a^{(3)}(2) &= \frac{1}{54} - \frac{16\log 2}{81}, \qquad -27\,a^{(4)}(2) = \frac{1}{4} - \frac{8\log 2}{3} + \frac{100\log^2 2}{9} - \frac{4}{9}Q_2,\\
-27\,a^{(5)}(2) &= -\frac{1}{8} + \frac{4\log 2}{3} - \frac{50\log^2 2}{9} + \frac{272\log^3 2}{27} - \frac{485}{108}\zeta(3)\\
&\quad + \Big(\frac{2}{9} - \frac{16\log 2}{9}\Big)Q_2 + \frac{4}{27}P_4,\\
-27\,a^{(6)}(2) &= \frac{1}{16} - \frac{2\log 2}{3} + \frac{25\log^2 2}{9} - \frac{136\log^3 2}{27} - \frac{28\log^4 2}{81}\\
&\quad + \frac{485}{216}\zeta(3) - \frac{2933}{324}\zeta(3)\log 2 - \frac{896}{81}\mathsf{G}^2\\
&\quad + \Big(\frac{350}{81}\mathsf{G} - \frac{56\log^2 2}{81} + \frac{8\log 2}{9} - \frac{1}{9}\Big)Q_2 - \frac{14}{81}Q_2^2 + \Big(\frac{70\log 2}{81} - \frac{2}{27}\Big)P_4 - \frac{16}{81}P_5,
\end{aligned} \tag{859}$$

with $a^{(2)}(2) = -\frac{1}{27}$. So $a^{(6)}(2)$ needs no constant beyond those of $C_5(2)$ in (272): $\Gamma(\frac{1}{4})$, π, $\log 2$, G, $\zeta(3)$, $F_\pm$ and the five values $\mathcal{Y}_r$. Its value is $a^{(6)}(2) = 6.0994679523858305821\ldots$

The thermal map at $d = 2$. The extra combination of (829) is

$$13Z_2 - \mathcal{L}_2 - 12\mathcal{K}_2 = 52\mathsf{G}Q_2 + 2\log 2\,P_4(2) - 384\mathsf{G}^2 + 336\,\zeta(3)\log 2 = 913.335423400404160696 29\ldots, \tag{860}$$

and (192), (269) give, with C_j standing for $C_j(2)$ in this paragraph,

$$Q_2 = 2\log^2 2 - \frac{27}{2}C_3, \qquad P_4(2) = \frac{81}{2}C_4 + 8\log^3 2 - 27\log 2\,C_3 + \frac{359}{16}\zeta(3). \tag{861}$$

Inserting these into (829) removes (860):

$$\begin{aligned}
\mathsf{I}_1(2) &= \frac{16\log 2}{3} - \frac{1}{2}, \qquad \mathsf{I}_2(2) = 6C_3 - 4\log^2 2 + \frac{1}{8},\\
\mathsf{I}_3(2) &= 6C_4 - 12\log 2\,C_3 + \frac{304}{81}\log^3 2 - \frac{7}{6}\zeta(3) - \frac{1}{24},\\
\mathsf{I}_4(2) &= 6C_5 - 11\log 2\,C_4 + \frac{46\log^2 2 - 26\mathsf{G}}{3}C_3 - 9C_3^2 - \frac{380}{81}\log^4 2\\
&\quad + \frac{104}{81}\mathsf{G}\log^2 2 - \frac{128}{27}\mathsf{G}^2 + \frac{1303}{216}\zeta(3)\log 2 + \frac{1}{64}.
\end{aligned} \tag{862}$$

At $d = 2$ the one-point coefficients through $a^{(6)}$ thus follow from $C_3(2)$, $C_4(2)$, $C_5(2)$ and the constants $\log 2$, G, $\zeta(3)$. Numerically $\mathsf{I}_1, \ldots, \mathsf{I}_4$ are 3.19678496299, -7.04591213159, 37.3758123361 and -277.339425824.

Checks at $d = 2$. The following were checked with code of their own (`ope_checks/d2/ope_d2_check.py`, record `ope_d2_check.json`), which takes every $d = 2$ input from this note. Exactly: the logarithm of (858), with $\mathsf{H}_1, \ldots, \mathsf{H}_4$ of (821), (822), the closed forms above and $C_3(2)$, $C_4(2)$, $C_5(2)$ from (269), gives (862), and also (829) at $d = 2$; (859) and the second form of (860) follow in the same way, and the two forms of $C_5(2)$, (269) and (272), agree. All residuals are zero. Numerically, at 60 digits: the closed-form masters reproduce the supplied package's $6Q_2$, $P_4(2)$, $P_5(2)$ and Z_2 to 2×10^{-58}, and (862) with the numerical $C_3(2)$, $C_4(2)$, $C_5(2)$ reproduces (825) to 10^{-59}. These checks take H_4 as given. They test the two-dimensional specialization and the shared masters, not the six-vertex calculation.

The five-loop coefficient b_5 enters $a^{(6)}$ on its own, not only through $D_5 = 3c_5 - b_5$ of (229) (Section 34.5). Our two-dimensional productions use the finite-MOM coefficients $\widehat{b}_L(z) = \sum_j b_{L,j} z^j$

of (693). The conversion of Section 34.12 (item "Five-loop split") to regulator-independent coefficients gives, explicitly,

$$b_5 = b_{5,0} + \frac{9}{4}b_{4,1} + \frac{9}{2}b_{3,2} + \frac{27}{4}b_{2,3} - \frac{1}{2}b_{2,1}b_{3,0} + \frac{1}{3}b_{2,0}b_{3,1} + \frac{1}{2}b_{2,0}b_{2,2} - \frac{3}{8}b_{2,1}^2, \tag{863}$$

where $b_{2,j}$ are the exact two-loop jets. With D_5 held at (229), the jets of the seventh-order production (Section 28.7) and of the eighth-order production (Section 31), which use independent random streams, give $b_5(2) = 135708.65 \pm 0.27$ and 135709.17 ± 0.42, together 135708.80 ± 0.23; (834) gives $135708.425\ldots$. Without that constraint the two productions give 135709.43 ± 0.55 and 135709.64 ± 0.96, because their five-loop jets fluctuate along D_5 (by -2.1 and -0.7 standard errors). The same conversion reproduces b_4 and c_4 of (226), (227) within 2.0 standard errors. Since $a^{(6)}(2)$ changes by 3.05×10^{-4} per unit of b_5, this checks the five-loop input of $a^{(6)}(2)$ at the level of 10^{-4} in $a^{(6)}(2)$, against data of this note only (`ope_checks/d2/ms_b5_from_jets.py`, records `ms_b5_c7d2.json`, `ms_b5_c8d2.json`).

34.9 One dimension

At $d = 1$ every master in (822) is a closed form proved in Parts II–III: $Q = \Gamma(\frac{1}{4})^4/\pi$, $P_4 = 13B_- - 3B_+$, P_5 in the five values $\mathcal{Y}_r$ (Section 1.1), $\mathcal{K}$ (76), $\mathcal{L}_1$ (541) and $Z_1 = QH$ (224). The coefficients $a^{(2)}(1), \ldots, a^{(6)}(1)$ of Section 34.7 are therefore exact expressions in the same fixed values as $C_5(1)$: $\Gamma(\frac{1}{4})$, π, $\log 2$, G, $\zeta(3)$, $F_\pm$ and the $\mathcal{Y}_r$. They are conditional only on the calculation itself (Section 34.1). The first two are

$$a_{\phi^2}(1,\epsilon) = -\frac{2\pi^{3/2}\Gamma(\frac{3}{4})}{27\,\Gamma(\frac{1}{4})}\,\epsilon^2\Big[1 + \Big(\frac{13\pi}{12} + \frac{16}{3}\log 2 - 1\Big)\epsilon\Big] + O(\epsilon^4). \tag{864}$$

Both agree with the general formulas of [8]: $a^{(2)}$ with its quadratic coefficient for the Ising case, Eq. (3.35), and $a^{(3)}$ with its cubic coefficient for the long-range $O(N)$ model at $N = 1$, Eq. (3.47). Read literally, its Ising cubic coefficient in Eq. (3.35) and the sign of its $O(N)$ quadratic coefficient in Eq. (3.47) disagree with these values and with its own Eq. (3.36).

Two checks exist only at $d = 1$. First, the package's reconstruction of all six-vertex kernels and five-loop forest sums at 60 digits, with rational regulator and reference directions other than those of the derivation and $\mathsf{c}_\times = 17$, runs at $d = 1$. It differs from (822) by 7.6×10^{-54} in $a^{(6)}(1)$, with residual poles below 1.4×10^{-52}; it shares the master identities and the subtraction framework (Section 34.1). Second, the split of D_5 into b_5 and c_5, which $a^{(6)}$ needs and (229) does not fix, is compared at $d = 1$ with numerical graph integrations that do not use the master reduction (Section 34.12). The residue certificate of Appendix A cannot serve for this: it omits the 28 five-loop classes of zero weight in $3c_5 - b_5$, which is correct for D_5 but not for b_5 alone. Since b_5 enters (856) as $-\frac{2}{243}b_5$, the coefficient $a^{(6)}(1)$ moves by $-\frac{2}{243}a^{(2)}(1) = 1.15 \times 10^{-3}$ per unit of b_5.

34.10 A one-dimensional functional cross-check

In one dimension the super sum rule of [10] gives a determination of the low-order thermal coefficients that is independent of the graphs. Its expansion parameter $\epsilon = 1 - 4\Delta_\phi$ is ours at $d = 1$. Writing $\Delta_{\mathcal{E}} = 2\Delta_\phi + \gamma^{(1)}\epsilon + \gamma^{(2)}\epsilon^2 + \cdots$, the functional $\widehat{\beta}_0 = -\frac{\epsilon}{4}\widetilde{\beta}_0$ gives $-\gamma^{(1)} + 3(\gamma^{(1)})^2 = 0$, so $\gamma^{(1)} = \frac{1}{3} = C_1 + \frac{1}{2}$ on the nontrivial branch. At the next order, assuming that only $\mathcal{E}$ contributes, a high-precision numerical evaluation of the functional gives $\gamma^{(2)} = \frac{1}{9}(\pi + 4\log 2) = C_2(1)$. The cross-check therefore concerns C_1 and $C_2(1)$; it does not test the one-point coefficients. At the next order the sums over double-trace operators and over the twist-four family, and the ϕ^4 operator, contribute at $O(\epsilon^3)$ although each term starts at $O(\epsilon^4)$, for example through

$$\epsilon^4 \sum_{n\geq 1} n^{-1-\nu\epsilon} = \frac{\epsilon^3}{\nu} + \gamma_{\mathrm{E}}\epsilon^4 + \cdots; \tag{865}$$

their cancellation requires an exponent that [10] does not determine, so [10] does not determine $C_3(1)$, and no functional determination of it is made here. In that reference a_{ϕ^2} denotes the squared defect OPE coefficient of ϕ^2 in the four-point function of ϕ, not the bulk one-point coefficient of this part.

34.11 Comparison with the numerical bootstrap

At fixed d and Δ_ϕ, the bootstrap of [8] constrains a region in the plane $(a_{\phi^2}, \Delta_{\mathcal{E}})$; a_{ϕ^2} is its horizontal coordinate. Figure 11 of [8] shows allowed regions at $d = 3$ for four external dimensions; their upper right corners (kinks) are the candidate long-range Ising points. The comparison below inserts the series of this note into one resummation prescription and places the result next to readings of that figure. It does not rerun the bootstrap, it has no access to the underlying numerical data, and it does not provide an excluding functional. The package draws both comparisons, with its pixel traces of the published boundaries and kinks, in `lri_ope_reproducibility/results/figures/`: `3d_sixth_order_comparison.png` and `2d_sixth_order_comparison.png`, and, for the successive truncations and the two resummations at the weakest coupling, `3d_weak_point_orders.png` and `2d_weak_point_orders.png`. These plots use the older C_6 inputs; `results/bootstrap/current/` holds the same plots with the current ones. The traces are approximate readings, not bootstrap data, and the plots are not exclusion certificates.

Prescription. For a truncated series $F(\epsilon) = \sum_k c_k \epsilon^k$ use the Borel transform $\sum_k c_k t^k/k!$, its $[L/M]$ Padé approximant, and the Laplace integral $\int_0^\infty e^{-t}[L/M](\epsilon t)\, dt$. Horizontally, resum $a_{\phi^2}/\epsilon^2 = a^{(2)} + \cdots + a^{(6)}\epsilon^4$ with [3/1] and [1/3], whose denominators have no positive real root at these points; [2/2] has one at every point and is not used. Vertically, resum $\Delta_{\mathcal{E}}$ through C_6 with [4/2]; the package also records [3/2] through C_5 as an order diagnostic. Factoring out ϵ^2 is part of the prescription. No bootstrap point and no short-range endpoint is fitted, and varying the Borel–Leroy parameter, the approximant or the factored power would add uncertainty that has not been quantified.

Three dimensions. The four external dimensions of Figure 11 are equally spaced between $\Delta_\phi = \frac{3}{4}$ and the short-range end $\Delta_\phi = 0.5181489$, the short-range Ising value used by [8] (Section 34.2), so $\epsilon = 3 - 4\Delta_\phi$:

Δ_ϕ	ϵ	a_{ϕ^2}, [3/1]	a_{ϕ^2}, [1/3]	$\Delta_{\mathcal{E}}$, [4/2]	kink of Fig. 11
0.70363	0.18548	−0.00135602	−0.00134382	1.4738699	(−0.00187, 1.4700)
0.65726	0.37096	−0.00797138	−0.00713938	1.4541700	(−0.00292, 1.4500)
0.61089	0.55644	−0.0253361	−0.0185657	1.4383505	(−0.0110, 1.4318)
0.56452	0.74192	−0.0613651	−0.0352226	1.4248531	(−0.0420, 1.4154)

The [4/2] column uses $C_6(3) = 33.5754$ of Section 26.8; with the older input 33.57 it is lower by 4.6×10^{-8}, 1.2×10^{-6}, 7.6×10^{-6} and 2.6×10^{-5}. The kinks are read from the vector data of the published figure (`ope_checks/d3/check_fig11.py`, record `ope_checks/d3/check_fig11.json`; Section 34.12). The figure file of the arXiv source, converted with `pdftocairo -svg`, draws each region boundary as a polyline; its vertices are mapped to data coordinates by the axis frame, and the kink is the vertex with the largest turning angle in axis-normalized coordinates. Vertices that should lie on round values, such as $a_{\phi^2} = -0.005$ or $\Delta_{\mathcal{E}} = 1.45$, are recovered to within 7×10^{-7}, so a kink is limited by the spacing of the polyline vertices around it, 3×10^{-4} to 4×10^{-3} in a_{ϕ^2}, not by the reading. The supplied package used pixel readings that agree to about one pixel.

At the weakest coupling, $\Delta_\phi = 0.70363$, the drawn region at $\Delta_{\mathcal{E}} = 1.4739$ requires $a_{\phi^2} \lesssim -0.0032$, while both resummations give -0.0013 to -0.0014: this resummed point lies outside the drawn

region. The truncated sums through ϵ^4, ϵ^5 and ϵ^6 are -0.00133384, -0.00136823 and -0.00134757, and the cubic-order point that [8] itself plots, $(-0.0013004, 1.473078)$, is also on the excluded side of its own boundary. The mismatch is therefore not created by the higher orders; equally, it does not test them. At this point the thermal terms $C_4\epsilon^4$, $C_5\epsilon^5$, $C_6\epsilon^6$ are about $+0.00145$, -0.00128, $+0.00137$, so the vertical coordinate is not known to six digits from apparent Padé stability.

What the mismatch can and cannot mean. An excluding functional must be positive on every admissible exchanged dimension and spin. The bounds of [8] impose the long-range OPE relations on a grid of exchanged dimensions (its App. A.2: 3000 points, and up to 12000 in its refined two-dimensional runs, where the kinks move up as the grid is refined). A functional positive on the grid can be negative between grid points, which would make a finite-grid boundary too restrictive. The two numerical truncations have different logical effects: keeping fewer crossing derivatives, with positivity still imposed on the continuum, only restricts the available functionals and weakens the exclusions, so it cannot by itself make a true solution inconsistent; sampling the exchanged dimension instead can miss a negative interval. Near the generalized-free end the spectrum approaches double-twist values at a rate set by ϵ^2; the package estimates the leading spin-2 shift as $-4\epsilon^2/[9d(d+2)] \approx -0.0010$ at the first point, smaller than a grid spacing of order 0.007 (this estimate was not checked here). This is a possible mechanism, not a demonstrated one. The decisive test would be to check the candidate functional between grid points, including the smallest eigenvalue of each matrix sector and the large-dimension tail. If positivity held and both coordinates had controlled errors, the discrepancy would have to be resolved in the perturbation theory, the resummation or the spectral assumptions. None of this has been done.

Two dimensions. Figure 7 of [8] shows, at $d = 2$ and eight external dimensions between $\Delta_\phi = 0.13333$ and 0.425, the kinks of runs with 3000, 6000 and 12000 grid points and the authors' own two-sided Padé estimates; it shows kinks, not allowed regions. With $\epsilon = 2 - 4\Delta_\phi$, the four weakest couplings give

Δ_ϕ	ϵ	a_{ϕ^2}, [3/1]	a_{ϕ^2}, [1/3]	$\Delta_\mathcal{E}$, [4/2]	kink of Fig. 7	$\Delta_\mathcal{E}$, [5/3] to C_8
0.425	0.3	-0.00654824	-0.00601150	0.9678196	$(-0.0060, 0.9600)$	0.9679055
0.35	0.6	-0.0414334	-0.0277046	0.9578452	$(-0.0430, 0.9602)$	0.9590956
0.275	0.9	-0.134446	-0.0621047	0.9629664	$(-0.1420, 0.9660)$	0.9680003
0.2	1.2	-0.325205	-0.105209	0.9803317	$(-0.3750, 0.9878)$	0.9929583

The [4/2] column uses $C_6(2) = 362.366$ of Section 26; with the older input 362.3 it is lower by 1.7×10^{-6}, 2.6×10^{-5}, 1.1×10^{-4} and 2.7×10^{-4}, and the allowance ± 0.004 of $C_6(2)$ moves the first row by 10^{-7}. As at $d = 3$, [2/2] has a positive Borel pole at every point. The kinks are those of the 12000-point runs. Each is the centre of its cross marker, read from the marker's placement in the vector figure file of the arXiv source of [8] (`ope_checks/d2/check_fig7.py`, record `check_fig7.json`), which places it to about 3×10^{-9} in a_{ϕ^2} and 2×10^{-10} in $\Delta_\mathcal{E}$; the supplied package's pixel readings agree to 2×10^{-4} in a_{ϕ^2}, except at $\Delta_\phi = 0.425$, where its -0.00676 is 7.6×10^{-4} off, and to 8×10^{-5} in $\Delta_\mathcal{E}$.

At the weakest coupling, $\Delta_\phi = 0.425$, the kink at $a_{\phi^2} = -0.0060$ lies between the two horizontal resummations, while the resummed $\Delta_\mathcal{E} = 0.96782$ lies 0.0078 above it. The truncated sums $\sum_{n \le k} a^{(n)}(2)\,\epsilon^n$ at this point are -0.00594925977, -0.00777593895 and -0.00332942681 for $k = 4, 5, 6$, so the last available term is not an error estimate. The kinks of the same figure move up as the grid is refined, from 0.9575 at 3000 points to 0.9592 and 0.9600, and the authors' own Padé estimates at this Δ_ϕ are 0.9695 with all their perturbative data and 0.9723 without the order ϵ^3; [8] itself notes that the kinks of best agreement lie noticeably below its perturbative line. So the gap was present before the orders computed here, as at $d = 3$. At the three stronger couplings the resummed points lie 0.0023 to 0.0074 below the kinks. The last column is a diagnostic only: it adds

$C_7(2)$ of Section 28 and $C_8(2)$ of Section 31 in [5/3], which has no positive Borel pole at the four points ([4/3], [3/4], [4/4] and [3/5] have one at $\epsilon = 0.3$). It moves $\Delta_{\mathcal{E}}$ at $\epsilon = 0.3$ by 8.6×10^{-5}, and by at most 7.4×10^{-6} more when $C_7(2)$ and $C_8(2)$ vary over their allowances; at the three stronger couplings it moves it by 1.3×10^{-3}, 5.0×10^{-3} and 1.3×10^{-2}, comparable to the distances from the kinks. Neither column is an error estimate (`ope_checks/d2/ope_d2_check.py`).

34.12 Checks made for this note

The following were done independently of the supplied code, with this note's inputs; the scripts and their outputs are in `ope_checks/`.

1. *Rerun.* All tasks of the package except the TEX build (low-order checks, symbolic reconstruction of the five- and six-vertex kernels and of the 129-class forest sums, the $d = 1$ alternative-regulator reconstruction at 60 digits, the four-vertex finite-regulator checks, the $d = 2, 3$ spectral masters, and the comparison plots) pass on a clean extraction on the symmetry cluster (Intel Xeon Gold 6148, Python 3.13.3, the pinned library versions). Every output equals the shipped one byte for byte, apart from recorded run times.

2. *Exact algebra* (32 identities, own code): the package's $C_2, \ldots, C_5$ equal (222)–(223); its b_3, c_3, b_4, c_4 equal (179), (226), (227); $3c_5 - b_5$ from (834)–(835) equals (229), and Lagrange inversion with these coefficients gives (223); (833) solves (831); (856) with (852)–(854) and (834) gives (822); (824), (826), (827), (829) and (830) are exact.

3. *Low orders.* $a^{(2)}$ and $a^{(3)}$ at $d = 2, 3$ equal Eq. (3.36) of [8], (840), to 35 digits. At $d = 1$ they equal the quadratic coefficient of Eq. (3.35) and the $N = 1$ cubic coefficient of Eq. (3.47) of [8], and the ratios $R^{(1)}$, $R^{(2)}$ of its Eq. (3.34), to 80 digits; the cubic coefficient of its Eq. (3.35) read literally gives -1.50734 instead of $a^{(3)}(1) = -0.850425$, and the quadratic coefficient of its Eq. (3.47) has the opposite sign (`ope_checks/d1/`, with its recorded run).

4. *Amplitudes.* With (837), (838), (839), (847), (850), the kernels (846), (849) and g_*, the amplitudes a_3, a_4, a_5 are finite and the re-expanded $\sum_n g_*^n \mathsf{a}_n$ reproduces $a^{(2)}, \ldots, a^{(5)}$ of (825) to 10^{-31} at $d = 3$ and at $d = 2.6$. The bubble expansion (838) and the closed form (836) (by direct quadrature at $d = 1$) were checked numerically.

5. *Kernels.* The symmetric kite (845) through ϵ^0 follows from (234) at $d = 1$, 2.6 and 3, including its R term; the four-vertex table with the exact chains (844) gives the pole and constant terms of (846); the five-vertex table sums to the finite part of (849). The crossed kite (845) enters the package through a two-fold Mellin–Barnes representation, whose finite-ϵ values reproduce the assembled $a^{(4)}$ (Section 34.6). At $d = 2$ this note's kite formula (674), evaluated at $\epsilon = 0.02, 0.04, 0.06, 0.1, 0.15$ (the slowly convergent ${}_3F_2(1)$ after Thomae's transformation), reproduces those finite-ϵ values of $\mathsf{K}_\times$ to 2×10^{-14} or better, and those of $\mathsf{K}_\parallel$ to 10^{-16}. At $d = 1$ the full-line Selberg formula of Section 5 gives $\mathsf{K}_\parallel$ to all 15 digits of the package's values; for $\mathsf{K}_\times$ at $d = 1$ only a double-precision direct quadrature was made, which agrees to 0.3%, within its own error (it misses the exact $\mathsf{K}_\parallel$ by 0.7%). Since $R = 0$ at $d = 2$, the $4R$ term of $\mathsf{K}_\times$ is tested only through the package's representation.

6. *Five-loop split.* At $d = 1$, b_5 and c_5 of (834)–(835) agree to about 10^{-12} relative with the two-channel five-loop graph sums of Section 14 at quadrature degree 18, renormalized with the degree-28 lower-loop inputs of the $C_6(1)$ calculation (records `amplitude5_maxgap_18.json` and `amplitude5_span_18.json` in `lri_numerical_resolution/verification/records/`, assembled in `LRI_C6_1D_HPC/reference/assembly_lower6.json`). These are numerical graph integrations that do not use the master reduction (Section 34.9). At $d = 2$, the M-dependent parts of (834) and (835) equal the minimal-subtraction power sums b_5^p and c_5^p of (678) to all 21 printed digits, and their b_3, b_4 parts equal b_3^p, b_4^p there. At $d = 2$

and $d = 3$ the complete b_5 was also extracted from the finite-scheme Taylor jets of the seventh- and eighth-order productions. A regular redefinition $g = g' + \sum_k \rho_k(z) g'^{k+1}$ with an operator normalization brings the finite-scheme β and γ_r to regulator-independent coefficients. Because of the $-zg'$ term this redefinition is unique, so the result is the minimal-subtraction form used here. It needs b_2 through z^3, b_3 through z^2, b_4 through z^1 and b_5 at z^0, and likewise for c_L. Holding D_5 at its proved value (229), the $C_7(3)$ and $C_8(3)$ productions give $b_5(3) = 18822.676 \pm 0.097$ and 18822.42 ± 0.28, against 18822.627 from (834). The same conversion reproduces b_3, c_3, b_4, c_4 of (179)–(227) within the sampling errors. At $d = 2$ it gives $b_5(2) = 135708.80 \pm 0.23$, against 135708.43. The individual b_5 sums of Section 24 cannot serve here, because they depend on trial core assignments.

7. *Values.* The table of Section 34.7 was computed from this note's master values, which reproduce $C_2, \dots, C_5$ of the coefficient table of Section 1 at $d = 1, 2, 3$. The log map with those $C_2, \dots, C_5$ and the master value of $13Z_d - \mathcal{L}_d - 12\mathcal{K}_d$ reproduces it to the precision of the displayed C_j.

8. *Bootstrap rows.* An own Borel–Padé evaluation reproduces the four $d = 3$ rows of Section 34.11 to all printed digits and finds the positive Borel pole of [2/2] at every point. The kinks and the cubic-order point of Figure 11 were taken from the vector paths of the figure file in the arXiv source of [8]: the plotted point lies at $(-0.0013004, 1.473078)$, the package's recomputation of it is $(-0.00130080, 1.473065)$, and the package's four kink readings agree with the vector kinks to within 9×10^{-5} in a_{ϕ^2} and 2×10^{-4} in $\Delta_{\mathcal{E}}$.

34.13 Reproduction

The package `lri_ope_reproducibility/` contains the low-order, four-, five- and six-vertex and relation engines, the graph inventories (`data/a6/graphs_n*.json`, `vertex_n*.json`), the master inputs, the bootstrap figure readings, and stored certificates. Its runner stages each engine in a temporary directory and does not overwrite inputs:

task	content (run time on the cluster above)
`quick`	low-order diagrams, the thermal map, the coefficient table, and a regression against stored certificates (4 s)
`symbolic`	fresh graph inventories, five- and six-vertex symbolic kernels, forest sums, pole cancellation, $3c_5 - b_5$, the compact result (2 min)
`numeric`	all sixth-order kernels and five-loop forest sums at 60 digits in $d = 1$ with other rational regulator directions and $\mathtt{c}_\times = 17$; agreement with (822) to about 10^{-53}, residual poles about 10^{-52} (5 min)
`integrals`	finite-regulator four-vertex kite and chain checks at $d = 1, 2, 3$, the $d = 2$ Selberg check, the five-vertex checks (1.5 min)
`masters`	$d = 2, 3$ spectral evaluation of $\mathcal{K}_d, \mathcal{L}_d, Z_d$ with varied quadrature (20 s)
`figures`	Borel–Padé rows and comparison plots, with both C_6 inputs (30 s)

The command `python run.py`, followed by the task names and `--output-dir DIR`, runs them; each writes `verification/run_TASK.json`. The numerical residuals of these tasks are consistency checks within the framework above, not uncertainties of the coefficients.

34.14 Open points

- An evaluation of selected five- and six-vertex finite parts by a method that does not use the pattern jets, for example sector decomposition with all local subtractions, would test the reconstruction beyond its alternative directions. The R term of the crossed-kite finite part (845) is checked only through the package's own Mellin–Barnes representation; an evaluation at $d \neq 2$ to many digits would close this.

- The resummation has no controlled error: varying the Borel–Leroy parameter, the approximants and the factored power, rejecting or treating positive Borel poles, and propagating the input uncertainties remain to be done.
- The bootstrap mismatch needs a continuum-positivity check of the candidate functional, not more coefficients.
- The apparent printed-formula inconsistencies in Eqs. (3.35) and (3.47) of [8] (Sections 34.6.1 and 34.9) should be checked against the authors' ancillary calculation or clarified with them. They should not be used to infer that the authors' numerical bootstrap code shares them.
- $a^{(7)}$ would need $\mathsf{a}_2, \ldots, \mathsf{a}_6$ one order further, a new finite a_7 and b_6; neither C_6 nor the pattern of (829) determines them, and new master derivatives may enter.

A Complete retained residue certificate

This appendix lists the residues that enter the one-dimensional assembly of Sections 11–13: the 26 non-wheel four-loop rows and 92 retained five-loop rows, as exact polynomials, so that the weighted sum can be inspected without the machine-readable certificate. The six wheel rows are in Section 12; the three primitive rows are determined by their periods; the 28 five-loop zero-weight classes contribute nothing to $3c_5 - b_5$. Together these disjoint sets exhaust the stated graph list.

For each row, i is the zero-based canonical class index and the word lists upper-triangular multiplicities in lexicographic vertex-pair order. The columns are *RG weights*, not raw Wick weights:

$$\omega_b = L(-1)^L w_4/|\operatorname{Aut} G|, \qquad \omega_c = L(-1)^L w_2/|\operatorname{Aut} G|.$$

A row with residue $R_j^{(L)}$ contributes $\omega_b R_j^{(L)}$ to b_L and $\omega_c R_j^{(L)}$ to c_L. Equal residues are identified to keep the certificate legible. All expressions use the same $A, Q, \mathsf{G}, \mathcal{K}$ as the main result. Their transcription from the retained JSON is checked by exact symbolic substitution, including $\log 2 = (A - \pi)/4$. No numerical fit enters this table.

A.1 4-loop rows and residue polynomials

Class	Edge word	ω_b	ω_c	Residue
0	0002011310	2	$\frac{2}{3}$	$R_0^{(4)}$
1	0002020220	$\frac{3}{4}$	$\frac{1}{4}$	$R_0^{(4)}$
2	0002021210	3	$\frac{1}{2}$	$R_0^{(4)}$
3	0002110211	3	1	$R_0^{(4)}$
4	0002111201	12	2	$R_0^{(4)}$
5	0002112200	3	$\frac{1}{2}$	$R_0^{(4)}$
6	0011011220	$\frac{3}{2}$	$\frac{1}{2}$	$R_1^{(4)}$
7	0011012210	6	1	$R_2^{(4)}$
8	0011101211	6	2	$R_2^{(4)}$
9	0011102201	6	1	$R_3^{(4)}$
10	0011102210	12	2	$R_4^{(4)}$
11	0011102300	4	$\frac{2}{3}$	$R_5^{(4)}$
12	0011111111	12	2	$R_6^{(4)}$
13	0011111120	24	4	$R_7^{(4)}$
14	0011201102	6	1	$R_8^{(4)}$

Class	Edge word	ω_b	ω_c	Residue
15	0011201111	24	4	$R_9^{(4)}$
16	0011201201	12	2	$R_3^{(4)}$
17	0011202110	3	1	$R_{10}^{(4)}$
18	0011202200	3	$\frac{1}{2}$	$R_{11}^{(4)}$
19	0011211110	6	1	$R_{12}^{(4)}$
20	0011301011	1	$\frac{1}{3}$	$R_0^{(4)}$
21	0011301101	4	$\frac{2}{3}$	$R_{13}^{(4)}$
22	0012102200	6	0	$R_{11}^{(4)}$
23	0012111110	24	0	$R_6^{(4)}$
24	0012111200	24	0	$R_4^{(4)}$
25	0012210101	3	0	$R_0^{(4)}$

$$R_0^{(4)} = 0.$$

$$R_1^{(4)} = -\frac{2A^3}{9} + \frac{\pi^3}{9} - 4\zeta(3) - \frac{2A\pi^2}{9}.$$

$$R_2^{(4)} = -\frac{23A^3}{36} + \frac{25\pi^3}{36} + 19\zeta(3) + \frac{A\pi^2}{36}.$$

$$R_3^{(4)} = -\frac{23A^3}{36} - \frac{11\pi^3}{36} - 9\zeta(3) + \frac{A\pi^2}{36}.$$

$$R_4^{(4)} = -\frac{5A^3}{18} + \frac{5\pi^3}{36} + 3\zeta(3) - \frac{A\pi^2}{9}.$$

$$R_5^{(4)} = 2A\pi - 2\pi^2 - 4\pi.$$

$$R_6^{(4)} = -\frac{3AQ}{2}.$$

$$R_7^{(4)} = -\frac{4A^3}{3} + \frac{5\pi^3}{3} + 48\zeta(3) + \frac{A\pi^2}{3}.$$

$$R_8^{(4)} = -\frac{A^3}{9} + \frac{\pi^3}{18} + 2\zeta(3) - \frac{A\pi^2}{9}.$$

$$R_9^{(4)} = -\frac{4A^3}{3} - \frac{\pi^3}{3} - 8\zeta(3) + \frac{A\pi^2}{3}.$$

$$R_{10}^{(4)} = 4\pi^2 + 8\pi.$$

$$R_{11}^{(4)} = -\frac{A^3}{12} - \frac{\pi^3}{12} - \zeta(3) - \frac{A\pi^2}{12}.$$

$$R_{12}^{(4)} = -\frac{5A^3}{9} + \frac{5\pi^3}{18} - 2\zeta(3) - \frac{2A\pi^2}{9}.$$

$$R_{13}^{(4)} = 2A\pi + 2\pi^2 + 4\pi.$$

A.2 5-loop rows and residue polynomials

Class	Edge word	ω_b	ω_c	Residue
5	000020021201200	$-\frac{15}{8}$	$-\frac{5}{16}$	$R_0^{(5)}$
6	000020021210110	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_0^{(5)}$
7	000020021301100	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_0^{(5)}$
8	000020022210100	$-\frac{15}{8}$	$-\frac{5}{16}$	$R_0^{(5)}$
13	000020111111200	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_0^{(5)}$
14	000020111120110	-15	$-\frac{5}{2}$	$R_0^{(5)}$
15	000020111121100	-15	$-\frac{5}{2}$	$R_0^{(5)}$
16	000020111130010	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_0^{(5)}$
17	000020111220001	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_0^{(5)}$
18	000020112120100	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_0^{(5)}$
19	000020112220000	$-\frac{15}{8}$	$-\frac{5}{16}$	$R_0^{(5)}$
20	000020120210011	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_0^{(5)}$
21	000020120211001	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_0^{(5)}$
22	000020120211010	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_0^{(5)}$
23	000021110110111	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_0^{(5)}$
27	000110012201200	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_1^{(5)}$
28	000110012210110	-15	$-\frac{5}{2}$	$R_2^{(5)}$
29	000110012301100	-5	$-\frac{5}{6}$	$R_3^{(5)}$
30	000110012310100	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_0^{(5)}$
31	000110013210100	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_0^{(5)}$
32	000110022201100	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_4^{(5)}$
37	000110102111200	-15	$-\frac{5}{2}$	$R_5^{(5)}$
38	000110102120110	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_6^{(5)}$
39	000110102121100	-15	$-\frac{5}{2}$	$R_7^{(5)}$
105	000120111310000	-5	0	$R_8^{(5)}$
106	000120120201001	-15	0	$R_9^{(5)}$
107	000120120202000	$-\frac{15}{4}$	0	$R_{10}^{(5)}$
108	000120120211000	$-\frac{15}{2}$	0	$R_{11}^{(5)}$
109	000120120301000	$-\frac{5}{2}$	0	$R_{12}^{(5)}$
110	000120121210000	$-\frac{15}{4}$	0	$R_{13}^{(5)}$

Class	Edge word	ω_b	ω_c	Residue
111	000121101101200	$-\frac{15}{2}$	0	$R_{14}^{(5)}$
112	000121101110101	-30	0	$R_{15}^{(5)}$
114	000121101110200	-30	0	$R_{16}^{(5)}$
115	000121101200101	-15	0	$R_{6}^{(5)}$
116	000121101210100	$-\frac{15}{2}$	0	$R_{17}^{(5)}$
117	000121102110100	-15	0	$R_{14}^{(5)}$
118	000121110110110	-15	0	$R_{18}^{(5)}$
119	000121110200011	-15	0	$R_{6}^{(5)}$
120	000121110200101	-15	0	$R_{0}^{(5)}$
121	000121120200001	$-\frac{15}{8}$	0	$R_{0}^{(5)}$
122	000131101110100	-5	0	$R_{19}^{(5)}$
127	001111011101200	-30	0	$R_{20}^{(5)}$
40	000110102130010	-5	$-\frac{5}{6}$	$R_{21}^{(5)}$
41	000110102210101	-15	$-\frac{5}{2}$	$R_{22}^{(5)}$
42	000110102211100	-15	$-\frac{5}{2}$	$R_{23}^{(5)}$
43	000110102220001	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{24}^{(5)}$
44	000110102220010	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{9}^{(5)}$
45	000110102220100	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{25}^{(5)}$
46	000110102310001	-5	$-\frac{5}{6}$	$R_{3}^{(5)}$
48	000110103210100	-5	$-\frac{5}{6}$	$R_{21}^{(5)}$
49	000110103220000	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_{12}^{(5)}$
52	000110111112100	-30	-5	$R_{26}^{(5)}$
53	000110111201101	-30	-5	$R_{27}^{(5)}$
54	000110111202100	-15	$-\frac{5}{2}$	$R_{28}^{(5)}$
55	000110111211001	-15	$-\frac{5}{2}$	$R_{29}^{(5)}$
56	000110111211010	-30	-5	$R_{30}^{(5)}$
57	000110111301001	-10	$-\frac{5}{3}$	$R_{31}^{(5)}$
59	000110112201100	-15	$-\frac{5}{2}$	$R_{32}^{(5)}$
60	000110112210001	-15	$-\frac{5}{2}$	$R_{33}^{(5)}$
61	000110112210010	-15	$-\frac{5}{2}$	$R_{7}^{(5)}$
62	000110112210100	-15	$-\frac{5}{2}$	$R_{34}^{(5)}$
63	000110112211000	-15	$-\frac{5}{2}$	$R_{35}^{(5)}$
64	000110112220000	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{11}^{(5)}$
66	000110112310000	-5	$-\frac{5}{6}$	$R_{8}^{(5)}$
67	000110201210002	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_{36}^{(5)}$
68	000110201211001	-15	$-\frac{5}{2}$	$R_{37}^{(5)}$
69	000110201220001	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{24}^{(5)}$
71	000110202220000	$-\frac{15}{8}$	$-\frac{5}{16}$	$R_{10}^{(5)}$
72	000110211211000	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_{38}^{(5)}$
76	000111101101201	-15	$-\frac{5}{2}$	$R_{39}^{(5)}$
77	000111101102200	-15	$-\frac{5}{2}$	$R_{16}^{(5)}$
79	000111101111110	-30	-5	$R_{20}^{(5)}$
80	000111101111200	-30	-5	$R_{40}^{(5)}$

Class	Edge word	ω_b	ω_c	Residue
81	000111101200102	-15	$-\frac{5}{2}$	$R_{41}^{(5)}$
82	000111101201101	-30	-5	$R_{42}^{(5)}$
83	000111101210101	-15	$-\frac{5}{2}$	$R_{43}^{(5)}$
84	000111102200101	-15	$-\frac{5}{2}$	$R_{22}^{(5)}$
86	000111102210100	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{17}^{(5)}$
88	000111111200002	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_{0}^{(5)}$
89	000111111200011	-30	-5	$R_{44}^{(5)}$
91	000111111201100	-15	$-\frac{5}{2}$	$R_{45}^{(5)}$
92	000111111300001	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_{0}^{(5)}$
94	000111201210001	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{46}^{(5)}$
95	000120012210100	$-\frac{15}{4}$	0	$R_{0}^{(5)}$
96	000120021201100	$-\frac{15}{2}$	0	$R_{47}^{(5)}$
97	000120102120100	-15	0	$R_{9}^{(5)}$
98	000120102220000	$-\frac{15}{8}$	0	$R_{10}^{(5)}$
100	000120111120010	-30	0	$R_{48}^{(5)}$
101	000120111210001	-30	0	$R_{34}^{(5)}$
102	000120111210010	-15	0	$R_{48}^{(5)}$
103	000120111211000	-15	0	$R_{49}^{(5)}$
104	000120111220000	-15	0	$R_{11}^{(5)}$

$$R_0^{(5)} = 0.$$

$$R_1^{(5)} = -\frac{19A^4}{60} + 16A\zeta(3) + \frac{19\pi^4}{900} - \frac{A^2\pi^2}{30} + \frac{8A\pi^3}{15}.$$

$$R_2^{(5)} = -\frac{499A^4}{720} + \frac{17A^2\pi^2}{360} - \frac{3\pi^4}{400} + \frac{191A\pi^3}{180} + \frac{141A\zeta(3)}{5}.$$

$$R_3^{(5)} = -\frac{4\pi^3}{3} - \frac{24\pi^2}{5} - \frac{48\pi}{5} + 2A^2\pi.$$

$$R_4^{(5)} = -\frac{47A^4}{720} - \frac{43A^2\pi^2}{360} + \frac{83\pi^4}{1200} + \frac{7A\pi^3}{180} - \frac{3A\zeta(3)}{5}.$$

$$R_5^{(5)} = -\frac{499A^4}{720} - \frac{4\mathcal{K}}{5} - \frac{3\pi^4}{400} + \frac{17A^2\pi^2}{360} + \frac{191A\pi^3}{180} + \frac{141A\zeta(3)}{5}.$$

$$R_6^{(5)} = -\frac{47A^4}{180} - \frac{A^2\pi^2}{18} - \frac{11\pi^4}{900} + \frac{2A\pi^3}{9} + \frac{24A\zeta(3)}{5}.$$

$$R_7^{(5)} = -\frac{499A^4}{720} + \frac{4\mathcal{K}}{5} - \frac{3\pi^4}{400} + \frac{17A^2\pi^2}{360} + \frac{227A\pi^3}{180} + \frac{169A\zeta(3)}{5}.$$

$$R_8^{(5)} = -\frac{4\pi^3}{15} - \frac{12\pi^2}{5} - \frac{24\pi}{5} + \frac{3A^2\pi}{5} + \frac{2A\pi^2}{5} + \frac{4A\pi}{5}.$$

$$R_9^{(5)} = -\frac{A^4}{16} - A\zeta(3) + \frac{3\pi^4}{400} - \frac{5A^2\pi^2}{72} - \frac{A\pi^3}{12}.$$

$$R_{10}^{(5)} = -\frac{A^4}{60} - \frac{4A\zeta(3)}{5} - \frac{19\pi^4}{900} - \frac{A^2\pi^2}{30} - \frac{A\pi^3}{15}.$$

$$R_{11}^{(5)} = -\frac{11A^4}{80} - \frac{47A^2\pi^2}{360} - \frac{3\pi^4}{400} + \frac{17A\pi^3}{60} + \frac{33A\zeta(3)}{5}.$$

$$R_{12}^{(5)} = \frac{4\pi^3}{3} + \frac{24\pi^2}{5} + \frac{48\pi}{5} + \frac{6A^2\pi}{5} - \frac{12A\pi^2}{5} - \frac{24A\pi}{5}.$$

$$R_{13}^{(5)} = -\frac{7A^4}{180} - \frac{A^2\pi^2}{18} - \frac{11\pi^4}{900} + \frac{2A\pi^3}{45} + \frac{8A\zeta(3)}{5}.$$

$$R_{14}^{(5)} = -\frac{9A^2Q}{10} - \frac{44\mathsf{G}Q}{5} - \frac{5Q\pi^2}{6}.$$

$$R_{15}^{(5)} = -\frac{39A^2Q}{20} - \frac{22\mathsf{G}Q}{5} - \frac{7Q\pi^2}{60}.$$

$$R_{16}^{(5)} = -\frac{53A^4}{180} + \frac{4\mathcal{K}}{5} + \frac{11\pi^4}{900} - \frac{A^2\pi^2}{6} + \frac{16A\pi^3}{45} + 8A\zeta(3).$$

$$R_{17}^{(5)} = -\frac{7A^4}{48} - A\zeta(3) + \frac{623\pi^4}{3600} - \frac{A^2\pi^2}{8} - \frac{A\pi^3}{12}.$$

$$R_{18}^{(5)} = -\frac{9A^2Q}{20} + \frac{22\mathsf{G}Q}{5} + \frac{7Q\pi^2}{60}.$$

$$R_{19}^{(5)} = \frac{24Q\pi}{5}.$$

$$R_{20}^{(5)} = -\frac{39A^2Q}{20} + \frac{74\mathsf{G}Q}{5} + \frac{89Q\pi^2}{60}.$$

$$R_{21}^{(5)} = \frac{4\pi^3}{15} + \frac{12\pi^2}{5} + \frac{24\pi}{5} + \frac{13A^2\pi}{5} - \frac{14A\pi^2}{5} - \frac{28A\pi}{5}.$$

$$R_{22}^{(5)} = -\frac{499A^4}{720} - 11A\zeta(3) - \frac{3\pi^4}{400} + \frac{17A^2\pi^2}{360} - \frac{61A\pi^3}{180}.$$

$$R_{23}^{(5)} = -\frac{A^4}{3} - \frac{2A^2\pi^2}{9} + \frac{2A\pi^3}{5} + \frac{16A\zeta(3)}{5}.$$

$$R_{24}^{(5)} = -\frac{19A^4}{60} - \frac{88A\zeta(3)}{5} + \frac{19\pi^4}{900} - \frac{A^2\pi^2}{30} - \frac{2A\pi^3}{3}.$$

$$R_{25}^{(5)} = -\frac{62\pi^3}{15} - 16\pi^2 - 32\pi - \frac{2A^2\pi}{5} + 4A\pi^2 + 8A\pi.$$

$$R_{26}^{(5)} = -\frac{3A^2Q}{2} + \frac{76\mathsf{G}Q}{5} + \frac{41Q\pi^2}{30}.$$

$$R_{27}^{(5)} = -\frac{3A^2Q}{2} - 4\mathsf{G}Q - \frac{7Q\pi^2}{30}.$$

$$R_{28}^{(5)} = -\frac{287A^4}{240} - \frac{8\mathcal{K}}{5} + \frac{3\pi^4}{400} + \frac{127A^2\pi^2}{360} + \frac{37A\pi^3}{20} + \frac{271A\zeta(3)}{5}.$$

$$R_{29}^{(5)} = -\frac{3A^2Q}{2} - \frac{52\mathsf{G}Q}{5} - \frac{23Q\pi^2}{30}.$$

$$R_{30}^{(5)} = -\frac{25A^4}{12} - \frac{16\mathcal{K}}{5} - \frac{11\pi^4}{900} + \frac{5A^2\pi^2}{6} + \frac{52A\pi^3}{15} + \frac{512A\zeta(3)}{5}.$$

$$R_{31}^{(5)} = \frac{4\pi^3}{15} + \frac{12\pi^2}{5} + \frac{24\pi}{5} + \frac{13A^2\pi}{5} + \frac{14A\pi^2}{5} + \frac{28A\pi}{5}.$$

$$R_{32}^{(5)} = -\frac{287A^4}{240} + \frac{4\mathcal{K}}{5} + \frac{3\pi^4}{400} + \frac{127A^2\pi^2}{360} + \frac{37A\pi^3}{20} + \frac{271A\zeta(3)}{5}.$$

$$R_{33}^{(5)} = -\frac{47A^4}{240} - \frac{11A^2\pi^2}{120} + \frac{3\pi^4}{400} + \frac{19A\pi^3}{60} + \frac{43A\zeta(3)}{5}.$$

$$R_{34}^{(5)} = -\frac{53A^4}{180} + 8A\zeta(3) + \frac{11\pi^4}{900} - \frac{A^2\pi^2}{6} + \frac{16A\pi^3}{45}.$$

$$R_{35}^{(5)} = -\frac{287A^4}{240} - \frac{4\mathcal{K}}{5} + \frac{3\pi^4}{400} + \frac{127A^2\pi^2}{360} + \frac{41A\pi^3}{20} + \frac{299A\zeta(3)}{5}.$$

$$R_{36}^{(5)} = -\frac{19A^4}{720} + A\zeta(3) - \frac{3\pi^4}{400} - \frac{23A^2\pi^2}{360} - \frac{A\pi^3}{180}.$$

$$R_{37}^{(5)} = -\frac{287A^4}{240} - \frac{177A\zeta(3)}{5} + \frac{3\pi^4}{400} + \frac{127A^2\pi^2}{360} - \frac{27A\pi^3}{20}.$$

$$R_{38}^{(5)} = -\frac{73A^4}{180} - 8A\zeta(3) - \frac{389\pi^4}{900} - \frac{7A^2\pi^2}{18} + \frac{26A\pi^3}{45}.$$

$$R_{39}^{(5)} = -\frac{3A^2Q}{2} + \frac{44\mathsf{G}Q}{5} + \frac{5Q\pi^2}{6}.$$

$$R_{40}^{(5)} = -\frac{25A^4}{12} + \frac{4\mathcal{K}}{5} - \frac{11\pi^4}{900} + \frac{5A^2\pi^2}{6} + \frac{52A\pi^3}{15} + \frac{512A\zeta(3)}{5}.$$

$$R_{41}^{(5)} = -\frac{13A^4}{180} - \frac{A^2\pi^2}{6} + \frac{11\pi^4}{900} + \frac{8A\pi^3}{45} + \frac{24A\zeta(3)}{5}.$$

$$R_{42}^{(5)} = -\frac{25A^4}{12} - \frac{4\mathcal{K}}{5} - \frac{11\pi^4}{900} + \frac{5A^2\pi^2}{6} + \frac{28A\pi^3}{15} + \frac{288A\zeta(3)}{5}.$$

$$R_{43}^{(5)} = -\frac{79A^4}{80} - \frac{79A\zeta(3)}{5} - \frac{623\pi^4}{3600} - \frac{43A^2\pi^2}{360} + \frac{A\pi^3}{60}.$$

$$R_{44}^{(5)} = -\frac{25A^4}{12} - 32A\zeta(3) - \frac{11\pi^4}{900} + \frac{5A^2\pi^2}{6} - \frac{4A\pi^3}{3}.$$

$$R_{45}^{(5)} = -\frac{79A^4}{80} - \frac{43A^2\pi^2}{360} - \frac{623\pi^4}{3600} + \frac{17A\pi^3}{12} + \frac{117A\zeta(3)}{5}.$$

$$R_{46}^{(5)} = \frac{62\pi^3}{15} + 16\pi^2 + 32\pi + \frac{2A^2\pi}{5} + 4A\pi^2 + 8A\pi.$$

$$R_{47}^{(5)} = -\frac{11A^4}{80} + A\zeta(3) - \frac{3\pi^4}{400} - \frac{47A^2\pi^2}{360} + \frac{A\pi^3}{12}.$$

$$R_{48}^{(5)} = -\frac{9A^2Q}{10} + 4\mathsf{G}Q + \frac{7Q\pi^2}{30}.$$

$$R_{49}^{(5)} = -\frac{9A^2Q}{10} + \frac{52\mathsf{G}Q}{5} + \frac{23Q\pi^2}{30}.$$

B Derivation of the special-function representation of $C_6(1)$

This appendix reproduces the derivation of the analytic calculation in the package `C6_research_note/`, written by OpenAI GPT-6 astra, adapted to the notation of this note. It starts from the reduced input (604) of Section 22.9 and proves the conversions stated there and in Section 1.2: the ordinary-period sectors (Section B.3), the wheel (Section B.4), the seventh-power sector (Section B.5), the primitive (Section B.6) and the polylogarithmic sector (Section B.7). Sections B.8 and B.9 give the numerical method and its error control. Definitions already given in Sections 1.2 and 22.9 are used without restating them. The package's records of these steps are the reduced setup `C6_research_note/inputs/upstream_reduced_setup.md` (extracted from `C6_research_note/inputs/source_snapshots/C6_central_binomial_wheel_expression.json`), the derivation snapshots `C6_research_note/inputs/source_snapshots/*_derivation.md` with their expression and check files, and the stage folders `C6_research_note/code/stages/` with their `DERIVATION.md` files, inputs and code; the steps below are written out where they are used, so these records are not cited again step by step, and the inputs that remain are named in Section B.1.

B.1 Scope, levels of statements and notation

Scope. The identity (604) is the supplied reduced analytic input of the method; the graph-to-period reduction that produced it is derived neither here nor in the package. The derivations below convert its sectors into the formula (2). The model, the expansion parameter, the conventions and the graph enumeration are those of this note (Sections 1–3), and Section 1.3 gives the status of each statement.

Three levels of statements. The package keeps three kinds of statement apart, and so does this appendix. *Physical identification*: the graph-to-period identities and the renormalization-group assembly behind (604) are inherited inputs; the downstream checks do not establish them independently. *Analytic identity*: a finite algebraic, differential or contour derivation may prove the equality of two representations, conditional on its stated input identities. *Numerical certification*: an outward-rounded interval plus an analytic tail bound encloses a specified expression; agreement of floating-point calculations alone is not a certificate. Input identities that are supplied with the reduced expression, rather than derived from it, are named as such where they are used. After the derivations below, the conversion to the nine values uses three of them, the elliptic-moment evaluations (911)–(913) on which the wheel's periodization identities rest (Section B.4.3), besides published identities. The interval certificate uses more: it encloses an earlier form of the reduced expression (Section B.8.5), which reaches (604) only through further exact identities of the package, among them the inherited wheel normalizations of $B_- + B_+$ and $B_- - B_+$. This appendix does not reproduce those identities; they are inputs of the enclosure.

Symbols. The package writes B_{pol}, B_{low}, B_6, B_8 for the sectors S_{pol}, S_{low}, S_6, S_8 of (604); U_4, U_6, U_8 and H_-, H_0, H_+ for the hypergeometric values $\varpi_4, \varpi_6, \varpi_8$ and Φ_-, Φ_0, Φ_+ of (607); K_n for κ_n of (608), whose sum M_7 it writes with a script letter; $h(x)$ for the Gauss function $w(x)$ of (609); G for G; ℓ for $\log 2$; and $\mathcal{H}_7^{(u)}$ for $\mathcal{H}_7$. It defines P_b by $S_6 = 24P_b$, and it writes $E_{\mathrm{real}} = E - \frac{476}{3}\pi^3(32\mathsf{G} + 9\pi^2)$ for the polynomial part of (2). To avoid clashes with symbols of this note, the following local symbols are also renamed: the Barnes functional (package $\mathcal{B}_c$) is $\mathfrak{B}_c$; the wheel sum that the package writes as a script S_7 is Σ_7; the contour kernel of the primitive (package $\mathcal{K}$) is $\widetilde{\mathcal{N}}$; the primitive Meijer kernels (package M_j) are $M_j^{(3)}$; the Barnes kernel of the lower block (package K) is $\widetilde{M}$; the wheel's parameter point and direction are $\mathbf{p}_0$, $\mathbf{v}$; the coefficient matrices of the primitive are M, N; and the Euler-weighted approximant (package E_N) is $\widehat{E}_N$. Repetition is written $a^{\times k}$, as in Section 1.2.

Harmonic remainders. The constants $A = \pi + 4\log 2$, $Q = \Gamma(\tfrac14)^4/\pi$, G and the sequences a_n, h_n, δ_n, y_n are those of Section 22.9, with $H_0^{(p)} = 0$. Equivalently,

$$a_n = \frac{(1/2)_n}{n!}, \qquad \delta_n = \frac{\pi^2}{3} - 4H_{2n}^{(2)} + 2H_n^{(2)} = \psi_1(n+\tfrac12) - \psi_1(n+1) > 0 \tag{866}$$

for every $n \geq 0$. For example $\delta_0 = \pi^2/3$, $\delta_1 = \pi^2/3 - 3$ and $\delta_2 = \pi^2/3 - 115/36$. The empty harmonic sum vanishes; the terms with $n > 0$ do not.

Polylogarithm and Meijer conventions. The multiple polylogarithm has the descending index order of Section 22.9. In the ordinary alphabet of harmonic polylogarithms, $H_0(z) = \log z$ and $H_1(z) = -\log(1-z)$, with differential letters dz/z and $dz/(1-z)$. Thus $H_{011}(z) = \mathrm{Li}_{2,1}(z,1)$ and $H_{01101}(z) = \mathrm{Li}_{2,1,2}(z,1,1)$. Principal branches continued from the stated convergence domains are used.

For an ordinary $G_{p,q}^{m,n}$ we use the inverse Mellin measure $ds/(2\pi\mathrm{i})$ and the argument factor x^{-s}. Its kernel is

$$\frac{\prod_{j=1}^{m}\Gamma(b_j+s)\prod_{j=1}^{n}\Gamma(1-a_j-s)}{\prod_{j=m+1}^{q}\Gamma(1-b_j-s)\prod_{j=n+1}^{p}\Gamma(a_j+s)}. \tag{867}$$

The first n upper and the first m lower parameters are in numerator groups. An identical numerical parameter in two different denominator groups does not cancel. The repetition notation $a^{\times k}$ means a list of k entries, not a power of a function [22, 18].

For contour manipulations we use the functional

$$\mathfrak{B}_c[f] = \frac{1}{2\pi\mathrm{i}}\int_{\mathrm{Re}\,s=c} f(s)\,ds, \tag{868}$$

with the vertical contour directed upward. The kernels and contour majorants are specified wherever this notation is used.

The restored Laurent coefficients. The correction of Section 22.9, $R_{147,-3} = R_{148,-3} = \frac{2}{5}$, contributes to $r_6 = 2187\,C_6(1)$

$$\begin{aligned}\Delta r_6^{\mathrm{repair}} = {}& \frac{1024}{25}A^5 - \frac{1216}{15}A^3\pi^2 + \frac{2672}{5}A^2\pi^3 + \frac{60864}{5}A^2\zeta(3)\\ & - \frac{2738}{75}A\pi^4 + \frac{5963}{15}\pi^5 - \frac{12046}{5}\pi^2\zeta(3) + \frac{711576}{5}\zeta(5).\end{aligned} \tag{869}$$

Divided by 2187 this is the contribution +659.278983 quoted in Section 22.9. It is already included in the polynomial E of (605) and must not be added again. The package describes it as a correction of the symbolic skeleton, not an adjustment of the independent numerical estimate.

B.2 Specification of the nine values

B.2.1 The four ordinary Meijer values

The values $\mathcal{G}_4, \mathcal{G}_6, \mathcal{G}_8, \mathcal{G}_7$ and their parameters are defined in Section 1.2. The package writes $X_4 = -C_d$ and $Y_4 = -C_s$, so that $\rho_4 = \sqrt{X_4^2 + Y_4^2}$ and $\alpha_4 = \pi^{-1}\arctan(Y_4/X_4)$. Both $X_4, Y_4 > 0$ for the specified constants (Section B.3.3); the real principal inverse tangent is used. For the eighth-power block,

$$d_8 = \frac{1}{2\pi}\arctan\frac{556}{1233 - 417\sqrt{2}}, \qquad e_8 = \frac{1}{2\pi}\operatorname{arcosh}\frac{1233 - 139\sqrt{2}}{\rho_8}, \qquad \alpha_{8,\pm} = d_8 \pm \mathrm{i}e_8. \tag{870}$$

All displayed real square roots and real inverse functions take their positive or principal values. The subscript 8 denotes eighth-power kernel content, not an eighth-order thermal coefficient. In the seventh block the $\eta_{7,\pm}$ use real principal inverse tangents and β_7 uses the principal complex inverse tangent; here $q_{7,2} > 0$, $q_{7,1} > 0$ and $q_{7,0} < 0$, as follows from $2/3 < \log 2 < 7/10$ and $3 < \pi < 22/7$. These are the retained lower-, sixth-, eighth- and seventh-sector reductions.

B.2.2 The bivariate convention: the kernel is part of the specification

The bivariate H convention is that of Andrade, Rathie and Farias [17] (its bivariate Definition B), as adopted by the package. All three values have arguments $(1, 1)$ and unit slopes. In a local variable s, the factor is

$$\Theta(s) = \frac{\prod_{j=1}^{m} \Gamma(b_j + B_j s) \prod_{j=1}^{n} \Gamma(1 - a_j - A_j s)}{\prod_{j=m+1}^{q} \Gamma(1 - b_j - B_j s) \prod_{j=n+1}^{p} \Gamma(a_j + A_j s)}. \tag{871}$$

For the shared group, the present cases have $m_0 = 0$. Its numerator factors are $\Gamma(1 - e_j - E_j s - F_j t)$ for the first n_0 upper entries; its denominator contains the remaining upper factors $\Gamma(e_j + E_j s + F_j t)$ and every lower factor $\Gamma(1 - f_j - A_j s - B_j t)$. The standard measure is $ds\, dt/(2\pi\mathrm{i})^2$, with $x^{-s}y^{-t}$. Ordered pair and triple lists and numerator counts fix each function without naming a residual integral as a new function. Both the lists and the simplified kernels are given, to prevent any ambiguity of convention; the kernels and contours are those displayed in Section 1.2.

B.2.3 The wheel value

With $\alpha_W = \pi^{-1} \arctan(5/8)$ put

$$\boldsymbol{a}_W = ((1/2, 1)^{\times 4}, (\alpha_W, 1)), \qquad \boldsymbol{b}_W = ((0, 1)^{\times 4}, (\alpha_W, 1)). \tag{872}$$

Then

$$\mathcal{H}_W = H^{0,2;2,3;2,2}_{2,2;5,5;2,2}\left[\begin{matrix} (1/2; 1, 1)^{\times 2} : \boldsymbol{a}_W; (1/2, 1), (0, 1) \\ (-1/2; 1, 1)^{\times 2} : \boldsymbol{b}_W; (0, 1), (1/2, 1) \end{matrix} \middle| 1, 1 \right]. \tag{873}$$

Its kernel is $N_{\alpha_W}(s)P(t)/(1/2 - s - t)^2$, the integrand of Section 1.2, where for any angle parameter α

$$\begin{aligned} N_\alpha(s) &= \frac{\Gamma(s)^2 \Gamma(1/2 - s)^3}{\Gamma(1 - s)^2 \Gamma(1/2 + s) \Gamma(s + \alpha) \Gamma(1 - s - \alpha)}, \\ P(t) &= \Gamma(t)\Gamma(t + 1/2)\Gamma(1/2 - t)\Gamma(1 - t), \end{aligned} \tag{874}$$

so that the kernel $N(s)$ of Section 1.2 is $N_{\alpha_W}(s)$. The upward lines $\mathrm{Re}\, s = 1/4$, $\mathrm{Re}\, t = 1/8$ separate the poles and give absolute convergence. The shared quotient is $\Gamma(1/2 - s - t)^2/\Gamma(3/2 - s - t)^2$. The normalization $W_0 = \sqrt{89}\, \mathcal{H}_W/\pi^2$, so that $-393\pi^3 W_0 = -393\pi\sqrt{89}\, \mathcal{H}_W$, is derived in Section B.4.4 from the wheel identity (901), which rests on the periodization identities (908) proved in Section B.4.3.

B.2.4 The unit-slope seventh-sector value

With $\beta = \beta_7$, define the lists

$$\begin{aligned} \boldsymbol{a}_7 &= ((-1/4, 1), (1/2, 1)^{\times 7}, (\beta, 1), (\bar{\beta}, 1)), \\ \boldsymbol{b}_7 &= ((0, 1)^{\times 6}, (1, 1), (3/4, 1), (\beta, 1), (\bar{\beta}, 1)), \\ \boldsymbol{c}_7 &= ((0, 1), (1/2, 1), (1/2, 1), (1/4, 1), (3/4, 1)), \\ \boldsymbol{d}_7 &= ((0, 1), (1/2, 1), (-1/2, 1), (1/4, 1), (3/4, 1)). \end{aligned}$$

Then

$$\mathcal{H}_7 = H^{0,1;3,8;2,3}_{1,1;10,10;5,5}\left[\begin{matrix}(1/2;1,1):\boldsymbol{a}_7;\boldsymbol{c}_7\\(-1/2;1,1):\boldsymbol{b}_7;\boldsymbol{d}_7\end{matrix}\middle|1,1\right], \tag{875}$$

with the simplified kernel $\mathcal{A}_7(s)\cot(2\pi t)/[(1/2-t)(1/2-s-t)]$ and the contours $\operatorname{Re} s = 1/4$, $\operatorname{Re} t = 1/16$ of Section 1.2. The second local gamma quotient equals $\cot(2\pi t)/(1/2-t)$; its parameter lists are those in (875). This is a bivariate function, not an ordinary Meijer G value. The normalization $24\pi^3 M_7 = 24\pi^{3/2}[\rho_7\mathcal{G}_7 + 196\pi\sqrt{185}\,\mathcal{H}_7]$ is proved in Section B.5.3.

B.2.5 The primitive value

The unique root $\lambda_* \in (15,16)$ of $4\lambda^3 - 112\lambda^2 + 909\lambda - 2250 = 0$ has the explicit form

$$\lambda_* = \frac{28 + \sqrt{409}\cos\left[\frac{1}{3}\arccos\frac{7298}{409\sqrt{409}}\right]}{3}. \tag{876}$$

In the definitions of e_*, χ_*, B_σ, C_σ and β_σ of Section 1.2, both real square-root radicands are positive, and the complex inverse tangent is principal. These parameters depend only on elementary and algebraic constants, not on a numerically fitted primitive value. Define

$$\boldsymbol{a}_\sigma = ((1/2,1)^{\times 4}, (\beta_\sigma,1), (\bar\beta_\sigma,1)),$$
$$\boldsymbol{b}_\sigma = ((0,1)^{\times 3}, (\beta_\sigma,1), (\bar\beta_\sigma,1)).$$

Then

$$\mathcal{H}_* = H^{0,0;3,3;3,3}_{0,1;6,5;6,5}\left[\begin{matrix}-:\boldsymbol{a}_+;\boldsymbol{a}_-\\(0;1,1):\boldsymbol{b}_+;\boldsymbol{b}_-\end{matrix}\middle|1,1\right]. \tag{877}$$

For each local group the first three upper and the first three lower parameters are numerator entries. The kernel is $\Psi_+(s)\Psi_-(t)/\Gamma(1-s-t)$ on the lines $\operatorname{Re} s = \operatorname{Re} t = 1/4$ (Section 1.2). The exact primitive normalizations are $P_a = 80\pi^3\mathcal{H}_*$ and $96P_a = 7680\pi^3\mathcal{H}_*$. The primitive normalization is unchanged. The global contour relation and the factorization producing this value are derived in Section B.6 from the stated Gauss-product input.

B.3 The ordinary-period sectors

B.3.1 The input periods and their weights

The three ordinary sectors of (604) have the hypergeometric input forms (607), with $S_6 = 24P_b$. They specify the analytic quantities that are converted into $\mathcal{G}_4$, $\mathcal{G}_6$ and $\mathcal{G}_8$; their graph identifications are part of the reduced input. The formerly separated contribution $-144Q^3/\pi$ belongs to the complete P_b and is inside S_6 there (it comes from the -1 in $2\Phi_- - 1$); it is not an additional term in E and must not be appended to E. The parameter excesses are positive, so the ordinary series converge at their unit arguments.

B.3.2 A common gamma expansion

The gamma recurrence and the integer and half-integer polygamma values give, for $n \geq 0$ and $\varepsilon \to 0$,

$$\frac{\Gamma(-n+\varepsilon)}{\Gamma(1/2-n+\varepsilon)} = \frac{a_n}{\sqrt{\pi}\,\varepsilon}\exp\left[2h_n\varepsilon + \frac{\delta_n - 2\pi^2/3}{2}\varepsilon^2 + O(\varepsilon^3)\right]. \tag{878}$$

For $n=0$, the first logarithmic coefficient is $2\log 2$ and the second derivative is $-\pi^2/3$. Moving to $-n$ adds $2(H_n - H_{2n})$ and $-4H^{(2)}_{2n} + 2H^{(2)}_n$ respectively. This gives (878) with exactly the remainders h_n and δ_n of Section 22.9 and (866). The same expansion fixes the logarithmic normalizations of the primitive and of the wheel residues.

B.3.3 The lower block: two fourth-order kernels and sine addition

On the upward line $\operatorname{Re} s = 1/4$, define

$$M_1(s) = \frac{\Gamma(s)\Gamma(1/2-s)^3}{\Gamma(1-s)^3\Gamma(1/2+s)},$$
$$M_2(s) = \frac{\Gamma(s)^2\Gamma(1/2-s)^2}{\Gamma(1-s)^2\Gamma(1/2+s)^2}. \tag{879}$$

Gamma reflection writes these as $R_4(s)\tan^3(\pi s)$ and $R_4(s)\tan^2(\pi s)$, where $R_4(s) = [\Gamma(s)/\Gamma(s+1/2)]^4$. Their contour values are

$$g_{13} = G^{1,3}_{4,4}\left(1 \,\middle|\, \begin{smallmatrix}(\frac{1}{2})^{\times 4}\\ 0^{\times 4}\end{smallmatrix}\right), \qquad g_{22} = G^{2,2}_{4,4}\left(1 \,\middle|\, \begin{smallmatrix}(\frac{1}{2})^{\times 4}\\ 0^{\times 4}\end{smallmatrix}\right). \tag{880}$$

The residues of M_1 at $s = -n$ are πa_n^4; hence $g_{13} = \pi\varpi_4$. To obtain the other normalization, the Bailey–Mishev Barnes and ${}_7F_6$ representations at $(1/2,1/2,1/2,1/2;1;1,1)$ [14], a published identity, give

$$\frac{\varpi_6}{2\pi} = \frac{1}{\pi^5}\mathfrak{B}_{1/4}[\widetilde{M}], \qquad \widetilde{M}(s) = \frac{\Gamma(s)^2\Gamma(1/2-s)^4}{\Gamma(1-s)^2}. \tag{881}$$

Here and below $\mathfrak{B}_c$ is the upward Barnes functional (868). Set $M_0(s) = \Gamma(1/2-s)^4/\Gamma(1-s)^4$. Reflection gives $\widetilde{M} = \pi^2(M_2 + M_0)$. The contribution $\mathfrak{B}_{1/4}[M_0]$ is zero: under $s \mapsto 1/2 - s$ it is the inverse Mellin value of R_4 at 1. The beta integral identifies $\Gamma(s)/\Gamma(s+1/2)$ as the Laplace transform of

$$b(t) = \frac{1}{\sqrt{\pi}\sqrt{1-e^{-t}}}, \qquad t > 0.$$

Since $b(t) = t^{-1/2}(1+O(t))/\sqrt{\pi}$, its fourfold additive convolution is $t(1+O(t))$ at 0 and has value zero there. Inversion is justified by the integrability of $R_4(1/4 + \mathrm{i}u)$. Therefore

$$\varpi_4 = g_{13}/\pi, \qquad \varpi_6 = 2g_{22}/\pi^2. \tag{882}$$

The gamma kernel N_α of (874) obeys

$$N_\alpha = \cos(\pi\alpha)M_1 + \sin(\pi\alpha)M_2. \tag{883}$$

Using $X_4 = -C_d$, $Y_4 = -C_s$ and the angle α_4, (882) and (883) yield

$$S_{\mathrm{low}} = 4\pi^2(C_d g_{13} + C_s g_{22}) = -4\pi^2\rho_4\mathcal{G}_4.$$

There is no contour boundary correction: the kernels are $O((1+|\operatorname{Im} s|)^{-2})$ on the common line. The branch can be checked without sampling the periods. The inequalities $\pi > 3$ and $\log 2 > 1/2$ give $A > 5$. Splitting Euler's integral for $\Gamma(1/4)$ at $t = 1$, and using $e^{-t} < 1 - t + t^2/2$ on $(0,1)$ and $t^{-3/4} < 1$ on $(1,\infty)$,

$$\Gamma(1/4) < \int_0^1 t^{-3/4}\Big(1 - t + \frac{t^2}{2}\Big)\,dt + \int_1^\infty e^{-t}\,dt < 4 - \frac{4}{5} + \frac{2}{9} + \frac{2}{5} < 4.$$

Thus $Q < 256/3$. These bounds imply

$$C_s < 5120 - \frac{105}{4}\,25 - 135 - \frac{2469}{4}\,9 = -\frac{2453}{2}, \qquad C_d < 8192 - 42\cdot 25 - 270 - 1227\cdot 9 = -4171.$$

Hence $X_4, Y_4 > 0$, as used above.

B.3.4 The sixth-power block

The sixth-power sector $S_6 = 24P_b$ of (607), with $P_b = \frac{6Q^3}{\pi}(2\Phi_- - 1) - 104\pi^5\Phi_0 + \frac{2^{18}\pi^5}{Q^3}\Phi_+$, has the absolutely convergent coefficient series

$$P_b = 24\pi^3 \sum_{n\geq 0} a_n^6(312h_n^2 + 13\delta_n + 88\pi h_n). \tag{884}$$

It is derived from the three hypergeometric values by a finite Fourier interpolation, not inferred from a numerical value. Define the proof-only function

$$\mathsf{f}_6(t) = \pi^3 \sum_{m\in\mathbb{Z}} \left[\Gamma(\tfrac{3}{4} + m + t)\Gamma(\tfrac{3}{4} - m - t)\right]^{-6}. \tag{885}$$

It is entire, even and one-periodic. Gamma reflection and ratio asymptotics [26, 27] give locally uniform convergence and, on a vertical strip, $\mathsf{f}_6(t) = O(e^{6\pi|\operatorname{Im} t|}(1 + |\operatorname{Im} t|)^{-2})$. Moving Fourier-coefficient contours vertically excludes the frequencies $|k| \geq 3$, including the boundary frequencies because of the factor $(1 + |\operatorname{Im} t|)^{-2}$. Hence f_6 lies in the span of 1, $\cos 2\pi t$ and $\cos 4\pi t$. For every function f in this three-dimensional space,

$$384f(0) - 104f(\tfrac{1}{4}) + 32f(\tfrac{1}{2}) = 312f(\tfrac{1}{4}) - \frac{176}{\pi}f'(\tfrac{1}{4}) + \frac{52}{\pi^2}f''(\tfrac{1}{4}). \tag{886}$$

Gamma reflection fixes the sample normalizations exactly:

$$\mathsf{f}_6(0) = \frac{Q^3}{64\pi^6}(2\Phi_- - 1), \qquad \mathsf{f}_6(\tfrac{1}{4}) = \Phi_0, \qquad \mathsf{f}_6(\tfrac{1}{2}) = \frac{8192}{Q^3}\Phi_+.$$

Consequently P_b equals π^5 times the left side of (886) with $f = \mathsf{f}_6$. At $t = \frac{1}{4}$ each negative-index term of (885) vanishes to order six. All derivatives needed here have order at most two, so this half contributes zero. On the nonnegative half, polygamma reflection and recurrence [32] yield

$$\mathsf{f}_6(\tfrac{1}{4}) = \sum a_n^6, \qquad \mathsf{f}_6'(\tfrac{1}{4}) = -12\sum a_n^6 h_n, \qquad \mathsf{f}_6''(\tfrac{1}{4}) = \sum a_n^6(144h_n^2 + 6\delta_n - 6\pi^2).$$

The constant 312 term cancels the $-6\pi^2$ derivative term, and π^5 times the right side of (886) is (884). This is a positive central-binomial series; it does not give separate finite evaluations of Φ_-, Φ_0 or Φ_+.

Introduce the proof kernels

$$M_j^{(6)}(s) = \left[\frac{\Gamma(s)}{\Gamma(s + 1/2)}\right]^6 \tan^{6-j}(\pi s), \qquad j = 1, \dots, 5.$$

Their reflection is $M_j^{(6)}(1/2 - s) = M_{6-j}^{(6)}(s)$. For the ordinary values

$$X_{35} = G_{6,6}^{3,5}\left(1 \,\middle|\, \begin{matrix}(\frac{1}{2})^{\times 6}\\ 0^{\times 6}\end{matrix}\right), \qquad X_{44} = G_{6,6}^{4,4}\left(1 \,\middle|\, \begin{matrix}(\frac{1}{2})^{\times 6}\\ 0^{\times 6}\end{matrix}\right),$$

the corresponding kernels satisfy

$$\begin{aligned} N_{35} &= \frac{\Gamma(s)^3\Gamma(1/2 - s)^5}{\Gamma(1 - s)^3\Gamma(1/2 + s)} = \pi^2(M_1^{(6)} + M_3^{(6)}),\\ N_{44} &= \frac{\Gamma(s)^4\Gamma(1/2 - s)^4}{\Gamma(1 - s)^2\Gamma(1/2 + s)^2} = \pi^2(M_2^{(6)} + M_4^{(6)}). \end{aligned} \tag{887}$$

Expanding by (878) and using the reflection of $M_4^{(6)}$ gives

$$X_{35} = 3\pi^2 \sum a_n^6(24h_n^2 + \delta_n), \qquad X_{44} = 24\pi^3 \sum a_n^6 h_n. \tag{888}$$

Multiplication by 13, 11 and 8π now reproduces every term in (884): $P_b = 8\pi(13X_{35} + 11X_{44})$. The kernel of $\mathcal{G}_6$ is

$$N_6(s) = \frac{\Gamma(s)^4\Gamma(1/2-s)^5}{\Gamma(1-s)^2\Gamma(1/2+s)\Gamma(s+\alpha_6)\Gamma(1-s-\alpha_6)}.$$

Sine addition gives $N_6 = \cos(\pi\alpha_6)N_{35} + \sin(\pi\alpha_6)N_{44}$. Because these two coefficients are $13/\sqrt{290}$ and $11/\sqrt{290}$, respectively, this proves $S_6 = 192\pi\sqrt{290}\,\mathcal{G}_6$. All kernels decay at least cubically on $\operatorname{Re} s = 1/4$, and their residue series converge absolutely at 1, justifying these operations.

B.3.5 The eighth-power block

The eighth-power sector of (607) has the coefficient series, with $y_n = n + 1/4$,

$$S_8 = 8\pi^4 \sum_{n\ge0} a_n^8 [(2466 - 834\sqrt{2})\pi^2 y_n - 1112\pi(1 - 16y_n h_n) + 139\sqrt{2}(256y_n h_n^2 - 32h_n + 8y_n\delta_n)]. \tag{889}$$

It is derived from ϖ_8, Y_1 and Y_7 as follows. Write $\mathfrak{Y} = (17 + 12\sqrt{2})Y_1 - (17 - 12\sqrt{2})Y_7$, so that $S_8 = \pi^6(4932 - 5004\sqrt{2})\varpi_8 + 556\pi^2\mathfrak{Y}$, and note that $Y_r = \sum_{n\ge0}(n+t)[\Gamma(n+t+\frac14)/\Gamma(n+t+\frac34)]^8$ with $t = r/8$; in particular $Y_2 = \frac{\pi^4}{4}\varpi_8$, so that $S_8 = \pi^2[(19728 - 20016\sqrt{2})Y_2 + 556\mathfrak{Y}]$. Define the proof-only function

$$\mathsf{f}_8(t) = \pi^4 \sum_{m\in\mathbb{Z}} \frac{m+t}{[\Gamma(\frac34 + m + t)\Gamma(\frac34 - m - t)]^8}. \tag{890}$$

It is odd and one-periodic. The same argument gives $\mathsf{f}_8(t) = O(e^{8\pi|\operatorname{Im} t|}(1 + |\operatorname{Im} t|)^{-2})$, so the frequencies $|k| \ge 4$ vanish, including $k = \pm4$. Thus f_8 lies in the span of $\sin 2\pi t$, $\sin 4\pi t$ and $\sin 6\pi t$, and interpolation on these three sine modes gives the exact identity

$$\mathsf{f}_8(\tfrac18) = \frac{5\sqrt{2}}{8}\mathsf{f}_8(\tfrac14) - \frac{1}{4\pi}\mathsf{f}_8'(\tfrac14) + \frac{\sqrt{2}}{32\pi^2}\mathsf{f}_8''(\tfrac14). \tag{891}$$

Gamma reflection, including the sign from the numerator of the negative half, gives

$$Y_2 = \pi^4\mathsf{f}_8(\tfrac14), \qquad \mathfrak{Y} = 64\pi^4\mathsf{f}_8(\tfrac18).$$

The factors 64 are fixed by $\cos^8(\pi/8) = (17 + 12\sqrt{2})/64$ and $\sin^8(\pi/8) = (17 - 12\sqrt{2})/64$. At the center the negative half now vanishes to order eight. Put

$$u_n^{(8)} = 1 - 16y_n h_n, \qquad v_n^{(8)} = 256y_n h_n^2 - 32h_n + 8y_n\delta_n - 8\pi^2 y_n.$$

Then

$$\mathsf{f}_8(\tfrac14) = \sum a_n^8 y_n, \qquad \mathsf{f}_8'(\tfrac14) = \sum a_n^8 u_n^{(8)}, \qquad \mathsf{f}_8''(\tfrac14) = \sum a_n^8 v_n^{(8)}.$$

Substituting these sums and (891) into the expression of S_8 through Y_2 and $\mathfrak{Y}$ gives (889). No eighth-shifted gamma value or parameter derivative occurs on the right. This converts the combined ordinary periods to explicit harmonic sums; it is not a finite elementary evaluation.

To convert this coefficient polynomial, let K_j be the contour value of

$$(1/4 - s)\left[\frac{\Gamma(s)}{\Gamma(s+1/2)}\right]^8 \tan^{8-j}(\pi s).$$

The factor $1/4 - s = \Gamma(5/4 - s)/\Gamma(1/4 - s)$ supplies an ordinary contiguous gamma pair. Equation (878) gives the first three residue polynomials, each multiplied by a_n^8:

$$\begin{aligned} K_1 &: \ \pi^3 y_n, \\ K_2 &: \ \pi^2(16y_n h_n - 1), \\ K_3 &: \ \pi[y_n(128h_n^2 + 4\delta_n - \pi^2) - 16h_n]. \end{aligned} \tag{892}$$

Set $a = 1233 - 278\sqrt{2}$, $b = 556$, $c = 139\sqrt{2}$. Comparing the full polynomial in (889) gives

$$S_8 = 16\pi^3(aK_1 + bK_2 + cK_3). \tag{893}$$

The angle parameters $\alpha_{8,\pm}$ satisfy the exact identity

$$\begin{aligned} a\sin^2(\pi s) + b\sin(\pi s)\cos(\pi s) + c\cos^2(\pi s) \\ = \rho_8 \sin\pi(s+\alpha_{8,+})\sin\pi(s+\alpha_{8,-}). \end{aligned} \tag{894}$$

Indeed $\rho_8^2 = (a-c)^2 + b^2$ and $4ac - b^2 = 685548\sqrt{2} - 618272 > 0$, fixing the real e_8 and the conjugate angles (870). Gamma reflection converts the full kernel to

$$\frac{\Gamma(s)^3\Gamma(1/2-s)^7\Gamma(5/4-s)}{\Gamma(1-s)^5\Gamma(1/2+s)\Gamma(1/4-s)\prod_{\sigma=\pm}\Gamma(s+\alpha_{8,\sigma})\Gamma(1-s-\alpha_{8,\sigma})}. \tag{895}$$

This is exactly the kernel of $\mathcal{G}_8$, proving $S_8 = 16\pi^3\rho_8\mathcal{G}_8$. The common contour has cubic decay, so the evaluation at 1 needs no regularization.

B.4 The wheel: one positive sequence and one bivariate value

B.4.1 Zero-index bookkeeping and direct convergence

The legacy positive wheel and its zero-included version are

$$\begin{aligned} W_c &= \sum_{n=1}^{\infty} a_n^4[8 + 5\pi(4n+1)a_n^2]\delta_n, \\ W_0 &= W_c + \frac{(8+5\pi)\pi^2}{3}. \end{aligned} \tag{896}$$

The zero-included convention is used in (604) and (2). The $n = 0$ contribution to $-393\pi^3 W_0$ is $-1048\pi^5 - 655\pi^6$ and is already present there. The recurrence

$$a_{n+1} = a_n\frac{2n+1}{2n+2}, \qquad \delta_{n+1} = \delta_n - \frac{4}{(2n+1)^2} + \frac{1}{(n+1)^2} \tag{897}$$

requires only one binomial sequence and one reciprocal-square accumulator.

For $x = n + 1/4$, the integral $\psi_1(z) = \int_0^\infty ue^{-zu}(1-e^{-u})^{-1}\,du$ [32] and (866) give $\delta_n = \int_0^\infty ue^{-(n+1/2)u}(1+e^{-u/2})^{-1}\,du$, that is,

$$\delta_n = \frac{1}{2}\int_0^\infty ue^{-xu}\operatorname{sech}(u/4)\,du, \qquad 0 < \delta_n < \frac{1}{2x^2}; \tag{898}$$

the bound follows from $0 < \operatorname{sech}(u/4) < 1$ for $u > 0$. Moreover $a_n^2 \le 1/(3n+1)$ for every $n \ge 0$: equality holds at $n = 0$, and since $a_{n+1}^2 = a_n^2(2n+1)^2/(2n+2)^2$ and $(2n+1)^2(3n+4) = (3n+1)(2n+2)^2 - n$, the bound passes from n to $n+1$. Hence $a_n^2 \le 1/(3x)$ and $(4n+1)a_n^2 \le 4/3$. Together these give, for the literal sum over $1 \le n < N$,

$$0 < W_c - W_{c,N} < \frac{8 + 20\pi/3}{54(N-3/4)^3}. \tag{899}$$

The direct-tail estimate applies to the literal finite sum; the stronger numerical construction of Section B.8.2 controls the corrected tail.

B.4.2 The global fourth/sixth-power identity

Put

$$\tau_n = 12\zeta(3) - 16H_{2n}^{(3)} + 4H_n^{(3)}, \quad D_4 = \sum_{n\geq 0} a_n^4\delta_n, \quad D_6 = \sum_{n\geq 0}(4n+1)a_n^6\delta_n. \tag{900}$$

The key identity of the calculation is

$$\sum_{n\geq 0} a_n^4(8h_n\delta_n + \tau_n) = \pi^2 D_6, \qquad W_0 = 8D_4 + 5\pi D_6. \tag{901}$$

It is a global summation identity, not a termwise identification of fourth and sixth binomial powers. The package derives it from a completed hypergeometric symmetry [14, 15] and the two periodization identities (908) below. The specialized symmetry calculation is given here, and the two periodization identities are proved in Section B.4.3.

Write

$$\begin{aligned} q_n &= (4n+1)a_n^6, \\ A_n &= 4H_{2n} - 3H_n - 2\log 2 + \frac{1}{4n+1}, \\ R_n &= 4H_{2n}^{(2)} - H_n^{(2)} + \frac{2}{(4n+1)^2}. \end{aligned} \tag{902}$$

For parameters satisfying $e+f+g-a-b-c-d = 1$, define the completed function $L(a,b,c,d;e;f,g)$ by the following ${}_7F_6$ representation. Put $\alpha = d+g-e$ and take $\mathrm{Re}(f-d) > 0$:

$$\begin{aligned} L = \frac{\Gamma(1+\alpha)}{\pi\Gamma(g)\Gamma(1+g-e)\Gamma(f-d)\Gamma(1+a+d-e)\Gamma(1+b+d-e)\Gamma(1+c+d-e)} \\ \times {}_7F_6\left(\begin{matrix}\alpha, 1+\alpha/2, g-a, g-b, g-c, d, 1+d-e \\ \alpha/2, 1+a+d-e, 1+b+d-e, 1+c+d-e, 1+g-e, g\end{matrix}; 1\right). \end{aligned} \tag{903}$$

The input symmetry is invariance under

$$\mathcal{C}(a,b,c,d;e;f,g) = (g-c, a, b, g-d; 1+a+b-f; g, 1+a+b-e). \tag{904}$$

Let $\omega = e^{\mathrm{i}\pi/4}$ and set

$$\begin{aligned} \mathbf{p}_0 &= (1/2, 1/2, 1/2, 1/2; 1; 1, 1), \\ \mathbf{v} &= (\omega^2, \omega, 1, 1-\omega+\omega^2; 1+\omega^2; \omega^2-\omega^3, 1+\omega^3). \end{aligned} \tag{905}$$

Direct substitution gives $\mathcal{C}(\mathbf{p}_0+x\mathbf{v}) = \mathbf{p}_0+\omega x\mathbf{v}$. Hence $L(\mathbf{p}_0+x\mathbf{v}) = L(\mathbf{p}_0+\omega x\mathbf{v})$, and the coefficient of x^2 vanishes. The parameter excess is one at $x=0$ and stays positive in a sufficiently small closed neighborhood; the twice-differentiated coefficients have a summable bound $n^{-2+\varepsilon}(1+\log^2 n)$ for some $\varepsilon < 1$.

Include the gamma prefactor in each summand $l_n(x)$ of (903). At the base point, $l_n(0) = q_n/(2\pi)$. With $\varrho = 1-\omega+\omega^3 = 1-\sqrt{2}$, the sums of slopes and of squared slopes needed for differentiation are

Gamma-factor group	Sum of slopes	Sum of squared slopes
Six half-parameter factors depending on n	4ϱ	$2\varrho^2$
Five unit-parameter denominator factors	2ϱ	0
Half-parameter normalization factors	2ϱ	$2\varrho^2$

The very-well-poised linear factor has slope ϱ. The polygamma recurrence then gives

$$(\log l_n)'(0) = 2\varrho A_n, \qquad (\log l_n)''(0) = -2\varrho^2 R_n.$$

Therefore

$$l_n''(0) = \frac{q_n}{2\pi} 4\varrho^2 (A_n^2 - R_n/2), \qquad \sum_{n\geq 0} q_n (A_n^2 - R_n/2) = 0. \tag{906}$$

Differentiating only the hypergeometric factor and omitting its gamma normalization would not give this identity.

To apply (906), define

$$\begin{aligned} L_n &= 8H_{2n} - 6H_n - 4\log 2 + \frac{2}{4n+1} = 2A_n, \\ M_n &= \frac{5\pi^2}{3} - 16H_{2n}^{(2)} + 6H_n^{(2)} - \frac{4}{(4n+1)^2}, \\ \Sigma_7 &= \sum_{n\geq 0} q_n (L_n^2 + M_n), \qquad V_w = \sum_{n\geq 0} q_n = \varpi_6. \end{aligned} \tag{907}$$

The pointwise identity

$$L_n^2 + M_n - (\pi^2 + 2\delta_n) = 4(A_n^2 - R_n/2)$$

implies $\Sigma_7 = \pi^2 V_w + 2D_6$. The two periodization identities specific to the calculation are, with $B_s = 2\pi^4 V_w$ and $\mathcal{T} = \sum a_n^4 h_n (8h_n^2 + \delta_n)$,

$$\begin{aligned} \Sigma_7 &= \frac{3B_s}{2\pi^2} - \frac{32}{\pi^2}\mathcal{T}, \\ \sum_{n\geq 0} a_n^4 (128 h_n^3 + 24 h_n \delta_n + \tau_n) &= B_s/2. \end{aligned} \tag{908}$$

They are proved in Section B.4.3 from a periodization of the gamma ratio $\Gamma(z+\frac{1}{2})/\Gamma(z+1)$ and three elliptic-moment evaluations that the package inherits; they do not follow from the symmetry alone. Subtracting $16\mathcal{T}$ from the second line gives

$$\begin{aligned} \sum a_n^4 (8h_n\delta_n + \tau_n) &= \frac{B_s}{2} - 16\mathcal{T} = \frac{\pi^2 \Sigma_7}{2} - \frac{B_s}{4} \\ &= \frac{\pi^2}{2}(\pi^2 V_w + 2D_6) - \frac{2\pi^4 V_w}{4} = \pi^2 D_6. \end{aligned}$$

This proves (901) from the symmetry and (908). Every series used in this elimination is absolutely convergent.

B.4.3 Proof of the two periodization identities

The quartic periodization. Put $\mathsf{g}(z) = \Gamma(z+\frac{1}{2})/\Gamma(z+1)$, so that $\mathsf{g}(n)^4 = \pi^2 a_n^4$, and $\mathsf{b}(t) = \sum_{n\geq 0} \mathsf{g}(n+t)^4$. Half-integer recurrence and special values of the polygamma functions [32] give

$$(\log \mathsf{g})'(n) = -2h_n, \qquad (\log \mathsf{g})''(n) = \delta_n, \qquad (\log \mathsf{g})'''(n) = -\tau_n. \tag{909}$$

Gamma reflection [26] gives $\mathsf{g}(-\frac{1}{2} - z) = -\cot(\pi z)\,\mathsf{g}(z)$. Consequently

$$\mathfrak{P}(t) = \cos^4(\pi t)\,\mathsf{b}(t) + \sin^4(\pi t)\,\mathsf{b}(\tfrac{1}{2} - t)$$

is the cosine-multiplied bilateral sum $\cos^4(\pi t)\sum_{n\in\mathbb{Z}} \mathsf{g}(n+t)^4$. It is entire and periodic of period one: the cosine factor removes all fourth-order half-integer poles. In a vertical strip, gamma-ratio asymptotics and reflection [26, 27] give $|\mathfrak{P}(\theta + \mathrm{i}y)| = O(e^{4\pi|y|}/|y|)$; indeed the one-sided sum

of four gamma ratios is bounded by a constant times $\sum_{n\ge0}|n+\theta+\mathrm{i}y|^{-2}=O(1/|y|)$. Shifting the contour of a Fourier coefficient vertically therefore excludes the frequencies ±2 as well as all larger ones. Reflection under $t\mapsto\frac12-t$, which leaves $\mathfrak{P}$ invariant, removes the cosine mode, so $\mathfrak{P}$ is a constant plus a multiple of $\sin(2\pi t)$. Since $\sin^4(\pi t)\mathsf{b}(\frac12-t)$ starts at order four and $\cos^4(\pi t)=1-2\pi^2t^2+O(t^4)$,

$$\mathfrak{P}'(0)=\mathsf{b}'(0),\qquad \mathfrak{P}'''(0)=\mathsf{b}'''(0)-12\pi^2\mathsf{b}'(0).$$

The finite Fourier form gives $\mathfrak{P}'''(0)=-4\pi^2\mathfrak{P}'(0)$. Hence

$$\mathsf{b}'''(0)=8\pi^2\mathsf{b}'(0). \tag{910}$$

Three inherited elliptic-moment evaluations. With $\mathsf{k}(t)={}_2F_1(\frac12,\frac12;1;t)=2K(t)/\pi$ in the parameter convention, put

$$\mathfrak{N}(a,b)=\int_0^1 t^{a-1}(1-t)^{b-1}\mathsf{k}(t)\mathsf{k}(1-t)\,dt,\qquad \mathfrak{M}(s)=\mathfrak{N}(s,1),\qquad \mathfrak{I}(s)=\mathfrak{N}(s,1-s),$$

and $\mathfrak{M}_2(a,b)=\int_0^1 t^{a-1}(1-t)^{b-1}\mathsf{k}(t)^2\,dt$. These integrals and their fixed-order logarithmic derivatives are holomorphic in their common positive-beta domain, and the endpoints have at most logarithmic growth. The package takes three evaluations of these moments from the earlier stages of the calculation and does not rederive them. First, the very-well-poised Mellin series, for $0<\operatorname{Re}s<1$,

$$\mathfrak{I}(s)=\frac{\Gamma(1+s)\Gamma(s)^2\Gamma(1-s)}{\Gamma(s+\frac12)^2}\,{}_7F_6\!\left(\begin{matrix}s,s,s,s,1+\frac{s}{2},\frac12,\frac12\\ \frac{s}{2},1,1,1,s+\frac12,s+\frac12\end{matrix};1\right). \tag{911}$$

Second, the mixed Mellin series

$$\mathfrak{M}(s)=\frac{\Gamma(s)^2}{\Gamma(s+\frac12)^2}\,{}_4F_3\!\left(\begin{matrix}\frac12,\frac12,s,s\\ 1,s+\frac12,s+\frac12\end{matrix};1\right). \tag{912}$$

Third, the upper-half-plane contour relation for the square moments, with $c=2-a-b$,

$$0=\mathfrak{M}_2(a,b)-e^{\mathrm{i}\pi a}\mathfrak{M}_2(a,c)-e^{-\mathrm{i}\pi b}\big[\mathfrak{M}_2(c,b)-\mathfrak{M}_2(b,c)+2\mathrm{i}\,\mathfrak{N}(c,b)\big], \tag{913}$$

whose arcs vanish when $\operatorname{Re}a,\operatorname{Re}b,\operatorname{Re}c>0$.

Consequences of the inputs. At $s=\frac12$ the n-th term of (911), gamma prefactor included, is $\frac{\pi^2}{2}q_n$, and its first and second logarithmic derivatives in s are L_n and M_n. Hence

$$\mathfrak{I}(\tfrac12)=\frac{\pi^2}{2}V_w=\frac{B_s}{4\pi^2},\qquad \Sigma_7=\frac{2}{\pi^2}\,\mathfrak{I}''(\tfrac12). \tag{914}$$

Taking $(a,b,c)=(s,1,1-s)$ in (913) gives $\mathfrak{M}_2(s,1-s)=2\mathfrak{M}(1-s)/\sin\pi s$. Taking $(a,b,c)=(1,s,1-s)$, and its imaginary part, then gives

$$\mathfrak{I}(s)=\frac{\mathfrak{M}(s)-\mathfrak{M}(1-s)}{\cos\pi s}. \tag{915}$$

The singularity at $s=\frac12$ is removable. All derivatives below include the derivative of this denominator, with its sign. By (912), $\mathfrak{M}(\frac12+t)=\pi^{-1}\sum_{n\ge0}\mathsf{g}(n)^2\mathsf{g}(n+t)^2$, and the differentiated series converge locally normally.

The second identity. At its removable point (915) gives $\mathfrak{M}'(\frac12) = -\frac{\pi}{2}\mathfrak{I}(\frac12)$, and termwise differentiation gives $\mathfrak{M}'(\frac12) = \mathsf{b}'(0)/(2\pi)$. With (914),

$$\mathsf{b}'(0) = -\frac{B_s}{4}, \qquad \text{equivalently} \qquad \sum_{n\ge0} a_n^4 h_n = \frac{B_s}{32\pi^2}. \tag{916}$$

By (909),

$$\frac{d^3}{dt^3}\mathsf{g}(n+t)^4\bigg|_{t=0} = -4\pi^2 a_n^4(128h_n^3 + 24h_n\delta_n + \tau_n).$$

Equations (910) and (916) now give $\sum_{n\ge0} a_n^4(128h_n^3 + 24h_n\delta_n + \tau_n) = -2\,\mathsf{b}'(0) = B_s/2$, the second line of (908).

The first identity. Write $\langle f\rangle = \sum_{n\ge0}\mathsf{g}(n)^4 f_n$ and $\mathsf{d}_n = (\log\mathsf{g})'(n) = -2h_n$. Expanding (915) about $s=\frac12$ gives $\mathfrak{I}''(\frac12) = -\frac{2}{3\pi}[\mathfrak{M}'''(\frac12) + \pi^2\mathfrak{M}'(\frac12)]$. The third derivative of the series for $\mathfrak{M}(\frac12+t)$ contains $\langle(\log\mathsf{g})'''\rangle$, which (910), written as $\langle 64\mathsf{d}^3 + 48\mathsf{d}\delta + 4(\log\mathsf{g})'''\rangle = 8\pi^2\langle 4\mathsf{d}\rangle$, eliminates. The result is

$$\mathfrak{I}''(\tfrac12) = \frac{16}{\pi^2}\langle\mathsf{d}^3\rangle + \frac{8}{\pi^2}\langle\mathsf{d}\delta\rangle - 12\langle\mathsf{d}\rangle.$$

With $\mathsf{g}(n)^4 = \pi^2 a_n^4$, $\mathsf{d}_n = -2h_n$, (916) and (914), this is the first line of (908):

$$\Sigma_7 = \frac{3B_s}{2\pi^2} - \frac{32}{\pi^2}\sum_{n\ge0} a_n^4 h_n(8h_n^2 + \delta_n).$$

What remains an input. The two periodization identities, and with them (901) and the wheel normalization of Section B.4.4, rest on the three inherited evaluations (911)–(913), besides gamma and polygamma identities and the published symmetry used above. The quartic periodization (910) is proved here. Both periodization identities were also confirmed numerically (Section 22.9); these are floating-point checks, not part of the proof.

B.4.4 From the common Barnes contour to one H value

Use the kernels M_1, M_2 of (879), and put $d(s) = \psi_1(1/2-s) - \psi_1(1-s)$. At $s=-n$, $d(-n) = \delta_n$ and $d'(-n) = \tau_n$. Equation (878) gives

$$\mathrm{Res}_{s=-n}(M_1 d) = \pi a_n^4\delta_n, \qquad \mathrm{Res}_{s=-n}(M_2 d) = a_n^4(8h_n\delta_n + \tau_n).$$

Thus (901) yields

$$W_0 = \frac{1}{\pi}\mathfrak{B}_{1/4}[(8M_1 + 5M_2)d] = \frac{\sqrt{89}}{\pi}\mathfrak{B}_{1/4}[N_{\alpha_W} d]. \tag{917}$$

The second equality uses (883) and $\cos(\pi\alpha_W) = 8/\sqrt{89}$, $\sin(\pi\alpha_W) = 5/\sqrt{89}$.

To remove the trigamma insertion, start with

$$d(s) = -4\int_0^1 \frac{x^{-2s}\log x}{1+x}\,dx, \qquad \mathrm{Re}\,s < 1/2.$$

The Mellin representation $(1+x)^{-1} = \mathfrak{B}_{1/4}[\Gamma(z)\Gamma(1-z)x^{-z}]$ and $\int_0^1 x^{a-1}(-\log x)\,dx = a^{-2}$ give

$$d(s) = 4\mathfrak{B}_{1/4}\left[\frac{\Gamma(z)\Gamma(1-z)}{(1-2s-z)^2}\right].$$

Substitute $z = 2t$ and use gamma duplication,

$$\Gamma(2t)\Gamma(1-2t) = \frac{\Gamma(t)\Gamma(t+1/2)\Gamma(1/2-t)\Gamma(1-t)}{2\pi} = \frac{P(t)}{2\pi}.$$

The Jacobian and the squared denominator then give

$$\mathfrak{B}_{1/8}\Big[\frac{P(t)}{(1/2-s-t)^2}\Big] = \pi d(s). \tag{918}$$

Substituting this identity proves $W_0 = \sqrt{89}\,\mathcal{H}_W/\pi^2$, with precisely the kernel of Section B.2.3. On the specified contours, $|1/2-s-t| \geq 1/8$, $P(t) = 2\pi^2/\sin(2\pi t)$ decays exponentially, and $N_{\alpha_W}(1/4+\mathrm{i}u) = O((1+|u|)^{-2})$. These give an integrable product majorant and justify the Mellin interchanges.

B.5 The seventh-power sector

B.5.1 A complete central series, not just a logarithmic fragment

The sum $M_7 = \sum_{n\geq 0}(-1)^n a_n^7 F_n$ and its coefficients are defined in (608). The $n=0$ term is included. The definition includes the complete ordinary and logarithmic seventh-sector contributions; no separate ordinary correction is added to it. For auditing the initial term,

$$\begin{aligned} F_0 = {} & 39886\log^3 2 - 10906\pi\log^2 2 - 22792\log^2 2 - 2188\pi^2\log 2 + 8156\pi\log 2 \\ & - \frac{1679}{6}\pi^3 + \frac{2330}{3}\pi^2 - 2156\zeta(3). \end{aligned} \tag{919}$$

B.5.2 Matching the complete coefficient with a Barnes kernel

For the Barnes proof, put

$$Y(s) = \tfrac{1}{4} - s, \quad R(s) = \Big[\frac{\Gamma(s)}{\Gamma(s+1/2)}\Big]^7, \quad z(s) = \tan(\pi s), \quad \Psi(s) = \psi(1-2s) + \gamma_{\mathrm{E}}. \tag{920}$$

Here γ_{E} is Euler's constant; it is used in the proof but is not a parameter of the final functions. Define $p(z) = 7z^2 + 13z + 11$ and $q(z) = q_{7,2}z^2 + q_{7,1}z + q_{7,0}$. The common kernel is

$$I(s) = Y(s)R(s)\sec(\pi s)z(s)^4[196p(z(s))\Psi(s) + q(z(s))]. \tag{921}$$

For $j = 1, 2, 3$, the kernels $M_j^{(7)}(s) = Y(s)R(s)\sec(\pi s)z(s)^{7-j}$ have the following residues at $s = -n$, with the common factor $(-1)^n a_n^7$ omitted:

$$\begin{aligned} M_1^{(7)} &: \quad \pi^{5/2} y_n, \\ M_2^{(7)} &: \quad \pi^{3/2}(14 y_n h_n - 1), \\ M_3^{(7)} &: \quad \sqrt{\pi}\,[y_n(98h_n^2 + 7\delta_n/2 - \pi^2/2) - 14h_n]. \end{aligned} \tag{922}$$

They follow from (878) and the tangent and secant expansions. The digamma insertion satisfies

$$\Psi(-n) = H_{2n}, \quad \Psi'(-n) = 2H_{2n}^{(2)} - \pi^2/3, \quad \Psi''(-n) = -8[\zeta(3) - H_{2n}^{(3)}]. \tag{923}$$

Taking the residues of the products with this Taylor expansion reproduces the complete polynomial F_n:

$$\mathrm{Res}_{s=-n} I(s) = \sqrt{\pi}\,(-1)^n a_n^7 F_n. \tag{924}$$

At negative half-integers, $R(s)$ has zeros of order seven. The tangent and secant remove $8-j$ of these orders in $M_j^{(7)}$, leaving order $j-1 \geq 0$; Ψ is analytic there. There is consequently no additional left-pole family. On the initial line $\mathrm{Re}\, s = 1/4$ the kernels decay as $O((1+|\mathrm{Im}\, s|)^{-5/2} e^{-\pi|\mathrm{Im}\, s|})$, with a logarithmic factor for Ψ. On the closing lines $\mathrm{Re}\, s = -N - 1/4$, reflection gives an integrated

bound $O(N^{-5/2}\log(N+2))$. The horizontal sides vanish exponentially. Thus closing to the left is justified and gives

$$M_7 = \pi^{-1/2}\mathfrak{B}_{1/4}[I]. \tag{925}$$

The parameters $\eta_{7,\pm}$ and $\beta_7, \bar{\beta}_7$ of Section 1.2 factor both quadratics:

$$\begin{aligned} p(\tan\theta) &= \sqrt{185}\,\frac{\sin(\theta+\pi\beta_7)\sin(\theta+\pi\bar{\beta}_7)}{\cos^2\theta}, \\ q(\tan\theta) &= \rho_7\,\frac{\sin(\theta+\pi\eta_{7,+})\sin(\theta+\pi\eta_{7,-})}{\cos^2\theta}. \end{aligned} \tag{926}$$

For example the tangent parameters in the first line have sum $13/7$, product $11/7$, and normalizing square $185/49$. Gamma reflection gives

$$\begin{aligned} YR\sec(\pi s)z^4 p(z) &= \frac{\sqrt{185}}{\pi}N_p(s), \\ YR\sec(\pi s)z^4 q(z) &= \frac{\rho_7}{\pi}N_7(s), \\ N_p(s) &= \frac{\Gamma(s)^3\Gamma(1/2-s)^7\Gamma(5/4-s)}{\Gamma(1-s)^4\Gamma(1/4-s)\prod_{b\in\{\beta_7,\bar{\beta}_7\}}\Gamma(s+b)\Gamma(1-s-b)}. \end{aligned} \tag{927}$$

The kernel N_7 is the same expression with $\beta_7, \bar{\beta}_7$ replaced by $\eta_{7,+}, \eta_{7,-}$, and hence is the ordinary kernel of $\mathcal{G}_7$. Also $\mathcal{A}_7(s) = (-s)N_p(s)$ by the gamma recurrence. These identities fix both the ordinary contribution and the outer kernel of the bivariate contribution.

B.5.3 The inner digamma transform and the unit-slope value

For $\operatorname{Re} s = 1/4$, set

$$J_\pm(s) = \mathfrak{B}_{1/8}\left[\frac{\Gamma(t)\Gamma(1-t)e^{\pm i\pi t}}{(1-t)(1-2s-t)}\right]. \tag{928}$$

Residues at $t=-k$ give $1/[(k+1)(k+1-2s)]$; their partial-fraction sum is

$$J_-(s) = \frac{\psi(1-2s)+\gamma_{\mathrm{E}}}{-2s} = \frac{\Psi(s)}{-2s}. \tag{929}$$

The phase and the alternating gamma residue cancel. Quadratic rational decay in the undamped direction and exponential decay in the other direction justify closing the contour. Gamma reflection gives

$$\Gamma(t)\Gamma(1-t)(e^{-i\pi t}-e^{i\pi t}) = -2\pi i.$$

Thus $J_- - J_+$ is $-2\pi i$ times the contour value of $[(1-t)(1-2s-t)]^{-1}$. Both poles of this rational function are on the right, and its quadratic decay permits closing to the left. The difference is zero. Averaging the two phases and substituting $t = 2v$ now gives

$$J_-(s) = \frac{\pi}{2}\mathfrak{B}_{1/16}\left[\frac{\cot(2\pi v)}{(1/2-v)(1/2-s-v)}\right]. \tag{930}$$

The factor is $\pi/2$: the measure contributes 2, and the two denominators contribute 4. Therefore

$$\mathfrak{B}_{1/16}\left[\frac{\cot(2\pi v)}{(1/2-v)(1/2-s-v)}\right] = -\frac{\Psi(s)}{\pi s}. \tag{931}$$

Gamma reflection at quarter arguments gives the unit-slope quotient

$$\cot(2\pi v) = \frac{\Gamma(v)\Gamma(v+1/2)\Gamma(1-v)\Gamma(1/2-v)}{\Gamma(v+1/4)\Gamma(v+3/4)\Gamma(1/4-v)\Gamma(3/4-v)}. \tag{932}$$

There is no extra π in this identity. Combining it with the two rational factors yields exactly the lists in (875). Inserting $\mathcal{A}_7 = (-s)N_p$ into (931) gives $\mathcal{H}_7 = \mathfrak{B}_{1/4}[N_p\Psi]/\pi$. Consequently

$$M_7 = \frac{\rho_7\mathcal{G}_7}{\pi^{3/2}} + \frac{196\sqrt{185}}{\sqrt{\pi}}\mathcal{H}_7,$$
$$24\pi^3 M_7 = 24\pi^{3/2}[\rho_7\mathcal{G}_7 + 196\pi\sqrt{185}\,\mathcal{H}_7]. \tag{933}$$

This proves the complete normalization of the seventh-power sector in (2).

For $s = 1/4 + \mathrm{i}u$ and $v = 1/16 + \mathrm{i}w$, $|\cot(2\pi v)| \le 1 + \sqrt{2}$. Cauchy–Schwarz bounds the integral of the absolute rational factors by

$$\int_{\mathbb{R}} \frac{dw}{\sqrt{(7/16)^2 + w^2}\sqrt{(3/16)^2 + (u+w)^2}} \le \frac{16\pi}{\sqrt{21}}.$$

The outer factor is $O((1+|u|)^{-3/2}e^{-\pi|u|})$. This gives a product majorant for the bivariate integral and justifies the interchanges. The logarithms of both final arguments $(1,1)$ are zero; conjugation of the parameter pairs and of the imaginary contour coordinates also shows that the specified value is real.

B.6 The primitive: from the Gauss pairing to one bivariate value

B.6.1 The Gauss-product input

The primitive input is (609), with $v(x) = {}_2F_1(\frac{1}{4},\frac{1}{4};1;x)$, $w(x) = {}_2F_1(\frac{1}{4},\frac{1}{4};\frac{1}{2};x)$, the products p_0, p_1, p_2 and the pairings $I_{ij} = I_{ji}$ defined there. The zero-balanced Gauss factor has at most a logarithmic endpoint singularity, so these pairings converge absolutely. The coefficient in r_6 is $96P_a$.

B.6.2 The contour relation used in the final reduction

On the open upper half-plane define

$$f(z) = v(z), \qquad g(z) = \frac{\sqrt{Q}}{\pi}w(1-z) - v(z), \qquad f_* = f(1-z), \quad g_* = g(1-z).$$

The star means reflection, not complex conjugation. Principal Gauss branches and separate principal square roots $\sqrt{z}, \sqrt{1-z}$ are used. The Gauss cut-jump and connection formulas give the exterior boundary data

Region	f	g	f_*	g_*
$x<0,\ z = x+\mathrm{i}0$	u	$r-\mathrm{i}u$	$a-\mathrm{i}b$	$a+\mathrm{i}b$
$x>1,\ z = x+\mathrm{i}0$	$a+\mathrm{i}b$	$a-\mathrm{i}b$	u	$r+\mathrm{i}u$

Here u, r, a, b are real local values. For $x<0$, the argument $1-z$ approaches the zero-balanced cut from below; this fixes the minus sign in $g = r - \mathrm{i}u$. For $x>1$, the connection formula

$$v(z) = \frac{\sqrt{Q}}{2\pi}w(1-z) - \frac{2}{\sqrt{Q}}\sqrt{1-z}\;{}_2F_1\left(\begin{matrix}3/4, 3/4\\ 3/2\end{matrix};1-z\right)$$

shows that f and g are conjugate on the upper bank. Reflection supplies the remaining entries.

Use the holomorphic numerator

$$\mathcal{N} = g^2g_*^2 + g^2f_*^2 + f^2g_*^2 - 4fgf_*g_* - 3f^2f_*^2$$
$$+ 4\mathrm{i}(fgf_*^2 - f^2f_*g_*), \qquad \widetilde{\mathcal{N}}(z) = \frac{\mathcal{N}(z)}{\sqrt{z}\sqrt{1-z}}. \tag{934}$$

Substitution of either boundary row gives the polynomial identity

$$\mathcal{N} = 2r(a-b)[(a+b)r - 2(a-b)u] \in \mathbb{R}.$$

On $x < 0$ and $x > 1$, the reciprocal square-root factor is respectively $-\mathrm{i}/\sqrt{(-x)(1-x)}$ and $\mathrm{i}/\sqrt{x(x-1)}$. Therefore $\mathrm{Re}\,\widetilde{\mathcal{N}}(x+\mathrm{i}0) = 0$ outside $[0,1]$. At infinity the Gauss connection expansion gives $\widetilde{\mathcal{N}}(z) = O(|z|^{-2}(1+\log|z|)^4)$. Near 0 and 1 the bounds are of inverse-square-root type with at most quadratic logarithms. The large semicircle and the two endpoint indentations all vanish. Taking the real part of Cauchy's theorem leaves $\int_0^1 \mathrm{Re}\,\widetilde{\mathcal{N}}(x)\,dx = 0$.

On the interior interval, $p_0 = f^2$, $p_1/\pi = f(f+g)$ and $p_2/\pi^2 = (f+g)^2$. Expanding the real numerator in these products, and using $I_{ij} = I_{ji}$, gives

$$\begin{aligned} I_{22} - 4I_{12} + 4I_{02} - 4I_{00} &= 0, \\ P_a &= \pi^4[60(I_{12} - I_{02}) + 24I_{11} + 56I_{00}]. \end{aligned} \tag{935}$$

Thus the logarithm-squared times logarithm-squared pairing I_{22} of (609) is removed. The contour proof, not an equality of individual power-series coefficients, establishes the relation.

B.6.3 Identification of the three ordinary Meijer solutions

All three p_i solve the symmetric-square equation

$$[\theta^3 - x(\theta + \tfrac{1}{2})^3]p = 0, \qquad \theta = x\frac{d}{dx}. \tag{936}$$

Define

$$g_j(x) = G^{j,3-j}_{3,3}\left(x \,\middle|\, \begin{matrix}(\frac{1}{2})^{\times 3} \\ 0^{\times 3}\end{matrix}\right), \quad j = 1,2,3. \tag{937}$$

Their inverse-Mellin kernels are

$$M_j^{(3)}(s) = \frac{\Gamma(s)^j\Gamma(1/2-s)^{3-j}}{\Gamma(1-s)^{3-j}\Gamma(1/2+s)^j} = \left[\frac{\Gamma(s)}{\Gamma(s+1/2)}\right]^3 \tan^{3-j}(\pi s). \tag{938}$$

The standard Meijer differential equation is also (936). To determine the complete normalization, expand at $s = 0$:

$$\left[\frac{\Gamma(s)}{\Gamma(s+1/2)}\right]^3 = \frac{1 + 6s\log 2 + (18\log^2 2 - \pi^2/2)s^2 + O(s^3)}{\pi^{3/2}s^3}.$$

Writing $L = -\log x$, the residues give

$$\begin{aligned} g_1(x) &= \sqrt{\pi} + O(x), \\ g_2(x) &= (L + 6\log 2)/\sqrt{\pi} + O(x|\log x|), \\ g_3(x) &= [(L+6\log 2)^2 - \pi^2]/(2\pi^{3/2}) + O(x\log^2 x). \end{aligned} \tag{939}$$

On the Gauss side, $v(x) = 1 + O(x)$ and the zero-balanced connection formula is $\sqrt{Q}\,w(1-x) = L + \pi + 6\log 2 + O(x|\log x|)$. Matching all constant and logarithmic coefficients gives

$$g_1 = \sqrt{\pi}p_0, \qquad g_2 = \frac{p_1 - \pi p_0}{\sqrt{\pi}}, \qquad g_3 = \frac{p_2 - 2\pi p_1}{2\pi^{3/2}}. \tag{940}$$

The Frobenius recurrence of (936) then fixes every higher coefficient. Matching the differential equation without these boundary polynomials would not suffice.

For the real numerical evaluator, introduce the elliptic coordinate

$$m(x) = \frac{x}{2(1+\sqrt{1-x})}, \qquad K_x = K(m(x)), \qquad K_x^c = K(1-m(x)), \tag{941}$$

with $K(m)$ in the parameter convention of Section 1.4. The argument is the elliptic parameter, not the modulus. The substitution $x = 4m(1-m)$ transforms the elliptic equation $m(1-m)Y'' + (1-2m)Y' - Y/4 = 0$ into $x(1-x)y'' + (1-3x/2)y' - y/16 = 0$. Its regular normalization gives $v(x) = 2K_x/\pi$. At zero the complete logarithmic normalization is $2(K_x + K_x^c) = -\log x + \pi + 6\log 2 + O(x|\log x|)$, which matches the zero-balanced connection formula above; the regular and logarithmic Frobenius data then give $\sqrt{Q}\, w(1-x) = 2(K_x + K_x^c)$. Therefore

$$g_1 = \frac{4}{\pi^{3/2}} K_x^2, \qquad g_2 = \frac{4}{\pi^{3/2}} K_x K_x^c, \qquad g_3 = \frac{2}{\pi^{3/2}} [(K_x^c)^2 - K_x^2]. \tag{942}$$

These are finite kernel identities used to check the primitive numerically. The package also contains a Wolfram implementation of them but claims no run of it; its numerical values come from the equivalent Python implementations.

B.6.4 Beta integration and the four intermediate pairings

Write

$$\langle f, g\rangle = \int_0^1 \frac{f(x)g(1-x)}{\sqrt{x(1-x)}}\, dx, \qquad H_{jk} = \langle g_j, g_k\rangle.$$

Inserting the two kernels (938) and applying Euler's beta integral gives the bivariate kernel

$$\frac{\Gamma(s)^j \Gamma(1/2-s)^{4-j} \Gamma(t)^k \Gamma(1/2-t)^{4-k}}{\Gamma(1-s)^{3-j} \Gamma(1/2+s)^j \Gamma(1-t)^{3-k} \Gamma(1/2+t)^k \Gamma(1-s-t)}. \tag{943}$$

The added factor is $\Gamma(1/2-s)\Gamma(1/2-t)/\Gamma(1-s-t)$. On $\operatorname{Re} s = \operatorname{Re} t = 1/4$, each $M_j^{(3)}$ is $O((1+|\operatorname{Im} s|)^{-3/2})$. Before integrating in x, the absolute three-variable integral is bounded by $B(1/4, 1/4)$ times the product of the two integrable local majorants. This justifies the interchanges and the absolute convergence of (943).

Inverting (940) gives

$$p_0 = g_1/\sqrt{\pi}, \quad p_1 = \sqrt{\pi}(g_1 + g_2), \quad p_2 = 2\pi^{3/2}(g_1 + g_2 + g_3).$$

Substituting these into (935) gives

$$\begin{aligned} P_a/\pi^3 = {} & 120(H_{11} + 2H_{12} + H_{13} + H_{22} + H_{23}) \\ & - 120(H_{11} + H_{12} + H_{13}) + 24(H_{11} + 2H_{12} + H_{22}) + 56H_{11}. \end{aligned}$$

Thus

$$P_a = 8\pi^3(10H_{11} + 21H_{12} + 18H_{22} + 15H_{23}), \qquad H_{33} + 2H_{13} - H_{22} = 0. \tag{944}$$

The second relation is the first line of (935) in the g_j basis. It is the relation used in the final compression.

B.6.5 From the four pairings to the final single value

The coefficient matrix and the zero-relation matrix are

$$\mathsf{M} = \begin{pmatrix} 10 & 21/2 & 0 \\ 21/2 & 18 & 15/2 \\ 0 & 15/2 & 0 \end{pmatrix}, \qquad \mathsf{N} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}. \tag{945}$$

Adding $\lambda\mathsf{N}$ leaves the definite pairing unchanged. The cubic $4\lambda^3 - 112\lambda^2 + 909\lambda - 2250$ is $4\det(\mathsf{M} + \lambda\mathsf{N})$. For its root λ_* and the parameters of $\mathcal{H}_*$, $v_\sigma = (1, B_\sigma, C_\sigma)^T$ satisfies

$$\mathsf{M} + \lambda_*\mathsf{N} = 5(v_+ v_-^T + v_- v_+^T). \tag{946}$$

With $f_\sigma = g_1 + B_\sigma g_2 + C_\sigma g_3$, symmetry of the pairing gives

$$10H_{11} + 21H_{12} + 18H_{22} + 15H_{23} = 10\langle f_+, f_-\rangle.$$

The kernel of f_σ contains $P_\sigma(z) = z^2 + B_\sigma z + C_\sigma$, $z = \tan(\pi s)$. Put $D_\sigma = \sqrt{(1-C_\sigma)^2 + B_\sigma^2}$. The angle parameters β_σ give

$$P_\sigma(\tan\theta) = D_\sigma \frac{\sin(\theta + \pi\beta_\sigma)\sin(\theta + \pi\bar{\beta}_\sigma)}{\cos^2\theta}.$$

Gamma reflection consequently converts f_σ/D_σ to one local Meijer kernel. The normalization is exact:

$$P_+(\mathrm{i})P_-(\mathrm{i}) = -\frac{4}{5} - \frac{3}{5}\mathrm{i}, \qquad |P_\sigma(\mathrm{i})| = D_\sigma, \qquad D_+D_- = 1.$$

After beta integration the local kernels are precisely Ψ_+, Ψ_- of Section 1.2, with the shared factor $1/\Gamma(1-s-t)$. It follows that $\langle f_+, f_-\rangle = \mathcal{H}_*$, and hence

$$P_a = 80\pi^3\mathcal{H}_*, \qquad 96P_a = 7680\pi^3\mathcal{H}_*. \tag{947}$$

No term is extracted from an endpoint or transferred into the known polynomial in this conversion.

B.7 Conversion to the real log-sine and Clausen values

The starting block is S_{pol} of Section 22.9, with the descending-index convention. For $l(\theta) = \log(2\sin(\theta/2))$ and $0 < \theta < 2\pi$, the principal logarithm is

$$\log(1 - e^{\mathrm{i}\theta}) = l(\theta) + \frac{\mathrm{i}}{2}(\theta - \pi). \tag{948}$$

The derivative

$$\frac{d}{dz}\operatorname{Li}_{2,1}(z,1) = \frac{\log^2(1-z)}{2z}$$

and $z = e^{\mathrm{i}\theta}$ imply

$$\frac{d}{d\theta}\operatorname{Im}\operatorname{Li}_{2,1}(e^{\mathrm{i}\theta},1) = \frac{1}{2}l(\theta)^2 - \frac{(\theta-\pi)^2}{8}. \tag{949}$$

The initial imaginary part at $\theta = 0$ is zero. The logarithmic singularity is integrable. Integrating to $\pi/2$ gives

$$D_3 = -\frac{1}{2}\operatorname{Ls}_3(\pi/2) - \int_0^{\pi/2}\frac{(\theta-\pi)^2}{8}\,d\theta = -\frac{1}{2}L_3 - \frac{7\pi^3}{192}.$$

This is a differential identity. For weight five, the real coefficient convention $\operatorname{Cl}_{2,1,2}(\sigma) = \sum_{n>m>k\geq 1}\cos(n\sigma)/(n^2mk^2)$ of Section 1.2 gives directly $D_5 = C_{212}$: at odd total weight the multiple Clausen convention used here is the cosine component of the descending multiple polylogarithm. This is a real-function convention, not a new elementary evaluation of D_5. Absolute convergence follows from

$$\sum_{n>m>k\geq 1}\frac{1}{n^2mk^2} \leq \zeta(2)\sum_{n\geq 1}\frac{H_{n-1}}{n^2} < \infty,$$

so the cosine component may be taken term by term. The definitions of L_3 and C_{212} fix established special functions [16]; they are not additional residual operations in the fixed-value formula (2).

Substitute these two identities into the entire block, not just into its individual special values:

$$\begin{aligned} S_{\mathrm{pol}} &= 4352\big[(32\mathsf{G} + 9\pi^2)(-L_3/2 - 7\pi^3/192) - 64C_{212}\big] \\ &= -2176\big[(32\mathsf{G} + 9\pi^2)L_3 + 128C_{212}\big] - \frac{476}{3}\pi^3(32\mathsf{G} + 9\pi^2). \end{aligned} \tag{950}$$

Adding the last polynomial to E gives the polynomial $E_{\mathrm{real}} = E - \frac{476}{3}\pi^3(32\mathsf{G} + 9\pi^2)$ of (2). The fully expanded E has eighteen monomials in $A, Q, \mathsf{G}, \pi, \zeta(3), \zeta(5)$, and only two of them change. Their new coefficients are

$$-\frac{544}{3} - \frac{476\cdot 32}{3} = -\frac{15776}{3} \quad (\mathsf{G}\pi^3), \qquad \frac{37631}{128} - \frac{476\cdot 9}{3} = -\frac{145153}{128} \quad (\pi^5).$$

There is no new physical contribution, and the Q-dependent block is unchanged. Together with the ordinary, wheel, seventh and primitive conversions, this completes the derivation of (2) from the stated reduced input.

The formula (2) has no explicit Re or Im operation. Some Meijer G and bivariate H parameter lists still contain complex-conjugate pairs. Removing the projection notation does not remove those parameters, and the multiple Clausen name does not constitute a reduction of arithmetic independence.

B.8 Numerical evaluation and error control

B.8.1 Evaluation of the final expression

A mathematically short special-function formula need not be the fastest way to evaluate the constant. The preferred numerical routes are a rational central-wheel tail correction, Euler-transformed seventh coefficients, a geometric primitive recurrence, and algorithms for harmonic polylogarithms and log-sine values for the real polylogarithmic sector. These compute the same specified values using the proved conversion identities, not by subtracting unknown parts from the target C_6.

B.8.2 Wheel certificate: a finite correction, not an infinite asymptotic sum

For $t \ge N$, put $x = t + 1/4$, $y = x^{-2}$ and

$$\Delta_r q(t) = q(t) - \left(\frac{t+1/2}{t+1}\right)^r q(t+1), \qquad d_j = \frac{E_{2j}(2j+1)}{2\,16^j}. \tag{951}$$

The rational bases are

$$\begin{aligned} \phi^{(4)}_{-1}(t) &= \frac{(t-1/4)^3}{(t-1/2)^4}, & \phi^{(4)}_{j}(t) &= \frac{t-1/4}{t^j(t-1/2)^{j+4}}, \\ \phi^{(6)}_{j}(t) &= \frac{(t-1/4)^6}{t^j(t-1/2)^{j+6}}. \end{aligned} \tag{952}$$

To make the matching and the bound explicit, define the Euler numbers by $\operatorname{sech} v = \sum_{j\ge 0} E_{2j} v^{2j}/(2j)!$ near zero. Write

$$q^{(4)}_M = \sum_{j=-1}^{M} \alpha_j \phi^{(4)}_j, \qquad q^{(6)}_L = \sum_{j=0}^{L} \beta_j \phi^{(6)}_j, \qquad L = M+1.$$

Choose their rational coefficients by matching

$$\Delta_4 q^{(4)}_M = \sum_{k=0}^{M+1} d_k y^{k+1} + O(y^{M+3}), \qquad \Delta_6 q^{(6)}_L = 4\sqrt{y}\sum_{k=0}^{L} d_k y^k + O(\sqrt{y}\,y^{L+1}).$$

The triangular leading multipliers are 3 for the fourth-power index -1, $2j+5$ for its indices $j \ge 0$, and $2j+3$ in the sixth-power construction. The initial coefficients are $\alpha_{-1} = 1/6$, $\alpha_0 = -77/480$ and $\beta_0 = 2/3$, $\beta_1 = -91/120$. The rational differences have the denominator polynomials

$$\begin{aligned} \mathcal{D}_4(y) &= (1-y/16)^M(1-9y/16)^{M+4}, & \Delta_4 q^{(4)}_M &= \mathcal{P}_4(y)/\mathcal{D}_4(y), \\ \mathcal{D}_6(y) &= (1-y/16)^L(1-9y/16)^{L+6}, & \Delta_6 q^{(6)}_L &= \sqrt{y}\,\mathcal{P}_6(y)/\mathcal{D}_6(y). \end{aligned}$$

Let $\mathcal{U}_4(y)=\sum_{k=0}^{M+1} d_k y^{k+1}$ and $\mathcal{U}_6(y)=4\sum_{k=0}^{L} d_k y^k$. Exact polynomial division gives

$$\mathcal{E}_4(y)=\frac{\mathcal{P}_4-\mathcal{D}_4\mathcal{U}_4}{y^{M+3}},\qquad \mathcal{E}_6(y)=\frac{\mathcal{P}_6-\mathcal{D}_6\mathcal{U}_6}{y^{L+1}}.$$

For $Y=(N+1/4)^{-2}$, the constants used below are

$$K_4=|d_{M+2}|+\frac{\sum_j |[y^j]\mathcal{E}_4(y)|Y^j}{\mathcal{D}_4(Y)},$$
$$K_6=4|d_{L+1}|+\frac{\sum_j |[y^j]\mathcal{E}_6(y)|Y^j}{\mathcal{D}_6(Y)}.$$

Both denominators are positive for $0\le y\le Y$ when $N\ge 2$. They imply, for $t\ge N$ and $x=t+1/4$,

$$|\delta(t)-\Delta_4 q_M^{(4)}(t)|\le K_4 x^{-2M-6},\qquad |4x\delta(t)-\Delta_6 q_L^{(6)}(t)|\le K_6 x^{-2L-3}.$$

Here $\delta(t)=\psi_1(t+1/2)-\psi_1(t+1)$ extends the integer remainder, and (898) holds with $x=t+1/4$. The global Taylor estimate used is

$$\left|\operatorname{sech} v-\sum_{k=0}^{K}\frac{E_{2k}v^{2k}}{(2k)!}\right|\le\frac{|E_{2K+2}|v^{2K+2}}{(2K+2)!}\quad(v\in\mathbb{R}).\tag{953}$$

It follows from the positive product $\cosh v=\prod_{k\ge1}(1+4v^2/((2k-1)^2\pi^2))$ [20]. Since $(1+\mu\varsigma)^{-1}$ is the Laplace transform at ς of an exponential variable with mean μ, $\operatorname{sech}\sqrt{\varsigma}=\mathbb{E}\,e^{-\varsigma \mathsf{X}}$, where X is the sum of independent exponential variables with means $4/((2k-1)^2\pi^2)$; these means are summable, with sum $\frac12$. Comparing Taylor coefficients at $\varsigma=0$ gives $\mathbb{E}\,\mathsf{X}^k=k!\,|E_{2k}|/(2k)!$. The finite remainder $|e^{-u}-\sum_{k=0}^{K}(-u)^k/k!|\le u^{K+1}/(K+1)!$ for $u\ge0$, applied at $u=v^2\mathsf{X}$ and averaged, gives (953). Integrated against (898), it gives $\delta(t)=\sum_{k=0}^{K} d_k x^{-2k-2}+\mathsf{r}_K$ with $|\mathsf{r}_K|\le|d_{K+1}|x^{-2K-4}$, which supplies the next-term bounds above. No infinite asymptotic expansion is summed at fixed t.

Rational triangular matching determines $q_M^{(4)}$ and $q_{M+1}^{(6)}$ so that their finite differences match the truncated expansions of $\delta(t)$ and $(4t+1)\delta(t)$. The constants K_4,K_6 above give the finite approximant

$$\begin{aligned}W_{N,M}=&\sum_{n=1}^{N-1}a_n^4[8+5\pi(4n+1)a_n^2]\delta_n\\&+8a_N^4q_M^{(4)}(N)+5\pi a_N^6q_{M+1}^{(6)}(N),\\|W_c-W_{N,M}|\le&\frac{(8/9)K_4+(110/189)K_6}{(2M+7)(N-3/4)^{2M+7}}.\end{aligned}\tag{954}$$

The added terms evaluate the two rational tails completely by telescoping, since $a_{n+1}^r=a_n^r[(n+\frac12)/(n+1)]^r$. The upper boundaries vanish: $q_M^{(4)}$ is $O(t^{-1})$ and $q_{M+1}^{(6)}$ is $O(1)$, whereas the respective gamma weights a_n^4 and a_n^6 are $O(n^{-2})$ and $O(n^{-3})$. Using $a_n^4\le1/(9x^2)$, $a_n^6\le1/(27x^3)$, $\pi<22/7$ and integral comparison then gives (954). Each approximant is a rational cubic in π, but its coefficients depend on N,M and the error is not identically zero. It does not prove that the exact wheel lies in $\mathbb{Q}+\mathbb{Q}\pi+\mathbb{Q}\pi^2+\mathbb{Q}\pi^3$.

The remainder is controlled by a finite global Taylor inequality for sech, obtained from the positive product for cosh, not by assuming convergence of an infinite asymptotic expansion at fixed t. The certificate independently bounds π by Machin's formula and uses exact rational arithmetic plus outward dyadic rounding.

B.8.3 Euler-weighted alternating sums

For alternating coefficients b_n, the evaluator uses the finite weights

$$w_{N,n} = 1 - 2^{-N} \sum_{k=0}^{n} \binom{N}{k}, \qquad \widehat{E}_N = \sum_{n=0}^{N-1} (-1)^n w_{N,n} b_n. \tag{955}$$

For the seventh-sector sum, $b_n = a_n^7 F_n$, write $\widehat{M}_{7,N}$ for $\widehat{E}_N$. The lemma below gives the tail bound

$$|M_7 - \widehat{M}_{7,N}| \le 2^{100-N}, \qquad N \ge 1. \tag{956}$$

Finite arithmetic must still be enclosed; the dyadic interval code does so. This bound applies to the Euler-weighted approximant, not to a raw partial sum. The elementary remainder kernel $2^{-N}(1-x)^N/(1+x)$ and beta-moment bounds underpin this family of evaluations; Section B.8.5 applies them to further alternating series.

Lemma (Euler tail of the seventh-sector sum). For $N \ge 1$, $|M_7 - \widehat{M}_{7,N}| \le 2^{99-N}$; in particular (956) holds with one bit of slack.

Proof. Restrict x and s to the closed bidisk $|x|, |s| \le \frac{1}{64}$, put $t = \frac{1}{4} + s$, $\mathsf{a} = \frac{1}{2} + s + x$, $\mathsf{d} = \frac{1}{2} - 2x$, and define

$$\mathfrak{Z}(x,s) = \left[\frac{\Gamma(1-x)\cos\pi(s+x)}{\Gamma(\frac{1}{2}+x)\cos\pi x}\right]^7 \sum_{n\ge 0} (-1)^n (n+t) \left[\frac{\Gamma(n+\mathsf{a})}{\Gamma(n+\mathsf{a}+\mathsf{d})}\right]^7.$$

By gamma reflection this is the nonnegative-index half of the finite-Fourier family with which the package derives (608) from its earlier five-shift form. Its terms are $O(n^{-5/2+14\,\mathrm{Re}\,x})$, so the series converges absolutely and uniformly on the bidisk, where $\mathfrak{Z}$ is analytic. At $x = s = 0$ the n-th term is $(-1)^n a_n^7 y_n$. Its local logarithmic derivatives there follow from polygamma reflection, recurrence and half-integer values [32]:

$$\partial_s :\ y_n^{-1} - 14h_n, \qquad \partial_s^2 :\ -y_n^{-2} - 7\pi^2 + 7\delta_n, \qquad \partial_x :\ 14H_{2n}, \qquad \partial_x\partial_s :\ R_n^{(7)}, \qquad \partial_x\partial_s^2 :\ S_n^{(7)},$$

with $R_n^{(7)} = -7(\frac{\pi^2}{3} + 4H_{2n}^{(2)})$ and $S_n^{(7)} = -112[\zeta(3) - H_{2n}^{(3)}]$. With u_n, v_n of (608), the six derivatives at $(0,0)$ needed below are

$$\begin{aligned} \mathfrak{Z} &= \textstyle\sum (-1)^n a_n^7 y_n, & \mathfrak{Z}_x &= \textstyle\sum (-1)^n a_n^7\, 14H_{2n} y_n, \\ \mathfrak{Z}_s &= \textstyle\sum (-1)^n a_n^7 u_n, & \mathfrak{Z}_{xs} &= \textstyle\sum (-1)^n a_n^7 (14H_{2n}u_n + y_n R_n^{(7)}), \\ \mathfrak{Z}_{ss} &= \textstyle\sum (-1)^n a_n^7 v_n, & \mathfrak{Z}_{xss} &= \textstyle\sum (-1)^n a_n^7 (14H_{2n}v_n + 2u_n R_n^{(7)} + y_n S_n^{(7)}). \end{aligned}$$

The linear combination with the coefficients

Derivative	Coefficient	Derivative	Coefficient
$\mathfrak{Z}_x$	$560\pi^2$	$\mathfrak{Z}$	$5920\pi^2\log 2 - 3224\pi^3$
$\mathfrak{Z}_{xs}$	-182π	$\mathfrak{Z}_s$	$1136\pi^2 - 1924\pi\log 2$
$\mathfrak{Z}_{xss}$	77	$\mathfrak{Z}_{ss}$	$814\log 2 - 360\pi$

reproduces F_n of (608) term by term (an identity of polynomials in π, $\log 2$, y_n, h_n, δ_n, H_{2n}, $H_{2n}^{(2)}$, $H_{2n}^{(3)}$ and $\zeta(3)$), so it equals M_7. Let $\mathfrak{Z}_N$ be $\mathfrak{Z}$ with the n-sum replaced by its Euler approximant with the weights $w_{N,n}$. The weights act termwise, so the same combination of the derivatives of $\mathfrak{Z}_N$ at $(0,0)$ is $\widehat{M}_{7,N}$.

Euler's beta integral [19] gives the sevenfold beta-moment representation

$$\mathfrak{Z}(x,s)=\mathfrak{c}(x,s)\int_{[0,1]^7}\Big[\sum_{n\ge0}(-1)^n(n+t)\xi^n\Big]\prod_{i=1}^{7}u_i^{\mathsf{a}-1}(1-u_i)^{\mathsf{d}-1}\,du_i,\qquad \xi=u_1\cdots u_7,$$

with

$$\mathfrak{c}(x,s)=\left[\frac{\Gamma(1-x)\cos\pi(s+x)}{\Gamma(\frac12+x)\cos\pi x\,\Gamma(\frac12-2x)}\right]^7.$$

The alternating kernel is $\sum_{n\ge0}(-1)^n(n+t)\xi^n=t/(1+\xi)-\xi/(1+\xi)^2$, and its exact remainder after N Euler coefficients is

$$2^{-N}\left[\frac{t(1-\xi)^N}{1+\xi}-\frac{N\xi(1-\xi)^{N-1}}{1+\xi}-\frac{\xi(1-\xi)^N}{(1+\xi)^2}\right].$$

Its absolute value is at most $(|t|+2)2^{-N}$, since $N\xi(1-\xi)^{N-1}\le1$. On the bidisk $\operatorname{Re}\mathsf{a},\operatorname{Re}\mathsf{d}\ge\frac{15}{32}$ and $B(\operatorname{Re}\mathsf{a},\operatorname{Re}\mathsf{d})<16$. Elementary integral bounds give $|\Gamma(1-x)|<3$; reflection and the same integral bounds give $|1/\Gamma(\frac12+x)|<2$ and $|1/\Gamma(\frac12-2x)|<2$. Also $|\cos\pi(s+x)|<2$ and $|1/\cos\pi x|<2$. Thus

$$|\mathfrak{c}(x,s)|\,B(\operatorname{Re}\mathsf{a},\operatorname{Re}\mathsf{d})^7<(48\cdot16)^7<2^{70},$$

and $|\mathfrak{Z}-\mathfrak{Z}_N|<2^{72-N}$ uniformly on the bidisk. Cauchy's coefficient bound yields

$$|\partial_x^\mu\partial_s^\nu(\mathfrak{Z}-\mathfrak{Z}_N)(0,0)|\le\mu!\,\nu!\,64^{\mu+\nu}2^{72-N},\qquad \mu=0,1,\quad \nu=0,1,2.$$

Using $\pi<4$ and $0<\log2<1$, the sum of the Cauchy-weighted absolute coefficients of the table is less than $2^{20}+2^{22}+2^{26}+2^{19}+2^{21}+2^{25}<2^{27}$. This proves the bound 2^{99-N}. The proof differentiates a bounded analytic remainder; it makes no assumption that an infinite asymptotic expansion converges.

B.8.4 The primitive's fast recurrence

The primitive evaluator uses integer and half-integer Frobenius branches in a nine-component system with logarithmic degree at most two. Section B.9 states both exact 9×9 matrices and all seed polynomials. The recurrence is

$$\begin{aligned}[(n+\lambda)I-A_0]z_{n,j}=C_0z_{n-1,j}+[(n-1+\lambda)I+A_0]z_{n-2,j}\\ +(j+1)(z_{n-2,j+1}-z_{n,j+1}),\quad \lambda\in\{0,1/2\}.\end{aligned}\tag{957}$$

The terminal log channel is zero, and j is processed in descending order. The first components are combined with elementary logarithmic moments at $\rho=3-2\sqrt2$. The complete-interval evaluation uses an explicit last-two-state norm majorant; its numerical values are recomputed for each run. The evaluation weights and the last-two-state bound are specified in Section B.9. They are implemented in the package's interval code; no previous primitive value is used to initialize the recurrence.

B.8.5 The complete interval certificate

What is enclosed. The interval of width below 1.183×10^{-145} quoted in Section 22.9 is built by the package's interval stage for the form of the reduced expression that this stage receives,

$$\begin{aligned}2187\,C_6^{\mathrm{rep}}(1)=E_{\mathrm{central}}+835584\Lambda_{3,2}+391680\mathsf{H}_5+13056\sqrt2(32\mathsf{G}+3\pi^2)\mathsf{H}_3\\ -393\pi^3W_c+96P_a-784\pi^{3/2}\mathsf{L}_7.\end{aligned}\tag{958}$$

Here W_c is (896) and P_a is (609); $\Lambda_{3,2}=\operatorname{Re}\operatorname{Li}_{3,2}(\frac{1+\mathrm{i}}{2},1-\mathrm{i})$, $\mathsf{H}_3=\sum_{n\ge0}a_n2^{-n}(2n+1)^{-3}$ and $\mathsf{H}_5=\sum_{n\ge0}2^{-n}(n+1)^{-5}=2\operatorname{Li}_5(\frac12)$. The sum L_7 is a fixed combination of the five alternating series

$\mathsf{L}(t)$ below, and the polynomial E_{central} contains, besides elementary constants, the ten ordinary series of the table below and the five alternating series $\mathsf{Z}(t)$. The coefficients of E_{central} and the weights of L_7 are those of the package's input expression for this stage. Exact identities established in later stages of the package carry (958) into (604). Among them are closed-form evaluations of two ordinary values in E_{central}, the identity that turns the complete seventh-power contribution into $24\pi^3 M_7$, the conversion of $\Lambda_{3,2}$, H_3 and H_5 into the values D_3, D_5 of S_{pol}, and the inherited wheel normalizations $B_- + B_+ = 2\pi^4\varpi_6$, $B_- - B_+ = 4\pi^3\varpi_4$ that write the lower block through ϖ_4 and ϖ_6. These identities are not reproduced in this appendix. Reassembling the later forms from the same intervals of the unchanged constants, with M_7 enclosed directly through (956), again gives a width below 1.183×10^{-145}; the values D_3 and D_5 receive no separate interval.

Alternating series of the earlier form. For the five shifts $t \in \{\frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}\}$ put $\mathsf{a}_t = t + \frac{1}{4}$, $\mathsf{b}_t = t + \frac{3}{4}$, $g_t = \Gamma(\mathsf{a}_t)/\Gamma(\mathsf{b}_t)$ and

$$\mathsf{Z}(t) = \sum_{n\ge 0} (-1)^n (n+t) \left[\frac{\Gamma(n+\mathsf{a}_t)}{\Gamma(n+\mathsf{b}_t)}\right]^7, \qquad \mathsf{L}(t) = \sum_{n\ge 0} (-1)^n (n+t) \left[\frac{\Gamma(n+\mathsf{a}_t)}{\Gamma(n+\mathsf{b}_t)}\right]^7 H_{2n}(2\mathsf{a}_t),$$

with $H_m(a) = \sum_{j=0}^{m-1} (j+a)^{-1}$. Their finite Euler approximants $\widehat{E}_N$ use the weights $w_{N,n}$ above; the equivalent form $w_{N,n} = \sum_{k=n}^{N-1} \binom{k}{n} 2^{-k-1}$ follows from finite binomial summation, and the first form avoids a forward-difference table. Euler's beta integral [19] writes $[(\mathsf{a}_t)_n/(\mathsf{b}_t)_n]^7 = \mathbb{E}\,\mathsf{S}^n$, where S is the product of seven independent $\mathrm{Beta}(\mathsf{a}_t, \frac{1}{2})$ variables. For fixed $0 \le \xi \le 1$ the alternating generating function of $(n+t)\xi^n$ is $t/(1+\xi) - \xi/(1+\xi)^2$, and its finite Euler remainder is, with $u = (1-\xi)/2$,

$$\mathsf{r}_N(\xi) = \frac{t u^N}{1+\xi} - \frac{\xi u^N}{(1+\xi)^2} - \frac{N\xi(1-\xi)^{N-1}}{2^N(1+\xi)}, \qquad |\mathsf{r}_N(\xi)| \le (t+2)2^{-N},$$

since $N\xi(1-\xi)^{N-1} \le 1$ for $N \ge 1$, endpoints included. Integrating against the positive beta measure proves

$$|\mathsf{Z}(t) - \widehat{E}_N[\mathsf{Z}(t)]| \le (t+2) g_t^7\, 2^{-N}. \tag{959}$$

For $\mathsf{L}(t)$ deform each beta density to $\mathrm{Beta}(\mathsf{a}_t + x, \frac{1}{2} - 2x)$. Its normalized n-th moment is $(\mathsf{a}_t + x)_n/(\mathsf{b}_t - x)_n$, and for seven copies

$$\frac{d}{dx}\left[\frac{(\mathsf{a}_t + x)_n}{(\mathsf{b}_t - x)_n}\right]^7\Bigg|_{x=0} = 14 \left[\frac{(\mathsf{a}_t)_n}{(\mathsf{b}_t)_n}\right]^7 H_{2n}(2\mathsf{a}_t),$$

by the duplication $H_{2n}(2a) = [H_n(a) + H_n(a+\frac{1}{2})]/2$. The logarithmic derivative of one density is $\varphi(\xi) - \mathbb{E}\,\varphi(\xi)$ with $\varphi(\xi) = \log\xi - 2\log(1-\xi)$; its mean absolute value is at most $2\,\mathbb{E}|\varphi|$. With the factor $1/14$, the seven copies give a signed measure whose total variation is at most $\mathbb{E}|\varphi| \le \psi(\mathsf{b}_t) - \psi(\mathsf{a}_t) + 2[\psi(\mathsf{b}_t) - \psi(\frac{1}{2})]$. The polygamma series and an integral comparison give $\psi_1(a) \le a^{-2} + a^{-1}$ [32]. Since $\mathsf{a}_t \ge \frac{3}{8}$, $\mathsf{b}_t \le \frac{13}{8}$ and $\psi_1(\frac{1}{2}) = \pi^2/2 < 5$,

$$\psi(\mathsf{b}_t) - \psi(\mathsf{a}_t) \le \frac{44}{9}, \qquad 2[\psi(\mathsf{b}_t) - \psi(\tfrac{1}{2})] \le \frac{45}{4}, \qquad \frac{44}{9} + \frac{45}{4} < 17.$$

Integrating r_N against this signed measure proves

$$|\mathsf{L}(t) - \widehat{E}_N[\mathsf{L}(t)]| \le 17(t+2) g_t^7\, 2^{-N}. \tag{960}$$

The beta parameters remain positive near $x = 0$ and logarithmic beta moments are integrable, which justifies the differentiation and the finite Euler operation. The derivative serves only the error bound; the evaluator contains none.

Ordinary positive series. The ten positive unit-argument series have rational term ratios $R(n) = t_{n+1}/t_n = \prod_i (n+a_i)/\prod_i (n+b_i)$, $t_0 = 1$, with all shifts positive, equal numerator and denominator degrees, and $\alpha = \sum_i b_i - \sum_i a_i > 1$; gamma-product asymptotics give $t_n = O(n^{-\alpha})$ [27, 22]. For a finite $M \geq 1$ choose $q_M(n) = \sum_{j=0}^{M} c_j n^{1-j}$ with $q_M(n) - R(n) q_M(n+1) = 1 + O(n^{-M-1})$. This is a finite triangular rational calculation: the leading multiplier for c_j is $\alpha + j - 1 > 0$. To turn it into a bound, put $y = 1/n$, $\mathcal{A}_R(y) = \prod_i (1 + a_i y)$, $\mathcal{B}_R(y) = \prod_i (1 + b_i y)$ and $\mathcal{C}_R(y) = \sum_{j=0}^{M} c_j y^j$. The exact polynomial

$$\mathcal{E}(y) = y(1+y)^{M-1}\mathcal{B}_R(y) - (1+y)^{M-1}\mathcal{B}_R(y)\mathcal{C}_R(y) + \mathcal{A}_R(y)\sum_{j=0}^{M} c_j y^j (1+y)^{M-j} \tag{961}$$

has zero coefficients through degree $M+1$, and the exact rational residual is

$$\mathsf{r}(n) = 1 - q_M(n) + R(n) q_M(n+1) = \frac{\mathcal{E}(y)}{y(1+y)^{M-1}\mathcal{B}_R(y)}.$$

Since the denominator factors other than y are at least 1 for $y \geq 0$, for $n \geq N$

$$|\mathsf{r}(n)| \leq \varepsilon_{N,M} = \sum_{k \geq M+2} |[y^k]\mathcal{E}(y)|\, N^{1-k},$$

a finite rational number, not an omitted asymptotic term. The identity $t_n R(n) = t_{n+1}$ telescopes, and $t_n q_M(n) \to 0$ because $\alpha > 1$. For $T_N = \sum_{n \geq N} t_n$ this gives $|T_N - t_N q_M(N)| \leq \varepsilon_{N,M} T_N$, so that, whenever $\varepsilon_{N,M} < 1$ and $q_M(N) > 0$ (both checked exactly),

$$\frac{t_N q_M(N)}{1 + \varepsilon_{N,M}} \leq T_N \leq \frac{t_N q_M(N)}{1 - \varepsilon_{N,M}}. \tag{962}$$

Adding the finite rational partial sum gives an exact rational interval for each series. The parameter ratios, including the factorial denominator, are

Series	Numerator shifts a_i	Denominator shifts b_i	α
${}_4F_3$ factor of B_-	$\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}$	$\frac{3}{4}, \frac{3}{4}, 1, 1$	2
${}_4F_3$ factor of B_+	$\frac{3}{4}, \frac{3}{4}, \frac{1}{2}, \frac{1}{2}$	$\frac{5}{4}, \frac{5}{4}, 1, 1$	2
${}_5F_4$ value	$\frac{1}{4}, (\frac{1}{2})^{\times 4}$	$\frac{5}{4}, 1^{\times 4}$	3
${}_6F_5$ value	$1, 1, (\frac{5}{4})^{\times 4}$	$2, (\frac{7}{4})^{\times 4}, 1$	3
Φ_-	$(\frac{1}{4})^{\times 6}$	$(\frac{3}{4})^{\times 6}$	3
Φ_0	$(\frac{1}{2})^{\times 6}$	$1^{\times 6}$	3
Φ_+	$(\frac{3}{4})^{\times 6}$	$(\frac{5}{4})^{\times 6}$	3
Y_r, $r = 1, 2, 7$, normalized	$t+1, (t+\frac{1}{4})^{\times 8}$	$t, (t+\frac{3}{4})^{\times 8}$	3

Here $t = r/8$, the Y_r are normalized by their $n = 0$ term, which is restored afterwards, $B_- = \Gamma(\frac{1}{4})^4\, {}_4F_3(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}; \frac{3}{4}, \frac{3}{4}, 1; 1)$, $B_+ = 64\pi^4 \Gamma(\frac{1}{4})^{-4}\, {}_4F_3(\frac{3}{4}, \frac{3}{4}, \frac{1}{2}, \frac{1}{2}; \frac{5}{4}, \frac{5}{4}, 1; 1)$, and the ${}_5F_4$ and ${}_6F_5$ values are the two ordinary values of E_{central} named above. Gamma prefactors and the linear combinations built from these series are evaluated with outward-rounded interval operations.

Elementary constants, polylogarithms and the wheel. All certificate arithmetic uses dyadic endpoints, integers divided by 2^P. Integer floors and ceilings enclose every multiplication and division, square roots use integer square roots, and additions and subtractions of endpoint integers are exact; no binary floating-point special-value call enters the construction. The constant π is enclosed by Machin's arctangent identity with alternating rational tails. Logarithms use range reduction and a finite artanh series with a geometric tail; exponentials use range reduction and a factorial Taylor remainder. For a positive gamma argument a, the argument is shifted to $a + K$, the finite Stirling expansion of $\log\Gamma$ is applied, and the logarithm of the exact product

$a(a+1)\cdots(a+K-1)$ is subtracted; the real-positive remainder has the sign of, and at most the magnitude of, its first omitted Bernoulli term [27]. Exponentiating the resulting interval encloses Γ. This uses a finite remainder theorem, not convergence of Stirling's formal series. Catalan's constant, $\zeta(3)$ and $\zeta(5)$ are enclosed by Euler-transformed alternating Hausdorff-moment sums with nonnegative tails at most 2^{-N}; the zeta values include the exact Dirichlet-eta prefactor, and $\zeta(2) = \pi^2/6$. The trigonometric eighth-angle constants are algebraic expressions in $\sqrt{2}$. The half-argument series H_3 and H_5 are positive, with tails bounded by twice their first omitted term. For $\Lambda_{3,2}$, the substitution $n = m + r$ in the strict double sum and $\frac{1+\mathrm{i}}{2}(1-\mathrm{i}) = 1$ give

$$\Lambda_{3,2} = \operatorname{Re}\sum_{r\ge1} z_\Lambda^r \Lambda_r, \qquad z_\Lambda = \frac{1+\mathrm{i}}{2},$$

$$\Lambda_r = \frac{\zeta(3) - H_r^{(3)}}{r^2} + \frac{3\zeta(2) - 2H_r^{(2)}}{r^3} - \frac{3H_r}{r^4} = \sum_{m\ge1} \frac{1}{m^2(m+r)^3}.$$

Since $\Lambda_r \le \zeta(2)/(r+1)^3$, $|z_\Lambda| < \frac{3}{4}$ and $\zeta(2)/(1-|z_\Lambda|) < 8$, the tail after N terms is less than $8(\frac{3}{4})^{N+1}/(N+2)^3$; the real and imaginary parts of the powers of z_Λ are generated by rational interval recurrences. The wheel W_c is recomputed with the certificate of Section B.8.2 at $N = 256$, $M = 40$ (orders 40 and 41 for the fourth- and sixth-power corrections), using the new π interval; earlier wheel endpoints are read only for an overlap check. The primitive uses (968).

Assembly. Each run evaluates the complete polynomial and all six further weighted contributions of (958) before division by 2187; the source repair is included once, in the polynomial. The three recorded configurations are

Configuration	Arithmetic bits	Ordinary N/M	Primitive N	Full interval width
smoke	640	128/40	140	$< 4.673 \times 10^{-67}$
release	1024	256/80	240	$< 1.986 \times 10^{-136}$
refined	1280	320/96	280	$< 1.183 \times 10^{-145}$

The three intervals are nested, as checked with their exact rational endpoints. All runs use the same wheel cutoff, so its finite tail bound dominates the last run. In the refined run the weighted widths are bounded by 1.1826×10^{-145} (wheel), 6.1870×10^{-166} (polynomial), 2.1146×10^{-205} (primitive), 2.7549×10^{-244} ($\Lambda_{3,2}$), 3.7886×10^{-280} (H_3), 7.1367×10^{-284} (H_5) and 4.9568×10^{-323} (L_7). These are outward-rounded upper bounds, not differences between floating-point approximations. The direct enclosure of M_7 by (956) initializes π by Machin's formula, $\log 2$ by a rational artanh series and $\zeta(3)$ by an Euler-transformed eta series, never starts from an earlier value of M_7, and gives nested intervals of width below 1.245×10^{-60}, 6.110×10^{-151} and 3.000×10^{-241} at $N = 300$, 600 and 900.

B.8.6 Recorded validation of the retained computation

The execution records of the package contain real-function checks, interval assemblies, residual enclosures, exact kernel transformations and separate Barnes comparisons. These records were produced for the consolidation of the package's sources. They are reruns of the supplied implementations, not independent rederivations of every analytic input. The final-method profile of the package (Section 33) selects the tasks needed for the method described here.

The real check used 105 working decimal digits for its 90-digit request. The formula (2) was then freshly assembled from the rerun values and the proved conversions. Its absolute discrepancy from the refined interval midpoint was about 4.2620×10^{-93}. The baseline assembly from complex polylogarithms and series differed by about 3.9697×10^{-94}. Using the independently rerun 40-digit

seventh Barnes block gave a discrepancy in the full coefficient of about 4.1439×10^{-37}. These are floating-point assembly checks, not new interval bounds for the bivariate Barnes integrals.

The refined full interval constructor of Section B.8.5 was also rerun. It uses 1280-bit outward interval arithmetic and returns a width below 1.183×10^{-145} for the retained analytic expression in the form (958). The subsequent assembly in the real basis uses those intervals through exact identities but is not itself advertised as a new nine-function interval implementation.

Quantity	Rerun numerical value (shortened)
L_3	$-2.033576506072054600912068969700518 2\ldots$
C_{212}	$0.032383901800869146760958099191565833\ldots$
W_0	$78.570312136547117160682748886719852 44\ldots$
M_7	$-15332.554194774575276978946534093031 57\ldots$
P_a	$334926.045248051837883222001240077786 61\ldots$
$\mathcal{H}_W$	$82.198372839948704474444779671977025\ldots$
$\mathcal{H}_7$	$-12.961075412927996857090472232198888\ldots$
$\mathcal{H}_*$	$135.023486011160924990848819782168 82\ldots$

The $\mathcal{H}$ entries are obtained through the proved beta, Barnes and series equivalents; they are not three independent fully two-dimensional numerical integrations. Their exact normalization maps are $W_0 = \sqrt{89}\,\mathcal{H}_W/\pi^2$, (933) and (947).

The three residual contributions, in C_6 rather than r_6 units, are

$$\begin{aligned} -\frac{393\pi^3 W_0}{2187} &= -437.77591446116892363503333212700 9\ldots, \\ \frac{24\pi^3 M_7}{2187} &= -5217.06905436347983930506748528098 7\ldots, \\ \frac{96 P_a}{2187} &= 14701.82914669089000310439511616253 7\ldots. \end{aligned} \tag{963}$$

Their sum is $9046.98417786624124016429429875454 1\ldots$. They are three normalization-consistent contributions, not nine additional unknown constants.

B.9 Primitive recurrence data

This subsection supplies the matrices, seeds, evaluation weights and the a posteriori bound used in (957). They are the computational input recorded in the supplied recurrence and interval implementation. The global constant A still means $\pi + 4\log 2$; the matrix A_0 below is a different object.

$$A_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 1 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{964}$$

$$C_0 = \begin{pmatrix} -1 & -4 & 0 & -4 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{4} & 2 & -2 & 0 & -4 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 5 & 0 & 0 & -4 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & 0 & 1 & -4 & 0 & -2 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{1}{4} & 4 & -2 & 0 & -2 & 0 \\ 0 & 0 & \frac{1}{4} & 0 & \frac{1}{2} & 7 & 0 & 0 & -2 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 3 & -4 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{4} & 6 & -2 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 9 \end{pmatrix}. \tag{965}$$

There are two branches $\lambda = 0, 1/2$. At each order n, the nine-component vectors $z_{n,j}^{(\lambda)}$ have logarithmic channels $j = 0, 1, 2$; set $z_{n,3}^{(\lambda)} = 0$. For $n \geq 2$, compute the channels in descending order using (957). The matrix on its left is upper triangular, so each solve is an exact back substitution.

The initial vectors are specified by finite polynomials. Let L be a formal variable, $x = L - A$, and define the three-vectors

$$\begin{aligned} \mathbf{e} &= (1, 0, 0), \\ \mathbf{a} &= (15x^2 - 12\pi x - \pi^2,\ 15x - 6\pi,\ 15), \\ \mathbf{b} &= \left(\frac{15(x^2 + 6x) - 12\pi(x + 3) - \pi^2}{8}, \right. \\ &\qquad \left. \frac{15(x^2 + 8x + 6) - 12\pi(x + 4) - \pi^2}{16}, \frac{15(x + 7) - 6\pi}{8} \right), \\ \mathbf{p} &= (\pi + 6x, 3, 0), \qquad \mathbf{q} = \left(\frac{\pi + 6x + 18}{8}, \frac{\pi + 6x + 24}{16}, \frac{3}{8} \right). \end{aligned}$$

Here $\otimes$ is the Kronecker product with the first vector's index outermost. Form the nine-component polynomials

$$\begin{aligned} U_0 &= Q\, \mathbf{a} \otimes \mathbf{e}, \\ U_1 &= Q[\mathbf{b} \otimes \mathbf{e} + \mathbf{a} \otimes (3/4, 1/8, 0)] - \frac{16\pi^4}{Q} \mathbf{e} \otimes (1, 1/2, 1/4), \\ V_0 &= 8\pi^2 \mathbf{p} \otimes (1, 1/4, 0), \\ V_1 &= 8\pi^2 [\mathbf{q} \otimes (1, 1/4, 0) + \mathbf{p} \otimes (1, 1/2, 1/16)]. \end{aligned}$$

With $[L^j]$ denoting coefficient extraction from these finite polynomials, the seeds are

$$\begin{aligned} z_{0,j}^{(0)} &= [L^j] U_0, & z_{1,j}^{(0)} &= [L^j](4U_1 - 3U_0 - 2\partial_L U_0), \\ z_{0,j}^{(1/2)} &= 2[L^j] V_0, & z_{1,j}^{(1/2)} &= [L^j](8V_1 - 8V_0 - 4\partial_L V_0). \end{aligned}$$

The half-integer branch has no degree-two logarithmic channel. The displayed derivatives are derivatives of explicit finite polynomials used to initialize the algorithm, not special-function parameter derivatives.

Put $\rho = 3 - 2\sqrt{2}$, and write $u_{n,j}$ and $v_{n,j}$ for the first components of $z_{n,j}^{(0)}$ and $z_{n,j}^{(1/2)}$. Define

$$J_j(s) = \rho^s \sum_{r=0}^{j} \binom{j}{r} \frac{(-1)^r r! \log^{j-r} \rho}{s^{r+1}}. \tag{966}$$

These are the elementary moments $\int_0^\rho x^{s-1} \log^j x\, dx$. The finite primitive evaluator, retaining $n < N$, is

$$P_{a,N} = 4 \sum_{n=0}^{N-1} \left[\sum_{j=0}^{2} u_{n,j} J_j(n + 1/2) + \sum_{j=0}^{1} v_{n,j} J_j(n + 1) \right]. \tag{967}$$

The recurrence represents P_a as the convergent limit of these approximants. No numerical primitive value is used as a seed.

For completeness, the error bound can be constructed from the final two states. The matrix norms are $\|A_0\|_\infty = 4$ and $\|C_0\|_\infty = 10$. On all logarithmic channels, the operator $(Dz)_j = (j+1)z_{j+1}$ has norm 2. Rearranging the recurrence and applying a Neumann inverse bound gives

$$\|z_n^{(\lambda)}\|_\infty \le \frac{10\|z_{n-1}^{(\lambda)}\|_\infty + (n+\lambda+5)\|z_{n-2}^{(\lambda)}\|_\infty}{n+\lambda-6}.$$

Let $U_n^{(\lambda)} = \rho^n \|z_n^{(\lambda)}\|_\infty$. Since $\rho < 1/5$, the two scalar coefficients are bounded by $\mathfrak{a}_n = 2/(n-6)$ and $\mathfrak{b}_n = (n+5)/(25(n-6))$. For $n \ge 48$,

$$\frac{\mathfrak{a}_n}{1/4} + \frac{\mathfrak{b}_n}{(1/4)^2} = \frac{16n+280}{25(n-6)} \le 1.$$

With $K_\lambda = \max\{U_{N-1}^{(\lambda)}, U_{N-2}^{(\lambda)}/4\}$, induction gives $\sum_{n\ge N} U_n^{(\lambda)} \le K_\lambda/3$. Also $|\log\rho| < 2$ and $j \le 2$ imply, for $s \ge N \ge 48$,

$$\sum_{j=0}^{2} |J_j(s)|/\rho^s \le 7/N + 5/N^2 + 2/N^3 < 8/N.$$

The resulting bound is

$$|P_a - P_{a,N}| \le \frac{32}{3N}(\sqrt{\rho}\,K_0 + \rho K_{1/2}), \qquad N \ge 48. \tag{968}$$

The code uses outward upper norms of the computed states in this inequality and encloses every finite operation in (967). Thus both the arithmetic uncertainty and the remaining infinite tail are included.

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Part II

Expansions about the short-range end of the long-range Ising model in one, two and three dimensions

The dual of the chain through eighth order, the crossover coefficients c_2 ($d = 2$) and c_1 ($d = 3$), their checks and their status

LRI project note

17 October 2026

Abstract

We collect the expansions of the thermal scaling dimension $\Delta_{\mathcal{E}}$ of the critical long-range Ising model about the point where it joins the short-range Ising universality class, in $d = 1, 2, 3$. In $d = 1$ the strong–weak dual of the chain of Benedetti, Lauria, Mazáč and van Vliet gives $\Delta_{\mathcal{E}} = 1 + \sum_n C_n^{\mathrm{SR}} \Delta_\sigma^{n/2}$; its published coefficients are $C_1^{\mathrm{SR}} = -2$ and $C_2^{\mathrm{SR}} = \frac{1}{2}$. In the matched two-coupling prescription defined in Sec. 2.1, the archived integral expressions for C_3^{SR} to C_7^{SR} reduce exactly to rational combinations of zeta values, C_5^{SR} to C_7^{SR} by exact word-algebra certificates. The step from the kink integrals to those expressions cannot be rerun, because 21 stage programs of the archive are missing; it is backed by an independent re-derivation of the inputs by Mellin–Hepp sector integration, which reproduces C_3^{SR} to C_7^{SR} within statistical errors of 3×10^{-14} to 3×10^{-5} (largest pull 0.9 over the 25 inputs of C_5^{SR} to C_7^{SR}). $C_8^{\mathrm{SR}} = 247.381204805\ldots$ is a numerical record printed to about 85 digits, of which a second evaluation of one input cross-checks about 57 decimal places; its rational part, 88, is exact, and a weight-seven zeta-value formula matches all printed digits but is not proved. The independent re-derivation gives $C_8^{\mathrm{SR}} = 242.8 \pm 2.4$ (one standard error), 1.9 to 4.0 standard errors below the record depending on the error estimator; the difference sits in one input integral. A third calculation, in a scheme defined by the magnetization at the integrable point $s = 1$, where the Bethe-ansatz solution gives the whole $s = 1$ part, shares no input with the other two and gives $C_8^{\mathrm{SR}} = 247.38118 \pm 0.13$, an empirical numerical uncertainty. It agrees with the record to 2.5×10^{-5}, reproduces C_3^{SR} to C_7^{SR} to 7×10^{-26}, and so indicates that the low independent value is the one in error; the re-derivation's own input comparison locates the difference in that integral. In $d = 2$, $c_1 = 0$, and $c_2 = -4.0662701481325\ldots$ is a closed form in Gamma values and three ${}_4F_3(1)$ constants whose identification with the physical coefficient rests on one unproved uniform-remainder step. A Monte Carlo integration of the regulated integral gives -4.075 ± 0.019 (one standard error), and fourth-order perturbation theory with the truncated cylinder Hamiltonian gives $-4.0670(24)$, where the error is a linear sum of systematic diagnostics; its relative difference from the closed form is 1.9×10^{-4}. In $d = 3$, $c_1 = 0.2732$, with the estimated range $[0.2690, 0.2760]$, conditional on assumed gaps in the omitted spectrum and on assumed ranges of the input data. Every result carries its status word and caveats. The appendices give in full the derivations, proofs, methods and checks of the supplied packages behind these results (the independent re-derivation, the Monte Carlo and the cylinder continuum analysis are described in the main text), including the formal arguments that two couplings close under renormalization of the dual and that its prescription extracts the local beta functions. Appendix H maps each number to the package and command that reproduce it, and a last appendix gives the magnetization-scheme calculation of C_8^{SR} that the paper quotes.

1 Introduction and conventions

1.1 The short-range end

The critical long-range Ising (LRI) model in d dimensions has couplings $J(r) \propto r^{-(d+s)}$. Its spin field has the exact dimension $\Delta_\sigma = (d-s)/2$ throughout the long-range regime, and its fixed point joins the short-range Ising (SRI) universality class at

$$s_* = d - 2\Delta_\sigma^{\mathrm{SR}}. \tag{1.1}$$

Near s_* the natural small parameter is

$$\delta = \frac{s_* - s}{2} = \Delta_\sigma - \Delta_\sigma^{\mathrm{SR}}. \tag{1.2}$$

This note collects what is known about the thermal dimension $\Delta_\mathcal{E}$ near that end in $d = 1, 2, 3$. These are the results quoted in Sec. 3.2, "Near the short-range end", of the accompanying paper [1] (its table of short-range coefficients, the closed forms of the dual of the chain, the c_2 row of its table of perturbative Hamiltonian truncation, and the closed-form appendix for c_2), with the derivations, checks and status of each. The paper's resummation of the expansion about the generalized-free-field end, $s = d/2$, also imposes these values at the other end of the long-range window, $s = s_*$.

Two different expansions apply. In $d = 1$ the short-range end is $s_* = 1$, where $\Delta_\sigma^{\mathrm{SR}} = 0$ (for $s > 1$ the chain has no finite-temperature transition); the expansion is that of a strong–weak dual and proceeds in $\sqrt{\Delta_\sigma}$ (Sec. 2). In $d = 2, 3$ the fixed point near s_* is the SRI conformal field theory (CFT) coupled to a generalized free field (GFF), and the expansion proceeds in integer powers of δ (Secs. 3 and 4). Table 1 lists the conventions.

d	s_*	$\Delta_\sigma^{\mathrm{SR}}$, $\Delta_\mathcal{E}^{\mathrm{SR}}$	variable	expansion
1	1	0, 1	$x = \Delta_\sigma = \frac{1-s}{2}$	$\Delta_\mathcal{E} = 1 + \sum_{n\geq 1} C_n^{\mathrm{SR}}\, x^{n/2}$
2	$\frac{7}{4}$	$\frac{1}{8}$, 1	$\delta = \frac{7/4-s}{2} = \Delta_\sigma - \frac{1}{8}$	$\Delta_\mathcal{E} = 1 + c_1\delta + c_2\delta^2 + O(\delta^3)$
3	1.963702388(48)	0.518148806(24), 1.41262528(29)	$\delta = \frac{s_*-s}{2}$	$\Delta_\mathcal{E} = \Delta_\mathcal{E}^{\mathrm{SR}} + c_1\delta + O(\delta^2)$

Table 1: Conventions. The $d = 3$ data are those of Ref. [15]. In $d = 1$ the variable δ of Eq. (1.2) is x; the dual of Sec. 2 uses $\bar\delta = 1 - s = 2x$, which its computational sources write δ (as in the input name N_δ). A coefficient of $(s_* - s)^n$ is $c_n/2^n$: the coefficient of $(7/4 - s)^2$ in $d = 2$ is $c_2/4$, and that of $s_* - s$ in $d = 3$ is $c_1/2$. The paper writes $C_n^{\mathrm{SR}}(d)$ for all three: $C_n^{\mathrm{SR}}(1) = C_n^{\mathrm{SR}}$ multiplies $\Delta_\sigma^{n/2}$, while $C_n^{\mathrm{SR}}(2) = c_n$ and $C_n^{\mathrm{SR}}(3) = c_n$ multiply δ^n (so $c_2 = C_2^{\mathrm{SR}}(2)$ and $c_1 = C_1^{\mathrm{SR}}(3)$).

The symbol b_0 follows the sources and has two unrelated meanings: in $d = 1$ it is the charge of the compact field of the dual (Sec. 2.1); in $d = 2$ and $d = 3$ it is the one-loop coefficient of the beta function of the crossover coupling (Secs. 3.2 and 4). A few other letters are reused, always with a local definition: h is the current coupling of the dual in Sec. 2 and the g^4 coefficient of $\gamma_\mathcal{E}$ or of the cylinder observable in Sec. 3 (h_0, $h(\lambda)$); W is an input of C_8^{SR}, the Gaussian source factor of Sec. 3 and, as W_k, the cylinder counterterms; κ is the reference inverse length of the dual (Appendix A), the finite redefinition parameter of Sec. 3.2 and, as κ_r, a counterterm coefficient of Appendix F; and χ is the generalized free field in $d = 2, 3$, while the bare current of the dual is χ_B and its renormalized current primary (Appendices A and C) is written χ there.

1.2 Sources and who produced them

The note rests on seven packages of the release. Four were written by another AI system, OpenAI GPT-6 astra (via ChatGPT), and supplied by the author: a research compendium for the $d = 1$ dual (its derivations, code, exact expressions, certificates and numerical records of C_3^{SR} to C_8^{SR}; "the compendium"; its original archive, of which the release ships a pruned copy, is "the archive"), a closed-form study of c_2 in $d = 2$ ("the study", also called the closed-form or final-method study), a base study of c_2 with the truncated cylinder Hamiltonian, and a package for c_1 in $d = 3$ ("the c_1 package"). Claude Code (Anthropic) sessions reviewed each of them ("the review" of each), re-derived every input of C_5^{SR} to C_8^{SR} independently, ran the Monte Carlo of c_2, and wrote the cluster layer and the continuum analysis of the cylinder calculation, and computed C_8^{SR} in the magnetization scheme (Sec. 2.7); OpenAI Codex reviewed the plans and the code of those two calculations. Two formal arguments about the dual, that two couplings close under its renormalization and that its prescription extracts the local beta functions, are also by GPT-6 astra.

The main text states each result with its basis and caveats; the appendices give the derivations behind them, edited to the notation of this note: the compendium in Appendices A (the model, the regulator and the finite-response renormalization), B (the third to sixth orders) and C (the exact certificates and the seventh and eighth orders); the two arguments about the dual in Appendix D; the study in Appendix E; the method of the cylinder calculation in Appendix F; and the c_1 package in Appendix G. Section 5 gives the authorship, and Appendix H the folder of each package. A third calculation of C_8^{SR}, in a scheme defined by the magnetization at the integrable point $s = 1$, was made for the project after the other packages: Sec. 2.7 summarizes it, Appendix I gives it in full, and its code and records are the package `integ_d1`.

1.3 Status words

Each quoted result carries one of the following status words.

literature a published value, cited where it is used and quoted as published; where this note re-derives it as a check, Table 2 says so;

proved closed form within the prescription ($d = 1$) an exact expression into which the archived integral expressions of the matched two-coupling prescription (Sec. 2.1) reduce exactly, with exact certificates where the reduction is computer-assisted; it covers neither the step from the kink integrals to those expressions nor the prescription itself;

closed form conditional on one unproved step (c_2) an exact expression whose evaluation is proved, but whose identification with the physical coefficient needs a step that is not proved;

numerical a number obtained by numerical integration, extrapolation or sampling, with the kind of its error stated: a statistical standard error, a linear sum of systematic diagnostics that is not a standard deviation, or, for a record without an error estimate, the precision to which it is cross-checked;

conjectured an exact expression supported numerically but without a proof;

conditional a statement or range that holds only under explicitly stated, unproved assumptions.

The file `results/sr_values.json` uses the short forms literature, closed form, numerical, conjectured and conditional, with the qualification in a note or caveat field; it lists c_1 in $d = 3$ as a numerical retained value and an estimated range with the status conditional. Errors are always labelled with their kind. A pull is (our value − claimed value)/(quoted standard error).

1.4 Results and their status

Table 2 gives every result of this note that the paper quotes, and three that it does not print (the largest pull of the recomputed inputs, the independent value of C_8^{SR} and the agreement of the two independent codes for c_1), with its status word, its basis and its caveats. The summaries and checks follow in Secs. 2 to 4, the derivations in Appendices A to G and I, and the authorship in Sec. 5.

Table 2: The results near the short-range end, with the status words of Sec. 1.3. In $d = 1$ "the prescription" is the matched two-coupling prescription of Sec. 2.1. The paper prints six significant digits; the longer strings are evaluations of the closed forms and of the C_8^{SR} record, rounded to 15 significant digits (all digits are in `results/sr_values.json`).

result	value	status	basis and caveats
$d = 1$			
C_1^{SR}, C_2^{SR}	-2, $\frac{1}{2}$	literature	Refs. [3, 4]; re-derived from the leading flow (Sec. 2.2).
C_3^{SR} to C_7^{SR}	Eq. (2.12):	proved closed form within the prescription	The archived integral expressions of the beta inputs reduce exactly to zeta values: C_3^{SR}, C_4^{SR} by analytic period evaluations (with four archived mixed response coefficients), C_5^{SR} to C_7^{SR} by exact word-algebra certificates (Sec. 2.4). The step from the kink integrals to these expressions cannot be rerun (Appendix H.5); the independent re-derivation reproduces C_3^{SR} to C_7^{SR} numerically (Sec. 2.6). Prescription caveat: Secs. 2.1 and 2.9, Appendix D.
C_3^{SR}	$5.72736813369645\ldots$		
C_4^{SR}	$-12.1961914926697\ldots$		
C_5^{SR}	$29.2378115809900\ldots$		
C_6^{SR}	$-58.6656974742253\ldots$		
C_7^{SR}	$51.0689886129281\ldots$		
$\pi_0 C_8^{\mathrm{SR}}$ (rational part)	88	proved closed form within the prescription	Exact rational projection, from two separate exact calculations (Sec. 2.5); the same two qualifications as C_3^{SR} to C_7^{SR}.
C_8^{SR}	$247.381204804994\ldots$	numerical	Record printed to about 85 digits (printing floor 2×10^{-85}); a second evaluation of its input P_6 cross-checks about 57 decimal places; six inputs are single numerical records; no error bound (Sec. 2.5).
Eq. (2.15)	weight-seven formula for C_8^{SR}	conjectured	Matches the record to 3.1×10^{-86}; no certificate; the record is its only high-precision support (the magnetization-scheme value agrees to 2.5×10^{-5}).
inputs of C_5^{SR} to C_7^{SR}, independent	largest pull 0.9	numerical	One standard error (the largest of three statistical estimators); 25 inputs, largest pull 0.915, rms 0.34; the coefficients within 0.78 (C_6^{SR}); the paper says only that the recomputed inputs match.

result	value	status	basis and caveats
C_8^{SR}, independent	242.8 ± 2.4	numerical	One standard error (largest of three estimators); 1.9 to 4.0 standard errors below the record, depending on the estimator; the difference sits in one input integral, $A_9[\varepsilon^0 \ell^1]$ (Sec. 2.6); the magnetization-scheme value (next row) indicates that this value is the one in error; not printed by the paper.
C_8^{SR}, magnetization scheme	247.38118 ± 0.13	numerical	Empirical numerical uncertainty, a linear sum of the differences between three runs and a validated roundoff estimate, not a standard deviation or a bound; no input shared with the two hard-core calculations, no nine-kink integral; reproduces C_3^{SR} to C_7^{SR} to 7×10^{-26}; 2.5×10^{-5} from the record; two check criteria amended after their data were seen; the paper quotes it as 247.38 ± 0.13 (its Sec. 3.2); package `integ_d1` (Sec. 2.7, Appendix I).
$d = 2$			
c_1	0	literature	BRRZ [12]; re-derived ($I_2(0) = 0$, Sec. 3).
c_2	$-4.06627014813252\ldots$	closed form conditional on one unproved step	The evaluation of $\mathcal{I}$ is proved; the identification with the physical coefficient needs the uniform-remainder step (Sec. 3.6); a formal result in critically tuned analytic minimal subtraction.
c_2, Monte Carlo	-4.075 ± 0.019	numerical	One standard error (statistical); 0.5 standard errors from the closed form (Sec. 3.7).
c_2, cylinder	$-4.0670(24)$	numerical	Linear sum of systematic diagnostics (64% regulator model), not a standard deviation; 1.9×10^{-4} relative from the closed form, which was known but is not an input; error rule revised after the first result; 11 of the 25 points are archived records of the base study (Sec. 3.8).
$d = 3$			
c_1	0.2732; estimated range [0.2690, 0.2760]	conditional	Retained 60-primary value $0.27315242\ldots$ (Sec. 4.2). The range holds if no omitted primary lies below the gaps 8 ($\mathbb{Z}_2$-even) and 7 ($\mathbb{Z}_2$-odd), the inputs lie within twice their quoted errors, a published constraint on $\lambda_{\epsilon\epsilon\epsilon'}$ applies and the chosen margins and allowances suffice; it is neither a bound nor a confidence interval, and mild relaxations extend it (Secs. 4.3 and 4.4).
c_1, two independent codes	agree to 10^{-12}	numerical	Absolute difference in c_1 of the review's two re-implementations from the package (Sec. 4.2); not printed by the paper.
c_1, BRRZ	0.27(4)	literature	Ref. [12].

2 One dimension: the dual of the chain

2.1 The dual, its two couplings and the prescription

Near $s = 1$ the domain walls of the critical chain form a dilute gas of alternating kinks and antikinks with a logarithmic interaction [6, 10]. Benedetti, Lauria, Mazáč and van Vliet recast this gas as a two-state system (a qubit) coupled to a compact Gaussian field $\phi \sim \phi + 2\pi/b_0$ of negative dimension [3, 4]. With $\bar\delta = 1 - s$ and $f_{\bar\delta}(r) = (r^{\bar\delta} - 1)/\bar\delta$, the field has covariance $-2f_{\bar\delta}(|X - Y|)$ on neutral test functions, and the path-ordered evolution

$$\mathrm{P}\exp\int \mathrm{d}X\,[g\,\mathcal{O}_g + h\,\mathcal{O}_h], \qquad \mathcal{O}_g = \hat\sigma_+ {:}e^{\mathrm{i}b_0\phi}{:} + \hat\sigma_- {:}e^{-\mathrm{i}b_0\phi}{:}, \qquad \mathcal{O}_h = \hat\sigma_3\chi_B, \qquad \chi_B = \tfrac{\mathrm{i}}{\sqrt{2}}\partial\phi, \tag{2.1}$$

reproduces the alternating gas because $\hat\sigma_\pm^2 = 0$ (Appendices A.2 and A.3 derive the gas and this representation). The bare current χ_B is a different field from the generalized free field χ of Secs. 3 and 4 (and from the renormalized current primary that Appendix A.6 writes χ); the symbol table of the dual is Table 4. The critical chain is the sector invariant under a $U(1)$ that shifts ϕ and rotates the qubit. The fugacity g and the current coupling h flow; the current can be resummed into the effective charge $b = b_0 - \sqrt{2}h$.

The matched two-coupling prescription. The coefficients C_3^{SR} to C_8^{SR} are computed in one fixed prescription, taken from the compendium and derived in Appendices A.4 to A.9:

(i) *Regulator.* Kinks at ordered positions $0 < X_1 < \cdots < X_n < 1$ on an interval of unit length, with a hard core $X_{i+1} - X_i \geq \varepsilon$ on consecutive kinks only (none between a kink and an endpoint), and $\ell = \log\varepsilon$ (Eq. (A.14)).

(ii) *Exact current resummation.* Before the hard core is imposed, a field-dependent change of frame removes the bulk current interaction. It shifts the wall charge to $b = b_0 - \sqrt{2}h$ and leaves charges $\pm h/\sqrt{2}$ at the two endpoints of the interval, whose contraction with the walls is an *endpoint source* $\eta = \sqrt{2}\,h\,b$ (Appendix A.4).

(iii) *Matching at the cutoff.* Normal ordering at the cutoff defines the charge $B = b\,\varepsilon^{\bar\delta/2}$ and the fugacity g, keeping the $\bar\delta$-dependent normal-ordering factor of the wall interaction exactly (finite-tail matching); the essential couplings are

$$u = g^2, \qquad v = B^2 = 1 + z, \qquad \beta_u = 2uF(u, z, \bar\delta), \qquad \beta_z = (1 + z)H(u, z, \bar\delta), \tag{2.2}$$

in units with $b_0 = 1$, so that $z = 0$ is the marginal point (Appendix A.5).

(iv) *Vacuum normalization.* The ordered moments of the regulated gas (Sec. 2.3) are divided by the vacuum moments, as complete series in ε and ℓ, before any coefficient is extracted (Appendix A.7).

(v) *Two projected responses.* The beta functions of u and z are solved, order by order, from two responses only: the odd (charged) response and the current response, each projected on its coefficient $[\varepsilon^0\ell^1]$, through a finite-response Jacobian that carries their finite parts $[\varepsilon^0\ell^0]$ (Appendix A.9).

"Matched" refers to (iii), and "two-coupling" to the assumption that (u, z) close under renormalization: no further coupling enters their flow at the orders computed. At leading order (Appendix A.6)

$$F = z + u + \cdots, \qquad H = -\bar\delta + 4u + \cdots, \qquad u_* = \frac{\bar\delta}{4} + O(\bar\delta^2), \quad z_* = -\frac{\bar\delta}{4} + O(\bar\delta^2), \tag{2.3}$$

so the fugacity at the fixed point is of order $\sqrt{\bar\delta}$, which is why the series runs in $\sqrt{\Delta_\sigma}$.

What "proved within the prescription" covers. In this prescription the beta inputs are fixed regulated integrals. A *proved closed form within the prescription* means that the archived exact expressions of these integrals reduce exactly to rational combinations of zeta values (Sec. 2.4). Two steps lie outside it. First, the step from the kink integrals to the archived expressions cannot be rerun, because 21 stage programs of the archive are missing (Appendix H.5); it is backed numerically by the independent re-derivation of Sec. 2.6. Second, whether the prescription computes the coefficients of the dual (that the dual describes the chain is the published duality proposal), which rests on the two-coupling closure and on the prescription extracting the local beta functions; two unrefereed arguments support both formally (Appendix D: the closure statement in Appendix D.2, the prescription statement in Appendix D.3). Their status is stated once, in the caveat "Two-coupling closure" of Sec. 2.9, together with the evidence that bears on it.

2.2 Stability invariants and the thermal series

At the fixed point $F = H = 0$, so the trace and determinant of the stability matrix in (u, z) are

$$T = 2u_*F_u + (1 + z_*)H_z, \qquad D = 2u_*(1 + z_*)(F_uH_z - F_zH_u), \tag{2.4}$$

with the derivatives taken before the fixed point is substituted. They are invariant under regular changes of couplings, although individual beta coefficients are not (Appendix A.10). With

$$T(\bar\delta) = \frac{\bar\delta}{2} + \sum_{m\geq 2} T_m\bar\delta^m, \qquad D(\bar\delta) = -2\bar\delta + \sum_{m\geq 2} D_m\bar\delta^m, \tag{2.5}$$

the two dimensions are

$$\Delta_\pm = 1 + \frac{T}{2} \pm \sqrt{\frac{T^2}{4} - D}, \qquad \Delta_\mathcal{E} = \Delta_-, \tag{2.6}$$

assuming, as in Ref. [1], that the relevant operator is not crossed by another level. The square root is $\sqrt{2\bar\delta}$ times a power series in $\bar\delta$, so the odd powers of $\sqrt{\Delta_\sigma}$ come from the eigenvalue splitting and the even powers from $T/2$. At leading order $\Delta_\pm = 1 \pm \sqrt{2\bar\delta} + \bar\delta/4 + O(\bar\delta^{3/2})$ [4], which is

$$C_1^{\mathrm{SR}} = -2, \qquad C_2^{\mathrm{SR}} = \frac{1}{2}. \tag{2.7}$$

Expanding Eq. (2.6) gives the exact conversions

$$\begin{array}{lll} C_3^{\mathrm{SR}} = D_2 - \frac{1}{16}, & C_4^{\mathrm{SR}} = 2T_2, & C_5^{\mathrm{SR}} = 2D_3 - \frac{T_2}{2} + \frac{1}{4}\big(D_2 - \frac{1}{16}\big)^2, \\ C_6^{\mathrm{SR}} = 4T_3, & C_7^{\mathrm{SR}} = 4D_4 - T_3 - T_2^2 + \frac{1}{2}C_3^{\mathrm{SR}}C_5^{\mathrm{SR}}, & C_8^{\mathrm{SR}} = 8T_4, \end{array} \tag{2.8}$$

and in general $C_{2m}^{\mathrm{SR}} = 2^{m-1}T_m$. These are algebraic identities; they do not determine an uncomputed invariant (C_9^{SR} needs D_5).

2.3 Beta inputs and finite-response renormalization

The invariants are polynomials in Taylor coefficients of F and H at the marginal point $v = 1$, $\bar\delta = 0$, which the computation calls beta inputs. They come from ordered moments of the regulated gas, odd moments A_{2m+1}, vacuum moments Z_{2m} and current moments Q_{2m}, with up to nine kinks and their derivatives in the charge v, the tail parameter $\bar\delta$ and the endpoint source η. After the vacuum normalization (iv), the coefficients $[\varepsilon^0\ell^0]$ and $[\varepsilon^0\ell^1]$ are taken, and the finite-response Jacobian (v) turns them into beta coefficients (Appendices A.7 to A.9; the inputs of each order, evaluated or given by their archived exact expressions, are in Appendices B and C). Five inputs (a, p, b, c, f)

fix C_3^{SR} and C_4^{SR}, 25 fix C_7^{SR}, and ten more $(W, S, U, J_9, L_3, M_{21}, M_{12}, N_2, N_\delta, K_v)$ enter C_8^{SR}. For example (the input $b = b_5$ is not the charge b)

$$\begin{aligned} &a = -4\zeta(2), \qquad c = -16, \qquad p = 8 - 2\zeta(2) - 7\zeta(3) - 8\log 2, \\ &b = -18 - 6\zeta(2) + \tfrac{28}{3}\zeta(3) + 32\log 2 - 12\log 3 - 4\log^2 2 - 8\,\mathrm{Li}_2(-\tfrac{1}{2}), \\ &f = 32 - \tfrac{136}{3}\zeta(3) - 64\log 2 + 48\log 3 + 16\log^2 2 + 32\,\mathrm{Li}_2(-\tfrac{1}{2}), \end{aligned} \tag{2.9}$$

and $D_2 = \frac{1}{2} - \frac{a}{2} - \frac{c}{8}$, $T_2 = -\frac{a}{8} + \frac{b}{4} - \frac{c}{32} + \frac{f}{16} + \frac{p}{2}$. The regulator constants $\log 2$, $\log 3$ and $\mathrm{Li}_2(-\frac{1}{2})$ are genuine parts of the individual inputs and cancel from the invariants. The eighth order is affine in the ten new inputs,

$$C_8^{\mathrm{SR}} = B_{\mathrm{old}} + 4W + 2S + \tfrac{3}{4}U + \tfrac{1}{4}J_9 + \tfrac{3}{32}L_3 - \tfrac{1}{4}M_{21} + \tfrac{1}{2}M_{12} - \tfrac{1}{16}N_2 + \tfrac{1}{8}N_\delta + \tfrac{1}{32}K_v, \tag{2.10}$$

where the polynomial B_{old} in the 25 earlier inputs (printed in Appendix C.12) has the value $-861.66356723547977920\ldots$.

2.4 Closed forms through the seventh order

The invariants reduce to

$$\begin{aligned} &D_2 = \tfrac{5}{2} + 2\zeta(2), \qquad T_2 = 2 - 2\zeta(2) - 4\zeta(3), \qquad D_3 = \tfrac{21}{4} - 13\zeta(2) + 11\zeta(3) + 11\zeta(4), \\ &T_3 = \tfrac{13}{2} - \tfrac{19}{2}\zeta(2) + 26\zeta(3) - 20\zeta(4) - 68\zeta(5) + 28\zeta(2)\zeta(3), \\ &D_4 = 13 - \tfrac{179}{8}\zeta(2) + \tfrac{47}{2}\zeta(3) - \tfrac{493}{8}\zeta(4) + 289\zeta(5) - 130\zeta(2)\zeta(3) + \tfrac{499}{4}\zeta(6) - 76\zeta(3)^2, \end{aligned} \tag{2.11}$$

and Eq. (2.8) gives

$$\begin{aligned} C_3^{\mathrm{SR}} &= \frac{39}{16} + 2\zeta(2), \qquad C_4^{\mathrm{SR}} = 4 - 4\zeta(2) - 8\zeta(3), \\ C_5^{\mathrm{SR}} &= \frac{11249}{1024} - \frac{361}{16}\zeta(2) + 24\zeta(3) + \frac{49}{2}\zeta(4), \\ C_6^{\mathrm{SR}} &= 26 - 38\zeta(2) + 104\zeta(3) - 80\zeta(4) - 272\zeta(5) + 112\zeta(2)\zeta(3), \\ C_7^{\mathrm{SR}} &= \frac{1798583}{32768} - \frac{90637}{1024}\zeta(2) + \frac{453}{4}\zeta(3) - \frac{16835}{64}\zeta(4) + 1224\zeta(5) \\ &\quad - 540\zeta(2)\zeta(3) + \frac{4335}{8}\zeta(6) - 320\zeta(3)^2. \end{aligned} \tag{2.12}$$

Their values are in Table 2. Each C_n^{SR} with $n \le 7$ lies in the rational span of multiple zeta values of weight at most $n - 1$.

How the closed forms are established. C_3^{SR} and C_4^{SR} follow from analytic evaluations of the four- and five-insertion periods, closed by a finite-part trilogarithm identity proved with explicit primitives and a real-branch Spence–Kummer identity; the four mixed response coefficients of Eq. (B.102) that attach the periods to the inputs are archived values, not derived here (Appendix B, which also assembles C_5^{SR} and C_6^{SR} from their inputs and evaluates the analytic inputs $A = \frac{1}{2}b_{3,vv}$, $E = b_{3,v\bar{\delta}}$ (Appendix B.1) and the two endpoint constants of C_5^{SR}; the other inputs of these orders are archived exact expressions). From C_5^{SR} on, the exact archived expressions of the beta inputs, rational functions of the cutoff multiplying iterated integrals, are lifted to a free rational vector space of words of regularized hyperlogarithm constants. The difference between the archived expression and the proposed closed form is then written as an explicit rational combination of shuffle products of proved zero relations (shuffle, stuffle, duality, a convergent confluence lemma and a beta-integral identity), and a standalone verifier checks it word by word in exact rational arithmetic, with no numerical period and no assumption of numerical independence (Appendix C.1):

target	weights	nonzero words	certificate rows
D_3 (hence C_5^{SR})	2–4	170	476
C_6^{SR}	2–5	1 945	56 381
D_4, C_7^{SR} (with all product terms)	2–6	25 212	588 878

A separate standard-library checker parses the archived input expressions, lifts them into the word algebra and verifies the trace assembly and the target coefficients, so the certificates are tied to the inputs. The certificates prove the reduction of the archived regulated-integral expressions to zeta values. They do not reintegrate the original kink kernels, and they do not test the prescription itself (Sec. 2.1).

2.5 The eighth order

Rational part. The formal weight-zero projection π_0, which removes positive-weight constant periods after integration, is computed exactly (Appendix C.4; the trace reduction of the eighth order is in Appendix C.3). A decomposition into four raw residues (of the six-kink and eight-kink current moments, the seven-kink tail and the nine-kink fugacity) gives, after the first two are combined with the known terms,

$$\pi_0 C_8^{\mathrm{SR}} = -\frac{111389}{24} - \frac{910891}{8} + \frac{1423087}{12} = 88, \tag{2.13}$$

and a separate literal full-Jacobian calculation gives $\pi_0 T(\bar\delta) = \frac{\bar\delta}{2} + 2\bar\delta^2 + \frac{13}{2}\bar\delta^3 + 11\bar\delta^4 + \cdots$, hence $\pi_0 C_8^{\mathrm{SR}} = 8 \times 11 = 88$. None of the raw integrals was inferred from the answer or from an earlier conjectured value, 160, which this result excludes.

Full value. Four of the ten new inputs were computed in four separate stages: the six-kink mixed logarithm $P_6 = 1261.944897843140491372852392951531767542124984174\ldots$ (an exact period expression with 66 404 terms, evaluated at 70 and 100 working digits), from which $N_\delta = -4P_6 + 11218.914021881803142\ldots$; the seven-kink tail coefficient $U = -1296.687960291526491656\ldots$; the eight-kink current derivative $K_v = -1486.962109278823406128\ldots$; and the ninth-order fugacity $J_9 = -273.724515999324646514\ldots$, a cancellation between pieces of order 6×10^5. The other six $(W, S, L_3, M_{21}, M_{12}, N_2)$ enter as *inherited numerical records*: decimal values from one recorded 100-digit evaluation, with no surviving exact expression and no second evaluation (Appendix C.5 describes all ten). The assembly reduces to $C_8^{\mathrm{SR}} = C_{\mathrm{fixed}} - \frac{1}{2}P_6$ with $C_{\mathrm{fixed}} = 878.353653726563794093781271\ldots$, and gives

$$C_8^{\mathrm{SR}} = 247.381204804993548407355074766099998652159151220 9\ldots. \tag{2.14}$$

How many digits. Four facts bear on the digits of Eq. (2.14); none is an interval bound.

- *Cross-check.* The two evaluations of P_6 give values of C_8^{SR} that differ by 2.2×10^{-57}, the precision of the 70-digit run. This cross-checks about 57 decimal places, and only through P_6.
- *Printing floor.* The record is assembled exactly from 36 numbers (the 35 inputs, with N_δ given as P_6 and its remainder), all decimal strings of 85 to 96 decimals except the exact $c = -16$. Changing each by half a unit in its last printed digit changes C_8^{SR} by about 2×10^{-85} in all, about 87 significant digits, quoted as about 85. This is the precision to which the inputs are written, not a demonstrated accuracy: the six inherited inputs have neither exact expressions nor second evaluations.
- *The formula.* The 100-digit value agrees with the conjectured Eq. (2.15) below to 3.1×10^{-86}. This is consistent with both, but the formula has no high-precision support other than this record, so the agreement is not independent evidence for the record's digits.

- *A third value.* The magnetization-scheme value of Sec. 2.7, 247.38118 ± 0.13, agrees with the record to 2.5×10^{-5}; it bears on the first few digits only.

The weight-seven candidate. An integer-relation search (Appendix C.6) on the 70-digit value, in the basis $\{\zeta(2), \zeta(3), \zeta(4), \zeta(5), \zeta(2)\zeta(3), \zeta(6), \zeta(3)^2, \zeta(7), \zeta(2)\zeta(5), \zeta(3)\zeta(4)\}$ with the rational part 88 subtracted, gives

$$C_8^{\mathrm{SR}} \stackrel{?}{=} 88 - 178\zeta(2) + 110\zeta(3) + 113\zeta(4) + 4892\zeta(5) - 2376\zeta(2)\zeta(3) - \frac{8630}{3}\zeta(6) + 1872\zeta(3)^2 - 12168\zeta(7) + 6048\zeta(2)\zeta(5) + 1536\zeta(3)\zeta(4). \tag{2.15}$$

The 100-digit evaluation differs from Eq. (2.15) by 3.1×10^{-86}, and a search with the rational part left free recovers 88. In the review of the compendium (Sec. 2.8) the relation stayed the only one when 26 further constants were added to the basis. A looser earlier search had produced a different relation that failed at higher precision. Equation (2.15) is therefore conjectured: the residual is strong numerical evidence, not an interval enclosure or an exact certificate. A proof would assemble every input of C_8^{SR} as an exact expression in one regularized-period convention (which requires recovering or regenerating the six inherited inputs), subtract Eq. (2.15) and certify the difference at every weight through seven, products included, as was done for C_7^{SR}.

2.6 Independent re-derivation of the inputs

The compendium's raw integral expressions had been replayed only with its own code. Every input of C_5^{SR} to C_8^{SR} was therefore recomputed from scratch.

- *Definitions.* The 35 inputs were written as Taylor coefficients of the auxiliary beta functions from the compendium's text alone (the definitions now in Appendices A.9 and C.3).
- *Renormalization map.* The response-to-beta map, the vacuum normalization, the fixed point and the invariants were derived again from first principles (symbolically); the map agrees with the compendium's (Eq. (A.47)). All required raw coefficients must be supplied; a missing one is an error, never a zero.
- *Engine.* A new numerical engine computes the ε expansion of the hard-core moments: a Mellin transform in the cutoff, Hepp sectors (total orderings of the gaps) with the divergent variables subtracted analytically, and rank-one lattice quasi-Monte Carlo with six independent random shifts per job.
- *Direct check.* The raw moments were also integrated directly at finite cutoffs (nested Gauss–Legendre, relative accuracy about 10^{-14}) and compared with the engine's series.
- *Independence.* No integration code or raw data of the compendium was used. Its numbers were read by a script only in a comparison step that refused to run before the independent results had been written and hashed. Some compendium numbers were visible earlier in task descriptions and reports; the result records disclose each case, and none entered a computation.

Three error estimators were computed: linear propagation with the replicate covariance, the standard error of six replicate assemblies (one per random shift), and a sector-pooled estimate from the scatter of the shifts within each integration sector, summed over sectors. The protocol, fixed before the comparison, quotes the largest of the three; all three are statistical estimates, not bounds.

Through the seventh order (production about 3.3×10^3 core-hours on the symmetry cluster). All 25 inputs agree with the compendium: the rms pull is 0.34, and the largest, 0.915, is that of

the input the compendium calls Ω (the coefficient of $u^3 z$ in F, Eq. (C.18)). The coefficients agree within 0.78 standard errors:

	independent	closed form	pull
C_5^{SR}	$29.237811563 \pm 1.35 \times 10^{-7}$	$29.2378115809900\ldots$	-0.13
C_6^{SR}	$-58.66569861 \pm 1.45 \times 10^{-6}$	$-58.6656974742253\ldots$	-0.78
D_4	$-2.53463048 \pm 7.85 \times 10^{-6}$	$-2.53462979953825\ldots$	-0.09
C_7^{SR}	$51.0689860 \pm 3.14 \times 10^{-5}$	$51.0689886129281\ldots$	-0.08

The errors are one standard error. Five checks use no compendium number. They are the proved low-order closed forms (17 quantities, among them C_3^{SR} and C_4^{SR} with errors 3.3×10^{-14} and 9.8×10^{-10}; largest pull 0.52), the cancellation of 702 negative powers of ε in the normalized responses (largest pull 2.37), 39 higher-logarithm predictions (largest pull 0.96), a renormalization-group second-logarithm test derived for this work (six responses, largest pull 1.09), and the direct finite-cutoff comparison. In the last, a moment passes when its two leading coefficients agree with the direct fit, that is, no pull reaches 3 and their joint two-degree-of-freedom χ^2 stays below 11.83; 44 of the 47 moments pass. The other three are flagged: two depend on the fit window, and one comes from a single production coefficient that is 4 replicate standard errors low and changes C_7^{SR} by 7×10^{-10}. C_7^{SR} and D_4 are effectively one test, dominated by the eight-kink current moment.

The eighth order (about 1.9×10^4 core-hours, on the symmetry cluster and a personal desktop). All ten new inputs and all 201 raw scalars are the project's own. Table 3 compares them with the compendium.

input	coefficient	compendium	independent	pull
W	4	-37.70811151495	$-37.70811151495 \pm 1.1 \times 10^{-10}$	-0.02
S	2	878.1837337477	$878.1837337543 \pm 3.1 \times 10^{-9}$	$+2.13$
U	$\frac{3}{4}$	-1296.687960	-1296.687551 ± 0.0015	$+0.27$
J_9	$\frac{1}{4}$	-273.72452	-291.94 ± 9.48	-1.92
L_3	$\frac{3}{32}$	-0.6386928106	$-0.6386928106 \pm 9 \times 10^{-11}$	-0.02
M_{21}	$-\frac{1}{4}$	-171.2355765656	$-171.2355765656 \pm 1.2 \times 10^{-9}$	-0.01
M_{12}	$\frac{1}{2}$	-422.7150579699	$-422.7150579699 \pm 6.1 \times 10^{-9}$	0.00
N_2	$-\frac{1}{16}$	189.742012494	$189.742009475 \pm 8.5 \times 10^{-6}$	-0.35
N_δ	$\frac{1}{8}$	6171.134430509	$6171.134427764 \pm 9.1 \times 10^{-6}$	-0.30
K_v	$\frac{1}{32}$	-1486.962109	-1487.004626 ± 0.158	-0.27
$C_8^{\mathrm{SR}} - \frac{1}{4} J_9$		315.812334	315.811301 ± 0.0051	-0.20
C_8^{SR}		247.381204805	242.825 ± 2.371	-1.92

Table 3: The new inputs of C_8^{SR}, with their coefficients in Eq. (2.10): compendium values and the independent re-derivation (one standard error, the largest of three estimators). The $+2.13$ of S is 7.5×10^{-12} relative and moves C_8^{SR} by 1.3×10^{-8}.

The result is

$$C_8^{\mathrm{SR}} = 242.83 \pm 2.37 \qquad \text{(independent; one standard error)}, \tag{2.16}$$

1.92 standard errors below the record with the quoted error (rounded, 242.8 ± 2.4; for the other estimators see the next paragraph). Everything except the J_9 term is tested sharply: $C_8^{\mathrm{SR}} - \frac{1}{4} J_9$ agrees at -0.20σ, which checks B_{old}, the seven inputs of the re-derivation's first tier (the six inherited ones and N_δ), U and K_v together to 2×10^{-5} of C_8^{SR}. The whole difference sits in J_9, and within it in one raw coefficient, $A_9[\varepsilon^0 \ell^1]$ of the nine-kink odd moment: independent minus compendium is -20.4 ± 9.3 (-2.19σ, 3.3×10^{-5} relative), while the other four pieces of J_9 agree.

How large the tension is. The size of the pull depends on the error estimator, because $\sigma(C_8^{\mathrm{SR}})$ comes almost entirely (93%) from $A_9[\varepsilon^0\ell^1]$. With the same central value 242.825:

estimator	$\sigma(C_8^{\mathrm{SR}})$	pull
sector-pooled (quoted, the largest of the three)	2.37	−1.92
per job the larger of the replicate and pooled projections	2.42	−1.89
chunk-pooled (the post-assembly statistics check)		−2.09
linear propagation	1.37	−3.32
replicate spread (five degrees of freedom)	1.15	−3.98

The independent value therefore lies 1.9 to 4.0 standard errors below the record, depending on the estimator. Through the seventh order the same rule, the largest of the three, gives an rms pull of 0.34 over the 25 inputs, so there it is conservative. For A_9 the replicate error is 0.52 of the sector-pooled one, and an archive-free comparison of production with an independent probe on a subsample of its sectors differs by +2.16 standard errors, so $\sigma(A_9)$, and with it $\sigma(C_8^{\mathrm{SR}})$, may be underestimated. Which value is right is indicated by the third calculation of Sec. 2.7, which agrees with the record. A second independent evaluation of $A_9[\varepsilon^0\ell^1]$ with fresh random shifts (about 9×10^3 core-hours) would show whether the low value is a fluctuation or a systematic error. A 1% measurement of C_8^{SR} cannot test the eleven rational coefficients of Eq. (2.15) in any case; that would need A_9 to about 10^{-9} relative, beyond the present method.

2.7 The eighth order from the integrable point

A third calculation of C_8^{SR} uses neither the matched two-coupling prescription nor any input of the two calculations above. At $s=1$ the wall gas is the Coulomb gas of the anisotropic Kondo model [5, 6], equivalently of the ohmic spin–boson model [7], which is solvable by Bethe ansatz [8, 9]. The calculation (Appendix I) has three parts.

- *Scheme.* The gas, without a hard core, is put in a uniform magnetic field, and its couplings $\hat{u}$ and $\hat{z}$ are defined from the magnetization (the magnetization scheme). The flow of $\hat{z}$ is then exact, $\hat{H}=4\hat{u}-\bar{\delta}$, and only one function $\hat{F}$ has to be computed; T and D follow as in Sec. 2.2, with a factor $1-\bar{\delta}/2$ that converts the field scale to a length scale.
- *The exact part at $s=1$.* The whole $\bar{\delta}=0$ part of $\hat{F}$, to all orders in the fugacity, follows from the magnetization at $s=1$, derived from the Bethe-ansatz equations by a Wiener–Hopf solution as a Mellin–Barnes integral (Eq. (I.4)). It contains the counterpart of J_9 exactly, so no nine-kink integral is needed.
- *Kink clusters.* The $\bar{\delta}$-dependent coefficients come from connected clusters of two to eight kinks with logarithm insertions, continued analytically in the charge v to $v=1$ by sector decomposition. Their poles at $v=1$ must cancel in $\hat{F}$. These cancellations, the exact clusters at $s=1$ and the closed forms of C_3^{SR} to C_7^{SR} are the checks.

The same assembly reproduces the closed forms of C_3^{SR} to C_7^{SR} to at most 7×10^{-26} relative. C_8^{SR} needs two new coefficients, a six-kink one, $f_{202}=172415.83018563\pm1.7\times10^{-7}$, and an eight-kink one, $f_{301}=-514275.75\pm0.17$, which enter as $2f_{202}+\frac{3}{4}f_{301}$. The result is

$$C_8^{\mathrm{SR}}=247.38118\pm0.13 \qquad \text{(magnetization scheme; empirical numerical uncertainty).} \tag{2.17}$$

The central value is given to five decimals because it is compared with the record below. The uncertainty is $\frac{3}{4}$ of that of f_{301}, a linear sum of the differences between three complete runs (two quadrature grids and two contour radii) and of a roundoff estimate validated by recomputing 56 sectors in double-double arithmetic. It is not a standard deviation or a bound, and it cannot exclude an error common to all three runs. The central value was frozen before any comparison; two criteria of the checks were amended after their data had been seen (Appendix I.7). The code

and records of this calculation are the package `code/integ_d1/`, whose replay rebuilds every record from the shipped engine outputs (Appendix H).

Comparison. Equation (2.17) agrees with the record (2.14) to 2.5×10^{-5} (10^{-7} relative, 0.0002 of its uncertainty) and lies 4.56 above the independent value, 1.9 of that value's standard error. It shares no input with either hard-core calculation and never uses A_9. Since $C_8^{\mathrm{SR}} - \frac{1}{4}J_9$ of the independent re-derivation agrees with the compendium (Sec. 2.6), the low independent value lies in its J_9, and within J_9 in $A_9[\varepsilon^0\ell^1]$, the one piece that deviates. Whether that deviation is a fluctuation with an underestimated error or a systematic error is not known. At 5×10^{-4} relative, Eq. (2.17) supports the leading digits of the record but can neither identify nor certify the rational coefficients of Eq. (2.15).

What the integrable point tests directly. Of the $s = 1$ coefficients of the flow up to the order that C_8^{SR} needs, only two combinations are invariant under changes of coupling, and both agree with the compendium; one of them, $b + \frac{f}{4} - \frac{a}{2} - \frac{c}{8} - A$, vanishes identically with the closed forms of Sec. 2.3 (Appendix I.3). All the others, J_9 among them, are coordinate choices of the hard-core scheme, so the $s = 1$ solution alone cannot test A_9.

2.8 Review of the compendium

The compendium was reviewed on 2 October 2026 by a multi-agent workflow (ten review dimensions and adversarial verification). It found no error in any coefficient or proof claim: the conversions and invariants of C_3^{SR} to C_7^{SR} were recomputed from their stated formulas, the C_5^{SR} to C_7^{SR} certificates were found sound (mutation tests: altered rows are rejected), and every task of the archive's reproduction script passed on Linux, including a fresh evaluation of the exact P_6 expression that reproduces it byte for byte. These tasks replay the archived expressions, certificates and records; none regenerates the raw kink integrals, which is impossible because 21 ordered-integration stage programs are missing from the archive (Appendix H.5). The other defects concern the archive as well: two files whose names differ only by case (they collide on case-insensitive file systems), a reconstruction step that does not recreate two historical paths, and bibliographic errors. None affects a quoted number.

2.9 Caveats

- *Two-coupling closure.* The expansion assumes that the renormalization of the dual involves only the couplings g and b, and that the prescription of Sec. 2.1 extracts their β functions. Two unrefereed arguments, given in full in Appendix D, support both formally, coefficientwise in g, h and $\bar{\delta}$ and so at every order. The first (Appendix D.2) argues that, for a symmetry-preserving local subtraction, every bulk ultraviolet counterterm lies in $\mathrm{span}\{\mathbf{1}, \mathcal{O}_g, \mathcal{O}_h\}$, so no third bulk coupling is needed (h and b are two coordinates on one direction). A literal two-operator truncation would nevertheless be wrong: finite irrelevant terms, such as $:\chi_B^2:$ from $\mathcal{O}_g\mathcal{O}_g$, feed the marginal β functions, and the full multi-kink integrals of the archive include them (Appendices D.6 and D.9). The second (Appendix D.3) argues that the prescription extracts the local β functions. A divergence at a fixed endpoint of the interval can only multiply the identity, so it cancels when the responses are divided by the vacuum moments (Appendix D.10), while the finite endpoint-source terms remain and are those of the archive's recurrence (Eq. (A.47), Appendix D.13); the two-response system then has the local β functions as its solution, up to a finite change of (g, b), which leaves T and D, and so the series, unchanged. Neither argument checks the historical high-order integrations, which are tested only numerically, by the independent re-derivation and by the magnetization scheme of Sec. 2.7 (the certificates concern the later reduction to zeta values), nor the identification

of the dual with the chain (Appendix D.14). The magnetization scheme, with a different regulator and different responses but the same two-coupling description, reproduces C_3^{SR} to C_7^{SR} to 7×10^{-26} and the record of C_8^{SR} within its uncertainty: numerical support for the prescription, not a test of the closure. Independently, the NRG (next item) confirms C_3^{SR} to C_5^{SR} without using the dual.

- *The NRG test in numbers.* The numerical renormalization group (NRG) of the accompanying paper [1, 2] does not use the dual. At $\Delta_\sigma = 0.01$, the residual of its $\Delta_{\mathcal{E}}$ after each added order of the series is 4.7×10^{-3}, 9.8×10^{-4}, 2.4×10^{-4} and 5.3×10^{-5} through C_2^{SR} to C_5^{SR}; the NRG error there is 4.7×10^{-5} (a linear sum of chain-depth, truncation and discretization parts, not a standard deviation), so the last residual is 1.1 times that error. The test is of the partial sums, and with them of C_3^{SR} to C_5^{SR}; the term of C_6^{SR} (5.9×10^{-5}) is of the size of the NRG error, and the higher terms are below it.
- *Level crossing.* $\Delta_{\mathcal{E}}$ is identified with Δ_-, assuming that no other level crosses it.
- C_8^{SR} *is not proved.* Its record is printed to about 85 digits and cross-checked to about 57 decimal places (Sec. 2.5), and the zeta-value form is conjectured. The magnetization-scheme value agrees with the record to 2.5×10^{-5}, within its empirical uncertainty of 0.13 (Sec. 2.7); the independent value lies 1.9 to 4.0 standard errors below, with the difference traced to one input integral.
- *Numerical errors are not bounds.* The compendium's high-precision evaluations are checked by repeated runs with different settings (Appendix C.7), not by interval arithmetic; the independent errors are statistical, and the magnetization-scheme uncertainty is an empirical numerical estimate.
- *Side results.* Appendices C.8 to C.11 also record general-formula tests and results derived from published data of the dual (the shadow relation, the logarithmic endpoint). They hold under the regularity assumptions stated there, are not quoted by the paper and are not results of Table 2.

3 Two dimensions: c_2

3.1 The crossover theory

In $d = 2$ the long-range fixed point meets the short-range Ising CFT at $s_* = 2 - 2\Delta_\sigma^{\mathrm{SR}} = 7/4$. Near s_*, Behan, Rastelli, Rychkov and Zan (BRRZ) describe it as the Ising CFT coupled to a generalized free scalar χ of dimension $\Delta_\chi = 15/8 - \delta$ through $g \int \sigma\chi$ [11, 12]; the relevant thermal coupling is tuned to the critical surface. With unit-normalized fields, $\Delta_\sigma^{\mathrm{SR}} = 1/8$, $\Delta_{\mathcal{E}}^{\mathrm{SR}} = 1$, $c = 1/2$, $C_{\sigma\sigma\epsilon} = 1/2$ and $C_{\epsilon\epsilon\epsilon} = 0$. The exact Ising correlators $\langle\sigma_1 \cdots \sigma_{2n}\rangle$ and $\langle\epsilon\epsilon\sigma_1 \cdots \sigma_{2n}\rangle$ needed here follow from a neutral-charge (bosonization) formula [13]. Appendix E derives every step of this section in full: the conventions in Appendix E.1 and the correlators in Appendix E.3.

First order. With AC the analytic continuation in the exponents (defined, with the circular finite part FP used below, in Appendix E.1), the first-order energy shift is proportional to

$$\begin{aligned} I_2(\delta) &= \mathrm{AC} \int_{\mathbb{C}} |1-z|^{-4+2\delta} \Big(\frac{|1+z|^2}{4|z|} - 1 \Big) \, \mathrm{d}^2 z \\ &= \frac{\pi\delta^2}{4(1-\delta)^2} \frac{\Gamma(\delta)\Gamma(\frac{1}{2}-\delta)}{\Gamma(1-\delta)\Gamma(\frac{1}{2}+\delta)} = \frac{\pi}{4} \big[\delta + (2 + 4\log 2)\delta^2 + \cdots \big], \end{aligned} \tag{3.1}$$

which vanishes at $\delta = 0$ (Appendix E.4). This reproduces the known cancellation $c_1 = 0$ [12], and the expansion starts at $c_2\delta^2$.

3.2 Renormalization-group reduction

In pole-only analytic minimal subtraction (Appendix E.2), $\beta(g) = -\delta g + b_0 g^3 + O(g^5)$ and $\gamma_{\mathcal{E}} = h_{\mathrm{MS}} g^4 + O(g^6)$, so $g_*^2 = \delta/b_0$ and

$$c_2 = \frac{h_{\mathrm{MS}}}{b_0^2} = -\frac{4 z_{4,1}}{b_0^2}, \tag{3.2}$$

where $z_{4,1}$ is the coefficient of g^4/δ in $\log Z_{\mathcal{E}}$. The one-loop coefficient is exact:

$$b_0 = \frac{9\pi^2}{8}\Big[\frac{\Gamma(3/4)}{\Gamma(1/4)}\Big]^2 = 1.26840430893914254379\ldots, \tag{3.3}$$

obtained from the four-spin correlator by a quadratic change of variables and the complex beta integral (Appendix E.4); it agrees with the BRRZ value 1.268404. The split between a g^2 and a g^4 term is prescription dependent: if $\gamma_{\mathcal{E}} = a_1 \delta g^2 + h g^4$, then $c_2 = a_1/b_0 + h/b_0^2$ is invariant under $\mathcal{E}_R \to e^{\kappa g^2}\mathcal{E}_R$, which shifts a_1 by 2κ and h by $-2\kappa b_0$. Minimal subtraction has $a_1 = 0$ and $h = h_{\mathrm{MS}} = c_2 b_0^2 \approx -6.54$.

3.3 Localized source, elliptic cover and the finite contact

The fourth-order logarithm is extracted with the coupling smeared by a localized source $f(w) = e^{-|w|^2}$ about the origin and with the two energy operators at 0 and ∞. Write S_4 for the four-spin correlator $\langle\sigma\sigma\sigma\sigma\rangle$ and H_4 for the same correlator with the two energies inserted, and S_{ij}, H_{ij} for the corresponding two-spin functions of spins i and j. The fourth-order term of the logarithm of the energy two-point function has the connected kernel $\mathcal{F} = H_4 - S_4 - H_{12}H_{34} + S_{12}S_{34}$ for the GFF pairing (12)(34), and the three equal pairings give $3/4! = 1/8$; its integral over the four spin positions is $A_4(\delta)$. The local coupling counterterm $b_0 g^3 f^3/(2\delta)$ contributes at the same order through the second-order integral $A_{a,b}(\delta)$, the two-spin integral with Gaussian weights $e^{-a|w_1|^2 - b|w_2|^2}$: at fourth order it gives $b_0 A_{3,1}(\delta)/\delta$ (Appendix E.5), so that

$$z_{4,1} = \operatorname*{Res}_{\delta=0} A_4 + b_0 B_{31}, \qquad B_{31} = A_{3,1}(0) = \frac{\pi^2}{96}(12 - \pi\sqrt{3} - 9\log 3) = -0.342239706924080236 55\ldots. \tag{3.4}$$

A Möbius map returns the energies to 0 and ∞ (Appendix E.6); the integral over the overall scale gives $A_4(\delta) = \frac{\pi}{8}\Gamma(2\delta) J(\delta)$, where $J(\delta)$ is the remaining integral over the four-spin cross ratio z and the two energy positions, weighted by $W^{2\delta}$, with W the Gaussian source factor carried through the map. The energy positions are uniformized by the elliptic curve E_z, the double cover of the t-plane branched at 0, z, 1 and ∞, on which the Ising kernel becomes an algebraic expression in Weierstrass functions, and the torus integration defines the elliptic finite part

$$\mathcal{I} = \mathrm{FP}\int_{\mathbb{C}} \mathrm{d}^2 z\, |z|^{-15/4} \frac{\mathcal{A}(z)^2}{4}\Big[w_+ |c_+ - 2q|^2 + w_- |c_- - 2q|^2 - w_0 |c_0 - 2q|^2\Big], \tag{3.5}$$

with the torus area $\mathcal{A} = 16\,\mathrm{Re}[\overline{K(z)}K(1-z)]$, the circular mean $q = E(z)/(4K(z)) - (2-z)/12 - \pi\overline{K(z)}/(K(z)\mathcal{A})$, the Ising block weights $w_\pm = |1 \pm \sqrt{1-z}|/(2|z|^{1/4}|1-z|^{1/4})$, $w_0 = |z|^{-1/4}$, and $c_\pm = (z - 2 \pm 3\sqrt{1-z})/6$, $c_0 = (2-z)/12$. In this notation the four-spin function at cross ratio z is $S_4(z) = w_+ + w_-$.

The flat torus finite part is not yet the limit of the source-weighted integral $J(\delta)$. When both energies approach the same branch point, the first derivative of the source factor $W^{2\delta}$ survives against a vanishing zeroth-order angular residue and leaves a finite contact A_{br} (Appendix E.7). It is related exactly to the lower Gaussian integral,

$$A_{\mathrm{br}} = 4B_{31} - \frac{\pi^2}{4}, \tag{3.6}$$

and at fixed modulus z the regulated energy integral equals the torus integrand plus $4A_{\mathrm{br}}[S_4(z) - |z|^{-1/4}]$. The energy-primary term of the shrinking spin pair cancels exactly in the connected logarithm, and no scalar with a logarithmic threshold remains at the modulus boundaries (Appendix E.8).

3.4 Assembly and the matching term

Integrating the fixed-modulus comparison over z, with $\mathrm{AC}\int |z|^{-15/4}[S_4 - |z|^{-1/4}]\,\mathrm{d}^2z = -b_0/\pi$, gives $J(0) = \mathcal{I} - 16b_0B_{31}/\pi + \pi b_0$ (Appendix E.9), so B_{31} cancels and

$$z_{4,1} = \frac{\pi\mathcal{I} + \pi^2 b_0}{16}, \qquad \boxed{c_2 = -\frac{\pi\mathcal{I}}{4b_0^2} - \frac{\pi^2}{4b_0}.} \tag{3.7}$$

The finite matching term $-\pi^2/(4b_0) = -1.94527965797909034563\ldots$ is comparable to the period term $-\pi\mathcal{I}/(4b_0^2) = -2.12099049015343120711\ldots$. The study's own source-contact checks (Appendix E.15) and its review showed that it does not depend on the shape of the source (analytically, and numerically for two other shapes), and that it equals a_1/b_0, where $a_1 = -\pi^2/4$ is the $O(\delta)$ slope of the leading-order formula of BRRZ with the δ-deformed χ propagator; the finite redefinition with $\kappa = -\pi^2/8$ maps minimal subtraction to that scheme. The cylinder calculation of Sec. 3.8 finds the same $a_1 = -\pi^2/4$ independently (Appendix F.2). Without the matching term c_2 would be about -2.12, which the Monte Carlo of Sec. 3.7 excludes by about 100 standard errors.

3.5 Closed form of the elliptic finite part

The integral $\mathcal{I}$ is reduced by a Landen substitution and a single-valued period pairing to periods of $K(t)^2$, $K(t)K(1-t)$ and $K(1-t)^2$ over $0 < t < 1$, then by contour reflection to symmetric elliptic moments, and finally, through summand-level rational telescoping certificates with their nonzero endpoint terms, to three convergent hypergeometric values (Appendices E.10 to E.14). With $R_\Gamma = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$, so that $b_0 = 9\pi^2/(8R_\Gamma^2)$, and

$$\mathsf{P} = {}_4F_3\left(\begin{matrix}-\frac{1}{2}, -\frac{1}{2}, \frac{3}{2}, \frac{3}{4} \\ 1, 2, \frac{1}{4}\end{matrix}; 1\right), \quad \mathsf{Q} = {}_4F_3\left(\begin{matrix}\frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \frac{9}{4} \\ 3, 3, \frac{7}{4}\end{matrix}; 1\right), \quad \mathsf{T} = {}_4F_3\left(\begin{matrix}\frac{3}{2}, -\frac{1}{4}, \frac{3}{4}, \frac{3}{4} \\ \frac{1}{4}, \frac{9}{4}, \frac{9}{4}\end{matrix}; 1\right), \tag{3.8}$$

the coefficient is

$$c_2 = \frac{16}{R_\Gamma^2} - \frac{R_\Gamma^2}{9} - \frac{\pi^2 R_\Gamma^2}{16}\Big(\mathsf{P} - \frac{5R_\Gamma^2}{384}\mathsf{Q}\Big)\Big(\mathsf{P} - \frac{5R_\Gamma^2}{128}\mathsf{Q} - \frac{512}{25\pi R_\Gamma^2}\mathsf{T}\Big), \tag{3.9}$$

which evaluates to

$$c_2 = -4.06627014813252155274002760857287648752990114118 66\ldots. \tag{3.10}$$

Each series has $\sum b_j - \sum a_i = 2$, so its terms decay as n^{-3}, it converges absolutely and its unaccelerated remainder is $O(N^{-2})$. The three constants need no analytic continuation, and the expression contains no unevaluated integral, parameter derivative or unresolved subtraction constant. No reduction to Gamma values alone has been established (an integer-relation search found none). The constants are

R_Γ	2.95867511918863889231	b_0	1.26840430893914254379
P	1.62562945952539344424	B_{31}	−0.34223970692408023655
Q	1.28812240964200676062	$\mathcal{I}$	4.34474465231888563075
T	0.76385352476101335168	$z_{4,1}$	1.63550416445718870488

Checks of the evaluation (Appendix E.15). The six-function and three-function expressions and a half-interval Frobenius series of the moments agree to 2.53×10^{-80} (the 80-digit arithmetic floor). An endpoint-subtracted quadrature with Carlson's integrals, which uses no Frobenius coefficients, agrees to 1.40×10^{-61}, a difference set by the finite upper limit of its last integral (doubling that limit lowers it to 7.4×10^{-77}). Exact tail inequalities for the three series give $-4.066270148133018 < c_2 < -4.066270148131205$ (floating-point propagation, not a certified rounding). The base study of the cylinder calculation (Sec. 3.8) contains an integer-interval evaluation that certifies the value of the closed expression to 51 decimal places (Appendix F.14; its code is not in the release, Appendix H.5). In the review, $\mathcal{I}$ was also evaluated directly from its defining integral, without the reduction, to 51 digits in one implementation and to 5×10^{-15} by z-plane integration in another, and every intermediate identity of the chain was checked at 30 to 80 digits.

3.6 The step that is not proved

Equation (3.7) needs one step that is not proved: exchanging the limit $\delta \to 0$ with the nested subtracted integrals of the global comparison. In each collision sector the comparison subtracts the local OPE and Taylor terms, restores powers by their Mellin continuation and spin terms by their circular finite parts, and then takes the regulator limit of the subtracted remainder. That exchange needs an integrable bound on the remainder, uniform near $\delta = 0$, including the degenerating shapes where sectors intersect. The study classifies the channels and gives a general subtraction argument (Appendix E.8), but this step is incomplete in two respects: not every sectorwise subtraction is written out, and the uniform bound is not proved. We count both as the one unproved step of the abstract and Table 2. Its weakest point is where a same-branch contact meets the corner $z \to 0$.

The review of the study (summarized in `docs/c2_closed_d2/REVIEWS.md`) looked for each mechanism by which an $O(1)$ term could arise there, and found none. There is no scalar with a logarithmic threshold at dimension two (the energy primary cancels), no dependence of the leading regulator on $\arg z$ at the corners, and no term above threshold from subleading regulator terms combined with spin-one terms. Two numerical tests bear on the step: the Monte Carlo of Sec. 3.7 integrates the full regulated integral, and the cylinder calculation of Sec. 3.8 uses a different regularization. The identification of Eq. (3.9) with the physical coefficient is therefore a formal perturbative result in the critically tuned analytic minimal-subtraction prescription, conditional on this step; the evaluation of $\mathcal{I}$ itself is proved.

3.7 Independent Monte Carlo of the regulated integral

The review integrated the complete regulated eight-dimensional integral over the four insertion positions by Monte Carlo, with the exact Ising correlators built by bosonization and no Möbius map, elliptic cover, contact derivation or sector subtraction. Because it integrates the whole regulated integral, it tests the global exchange step numerically. Three configurations with different channel weights were run, two on the symmetry cluster and one on a personal desktop, with 4.26×10^{10} samples in all (142 chunks, about 135 core-hours); all three enter the result. Their inverse-variance combination gives

quantity	Monte Carlo	closed-form route
$J(0)$	10.559 ± 0.039	10.5404
$z_{4,1}$	1.6391 ± 0.0076	1.635504
c_2	-4.075 ± 0.019	-4.066270

The errors are statistical (one standard error). The recombination gives $c_2 = -4.0752 \pm 0.0189$, quoted as -4.075 ± 0.019; it differs from the closed form by -0.009, 0.5 standard errors. The two configurations run on the symmetry cluster alone give -4.0724 ± 0.0197. The three configurations

agree on the integral itself ($\chi^2 = 1.29$ for two degrees of freedom). A source-shape control, the difference between a quartic and a Gaussian source, reproduces its prediction when the three are combined (16.185 ± 0.053 against 16.150), but the three configurations disagree among themselves on it ($\chi^2 = 7.6$ for two degrees of freedom). The control does not enter c_2; the disagreement is unexplained and left open. The same code reproduces b_0, the vanishing first-order shift, B_{31}, and $z_{2,1} = -b_0/2$, which gives $\Delta_\sigma = \frac{1}{8} + \delta$.

3.8 Fourth-order perturbation theory on the cylinder

Setup. A separate calculation (its method in full in Appendix F) does Rayleigh–Schrödinger perturbation theory for the Ising CFT times the GFF on the radial cylinder, with $H_0 = D_{\text{Ising}} + D_\chi$ and $V = \int_0^{2\pi} \mathrm{d}\theta\, \sigma\chi$. It shares with the closed-form route only the crossover description, the exact Ising data and b_0; it uses none of $\mathcal{I}$, the hypergeometric form, the branch contact or the proposed value. The observable is $X = 2E_{\mathcal{E}} - E_d - E_0$, with $|d\rangle = L_{-1}|\mathcal{E}\rangle$, which equals $\Delta_{\mathcal{E}} - 1$ at a conformal fixed point and removes the direct fourth-order local terms (Appendix F.7). In the scheme defined by the cylinder Hamiltonian with its locally derived counterterms, $H = H_0 + gV + \sum_{k\geq 2} g^k W_k$, write $X = a_1\delta g^2 + hg^4 + \cdots$. With h_0 the value of h once the regulator is removed, the invariant combination of Sec. 3.2 gives

$$c_2 = \frac{a_1}{b_0} + \frac{h_0}{b_0^2}; \tag{3.11}$$

here $h_0 \approx -3.41$, while minimal subtraction has $a_1 = 0$ and $h_{\text{MS}} \approx -6.54$. The second-order coefficient is an exact meromorphic continuation (Appendix F.2), $a(\delta) = \pi^2\Gamma(1 - \frac{\delta}{2})^2\Gamma(\delta - 1)/[\Gamma(\frac{\delta}{2})^2\Gamma(2-\delta)] = -\frac{\pi^2}{4}\delta + O(\delta^2)$, so $a_1 = -\pi^2/4$ and $a_1/b_0 = -1.94527965797909\ldots$. The fourth-order coefficient h is computed at $\delta = 0$ with a family of smooth positive oscillator regulators of order $r = 1, 2, 3$ and length λ, locally derived thermal and spin-two counterterms, and exact chiral conformal blocks as spectral generating functions (Appendices F.6, F.7 and F.8). The spectral sums run over intermediate states labelled by three level indices and are truncated at a cube $0 \leq n, m, \ell < N$, with $N \leq 1024$; they are extrapolated first in N at fixed (r, λ) and then in $\lambda \to 0$. No fit has the physical δ as its variable. As a lower-order check, the renormalized magnetic gap gives $[g^2](E_\sigma - E_0) = 1.268404004672631$ at $r = 3$, $\lambda = 1/32$, which differs from b_0 by -3.04×10^{-7}; it was not used to tune the counterterms (Appendix F.12).

Continuum analysis. The base study, with $N \leq 384$, gave $c_2 \simeq -4.07$ (representative extrapolations -4.0669, -4.0680 and -4.0713 for $r = 1, 2, 3$). The continuum analysis uses 25 fixed-regulator points (r, λ), for each the record with the largest N. Fourteen are new production records with $N = 768$ or 1024, computed on SDSC Expanse and the symmetry cluster ($\lambda = 0.05$ to 0.175). The other eleven are archived records of the base study, computed by GPT-6 astra (Sec. 5): $N = 256$ at $\lambda = 0.125$ to 0.25 for $r = 1$ and 0.175 to 0.35 for $r = 2$, and $N = 384$, 256, 256 and 128 at $\lambda = 0.2$, 0.25, 0.35 and 0.5 for $r = 3$. Their spectral sums are taken as archived and extrapolated under the same rules; their auxiliary sums were recomputed on the symmetry cluster at the common cutoffs of all points (below). Nine of them are among the 23 points of the joint fit. The rules were all fixed in code before the final numbers were computed and are copied into every output:

- *Large-N limit at fixed (r, λ).* The leading large-N correction, a power $N^{-7/4}$ from the faces of the truncation cube with an analytically derived coefficient (Appendix F.18), is subtracted; the remaining tail is fitted with sets of powers and exponentials fixed in advance, on windows that are fixed fractions of N, and the median of the selected fits is $h(\lambda)$.
- *Checks of that limit.* The same extrapolation is repeated on the sums truncated at $N' = N/2, \ldots, N$, and records with different N are compared. Points with λN below a minimum (50, 60 and 75 for $r = 1, 2, 3$) are flagged, since the cutoff must resolve the regulator; the five flagged points, all archived, stay in the fits, and only records above the minimum enter

the cross-N comparison. The spectral coefficients, computed by discrete Cauchy integrals on a grid of points on a torus (Appendix F.8), are extracted again on a second grid with a different radius, for every record, and roundoff is estimated. Three auxiliary truncations (of the mixed thermal sums, of the collapsed sums and of the two-index leading-tail calculation; Appendices F.9 and F.18) are set to common values for all points (16384, 800000 and 8192), and their shifts under a change (to 400000 and 2048 for the last two) are propagated.

- *Regulator limit* (Appendix F.10). Per order, $h(\lambda) = h_0 + k_1\lambda^2 + k_2\lambda^2 \log\lambda + k_3\lambda^3$, the preferred model fixed in advance, on $\lambda \le 0.25$ for $r = 1, 2$ and $\lambda \le 0.35$ for $r = 3$ (the smallest λ is 0.05, 0.0625 and 0.1; the points $(2, 0.35)$ and $(3, 0.5)$ lie outside these windows). Alternative models and windows are admitted by fixed rules, and the regulator-model error is the largest deviation of an admissible fit.
- *Choice of estimator.* A rule fixed in advance chooses between the per-order results and a joint fit. Here the three orders are consistent (each pairwise difference of h_0, at most 1.9×10^{-3}, is below the sum of the two errors), and the joint fit with a shared h_0 is accepted (23 of the 25 fixed-regulator points enter it).

The closed form was never an input, a prior or a selection criterion. The per-order and joint results are

	points	h_0	c_2	error in c_2
$r = 1$	9	-3.414150	-4.06739	0.01006
$r = 2$	7	-3.412774	-4.06653	0.00417
$r = 3$	7	-3.412269	-4.06622	0.00346
joint	23	-3.413609 ± 0.003794	-4.0670498	0.0023580

so that

$$c_2 = -4.0670498 \pm 0.0023580, \qquad \text{quoted } -4.0670(24). \tag{3.12}$$

The error is a linear sum of systematic numerical diagnostics, not a standard deviation and not a rigorous bound. Its components in c_2 are: the choice of regulator model 1.52×10^{-3} (64%); the large-N extrapolation of the points, propagated through the fit, 4.04×10^{-4} (17%); the cutoff and model spread of the mixed thermal sums 2.14×10^{-4} (9%); roundoff 8.7×10^{-5} (4%); coefficient extraction, from the comparison of the two Cauchy grids, 7.5×10^{-5} (3%); the cutoff of the collapsed sums 6.2×10^{-5} (3%); and the cutoff of the leading-tail calculation 2.5×10^{-9}. The difference from the closed form is -7.8×10^{-4}, or $|-4.0670498 + 4.0662701|/4.06627 = 1.9 \times 10^{-4}$ relative, 0.33 of the quoted error.

Caveats, as the paper states them.

- The closed form was known before this analysis, but it enters the analysis in no way.
- The error rule was revised after the first result had been seen. The first complete analysis gave the same central value with $E(c_2) = \pm 2.21$ (absolute, in c_2). Of that, 99.9% was the coefficient-extraction term, bounded by a worst case over sign patterns: the weights $\sum_n |v_n|$ of the large-N fits on the raw sums, up to about 2.4×10^5, times differences between two coefficient grids of 5×10^{-7} to 7×10^{-5}, measured at six of the 25 fit points and transported to the others. The revised rule, written after that result and approved in the Codex review, measures the grid difference directly at all 25 points (20 new extraction-check records), carries its smooth part with its measured sign and keeps only the rough residual at worst case; it gives 7.5×10^{-5} in c_2. Every other component is identical in both analyses (2.28×10^{-3} in all). The quoted error therefore fell from ± 2.21 to ± 0.00236, about 940-fold, entirely through the replaced extraction bound, and the central value is unchanged by construction. The superseded result is archived.

- The two $N = 1024$ coefficient grids were rebuilt on the symmetry cluster, and the last nine extraction-check computations ran there, because SDSC Expanse's file system stopped serving two small files of the grids built there (an author-approved change of the plan). The grids are a deterministic function of their parameters; a pilot at $N = 512$ built on both machines agreed to 6.9×10^{-11} in the raw sums.
- A difference between two Cauchy grids measures extraction sensitivity; it does not bound an error that both share. The regulator extrapolation dominates the error and is not rigorously bounded.

3.9 Summary for $d = 2$

The closed form is $c_2 = -4.06627014813\ldots$, conditional on one unproved step. The Monte Carlo of the regulated integral gives -4.075 ± 0.019, one standard error (0.5% relative), and differs from it by 0.5 standard errors. The cylinder Hamiltonian gives $-4.0670(24)$, where the error is a linear sum of diagnostics with no probability meaning, and differs from it by 1.9×10^{-4} relative, 0.33 of that error. The Monte Carlo integrates the full regulated integral without the exchange step, and the cylinder calculation uses a different regularization; both test the unproved step numerically. No published $O(\delta^2)$ value exists in two dimensions to compare with.

4 Three dimensions: c_1

4.1 The first-order coefficient as a ratio of integrated correlators

In $d = 3$ the same crossover theory applies with $\Delta_\chi = 3 - \Delta_\sigma^{\mathrm{SR}} - \delta$. With $\beta(g) = -\delta g + b_0 g^3$ (b_0 again the one-loop coefficient, now in $d = 3$) and $\gamma_{\mathcal{E}} = a_0 g^2$,

$$c_1 = \frac{a_0}{b_0}, \qquad a_0 = -2\pi I_\epsilon, \quad b_0 = -2\pi I_\sigma, \tag{4.1}$$

where, with one external state at infinity,

$$\begin{aligned} I_O &= \mathrm{AC} \int_{\mathbb{R}^3} \mathrm{d}^3 y \, |e - y|^{-2(3-\Delta_\sigma^{\mathrm{SR}})} \big[H_O(y) - |e - y|^{-2\Delta_\sigma^{\mathrm{SR}}} \big], \\ H_O(y) &= \lim_{R \to \infty} R^{2\Delta_O} \langle O(0)\sigma(y)\sigma(e)O(Re)\rangle, \qquad O \in \{\sigma, \epsilon\}, \end{aligned} \tag{4.2}$$

with e a unit vector and AC the analytic continuation through the power-divergent short-distance terms. The denominator comes from the protected spin dimension, $a_\sigma = b_0$ (Appendices G.2 and G.3). Both integrals are evaluated on a folded region with conformal-block sums of the 3d Ising CFT: 60 primaries from the spectrum of Simmons-Duffin [14] (the stable entries with $\Delta < 24$), with $\Delta_\sigma^{\mathrm{SR}}$, $\Delta_{\mathcal{E}}^{\mathrm{SR}}$ and the stress-tensor couplings from Ref. [15] and the Ward identity, radial and small-z block recurrences, and analytic continuation of the power-divergent terms. Appendix G derives the folding (Appendix G.4), the blocks and the integration (Appendices G.5 and G.15) and lists the retained spectrum (Table 6).

4.2 Retained-spectrum value

With the retained 60 primaries,

$$a_0 = 3.3474594916917857, \qquad b_0 = 12.254914105481044, \qquad c_1 = 0.27315242382601645, \tag{4.3}$$

quoted as 0.2732. The long strings are numerical regression targets, not physical digits: the uncertainty of c_1 is set by the omitted spectrum, through the estimated range of Sec. 4.3 (the three

meanings of precision are separated in Appendix G.1). The numbers agree with BRRZ [11, 12] component by component: $I_\sigma = -b_0/(2\pi) = -1.950430$ against $-1.950(5)$, $b_0 = 12.2549$ against $12.26(3)$, $a_0 = 3.3475$ against $3.3(5)$, and c_1 against their estimate $0.27(4)$; later bootstrap studies found 0.269 [21] and 0.27 [22].

Implementations. The package's full workflow, rerun from scratch, reproduces every shipped number bit for bit in the review's run and on the Intel nodes of the symmetry cluster, and within the replay's tolerances on its AMD nodes (5×10^{-10} for a_0, b_0 and c_1). The review of the package wrote two independent implementations, A and B, each with its own conformal blocks, quadrature and analytic continuation, neither importing the package; they agree on c_1 to 10^{-12}, including all 60 per-family contributions. Both are included in `review_indep/` of the package and rerun by its replay on the retained spectrum: on the symmetry cluster (Intel and AMD nodes) A differs from the package's c_1 by at most 6×10^{-14}, B by at most 5×10^{-13}, and A from B by at most 4×10^{-13}, all absolute (the replay tolerance is 10^{-12}). For a_0 and b_0 the agreement is only at the level of 2×10^{-12} relative, so the claim of 10^{-12} is made for c_1 alone. The package's own two integrated forms (`blocks` and `kernels`) agree to about 1.5×10^{-13} but share the folded domain, so they are a consistency check, not an independent one. The blocks satisfy the Casimir equation and agree with the Zamolodchikov-type recursion of implementation B to 6×10^{-13}; the package's own block checks, against a small-z series and a closed form on the diagonal, are in Appendix G.5.

4.3 Estimated range

The omitted spectrum is controlled only under explicit hypotheses:

- every primary missing from the retained list and exchanged in $\sigma \times \sigma$, $\sigma \times \epsilon$ or $\epsilon \times \epsilon$ has dimension at least 8 if it is $\mathbb{Z}_2$-even and at least 7 if it is $\mathbb{Z}_2$-odd (published spectra motivate these gaps but do not prove them [14, 18, 19, 20]);
- each retained input lies within twice its quoted error (the input box). The errors of the higher families are spreads over spectra at fixed derivative order, not rigorous; of the five low inputs from Ref. [15], $\Delta_\sigma^{\mathrm{SR}}$ and $\Delta_{\mathcal{E}}^{\mathrm{SR}}$ have rigorous errors there, the two OPE coefficients and c do not (Appendix G.6);
- the conditional bootstrap result $\lambda_{\epsilon\epsilon\epsilon'} = 1.5362(12)$ of Ref. [17], with its sparsity assumptions, applies;
- the margins added to the linearized crossing constraints and to the integrated ratio, which cover nonlinear terms and sampled remainders, and the sampled allowances for the denominator ($|\Delta b_{\mathrm{in}}| \le 0.001428$), for the reference bounds (a factor 1.001) and for the quadrature, are large enough.

The hypotheses are discussed in Appendices G.9 (gaps and $\lambda_{\epsilon\epsilon\epsilon'}$), G.8 (the input box), and G.10 and G.14 (the margins and their sampled tests). Under them, an identical-channel positive remainder, a Cauchy–Schwarz bound on the even mixed remainder, a sign-aware integration of the odd remainder and an affine linear program over crossing constraints and inputs (Appendices G.9 to G.12) give the nominal interval [0.2697959, 0.2752135] (ten constraint updates); doubling the crossing margins gives [0.2692738, 0.2754104] (the package's recorded run, eight updates; ten give [0.26927, 0.27541]). The outward-rounded estimated range is

$$c_1 \in [0.2690,\ 0.2760] \qquad \text{(conditional)}, \tag{4.4}$$

to be read next to the retained value 0.2732, about -0.0042 and $+0.0028$ from it. It is neither a certified bound nor a confidence interval. A weaker-gap scenario (6 and 6) gives [0.26400, 0.27842] (recorded run, eight updates; ten give [0.26420, 0.27834]).

4.4 How conditional the range is

The review found, and the revised package now shows, that the range is set by the assumed input box and the gap hypotheses rather than by numerical precision.

- *Feasibility* (Appendix G.10). With all inputs at their central values, the final linear program is infeasible: 68 of its 552 constraints, all in the spin four-point channel, are violated. With the nominal program held fixed, the smallest feasible displacement is 0.574 of the box half-width, that is, 1.15 quoted errors for the input that moves most (it is neither a common displacement of all inputs nor a significance). Underestimated spectrum errors and a missing contribution below dimension 8 are natural explanations, and the chosen numerical relaxation may contribute (Appendix G.10); the data cannot tell which.
- *Box size* (Appendix G.13). Replacing each quoted error e_k by me_k, so that the inputs range over $|u_k - u_k^{(0)}| \le 2me_k$, and rebuilding the program (sensitivities, crossing secants and margins; the $\lambda_{\epsilon\epsilon\epsilon'}$ window held fixed) gives, with ten constraint updates each,

m	≤ 0.6	0.8	1 (nominal)	2
interval	infeasible	$[0.27027, 0.27492]$	$[0.26980, 0.27521]$	$[0.26708, 0.27684]$

  The infeasibility at $m = 0.6$ (± 1.2 quoted errors) does not contradict the 1.15 above: that minimization kept the nominal program fixed, whereas this scan rebuilds it.
- *Gaps* (Appendices G.9 and G.13). Mild relaxations of the gap hypotheses extend the interval outside the estimated range $[0.2690, 0.2760]$, while every interval still contains the retained value:

(even, odd) gaps	$(8, 6.5)$	$(7, 7)$	$(8, 6)$	$(6, 5.5)$
interval	$[0.26890, 0.27541]$	$[0.26891, 0.27629]$	$[0.26720, 0.27576]$	$[0.25839, 0.27948]$

  Lowering the odd gap mainly lowers the lower end, because the odd remainder enters with a definite sign; lowering the even gap, as in $(7, 7)$, widens both ends. Unlisted odd primaries are estimated at about 7.3 to 7.7, only 0.3 to 0.7 above the assumed gap: spin six at about 7.3 [18] and 7.73(5) [20], spin three at 7.53(6) [20].

The package contains its review revision of 8 October 2026, which adds these feasibility and sensitivity tests, the missing derivations (now in Appendix G) and explicit guards, with the numerical values unchanged; its `replay-full` reruns the feasibility diagnostics and the gap, margin and box scenarios. The three-dimensional Monte Carlo of the accompanying paper [1] cannot test c_1: its only accepted point, at $s = 1.6$ ($\delta = 0.18$), is too far from s_* to isolate the linear term.

5 Authorship and provenance

The author of the project chose the problems, approved plans and production runs, and decided between the options the agents presented; the author wrote none of the computations below. They were done as follows.

- C_1^{SR}, C_2^{SR}: published by Benedetti, Lauria, Mazáč and van Vliet.
- C_3^{SR} *to* C_8^{SR}: computed by another AI system, OpenAI GPT-6 astra (via ChatGPT), and supplied by the author as a research compendium with its derivations, code, exact expressions, certificates and numerical records. It was reviewed by Claude Code (Anthropic) on 2 October 2026 (Sec. 2.8). The derivations of Appendices A to C are GPT-6 astra's.

- *Independent re-derivation of the inputs of C_5^{SR} to C_8^{SR}*: Claude Code sessions, 2 to 7 October 2026, with adversarial verification by separate agents for independence, statistics and mathematics; production on the symmetry cluster, part of the eighth-order work on a personal desktop.
- *The two-coupling closure and prescription arguments* (Appendix D): written by OpenAI GPT-6 astra (via ChatGPT) and supplied by the author on 15 October 2026, the second with its checker; not refereed. Claude Code assessed both at the author's request, checked the finite algebra of the first independently, reran the checker of the second and re-derived its beta recurrence independently (Appendix D.15; `docs/closure_d1/REVIEWS.md`).
- *c_2 closed form*: derived by OpenAI GPT-6 astra (via ChatGPT), supplied by the author (the final-method study; its derivation is Appendix E). Reviewed by Claude Code on 8 October 2026; the review wrote and ran the Monte Carlo of Sec. 3.7.
- *Cylinder Hamiltonian for c_2*: the base study was written by OpenAI GPT-6 astra (via ChatGPT) and supplied by the author; its method is Appendix F. All four archives it consolidates are GPT-6 astra's: the final-method study, the independent formula, elliptic and local checks (with the integer-interval certificate of the closed expression), and the two Hamiltonian packages (the solver). Its 17 archived raw records ($N = 128$ to 384) were computed by GPT-6 astra in its own environment; their machine, dates and software versions are not recorded. Eleven of them are points of the continuum analysis (nine of the 23 in its joint fit; Sec. 3.8), with their auxiliary sums recomputed on the symmetry cluster. Claude Code sessions wrote the cluster layer for SDSC Expanse and the symmetry cluster, ran the production and wrote the arbitrary-N continuum analysis. OpenAI Codex reviewed the analysis plan in seven rounds, approving it at revision 7, and inspected the cluster layer and the analysis code twice each. The fixes made after the last inspection of each were not re-inspected (for the analysis, four hardening findings with no numerical change, by the author's choice).
- *c_1 in $d = 3$*: the package was written by OpenAI GPT-6 astra (via ChatGPT), supplied by the author and dated 7 October 2026; its derivation is Appendix G. It was reviewed by Claude Code on 7 October 2026; the two independent re-implementations of the review (`review_indep/`) were written by Claude Code review agents. The package contains its review revision of 8 October 2026 (UTC; 7 October local time; a full rerun with explicit guards, the missing derivations and the sensitivity tests; numbers unchanged), which OpenAI GPT-6 astra made after our review and the author supplied; the four earlier archives it was built from are also GPT-6 astra's. Claude Code checked the revision on 8 October 2026.
- *Magnetization-scheme value of C_8^{SR}* (Sec. 2.7, Appendix I): Claude Code sessions, 14 to 17 October 2026, on the symmetry cluster (about 2.7×10^4 core-hours, including pilots and failed jobs); OpenAI Codex reviewed the plan and inspected the code (Appendix I.7). Its code and records ship as `code/integ_d1/` (commit 8bd5a22), with `public.mk` and `public/` added by the same session on 18 October 2026.
- *This note*: written by a Claude Code session on 15 October 2026 from the sources above, and revised the same day after an independent readability and precision review. On 16 October 2026, at the author's request that each direction of the release have a single note, the derivations of the sources, which had accompanied the packages as six separate notes by GPT-6 astra, were brought into Appendices A to G by Claude Code sessions, edited to the notation and status words of this note and checked against the sources (A to C from the compendium, D from the two notes on the closure and the prescription, E from the study, F from the base study, G from the c_1 package); short explanatory steps were added in the merge and checked, and none changes a result. The separate notes are not part of the release, nor are the scripts that only build them; the few TeX blocks of the compendium that the checks parse are kept verbatim in `code/dual_d1/public/compendium_extract.json` (Appendix H). On

17 October 2026, at the author's request, the magnetization-scheme calculation was added as Sec. 2.7 and Appendix I, from its records and documentation, which are not in the release.

6 Open problems

- A proof of Eq. (2.15): exact expressions for all inputs of C_8^{SR} and a weight-seven certificate.
- The cause of the low independent value of C_8^{SR}, which Sec. 2.7 indicates is the one in error and the re-derivation's own input comparison locates in $A_9[\varepsilon^0\ell^1]$: a rerun with fresh random shifts would show whether it is a fluctuation with an underestimated error or a systematic error.
- Beyond the magnetization scheme of Sec. 2.7, which checks the prescription argument of Appendix D numerically with a second regulator and different responses but assumes the same two-coupling description: a check that does not assume it, for example a higher-order analytic bootstrap (a ring also needs the monodromy sector of the compact boson), and a source-level audit of the historical high-order integrations, whose 21 stage programs are missing from the archive.
- C_9^{SR}, which needs the next determinant coefficient D_5 (a rational-part guess for it, $D_5^{(0)} = \frac{141}{4}$, is untested; Appendix C.8).
- Interval enclosures of the compendium's high-order numerical integrals (Appendix C.7), and a proof of the weight bound that the closed forms through C_7^{SR} obey (Appendix C.8).
- In $d = 2$, the uniform remainder bound that would make Eq. (3.9) a theorem within the prescription.
- In $d = 2$, the disagreement of the three Monte Carlo configurations on the source-shape control ($\chi^2 = 7.6$ for two degrees of freedom), which does not affect c_2.
- In $d = 3$, gap hypotheses with independent support, or input errors with a calibrated meaning, so that the range of c_1 stops depending on chosen margins.

A The dual of the chain: model, regulator and finite-response renormalization

This appendix gives the derivations of the compendium (written by OpenAI GPT-6 astra; Sec. 1.2) behind Secs. 2.1 to 2.3, edited to the notation of this note. It starts from the lattice model and derives the alternating domain-wall gas and its matrix representation (Appendices A.2 and A.3). It then defines the regulated finite-interval responses of the matched two-coupling prescription (Appendices A.4 to A.8) and the finite-response Jacobian and subtraction recurrence that turn them into beta coefficients (Appendix A.9). The last part explains why the invariants T and D need the higher-insertion integrals (Appendix A.10). The beta inputs of the individual orders and their evaluation are in Appendices B and C.

Apart from the published results quoted in Appendix A.6 and the continuum derivation of the wall gas from the lattice model (Appendices A.2 and A.3), everything here either defines the prescription or derives exact relations within it. Two things are not established here: that the dual describes the critical chain, which is the published duality proposal [3, 4], and that the renormalization of the dual involves only the couplings g and b and the prescription extracts their local beta functions, which is the subject of the caveat "Two-coupling closure" in Sec. 2.9 and of Appendix D.

A.1 Notation

As in the main text, $\bar{\delta} = 1 - s$ and $x = \Delta_\sigma = \bar{\delta}/2$; the compendium writes δ for $\bar{\delta}$ (record and input names such as N_δ keep their spelling), C_n for C_n^{SR}, b_{dim} for the resummed charge b and D_{2m} for Ξ_{2m}. The hard core is ε and $\ell = \log\varepsilon$; the thermal operator is $\mathcal{E}$, with dimension $\Delta_\mathcal{E}$. Table 4 collects the symbols of the dual that this appendix and the two following ones use.

Table 4: Symbols of the dual of the chain. The calligraphic $\mathcal{C}_n$ and $\mathcal{K}_n$ are normalized responses (archived programs call them C_n and K_n); they are never the thermal coefficients C_n^{SR}.

symbol	meaning
X, X_i, q_i	coordinate along the chain; ordered kink (wall) positions; wall charges ±1, alternating, $q_i = (-1)^{i+1}$
ε, ℓ, $\Omega_n(\varepsilon)$	dimensionless hard core between consecutive kinks; $\ell = \log\varepsilon$; ordered domain of n kinks, Eq. (A.14)
κ, $f_{\bar{\delta}}$	reference inverse length (set to 1 from Appendix A.4 on); wall interaction function, Eq. (A.5)
ϕ, φ	Gaussian field of the dual; coarse-grained Ising field (different fields)
$\hat{\sigma}_3$, $\hat{\sigma}_\pm$, σ	qubit matrices; spin primary (unit normalized)
$\mathcal{O}_g$, $\mathcal{O}_h$, χ_B, χ	kink operator and current operator of Eq. (2.1); bare current $\frac{\mathrm{i}}{\sqrt{2}}\partial\phi$ (written χ in Eq. (2.1)); renormalized unit-normalized current primary
g, h, b_0, b	auxiliary wall fugacity and current coupling; reference charge; resummed charge $b = b_0 - \sqrt{2}h$
η, ∂_η	endpoint source $\sqrt{2}hb$, Eq. (A.17); derivative with respect to η at $\eta = 0$ (the compendium's script-E operator)
B, v, z; Y, u	essential charge, $v = B^2$, $z = v - 1$; essential wall fugacity (renamed g after matching), $u = g^2$
F, H	essential beta functions divided by $2u$ and by $1 + z$, Eq. (A.23)
T, D; T_m, D_m	trace and determinant of the stability matrix in (u, z); their coefficients of $\bar{\delta}^m$
W_n, Ξ_{2m}	wall weight of n ordered kinks; total length of the reversed domains, Eq. (A.31)
A_{2m+1}, Z_{2m}, Q_{2m}	odd, vacuum and current moments, Eq. (A.31)
$\mathcal{C}_{2m+1}$, $\mathcal{K}_{2m}$	vacuum-normalized odd and current responses, Eq. (A.37)
I_0, I_1	coefficients $[\varepsilon^0\ell^0]I$ and $[\varepsilon^0\ell^1]I$ of a response I; e.g. $\mathcal{C}_{3,1}$, $\mathcal{K}_{4,0}$, $Q_{2,0}$
S_j	charge-weighted sums of $\log^j$ of the kink separations, Eq. (A.41)
j_n, λ_{2m}	entries of the finite-response Jacobian, Eq. (A.42)
b_n (n odd), h_n (n even)	coefficients of the auxiliary fugacity and current beta functions, Eq. (A.43) (archived scripts write η_4, η_6 for h_4, h_6); unrelated to b_0, b, h and η
$a, p, b, c, f, \ldots$	beta inputs: Taylor coefficients of F and H at $v = 1$, $\bar{\delta} = 0$ (Sec. 2.3); the input b is b_5, not the charge

A.2 The classical model and the short-range end

The object of interest is the thermal dimension of the *classical* one-dimensional long-range Ising critical family, not the spectrum of the quantum transverse-field Ising chain. A microscopic representative is the chain of spins $m_i = \pm1$ with reduced Hamiltonian (energy divided by temperature)

$$\mathcal{H}_{\mathrm{red}} = -K_{\mathrm{nn}} \sum_i m_i m_{i+1} - K_{\mathrm{lr}} \sum_{i<j} \frac{m_i m_j}{|i-j|^{1+s}}, \qquad K_{\mathrm{lr}} > 0. \tag{A.1}$$

The nearest-neighbour coupling changes the core energies of the walls and the nonuniversal critical temperature; it need not be present separately. The long-range critical regime of interest is $\frac{1}{2} < s < 1$.

The endpoint $s = 1$ is marginal, and for $s > 1$ there is no finite-temperature transition [3, 4].

The two ends of the long-range window. The familiar coarse-grained action is

$$S[\varphi] = \frac{c_s}{4} \int \mathrm{d}X\, \mathrm{d}Y\, \frac{[\varphi(X) - \varphi(Y)]^2}{|X - Y|^{1+s}} + \int \mathrm{d}X \Big(\frac{r_\varphi}{2} \varphi^2 + \frac{\lambda_\varphi}{4} \varphi^4 \Big), \qquad c_s > 0. \tag{A.2}$$

Its nonanalytic quadratic kernel is proportional to $|p|^s$ in momentum space and gives $\Delta_\varphi = (1-s)/2$, which is Δ_σ. The quartic coupling has engineering dimension $1 - 4\Delta_\varphi = 2s - 1$. The expansion about the generalized free field (GFF) is therefore weak near $s = \frac{1}{2}$, whereas the dual used here is weak near $s = 1$. These are opposite ends of the same critical family, not the same perturbative expansion [4, 23].

What sits at the short-range end in $d = 1$. For the nearest-neighbour chain alone, the transfer matrix is

$$\mathsf{T}_{\mathrm{nn}} = \begin{pmatrix} e^{K_{\mathrm{nn}}} & e^{-K_{\mathrm{nn}}} \\ e^{-K_{\mathrm{nn}}} & e^{K_{\mathrm{nn}}} \end{pmatrix} = \mathcal{N} \exp(\gamma \hat{\sigma}_1), \qquad \tanh \gamma = e^{-2K_{\mathrm{nn}}}, \tag{A.3}$$

with a constant $\mathcal{N} > 0$. Its transfer Hamiltonian is a single two-state system, $\hat{H} = -\gamma \hat{\sigma}_1$ up to an additive constant, and the inverse correlation length is $\xi^{-1} = 2\gamma = \log \coth K_{\mathrm{nn}}$. At zero temperature $\gamma \to 0$, and the symmetric normalized trace gives $\frac{1}{2} \operatorname{tr} \hat{\sigma}_3^2 = 1$: the spin $\hat{\sigma}_3$ then has dimension zero. This endpoint spin sector plays, in $d = 1$, the role of the short-range fixed point ($\Delta_\sigma^{\mathrm{SR}} = 0$, Table 1); it is not an ordinary finite-temperature one-dimensional conformal field theory. In $d = 2, 3$ the crossover theory couples the short-range fixed point to a generalized free field through the spin (Sec. 3). In $d = 1$ the analogous diagonal coupling of the dual, $h\, \mathcal{O}_h = h \hat{\sigma}_3 \chi_B$, commutes with $\hat{\sigma}_3$ and cannot flip the spin, so operators that create kinks are essential. The higher-dimensional crossover construction therefore has a special one-dimensional realization, the dual of Sec. 2.1 [4, 11, 12].

A.3 Alternating domain walls, the Gaussian field and the matrices

Domain walls. Let $m(X) = \pm 1$ be piecewise constant along the chain, with walls at X_i of charge q_i:

$$\partial_X m(X) = 2 \sum_i q_i\, \delta(X - X_i), \qquad q_i = \pm 1, \tag{A.4}$$

where $\delta(X)$ is the Dirac distribution. Ising walls necessarily alternate in sign, and a configuration that returns to its initial spin has an even number of walls and zero total charge. With a reference inverse length κ, define

$$f_{\bar{\delta}}(r) = \frac{r^{\bar{\delta}} - \kappa^{-\bar{\delta}}}{\bar{\delta}}, \qquad f_0(r) = \log(\kappa r), \qquad f_{\bar{\delta}}''(r) = -(1 - \bar{\delta})\, r^{-2+\bar{\delta}} \quad (r > 0). \tag{A.5}$$

From Appendix A.4 on $\kappa = 1$, so that $f_{\bar{\delta}}(r) = (r^{\bar{\delta}} - 1)/\bar{\delta}$ as in Sec. 2.1. Take the continuum tail action, relative to the uniform configuration, to be

$$S_{\mathrm{tail}}[m] = \frac{\kappa_s}{4} \int \mathrm{d}X\, \mathrm{d}Y\, \frac{[m(X) - m(Y)]^2}{|X - Y|^{2-\bar{\delta}}}, \tag{A.6}$$

the long-range term of Eq. (A.1) with $1 + s = 2 - \bar{\delta}$. Since $\partial_X \partial_Y f_{\bar{\delta}}(|X - Y|) = (1 - \bar{\delta})|X - Y|^{-2+\bar{\delta}}$ away from contact points, two integrations by parts give

$$S_{\mathrm{tail}} = -\frac{\kappa_s}{2(1 - \bar{\delta})} \int \mathrm{d}X\, \mathrm{d}Y\, m'(X)\, m'(Y)\, f_{\bar{\delta}}(|X - Y|) + \text{core and boundary terms}. \tag{A.7}$$

Inserting the wall charges, removing the self-energies and absorbing the positive coefficient into $b_{\mathrm{w}}^2 = 2\kappa_s/(1-\bar{\delta})$ gives the wall action

$$S_{\mathrm{walls}} = -2b_{\mathrm{w}}^2 \sum_{i<j} q_i q_j \, f_{\bar{\delta}}(X_j - X_i) + N_{\mathrm{w}} E_{\mathrm{core}} + \text{boundary terms}, \tag{A.8}$$

with N_{w} the number of walls and E_{core} their core energy. As a check of the sign, an isolated pair of opposite charges at separation r has interaction energy $2b_{\mathrm{w}}^2 f_{\bar{\delta}}(r)$, which increases with r and is positive beyond the reference length $1/\kappa$. Short-distance details are encoded by a fugacity and by local core counterterms.

On a finite interval the regulated neutral gas is

$$Z_{\mathrm{gas}} = \sum_{m\geq 0} g^{2m} \int_{\Omega_{2m}(\varepsilon)} \prod_i \mathrm{d}X_i \, \exp\Big[2b_{\mathrm{w}}^2 \sum_{i<j} q_i q_j \, f_{\bar{\delta}}(X_j - X_i)\Big], \tag{A.9}$$

up to the overall two-state trace and reference-length normalizations, with the ordered hard-core domain $\Omega_n(\varepsilon)$ of Eq. (A.14). The positions are ordered, so there is no additional divisor $(2m)!$. The odd projected responses of Appendix A.7 carry a specified compensating external charge; they are not neutral partition functions.

The Gaussian field and its zero mode. The wall weight is reproduced by vertex operators of a Gaussian field. Choose a real Gaussian field modulo constant shifts, with covariance on neutral test functions

$$C_{\bar{\delta}}(X - Y) = -2 f_{\bar{\delta}}(|X - Y|), \tag{A.10}$$

the covariance of Sec. 2.1. With a finite-volume momentum cutoff and the zero mode removed, it is defined by the positive quadratic action

$$S_0[\phi] = \frac{\mathcal{A}_{\bar{\delta}}}{2} \int \frac{\mathrm{d}p}{2\pi} \, |p|^{1+\bar{\delta}} \, |\widetilde{\phi}(p)|^2, \qquad \mathcal{A}_{\bar{\delta}} > 0. \tag{A.11}$$

The constant in $C_{\bar{\delta}}$ is immaterial for neutral observables, and ϕ itself is not a shift-invariant local primary. Its increments have nonnegative variance. The formal scaling dimension of ϕ is $-\bar{\delta}/2$, but this does not license treating its unphysical zero mode as a negative-norm primary.

Compactifying the constant mode with period $2\pi/b_0$ implements charge neutrality for the vertices $V_\pm = \kappa \, {:}e^{\pm \mathrm{i} b_0 \phi}{:}$. For $\sum_i q_i = 0$, Wick's rule gives

$$\Big\langle \prod_i {:}e^{\mathrm{i} b_0 q_i \phi(X_i)}{:} \Big\rangle_0 = \exp\Big[-b_0^2 \sum_{i<j} q_i q_j \, C_{\bar{\delta}}(X_i - X_j)\Big] = \exp\Big[2b_0^2 \sum_{i<j} q_i q_j \, f_{\bar{\delta}}(X_j - X_i)\Big], \tag{A.12}$$

which is the nonlocal wall weight of Eq. (A.8) with $b_0^2 = b_{\mathrm{w}}^2$. At $\bar{\delta} = 0$ the two-point function of opposite charges is proportional to $r^{-2b_0^2}$.

Matrices enforce the Ising constraint. With these vertices the operators of Eq. (2.1) are $\mathcal{O}_g = \hat{\sigma}_+ V_+ + \hat{\sigma}_- V_-$ and $\mathcal{O}_h = \hat{\sigma}_3 \chi_B$, with $\chi_B = \frac{\mathrm{i}}{\sqrt{2}} \partial\phi$ the bare current (the subscript distinguishes it from the renormalized primary χ of Appendix A.6). Their path-ordered evolution from U to V is

$$\mathcal{D}(V, U) = \mathrm{P} \exp \int_U^V \mathrm{d}X \, \big[g \, \mathcal{O}_g(X) + h \, \mathcal{O}_h(X)\big]. \tag{A.13}$$

Because $\hat{\sigma}_+^2 = \hat{\sigma}_-^2 = 0$, only alternating charges survive in its expansion, so the traced matrix expansion matches the regulated Ising gas, whereas an unconstrained cosine interaction would also admit nonalternating walls. Spin correlators are obtained by inserting $\hat{\sigma}_3$ in the ordered trace. Two insertions multiply a wall history by $(-1)^{N_{\mathrm{between}}}$, with N_{between} the number of walls between them, which is exactly the product of the two Ising spins.

The auxiliary model also contains charged objects that are not physical operators of the chain. Constant shifts of ϕ compensated by rotations of $\hat{\sigma}_\pm$ define a $U(1)$ action, and the physical operator algebra is its invariant sector (Sec. 2.1). The equality of consistently regulated wall weights and matrix histories is a direct algebraic statement. The identification of the critical continuum limit with the long-range Ising CFT is the published duality proposal and its tested consequences [4]; this calculation supplies no all-orders theorem on the lattice scaling limit.

A.4 The regulator and exact current resummation

Work on an interval of length one, with $\kappa = 1$. All ordered integrations use

$$\Omega_n(\varepsilon) = \{0 < X_1 < \cdots < X_n < 1 : \ X_{i+1} - X_i \geq \varepsilon\}, \qquad \ell = \log \varepsilon. \tag{A.14}$$

Only consecutive kink separations have a hard core; there is no separate hard core between a wall and an endpoint. Current insertions are resummed *before* this regulator is imposed. An independent hard core on unresolved current insertions would change the finite parts of the responses, and could not be combined with the formulas below.

Change of frame. The resummation is a field-dependent change of frame,

$$R(X) = \exp\big[\mathrm{i}h\,\hat{\sigma}_3\,\phi(X)/\sqrt{2}\big], \qquad \mathcal{D}(V,U) = R(V)\,\widetilde{\mathcal{D}}(V,U)\,R(U)^{-1}, \tag{A.15}$$

which removes the bulk derivative interaction $h\,\mathcal{O}_h$. Conjugation by R multiplies $\hat{\sigma}_\pm$ by $e^{\mp\mathrm{i}\sqrt{2}h\phi}$ and so shifts the bulk wall charge to

$$b = b_0 - \sqrt{2}h, \tag{A.16}$$

the effective charge of Sec. 2.1 (the beta input b of Sec. 2.3 is a different quantity). The endpoint matrices $R(1)$ and $R(0)^{-1}$ remain. Equivalently, one integrates by parts on the interval I, $\int m\,\partial\phi = m\phi|_{\partial I} - \int m'\phi$. The endpoint charges have magnitude $h/\sqrt{2}$, and their contraction with a wall produces the *endpoint source*

$$\eta = \sqrt{2}\,h\,b = \sqrt{2}\,b_0 h - 2h^2. \tag{A.17}$$

The coefficient of h^2 is -2. An intermediate higher-logarithm check in the archive used -1; the beta coefficients below use only first h derivatives at $h = 0$ and are unaffected, but checks that involve second or higher source derivatives are.

Endpoint factors. With alternating charges $q_i = (-1)^{i+1}$, the endpoint factors of n kinks are

$$\begin{aligned} U_n^{\mathrm{odd}}(\eta) &= \exp\Big\{\eta \sum_i q_i \big[f_{\bar{\delta}}(X_i) + f_{\bar{\delta}}(1 - X_i)\big]\Big\} \quad (n \text{ odd}), \\ U_n^{\mathrm{even}}(\eta) &= \exp\Big\{\eta \sum_i q_i \big[f_{\bar{\delta}}(X_i) - f_{\bar{\delta}}(1 - X_i)\big]\Big\} \quad (n \text{ even}). \end{aligned} \tag{A.18}$$

We write ∂_η for the derivative with respect to η at $\eta = 0$ (the compendium writes it as a script E). By the reflection $X \to 1 - X$, a *first* source derivative may be represented after integration by $2\sum_i q_i f_{\bar{\delta}}(X_i)$; this replacement is not valid before squaring, for a second derivative. At the reference charge $v = b_0^2$ the coupling h enters the moments through the squared charge $v = b^2 = (b_0 - \sqrt{2}h)^2$ of the wall weight W_n (Eq. (A.31); it equals the essential B^2 of Appendix A.5 at $\bar{\delta} = 0$) and through η, so that

$$\partial_h\big|_{h=0} = \sqrt{2}\,b_0\,(-2\partial_v + \partial_\eta). \tag{A.19}$$

For example, for a single kink at $\bar{\delta} = 0$ and $b_0 = 1$ the odd endpoint factor is $X^\eta(1 - X)^\eta$, so the one-kink moment is Euler's beta function, $A_1(h) = \Gamma(1 + \eta)^2/\Gamma(2 + 2\eta)$, and

$$A_1(h) = 1 - 2\sqrt{2}\,h + (12 - 2\zeta(2))h^2 + O(h^3). \tag{A.20}$$

A.5 Essential couplings and finite-tail matching

Normal ordering at the cutoff and the charge shift give the essential charge B and the essential wall fugacity Y,

$$B = (b_0 - \sqrt{2}h)\,\varepsilon^{\bar\delta/2}, \qquad Y = g_{\rm aux}\,\kappa\varepsilon\,\exp\Big[\frac{b_0^2(\kappa^{-\bar\delta} - \varepsilon^{\bar\delta})}{\bar\delta}\Big], \tag{A.21}$$

where $g_{\rm aux}$ is the fugacity of the auxiliary (unmatched) theory. The h dependence cancels from Y once both normal-ordering factors are included. At unit reference length and $b_0 = 1$,

$$Y = g_{\rm aux}\exp\Big[\ell + \frac{1 - e^{\bar\delta\ell}}{\bar\delta}\Big] = g_{\rm aux}\Big[1 - \frac{\bar\delta}{2}\ell^2 + \bar\delta^2\Big(\frac{\ell^4}{8} - \frac{\ell^3}{6}\Big) + \cdots\Big]. \tag{A.22}$$

Charge derivatives require keeping b_0 variable until they are taken; setting $b_0 = 1$ prematurely loses terms. With Y renamed g, the essential flow is written as in Sec. 2.1,

$$u = g^2, \qquad v = B^2 = 1 + z, \qquad \beta_u = 2uF, \qquad \beta_z = (1+z)H, \qquad \beta = \mu\frac{\mathrm{d}}{\mathrm{d}\mu} = -\frac{\mathrm{d}}{\mathrm{d}\ell}. \tag{A.23}$$

The finite $\bar\delta$ factors of Eq. (A.22) must be matched together with the endpoint integrals. It is not consistent to compute at $\bar\delta = 0$ in one scheme and append the classical-scaling term in unrelated coordinates. The finite-tail factor is needed, for example, for the input p of Sec. 2.3 (Appendix B.9).

A.6 Leading spectrum and operator dictionary

This subsection collects the published leading-order results [3, 4] in the notation of this appendix. The published field-theory and bootstrap methods agree at their stated orders. They give

$$\Delta_\pm = 1 \pm \sqrt{2\bar\delta} + \frac{\bar\delta}{4} + O(\bar\delta^{3/2}), \qquad \Delta_\sigma = \frac{\bar\delta}{2}, \qquad \Delta_\chi = 1 - \frac{\bar\delta}{2}. \tag{A.24}$$

The coefficients $C_3^{\rm SR}$ and beyond are not quoted from those papers; they belong to the computation described in this and the following appendices and carry its conditions (Sec. 2.1).

OPE and leading flow. At $\bar\delta = 0$ and $b_0 = 1$ the ordered OPE of two kink operators is

$$\mathcal{O}_g(X)\,\mathcal{O}_g(0) = X^{-2}\,\mathbb{1} + \sqrt{2}\,X^{-1}\,\mathcal{O}_h(0) + \cdots, \qquad X > 0, \tag{A.25}$$

and the two ordered mixed OPEs, of $\mathcal{O}_g$ with $\mathcal{O}_h$ in either order, each contain $\sqrt{2}\,\mathcal{O}_g/X$. The beta functions of the dual through the published order are

$$\beta_g = -2\sqrt{2}\,gh + 2gh^2 + g^3 + \cdots, \qquad \beta_h = -\sqrt{2}\,g^2 + 2hg^2 + \cdots. \tag{A.26}$$

In the essential couplings, with $z = b^2 - 1 = -2\sqrt{2}h + 2h^2$ at $b_0 = 1$ and $\bar\delta = 0$, they become $\beta_g = gF$ and $\beta_z = 2b\,\partial_h b\,\beta_h = (1+z)H$ with $F = z + u + \cdots$ and $H = 4u + \cdots$; at $\bar\delta \neq 0$ the classical scaling of $B = b\,\varepsilon^{\bar\delta/2}$ adds $-\bar\delta$ to H, so that $H = -\bar\delta + 4u + \cdots$, whose fixed point $u_* = \bar\delta/4$, $z_* = -\bar\delta/4$ is given in Sec. 2.1. This explains the expansion in $\sqrt{\bar\delta}$: the fugacity itself is of order $\sqrt{\bar\delta}$, although u and z are of order $\bar\delta$. In the coordinates (g, v) the leading stability matrix is

$$M_* = \begin{pmatrix} \bar\delta/2 & g_* \\ 8(1 - \bar\delta/4)\,g_* & 0 \end{pmatrix} + \text{higher-order terms}. \tag{A.27}$$

A linear beta eigenvalue λ corresponds to the dimension $1 + \lambda$; with $g_*^2 = \bar\delta/4$ the eigenvalues of M_* give $\Delta_\pm$ of Eq. (A.24). The relevant branch is the thermal primary $\mathcal{E} = \mathcal{O}_-$, so $\nu^{-1} = 1 - \Delta_\mathcal{E}$, and the other eigenvalue gives the leading correction exponent $\omega = \Delta_+ - 1$. Only the orders shown in Eq. (A.24) are controlled by this truncation. Expanding the small matrix to more orders does not compute the higher coefficients (Appendix A.10).

Spin dimension. A direct wall-gas check gives, for the spin two-point function at separation r, $G_\sigma(r) - 1 = -4g^2 \log(r/\varepsilon) + \cdots$, because a kink pair changes the spin product only when one wall lies between the insertions. Thus $\gamma_\sigma = 2g^2 + \cdots$, which reproduces $\Delta_\sigma = \bar{\delta}/2$ at the fixed point.

Equation of motion and the shadow pair. The covariant derivative of an insertion is $D_X\mathcal{O} = \partial_X\mathcal{O} - [g\,\mathcal{O}_g + h\,\mathcal{O}_h, \mathcal{O}]$. In particular

$$D_X\hat{\sigma}_3 = 2g(\hat{\sigma}_+V_+ - \hat{\sigma}_-V_-), \qquad C_{\bar{\delta}}^{-1}\phi = \frac{\mathrm{i}b}{2}\,D_X\hat{\sigma}_3 \tag{A.28}$$

away from contact terms, where $C_{\bar{\delta}}^{-1}$ is the inverse of the covariance (A.10). The nonanalytic Gaussian kernel and this equation of motion give the protected shadow pair of Eq. (A.24), $\Delta_\sigma + \Delta_\chi = 1$: the antisymmetric flip operator $\hat{\sigma}_+V_+ - \hat{\sigma}_-V_-$ becomes the descendant of σ. The bare current χ_B can mix with σ through power-divergent counterterms and needs a finite normalization; all CFT data below refer to the renormalized, unit-normalized primary χ, not to an unsubtracted current [4].

Published OPE coefficients. For completeness, with $\tau = \sqrt{\bar{\delta}} = \sqrt{2\Delta_\sigma}$, with $\pm$ labelling the operators $\mathcal{O}_\pm$ of dimensions $\Delta_\pm$, and with the published operator signs [4, eqs. (5.115) and (5.120)],

$$\begin{aligned}
c_{\sigma\chi\pm} &= \frac{1}{\sqrt{2}} \mp \frac{\tau}{16} - \frac{\tau^2}{256\sqrt{2}} + O(\tau^3),\\
c_{\sigma\sigma\pm} &= \pm\frac{\tau}{\sqrt{2}} - \frac{15}{16}\tau^2 \pm \frac{1}{\sqrt{2}}\Big(\frac{543}{256} - \frac{\pi^2}{6}\Big)\tau^3 + O(\tau^4),\\
c_{\chi\chi\pm} &= -\frac{\pi^2}{2}\tau^2 \mp \frac{11\pi^2}{16\sqrt{2}}\tau^3 + O(\tau^4),\\
c_{+++} &= \frac{3}{2} + \frac{39}{16\sqrt{2}}\tau + O(\tau^2), \qquad c_{---} = \frac{3}{2} - \frac{39}{16\sqrt{2}}\tau + O(\tau^2),\\
c_{++-} &= -\frac{1}{2} + \frac{\tau}{16\sqrt{2}} + O(\tau^2), \qquad c_{+--} = -\frac{1}{2} - \frac{\tau}{16\sqrt{2}} + O(\tau^2).
\end{aligned} \tag{A.29}$$

These OPE coefficients, including their π^2 terms, are published inputs, not loop results of the compendium. The compendium takes its published inputs from Ref. [4], Secs. 2–3 and App. B, Secs. 4.1–4.4, Eqs. (5.48), (5.53)–(5.54) and (5.115)–(5.120), and App. E.

A.7 Ordered moments and vacuum normalization

Moments. At unit reference length and with $q_i = (-1)^{i+1}$, define the wall weight of n ordered kinks and, for an even number of kinks, the total length of the reversed domains,

$$W_n(v, \bar{\delta}; \boldsymbol{X}) = \exp\Big[2v \sum_{i<j} q_i q_j \,\frac{(X_j - X_i)^{\bar{\delta}} - 1}{\bar{\delta}}\Big], \qquad \Xi_{2m}(\boldsymbol{X}) = \sum_{j=1}^{m}(X_{2j} - X_{2j-1}), \tag{A.30}$$

and the odd, vacuum and current moments

$$\begin{aligned}
A_{2m+1} &= \int_{\Omega_{2m+1}(\varepsilon)} W_{2m+1} \prod_i \mathrm{d}X_i,\\
Z_{2m} = \int_{\Omega_{2m}(\varepsilon)} W_{2m} \prod_i \mathrm{d}X_i, &\qquad Q_{2m} = \int_{\Omega_{2m}(\varepsilon)} \Xi_{2m} W_{2m} \prod_i \mathrm{d}X_i.
\end{aligned} \tag{A.31}$$

When the endpoint source is switched on, the integrands also carry the endpoint factors of Eq. (A.18), on which all ∂_η derivatives act (as in the example of A_1, Eq. (A.20)). The A moments are coefficients of charged projected responses, with their compensating external operator understood; they do not contradict the zero-mode neutrality of Z. At zero endpoint source $A_1 = 1$.

Gap variables and the first kernels. Integrating the first position gives a translation factor $1-R$ in the gap variables $r_i = X_{i+1} - X_i \geq \varepsilon$, with $R = \sum_i r_i \leq 1$. At $v = 1$ and $\bar{\delta} = 0$, $W_n = \prod_{i<j}(X_j - X_i)^{2q_iq_j}$; naming the consecutive gaps r, t, w, y, the first kernels are

$$W_3 = \frac{(r+t)^2}{r^2t^2}, \tag{A.32}$$

$$W_4 = \frac{(r+t)^2(t+w)^2}{r^2t^2w^2(r+t+w)^2}, \tag{A.33}$$

$$W_5 = \frac{(r+t)^2(t+w)^2(w+y)^2(r+t+w+y)^2}{r^2t^2w^2y^2(r+t+w)^2(t+w+y)^2}. \tag{A.34}$$

(The compendium names the second gap t'; it is unrelated to the trace T.) At $v = 1$, $\bar{\delta} = 0$ the two-kink moments are elementary, $Z_2 = \int_\varepsilon^1 (1-r)r^{-2}\,\mathrm{d}r$ and $Q_2 = \int_\varepsilon^1 (1-r)r^{-1}\,\mathrm{d}r$:

$$Z_2 = \varepsilon^{-1} + \ell - 1, \qquad Q_2 = -\ell - 1 + \varepsilon. \tag{A.35}$$

These elementary expressions fix the signs of the finite parts of the responses.

Vacuum normalization before coefficient extraction. The responses are divided by the vacuum before any coefficient is extracted. One forms generating functions in the fugacity and divides the *integrated* numerator by the integrated vacuum denominator,

$$\sum_{m\geq 0} \mathcal{C}_{2m+1}\, g^{2m} = \frac{\sum_{m\geq 0} A_{2m+1} g^{2m}}{\sum_{m\geq 0} Z_{2m} g^{2m}}, \qquad \sum_{m\geq 1} \mathcal{K}_{2m}\, g^{2m} = \frac{\sum_{m\geq 1} Q_{2m} g^{2m}}{\sum_{m\geq 0} Z_{2m} g^{2m}}, \qquad Z_0 = 1, \tag{A.36}$$

so that $\mathcal{C}_1 = A_1$ and $\mathcal{K}_2 = Q_2$. At zero source ($A_1 = 1$) this gives, through the orders used,

$$\begin{aligned}
\mathcal{C}_3 &= A_3 - Z_2, \\
\mathcal{C}_5 &= A_5 - A_3 Z_2 + Z_2^2 - Z_4, \\
\mathcal{C}_7 &= A_7 - A_5 Z_2 + A_3(Z_2^2 - Z_4) - Z_2^3 + 2Z_2Z_4 - Z_6, \\
\mathcal{K}_4 &= Q_4 - Q_2 Z_2, \\
\mathcal{K}_6 &= Q_6 - Q_4 Z_2 + Q_2(Z_2^2 - Z_4), \\
\mathcal{K}_8 &= Q_8 - Q_6 Z_2 + Q_4(Z_2^2 - Z_4) + Q_2(-Z_2^3 + 2Z_2Z_4 - Z_6).
\end{aligned} \tag{A.37}$$

Source derivatives differentiate *every* factor, including A_1; for example

$$\partial_\eta \mathcal{C}_3 = \partial_\eta A_3 - (\partial_\eta A_1)\, Z_2 - \partial_\eta Z_2. \tag{A.38}$$

For a response I define

$$I_0 = [\varepsilon^0 \ell^0]\, I, \qquad I_1 = [\varepsilon^0 \ell^1]\, I, \tag{A.39}$$

so that, for instance, $\mathcal{C}_{3,1}$ is the single-logarithm coefficient of $\mathcal{C}_3$ and $Q_{2,0} = -1$ the finite part of Q_2. The products in Eq. (A.37) must be formed, as complete series in ε and ℓ, before these projections are applied: since $Z_2 \sim \varepsilon^{-1}$, a term such as $\varepsilon\ell$ in another factor contributes to I_1. Multiplying only finite parts is not a valid normalization rule. Derivatives are likewise taken before the projection: $\partial_v \mathcal{C}_{n,0}$ and $\partial_\eta \mathcal{C}_{n,0}$ mean the finite parts of $\partial_v \mathcal{C}_n$ and $\partial_\eta \mathcal{C}_n$.

A.8 Charge and tail insertions

Derivatives of the wall weight in the charge v and the tail parameter $\bar{\delta}$ insert logarithms of the kink separations. Let

$$S_j = \sum_{i<k} q_i q_k \log^j(X_k - X_i). \tag{A.40}$$

Since $[(X_k - X_i)^{\bar{\delta}} - 1]/\bar{\delta} = \log(X_k - X_i) + \frac{\bar{\delta}}{2}\log^2(X_k - X_i) + \frac{\bar{\delta}^2}{6}\log^3(X_k - X_i) + \cdots$, the derivatives at $v = 1$, $\bar{\delta} = 0$ are

$$\begin{aligned} \partial_v W_n &= 2S_1 W_n, & \partial_v^2 W_n &= 4S_1^2 W_n, \\ \partial_{\bar{\delta}} W_n &= S_2 W_n, & \partial_v \partial_{\bar{\delta}} W_n &= (S_2 + 2S_1 S_2) W_n, \\ [\bar{\delta}^2]\, W_n &= \Big(\frac{1}{3}S_3 + \frac{1}{2}S_2^2\Big) W_n. \end{aligned} \tag{A.41}$$

The second term of the mixed derivative and both terms of the second tail insertion are essential. Tail derivatives also act on the endpoint factors of Eq. (A.18), through $f_{\bar{\delta}}$. A Taylor coefficient such as $[\bar{\delta}^2]$ includes the factor $1/2!$; it is not the second derivative itself.

A.9 The finite-response Jacobian and the subtraction recurrence

The beta functions are extracted from two vacuum-normalized responses at $h = 0$: the odd response, the projection of the ordered evolution on the odd spin-flip vertex, with finite parts built from the $\mathcal{C}_{2m+1}$, and the current response, its projection on the current, built from the $\mathcal{K}_{2m}$ (with $\mathcal{K}_2 = Q_2$). The regulator convention subtracts logarithmic coupling counterterms with no additional finite coupling redefinition in these coordinates. The finite parts of the responses are nevertheless required: the single logarithm of a normalized response is not by itself a beta coefficient, because finite parts of lower orders, multiplied by lower beta coefficients, also contribute to it.

Jacobian entries. Set

$$\begin{aligned} j_1 &= -\frac{2}{1+\bar{\delta}}, \qquad j_n = -2\partial_v \mathcal{C}_{n,0} + \partial_\eta \mathcal{C}_{n,0} \quad (n = 3, 5), \\ \lambda_2 &= -2Q_{2,0} - 4v\,\partial_v Q_{2,0} + 2v\,\partial_\eta Q_{2,0}, \\ \lambda_4 &= -2\mathcal{K}_{4,0} - 4v\,\partial_v \mathcal{K}_{4,0} + 2v\,\partial_\eta \mathcal{K}_{4,0}. \end{aligned} \tag{A.42}$$

The integral defining j_1 is $\int_0^1 [f_{\bar{\delta}}(X) + f_{\bar{\delta}}(1-X)]\,\mathrm{d}X$, the source derivative $\partial_\eta A_1$ of the one-kink moment. The same definition of j_7 is used at the eighth order (Appendix C.3).

Auxiliary beta coefficients. Write the beta functions of the auxiliary couplings at $h = 0$ as

$$\beta_g = b_3 g^3 + b_5 g^5 + b_7 g^7 + \cdots, \qquad \frac{\beta_h}{\sqrt{2}\,b_0} = -g^2 + h_4 g^4 + h_6 g^6 + \cdots, \tag{A.43}$$

so that b_n (n odd) are the odd fugacity coefficients and h_n (n even, $h_2 = -1$) the normalized current coefficients. Archived scripts call h_4 and h_6 by η_4 and η_6; these are not the endpoint source η. The coefficients depend on the charge $v = b_0^2$ and on $\bar{\delta}$, and their Taylor coefficients at $v = 1$, $\bar{\delta} = 0$, after the matching factors of Appendix A.5 are restored, are the beta inputs of Sec. 2.3. For example $b_3 = 1$ and $h_2 = -1$ at that point give the leading terms u of F and $4u$ of H, while $b = b_5$, $c = -4h_4$ and $k = -4h_6$ are the coefficients of u^2 in F and of u^2 and u^3 in H; the complete dictionary is in Appendices B.1 and C.3.

The finite-response Jacobian. Through the required orders the Jacobian is

$$\mathcal{J} = \begin{pmatrix} 1 + 3\mathcal{C}_{3,0}g^2 + 5\mathcal{C}_{5,0}g^4 & \sqrt{2}\, b_0\,(j_1 g + j_3 g^3 + j_5 g^5) \\ \sqrt{2}\, b_0\,(2Q_{2,0}g + 4\mathcal{K}_{4,0}g^3) & 1 + \lambda_2 g^2 + \lambda_4 g^4 \end{pmatrix}, \tag{A.44}$$

and the vector of single-logarithm coefficients is

$$R_1 = \begin{pmatrix} \mathcal{C}_{3,1}g^3 + \mathcal{C}_{5,1}g^5 + \mathcal{C}_{7,1}g^7 \\ \sqrt{2}\, b_0\,(-g^2 + \mathcal{K}_{4,1}g^4 + \mathcal{K}_{6,1}g^6) \end{pmatrix}. \tag{A.45}$$

The entries have a direct reading. The first column holds the g derivatives of the finite parts of the odd response, $g + \mathcal{C}_{3,0}g^3 + \mathcal{C}_{5,0}g^5$, and of the current response, $h + \sqrt{2}\, b\,(Q_{2,0}g^2 + \mathcal{K}_{4,0}g^4)$. The second column holds their h derivatives at $h = 0$, taken with Eq. (A.19); the terms $-2Q_{2,0}$ and $-2\mathcal{K}_{4,0}$ of λ_2 and λ_4 are the derivative of the charge $b = b_0 - \sqrt{2}h$ that multiplies the current moments. The leading $-g^2$ in the second row of R_1 is $Q_{2,1} = -1$, from Eq. (A.35).

The subtraction recurrence. The prescription solves

$$\mathcal{J} \begin{pmatrix} \beta_g \\ \beta_h \end{pmatrix} = R_1 \tag{A.46}$$

order by order in g. With $v = b_0^2$ (the factors $2v$ are $(\sqrt{2}b_0)^2$) this gives

$$\begin{aligned} b_3 &= \mathcal{C}_{3,1} + 2vj_1, \\ h_4 &= \mathcal{K}_{4,1} - 2Q_{2,0}b_3 + \lambda_2, \\ b_5 &= \mathcal{C}_{5,1} - 3\mathcal{C}_{3,0}b_3 - 2vj_1h_4 + 2vj_3, \\ h_6 &= \mathcal{K}_{6,1} - 2Q_{2,0}b_5 - 4\mathcal{K}_{4,0}b_3 - \lambda_2 h_4 + \lambda_4, \\ b_7 &= \mathcal{C}_{7,1} - 3\mathcal{C}_{3,0}b_5 - 5\mathcal{C}_{5,0}b_3 - 2vj_1h_6 - 2vj_3h_4 + 2vj_5. \end{aligned} \tag{A.47}$$

The classical-scaling and matching contributions of Eq. (A.21) are then restored to pass from the auxiliary couplings (g_{aux}, b) to the essential couplings (u, z). This derives the recurrence; it is not an assignment of a raw single logarithm to a physical beta function. The recurrence uses only first h derivatives at $h = 0$, which is why the coefficient of h^2 in Eq. (A.17) does not enter it. Its extension through the ninth order (h_8 and b_9, with $\mathcal{C}_9$) is in Appendix C.3, Eqs. (C.29) and (C.30).

The lowest order as a check of signs. The normalized three-kink response has $\mathcal{C}_{3,1} = 5$ (its expansion is in Appendix B.2), and $j_1 = -2$ at $\bar{\delta} = 0$, so $b_3 = 5 - 4 = 1$. With $\mathcal{K}_{4,1} = -8 + 4\zeta(2)$, $Q_{2,0} = -1$, $\partial_v Q_{2,0} = -2$ and $\partial_\eta Q_{2,0} = -2\zeta(2)$, one has $\lambda_2 = 10 - 4\zeta(2)$ and $h_4 = 4$, that is $c = -4h_4 = -16$ (Sec. 2.3).

Scope. Equation (A.46) defines the beta functions of the prescription. That its solution is the local beta function of the dual, up to a finite change of (g, b) that leaves T and D unchanged, is the formal argument of Appendix D (Appendices D.3 and D.13), summarized in the caveat "Two-coupling closure" of Sec. 2.9.

A.10 Stability invariants and the limits of the leading flow

Trace and determinant. Linearizing $(\beta_u, \beta_z) = (2uF, (1+z)H)$ about a fixed point with $F = H = 0$, the terms $2F$ and H of the derivatives vanish, and the stability matrix in (u, z) is

$$\begin{pmatrix} 2u_* F_u & 2u_* F_z \\ (1+z_*)H_u & (1+z_*)H_z \end{pmatrix}, \tag{A.48}$$

with the derivatives taken before the fixed point is substituted. Its trace and determinant are the T and D of Sec. 2.2, and its eigenvalues $\lambda_\pm = T/2 \pm (T^2/4 - D)^{1/2}$ give the dimensions $\Delta_\pm = 1 + \lambda_\pm$ of Eq. (2.6). A regular change of couplings transforms the exact stability matrix by similarity, so T and D are invariant although individual beta coefficients are not. The even coefficients need only the trace, $C^{\mathrm{SR}}_{2m} = 2^{m-1} T_m$ (Eq. (2.8)): a new even coefficient needs the next trace coefficient but not the next determinant coefficient.

Why the leading flow cannot be extrapolated. If one regarded the leading $F = z + u$ and $H = -\bar{\delta} + 4u$ as exact, the invariants would be $T = 2u_* = \bar{\delta}/2$ and $D = -8u_*(1+z_*) = -2\bar{\delta} + \bar{\delta}^2/2$, so $D_2 = \frac{1}{2}$ and $T_2 = 0$, and Eq. (2.8) would give $C^{\mathrm{SR}}_3 = \frac{7}{16}$ and $C^{\mathrm{SR}}_4 = 0$ instead of the values of Eq. (2.12). Terms omitted in that truncation change both results. For instance, a term azu in F is of order $\bar{\delta}^2$ at the fixed point, but it adds au_*, of order $\bar{\delta}$, to the derivative F_z and so changes D_2 (indeed $D_2 = \frac{1}{2} - \frac{a}{2} - \frac{c}{8}$, Sec. 2.3). A term zu^2 in H has a value of order $\bar{\delta}^3$, but its derivative H_z is of order $\bar{\delta}^2$. Power counting must be applied *after* differentiation. This is the distinction between a higher-insertion calculation and a formal re-expansion of an already truncated flow, and it is why the beta inputs include derivatives in z and $\bar{\delta}$ of integrals with more kinks than the order would suggest. For the same reason, expanding the leading matrix M_* of Eq. (A.27) to more orders does not compute the higher coefficients.

B The dual of the chain: the third to sixth orders

This appendix gives the derivations of the compendium (written by OpenAI GPT-6 astra; Sec. 1.2) for the coefficients C^{SR}_3 to C^{SR}_6, edited to the notation of this note. It builds on the model, the regulator, the moments and the subtraction recurrence of Appendix A, and gives the calculations that Secs. 2.3 and 2.4 summarize for these orders; the inputs that rest on archived exact expressions are marked as such below.

Appendices B.2 to B.4 assemble the invariants D_2, T_2, D_3 and T_3 from the beta inputs. The remaining subsections evaluate the inputs: the Mellin representation of the regulated moments and the rules for finite parts, with the charge-derivative period $\mathcal{R}_4$ (Appendix B.5); a trilogarithm lemma and the Spence–Kummer identity on which it rests (Appendix B.6); the closed forms of the periods $\mathcal{P}_4$ and $\mathcal{S}_5$ (Appendix B.7); the counterterm calculation that attaches the periods to the inputs (Appendix B.8); the charge and tail derivatives at three insertions (Appendix B.9); and the three-kink and endpoint inputs of D_3 (Appendix B.10). What these derivations establish differs by order:

- C^{SR}_3 and C^{SR}_4. The three periods $\mathcal{P}_4$, $\mathcal{R}_4$ and $\mathcal{S}_5$ are evaluated in closed form with explicit primitives, a table of elementary finite parts and the trilogarithm lemma, without numerical recognition. Their attachment to the five inputs a, p, b, c, f goes through response coefficients and counterterms: the mixed response coefficients of Eq. (B.102) are taken from the archive without derivation, the counterterm algebra is displayed here in full, and the compendium checks the charge derivative of the fourth-order counterterm and the three-insertion derivatives with its code.
- C^{SR}_5. The inputs A and E and two endpoint constants are evaluated here. The inputs G, q, k are archived exact period expressions, and the combination of them that enters D_3 is

reduced to zeta values by the exact certificate of D_3 (Appendix C.1; Sec. 2.4).

- C_6^{SR}. The six new inputs are archived exact period expressions, and the closed form of C_6^{SR} is established by the exact certificate at weights two to five (same appendix). This appendix gives the assembly, a convergent check of one input and checks that do not use the new single logarithms.

Everything holds within the matched two-coupling prescription of Sec. 2.1, with the caveats stated there and in Sec. 2.9.

B.1 Conventions and the beta inputs

The notation is that of Table 4. In particular $\bar\delta = 1 - s$; ε is the hard core and $\ell = \log\varepsilon$; $\mathcal{C}_{2m+1}$ and $\mathcal{K}_{2m}$ are the vacuum-normalized responses of Eq. (A.37); a second subscript 0 or 1 denotes the coefficient $[\varepsilon^0\ell^0]$ or $[\varepsilon^0\ell^1]$ (Eq. (A.39)); ∂_η is the derivative with respect to the endpoint source at $\eta = 0$; and b_n (n odd) and h_n (n even) are the auxiliary beta coefficients of Eq. (A.43), determined by the recurrence (A.47). Derivatives in v and $\bar\delta$ are taken at the marginal point $v = 1$, $\bar\delta = 0$, before any fixed point is substituted. We write $\log^2 2$ for $(\log 2)^2$, and similarly for other powers.

The beta inputs of orders three to six. The beta inputs are the Taylor coefficients of F and H at the marginal point. In the compendium's dictionary, with the matching factors of Eq. (A.22) restored as described in Appendix A.9, and with a subscript v or $\bar\delta$ denoting a derivative of the auxiliary coefficient, the inputs used here are

$$\begin{gathered} a = b_{3,v}, \qquad p = b_{3,\bar\delta}, \qquad A = \tfrac{1}{2}b_{3,vv}, \qquad E = b_{3,v\bar\delta}, \qquad V = \tfrac{1}{2}b_{3,\bar\delta\bar\delta}, \\ b = b_5, \qquad G = b_{5,v}, \qquad R = b_{5,\bar\delta}, \qquad J = b_7, \\ c = -4h_4, \qquad f = -4h_{4,v}, \qquad q = -4h_{4,\bar\delta}, \qquad L = -2h_{4,vv}, \qquad M = -4h_{4,v\bar\delta}, \\ k = -4h_6, \qquad N = -4h_{6,v}. \end{gathered} \tag{B.1}$$

Taylor factors appear only for repeated derivatives; the mixed derivatives E and M carry none. The input $b = b_5$ is not the resummed charge $b_0 - \sqrt{2}h$, and the input q is not a kink charge or a Mellin variable.

Local symbols. From Appendix B.5 on, the letters e, t, w, x, y and z are integration or auxiliary variables defined where they are used; there x and z are never Δ_σ or the coupling z. Where the compendium uses one letter for two quantities, the local quantity is renamed; the renamed symbols are defined at first use.

B.2 Third and fourth orders: D_2 and T_2

Beta functions. To the first new orders the matched beta functions have the form

$$F = z + (1 + az + p\bar\delta)u + bu^2 + \cdots, \qquad H = -\bar\delta + 4u + cu^2 + fzu^2 + \cdots. \tag{B.2}$$

The one-pair calculation fixes the term $4u$ of H, with no further terms zu, z^2u, $\bar\delta u$ or $\bar\delta zu$ at these orders in this prescription. Indeed the one-pair current moment is

$$Q_2(v, \bar\delta; \varepsilon) = \int_\varepsilon^1 r(1-r)\, e^{-2vf_{\bar\delta}(r)}\, \mathrm{d}r, \tag{B.3}$$

with $f_{\bar\delta}$ of Eq. (A.5) at $\kappa = 1$. At the marginal point its single logarithm is $-\ell$ (Eq. (A.35)), while derivatives in the charge or the tail parameter add powers of $\log r$ to the integrand and so produce higher powers of ℓ, not a new single logarithm (Appendix B.8). The tree coefficient of the fugacity in the essential variables is exactly z.

Inputs and periods. The five inputs a, p, b, c, f are listed in Sec. 2.3. They are fixed by elementary constants and three regulator periods: $\mathcal{P}_4$ and $\mathcal{R}_4$, Laurent coefficients of the angular Mellin function of the four-kink current moment and of its charge derivative, and $\mathcal{S}_5$, a Laurent coefficient of the angular Mellin function of the five-kink odd moment (Eq. (B.36)). The derivations of Appendices B.8 and B.9, and of Appendix A.9 for c, give

$$\begin{aligned} a &= -4\zeta(2), \qquad c = -16, \qquad p = 8 - 2\zeta(2) - 7\zeta(3) - 8\log 2, \\ b &= -21 + 38\zeta(2) + 60\zeta(3) + 32\log 2 - \mathcal{S}_5, \qquad f = 16\mathcal{P}_4 + 4\mathcal{R}_4 - 32 + 16\zeta(2) + 32\zeta(3), \end{aligned} \tag{B.4}$$

and Appendices B.5 and B.7 evaluate the periods,

$$\begin{aligned} \mathcal{P}_4 &= 4\zeta(2) + \tfrac{8}{3}\zeta(3) - 2 + 3\log 3 + \log^2 2 + 2\,\mathrm{Li}_2(-\tfrac{1}{2}), \\ \mathcal{R}_4 &= 24 - 20\zeta(2) - 30\zeta(3) - 16\log 2, \\ \mathcal{S}_5 &= 44\zeta(2) + \tfrac{152}{3}\zeta(3) - 3 + 12\log 3 + 4\log^2 2 + 8\,\mathrm{Li}_2(-\tfrac{1}{2}). \end{aligned} \tag{B.5}$$

In particular

$$\mathcal{S}_5 - 4\mathcal{P}_4 = 5 + 28\zeta(2) + 40\zeta(3). \tag{B.6}$$

With Eq. (B.5), Eq. (B.4) gives the values of b and f in Sec. 2.3. The regulator constants $\log 2$, $\log 3$, $\log^2 2$ and $\mathrm{Li}_2(-\frac{1}{2})$ are genuine parts of the individual inputs. In T_2 below, b and f enter as

$$\frac{b}{4} + \frac{f}{16} = -\frac{29}{4} + \frac{21}{2}\zeta(2) + 17\zeta(3) + 8\log 2 - \frac{\mathcal{S}_5 - 4\mathcal{P}_4}{4} + \frac{\mathcal{R}_4}{4}, \tag{B.7}$$

so the periods enter only through $\mathcal{R}_4$ and the combination (B.6), and the regulator constants cancel. The evaluation of $\mathcal{P}_4$ and $\mathcal{S}_5$ turns on one finite part,

$$J_L = \lim_{T\to\infty}\Big[\int_2^T \frac{\log(t-1)\log(t+1)}{t}\,\mathrm{d}t - \frac{1}{3}\log^3 T\Big] = -\frac{1}{3}\zeta(3), \tag{B.8}$$

which Appendix B.6 proves with explicit primitives and a real-branch Spence–Kummer identity, not by numerical recognition.

Normalized responses. With $\ell = \log\varepsilon$, the normalized low-order responses at $v = 1$, $\bar\delta = 0$ begin

$$\mathcal{C}_3 = 2\ell^2 + 5\ell + 3 - 2\zeta(2) + \cdots, \tag{B.9}$$

$$\mathcal{K}_4 = -\frac{4}{3}\ell^3 - 6\ell^2 + [-8 + 4\zeta(2)]\ell + \mathcal{P}_4 + 4\zeta(2) - 10 + \cdots, \tag{B.10}$$

$$\mathcal{C}_5 = \frac{10}{3}\ell^4 + \frac{62}{3}\ell^3 + \Big(\frac{95}{2} - 24\zeta(2)\Big)\ell^2 + [96 - 40\zeta(2) - \mathcal{S}_5]\ell + \cdots, \tag{B.11}$$

where the dots stand for terms that vanish as $\varepsilon \to 0$ and, in $\mathcal{C}_5$, for its finite part, which is not needed here. Their substitution into the subtraction recurrence (A.47), together with the finite parts and the charge and source derivatives of the lower responses, yields the inputs; the raw single logarithms alone do not. The finite-tail factor of Eq. (A.22) is needed for p (Appendix B.9).

Invariant assembly. Solving $H = 0$ and $F = 0$ order by order gives $u_* = \bar\delta/4 - c\,\bar\delta^2/64 + O(\bar\delta^3)$ and $z_* = -\bar\delta/4 + O(\bar\delta^2)$. In $D = 2u(1+z)(F_uH_z - F_zH_u)$ the derivative $H_z = fu^2 + \cdots$ is of order $\bar\delta^2$, so through order $\bar\delta^2$, with $F_z = 1 + au + \cdots$ and $H_u = 4 + 2cu + \cdots$,

$$D = -2u(1+z)F_zH_u + O(\bar\delta^3) = -8u\Big(1 + z + au + \frac{c}{2}u\Big) + O(\bar\delta^3). \tag{B.12}$$

In $T = 2uF_u + (1+z)H_z$, with $F_u = 1 + az + p\bar\delta + 2bu + \cdots$,

$$T = 2u\big(1 + az + p\bar\delta + 2bu\big) + fu^2 + O(\bar\delta^3). \tag{B.13}$$

Inserting the fixed point gives $D = -2\bar{\delta} + D_2\bar{\delta}^2 + \cdots$ and $T = \bar{\delta}/2 + T_2\bar{\delta}^2 + \cdots$ with

$$D_2 = \frac{1}{2} - \frac{a}{2} - \frac{c}{8}, \qquad T_2 = -\frac{a}{8} + \frac{b}{4} - \frac{c}{32} + \frac{f}{16} + \frac{p}{2}, \tag{B.14}$$

as in Sec. 2.3. With Eqs. (B.4) to (B.6) these are the closed forms $D_2 = \frac{5}{2} + 2\zeta(2)$ and $T_2 = 2 - 2\zeta(2) - 4\zeta(3)$ of Sec. 2.4, and Eq. (2.8) gives $C_3^{\mathrm{SR}} = D_2 - \frac{1}{16}$ and $C_4^{\mathrm{SR}} = 2T_2$ of Eq. (2.12). The highest insertion order needed for C_4^{SR} is five (the period $\mathcal{S}_5$); an insertion count is not a Feynman-loop count.

B.3 Fifth order: D_3

The determinant at order $\bar{\delta}^3$ needs the further sectors

$$\begin{aligned} F &= z + (1 + az + p\bar{\delta} + Az^2 + E\bar{\delta}z)u + (b + Gz)u^2 + \cdots, \\ H &= -\bar{\delta} + 4u + (c + fz + q\bar{\delta})u^2 + ku^3 + \cdots. \end{aligned} \tag{B.15}$$

The three-kink inputs A and E. They have convergent representations and closed forms,

$$\begin{aligned} A &= -8 - 4\zeta(2) - 2\zeta(3) + 16\log 2, \\ E &= -16 + 14\zeta(2) - 22\zeta(3) + 12\zeta(4) + 8\log 2 - 8\log^2 2 - 16\,\mathrm{Li}_4(\tfrac{1}{2}) \\ &\quad - 14\zeta(3)\log 2 + 4\zeta(2)\log^2 2 - \tfrac{2}{3}\log^4 2. \end{aligned} \tag{B.16}$$

For A one may evaluate

$$A = -4\int_0^{1/2} \frac{\log^2(1-y)}{y^2(1-y)^2}\,\mathrm{d}y, \tag{B.17}$$

and for E an independent convergent representation is

$$E = p + 4\int_0^{1/2} \frac{\log^2(1-y)\big[2\log y - \log(1-y)\big]}{y^2(1-y)^2}\,\mathrm{d}y. \tag{B.18}$$

Appendix B.10 reduces E to the weight-four constants of Eq. (B.16); the key step is

$$\int_0^1 \frac{\log t\,\log^2(1+t)}{t}\,\mathrm{d}t = \frac{2}{3}\int_0^1 \frac{\log^3(1+t)}{t}\,\mathrm{d}t - \frac{1}{4}\zeta(4). \tag{B.19}$$

Endpoint constants. Appendix B.10 also derives the finite parts of two source derivatives of raw moments,

$$\partial_v\partial_\eta A_{3,0} = -108 + 60\zeta(2) + 22\zeta(3) - 34\zeta(4) + 16\log 2, \qquad \partial_\eta Q_{4,0} = 6 - 12\zeta(2) - 54\zeta(3) - 25\zeta(4). \tag{B.20}$$

Source derivatives of this kind enter the subtraction recurrence through the Jacobian entries j_n and λ_{2m} of Eq. (A.42). The precise moment convention is given in Appendix B.10. In particular, the conversion from angular to radial variables must keep $\log(\varepsilon/R) = \ell - \log R$ in full (R the span of the kinks); dropping the radial logarithms changes finite parts.

The inputs G, q, k. The remaining inputs, $G = b_{5,v}$, $q = -4h_{4,\bar{\delta}}$ and $k = -4h_6$, are archived exact period expressions (`code/dual_d1/data/archived/C6_closed_form/input_expressions/`). Their values are

$$\begin{aligned} G &= 32.763409276837655798312474117 4\ldots, \\ q &= -103.560357927042746213210123654 5\ldots, \\ k &= 104.672392743212637331086084748 7\ldots. \end{aligned} \tag{B.21}$$

The exact expressions, not these decimals, enter the assembly.

Fixed point and determinant. Through the orders needed,

$$\begin{aligned} u_* &= \frac{\bar\delta}{4} - \frac{c}{64}\bar\delta^2 + \Big(\frac{c^2}{512} + \frac{f-k}{256} - \frac{q}{64}\Big)\bar\delta^3 + \cdots, \\ z_* &= -\frac{\bar\delta}{4} + \Big(\frac{a-b}{16} + \frac{c}{64} - \frac{p}{4}\Big)\bar\delta^2 + \cdots, \end{aligned} \tag{B.22}$$

and the determinant gives

$$D_3 = \frac{16A - 32E - 8G + 8b + c^2 + 4f - 4k + 32p - 8q}{64}. \tag{B.23}$$

With $\mathcal{R} = 2G + k + 2q - 2b - f$ and $c^2/64 = 4$, this is $D_3 = \frac{A}{4} - \frac{E}{2} + 4 + \frac{p}{2} - \frac{\mathcal{R}}{16}$. The exact certificate of D_3 (Appendix C.1; the table of Sec. 2.4) proves

$$\begin{aligned} \mathcal{R} = 140 &+ 64\zeta(2) - 64\zeta(3) - 272\zeta(4) - 64\log 2 + 64\log^2 2 \\ &+ 112\zeta(3)\log 2 - 32\zeta(2)\log^2 2 + \tfrac{16}{3}\log^4 2 + 128\,\mathrm{Li}_4(\tfrac{1}{2}). \end{aligned} \tag{B.24}$$

With Eq. (B.16) and the value of p, all regulator constants other than zeta values cancel, and $D_3 = \frac{21}{4} - 13\zeta(2) + 11\zeta(3) + 11\zeta(4) = 8.99403863655111357\ldots$, the closed form of Sec. 2.4. Equation (2.8) then gives C_5^{SR} of Eq. (2.12), $29.2378115809900396\ldots$.

B.4 Sixth order: T_3

To determine T_3, retain

$$\begin{aligned} F &= z + u(1 + az + p\bar\delta + Az^2 + E\bar\delta z + V\bar\delta^2) + u^2(b + Gz + R\bar\delta) + Ju^3 + \cdots, \\ H &= -\bar\delta + 4u + u^2(c + fz + q\bar\delta + Lz^2 + M\bar\delta z) + u^3(k + Nz) + \cdots. \end{aligned} \tag{B.25}$$

The six new inputs of Eq. (B.1),

$$V = \tfrac{1}{2}b_{3,\bar\delta\bar\delta}, \quad R = b_{5,\bar\delta}, \quad J = b_7, \quad L = -2h_{4,vv}, \quad M = -4h_{4,v\bar\delta}, \quad N = -4h_{6,v}, \tag{B.26}$$

are evaluated at the marginal point $v = 1$, $\bar\delta = 0$, not at the interacting fixed point. The term Nzu^3 is of order $\bar\delta^4$ in H at the fixed point but contributes at order $\bar\delta^3$ after a derivative in z, so it cannot be dropped.

input	largest new moment sector	value
V	three-kink second tail	-19.518741011574465808
R	five-kink tail	-1.148356985726247710
J	seven-kink odd response	32.280324973629037003
L	four-kink second charge	-7.677303759056569914
M	four-kink charge–tail	13.315217322864044765
N	six-kink charge derivative	210.030599248618268410

These are beta inputs of the prescription, not universal exponents. Their exact expressions are archived with those of G, q, k (`code/dual_d1/data/archived/C6_closed_form/input_expressions/`).

A convergent check of V. For three kinks, put $l = \log y$, $\bar l = \log(1-y)$ and

$$P_V(y) = -l^2\bar l + l\bar l^2 - \tfrac{1}{3}\bar l^3 + 2l^2\bar l^2 - 2l\bar l^3 + \tfrac{1}{2}\bar l^4. \tag{B.27}$$

The direct regulated calculation reduces to

$$V = \frac{2}{3}\log^3 2 - 2\int_0^{1/2}\Big[\frac{P_V(y)}{y^2(1-y)^2} - \frac{\log^2 y}{y}\Big]\,\mathrm{d}y. \tag{B.28}$$

The subtraction removes the endpoint singularity. Both the cubic-logarithm term $\frac{1}{3}S_3$ and the square $\frac{1}{2}S_2^2$ of the first tail insertion in Eq. (A.41) are needed. The finite hard-core endpoint contributes $2\int_1^\infty \log^2 w\, w^{-2}\,\mathrm{d}w = 4$, which cancels a -4 in the beta assembly (the coefficient of $\bar\delta^2$ in $2vj_1 = -4/(1+\bar\delta)$ at $v=1$, Eqs. (A.42) and (A.47)). This finite piece is derived with the regulator kept; it is not an adjustable counterterm.

Trace assembly. With the fixed point (B.22), the trace gives

$$C_6^{\mathrm{SR}} = \mathcal{B}_{\mathrm{old}}^{(6)} + 2V + R + \frac{3}{8}J - \frac{1}{8}L + \frac{1}{4}M + \frac{1}{16}N, \tag{B.29}$$

where the part fixed by the inputs of the lower orders is

$$\begin{aligned}\mathcal{B}_{\mathrm{old}}^{(6)} = {}& \frac{A}{8} - \frac{E}{2} - \frac{G}{4} + \frac{a^2-ab}{8} + \frac{ac}{16} - \frac{ap}{2} - \frac{bc}{8} + \frac{c^2}{64} \\ & - \frac{cf}{32} - \frac{cp}{8} - \frac{f}{32} - \frac{k}{32} - \frac{q}{8},\end{aligned} \tag{B.30}$$

with the archived value $\mathcal{B}_{\mathrm{old}}^{(6)} = -48.0003600840977427059\ldots$. The complete regulated expression gives $C_6^{\mathrm{SR}} = -58.665697474225308949717146128630 82\ldots$. The exact certificate at weights two to five (Appendix C.1) establishes the closed form of C_6^{SR} in Eq. (2.12), and $T_3 = C_6^{\mathrm{SR}}/4$ is the expression of Sec. 2.4.

Checks that do not use the new single logarithms. The lower flow predicts, for the normalized seven-kink response,

$$[\ell^6]\,\mathcal{C}_7 = \frac{244}{45}, \qquad [\ell^5]\,\mathcal{C}_7 = \frac{170}{3}, \qquad [\ell^4]\,\mathcal{C}_7 = 247 - \frac{316}{3}\zeta(2), \tag{B.31}$$

and the archived exact integrations reproduce these coefficients without using J. For the finite-tail correction to the five-kink response, the matching of Eq. (A.22) predicts

$$[\ell^6]\,\partial_{\bar\delta}\mathcal{C}_5 = -\frac{142}{45}, \qquad [\ell^5]\,\partial_{\bar\delta}\mathcal{C}_5 = -\frac{71}{5}, \qquad [\ell^4]\,\partial_{\bar\delta}\mathcal{C}_5 = \frac{14}{3}\zeta(2) - \frac{55}{3}, \tag{B.32}$$

which do not use R. The archive contains further higher-logarithm checks, exact derivatives of the primitives and direct finite-cutoff numerical integrations of the original kernels. Some period residuals in the higher-logarithm tests were checked numerically, not by exact certificates, and the verification reports keep that distinction.

Two archived full period evaluations used 80 and 100 decimal digits, matching points 100 and 120, and different continuation steps and series orders; they agreed in all 65 output digits of the first. This is a convergence check, not an interval bound; the closed form of C_6^{SR} rests on the rational certificate, not on numerical agreement.

B.5 Mellin representation, finite parts and the period $\mathcal{R}_4$

This and the following subsections give the low-insertion calculations behind Appendix B.2, with their subtractions and branch conventions. The normalized responses are those of Eq. (A.37), and the gap variables are those of Appendix A.7.

Separating the radial integral. Let F be homogeneous of degree d_F in m gaps, and put $k_F = m + d_F$. Define, with the translation factor of Appendix A.7,

$$I_F(\varepsilon) = \int_{r_i\ge\varepsilon,\ \sum r_i\le 1} \Big(1 - \sum_i r_i\Big) F(\boldsymbol{r})\ \mathrm{d}^m r. \tag{B.33}$$

First integrate ε up to $\min_i r_i$, then write $r_i = Rx_i$ with $\sum_i x_i = 1$. This gives

$$\int_0^\infty \varepsilon^{q-1} I_F(\varepsilon)\,\mathrm{d}\varepsilon = \frac{\mathcal{A}_F(q)}{q(q+k_F)(q+k_F+1)}, \qquad \mathcal{A}_F(q) = \int_{\sum x_i=1} F(\boldsymbol{x})(\min_i x_i)^q\,\mathrm{d}^{m-1}x. \quad \text{(B.34)}$$

Here q is the Mellin variable, not the input q. The identity is first used in a half-plane where it converges and is then continued after explicit subtraction; no divergent integral is evaluated naively at $q = 0$. The pole of order n at $q = 0$ gives the ε^0 terms of I_F through

$$q^{-n} \;\longleftrightarrow\; \frac{(-\ell)^{n-1}}{(n-1)!}, \quad \text{(B.35)}$$

so the coefficient of q^{-1} is the finite part; power divergences give poles away from $q = 0$.

The needed Laurent coefficients of the angular functions define the regulator periods:

$$\begin{aligned}
\mathcal{A}_{Q_4}(q)\big|_{v=1} &= \frac{8}{q^2} - \frac{6}{q} - 6 - 4\zeta(2) + \mathcal{P}_4\,q + O(q^2),\\
\partial_v\mathcal{A}_{Q_4}(q)\big|_{v=1} &= \frac{32}{q^3} - \frac{4}{q^2} + \frac{4+16\zeta(2)}{q} + \mathcal{R}_4 + O(q),\\
\mathcal{A}_{Z_4}(q) &= \frac{12}{q^2} - \frac{3}{q} - 5 - 6\zeta(2) + O(q),\\
\mathcal{A}_{A_5}(q) &= \frac{80}{q^3} - \frac{68}{q^2} - \frac{56+48\zeta(2)}{q} + \mathcal{S}_5 + O(q).
\end{aligned} \quad \text{(B.36)}$$

The subscript names the moment whose kernel is meant: $\Xi_4 W_4$ for Q_4, W_4 for Z_4 and W_5 for A_5 (the compendium writes G_5 for the last). For Q_4, $k_F = 4 - 4v$, so a charge derivative of Eq. (B.34) also differentiates its radial denominator; that contribution must be included (Appendix B.8). For example, at $v = 1$ ($k_F = 0$) the first line and Eq. (B.35) give the ε^0 part of Q_4, $-\frac{4}{3}\ell^3 - 7\ell^2 + [-8 + 4\zeta(2)]\ell + \mathcal{P}_4 + 4\zeta(2) - 8$; subtracting the ε^0 part of $Q_2 Z_2$, which is $2 - \ell^2$ by Eq. (A.35), gives $\mathcal{K}_4$ of Eq. (B.10).

Hard-cut angular functions. Define

$$\mathcal{H}_F(e) = \int_{x_i \ge e,\ \sum x_i = 1} F(\boldsymbol{x})\,\mathrm{d}^{m-1}x, \qquad \mathcal{A}_F(q) = q\int_0^{1/m} e^{q-1}\mathcal{H}_F(e)\,\mathrm{d}e. \quad \text{(B.37)}$$

The constant term ($e^0 \log^0 e$) of $\mathcal{H}_{A_5}$ is $\mathcal{S}_5$, and that of $\partial_v \mathcal{H}_{Q_4}$ is $\mathcal{R}_4$. Power divergences produce poles away from $q = 0$ and do not change these identifications. The period $\mathcal{P}_4$ is instead a finite part of $\int \mathcal{H}_{Q_4}(e)\,\mathrm{d}e/e$ (Appendix B.7).

Finite parts. Throughout, FP denotes the constant coefficient that remains after all displayed powers and logarithms of the endpoint cutoff are subtracted; for instance,

$$\mathrm{FP}\int_2^\infty f(t)\,\mathrm{d}t = \text{constant term of } \int_2^T f(t)\,\mathrm{d}t \text{ as } T \to \infty. \quad \text{(B.38)}$$

On the interval $(0, 1)$ the cutoffs are ϑ and $1 - \vartheta$. Finite parts are not invariant under a change of cutoff variable unless the corresponding conversion terms are kept; an example occurs in the evaluation of $\mathcal{P}_4$.

To extract the constant of a one-dimensional angular integral with moving endpoints, one uses three regions: the bulk, a lower endpoint region $w = et$ and an upper endpoint region $w = 1 - et$. The integrand, *including the Jacobian*, is expanded in each region, and its coefficient of $e^0 \log^0 e$ is kept; the desired constant is the sum of the three finite-part integrals. To justify this, split the original domain at fixed ϑ and $1 - \vartheta$, take $e \ll \vartheta$ first, and cancel the powers and logarithms of the overlaps before taking $\vartheta \to 0$; the constant that remains is the sum stated. This procedure keeps finite boundary contributions that an expansion of the bulk alone would miss. In symbolic software, a term such as $1/[e(t+1)]$ must be recognized as $e^{-1}/(t+1)$ before a Laurent coefficient is read off; reading a coefficient from an uncollected rational denominator can silently drop an endpoint term.

Elementary finite parts. With $l = \log w$ and $\bar{l} = \log(1-w)$, the nontrivial finite parts over $(0,1)$ needed below are (convergent integrals take their ordinary values):

integrand	FP $\int_0^1$	integrand	FP $\int_0^1$
$l\bar{l}$	$2-\zeta(2)$	$\bar{l}/w$	$-\zeta(2)$
$l\bar{l}/w$	$\zeta(3)$	$\bar{l}/(w-1)$	0
$l\bar{l}/(w-1)$	$-\zeta(3)$	$\bar{l}/(w-1)^2$	-1
$l\bar{l}/w^2$, $l\bar{l}/(w-1)^2$	$\zeta(2)-1$	$\bar{l}/(w-1)^3$	$\frac{1}{4}$
l	-1	l^2/w	0
l/w	0	l^2/w^2	-2
l/w^2	-1	$l^2/(w+1)$	$\frac{3}{2}\zeta(3)$
$l/(w-1)$	$\zeta(2)$	$l^2/(w+1)^2$	$\zeta(2)$
$l/(w-1)^2$	-1	$\bar{l}^2$, $\bar{l}^2/(w-1)$	2, 0
$l/(w+1)$	$-\frac{1}{2}\zeta(2)$	$\bar{l}^2/w$, $\bar{l}^2/w^2$	$2\zeta(3)$, $2\zeta(2)$
$l/(w+1)^2$	$-\log 2$	$1/(w+1)$, $1/(w+1)^2$	$\log 2$, $\frac{1}{2}$
$l/(w+1)^3$	$-\frac{1}{2}\log 2-\frac{1}{4}$	$1/w$, $1/(w-1)$	0, 0
		$1/w^2$, $1/(w-1)^2$	-1, -1
		$1/(w-1)^3$	$\frac{1}{2}$

For example,

$$\mathrm{FP}\int_0^1 \frac{\log w \log(1-w)}{w^2}\,\mathrm{d}w = \sum_{n=2}^{\infty}\frac{1}{n(n-1)^2} = \zeta(2)-1, \tag{B.39}$$

where the term $n=1$ of the expansion of $\log(1-w)$ gives a pure logarithmic divergence with zero finite constant. Likewise

$$\int_0^1 \frac{\log^2 w}{1+w}\,\mathrm{d}w = 2\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^3} = \frac{3}{2}\zeta(3). \tag{B.40}$$

Repeated denominators follow by integration by parts, and pure powers and logarithms can be integrated directly; these procedures derive the whole table.

The period $\mathcal{R}_4$. On the angular simplex $r+t+w=1$ of three gaps, Eq. (A.33) simplifies to

$$W_4 = \frac{1}{r^2w^2} + \frac{2}{rtw} + \frac{1}{t^2}, \tag{B.41}$$

and the current kernel is $\Xi_4 W_4 = (r+w)W_4$. For a fixed outer gap r, let $\mathcal{M}_{\mathrm{out}}(r,e)$ be the integral of $\Xi_4 W_4$ over the other outer gap, and for a fixed middle gap r let $\mathcal{M}_{\mathrm{mid}}(r,e)$ be the integral over an outer gap. Elementary partial fractions give, with $\rho_r = e^{-1} - (1-r-e)^{-1}$ and $L_r = \log[(1-r-e)/e]$,

$$\begin{aligned}
\mathcal{M}_{\mathrm{out}}(r,e) &= (1+r^{-1})\rho_r + \frac{(r+1)(r^2+1)}{r^2(1-r)}L_r,\\
\mathcal{M}_{\mathrm{mid}}(r,e) &= \frac{2\rho_r}{1-r} + \frac{4(r^2-r+1)}{r(1-r)^2}L_r + \frac{(1-r)(1-r-2e)}{r^2}.
\end{aligned} \tag{B.42}$$

On the simplex $\log W_4 = 2\log[(1-r)/r] + 2\log[(1-w)/w] - 2\log t$, and $\partial_v W_4^v|_{v=1} = W_4 \log W_4$. The symmetry between the two outer gaps then gives the exact one-dimensional reduction

$$\partial_v \mathcal{H}_{Q_4}(e,v)\big|_{v=1} = \int_e^{1-2e}\Big[4\mathcal{M}_{\mathrm{out}}(r,e)\log\frac{1-r}{r} - 2\mathcal{M}_{\mathrm{mid}}(r,e)\log r\Big]\,\mathrm{d}r. \tag{B.43}$$

Its three regions are the bulk (at fixed r, renamed w), the lower region $r = et$ with $t \geq 1$, and the upper region $r = 1 - et$ with $t \geq 2$. With $l = \log w$ and $\bar{l} = \log(1 - w)$, the constant integrands are

$$B_4(w) = l\bar{l}\Big(4 + \frac{16}{w-1} - \frac{8}{(w-1)^2} - \frac{16}{w} - \frac{4}{w^2}\Big) + l\Big(-2 - \frac{8}{w-1} + \frac{4}{(w-1)^2} + \frac{8}{w} - \frac{2}{w^2}\Big) + \bar{l}^2\Big(-4 - \frac{16}{w-1} + \frac{8}{w} + \frac{4}{w^2}\Big) + \bar{l}\Big(\frac{8}{w-1} - \frac{4}{w}\Big), \tag{B.44}$$

$$L_4(t) = \Big(-8 + \frac{12}{t} + \frac{8}{t^2}\Big)\log t - 4, \qquad t \geq 1, \tag{B.45}$$

$$U_4(t) = \frac{16}{t}\log t \log(t-1) + \Big(8 - \frac{8}{t-1}\Big)\log t + \frac{8}{t}\log(t-1) + 4 - \frac{4}{t-1}, \qquad t \geq 2. \tag{B.46}$$

The finite-part table gives, group by group,

group of B_4	$l\bar{l}$	l	$\bar{l}^2$	$\bar{l}$
FP $\int_0^1$	$20 - 16\zeta(2) - 32\zeta(3)$	$-8\zeta(2)$	$-8 + 8\zeta(2) + 16\zeta(3)$	$4\zeta(2)$

so that

$$\mathrm{FP}\int_0^1 B_4(w)\,\mathrm{d}w = 12 - 12\zeta(2) - 16\zeta(3), \qquad \mathrm{FP}\int_1^\infty L_4(t)\,\mathrm{d}t = 4. \tag{B.47}$$

In the upper integral, the term with two logarithms is

$$J_U = \mathrm{FP}\int_2^\infty \frac{\log t \log(t-1)}{t}\,\mathrm{d}t = -\frac{\log^3 2}{3} - \log 2\,\mathrm{Li}_2(\tfrac{1}{2}) - \mathrm{Li}_3(\tfrac{1}{2}) = -\frac{7}{8}\zeta(3), \tag{B.48}$$

where the last step uses the values (B.56) at $\frac{1}{2}$. The remaining terms have the primitive

$$8t\log t - 4t + 8\,\mathrm{Li}_2(1-t) + 4\log^2 t + 8\,\mathrm{Li}_2(1/t) - 4\log(t-1), \tag{B.49}$$

whose constant at infinity is $-8\zeta(2)$ and whose value at $t = 2$ is $16\log 2 - 8$. Hence

$$\mathrm{FP}\int_2^\infty U_4(t)\,\mathrm{d}t = 8 - 8\zeta(2) - 14\zeta(3) - 16\log 2. \tag{B.50}$$

Adding the three regions gives

$$\mathcal{R}_4 = 24 - 20\zeta(2) - 30\zeta(3) - 16\log 2, \tag{B.51}$$

with no numerical reconstruction.

B.6 The trilogarithm lemma and the Spence–Kummer identity

Polylogarithms. For $|z| < 1$, $\mathrm{Li}_n(z) = \sum_{k\geq 1} z^k/k^n$, continued by $\mathrm{d}\,\mathrm{Li}_n(z)/\,\mathrm{d}z = \mathrm{Li}_{n-1}(z)/z$ with $\mathrm{Li}_1(z) = -\log(1-z)$. Real principal values are used for $z < 1$; where an argument exceeds one, the real part is indicated explicitly. The identities needed are

$$\mathrm{Li}_2(z) + \mathrm{Li}_2(1-z) = \zeta(2) - \log z \log(1-z), \tag{B.52}$$

$$\mathrm{Li}_2\Big(\frac{z}{z-1}\Big) = -\,\mathrm{Li}_2(z) - \frac{1}{2}\log^2(1-z), \tag{B.53}$$

$$\mathrm{Li}_n(z) + \mathrm{Li}_n(-z) = 2^{1-n}\,\mathrm{Li}_n(z^2), \tag{B.54}$$

$$\mathrm{Li}_3(z) + \mathrm{Li}_3(1-z) + \mathrm{Li}_3\Big(-\frac{z}{1-z}\Big) = \zeta(3) + \zeta(2)\log(1-z) - \frac{1}{2}\log z\log^2(1-z) + \frac{1}{6}\log^3(1-z). \tag{B.55}$$

The first two follow by differentiation and evaluation at an endpoint, the third from the power series, and the fourth by differentiation using the first two. At $z = \frac{1}{2}$ they give

$$\mathrm{Li}_2(\tfrac{1}{2}) = \frac{\zeta(2)}{2} - \frac{\log^2 2}{2}, \qquad \mathrm{Li}_3(\tfrac{1}{2}) = \frac{7}{8}\zeta(3) - \frac{\zeta(2)\log 2}{2} + \frac{\log^3 2}{6}. \tag{B.56}$$

The real-domain Spence–Kummer identity. For real $0 < x < y < 1$ define

$$\begin{aligned} f_1 &= \frac{x(1-y)^2}{y(1-x)^2}, & f_2 &= xy, & f_3 &= x/y, \\ f_4 &= \frac{x(1-y)}{y(1-x)}, & f_5 &= -\frac{x(1-y)}{1-x}, & f_6 &= \frac{1-y}{1-x}, \\ f_7 &= -\frac{1-y}{y(1-x)}, & f_8 &= x, & f_9 &= y. \end{aligned} \tag{B.57}$$

All positive arguments are smaller than one, and the two negative ones are on the real principal branch. The identity is

$$\mathrm{Li}_3(f_1)+\mathrm{Li}_3(f_2)+\mathrm{Li}_3(f_3)-2\sum_{i=4}^{9}\mathrm{Li}_3(f_i)+2\zeta(3) = \log^2 y\,\log\frac{1-y}{1-x}-2\zeta(2)\log y-\frac{1}{3}\log^3 y. \tag{B.58}$$

The branch choices are the real-domain specialization of the path formulation of Ref. [24].

A differential verification uses the five-term relation of the dilogarithm,

$$\mathrm{Li}_2(a)+\mathrm{Li}_2(b) = \mathrm{Li}_2(ab)+\mathrm{Li}_2\Big(\frac{a(1-b)}{1-ab}\Big)+\mathrm{Li}_2\Big(\frac{b(1-a)}{1-ab}\Big)+\log\frac{1-a}{1-ab}\log\frac{1-b}{1-ab}, \tag{B.59}$$

together with Eqs. (B.52) and (B.53); Eq. (B.59) itself follows by differentiation in a and b and evaluation at $a=0$. Differentiating Eq. (B.58) with respect to x and multiplying by $x(1-x)$ leaves the dilogarithm equality

$$\begin{aligned} &(1+x)\,\mathrm{Li}_2(f_1)+(1-x)\big[\mathrm{Li}_2(f_2)+\mathrm{Li}_2(f_3)\big]-2\big[\mathrm{Li}_2(f_4)+\mathrm{Li}_2(f_5)\big] \\ &\quad -2x\big[\mathrm{Li}_2(f_6)+\mathrm{Li}_2(f_7)\big]-2(1-x)\,\mathrm{Li}_2(x) = x\log^2 y. \end{aligned} \tag{B.60}$$

Substituting the rational f_i into Eq. (B.59), followed by Eqs. (B.52) and (B.53), verifies this equality. The integration constant is fixed by $x \to 0$, where Eq. (B.58) reduces exactly to the Landen relation (B.55) at $1-y$. This proves the real-domain identity without a numerically inferred special value.

At $x = \frac{1}{2}$, $y = \frac{2}{3}$ the nine arguments are

$$(f_1,\dots,f_9) = \Big(\frac{1}{3},\frac{1}{3},\frac{3}{4},\frac{1}{2},-\frac{1}{3},\frac{2}{3},-1,\frac{1}{2},\frac{2}{3}\Big). \tag{B.61}$$

Since $\mathrm{Li}_3(-1) = -\frac{3}{4}\zeta(3)$, Eq. (B.58) becomes

$$2\,\mathrm{Li}_3(\tfrac{1}{3})+\mathrm{Li}_3(\tfrac{3}{4})-4\,\mathrm{Li}_3(\tfrac{1}{2})-2\,\mathrm{Li}_3(-\tfrac{1}{3})-4\,\mathrm{Li}_3(\tfrac{2}{3})+\frac{7}{2}\zeta(3) = \frac{2}{3}\log^3\tfrac{2}{3}-2\zeta(2)\log\tfrac{2}{3}. \tag{B.62}$$

All its arguments are real and at most one, so no continuation across a branch cut of the trilogarithm is needed. The compendium's script `code/dual_d1/code/archived/LRI_thermal_closed_form/code/verify_closed_forms.py` also checks this specialization and the Landen eliminations that follow.

Lemma (a finite part with an exact zeta value). With the finite-part convention of Eq. (B.38),

$$J_L = \lim_{T\to\infty}\Big[\int_2^T \frac{\log(t-1)\log(t+1)}{t}\,\mathrm{d}t-\frac{1}{3}\log^3 T\Big] = -\frac{1}{3}\zeta(3). \tag{B.63}$$

Proof. Define the convergent integral

$$\Lambda(z) = \int_0^z \frac{\log(1-t)\log(1+t)}{t}\,\mathrm{d}t. \tag{B.64}$$

The substitution $t \mapsto 1/t$ in Eq. (B.63) gives the absolutely convergent representation

$$J_L = \Lambda(\tfrac{1}{2}) - \frac{1}{2}\log 2\ \mathrm{Li}_2(\tfrac{1}{4}) - \frac{1}{4}\mathrm{Li}_3(\tfrac{1}{4}) - \frac{1}{3}\log^3 2. \tag{B.65}$$

For an explicit reduction introduce

$$\Theta(z) = \int_0^z \frac{\log^2(1-t)}{t}\,\mathrm{d}t = \log z \log^2(1-z) + 2\log(1-z)\,\mathrm{Li}_2(1-z) - 2\,\mathrm{Li}_3(1-z) + 2\zeta(3); \tag{B.66}$$

the equality follows by differentiation and the limit $z \to 0$. Squaring $\log(1-t^2) = \log(1-t)+\log(1+t)$ gives

$$\Lambda(z) = \frac{1}{4}\Theta(z^2) - \frac{1}{2}\Theta(z) - \frac{1}{2}\Theta(-z). \tag{B.67}$$

In $\Theta(-\frac{1}{2})$ the real parts of Eq. (B.66) are taken. Equivalently, for $a > 0$ the manifestly real formula

$$\begin{aligned}\Theta(-a) = &-\log a \log^2(1+a) - 2\zeta(2)\log(1+a) - 2\log(1+a)\,\mathrm{Li}_2(-a)\\ &- 2\,\mathrm{Li}_3\Big(\frac{1}{1+a}\Big) + \frac{1}{3}\log^3(1+a) + 2\zeta(3)\end{aligned} \tag{B.68}$$

holds; it follows from Eq. (B.52) and from $\mathrm{Re}\,\mathrm{Li}_3(\beta) = \mathrm{Li}_3(1/\beta) - \frac{1}{6}\log^3\beta + 2\zeta(2)\log\beta$ for real $\beta > 1$. Let

$$\Pi = \log^3 3 - 3\log^2 3\log 2 + 6\log 3\log^2 2 - 5\log^3 2 - 6\zeta(2)\log 3 + 6\zeta(2)\log 2. \tag{B.69}$$

Substituting Eq. (B.66) and the formula for $\Theta(-a)$ into Eq. (B.65), and using Eqs. (B.52) to (B.54), gives

$$J_L = \mathrm{Li}_3(\tfrac{2}{3}) - \frac{1}{2}\mathrm{Li}_3(\tfrac{3}{4}) - \mathrm{Li}_3(-\tfrac{1}{2}) - \frac{3}{2}\zeta(3) - \frac{\Pi}{6}. \tag{B.70}$$

The dilogarithms are eliminated in favour of $\mathrm{Li}_2(\frac{1}{3})$,

$$\begin{gathered}\mathrm{Li}_2(-\tfrac{1}{2}) = -\mathrm{Li}_2(\tfrac{1}{3}) - \tfrac{1}{2}\log^2\tfrac{3}{2}, \qquad \mathrm{Li}_2(\tfrac{1}{4}) = \zeta(2) - \log^2 2 - 2\,\mathrm{Li}_2(\tfrac{1}{3}) - \log^2\tfrac{3}{2},\\ \mathrm{Li}_2(\tfrac{3}{4}) = \zeta(2) - \mathrm{Li}_2(\tfrac{1}{4}) + 2\log 2\,\log\tfrac{3}{4},\end{gathered} \tag{B.71}$$

and all coefficients of $\mathrm{Li}_2(\frac{1}{3})$ cancel. Next, Eq. (B.55) at $z = \frac{1}{3}$ and $z = \frac{1}{4}$ eliminates $\mathrm{Li}_3(\frac{1}{3})$ and $\mathrm{Li}_3(-\frac{1}{3})$ from Eq. (B.62). With $\mathrm{Li}_3(\frac{1}{4}) = 4[\mathrm{Li}_3(\frac{1}{2}) + \mathrm{Li}_3(-\frac{1}{2})]$ and Eq. (B.56) the result is

$$\mathrm{Li}_3(\tfrac{2}{3}) - \frac{1}{2}\mathrm{Li}_3(\tfrac{3}{4}) - \mathrm{Li}_3(-\tfrac{1}{2}) = \frac{7}{6}\zeta(3) + \frac{\Pi}{6}. \tag{B.72}$$

Substitution in Eq. (B.70) gives $J_L = -\zeta(3)/3$. Every functional identity is used with its branch specified, and the final cancellation is symbolic, not an integer-relation fit. □

B.7 Closed forms of the four- and five-insertion periods

The four-insertion period $\mathcal{P}_4$. The decomposition (B.41) allows the angular current kernel to be integrated directly. Put

$$L_e = \log\frac{1-2e}{e}, \qquad z_e = \frac{e}{1-e}, \qquad 0 < e < \frac{1}{3}. \tag{B.73}$$

The result is

$$\mathcal{H}_{Q_4}(e) = \frac{2L_e+1}{e} - \frac{4L_e}{1-e} - 2(1-e)L_e - 2 - 3e + 4\log^2 z_e - 4\zeta(2) + 8\,\mathrm{Li}_2(z_e), \tag{B.74}$$

with the small-e expansion

$$\mathcal{H}_{Q_4}(e) = \frac{-2\log e + 1}{e} + 4\log^2 e + 6\log e - 6 - 4\zeta(2) + O(e\log e). \tag{B.75}$$

The finite part that corresponds to the first line of Eq. (B.36) is therefore

$$\mathcal{P}_4 = \lim_{\vartheta\to 0}\Big[\int_{\vartheta}^{1/3} \frac{\mathcal{H}_{Q_4}(e)}{e}\,\mathrm{d}e + \frac{2\log\vartheta+1}{\vartheta} + \frac{4}{3}\log^3\vartheta + 3\log^2\vartheta - \big(6+4\zeta(2)\big)\log\vartheta\Big]. \tag{B.76}$$

An explicit antiderivative evaluates it. Change variables to $z = e/(1-e)$, so that the upper endpoint is $z = \frac{1}{2}$, and write $\lambda(z) = \log(1-z) - \log z$. The integrand becomes

$$A_P(z)\lambda(z) + B_P(z) + \frac{4\log^2 z - 4\zeta(2) + 8\,\mathrm{Li}_2(z)}{z(1+z)}, \tag{B.77}$$

with

$$A_P(z) = \frac{2}{1+z} + \frac{2}{(1+z)^2} - \frac{6}{z} + \frac{2}{z^2}, \qquad B_P(z) = \frac{2}{1+z} - \frac{3}{(1+z)^2} - \frac{2}{z} + \frac{1}{z^2}. \tag{B.78}$$

A primitive is $K_P = K_{P,\le 2} + K_{P,3}$, with

$$\begin{aligned} K_{P,\le 2}(z) &= 2\Big[(\log 2 - \log z)\log(1+z) - \mathrm{Li}_2\Big(\frac{1+z}{2}\Big) - \mathrm{Li}_2(-z)\Big] - 2\lambda(z)\Big(\frac{1}{z} + \frac{1}{1+z}\Big) \\ &\quad + \frac{1}{z} + \frac{3}{1+z} + 6\,\mathrm{Li}_2(z) + 3\log^2 z - 6\log z + 3\log(1-z^2), \end{aligned} \tag{B.79}$$

$$\begin{aligned} K_{P,3}(z) &= \frac{4}{3}\log^3 z - 4\log^2 z\log(1+z) - 8\log z\,\mathrm{Li}_2(-z) + 8\,\mathrm{Li}_3(-z) + 8\,\mathrm{Li}_3(z) \\ &\quad - 8\,\mathrm{Li}_2(z)\log(1+z) - 8\Lambda(z) - 4\zeta(2)\log\frac{z}{1+z}, \end{aligned} \tag{B.80}$$

which is verified directly by differentiation, with $\Lambda'(z) = \log(1-z)\log(1+z)/z$ (Λ as in Appendix B.6). The lower endpoint is $z = \vartheta/(1-\vartheta)$, not $z = \vartheta$, and there

$$K_P\Big(\frac{\vartheta}{1-\vartheta}\Big) = \frac{2\log\vartheta+1}{\vartheta} + \frac{4}{3}\log^3\vartheta + 3\log^2\vartheta - \big(6+4\zeta(2)\big)\log\vartheta + 6 - 2\,\mathrm{Li}_2(\tfrac{1}{2}) + o(1). \tag{B.81}$$

Consequently

$$\mathcal{P}_4 = K_P(\tfrac{1}{2}) - 6 + 2\,\mathrm{Li}_2(\tfrac{1}{2}). \tag{B.82}$$

The representation (B.65), Euler reflection and duplication reduce this to

$$\mathcal{P}_4 = 4\zeta(2) - 2 + 3\log 3 + \log^2 2 + 2\,\mathrm{Li}_2(-\tfrac{1}{2}) - 8J_L, \tag{B.83}$$

and the lemma (B.63) gives

$$\mathcal{P}_4 = 4\zeta(2) + \frac{8}{3}\zeta(3) - 2 + 3\log 3 + \log^2 2 + 2\,\mathrm{Li}_2(-\tfrac{1}{2}). \tag{B.84}$$

The logarithms and the dilogarithm are allowed in this regulator-dependent period; they cancel from the stability invariants (Appendix B.2).

The five-insertion period $\mathcal{S}_5$: the middle vertex. On the angular simplex of five kinks, fix the first and last kinks at 0 and 1 and write the intermediate positions as $0 < \alpha < \xi < \beta < 1$, with every consecutive gap at least e. Let $w = \beta - \alpha$ and $P_0(x) = x^2(1-x)^2$. Then

$$W_5 = \frac{w^2}{P_0(\alpha)P_0(\beta)}\,\frac{P_0(\xi)}{(\xi-\alpha)^2(\beta-\xi)^2}. \tag{B.85}$$

With $\xi = \alpha + t$, the rational identity

$$\frac{P_0(\alpha+t)}{t^2(w-t)^2} = 1 + \frac{\mu_1}{t} + \frac{\mu_2}{w-t} + \frac{\nu_1}{t^2} + \frac{\nu_2}{(w-t)^2} \tag{B.86}$$

has the coefficients

$$\nu_1 = \frac{P_0(\alpha)}{w^2}, \quad \nu_2 = \frac{P_0(\beta)}{w^2}, \quad \mu_1 = \frac{P_0'(\alpha)}{w^2} + \frac{2P_0(\alpha)}{w^3}, \quad \mu_2 = -\frac{P_0'(\beta)}{w^2} + \frac{2P_0(\beta)}{w^3}. \tag{B.87}$$

The t integration runs from e to $w-e$ and leaves

$$\frac{w^2(w-2e) + \mathcal{N}_1 \log[(w-e)/e] + \mathcal{N}_2\big[e^{-1} - (w-e)^{-1}\big]}{P_0(\alpha)P_0(\beta)}, \tag{B.88}$$

where $\mathcal{N}_2 = P_0(\alpha) + P_0(\beta)$ and $\mathcal{N}_1 = P_0'(\alpha) - P_0'(\beta) + 2\mathcal{N}_2/w$. The integration over α, from e to $1-e-w$, is again elementary, and leaves

$$\mathcal{H}_{A_5}(e) = \int_{2e}^{1-2e} I_5(w,e)\,\mathrm{d}w. \tag{B.89}$$

With

$$\begin{gathered} L_0 = \log\frac{w-e}{e}, \qquad \rho_0 = \frac{1}{e} - \frac{1}{w-e}, \qquad \rho_- = \frac{1}{e} - \frac{1}{1-e-w}, \qquad \rho_+ = \frac{1}{w+e} - \frac{1}{1-e}, \\ L_- = \log\frac{1-e-w}{e}, \qquad L_+ = \log\frac{1-e}{w+e}, \end{gathered} \tag{B.90}$$

the exact remaining kernel is

$$\begin{aligned} I_5(w,e) = {}& \rho_-\Big[\frac{2(w-2e)}{(w-1)^2} - \frac{4L_0}{w-1} + 2\rho_0\Big] + \rho_+\Big[\frac{2(w-2e)}{(w+1)^2} - \frac{4L_0}{w+1} + 2\rho_0\Big] \\ & + L_-\Big[\frac{4(w^2-3w+1)(w-2e)}{w(w-1)^3} + \Big(-\frac{8}{w} + \frac{8}{w^2(w-1)^2}\Big)L_0 + 4\rho_0\Big] \\ & + L_+\Big[\frac{4(w^2+3w+1)(w-2e)}{w(w+1)^3} + \Big(-\frac{8}{w} - \frac{8}{w^2(w+1)^2}\Big)L_0 + 4\rho_0\Big]. \end{aligned} \tag{B.91}$$

Its equality with the α integral of Eq. (B.88) can be checked by rational partial fractions in α; the compendium's code also compares both sides numerically at finite positive cutoffs, before any subtraction or expansion. This makes the reduction reproducible without a sector-decomposition package.

The five-insertion period $\mathcal{S}_5$: bulk and endpoint finite parts. The three regions are obtained by expanding $I_5(w,e)$, $e\,I_5(et,e)$ and $e\,I_5(1-et,e)$ and keeping the coefficient of $e^0\log^0 e$. The lower and upper ranges begin at $t=2$, because the middle separation satisfies $w \geq 2e$ and the outer complement $1-w \geq 2e$. With $l = \log w$ and $\bar{l} = \log(1-w)$, the bulk integrand is

$$\begin{aligned} B_5(w) = {}& l^2\Big(\frac{16}{w+1} + \frac{8}{(w+1)^2} - \frac{8}{w} + \frac{8}{w^2}\Big) + l\bar{l}\Big(-\frac{16}{w-1} + \frac{8}{(w-1)^2} + \frac{8}{w} + \frac{8}{w^2}\Big) \\ & + l\Big(\frac{4}{w+1} - \frac{4}{(w+1)^2} + \frac{4}{(w+1)^3} - \frac{4}{(w-1)^2}\Big) + \bar{l}\Big(\frac{4}{w-1} - \frac{4}{(w-1)^2} - \frac{4}{(w-1)^3} - \frac{4}{w}\Big) \\ & - 6 - \frac{2}{w+1} + \frac{4}{(w+1)^2} + \frac{6}{w-1} - \frac{4}{(w-1)^2} + \frac{2}{(w-1)^3} - \frac{4}{w} - \frac{6}{w^2}. \end{aligned} \tag{B.92}$$

The lower region $w = et$ gives

$$L_5(t) = -\frac{8}{t}\log(t-1)\log(t+1) + \Big(4 - \frac{4}{t+1} - \frac{8}{t}\Big)\log(t-1) + \Big(-4 + \frac{4}{t-1}\Big)\log(t+1) - 4 + \frac{4}{t-1}. \tag{B.93}$$

The term $-4\log(t-1)/(t+1)$ is essential; omitting it changes the finite part. The upper region $w = 1-et$ gives

$$U_5(t) = \Big(4 - \frac{12}{t} - \frac{8}{t^2}\Big)\log(t-1) - 6 + \frac{10}{t-1} - \frac{8}{t}. \tag{B.94}$$

These are the coefficients of $e^0\log^0 e$ after the Jacobians of the endpoint regions are included.

The finite-part table evaluates the bulk integral group by group,

group of B_5	l^2	$l\bar{l}$	l	$\bar{l}$	rational
FP $\int_0^1$	$24\zeta(3)+8\zeta(2)-16$	$24\zeta(3)+16\zeta(2)-16$	$-2\zeta(2)+2\log 2+3$	$4\zeta(2)+3$	$7-2\log 2$

so that

$$\mathrm{FP}\int_0^1 B_5(w)\,\mathrm{d}w = 48\zeta(3)+26\zeta(2)-19. \tag{B.95}$$

For the upper region a primitive is

$$(4t-2)\log(t-1)-10t+4-6\log^2 t-12\,\mathrm{Li}_2(1/t)+\frac{8}{t}\log(t-1); \tag{B.96}$$

its finite constant at infinity is zero and its value at $t=2$ is $-16-6\zeta(2)$. Thus

$$\mathrm{FP}\int_2^\infty U_5(t)\,\mathrm{d}t = 16+6\zeta(2). \tag{B.97}$$

The lower region contains the only nontrivial weight-three endpoint integral, $-8J_L$, from its first term. For the remaining terms one primitive is

$$\begin{aligned} K_L(t) = {}& 4\big[(t-1)\log(t-1)-(t+1)\log(t+1)+2\big]-2\log^2(t+1)-4\,\mathrm{Li}_2\Big(\frac{2}{t+1}\Big) \\ & +4\log 2\log(t-1)-4\,\mathrm{Li}_2\Big(-\frac{t-1}{2}\Big)-4\log^2 t-8\,\mathrm{Li}_2(1/t)-4t+4\log(t-1), \end{aligned} \tag{B.98}$$

whose finite part at infinity is $4\zeta(2)+2\log^2 2$. Its value at $t=2$, followed by Euler reflection and the inversion formula of the dilogarithm at negative argument, gives

$$\mathrm{FP}\int_2^\infty L_5(t)\,\mathrm{d}t = -8J_L+12\zeta(2)+12\log 3+4\log^2 2+8\,\mathrm{Li}_2(-\tfrac{1}{2}). \tag{B.99}$$

Combining Eqs. (B.95) to (B.99) with the lemma (B.63) gives

$$\mathcal{S}_5 = 44\zeta(2)+\frac{152}{3}\zeta(3)-3+12\log 3+4\log^2 2+8\,\mathrm{Li}_2(-\tfrac{1}{2}), \tag{B.100}$$

and subtracting four times Eq. (B.84) gives Eq. (B.6) without any numerical inference. With Eq. (B.51), this completes the closed-form evaluation of the three periods.

B.8 Counterterms and the inputs b, c and f

The period evaluation is one part of the calculation of the exponents. This subsection records the response functions, finite terms and counterterm insertions that attach the periods to the beta inputs.

Projected responses. The two essential counterterms can be extracted by projecting the finite-interval ordered exponential onto the odd spin-flip vertex and onto the even current operator. At the endpoint the odd projection can equivalently be neutralized by a compensating vertex at infinity, with its free two-point normalization divided out; the current projection is the corresponding normalized dipole response. At finite $\bar{\delta}$ the calculation uses the regulated finite-interval responses and their kernel derivatives; it does not assume a charged conformal limit of the auxiliary theory. Displaying the terms needed, the odd and even responses are

$$\begin{aligned} \Gamma_g &= g+\Gamma_{gh}\,gh+\Gamma_{gh^2}\,gh^2+\mathcal{C}_3\,g^3+\Gamma_{g^3h}\,g^3h+\mathcal{C}_5\,g^5+\cdots, \\ \Gamma_h &= h+\sqrt{2}\,b_0\,g^2Q_2+\Gamma_{hg^2}\,hg^2+\sqrt{2}\,b_0\,g^4\mathcal{K}_4+\cdots, \end{aligned} \tag{B.101}$$

with the normalized responses of Eq. (A.37), not raw moments. Contractions involving the current coupling are generated by differentiating the exact current-resummed charges, keeping the endpoint factors of Eq. (A.18). At $b_0 = 1$, $\bar{\delta} = 0$ the mixed coefficients needed are

$$\begin{aligned}
\Gamma_{gh} &= -2\sqrt{2}\,(\ell+1),\\
\Gamma_{hg^2} &= 4\ell^2 + 10\ell + 10 - 4\zeta(2),\\
\Gamma_{gh^2} &= 4\ell^2 + 10\ell + 12 - 2\zeta(2),\\
\frac{\Gamma_{g^3h}}{\sqrt{2}} &= -\frac{20}{3}\ell^3 - 28\ell^2 + \big[-48 + 24\zeta(2)\big]\ell + \Gamma_{40}, \qquad \Gamma_{40} = -62 + 36\zeta(2) + 30\zeta(3) + 16\log 2.
\end{aligned} \tag{B.102}$$

These finite constants belong to the kink-only prescription, not to one that regulates current–kink separations independently. For example, Γ_{40} multiplies the counterterm of h at second order in the renormalized fugacity g_R and contributes to the fifth-order single logarithm. (The compendium writes F_G, F_H, F_{gh}, … for these responses and coefficients.)

Lower counterterms. The two-kink responses of Eq. (A.35) give, at $h_R = 0$,

$$h_{\text{bare}} = \sqrt{2}\,g_R^2\ell + O(g_R^4), \qquad g_{\text{bare}} = g_R + g_R^3(2\ell^2 - \ell) + O(g_R^5), \tag{B.103}$$

where g_R and h_R are the renormalized couplings. The odd subtraction can be checked directly: $\mathcal{C}_3 + \Gamma_{gh}\,(\sqrt{2}\ell) + (2\ell^2 - \ell)$ has no logarithm. Charge differentiation of the one-pair integral generates only higher powers of ℓ, not a new charge-dependent single logarithm, and its finite-$\bar{\delta}$ derivative likewise gives the cubic logarithm of Appendix B.9. This is the one-pair origin of the absent terms stated below Eq. (B.2).

Fourth and fifth insertions. At $h_R = 0$ and $\bar{\delta} = 0$ the counterterms are

$$\begin{aligned}
h_{\text{bare}} &= \sqrt{2}\,g_R^2\ell + \sqrt{2}\,g_R^4\Big(\frac{4}{3}\ell^3 - 2\ell^2 - 4\ell\Big) + \cdots,\\
g_{\text{bare}} &= g_R + g_R^3(2\ell^2 - \ell) + g_R^5\Big[\frac{10}{3}\ell^4 - \frac{22}{3}\ell^3 - \Big(\frac{13}{2} + 8\zeta(2)\Big)\ell^2 - b\,\ell\Big] + \cdots.
\end{aligned} \tag{B.104}$$

To make the subtraction reproducible, set

$$h_2^{\text{ct}} = \sqrt{2}\,\ell, \qquad g_3^{\text{ct}} = 2\ell^2 - \ell, \qquad h_4^{\text{ct}} = \sqrt{2}\Big(\frac{4}{3}\ell^3 - 2\ell^2 - 4\ell\Big). \tag{B.105}$$

Then the coefficient of g_R^5 in the bare fugacity is

$$g_5^{\text{ct}} = -\Big[\mathcal{C}_5 + \Gamma_{gh}(h_4^{\text{ct}} + g_3^{\text{ct}}h_2^{\text{ct}}) + 3\,\mathcal{C}_3\,g_3^{\text{ct}} + \Gamma_{gh^2}(h_2^{\text{ct}})^2 + \Gamma_{g^3h}\,h_2^{\text{ct}}\Big]_{\text{logs}}, \tag{B.106}$$

where $[\,\cdot\,]_{\text{logs}}$ keeps the terms of order ε^0 with a positive power of ℓ. All quantities on the right are specified in Eqs. (B.9), (B.11) and (B.102). The coefficient of the single logarithm gives

$$b = -21 + 38\zeta(2) + 60\zeta(3) + 32\log 2 - \mathcal{S}_5. \tag{B.107}$$

The fourth-insertion charge counterterm gives $c = -16$ (the value $h_4 = 4$ of Appendix A.9).

The charge derivative that enters the trace. Write the fourth-order auxiliary charge counterterm as $h_4^{\text{ct}} = \sqrt{2}\,b_0\,c_4^{\text{ct}}(v_0)$ with $v_0 = b_0^2$. Its derivative at $v_0 = 1$ is

$$c_4^{\text{ct}\,\prime} = -\frac{8}{3}\ell^4 + \frac{16}{3}\ell^3 + (8 + 4\zeta(2))\ell^2 + \big[4\mathcal{P}_4 + \mathcal{R}_4 - 8 + 4\zeta(2) + 8\zeta(3)\big]\ell. \tag{B.108}$$

Consequently

$$f = 16\mathcal{P}_4 + 4\mathcal{R}_4 - 32 + 16\zeta(2) + 32\zeta(3). \tag{B.109}$$

The lower endpoint integral needed in this derivative is

$$\begin{aligned} J_{\mathrm{end}}(v) &= 2\int_0^1 r^{1-2v}\big[(1-r)\log(1-r) + r\log r\big]\,\mathrm{d}r, \\ J_{\mathrm{end}}(1) &= -2\zeta(2), \qquad J'_{\mathrm{end}}(1) = -4\big(\zeta(2)+\zeta(3)\big), \end{aligned} \tag{B.110}$$

and it gives

$$\partial_v \Gamma_{hg^2}\big|_{v=1} = -\frac{8}{3}\ell^3 + 2\ell^2 + 24\ell + 44 - 12\zeta(2) - 8\zeta(3). \tag{B.111}$$

Equation (B.108) can be reconstructed algebraically. Let $Q_2 \to -\ell - 1$ and $\partial_v Q_2 \to \ell^2 - 2$ denote the ε^0 parts of the one-pair current moment and of its charge derivative, and $\partial_v g_3^{\mathrm{ct}} = -\frac{4}{3}\ell^3 + 3\ell^2 + 4\zeta(2)\ell$. Differentiating Eq. (B.34) at $k_F = 4 - 4v = 0$ gives

$$\partial_v \mathcal{M}_{Q_4} = \frac{\partial_v \mathcal{A}_{Q_4}}{q^2(q+1)} + \frac{4(2q+1)\,\mathcal{A}_{Q_4}}{q^3(q+1)^2}, \tag{B.112}$$

where $\mathcal{M}_{Q_4}$ is the Mellin transform of Q_4. Its poles give the ε^0 part of $\partial_v Q_4$ through Eq. (B.35). Subtracting the derivative of $Q_2 Z_2$ forms $\partial_v \mathcal{K}_4$; with $Z_2 = \int_\varepsilon^1 (1-r) r^{-2v}\,\mathrm{d}r$ and $Q_2 = \int_\varepsilon^1 (1-r) r^{1-2v}\,\mathrm{d}r$ at $\bar\delta = 0$, the ε^0 part of $\partial_v(Q_2 Z_2)$ at $v = 1$ is $2\ell^3 - 8\ell$. Then

$$c_4^{\mathrm{ct}\,\prime} = -\partial_v \mathcal{K}_4\big|_{\mathrm{logs}} - 2(\partial_v Q_2)\, g_3^{\mathrm{ct}} - 2Q_2\, \partial_v g_3^{\mathrm{ct}} - (\partial_v \Gamma_{hg^2})\,\ell + \Gamma_{hg^2}\,\ell^2. \tag{B.113}$$

Equations (B.36) and (B.112) fix every term of this expression; the compendium checks it symbolically in its code.

B.9 Charge and finite-$\bar\delta$ derivatives at three insertions: a and p

An incomplete-beta representation. For three kinks with gaps r and t, take the kernel $r^{-2p_{12}} t^{-2p_{23}} (r+t)^{2p_{13}}$ with general powers (the subscripts name the kink pairs). Its Mellin integral is

$$\mathcal{M}_3(q; p_{12}, p_{23}, p_{13}) = \frac{\mathrm{B}_{1/2}(q+1-2p_{12},\, 1-2p_{23}) + \mathrm{B}_{1/2}(q+1-2p_{23},\, 1-2p_{12})}{q(q+k_3)(q+k_3+1)}, \tag{B.114}$$

with $k_3 = 2 - 2p_{12} - 2p_{23} + 2p_{13}$ and the incomplete beta function $\mathrm{B}_x(a,b) = \int_0^x y^{a-1}(1-y)^{b-1}\,\mathrm{d}y$, continued meromorphically from its domain of convergence. The formula follows by writing $r = Ry$ and $t = R(1-y)$, splitting at $y = \frac{1}{2}$ and doing the radial integral. The kernel W_3 at $v = 1$, $\bar\delta = 0$ corresponds to $p_{12} = p_{23} = p_{13} = 1$.

The charge derivative and a. The charge derivative varies all three powers together. The normalized three-kink response then has

$$\partial_v \mathcal{C}_3\big|_{v=1} = -\frac{8}{3}\ell^3 - 3\ell^2 + (4 - 4\zeta(2))\ell + 18 - 14\zeta(2) - 15\zeta(3) - 8\log 2 + \cdots. \tag{B.115}$$

Combining it with the lower mixed response and counterterms gives

$$a = -4\zeta(2). \tag{B.116}$$

Equation (B.114) is a compact exact representation from which the logarithmic and finite terms of Eq. (B.115) can be regenerated.

The finite-tail deformation of the kernel and p. For separated charges the Gaussian weight expands as

$$\exp\Big[\frac{2v}{\bar\delta}\sum_{i<j}q_iq_j(r_{ij}^{\bar\delta}-1)\Big]=W^v\Big[1+v\,\bar\delta\sum_{i<j}q_iq_j\log^2 r_{ij}+O(\bar\delta^2)\Big], \tag{B.117}$$

where r_{ij} is the separation of kinks i and j and W^v the weight at $\bar\delta=0$. Thus the three-kink Mellin derivative is

$$\partial_{\bar\delta}\mathcal{M}_3|_{\bar\delta=0}=\frac{1}{4}(\partial_{p_{13}}^2-\partial_{p_{12}}^2-\partial_{p_{23}}^2)\mathcal{M}_3\big|_{p_{12}=p_{23}=p_{13}=1}. \tag{B.118}$$

The positive sign belongs to the like-charge outer pair, the negative signs to the two neighbouring opposite-charge pairs. Expanding Eq. (B.118) and normalizing by the two-kink moment gives

$$\begin{gathered}\partial_{\bar\delta}\mathcal{C}_3|_{\bar\delta=0}=-\ell^4-\ell^3-2\zeta(2)\ell^2+(p-4)\ell+O(1),\qquad \partial_{\bar\delta}\Gamma_{gh}|_{\bar\delta=0}=\sqrt{2}\,(2-\ell^2),\\ p=8-2\zeta(2)-7\zeta(3)-8\log 2.\end{gathered} \tag{B.119}$$

The $\mathrm{B}_{1/2}$ formula, the explicit differential operator and the lower moments fix this single logarithm independently of any fit to thermal exponents. The compendium's three-vertex residue script evaluates these derivatives as a separate check.

The finite-$\bar\delta$ current counterterm at order g_R^2 is

$$h_{\mathrm{bare}}|_{\bar\delta g_R^2}=-\frac{\sqrt{2}}{3}\,\bar\delta\,g_R^2\ell^3. \tag{B.120}$$

Substitution into the odd response gives

$$g_{\mathrm{bare}}|_{\bar\delta g_R^3}=\bar\delta\,g_R^3\Big[-\frac{1}{3}\ell^4+\frac{5}{3}\ell^3+2\zeta(2)\ell^2-p\,\ell\Big]. \tag{B.121}$$

This auxiliary fugacity is not yet the essential wall fugacity. Multiplication by the factor of Eq. (A.22) changes it to

$$Y_{\mathrm{bare}}|_{\bar\delta g_R^3}=\bar\delta\,g_R^3\Big[-\frac{4}{3}\ell^4+\frac{13}{6}\ell^3+2\zeta(2)\ell^2-p\,\ell\Big]. \tag{B.122}$$

The difference in the terms ℓ^4 and ℓ^3 is the finite-$\bar\delta$ matching, and it is indispensable.

B.10 Analytic three-kink and endpoint inputs

This subsection gives the analytic inputs of Appendix B.3. These reductions of lower inputs do not assume the higher-kink identity (B.24) that the certificate proves. Here $\partial_\eta A_3$ denotes the integrated first source derivative of the raw odd three-kink moment, not the derivative of an already normalized CFT correlator; the normalization and derivative factors of the subtraction recurrence must still be applied. At $\bar\delta=0$ the endpoint factors of Eq. (A.18) are

$$U_n^{\mathrm{odd}}(\eta)=\prod_i[X_i(1-X_i)]^{\eta q_i},\qquad U_n^{\mathrm{even}}(\eta)=\prod_i[X_i/(1-X_i)]^{\eta q_i}, \tag{B.123}$$

and

$$\partial_\eta A_3=\partial_\eta\int W_3^vU_3^{\mathrm{odd}}\Big|_{\eta=0},\qquad \partial_\eta Q_4=\partial_\eta\int\Xi_4W_4^vU_4^{\mathrm{even}}\Big|_{\eta=0}, \tag{B.124}$$

with the integrals over $\Omega_3(\varepsilon)$ and $\Omega_4(\varepsilon)$; the endpoints of the interval are not cut. The charge-curvature input has the convergent representation

$$A=-4\int_0^{1/2}\frac{\log^2(1-t)}{t^2(1-t)^2}\,\mathrm{d}t=-8-4\zeta(2)-2\zeta(3)+16\log 2, \tag{B.125}$$

obtained by partial fractions and integration by parts.

The mixed three-kink coefficient E. The representation (B.18) reads

$$E = p - 4\,\mathsf{I}_3 + 8\,\mathsf{I}_{12}, \qquad \mathsf{I}_3 = \int_0^{1/2} \frac{\log^3(1-y)}{y^2(1-y)^2}\,\mathrm{d}y, \qquad \mathsf{I}_{12} = \int_0^{1/2} \frac{\log y \log^2(1-y)}{y^2(1-y)^2}\,\mathrm{d}y, \tag{B.126}$$

and it can be integrated exactly. The result is the expression of E in Eq. (B.16).

Proof. Put $t = y/(1-y)$. Then $0 < t < 1$, $\log(1-y) = -\log(1+t)$, and the rational measure becomes $(1 + 2/t + 1/t^2)\,\mathrm{d}t$. Define

$$\mathsf{B}_3 = \int_0^1 \frac{\log^3(1+t)}{t}\,\mathrm{d}t, \qquad \mathsf{B}_{12} = \int_0^1 \frac{\log t \log^2(1+t)}{t}\,\mathrm{d}t. \tag{B.127}$$

The elementary reductions by integration by parts are

$$\begin{aligned} \int_0^1 \Big(1 + \frac{2}{t} + \frac{1}{t^2}\Big) \log^3(1+t)\,\mathrm{d}t &= -6\log^2 2 + 12\log 2 - 6 + \frac{3}{4}\zeta(3) + 2\mathsf{B}_3, \\ \int_0^1 \Big(1 + \frac{2}{t} + \frac{1}{t^2}\Big) \log t \log^2(1+t)\,\mathrm{d}t &= 2\mathsf{B}_{12} + 2\zeta(2) - 4\log^2 2 + 8\log 2 - 6 - \frac{3}{2}\zeta(3). \end{aligned} \tag{B.128}$$

Thus

$$E = -16 + 14\zeta(2) - 22\zeta(3) - 8\log^2 2 + 8\log 2 + 16\mathsf{B}_{12} - 8\mathsf{B}_3. \tag{B.129}$$

To evaluate B_3, set $u = (1+t)^{-1}$ and expand $(1-u)^{-1}$. A primitive needed for the logarithmic part is

$$\int \frac{\log^3 u}{1-u}\,\mathrm{d}u = -\log^3 u \log(1-u) - 3\log^2 u\,\mathrm{Li}_2(u) + 6\log u\,\mathrm{Li}_3(u) - 6\,\mathrm{Li}_4(u). \tag{B.130}$$

Evaluation between $u = \frac{1}{2}$ and 1, with Eq. (B.56), gives

$$\mathsf{B}_3 = 6\zeta(4) - 6\,\mathrm{Li}_4(\tfrac{1}{2}) - \frac{21}{4}\zeta(3)\log 2 + \frac{3}{2}\zeta(2)\log^2 2 - \frac{1}{4}\log^4 2. \tag{B.131}$$

For B_{12} introduce

$$\mathsf{K}(\alpha,\beta) = \int_0^1 t^{\alpha-1}(1+t)^{\beta}\,\mathrm{d}t. \tag{B.132}$$

Splitting the complete integral over $(0,\infty)$ at one and inverting t in the second half proves, first in a common domain of convergence and then by meromorphic continuation,

$$\mathsf{K}(\alpha,\beta) + \mathsf{K}(-\alpha-\beta,\beta) = \mathrm{B}(\alpha,-\alpha-\beta), \tag{B.133}$$

with B Euler's beta function. The cubic regular part of the right side, after its pole, is

$$\frac{7}{4}\zeta(4)(\alpha^2\beta + \alpha\beta^2) + \zeta(4)\beta^3, \tag{B.134}$$

which follows from the expansion

$$\mathrm{B}(\alpha,-\alpha-\beta) = \frac{\beta}{\alpha(\alpha+\beta)} \exp \sum_{n\geq 2} \frac{(-1)^n\zeta(n)}{n}\big[\alpha^n + (-\alpha-\beta)^n - (-\beta)^n\big]; \tag{B.135}$$

the term of degree four in the exponential, including its quadratic $\zeta(2)^2$ contribution, supplies the cubic regular part, with $\zeta(2)^2 = \frac{5}{2}\zeta(4)$. The coefficient of β^3 on the left of Eq. (B.133) is

$$\frac{\mathsf{B}_3}{3} - \frac{\mathsf{B}_{12}}{2} + \frac{1}{2}\int_0^1 \frac{\log^2 t \log(1+t)}{t}\,\mathrm{d}t = \frac{\mathsf{B}_3}{3} - \frac{\mathsf{B}_{12}}{2} + \frac{7}{8}\zeta(4). \tag{B.136}$$

Therefore

$$\mathsf{B}_{12} = \frac{2}{3}\mathsf{B}_3 - \frac{1}{4}\zeta(4). \tag{B.137}$$

Substituting Eqs. (B.131) and (B.137) into Eq. (B.129) gives the closed form of E. □

An independent quadrature of Eq. (B.126), which does not use the functional identity, gives

$$E = -21.663876674184610081020147908761053559049257064774\ldots, \tag{B.138}$$

in agreement with the closed form to more than eighty decimal places. This precision refers to a convergent one-dimensional check, not to the full quadrature of D_3.

The three-kink endpoint derivative. For the uncut interval endpoints and the consecutive-kink hard core, the finite part of the charge-differentiated source derivative of the odd three-kink moment is

$$\partial_v\partial_\eta A_{3,0} = -108 + 60\zeta(2) + 22\zeta(3) - 34\zeta(4) + 16\log 2. \tag{B.139}$$

Its reduction is explicit. Put

$$j(r) = (1-r)\log(1-r) - r\log r. \tag{B.140}$$

For the two gaps r and t with span $R = r + t$, the integral over the first position gives the odd endpoint factor

$$\Phi_{\mathrm{odd}}(r,t) = 2j(R) - j(r) - j(t) - 2(1-R). \tag{B.141}$$

Consequently

$$\partial_v\partial_\eta A_3(\varepsilon) = 2\int_{2\varepsilon}^{1} \big[j(R) - (1-R)\big] I_{R,v}(R,\varepsilon)\,\mathrm{d}R - 2\int_{\varepsilon}^{1-\varepsilon} j(r)\, I_{r,v}(r,\varepsilon)\,\mathrm{d}r. \tag{B.142}$$

Both inner integrals, $I_{R,v}$ at fixed span and $I_{r,v}$ at fixed gap, have elementary dilogarithmic primitives. Define

$$\begin{aligned} J_0(e) &= \frac{2}{e} - \frac{2}{1-e} - 4\log e + 4\log(1-e), \\ \Phi_{\log}(y) &= -\frac{\log y + 1}{y} + \frac{\log y}{1-y} - \log y + \log(1-y) + \log^2 y - 2\log y\log(1-y) - 2\,\mathrm{Li}_2(y), \\ J_1(e) &= -4\big[\Phi_{\log}(1-e) - \Phi_{\log}(e)\big]. \end{aligned} \tag{B.143}$$

Then

$$I_{R,v} = R^{-1}\big[-2\log R\, J_0(\varepsilon/R) + J_1(\varepsilon/R)\big], \tag{B.144}$$

and indeed $\Phi'_{\log}(y) = \log y/[y^2(1-y)^2]$. For the second integral let

$$\begin{aligned} \Psi_0(t) &= t + 2\log t - \frac{1}{t}, \\ \Psi_1(t) &= t\big[\log(1+t) - \log t\big] - 2\,\mathrm{Li}_2(-t) - \log^2 t + \frac{\log t + 1 - \log(1+t)}{t} + \log t. \end{aligned} \tag{B.145}$$

The derivative of Ψ_1 is $(1 + 2/t + 1/t^2)[\log(1+t) - \log t]$, and

$$I_{r,v} = \frac{2}{r}\Big[\Psi_1\Big(\frac{1-r}{r}\Big) - \Psi_1\Big(\frac{\varepsilon}{r}\Big) - \log r\Big\{\Psi_0\Big(\frac{1-r}{r}\Big) - \Psi_0\Big(\frac{\varepsilon}{r}\Big)\Big\}\Big]. \tag{B.146}$$

The finite part of Eq. (B.142) consists of two bulk terms and the lower-span region $R = \varepsilon t$; the second integral has no additional finite boundary term. The pieces are

piece	finite contribution
first bulk	$-144 + 64\zeta(2) + 32\zeta(3) - 24\zeta(4)$
second bulk	$60 - 16\zeta(2) - 40\zeta(3) - 10\zeta(4)$
lower-span boundary	$-24 + 16\log 2 + 12\zeta(2) + 30\zeta(3)$

and their sum is Eq. (B.139). The first bulk is

$$2\ \mathrm{FP}\int_0^1 \frac{j(r) - (1-r)}{r}\big[-4\log^2 r + 8\log r + 8 + 8\zeta(2)\big]\,\mathrm{d}r. \tag{B.147}$$

For the second bulk set $l = \log r$, $\bar{l} = \log(1-r)$, $\tau = (1-r)/r$ and

$$\Upsilon = -2\tau(\bar{l} + l) + 4\,\mathrm{Li}_2(1-r) - 2\bar{l}^2 + 2l^2 + \frac{2(\bar{l} + l + 1)}{\tau} + 2\bar{l} + 2. \tag{B.148}$$

The integrand is $-2(\tau\bar{l}-l)\Upsilon$; its part with logarithms only integrates to $12+8\zeta(2)-24\zeta(3)-16\zeta(4)$, and its dilogarithm part to $48-24\zeta(2)-16\zeta(3)+6\zeta(4)$. The lower-span contribution is

$$2\ \mathrm{FP}\int_0^{1/2}\frac{-2\log e\,J_0(e)-J_1(e)}{e}\,\mathrm{d}e. \tag{B.149}$$

Euler's identity (B.52) reduces its integrand, with $l=\log e$ and $\bar{l}=\log(1-e)$, to

$$\frac{8(\bar{l}+1)}{e^2}-\frac{8(\bar{l}+1)}{e(1-e)}+\frac{16(l-\bar{l})}{e}+\frac{8(\bar{l}^2+l^2)}{e}-\frac{16\zeta(2)}{e}+\frac{32\,\mathrm{Li}_2(e)}{e}, \tag{B.150}$$

whose six finite integrals over $(0,\frac{1}{2})$ are, in order,

$$-24+16\log 2,\quad 4\zeta(2),\quad 8\zeta(2),\quad 2\zeta(3)-\tfrac{16}{3}\log^3 2,\quad 16\zeta(2)\log 2,\quad 28\zeta(3)-16\zeta(2)\log 2+\tfrac{16}{3}\log^3 2. \tag{B.151}$$

This gives the boundary entry of the table directly. The compendium checks all primitives and rational-logarithm moment operations symbolically (`code/dual_d1/code/archived/LRI_1d_complete/stages/D3_exact/reproducibility/`).

The four-kink endpoint finite part. The current endpoint response has the expansion

$$\begin{aligned}\partial_\eta Q_4(\varepsilon)=&-\frac{2\zeta(2)}{\varepsilon}+\ell^3-(1+8\zeta(2))\ell^2-(2+32\zeta(2)+28\zeta(3))\ell\\&+6-12\zeta(2)-54\zeta(3)-25\zeta(4)+o(1),\end{aligned} \tag{B.152}$$

so that

$$\partial_\eta Q_{4,0}=6-12\zeta(2)-54\zeta(3)-25\zeta(4). \tag{B.153}$$

Let $\mathsf{f}(t)=t\log t+(1-t)\log(1-t)$, with continuous values at the endpoints. For the gaps r,t,w and span $R=r+t+w$, the integral of the endpoint logarithm over the first position is

$$\Phi_4=2\mathsf{f}(R)+\mathsf{f}(r)+\mathsf{f}(w)-\mathsf{f}(R-r)-\mathsf{f}(R-w). \tag{B.154}$$

The rational four-kink kernel is W_4 of Eq. (A.33). At fixed span put $y=r/R$ and $e=\varepsilon/R$; the integration over the other angular gap is elementary. With

$$\mathcal{A}(y)=\frac{(y+1)(y^2+1)}{y^2(1-y)}=-1+\frac{4}{1-y}+\frac{2}{y}+\frac{1}{y^2},\qquad \mathsf{F}_R(y)=\mathsf{f}(R)+\mathsf{f}(Ry)-\mathsf{f}\big(R(1-y)\big), \tag{B.155}$$

and the function $\mathcal{M}_{\mathrm{out}}(y,e)=(1+1/y)[e^{-1}-(1-y-e)^{-1}]+\mathcal{A}(y)\log[(1-y-e)/e]$ of Eq. (B.42), one obtains the exact two-dimensional formula

$$\partial_\eta Q_4(\varepsilon)=2\int_{3\varepsilon}^1\frac{\mathrm{d}R}{R}\int_{\varepsilon/R}^{1-2\varepsilon/R}\mathcal{M}_{\mathrm{out}}(y,\varepsilon/R)\,\mathsf{F}_R(y)\,\mathrm{d}y, \tag{B.156}$$

which is also integrated directly at finite cutoff as a separate check.

Angular finite part and the radial logarithms. Let $\varkappa(R)=2\log R-\log(1-R)-1$ and $\mathsf{F}_1=2\mathsf{f}(R)$. Near $y=0$, $\mathsf{F}_R(y)=Ry[\log y+\varkappa(R)]+O(y^2)$; near $y=1$, $\mathsf{F}_R(y)\to\mathsf{F}_1$. Define

$$\mathcal{H}(e,R)=\int_e^{1-2e}\mathcal{M}_{\mathrm{out}}(y,e)\,\mathsf{F}_R(y)\,\mathrm{d}y,\qquad J_{\mathrm{FP}}(R)=\mathrm{FP}\int_0^1\mathcal{A}(y)\,\mathsf{F}_R(y)\,\mathrm{d}y. \tag{B.157}$$

Differentiating $\mathcal{H}$ with respect to $\log e$, including the moving boundaries, gives its part of order e^0,

$$\partial_{\log e}\mathcal{H}\big|_{e^0}=\frac{3R}{2}\log^2 e+(2R\varkappa+4\mathsf{F}_1)\log e-J_{\mathrm{FP}}-R+2\mathsf{F}_1. \tag{B.158}$$

Hence

$$\mathcal{H}\big|_{e^0} = \frac{R}{2}\log^3 e + (R\varkappa + 2\mathsf{F}_1)\log^2 e + (-J_{\mathrm{FP}} - R + 2\mathsf{F}_1)\log e + C_0(R), \tag{B.159}$$

where the constant is

$$C_0(R) = \mathrm{FP}\int_0^1 [-\mathcal{B}(y) + \mathcal{A}(y)\log(1-y)]\mathsf{F}_R(y)\,\mathrm{d}y + R(1-\varkappa) - (4 + 2\zeta(2))\mathsf{F}_1, \tag{B.160}$$

with $\mathcal{B}(y) = (1 + 1/y)/(1 - y)$. These expressions keep both angular endpoints.

It is essential to insert $\log e = \ell - \log R$ in the *whole* polynomial; integrating only $C_0(R)$ would miss finite terms that come from powers of $\log e$. The radial moments are

$$\begin{aligned}
\mathsf{U}(y) &= \int_0^1 \mathsf{F}_R(y)\frac{\mathrm{d}R}{R} = -2\,\mathrm{Li}_2(y) - \log y \log(1-y), \\
\mathsf{V}(y) &= \int_0^1 \log R\,\mathsf{F}_R(y)\frac{\mathrm{d}R}{R} = \zeta(3) - \mathsf{U}(y) + \mathrm{Li}_3(y) - \mathrm{Li}_3(1-y),
\end{aligned} \tag{B.161}$$

which follow, for example, from

$$\int_0^1 \mathsf{f}(cR)\frac{\mathrm{d}R}{R} = \mathsf{f}(c) - \mathrm{Li}_2(c), \qquad \int_0^1 \log R\,\mathsf{f}(cR)\frac{\mathrm{d}R}{R} = -\mathsf{f}(c) + \mathrm{Li}_2(c) + \mathrm{Li}_3(c). \tag{B.162}$$

The remaining bulk integral is

$$2\ \mathrm{FP}\int_0^1 \Big[-\mathcal{B}\,\mathsf{U} + \mathcal{A}\big\{\log(1-y)\,\mathsf{U} + \mathsf{V}\big\}\Big]\,\mathrm{d}y = 12 - 26\zeta(3) - 29\zeta(4). \tag{B.163}$$

One integration by parts removes the trilogarithms. A primitive of $\mathcal{A}$ is

$$\Pi_{\mathcal{A}}(y) = -y + 2\log y - 4\log(1-y) - \frac{1}{y}, \tag{B.164}$$

and the finite endpoint term for $\mathcal{D}(y) = \zeta(3) + \mathrm{Li}_3(y) - \mathrm{Li}_3(1-y)$ is

$$\big[\Pi_{\mathcal{A}}\mathcal{D}\big]_0^1 = 1 + \zeta(2) - 4\zeta(3). \tag{B.165}$$

The resulting rational-logarithm and dilogarithm moments are elementary derivatives of beta integrals.

The other endpoint constants contribute $4+16\zeta(2)+20\zeta(4)$. The shift from the radial logarithms is

$$\Delta_{\mathrm{rad}} = 2\int_0^1 \Big[-\frac{2\mathsf{F}_1}{R}\log R + \Big(\varkappa + \frac{2\mathsf{F}_1}{R}\Big)\log^2 R - \frac{1}{2}\log^3 R\Big]\,\mathrm{d}R = -10 - 28\zeta(2) - 28\zeta(3) - 16\zeta(4), \tag{B.166}$$

which follows from

$$\begin{gathered}
\int_0^1 \mathsf{f}(R)\frac{\log R}{R}\,\mathrm{d}R = \zeta(2) + \zeta(3), \qquad \int_0^1 \mathsf{f}(R)\frac{\log^2 R}{R}\,\mathrm{d}R = -2\big(\zeta(2) + \zeta(3) + \zeta(4)\big), \\
\int_0^1 \varkappa(R)\log^2 R\,\mathrm{d}R = -8 - 2\zeta(2) - 2\zeta(3).
\end{gathered} \tag{B.167}$$

Adding Eq. (B.163), the other endpoint constants and Δ_{rad} gives Eq. (B.153), and collecting the remaining logarithms gives Eq. (B.152).

C The dual of the chain: exact certificates, the seventh and eighth orders

This appendix gives the derivations of the compendium (written by OpenAI GPT-6 astra; Sec. 1.2) behind Secs. 2.4 and 2.5, edited to the notation of this note. As in Appendices A and B, everything here holds within the matched two-coupling prescription of Sec. 2.1: the certificates and the rational part of C_8^{SR} prove reductions of the archived expressions of that prescription, and the numerical values are evaluations within it (Sec. 2.1, "What 'proved within the prescription' covers"). Appendix C.1 sets up the exact integration and the word algebra in which the certificates of C_5^{SR} to C_7^{SR} are written, and Appendix C.2 completes the seventh order. Appendices C.3 to C.7 treat the eighth order: its trace reduction, the exact rational part, the four computational stages of the full value, the assembly with the zeta-value candidate, and the numerical evaluation of periods. Appendices C.8 and C.9 record the general formulas that were tested and the relation to the other end of the long-range window; Appendices C.10 and C.11 give two results that use only published data of the dual; Appendix C.12 prints the polynomial B_{old} of Eq. (2.10).

Notation. As in Sec. 2.1, $\bar\delta = 1 - s$, $x = \Delta_\sigma = \bar\delta/2$, ε is the hard core and $\ell = \log\varepsilon$; $u = g^2$ and $v = B^2 = 1 + z$ are the essential couplings, with $\beta_u = 2uF$ and $\beta_z = (1+z)H$. The beta inputs are Taylor coefficients of F and H at the marginal reference point $v = 1$, $\bar\delta = 0$, not at the fixed point. Their names $(a, p, b, c, f, \dots, J_9, K_v)$ are the compendium's and denote coefficients only. Several of these letters also have unrelated uses here, distinguished by subscripts or context: $\Omega_n(\varepsilon)$ is the hard-core domain, W_n a wall weight and $W_N\mathcal{Z}$ a span of MZVs, Q_{2m} a current moment, R a sum of gaps and R_w a regularized constant, S_j an insertion, P_6, P_8, $P_{w,k}$ periods and polynomials, L a logarithmic variable, and B, B_j a boundary derivative and its face integrals (Appendices C.4 and C.5), not the essential charge. The input N_δ keeps the compendium's name; the δ in it is $\bar\delta$. For a regulated response I we write $I_0 = [\varepsilon^0\ell^0]I$ and $I_1 = [\varepsilon^0\ell^1]I$, and more generally $(I)_m = [\varepsilon^0\ell^m]I$. A subscript v or $\bar\delta$ after a comma denotes a derivative at the reference point, as in $b_{3,v\bar\delta} = \partial_v\partial_{\bar\delta}b_3$, while a final subscript 0 or 1 after a comma is the projection I_0 or I_1, as in $Q_{2,0}$, $\mathcal{K}_{8,1}$ and $A_{9,1} = (A_9)_1$, the coefficient written $A_9[\varepsilon^0\ell^1]$ in Sec. 2.6; thus $(\partial_v A_7)_1$ is the coefficient of $\varepsilon^0\ell^1$ in $\partial_v A_7$. The symbol ∂_η is the derivative with respect to the endpoint source η of Sec. 2.1, item (ii), at $\eta = 0$. The other symbols of the dual are those of Appendix A.1: the ordered moments A_{2m+1}, Z_{2m} and Q_{2m} with the current weight Ξ_{2m} (Eq. (A.31)), the vacuum-normalized odd and current responses $\mathcal{C}_{2m+1}$ and $\mathcal{K}_{2m}$ (Eq. (A.37)), the charge and tail insertions $S_j = \sum_{i<k} q_iq_k\log^j(X_k - X_i)$ (Eq. (A.41)), the Jacobian entries j_n and λ_{2m} (Eq. (A.42)) and the subtraction recurrence (Eq. (A.47)). The formal projection π_0 is defined in Appendix C.4; a superscript (0) denotes it, as in $U^{(0)} = \pi_0 U$.

C.1 Exact integration, regularized periods and the certificates

The numerical tables of the compendium are not the definitions of its results. The primary exact data are rational functions of the cutoff multiplying iterated integrals. This subsection explains how they are integrated and reduced, including the constants that a symbol calculation does not see.

Keeping the hard core during exact integration. After the integral over the overall translation, use cumulative coordinates and shift $X_j = j\varepsilon + \tilde X_j$. The remaining simplex is $0 < \tilde X_1 < \cdots < \tilde X_{n-1} < 1 - (n-1)\varepsilon$. At $v = 1$, $\bar\delta = 0$ the kernel is rational, and the charge and tail derivatives insert polynomials in logarithms. Partial fractions, integration by parts and shuffle products reduce the successive integrations to hyperlogarithms, defined with the outermost letter first by

$$G(a_1, \dots, a_n; \Lambda) = \int_0^\Lambda \frac{\mathrm{d}t'}{t' - a_1}\, G(a_2, \dots, a_n; t'), \qquad G(;\Lambda) = 1 \tag{C.1}$$

(the compendium writes T for the argument Λ; here T is the stability trace). For zero letters the tangential-base convention $G(0^n;\Lambda) = \log^n \Lambda/n!$ is used. The total differential is

$$\mathrm{d}G(a_1,\dots,a_n;\Lambda) = \sum_{i=1}^{n} G(a_1,\dots,\widehat{a_i},\dots,a_n;\Lambda)\ \mathrm{d}\log\frac{a_{i-1}-a_i}{a_{i+1}-a_i}, \qquad a_0=\Lambda, \quad a_{n+1}=0, \tag{C.2}$$

with coincident letters taken as limits. For affine letters it gives an exact recurrence in rational kernels. Antiderivatives are checked by differentiation, and their constants are fixed by the specified base point; differentiation alone does not fix a constant.

For a word w of nonnegative letters write $-w$ for the word with every letter negated. At infinity

$$G(-w;\Lambda) = \sum_{n\ge 0} \Lambda^{-n} P_{w,n}(\log\Lambda), \qquad P_{w,0}(L) = \sum_{j=0}^{|w|} \frac{L^j}{j!}\, R_{w_{j+1}\cdots w_{|w|}}, \tag{C.3}$$

with the constants R defined below. The inverse-power coefficients follow from the differential equation. A rational prefactor can grow as a power of Λ, so enough inverse powers must be kept before a finite coefficient is taken. In the same way, positive powers of ε are kept whenever a vacuum denominator can lower their degree.

The rational shuffle algebra. For a word $w = (a_1,\dots,a_n)$ of nonnegative letters define the regularized constant at infinity

$$R_w = \mathrm{Reg}_{\Lambda\to\infty}\, G(-a_1,\dots,-a_n;\Lambda), \qquad R_\varnothing = 1, \tag{C.4}$$

where Reg takes the coefficient of $\Lambda^0(\log\Lambda)^0$. The hyperlogarithms themselves have no positive powers of Λ, so the constants obey the shuffle product,

$$R_w R_{w'} = \sum_{w''\in w \amalg w'} R_{w''}, \tag{C.5}$$

with shuffle multiplicities. The basic values are $R_0 = R_1 = 0$ and $R_a = -\log a$ for $a > 0$. A subscript ab that denotes the single integer product satisfies $R_{ab} = R_a + R_b$; it must not be confused with the two-letter word (a,b).

Lemma (repeated-letter values). For $b > 0$, $a \ge 0$ and $n \ge 2$,

$$R_{a,b^{n-1}} = \frac{(-\log b)^n}{n!} + \mathrm{Li}_n(1-a/b), \tag{C.6}$$

where b^{n-1} is the letter b repeated $n-1$ times.

Proof. The inner repeated-letter integral is $\log^{n-1}(1+t'/b)/(n-1)!$. Set $\vartheta = \log(1+t'/b)$ in the outer integration. The kernel splits into a constant 1 and $(1-a/b)/(e^\vartheta - (1-a/b))$. The first term integrates to the logarithmic polynomial. The remaining convergent geometric-series integral is $\mathrm{Li}_n(1-a/b)$, extended along the nonsingular real domain; its integrable limit at $a = 0$ is $\zeta(n)$. □

Hence

$$\zeta(n) = -\frac{2^{n-1}}{2^{n-1}-1}\, R_{2,1^{n-1}}, \qquad \mathrm{Li}_n(r) = R_{a,b^{n-1}} - R_{b^n} \quad (1-r = a/b > 0). \tag{C.7}$$

These identities, together with the prime factorization of rational logarithms, give a deterministic lift of the exact beta expressions to rational vectors of words. For words whose last letter is positive, uniform scaling gives

$$R_{\kappa w} = \sum_{j=0}^{|w|} R_{\kappa^j}\, R_{w_{j+1}\cdots w_{|w|}}, \qquad \kappa > 0, \tag{C.8}$$

where κw multiplies every letter by κ and κ^j is the word of j letters κ. The restriction on the last letter prevents silently rescaling a trailing-zero tangential logarithm.

Confluence fixes the full constants. *Lemma (convergent confluence).* Let $e = (e_1, \dots, e_n)$ be a binary word with $e_1 = 0$ and $e_n = 1$. Choose $a_i \geq 0$ for a zero bit and $a_i > 0$ for a one bit, and set $\lambda_i(\Lambda) = -a_i$ if $e_i = 0$ and $\lambda_i(\Lambda) = \Lambda + a_i$ if $e_i = 1$. Then

$$\lim_{\Lambda\to\infty} G\big(\lambda_1(\Lambda), \dots, \lambda_n(\Lambda); \Lambda\big) = G(e_1, \dots, e_n; 1). \tag{C.9}$$

Proof. Rescale every integration variable by Λ. The two kinds of kernels become $\mathrm{d}\upsilon/(\upsilon + a_i/\Lambda)$ and $-\mathrm{d}\upsilon/(1-\upsilon+a_i/\Lambda)$, whose absolute values are bounded by $\mathrm{d}\upsilon/\upsilon$ and $\mathrm{d}\upsilon/(1-\upsilon)$. The product of these bounds is integrable on the ordered simplex because the binary word begins with 0 and ends with 1. Dominated convergence gives the limit of the actual integral, including its constant. Fixed zero intercepts do not invalidate the bound. □

Applying Eq. (C.2) to the affine letters, every nonconstant logarithmic derivative has a fixed nonpositive pole, and deleting a letter produces a finite recursive expansion

$$G\big(\lambda_1(\Lambda), \dots, \lambda_n(\Lambda); \Lambda\big) = \sum_w f_w\, G(-w; \Lambda), \qquad f_w \in \mathbb{Z}. \tag{C.10}$$

The original convergent word vanishes at $\Lambda \to 0$, while pure zero subwords keep their tangential logarithm; these conditions fix the constants. Taking Reg yields the exact relations $\sum_w f_w R_w = G(e; 1)$. For example,

$$-R_{a,b} - R_{b,a} + R_{a+b,b} + R_{a+b,a} = -\zeta(2) \qquad (a, b > 0), \tag{C.11}$$

and the zero-intercept case is $-R_{0,b} + R_{b,b} = -\zeta(2)$. This zero-letter extension is needed by the target of C_6^{SR}; treating all letters as strictly positive would not suffice.

The weight-five constants. Multiple zeta values (MZVs) are written with decreasing indices,

$$\zeta(n_1, \dots, n_r) = \sum_{k_1 > \cdots > k_r \geq 1} \frac{1}{k_1^{n_1} \cdots k_r^{n_r}}, \qquad n_1 \geq 2. \tag{C.12}$$

For convergent words the right-hand sides of Eq. (C.9) are MZVs, with a sign set by the number of one letters. Through weight four their reduction uses shuffle, stuffle, duality and $\zeta(2)^2 = \frac{5}{2}\zeta(4)$. At weight five the required height-one identity follows from

$$1 - \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-u-v)} = u \int_0^1 t'^{-u-1}\big[(1-t')^{-v} - 1\big]\,\mathrm{d}t' = \sum_{m,n\geq 1} \zeta(m+1, 1^{n-1})\, u^m v^n, \tag{C.13}$$

where u and v are formal variables here. The Taylor expansion of $\log\Gamma$ and the coefficient of u^3v^2 give $\zeta(4,1) = 2\zeta(5) - \zeta(2)\zeta(3)$. The convergent stuffle and shuffle products give respectively

$$\zeta(2)\zeta(3) = \zeta(2,3) + \zeta(3,2) + \zeta(5), \qquad \zeta(2)\zeta(3) = \zeta(2,3) + 3\zeta(3,2) + 6\zeta(4,1), \tag{C.14}$$

hence

$$\zeta(3,2) = 3\zeta(2)\zeta(3) - \frac{11}{2}\zeta(5), \qquad \zeta(2,3) = \frac{9}{2}\zeta(5) - 2\zeta(2)\zeta(3). \tag{C.15}$$

Duality supplies the remaining binary words of weight five. No assumption of numerical independence is needed. The iterated-integral and MZV framework is standard [25, 27]; the programs of the compendium implement the explicit conventions above.

The certificates and their logical scope. The proposed closed expression is subtracted from the exact archived expression of the renormalization-group (RG) assembly, and the difference is lifted to the word algebra. The following finite identities then hold in the free rational vector space of words (C_7^{SR} is in Appendix C.2):

target	weight	nonzero words	certificate rows
$D_3 - D_3^{\text{closed}}$	2	14	10
	3	54	47
	4	102	419
$C_6^{\text{SR}} - C_6^{\text{SR closed}}$	2	26	21
	3	169	142
	4	608	3 879
	5	1 142	52 339

The totals, 170 words and 476 rows for D_3 and 1 945 words and 56 381 rows for C_6^{SR}, are those of Sec. 2.4. For each weight the target X is written explicitly as

$$\mathsf{X} = \sum_j r_j (\mathsf{B}_j \sqcup\!\sqcup \langle v_j \rangle), \qquad r_j \in \mathbb{Q}, \tag{C.16}$$

where $\langle v_j \rangle$ is a word and every B_j is one of the analytically established zero relations above (shuffle, stuffle, duality, the confluence relations and the height-one identity). The verifier regenerates these relations from their labels and checks the rational coefficients word by word. Since the relations evaluate to zero, so does the target.

The electronic certificate is part of the proof of the reduction, not optional numerical support: it contains all coefficients, and none is inferred from decimal agreement. Both certificates are plain JSON embedded in standalone Python verifiers (for D_3, `code/dual_d1/code/archived/D3_analytic_proof/verify_D3_proof_standalone.py`). Modular elimination was used to discover the certificates, but the final verification uses exact rational arithmetic and no numerical period. The proof does not require the chosen relations to span all MZV identities. Both certificates replay successfully (Sec. 2.8). In addition, a standard-library checker (`code/dual_d1/code/archived/C7_closed_form_proof/code/verify_C6_input_map.py`) parses the archived beta expressions as a restricted syntax tree, lifts the individual beta coefficients directly into the shuffle algebra and verifies the trace assembly, the archived C_6^{SR} expression, its difference from the closed form and all 1 945 target coefficients, so that the certificate is tied to the RG inputs. It does not reintegrate the original kink kernels.

C.2 The seventh order and the weight-six certificate

By Eq. (2.8), $C_7^{\text{SR}} = 4D_4 - T_3 - T_2^2 + \frac{1}{2} C_3^{\text{SR}} C_5^{\text{SR}}$ needs a new determinant coefficient but no new trace coefficient. This follows from expanding the characteristic polynomial, not from extrapolating a truncated beta function. The closed form of D_4 is in Sec. 2.4, and

$$D_4 = -2.534629799538250115260606412 4\ldots, \qquad C_7^{\text{SR}} = 51.068988612928108831 18\ldots. \tag{C.17}$$

New derivatives. At this order the required jets are

$$\begin{aligned} F &= z + u(1 + az + p\bar\delta + Az^2 + E\bar\delta z + V\bar\delta^2 + \alpha z^3 + \beta z^2\bar\delta + \gamma z\bar\delta^2) \\ &\quad + u^2(b + Gz + R\bar\delta + \Phi z^2 + \Psi z\bar\delta) + u^3(J + \Omega z) + \cdots, \\ H &= -\bar\delta + 4u + u^2(c + fz + q\bar\delta + Lz^2 + Mz\bar\delta + Q\bar\delta^2) + u^3(k + Nz + P\bar\delta) + Ku^4 + \cdots, \end{aligned} \tag{C.18}$$

and the determinant combination is

$$D_4 = B_{\text{old}}^{D_4} - \frac{3\alpha}{32} + \frac{\beta}{4} - \frac{\gamma}{2} + \frac{\Phi}{16} - \frac{\Psi}{8} - \frac{\Omega}{32} - \frac{Q}{8} - \frac{P}{16} - \frac{3K}{128}. \tag{C.19}$$

The polynomial $B_{\text{old}}^{D_4}$ involves only earlier inputs; it is regenerated by `code/dual_d1/code/archived/C7_closed_form_proof/code/derive_D4.py`. Differentiation must precede the fixed-point

substitution, since a term that is small at the fixed point can contribute to a derivative at the required order. The derivatives that define the new inputs are listed in Appendix C.3.

The two largest new integrations were the eight-kink current response and the charge derivative of the seven-kink odd response. The latter enters

$$\Omega = \partial_v b_7\big|_{v=1,\bar\delta=0} = -234.421455288963687526514445252 3\ldots. \tag{C.20}$$

Every term of b_7 in the subtraction recurrence, Eq. (A.47), is differentiated, including the endpoint terms and the explicit factors v, and so is every product in the normalized response $\mathcal{C}_7 = A_7 - A_5Z_2 + A_3(Z_2^2 - Z_4) - Z_2^3 + 2Z_2Z_4 - Z_6$; the raw integral $\partial_v A_7$ alone is not Ω. An early partial calculation of C_7^{SR} had used an incorrect numerical value of the second-tail input; the rebuilt value is $Q = -107.065481495547639404572681316 5\ldots$, and with all other inputs fixed $C_7^{\mathrm{SR}} = 21.76630670180764789\ldots - \Omega/8$. The certificate below is tied to the rebuilt input expressions.

Higher-logarithm checks. The rebuilt numerical integration passed the following higher-logarithm tests, which do not use the new single-logarithm coefficient:

$$\begin{gathered}[\ell^8]\partial_v\mathcal{C}_7 = 0, \quad [\ell^7]\partial_v\mathcal{C}_7 = -\frac{6416}{315}, \quad [\ell^6]\partial_v\mathcal{C}_7 = -\frac{1622}{9}, \quad [\ell^5]\partial_v\mathcal{C}_7 = \frac{3848\zeta(2) - 8824}{15}, \\ [\ell^7]\mathcal{K}_8 = -\frac{1088}{315}, \qquad [\ell^6]\mathcal{K}_8 = -\frac{664}{15}, \qquad [\ell^5]\mathcal{K}_8 = 96\zeta(2) - \frac{3656}{15}.\end{gathered} \tag{C.21}$$

A direct finite-cutoff test compares the whole of $\partial_v A_7$ at $\varepsilon = 1/10$ with a separate cubature of the original kernels, not with a fitted asymptotic polynomial.

What the certificate establishes. All 25 archived beta inputs of C_3^{SR} to C_7^{SR} are parsed directly from the archived expressions and assembled into D_4 and C_7^{SR}. Subtracting the proposed zeta-value result and lifting logarithms and polylogarithms to the word algebra gives a target with the homogeneous components

weight	target words	certificate rows	final residual words
2	37	33	0
3	337	284	0
4	1 810	2 880	0
5	7 428	54 363	0
6	15 600	531 318	0
total	25 212	588 878	0

The weight-six identity includes the products of lower-weight periods; a reduction modulo products would not suffice. In the free rational word space the certificate has the form

$$\mathsf{X}_6 = \sum_j r_j\, \mathsf{B}_j \sqcup\!\sqcup \mathsf{M}_j, \qquad r_j \in \mathbb{Q}, \tag{C.22}$$

where X_6 is the weight-six target, every B_j is a proved zero relation and every multiplier M_j is expanded with its full shuffle multiplicities, so that evaluation annihilates the right-hand side. The verifier does not use the Lyndon-polynomial representation employed during discovery; it expands ordinary words with exact integers after clearing denominators.

All 16 convergent binary boundary values of weight six are derived from 31 rational equations, using the shuffle and stuffle products, duality and the height-one identity (C.13). Examples are

$$\begin{aligned}\zeta(5,1) &= \tfrac{3}{4}\zeta(6) - \tfrac{1}{2}\zeta(3)^2, \qquad \zeta(4,2) = -\tfrac{4}{3}\zeta(6) + \zeta(3)^2, \\ \zeta(3,3) &= -\tfrac{1}{2}\zeta(6) + \tfrac{1}{2}\zeta(3)^2, \quad \zeta(2,4) = \tfrac{25}{12}\zeta(6) - \zeta(3)^2.\end{aligned} \tag{C.23}$$

No assumption of numerical independence of MZVs is used. The final certificates and the target suffice for verification; the large discovery matrices and redundant intermediate certificates are not part of the proof input.

C.3 The eighth order: trace reduction and recurrence

The integer powers of x in $\Delta_{\mathcal{E}}$ (the even powers of $\sqrt{x}$) come entirely from the half-trace $T/2$, so $C_8^{\mathrm{SR}} = 8T_4$ (Eq. (2.8)). No new determinant coefficient D_5 is needed, but T_4 needs new derivatives that D_4 does not fix. The jets required for this trace, keeping only the sectors that can contribute after differentiation, extend Eq. (C.18) by ten coefficients:

$$\begin{aligned} F = {} & z + u(1 + az + p\bar{\delta} + Az^2 + E\bar{\delta}z + V\bar{\delta}^2 + \alpha z^3 + \beta\bar{\delta}z^2 + \gamma\bar{\delta}^2 z + W\bar{\delta}^3) \\ & + u^2(b + Gz + R\bar{\delta} + \Phi z^2 + \Psi\bar{\delta}z + S\bar{\delta}^2) + u^3(J + \Omega z + U\bar{\delta}) + J_9 u^4 + \cdots , \\ H = {} & -\bar{\delta} + 4u + u^2(c + fz + q\bar{\delta} + Lz^2 + M\bar{\delta}z + Q\bar{\delta}^2 + L_3 z^3 + M_{21}\bar{\delta}z^2 + M_{12}\bar{\delta}^2 z) \\ & + u^3(k + Nz + P\bar{\delta} + N_2 z^2 + N_\delta \bar{\delta}z) + u^4(K + K_v z) + \cdots . \end{aligned} \tag{C.24}$$

These are not the exact beta functions beyond the displayed order. For example, the term $K_v z u^4$ of H must be kept, because H_z contributes at order $\bar{\delta}^4$.

Derivative dictionary. All derivatives are taken at $v = 1$, $\bar{\delta} = 0$. The normalized auxiliary current beta function is $\beta_h/(\sqrt{2}b_0) = -g^2 + h_4 g^4 + h_6 g^6 + h_8 g^8 + \cdots$ (so $h_2 = -1$), and b_3, b_5, b_7, b_9 are the odd fugacity coefficients of the subtraction recurrence. The lower jets are

$$\begin{aligned} & a = b_{3,v}, && p = b_{3,\bar{\delta}}, && A = \tfrac{1}{2}b_{3,vv}, && E = b_{3,v\bar{\delta}}, && V = \tfrac{1}{2}b_{3,\bar{\delta}\bar{\delta}}, \\ & \alpha = \tfrac{1}{6}b_{3,vvv}, && \beta = \tfrac{1}{2}b_{3,vv\bar{\delta}}, && \gamma = \tfrac{1}{2}b_{3,v\bar{\delta}\bar{\delta}}, \\ & b = b_5, && G = b_{5,v}, && R = b_{5,\bar{\delta}}, && \Phi = \tfrac{1}{2}b_{5,vv}, && \Psi = b_{5,v\bar{\delta}}, \\ & J = b_7, && \Omega = b_{7,v}, \\ & c = -4h_4, && f = -4h_{4,v}, && q = -4h_{4,\bar{\delta}}, && L = -2h_{4,vv}, && M = -4h_{4,v\bar{\delta}}, \\ & Q = -2h_{4,\bar{\delta}\bar{\delta}}, && k = -4h_6, && N = -4h_{6,v}, && P = -4h_{6,\bar{\delta}}, && K = -4h_8, \end{aligned} \tag{C.25}$$

and the ten additional inputs of C_8^{SR} are

$$\begin{aligned} & W = \tfrac{1}{6}b_{3,\bar{\delta}\bar{\delta}\bar{\delta}}, && S = \tfrac{1}{2}b_{5,\bar{\delta}\bar{\delta}}, && U = b_{7,\bar{\delta}}, && J_9 = b_9, && L_3 = -\tfrac{2}{3}h_{4,vvv}, \\ & M_{21} = -2h_{4,vv\bar{\delta}}, && M_{12} = -2h_{4,v\bar{\delta}\bar{\delta}}, && N_2 = -2h_{6,vv}, && N_\delta = -4h_{6,v\bar{\delta}}, && K_v = -4h_{8,v}. \end{aligned} \tag{C.26}$$

A mixed first derivative carries no factor $\frac{1}{2}$; Taylor factors appear only for repeated derivatives, as displayed.

Finite-response recurrence through ninth order. The eight-kink current response is

$$\mathcal{K}_8 = Q_8 - Q_6 Z_2 + Q_4(Z_2^2 - Z_4) + Q_2(-Z_2^3 + 2Z_2 Z_4 - Z_6), \tag{C.27}$$

as in Eq. (A.37), and the nine-kink odd response, the coefficient of g^8 in the ratio of the odd generating function to the vacuum one (Appendix A.7) at zero source ($A_1 = 1$), is

$$\mathcal{C}_9 = A_9 - A_7 Z_2 + A_5(Z_2^2 - Z_4) + A_3(-Z_2^3 + 2Z_2 Z_4 - Z_6) + Z_2^4 - 3Z_2^2 Z_4 + Z_4^2 + 2Z_2 Z_6 - Z_8. \tag{C.28}$$

The Jacobian entries needed next are

$$j_7 = -2\partial_v \mathcal{C}_{7,0} + \partial_\eta \mathcal{C}_{7,0}, \qquad \lambda_6 = -2\mathcal{K}_{6,0} - 4v\,\partial_v \mathcal{K}_{6,0} + 2v\,\partial_\eta \mathcal{K}_{6,0}, \tag{C.29}$$

with the same rule as j_3, j_5 and λ_2, λ_4 in Eq. (A.42). The compendium defines j_7 in this form but uses λ_6 without writing it; Eq. (C.29) gives λ_6 by that rule. The new counterterm recurrences, which continue Eq. (A.47), are

$$\begin{aligned} h_8 &= \mathcal{K}_{8,1} - 2Q_{2,0}b_7 - 4\mathcal{K}_{4,0}b_5 - 6\mathcal{K}_{6,0}b_3 - \lambda_2 h_6 - \lambda_4 h_4 + \lambda_6, \\ b_9 &= \mathcal{C}_{9,1} - 3\mathcal{C}_{3,0}b_7 - 5\mathcal{C}_{5,0}b_5 - 7\mathcal{C}_{7,0}b_3 - 2vj_1 h_8 - 2vj_3 h_6 - 2vj_5 h_4 + 2vj_7. \end{aligned} \tag{C.30}$$

Products and derivatives are formed before the finite parts are extracted. An unavailable raw scalar is an error, not a zero coefficient, and the compendium's full scalar-aware convolution code enforces this.

Solving $F = H = 0$ with Eq. (C.24) through $\bar{\delta}^4$ and expanding $T = 2uF_u + (1+z)H_z$ gives Eq. (2.10): C_8^{SR} is B_{old} plus a fixed linear combination of the ten new inputs. The polynomial B_{old} involves only the 25 earlier inputs; it is printed in Appendix C.12. Its value is $-861.663567235479779201497190992 3\ldots$, and its rational projection is $\frac{1153}{6}$. Replacing it by its rational projection before the assembly would lose genuine terms.

C.4 The rational part of the eighth order

The formal weight-zero map π_0 acts on the graded representation of an integrated quantity: it removes the positive-weight constant periods. It is not a numerical limit in which the real zeta values vanish, and variable logarithms cannot be discarded before integration; for example,

$$\int_0^\infty \frac{\log(1+t')}{(1+t')^2}\,\mathrm{d}t' = 1. \tag{C.31}$$

Products, coefficient convolutions and power divergences must already have been treated consistently when π_0 is applied.

Four raw residues. Define

$$\begin{aligned} \mathcal{R}_6 &= \pi_0(\partial_v\partial_{\bar{\delta}}Q_6)_1, & \mathcal{R}_8 &= \pi_0(\partial_v Q_8)_1, \\ \mathcal{R}_{\mathrm{tail}} &= \pi_0(\partial_{\bar{\delta}}A_7)_1 - \pi_0(\partial_{\bar{\delta}}Z_6)_1, & \mathcal{R}_9 &= \pi_0(A_9)_1 - \pi_0(Z_8)_1 + 2\pi_0(\partial_\eta A_7)_0, \end{aligned} \tag{C.32}$$

the residues of the six-kink and eight-kink current moments, of the seven-kink tail and of the nine-kink fugacity. Before any of them was inserted, the exact trace and subtraction reduction gave

$$\pi_0 C_8^{\mathrm{SR}} = \frac{5345}{12} - \frac{1}{2}\mathcal{R}_6 - \frac{1}{8}\mathcal{R}_8 + \frac{3}{4}\mathcal{R}_{\mathrm{tail}} + \frac{1}{4}\mathcal{R}_9. \tag{C.33}$$

The completed values are

$$\mathcal{R}_6 = \frac{399343}{2}, \qquad \mathcal{R}_8 = -757993, \qquad \mathcal{R}_{\mathrm{tail}} = -\frac{910891}{6}, \qquad \mathcal{R}_9 = \frac{1423087}{3}, \tag{C.34}$$

and after the first two are inserted the known term is $-\frac{111389}{24}$, which gives the sum $\pi_0 C_8^{\mathrm{SR}} = 88$ of Sec. 2.5. None of the raw integrals was inferred from that answer or from the earlier conjecture 160. For reconstruction, the last five raw scalars are

raw scalar	rational value	raw scalar	rational value
$\pi_0(\partial_{\bar{\delta}}A_7)_1$	$-\frac{468683}{3}$	$\pi_0(Z_8)_1$	$\frac{23294}{3}$
$\pi_0(\partial_{\bar{\delta}}Z_6)_1$	$-\frac{8825}{2}$	$\pi_0(\partial_\eta A_7)_0$	$-\frac{184413}{2}$
$\pi_0(A_9)_1$	666540		

The rational inputs and the full-Jacobian check. A separate finite-response reconstruction gives

$$U^{(0)} = \frac{16534}{3}, \qquad J_9^{(0)} = -\frac{10192}{3}, \qquad N_\delta^{(0)} = -20848, \qquad K_v^{(0)} = \frac{77920}{3}, \tag{C.35}$$

and the other six new rational inputs are

$$W^{(0)} = 168, \quad S^{(0)} = -1751, \quad L_3^{(0)} = 128, \quad M_{21}^{(0)} = -384, \quad M_{12}^{(0)} = 2720, \quad N_2^{(0)} = 3712. \tag{C.36}$$

A separate literal full-Jacobian calculation then yields $T^{(0)}(\bar{\delta}) = \frac{\bar{\delta}}{2} + 2\bar{\delta}^2 + \frac{13}{2}\bar{\delta}^3 + 11\bar{\delta}^4 + O(\bar{\delta}^5)$, which reproduces $\pi_0 C_8^{\mathrm{SR}} = 8 \times 11$ without using the residue assembly above.

Boundary differentiation and its checks. If $M(\varepsilon)=\sum_{p,m} c_{p,m}\varepsilon^p\ell^m$ and $B=-\varepsilon\partial_\varepsilon M$, then

$$[\varepsilon^p\ell^0]B=-p\,c_{p,0}-c_{p,1},\qquad [\varepsilon^0\ell^1]M=-[\varepsilon^0\ell^0]B. \tag{C.37}$$

Differentiating a consecutive-gap lower limit removes one integration. For $n=2m$ kinks the reflection weights of the $2m-1$ faces are $(2,\dots,2,1)$; for odd n the faces are all paired. These factors and the minus sign in Eq. (C.37) are essential. The negative powers of ε are predicted from lower moments. For $Q_{6,v\bar\delta}$ the predictions are $(c_{-2,0},c_{-2,1},c_{-1,0},c_{-1,1})^{(0)}=(-7,-32,2193,555)$; for $Q_{8,v}$ they are

power p	$c_{p,0}^{(0)}$	$c_{p,1}^{(0)}$
-3	$\frac{2}{3}$	2
-2	$-\frac{104}{3}$	$-\frac{52}{3}$
-1	$-\frac{80255}{12}$	$-\frac{20429}{6}$

All comparisons in the rational calculation are exact. They are generated from lower moments and do not use the target finite-power residue. The tail and ninth-order groups also pass complete comparisons of their negative-power logarithmic polynomials, and further direct finite-cutoff and alternative endpoint-representation tests are kept in their stage records.

C.5 The full eighth order: the four computational stages

The rational result does not give the full coefficient. Restoring the constant periods requires undoing every earlier projection, keeping the integer-offset logarithms and lower-endpoint constants, and keeping the full counterterm contributions.

The six inherited inputs. Six of the ten new inputs have recorded numerical evaluations only:

input	full numerical value	rational projection
W	$-37.708111514952555\dots$	168
S	$878.183733747745103\dots$	-1751
L_3	$-0.638692810583406\dots$	128
M_{21}	$-171.235576565588699\dots$	-384
M_{12}	$-422.715057969894520\dots$	2720
N_2	$189.742012494084223\dots$	3712

They enter the assembly as inherited numerical records: decimal values from one recorded 100-digit evaluation, with no surviving exact expression and no second evaluation (Sec. 2.5). This is one reason why a complete analytic proof for C_8^{SR} is a further task. The other four new inputs were computed in four stages, F1 (N_δ), F2 (U), F3 (K_v) and F4 (J_9). Their definitions and subtraction factors are fixed by Eq. (2.10) and the recurrence; they are not adjustable parameters.

F1a: the six-kink mixed logarithm. Define

$$P_6=(Q_{6,v\bar\delta})_1\big|_{v=1,\bar\delta=0}=[\varepsilon^0\ell^1]\int_{\Omega_6(\varepsilon)}\Xi_6W_6\,(S_2+2S_1S_2)\,\mathrm{d}^6X, \tag{C.38}$$

where $\Omega_6(\varepsilon)$ is the ordered hard-core domain of six kinks, W_6 the six-kink weight, $\Xi_6=\sum_{j=1}^3(X_{2j}-X_{2j-1})$ the current weight, and $(S_2+2S_1S_2)W_6=\partial_v\partial_{\bar\delta}W_6$ the mixed insertion, with both terms and no Taylor factor. Early quadrature estimates around $1261.9448\dots$ were not accurate enough for an integer-relation search with many constants. The final computation instead integrated the

complete cutoff-boundary source with all constant periods kept. With the five gaps written r_i and $R = \sum_i r_i$, the face integrals are

$$B_j = \varepsilon \int_{\substack{r_i \geq \varepsilon \ (i \neq j) \\ R<1}} (1-R)\, \Xi_6 W_6 \,(S_2 + 2S_1 S_2) \prod_{i \neq j} \mathrm{d}r_i, \qquad r_j = \varepsilon, \tag{C.39}$$

with $B = 2B_1 + 2B_2 + B_3$ (the reflection weights of the five faces) and $P_6 = -[\varepsilon^0 \ell^0] B$ by Eq. (C.37). The final record includes all 21 first-stage blocks and all 64 terminal outputs; exact collection gives 66 404 period terms, and the rational projection is $\frac{399343}{2} = \mathcal{R}_6$. The value of P_6 is given in Sec. 2.5. The two fresh evaluations at 70 and 100 working digits differ in the matching point, the step size, the Taylor order and the asymptotic order. A complete four-kink mixed control agrees exactly with the archived expression; a comparison of the full stage in moving-letter and in ordinary-word form agrees in 196 169 rational coefficients; and all 23 816 rational prefactors are reconstructed exactly from their partial fractions.

On a face all separations carry a factor ε, and the scale is restored by

$$S_1 = -3\ell + s_1, \qquad S_2 = -3\ell^2 + 2s_1\ell + s_2, \tag{C.40}$$

where s_1 and s_2 are the insertions of the rescaled configuration (the six alternating charges give $\sum_{i<k} q_i q_k = -3$). The logarithm-free boundary coefficient uses $s_2 + 2s_1 s_2$, but its variable and constant logarithms remain. The full positive-weight divergence differences at the powers ε^{-2} and ε^{-1} are numerically consistent with zero at about 10^{-88} and 10^{-87}; these are not exact period certificates.

F1b: vacuum normalization. The six-kink current response is $\mathcal{K}_6 = Q_6 - Q_4 Z_2 + Q_2 Z_2^2 - Q_2 Z_4$, so that

$$(\partial_v \partial_{\bar\delta} \mathcal{K}_6)_1 = P_6 + D_{\mathrm{norm}}, \qquad D_{\mathrm{norm}} = [\varepsilon^0 \ell^1]\, \partial_v \partial_{\bar\delta} (-Q_4 Z_2 + Q_2 Z_2^2 - Q_2 Z_4)\big|_{v=1, \bar\delta = 0}. \tag{C.41}$$

The three product contributions are

$$-1300.763962200486633\ldots, \qquad -108 \text{ (exact)}, \qquad -1261.404579171677298\ldots, \tag{C.42}$$

which give

$$D_{\mathrm{norm}} = -2670.168541372163931149878165554559\ldots, \qquad \pi_0 D_{\mathrm{norm}} = -\frac{8719}{2}. \tag{C.43}$$

All 13 Leibniz terms are included, with their repeated-factor multiplicities. The terms with zero cutoff powers alone would give $-1497.3871181439\ldots$, so the products of suppressed and divergent powers are substantial. The input map uses 32 complete scalar coefficients; the 16 two-kink ones can be rederived by elementary integration, and the four-kink finite expressions are inherited exact inputs, not integrated afresh in this step.

F1c: finite-response correction. Define

$$D_{\mathrm{ct}} = \partial_v \partial_{\bar\delta} \big[-2Q_{2,0} b_5 - 4\mathcal{K}_{4,0} b_3 - \lambda_2 h_4 + \lambda_4\big]\big|_{v=1, \bar\delta = 0}, \tag{C.44}$$

the derivative of the counterterm terms of h_6 in the recurrence. The complete result is

$$D_{\mathrm{ct}} = -134.559964098286854425679630067592\ldots, \qquad \pi_0 D_{\mathrm{ct}} = -190100. \tag{C.45}$$

With $k_4 = \mathcal{K}_{4,0}$ and $e_4 = \partial_\eta \mathcal{K}_{4,0}$, the explicit coupling factors of λ_4 give

$$(\lambda_4)_{v\bar\delta} = -6k_{4,v\bar\delta} - 4k_{4,vv\bar\delta} + 2e_{4,\bar\delta} + 2e_{4,v\bar\delta}; \tag{C.46}$$

the extra terms would be lost if the prefactors were treated as constants. Two new complete four-kink finite parts are

$$\begin{aligned}(Q_{4,vv\bar\delta})_0 &= 17290.158097306598287876627149758 44\ldots,\\ (\partial_\eta Q_{4,v\bar\delta})_0 &= 34463.595640455961750271581755932 35\ldots.\end{aligned} \tag{C.47}$$

Since $N_\delta = -4h_{6,v\bar\delta}$ and $h_{6,v\bar\delta} = P_6 + D_{\mathrm{norm}} + D_{\mathrm{ct}}$, the three F1 stages combine into

$$N_\delta = -4P_6 + 11218.914021881803142302231182488606\ldots, \tag{C.48}$$

where the ellipsis abbreviates an archived exact period expression, not an unknown contribution. The rational check is $\frac{399343}{2} - \frac{8719}{2} - 190100 = 5212$, hence $N_\delta^{(0)} = -20848$.

F2: the seven-kink tail coefficient. The derivative of the b_7 recurrence with respect to $\bar\delta$ isolates

$$U = (\partial_{\bar\delta} A_7)_1 - (\partial_{\bar\delta} Z_6)_1 + 2(\partial_\eta \partial_{\bar\delta} A_5)_0 - 2(\partial_\eta \partial_{\bar\delta} Z_4)_0 + D_{\mathrm{lower}}, \tag{C.49}$$

where D_{lower} collects the remaining lower-order terms of the differentiated recurrence. The lower shift terms, the vacuum tail and the endpoint terms are kept unprojected. Collecting everything but the first term into D_{known},

$$\begin{aligned}(\partial_{\bar\delta} A_7)_1 &= 315459.45115269022844857671653830079 7\ldots,\\ D_{\mathrm{known}} &= -316756.13911298175494023311031164286 6\ldots,\end{aligned} \tag{C.50}$$

and their sum is the value of U in Sec. 2.5, $U = -1296.68796029152649165639377334206 9\ldots$, with $U^{(0)} = \frac{16534}{3}$. The seven-kink expression has 66 476 terms and the final U 81 495; their exact scalar relation and the rational projection are checked, and full evaluations at 80 and 110 digits cover 57 346 periods through weight seven. A separately derived leading logarithm,

$$[\varepsilon^0 \ell^8]\partial_{\bar\delta} A_7 = -\frac{2362}{315}, \qquad [\varepsilon^0 \ell^9]\partial_{\bar\delta} A_7 = 0, \tag{C.51}$$

agrees with the leading-logarithm recursion of the lower flow; this is not a replay of the whole subleading logarithmic polynomial. The historical coverage record of F2 names 258 intermediate files that are not among the surviving files. The complete period expression of U and its numerical replay survive, but an independent replay from the full kernels would require regenerating those source stages.

F3: the eight-kink logarithm and six-kink finite responses. Define the full, unprojected scalars

$$P_8 = (Q_{8,v})_1, \qquad X_{vv} = (Q_{6,vv})_0, \qquad X_E = (\partial_v \partial_\eta Q_6)_0. \tag{C.52}$$

The finite-response calculation of $K_v = -4h_{8,v}$ gives

$$K_v = -4P_8 + 16X_{vv} - 8X_E + \mathcal{S}_3, \qquad \mathcal{S}_3 = 24144.92587771278969209842737284526 3\ldots, \tag{C.53}$$

with

$$\begin{aligned}X_{vv} &= 8059.73610218661612575272790112098 5\ldots,\\ X_E &= -88077.17236355213705219767553829123 6\ldots,\\ P_8 &= 214801.26113259864188196280525361862 7\ldots,\\ K_v &= -1486.96210927882340612774291736359 3\ldots.\end{aligned} \tag{C.54}$$

The complete period expression of P_8 has 120 411 terms, and all 37 original source blocks are included through the weighted boundary calculation. Its rational projection is $-757993 = \mathcal{R}_8$, a separate check of the rational calculation.

For certain six-kink finite endpoint responses, scaling gives an overall factor ε multiplying the last integral; proper rational terms then grow only logarithmically after integration and vanish as $\varepsilon \log^k \varepsilon$, so only the part with polynomial growth contributes to the finite constant. This shortcut is not applied to P_8, whose scaling differs and whose proper rational terms contribute to the residue; both endpoints and the corresponding regularized constants are kept. The complete expression of $Q_{6,vv}$ at $\varepsilon = 1/10$ equals $425.5449632473382428\ldots$ and agrees with direct quadrature of the original kernels. The full source of P_8 passes the six rational divergence checks of the table in Appendix C.4, and a positive-weight divergence comparison is numerically below 10^{-85}. Exact source coverage and numerical period checks are distinct kinds of check.

F4: the ninth-order fugacity. The last new combination is

$$J_9 = A_{9,1} - Z_{8,1} + 2(\partial_\eta A_7)_0 - 2(\partial_\eta Z_6)_0 + D_{9,\text{lower}}, \tag{C.55}$$

with the full inputs

quantity	numerical value
$A_{9,1}$	$-613817.497451118368344265\ldots$
$Z_{8,1}$	$11059.024476469114411035\ldots$
$(\partial_\eta A_7)_0$	$-14142.938944879185749695\ldots$
$(\partial_\eta Z_6)_0$	$305.264828822013665772\ldots$
$D_{9,\text{lower}}$	$653499.204958990556939720\ldots$

They give $J_9 = -273.72451599932464651407791039864 7\ldots$ and $J_9^{(0)} = -\frac{10192}{3}$; the value is a cancellation between terms of order 6×10^5. The complete expression has 84 928 terms. Some base-integral stages were restored from rational-stage files only after an exact audit of the omitted constant terms, and the endpoint-offset logarithms were restored separately. An intermediate error in an endpoint-vacuum multiplicity was corrected, and the affected outputs were regenerated before the accepted evaluations; only the corrected scalar enters the assembly. The independent re-derivation of Sec. 2.6 differs from this record in the first entry of the table, $A_{9,1}$, and agrees on the other four pieces of J_9.

C.6 The full numerical assembly and the zeta-value candidate

The six inherited inputs combined with B_{old} give

$$B_{\text{done}} = B_{\text{old}} + 4W + 2S + \tfrac{3}{32}L_3 - \tfrac{1}{4}M_{21} + \tfrac{1}{2}M_{12} - \tfrac{1}{16}N_2 = 563.4040661247776631183091472046 16\ldots, \tag{C.56}$$

and Eq. (2.10) becomes the scalar identity

$$C_8^{\text{SR}} = B_{\text{done}} + \frac{N_\delta}{8} + \frac{3U}{4} + \frac{K_v}{32} + \frac{J_9}{4}. \tag{C.57}$$

With Eq. (C.48) it reduces to $C_8^{\text{SR}} = C_{\text{fixed}} - \frac{1}{2}P_6$, with the C_{fixed} and the record of Eq. (2.14) given in Sec. 2.5. Every additive ingredient has a numerical value; there is no missing F3 or F4 term, and the exact P_6 integration removes the earlier 10^{-4} discrepancy of the quadrature estimates of P_6 (Appendix C.7). A separate literal-polynomial implementation solves $F = H = 0$ with Eq. (C.24) through $\bar{\delta}^4$, differentiates the full beta functions and obtains the same $8T_4$. The exact equality of this result, of the literal trace polynomial and of $C_{\text{fixed}} - \frac{1}{2}P_6$, with the supplied decimal strings treated as fractions, checks the assembly; it does not certify the numerical integration errors of those strings.

Recognition. The first fresh evaluation, at 70 digits, was used for an integer-relation search in the basis of Sec. 2.5, $\{\zeta(2), \zeta(3), \zeta(4), \zeta(5), \zeta(2)\zeta(3), \zeta(6), \zeta(3)^2, \zeta(7), \zeta(2)\zeta(5), \zeta(3)\zeta(4)\}$. The relation found for ($C_8^{\mathrm{SR}} - 88,$ basis) is

$$(3,\ 534,\ -330,\ -339,\ -14676,\ 7128,\ 8630,\ -5616,\ 36504,\ -18144,\ -4608), \tag{C.58}$$

which is Eq. (2.15). The second evaluation, at 100 digits and with the candidate held fixed, differs from it by about 3.0854×10^{-86}; a separate search with the rational intercept free recovers 88. A looser earlier search had produced a different relation that failed at higher precision; its rejected record is kept as a warning against overinterpreting integer relations. The highest-weight part of the candidate, $-12168\zeta(7) + 6048\zeta(2)\zeta(5) + 1536\zeta(3)\zeta(4)$, is compatible with the maximal weight seven (Appendix C.8); compatibility proves neither this coefficient identity nor a period-class theorem to all orders. With the even zeta values written out, the same candidate is

$$\begin{aligned} C_8^{\mathrm{SR}} \stackrel{?}{=} 88 - \frac{89\pi^2}{3} + \frac{113\pi^4}{90} - \frac{1726\pi^6}{567} + \Big(110 - 396\pi^2 + \frac{256\pi^4}{15}\Big)\zeta(3) + (4892 + 1008\pi^2)\zeta(5) \\ + 1872\zeta(3)^2 - 12168\zeta(7), \end{aligned} \tag{C.59}$$

algebraically identical to Eq. (2.15).

Status. The record is numerical: its digits are cross-checked only through P_6, to about 57 decimal places; six of its inputs are single inherited records; and it carries no error bound (Sec. 2.5). Equation (2.15) is conjectured: it has no certificate, and the record is its only high-precision support, so its agreement with the record is not independent evidence for the record's digits. The independent re-derivation of Sec. 2.6 gives $C_8^{\mathrm{SR}} = 242.8 \pm 2.4$ (one standard error), 1.9 to 4.0 standard errors below the record depending on the error estimator, with the difference in $A_{9,1}$. The magnetization-scheme calculation of Appendix I, which shares no input with the compendium, gives 247.38118 ± 0.13, 2.5×10^{-5} from the record; it supports the leading digits of the record but not the coefficients of Eq. (2.15). A proof of Eq. (2.15) would follow the route stated in Sec. 2.5; its verifier would also have to regenerate each source relation and check the complete rational vector, not only its primitive quotient or its numerical value. No such certificate exists.

C.7 Numerical evaluation of periods and error control

The numerical value of a regulated period is not obtained by substituting a large cutoff into a divergent integral; it is the constant term of a controlled expansion in logarithms and inverse powers. For a fixed nonnegative letter a the differential system is triangular,

$$\frac{\mathrm{d}}{\mathrm{d}\Lambda} G(-a, -w; \Lambda) = \frac{G(-w; \Lambda)}{\Lambda + a}, \tag{C.60}$$

where $(-a, -w)$ is the word $-w$ with the letter $-a$ prefixed. The empty word is one, all-zero words have the tangential value $\log^n \Lambda / n!$, and words beginning with a zero letter need their regular and singular parts treated explicitly.

For $a > 0$ and a regular Taylor expansion $G(-w; \Lambda) = \sum_n c_{w,n} \Lambda^n$, the coefficients $c_{a,w,n}$ of $G(-a, -w; \Lambda)$ near the origin obey

$$c_{a,w,n} = \frac{1}{n} \sum_{j=0}^{n-1} (-1)^j a^{-j-1} c_{w,n-1-j}, \qquad n \geq 1. \tag{C.61}$$

The local Taylor expansion is continued along the positive axis in steps smaller than the distance to the nearest singularity. At a large enough matching point the expansion (C.3), $G(-w; \Lambda) =$

$\sum_{k\geq 0}\Lambda^{-k}P_{w,k}(\log\Lambda)$, is used. For the word (a,w) the leading polynomial satisfies $P'_{a,w,0}=P_{w,0}$, with the constant fixed by matching, and the higher polynomials solve

$$(\partial_L-k)P_{a,w,k}(L)=\sum_{j=0}^{k}(-a)^jP_{w,k-j}(L),\qquad k\geq 1. \tag{C.62}$$

The regularized period is $R_w=P_{w,0}(0)$. Ordering the suffixes depth first lowers the peak memory without changing the recurrence. A C++ implementation with the GMP library supplies the arithmetic; a separate Python implementation with decimal arithmetic checks sampled periods and closed controls.

Every accepted run records its working precision, matching point, Taylor order, asymptotic order, step fraction, input hashes and backend. Repeating a run with more printed digits only is not an independent accuracy test; the comparisons of the compendium change several settings at once. Their stability and the controls on lower integrals support the quoted values, but no rigorous global error bound for the high-order assembly exists (Sec. 2.5, "How many digits").

The superseded Mellin quadrature. For a cutoff moment $M(\varepsilon)$ the Mellin transform $\mathcal{M}(\varsigma)=\int_0^{\varepsilon_{\max}}\varepsilon^{\varsigma-1}M(\varepsilon)\,\mathrm{d}\varepsilon$ converges in a right half-plane and is continued analytically; a term $\varepsilon^0\log\varepsilon$ contributes $-\varsigma^{-2}$. Sector decomposition with analytic power and logarithm moments evaluates the continued transform without integrating divergent powers. The compendium's six-kink runs of this kind covered all ordered-gap sectors with compensated summation, but its finest runs of P_6 still differed at about 10^{-4}. They are kept only as historical diagnostics and are not used in the high-precision test of the candidate; the exact boundary calculation of Appendix C.5 replaces them.

C.8 General-formula conjectures and what survived

Matching a finite series and deriving the next coefficient from an independent integral are different tests. The compendium used the latter to reject several attractive extrapolations; none of the general generating formulas it tried is validated.

Exact recursion versus an all-orders solution. Define

$$Q_{\mathrm{root}}(\bar\delta)=\frac{T(\bar\delta)^2/4-D(\bar\delta)}{2\bar\delta}=1+\sum_{m\geq 1}q_m\bar\delta^m,\qquad \sqrt{Q_{\mathrm{root}}}=1+\sum_{m\geq 1}s_m\bar\delta^m. \tag{C.63}$$

Then, with $T_1=\frac12$ and $s_0=1$,

$$\begin{gathered}q_m=\frac18\sum_{i=1}^{m}T_iT_{m+1-i}-\frac12D_{m+1},\qquad s_m=\frac12\Big(q_m-\sum_{i=1}^{m-1}s_is_{m-i}\Big),\\ C^{\mathrm{SR}}_{2m}=2^{m-1}T_m,\qquad C^{\mathrm{SR}}_{2m+1}=-2^{m+1}s_m.\end{gathered} \tag{C.64}$$

These equations are exact algebra; they do not determine uncalculated invariants. For instance, C^{SR}_9 needs the new determinant coefficient D_5, which T_4 does not fix.

Filtered MZV structure. Give $\zeta(n_1,\dots,n_r)$ the weight $\sum_i n_i$, with weights adding under products, and let $W_N\mathcal{Z}$ be the rational span of MZVs of weight at most N. The structural conjecture that remains viable is

$$C^{\mathrm{SR}}_n\in W_{n-1}\mathcal{Z}. \tag{C.65}$$

It is not uniform transcendentality, and it does not specify the rational coefficients. The closed forms through C^{SR}_7 obey it, and the candidate (2.15) is consistent with it; its unproved status must

not be upgraded into a theorem to all orders. The geometry of ordered points on the line (genus zero) motivates the period class [26, 25]. A conditional angular-counting argument suggests

$$[u^k z^j \bar\delta^r]F \in W_{2k+j+2r-1}\mathcal{Z}, \qquad [u^k z^j \bar\delta^r]H \in W_{2k+j+2r-2}\mathcal{Z}. \tag{C.66}$$

The individual inputs of the prescription contain the regulator constants $\log 2$, $\log 3$, $\mathrm{Li}_2(-\frac{1}{2})$ and $\mathrm{Li}_4(\frac{1}{2})$, which are not written as MZVs and which cancel only from the invariants (Sec. 2.3; $\mathrm{Li}_4(\frac{1}{2})$ enters through the input E, Appendix B.3); the compendium does not say how the bound is to be read for them. If subtraction and matching preserve these bounds and close in the two essential couplings, the fixed-point recursion divides only by the rational determinant -4 of the leading Jacobian of (F, H) in (u, z), and the bounds imply Eq. (C.65). Proving that every required renormalization operation preserves them is the missing step. At weight eight the motivic MZV algebra admits a new depth-two direction, representable by $\zeta(5,3)$, beyond products of single zeta values [27]. Under the conjectured bound it could first enter C_9^{SR}, and it need not have a nonzero coefficient; this is not a claim of numerical independence of the corresponding real constants.

Specific guesses and direct tests. For the inverse curve $y = 1 - \Delta_{\mathcal{E}}$ write $x = y^2\mathcal{G}(y)/4$. A specified trial function $\widehat{\mathcal{G}}$ predicts, by Lagrange inversion,

$$\widehat{C}_n = -\frac{2^n}{n}[y^{n-1}]\,\widehat{\mathcal{G}}(y)^{-n/2}. \tag{C.67}$$

Two rational–digamma trials, with scales $(1,2,3)$ and $(1,2,4)$, fit C_1^{SR} to C_6^{SR} exactly but predicted $C_7^{\mathrm{SR}} \simeq 17.91784$ and 247.18113; the completed value $51.06899\ldots$ rejects both fixed generators. A later implicit Gamma family used

$$\mathcal{K}_{ab}(y,t) = 1 - \frac{\Gamma(1+ay)^2}{\Gamma(1+ay+bt)\,\Gamma(1+ay-bt)}, \qquad \mathcal{S} = y^2 - 4t^2 + t^2\Big[P(y) + \sum P_{ab}(y)\,\mathcal{K}_{ab}\Big], \tag{C.68}$$

with $t = \sqrt{x}$. It fits all seven supplied coefficients but predicts the wrong rational part of C_8^{SR}. Quadratic log-Gamma and Borel–Laplace variants also fit their inputs but fail the same direct test. The rational-invariant Padé prescription $\widehat{C}_8 = C_6^{\mathrm{SR}2}/C_4^{\mathrm{SR}}$ fails after clearing denominators and taking weight zero:

fixed hypothesis	prediction tested	outcome
geometric rational trace	$\pi_0 C_8^{\mathrm{SR}} = 160$	false: 88
implicit Gamma fit	$\pi_0 C_8^{\mathrm{SR}} = -\frac{55777}{64}$	false
quadratic log-Gamma and Borel–Gamma fits	$\pi_0 C_8^{\mathrm{SR}} = -\frac{78669187}{917504}$	false
rational-invariant Padé	$\pi_0 C_8^{\mathrm{SR}} = 26^2/4 = 169$	false
rational–digamma fits	$C_7^{\mathrm{SR}} \simeq 17.92$ or 247.18	false
filtered MZV class and weight bound	constants allowed at each order	unproved; not contradicted
determinant-only rational guess	$D_m^{(0)} = (3^m + 2^m + 7)/8$ $(m \geq 2)$	next value $D_5^{(0)} = \frac{141}{4}$ untested

The rational trace guess $T_m^{(0)} = (3^m - 1)/4$ is excluded by $T_4^{(0)} = 11 \neq 20$; this alone does not exclude its determinant-only companion. Adjusting a new parameter after learning C_8^{SR} is a new fit, not a successful prior prediction.

Gamma-product and Euler-kernel obstructions. The logarithm of a finite product of Gamma functions with rational linear arguments consists of single-zeta terms, whereas the inverse series contains

$$[y^5]\log\mathcal{G}(y)\big|_{\text{weight }5} = -\frac{17}{2}\zeta(5) + 2\zeta(2)\zeta(3). \tag{C.69}$$

This excludes the simple Gamma-product ansatz as a formal (motivic) identity, but not every nonlinear implicit Gamma equation. A broader single-product trial,

$$K_{\rho,e}(y) = \sum_{n\geq 1} \frac{1}{n(n+ey)} \prod_{k<n} \rho(y/k), \qquad \mathcal{G}_{\max}(y) = 1 + \frac{y^2}{2} K_{\rho,e}(y), \tag{C.70}$$

was tested with an arbitrary analytic ρ, $\rho(0) = 1$. Formal matching of the product terms requires both

$$8e^2 - 16e + 23 = 0, \qquad 864e^4 - 4176e^3 + 16504e^2 - 21884e + 14095 = 0, \tag{C.71}$$

whose resultant is $401375222784 \neq 0$. This exact exclusion applies to the stated family, not to all possible spectral kernels; the code derives the elimination equations rather than loading them as empirical numerical constraints.

C.9 The two ends are not a coefficient-by-coefficient duality

Both perturbative descriptions concern the same long-range family. The expansion of Sec. 2 is about $s = 1$, with $x \to 0$; the expansion about the generalized free field (GFF) with a quartic coupling is about $s = \frac{1}{2}$, with

$$\varepsilon_{\mathrm{G}} = 2s - 1 = 1 - 4x \to 0. \tag{C.72}$$

The strong–weak duality compares two descriptions at the same s; it is not a symmetry that exchanges the two ends [4]. A finite GFF expansion is not controlled at $\varepsilon_{\mathrm{G}} = 1$ and cannot be substituted there to compute a coefficient C_n^{SR}.

In one dimension $\Delta_{\mathcal{E}} = 1 - \nu^{-1}$, and the GFF expansion gives

$$\Delta_{\mathcal{E}} = \frac{1}{2} - \frac{1-4x}{6} + \frac{\pi + 4\log 2}{9}(1-4x)^2 + O\big((1-4x)^3\big), \tag{C.73}$$

where the three-loop diagram must be the corrected tetrahedron value, not the superseded one [23, 28]. With $t = \sqrt{x}$ and $f(t) = \Delta_{\mathcal{E}}(t^2)$, this gives the local derivatives at the GFF end

$$f(\tfrac{1}{2}) = \frac{1}{2}, \qquad f'(\tfrac{1}{2}) = \frac{2}{3}, \qquad f''(\tfrac{1}{2}) = \frac{4}{3} + \frac{32}{9}(\pi + 4\log 2). \tag{C.74}$$

They are not relations derived from C_1^{SR} to C_7^{SR}. Only under extra convergence and termwise-continuation assumptions could one write, for the GFF coefficients a_m of $\varepsilon_{\mathrm{G}}^m$,

$$a_m = (-1)^m \sum_{n\geq 1} \frac{C_n^{\mathrm{SR}}}{2^n} \binom{n/2}{m}, \qquad m \geq 1, \tag{C.75}$$

and even then each a_m mixes infinitely many orders of the expansion about $s = 1$. For an asymptotic series alone the identity is not justified: a flat term $e^{-1/t^2} h(t)$ changes the data at the GFF end without changing any formal coefficient C_n^{SR}. This is a logical example, not a proposed physical correction.

An independent global exclusion. A specified global Gamma trial fails the value at the GFF end outright. Its one-parameter repair enforces the value and the slope, and matching the curvature fixes its last parameter; the resulting third derivative is less than 927. With $A_{\mathrm{G}} = \pi + 4\log 2$, the corrected cubic coefficient b_{G} of the GFF expansion of $\nu^{-1} = 1 - \Delta_{\mathcal{E}}$ (the coefficient of $\varepsilon_{\mathrm{G}}^3$; the compendium calls it the cubic mass coefficient) and the third derivative are

$$b_{\mathrm{G}} = \frac{\mathfrak{a}}{81} - \frac{A_{\mathrm{G}}^2 + \pi^2/3}{108}, \qquad f'''(\tfrac{1}{2}) = \frac{64 A_{\mathrm{G}}}{3} + 384 b_{\mathrm{G}}, \tag{C.76}$$

where, in the corrected integral normalization [28],

$$\mathfrak{a} = 2\int_{\mathbb{R}^2} \frac{\mathrm{d}u\ \mathrm{d}v}{\sqrt{|u(1-u)v(1-v)(u-v)|}} \tag{C.77}$$

(u and v are integration variables here). Reflection and inversion divide the integral into three equal diagonal and six equal off-diagonal sectors; with their integrals $\mathcal{D}$ and $\mathcal{O}$, $\mathfrak{a} = 6\mathcal{D} + 12\mathcal{O}$. The off-diagonal substitution $v \to 1/v$ gives $\mathcal{O} \geq \pi^2$. For independent arcsine variables U_1, U_2 on $[0,1]$, $\mathcal{D} = \pi^2\,\mathbb{E}|U_1 - U_2|^{-1/2}$ and $\mathbb{E}|U_1 - U_2| = 4/\pi^2$, so Jensen's inequality gives

$$\mathcal{D} \geq \frac{\pi^3}{2}, \qquad \mathfrak{a} \geq 3\pi^3 + 12\pi^2 > 211. \tag{C.78}$$

This implies $f'''(\frac{1}{2}) > 992$, which contradicts the repaired global trial; the archived proof gives exact rational interval enclosures for the strict inequalities. The argument excludes that global function family; without further analyticity or uniqueness assumptions it does not by itself exclude a formal rule for the coefficients C_n^{SR}. The rational tests of C_8^{SR} in Appendix C.8 do not need that global assumption.

C.10 Further consequences of the shadow relation

The results of this subsection are separate from the thermal series. They use the published shadow identity and leading data of operator product expansion (OPE) coefficients of the dual [4], not the integrals of many kinks. Unit-normalized primaries obey $\langle \mathcal{O}_i(X)\mathcal{O}_j(0)\rangle = \delta_{ij}|X|^{-2\Delta_i}$, and χ denotes the renormalized, unit-normalized primary of dimension $\Delta_\chi = 1 - x$ obtained from the current of Eq. (2.1), not the unsubtracted current. Here $x = \Delta_\sigma = \bar{\delta}/2$ as throughout.

The gamma-function factor. A scalar shadow transform is proportional to

$$\mathsf{S}_x f(X) = \mathcal{N}_x \int \frac{f(Y)\ \mathrm{d}Y}{|X-Y|^{2(1-x)}}, \tag{C.79}$$

defined by continuation of homogeneous distributions with contact subtractions. For three-point functions at separated points the factor that depends on the pair follows from

$$\int \mathrm{d}X\, e^{-\mathrm{i}pX}|X|^{-\alpha} = 2^{1-\alpha}\sqrt{\pi}\,\frac{\Gamma((1-\alpha)/2)}{\Gamma(\alpha/2)}\,|p|^{\alpha-1}. \tag{C.80}$$

Putting one spectator at zero and one at infinity and convolving the remaining power law gives

$$q(x;\Delta_i,\Delta_j) = \frac{\Gamma\big((1-x+\Delta_i-\Delta_j)/2\big)\Gamma\big((1-x-\Delta_i+\Delta_j)/2\big)}{\Gamma\big((x+\Delta_i-\Delta_j)/2\big)\Gamma\big((x-\Delta_i+\Delta_j)/2\big)}. \tag{C.81}$$

The published relation has the form $c_{\chi ij}/c_{\sigma ij} = \mathcal{N}(\bar{\delta})\,q(x;\Delta_i,\Delta_j)$, with a normalization independent of the spectator pair (Ref. [4], Eq. (5.116)). For the leading operator $\mathcal{O}_+$, of dimension Δ_+ (Eq. (2.6)), the published data give $c_{\sigma\chi+}/c_{\sigma\sigma+} \sim 1/\sqrt{\bar{\delta}}$, while $q(x;x,\Delta_+) \sim 1/(\sqrt{2}\pi\sqrt{\bar{\delta}})$; hence $\mathcal{N}(0) = \sqrt{2}\pi$. Applying the relation twice to an even primary $\mathcal{O}$ of dimension Δ gives

$$\frac{c_{\chi\chi\mathcal{O}}}{c_{\sigma\sigma\mathcal{O}}} = \mathcal{N}(\bar{\delta})^2 P(x,\Delta), \qquad P(x,\Delta) = \frac{\Gamma\big((1-2x+\Delta)/2\big)\Gamma\big((2-2x-\Delta)/2\big)}{\Gamma\big((2x-\Delta)/2\big)\Gamma\big((2x-1+\Delta)/2\big)}. \tag{C.82}$$

The isolated dimension-two primary. At the endpoint the unique primary at dimension two that is even under parity and under the Ising symmetry is $\rho \to :\chi^2:/\sqrt{2}$. Its published leading data are $c_{\chi\chi\rho} = \sqrt{2} + \cdots$ and $c_{\sigma\sigma\rho} = \bar{\delta}/\sqrt{2} + \cdots$, so the double-shadow ratio diverges as $2/\bar{\delta}$. A generic

displacement $\Delta = 2 + a'\bar{\delta} + \cdots$ leaves Eq. (C.82) finite; the required pole lies at $\Delta = 2 - \bar{\delta} + e$ with $e = o(\bar{\delta})$, where

$$P\big(\tfrac{\bar{\delta}}{2}, 2 - \bar{\delta} + e\big) = \frac{\Gamma(\frac{3}{2} - \bar{\delta} + \frac{e}{2})\,\Gamma(-\frac{e}{2})}{\Gamma(-1 + \bar{\delta} - \frac{e}{2})\,\Gamma(\frac{1}{2} + \frac{e}{2})} \sim \frac{\bar{\delta}}{e}. \tag{C.83}$$

Matching $2/\bar{\delta} \sim 2\pi^2\bar{\delta}/e$ gives

$$\Delta_\rho = 2 - \bar{\delta} + \pi^2\bar{\delta}^2 + O(\bar{\delta}^3) = 2 - 2x + 4\pi^2x^2 + O(x^3). \tag{C.84}$$

For the same solution $q(x; x, \Delta_\rho) = \pi\bar{\delta} + O(\bar{\delta}^2)$, hence

$$c_{\sigma\chi\rho} = \pi^2\bar{\delta}^2 + O(\bar{\delta}^3). \tag{C.85}$$

The sign assumes $c_{\chi\chi\rho} > 0$ and the stated sign of χ. The integer-power remainders assume analytic continuation of the isolated even branch; without that regularity the last justified term is followed by $o(\bar{\delta}^2)$. Conjugation by $\hat{\sigma}_3$ implements $g \mapsto -g$, which explains why an isolated even analytic branch has no odd powers of g. No new fully renormalized four-point integral was used.

A weighted statement, not a resolved higher spectrum. Let $N = 2, 4, 6, \ldots$, and let α label the branches that approach dimension N, with v_α the value of $c_{\chi\chi\alpha}$ at $\bar{\delta} = 0$ and $u_\alpha = \lim_{\bar{\delta}\to 0} c_{\sigma\sigma\alpha}/\bar{\delta}$. For regular branches visible to the leading $\chi\chi$ OPE write $\Delta_\alpha = N - \bar{\delta} + \gamma_\alpha^{[2]}\bar{\delta}^2 + o(\bar{\delta}^2)$. The shadow residue implies

$$v_\alpha\gamma_\alpha^{[2]} = \pi^2 N(N-1)\,u_\alpha, \tag{C.86}$$

with a zero u_α interpreted by continuity. The published mixed-correlator weight and the free χ block coefficient are

$$A_N = \sum_\alpha u_\alpha v_\alpha = \frac{4(N-1)[(N-2)!]^2}{N(2N-2)!}, \qquad a_N = \sum_\alpha v_\alpha^2 = \frac{2N(N-1)[(N-1)!]^2}{(2N-2)!}; \tag{C.87}$$

the second follows by matching $1 + z_c^2 + [z_c/(1-z_c)]^2$ to $1 + \sum_N a_N z_c^N\, {}_2F_1(N, N; 2N; z_c)$, with z_c the cross ratio. Therefore

$$\frac{\sum_\alpha v_\alpha^2\gamma_\alpha^{[2]}}{\sum_\alpha v_\alpha^2} = \frac{2\pi^2}{N(N-1)}. \tag{C.88}$$

For $N = 2$ the primary is unique. For higher N this is only an average weighted by the OPE: it does not assign the same correction to every degenerate branch, and it does not determine branches invisible to the leading OPE.

C.11 The logarithmic endpoint at $s = 1$

At $\bar{\delta} = 0$, with $b_0 = 1$, the published beta functions of the dual are those of Eq. (A.26) [4]. Choose the marginally irrelevant branch $g > 0$, $h < 0$, and write its critical separatrix as $h = -g/\sqrt{2} + a_2g^2 + O(g^3)$. The cubic beta functions give $\beta_h - h'(g)\beta_g = -6a_2g^3 + O(g^4)$, hence $a_2 = 0$. With $L = \log(r/r_0)$ (not the input L),

$$\frac{\mathrm{d}g}{\mathrm{d}L} = -2g^2 - 2g^3 + O(g^4), \qquad g(L) = \frac{1}{2L} - \frac{\log L}{4L^2} + \frac{C}{L^2} + O\Big(\frac{\log^2 L}{L^3}\Big). \tag{C.89}$$

Let $M_\sigma^2 = \lim_{r\to\infty}\langle\sigma(r)\sigma(0)\rangle$ on this trajectory. The wall gas gives $G_\sigma(r) - 1 = -4g^2\log(r/\varepsilon) + \cdots$, because a kink pair changes the spin product only when one wall lies between the insertions; so $\gamma_\sigma = 2g^2 + \cdots$, and

$$\log\frac{G_\sigma(r)}{M_\sigma^2} = 2\int_L^\infty \gamma_\sigma(g(u'), h(u'))\,\mathrm{d}u' + \log F_\sigma(g(L), h(L)). \tag{C.90}$$

The finite matching function satisfies $F_\sigma = 1 + O(g^2)$: at $g = 0$ the spin is conserved for every h, and a nonzero spin-flip correction needs two flips, so it cannot change the term in $\log L$. Inserting the running coupling and exponentiating gives

$$\frac{G_\sigma(r)}{M_\sigma^2} = 1 + \frac{1}{L} - \frac{\log L}{2L^2} + \frac{K_{\mathrm{match}}}{L^2} + O\Big(\frac{\log^2 L}{L^3}\Big). \tag{C.91}$$

The coefficient $-\frac{1}{2}$ is fixed by the cubic flow; K_{match} depends on the matching convention and on the definition of r_0. This extends the leading endpoint logarithm of Ref. [4] (its Appendix E). The difference $G_\sigma(r) - M_\sigma^2$ is best called the plateau-subtracted correlator: in the symmetric mixture $\langle\sigma\rangle = 0$ while the plateau is nonzero, so it need not equal the usual connected subtraction. Neither the results of Appendices C.10 and C.11 nor the thermal coefficients rely on a stronger four-point statement that the compendium's earliest record had argued from a formal principal-value cancellation in a current integral; such an identity alone does not determine composite-operator counterterms or the fully renormalized four-point function.

C.12 The polynomial B_{old} of the eighth order

The coefficient names are those of the derivative dictionary of Appendix C.3. All terms are fixed by the 25 inputs of C_3^{SR} to C_7^{SR}; the ten new inputs enter only through the linear terms of Eq. (2.10). This is the unprojected expression: its rational projection is $\frac{1153}{6}$, and its value is $-861.663567235479779201497190992 3\ldots$.

$$\begin{aligned} B_{\mathrm{old}} = {} & -\gamma - \frac{5c^3}{1024} - \frac{3\Omega}{16} - \frac{\Psi}{2} - \frac{Q}{4} - \frac{M}{16} - \frac{P}{16} \\ & - \frac{\alpha}{16} - \frac{a^3}{16} - \frac{K}{64} - \frac{N}{64} + \frac{\beta}{4} + \frac{\Phi}{8} + \frac{f^2}{64} \\ & + \frac{3L}{64} + \frac{Ap}{2} + \frac{Ea}{2} - Ep - Va - \frac{9Jc}{64} - \frac{5ac^2}{256} \\ & - \frac{3Aa}{16} - \frac{3Ac}{64} - \frac{3a^2c}{64} - \frac{3Nc}{128} - \frac{Gp}{2} - \frac{Eb}{4} - \frac{Lp}{4} \\ & - \frac{Ra}{4} - \frac{Rc}{4} - \frac{Vc}{4} - \frac{bq}{4} - \frac{pq}{4} - \frac{Gb}{8} - \frac{Ja}{16} \\ & - \frac{Lb}{16} - \frac{Mc}{16} - \frac{bk}{16} - \frac{fq}{16} - \frac{kp}{16} - \frac{af}{64} - \frac{fk}{64} \\ & + \frac{a^2p}{4} + \frac{Ab}{8} + \frac{Ec}{8} + \frac{aq}{8} + \frac{La}{16} + \frac{a^2b}{16} + \frac{cq}{16} \\ & + \frac{ak}{32} + \frac{c^2p}{32} + \frac{cf}{256} + \frac{3Ga}{16} + \frac{3Gc}{32} + \frac{3Lc}{64} + \frac{3bf}{64} \\ & + \frac{5bc^2}{128} + \frac{5ck}{256} + \frac{5c^2f}{512} + \frac{acp}{8} + \frac{3abc}{64}. \end{aligned} \tag{C.92}$$

The machine-readable version is the field `old_part` of `code/dual_d1/data/archived/C8_full_work/full_trace.json`. The separate full-Jacobian program recomputes this expression from the beta functions; it does not read an expected value of C_8^{SR} to obtain it.

D The two-coupling closure and the prescription

This appendix gives two formal arguments about the dual of Sec. 2.1, both written by OpenAI GPT-6 astra (via ChatGPT) and supplied by the author on 15 October 2026 (Secs. 1.2 and 5), merged and edited to the notation of this note: the *closure argument*, on the bulk counterterms of the dual, and the *prescription argument*, on what the matched two-coupling prescription computes. Neither has been refereed. This note counts both as formal, unrefereed arguments (Secs. 2.1 and 2.9). Where an argument states a proposition or a theorem, it is given below as the statement that the argument establishes formally, coefficientwise in g, h and $\bar{\delta}$; it is not a proved result of this note. Appendix D.15 lists what the review checked.

Throughout, $\bar{\delta} = 1 - s$ and $x = \Delta_\sigma = \bar{\delta}/2$. The dual, its operators $\mathcal{O}_g$ and $\mathcal{O}_h$ and the charge $b = b_0 - \sqrt{2}h$ are those of Eq. (2.1). The bare current $\chi_B = \frac{\mathrm{i}}{\sqrt{2}}\partial\phi$ is the χ of Eq. (2.1); as in Appendix A.3, the subscript distinguishes it from the renormalized current primary, and neither is the generalized free field χ of Secs. 3 and 4. The regulator, the current resummation, the finite-tail matching, the vacuum normalization, and the finite-response Jacobian and recurrence are given as the compendium states them in Appendices A.4, A.5, A.7 and A.9, whose notation is used here (in particular ∂_η is the derivative with respect to the endpoint source η at $\eta = 0$). Where the prescription argument re-derives one of these objects, its derivation is given here and the formula is referred to there.

D.1 The question, three claims and the common framework

The thermal series of Sec. 2 rests on two assumptions (Sec. 2.1): the renormalization of the dual involves only the two couplings (g, b), and the matched prescription extracts their β functions. Three claims must be kept apart.

(1) *The full Wilsonian action at every finite cutoff contains only two interactions.* This is false in an ordinary fixed-field cutoff scheme (Appendix D.6).

(2) *Only two nontrivial independent bulk counterterm structures are needed in the formal perturbation theory about the endpoint.* This is the restricted statement of the closure argument (Appendix D.2).

(3) *The archived subtraction algorithm implements that renormalization, so that its computed coefficients are those of the dual.* This follows neither from operator counting nor from the exact period certificates of Sec. 2.4. The prescription argument addresses the analytic recipe behind it (Appendix D.3); whether the historical integrations follow that recipe is a separate question (Sec. 2.6 and Appendix H.5).

The distinction between (1) and (2) is not semantic: finite irrelevant interactions can contribute to higher-order marginal β functions even when no independent irrelevant counterterm is required (Appendix D.9).

Claim (2) does not imply claim (3). The argument "there are two counterterms, and two response equations determine two numbers" is insufficient, because for arbitrary responses the two numbers can absorb an endpoint or insertion anomalous dimension. Schematically, the single-logarithm equation of two responses could be

$$R_1 = \mathcal{J}\beta + \Gamma_\partial R_0 + \mathcal{A}_{\mathrm{probe}}, \tag{D.1}$$

where R_0 and R_1 are the finite and single-logarithm parts of the responses, $\mathcal{J} = D_c R_0$ is the Jacobian of their finite parts (Appendix D.12; its tree-level value is the identity), Γ_∂ an endpoint anomalous dimension and $\mathcal{A}_{\mathrm{probe}}$ a renormalization of the probe. Inverting $\mathcal{J}$ does not show that the last two terms vanish. The prescription argument argues that both are absent for the specific normalized, separated-probe prescription of Sec. 2.1, and only then inverts $\mathcal{J}$.

Common framework. Both arguments work coefficientwise in a joint formal series in g, h and $\bar{\delta}$ about $g = h = \bar{\delta} = 0$ (and, where needed, in the reference charge about $b_0 = 1$), at zero magnetic field. They use a fixed compact target coordinate and a reference normalization of the Wick products. Distances in the Gaussian formulas are dimensionless; a reference inverse length κ is restored by replacing r by κr. An infrared regulator, or a smooth compactly supported switching function of the interaction, is kept throughout. The subject is ultraviolet collisions, not an infrared limit and not the renormalization of arbitrary composite insertions.

The published analysis identifies two symmetry-invariant marginal perturbations at $\bar{\delta} = 0$ but distinguishes this explicitly from an all-orders renormalizability result for its non-scaling perturbation at $\bar{\delta} > 0$ [4, Secs. 3.2.3, 4 and 6]; the compendium likewise qualifies its higher coefficients by the two-coupling and matching assumptions.

D.2 The closure statement

The closure argument, a formal and unrefereed argument written by OpenAI GPT-6 astra (see the head of this appendix; its scope is in Appendix D.14 and its checks in Appendix D.15), establishes formally the following statement, which it states as a proposition.

> *Formal bulk counterterm class.* For the matrix-valued Gaussian model of Eq. (2.1), impose its diagonal shift symmetry, internal reflection and parity (Appendix D.4). Work coefficientwise in g, h and $\bar{\delta}$ about the marginal endpoint, using the Wick products of the deformed covariance (Appendix D.7), with a fixed infrared cutoff. Then local symmetry-preserving subtractions can be constructed recursively such that every required bulk action counterterm for constant couplings lies in
>
> $$\operatorname{span}\{\mathbf{1}, \mathcal{O}_g, \mathcal{O}_h\}, \tag{D.2}$$
>
> modulo total derivatives. Only two nontrivial independent bulk coupling counterterms are therefore required. Field-insertion and endpoint counterterms are not included in this count.

The argument has three parts: the exact selection rules and the classification of local operators (Appendix D.5), power counting (Appendix D.7), and the recursive subtraction of all partial coincidences (Appendix D.8). Since h and $b = b_0 - \sqrt{2}h$ are two coordinates on one coupling direction (Appendix D.11), the statement supplies a counterterm-class argument for the two essential coordinates (g, b). The closure argument sums it up as follows: in the zero-field, symmetry-preserving formal expansion about the marginal endpoint, two nontrivial independent bulk ultraviolet counterterm directions suffice, in addition to vacuum and external-insertion or endpoint renormalization; a finite-cutoff Wilsonian action nevertheless contains irrelevant terms, and their finite contributions must be incorporated in any correct high-order calculation.

The statement uses the standard local extension construction explicitly. It is not a claim that any procedure that cancels two chosen scalar divergences implements that construction; that is the subject of the prescription argument.

D.3 The prescription statement

The prescription argument, a formal and unrefereed argument written by OpenAI GPT-6 astra (see the head of this appendix; its scope is in Appendix D.14 and its checks in Appendix D.15), establishes formally the following statement, which it states as a theorem. Its hypotheses are:

(a) the reference Gaussian and matrix operator algebra of Appendix D.4, at zero magnetic field;

(b) a coefficientwise expansion in g, h and $\bar{\delta}$ and, when needed, in the reference charge about $b_0 = 1$;

(c) the diagonal $O(2)$ symmetry (the $U(1)$ of Sec. 2.1 and the internal reflection) is preserved;

(d) the interaction is restricted to a finite interval, with a hard core on successive kink positions only;

(e) current products are defined by exact Gaussian current resummation, retaining the specified Wick-ordering and endpoint factors;

(f) every response is normalized by the same vacuum expectation of that interval operator;

(g) the reference-theory probe of each response lies outside the interaction support, and any projection to an infinitely distant probe is taken only afterwards.

> *Formal normalized-interval prescription.* Within local, scaling-degree-preserving position-space renormalization, the bulk counterterms are $\mathbf{1}$, $\mathcal{O}_g$ and $\mathcal{O}_h$, and all ultraviolet counterterms localized at either endpoint of the interval are scalar multiples of $\mathbf{1}$. Consequently the normalized responses require no additional endpoint or probe counterterm. If their tree-level two-by-two response matrix is invertible, their single-logarithm equation determines the local logarithmic β function. Relative to any other local normalization in the same class, the result is related by a regular finite change of the two nontrivial bulk couplings, so the fixed-point stability eigenvalues are identical.

As defined in Sec. 2.1, the matched two-coupling prescription has the form of hypotheses (d) to (f): (d) is its item (i), (e) its items (ii) and (iii), and (f) its item (iv). The prescription argument identifies its two responses with the far-field projections of separated reference probes, as hypothesis (g) requires (Appendix D.10); that projection is stated rather than shown in detail (Appendix D.15). Identical eigenvalues mean identical invariants T and D (Sec. 2.2), and so an identical series $\Delta_{\mathcal{E}} = 1 + \sum_n C_n^{\mathrm{SR}} x^{n/2}$.

The statement concerns the prescription as an analytic recipe, not whether a particular computer output implemented every part of it without error. The complete normalized moments, the first endpoint-source derivatives and the exact normal-ordering map are essential hypotheses, not optional approximations; the argument compares them with the compendium's formulas (Appendices D.11 and D.12). It uses standard local distribution-extension and finite-renormalization results [29, 30]; the model-specific steps are the collision bounds, the endpoint classification, the spectator factorization and the audit of the current and normal-ordering factors. These steps do not use Lorentzian locality of the nonlocal Gaussian action: they concern the explicit Euclidean Wick kernels, which are smooth away from coincidence diagonals.

The order of the steps is what makes the argument non-circular: the local renormalization is built for arbitrary spectators first (Appendix D.8), the endpoint cancellation and the absence of probe renormalization follow (Appendix D.10), and the invertibility of the response matrix is used only at the end, for uniqueness (Appendix D.13).

D.4 The reference theory, its symmetries and the regulated responses

Reference theory. At $\bar{\delta} = 0$ the field normalization and compactification are chosen so that

$$\langle \phi(X)\phi(0)\rangle_0 = -2\log(\kappa|X|), \qquad V_\pm = \kappa :e^{\pm \mathrm{i} b_0 \phi}:, \qquad \chi_B = \frac{\mathrm{i}}{\sqrt{2}}\partial\phi, \qquad b_0 = 1. \tag{D.3}$$

Thus $V_\pm$ and χ_B have reference dimension one. The deformed covariance, before any expansion in $\bar{\delta}$, is

$$C_{\bar{\delta}}(r) = \frac{2(\kappa^{-\bar{\delta}} - |r|^{\bar{\delta}})}{\bar{\delta}} = -2 f_{\bar{\delta}}(|r|), \qquad f_{\bar{\delta}}(r) = \frac{r^{\bar{\delta}} - \kappa^{-\bar{\delta}}}{\bar{\delta}}, \tag{D.4}$$

which at $\kappa = 1$ is the covariance of Sec. 2.1. The zero mode is removed, and complete charged observables carry the appropriate neutralizing probe. With $\hat{\sigma}_\pm = (\hat{\sigma}_1 \pm \mathrm{i}\hat{\sigma}_2)/2$, the interactions are $\mathcal{O}_g = \hat{\sigma}_+ V_+ + \hat{\sigma}_- V_-$ and $\mathcal{O}_h = \hat{\sigma}_3 \chi_B$, and the evolution over an interval I is

$$\mathcal{D}_I(c) = \mathrm{P} \exp \int_I \mathrm{d}X \, [g\mathcal{O}_g + h\mathcal{O}_h], \qquad c = (g, h). \tag{D.5}$$

Exact symmetries. With A a matrix of the qubit, the global symmetry includes

$$\phi \mapsto \phi + \alpha/b_0, \quad A \mapsto e^{-\mathrm{i}\alpha\hat{\sigma}_3/2} A e^{\mathrm{i}\alpha\hat{\sigma}_3/2}; \qquad \phi \mapsto -\phi, \quad A \mapsto \hat{\sigma}_1 A \hat{\sigma}_1. \tag{D.6}$$

The first is the $U(1)$ of Sec. 2.1, the second the internal reflection; together they form the diagonal $O(2)$. Parity combines $X \mapsto -X$, $\phi(X) \mapsto -\phi(-X)$, the reversal of the path ordering and matrix transposition. These transformations preserve the regulated matrix expansion [4, Eqs. (3.31)–(3.35)]. At zero magnetic field the reflection is not broken. The internal reflection, not spatial parity alone, is what excludes a local $\hat{\sigma}_3$ counterterm at a single endpoint (Appendix D.7).

The regulated responses. The interaction is restricted to $I = [0, L]$, and the distinct-kink integrations run over

$$0 < X_1 < \cdots < X_n < L, \qquad X_{i+1} - X_i \geq \varepsilon, \tag{D.7}$$

with no hard core between a kink and an endpoint; $\ell = \log \varepsilon$ (or $\log(\varepsilon\mu)$ at a reference scale μ). The compendium works at $L = \kappa^{-1} = 1$ (Eq. (A.14)). At a finite regulator it is useful first to keep a smoothed Gaussian covariance, for which the matrix change-of-frame identity of Appendix D.11 is exact, and then to pass to normal-ordered products. Changes of the smoothing at separations of order ε are local regulator changes, handled by the forest construction of Appendix D.8. The coincident Wick convention used in the archive is $C(0) \to C_{\bar{\delta}}(\varepsilon)$.

For a reference probe $\mathcal{X}$ separated from I, the normalized response is

$$\mathcal{R}_{\mathcal{X}}(\varepsilon; c) = \frac{\frac{1}{2}\operatorname{tr}\langle \mathcal{X}\, \mathcal{D}_{I,\varepsilon}(c)\rangle_0}{\frac{1}{2}\operatorname{tr}\langle \mathcal{D}_{I,\varepsilon}(c)\rangle_0}, \tag{D.8}$$

with the same interval and regulator in numerator and denominator. The probes used for the two coordinates are the reference vertex-and-matrix and current-and-matrix probes, normalized by their free two-point functions. They are *not* the fully interacting, renormalized primaries inserted inside the interacting interval.

D.5 Matrix algebra and the local operators below threshold

Matrix algebra excludes higher-charge vertices. The exact relations

$$\hat{\sigma}_+^2 = \hat{\sigma}_-^2 = 0, \qquad \hat{\sigma}_+\hat{\sigma}_- = \frac{\mathbf{1} + \hat{\sigma}_3}{2}, \qquad \hat{\sigma}_-\hat{\sigma}_+ = \frac{\mathbf{1} - \hat{\sigma}_3}{2} \tag{D.9}$$

force the sequence of nonzero flips to alternate. Inserting diagonal $\hat{\sigma}_3$ factors, as the current interaction does, does not change this, since $\hat{\sigma}_\pm \hat{\sigma}_3^k \hat{\sigma}_\pm = 0$. Consequently every nonzero contiguous matrix word M has vertex charge $Q \in \{0, +1, -1\}$ and satisfies $[\hat{\sigma}_3, M] = 2QM$: an even number of flips gives $Q = 0$ and a diagonal matrix, an odd number gives $Q = \pm 1$ and a matrix proportional to $\hat{\sigma}_\pm$. This holds for words of every length, not only for a two-vertex product. In particular, no ordering of the interactions generates a bulk counterterm of charge two.

Equivalently, the complete shift-invariant local Wick algebra generated by these fields consists of

$$P_{\mathbf{1}}(\partial\phi, \partial^2\phi, \ldots)\, \mathbf{1}, \qquad P_3(\partial\phi, \partial^2\phi, \ldots)\, \hat{\sigma}_3, \qquad P_+(\partial\phi, \ldots)\, \hat{\sigma}_+ V_+, \qquad P_-(\partial\phi, \ldots)\, \hat{\sigma}_- V_-, \tag{D.10}$$

with $P_{\mathbf{1}}, P_3, P_\pm$ polynomials in the derivatives of ϕ. This is the perturbative local algebra: disorder or boundary-condition-changing fields outside it are not adjoined to the theory. An undifferentiated polynomial in ϕ is not an extra shift-invariant local operator (ϕ itself is the unobservable zero-mode field); expanding a vertex in powers of ϕ and giving those powers independent couplings would break the exact diagonal shift symmetry.

Classification. The reference dimension of a derivative monomial is

$$d\Big(\prod_{j\geq 1}(\partial^j\phi)^{n_j}\Big) = \sum_{j\geq 1} j n_j, \tag{D.11}$$

and a charged vertex sector adds one. These are dimensions at $\bar{\delta} = 0$; they do not claim that the deformed vertex is a scaling operator at $\bar{\delta} \neq 0$. Through dimension one the complete list of candidates is that of Table 5. The two even entries of dimension one also have the required parity.

candidate	reference dimension	internal reflection
$\mathbf{1}$	0	even
$\hat{\sigma}_3$	0	odd
$\partial\phi$	1	odd
$\hat{\sigma}_3\partial\phi$	1	even
$\hat{\sigma}_+V_+ + \hat{\sigma}_-V_-$	1	even
$\hat{\sigma}_+V_+ - \hat{\sigma}_-V_-$	1	odd

Table 5: The $U(1)$-invariant local operators of the algebra (D.10) through reference dimension one, with their behaviour under the internal reflection of Eq. (D.6).

The only nontrivial symmetry-invariant local action densities at dimension one are therefore $\mathcal{O}_g$ and $\mathcal{O}_h = \frac{\mathrm{i}}{\sqrt{2}}\hat{\sigma}_3\partial\phi$. The identity changes the vacuum normalization. A source for $\hat{\sigma}_3$ (a magnetic field) would be an additional relevant coupling; it is excluded by the zero-field symmetry of the problem considered.

At dimension zero the invariant sector is $\mathbb{R}\,\mathbf{1}$: a neutral constant matrix is diagonal, and the internal reflection equates its two entries ($\hat{\sigma}_3$ is odd, and the bare $\hat{\sigma}_\pm$ are charged). This is the input of the endpoint classification of Appendix D.7.

D.6 The product $\mathcal{O}_g\mathcal{O}_g$: no literal two-operator closure

At a symmetric pair of points with $r > 0$, Wick's theorem with Eq. (D.3) gives

$$\begin{aligned}\mathcal{O}_g(X+\tfrac{r}{2})\,\mathcal{O}_g(X-\tfrac{r}{2}) &= r^{-2}\Big[\mathbf{1} + \mathrm{i}r\hat{\sigma}_3\partial\phi - \frac{r^2}{2}{:}(\partial\phi)^2{:} + O(r^3)\Big] \\ &= \frac{\mathbf{1}}{r^2} + \frac{\sqrt{2}}{r}\mathcal{O}_h(X) + {:}\chi_B^2{:}(X) + O(r).\end{aligned} \tag{D.12}$$

Only $\hat{\sigma}_+\hat{\sigma}_-$ and $\hat{\sigma}_-\hat{\sigma}_+$ contribute; they multiply $r^{-2}{:}e^{\pm\mathrm{i}(\phi(X+r/2)-\phi(X-r/2))}{:}$, and $\phi(X+\frac{r}{2}) - \phi(X-\frac{r}{2}) = r\partial\phi + O(r^3)$. The midpoint convention removes a spurious first derivative of the expansion point. The coefficient of ${:}\chi_B^2{:}$ is one, not zero. Integrating a short-distance shell $\varepsilon < r < \varepsilon e^{\mathrm{d}\ell}$ in a standard real-space shell prescription therefore produces, besides the two lower-dimensional terms, an irrelevant term proportional to

$$g^2\varepsilon\;\mathrm{d}\ell\int \mathrm{d}X\,{:}\chi_B^2{:}(X). \tag{D.13}$$

Its detailed Wilsonian coefficient depends on the scheme; its appearance as an allowed term of the operator product expansion does not.

The current-squared channel is finite as $\varepsilon \to 0$ in this two-insertion integral (its radial integrand is r^0). This is why its generation does not refute the closure of the ultraviolet counterterm class (Appendix D.2). It does refute the unqualified statement that no other operator coefficient is generated at finite cutoff, that is, claim (1) of Appendix D.1. Some such terms can be removed by redundant transformations in particular schemes, but that is not the same as their absence from a fixed-field Wilsonian action. How these finite terms enter the marginal β functions is the subject of Appendix D.9.

D.7 Power counting in the bulk and at the endpoints

Bulk clusters. Consider $N \geq 2$ insertions of reference dimension one collapsing toward one interior point. Removing the centre coordinate leaves $N-1$ relative coordinates, written as a scale ρ and shape variables, with radial measure $\rho^{N-2}\,\mathrm{d}\rho$. If the product produces a local field $\mathcal{A}$ of reference dimension $d_{\mathcal{A}}$, its coefficient scales as $\rho^{d_{\mathcal{A}}-N}$, up to logarithms introduced by subtractions, and the radial integral has the form

$$\int_0^{\rho_0} \mathrm{d}\rho\, \rho^{N-2}\rho^{d_{\mathcal{A}}-N}(\log\rho)^M = \int_0^{\rho_0} \mathrm{d}\rho\, \rho^{d_{\mathcal{A}}-2}(\log\rho)^M. \tag{D.14}$$

The threshold does not depend on N:

$$d_{\mathcal{A}} = 0:\ \text{power divergence}; \qquad d_{\mathcal{A}} = 1:\ \text{logarithmic divergence}; \qquad d_{\mathcal{A}} \geq 2:\ \text{integrable}, \tag{D.15}$$

and for any finite M the logarithms do not change it. By Table 5, symmetry leaves $\mathbf{1}$, $\mathcal{O}_g$ and $\mathcal{O}_h$ below the threshold.

Alternating vertices. The same follows directly from Wick's theorem. For n vertex insertions with charges $q_i = \pm 1$ and total charge Q, at $b_0 = \kappa = 1$,

$$\prod_i V_{q_i}(X_i) = \prod_{i<j} |X_i - X_j|^{2q_iq_j} : \exp\Big(\mathrm{i}\sum_i q_i\phi(X_i)\Big):, \qquad 2\sum_{i<j} q_iq_j = Q^2 - n. \tag{D.16}$$

For a cluster $X_i = X + \rho r_i$ the normal product is organized exactly as

$$\prod_i V_{q_i}(X_i) = \mathcal{W}(\rho, \boldsymbol{r}) : \exp\Big[\mathrm{i}b_0 Q\phi(X) + \mathrm{i}b_0 \sum_{k\geq 1} \frac{\rho^k m_k}{k!}\partial^k\phi(X)\Big]:, \qquad m_k = \sum_i q_i r_i^k, \tag{D.17}$$

where the prefactor $\mathcal{W}$ has radial degree $Q^2 - n$ at the reference point. A descendant of derivative level k adds a factor ρ^k, so the radial power is

$$\rho^{n-2}\rho^{Q^2-n+k} = \rho^{Q^2+k-2}. \tag{D.18}$$

For even (neutral) clusters, $Q = 0$, only $k \leq 1$ diverges; for odd clusters, $Q = \pm 1$, only $k = 0$; all higher derivatives are integrable. The operator content follows without reference to any probe: summing the two orientations of an even cluster gives the identity at level zero and a multiple of $\hat{\sigma}_3\partial\phi$ at level one, while an odd cluster gives $\mathcal{O}_g$ at level zero and its first-derivative descendants already have dimension two. Current insertions obey the same bound, because χ_B has reference dimension one and its contractions use one or two derivatives of the logarithmic covariance. Simultaneous collapse is not the whole argument, since the shape variables still have partial-collision singularities; these are treated in Appendix D.8.

Endpoint clusters: the degree differs by one. At a fixed endpoint of I the centre of the cluster also approaches the endpoint. There are then N, not $N-1$, scaled coordinates, and

$$\int_0 \mathrm{d}\rho\, \rho^{N-1}\rho^{d_{\mathcal{A}}-N}(\log\rho)^M = \int_0 \mathrm{d}\rho\, \rho^{d_{\mathcal{A}}-1}(\log\rho)^M. \tag{D.19}$$

An endpoint divergence therefore requires $d_{\mathcal{A}} \leq 0$. The invariant dimension-zero sector is $\mathbb{R}\,\mathbf{1}$ (Appendix D.5), hence

every primitive ultraviolet counterterm at an endpoint is a scalar multiple of $\mathbf{1}$. (D.20)

This holds separately at the left and at the right endpoint; their scalar coefficients need not be assumed equal. The classification is made in the original invariant (g, h) frame. In the frame in which the current is resummed, the endpoint vertices are dressed and can carry non-integer auxiliary charges; they must be transformed together with the bulk charge and the Wick factors, and treating them as unrelated new endpoint interactions would not describe the same theory. The estimates (D.14) and (D.19) apply after proper collision subgraphs have been subtracted; they must not be applied separately to unprepared products of a power-divergent counterterm and an omitted remainder.

The long-range deformation does not enlarge the threshold. At fixed reference scale ($\kappa = 1$) the deformation of the covariance (D.4) is

$$f_{\bar{\delta}}(r) = \frac{r^{\bar{\delta}} - 1}{\bar{\delta}} = \sum_{j \geq 0} \frac{\bar{\delta}^{\,j} \log^{j+1} r}{(j+1)!} = \log r + \frac{\bar{\delta}}{2} \log^2 r + \frac{\bar{\delta}^2}{6} \log^3 r + \cdots . \tag{D.21}$$

At each fixed order in $\bar{\delta}$, a Gaussian vertex coefficient is the power-law coefficient at $\bar{\delta} = 0$ times a finite polynomial in logarithms. Spatial derivatives of $\log^j r$ have the same inverse powers of the distance as the same derivatives of $\log r$, multiplied by logarithms and constants, so they do not raise the superficial degree of divergence at that order. This is visible before the outgoing operator is expanded: the Wick normal product has the exact form

$$:\exp\Big(\mathrm{i} \sum_i q_i \phi(X_i)\Big):_{\bar{\delta}} = :\exp\Big(\mathrm{i} Q \phi(X) + \mathrm{i} \sum_{k \geq 1} \frac{\rho^k m_k}{k!} \partial^k \phi(X)\Big):_{\bar{\delta}}, \tag{D.22}$$

with the moments m_k of Eq. (D.17). The deformation changes the contraction coefficients and the normal-ordering convention, but not the list of possible outgoing charges or derivative levels. The normal ordering is varied consistently with the contractions, rather than differentiating the contraction and forgetting the definition of the local field. Consequently, for every fixed coefficient of $g^{n_1} h^{n_2} \bar{\delta}^{\,n_3}$, the coefficient $t_{\mathcal{A}}$ of an outgoing field obeys the reference scaling-degree bound

$$\operatorname{sd} t_{\mathcal{A}} \leq N - d_{\mathcal{A}}, \tag{D.23}$$

with possibly higher powers of logarithms, so that the degree of divergence is $\omega_{\mathcal{A}} = \operatorname{sd} t_{\mathcal{A}} - (N-1) \leq 1 - d_{\mathcal{A}}$ in the bulk and $\omega_{\mathcal{A}} \leq -d_{\mathcal{A}}$ at an endpoint. Derivatives in the charge, that is an expansion in the resummed charge $b^2 - 1$, likewise insert only logarithms, provided the expansion is taken order by order about the reference value and is not a statement uniform in an arbitrary fixed b.

Order of limits. The bound (D.23) concerns a formal coefficient before all powers of $\bar{\delta}$ are summed. It does not justify interchanging that infinite sum with $\varepsilon \to 0$. At fixed $\bar{\delta} > 0$, $r^{\bar{\delta}}$ tends to zero as $r \to 0$, while each fixed Taylor coefficient in Eq. (D.21) grows logarithmically; these are different ultraviolet organizations. The existence of coefficientwise counterterms establishes neither convergence, nor Borel summability, nor a unique identification of the formal series with a theory at fixed $\bar{\delta}$; neither argument asserts such an interchange.

D.8 The local forest construction

This construction supplies the renormalized family *before* any response condition is used, which is what keeps the prescription argument from being circular. It is the recursive position-space (Epstein–Glaser) construction, with the matrix multiplication kept in its original order.

The extension principle. A distribution in m relative variables, defined away from a coincidence point and of finite scaling degree sd, has an extension of the same scaling degree. If $\mathrm{sd} < m$ the extension is unique; otherwise its local ambiguity is a finite sum of derivatives, of order at most $\lfloor \mathrm{sd} - m \rfloor$, of the Dirac distribution at the coincidence point. A finite logarithmic scaling anomaly does not change that bound [29, 32]; Euclidean recursive versions are also available [31]. The arguments apply these results to the explicit finite-order Gaussian and operator-product coefficient distributions of the dual; the cited works do not establish the nonperturbative impurity theory.

Structure at separated points. At separated points every fixed perturbative coefficient is a finite sum of products of rational powers of differences and finite powers of logarithms, multiplying multilocal normal-ordered exponentials and derivative polynomials with Pauli matrices. Only coincidence loci can be singular, and all contractions with spectators a fixed distance away are smooth functions of the cluster coordinates.

The expansion is operator valued and made before projection. The collision of a cluster is expanded with Eq. (D.17) into the local operators of Eq. (D.10), as an operator statement, with no reference to the two probes used in the computation. To see that its coefficients do not depend on distant spectators, take external charges Q_α at points y_α a fixed positive distance from the cluster. In a chamber where the signed distances are fixed, their Wick factors expand as

$$\prod_{i,\alpha} |y_\alpha - X - \rho r_i|^{2q_i Q_\alpha} = \prod_{\alpha} |y_\alpha - X|^{2QQ_\alpha} \exp\Big[-2 \sum_{k\geq 1} \frac{\rho^k m_k}{k} \sum_{\alpha} \frac{Q_\alpha}{(y_\alpha - X)^k}\Big]. \tag{D.24}$$

The level-zero and level-one terms are exactly the Wick correlators of the local operators of Eq. (D.17) with those spectators; derivative spectators follow by differentiation. Spectator dependence therefore multiplies universal local collision coefficients and does not enter their definition.

A finite Taylor jet suffices. The expansion into local output fields is in general infinite, but only a finite Taylor jet is needed: neutral clusters need the constant and first-derivative jets, and charged clusters the leading vertex jet. The next Taylor remainder gains at least two powers of ρ in the neutral channel and one in the charged channel. Tested against separated smooth spectators, it obeys the improved bound in the remaining shape sectors, so the argument does not interchange an unproved infinite operator-product sum with a divergent integral.

Induction over clusters. Suppose that the products of fewer than N insertions have been extended at all their coincidence loci. The N-point configuration space is subdivided by relative cluster scales. In one dimension a collapsing subset of an ordered chamber is consecutive, and compatible scale hierarchies consist of nested or disjoint clusters, that is, forests. On sectors where some clusters have already collapsed, the lower-order local subtractions are inserted. This defines the renormalized product away from the total diagonal. Overlapping candidate subclusters occur in different hierarchy sectors, so they are never subtracted twice in one forest, nor treated as independent simultaneous subtractions.

For the remaining total collision of a cluster γ, let $t_{\gamma,\mathcal{A}}$ be the prepared coefficient distribution of its output field $\mathcal{A}$, in m relative coordinates. By Eq. (D.23) its excess of scaling degree obeys

$$\omega_{\gamma,\mathcal{A}} \leq \begin{cases} 1 - d_{\mathcal{A}} & \text{(bulk)}, \\ -d_{\mathcal{A}} & \text{(fixed endpoint)}. \end{cases} \tag{D.25}$$

Choose a smooth compactly supported function w equal to one near zero, at a reference scale μ. For $\omega \geq 0$ a test function f is replaced by

$$(\mathcal{T}_\omega f)(\xi) = f(\xi) - w(\xi) \sum_{|\alpha|\leq\omega} \frac{\xi^\alpha}{\alpha!}\, \partial^\alpha f(0). \tag{D.26}$$

The extra vanishing order makes $t_{\gamma,\mathcal{A}}(\mathcal{T}_\omega f)$ integrable; for $\omega < 0$ nothing is subtracted. Finite local normalization constants are then added to specify the chosen extension. The subtractions are local by Eq. (D.26), and their coefficients are independent of spectators by Eq. (D.24). In channels with $d_\mathcal{A} \geq 2$ the scaling-degree-preserving extension is unique once the subdivergences have been removed, so no new irrelevant bulk counterterm is needed. Only fields of dimension at most one can carry a counterterm ambiguity, and Table 5 leaves $\mathbf{1}$, $\mathcal{O}_g$ and $\mathcal{O}_h$. Derivatives of the Dirac distribution in a dimension-zero channel act on switching functions or produce total derivatives; they are not extra bulk interactions with constant couplings. Counterterms of the identity give the vacuum normalization.

Stability under counterterm insertions. An inserted counterterm is an effective vertex of dimension at most one. With N' effective vertices of dimensions $d_i \leq 1$, the new degree of divergence is bounded by

$$\omega_\mathcal{A} \leq \sum_{i=1}^{N'} d_i - d_\mathcal{A} - (N' - 1) \leq 1 - d_\mathcal{A} \tag{D.27}$$

in the bulk, with one further unit of improvement at an endpoint. All proper collisions therefore remain subtractable within the same class. Finite powers of logarithms from $\bar{\delta}$ derivatives and from earlier subtractions do not change the power bound. This completes the induction at every finite order, rather than assuming only simultaneous collapse.

Symmetric extensions. Symmetric extensions can be chosen: the original kernels are invariant, and averaging an extension over the compact internal symmetry and the finite group of reflection and parity preserves its scaling bound and its factorization, once the lower extensions have been chosen invariantly. No anomalous symmetry breaking is forced by this finite-cutoff matrix and Gaussian construction.

The hard-core regulator. For a coefficient regulated by the hard core, the difference between its integral and the Taylor-subtracted one is a finite sum of local Taylor jets,

$$\langle t_{\gamma,\mathcal{A};\varepsilon}, f\rangle - \langle t_{\gamma,\mathcal{A};\varepsilon}, \mathcal{T}_\omega f\rangle = \sum_{|\alpha|\leq\omega} c_{\gamma,\mathcal{A},\alpha}(\varepsilon,\mu)\, \partial^\alpha f(0), \tag{D.28}$$

where the coefficients are regulated moments of $t_{\gamma,\mathcal{A};\varepsilon}\, w(\xi)\, \xi^\alpha/\alpha!$. Their divergent and finite parts define the cutoff counterterms recursively. Replacing the cutoff for an already collapsed effective vertex changes only a shrinking collision region, and the next step of the same Taylor construction absorbs its local difference. Thus no exact finite-cutoff factorization of the exclusion region of a collapsed kink is assumed.

Identity terms and sharp endpoints. The identity and its first test-function derivatives are vacuum and switching-function counterterms. For a constant coupling on the interval they give a bulk vacuum energy and scalar endpoint factors; the endpoint Taylor ambiguities have order zero and multiply only $\mathbf{1}$. The construction can first be formulated with a smooth switching function and then taken to the sharp-edge limit. The renormalization is local in the coupling profile, and at finite smoothing width a profile is not assumed equal to its square. In the sharp limit all nontrivial bulk terms have the common plateau coupling, while the additional ultraviolet terms at either edge obey Eq. (D.19) and are scalar. Finite higher-dimensional regulator remainders are kept in the prepared integrals; dropping them by hand would not be this construction.

No new nonlocal Gaussian counterterm. A primitive ultraviolet counterterm is supported on a collision diagonal. In momentum space it is therefore polynomial in the external momenta, not

a new nonanalytic kernel such as $|p|^{1+\bar{\delta}}$ or one of its logarithmic Taylor coefficients. The only local quadratic derivative term that could occur has reference dimension at least two and is excluded from the required counterterms by the same power counting. This concerns divergent counterterms; finite self-energy effects and anomalous dimensions are not excluded.

D.9 What the closure statement does not mean

Irrelevant operators feed the marginal β functions. A simple formal example has a marginal coupling g and an irrelevant dimensionless coupling ϑ, with (in the $\mu\partial_\mu$ convention)

$$\beta_g = e_1 g^3 + e_2 g\vartheta + \cdots , \qquad \beta_\vartheta = \vartheta - e_3 g^2 + \cdots . \tag{D.29}$$

An invariant perturbative trajectory has $\vartheta = e_3 g^2 + O(g^4)$, so that

$$\beta_g^{\text{reduced}} = (e_1 + e_2 e_3) g^3 + O(g^5). \tag{D.30}$$

Setting $\vartheta = 0$ blindly loses a cubic contribution, although ϑ need not be an independent continuum parameter. In the dual such finite effects, among them the $:\chi_B^2:$ term of Eq. (D.12), occur inside the fully integrated multi-kink kernels and their finite subtractions. The absence of an independent counterterm does not license truncating every intermediate operator product to its singular terms when finite higher-order coefficients are computed. The coefficient of $:\chi_B^2:$ at level two is locally integrable in the expansion about the endpoint; all such finite terms are retained in the full cutoff integrals and finite responses of the prescription. The prescription argument would fail for an algorithm that replaced every cluster by its two marginal channels and discarded the integrable remainder before the higher integrations.

The stability matrix is tangent to the renormalized family. Suppose a full Wilsonian description has coordinates (q^a, ϑ^i), with q the two retained couplings, and a two-parameter invariant surface $\vartheta^i = \Theta^i(q)$ with slope $\Theta' = \partial\Theta/\partial q$. At its fixed point write the full linearization in blocks as

$$M_{\text{full}} = \begin{pmatrix} M_{qq} & M_{q\vartheta} \\ M_{\vartheta q} & M_{\vartheta\vartheta} \end{pmatrix}. \tag{D.31}$$

The reduced matrix and the invariance condition are

$$M_{\text{red}} = M_{qq} + M_{q\vartheta}\Theta', \qquad M_{\text{full}} \begin{pmatrix} \mathbf{1} \\ \Theta' \end{pmatrix} = \begin{pmatrix} \mathbf{1} \\ \Theta' \end{pmatrix} M_{\text{red}}. \tag{D.32}$$

The two eigenvalues of M_{red} are genuine tangent eigenvalues of the full flow; the block M_{qq} alone is in general not the correct matrix. This identity explains what a correctly renormalized two-coupling description computes; it does not construct an infinite-dimensional Wilsonian invariant manifold.

The statement is formal, not uniform in a finite charge displacement. Even at $\bar{\delta} = 0$, resumming a charge displacement $b^2 = 1 + z$ before expanding can change the ultraviolet power counting. For an even cluster of n flips the Wick prefactor scales as $\rho^{(1+z)(Q^2-n)} = \rho^{-n(1+z)}$, so the level-two channel has radial behaviour $\rho^{n-2}\rho^{-n(1+z)}\rho^2 = \rho^{-nz}$. It is integrable for $nz < 1$, while each of its fixed Taylor coefficients in z is an integrable polynomial in $\log\rho$. A formal all-order statement about coefficients is therefore not a uniform statement for an arbitrary fixed b and an arbitrarily large number of insertions. The analogous non-uniformity in $\bar{\delta}$ is the order of limits of Appendix D.7.

Identity, endpoints and composite insertions. The identity counterterm cancels vacuum divergences and may depend on the couplings. It drops out of normalized bulk correlators, but it cannot be ignored before the normalization is formed correctly. Sharp interval endpoints and external operator insertions can require their own local renormalizations. Such constants are not new translation-invariant couplings of the action, but they can affect an extraction prescription based on integrated responses on a finite interval; the endpoint sources created by the current resummation belong to this separate bookkeeping. This is the gap that the prescription argument addresses (Appendices D.10 to D.13).

D.10 Endpoint cancellation and the separated probes

Factorization and cancellation. In correlators with separated probes, the operator-valued forest construction gives the asymptotic factorization

$$\mathcal{D}_{I,\varepsilon}(c_B) = \mathcal{V}(\varepsilon, c_B, L)\, \mathcal{Z}_0(\varepsilon, c_B)\, \mathcal{Z}_L(\varepsilon, c_B)\, \mathcal{D}_I^{\mathrm{loc}}(c_R) + o(1), \tag{D.33}$$

after the same nontrivial local bulk redefinition $c_B = \mathcal{B}(c_R, \varepsilon\mu, \bar{\delta})$ at each order. Here c_B and c_R are the bare and renormalized couplings, $\mathcal{D}_I^{\mathrm{loc}}$ is the locally renormalized evolution, $\mathcal{V}$ is the bulk identity normalization, and $\mathcal{Z}_0$, $\mathcal{Z}_L$ are the scalar factors of the two endpoints (Eq. (D.20)). A finite scalar factor can be moved to either side without changing the conclusion. Equation (D.33) holds coefficientwise, after the proper subdivergences are prepared; it is not a uniform nonperturbative limit in operator norm.

For any separated reference probe in the numerator and the vacuum in the denominator, the three scalar factors cancel:

$$\mathcal{R}_{\mathcal{X},\varepsilon}(\mathcal{B}(c_R)) \longrightarrow \frac{\mathrm{tr}\langle \mathcal{X}\, \mathcal{D}_I^{\mathrm{loc}}(c_R)\rangle_0}{\mathrm{tr}\langle \mathcal{D}_I^{\mathrm{loc}}(c_R)\rangle_0}. \tag{D.34}$$

The cancellation happens before any differentiation with respect to parameters, so derivatives of coupling-dependent scalar endpoint normalizations cancel as well.

Only the ultraviolet normalization factors cancel. The finite endpoint-source interactions created by the current resummation do not; they remain in the finite responses and in the Jacobian of Appendix D.12. Ignoring them would be a different, unjustified prescription.

The first endpoint logarithm appears at two kinks. On an interval of length L with hard core ε, the two-kink vacuum and current integrals at $\bar{\delta} = 0$, $b_0 = 1$ are

$$Z_2(L, \varepsilon) = \int_\varepsilon^L \frac{L - r}{r^2}\, \mathrm{d}r = \frac{L}{\varepsilon} + \log\frac{\varepsilon}{L} - 1, \tag{D.35}$$

$$Q_2(L, \varepsilon) = \int_\varepsilon^L \frac{L - r}{r}\, \mathrm{d}r = L \log\frac{L}{\varepsilon} - L + \varepsilon, \tag{D.36}$$

which at $L = 1$ are the moments of Appendix A.7. In Z_2 the term L/ε is the bulk identity normalization and the logarithm, which does not scale with L, is an endpoint identity contribution; it is not zero. Both multiply any distant numerator by the same identity coefficient and cancel when it is divided by the vacuum moment. In Q_2 the logarithm is proportional to L: it multiplies the bulk operator $\mathcal{O}_h$ and survives the normalization. With the operator product $\mathcal{O}_g(X + r)\mathcal{O}_g(X) = r^{-2}\mathbf{1} + \sqrt{2}r^{-1}\mathcal{O}_h(X) + O(1)$ it gives the standard leading $\beta_h = -\sqrt{2}g^2$. The two integrals exhibit, rather than assume, the difference between an endpoint logarithm and a bulk one.

Separated reference probes need no ultraviolet counterterm. Fix a probe at y, a positive distance from I. There is no collision diagonal involving y and an integrated point, so its fixed free normal ordering suffices. Counterterms inside I act on any spectator through the universal local expansion (D.24). A qubit matrix attached to the probe may be transported freely through the

free exterior; the colliding interval product still multiplies it by the same scalar endpoint coefficient, and transporting the matrix creates no new Gaussian contraction.

At $\bar{\delta} > 0$ the projection to a distant probe is normalized by the free two-point function, as in [4, Eq. (4.18)]. For a vertex probe the remaining position-dependent external Wick factors contain $C(y - X) - C(y)$. On a fixed compact interval these tend to zero uniformly, like $O(y^{\bar{\delta}-1})$ for $\bar{\delta} < 1$, or coefficientwise like y^{-1} times finite polynomials in logarithms. For a neutral current probe the monopole term cancels and the first moment survives after the free normalization: it is the total length Ξ_{2m} of the reversed domains, which weights the current moments Q_{2m} (Appendix A.7), plus the exact endpoint current term. This yields the archived odd responses A_{2m+1} and current moments Q_{2m}. The limit is taken after the local ultraviolet subtractions and introduces no further cutoff renormalization of the probe.

An interacting primary *inside* the interval is different: it needs its own insertion renormalization, and it cannot replace the exterior reference probe in the same two-response formula.

D.11 The current resummation and the normal-ordering factors

From the longitudinal coupling to the charge. For smooth regulated fields, the change of frame with $R_h(X) = \exp[ih\hat{\sigma}_3\phi(X)/\sqrt{2}]$, $\mathcal{D}_h(L,0) = R_h(L)\widetilde{\mathcal{D}}(L,0)R_h(0)^{-1}$ (Appendix A.4), removes the derivative interaction from the interior and shifts the kink charge to $b = b_0 - \sqrt{2}h$. It follows either by differentiating the identity or by integrating the piecewise-constant spin history $m(X) = \pm 1$ by parts, $\int m\,\partial\phi = m\phi|_{\partial I} - \int m'\phi$. It is not a change of the Gaussian integration variable and needs no path-integral Jacobian. The endpoint matrices remain and must not be omitted [4, Sec. 3.2.1]. With the reference normalization fixed, h and b are therefore two coordinates on the same coupling direction, not independent third and fourth parameters; at $\bar{\delta} \neq 0$ the classical scaling must also be included in passing to a dimensionless charge. This is why the closure statement in (g, h) is a statement about the two essential coordinates (g, b).

Wick-ordering factors. Wick ordering supplies a local factor at each kink [4, Eqs. (3.20)–(3.23)],

$$g_h = g\exp\big[(\sqrt{2}b_0h - h^2)\,C(\varepsilon)\big], \tag{D.37}$$

with $C = C_{\bar{\delta}}$ of Eq. (D.4). The endpoint charges have magnitude $h/\sqrt{2}$. Their two self-contractions give the common scalar $e^{-h^2C(\varepsilon)/2}$, identical in numerator and denominator. The finite endpoint-to-endpoint and endpoint-to-kink contractions must be kept. The endpoint-to-kink source parameter is the $\eta = \sqrt{2}hb = \sqrt{2}b_0h - 2h^2$ of Eq. (A.17), and for alternating histories with $q_i = (-1)^{i+1}$ the finite endpoint factors are the U_n^{odd} and U_n^{even} of Eq. (A.18), with $f_{\bar{\delta}}(L - X_i)$ in place of $f_{\bar{\delta}}(1 - X_i)$ on an interval of length L. At $L = \kappa^{-1} = 1$ the finite endpoint-pair covariance vanishes ($C_{\bar{\delta}}(\kappa^{-1}) = 0$); a general L retains it as part of the finite response, not as a new bulk coupling. The first source derivative may be combined by reflection, as in the archive; that replacement cannot be made before squaring for a second source derivative.

The essential charge and fugacity. The dimensionless essential charge and fugacity are

$$B = (b_0 - \sqrt{2}h)\,\varepsilon^{\bar{\delta}/2}, \qquad Y = g_h\,\kappa\varepsilon\,e^{b^2C(\varepsilon)/2} = g\,\kappa\varepsilon\,e^{b_0^2C(\varepsilon)/2}, \tag{D.38}$$

where the second equality is the exact cancellation

$$(\sqrt{2}b_0h - h^2) + \tfrac{1}{2}(b_0 - \sqrt{2}h)^2 = \tfrac{1}{2}b_0^2. \tag{D.39}$$

With Eq. (D.4), $Y = g\kappa\varepsilon\exp[b_0^2(\kappa^{-\bar{\delta}} - \varepsilon^{\bar{\delta}})/\bar{\delta}]$; this is precisely the finite-tail matching (A.21) of the compendium (expanded in Eq. (A.22)), where g is called g_{aux} and Y is renamed g after matching (Sec. 2.1). The known explicit scale factors must be differentiated when the essential β functions

are formed. On the reference section $h = 0$, with the couplings identified at the reference matching scale and $v = B^2$, the chain rule applied to Eq. (D.38) gives

$$\frac{\beta_Y}{Y} = v - 1 + \frac{\beta_g^{\log}}{g}, \qquad \frac{\beta_v}{v} = -\bar{\delta} - \frac{4\beta_h^{\log}}{\sqrt{2}b_0}, \tag{D.40}$$

where $\beta^{\log}$ denotes the logarithmic part extracted from the responses. This explains the classical-scaling terms $v - 1$ and $-\bar{\delta}$, rather than appending them in an unrelated scheme. With $u = Y^2$ and $z = v - 1$ these are F and H of Sec. 2.1 (Eq. (A.23)); for instance $\beta_h^{\log}/(\sqrt{2}b_0) = -g^2 + \cdots$ gives $H = -\bar{\delta} + 4u + \cdots$.

An exact check at $g = 0$. With $J_I = \int_I \chi_B$,

$$\frac{\mathrm{tr}\langle \hat{\sigma}_3 \chi_B(y)\, e^{h\hat{\sigma}_3 J_I}\rangle_0}{\mathrm{tr}\langle e^{h\hat{\sigma}_3 J_I}\rangle_0} = h\,\langle \chi_B(y) J_I\rangle_0, \tag{D.41}$$

since the qubit traces give $e^{hJ_I} \mp e^{-hJ_I}$, and for a Gaussian field

$$\langle \chi_B(y)\, e^{\pm hJ_I}\rangle_0 = \pm h\,\langle \chi_B(y) J_I\rangle_0\,\langle e^{hJ_I}\rangle_0. \tag{D.42}$$

Every Gaussian endpoint normalization factor cancels. The normalized far current response is exactly hL, so an endpoint anomalous factor cannot generate a spurious pure-h quantum β function.

D.12 The response Jacobian

Dividing the integrated odd and current numerators by the same integrated vacuum series gives, at $h = 0$ and on the unit interval, the two normalized responses

$$R_g = g + \mathcal{C}_3 g^3 + \mathcal{C}_5 g^5 + \cdots, \qquad R_h = \sqrt{2}b_0(Q_2 g^2 + \mathcal{K}_4 g^4 + \mathcal{K}_6 g^6 + \cdots), \tag{D.43}$$

and away from that section R_h also contains the tree term h. The coefficients $\mathcal{C}_{2m+1}$ and $\mathcal{K}_{2m}$ are the vacuum-normalized moments of Eq. (A.37), for example $\mathcal{C}_3 = A_3 - Z_2$ and $\mathcal{K}_4 = Q_4 - Q_2 Z_2$. They are the coefficients of the normalized correlators (D.8), not two functions selected after integration. The products are formed before $[\varepsilon^0 \ell^0]$ or $[\varepsilon^0 \ell^1]$ is extracted, since a term of order ε can multiply one of order $1/\varepsilon$; the common-scalar cancellation (D.34) requires this, and it is not a dispensable numerical detail.

The projections $\mathcal{C}_{n,0}$, $\mathcal{C}_{n,1}$, $\mathcal{K}_{2m,0}$, $\mathcal{K}_{2m,1}$ and $Q_{2,0}$ (finite part and single-logarithm coefficient) are those of Appendix A.7, and $v = b_0^2$ on the reference section. Exact resummation gives, on the normalized moment part at $h = 0$, the source chain rule $\partial_h = \sqrt{2}b_0(-2\partial_v + \partial_\eta)$ of Eq. (A.19), because h enters only through the charge $b^2 = v$ and the endpoint source η. The local factor (D.37) has no finite constant at $\kappa\varepsilon = 1$ (there $C(\varepsilon) = 0$); its logarithms are accounted for by the matching. Differentiating also the explicit current prefactor $\sqrt{2}b$, with $\partial_h(\sqrt{2}b) = -2$, gives the finite source and current entries

$$\begin{aligned} j_1 &= -\frac{2}{1+\bar{\delta}}, \qquad j_{2m+1} = -2\partial_v \mathcal{C}_{2m+1,0} + \partial_\eta \mathcal{C}_{2m+1,0}, \\ \lambda_{2m} &= -2\mathcal{K}_{2m,0} - 4v\partial_v \mathcal{K}_{2m,0} + 2v\partial_\eta \mathcal{K}_{2m,0}, \end{aligned} \tag{D.44}$$

with $\mathcal{K}_2 = Q_2$; for $m = 1, 2$ these are the entries of Eq. (A.42). Here j_1 is the case $m = 0$, with $\mathcal{C}_1 = A_1$: the one-kink response has no charge dependence, and j_1 is its endpoint integral $\int_0^1 [f_{\bar{\delta}}(X) + f_{\bar{\delta}}(1-X)]\,\mathrm{d}X = 2(\frac{1}{1+\bar{\delta}} - 1)/\bar{\delta}$. The coupling and source derivatives thus produce exactly the compendium's finite-response matrix $\mathcal{J} = D_c R_0$ of Eq. (A.44), continued to the orders needed: ∂_g of the finite parts of R_g and R_h gives its first column, and ∂_h, through Eq. (D.44), its second. Its free value is the identity, so it is formally invertible; the invertibility is used only *after* locality and the endpoint cancellation have been established.

D.13 Uniqueness of the two-response solution, the recurrence and scheme independence

Uniqueness. Keep the known engineering and normal-ordering factors separate for the moment, and write the normalized bare responses, in the common reference coordinates c, as

$$R(c;\varepsilon) = R_0(c) + \ell R_1(c) + \ell^2 R_2(c) + \cdots + o(1), \qquad \mathcal{J} = D_c R_0. \tag{D.45}$$

The absence of residual power divergences follows from the operator-valued local construction and the scalar normalization, not from dropping such terms in the raw moments. The independently constructed local scheme has a bare map

$$c_B = \mathcal{B}(\tilde{c},\ell), \qquad \mathcal{B}(\tilde{c},0) = \mathcal{G}(\tilde{c}), \qquad \mathcal{G}(\tilde{c}) = \tilde{c} + O(\tilde{c}^2), \tag{D.46}$$

whose coefficients are finite formal coefficients: they are the finite local Taylor moments of Eq. (D.28), recursively including forests. Local finite normalizations can be chosen so that $\mathcal{G}$ is the identity, and no response condition is needed for that choice; for an arbitrary local normalization $\mathcal{G}$ is kept explicit. At fixed bare couplings, with $\ell = \log(\varepsilon\mu)$, the local logarithmic renormalization-group identity is

$$\partial_\ell \mathcal{B} + D_{\tilde{c}}\mathcal{B}\,\tilde{\beta}^{\log} = 0. \tag{D.47}$$

Every normalized spectator response is finite in the renormalized couplings, so its ℓ derivative at fixed $\tilde{c}$ vanishes. At first order in ℓ this gives

$$R_1(\mathcal{G}(\tilde{c})) = DR_0(\mathcal{G}(\tilde{c}))\, D\mathcal{G}(\tilde{c})\, \tilde{\beta}^{\log}(\tilde{c}), \tag{D.48}$$

and therefore

$$\beta^{\log}_{\text{archive}}(\mathcal{G}(\tilde{c})) = D\mathcal{G}(\tilde{c})\, \tilde{\beta}^{\log}(\tilde{c}), \qquad \beta^{\log}_{\text{archive}} = \mathcal{J}^{-1} R_1. \tag{D.49}$$

The known explicit classical-scaling and normal-ordering maps are then applied on both sides, including their scale derivatives as in Eq. (D.40); this gives the corresponding relation for the full essential vector fields. The charge is not held fixed when a change of scale physically changes B.

The argument is not circular. It constructs the local renormalization and the finiteness for all spectators in Appendices D.7 to D.10, independently of the response equations; Equation (D.48) then makes the local β function a solution, and the invertibility of $\mathcal{J}$ makes it the *unique* solution in the specified coordinates. No independent anomalous term of Eq. (D.1) remains: the endpoint factors cancel before differentiation, and the exterior reference probe does not collide with an integrated operator. If either fact were false, the uniqueness argument would not apply.

The recurrence at every required power. With the auxiliary coefficients b_{2m+1} and h_{2m} of Appendix A.9 ($\beta_g = b_3 g^3 + b_5 g^5 + \cdots$, $\beta_h/(\sqrt{2}b_0) = -g^2 + h_4 g^4 + \cdots$), solving Eq. (D.49) on the section $h = 0$ means solving the system $\mathcal{J}\beta = R_1$ of Eq. (A.46), with R_1 of Eq. (A.45) continued to the orders needed. Order by order in g it is triangular: the coefficient of g^{2m+1} in the first row contains b_{2m+1} with coefficient one and otherwise only lower coefficients, and the coefficient of g^{2m} in the second row does the same for h_{2m} (the factor $\sqrt{2}b_0$ of $\mathcal{J}_{12}$ times the factor $\sqrt{2}b_0$ of β_h gives the $2v$ of the recurrence; in the second row the factor $\sqrt{2}b_0$ of $\mathcal{J}_{21}$ cancels against that of R_1). The solution is the compendium's recurrence: Eq. (A.47) through b_7, and its continuation to h_8 and b_9 in Appendix C.3 (Eq. (C.30)). All finite and source terms are retained; it is not a new fit to the final C_n^{SR}. It was reproduced by the prescription argument's symbolic checker and re-derived independently by the review (Appendix D.15).

The lowest order, also the check of signs of Appendix A.9, shows what the finite endpoint term does. At $\bar{\delta} = 0$ and $v = 1$ the archived exact normalization gives

$$\mathcal{C}_3 = 2\ell^2 + 5\ell + 3 - 2\zeta(2) + o(1), \qquad j_1 = -2, \tag{D.50}$$

so the first line of the recurrence, $b_3 = \mathcal{C}_{3,1} + 2vj_1$, gives $b_3 = 5 - 4 = 1$, while $h_2 = -1$ follows from Eq. (D.36). This is the leading $F = z + u + \cdots$ of Sec. 2.1. Discarding the finite endpoint derivative j_1 would give $b_3 = 5$, a wrong cubic coefficient, even though the ultraviolet endpoint factor itself has cancelled.

Scheme independence. For the full essential vector fields, Eq. (D.49) is the usual transformation under a finite change of coordinates. At a fixed point c_*,

$$M_{\text{archive}}(\mathcal{G}(c_*)) = D\mathcal{G}(c_*)\, M_{\text{local}}(c_*)\, D\mathcal{G}(c_*)^{-1}; \tag{D.51}$$

the term with second derivatives of $\mathcal{G}$ vanishes because the β function itself vanishes there. The trace, the determinant and the two scaling eigenvalues therefore coincide coefficientwise. The finite map is regular in the formal $\bar{\delta}$ expansion, so substituting the fixed-point series in $\sqrt{\bar{\delta}}$ does not introduce a hidden change of the physical parameter s. This gives the formal scheme statement at any finite computed order. It needs no separately evaluated integral on a ring, and none is claimed: a ring in the compact boson theory also has a monodromy sector that must be treated correctly, and deleting the endpoints of an existing interval integral is not automatically a second prescription [4].

Other observables. For any other separated observable the same local forest construction and the same bare map apply: Eq. (D.24) identifies its local collision coefficients before projection, and Eq. (D.34) removes its scalar endpoint factors. There is therefore no spectator-dependent bulk counterterm. Physical operator insertions inside the support are renormalized separately as composite operators, and their mixing is not a third bulk coupling.

D.14 What the arguments establish and what they do not

For the operator content. The closure argument improves the status of the operator-content assumption: at the finite orders used for the expansion about the endpoint, no third symmetry-preserving bulk operator that needs an ultraviolet counterterm is expected or required in the formal local subtraction construction, and the same argument applies at each finite order rather than stopping at a two-insertion test. It does not certify the archived numerical β coefficients. The exact certificates of Sec. 2.4 prove identities between supplied integral expressions; they do not show that the finite-response prescription assembled the correct local counterterms for all collision configurations.

For the prescription. To connect the archive with the closure statement, the closure argument named a concrete audit in three steps: (i) construct the collision subtractions of the regulated, path-ordered kernels as operator-valued products, including every proper cluster and not only the projections that define the two response coordinates; (ii) show that the archive's vacuum normalization, current resummation, endpoint prescription and finite-$\bar{\delta}$ matching implement those subtractions, or differ from them by an explicit finite redefinition of the two essential couplings and separately specified operator and endpoint normalizations; (iii) check an independent family of non-projected correlators after inserting the same counterterms, whose coefficients in the $\mathbf{1}$, $\mathcal{O}_g$ and $\mathcal{O}_h$ channels must not depend on the distant spectator configuration, with locally integrable remainders of derivative order two and higher. In this note's assessment (Appendix D.15), the prescription argument supplies steps (i) and (ii) at the formal level (Appendices D.8 to D.13). The prescription argument addresses the analytic content of step (iii) through the spectator expansion (D.24) and the paragraph "Other observables" of Appendix D.13; the computation itself has not been done. A numerical check of the prescription with a second regulator and different responses, within the same two-coupling description, is the magnetization-scheme calculation of Sec. 2.7, which reproduces C_3^{SR} to C_7^{SR} and the record of C_8^{SR} (Sec. 6). Given (i) and (ii), regular finite coupling

redefinitions transform the stability matrix by similarity, so its eigenvalues do not depend on the chosen two-coupling coordinates.

What is not established. Neither argument proves the lattice duality, the identification of the dual and of its thermal branch with the critical chain, or the infinite-volume limit; these keep the scope of the published proposal [3, 4]. Neither proves convergence or resummability at fixed $\bar{\delta}$, nor a non-perturbative renormalizability statement. Neither recomputes C_3^{SR} to C_7^{SR} in a second regulator or audits the arithmetic of the historical high-order integrations, whose 21 stage programs are missing (Appendix H.5). Their fidelity to the recipe is tested only numerically: by the independent re-derivation (Sec. 2.6) and, through the invariants they produce, by the magnetization-scheme calculation in a second regulator (Sec. 2.7); the exact certificates (Sec. 2.4) concern the later reduction of the archived expressions to zeta values. Neither bears on the conjectured zeta-value form of C_8^{SR} (Eq. (2.15)) or on the statistical claims of the NRG or Monte Carlo comparisons, for which no raw data were supplied to them. The statement of the prescription argument concerns the mathematically specified recipe: implementations must still be checked to follow it, and its formula audit covers the normalization, the current and endpoint identity, the Wick factors, the finite-response Jacobian and its recurrence. The arguments are also narrower than a proof of the fixed-s duality, and they avoid the opposite mistake of treating every generated irrelevant Wilsonian coefficient as a new independent tuning parameter.

Status in this note. Both arguments are formal and unrefereed; this note uses them as such. The status word "proved closed form within the prescription" (Sec. 1.3) is unchanged by them: they bear on whether the prescription computes the coefficients of the dual, not on the reduction of the archived expressions to zeta values. The evidence that does not use the dual at all is the NRG test of C_3^{SR} to C_5^{SR} (Sec. 2.9).

D.15 Assessment and checks by the review

Claude Code assessed both arguments at the author's request on 15 October 2026 (Sec. 5); the caveat "Two-coupling closure" of Sec. 2.9 is the summary that the author approved afterwards.

The closure argument. Verdict: narrowed; the operator content is closed formally, and the matching of the archived prescription was left open. The argument is sound as a standard perturbative power-counting (Epstein–Glaser) argument, and no gap was found that changes its conclusion. It is a careful sketch: the uniform scaling bounds in the shape sectors and the sufficiency of a finite Taylor jet are asserted, not estimated in detail, and the extension step is invoked from the cited theorems. Checked independently: the classification through dimension one under the shift symmetry, the internal reflection and parity (Table 5); the charge selection rule; the radial-power identity; the product $\mathcal{O}_g\mathcal{O}_g$ (D.12); the coefficientwise logarithmic deformation (D.21); symmetric extensions by group averaging; and the frame change between h and b, identical to the compendium's. The assessment found that the archive's prescription did not obviously implement the local subtraction, and named the risks: (i) the β functions are solved from exactly two projected responses, a square system that would absorb a mis-assigned endpoint or insertion logarithm silently; (ii) all responses live on one unit interval, with the current resummed into endpoint sources and no endpoint hard core, so endpoint and bulk logarithms are not separated by construction; (iii) the finite-$\bar{\delta}$ matching is the exact normal-ordering factor, coefficientwise a regular redefinition, but its equivalence to the local scheme was not shown; (iv) the current resummation and (v) the vacuum normalization were judged low risk. The prescription argument answers this open link: (i) by the scalar endpoint factors, the separated probes and the uniqueness argument (Appendices D.10 and D.13); (ii) by the endpoint power counting and the two-kink integrals (D.35) and (D.36); (iii) by the normal-ordering audit (Appendix D.11).

The prescription argument. Verdict: the link to the prescription is closed at the formal level, at the same level of rigour as the closure argument. Checked: the endpoint power counting (one fewer free coordinate, so endpoint divergences only at reference dimension zero or below); the endpoint classification (the invariant dimension-zero sector is the identity, so endpoint ultraviolet factors are scalars that cancel in the normalized responses); that the finite endpoint-source terms survive and are the compendium's j and λ terms ($b_3 = 1$, against 5 without j_1); the cancellation (D.39) and $\eta = \sqrt{2}b_0 h - 2h^2$; the two-kink integrals at general L and ε; and the recurrence for b_3 to b_9 and h_4 to h_8, re-derived from $\mathcal{J}\beta = R_1$ and equal to the compendium's formulas. The logic is not circular: local renormalization for arbitrary spectators is constructed first, and the invertibility of the response system is used only for uniqueness. Stated rather than shown in detail: the sharp-edge limit from smooth switching, the recursion relating the hard-core regulator to the local subtraction when exclusion regions do not factorize (Eq. (D.28)), and the projection to a distant probe.

Code. `code/closure_d1/public/check_closure.py`, written by the review and run by the package's `test` and `replay` targets (standard library and sympy, a few seconds), checks the finite algebra:

- all $3 + 3^2 + \cdots + 3^9 = 29\,523$ words of length one to nine in $\hat{\sigma}_+$, $\hat{\sigma}_-$, $\hat{\sigma}_3$, in exact integer arithmetic: the $2\,035$ nonzero words all have $Q \in \{0, \pm 1\}$ and $[\hat{\sigma}_3, M] = 2QM$, with no violation;
- the radial-power identity $2\sum_{i<j} q_i q_j = Q^2 - n$ for alternating charges, $n = 1$ to 11;
- the coefficients 1, $\sqrt{2}$, 1 of the midpoint product (D.12) (to 10^{-15});
- a rerun of the prescription argument's own checker, `code/closure_d1/prescription/code/check_prescription.py` (sympy): all 22 of its checks are true and equal to its shipped record. They cover the matrix words, the scalar invariant matrix at dimension zero, the bulk and endpoint radial powers, the cancellation (D.39) and η, the source chain rule of λ_{2m}, the integral j_1, every coefficient equation of the response system through g^9, both two-kink integrals at general L and ε, the spectator expansion at zeroth and first order, a toy response-generator identity, and the cancellation of a common scalar factor;
- independently of that checker, the recurrence $b_3, \ldots, b_9$, $h_4, \ldots, h_8$ solved order by order from $\mathcal{J}\beta = R_1$ (sympy), which equals the compendium's formulas exactly and gives $b_3 = 1$ at $v = 1$, $\mathcal{C}_{3,1} = 5$, $j_1 = -2$.

The export tests on the Intel and AMD nodes of the symmetry cluster give the same table.

What the checks establish. They establish the finite algebra on which the arguments rest: the selection rule for words up to length nine (the general rule is argued for every length independently of the enumeration), the pair product, the radial powers, the normal-ordering identities, the two-kink integrals, and that the compendium's recurrence is exactly what the response system $\mathcal{J}\beta = R_1$ implies. They do not check the analytic induction, the uniform bounds in the shape sectors, the sharp-edge limit, the hard-core recursion or the far-field projection; they recompute no archived β coefficient; and they are not a numerical test of the prescription. The closure argument describes its own script of finite checks (the same algebra, a bound on the logarithmic degree through order nine and an exact example of the tangent-stability identity); the script was not supplied and is not part of the release, and the review's checker above replaces its first three items.

E The closed form of c_2: derivation

This appendix gives the derivation of the closed form of c_2 in the closed-form study ("the study", written by OpenAI GPT-6 astra; Sec. 1.2), edited to the notation of this note.

Section 3 states the steps and their results; here each step is derived, in the order in which the results depend on one another. Appendices E.1 to E.9 reduce the coefficient to the elliptic finite part $\mathcal{I}$ of Eq. (3.5) and give Eq. (3.7); Appendices E.10 to E.14 evaluate $\mathcal{I}$ in closed form; and Appendix E.15 describes the numerical checks of that evaluation. The status is that of Table 2: the evaluation of $\mathcal{I}$ is proved, and c_2 is a closed form conditional on one unproved step, the uniform-remainder exchange (Appendix E.8 and Sec. 3.6) on which the identification of Eq. (3.9) with the physical coefficient rests.

Notation. We write c_2 for the study's C_2, h_{MS} for the fourth-order coefficient of the anomalous dimension in minimal subtraction (the study's h_0), κ for the parameter of a finite operator redefinition (the study's q), and $\mathsf{P}, \mathsf{Q}, \mathsf{T}$ for its three ${}_4F_3(1)$ constants. Auxiliary symbols are renamed where the study's symbol would clash with another one in this note: the Möbius scale Λ (study: a), the factor Υ (Y), the kernel combination $\Sigma(x, y; z)$ (S), the weights Ω_1, Ω_0 (U, W), the constant matrices $\mathsf{J}, \mathsf{L}, \mathsf{C}_\pm, \mathsf{B}_-, \mathsf{H}, \mathsf{D}$ ($J, L, C_\pm, B_-, H, D$), the charge factors $\mathcal{W}_{\boldsymbol{e}}, \mathcal{P}_{\boldsymbol{e}}$ ($W_{\boldsymbol{e}}, P_{\boldsymbol{e}}$), the moment combinations m_U, m_V (f, g) and d_O (d), the polynomial ϖ (p), the normalized integral $\hat{K} = 2K/\pi$ (F), the hypergeometric functions Φ_A, Φ_D (A, D_h), the period vectors $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{W}$ (X_v, Y_v, Z_v, W_v), the derivative combinations $\mathcal{D}_1, \mathcal{D}_0$ (D_1, D_0), and the Mellin integrals j_0, j_1, j_2 (J_0, J_1, J_2). The complex beta integral is written throughout as

$$\mathcal{B}(a, b) = \mathrm{AC}\int_{\mathbb{C}} |z|^{2a-2}|1-z|^{2b-2}\,\mathrm{d}^2 z = \pi\frac{\Gamma(a)\Gamma(b)\Gamma(1-a-b)}{\Gamma(1-a)\Gamma(1-b)\Gamma(a+b)}, \tag{E.1}$$

which the study writes as $J(a-1, b-1)$ in its lower-order steps.

E.1 Observable, regulator and continuation

The crossover theory is that of Sec. 3: the Ising CFT times a generalized free scalar χ, with unit-normalized fields, $\langle\sigma(x)\sigma(0)\rangle = |x|^{-1/4}$ and $\langle\epsilon(x)\epsilon(0)\rangle = |x|^{-2}$, and the Ising data listed there. An insertion at infinity means $O(\infty) = \lim_{|x|\to\infty} |x|^{2\Delta_O} O(x)$, so that $\langle O(0)O(\infty)\rangle = 1$. The fusion rules are $\sigma \times \sigma = \mathbb{1} + \epsilon$ and $\epsilon \times \epsilon = \mathbb{1}$, each family including all its Virasoro descendants. The generalized free field is fixed by Wick's theorem and

$$\langle\chi(x)\chi(y)\rangle = |x-y|^{-15/4+2\delta}, \tag{E.2}$$

so that $\Delta_\chi = 15/8 - \delta$ and the perturbing operator $\sigma\chi$ has dimension $2-\delta$. The bare action is

$$S = S_{\mathrm{Ising}} + S_\chi + \int g_0(x)\,\sigma\chi(x)\,\mathrm{d}^2 x + \int t_0(x)\,\epsilon(x)\,\mathrm{d}^2 x, \tag{E.3}$$

where g_0 has mass dimension δ and the thermal source t_0 is tuned to the critical surface. The thermal source cannot be left out of the definition of the prescription, because the short-distance product contains

$$\sigma\chi(x)\,\sigma\chi(0) \supset |x|^{-4+2\delta}\mathbb{1} + \tfrac{1}{2}|x|^{-3+2\delta}\epsilon(0). \tag{E.4}$$

The second term produces a thermal power divergence. It is retained by analytic continuation, as appropriate to the critically tuned theory.

Observable. The observable is the dimension $\Delta_{\mathcal{E}}$ of the critical thermal eigenoperator continued from ϵ,

$$\Delta_{\mathcal{E}} = 1 + c_2\delta^2 + O(\delta^3); \tag{E.5}$$

the absence of a linear term is derived in Appendix E.4. In terms of s, $\Delta_{\mathcal{E}} = 1 - 1.01656753703313038819\ldots\,(7/4-s)^2 + O((7/4-s)^3)$, the coefficient being $c_2/4$ (Table 1). Since $1/\nu = 2 - \Delta_{\mathcal{E}}$, also $\nu = 1 + c_2\delta^2 + O(\delta^3)$.

Regulator and continuation. All complex integrals use $\mathrm{d}^2 z = \mathrm{d}(\mathrm{Re}\, z)\,\mathrm{d}(\mathrm{Im}\, z)$. The renormalized coupling is localized by the source profile $f(w) = e^{-|w|^2}$; its width and the renormalization scale μ are set to one after the renormalization identities have been derived. The Gaussian controls the infrared. Ultraviolet integrals are first defined in a domain of convergence and then continued meromorphically in δ and, where needed, in auxiliary exponents; AC denotes this continuation. In particular,

$$\mathrm{AC}\int_0^R r^{p-1+b\delta}\,\mathrm{d}r = \frac{R^{p+b\delta}}{p+b\delta}. \tag{E.6}$$

For $p \neq 0$ the value at $\delta = 0$ need not vanish; for $p = 0$ the integral has a logarithmic pole. A circular finite part, FP, is defined with rotationally symmetric exclusions about each specified singularity, and the subtracted local terms are restored in the same continuation. This convention fixes the angular contacts used below.

E.2 Renormalization-group reduction

This subsection explains why a fourth-order residue determines the quadratic coefficient. In pole-only analytic minimal subtraction let

$$\beta(g,\delta) = \mu\frac{\mathrm{d}g}{\mathrm{d}\mu} = -\delta g + b_0 g^3 + O(g^5), \qquad \gamma_{\mathcal{E}}(g) = h_{\mathrm{MS}} g^4 + O(g^6). \tag{E.7}$$

The interacting fixed point has $g_*^2 = \delta/b_0 + O(\delta^2)$, and the dimension there is $\Delta_{\mathcal{E}} = 1 + \gamma_{\mathcal{E}}(g_*)$, so $c_2 = h_{\mathrm{MS}}/b_0^2$. The thermal operator renormalizes multiplicatively, $\mathcal{E}_0 = Z_{\mathcal{E}}\mathcal{E}_R$, with $\gamma_{\mathcal{E}} = \beta\,\partial_g \log Z_{\mathcal{E}}$. There is no energy pole at order g^2: the second-order integral is finite at $\delta = 0$ because the first-order shift vanishes (Appendices E.4 and E.5). Therefore

$$\log Z_{\mathcal{E}} = \frac{z_{4,1}}{\delta} g^4 + O(g^6), \qquad \gamma_{\mathcal{E}} = \Big(-\delta g + \cdots\Big)\frac{4 z_{4,1} g^3}{\delta} = -4 z_{4,1} g^4 + O(g^6), \tag{E.8}$$

which gives $h_{\mathrm{MS}} = -4 z_{4,1}$ and the relation $c_2 = -4 z_{4,1}/b_0^2$ of Sec. 3.2.

The factor four relies on there being a single renormalization factor. The source is localized at the origin, so in the correlator with the energies at 0 and ∞ the only local ultraviolet factor is the $Z_{\mathcal{E}}$ at 0; since $f(\infty) = 0$, the energy at infinity supplies no second source-dependent ultraviolet factor.

The restriction to pole-only subtraction is substantive. The split of c_2 between a δg^2 term and a g^4 term of $\gamma_{\mathcal{E}}$ depends on the scheme (Sec. 3.2): the finite redefinition $\mathcal{E}_R \to e^{\kappa g^2}\mathcal{E}_R$ moves weight from one to the other. The finite lower-order source contact and the fourth-order pole must therefore be matched in one and the same convention, which is what Appendices E.5 to E.9 do.

E.3 Exact Ising correlators

For $2n = 2, 4$ distinct spin positions z_i, sum over the neutral charges $e_i = \pm 1$ with $\sum_i e_i = 0$, and define

$$\mathcal{W}_{\boldsymbol{e}} = \prod_{i<j} |z_i - z_j|^{e_i e_j/2}, \qquad \mathcal{P}_{\boldsymbol{e}} = 1 - \frac{1}{4}\Big(\sum_i e_i z_i\Big)\Big(\sum_i \frac{e_i}{z_i}\Big). \tag{E.9}$$

The Euclidean correlators needed are [13]

$$S_{2n} = \langle \sigma_1 \cdots \sigma_{2n}\rangle = \Big[2^{-n}\sum_{\boldsymbol{e}} \mathcal{W}_{\boldsymbol{e}}\Big]^{1/2}, \qquad H_{2n} = \langle \epsilon(0)\epsilon(\infty)\sigma_1 \cdots \sigma_{2n}\rangle = \Big[2^{-n}\sum_{\boldsymbol{e}} \mathcal{W}_{\boldsymbol{e}}|\mathcal{P}_{\boldsymbol{e}}|^2\Big]^{1/2}. \tag{E.10}$$

The positive Euclidean square root is fixed by the identity OPE. These formulas follow by combining two independent Ising copies, $\sigma^{(1)}\sigma^{(2)} = \sqrt{2}\cos(\varphi/2)$ with $\langle\varphi(z)\varphi(w)\rangle = -2\log|z-w|$, and representing the product of energies by currents. Charge neutrality gives the displayed sum; taking the square root removes the second copy. For energies at finite positions x, y, $\mathcal{P}_e$ is replaced by

$$\frac{1}{(x-y)^2} + \frac{1}{4}\Big(\sum_i \frac{e_i}{x-z_i}\Big)\Big(\sum_i \frac{e_i}{y-z_i}\Big). \tag{E.11}$$

For two spins (energies at 0 and ∞) this gives

$$S_2(a,b) = |a-b|^{-1/4}, \qquad H_2(a,b) = \frac{|a+b|^2}{4|ab|\,|a-b|^{1/4}}. \tag{E.12}$$

E.4 The vanishing first-order shift and the exact b_0

First order. The first-order energy shift is proportional to the integral $I_2(\delta)$ of Sec. 3, whose integrand is $|1-z|^{-4+2\delta}$ times $H_2(z,1)/S_2(z,1) - 1$ from Eq. (E.12). The combined integrand is absolutely integrable for $0 < \operatorname{Re}\delta < 1/2$. The term -1 (the subtracted spin two-point function) gives a scaleless integral, which vanishes in the continuation. With $|1+z|^2 = 2(1+|z|^2) - |1-z|^2$, the rest is three complex beta integrals (E.1):

$$I_2(\delta) = \tfrac{1}{2}\mathcal{B}(\tfrac{1}{2}, \delta-1) + \tfrac{1}{2}\mathcal{B}(\tfrac{3}{2}, \delta-1) - \tfrac{1}{4}\mathcal{B}(\tfrac{1}{2}, \delta) = \frac{\pi\delta^2}{4(1-\delta)^2}\frac{\Gamma(\delta)\Gamma(\frac{1}{2}-\delta)}{\Gamma(1-\delta)\Gamma(\frac{1}{2}+\delta)}. \tag{E.13}$$

Expanding the Gamma ratios gives $I_2(\delta) = \frac{\pi}{4}[\delta + (2+4\log 2)\delta^2 + O(\delta^3)]$. Thus $I_2(0) = 0$, which reproduces the leading cancellation ($c_1 = 0$), while $I_2'(0) = \pi/4$. The finite derivative is retained in the source matching of Appendix E.5; it is not a separate contribution to be added to $c_2 = h_{\mathrm{MS}}/b_0^2$.

The exact normalization of b_0. In the same field normalization the leading coefficient of the beta function is

$$b_0 = -\pi I_\beta, \qquad I_\beta = \mathrm{FP}\int_{\mathbb{C}} |z|^{-15/4}\big[S_4(0,z,1,\infty) - |z|^{-1/4}\big]\,\mathrm{d}^2 z, \tag{E.14}$$

with S_4 from Eq. (E.10). The quadratic map $z = 4w/(1+w)^2$, followed by $u = w^2$, with its multiplicity and Jacobian, reduces the integral to

$$I_\beta = \frac{1}{64}\,\mathrm{AC}\int_{\mathbb{C}} (|u|^{-3} + |u|^{-5/2})|1-u|^{3/2}\,\mathrm{d}^2 u. \tag{E.15}$$

Inversion, $u \mapsto 1/u$, interchanges the two terms. Equation (E.1) therefore gives

$$I_\beta = \frac{1}{32}\mathcal{B}(-\tfrac{1}{2}, \tfrac{7}{4}) = -\frac{9\pi}{8}\Big[\frac{\Gamma(3/4)}{\Gamma(1/4)}\Big]^2, \tag{E.16}$$

which is the exact $b_0 = \frac{9\pi^2}{8}[\Gamma(\frac{3}{4})/\Gamma(\frac{1}{4})]^2$ of Sec. 3.2. All power terms in this reduction use Eq. (E.6).

E.5 The localized source and the fourth-order logarithm

The connected kernel. With the energies at 0 and ∞, write $S_{ij} = S_2(w_i, w_j)$ and $H_{ij} = H_2(w_i, w_j)$ from Eq. (E.12), and S_4, H_4 for the four-spin functions of Eq. (E.10). The fourth-order coefficient of a logarithm is its fourth-order moment minus one half the square of its second-order moment. Applied to the logarithm of the vacuum-normalized energy two-point function, for the fixed generalized-free pairing (12)(34), this gives the connected kernel $\mathcal{F} = H_4 - S_4 - H_{12}H_{34} + S_{12}S_{34}$ of Sec. 3; its last two terms are fixed by the logarithm. The three Wick pairings have equal integrals by permutation symmetry, so $3/4! = 1/8$ and

$$A_4(\delta) = \frac{1}{8}\,\mathrm{AC}\!\int_{\mathbb{C}^4} \prod_{i=1}^{4} \mathrm{d}^2 w_i\, e^{-\sum_i |w_i|^2}\, |w_{12}|^{-15/4+2\delta} |w_{34}|^{-15/4+2\delta}\, \mathcal{F}(w_1, w_2, w_3, w_4), \tag{E.17}$$

with $w_{ij} = w_i - w_j$. Connectedness alone does not make Eq. (E.17) an ordinary ultraviolet-convergent integral.

Why the finite Gaussian two-point term is needed. Locality of the coupling counterterm gives

$$g_0(w) = \mu^\delta \Big[g\, f(w) + \frac{b_0}{2\delta} g^3 f(w)^3 + O(g^5) \Big]. \tag{E.18}$$

The power f^3 follows from the three perturbing insertions in the local coupling logarithm. It cannot be replaced by f inside a finite lower-order integral. Define the second-order integral with two Gaussian weights,

$$A_{a,b}(\delta) = \frac{1}{2}\,\mathrm{AC}\!\int_{\mathbb{C}^2} \mathrm{d}^2 w_1\, \mathrm{d}^2 w_2\, e^{-a|w_1|^2 - b|w_2|^2}\, |w_{12}|^{-15/4+2\delta} (H_{12} - S_{12}). \tag{E.19}$$

Inserting Eq. (E.18) into the second-order term gives two equivalent placements of the cubic counterterm, whose total fourth-order contribution is $b_0 A_{3,1}(\delta)/\delta$. Hence $z_{4,1} = \mathrm{Res}_{\delta=0}\, A_4 + b_0 B_{31}$ with $B_{31} = A_{3,1}(0)$, as in Sec. 3. There is one ultraviolet external-operator factor: the localized source produces collisions at the finite energy insertion and is suppressed near infinity, so there is no extra factor of two from the second energy.

Evaluation of B_{31}. Set $w_1 = \Lambda z$ and $w_2 = \Lambda$ in Eq. (E.19), with Λ the overall complex scale. The integral over Λ gives

$$A_{a,b}(\delta) = \frac{\pi}{2}\Gamma(\delta)\,\mathrm{AC}\!\int_{\mathbb{C}} \mathrm{d}^2 z\, (a|z|^2 + b)^{-\delta} F_\delta(z), \qquad F_\delta(z) = |1-z|^{-4+2\delta}\Big(\frac{|1+z|^2}{4|z|} - 1\Big). \tag{E.20}$$

By Appendix E.4, $\mathrm{AC}\!\int F_\delta(z)\, \mathrm{d}^2 z = I_2(\delta) = \frac{\pi}{4}\delta + O(\delta^2)$, so the vanishing two-point residue leaves a finite contribution when it is multiplied by the scale factor $\Gamma(\delta)$. For smooth radial test functions h with the required decay, and by continuation for the logarithmic weights below, the circular finite part about $z = 1$ obeys the distributional identity

$$\mathrm{FP}\!\int_{\mathbb{C}} \mathrm{d}^2 z\, F_0(z)\, h(|z|) = \frac{\pi}{2}\int_0^\infty \frac{|r-1|}{(1+r)^3} h(r)\, \mathrm{d}r - \frac{\pi}{4} h(1). \tag{E.21}$$

For $r \neq 1$ the first term follows from the angular integral. The second is the local term from the circular exclusion about $z = 1$. Its coefficient is also fixed by $I_2(0) = 0$, since the bulk radial integral for $h = 1$ is $\pi/4$. It must be retained when the source weight is differentiated. Expanding Eq. (E.20) at $\delta = 0$ gives

$$A_{a,b}(0) = \frac{\pi^2}{8} - \frac{\pi}{2}\,\mathrm{FP}\!\int_{\mathbb{C}} \mathrm{d}^2 z\, F_0(z) \log(a|z|^2 + b). \tag{E.22}$$

At $(a,b)=(3,1)$, inversion of the part $r>1$ and an integration by parts reduce the remaining finite part to an ordinary integral:

$$B_{31}=\frac{\pi^2}{8}-\frac{\pi^2}{2}\int_0^1\frac{\mathrm{d}r}{(1+r)^2}\Big(\frac{1}{1+3r^2}-\frac{r^2}{r^2+3}\Big),$$
$$\int_0^1\frac{\mathrm{d}r}{(1+r)^2}\Big(\frac{1}{1+3r^2}-\frac{r^2}{r^2+3}\Big)=\frac{\pi\sqrt{3}}{48}+\frac{3\log 3}{16},\tag{E.23}$$

which gives $B_{31}=\frac{\pi^2}{96}(12-\pi\sqrt{3}-9\log 3)$, the value quoted in Sec. 3. It is the combination $\mathrm{Res}_{\delta=0}\,A_4+b_0B_{31}$, not this source-dependent finite number alone, that determines the exponent.

E.6 The regulated conformal map and the elliptic energy integral

Keeping the source factor through the coordinate change. Put the four spins at the reference positions $(0,z,1,\infty)$ and the energies at (x,y). The Möbius map

$$M(t)=\Lambda\frac{t-x}{t-y},\qquad (w_1,w_2,w_3,w_4)=\big(M(0),M(z),M(1),M(\infty)\big),\tag{E.24}$$

returns the energies to $(0,\infty)$; Λ is again an overall complex scale. Its complex Jacobian determinant is

$$\det\frac{\partial(w_1,w_2,w_3,w_4)}{\partial(\Lambda,x,y,z)}=-\frac{\Lambda^3(x-y)^2}{y^2(y-z)^2(y-1)^2}.\tag{E.25}$$

Hats denote correlators with the spins at these reference positions. In particular $\widehat{H}_4=\langle\epsilon(x)\epsilon(y)\sigma(0)\sigma(z)\sigma(1)\sigma(\infty)\rangle$, and $\widehat{H}_{12},\widehat{H}_{34}$ each include both energy insertions; their energy two-point function is not divided out. The transformed connected kernel is

$$\mathcal{G}(x,y;z)=\frac{\widehat{H}_4(x,y;z)}{|x-y|^2}-\frac{\widehat{S}_4(z)}{|x-y|^4}-\widehat{H}_{12}(x,y)\widehat{H}_{34}(x,y)+\frac{|z|^{-1/4}}{|x-y|^4},\tag{E.26}$$

where, from Eqs. (E.10) and (E.11),

$$\widehat{H}_{12}(x,y)=|z|^{-1/4}\Big|\frac{1}{(x-y)^2}+\frac{z^2}{4xy(x-z)(y-z)}\Big|,\qquad \widehat{H}_{34}(x,y)=\Big|\frac{1}{(x-y)^2}+\frac{1}{4(x-1)(y-1)}\Big|.\tag{E.27}$$

Define the exact Gaussian factors

$$R(x,y,z)=\Big|\frac{x}{y}\Big|^2+\Big|\frac{z-x}{z-y}\Big|^2+\Big|\frac{1-x}{1-y}\Big|^2+1,$$
$$\Upsilon(y,z)=|y(y-z)(y-1)|,\qquad W(x,y,z)=\frac{|x-y|^2}{\Upsilon(y,z)R(x,y,z)}.\tag{E.28}$$

Combining the conformal weights, the Wick kernels and the Jacobian gives the full density, apart from the factor $1/8$,

$$|\Lambda|^{-2+4\delta}e^{-|\Lambda|^2R}\,|z|^{-15/4+2\delta}\Big(\frac{|x-y|^2}{\Upsilon}\Big)^{2\delta}\mathcal{G}.\tag{E.29}$$

The scale integral $\int_{\mathbb{C}}|\Lambda|^{-2+4\delta}e^{-R|\Lambda|^2}\,\mathrm{d}^2\Lambda=\pi\Gamma(2\delta)R^{-2\delta}$ therefore yields

$$A_4(\delta)=\frac{\pi}{8}\Gamma(2\delta)J(\delta),\qquad J(\delta)=\mathrm{AC}\!\int_{\mathbb{C}}\mathrm{d}^2z\,|z|^{-15/4+2\delta}\int_{\mathbb{C}^2}\mathrm{d}^2x\,\mathrm{d}^2y\,W(x,y,z)^{2\delta}\,\mathcal{G}(x,y;z).\tag{E.30}$$

This is an equality before the endpoint $\delta\to 0$ is taken, and hence an equality of the prescribed meromorphic continuations. The source factor $W^{2\delta}$ (the W of Sec. 3) is responsible for the matching contact of Appendix E.7.

An explicit algebraic form of the lifted Ising kernel. The energy positions are uniformized by the elliptic curve E_z, the double cover of the t-plane branched at the zeros of $f_z(t) = t(t-z)(t-1)$ and at infinity, with Abel coordinate $\mathrm{d}u = \mathrm{d}t/\sqrt{f_z(t)}$. The complete elliptic integrals use the parameter convention

$$K(z) = \int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{1 - z\sin^2\theta}}, \qquad E(z) = \int_0^{\pi/2} \sqrt{1 - z\sin^2\theta}\,\mathrm{d}\theta, \tag{E.31}$$

continued consistently away from the real unit interval. The periods of E_z are $4K(z)$ and $4\mathrm{i}K(1-z)$; the torus area $\mathcal{A}(z)$, the circular mean parameter $q(z)$, the Ising block weights $w_\pm, w_0$ and the constants $c_\pm, c_0$ are those defined after Eq. (3.5). In particular $\widehat{S}_4 = w_+ + w_-$; changing the sign of $\sqrt{1-z}$ interchanges the two block labels and leaves the sum invariant.

The passage from the exact Ising correlators to the elliptic kernel is an algebraic identity. Set $s_z = (1+z)/3$ and

$$\Sigma(x,y;z) = -\frac{x+y-2s_z}{2} + \frac{f_z(x)+f_z(y)}{2(x-y)^2}, \qquad D^2(x,y;z) = \frac{|f_z(x)f_z(y)|}{|x-y|^4}. \tag{E.32}$$

Substituting Eq. (E.10) and its finite-energy variant (E.11) into Eq. (E.26) gives

$$\mathcal{G} = \frac{w_+|\Sigma + c_+|^2 + w_-|\Sigma + c_-|^2 - w_0|\Sigma + c_0|^2 - (w_+ + w_- - w_0)D^2}{|f_z(x)f_z(y)|}. \tag{E.33}$$

Every quantity in this identity is explicit; no unknown conformal-block coefficient remains.

The uniformization is $x = 4\wp(u) + s_z$, $y = 4\wp(v) + s_z$, with $\wp$ the Weierstrass function of E_z. The addition formulas give

$$\Sigma(x,y;z) = \wp(u-v) + \wp(u+v), \qquad D^2(x,y;z) = |\wp(u-v) - \wp(u+v)|^2; \tag{E.34}$$

for example $f_z(x) = 16\wp'(u)^2$ and $\wp(u-v)-\wp(u+v) = \wp'(u)\wp'(v)/(\wp(u)-\wp(v))^2$. The denominator of Eq. (E.33) is cancelled by $\mathrm{d}^2x\,\mathrm{d}^2y = |f_z(x)f_z(y)|\,\mathrm{d}^2u\,\mathrm{d}^2v$. Each energy plane has two preimages on the curve, so its integral is one half of the integral over the cover, and the two energies give an overall factor $1/4$.

Circular torus integration. In the circular torus prescription

$$\mathrm{FP}\!\int_{E_z} \wp(u)\,\mathrm{d}^2u = -\mathcal{A}\,q. \tag{E.35}$$

One way to obtain Eq. (E.35) is from the periodic completion $\zeta_{\mathrm{W}}(u) - qu - (\pi/\mathcal{A})\bar{u}$ of the Weierstrass zeta function ζ_{W}. For the half-period $\omega_1 = 2K(z)$, with quasi-period $\eta_1 = \zeta_{\mathrm{W}}(\omega_1)$, its periodicity gives $q = \eta_1/\omega_1 - \pi\bar{\omega}_1/(\omega_1\mathcal{A})$, and the elliptic-integral expression for η_1/ω_1 gives the formula for q after Eq. (3.5). Integrating the u derivative of the completion over a torus with a circular hole gives Eq. (E.35), because the spin-two principal part has zero circular boundary average.

Write $\wp_\mp = \wp(u \mp v)$. Subtracting the squared kernels before integrating gives

$$|\wp_- + \wp_+ + c|^2 - |\wp_- - \wp_+|^2 = 4\,\mathrm{Re}(\wp_-\bar{\wp}_+) + 2\,\mathrm{Re}\big[c(\bar{\wp}_- + \bar{\wp}_+)\big] + |c|^2. \tag{E.36}$$

The map $(u,v) \mapsto (u-v, u+v)$ is a finite covering of E_z^2 and preserves the normalized Haar measure. With the circular finite parts understood,

$$\mathrm{FP}\!\int_{E_z^2} \wp_-\bar{\wp}_+\,\mathrm{d}^2u\,\mathrm{d}^2v = \mathcal{A}^2|q|^2, \qquad \mathrm{FP}\!\int_{E_z^2} \wp_-\,\mathrm{d}^2u\,\mathrm{d}^2v = \mathrm{FP}\!\int_{E_z^2} \wp_+\,\mathrm{d}^2u\,\mathrm{d}^2v = -\mathcal{A}^2 q, \tag{E.37}$$

and therefore

$$\frac{1}{4}\,\mathrm{FP}\!\int_{E_z^2} \mathrm{d}^2u\,\mathrm{d}^2v\,\big(|\wp_- + \wp_+ + c|^2 - |\wp_- - \wp_+|^2\big) = \frac{\mathcal{A}^2}{4}|c - 2q|^2. \tag{E.38}$$

Applying Eq. (E.38) to Eq. (E.33), with $c = c_\pm, c_0$, gives

$$\mathcal{T}(z) = \frac{\mathcal{A}(z)^2}{4}\Big[w_+|c_+ - 2q|^2 + w_-|c_- - 2q|^2 - w_0|c_0 - 2q|^2\Big], \tag{E.39}$$

and $\mathcal{I} = \mathrm{FP}\int \mathrm{d}^2 z\,|z|^{-15/4}\mathcal{T}(z)$ is the reduced period integral of Eq. (3.5). This operation defines the flat torus finite part. It is not yet the limit of the source-weighted integral $J(\delta)$ of Eq. (E.30); the comparison follows.

E.7 The finite contact

The four same-branch contacts. Fix a nondegenerate modulus z. A single energy approaching a branch point is locally integrable, and so are energies approaching distinct branch points. Coincident energies away from a branch point leave a spin-two term with a radial leading regulator and zero scalar contact. The remaining case has both energies approaching the same branch point. In Abel coordinates u, v that vanish at that point, including the chart at infinity, the exact source factor (E.28) has the leading form

$$W = \frac{|u-v|^2|u+v|^2|v|^2}{4(|u|^4+3|v|^4)}\big[1 + O(|u|+|v|)\big]. \tag{E.40}$$

With $\alpha_1 = u - v$ and $\alpha_2 = u + v$, the leading density, including the cover and linear-coordinate factors, is

$$\frac{\widehat{S}_4 - w_0}{4}\,\mathrm{Re}\frac{1}{\alpha_1^2\bar{\alpha}_2^2}\,\mathrm{d}^2\alpha_1\,\mathrm{d}^2\alpha_2. \tag{E.41}$$

Now put $\alpha_1 = \alpha_2\zeta$ and $\alpha_2 = re^{\mathrm{i}\theta}$. Then

$$W = r^2|\zeta|^2\frac{|1-\zeta|^2}{|1+\zeta|^4+3|1-\zeta|^4} + o(r^2). \tag{E.42}$$

The radial integral up to an auxiliary radius ρ is $\rho^{4\delta}/(4\delta)$. Its zeroth-order angular coefficient is zero, but the first derivative of $W^{2\delta}$ survives and produces, per branch point, the contact $A_{\mathrm{br}}[\widehat{S}_4 - w_0]$ with

$$A_{\mathrm{br}} = \frac{\pi}{4}\,\mathrm{FP}\int_{\mathbb{C}} \mathrm{d}^2\zeta\,\mathrm{Re}(\zeta^{-2})\log\frac{|1-\zeta|^2}{|1+\zeta|^4+3|1-\zeta|^4}. \tag{E.43}$$

The term $2\log|\zeta|$ of $\log W$ has zero angular contribution. The auxiliary radius ρ also drops out, because it multiplies a vanishing zeroth-order residue. There are four branch points (0, z, 1 and ∞); the sheet factors are already included, so there is no further multiplicity.

The contact and the lower Gaussian integral. The contact is related exactly to the integral $A_{a,b}$ of Eq. (E.19). Let $B_{M1} = A_{M,1}(0)$ (so that B_{31} is the case $M = 3$), and let $A_{\mathrm{br}}(M)$ be Eq. (E.43) with 3 replaced by $M \geq 0$. The distributional change of variables $\tilde{w} = (1+\zeta)/(1-\zeta)$ gives

$$\mathrm{FP}\int_{\mathbb{C}} \mathrm{d}^2\zeta\,\mathrm{Re}(\zeta^{-2})\,h\Big(\Big|\frac{1+\zeta}{1-\zeta}\Big|^2\Big) = 8\,\mathrm{FP}\int_{\mathbb{C}} \mathrm{d}^2 z\,F_0(z)\,h(|z|). \tag{E.44}$$

Indeed, the transformed bulk kernel is $4\,\mathrm{Re}[(\tilde{w}-1)^{-2}(\bar{\tilde{w}}+1)^{-2}]$. Its angular integral, in the variable $\varrho = |\tilde{w}|^2$, is $4\pi|\varrho-1|/(1+\varrho)^3$, with local term $-2\pi h(1)$; this is eight times Eq. (E.21). Differentiating $A_{\mathrm{br}}(M)$ and $4B_{M1}$ in M, and using Eq. (E.44) and inversion, gives equal derivatives. The endpoint values $A_{\mathrm{br}}(0) = \pi^2/4$ and $B_{01} = \pi^2/8$ (from Eq. (E.22)) fix the constant, $A_{\mathrm{br}}(M) = 4B_{M1} - \pi^2/4$, which at $M = 3$ is the relation $A_{\mathrm{br}} = 4B_{31} - \pi^2/4$ of Sec. 3. These $M = 0$ values are the prescribed continued limits; no ordinary infrared convergence is asserted for the unweighted limit.

The fixed-modulus comparison. For $z \notin \{0, 1, \infty\}$ the comparison is therefore

$$\lim_{\delta\to 0} \mathrm{AC} \int \mathrm{d}^2x\, \mathrm{d}^2y\, W^{2\delta}\mathcal{G} = \mathcal{T}(z) + 4A_{\mathrm{br}}\big[\widehat{S}_4(z) - w_0(z)\big], \tag{E.45}$$

with $\mathcal{T}(z)$ the torus integrand of Appendix E.6. The source-dependent first derivative of the regulator is necessary even though the corresponding pointwise zeroth-order angular residue vanishes.

The external-primary cancellation. Extending Eq. (E.45) to the modulus boundaries uses a stronger fact than the integrated equality $I_2(0) = 0$. For fixed noncoincident a, b, c, d (energies at 0 and ∞), the exact simultaneous OPE gives

$$\begin{aligned} H_4(\lambda a, \lambda b, c, d) &= \lambda^{-5/4}\frac{|a-b|^{3/4}|c-d|^{3/4}}{4|ab|} + \lambda^{-1/4}H_2(a,b)H_2(c,d) + O(\lambda^{3/4}), \\ S_4(\lambda a, \lambda b, c, d) &= \lambda^{-1/4}S_2(a,b)S_2(c,d) + O(\lambda^{3/4}). \end{aligned} \tag{E.46}$$

Consequently

$$\mathcal{F}(\lambda a, \lambda b, c, d) = \lambda^{-5/4}\frac{|a-b|^{3/4}|c-d|^{3/4}}{4|ab|} + O(\lambda^{3/4}) : \tag{E.47}$$

the entire $\lambda^{-1/4}$ energy-primary term cancels in the connected logarithm. Multiplying by the internally contracted generalized-free pair changes the leading degree to $-5 + 2\delta$ in four real coordinates, which produces $\mathrm{d}\lambda\, \lambda^{-2+2\delta}$. Its Mellin denominator is $-1+2\delta$, which does not vanish at the endpoint. The absent term would have given $\mathrm{d}\lambda\, \lambda^{-1+2\delta}$, a logarithm. The cancellation persists for nonzero δ, since the same generalized-free kernel multiplies all terms of $\mathcal{F}$. If the shrinking pair is not internally contracted in the fixed pairing, the remaining generalized-free fields raise the output dimension beyond this threshold.

E.8 Modulus boundaries, subtractions and the qualification

Thresholds at the modulus boundaries. In the fixed pairing the contracted degeneration is $z \to 0$; the pair degenerations at $z \to 1$ and $z \to \infty$ lack this internal generalized-free contraction. At $r = |z| \to 0$, let Δ be the exchanged Ising dimension. Including the measures of all energy coordinates that shrink with the pair gives the scalar thresholds

shrinking cluster	radial measure	logarithmic threshold
two spins, neither energy	$\mathrm{d}r\, r^{\Delta-3+2\delta}$	$\Delta = 2$
two spins and one energy	$\mathrm{d}r\, r^{\Delta-2+2\delta}$	$\Delta = 1$
two spins and both energies	$\mathrm{d}r\, r^{\Delta-3+2\delta}$	$\Delta = 2$

The disconnected identity terms cancel in the connected expression; an identity-channel power term with denominator $-1 + 2\delta$ is retained and analytically continued. The energy primary of the middle row cancels by Eq. (E.47). At dimension two, the even Ising identity and energy families contain descendants of nonzero spin but no scalar. The remaining thermal powers are restored by Eq. (E.6); their regular denominators must not be dropped.

Spin residues and coupling logarithms. The regulator also has to preserve the vanishing spin residues. In the joint corner $x = z\xi$, $y = z\eta$ its leading term is

$$W \longrightarrow \frac{|\xi-\eta|^2}{|\eta(\eta-1)|\big(|\xi/\eta|^2 + |(1-\xi)/(1-\eta)|^2 + 2\big)}. \tag{E.48}$$

This is independent of $\arg z$, so a leading term of nonzero spin keeps zero angular residue for nonzero δ. The corners with zero and with one energy have the same property. The only proper

scalar coupling logarithm left is a cluster of three insertions of $\sigma\chi$, which is already represented by the counterterm (E.18). Two distinct triples among four perturbing fields overlap, so they cannot both be separate proper nested or disjoint members of one forest. Thermal pairs inside a triple are included in the continued lower-order coupling projection.

The global comparison and its unproved step. The global comparison uses local OPE and Taylor subtractions in each nested sector, in the distribution-extension framework of Ref. [29]: subtract the nonintegrable powers and spin terms, restore the powers by their Mellin continuation and the spin terms by their circular finite parts, and then interchange the regulator limit with the subtracted remainder integral. This last step requires an integrable dominating bound, uniform near $\delta = 0$, including the degenerating tangent shapes at intersections of sectors. The study gives the channel classification above and a general subtraction argument for this step. An audit stage that is part of the study itself (same authorship; not the review of Sec. 1.2) checked the exact algebra, the local contacts and representative simultaneous limits, but did not reconstruct every sectorwise subtraction or prove the uniform remainder bound. This is the one unproved step of Sec. 3.6, and the identification of Eq. (3.9) with the physical coefficient is a formal perturbative result in the critically tuned analytic minimal-subtraction prescription, conditional on it. The study adds that the qualification is not evidence for a missing contact, and that neither a fixed-modulus check nor a high-precision evaluation of $\mathcal{I}$ can replace the bound: they establish different parts of the method.

E.9 Assembly

Under the global comparison just described, Eq. (E.45) is integrated over z with the weight $|z|^{-15/4}$. The term $4A_{\mathrm{br}}[\widehat{S}_4 - w_0]$ then gives $4A_{\mathrm{br}}I_\beta$, with the coupling integral of Eq. (E.14), $I_\beta = \mathrm{AC}\int \mathrm{d}^2z\,|z|^{-15/4}[\widehat{S}_4(z) - |z|^{-1/4}] = -b_0/\pi$. With Eq. (E.30), in which $\Gamma(2\delta) = 1/(2\delta) + O(1)$, and the contact relation of Appendix E.7,

$$\begin{aligned} J(0) &= \mathcal{I} + 4A_{\mathrm{br}}I_\beta = \mathcal{I} - \frac{16b_0B_{31}}{\pi} + \pi b_0, \\ \operatorname*{Res}_{\delta=0} A_4 &= \frac{\pi}{16}J(0) = \frac{\pi}{16}\mathcal{I} - b_0B_{31} + \frac{\pi^2 b_0}{16}, \end{aligned} \tag{E.49}$$

and $z_{4,1} = \mathrm{Res}_{\delta=0}\,A_4 + b_0B_{31}$ is Eq. (3.7). The finite source constant B_{31} cancels between the bare residue and its source counterterm. The remaining $-\pi^2/(4b_0)$ in c_2 is the finite matching contribution; its size is comparable to that of the period term (Sec. 3). The evaluation that follows concerns $\mathcal{I}$ alone; it supplies an explicit value once this renormalization comparison has been specified.

E.10 Complex moments and their reduction to interval periods

From here to Appendix E.14 the target is $\mathcal{I}$ in $c_2 = -\pi\mathcal{I}/(4b_0^2) - \pi^2/(4b_0)$, with $b_0 = 9\pi^2/(8R_\Gamma^2)$ and $R_\Gamma = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$. The continuation and finite-part prescriptions remain those of Eq. (3.5). In particular, an integral with a negative endpoint exponent denotes its meromorphic continuation, not an ordinary improper integral.

Interval periods and complex moments. With $K(t)$ of parameter t, put

$$\begin{aligned} \mathbf{k}(t) &= \begin{pmatrix} K(t)^2 \\ K(t)K(1-t) \\ K(1-t)^2 \end{pmatrix}, \qquad \mathsf{J} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \\ \mathbf{P}(a,b) &= \mathrm{AC}\int_0^1 t^{a-1}(1-t)^{b-1}\mathbf{k}(t)\,\mathrm{d}t, \qquad \mathbf{p}(a,b) = \frac{4}{\pi^2}\mathbf{P}(a,b). \end{aligned} \tag{E.50}$$

The vector $\mathbf{k}$ reverses under $t \mapsto 1-t$; in particular the third component of $\mathbf{p}(a,b)$ is the first component of $\mathbf{p}(b,a)$. Write

$$\mathbf{p}(a,b) = \big(\mathcal{U}(a,b),\, \mathcal{V}(a,b),\, \mathcal{U}(b,a)\big)^{\mathsf{T}}. \tag{E.51}$$

The complex area moment is

$$\mathcal{M}(a,b) = \mathrm{AC}\int_{\mathbb{C}} \mathrm{d}^2 t\, |t|^{2a-2}|1-t|^{2b-2}\mathcal{A}(t)^2, \tag{E.52}$$

and $\mathcal{B}(a,b)$ is Eq. (E.1). The single-valued period pairing uses the elliptic connection formulas [34] and the bilinear contour construction of Ref. [33]. In the present normalization it reads

$$\mathcal{M}(a,b) = 64\,\mathbf{P}(a,b)^{\mathsf{T}}\mathsf{L}(b)\,\mathbf{P}(2-a-b,b), \qquad \mathsf{L}(b) = \begin{pmatrix} 0 & 0 & \sin\pi b \\ 0 & 2\sin\pi b & -2\cos\pi b \\ \sin\pi b & -2\cos\pi b & -\sin\pi b \end{pmatrix}. \tag{E.53}$$

Contour conventions. On $0<t<1$ both $K(t)$ and $K(1-t)$ are real and positive. They are continued from this interval to the slit plane, with the upper and lower boundary values used separately. For real $t>1$ they obey

$$K(t \pm \mathrm{i}0) = t^{-1/2}\big[K(1/t) \pm \mathrm{i}K(1-1/t)\big], \qquad K(1-t) = t^{-1/2}K(1-1/t), \tag{E.54}$$

so that

$$\mathbf{k}(t \pm \mathrm{i}0) = t^{-1}\mathsf{C}_{\pm}\mathbf{k}(1/t), \qquad \mathsf{C}_{\pm} = \begin{pmatrix} 1 & \pm 2\mathrm{i} & -1 \\ 0 & 1 & \pm\mathrm{i} \\ 0 & 0 & 1 \end{pmatrix}. \tag{E.55}$$

These are the symmetric-square connection matrices of the elliptic equation. Moreover $\mathcal{A}(t)^2 = 64\,\mathbf{k}(t)^{\mathsf{T}}\mathsf{J}\,\overline{\mathbf{k}(t)}$. To obtain Eq. (E.53), take a primitive of the holomorphic one-form $t^{a-1}(1-t)^{b-1}\mathbf{k}(t)\,\mathrm{d}t$ in this single-valued pairing and apply Stokes' theorem on the cut plane. The two banks of the cut $[1,\infty)$ give

$$\frac{e^{\mathrm{i}\pi b}\mathsf{J}\mathsf{C}_- - e^{-\mathrm{i}\pi b}\mathsf{J}\mathsf{C}_+}{2\mathrm{i}} = \mathsf{L}(b). \tag{E.56}$$

The substitution $t = 1/\tau$ on that cut changes the power to $\tau^{1-a-b}(1-\tau)^{b-1}$ and so gives the second interval period $\mathbf{P}(2-a-b,b)$. This accounts for the parameter $2-a-b$ and for every entry of $\mathsf{L}(b)$. The contour identity is first applied with convergent endpoint regulators and then continued meromorphically; small-circle terms at nonintegrable powers are continued with the same prescription, not discarded. This is the standard single-valued period-pairing step of the calculation, not a separate numerical integration of the four-point correlator.

E.11 Landen substitution and the six complex moments

The quadratic substitution and the area Wronskian are

$$z = \frac{4w}{(1+w)^2}, \quad t = w^2, \qquad \mathcal{A}\Big(\frac{4w}{(1+w)^2}\Big) = \frac{|1+w|^2}{2}\mathcal{A}(w^2), \qquad |\partial_t\mathcal{A}|^2 = \frac{1}{4}\partial_t\partial_{\bar t}(\mathcal{A}^2) + \frac{2\pi^2}{|t(1-t)|^2}. \tag{E.57}$$

They are inserted in the three-term integrand of Eq. (3.5) as follows. The derivative of the area and Legendre's relation give

$$q(z) = \frac{z(1-z)}{2}\frac{\partial_z\mathcal{A}(z)}{\mathcal{A}(z)} + \frac{1-2z}{12}. \tag{E.58}$$

For $t = w^2$, abbreviate (in this subsection only) $A = \mathcal{A}(t)$ and $A_t = \partial_t A$. Using $\sqrt{1-z} = (1-w)/(1+w)$ on a consistent sheet, the three factors of the integrand become

$$\begin{aligned}\mathcal{A}(z)(c_+ - 2q) &= -t(1-w)(1+\bar w)A_t,\\ \mathcal{A}(z)(c_- - 2q) &= -(1-w)(1+\bar w)(A/2 + tA_t),\\ \mathcal{A}(z)(c_0 - 2q) &= \frac{t(1+\bar w)}{1+w}\big[A/2 - (1-t)A_t\big].\end{aligned} \tag{E.59}$$

The exchange of the two square-root sheets interchanges the first two and leaves their sum unchanged. The Jacobian $|\,\mathrm{d}z/\,\mathrm{d}w|^2 = 16|1-w|^2/|1+w|^6$ and the multiplicities of the degree-two maps $w \mapsto z$ (w and $1/w$ have the same image) and $w \mapsto t$ give

$$\mathcal{I} = \frac{1}{256}\,\mathrm{AC}\!\int_{\mathbb{C}} \mathrm{d}^2 t \left\{ \frac{|1-t|^{7/2}}{|t|}|A_t|^2 + \frac{|1-t|^{7/2}}{|t|^{5/2}}|A/2 + tA_t|^2 - \frac{|1-t|^2}{|t|}|A/2 - (1-t)A_t|^2 \right\}. \tag{E.60}$$

Under inversion $\mathcal{A}(1/t) = |t|\mathcal{A}(t)$, so the first two integrals are equal. Put

$$\Omega_1 = |t|^{-1}|1-t|^{7/2}, \qquad \Omega_0 = |t|^{-1}|1-t|^2. \tag{E.61}$$

The two remaining integrals are evaluated by integration by parts. For the first, the Wronskian identity gives

$$\mathrm{AC}\!\int \Omega_1 |A_t|^2\,\mathrm{d}^2 t = \mathrm{AC}\!\int \Big[\frac{1}{4}(\partial_t\partial_{\bar t}\Omega_1)A^2 + \frac{2\pi^2\Omega_1}{|t(1-t)|^2}\Big]\,\mathrm{d}^2 t, \qquad \frac{1}{4}\partial_t\partial_{\bar t}\Omega_1 = \frac{|t|^{-3}|1-t|^{3/2}}{64}|2+5t|^2, \tag{E.62}$$

with $|2+5t|^2 = 14 + 35|t|^2 - 10|1-t|^2$. In the third integral the cross term is also integrated by parts. The resulting coefficient of A^2 is

$$\frac{\Omega_0}{4} + \frac{1}{4}\Big\{\partial_t\big[\Omega_0(1-t)\big] + \partial_{\bar t}\big[\Omega_0(1-\bar t)\big]\Big\} + \frac{1}{4}\partial_t\partial_{\bar t}\big[\Omega_0|1-t|^2\big] = \frac{|t|^{-3}|1-t|^2}{16}|1+t|^2, \tag{E.63}$$

with $|1+t|^2 = 2 + 2|t|^2 - |1-t|^2$, and its Wronskian term is $2\pi^2\mathcal{B}(-\frac{1}{2},2) = 0$, a zero of the reciprocal Gamma function $1/\Gamma(1-b)$ in Eq. (E.1). These integrations by parts are justified by first replacing the coefficient of $|A_t|^2$ by an auxiliary weight $|t|^{2a}|1-t|^{2b}$ with $\mathrm{Re}\,a > 0$, $\mathrm{Re}\,b > 0$ and $\mathrm{Re}(a+b) < 1$, where the boundary terms vanish, and then continuing to the stated exponents. They therefore preserve the finite-part boundary prescription. The result, written so as to keep every endpoint contribution, is

$$\begin{aligned}\mathcal{I} &= \frac{\mathcal{D}_1}{128} - \frac{\mathcal{D}_0}{1024},\\ \mathcal{D}_1 &= -\frac{5}{32}\mathcal{M}(-\tfrac12,\tfrac{11}{4}) + \frac{35}{64}\mathcal{M}(\tfrac12,\tfrac74) + \frac{7}{32}\mathcal{M}(-\tfrac12,\tfrac74) + 2\pi^2\mathcal{B}(-\tfrac12,\tfrac74),\\ \mathcal{D}_0 &= \frac{\mathcal{M}(\frac12,2) + \mathcal{M}(-\frac12,2)}{2} - \frac14\mathcal{M}(-\tfrac12,3).\end{aligned} \tag{E.64}$$

The derivative reduction thus gives six complex area moments and one elementary complex beta function, which evaluates exactly to $\mathcal{B}(-\frac12,\frac74) = -32b_0/\pi$.

The vanishing of $\mathcal{D}_0$. Behind $\mathcal{D}_0 = 0$ is an endpoint identity. The symmetric-square elliptic equation gives

$$\mathbf{P}(-\tfrac12,2) + \mathbf{P}(\tfrac12,2) = (-4,0,0)^{\mathsf{T}}, \tag{E.65}$$

an identity of continued periods that includes the Mellin boundary term at $t = 0$. It has an explicit differential certificate. For any component y of $\mathbf{k}(t)$ define

$$\mathcal{C}[y] = -8t^{3/2}(1-t)^2 y'' + 4\sqrt{t}(1-t)(5t-3)y' + \frac{2(1-t)(3t-1)}{\sqrt{t}}y. \tag{E.66}$$

The symmetric-square differential equation, $y''' + 3\rho_1 y'' + (\rho_1' + 2\rho_1^2 + 4\rho_0)y' + (2\rho_0' + 4\rho_1\rho_0)y = 0$ with $\rho_1 = (1-2t)/[t(1-t)]$ and $\rho_0 = -1/[4t(1-t)]$, gives, by direct substitution,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{C}[y] = t^{-3/2}(1-t^2)\,y. \tag{E.67}$$

At $t=0$ the continued constant boundary term is zero. At $t=1$ the three boundary values of $\mathcal{C}[y]$ are $-4, 0, 0$, from $K(t) \sim \frac{1}{2}\log[16/(1-t)]$ and $K(1-t) \sim \pi/2$. This proves Eq. (E.65) with its finite endpoint term. To see how it enters, let $\mathbf{u} = \mathbf{P}(-\frac{1}{2}, 2)$ and $\mathbf{v} = \mathbf{P}(\frac{1}{2}, 2)$. Since $\mathsf{L}(2)(\mathbf{u}+\mathbf{v}) = 0$, $\mathsf{L}(3) = -\mathsf{L}(2)$ and $\mathbf{P}(-\frac{1}{2}, 3) = \mathbf{u} - \mathbf{v}$ (by Eq. (E.70) below), the pairing gives

$$\mathcal{M}(-\tfrac{1}{2}, 3) = 4\mathcal{M}(\tfrac{1}{2}, 2) = 4\mathcal{M}(-\tfrac{1}{2}, 2), \qquad \mathcal{D}_0 = 0. \tag{E.68}$$

Reduction to four interval periods. Abbreviate

$$\mathbf{X} = \mathbf{P}(-\tfrac{1}{2}, \tfrac{7}{4}), \quad \mathbf{Y} = \mathbf{P}(\tfrac{1}{2}, \tfrac{7}{4}), \quad \mathbf{Z} = \mathbf{P}(-\tfrac{1}{4}, \tfrac{7}{4}), \quad \mathbf{W} = \mathbf{P}(\tfrac{3}{4}, \tfrac{7}{4}). \tag{E.69}$$

The contiguous identity

$$\mathbf{P}(a, b+1) = \mathbf{P}(a, b) - \mathbf{P}(a+1, b) \tag{E.70}$$

and $\mathsf{L}(\frac{11}{4}) = -\mathsf{L}(\frac{7}{4})$ turn the part of $\mathcal{D}_1$ without the beta function into

$$10(\mathbf{X}-\mathbf{Y})^{\mathsf{T}}\mathsf{L}(\tfrac{7}{4})(\mathbf{Z}-\mathbf{W}) + 35\,\mathbf{Y}^{\mathsf{T}}\mathsf{L}(\tfrac{7}{4})\mathbf{Z} + 14\,\mathbf{X}^{\mathsf{T}}\mathsf{L}(\tfrac{7}{4})\mathbf{W} = (2\mathbf{X}+5\mathbf{Y})^{\mathsf{T}}\mathsf{L}(\tfrac{7}{4})(5\mathbf{Z}+2\mathbf{W}). \tag{E.71}$$

Consequently

$$\mathcal{I} = \frac{1}{128}(2\mathbf{X}+5\mathbf{Y})^{\mathsf{T}}\mathsf{L}(\tfrac{7}{4})(5\mathbf{Z}+2\mathbf{W}) - \frac{\pi b_0}{2}. \tag{E.72}$$

The constant here is the complex-beta term of $\mathcal{D}_1$. It is separate from the finite regulator-matching term $-\pi^2/(4b_0)$ of Eq. (3.7).

E.12 Reduction to symmetric elliptic moments

Use normalized period vectors and set

$$\mathbf{a} = 2\mathbf{p}(-\tfrac{1}{2}, \tfrac{7}{4}) + 5\mathbf{p}(\tfrac{1}{2}, \tfrac{7}{4}), \quad \mathbf{b} = 5\mathbf{p}(-\tfrac{1}{4}, \tfrac{7}{4}) + 2\mathbf{p}(\tfrac{3}{4}, \tfrac{7}{4}), \quad \mathsf{H} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 2 & -1 \end{pmatrix}, \quad \mathsf{L}(\tfrac{7}{4}) = -\frac{\mathsf{H}}{\sqrt{2}}. \tag{E.73}$$

Equations (3.7) and (E.72) give

$$c_2 = -\frac{R_\Gamma^2}{9} + \frac{\pi R_\Gamma^4}{10368\sqrt{2}}\,\mathbf{a}^{\mathsf{T}}\mathsf{H}\,\mathbf{b}. \tag{E.74}$$

Contour reflection. In the same boundary convention, contour reflection of these particular interval periods gives

$$\mathbf{p}(-\tfrac{1}{2}, \tfrac{7}{4}) = \mathsf{D}\,\mathbf{p}(\tfrac{3}{4}, \tfrac{7}{4}), \qquad \mathbf{p}(\tfrac{1}{2}, \tfrac{7}{4}) = \mathsf{D}\,\mathbf{p}(-\tfrac{1}{4}, \tfrac{7}{4}), \qquad \mathsf{D} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -2 & -1 \\ -1 & -1 & 0 \\ 0 & -2 & -1 \end{pmatrix}. \tag{E.75}$$

These relations use the same connection matrices and continued endpoint contours as Eq. (E.53); they are not identities obtained by naive substitution in divergent real integrals. They can be derived without assuming D. For real $a, b > 0$ with $a' = 2-a-b > 0$, integrate $t^{a-1}(1-t)^{b-1}\mathbf{k}(t)\,\mathrm{d}t$

along the real axis, closed in the upper half-plane; the small and large arcs vanish in this domain of convergence. On the negative interval set $t = -\tau/(1-\tau)$; its connection matrix is

$$\mathsf{B}_- = \begin{pmatrix} 1 & 0 & 0 \\ -\mathrm{i} & 1 & 0 \\ -1 & -2\mathrm{i} & 1 \end{pmatrix}. \tag{E.76}$$

The three real intervals give the vector identity

$$\mathbf{P}(a,b) = e^{\mathrm{i}\pi a}\mathsf{B}_-\mathbf{P}(a,a') + e^{-\mathrm{i}\pi b}\mathsf{C}_+\mathbf{P}(a',b), \tag{E.77}$$

which continues to the parameters of Eq. (E.75). Let $\mathsf{A}_c = e^{\mathrm{i}\pi a}\mathsf{B}_-$ and $\mathsf{E}_c = e^{-\mathrm{i}\pi b}\mathsf{C}_+$. The left side and the two interval vectors are real, so eliminating $\mathbf{P}(a,a')$ between the imaginary and real parts gives

$$\mathbf{P}(a,b) = [\operatorname{Re}\mathsf{E}_c - \operatorname{Re}\mathsf{A}_c(\operatorname{Im}\mathsf{A}_c)^{-1}\operatorname{Im}\mathsf{E}_c]\mathbf{P}(a',b). \tag{E.78}$$

For $a = \pm\frac{1}{2}$ and $b = \frac{7}{4}$, $e^{\mathrm{i}\pi a} = \pm\mathrm{i}$ and $e^{-\mathrm{i}\pi b} = (1+\mathrm{i})/\sqrt{2}$, and the bracket evaluates exactly to D for either sign. This proves Eq. (E.75), and hence $\mathbf{a} = \mathsf{D}\,\mathbf{b}$.

Symmetric moments. Define

$$U(a) = \mathcal{U}(a,a), \qquad V(a) = \mathcal{V}(a,a), \qquad m_U = \tfrac{5}{2}U(-\tfrac{1}{4}) - 4U(\tfrac{3}{4}), \qquad m_V = \tfrac{5}{2}V(-\tfrac{1}{4}) - 4V(\tfrac{3}{4}). \tag{E.79}$$

For $a = -\frac{1}{4}$ the weight in $\mathbf{b}$ is $[t(1-t)]^{a-1}\varpi(t)$, where

$$\varpi(t) = (5+2t)(1-t)^2, \quad \frac{\varpi(t)+\varpi(1-t)}{2} = \frac{5}{2} - 4t(1-t), \quad \varpi(t) - \varpi(1-t) = (1-2t)\{5+2t(1-t)\}. \tag{E.80}$$

For the components $[\mathbf{b}]_i$ of $\mathbf{b}$ it follows directly that $([\mathbf{b}]_1 + [\mathbf{b}]_3)/2 = m_U$ and $[\mathbf{b}]_2 = m_V$.

The antisymmetric moment. The remaining antisymmetric moment has an exact Gamma evaluation:

$$O(a) := \mathcal{U}(a,a+1) - \mathcal{U}(a+1,a) = -\frac{2^{1-2a}}{\sqrt{\pi}}\Big[\frac{\Gamma(a)}{\Gamma(a+\frac{1}{2})}\Big]^3. \tag{E.81}$$

The following derivation also fixes its sign. Set $\hat{K}(t) = 2K(t)/\pi$ and $x = 4t(1-t)$ on $0 < t < \frac{1}{2}$, and introduce the Meijer function

$$G_0(x) = G^{3,0}_{3,3}\left(x \,\middle|\, \begin{matrix} \frac{1}{2},\frac{1}{2},\frac{1}{2} \\ 0,0,0 \end{matrix}\right), \qquad \int_0^1 x^{s-1}G_0(x)\,\mathrm{d}x = \frac{\Gamma(s)^3}{\Gamma(s+\frac{1}{2})^3}. \tag{E.82}$$

The elliptic symmetric-square equation $[\vartheta^3 - x(\vartheta+\frac{1}{2})^3]Y = 0$, with $\vartheta = x\partial_x$, together with its endpoint normalization, gives

$$\hat{K}(1-t)^2 - \hat{K}(t)^2 = \frac{2}{\sqrt{\pi}}G_0(4t(1-t)). \tag{E.83}$$

The normalization can be checked directly, without an unspecified constant. The triple pole at $s = 0$ of the Mellin transform (E.82) gives

$$\frac{2}{\sqrt{\pi}}G_0(x) = \frac{1}{\pi^2}\log^2\frac{64}{x} - 1 + O(x\log^2 x), \tag{E.84}$$

which agrees with $\hat{K}(1-t)^2 - \hat{K}(t)^2$ because $t = x/4 + O(x^2)$, $\hat{K}(t) = 1 + O(t)$ and $\hat{K}(1-t) = \pi^{-1}\log(16/t) + O(t\log t)$. The coefficients of the squared logarithm, the logarithm and the constant fix the three local solutions of the third-order equation. Folding the odd moment about $\frac{1}{2}$ gives

$$O(a) = \int_0^{1/2}[t(1-t)]^{a-1}(1-2t)\{\hat{K}(t)^2 - \hat{K}(1-t)^2\}\,\mathrm{d}t = -\frac{2\cdot 4^{-a}}{\sqrt{\pi}}\int_0^1 x^{a-1}G_0(x)\,\mathrm{d}x, \tag{E.85}$$

which proves Eq. (E.81) where the integrals converge, and then by meromorphic continuation. With the decomposition of ϖ,

$$d_O := [\mathbf{b}]_1 - [\mathbf{b}]_3 = 5O(-\tfrac{1}{4}) + 2O(\tfrac{3}{4}) = \frac{576\sqrt{2}}{\sqrt{\pi}R_\Gamma^3}, \qquad \mathbf{b} = \begin{pmatrix} m_U + d_O/2 \\ m_V \\ m_U - d_O/2 \end{pmatrix}. \tag{E.86}$$

Direct matrix multiplication, with no further integral identity, gives

$$\frac{(\mathsf{D}\mathbf{b})^{\mathsf{T}}\mathsf{H}\,\mathbf{b}}{\sqrt{2}} = \frac{d_O^2}{4} - (m_U + m_V)(m_U + 3m_V), \tag{E.87}$$

and therefore

$$c_2 = \frac{16}{R_\Gamma^2} - \frac{R_\Gamma^2}{9} - \frac{\pi R_\Gamma^4}{10368}(m_U + m_V)(m_U + 3m_V). \tag{E.88}$$

The term $16/R_\Gamma^2$ comes from the evaluated antisymmetric period. Its origin differs from that of both constants of the residue and complex-beta reductions.

E.13 Ordinary hypergeometric evaluation of the moments

The generalized hypergeometric series is [34]

$${}_pF_q\left(\begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix}; x\right) = \sum_{n=0}^{\infty} \frac{(a_1)_n \cdots (a_p)_n}{(b_1)_n \cdots (b_q)_n} \frac{x^n}{n!}, \qquad (a)_n = \frac{\Gamma(a+n)}{\Gamma(a)}, \tag{E.89}$$

the convention of $\mathsf{P}, \mathsf{Q}, \mathsf{T}$ in Sec. 3.5. Define

$$\Phi_A(a) = {}_4F_3\left(\begin{matrix} \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, a \\ 1, 1, a + \frac{1}{2} \end{matrix}; 1\right), \qquad \Phi_D(a) = {}_4F_3\left(\begin{matrix} \frac{1}{2}, a, a, a \\ a + \frac{1}{2}, a + \frac{1}{2}, a + \frac{1}{2} \end{matrix}; 1\right). \tag{E.90}$$

Both series have parametric excess one at every nonresonant parameter used below. The symmetric moments are

$$U(a) = 2^{1-2a}\left[\sqrt{\pi}\frac{\Gamma(a)}{\Gamma(a+\frac{1}{2})}\Phi_A(a) + \frac{1}{\sqrt{\pi}}\Big(\frac{\Gamma(a)}{\Gamma(a+\frac{1}{2})}\Big)^3\Phi_D(a)\right], \tag{E.91}$$

$$V(a) = 2^{1-2a}\left[\sqrt{\pi}\frac{\Gamma(a-\frac{1}{2})}{\Gamma(a)}\Phi_A(1-a) + \frac{\tan \pi a}{\sqrt{\pi}}\Big(\frac{\Gamma(a)}{\Gamma(a+\frac{1}{2})}\Big)^3\Phi_D(a)\right]. \tag{E.92}$$

These are meromorphic identities in a. At $a = -\frac{1}{4}$ and $a = \frac{3}{4}$ no limiting cancellation of Gamma poles is needed in the displayed summands.

The square moment. The quadratic transformation and Clausen's identity give

$$\hat{K}(t)^2 = {}_3F_2\left(\begin{matrix} \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \\ 1, 1 \end{matrix}; x\right), \qquad x = 4t(1-t), \quad 0 < t < \tfrac{1}{2}. \tag{E.93}$$

Combine this with Eq. (E.83) and fold the interval. Since $[t(1-t)]^{a-1}\,\mathrm{d}t = 4^{-a}x^{a-1}(1-x)^{-1/2}\,\mathrm{d}x$,

$$U(a) = 2\cdot 4^{-a}\left[\int_0^1 x^{a-1}(1-x)^{-1/2}\,{}_3F_2\left(\begin{matrix} \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \\ 1, 1 \end{matrix}; x\right)\,\mathrm{d}x + \frac{1}{\sqrt{\pi}}\int_0^1 x^{a-1}(1-x)^{-1/2}G_0(x)\,\mathrm{d}x\right]. \tag{E.94}$$

The first integral is the Euler beta integral for $\Phi_A(a)$. Expanding $(1-x)^{-1/2}$ and applying the Mellin transform (E.82) to each term gives $\Phi_D(a)$ for the second. This proves Eq. (E.91).

The mixed moment. Its Mellin–Barnes representation follows from the inverse Mellin formula, on $0 < x < 1$,

$$\hat{K}(t_x)\hat{K}(1-t_x) = \frac{1}{2\pi\mathrm{i}\sqrt{\pi}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{\Gamma(s)^2\Gamma(\frac{1}{2}-s)}{\Gamma(s+\frac{1}{2})^2\Gamma(1-s)} x^{-s}\,\mathrm{d}s, \qquad t_x = \frac{1-\sqrt{1-x}}{2}, \quad 0 < c < \tfrac{1}{2}. \tag{E.95}$$

One way to verify it: its Mellin kernel $m(s)$ satisfies $m(s+1)/m(s) = s^3/(s+\frac{1}{2})^3$, so it solves the same symmetric-square equation. The double pole at zero has principal part

$$\frac{\Gamma(s)^2\Gamma(\frac{1}{2}-s)}{\sqrt{\pi}\,\Gamma(s+\frac{1}{2})^2\Gamma(1-s)} = \frac{1}{\pi}\Big[\frac{1}{s^2} + \frac{6\log 2}{s} + O(1)\Big]. \tag{E.96}$$

The resulting local behaviour is $\pi^{-1}\log(64/x)$ with no squared logarithm, exactly that of the left side, and these Frobenius data determine the identity. Folding the interval in $V(a)$ and inserting Eq. (E.95) adds the Euler beta factor $B(a-s,\frac{1}{2}) = \sqrt{\pi}\,\Gamma(a-s)/\Gamma(a+\frac{1}{2}-s)$. Consequently

$$V(a) = 2\cdot 4^{-a}\frac{1}{2\pi\mathrm{i}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{\Gamma(s)^2\Gamma(\frac{1}{2}-s)\Gamma(a-s)}{\Gamma(s+\frac{1}{2})^2\Gamma(1-s)\Gamma(a+\frac{1}{2}-s)}\,\mathrm{d}s, \qquad 0 < c < \operatorname{Re} a < \tfrac{1}{2}. \tag{E.97}$$

Close the contour to the right, with its clockwise orientation. The two pole families, $s = \frac{1}{2}+n$ and $s = a+n$, give respectively

$$\sqrt{\pi}\frac{\Gamma(a-\frac{1}{2})}{\Gamma(a)}\Phi_A(1-a), \qquad \frac{\Gamma(a)^2\Gamma(\frac{1}{2}-a)}{\sqrt{\pi}\,\Gamma(a+\frac{1}{2})^2\Gamma(1-a)}\Phi_D(a). \tag{E.98}$$

The residues of $\Gamma(\frac{1}{2}-s)$ and $\Gamma(a-s)$ have the opposite sign to ordinary left-pole residues; the clockwise closure cancels that sign. Gamma reflection reduces the second prefactor to $(\tan\pi a/\sqrt{\pi})[\Gamma(a)/\Gamma(a+\frac{1}{2})]^3$. Restoring $2\cdot 4^{-a}$ proves Eq. (E.92) and allows its continuation to the required values of a. Both pole families are part of the derivation.

The six-function form. At the two values of a in Eq. (E.79) exactly six ordinary ${}_4F_3(1)$ values occur. They enter only through

$$X = 5\Phi_A(-\tfrac{1}{4}) + 2\Phi_A(\tfrac{3}{4}), \qquad Y = 5\Phi_A(\tfrac{5}{4}) - 6\Phi_A(\tfrac{1}{4}), \qquad Z = 5\Phi_D(-\tfrac{1}{4}) + 2\Phi_D(\tfrac{3}{4}). \tag{E.99}$$

Gamma recursion in Eqs. (E.91) and (E.92) gives

$$m_U = -\frac{4\sqrt{2\pi}}{R_\Gamma}X - \frac{64\sqrt{2}}{\sqrt{\pi}R_\Gamma^3}Z, \qquad m_V = \frac{\sqrt{2\pi}R_\Gamma}{3}Y + \frac{64\sqrt{2}}{\sqrt{\pi}R_\Gamma^3}Z. \tag{E.100}$$

In particular Z cancels in $m_U + m_V$, but it contributes to $m_U + 3m_V$. Equivalently,

$$c_2 = \frac{16}{R_\Gamma^2} - \frac{R_\Gamma^2}{9} - \frac{\pi^2}{15552}(12X - R_\Gamma^2 Y)\Big[R_\Gamma^2(4X - R_\Gamma^2 Y) - \frac{128Z}{\pi}\Big]. \tag{E.101}$$

This six-function expression is an intermediate identity. The next step eliminates three independent combinations by exact telescoping, including their boundary values.

E.14 Telescoping proof of the three-function closed form

Let $\mathsf{P},\mathsf{Q},\mathsf{T}$ be the convergent series of Sec. 3.5. The closed form (3.9) follows from Eq. (E.101) by the three exact reductions

$$X=\frac{9}{2}\mathsf{P}-\frac{2R_\Gamma}{\pi^{3/2}},\qquad Y=\frac{45}{64}\mathsf{Q}-\frac{24}{\pi^{3/2}R_\Gamma},\qquad Z=\frac{72}{25}\mathsf{T}+\frac{R_\Gamma^3}{8\sqrt{\pi}},\tag{E.102}$$

which are proved here with every telescoping endpoint retained. Take the base summands

$$a_n=\frac{(\frac12)_n^3(\frac34)_n}{(n!)^3(\frac54)_n},\qquad b_n=\frac{(\frac12)_n^3(\frac14)_n}{(n!)^3(\frac34)_n},\qquad d_n=\frac{(\frac12)_n(\frac34)_n^3}{n!\,(\frac54)_n^3}.\tag{E.103}$$

The summands of X,Y,Z in Eq. (E.99) are respectively

$$-\frac{12n+7}{4n-1}a_n,\qquad \frac{3(12n-1)}{4n+3}b_n,\qquad \Big[2-5\Big(\frac{4n+1}{4n-1}\Big)^3\Big]d_n.\tag{E.104}$$

Let p_n,q_n,t_n denote the summands of $\mathsf{P},\mathsf{Q},\mathsf{T}$, and define the certificates

$$\begin{aligned}H_X(n)&=-\frac{4n^2(2n-3)(4n+1)^2}{(2n-1)^2(4n-1)},\qquad H_Y(n)=-\frac{3n(16n^3+44n^2+34n+15)}{2(n+1)^2},\\ H_Z(n)&=\frac{2n(4n+1)(64n^3+16n^2-20n+43)}{(4n-1)^3}.\end{aligned}\tag{E.105}$$

Direct rational simplification proves

$$\begin{aligned}-\frac{12n+7}{4n-1}a_n-\frac{9}{2}p_n&=H_X(n+1)a_{n+1}-H_X(n)a_n,\\ \frac{3(12n-1)}{4n+3}b_n-\frac{45}{64}q_n&=H_Y(n+1)b_{n+1}-H_Y(n)b_n,\\ \Big[2-5\Big(\frac{4n+1}{4n-1}\Big)^3\Big]d_n-\frac{72}{25}t_n&=H_Z(n+1)d_{n+1}-H_Z(n)d_n.\end{aligned}\tag{E.106}$$

For example, divide the first identity by a_n and use

$$\frac{a_{n+1}}{a_n}=\frac{(n+\frac12)^3(n+\frac34)}{(n+1)^3(n+\frac54)},\qquad \frac{p_n}{a_n}=\frac{(2n+1)(4n+1)}{(2n-1)^2(n+1)}.\tag{E.107}$$

It becomes an identity of rational functions of n, so checking it depends neither on numerical precision nor on summing the series. The other two identities are checked in the same way.

Every $H(0)$ vanishes, but the upper boundary terms do not:

$$\begin{aligned}\lim_{n\to\infty}\frac{(H_X,H_Y,H_Z)}{n^2}&=(-8,-24,8),\\ \lim_{n\to\infty}n^2a_n=\frac{R_\Gamma}{4\pi^{3/2}},\qquad \lim_{n\to\infty}n^2b_n&=\frac{1}{\pi^{3/2}R_\Gamma},\qquad \lim_{n\to\infty}n^2d_n=\frac{R_\Gamma^3}{64\sqrt{\pi}}.\end{aligned}\tag{E.108}$$

Summing Eq. (E.106) from $n=0$ to N and letting $N\to\infty$ gives exactly the three Gamma terms of Eq. (E.102). Substituting Eq. (E.102) into Eq. (E.101), the boundary constants cancel from both factors, and the result is exactly the closed form (3.9).

E.15 Numerical evaluation and the scope of the checks

Default evaluation. The study evaluates Eq. (3.9) at 80 working digits, and checks it against the six-function form (E.101) and against the endpoint series below, with 400 terms in each half-interval series; Sec. 3.5 quotes their agreement. With N_{h} terms in each half-interval series, the absolute difference $|\Delta c_2|$ between the endpoint-series value and the three-function value is

N_{h}	$\lvert\Delta c_2\rvert$	N_{h}	$\lvert\Delta c_2\rvert$	N_{h}	$\lvert\Delta c_2\rvert$	N_{h}	$\lvert\Delta c_2\rvert$
24	5.2×10^{-7}	64	2.8×10^{-19}	144	1.5×10^{-43}	320	2.5×10^{-80}
40	6.0×10^{-12}	96	5.1×10^{-29}	224	9.4×10^{-68}	400	2.5×10^{-80}

and it reaches the 80-digit arithmetic floor at 320 terms. This is numerical consistency, not an error bound. The values are in `code/c2_closed_d2/outputs/raw/three_4f3_80dps.json` and `code/c2_closed_d2/outputs/raw/convergence.json`.

Endpoint-series evaluation. An evaluation independent of the ${}_4F_3$ identities uses the Frobenius expansions

$$\begin{aligned}\hat K(t)&=\sum_{n\ge0}\phi_n t^n,\qquad \phi_n=\Big[\frac{(\frac12)_n}{n!}\Big]^2,\\ G(t)&=4\sum_{n\ge0}\phi_n(\mathsf h_{2n}-\mathsf h_n)t^n,\qquad K(1-t)=\tfrac12\{\log(16/t)\hat K(t)-G(t)\},\end{aligned}\tag{E.109}$$

with the harmonic numbers $\mathsf h_0=0$, $\mathsf h_n=\sum_{k=1}^n 1/k$. Each period is split at $t=\frac12$, its second half is reflected, and $(1-t)^{b-1}\hat K^2$, $(1-t)^{b-1}\hat K G$ and $(1-t)^{b-1}G^2$ are expanded in power series of t, with coefficient arrays $\alpha_n,\beta_n,\gamma_n$. For each coefficient the only Mellin integrals required are, with $p=a+n$ and $\ell_{32}=\log 32$,

$$\begin{aligned}j_0(p)&=\mathrm{AC}\int_0^{1/2}t^{p-1}\,\mathrm dt=\frac{2^{-p}}{p},\qquad j_1(p)=\mathrm{AC}\int_0^{1/2}t^{p-1}\log(16/t)\,\mathrm dt=2^{-p}\Big(\frac{\ell_{32}}{p}+\frac{1}{p^2}\Big),\\ j_2(p)&=\mathrm{AC}\int_0^{1/2}t^{p-1}\log^2(16/t)\,\mathrm dt=2^{-p}\Big(\frac{\ell_{32}^2}{p}+\frac{2\ell_{32}}{p^2}+\frac{2}{p^3}\Big).\end{aligned}\tag{E.110}$$

These hold first at $\operatorname{Re}p>0$ and continue analytically to the nonzero negative p needed here. The three unnormalized half-period sums (for $K(t)^2$, $K(t)K(1-t)$ and $K(1-t)^2$) are

$$\sum_n\frac{\pi^2}{4}\alpha_n j_0,\qquad \sum_n\frac{\pi}{4}(\alpha_n j_1-\beta_n j_0),\qquad \sum_n\frac14(\alpha_n j_2-2\beta_n j_1+\gamma_n j_0),\tag{E.111}$$

and adding the reflected half gives the full vector. Because the expansion is restricted to $0\le t\le\frac12$, this is an exponentially convergent numerical check of the continued periods that uses neither the six- nor the three-function formula. Agreement is a reproducible numerical consistency check, not an interval enclosure of the final digits.

A separate endpoint-subtracted quadrature. A second implementation checks the same moments by quadrature, with the elliptic integrals evaluated through Carlson's symmetric integral,

$$K(t)=R_F(0,1-t,1),\qquad K(1-t)=R_F(0,t,1),\qquad R_F(x,y,z)=\frac12\int_0^\infty\frac{\mathrm dv}{\sqrt{(v+x)(v+y)(v+z)}},\tag{E.112}$$

for positive arguments, continued to the endpoint limits shown. This representation does not use the Frobenius coefficients. To make the negative moment at $a=-\frac14$ integrable, each symmetric

moment is folded onto $0 < t < \frac{1}{2}$ and its leading endpoint terms are subtracted. With $u = -\log t$ and $\ell = \log 16 + u$, define

$$\begin{aligned} R_U(u;a) &= (1-e^{-u})^{a-1}\{K(e^{-u})^2 + K(1-e^{-u})^2\} - \frac{\pi^2+\ell^2}{4}, \\ R_V(u;a) &= (1-e^{-u})^{a-1}K(e^{-u})K(1-e^{-u}) - \frac{\pi\ell}{4}. \end{aligned} \tag{E.113}$$

The continued Mellin integrals $j_k(a)$ then give

$$\begin{aligned} U(a) &= \frac{4}{\pi^2}\Big[\int_{\log 2}^{\infty} e^{-au} R_U(u;a)\,\mathrm{d}u + \frac{\pi^2 j_0(a) + j_2(a)}{4}\Big], \\ V(a) &= \frac{8}{\pi^2}\Big[\int_{\log 2}^{\infty} e^{-au} R_V(u;a)\,\mathrm{d}u + \frac{\pi j_1(a)}{4}\Big]. \end{aligned} \tag{E.114}$$

The remainders are $O(e^{-u}u^2)$, so the integrals converge for $\operatorname{Re} a > -1$, including $a = -\frac{1}{4}$ and $a = \frac{3}{4}$. The implementation uses Gauss–Legendre quadrature on the intervals with endpoints $\log 2, 2, 5, 10, 20, 40, 80, 120, 200$, evaluates the integrand at 100 decimal digits, and restores the analytic leading terms exactly as in Eq. (E.114). The finite upper limit $u = 200$ leaves a rapidly decaying tail. Inserted in Eq. (E.88), these moments reproduce the closed form to 1.40×10^{-61} in c_2 (record value 1.4045×10^{-61}; Sec. 3.5). This is an observed difference; the script supplies no rigorous bound on the quadrature error or on the cutoff (record: `code/c2_closed_d2/outputs/raw/independent_c2_80dps.json`).

Exact bounds on the series tails. The three final series also admit direct remainder inequalities. For any of them let v_n be the absolute value of its nth summand, $n \geq 1$. Its positive ratio is the rational function

$$r_n = \frac{v_{n+1}}{v_n} = \frac{\prod_{j=1}^{4}(n+u_j)}{(n+1)\prod_{j=1}^{3}(n+l_j)}, \tag{E.115}$$

with the upper parameters u_j and lower parameters l_j of that series. Set $\phi_c(n) = n/2 + c$ and $E_c(n) = \phi_c(n) - 1 - \phi_c(n+1)r_n$. For P and T take $(c_-, c_+) = (0, 1)$, and for Q take $(1, 2)$. Then $E_{c_-}(n) \leq 0$ and $E_{c_+}(n) \geq 0$ for all integers $n \geq 1$; these signs are verified exactly by replacing n with $m + 1$ and checking that the numerator and denominator of the suitably signed rational expression have nonnegative coefficients. Multiplying by v_n and summing telescopically gives

$$\Big(\frac{N}{2} + c_-\Big)v_N \leq \sum_{n=N}^{\infty} v_n \leq \Big(\frac{N}{2} + c_+\Big)v_N, \tag{E.116}$$

where the upper boundary term vanishes because $v_n = O(n^{-3})$. All terms of P and Q are positive; for T the terms with $n \geq 1$ are negative. These signs decide which tail is added to or subtracted from a partial sum. With $N = 20000$, propagation of these bounds gives the envelope of c_2 quoted in Sec. 3.5. The inequalities for the mathematical series are exact, but the arithmetic that evaluates their endpoints is ordinary arbitrary-precision floating point, so the envelope is an independent check, not a certified rounding interval.

Scope of the checks. The study's exact checks cover the three rational telescoping identities with their nonzero endpoint terms, the complex coordinate Jacobians, the lower-order Gamma reductions and the scheme covariance of c_2 under $\mathcal{E}_R \to e^{\kappa g^2}\mathcal{E}_R$. A separate check of the analytic bridges verifies the Landen coefficient reduction, the cut-jump and reflection matrices, and the differential certificate used for $\mathcal{D}_0 = 0$. The study's numerical checks test the elliptic-cover identity (E.33) at noncoincident points, the Gaussian finite term B_{31}, the contact relation, and representative simultaneous OPE limits; the source-contact checks test the dependence of the finite contact on the smooth infrared cutoff and its cancellation against the local counterterm. None of these checks

evaluates every global collision sector. The discrepancies quoted above are neither statistical error bars nor statements about the physical regulator comparison of Appendix E.8. The checks made by the review of the study are those of Secs. 3.5 and 3.7.

E.16 Further conditions stated by the study

Beyond Secs. 3.6 and 6, the study states three conditions on what it establishes.

- *Endpoint terms are part of the value.* The evaluation requires the finite source contact, the interval-period pairing with its continued endpoint terms, and the nonzero endpoint terms of the telescoping proof; every integral with a negative endpoint exponent is a meromorphic continuation, not an ordinary improper integral. The endpoint values $A_{\mathrm{br}}(0) = \pi^2/4$ and $B_{01} = \pi^2/8$ of the contact relation are prescribed continued limits; no ordinary infrared convergence is asserted for them.
- *The digits are checked, not certified, within the study.* Its long decimal value rests on high-precision evaluations by different routes (the six- and three-function forms, the endpoint series and the Carlson quadrature); it is not presented as a certified rounding theorem. A certified enclosure would combine interval arithmetic with the analytic tail bounds above, or bound both the Carlson quadrature error and its tail beyond $u = 200$. (The integer-interval evaluation to 51 decimal places mentioned in Sec. 3.5 is part of the cylinder base study (Appendix F.14), not of this study.)
- *The local checks do not bear on the global step.* The fixed-modulus contact calculation and the high-precision agreement of the special-function evaluations establish different parts of the method; neither supplies the uniform remainder bound of Appendix E.8.

F c_2 from fourth-order perturbation theory on the cylinder: method

This appendix gives the method of the base study of c_2 with the truncated cylinder Hamiltonian (written by OpenAI GPT-6 astra; Sec. 1.2), edited to the notation of this note.

Section 3.8 summarises the setup and gives the continuum analysis that produced the quoted value $c_2 = -4.0670(24)$, whose error is a linear sum of systematic diagnostics, not a standard deviation. That analysis was done later, with $N \leq 1024$ and with fit and selection rules fixed in code before the final numbers were computed (its error rule was revised once, after the first result had been seen; Sec. 3.8); it evaluates the quantities defined here. The appendix gives the normalization and the exact second-order coefficient (Appendices F.1 to F.3), the fourth-order Rayleigh–Schrödinger formula, its Hilbert space, regulator and counterterms (Appendices F.4 to F.7), the spectral generating functions that evaluate it (Appendices F.8 and F.9), the large-N tails and the regulator limit (Appendix F.10), the checks (Appendices F.11 to F.14), and the explicit formulas behind them (Appendices F.15 to F.19). The calculation is numerical: no step below bounds the remaining spectral or regulator extrapolation. The base study's checks of the closed form (Appendix F.14) certify the value of the closed expression, Eq. (3.9), not its identification with the physical coefficient, which remains conditional on the step of Sec. 3.6.

Notation: δ is that of Table 1 ($d = 2$), and c_2 is the coefficient the base study writes C_2. As in Sec. 3.8, $|\mathcal{E}\rangle$ is the state of the Ising energy field ϵ at the ultraviolet point, $E_{\mathcal{E}}$ its energy, and a subscript $\mathcal{E}$ on a state-dependent quantity refers to it.

F.1 Continuum setup and normalization

The short-range description. The theory is the crossover description of Sec. 3: the diagonal Ising CFT, with central charge $c = \frac{1}{2}$, tensored with a generalized free scalar χ, with

$$\Delta_\sigma = \frac{1}{8}, \qquad \Delta_\epsilon = 1, \qquad \Delta_\chi = \frac{15}{8} - \delta, \qquad C_{\sigma\sigma\epsilon} = \frac{1}{2}, \qquad C_{\epsilon\epsilon\epsilon} = 0, \tag{F.1}$$

and unit-normalized plane two-point functions, $\langle\sigma(x)\sigma(0)\rangle = |x|^{-1/4}$, $\langle\epsilon(x)\epsilon(0)\rangle = |x|^{-2}$ and $\langle\chi(x)\chi(0)\rangle = |x|^{-2\Delta_\chi}$. The higher correlators of χ are Wick-factorized. The perturbation is $\sigma\chi$, of dimension $p = 2 - \delta$. Critical thermal tuning is understood: the relevant energy coupling generated by the perturbation is removed along the critical trajectory. The limit is $\delta \to 0^+$, from the long-range side.

The cylinder Hamiltonian. On the radial cylinder of unit radius, with the unperturbed vacuum energy subtracted,

$$H_0 = D_{\text{Ising}} + D_\chi, \qquad V = \int_0^{2\pi} \mathrm{d}\theta\, \sigma(\theta)\chi(\theta), \qquad V_\epsilon = \int_0^{2\pi} \mathrm{d}\theta\, \epsilon(\theta), \tag{F.2}$$

where D_{Ising} and D_χ are the dilatation operators and V_ϵ is the integrated thermal operator (the base study writes $\mathcal{E}$ for it). The angular integral projects onto equal total spin and contributes a factor 2π at every vertex, so a chain of four vertices carries $(2\pi)^4$. These factors are part of the coupling convention and must not be changed independently of b_0.

Fixed point and coefficient extraction. The beta function is $\beta(g) = -\delta g + b_0 g^3 + O(g^5)$, with the exact b_0 of Sec. 3.2. This normalization agrees with the lower-order crossover calculation [12], and it is checked independently by the magnetic gap (Appendix F.12). The infrared fixed point has $g_*^2 = \delta/b_0 + O(\delta^2)$. Let $|d\rangle = L_{-1}|\mathcal{E}\rangle$ be the left-derivative state at the ultraviolet point. Since the energy field has holomorphic weight $\frac{1}{2}$, its norm is $2 \cdot \frac{1}{2} = 1$; its unperturbed dimension is two and its spin one. Following the energy eigenstates continuously connected to these states, define

$$X = 2(E_\mathcal{E} - E_0) - (E_d - E_0) = 2E_\mathcal{E} - E_d - E_0. \tag{F.3}$$

At a conformal fixed point $E_\mathcal{E} - E_0 = \Delta_\mathcal{E}$ and $E_d - E_0 = \Delta_\mathcal{E} + 1$, hence $X = \Delta_\mathcal{E} - 1$; at the ultraviolet point $X = 0$. In a consistently matched Hamiltonian scheme,

$$X = a_1\delta g^2 + h g^4 + O(g^6, \delta^2 g^2, \delta g^4), \tag{F.4}$$

and substituting g_* gives the relation $c_2 = a_1/b_0 + h_0/b_0^2$ of Sec. 3.8, with h_0 the value of h once the regulator is removed (Eq. (F.5) below). Only the leading fixed-point coupling is needed at this order, because there is no term $\delta^0 g^2$. The split between the two terms depends on the prescription (Sec. 3.2): the Hamiltonian h is not the pole-only coefficient h_{MS} of minimal subtraction, and dropping a_1/b_0 would be wrong in this convention. A shift Δh_0 moves c_2 by $\Delta h_0/b_0^2 \simeq 0.62156\,\Delta h_0$ ($b_0^2 \simeq 1.60885$); this is the conversion by which errors in h_0 are quoted in c_2.

Why the coefficients are computed at $\delta = 0$. Inferring a curvature from dimensions computed at finite δ would magnify an error in a dimension by δ^{-2}. The method instead computes the g^4 coefficient at $\delta = 0$ and the analytic δ derivative of the g^2 coefficient. The remaining fits are numerical fits in auxiliary cutoffs and in the regulator length, never in the physical δ. What the calculation shares with the closed-form route, and what it does not use, is listed in Sec. 3.8.

Notation and the three separate limits.

symbol	meaning
δ	physical distance from the short-range end; the fourth-order sums are evaluated at $\delta = 0$
$\Delta_{\chi,0} = \frac{15}{8}$	endpoint dimension of χ (not an oscillator index)
$r = 1, 2, 3$	order of the positive, polynomial-improved oscillator regulator (Appendix F.6)
$\lambda > 0$	length of that smooth ultraviolet regulator on the unit cylinder
N	cubical coefficient cutoff, $0 \le n, m, \ell < N$; it is not the number of Hilbert-space states
N_{mix}	two-index cutoff of the mixed thermal sums (4096 in the base study)
N_{col}	cutoff of the collapsed sums (200 000 and 800 000 in the base study)

A third auxiliary cutoff, besides N_{mix} and N_{col}, is the two-index cutoff of the leading-tail calculation (Appendix F.18). The auxiliary cutoffs are not interchangeable with N. The numerical limits are taken in the order

$$h_0 = \lim_{\lambda \to 0^+} \Big[\lim_{N \to \infty} h_N(\lambda) \Big], \tag{F.5}$$

with N_{mix}, N_{col} and the tail cutoff removed together with N at fixed λ (in the analysis of Sec. 3.8 they are held at common values for all points and their shifts under a change are propagated); $h_N(\lambda)$ is the assembled fourth-order quantity of Eq. (F.42) with the chain sums truncated at N. None of these limits samples the physical δ.

F.2 Exact second-order spectral continuation

Vacuum and thermal sums. Set $p = 2 - \delta$ and

$$A_n = \frac{(p)_n}{n!}, \qquad A_{-1} = A_{-2} = 0. \tag{F.6}$$

Expanding the exact cylinder correlators and projecting on angles gives

$$E_0^{(2)} = -4\pi^2 \sum_{n=0}^{\infty} \frac{A_n^2}{p + 2n}, \tag{F.7}$$

$$E_{\mathcal{E}}^{(2)} = -\pi^2 \sum_{n=0}^{\infty} \frac{(A_n + A_{n-1})^2}{p - 1 + 2n}. \tag{F.8}$$

The denominators are measured relative to the external state. The sums converge in an ordinary domain, for example $\frac{1}{2} < p < 1$ away from level crossings, and are then continued meromorphically; they are not convergent positive sums at the physical endpoint. With $\tilde{A}_n = (p-1)_n/n!$ (the base study's c_n),

$$A_n + A_{n-1} = \frac{2n + p - 1}{p - 1} \tilde{A}_n, \tag{F.9}$$

so Eq. (F.8) reduces to a linear combination of two Gauss sums,

$$\sum_{n \ge 0} \tilde{A}_n^2 = \frac{\Gamma(3 - 2p)}{\Gamma(2 - p)^2}, \qquad \sum_{n \ge 0} n \tilde{A}_n^2 = (p-1)^2 \frac{\Gamma(2 - 2p)}{\Gamma(2 - p)^2}. \tag{F.10}$$

With $\Gamma(3 - 2p) = (2 - 2p)\Gamma(2 - 2p)$ the required combination vanishes exactly:

$$E_{\mathcal{E}}^{(2)} = 0 \qquad \text{as a meromorphically continued function.} \tag{F.11}$$

The vacuum sum is a well-poised hypergeometric sum. Written as $p^{-1}\, {}_3F_2(p, p, \frac{p}{2}; 1, 1 + \frac{p}{2}; 1)$, it reduces to the standard well-poised summation, followed by Gamma reflection and recursion:

$$-E_0^{(2)} = \pi^2 \frac{\Gamma(p/2)^2\, \Gamma(1 - p)}{\Gamma(1 - p/2)^2\, \Gamma(p)}. \tag{F.12}$$

This fixes the analytic continuation used here; it does not assign a finite part to an endpoint cutoff sum.

The derivative state and the slope of the observable. Define

$$D_n = A_{n+1} + 3A_n + 3A_{n-1} + A_{n-2}. \tag{F.13}$$

The absolute second-order correction of the derivative state is

$$E_d^{(2)} = -\frac{\pi^2}{4(p-1)} \sum_{n\geq 0} \tilde{A}_n D_n. \tag{F.14}$$

For $\Sigma_j = \sum_{n\geq 0} \tilde{A}_n A_{n+j}$ and $\mathcal{G} = \Gamma(2-2p)/\Gamma(2-p)^2$ the continued shifted sums are

$$\frac{\Sigma_1}{\mathcal{G}} = \frac{p(1-p)}{2-p}, \qquad \frac{\Sigma_0}{\mathcal{G}} = 1-p, \qquad \frac{\Sigma_{-1}}{\mathcal{G}} = p-1, \qquad \frac{\Sigma_{-2}}{\mathcal{G}} = \frac{p(p-1)}{2-p}. \tag{F.15}$$

Consequently $\Sigma_1 + 3\Sigma_0 + 3\Sigma_{-1} + \Sigma_{-2} = 0$, and $E_d^{(2)} = 0$ as well. The coefficient of g^2 in X is therefore $-E_0^{(2)}$ at $p = 2-\delta$:

$$a(\delta) = \pi^2 \frac{\Gamma(1-\delta/2)^2\,\Gamma(\delta-1)}{\Gamma(\delta/2)^2\,\Gamma(2-\delta)} = -\frac{\pi^2}{4}\delta - \frac{\pi^2}{2}\delta^2 + O(\delta^3), \tag{F.16}$$

so that

$$a_1 = -\frac{\pi^2}{4}, \qquad \frac{a_1}{b_0} = -1.945279657979090345 63\ldots, \tag{F.17}$$

which is the finite matching term of Eq. (3.7). A complementary check (`code/c2_cylinder_d2/code/hamiltonian/slope_check.py`) differentiates the regulated Pochhammer coefficients, the regulator factors and the resolvent denominators analytically in δ; it does not finite-difference dimensions at several values of δ.

F.3 A finite endpoint-regulator anomaly

Setting $p = 2$ first and multiplying each second-order resolvent term by $e^{-\tau(E_n - E_a)}$, where E_n is the energy of the intermediate state, E_a that of the external state and $\tau > 0$ a Euclidean time cutoff, direct geometric summation gives

$$E_0^{(2)}(\tau) = -\frac{\pi^2}{2\sinh^2\tau}, \qquad E_{\mathcal{E}}^{(2)}(\tau) = -\frac{\pi^2\cosh\tau}{2\sinh^2\tau}, \tag{F.18}$$

hence

$$E_{\mathcal{E}}^{(2)}(\tau) - E_0^{(2)}(\tau) = -\frac{\pi^2}{4\cosh^2(\tau/2)} \xrightarrow{\tau\to 0} -\frac{\pi^2}{4}. \tag{F.19}$$

The analytically continued endpoint gap, however, is $a(0) = 0$. Even a finite difference of cutoff sums can therefore have the wrong finite part. This is the failure of an attempted endpoint prescription, not an error in the closed-form study; it motivates the explicitly matched Hermitian regulator of Appendix F.6. General fourth-order difficulties of Hamiltonian cutoffs, including nonlocal cutoff effects and disconnected cancellations, are discussed in Refs. [35, 36]. An earlier explicit-basis calculation that fixed the counterterms by trial matching to finite spectral gaps gave strongly cutoff-dependent fourth-order values; it is not used.

F.4 Fourth-order Rayleigh–Schrödinger perturbation theory

Reduced resolvents and the normalization subtraction. For an external state $|a\rangle$, work in its conserved-spin sector and use

$$Q_a = 1 - |a\rangle\langle a|, \qquad R_a = \frac{Q_a}{E_a^{(0)} - H_0}, \qquad |w_a\rangle = R_a V|a\rangle, \qquad |z_a\rangle = Q_a V|w_a\rangle. \tag{F.20}$$

The relevant external levels are nondegenerate within their symmetry sectors at the endpoint. Since $V_{aa} = 0$, the second- and fourth-order energy coefficients are

$$E_a^{(2)} = \langle a|VR_aV|a\rangle, \tag{F.21}$$

$$E_a^{(4)} = \langle z_a|R_a|z_a\rangle - E_a^{(2)}\langle w_a|w_a\rangle. \tag{F.22}$$

The first term, $\langle a|VR_aVR_aVR_aV|a\rangle$, is the chain. The second comes from the perturbative normalization of the state and is essential. It must use the same regulator as the chain; assigning unrelated finite parts to the two separately can destroy their cancellation.

Because V is linear in χ, four insertions that begin and end in the χ vacuum pass through the χ occupations $0 \to 1 \to 0 \to 1 \to 0$ or $0 \to 1 \to 2 \to 1 \to 0$. The explicit-basis implementation includes the factors $\sqrt{2}$ for double occupation of a mode; the generating-function implementation (Appendix F.8) includes the same multiplicities through the three Wick pairings. Finite-box tests compare the two (Appendix F.11).

Counterterms enter at the same perturbative order. Write the matched Hamiltonian as

$$H = H_0 + gV + g^2W_2 + g^3W_3 + g^4W_4 + \cdots . \tag{F.23}$$

For any insertion W of order g^2 define its mixed and quadratic fourth-order contributions

$$\begin{aligned} L_a(W) &= \langle w_a|(W - W_{aa})|w_a\rangle + 2\,\mathrm{Re}\,\langle Q_aWa|R_a|z_a\rangle, \\ \mathcal{Q}_a(W) &= \langle Q_aWa|R_a|Q_aWa\rangle, \end{aligned} \tag{F.24}$$

where $|Q_aWa\rangle$ means $Q_aW|a\rangle$ (not an independent state). For every state-dependent quantity Y write $Y_X = 2Y_{\mathcal{E}} - Y_d - Y_0$. For $W_2 = tV_\epsilon + kD_{\mathrm{Ising}}$ the quadratic contribution of the thermal term to X is exactly

$$\mathcal{Q}_X(tV_\epsilon) = \frac{32\pi^2}{3}\,t^2 : \tag{F.25}$$

a free Majorana thermal deformation gives the second-order gap coefficients $8\pi^2$ and $16\pi^2/3$ for $|\mathcal{E}\rangle$ and $|d\rangle$, and $2 \cdot 8\pi^2 - 16\pi^2/3 = 32\pi^2/3$. Quadratic terms involving D_{Ising} vanish because it is diagonal; its *mixed* fourth-order contribution $L_X(D_{\mathrm{Ising}})$ does not. The observable X was chosen so that the direct local fourth-order terms cancel (Appendix F.7); the mixed terms of Eq. (F.24) cannot be omitted on that basis.

F.5 Hilbert space, matrix elements and oscillators

Ising states. The validation implementation uses an orthonormal Majorana basis of Neveu–Schwarz (NS) and Ramond (R) states. In the NS sector the occupied chiral modes are positive half-integers, in the R sector positive integers. Chiral R energies carry the vacuum shift $\frac{1}{16}$, so the full R shift is $\frac{1}{8}$, and the diagonal Ising theory retains NS left and right occupations of equal parity. The vacuum, energy, derivative and spin states are

$$|0\rangle = (\varnothing, \varnothing)_{\mathrm{NS}}, \qquad |\mathcal{E}\rangle = (\tfrac{1}{2}, \tfrac{1}{2})_{\mathrm{NS}}, \qquad |d\rangle = (\tfrac{3}{2}, \tfrac{1}{2})_{\mathrm{NS}}, \qquad |\sigma\rangle = (\varnothing, \varnothing)_{\mathrm{R}}. \tag{F.26}$$

Products of chiral spin-field form factors give the full Ising σ matrix elements, and the angular integration then enforces conservation of total spin. Appendix F.15 gives the product formula and the fermion signs; the representation is also available in Ref. [37].

Generalized-free oscillators. For $\tau > 0$ the unit-cylinder correlator is

$$\langle \chi(\tau,\theta)\chi(0,0)\rangle = \big[2\cosh\tau - 2\cos\theta\big]^{-\Delta_\chi}. \tag{F.27}$$

Expanding its two chiral factors gives modes $m, n \geq 0$ with

$$\omega_{mn} = \Delta_\chi + m + n, \qquad J_{mn} = m - n, \qquad \gamma_{mn} = \sqrt{\frac{(\Delta_\chi)_m (\Delta_\chi)_n}{m!\, n!}}, \tag{F.28}$$

energy ω_{mn}, spin J_{mn} and oscillator coefficient γ_{mn} (the base study's c_{mn}), so that $D_\chi = \sum_{m,n} \omega_{mn} a^\dagger_{mn} a_{mn}$. These modes implement the radial Hamiltonian of the isotropic theory; no identification with a spatially long-range quantum spin chain is assumed.

F.6 The regulator family

At the endpoint set $\Delta_{\chi,0} = \frac{15}{8}$ and define

$$P_r(x) = \sum_{j=0}^{r} \frac{x^j}{j!}, \qquad q = e^{-\lambda} \tag{F.29}$$

(q is unrelated to the circular mean of Eq. (3.5)). The oscillator coefficient is replaced by

$$\tilde\gamma_{mn} = q^{(\Delta_{\chi,0}+m+n)/2} \sqrt{\frac{(\Delta_{\chi,0})_m (\Delta_{\chi,0})_n}{m!\, n!}\, P_r\big(\lambda(m + \tfrac{\Delta_{\chi,0}}{2})\big)\, P_r\big(\lambda(n + \tfrac{\Delta_{\chi,0}}{2})\big)}. \tag{F.30}$$

All factors under the square root are positive. Applying the same real coefficient to the creation and annihilation parts gives a Hermitian interaction at fixed $\lambda > 0$. The un-improved regulator is $r = 0$; the analysis uses $r = 1, 2, 3$. Excluding a common factor $q^{\Delta_{\chi,0}/2}$, the chiral contraction series is

$$H(z) = \sum_{n\geq 0} H_n z^n, \qquad H_n = \frac{(\Delta_{\chi,0})_n}{n!}\, q^n\, P_r\big(\lambda(n + \tfrac{\Delta_{\chi,0}}{2})\big). \tag{F.31}$$

Since $e^{-x} P_r(x) = 1 + O(x^{r+1})$, low-frequency regulator artifacts are suppressed. This improvement does not remove the need for local counterterms. At fixed λ the χ modes are exponentially damped and the remaining Ising collision singularities of the sums are integrable. The cubical spectral cutoff N is therefore a numerical truncation of a regulated theory, not the physical ultraviolet prescription.

F.7 Local counterterms and the improved observable

Thermal and spin-two terms from the short-distance kernel. The Ising operator product expansion contains

$$\sigma(z)\sigma(0) = |z|^{-1/4}\Big[1 + \frac{1}{2}|z|\,\epsilon(0) + \frac{1}{4} z^2 T(0) + \frac{1}{4}\bar z^2 \bar T(0) + \cdots\Big], \tag{F.32}$$

with T, $\bar T$ the stress tensor. The ϵ term gives a thermal power subtraction. The regulator distinguishes cylinder time from angle at short distances, so the spin-two terms generate a finite anisotropic subtraction. With the normalization of Eq. (F.2) the counterterm is

$$W_2 = t_r(\lambda)\, V_\epsilon - \kappa_r\, D_{\mathrm{Ising}}. \tag{F.33}$$

It is derived from the regulated kernel, not by setting the fourth-order answer or by fitting an energy gap to a desired value. For a local coordinate $z = \tau + \mathrm{i}x$ and unit regulator length define

$$\mathfrak{h}_{\alpha,r}(z) = \sum_{j=0}^{r} \frac{(\alpha)_j}{j!} (z+1)^{-\alpha-j}. \tag{F.34}$$

The planar regulated χ kernel is $|\mathfrak{h}_{\Delta_{\chi,0},r}(z)|^2$. Its thermal moment is

$$c_{t,r} = \frac{1}{2}\int_{\tau>0} \mathrm{d}\tau\,\mathrm{d}x\,(\tau^2+x^2)^{3/8}\,|\mathfrak{h}_{\Delta_{\chi,0},r}(\tau+\mathrm{i}x)|^2, \qquad t_r(\lambda) = \frac{c_{t,r}}{\lambda} + \cdots . \tag{F.35}$$

For the un-improved regulator

$$\begin{aligned} c_{t,0} &= {}_3F_2\left(\begin{matrix}-\frac{3}{8},1,1\\ \frac{3}{2},\frac{3}{2}\end{matrix};1\right) = 0.76212879108984270790\ldots, \\ \kappa_0 &= \frac{\pi^2}{4}\,\frac{\Delta_{\chi,0}-\rho_\chi}{\Delta_{\chi,0}-1}, \qquad \rho_\chi = \frac{\Gamma(\Delta_{\chi,0}+\frac{1}{2})}{\Gamma(\frac{3}{2})\Gamma(\Delta_{\chi,0})}, \qquad \kappa_0 = 1.20829489911072968942\ldots. \end{aligned} \tag{F.36}$$

For $r>0$ the difference of the spin-two moments of the improved and un-improved kernels converges and determines $\kappa_r - \kappa_0$. Appendix F.16 gives the compact finite-domain integrals used for these constants, so that they can be recomputed without any saved number.

Removing a leading cylinder artifact. The holomorphic cylinder kernel obeys

$$\big[2\sinh(z/2)\big]^{-\Delta_{\chi,0}} = z^{-\Delta_{\chi,0}} - \frac{\Delta_{\chi,0}}{24} z^{2-\Delta_{\chi,0}} + \cdots . \tag{F.37}$$

Applying the same regulator to this expansion produces a thermal term of order λ. For $r \geq 1$ the subtracted moment converges:

$$c_{\mathrm{cyl},r} = -\frac{\Delta_{\chi,0}}{24}\int_{\tau>0} \mathrm{d}^2z\,|z|^{3/4}\Big[\mathrm{Re}\big(\mathfrak{h}_{\Delta_{\chi,0}-2,r}(z)\,\overline{\mathfrak{h}_{\Delta_{\chi,0},r}(z)}\big) - \mathrm{Re}(z^2)\,|z|^{-2\Delta_{\chi,0}}\Big]. \tag{F.38}$$

The calculation therefore uses

$$\boxed{t_r(\lambda) = \frac{c_{t,r}}{\lambda} + c_{\mathrm{cyl},r}\,\lambda.} \tag{F.39}$$

The second term vanishes in the continuum. It is an explicitly derived regulator improvement, not an adjustable finite fourth-order constant, and it is checked separately by the protected magnetic coefficient (Appendix F.12). The constants are

r	$c_{t,r}$	κ_r	$c_{\mathrm{cyl},r}$
1	1.96478977995214	1.78455951314437	−0.11068091195498
2	3.31217153117521	2.20281564903759	−0.05396644709415
3	4.74456159798846	2.54264875945082	−0.03541592348370

They are ordinary quadrature values, refined as described in Appendix F.16, not interval enclosures.

Why the direct fourth-order local terms cancel in X. After the auxiliary spectral cutoff has been removed at fixed λ, short-distance power counting and the separate Ising and χ parity selection rules restrict the nonvanishing relevant or marginal local counterterms of even order to the identity, the thermal operator and the parity-even Ising spin-two term. Odd-spin total derivatives do not contribute. In particular χ^2 has dimension $\frac{15}{4} > 2$ at the endpoint and is not an additional relevant or marginal subtraction. The direct fourth-order expectations of these operators in X vanish:

operator in W_4	$2(W_4)_{\mathcal{E}\mathcal{E}} - (W_4)_{dd} - (W_4)_{00}$
$\nVdash$	$2-1-1=0$
V_ϵ	0 (each diagonal element vanishes)
D_{Ising}	$2\cdot 1 - 2 - 0 = 0$

(F.40)

The diagonal thermal elements vanish by Ising fusion ($C_{\epsilon\epsilon\epsilon}=0$) and remain zero for the descendant. This cancellation does not remove the mixed effects of W_2, which are included explicitly.

At third order the coupling counterterm $W_3 = \mu_3(\lambda)V$ contributes $2\,\mathrm{Re}\,\langle a|W_3 R_a V|a\rangle = 2\mu_3(\lambda)E_a^{(2)}$, and thus $2\mu_3(\lambda)X_2(\lambda)$ to X, where $X_2(\lambda)$ is the second-order observable at finite λ and $\delta = 0$ (Appendix F.9). The coefficient μ_3 grows logarithmically, whereas $X_2(\lambda)$ tends to zero as a positive power of λ, so this product supplies no unknown finite constant to the limiting h. The analysis uses the convenient choice

$$\mu_3(\lambda) = -b_0 \log\lambda, \qquad \Delta X_4 = -2b_0 \log\lambda\, X_2(\lambda). \tag{F.41}$$

Omitting this term, or shifting the reference scale of the logarithm, is a sensitivity test at finite regulator.

This is an argument about the local short-distance expansion of a specified, smoothly regulated theory, made *after* the numerical spectral limit. It does not license arbitrary finite counterterms at a fixed hard Hilbert-space cutoff, and the numerical calculation does not supply a uniform treatment of every subleading remainder.

The assembled fourth-order quantity. Combining all terms,

$$\begin{aligned} h(\lambda) = X_{4,\mathrm{chain}} + X_{4,\mathrm{norm}} + t_r(\lambda)\, L_X(V_\epsilon) - \kappa_r\, L_X(D_{\mathrm{Ising}}) \\ + \frac{32\pi^2}{3} t_r(\lambda)^2 - 2b_0 \log\lambda\, X_2(\lambda), \end{aligned} \tag{F.42}$$

where every sum on the right must first be converged at fixed λ, and $h_0 = \lim_{\lambda\to 0} h(\lambda)$. Here $X_{4,\mathrm{chain}}$ is the chain term of Eq. (F.22) combined into X (Appendix F.8), $X_{4,\mathrm{norm}}$ its normalization term and $L_X(D_{\mathrm{Ising}})$ the mixed spin-two term (both Appendix F.9), and $L_X(V_\epsilon)$ the mixed thermal term (Appendix F.17). Large terms cancel in Eq. (F.42), and its individual pieces are not physical coefficients; this is one reason the collapsed and mixed sums need their own convergence checks. The continuum analysis of Sec. 3.8 evaluates this assembly at each (r, λ).

F.8 Exact chiral blocks as spectral generating functions

Replacing state enumeration. The explicit Majorana basis serves for validation but grows rapidly. The production implementation sums the level degeneracies analytically with chiral blocks. A Taylor coefficient of a radially ordered four-point block is the sum, over intermediate states at specified levels, of products of spin-field matrix elements. Dividing these coefficients by the corresponding three energy denominators therefore evaluates the same Rayleigh–Schrödinger sum as Eq. (F.22). This is not a return to the elliptic-integral calculation of the closed-form study: the blocks supply discrete spectral weights, and the operations that follow are coefficient extraction, angular projection and Hamiltonian denominators. The underlying Ising correlators are known independently [13].

Vacuum, thermal and derivative blocks. Place four chiral spin insertions at

$$z_1 = 1, \qquad z_2 = u, \qquad z_3 = uv, \qquad z_4 = uvw, \tag{F.43}$$

and define, for $|u|, |v|, |w| < 1$,

$$\begin{gathered} S = \sqrt{1-uv}\sqrt{1-vw}, \qquad T = \sqrt{1-v}\sqrt{1-uvw}, \qquad \mathcal{D} = \prod_{x\in\{u,v,w,uv,vw,uvw\}} (1-x)^{-1/8}, \\ \mathcal{R} = \sqrt{(S+T)/2}, \qquad \mathcal{U} = \sqrt{(1-u)(1-w)} \end{gathered} \tag{F.44}$$

(the base study writes $\mathcal{Q}$ for $\mathcal{U}$). All roots are continued analytically from the origin, with positive values for small positive real arguments. After removal of simple radial monomials the two

middle-fusion channels ($\eta = 0, 1$) of the vacuum and thermal blocks are

$$\begin{aligned} g_0 &= \mathcal{D}\mathcal{R}, & g_1 &= \frac{\mathcal{D}\mathcal{U}}{2\mathcal{R}}, \\ e_0 &= \frac{\mathcal{D}}{4\mathcal{R}}\big[S(1+uw)+T(u+w)\big], & e_1 &= \frac{\mathcal{D}\mathcal{R}}{2\mathcal{U}}\big[S(1+uw)-T(u+w)\big]. \end{aligned} \tag{F.45}$$

For a partition (A, B) of the four position labels set

$$K_{AB} = uvw\Big[1 - \frac{1}{4}\Big(\sum_{i\in B} z_i - \sum_{i\in A} z_i\Big)\Big(\sum_{i\in B} z_i^{-1} - \sum_{i\in A} z_i^{-1}\Big)\Big], \tag{F.46}$$

and let $K_{13} = K_{(1,3),(2,4)}$ and $K_{14} = K_{(1,4),(2,3)}$. The derivative blocks are

$$d_0 = \frac{\mathcal{D}}{4\mathcal{R}}\big[S(1+uw)K_{13} + T(u+w)K_{14}\big], \qquad d_1 = \frac{\mathcal{D}\mathcal{R}}{2\mathcal{U}}\big[S(1+uw)K_{13} - T(u+w)K_{14}\big]. \tag{F.47}$$

The apparent inverse powers in K_{13} and K_{14} cancel against the displayed uvw and the block factors, so the series are analytic at the origin. Their analytic branches are tested at complex arguments against explicit NS and R form factors (Appendix F.11).

Wick pairings, angular projection and denominators. For each fusion label η, the block is multiplied by the chiral χ factor of one of the three Wick pairings, with the series H of Eq. (F.31):

pairing	chiral factor	middle occupation o
adjacent	$H(u)H(w)$	0
crossed	$H(uv)H(vw)$	1
nested	$H(uvw)H(v)$	1

Here $o = 1$ means a middle state with two χ quanta and $o = 0$ one with none. The real Taylor coefficients of the products are written $G_{nm\ell}$, $E_{nm\ell}$ and $D_{nm\ell}$ for the vacuum, thermal and derivative blocks. The angular projection gives squares for the vacuum and thermal states; for the spin-one derivative state it gives the shifted product $D_{n+1,m+1,\ell+1}E_{nm\ell}$. For a given pairing and fusion label the chain energies are

$$\begin{aligned} E^{(4),\mathrm{chain}}_{0,\eta,o} &= -(2\pi)^4 q^{2\Delta_{\chi,0}} {\sum_{n,m,\ell\geq 0}}' \frac{G^2_{nm\ell}}{(2n+2)(2m+\eta+2\Delta_{\chi,0}o)(2\ell+2)}, \\ E^{(4),\mathrm{chain}}_{\mathcal{E},\eta,o} &= -(2\pi)^4 q^{2\Delta_{\chi,0}} {\sum_{n,m,\ell\geq 0}}' \frac{E^2_{nm\ell}}{(2n+1)(2m-\eta+2\Delta_{\chi,0}o)(2\ell+1)}, \\ E^{(4),\mathrm{chain}}_{d,\eta,o} &= -(2\pi)^4 q^{2\Delta_{\chi,0}} {\sum_{n,m,\ell\geq 0}}' \frac{D_{n+1,m+1,\ell+1}E_{nm\ell}}{(2n+1)(2m-\eta+2\Delta_{\chi,0}o)(2\ell+1)}. \end{aligned} \tag{F.48}$$

The prime excludes only a zero denominator, which corresponds to the external-state projector. A negative denominator, such as that of the lower vacuum intermediate state in a thermal channel, must be retained. Summing over both fusion labels and all three pairings and forming $2E_{\mathcal{E}} - E_d - E_0$ gives $X_{4,\mathrm{chain}}$. The common negative sign reflects three resolvents written with positive excitation-energy denominators. At cutoff N the indices run from 0 to $N-1$; because the derivative block is shifted by one, the coefficient array has shape $(6, N+1, N+1, N+1)$, with the six channels ordered $(g_0, g_1, e_0, e_1, d_0, d_1)$.

Coefficient extraction and convolution. The coefficients of the unregulated Ising blocks are extracted by a discrete Cauchy (Fourier) transform on a three-torus. With M angular points per direction and radius $\varrho = e^{-\varsigma/(N+1)}$,

$$[f]_{nm\ell} \simeq \frac{\varrho^{-n-m-\ell}}{M^3} \sum_{j_1,j_2,j_3=0}^{M-1} f\big(\varrho e^{2\pi \mathrm{i} j_1/M}, \varrho e^{2\pi \mathrm{i} j_2/M}, \varrho e^{2\pi \mathrm{i} j_3/M}\big)\, e^{-2\pi \mathrm{i}(j_1 n + j_2 m + j_3 \ell)/M}. \tag{F.49}$$

M is an even FFT length of about the oversampling factor times $N+1$, and ς is the radius cost. Oversampling controls aliasing; decreasing the radius reduces aliasing but amplifies roundoff through $\varrho^{-n-m-\ell}$. The main grids of the base study use oversampling five and radius cost four at $N=256$, and oversampling four and radius cost 4.5 at $N=384$. These are numerical choices, not physical regulators; the extraction check of Sec. 3.8 compares two grids that differ in the radius. The χ factors are then applied by one-dimensional convolutions along lines parallel to the monomials u, w, uv, vw, uvw or v, and prefix sums of the weighted three-index arrays return every cubical partial sum at once. The prefix test of Sec. 3.8, which repeats the large-N extrapolation on the sums truncated at $N'<N$, uses these partial sums. Independent versions of the convolution and of the coefficient grid are kept for validation.

F.9 Collapsed and mixed contributions

One-dimensional two-point contractions. The second-order energies, the norms of the normalization subtraction and the mixed spin-two term reduce to one-index sums. Let

$$p_0(z)=1, \qquad p_{\mathcal{E}}(z)=\frac{1+z}{2}, \qquad p_d(z)=\frac{(1+z)^3}{8}, \tag{F.50}$$

and $a_f(j)=[z^j](1-z)^{-1/8}p_f(z)$ for $f\in\{0,\mathcal{E},d\}$, with coefficients of negative index zero. With the regulated sequence H_j of Eq. (F.31), set

$$B_f(n)=\sum_{j=0}^{n} a_f(j)H_{n-j}, \qquad \hat{B}_f(n)=\sum_{j=0}^{n} j\,a_f(j)H_{n-j} \tag{F.51}$$

(the base study writes W_f for $\hat{B}_f$). The external states are labelled by

state a	left block l	right block l'	shift s_a	α_a
0	0	0	0	2
$\mathcal{E}$	$\mathcal{E}$	$\mathcal{E}$	0	1
d	d	$\mathcal{E}$	1	1

With $F_2=(2\pi)^2 q^{\Delta_{\chi,0}}$ and $\nu_n=2n+\alpha_a$,

$$E_a^{(2)}=-F_2\sum_{n\geq 0}\frac{B_l(n+s_a)B_{l'}(n)}{\nu_n}, \qquad \mathcal{N}_a:=\langle w_a|w_a\rangle=F_2\sum_{n\geq 0}\frac{B_l(n+s_a)B_{l'}(n)}{\nu_n^2},$$
$$L_a(D_{\mathrm{Ising}})=F_2\sum_{n\geq 0}\frac{1}{\nu_n^2}\Big[(\tfrac{1}{8}-E_a^{(0)})B_l(n+s_a)B_{l'}(n)+\hat{B}_l(n+s_a)B_{l'}(n)+B_l(n+s_a)\hat{B}_{l'}(n)\Big]. \tag{F.52}$$

The finite-regulator second-order observable is $X_2(\lambda)=2E_{\mathcal{E}}^{(2)}-E_d^{(2)}-E_0^{(2)}$ evaluated with these sums, and the normalization contribution is

$$X_{4,\mathrm{norm}}=-2E_{\mathcal{E}}^{(2)}\mathcal{N}_{\mathcal{E}}+E_d^{(2)}\mathcal{N}_d+E_0^{(2)}\mathcal{N}_0. \tag{F.53}$$

These sums are evaluated at much larger cutoffs than the three-index chain: $N_{\mathrm{col}}=200\,000$ in the base study's $N=256$ records and 800 000 in its larger-cutoff records, with the small change between them recorded separately.

Two-dimensional thermal insertions. The mixed thermal term $L_X(V_\epsilon)$ has the insertion V_ϵ either between the two V vertices (central placement) or outside them (side placement). The basic chiral generating functions are

$$G_c(u,v)=(1-u)^{-1/2}(1-v)^{-1/2}(1-uv)^{3/8}H(uv),$$
$$G_s(u,v)=(1-u)^{-1/2}(1-v)^{3/8}H(v)(1-uv)^{-1/2}. \tag{F.54}$$

For either placement the vacuum, energy and derivative factors are $G/\sqrt{2}$, $GP_{\mathcal{E}}/(2\sqrt{2})$ and $GP_d/(2\sqrt{2})$; the polynomials $P_{\mathcal{E},b}$ and $P_{d,b}$, the denominators and the shifted derivative projection are in Appendix F.17. The common factor is $(2\pi)^3 q^{\Delta_{\chi,0}}$, and

$$L_X(V_\epsilon) = X_c + 2X_s, \tag{F.55}$$

where the factor two counts the two side placements. All square partial sums are obtained through N_{mix} (4096 in the base study), and the large-level limit is extrapolated separately (Appendix F.10).

F.10 High-state tails and the regulator limit

The independently determined leading tail. At fixed λ the leading remaining Ising singularity on a face of the coefficient box is $(1-u)^{-1/8}$. Its coefficients $a_0(n) = (\frac{1}{8})_n/n!$ (Appendix F.9) behave as $a_0(n) \sim n^{-7/8}/\Gamma(\frac{1}{8})$, so

$$\sum_{n\geq N} \frac{a_0(n)^2}{2n} \sim \frac{2}{7\Gamma(\frac{1}{8})^2} N^{-7/4}; \tag{F.56}$$

the factor $\frac{2}{7}$ follows by integrating $n^{-11/4}/2$. Exact limits of the generating functions on the faces of the coefficient box determine the coefficient of this tail, with the sign convention

$$X_{4,\mathrm{chain}}(\infty) - X_{4,\mathrm{chain}}(N) = A_{\mathrm{face}}(\lambda, r)\, N^{-7/4} + \text{subleading terms}. \tag{F.57}$$

Appendix F.18 gives the two-index construction of A_{face}. It uses the block singularities and the collapsed norms, not the observed N sequence or any value of c_2. This is the analytically derived coefficient of the face tail that the continuum analysis of Sec. 3.8 subtracts.

What is still fitted at fixed regulator. The subleading powers include the one-face corrections $N^{-11/4}, N^{-15/4}, \ldots$ and overlapping-face contributions such as $N^{-7/2}$. The regulated χ kernel has its next singularities at e^{λ}, which generate terms exponentially small in $N\lambda$; on finite windows they can compete with the power terms. A reasonably large λN is therefore desirable, but this is a planning criterion, not a remainder bound (the analysis of Sec. 3.8 flags points with λN below a minimum). Let $h_n(\lambda)$ be the assembled quantity, Eq. (F.42), with the chain truncated to the cube $[0,n)^3$. With the leading tail held fixed, the corrected sequence is fitted on $n_{\min} \leq n \leq N$ as

$$h_n(\lambda) + A_{\mathrm{face}}\, n^{-7/4} = h(\lambda) + \sum_j \xi_j \Big(\frac{n}{N}\Big)^{-p_j} + \text{exponential terms}, \tag{F.58}$$

with the power sets

$$\{2.75, 3.75\}, \quad \{2.75, 3.75, 4.75\}, \quad \{2.75, 3.5, 3.75, 4.75\}, \quad \{2.75, 3.5, 3.75, 4.75, 5.5\} \tag{F.59}$$

and, for exponential order $n_e = 0, 1, 2, 3$, the optional columns

$$e^{-\lambda(n-n_{\min})}\Big(\frac{n}{N}\Big)^{r-1-j}, \qquad e^{-2\lambda(n-n_{\min})}\Big(\frac{n}{N}\Big)^{2r+3/4-j}, \qquad j = 0, \ldots, n_e - 1. \tag{F.60}$$

The base study's selected ensemble uses at least three power terms (the analysis of Sec. 3.8: three or four), $n_e = 1, 2$, and the two lower endpoints $n_{\min} = 240, 288$ at $N = 384$ and $160, 192$ at $N = 256$, that is $n_{\min} = 5N/8$ and $3N/4$. The fixed-regulator estimate $h(\lambda)$ is the median of the selected limits, and half the difference between the largest and the smallest is the *spectral-model half-spread*, a model diagnostic, neither a standard deviation nor a truncation bound. Small residuals alone do not prove the tail model correct. The continuum analysis of Sec. 3.8 uses these power sets, exponential columns and selection, with the windows $n_{\min} = 5N/8$ and $3N/4$ at every N. That

these windows reproduce the base study's relative windows does not by itself show the same tail model valid at smaller λ; the prefix and cross-N checks of Sec. 3.8 test it.

The mixed thermal sums have their own tail fits in N_{mix}, with powers drawn from $\{1, 2, \frac{11}{4}, 3, 4\}$; the chosen mixed limit is the median of the models with at least three powers and $n_{\min} \geq N_{\mathrm{mix}}/2$ (2048 in the base study). Its spread is recorded separately and not identified with the error of the chain.

Removing the smooth regulator. After the spectral limit, the regulator model is $h(\lambda) = h_0 + k_1\lambda^2 + k_2\lambda^2 \log\lambda + k_3\lambda^3$, the preferred model of Sec. 3.8. The improved low-frequency filter and the removal of the leading cylinder thermal artifact motivate this form; higher terms are not rigorously bounded. The base study varied the regulator order, the λ window, the spectral-tail model, the weighting and the last power (λ^4 instead of λ^3), and it tested omission and scale shifts of the term (F.41). No fit has the physical δ as its variable, but the uncertainty remains an extrapolation uncertainty. A four-parameter model fitted to four regulator points is saturated, and a near-zero residual is then no evidence of accuracy; two of the base study's three regulator families had only four points, which is why more regulator points were needed. The base study's own extrapolations ($N \leq 384$), quoted in Sec. 3.8, were superseded by the analysis given there; its archived records were not: eleven of them are points of that analysis.

For a fixed linear extrapolation $\hat{h}_0 = \sum_i w_i \hat{h}(\lambda_i)$ with input diagnostics ε_i and a regulator-model term $\varepsilon_{\mathrm{model}}$, the error in c_2 is propagated as

$$\frac{1}{b_0^2}\Big(\sum_i |w_i|\varepsilon_i + \varepsilon_{\mathrm{model}}\Big). \tag{F.61}$$

This is the form of the linear sum of Sec. 3.8. It would be a bound only if the ε_i and $\varepsilon_{\mathrm{model}}$ were valid bounds, which the half-spreads and model spreads are not. It also shows why the fixed-λ inputs need better precision than the desired final error. As λ decreases, cancellations between large raw terms and the counterterms make roundoff more important.

F.11 Algebraic and finite-sum checks

The validation suite tests different failure modes:

check	largest reported discrepancy
chiral Virasoro Ward identities (252 identities)	4.22×10^{-16}
exact second-order shell weights (vacuum, energy and derivative correlators)	below 5.4×10^{-16} relative
explicit basis against blocks, fourth-order finite sums	below 2×10^{-13}
blocks at complex arguments against finite-level form factors	about 1.16×10^{-10} (scaled)
alternative finite convolution and sequence implementations	below 2×10^{-11}

The suite also checks the four-spin block sums and the fourth-order formula (F.22) against an independent finite-matrix recursion for the perturbed eigenvector. The discrepancy of the complex-argument blocks includes the truncation of the explicit form factors. The entries are precisions of cross-checks on finite problems. None of these identities bounds the omitted continuum tail; they establish that signs, mode occupations, derivative shifts and normalization subtractions agree between independent implementations on the tested finite problems.

F.12 The protected magnetic dimension as a matching check

At the fixed point the magnetic dimension is protected, $\Delta_\sigma = \frac{1}{8} + \delta$ in this convention [12]. With $g_*^2 = \delta/b_0$, the leading renormalized Hamiltonian gap coefficient must therefore satisfy

$$[g^2](E_\sigma - E_0) = b_0. \tag{F.62}$$

The independently derived local counterterms, including the cylinder improvement of Eq. (F.39), give at $r = 3$ and $\lambda = 1/32$ the value 1.268404004672631 quoted in Sec. 3.8, which differs from b_0 by -3.04×10^{-7}. This value was not used to tune the counterterms. It is a lower-order check of the normalization, not a fourth-order error bound.

F.13 What is exact and what is numerical

The second-order meromorphic continuation (Appendix F.2) and the finite-basis identities are exact analytic or algebraic results. The counterterm classification and the cancellation (F.40) remove the unspecified direct local fourth-order constants within the stated smoothly regulated prescription; this is an argument, not a proof. The fourth-order continuum number is a numerical inference, supported by independent regulator families, the analytically derived leading tail and the lower-order normalization checks; its remaining uncertainty is systematic, not statistical. What is not supplied is a rigorous uniform bound on every nested spectral and regulator remainder, or an independent many-digit enclosure of h_0.

F.14 Checks of the closed form made in the base study

The base study also consolidates a set of formula, elliptic and local checks of the closed-form study (Secs. 3.5 and 3.6), likewise written by GPT-6 astra. Their code is not in the release (Appendix H.5); the value of the closed expression is reproduced, not as an interval enclosure, by `c2_closed_d2`. None of these quantities is an input to the Hamiltonian sums. They certify the value of the closed expression (3.9) and test local steps of the route to Eq. (3.7); they do not prove the identification of the expression with the physical coefficient, which rests on the step of Sec. 3.6.

The certified expression. The expression is Eq. (3.9), with ${}_pF_q(a_1, \dots, a_p; b_1, \dots, b_q; z) = \sum_n \prod_i (a_i)_n / \prod_j (b_j)_n \, z^n/n!$. All three series have excess two and absolutely convergent summands of order n^{-3}. An integer-interval implementation that uses no Gamma or generalized-hypergeometric evaluator and imports no code from the closed-form study encloses the value, with the tail construction below at $n_0 = 4000$, in an interval of width about $2.69823801745 \times 10^{-53}$, and so proves the rounding

$$c_2^{\text{closed}} = -4.06627014813252155274002760857287648752990114118661 8 \tag{F.63}$$

to 51 decimal places. This is a computer-assisted proof about the specified expression. It is not a physical error interval for the Hamiltonian extrapolation, nor a proof of the regulator-identification step of the closed-form study.

Exact tail inequalities and elementary constant bounds. For the absolute value v_n of a term of one of the three series, the ratio is rational,

$$\rho_n = \frac{v_{n+1}}{v_n} = \frac{\prod_{j=1}^{4}(n + a_j)}{(n+1)\prod_{j=1}^{3}(n + b_j)}. \tag{F.64}$$

Rational functions $\phi(n) = \frac{n}{2} + \phi_0 + \phi_1/n + \cdots + \phi_{12}/n^{12}$ are constructed by exact cancellation of the leading powers in $\Xi(n) = \phi(n) - 1 - \rho_n \phi(n+1)$; changing the last coefficient by -1 and $+1$ gives

ϕ_- and ϕ_+. After a positive denominator is cleared and $n = 4000 + m$ substituted, nonnegative polynomial coefficients prove

$$\phi_-(n) - 1 - \rho_n\phi_-(n+1) \le 0, \qquad \phi_+(n) - 1 - \rho_n\phi_+(n+1) \ge 0 \qquad \text{for all } m \ge 0. \tag{F.65}$$

A second exact polynomial inequality bounds $v_n \le v_{n_0}(n_0/n)^2$ and controls the telescoping boundary at infinity. Hence

$$\phi_-(n_0)\, v_{n_0} \le \sum_{n=n_0}^{\infty} v_n \le \phi_+(n_0)\, v_{n_0}. \tag{F.66}$$

The terms of T after index zero are negative, so its enclosed tail is subtracted with the endpoints reversed. For π the implementation uses the alternating rational sums of Machin's identity $\pi = 16\arctan\frac{1}{5} - 4\arctan\frac{1}{239}$. For R_Γ^2 it uses $B(\frac14,\frac12)^2/\pi$ and the convergent split beta integral

$$B(\tfrac14,\tfrac12) = 2^{-1/4}\sum_{n\ge0}\frac{(\frac12)_n}{n!}\frac{2^{-n}}{n+\frac14} + 2^{-1/2}\sum_{n\ge0}\frac{(\frac34)_n}{n!}\frac{2^{-n}}{n+\frac12}. \tag{F.67}$$

Both positive term ratios are below $\frac12$, so after 420 terms each remainder lies between its first omitted term and twice that term. Integer root inequalities enclose the powers of two. All arithmetic uses outward-rounded integer endpoints at scale 10^{110}, and assertions check the exact sign certificates. The closed-form study's own tail check behind the envelope quoted in Sec. 3.5 (Appendix E.15) is of the same type, with $\phi = \frac{n}{2} + \text{const}$ and floating-point endpoints; the higher-order ϕ and the integer arithmetic are what make this enclosure a certified rounding.

Elliptic integration from the period integrands. This check starts from the period integrands, not from the final series. With $K(t) = \int_0^{\pi/2} \mathrm{d}\theta\,(1 - t\sin^2\theta)^{-1/2}$ (parameter convention) and $\mathbf{k}(t) = \big(K(t)^2,\, K(t)K(1-t),\, K(1-t)^2\big)^{\mathsf{T}}$, define

$$\mathbf{P}(a,b) = \mathrm{AC}\int_0^1 t^{a-1}(1-t)^{b-1}\mathbf{k}(t)\,\mathrm{d}t, \qquad \mathbf{p}(a,b) = \frac{4}{\pi^2}\mathbf{P}(a,b), \tag{F.68}$$

where negative exponents mean meromorphic continuation, not a divergent improper integral. Near zero

$$\mathbf{k}(t) \sim \Big(\frac{\pi^2}{4},\ \frac{\pi}{4}\log\frac{16}{t},\ \frac14\log^2\frac{16}{t}\Big)^{\mathsf{T}}. \tag{F.69}$$

The quadrature splits at $\frac12$, subtracts these endpoint terms, restores their elementary continued Mellin integrals and integrates the convergent remainder after the substitution $t = u^{12}/2$; Carlson's R_F supplies K independently of the Frobenius arrays of the closed-form study. One nonsymmetric assembly is

$$\begin{gathered}\mathbf{a} = 2\mathbf{p}(-\tfrac12,\tfrac74) + 5\mathbf{p}(\tfrac12,\tfrac74), \qquad \mathbf{b} = 5\mathbf{p}(-\tfrac14,\tfrac74) + 2\mathbf{p}(\tfrac34,\tfrac74), \qquad \mathsf{H} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 2 & -1 \end{pmatrix}, \\ c_2 = -\frac{R_\Gamma^2}{9} + \frac{\pi R_\Gamma^4}{10368\sqrt{2}}\,\mathbf{a}^{\mathsf{T}}\mathsf{H}\,\mathbf{b},\end{gathered} \tag{F.70}$$

and a separate assembly uses symmetric moments. The observed differences from the enclosed value are

quadrature order	nonsymmetric periods	symmetric moments
96	1.74×10^{-19}	4.78×10^{-29}
192	5.24×10^{-23}	2.28×10^{-34}
384	1.51×10^{-26}	1.04×10^{-39}

These are observed agreements under refinement, not rigorous quadrature-error bounds. The closed-form study's own endpoint-subtracted quadrature with Carlson's integrals (Sec. 3.5, Appendix E.15) is a check of the same kind, implemented separately.

Complex-plane normalization. With the single-valued area pairing $\mathcal{A}(t) = 8\big[K(t)\overline{K(1-t)} + \overline{K(t)}K(1-t)\big]$ (the torus area of Eq. (3.5)) and the continued complex moment

$$\mathcal{M}(a,b) = \mathrm{AC}\int_{\mathbb{C}} \mathrm{d}^2 t\, |t|^{2a-2}|1-t|^{2b-2}\mathcal{A}(t)^2, \tag{F.71}$$

the tested period relation is

$$\mathcal{M}(a,b) = 64\,\mathbf{P}(a,b)^{\mathsf{T}}\mathsf{L}(b)\,\mathbf{P}(2-a-b,b), \qquad \mathsf{L}(b) = \begin{pmatrix} 0 & 0 & \sin\pi b \\ 0 & 2\sin\pi b & -2\cos\pi b \\ \sin\pi b & -2\cos\pi b & -\sin\pi b \end{pmatrix}. \tag{F.72}$$

The ordinary convergent moment $\mathcal{M}(\frac{2}{3},\frac{2}{3})$ was evaluated directly on the plane and through the real-period pairing, with the values 24181.76549557689 and 24181.76549557647, a relative difference of 1.72×10^{-14}. Generic convergent moments also agreed. Symbolic checks gave zero residual for the endpoint differential identity, the cut-jump matrix, the reflection reduction and the final symmetric matrix reduction. These tests support the period identity and the normalization in the regimes checked; they do not supply a uniform proof for the physical collisions.

Local contacts. The finite localized-source term B_{31} of Sec. 3 was checked with an elementary-integral residual of about 3.17×10^{-71} at 70 working digits. Direct polar quadrature of the local branch contact, at the source parameter $M=3$, agreed with

$$A_{\mathrm{br}} = 4B_{31} - \frac{\pi^2}{4} = -3.8363599279686606009\ldots \tag{F.73}$$

to about 1.05×10^{-14}; other source parameters gave comparable local agreement. Fixed-shape neutral-charge checks supported the cancellation of the energy-primary term of the shrinking spin pair. In the minimal-subtraction assembly, Eq. (3.7), the local calculation supports the factor $3/4! = \frac{1}{8}$ and the two placements of the coupling counterterm. The two contributions to c_2 there are $-2.12099049015343\ldots$ and $-1.94527965797909\ldots$; the agreement of the second with a_1/b_0 of Appendix F.2 is a comparison made after both calculations, not an input to Eq. (F.42).

The global qualification remains separate. The closed-form study states that its audit does not reconstruct every sectorwise subtraction or prove a regulator-uniform remainder bound across nested collisions (Sec. 3.6). Agreement of local contact terms at fixed nondegenerate modulus does not by itself control the degeneration regions at 0, 1 and ∞, or a simultaneous shrinking of relative scales. The base study's own review of these checks (GPT-6 astra; not the review of Sec. 1.2) reached the same boundary: no error that changes c_2 was found in the tested expressions, but the global regulator comparison was not proved independently. It also noted a point of exposition: the statement that identity projections cancel should distinguish disconnected identity terms from an identity-channel power term with denominator $-1+2\delta$, which is retained and analytically continued; this is not a numerical error. The Hamiltonian result is a different physical consistency check on Eq. (3.7); it does not turn these local checks into a proof of the global comparison.

F.15 Explicit Majorana form factors

Let $\mathbf{m} = (m_1,\ldots,m_{n_{\mathrm{R}}})$ be increasing positive integer R occupations and $\mathbf{s} = (s_1,\ldots,s_{n_{\mathrm{NS}}})$ increasing positive half-integer NS occupations, and

$$f_{\mathrm{R}}(n) = \binom{2n}{n}4^{-n}, \qquad f_{\mathrm{NS}}(s) = \binom{2s-1}{s-\frac{1}{2}}4^{-(s-1/2)}. \tag{F.74}$$

In the real chiral gauge of the implementation (`code/c2_cylinder_d2/code/hamiltonian/ising_basis.py`) the chiral spin-field element between these states is

$$\mathcal{S}(\mathbf{m},\mathbf{s}) = (-1)^{(n_{\mathrm{NS}} \bmod 2)\, n_{\mathrm{R}}}\, 2^{-(n_{\mathrm{R}}+n_{\mathrm{NS}})/2} \prod_{i=1}^{n_{\mathrm{R}}} f_{\mathrm{R}}(m_i) \prod_{j=1}^{n_{\mathrm{NS}}} f_{\mathrm{NS}}(s_j) \times \prod_{i<j} \frac{m_i - m_j}{m_i + m_j} \prod_{i<j} \frac{s_i - s_j}{s_i + s_j} \prod_{i,j} \frac{m_i + s_j}{m_i - s_j}, \tag{F.75}$$

with empty products equal to one. The full spin-field element is the product of the left- and right-chiral factors. This representation, including its phase convention, is checked by chiral Virasoro Ward identities; it gives $\langle\sigma|\sigma|0\rangle = 1$ and $C_{\sigma\sigma\epsilon} = \frac{1}{2}$ in the chosen normalization. Related scaling-limit product formulas are in Ref. [37].

For a fermion mode that creates or removes a positive occupied mode m, the sign is $(-1)^{\#\{k<m\}}$, where $\#\{k < m\}$ counts the occupied modes below m. The diagonal element of the Ramond zero mode is $(-1)^{\#\text{occupied}}/\sqrt{2}$. Products of left and right fermion elements give the thermal field. Angular integration forces equal signed energy changes on the two sides, so the integrated thermal matrix connects pair creation and annihilation and the diagonal of the Ramond zero mode; it is implemented as a sparse matrix, whose angular factor 2π enters only when the Hamiltonian insertion is assembled. A basis state with left and right occupations $\boldsymbol{\ell}$, $\bar{\boldsymbol{\ell}}$ has energy and spin

$$E = \sum_i \ell_i + \sum_j \bar{\ell}_j + \begin{cases} 0 & \text{NS}, \\ \frac{1}{8} & \text{R}, \end{cases} \qquad J = \sum_i \ell_i - \sum_j \bar{\ell}_j, \tag{F.76}$$

and in NS only equal left and right parity is retained. The finite-basis checks built on these elements are those of Appendix F.11; they detect basis-sign and projection errors without assuming the continuum answer.

F.16 Compact quadratures for the local counterterms

The constants of Appendix F.7 are computed from the following finite-domain integrals (`code/c2_cylinder_d2/code/hamiltonian/improved_regulator.py`). Let

$$\mathcal{P}_{\alpha,r}(z) = \sum_{j=0}^{r} \frac{(\alpha)_j}{j!} z^j. \tag{F.77}$$

For $a, b \in (0,1)$ put $Z = (1-a)\big[(1-b^2) + \mathrm{i}b\sqrt{1-b^2}\big]$. The thermal coefficient is

$$c_{t,r} = \int_0^1 \mathrm{d}a \int_0^1 \mathrm{d}b\, (a^2 + b^2 - a^2 b^2)^{3/8}\, |\mathcal{P}_{\Delta_{\chi,0},r}(Z)|^2. \tag{F.78}$$

For $t \in (0,1)$ and $\varphi \in (0, \pi/2)$ put

$$D_t = t^2 + 2t(1-t)\cos\varphi + (1-t)^2, \qquad Z_t = \frac{1-t}{1-t+te^{\mathrm{i}\varphi}}. \tag{F.79}$$

The finite spin-two change is

$$\kappa_r - \kappa_0 = -\pi \int_0^1 \mathrm{d}t \int_0^{\pi/2} \mathrm{d}\varphi\, \frac{\cos(2\varphi)\, t^{11/4}}{(1-t)\, D_t^{\Delta_{\chi,0}}} \Big(|\mathcal{P}_{\Delta_{\chi,0},r}(Z_t)|^2 - 1 \Big). \tag{F.80}$$

For the cylinder term put $y = (1-t)/t$ and

$$F(t,\varphi) = \frac{\mathrm{Re}\big[(e^{\mathrm{i}\varphi} + y)^2\, \mathcal{P}_{\Delta_{\chi,0}-2,r}(Z_t)\, \overline{\mathcal{P}_{\Delta_{\chi,0},r}(Z_t)}\big]}{(1 + 2y\cos\varphi + y^2)^{\Delta_{\chi,0}}}; \tag{F.81}$$

then

$$c_{\mathrm{cyl},r} = -\frac{\Delta_{\chi,0}}{12}\int_0^1 \mathrm{d}t \int_0^{\pi/2} \mathrm{d}\varphi\, \frac{F(t,\varphi)-\cos(2\varphi)}{(1-t)^2}, \qquad r \geq 1. \tag{F.82}$$

The factor two relative to the half-plane expression comes from reflection in x. Tensor Gauss–Legendre rules evaluate these integrals after their domains are mapped to the unit interval; orders 128, 256 and 512 are used for refinement, and the constants are evaluated with 512 points in each direction. The subtraction stays inside the integrand: numerically subtracting separately divergent moments would not define the same calculation.

F.17 Mixed thermal polynomials and denominators

For the central and side kernels of Appendix F.9, the energy polynomials are

$$P_{\mathcal{E},c} = 1 + uv - 2u - 2v, \qquad P_{\mathcal{E},s} = -2 + u + uv - 2u^2 v, \tag{F.83}$$

and the derivative polynomials are

$$\begin{aligned} P_{d,c} &= \frac{uv+1}{4}\big(u^2v^2 + 2u^2v - 4u^2 + 2uv^2 - 6uv + 2u - 4v^2 + 2v + 1\big), \\ P_{d,s} &= -\frac{v+1}{4}\big(4u^4v^2 - 2u^3v^2 - 2u^3v - u^2v^2 + 6u^2v - u^2 - 2uv - 2u + 4\big). \end{aligned} \tag{F.84}$$

For placement $b = c, s$ the coefficient arrays are

$$V^b_{nm} = [u^n v^m]\frac{G_b}{\sqrt{2}}, \qquad E^b_{nm} = [u^n v^m]\frac{G_b P_{\mathcal{E},b}}{2\sqrt{2}}, \qquad D^b_{nm} = [u^n v^m]\frac{G_b P_{d,b}}{2\sqrt{2}}, \tag{F.85}$$

and the derivative angular projection uses $D^b_{n+1,m+1}E^b_{nm}$. The denominators are

placement	vacuum, $d^b_0(n,m)$	energy and derivative, $d^b_{\mathcal{E}}(n,m)$
central	$(2n+2)(2m+2)$	$(2n+1)(2m+1)$
side	$(2n+1)(2m+2)$	$(2n-1)(2m+1)$

so that

$$X_b = (2\pi)^3 q^{\Delta_{\chi,0}} \sum_{n,m\geq 0}\Big[\frac{(2E^b_{nm} - D^b_{n+1,m+1})E^b_{nm}}{d^b_{\mathcal{E}}(n,m)} - \frac{(V^b_{nm})^2}{d^b_0(n,m)}\Big]. \tag{F.86}$$

The side denominator is negative at $n = 0$ and must be kept. The total mixed term is $L_X(V_\epsilon) = X_c + 2X_s$. A function of the form $A(u)B(v)C(uv)$ is generated by diagonal convolutions,

$$[u^n v^m]A(u)B(v)C(uv) = \sum_{j=0}^{\min(n,m)} A_{n-j}B_{m-j}C_j, \tag{F.87}$$

and the finite polynomials are applied by shifts. This reduces the mixed calculation to two-index arrays and permits the much larger N_{mix}.

F.18 The leading face-tail coefficient in full

The coefficient A_{face} of Eq. (F.56) is fixed independently of any fitted cutoff sequence. It is derived from exact face limits of the blocks but evaluated numerically through convergent two-index sums (`code/c2_cylinder_d2/code/hamiltonian/exact_leading_tail.py`; "exact" in the file name describes the source of the asymptotic formula, not an interval enclosure of its floating-point evaluation). With the regulated H of Eq. (F.31) define

$$\Phi_B(u,v) = H(u)(1-v)^{-1/8}H(uv), \qquad \Phi_A(u,v) = H(u)H(v)(1-uv)^{-1/8}, \tag{F.88}$$

with Taylor coefficients B_{ij} and A_{ij}, and set

$$\begin{aligned} E^B_{ij} &= [u^i v^j]\frac{1+v}{2}\Phi_B, \quad & D^B_{ij} &= [u^i v^j]\frac{(1+v)^3}{8}\Phi_B, \\ E^A_{ij} &= [u^i v^j]\frac{1+uv}{2}\Phi_A, \quad & D^A_{ij} &= [u^i v^j]\frac{(1+uv)^3}{8}\Phi_A. \end{aligned} \tag{F.89}$$

The two outer faces are organized by

$$O = \sum_{i,j\ge 0} \frac{1}{2i+2\Delta_{\chi,0}}\Big[\frac{(2E^B_{ij} - D^B_{i,j+1})E^B_{ij}}{2j+1} - \frac{B^2_{ij}}{2j+2}\Big], \tag{F.90}$$

the adjacent middle face by

$$M_{\mathrm{adj}} = \sum_{i,j\ge 0}\Big[\frac{(2E^A_{ij} - D^A_{i+1,j+1})E^A_{ij}}{(2i+1)(2j+1)} - \frac{A^2_{ij}}{(2i+2)(2j+2)}\Big], \tag{F.91}$$

and the nested middle face by the collapsed norms of Eq. (F.52),

$$\mathcal{N}_X = 2\mathcal{N}_{\mathcal{E}} - \mathcal{N}_d - \mathcal{N}_0, \qquad M_{\mathrm{nest}} = \frac{H(1)^2\mathcal{N}_X}{(2\pi)^2 q^{\Delta_{\chi,0}}}. \tag{F.92}$$

The leading coefficient is

$$\boxed{A_{\mathrm{face}} = -(2\pi)^4 q^{2\Delta_{\chi,0}}\,\frac{2}{7\Gamma(\frac{1}{8})^2}\,(4O + 2M_{\mathrm{adj}} + M_{\mathrm{nest}}).} \tag{F.93}$$

The base study evaluates it with two-index cutoff 2048 and records O, M_{adj}, M_{nest}, their combination and the sign convention; refinement at smaller and larger auxiliary cutoffs and a comparison with independently fitted leading tails are kept as diagnostics. The continuum analysis of Sec. 3.8 sets this two-index cutoff to a common value for all points and propagates its shift under a change (the "cutoff of the leading-tail calculation" in its error budget). Once Eq. (F.93) is fixed, the residual spectral models of Appendix F.10 account for the subleading powers and exponentials.

F.19 Minimal end-to-end algorithm

For an implementation in another language the logical order is:

1. Fix the unit-field normalization, $\Delta_{\chi,0} = \frac{15}{8}$, $a_1 = -\pi^2/4$ and the common lower-order b_0. Choose r and $\lambda > 0$. No value of c_2 enters the solver.
2. Generate the six Ising block coefficient arrays of Eqs. (F.45) and (F.47) through one index beyond the desired cube; check branches, reality and selected low coefficients against explicit form factors.
3. Construct H_n from Eq. (F.31), apply the three Wick-pairing convolutions, evaluate Eq. (F.48) for each pairing and channel, and return every cubical partial sum.

4. Compute the collapsed energies, norms and spin-two mixed terms of Eq. (F.52) at a much larger cutoff, including the normalization subtraction.
5. Generate the two-index thermal kernels and polynomials of Appendix F.17 and extrapolate their square cutoff separately.
6. Evaluate the compact counterterm integrals of Appendix F.16, form $t_r(\lambda)$ and $-\kappa_r$, and assemble Eq. (F.42), including the exact thermal-squared term.
7. Compute A_{face} from Appendix F.18 and extrapolate the corrected chain sequence, Eq. (F.58), over the predeclared power sets, exponential terms and windows.
8. Repeat at several λ for each regulator order, take the regulator limit only after the spectral limits, and keep model sensitivity and fits with residual degrees of freedom.
9. Convert h_0 to $c_2 = a_1/b_0 + h_0/b_0^2$, and compare with the closed expression only at this final stage.

A faithful reproduction also tests the second-order endpoint anomaly (Appendix F.3), the magnetic normalization (Appendix F.12), the finite-box equivalence of the two implementations (Appendix F.11) and the sensitivity to the coefficient grid; matching a final decimal number alone is not an adequate test.

G c_1 in three dimensions: derivation and conditional control of the omitted spectrum

This appendix gives the derivation and the method of the c_1 package (written by OpenAI GPT-6 astra, in its review revision of 8 October 2026; Sec. 1.2), edited to the notation of this note. Section 4 states the ratio formula, the retained value and its comparison with BRRZ, the implementations, the hypotheses of the estimated range, and the feasibility, box and gap tests; they are not repeated here. Throughout this appendix

$$S \equiv \Delta_\sigma^{\mathrm{SR}} = 0.518148806(24), \qquad E \equiv \Delta_\mathcal{E}^{\mathrm{SR}} = 1.41262528(29), \qquad \delta = \frac{s_* - s}{2}, \tag{G.1}$$

as in Table 1, and ϵ is the energy operator of the 3d Ising CFT. Sans-serif letters (S, E, M, V) denote block sums, never these two dimensions. All long decimal strings below are numerical regression targets of the package, not physical significant digits.

The coefficient c_1 is the right derivative of $\Delta_\mathcal{E}$ with respect to δ at $\delta = 0$; the long-range side has $\delta > 0$. With the retained value of Sec. 4.2 the expansion reads

$$\Delta_\mathcal{E} \simeq 1.41262528 + 0.27315242\,\delta = 1.41262528 + 0.13657621\,(s_* - s), \tag{G.2}$$

with the remainder of the expansion understood. It describes the local slope at s_* only, not a controlled extrapolation over a large range of s.

G.1 Three meanings of precision

Three kinds of uncertainty are kept apart throughout.

- *Numerical.* For a fixed finite list of primary operators with fixed dimensions and operator-product-expansion (OPE) coefficients, the integrals can be evaluated to many digits. Numerical stability then tests the descendant truncation, the quadrature and the subtraction. By itself it says nothing about omitted primaries or about errors of the input correlator.

- *Parametric.* Letting the retained dimensions and couplings vary gives a second uncertainty. The spectrum tables do not provide a calibrated joint probability distribution for their error estimates, so the calculation uses explicit ranges and crossing constraints, not a statistical error model.
- *Physical.* A physical interval must also control the missing conformal families and the possible incompleteness of the selected low-dimensional spectrum. The interval of Sec. 4.3 is conditional on explicit hypotheses about that missing spectrum and on numerical allowances for nonlinear effects. Whenever an interval for c_1 is written, its assumptions are part of the statement.

G.2 Reduction to short-range conformal perturbation theory

The product theory and the perturbing operator. Near the crossover the fixed point is described by the product of the short-range Ising CFT and a generalized free scalar χ [11, 12]. All scalar two-point functions are normalized to unity,

$$\langle\sigma(x)\sigma(0)\rangle = |x|^{-2S}, \qquad \langle\epsilon(x)\epsilon(0)\rangle = |x|^{-2E}, \qquad \langle\chi(x)\chi(0)\rangle = |x|^{-2\Delta_\chi}, \tag{G.3}$$

with

$$\Delta_\chi = 3 - S - \delta, \qquad \mathcal{O} = \sigma\chi, \qquad \Delta_{\mathcal{O}} = 3 - \delta. \tag{G.4}$$

The deformation is $g\mu^\delta \int \mathrm{d}^3x\, \sigma\chi$, with μ the renormalization scale, and the relevant thermal coupling is tuned to the critical trajectory. The generalized free field has Wick-factorized correlators; the Ising factors do not. With a dimensionless renormalized coupling g,

$$\beta(g) = -\delta g + b_0 g^3 + O(g^5), \qquad \gamma_{\mathcal{E}}(g) = a_0 g^2 + O(g^4). \tag{G.5}$$

There is no g^2 term in β, because the undeformed product theory forbids the relevant three-point function ($\langle\mathcal{O}\mathcal{O}\mathcal{O}\rangle = \langle\sigma\sigma\sigma\rangle\langle\chi\chi\chi\rangle = 0$). To first order in δ the infrared fixed point has $g_*^2 = \delta/b_0 + O(\delta^2)$, so $\Delta_{\mathcal{E}} = E + \gamma_{\mathcal{E}}(g_*) = E + (a_0/b_0)\,\delta + O(\delta^2)$, which is $c_1 = a_0/b_0$. Rescaling g by a constant rescales a_0 and b_0 by the same factor and leaves the ratio unchanged.

Why the spin channel supplies the denominator. In the crossover description the long-range spin dimension is protected [12]: $\Delta_\sigma = (3-s)/2 = S + \delta$. If $\gamma_\sigma(g) = a_\sigma g^2 + O(g^4)$, the fixed point gives $\Delta_\sigma = S + a_\sigma\delta/b_0 + O(\delta^2)$, and matching to $S+\delta$ gives $a_\sigma = b_0$. Both coefficients are therefore first-order anomalous-dimension integrals of short-range four-point functions: the energy external state supplies a_0 and the spin external state supplies b_0.

G.3 The integrated four-point functional

Normalization and scale extraction. With e a unit vector, recall from Sec. 4 the four-point function with one external state at infinity, H_O, and write the integral I_O as

$$\begin{aligned} H_O(y) &= \lim_{R\to\infty} R^{2\Delta_O}\langle O(0)\sigma(y)\sigma(e)O(Re)\rangle, \qquad O \in \{\sigma, \epsilon\}, \\ F_O(y) &= |e-y|^{-2(3-S)}\big[H_O(y) - |e-y|^{-2S}\big], \qquad I_O = \mathrm{AC}\int_{\mathbb{R}^3} \mathrm{d}^3y\, F_O(y). \end{aligned} \tag{G.6}$$

The vacuum term removes the identity contraction of the two inserted spins. The symbol AC denotes analytic continuation through the power-divergent short-distance terms, with the relevant deformation tuned critically; it is not a prescription to integrate an unregulated divergent expression numerically. The normalization $a_0 = -2\pi I_\epsilon$, $b_0 = -2\pi I_\sigma$ follows from a second-order amplitude with a Gaussian source. Let

$$D_O(x_1, x_2) = \langle O(\infty)\sigma(x_1)\sigma(x_2)O(0)\rangle - |x_1 - x_2|^{-2S}, \tag{G.7}$$

where the state at infinity has unit overlap with the unperturbed state at the origin, and define

$$A_{O,2}(\delta) = \frac{1}{2}\,\mathrm{AC}\int \mathrm{d}^3x_1\,\mathrm{d}^3x_2\,e^{-|x_1|^2-|x_2|^2}\,|x_1-x_2|^{-2(3-S-\delta)}D_O(x_1,x_2). \tag{G.8}$$

Set $x_2 = te$, $x_1 = ty$ and integrate over the direction of e. The measure is $4\pi t^5\,\mathrm{d}t\;\mathrm{d}^3y$, the correlator scales as $D_O(ty,te) = t^{-2S}D_O(y,e)$, and the kernel supplies $t^{-6+2S+2\delta}$. With the factor $\frac{1}{2}$ the remaining scale integral is

$$2\pi\int_0^\infty \mathrm{d}t\,t^{2\delta-1}e^{-t^2(1+|y|^2)} = \pi\Gamma(\delta)(1+|y|^2)^{-\delta}, \tag{G.9}$$

so that $A_{O,2}(\delta) = \pi\Gamma(\delta)\,I_O(\delta;1)$ with the source-dependent integral

$$I_O(\delta;1) = \mathrm{AC}\int \mathrm{d}^3y\,(1+|y|^2)^{-\delta}|e-y|^{-2(3-S-\delta)}\big[H_O(y) - |e-y|^{-2S}\big], \qquad I_O(0;1) = I_O. \tag{G.10}$$

The second argument records the equal Gaussian weights. This amplitude probes the local insertion at the origin against a fixed external state; the source suppresses the second short-distance region, at infinity, so its pole is cancelled by a *single* local operator-renormalization factor, not by twice that factor. With $O_R = Z_O O_{\mathrm{bare}}$ and

$$Z_O = 1 + \frac{a_O}{2\delta}g^2 + O(g^4), \qquad \gamma_O = -\beta(g)\,\partial_g \log Z_O = a_O g^2 + O(g^4), \tag{G.11}$$

finiteness of $Z_O[1 + g^2A_{O,2} + \cdots]$ requires

$$A_{O,2}(\delta) = -\frac{a_O}{2\delta} + O(1), \qquad a_\sigma = b_0, \quad a_\epsilon = a_0. \tag{G.12}$$

Since $\pi\Gamma(\delta)I_O(\delta;1) = \pi I_O/\delta + O(1)$, this gives $a_O = -2\pi I_O$, the normalization of Sec. 4. The common factor -2π cancels from c_1, but keeping it makes the comparison with the published normalization possible. The finite, source-dependent term is not needed at first order. All these identities use the same critical analytic continuation; the Gaussian source does not replace the subtraction of the short-distance power terms.

The folded integration region. Rotational invariance reduces the vector integral to a radius $r = |y|$ and a cosine $\eta = \widehat{y}\cdot e$. Write

$$z = r(\eta + \mathrm{i}\sqrt{1-\eta^2}), \qquad v = |1-z|^2 = 1 - 2r\eta + r^2, \qquad \mathrm{d}\mu = 2\pi r^2\,\mathrm{d}r\;\mathrm{d}\eta. \tag{G.13}$$

Here z only encodes the rotational invariants of a three-dimensional configuration; the integral is not a two-dimensional CFT integral. The conformal maps and the channel identity of Appendix G.4 fold the full integration into

$$\mathcal{R} = \{|z|<1,\ |z|<|1-z|\} = \{|z|<1,\ \mathrm{Re}\,z < \tfrac{1}{2}\}, \tag{G.14}$$

that is, $0<r<1$ and $-1<\eta<\min\{1,(2r)^{-1}\}$. The upper η limit changes at $r=\frac{1}{2}$, which is a breakpoint of the numerical integration.

Block sums. Let $g^{d_a,d_b}_{\Delta,\ell}(z,\bar z)$ be the conformal block with leading OPE behaviour $r^\Delta P_\ell(\eta)$ as $z\to 0$, where P_ℓ is the Legendre polynomial. For the ordered four-point function the external-dimension differences are $d_a = \Delta_1 - \Delta_2$ and $d_b = \Delta_3 - \Delta_4$; the mixed ordering $(\epsilon,\sigma,\sigma,\epsilon)$ has $(d_a,d_b) = (E-S, S-E)$. In the f normalization of the published spectrum (Appendix G.6), with the rising Pochhammer symbol $(a)_\ell = \Gamma(a+\ell)/\Gamma(a)$, define

$$\begin{aligned} w_\ell &= \frac{\ell!}{(1/2)_\ell}, \qquad \mathsf{S}(z) = \sum_{A\neq\mathbf{1},\ \mathbb{Z}_2\text{-even}} w_{\ell_A} f^2_{\sigma\sigma A}\, g^{0,0}_{\Delta_A,\ell_A}(z), \\ \mathsf{E}(z) &= \sum_{A\neq\mathbf{1},\ \mathbb{Z}_2\text{-even}} w_{\ell_A} f_{\sigma\sigma A} f_{\epsilon\epsilon A}\, g^{0,0}_{\Delta_A,\ell_A}(z), \qquad \mathsf{M}(z) = \sum_{B,\ \mathbb{Z}_2\text{-odd}} w_{\ell_B} f^2_{\sigma\epsilon B}\, g^{E-S,S-E}_{\Delta_B,\ell_B}(z). \end{aligned} \tag{G.15}$$

For example, the direct (odd-channel) representation of the mixed correlator is $H_\epsilon(z) = r^{-S-E}\mathsf{M}(z)$, and its crossed (even-channel) representation is $|1-z|^{-2S}[1+\mathsf{E}(1-z)]$. The two coincide for an exact crossing-symmetric correlator, not necessarily for independently truncated sums.

G.4 Derivation of the folding and its assumptions

Three steps take the integral I_O of Eq. (G.6) over $\mathbb{R}^3$ to the region $\mathcal{R}$ and account for the coefficients and the extra term of the folded expression.

Inversion about the origin. For $y = x/|x|^2$ the Jacobian is $|x|^{-6}$, and

$$H_O\Big(\frac{x}{|x|^2}\Big) = |x|^{2S} H_O(x), \qquad \Big|e - \frac{x}{|x|^2}\Big| = \frac{|e-x|}{|x|}. \tag{G.16}$$

The first identity uses the identical operators O at zero and infinity, which the inversion exchanges. Since $2(3-S)+2S = 6$, the kernel and correlator factors cancel the Jacobian exactly, and

$$I_O = 2\,\mathrm{AC}\int_{|y|<1} \mathrm{d}^3y\, F_O(y). \tag{G.17}$$

The cancellation uses $\Delta_\chi = 3 - S$ at $\delta = 0$; it is not an identity between integrands at finite δ.

The crossed representation on the rest of the ball. Split $|y| < 1$ into $\mathcal{R}$ and $\mathcal{C} = \{|e-y| < |y| < 1\}$, and set $w = e - y$ on $\mathcal{C}$. The exact correlator admits

$$H_O(e-w) = |w|^{-2S}\big[1 + X_O(w)\big], \qquad X_\sigma = \mathsf{S}, \quad X_\epsilon = \mathsf{E}. \tag{G.18}$$

For $O = \epsilon$ this equality selects the even $\sigma\sigma$–$\epsilon\epsilon$ OPE channel; it is *not* a permutation symmetry that interchanges σ and ϵ. The vacuum term cancels, $F_O(e-w) = |w|^{-6}X_O(w)$, and the new domain is $\mathcal{D} = \{|w| < |e-w| < 1\} \subset \mathcal{R}$. Hence

$$I_O = 2\,\mathrm{AC}\int_{\mathcal{R}} \mathrm{d}^3y\, F_O(y) + 2\,\mathrm{AC}\int_{\mathcal{D}} \mathrm{d}^3w\, |w|^{-6} X_O(w). \tag{G.19}$$

Inversion about the second spin insertion. Let $v = |e-w|^2$ and

$$T(w) = e + \frac{w-e}{|w-e|^2}, \qquad |T(w)|^2 = \frac{|w|^2}{v}, \qquad |e - T(w)|^2 = \frac{1}{v}, \qquad |\det \partial T| = v^{-3}. \tag{G.20}$$

The map T sends $\mathcal{R}$ to itself (it exchanges the two conditions $|w| < 1$ and $|w| < |e-w|$) and exchanges its parts with $v < 1$ and $v > 1$, namely $\mathcal{D}$ and its complement, up to boundaries of zero measure. In the complex representative it is $w \mapsto w/(w-1)$ up to conjugation. Both X_O contain pairs of equal external dimensions and only even-spin exchanges, so the pair-exchange factor $(-1)^\ell$ is one and $X_O(Tw) = X_O(w)$. Moreover $|T(w)|^{-6}|\det \partial T| = |w|^{-6}$. The two parts of the last integral of Eq. (G.19) are therefore equal, and

$$\boxed{I_O = \mathrm{AC}\int_{\mathcal{R}} \mathrm{d}^3y\, \big[2F_O(y) + |y|^{-6} X_O(y)\big].} \tag{G.21}$$

The changes of variables are applied with consistent subtractions and analytic continuation, not to bare divergent integrals. They establish Eq. (G.21) for the exact correlator. Replacing its channel sums by independently truncated data introduces a correlator approximation that no change of variables can remove.

Folded integrands and per-family kernels. With $H_\sigma = r^{-2S}(1+\mathsf{S})$ in the direct channel and $H_\epsilon = r^{-S-E}\mathsf{M}$, the implemented linear functionals are

$$I_\sigma = \mathrm{AC}\int_{\mathcal{R}} \mathrm{d}\mu \left\{2v^{-(3-S)}\left[r^{-2S}(1+\mathsf{S}) - v^{-S}\right] + r^{-6}\mathsf{S}\right\}, \tag{G.22}$$

$$I_\epsilon = \mathrm{AC}\int_{\mathcal{R}} \mathrm{d}\mu \left\{2v^{-(3-S)}\left[r^{-S-E}\mathsf{M} - v^{-S}\right] + r^{-6}\mathsf{E}\right\}. \tag{G.23}$$

Both the identity subtraction and the signs of the mixed products must be kept; in particular the energy integrand is not a sum of positive even-channel coefficients. The per-family kernels are

$$\begin{aligned} K_b(\Delta,\ell) &= -2\pi w_\ell\,\mathrm{AC}\int_{\mathcal{R}} \mathrm{d}\mu\,[2v^{-(3-S)}r^{-2S} + r^{-6}]g^{0,0}_{\Delta,\ell}, \qquad K_e(\Delta,\ell) = -2\pi w_\ell\,\mathrm{AC}\int_{\mathcal{R}} \mathrm{d}\mu\, r^{-6} g^{0,0}_{\Delta,\ell}, \\ K_o(\Delta,\ell) &= -4\pi w_\ell \int_{\mathcal{R}} \mathrm{d}\mu\, v^{-(3-S)} r^{-S-E} g^{E-S,S-E}_{\Delta,\ell}, \end{aligned} \tag{G.24}$$

so that, with $f_A = f_{\sigma\sigma A}$, $h_A = f_{\epsilon\epsilon A}$ and $k_B = f_{\sigma\epsilon B}$,

$$b_0 = b_{\mathbf{1}} + \sum_A K_b(\Delta_A,\ell_A) f_A^2, \qquad a_0 = a_{\mathbf{1}} + \sum_A K_e(\Delta_A,\ell_A) f_A h_A + \sum_B K_o(\Delta_B,\ell_B) k_B^2. \tag{G.25}$$

The nonspectral constants come from the remaining terms of Eqs. (G.22) and (G.23),

$$b_{\mathbf{1}} = -4\pi\int_{\mathcal{R}} \mathrm{d}\mu\, v^{-(3-S)}\left[r^{-2S} - v^{-S}\right] \approx -26.0638858592, \qquad a_{\mathbf{1}} = 4\pi\int_{\mathcal{R}} \mathrm{d}\mu\, v^{-3} \approx 44.4132198049 \tag{G.26}$$

(both integrals converge: $v > \frac{1}{4}$ on $\mathcal{R}$, and r^{-2S} is integrable at the origin because $2S < 3$). They are bookkeeping terms, not separate physical observables.

G.5 Conformal blocks and numerical integration

Radial expansion. With the conformal radial coordinate

$$\rho(z) = \frac{z}{(1+\sqrt{1-z})^2}, \qquad \varrho = |\rho|, \qquad \xi = \frac{\mathrm{Re}\,\rho}{|\rho|}, \tag{G.27}$$

where the branch of the square root is continued from $z = 0$, the blocks are evaluated as

$$g^{d_a,d_b}_{\Delta,\ell}(z,\bar z) = (4\varrho)^\Delta \sum_{n=0}^{N} \varrho^n \sum_j B_{n,j} P_j(\xi), \qquad B_{0,j} = \delta_{j\ell}. \tag{G.28}$$

The coefficients solve the quadratic Casimir equation; the matrix recurrence evaluated is given in Appendix G.15. The organization in the radial coordinate follows Ref. [38]; it converges fast in favourable OPE configurations. On the closure of $\mathcal{R}$ the largest radial coordinate is $\max|\rho| = 2-\sqrt{3} \simeq 0.267949$, reached at the corner $z = e^{\mathrm{i}\pi/3}$. This small maximum is why the folded domain is used: equivalent domains with a larger maximum radial coordinate magnify omitted-family and descendant effects, so their truncated values should not be averaged as independent measurements.

A primary is the lowest state of a conformal family, and the terms with $n > 0$ account for its descendants. Increasing N completes a retained family but does not add a primary. Descendant convergence and completeness of the primary spectrum are therefore different error questions.

Analytic continuation at small radius. The r^{-6} terms of Eqs. (G.22) and (G.23) are power divergent at the origin. Choose a subtraction radius $r_0 < \frac{1}{2}$, normally $r_0 = 0.2$, so that the small ball has the complete angular range. On it a separate small-z series is used,

$$g^{0,0}_{\Delta,\ell}(z) = \sum_{n,j} A_{n,j} r^{\Delta+n} P_j(\eta) \tag{G.29}$$

(Appendix G.15). The angular integral keeps only $j = 0$, and gives the explicit small-ball contribution

$$\mathrm{AC}\int_{r<r_0} \mathrm{d}\mu\, r^{-6} g^{0,0}_{\Delta,\ell} = 4\pi \sum_n A_{n,0}\, \frac{r_0^{\Delta+n-3}}{\Delta+n-3}. \tag{G.30}$$

The sum uses 32 levels of the z-series. Terms whose scalar coefficient vanishes are not evaluated as artificial 0/0 expressions. A genuine resonance with a nonzero coefficient would need a separate logarithmic treatment; the implementation rejects an active scalar-series term with $|\Delta+n-3| < 10^{-10}$. This is a diagnostic threshold, not a way to assign a finite part. It is inactive for the retained inputs, where the nearest active power is about 0.41 from resonance. At the conserved stress tensor the null-state consistency checks enforce the required cancellation.

For the rest of the integral, Gauss–Legendre quadrature is used in η, and r is split into $(0, r_0)$, $(r_0, \frac{1}{2})$ and $(\frac{1}{2}, 1)$, with the map $r = r_0 u^4$ on the first interval to resolve the integrable endpoint and linear maps on the others. The principal run uses $N = 24$ and 100×100 nodes in each radial and angular piece. The r^{-6} contribution is left out of the small-ball quadrature, because Eq. (G.30) already includes it. The critical analytic continuation, rather than a cutoff-dependent finite part with unspecified thermal tuning, is part of the definition of the observable. The relevant test is therefore agreement between equivalent analytically continued representations, not agreement between a cutoff integral and the intended critical quantity.

Numerical checks and their scope. The radial block is compared with the independent z-series at complex points and, for scalar exchange with equal external dimensions, with the diagonal hypergeometric expression

$$g_{\Delta,0}(z,z) = \Big(\frac{z^2}{1-z}\Big)^{\Delta/2} {}_3F_2\left(\begin{matrix}\Delta/2,\ \Delta/2,\ \Delta/2-1/2\\ (\Delta+1)/2,\ \Delta-1/2\end{matrix}; \frac{z^2}{4(z-1)}\right). \tag{G.31}$$

The largest discrepancies are 5.6×10^{-17} against the z-series and 2.3×10^{-16} against Eq. (G.31), floating-point effects at fixed input data. The package's two integrated forms (Sec. 4.2) agree on the retained coefficients; this checks the algebra and the bookkeeping on the *same* folded domain, including the spin-channel identity that reduces one form to the other, and is not an independent validation of the folding.

G.6 Input data and tensor conventions

The selected 60-primary snapshot. The retained selection transcribes the stable extracted operators of dimension below 24 in Tables 3 to 7 of Ref. [14]: 37 internally $\mathbb{Z}_2$-even and 23 internally $\mathbb{Z}_2$-odd primaries (Table 6). It is a selection of published stable entries, not a statement that every exchanged primary below 24 is present. The leading data are replaced coherently by the stress-tensor bootstrap inputs of Ref. [15]:

S	E	$\lambda_{\sigma\sigma\epsilon}$	$\lambda_{\epsilon\epsilon\epsilon}$	$c = c_T/c_B$
0.518148806(24)	1.41262528(29)	1.05185373(11)	1.53244304(58)	0.946538675(42)

Here c_B is the stress-tensor two-point normalization of one free real scalar, and $c = c_T/c_B$ is the dimensionless normalization used in the Ward identities. Shared inputs stay shared under variations: the same $\lambda_{\sigma\sigma\epsilon}$ enters the even ϵ channel and the odd σ channel. The stress-tensor couplings are derived from S, E and c, not treated as independent measured numbers:

$$f_{\sigma\sigma T} = S\sqrt{\frac{3}{8c}}, \qquad f_{\epsilon\epsilon T} = E\sqrt{\frac{3}{8c}}, \tag{G.32}$$

so the Ward ratio is $f_{\epsilon\epsilon T}/f_{\sigma\sigma T} = E/S$. A simultaneous sign change of both stress couplings is an operator convention and does not change the products that enter.

What the printed errors mean. In Table 1 of Ref. [15], the rigorous entries among these five inputs are S and E, under that bootstrap setup; the printed errors of the two OPE coefficients and of c are not established there as rigorous bounds. The errors of the higher families in Ref. [14] describe spreads of extracted data over 60 spectra at the fixed derivative cutoff $\Lambda = 43$; they contain no certified $\Lambda \to \infty$ uncertainty. For example, Ref. [18] reports an extraction near 7.297 at $\Lambda = 30$ for the even scalar listed at 7.2535(51) in the retained table. This does not determine the true dimension, but it shows why the printed spreads cannot be read as complete errors. The strong sensitivity of c_1 to the ϵ' family and to the odd spin-three coupling near $\Delta = 4.638$ makes this limitation relevant to the final interval; the package supplies no calibrated independent cross-check of the quoted error of that coupling.

Normalization conversion and signs. The λ convention meant here is the tensor convention of Kos, Poland and Simmons-Duffin [16] and of Reehorst [17], related to the table convention by $\lambda = 2^{\ell/2} f$; the symbol is not a convention-independent OPE normalization. The block contraction factor in the λ convention is $\ell!/[2^\ell (1/2)_\ell]$; multiplied by the conversion 2^ℓ of the two couplings, it gives $w_\ell = \ell!/(1/2)_\ell$ in Eq. (G.15). For scalar exchange the conventions agree.

The signs of $f_A h_A$ are physical relative signs. Nine rows, all in the selected $[\epsilon\epsilon]_0$ trajectory (family `ee0`), have a negative product; replacing all couplings by absolute values would change the mixed correlator and the answer. The overall sign of one exchanged primary cancels from its product and is immaterial.

Odd spins in the reflected mixed ordering. The lightcone-normalized blocks of Ref. [14] behave as $g^{\mathrm{SD}}_{\Delta,\ell} \sim (-1)^\ell w_\ell (4\varrho)^\Delta P_\ell(\xi)$, whereas the blocks of Eq. (G.28) have leading coefficient $+1$. In the ordering $(\epsilon, \sigma, \sigma, \epsilon)$ the two OPE pairs are reflected, and reversing the ordered scalar pair contributes a second factor $(-1)^\ell$ in the OPE product, which cancels the sign of the lightcone block. The coefficient of the mixed block is therefore $+w_\ell f^2_{\sigma\epsilon B}$, as in M, with no further $(-1)^\ell$. For $0 < z < 1$ the reflected radial-state decomposition must have positive leading squared amplitudes; its all-descendant positivity is formulated with the prefactor Q of Eq. (G.42), not asserted for every coefficient of the raw mixed block. The generalized-free prefactor and positive-state tests of Appendix G.14 check this convention. As an independent calibration of the stress family, a free real scalar has $S = \frac{1}{2}$, $c = 1$, and Eq. (G.32) gives $w_2 f^2_{\sigma\sigma T} = \frac{8}{3} \cdot \frac{3}{32} = \frac{1}{4}$.

G.7 Retained-spectrum evaluation

The package stores the per-family kernels and contributions, not only the final ratio, which exposes normalization changes and large cancellations. For example, the even ϵ family contributes about 108.0654 to a_0 and the odd σ family about -147.3028, before the other families and the nonspectral constant $a_{\mathbf{1}}$ are added; the comparatively small a_0 of Sec. 4.2 should not be reconstructed from rounded individual terms.

Numerical convergence at fixed inputs. Raising the descendant order from 24 to 28 and the quadrature from 100^2 to 128^2 nodes changes the ratio by about 1.7×10^{-12}; moving r_0 from 0.2 to 0.15 changes it by about 3×10^{-14}. Quadrature refinement is therefore not the route to a substantially smaller physical error.

Comparison with the published normalization. The component-by-component agreement with BRRZ in Sec. 4.2 uses the same normalization $a_0 = -2\pi I_\epsilon$, $b_0 = -2\pi I_\sigma$. These comparisons are compatible at the published accuracy; they are not independent measurements to be combined into a smaller error. The precision of the retained integral and the uncertainty of the full CFT remain separate.

G.8 Retained-input variations

In this and the following subsections a_0, b_0 denote the retained reference coefficients of Sec. 4.2, $c_1^{(0)} = a_0/b_0$, and Δa, Δb denote shifts away from them.

Linearized integrated objective. The first variation is

$$\mathrm{d}c_1 = \frac{\mathrm{d}a_0 - c_1^{(0)}\,\mathrm{d}b_0}{b_0}. \tag{G.33}$$

By Eq. (G.25) the coupling derivatives are elementary: for an even primary the variations with f_A and h_A are proportional to $K_e h_A - 2c_1^{(0)} K_b f_A$ and $K_e f_A$, and for an odd primary the derivative is $2K_o k_B$. Dimension derivatives are finite differences of the family kernels. The five shared low inputs S, E, $\lambda_{\sigma\sigma\epsilon}$, $\lambda_{\epsilon\epsilon\epsilon}$ and c are varied coherently through Eq. (G.32): varying an external dimension changes the powers in the integrands and all affected blocks, as well as the associated exchanged operator, so these inputs are varied globally and not counted again as independent entries of the σ, ϵ or T rows.

For every other input u_k with quoted error e_k, the scaled displacement is $x_k = (u_k - u_k^{(0)})/(2e_k)$. The range $|x_k| \le 1$ is a twice-quoted-error hypothesis, not a two-standard-deviation statement. The coupling derivatives are analytic. For an exchanged dimension, the coefficient of the integrated objective is the centred kernel difference at $\Delta \pm e_k$ multiplied by the full scale $2e_k$. For a shared low input it is the centred difference of the full ratio at $u_k^{(0)} \pm 2e_k$. Nonlinear corrections to this affine objective are allowed separately (Appendix G.10).

Retained-input bound for the denominator. The conversion of a numerator range into a ratio interval also needs a bound on the denominator. Every retained independent input is placed within twice its quoted error, keeping shared quantities coherent. For each nonleading family the dimension is sampled at $\Delta + n e_\Delta$, $n = -2, \dots, 2$, together with the coupling corners $f \pm 2e_f$ and, for an even family, $h \pm 2e_h$; if the interval of f contained zero, $f = 0$ would be added to the candidates, because b_0 depends on f^2 (no retained row triggers this). The smallest and largest sampled shifts of each family's contribution to b_0 are kept and summed over families; the shared low inputs are evaluated at their coherent $\pm 2e_k$ endpoints and added. This calculation uses descendant order 20 and 80×80 nodes. The sampled envelope is

$$-0.00142738811843 \lesssim \Delta b_{\mathrm{in}} \lesssim 0.00142355554030, \tag{G.34}$$

rounded outward to $|\Delta b_{\mathrm{in}}| \le 0.001428$. It does not impose crossing or the $h_{\epsilon'}$ restriction of Appendix G.9, so it remains applicable after those constraints shrink the input region. It is a sampled numerical allowance, not an interval-arithmetic extremum over every continuous parameter value, and its adequacy is one of the assumptions of the estimated range.

G.9 Conditional control of the omitted spectrum

The hypotheses and their limitations. The main scenario assumes the gaps

$$G_{\mathrm{even}} = 8, \qquad G_{\mathrm{odd}} = 7 \tag{G.35}$$

of Sec. 4.3: every unlisted primary exchanged in the relevant scalar OPEs has dimension at least the corresponding G. This allows arbitrary omitted spectral weight above the thresholds, subject to crossing and positivity; it does not assert that the retained list is complete up to its largest dimension, 24. Neither threshold is proved, and additional unlisted low-dimensional operators remain a possible cause of failure of the narrow interval. The input ranges are hypotheses as well: the actual retained CFT data are assumed to lie inside the twice-quoted-error box. Exact crossing and reflection positivity are imposed on the full theory, not on the finite retained approximation alone.

Evidence motivating the thresholds, not proving them. The gap hypotheses refer to spatial-parity-even primaries in the indicated internal $\mathbb{Z}_2$ sectors. Spatial and internal parity are different labels: a spatial-parity-odd state does not contribute to these two-scalar OPEs merely because its internal parity is appropriate. The selected list has 11 even primaries below dimension 8. Published spectral extractions and fuzzy-sphere studies [14, 18, 19, 20] give the following nearby unlisted families; the entries are estimates, not gap bounds.

internal parity	spin	estimated dimension	evidence and status
odd	3	7.53(6)	fuzzy sphere [20]; an additional trajectory is also discussed in extremal spectra
odd	6	7.73(5); about 7.3	Ref. [20]; lower extremal-functional estimate in Ref. [18]
even	0	about 8.55 to 8.92	differing extremal extractions compared in Ref. [18]; not a certified interval
even	2	about 8.6	higher trajectory in the spectral plots of Ref. [18]; approximate reading
even	6	about 8.78	described as an unstable extraction in Ref. [18]; tentative

The fuzzy-sphere results of Ref. [20] explain why an odd threshold of 8 cannot simply be kept: allowed unlisted odd states are estimated below 8. A threshold of 7 leaves only a margin of about 0.3 to 0.7 to the nearby spin-six estimates, and the associated OPE data do not justify assuming that missing states just above the threshold have negligible weight. None of these approximate couplings is used as a precision input; the remainder bounds must accommodate them. The comparisons certify neither gap. In particular, Figures 1 and 2 of Ref. [14] contain unstable extracted entries, and the main scenario assumes that no such unretained entry below its threshold is an additional genuine exchanged primary; calling an entry an artifact does not establish that assumption. The evidence motivates testing $(8, 7)$, and the feasibility and sensitivity tests (Sec. 4.4 and Appendix G.13) expose its tension and its consequences.

The published constraint on $\lambda_{\epsilon\epsilon\epsilon'}$. Let ϵ' be the next retained internally even scalar after ϵ, with reference dimension 3.82968 and $h_{\epsilon'} = f_{\epsilon\epsilon\epsilon'} = 1.5360$; for a scalar the f and λ conventions coincide. Reehorst's conditional bootstrap analysis gives $\lambda_{\epsilon\epsilon\epsilon'} = 1.5362(12)$ [17]. Its assumptions include an isolated nondegenerate ϵ', the next even scalar above dimension 6 and the next odd scalar above dimension 3, which are compatible with the spectral assumptions used here. To allow for the rounding of the printed bound, the retained-input box is intersected with

$$1.5349 \le h_{\epsilon'} \le 1.5375. \tag{G.36}$$

This applies a published conditional constraint; it is not a new semidefinite-programming computation, nor a weighted average of independent experiments. If its sparsity or nondegeneracy hypotheses are not assumed, the restriction must be removed and the envelope recomputed.

The identical-channel positive remainder. For an identical external scalar of dimension $D \in \{S, E\}$, write its reduced correlator as $P + R$, where P is the retained block sum including the identity and R the missing sum. In radial variables, reflection positivity gives nonnegative expansion coefficients on the collinear axis, $R(r, 1) = \sum_p c_p r^p$ with $c_p \ge 0$; each descendant adds a nonnegative integer level to the primary dimension, so with a missing-primary gap G every $p \ge G$. For $0 < r_1 < r_2$ this gives

$$0 \le R(r_1, 1) \le \Big(\frac{r_1}{r_2}\Big)^G R(r_2, 1), \tag{G.37}$$

and the same gap controls the full omitted family. Choose a real $0 < x < \frac{1}{2}$ and let $A = [x/(1-x)]^{2D}$, $r_1 = \rho(x)$, $r_2 = \rho(1-x)$. Crossing gives $P(x) + R(x) = A[P(1-x) + R(1-x)]$, with $R(x)$ evaluated

at radius r_1 and $R(1-x)$ at r_2. Solving for the remainder at the larger radius and using Eq. (G.37),

$$R(r_2,1) = A^{-1}R(r_1,1) + A^{-1}P(x) - P(1-x) \le A^{-1}\Big(\frac{r_1}{r_2}\Big)^G R(r_2,1) + A^{-1}P(x) - P(1-x), \quad \text{(G.38)}$$

and therefore

$$R(r_2,1) \le \frac{A^{-1}P(x) - P(1-x)}{1 - A^{-1}(r_1/r_2)^G}, \quad \text{(G.39)}$$

provided the denominator is positive. The numerator must be maximized over the allowed retained inputs and include the nonlinear and numerical allowances. A negative upper bound is a consistency failure, not permission to replace a sizeable negative number by zero. Different reference points give valid bounds at different radii: if B_i bounds the reference norm at r_i, then

$$R(r,1) \le \min_{i:\, r \le r_i} B_i \Big(\frac{r}{r_i}\Big)^G. \quad \text{(G.40)}$$

The minimum is justified because each reference bound is separately necessary; it is not a statistical combination. The radial-expansion framework is that of Ref. [38].

Even mixed and odd remainders. The even mixed remainder is not positive, but the Cauchy–Schwarz inequality bounds it by the two identical-channel norms,

$$\big|R_{\sigma\sigma;\epsilon\epsilon}(r,\xi)\big| \le \sqrt{R_{\sigma^4}(r,1)\, R_{\epsilon^4}(r,1)}. \quad \text{(G.41)}$$

This is why the analysis uses all three scalar correlators, including $\langle\epsilon\epsilon\epsilon\epsilon\rangle$, although that correlator does not enter the slope functional directly. In the odd mixed channel positivity applies to

$$Q(\rho,\bar\rho)\, R_{\rm odd}, \qquad Q(\rho,\bar\rho) = \Big|\frac{1+\rho}{1-\rho}\Big|^{2(E-S)}, \quad \text{(G.42)}$$

not to the raw mixed-block coefficients. At real odd-channel reference points $x > \frac{1}{2}$, mixed crossing and Eq. (G.41) bound this normalized odd norm; radial suppression is applied to the normalized quantity, and the bound at a complex integration point is divided by Q again. Applying positivity directly to the raw series, without this prefactor, would be invalid.

Let $U_\sigma(u)$ and $U_\epsilon(u)$ be the identical-channel envelopes of Eq. (G.40), and $d_m^{\rm min}(x)$ the smallest allowed retained mixed residual, with the direct-minus-crossed normalization of Eq. (G.46) and including its nonlinear allowance. At a real odd reference point $x > \frac{1}{2}$ mixed crossing reads

$$0 = d_m(x) + x^{-S-E} R_{\rm odd}(x) - (1-x)^{-2S} R_{\sigma\sigma;\epsilon\epsilon}(1-x). \quad \text{(G.43)}$$

Before the common numerical padding (Appendix G.10), the normalized odd reference norm at $u_x = \rho(x)$ is therefore bounded by

$$B_{\rm odd}(x) = x^{S+E}\, Q(u_x,u_x)\Big[-d_m^{\rm min}(x) + (1-x)^{-2S}\sqrt{U_\sigma(\rho(1-x))\, U_\epsilon(\rho(1-x))}\Big]. \quad \text{(G.44)}$$

Each valid odd reference supplies a radial bound, and the envelope used at radius u is

$$U_o(u) = \min_{x:\, u \le \rho(x)} B_{\rm odd}(x) \Big(\frac{u}{\rho(x)}\Big)^{G_{\rm odd}}. \quad \text{(G.45)}$$

The raw odd remainder at a complex point is bounded by $U_o(|\rho|)/Q(\rho,\bar\rho)$. As in the identical channel, a materially negative upper reference norm is a consistency failure. The identical-channel inequality and the mixed normalization were tested on exact generalized-free correlators; these tests check the algebra of the envelope and its implementation, not the completeness assumptions about the Ising spectrum.

G.10 Joint crossing and input optimization

The crossing residuals. The residuals are unnormalized differences of the direct and crossed correlators, not divided by an average correlator magnitude. With the further sum $\mathsf{V}(z) = \sum_{A\neq\mathbf{1},\ \mathbb{Z}_2\text{-even}} w_{\ell_A} h_A^2 g^{0,0}_{\Delta_A,\ell_A}(z)$, they are, for each retained input configuration,

$$\begin{aligned} d_\sigma(z) &= |z|^{-2S}\big[1+\mathsf{S}(z)\big] - |1-z|^{-2S}\big[1+\mathsf{S}(1-z)\big], \\ d_\epsilon(z) &= |z|^{-2E}\big[1+\mathsf{V}(z)\big] - |1-z|^{-2E}\big[1+\mathsf{V}(1-z)\big], \\ d_m(z) &= |z|^{-S-E}\mathsf{M}(z) - |1-z|^{-2S}\big[1+\mathsf{E}(1-z)\big]. \end{aligned} \tag{G.46}$$

The identity contributions are explicit, and the external dimensions and all occurrences of shared couplings vary coherently. These definitions fix the signs and normalization of the reference bounds, the Jacobian and the affine constraints below.

The affine necessary-constraint problem. Crossing constrains which retained-input shifts and omitted contributions can coexist. The retained inputs enter through the scaled variables x_k of Appendix G.8; except for the intersected $h_{\epsilon'}$ range, $-1 \le x_k \le 1$ is a displacement of at most twice a quoted error. Shared low inputs are parametrized once, and there are 154 variables. For residual type $q \in \{\sigma^4, \epsilon^4, \text{mixed}\}$ at point z_j,

$$d_{qj}(x) = d^{(0)}_{qj} + \sum_k J_{qjk} x_k + e_{qj}(x), \qquad |e_{qj}(x)| \le p_{qj}. \tag{G.47}$$

The columns of J are centred secants of the retained residual at $x_k = \pm 1$, with the other inputs at their reference values. The bound on e_{qj} is a sampled numerical allowance, not a proved uniform bound. For each nonleading family the affine remainder is evaluated at scaled dimension displacements $-1, -\frac{1}{2}, 0, \frac{1}{2}, 1$ and scaled coupling displacements $-1, 0, 1$; the largest absolute remainder at each crossing point is multiplied by 1.5 and summed over families. For each shared low input, twice the absolute difference between the mean of its two endpoint residuals and the reference residual is added. Finally the floor $2\times10^{-10}(1+|d^{(0)}|)$ is added. The factors 1.5 and 2 and the floor are chosen allowances, not consequences of a proved uniform Hessian bound. (The "margins" of Sec. 4.3 are these paddings, the objective allowance 5×10^{-5} and the sampled denominator, reference and quadrature allowances.) Adding separate per-family and low-input samples also does not enclose cross terms between a shared low input and a family input uniformly; the nonlinear tests of Appendix G.14 sample such combined shifts but do not close this logical gap.

The array has 92 point entries: 35 equally spaced real identical-channel references from 0.15 to 0.49, 15 equally spaced odd references from 0.64 to 0.85, and the grid $z = 0.2 + 0.1 i_1 + 0.12 i_2\,\mathrm{i}$ with $i_1 = 0,\dots,6$ and $i_2 = 0,\dots,5$. Some configurations serve more than one role, so 92 counts entries, not independent constraints. Blocks are evaluated at descendant level 40 for the principal optimization.

Exact crossing requires a nonzero omitted contribution t_{qj} to cancel the retained residual. With a conditional bound $t_{qj} \in [-B^-_{qj}, B^+_{qj}]$, Eq. (G.47) becomes two affine inequalities once the nonlinear padding is included. Real-axis positivity gives one-sided improvements; complex-point bounds use magnitudes. The truncated correlator is never forced to satisfy crossing with its tail set to zero.

The allowed residual ranges follow from the reference envelopes. Put $u = |\rho(z)|$, $u' = |\rho(1-z)|$ and $Q_z = Q(\rho(z), \overline{\rho(z)})$. For an identical external scalar of dimension $D \in \{S, E\}$, with $U_S = U_\sigma$ and $U_E = U_\epsilon$, define

$$T^d_D = |z|^{-2D} U_D(u), \qquad T^c_D = |1-z|^{-2D} U_D(u'). \tag{G.48}$$

In general the retained residual lies in $[L_D, U^{\text{res}}_D] = [-(T^d_D + T^c_D),\ T^d_D + T^c_D]$. On the real axis a one-sided improvement needs a sign condition: for $x = \min\{z, 1-z\} < \frac{1}{2}$ put

$$q_D(x; G) = \Big[\frac{\rho(x)}{\rho(1-x)}\Big]^G \Big(\frac{1-x}{x}\Big)^{2D} \le 1. \tag{G.49}$$

Then the direct positive tail on the side of the smaller radius is at most q_D times the crossed one, and the exact retained residual there, $T^c - T^d$ in terms of the actual tails, is nonnegative. At the nominal points the largest values are about 0.6629 for $D = S$ and 0.7121 for $D = E$, excluding the symmetric point $z = \frac{1}{2}$, where the residual vanishes identically. The implementation checks Eq. (G.49) and stops if it fails, so a different gap or point selection cannot silently keep the sharpening. Under this condition the ranges are

$$[L_D, U_D^{\mathrm{res}}] = \begin{cases} [0,\, T_D^c], & 0 < z < \frac{1}{2}, \\ [-T_D^d,\, 0], & \frac{1}{2} < z < 1. \end{cases} \tag{G.50}$$

For the mixed residual put

$$T_m^o = |z|^{-S-E} \frac{U_o(u)}{Q_z}, \qquad T_m^e = |1-z|^{-2S} \sqrt{U_\sigma(u')\, U_\epsilon(u')}. \tag{G.51}$$

The general range is $[-T_m^o - T_m^e,\, T_m^o + T_m^e]$; on the real axis its upper end is reduced to T_m^e, because the direct odd remainder is nonnegative. These bounds constrain the retained residuals, not the missing residual of opposite sign. Writing any allowed range as $[L_{qj}, U_{qj}]$, the necessary inequalities on the affine part are

$$J_{qj} \cdot x \le U_{qj} - d_{qj}^{(0)} + p_{qj}, \qquad -J_{qj} \cdot x \le -L_{qj} + d_{qj}^{(0)} + p_{qj}. \tag{G.52}$$

Each pair is divided by $\max\{|d_{qj}^{(0)}|, \max_k |J_{qjk}|, p_{qj}, 10^{-10}\}$ before it is passed to the solver, which does not change the feasible region. The result is a linear program, solved with the HiGHS backend of SciPy. It is not a full semidefinite bootstrap calculation, and feasibility of these finitely many necessary inequalities is not an existence proof of a CFT.

Iterating input constraints and reference bounds. Starting from the input box and Eq. (G.36), each reference numerator of Eq. (G.39) is maximized over the current feasible input region. Equations (G.39) to (G.45) then give omitted-tail bounds, which tighten the allowed crossing residuals and hence the input region; the references are recomputed in the smaller region and the step is repeated. New inequalities are intersected with the existing ones, not averaged. The nominal calculation makes ten constraint updates; the archived doubled-padding run makes eight (Appendix G.13 compares matched ten-update runs). A fixed number of updates is not a convergence criterion. The intermediate intervals of the iteration use symmetric magnitude envelopes for the tails; they are stages of the construction, not the reported result, which uses the sign-aware odd integral of Appendix G.11.

The reference bounds include a chosen multiplicative factor 1.001, intended to allow for tiny external-dimension effects; this too is a numerical allowance, not an interval-certified treatment of all continuous dependence on the dimensions. The final problem has 552 scaled inequalities in 154 variables, plus the variable bounds. For the contribution of the retained inputs the objective is

$$F_{\mathrm{in}} = \frac{\Delta a_{\mathrm{retained}} - c_1^{(0)} \Delta b_{\mathrm{retained}}}{b_0}, \tag{G.53}$$

and its final linear extrema are

$$-0.001584618275 \lesssim F_{\mathrm{in,lin}} \lesssim 0.001256144319. \tag{G.54}$$

A further 5×10^{-5} is allowed on each side for the nonlinearity of the integrated objective; the separately sampled sum of objective remainders is about 2.3952×10^{-5}, below this allowance. No covariance matrix is inferred by fitting crossing residuals, and no independent-Gaussian interpretation is given to the box.

The central input vector is excluded. Inserting $x = 0$ into the *final* nominal constraints $Ax \le B$ violates the 68 rows of Sec. 4.4 (those with $B_i < -10^{-9}$, all in the $\sigma\sigma\sigma\sigma$ channel); the smallest scaled right-hand side is about -0.727632. The central tabulated data are therefore not a feasible correlator input for the nominal relaxation. This concerns the vector of input data; its integrated value $c_1^{(0)}$ can, and does, lie inside the final interval. The displacement needed for this fixed problem is measured by

$$t_* = \min_{x,t} t \quad \text{subject to} \quad Ax \le B, \quad -t \le x_k \le t, \quad t \ge 0, \tag{G.55}$$

with the $h_{\epsilon'}$ restriction kept. The saved nominal matrices give $t_* = 0.5735097802$, that is $2t_* = 1.1470195604$ quoted errors. This is the smallest *maximum* displacement over the inputs; it does not say that every input moves by that amount and has no Gaussian interpretation. Equation (G.55) freezes the nominal reference bounds, secants and padding, so it does not give the feasibility threshold of a rebuilt input box, whose arrays and iteration change with the box (Sec. 4.4). The exclusion shows a tension among the central extracted spectrum, the missing-spectrum hypotheses and the chosen numerical relaxation. Under a valid remainder construction, underestimated tabulated errors or missing low-dimensional spectral weight are natural explanations; the calculation does not decide between them, does not establish a new operator, and does not prove that these possibilities exhaust the limitations of the numerical envelopes. Nor does it turn an empirical error spread into a rigorous physical range by shifting central values until crossing is feasible.

G.11 Sign-aware integration of the odd missing contribution

Why absolute values lose information. Integrating an absolute pointwise tail bound gives a valid conditional magnitude estimate, but it can be loose. In the normalized odd channel

$$Q\, R_{\mathrm{odd}}(\varrho, \xi) = \sum_{p,j} c_{p,j}\, \varrho^p P_j(\xi), \qquad c_{p,j} \ge 0, \quad p \ge G_{\mathrm{odd}}, \tag{G.56}$$

where the radial-state energy p is a primary dimension plus an integer descendant level, so it is not restricted to integers. A reference norm bounds $\sum_{p,j} c_{p,j} r_{\mathrm{ref}}^p$. For one such state, the Legendre polynomial is integrated against the actual odd contribution to a_0, whose kernel carries an overall minus sign. Separating a positive integration weight W, which includes $4\pi\, \mathrm{d}\mu\, v^{-(3-S)} r^{-S-E}/Q$, define

$$M_j(p; r_{\mathrm{ref}}) = -\int_{\mathcal{R}} W(z) \Big(\frac{|\rho(z)|}{r_{\mathrm{ref}}}\Big)^p P_j\Big(\frac{\mathrm{Re}\,\rho}{|\rho|}\Big). \tag{G.57}$$

A reference norm B then bounds the integrated omitted contribution by B times the extrema of $M_j(p; r_{\mathrm{ref}})$ over the allowed states. The angular integration is done before the worst case is taken, and the positive and negative extrema are in general unequal.

Coverage of spins and continuous exponents. The main calculation evaluates $0 \le j \le 64$, radial energies from G_{odd} to $G_{\mathrm{odd}} + 60$ in steps of 0.1, and 100 quadrature nodes in each variable. Three analytic bounds cover the rest. For higher spins, the Legendre bound implied by Eq. 18.14.7 of Ref. [34],

$$|P_j(\xi)| \le \min\Big\{1,\ \sqrt{\frac{2}{\pi(j+1/2)\sqrt{1-\xi^2}}}\Big\}, \tag{G.58}$$

is used with $j + \frac{1}{2}$ replaced by $J + \frac{3}{2}$, a common upper bound for every $j \ge J + 1$; the cap at 1 prevents an artificial divergence of the angular estimate at the endpoints. Between sampled exponents with step h, the second derivative with respect to p brings down $\log^2(|\rho|/r_{\mathrm{ref}})$, and since $|\rho|/r_{\mathrm{ref}} < 1$ a uniform interpolation allowance is

$$\frac{h^2}{8} \int_{\mathcal{R}} W(z) \Big(\frac{|\rho|}{r_{\mathrm{ref}}}\Big)^{G_{\mathrm{odd}}} \log^2\Big(\frac{|\rho|}{r_{\mathrm{ref}}}\Big). \tag{G.59}$$

Above the last sampled exponent, the absolute kernel is bounded by its magnitude at the gap times $[(2-\sqrt{3})/r_{\mathrm{ref}}]^{60}$; only reference radii larger than $2-\sqrt{3}$ are used for this bound. These are analytic envelopes evaluated by floating-point quadrature. They do not certify the input or nonlinear assumptions, but they control the tails in the state indices instead of stopping a scan at a convenient value.

Numerical effect. In the nominal scenario the symmetric magnitude estimate of the odd contribution to a_0 is 0.01652521405, while the sign-aware upper end is only 0.004677262775:

$$\Delta a_{\mathrm{odd}} \in [-0.01652521405,\ +0.004677262775] \tag{G.60}$$

under the hypotheses, about $[-0.0013485,\ +0.0003817]$ after division by b_0. The best upper reference in this run is $x = 0.76$, with $r_{\mathrm{ref}} \simeq 0.34237382$, and the sampled maximum occurs at $p = 7$, $j = 2$. Raising the quadrature to 160 nodes and the explicit spin cutoff to 96 and halving the exponent step changes the positive allowance to about 0.004672754193; the sampled maximum itself is stable to about 5×10^{-15}, and most of the change is the intentionally reduced interpolation allowance. The main interval uses the larger, original allowance. Because the odd remainder enters with this definite sign, lowering the odd gap mainly lowers the lower end of the range (Sec. 4.4).

G.12 The conditional result and its error budget

Combining numerator and denominator corrections. The nominal tail estimates are

$$|\Delta b_{\mathrm{tail}}| \le 7.02687010 \times 10^{-5}, \qquad |\Delta a_{\mathrm{even}}| \le 0.004552089294, \tag{G.61}$$

together with the asymmetric odd interval of Eq. (G.60). With Eq. (G.54), the allowances in units of c_1 (numerator contributions divided by b_0) are

contribution	lower	upper
crossing-constrained linear input variation	−0.00158462	+0.00125614
nonlinear integrated-input allowance	−0.00005000	+0.00005000
missing even contribution to a_0	−0.00037145	+0.00037145
missing odd contribution to a_0	−0.00134846	+0.00038166
missing contribution to b_0, times $-c_1^{(0)}$	−0.00000157	+0.00000157

These are allowed ranges, not independent standard deviations, and they are added linearly. The denominator is not frozen in the conversion to the ratio; exactly,

$$c_1 - c_1^{(0)} = \frac{\Delta a - c_1^{(0)} \Delta b}{b_0 + \Delta b}. \tag{G.62}$$

The retained-input bound $|\Delta b_{\mathrm{in}}| \le 0.001428$ of Eq. (G.34) plus the tail bound keeps the denominator away from zero. With the total scaled numerator range $[F_-, F_+]$ and $d_b = 0.001428 + |\Delta b_{\mathrm{tail}}|$, the conversion is

$$\Big[c_1^{(0)} + \frac{F_-}{1 - d_b/b_0},\ c_1^{(0)} + \frac{F_+}{1 - d_b/b_0}\Big], \qquad \text{provided} \quad F_- \le 0 \le F_+, \quad 0 \le d_b < b_0. \tag{G.63}$$

The nominal range is $[F_-, F_+] = [-0.003356090758,\ 0.002060824936]$, so both conditions hold. Taking the smaller allowed denominator for both ends is conservative only because the two ends have opposite signs; it is not a general rule for converting ratios. The code checks both conditions with explicit exceptions and does not use a first-order expansion of the denominator for the quoted range. The result is the nominal interval [0.269795922707, 0.275213500746]; the archived doubled-padding run with eight updates gives [0.269273823430, 0.275410362633] (Sec. 4.3 quotes both to seven digits).

Nominal and enlarged padding. The outward-rounded estimated range $[0.2690, 0.2760]$ of Sec. 4.3 covers these two runs. The retained value and the estimated range are reported separately. The range extends about -0.0042 and $+0.0028$ from the rounded retained value 0.2732, not symmetrically; its rounded midpoint, 0.2725, is not a new best fit or a feasible optimized correlator. The range has no claimed confidence level, and its rounding covers the specified doubled-padding test, not arbitrary changes of the input box, the gaps or the numerical allowances. All endpoints remain conditional on the omitted-primary gaps, on the validity of the retained-input ranges and of the nondegenerate-ϵ' constraint, and on the adequacy of the sampled nonlinear, denominator and quadrature allowances. The finite crossing relaxation supplies necessary conditions only; its feasible points and extrema need not be realized by an exact CFT. No assumption-independent physical interval is established.

G.13 Sensitivity scans: method and further rows

The scans of Sec. 4.4 rerun the same final method with ten constraint updates each; they are not alternative estimators to average. In the box scan each quoted error e_k is replaced by me_k, and the integrated sensitivities, the sampled denominator box, the crossing secants and padding, and the reference-bound iteration are all recomputed. The physical $h_{\epsilon'}$ window of Eq. (G.36) is held fixed, not scaled with the table errors, so its limits in the scaled variable are recomputed from the changed parameter scale. In the gap scan the central spectrum and the twice-quoted-error box are unchanged; the same level-40 basis is reused, while the radial suppression, the reference bounds, the crossing constraints and the integrated tails are recomputed at each threshold. A third scan multiplies every crossing padding by a constant with ten updates, which isolates the effect of the padding from that of the iteration count. The rows not shown in Sec. 4.4 are:

scan	setting	ten-update interval
box, m (box radius $\pm 2m$ quoted errors)	0.5	infeasible
	0.7	[0.27050311, 0.27455152]
	0.9	[0.27003354, 0.27506550]
	1.5	[0.26848108, 0.27600079]
	3	[0.26380302, 0.27869935]
gaps $(G_{\text{even}}, G_{\text{odd}})$	$(6.5, 7)$	[0.26850350, 0.27679517]
	$(7, 6)$	[0.26573717, 0.27697080]
	$(6, 6)$	[0.26419915, 0.27833854]
	$(8, 5.5)$	[0.26156563, 0.27688176]
crossing-padding multiplier	1 (nominal)	[0.26979592, 0.27521350]
	2	[0.26927464, 0.27540968]
	3	[0.26887007, 0.27557227]

Every feasible range listed contains the retained value $c_1^{(0)}$, but that does not remove the infeasibility of the central input vector. The scans keep the chosen 5×10^{-5} allowance for the integrated objective while recomputing the other quantities; enlarging the box does not certify this fixed allowance on the enlarged domain, so the wider-box outputs are numerical stress diagnostics, not validated confidence intervals. The sampled nonlinear maxima, the incomplete control of mixed input cross terms and the absence of interval arithmetic apply throughout.

The iterations are finite and not claimed to be converged. At the end of the original histories the nominal lower end still changed by order 10^{-6} per update, and the eight-update $(6, 6)$ history by order 1.5×10^{-4}, so the ten-update $(6, 6)$ row above differs correspondingly from that archived history. Assuming valid outer allowances, further intersections can tighten the feasible region; stopping earlier is no reason to claim a sharper result. Improving the precision requires resolving the tension between the spectrum and the inputs and certifying the numerical envelopes, not rounding the nominal endpoints more finely.

G.14 Validation

Nonlinear crossing and solver checks. The nonlinear test uses a fixed pseudorandom seed, 6087, and 34 configurations: sampled interior points, parameter corners and extrema of the linear optimization. It recomputes the full nonlinear retained residual and compares the affine error with its allocated padding. The largest ratio is 0.666645: below one, but about two thirds of the allowance, so the chosen factor 1.5 provides essentially the sampled margin rather than a large safety reserve. This is evidence that the padding is adequate on the tested configurations, not a proof that it covers every point of the continuous box. The test of the integrated objective samples nine dimension values and the coupling corners family by family, with a separate treatment of the global low inputs; the summed sampled remainder is 2.3952×10^{-5}, against the allowance 5×10^{-5}. Summing such samples cannot by itself give a certified statement. The two final objective linear programs have primal–dual gaps of order 10^{-18} and largest scaled constraint violations of about 8.87×10^{-10}. These tests establish solver consistency for the floating-point affine problem; they do not certify that the problem contains every physical correlator consistent with the stated gaps.

Descendants, integration and positive radial states. Raising the descendant order of the crossing basis from 40 to 48 changes the largest central residual by 5.75×10^{-11}, a Jacobian entry by 3.30×10^{-14} and a padding entry by 7.10×10^{-14}; the bases at orders 32, 40 and 48 are all kept. Changing the quadrature of the integrated envelopes from 80 to 220 nodes changes the even contribution to a_0 by about 1.01×10^{-7}, or 8.3×10^{-9} after division by b_0; the spin and odd magnitudes are much more stable. Separate low-level checks of the normalized odd radial expansion support the use of Eq. (G.42); they supplement the positivity argument, they do not replace it.

Executable guards and the remaining logical limits. The final runner reads $c_1^{(0)}$ and b_0 from the current retained-spectrum output, derives the outward-rounded denominator allowance from the input-box output, and derives the scaled $h_{\epsilon'}$ limits from the actual parameter scale; the chosen objective allowance and the physical $h_{\epsilon'}$ window are explicit settings in one configuration file. The numerical guards raise exceptions that stay active when Python assertions are disabled, among them the sign condition of Eq. (G.49) and the conditions of Eq. (G.63), and the table of the estimated range is generated only after checking that it contains the computed nominal and doubled-padding intervals. The verification script tests the level-40 and level-48 arrays (residual, Jacobian and padding), the normalized radial-state output, the generalized-free remainder tests, the integrated-tail quadrature and the angular and exponent envelopes, as well as the central values and the optimality of the linear programs. These are numerical regression checks: no passing test establishes completeness of the spectrum, a uniform bound on the nonlinearity in the inputs, or the existence of a CFT at an extremum of the linear program.

One limitation of the even mixed-tail construction is its dependence, through Eq. (G.41), on the positive ϵ^4 remainder. The retained $\langle\epsilon\epsilon\epsilon\epsilon\rangle$ correlator can have a substantially larger crossing mismatch than $\langle\sigma\sigma\sigma\sigma\rangle$ or the mixed correlator. It does not enter the slope functional directly, but it weakens the auxiliary norm bound, so a small mixed residual alone cannot replace the full remainder analysis.

G.15 The implemented conformal-block recurrences

The small-z series. For external differences d_a, d_b write

$$g_{\Delta,\ell}^{d_a,d_b}(z) = \sum_{n\geq 0}\sum_{j\geq 0} A_{n,j} r^{\Delta+n} P_j(\eta), \qquad A_{0,j} = \delta_{j\ell}. \tag{G.64}$$

Let $C_{\Delta,\ell} = \Delta(\Delta-3) + \ell(\ell+1)$ and $D_{n,j} = (\Delta+n)(\Delta+n-3) + j(j+1) - C_{\Delta,\ell}$. The coefficients obey

$$\begin{aligned} D_{n,j}A_{n,j} = {} & \frac{j}{2j-1}(\Delta+n+j-2-d_a)(\Delta+n+j-2+d_b)A_{n-1,j-1} \\ & + \frac{j+1}{2j+3}(\Delta+n-j-3-d_a)(\Delta+n-j-3+d_b)A_{n-1,j+1}, \end{aligned} \tag{G.65}$$

where the first term is absent for $j = 0$ and out-of-range coefficients vanish. At a null-state denominator the numerator must vanish consistently; otherwise the routine raises an error. Only the scalar angular coefficient $A_{n,0}$ enters the analytic continuation of Eq. (G.30); for block validation the series is also evaluated at complex points with all angular components.

The radial matrix recurrence. The coefficients $B_{n,j}$ of Eq. (G.28) are computed by matrices acting on vectors of Legendre coefficients (matrices carry hats). Let $\hat{M}$ represent multiplication by ξ and $\hat{D}$ differentiation with respect to ξ:

$$\hat{M}_{j+1,j} = \frac{j+1}{2j+1}, \qquad \hat{M}_{j-1,j} = \frac{j}{2j+1}, \qquad \hat{D}_{k,j} = \begin{cases} 2k+1, & k = j-1,\, j-3,\, \cdots \geq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{G.66}$$

With the Chebyshev matrix polynomials $T_m(\hat{M})$ and $U_m(\hat{M})$, built by the usual second-order recurrence, define

$$\hat{A} = 2\hat{M}^2 - \hat{I}, \qquad \hat{B} = 2\hat{M}(\hat{I} - \hat{M}^2)\hat{D}, \qquad a = -\frac{d_a}{2}, \quad b = \frac{d_b}{2}, \quad c = a+b, \quad p = ab. \tag{G.67}$$

(These a, b, c, p and the u below are local symbols: here c is not c_T/c_B, and p and u are not the radial energy and radius of the preceding subsections.) Let $\boldsymbol{B}_n$ be the vector of $B_{n,j}$; $\boldsymbol{B}_0$ has only its entry ℓ equal to one. For each $1 \leq m \leq n$ put $u = \Delta + n - m$ and accumulate a right-hand side acting on $\boldsymbol{B}_{n-m}$:

- for even $m = 2(k+1)$, add $-4\big[u(-\hat{A}U_k(\hat{A}) + U_{k-1}(\hat{A}) - \frac{1}{2}\hat{I}) + U_k(\hat{A})\hat{B}\big]\boldsymbol{B}_{n-m}$, with $U_{-1} = 0$;
- for odd m, add $8c\big[u\,T_m(\hat{M}) + (\hat{M}^2 - \hat{I})U_{m-1}(\hat{M})\hat{D}\big]\boldsymbol{B}_{n-m}$;
- for every m, add $16p\,(-1)^{m-1} m\,T_m(\hat{M})\boldsymbol{B}_{n-m}$.

The jth component of the total right-hand side, divided by $D_{n,j}$, is $B_{n,j}$. The same quadratic Casimir eigenvalue governs both series, but the recurrences and the evaluation coordinates differ. The implementation uses matrices of size $\ell + N + 8$ and returns angular spins up to $\ell + N$. Denominators smaller than 10^{-9} are treated as null candidates, whose numerator residual must be below 10^{-8}; these are floating-point consistency thresholds, not regulators of a genuine nonzero null-state obstruction, and the code fails rather than keep an inconsistent coefficient.

G.16 The retained input spectrum

Table 6 lists the central values actually evaluated, grouped by internal $\mathbb{Z}_2$ parity and by the trajectory labels `ss0`, `ee0`, `ss1`, `se0` and `other`. These labels are bookkeeping inherited from the published selection; they do not prove a classification of the CFT spectrum. All rows come from the stable-spectrum tables of Ref. [14], except that the leading scalars and the stress tensor (marked †) are replaced by the inputs of Ref. [15] and by Eq. (G.32). For even rows the columns are $(f, h) = (f_{\sigma\sigma A}, f_{\epsilon\epsilon A})$, and for odd rows the single coefficient is $f_{\sigma\epsilon B}$, all in the f normalization of Appendix G.6. The overall sign of a primary is conventional, but the signs of the even products are kept. The decimal places specify the numerical inputs, not a guaranteed physical accuracy. The package's input files also hold every quoted dimension and coupling error, the values before replacement and the source table of each row; the stress-tensor row has zero independent errors because its variations are inherited from the shared low inputs, which is not a claim that the stress couplings are exact.

Table 6: The retained input spectrum of the c_1 package: 37 internally $\mathbb{Z}_2$-even and 23 internally $\mathbb{Z}_2$-odd primaries, with central values. †: replaced by Ref. [15] (ϵ, σ) or by the Ward identity, Eq. (G.32) (T, at $\Delta = 3$).

family	ℓ	Δ	f	h
internally $\mathbb{Z}_2$*-even*: $f = f_{\sigma\sigma A}$, $h = f_{\epsilon\epsilon A}$				
`ss0`†	2	3	0.3261376622	0.8891467104
`ss0`	4	5.022665	0.069076	0.24792
`ss0`	6	7.028488	0.0157416	0.066136
`ss0`	8	9.031023	0.003685	0.017318
`ss0`	10	11.0324141	0.00087562	0.0044811
`ss0`	12	13.033286	0.00020992	0.00115174
`ss0`	14	15.033838	5.065×10^{-5}	0.00029484
`ss0`	16	17.034258	1.228×10^{-5}	7.517×10^{-5}
`ss0`	18	19.034564	2.98935×10^{-6}	1.91408×10^{-5}
`ss0`	20	21.0347884	7.2954×10^{-7}	4.8632×10^{-6}
`ss0`	22	23.034983	1.78412×10^{-7}	1.23201×10^{-6}
`ee0`	4	6.42065	0.0019552	−0.110247
`ee0`	6	8.4957	0.000472	−0.0431
`ee0`	8	10.562	0.0001084	−0.0139
`ee0`	10	12.5659	2.598×10^{-5}	−0.004437
`ee0`	12	14.633	6.1×10^{-6}	−0.001224
`ee0`	14	16.6174	1.417×10^{-6}	−0.0003791
`ee0`	16	18.678	3.547×10^{-7}	-9.72×10^{-5}
`ee0`	18	20.654	7.99×10^{-8}	-2.84×10^{-5}
`ee0`	20	22.651	1.83×10^{-8}	-7.58×10^{-6}
`ss1`	0	3.82968	0.053012	1.536
`ss1`	2	5.50915	0.0105745	0.69023
`ss1`	4	7.38568	0.00237745	0.22975
`ss1`	6	9.32032	0.00055657	0.06949
`ss1`	8	11.2751	0.00013251	0.0198
`ss1`	10	13.241	3.234×10^{-5}	0.005459
`ss1`	12	15.2301	7.64×10^{-6}	0.001538
`ss1`	14	17.1944	1.93×10^{-6}	0.000386
`ss1`	16	19.195	4.568×10^{-7}	0.0001107
`ss1`	18	21.172	1.153×10^{-7}	2.798×10^{-5}
`ss1`	20	23.167	2.74×10^{-8}	7.45×10^{-6}
`other`†	0	1.41262528	1.05185373	1.53244304
`other`	2	7.0758	0.0004773	0.21882
`other`	4	8.941	0.0001173	0.08635

family	ℓ	Δ	f	h
other	6	10.975	2.437×10^{-5}	0.02775
other	0	6.8956	0.0007338	0.1279
other	0	7.2535	0.000162	0.1874
internally $\mathbb{Z}_2$*-odd*: $f = f_{\sigma\epsilon B}$				
se0	2	4.180305	0.38915941	
se0	3	4.63804	0.1385	
se0	4	6.112674	0.1077052	
se0	5	6.709778	0.04191549	
se0	6	8.08097	0.0286902	
se0	7	8.747293	0.01161255	
se0	8	10.0623	0.00745	
se0	9	10.77075	0.003115	
se0	10	12.0492	0.00194	
se0	11	12.787668	0.000823634	
se0	12	14.0383	0.0004983	
se0	13	14.80006	0.000215	
se0	14	16.0305	0.0001291	
se0	15	16.81009	5.587×10^{-5}	
se0	16	18.025	3.13×10^{-5}	
se0	17	18.81794	1.44219×10^{-5}	
se0	18	20.01947	8.442×10^{-6}	
se0	19	20.8246	3.69×10^{-6}	
se0	20	22.0152	2.131×10^{-6}	
se0	21	22.83035	9.512×10^{-7}	
other†	0	0.518148806	1.05185373	
other	0	5.2906	0.057235	
other	2	6.9873	0.017413	

H Reproduction

Everything in this note is reproduced from the folder `SRI/` of the public repository; paths below are relative to it unless they are said to be in a package.

H.1 Layout

- `code/<pkg>/`: the code and records of each package. Production code is included verbatim, at its git commit or as a dated snapshot with a sha256 list (`public/SNAPSHOT.sha256`); the tools added for the public replay are in `public/` and `public.mk`.
  - `dual_d1`: the compendium of the dual of the chain (C_3^{SR} to C_8^{SR}, the certificates, the record of C_8^{SR}), pruned to what the replay and the cited checks need;
  - `indep_d1`: the independent re-derivation of the inputs of C_5^{SR} to C_8^{SR};
  - `integ_d1`: the magnetization-scheme calculation of C_8^{SR} (Appendix I): code, frozen records and the engine outputs they are rebuilt from;
  - `c2_closed_d2`: the final-method study of c_2 and the three Monte Carlo configurations of the review;
  - `c2_cylinder_d2`: the cylinder-Hamiltonian solver, its cluster records and the continuum analysis;
  - `c1_d3`: the c_1 package (review revision of 8 October 2026) with the review's two re-implementations in `review_indep/`;

  - `closure_d1`: the checks of the two arguments of Appendix D (no quoted number): an independent check of the closure argument's finite algebra (matrix-word charges, the radial-power identity, the $\mathcal{O}_g\mathcal{O}_g$ product), and in `prescription/` the prescription argument's own checker (rerun by `make test`) with an independent re-derivation of the β recurrence.

- `docs/<pkg>/`: the documentation of each package (README, REPRODUCE, PROVENANCE, DATA, REVIEWS, requirements, `results_map.md` with every quoted number, its record, rule and tolerance, and TESTS with the export tests on the symmetry cluster).
- `results/quoted_numbers.json`: every quoted number of the paper with its macro and its text as printed, the TeX of the two closed-form equations of the dual, and the cells of the table of short-range coefficients; `results/sr_values.json`: machine-readable values for other folders (Sec. H.4).
- `verify/`: `sri_lib.py` (the replay helper: PASS/FAIL/INFO/SKIP table, rounding rule) and `check_sr_values.py`; `note/`: this note and `tools/check_note.py`.

H.2 Commands and cost

From `SRI/`:

```
make test          # fast tests of every package              about 12 min
make replay        # every quoted number from shipped records about 17 min
make fetch-data    # the hosted inputs (Zenodo data record; see below)
make replay-full   # the heavier reruns                       about 35 min
make check-values  # results/sr_values.json against the tree
make note          # tools/check_note.py of this note
make -C code/<pkg> -f public.mk replay   # one package (also test, replay-full)
```

Measured on the symmetry cluster (seconds, Intel Xeon Gold 6148 / AMD EPYC 7532 nodes, Python 3.12, unpacked in node-local storage; `docs/<pkg>/TESTS_<pkg>.md`):

package	`test`	`replay`	`replay-full`
`dual_d1`	23 / 26	65 / 70	544 / 556 (a)
`indep_d1`	258 / 308 (b)	229 / 264	(c)
`integ_d1`	249 / 249 (b)	78 / 85	(c)
`c2_closed_d2`	4 / 4	39 / 41	–
`c2_cylinder_d2`	170 / 152	3 / 3	10 / 9
`c1_d3`	16 / 14	469 / 484 (d)	1474 / 1486

(a) After `fetch-data`. (b) On 8 cores. (c) Prints the production rerun. (d) Plus 66 s (Intel) for the two re-implementations in `review_indep/`. Further targets: `indep_d1 replay-raw RAW=<dir>` (349 s; the assembly from the raw per-sector engine outputs), `c2_closed_d2 rerun-mc-smoke` (24 s) and `mc-batch-check` (169 s; reruns the first batch of one Monte Carlo chunk with the production flags; bitwise identical on AMD, within 1.6×10^{-11} of the largest entry on Intel).

Rule. A replay recomputes each quoted number of its package from shipped records, rounds it half-up to the precision at which the paper prints it, and compares it with the printed text in `results/quoted_numbers.json` (for a value with its uncertainty in parentheses, the uncertainty is in units of the last digit). It prints a PASS/FAIL table and exits non-zero on any FAIL. Numbers that the note prints but the paper does not, and literature values, give INFO rows. The regenerated records are compared with the shipped ones within the tolerances of Table 7.

H.3 Map of the quoted numbers

Table 7 gives, for each result, the package, the record (path in the package), the target that checks it and the tolerance beyond the rounding rule; `results/quoted_numbers.json` gives the paper's macro for each. In the table and below, G stands for `results/continuum_2026-10/generations/20261015T000632Z_99019_1e66b9`, the current generation of the cylinder analysis (named by `results/continuum_2026-10/CURRENT`).

Table 7: Reproduction map. Target: `make -C code/<package> -f public.mk <target>`, or `make <target>` from `SRI/`.

result	package, target	record (in the package)	tolerance
C_1^{SR}, C_2^{SR}	`dual_d1`, replay	Refs. [3, 4]; registry `data/coefficients.json`	INFO (literature)
C_3^{SR} to C_7^{SR}	`dual_d1`, test, replay	Eq. (2.12) with mpmath at 50 digits; `data/coefficients.json`	rounding
Eq. (2.12)	`dual_d1`, test, replay	`data/coefficients.json`, the closed forms as the compendium's TeX states them (verbatim blocks in `public/compendium_extract.json`), the candidates proved by the C_6^{SR}, C_7^{SR} input maps	exact equality of every rational coefficient
certificates of C_5^{SR} to C_7^{SR}	`dual_d1`, replay; replay-full	C_5^{SR} (D_3) and C_6^{SR} certificates, C_6^{SR} and C_7^{SR} input maps, C_7^{SR} certificate at weights 2 to 5 (replay); the complete certificate task, with the hosted weight-6 part (replay-full)	exact zero residual (exit 0); input-map reports equal the recorded ones
$\pi_0 C_8^{\mathrm{SR}} = 88$	`dual_d1`, test, replay	`data/archived/C8_rational_completion/result.json /C8_weight_zero`	exact: the record, the full-Jacobian script on the 35 rational inputs and two separate rational sums agree
C_8^{SR}	`dual_d1`, test, replay	`data/archived/C8_zeta_candidate/result.json /C8_assembled_without_candidate`	rounding; recomputed from its 36 inputs within 10^{-95}
"about 85 digits"	`dual_d1`, test	the record and its inputs	printing floor about 2×10^{-85} (INFO)
Eq. (2.15)	`dual_d1`, test, replay	`data/coefficients.json` ($n = 8$), `result.json`	exact coefficients; residual 3.085×10^{-86} below the printing floor $10^{-(\mathrm{digits}-15)}$
fresh P_6 evaluations	`dual_d1`, replay-full	the 66 404-term P_6 expression at 70 and 100 digits	
largest input pull through C_7^{SR}	`indep_d1`, replay	`RESULTS_independent.json` against `spec/comparison/archive_comparison.json`; `RESULTS_archive_comparison.json`	INFO (not printed by the paper; $0.915 \to 0.9$); regenerated records below
C_8^{SR} independent and its pull	`indep_d1`, replay (replay-raw)	`RESULTS_C8_independent.json /step1/outputs/C8`; `RESULTS_C8_archive_comparison.json`; `public/pooled_sector_record.json`	INFO (not printed by the paper); central values 10^{-24} relative, linear and replicate errors 10^{-12}, sector-pooled errors 10^{-6} of the quoted error

result	package, target	record (in the package)	tolerance
C_8^{SR}, magnetization scheme (Eq. (2.17))	`integ_d1`, replay	`results/C8_record_v2.json` `/C8`, `/C8_error`; every record from two bubbles to C_8^{SR} rebuilt from `results/m*_runs/` by the verbatim gates	rounding; records byte-identical (sha256)
numbers of Appendix I (inputs, gates, comparison)	`integ_d1`, replay	the records and the outputs of the verbatim tools	25-digit inputs equal to the record; "to 10^{-n}" means in $[10^{-n-1/2}, 10^{-n+1})$; counts equal
engine builds and production sectors	`integ_d1`, test	`results/builds.json`; sectors of four production chunks rerun	build ids equal; $\lvert\mathrm{rerun} - \mathrm{shipped}\rvert \le 4\varepsilon B$ per coefficient (B the engine's bound)
c_2 closed form (Eq. (3.9))	`c2_closed_d2`, test, replay	`outputs/summary.json` `/C2`; the study's eight-stage rerun	rounding; 50-digit evaluation within 10^{-45}
c_2 Monte Carlo	`c2_closed_d2`, test, replay	`mc/run_{A,B,C}/out_*.txt` (142 chunks); `public/mc_combined.json` `/combined/C2`, `/combined/C2_se`	rounding; recombination 10^{-12} relative; `summ.py` output equal to the production logs
c_2 cylinder, its relative difference from the closed form and $N_{\max}$ (also the c_2 row of the paper's table of perturbative Hamiltonian truncation)	`c2_cylinder_d2`, replay	`G/C2_estimate.json` `/final/C2`, `/final/dC2`, from `G/continuum_points.json`	rounding ($N_{\max}$: INFO, not printed by the paper); fits recomputed twice (verbatim and independent) within 10^{-9} relative
same, from the raw records	`c2_cylinder_d2`, replay-full	16 production, 25 extraction-check and 17 archived records, auxiliary inputs	discrete choices equal; chain 10^{-7} absolute; fit diagnostics $10^{-6} + 10^{-3}\lvert x\rvert$
c_1 ($d = 3$)	`c1_d3`, replay	`outputs/tables/current_summary.json` `/retained/C1`; the 20-stage workflow rerun from scratch	rounding; rerun within 5×10^{-10}
c_1 estimated range	`c1_d3`, replay	`refinement/final_8_7_pad1_summary.json` and `refinement/final_8_7_pad2_summary.json` (`/interval`) of the rerun; `current_summary.json` `/recommended_working_summary/interval`	rounding; envelopes within 2×10^{-8}
two independent implementations of c_1 (Table 2)	`c1_d3`, replay	`review_indep/` (two scripts) on `data/inputs/retained_spectrum.json`	each within 10^{-12} of the package; their agreement, a bound, INFO (not printed by the paper)
feasibility and sensitivity (Sec. 4)	`c1_d3`, replay-full	the review-revision diagnostics and the 11 gap and margin and 8 box scenarios	status and endpoints within 2×10^{-8}
c_1 BRRZ	`c1_d3`, replay	Ref. [12]	INFO (literature)

result	package, target	record (in the package)	tolerance
numbers of orders (eight computed, seven in closed form, two published)	`dual_d1`, test	`data/coefficients.json`	equal; INFO for those the paper does not print

H.4 Values for other folders

`results/sr_values.json` publishes each value with all the digits of its record, its status, and the file and JSON pointer where it lives in the tree; `make check-values` resolves every entry. The stable pointers are

pointer	quantity	file (in `SRI/code/`)
`/d1/C/1` to `/d1/C/7`	C_1^{SR} to C_7^{SR}	`dual_d1/data/coefficients.json`
`/d1/C/8`	C_8^{SR} (record)	`dual_d1/data/archived/C8_zeta_candidate/result.json`
`/d1/magnetization/C8`	C_8^{SR}, magnetization scheme	`integ_d1/results/C8_record_v2.json`
`/d2/c2/closed`	c_2, closed form	`c2_closed_d2/outputs/summary.json`
`/d2/c2/mc`	c_2, Monte Carlo	`c2_closed_d2/public/mc_combined.json`
`/d2/c2/cylinder`	c_2, cylinder	`c2_cylinder_d2/G/C2_estimate.json`
`/d3/c1/value`	c_1	`c1_d3/outputs/tables/current_summary.json`
`/d3/c1/range`	estimated range of c_1	`c1_d3/outputs/tables/current_summary.json`
`/d3/S`, `/d3/E_SR`	$\Delta_\sigma^{\mathrm{SR}}$, $\Delta_\mathcal{E}^{\mathrm{SR}}$ ($d = 3$)	`c1_d3/data/inputs/low_inputs.json`

The resummation folder reads C_1^{SR} to C_8^{SR}, c_2 (closed form), c_1 with its estimated range, and the two $d = 3$ inputs at these pointers.

H.5 What cannot be reproduced

- *The compendium's raw integrals cannot be regenerated*, from this folder or from the compendium's original archive: 21 ordered-integration stage programs are missing from the archive, so the documented raw regenerations stop at the first missing stage. What is offered instead is the replay of every quoted number from the shipped records, the exact certificate checks of C_5^{SR} to C_7^{SR}, and the independent re-derivation of every input of C_5^{SR} to C_8^{SR} (`indep_d1`).
- Six inputs of C_8^{SR} ($W, S, L_3, M_{21}, M_{12}, N_2$) are single numerical records with no exact expression; no C_8^{SR} certificate exists.
- Hosted inputs: the weight-6 C_7^{SR} certificate (57 MB, needed only by `dual_d1 replay-full`) and the raw per-sector outputs of `indep_d1` (131 MB, as a 31 MB archive; needed only by `replay-raw`) are in the project's Zenodo data record, DOI 10.5281/zenodo.23466766; `make fetch-data` downloads and checks them (the certificate can also be taken from the compendium's original archive).
- `dual_d1` needs a case-sensitive work directory: the compendium has two input files whose names differ only in case.
- The base study's checks of the closed expression of c_2 (the integer-interval evaluation to 51 decimal places, its tail inequalities, elliptic integration and local checks; Appendix F.14) are described but their code is not in the release; the value of the closed expression is reproduced

by `c2_closed_d2` (50 digits, not as an interval enclosure). The closure argument's own script of finite checks was not supplied and is not part of the release (Appendix D.15).

- The release has no git history, so it cannot show the order in which the magnetization-scheme calculation committed its designs and its data (Appendix I.7): `integ_d1` checks that the files its tools ask git about equal the committed blobs, and the code inspections checked the order on the history.
- Production reruns (not needed for any replay): `indep_d1` about 2.2×10^4 core-hours on the symmetry cluster; the engine runs of `integ_d1` about 1.6×10^4 core-hours on the symmetry cluster; the Monte Carlo of `c2_closed_d2` about 135 core-hours; the cylinder records of `c2_cylinder_d2` about 1.5 node-hours on the symmetry cluster, about 470 service units on SDSC Expanse, and, again on the symmetry cluster, the two rebuilt $N = 1024$ grids (2.6 to 2.7 hours each) and twelve extraction-check jobs (0.3 to 1.6 hours each). A Monte Carlo rerun is statistically equivalent, not identical, unless the same binary, weights and seeds are used.

H.6 Numbers of this note that the paper does not quote

The decimal strings of Table 2 and Sec. 3.5 are evaluations of the closed forms (mpmath, 60 digits; the same digits as `results/sr_values.json`). The inputs of Table 3, the pulls, the estimators and the checks of Sec. 2.6 are in the result records of `code/indep_d1/` (and its `RESULTS.md`, `RESULTS_C8.md`); the eighth-order sub-results of Sec. 2.5 in those of `code/dual_d1/`; the constants of Sec. 3.5 in `code/c2_closed_d2/outputs/summary.json`; the per-order table and the error budget of Sec. 3.8, and the superseded first analysis, in `G/C2_estimate.json`, `G/REPORT.md` and `results/continuum_2026-10/evidence/rev4/` of `code/c2_cylinder_d2/`; and the intervals and sensitivity tables of Sec. 4 in `code/c1_d3/outputs/tables/` (the box scan of Sec. 4 at $m = 2$ is the package's; the review's own scan gave $[0.26698, 0.27693]$). The NRG residuals of Sec. 2.9 are checked in the NRG folder of the repository [2]. The magnetization-scheme values of Sec. 2.7 and Appendix I other than the one the paper quotes are in the records of `code/integ_d1/results/` (Appendix I.8), and the replay of `integ_d1` checks each of them.

I The eighth order from the integrable point: the magnetization scheme

This appendix gives the third calculation of C_8^{SR} summarized in Sec. 2.7: its derivation, method, checks and records. It starts from the wall gas of Appendix A.3, with the wall weight W_n of Eq. (A.31), but it uses nothing of the matched two-coupling prescription of Sec. 2.1 (no hard core, no finite interval, no endpoint source) and no input of the compendium or of the independent re-derivation. Its code and records are the package `code/integ_d1/` (Appendix H). Its result is numerical in the sense of Sec. 1.3.

Notation. As elsewhere $\bar\delta = 1 - s$; v is the charge of the wall weight W_n (at $\bar\delta = 0$ a pair of opposite walls at distance r has weight r^{-2v}), and $w = 1 - v$. In this appendix h_{m} is a uniform magnetic field (not the current coupling h of the dual), $\bar m$ is the magnetization, the mean of $m(X)$, ξ is the expansion variable of the magnetization, $\hat u$, $\hat z$ and $\hat F$ are the couplings and the beta function of the magnetization scheme (Appendix I.4), and f_{ijk} is the coefficient of $\hat u^i \hat z^j \bar\delta^k$ in $\hat F$. A "bubble" is a pair of consecutive opposite walls, which bounds one reversed domain.

I.1 The integrable point

At $s = 1$ the wall gas, with the weights $\prod_{i<j}(X_j - X_i)^{2vq_iq_j}$, is the Coulomb gas of Anderson, Yuval and Hamann for the anisotropic Kondo model [5, 6], and equivalently that of the ohmic spin–boson model with dissipation strength $\alpha = v$ [7]. These models are solvable by Bethe ansatz [8, 9]. For $\bar\delta > 0$ the interaction $f_{\bar\delta}$ of Eq. (A.5) breaks the integrability, and the long-range fixed point lies at $\bar\delta > 0$. The integrable point can therefore supply the $\bar\delta = 0$ part of the flow exactly, to all orders in the wall fugacity, while the dependence on $\bar\delta$ needs perturbation theory in kink clusters.

The observable is the magnetization of the gas on the infinite line in a uniform field $h_{\mathrm m}$ coupled to $m(X)$. Relative to the all-up configuration, a configuration of walls carries the extra weight $e^{-2h_{\mathrm m}\Xi}$, where Ξ is the total length of the reversed domains (as in Ξ_{2m} of Eq. (A.31)). No hard core is imposed: the short-distance divergences near $v = 1$ are regulated by analytic continuation in v (Appendix I.5).

I.2 The magnetization at $s = 1$ from the Bethe ansatz

Bethe-ansatz equations. The spin sector of the anisotropic Kondo model is an inhomogeneous six-vertex (XXZ) chain in its gapless regime, with anisotropy γ, N bulk sites at rapidity 0 and the impurity at rapidity Θ; $\Theta \to \infty$ is the weak-coupling scaling limit [8, 9]. With the kernels

$$a_n(\theta) = \frac{1}{2\pi}\,\frac{\gamma\sin(n\gamma)}{\cosh(\gamma\theta) - \cos(n\gamma)}, \qquad \hat a_n(\omega) = \int a_n(\theta)\,e^{\mathrm{i}\omega\theta}\,\mathrm{d}\theta = \frac{\sinh\bigl((\pi - n\gamma)\omega/\gamma\bigr)}{\sinh(\pi\omega/\gamma)}, \tag{I.1}$$

the roots fill $\theta < \theta_F$ in a field, and the impurity root density σ_{imp} solves

$$\sigma_{\mathrm{imp}}(\theta) + \int_{-\infty}^{\theta_F} a_2(\theta - \theta')\,\sigma_{\mathrm{imp}}(\theta')\,\mathrm{d}\theta' = a_1(\theta - \Theta) \quad (\theta < \theta_F), \qquad \bar m = 1 - 2\int_{-\infty}^{\theta_F}\sigma_{\mathrm{imp}}(\theta)\,\mathrm{d}\theta. \tag{I.2}$$

The impurity magnetization is $\frac12 - \int\sigma_{\mathrm{imp}}$, and $\bar m \in [0,1]$ is twice it. The zero-field bulk density is $1/(4\cosh(\pi\theta/2))$, so spinon energies scale as $e^{-\pi\theta/2}$, and $h_{\mathrm m}/T_K \propto e^{-\pi L/2}$ with $L = \theta_F - \Theta$ and T_K the Kondo scale.

Wiener–Hopf solution. With $x = \theta_F - \theta$ and $\varsigma(x) = \sigma_{\mathrm{imp}}(\theta_F - x)$, Eq. (I.2) becomes $\varsigma(x) + \int_0^\infty a_2(x-y)\varsigma(y)\,\mathrm{d}y = a_1(L - x)$ for $x > 0$. With $p = \pi/\gamma$ the kernel factorizes,

$$\begin{aligned} 1 + \hat a_2(\omega) &= \frac{2\sinh((p-1)\omega)\cosh\omega}{\sinh(p\omega)} = \Phi_+(\omega)\Phi_-(\omega),\\ \Phi_-(\omega) &= \sqrt{\frac{2\pi(p-1)}{p}}\,\frac{\Gamma(1 + \mathrm{i}p\omega/\pi)\,e^{-\mathrm{i}\omega c/\pi}}{\Gamma(1+\mathrm{i}(p-1)\omega/\pi)\,\Gamma(\frac12 + \mathrm{i}\omega/\pi)}, \end{aligned} \tag{I.3}$$

with $c = p\log p - (p-1)\log(p-1)$ and $\Phi_+(\omega) = \Phi_-(-\omega)$, which tends to 1 at infinity. The standard argument gives $\int_0^\infty\varsigma = [e^{\mathrm{i}\omega L}\hat a_1/\Phi_-]_+(0)/\Phi_+(0)$, where $[\cdot]_+$ is the part analytic in the upper half plane. With $\omega = \mathrm{i}\pi t/p$ the Gamma functions combine into one Mellin–Barnes integral,

$$\bar m = 1 + \frac{1}{\sqrt\pi}\,\frac{1}{2\pi\mathrm{i}}\int_{c_0-\mathrm{i}\infty}^{c_0+\mathrm{i}\infty}\frac{\Gamma(t)\,\Gamma\bigl(\frac12 - (1-v)t\bigr)}{\Gamma(1+vt)}\,\xi^{-t}\,\mathrm{d}t, \qquad -1 < c_0 < 0, \tag{I.4}$$

with $v = 1 - \gamma/\pi$ and $\xi = e^{\gamma(L + c/\pi)} \propto (h_{\mathrm m}/T_K)^{-2(1-v)}$. The identification $v = 1 - \gamma/\pi$ is forced by the exponent: perturbation theory in the wall fugacity y proceeds in powers of $y^2 h_{\mathrm m}^{-2(1-v)}$, the Bethe ansatz in powers of $h_{\mathrm m}^{-2\gamma/\pi}$. The free-fermion point $\gamma = \pi/2$ is then the Toulouse point $v = \frac12$, and the isotropic limit $\gamma \to 0$ is $v \to 1$.

The two expansions. Closing the contour to the left picks up the poles of $\Gamma(t)$ at $t=-n$ and gives the large-field series

$$\bar{m} = 1 - \frac{1}{\sqrt{\pi}} \sum_{n\geq 1} \frac{(-1)^{n+1}\,\Gamma\big(\frac{1}{2}+n(1-v)\big)}{n!\,\Gamma(1-nv)}\,\xi^{n}. \tag{I.5}$$

Closing to the right picks up the pole of $\Gamma(t)$ at $t=0$, which cancels the 1, and the poles at $t=s_m=(m+\frac{1}{2})/(1-v)$, and gives a series odd in h_{m},

$$\bar{m} = \frac{1}{\sqrt{\pi}} \sum_{m\geq 0} \frac{(-1)^{m}\,\Gamma(1+s_m)}{m!\,(m+\frac{1}{2})\,\Gamma(1+vs_m)}\,\xi^{-s_m}. \tag{I.6}$$

At $v=0$ Eq. (I.5) sums to $\bar{m}=(1+\xi)^{-1/2}$, the free spin. At $v=\frac{1}{2}$ only odd powers survive, since $1/\Gamma(1-\frac{n}{2})=0$ for even n, and $\bar{m}=\frac{2}{\pi}\arctan(h_{\mathrm{m}}/\Gamma_0)$, the resonant level of width Γ_0.

Checks.

- *The Wiener–Hopf algebra.* The integral equation (I.2) was solved numerically by Nyström discretization and compared with Eq. (I.4), including the shift c. At $\gamma=0.4$, 0.7 and 1.2 ($v=0.873$, 0.777, 0.618) and $L=-6$, -2, 0, 2 and 6, where $\bar{m}$ runs from 0.9996 to 4×10^{-5}, the differences in $\bar{m}$ are at most 8×10^{-13}.
- *Against the wall gas, with no Bethe ansatz.* The ratio $R=c_2/c_1^2$ of the coefficients of ξ^2 and ξ in Eq. (I.5) does not depend on the normalization of ξ:

$$R_{\mathrm{BA}} = \frac{\sqrt{\pi}\,\Gamma(\frac{5}{2}-2v)\,\Gamma(1-v)^2}{2\,\Gamma(1-2v)\,\Gamma(\frac{3}{2}-v)^2}. \tag{I.7}$$

  On the gas side R is the connected two-bubble cluster with the field weight,

$$R_{\mathrm{gas}} = \frac{(4v-3)\,A_2}{2\,\Gamma(2-2v)^2}, \qquad A_2 = \Gamma(3-4v)\int_0^1 \big(t(1-t)\big)^{-2v}\Big[I(t)-\frac{1}{2}\Big]\,\mathrm{d}t,$$
$$I(t) = \int_0^\infty \Big[\frac{(t+g)^{2v}(g+1-t)^{2v}}{g^{2v}(1+g)^{2v}} - 1\Big]\,\mathrm{d}g, \tag{I.8}$$

  which converges for $v<\frac{1}{2}$ (where R_{BA} is the analytic continuation of the formula). The two agree exactly at $v=0$ ($R=\frac{3}{2}$) and to 2×10^{-13}, 3×10^{-13}, 4×10^{-12}, 4×10^{-10} and 1×10^{-9} at $v=0.1$, 0.2, 0.3, 0.4 and 0.45, each within the quadrature error of the gas integral.
- *Higher clusters.* The later stages reproduce the exact two-, three- and four-bubble clusters at $\bar{\delta}=0$ (Appendix I.6).

The derivation covers $\frac{1}{2}<v<1$ directly (real roots, $0<\gamma<\frac{\pi}{2}$); for $v<\frac{1}{2}$ the formula is the analytic continuation in v, which the second check tests. The Bethe-ansatz equations and the identification of the gas with the anisotropic Kondo model are taken from the literature; everything from the impurity integral equation on is derived here.

I.3 What the integrable point can and cannot test

At $\bar{\delta}=0$ the change $u\to H/4$ brings the flow of Sec. 2.2 to coordinates in which $H=4u$ exactly. The freedom that remains is the shifts $z\to z+u\,\chi(u,z)$, with u readjusted so that $H=4u$ still holds. Write F_{ij} for the coefficient of $u^i z^j$ of F in these coordinates; $F(0,z)=z$. Through total degree four in (u,z), the order that C_8^{SR} needs, F has ten coefficients with $i\geq 1$. The linearized action of the shifts on them has rank eight (in exact arithmetic, at random flows), so exactly two combinations are invariant:

$$F_{10}=1, \qquad I_2 = F_{20} - \tfrac{1}{2}F_{11} - F_{12}. \tag{I.9}$$

In the inputs of the hard-core scheme (Sec. 2.3), $I_2 = b + \frac{f}{4} - \frac{a}{2} - \frac{c}{8} - A$. With the closed forms of a, b, c, f (Sec. 2.3) and $A = -8 - 4\zeta(2) - 2\zeta(3) + 16\log 2$ (Appendix B.3) it vanishes identically: the rational, $\zeta(2)$, $\zeta(3)$ and $\log 2$ parts cancel separately, and $\log 3$, $\log^2 2$ and $\mathrm{Li}_2(-\frac{1}{2})$ cancel between b and $f/4$. The Bethe-ansatz flow of Appendix I.4, whose $\bar\delta = 0$ part is already in these coordinates, gives $F_{10} = f_{100} = 1$ and $I_2 = f_{200} - \frac{1}{2}f_{110} - f_{120} = 0$ (to 10^{-54} in the first fit and 10^{-63} in the frozen matching). The agreement is a check of the compendium's $s = 1$ sector by integrability.

All other $s = 1$ coefficients through degree four can be changed by the shifts: among them J_9, and with it $A_9[\varepsilon^0\ell^1]$, and the current-type inputs L_3, N_2 and K_v of C_8^{SR} (the direction of J_9 lies in the span of the linearized action). They are coordinate choices of the hard-core scheme, and only C_8^{SR} as a whole is physical. The $s = 1$ solution alone therefore cannot test A_9 or J_9; the calculation below instead computes C_8^{SR} in its own scheme.

I.4 The magnetization scheme

Couplings. For the gas of bare charge v in the field h_{m}, at $0 < \bar\delta < 1$, define

$$K_h = v\,(2h_{\mathrm{m}})^{-\bar\delta}, \qquad 1 + \hat z = K_h\,\bar m^{\,2-\bar\delta}, \qquad \hat u = \frac{2-\bar\delta}{4}\,\frac{\mathrm{d}\log\bar m}{\mathrm{d}\log h_{\mathrm{m}}}. \tag{I.10}$$

Then, identically by these definitions,

$$h_{\mathrm{m}}\frac{\mathrm{d}\hat z}{\mathrm{d}h_{\mathrm{m}}} = (1+\hat z)\,\hat H, \qquad \hat H = 4\hat u - \bar\delta, \tag{I.11}$$

and the only function to be computed is $\hat F$ in

$$h_{\mathrm{m}}\frac{\mathrm{d}\hat u}{\mathrm{d}h_{\mathrm{m}}} = 2\hat u\,\hat F(\hat u,\hat z,\bar\delta), \qquad \hat F = \hat z + \sum_{i+j+k\ge 1} f_{ijk}\,\hat u^i\hat z^j\bar\delta^k, \qquad f_{0j0} = 0, \quad f_{100} = 1. \tag{I.12}$$

The flow is in $\log h_{\mathrm{m}}$, and Eqs. (I.11) and (I.12) have the form of Eq. (A.23) with $(\hat u,\hat z,\hat F,\hat H)$ in place of (u,z,F,H).

Fixed point and invariants. $\hat H = 0$ gives $\hat u_* = \bar\delta/4$ exactly, and $\hat F(\bar\delta/4,\hat z_*,\bar\delta) = 0$ fixes $\hat z_*$. Since $\hat H_{\hat z} = 0$ and $\hat H_{\hat u} = 4$, the trace and determinant of Eq. (A.48) per unit $\log h_{\mathrm{m}}$ are $T_h = \frac{\bar\delta}{2}\hat F_{\hat u}$ and $D_h = -2\bar\delta(1+\hat z_*)\hat F_{\hat z}$ at the fixed point. The field has dimension $1 - \Delta_\sigma = 1 - \bar\delta/2$, so per unit logarithm of length

$$T = \Big(1 - \frac{\bar\delta}{2}\Big)\frac{\bar\delta}{2}\,\hat F_{\hat u}, \qquad D = -\Big(1 - \frac{\bar\delta}{2}\Big)^2 2\bar\delta\,(1+\hat z_*)\,\hat F_{\hat z}, \tag{I.13}$$

and Eqs. (2.6) and (2.8) give the C_n^{SR}. The leading $\hat F = \hat z + \hat u$ gives $T = \bar\delta/2 + O(\bar\delta^2)$ and $D = -2\bar\delta + O(\bar\delta^2)$, hence $C_1^{\mathrm{SR}} = -2$, $C_2^{\mathrm{SR}} = \frac{1}{2}$. The next orders are, for example,

$$C_3^{\mathrm{SR}} = \frac{39}{16} + 2f_{001} - 2f_{011} - \frac{1}{2}f_{110}, \qquad C_4^{\mathrm{SR}} = f_{101} + \frac{1}{2}f_{200} - f_{001}f_{110} - \frac{1}{4}f_{110} - \frac{1}{2}, \tag{I.14}$$

and C_8^{SR} is affine in its two new coefficients, $\partial C_8^{\mathrm{SR}}/\partial f_{202} = 2$ and $\partial C_8^{\mathrm{SR}}/\partial f_{301} = \frac{3}{4}$; the $\bar\delta = 0$ coefficient f_{400} enters as $f_{400}/4$, the counterpart of $J_9/4$ in Eq. (2.10).

Which coefficients, from how many kinks. $\hat u \propto y^2$ counts bubbles, so f_{ijk} comes from connected clusters of $i+1$ bubbles, that is $2i+2$ kinks; k is the number of logarithm insertions from expanding $f_{\bar\delta}$, and the dependence on $\hat z$ comes from the expansion in w about $v = 1$. C_8^{SR} needs the f_{ijk} with $i+j+k \le 4$, 29 of them:

coefficients	content	kinks (integration dimension)	first needed for
f_{ij0}	$\bar\delta = 0$, from Eq. (I.5)	none	C_3^{SR}
f_{0jk}, $k \geq 1$	one bubble, all orders in $\bar\delta$	2 (closed form in derivatives of Γ)	C_3^{SR}
f_{101}; f_{111}; $f_{121}, f_{102}, f_{112}, f_{103}$	two bubbles, one to three logarithms	4 (2D)	C_4^{SR}; C_5^{SR}; C_6^{SR} to C_8^{SR}
f_{201}, f_{211}	three bubbles, one logarithm	6 (4D)	C_6^{SR}, C_7^{SR}
f_{202}	three bubbles, two logarithms	6 (4D)	C_8^{SR}
f_{301}	four bubbles, one logarithm	8 (6D)	C_8^{SR}

The $\bar\delta = 0$ coefficients follow from Eq. (I.5): with $K_h = v$ at $\bar\delta = 0$, Eq. (I.10) expresses $\hat u$ and $\hat z$ through ξ at $v = 1 - w$, and eliminating ξ gives $\hat F(\hat u, \hat z, 0)$ to all orders; through degree four the overdetermined matching has residual 10^{-63}. They include f_{400}: the $s = 1$ part of C_8^{SR} is exact in this scheme, and no nine-kink integral is needed. No current-type input is needed either, since $\hat H$ is exact.

I.5 Kink clusters and their continuation to $v = 1$

Clusters. The $\bar\delta$-dependent coefficients come from the connected Mayer clusters of n bubbles per unit length (field scaled to $2h_{\mathrm{m}} = 1$),

$$J_n = \frac{1}{n!} \int \prod_{a=1}^{n} \mathcal{W}_a \sum_{\text{connected graphs}} \prod_{(ab)} \varphi_{ab}, \tag{I.15}$$

where $\mathcal{W}_a$ is the weight of bubble a (its pair factor and field factor), and the bond φ_{ab} is $C_{ab} - 1$ for disjoint bubbles, with C_{ab} the cross-pair factor of W_n between them, and -1 for overlapping ones, which alternation excludes. The expansion of $f_{\bar\delta}$ to order k gives k logarithm insertions. A symbolic map, derived once per order, turns the Laurent coefficients of the J_n at $v = 1$ into the f_{ijk}.

Radial factor and sectors. Every monomial of the integrand is homogeneous of degree $-2nv$ in the gaps (the cross-pair factors are scale invariant), so after the overall size is scaled out the radial integral is exact: $J_n^{(k)} = \sum_j \Gamma^{(j)}(\alpha_n) A_j^{(k)}$ with $\alpha_n = 2n - 1 - 2nv$, the logarithm insertions giving the derivatives $\Gamma^{(j)}$, and $A_j^{(k)}$ angular integrals over a simplex of dimension $2n - 2$. The angular domain is split into primary Cheng–Wu sectors (the largest gap set to 1), refined by iterated Binoth–Heinrich splitting until every sum of gaps is a monomial times a positive function: 187 sectors for three bubbles and 4777 for four, of which 3445 are computed, using a mirror symmetry of the bubble patterns.

Exponents. In a sector every term is $\prod_k t_k^{a_k + b_k w}$ times an analytic function. A set G of gaps that shrinks together at rate ρ, possibly as several disjoint contacts, contributes $\rho^{a_G + b_G w}$ with $a_G = (|G| - 1) + 2S_G$ at $v = 1$, where S_G is the sum of $q_i q_k$ over all pairs whose separation shrinks with ρ. A single kink–antikink contact gives ρ^{-2+2w}; two simultaneous contacts give ρ^{-3+4w} and three ρ^{-4+6w}. Every Taylor term whose integrated exponent is at most -1 at $w = 0$ is continued analytically, which produces poles in w. No term has $a_k \leq -1$ with $b_k = 0$ (a divergence independent of v) for $n \leq 4$.

Continuation. Each sector variable has $[0, 1] = [0, \tau] \cup [\tau, 1]$. On $[\tau, 1]$ the integral is done by Gauss–Legendre quadrature in $\log t$. On $[0, \tau]$ the Taylor coefficients of the analytic part are

obtained by the trapezoid rule on the circle $|t| = \tau$ and integrated analytically against $t^{c+n} \log^p t$, which continues every non-integrable power. The radius τ is set by the analyticity radius of the regular part: sampled for three bubbles, and for four bubbles a rigorous radius from Rouché's theorem with a joint polydisc check. All quantities are truncated Laurent series in w, so the Laurent coefficients of the clusters at $v = 1$ come out directly. The engine is written in C++, in double-double arithmetic (about 32 digits) for three bubbles; for four bubbles it runs in double precision with double-double accumulators, the largest-bound sectors entirely in double-double. It gives a first-order roundoff bound for every sector.

What the poles test. The raw Laurent coefficients carry information in their pole parts as well as their finite parts, and all of them are computed. The checks are that the f_{ijk} have no negative powers of w (the analyticity relations, listed symbolically for each order), the exact $\bar{\delta} = 0$ clusters from Eq. (I.5), and the closed forms of C_3^{SR} to C_7^{SR}.

I.6 Gates and results

The plan fixed the stages, their checks and their stopping rules before any multi-bubble cluster with logarithm insertions was computed; each stage gated the next.

- *One bubble.* The poles cancel in $\hat{F}(0, \hat{z}, \bar{\delta})$ through $\bar{\delta}^3$, and $f_{001} = -\gamma_E$.
- C_3^{SR}. With $f_{001} = -\gamma_E$, $f_{011} = -\gamma_E - 2\zeta(2)$ and $f_{110} = 4\zeta(2)$, $C_3^{\mathrm{SR}} = \frac{39}{16} + 2\zeta(2)$, to 2×10^{-38} (absolute) numerically.
- *Two bubbles (C_4^{SR}, C_5^{SR}).* The cluster at $\bar{\delta} = 0$, continued to $v = 1$, matches the Bethe ansatz to 4×10^{-32}, and all 36 analyticity relations hold to 10^{-28}. The value of f_{101} implied by C_4^{SR} was printed in advance as a blind check; the computation gave $f_{101} = -36.49853744243762\ldots$. C_4^{SR} agrees to 3×10^{-28} and C_5^{SR} to 10^{-26} (absolute).
- *Three bubbles, one logarithm (C_6^{SR}, C_7^{SR}).* The engine matches the exact three-bubble cluster to 10^{-23} at real v and 10^{-28} at complex v, and the analyticity relations hold to 10^{-27}. With the results hashed before the comparison, C_6^{SR} agrees to 4×10^{-24} and C_7^{SR} to 2×10^{-24} (absolute).
- *Three bubbles, two logarithms (f_{202}).* The $\bar{\delta}^0$ part matches the Bethe ansatz to 4×10^{-23}, the $\bar{\delta}^1$ part the frozen one-logarithm record to 9×10^{-24}, and all twelve $\bar{\delta}^2$ analyticity relations hold to 4×10^{-24}. $f_{202} = 172415.83018563 \pm 1.7 \times 10^{-7}$, the difference between two quadrature grids.
- *Four bubbles, one logarithm (f_{301}).* Three complete runs: a fine and a coarse grid at contour-radius ratio 0.2, and a radius-check run at 0.3; in each, the sectors with the largest roundoff bounds were recomputed entirely in double-double arithmetic (8 at ratio 0.2, 14 at 0.3). The $\bar{\delta}^0$ part matches the exact four-bubble cluster to 4×10^{-11}, and all twelve $\bar{\delta}^1$ analyticity relations hold to 7×10^{-12}. A confirmatory check of the $\bar{\delta}^0$ part at two complex values of v gives 1.7×10^{-14} and 1.6×10^{-12} (the second under amendment A of Appendix I.7).
- *Roundoff, measured.* 56 further sectors were recomputed in double-double arithmetic at the fine grid: the 16 with the next-largest bounds, 24 drawn by bound decile, and 16 from the two top deciles as a held-out sample fixed in advance. The difference between double and double-double is at most 2.5×10^{-3} of the first-order bound in the top 16 sectors and the top decile, which together carry about 98% of the bound, at most 1.3×10^{-2} in the next decile, and reaches 0.41 only in sectors whose bound is negligible.
- *Result.* $f_{301} = -514275.75 \pm 0.17$; the uncertainty, 0.174, is the linear sum of the fine–coarse difference 0.016, the fine–radius difference 0.022 and the validated roundoff estimate 0.137.

The inputs were hashed and frozen before C_8^{SR} was assembled. The same assembly reproduces the closed forms of Eq. (2.12):

	magnetization scheme	closed form	relative difference
C_3^{SR}	5.72736813369645287294483	5.72736813369645287294483	3×10^{-39}
C_4^{SR}	−12.19619149266966002908757	−12.19619149266966002908757	2×10^{-29}
C_5^{SR}	29.23781158099003956882694	29.23781158099003956882694	3×10^{-28}
C_6^{SR}	−58.66569747422530894971715	−58.66569747422530894971715	7×10^{-26}
C_7^{SR}	51.06898861292810883118393	51.06898861292810883118394	4×10^{-26}

and gives

$$C_8^{\mathrm{SR}} = 247.381179745789 \pm 0.131, \tag{I.16}$$

quoted as 247.38118 ± 0.13 in Eq. (2.17). The uncertainty is the linear sum of the propagated input uncertainties: $\frac{3}{4}\times 0.174 = 0.13$ from f_{301}, 3.3×10^{-7} from f_{202} and 3.6×10^{-15} from all other inputs. Table 8 lists the 29 inputs, from which Eq. (I.13) reassembles C_3^{SR} to C_8^{SR}: from the printed 25-digit values C_8^{SR} equals the record to all its printed digits, and C_3^{SR} to C_7^{SR} come out to 10^{-25} to 2×10^{-22} relative (the table above used the unrounded inputs). The computation used about 2.7×10^4 core-hours on the symmetry cluster, including pilots and failed jobs; the engine runs behind the records, which a rerun repeats, took about 1.6×10^4.

Table 8: The 29 inputs of C_8^{SR} in the magnetization scheme, as frozen in the record of C_8^{SR} (Appendix I.8). Index ijk: $\hat{u}^i\hat{z}^j\bar{\delta}^k$, with $i+1$ bubbles and k logarithms. The $\bar{\delta}^0$ entries come from the exact Bethe-ansatz formula (matching residual 10^{-63}); the one-bubble entries are exact or high-precision numerics; the two-bubble entries are good to about 10^{-26} relative and the three-bubble $\bar{\delta}^1$ entries to about 10^{-22}. (a) f_{202} and f_{301} are printed in full only for exact reassembly; they are accurate only to $\pm1.7\times10^{-7}$ and ±0.174.

	monomial	value	source	closed form
f_{100}	$\hat{u}$	1	Bethe ansatz (exact)	1
f_{110}	$\hat{u}\hat{z}$	6.57973626739290574588966 1	Bethe ansatz (exact)	$4\zeta(2)$
f_{120}	$\hat{u}\hat{z}^2$	42.00883821061607434137304	Bethe ansatz (exact)	
f_{130}	$\hat{u}\hat{z}^3$	174.589640592463450382915 8	Bethe ansatz (exact)	
f_{200}	$\hat{u}^2$	45.29870634431252721431787	Bethe ansatz (exact)	
f_{210}	$\hat{u}^2\hat{z}$	777.007521055093297555226 2	Bethe ansatz (exact)	
f_{220}	$\hat{u}^2\hat{z}^2$	8099.313348527777075847784	Bethe ansatz (exact)	
f_{300}	$\hat{u}^3$	3848.445467772377268238629	Bethe ansatz (exact)	
f_{310}	$\hat{u}^3\hat{z}$	115632.5319783609240553502	Bethe ansatz (exact)	
f_{400}	$\hat{u}^4$	545193.0244029872263588049	Bethe ansatz (exact)	
f_{001}	$\bar{\delta}$	−0.57721566490153286060651 21	one bubble	$-\gamma_E$
f_{011}	$\hat{z}\bar{\delta}$	−3.86708379859798573355134 2	one bubble	$-\gamma_E - 2\zeta(2)$
f_{021}	$\hat{z}^2\bar{\delta}$	−8.09809574633483001454378 3	one bubble	
f_{031}	$\hat{z}^3\bar{\delta}$	−13.46681348232748267372698	one bubble	
f_{002}	$\bar{\delta}^2$	5.292133223877124175286	one bubble	
f_{012}	$\hat{z}\bar{\delta}^2$	43.26016741947440932389841	one bubble	
f_{022}	$\hat{z}^2\bar{\delta}^2$	194.2323046570014639460987	one bubble	
f_{003}	$\bar{\delta}^3$	−61.14566451935350750983253	one bubble	
f_{013}	$\hat{z}\bar{\delta}^3$	−767.6867345496617072942533	one bubble	
f_{101}	$\hat{u}\bar{\delta}$	−36.4985374424376232979563 4	two bubbles	
f_{111}	$\hat{u}\hat{z}\bar{\delta}$	−404.1590010240865956167455	two bubbles	
f_{121}	$\hat{u}\hat{z}^2\bar{\delta}$	−2982.066378636498938272215	two bubbles	
f_{102}	$\hat{u}\bar{\delta}^2$	810.9918099349166764628352	two bubbles	
f_{112}	$\hat{u}\hat{z}\bar{\delta}^2$	14885.335418594594510562 8608	two bubbles	
f_{103}	$\hat{u}\bar{\delta}^3$	−23498.571644954480587913 84	two bubbles	
f_{201}	$\hat{u}^2\bar{\delta}$	−3157.008605154717628460989	three bubbles	

	monomial	value	source	closed form
f_{211}	$\hat{u}^2\hat{z}\bar{\delta}$	$-75994.36349692199322461628$	three bubbles	
f_{202}	$\hat{u}^2\bar{\delta}^2$	$172415.8301856328384692596$	three bubbles	(a)
f_{301}	$\hat{u}^3\bar{\delta}$	$-514275.7548978331544915535$	four bubbles	(a)

I.7 Comparison, disclosures and review

Comparison. The comparison stage was run after the last code inspection had approved, and it refused to run unless the record of C_8^{SR} was the current frozen one, bound to the current records of its inputs and to a passing confirmatory check. With Δ the magnetization-scheme value minus the reference:

reference	value	Δ	$\Delta/0.131$	Δ/own error
record, Eq. (2.14)	$247.3812048049935\ldots$	-2.506×10^{-5}	-0.0002	–
independent, Sec. 2.6	242.825 ± 2.371	$+4.556$	$+34.8$	$+1.92$

(The comparison stage of the calculation used the rounded 242.83 ± 2.37, which gives $\Delta=+4.551$.)

Disclosures. The review of the code approved quoting the value on condition that the following are stated.

- The central value was frozen, by hash, before any comparison. A first version, 247.38118 ± 0.077 with the same central value, is superseded because its roundoff estimate had not been validated.
- The session coordinating the calculation saw the comparison when that first version was frozen. The session running the computation was exposed to it later: after the central values and the design of the additional runs for the error bar were frozen, but before the first record of the confirmatory check was computed (from finished runs, against its fixed criterion, which it failed) and before the error bar was evaluated.
- Two stopping rules fixed in advance fired during that work. The author decided both amendments after seeing the data; the failing first records stay frozen.

  A. *Confirmatory check.* The second complex point missed its criterion 10^{-12} by roundoff (1.6×10^{-12}, against that run's own first-order roundoff bound 1.7×10^{-9}). A double-double rerun (about 1.7×10^{4} core-hours) was not made; the criterion was amended to at most 10^{-11} and within the run's own roundoff bound.

  B. *Roundoff validation.* The global rule, a stop if in any sampled sector the difference between the double and double-double results exceeds 0.1 of the first-order bound, fired on sectors of negligible bound; applied as written it would have made the uncertainty of C_8^{SR} about 16. The rule was amended to apply the same factor-ten margin decile by decile, with a stop only for a ratio above 0.1 among the sectors ranked 9 to 24 by bound (the top 8, recomputed entirely in double-double, are checked separately) or in the two top deciles, and it was tested on the fresh held-out sample, fixed before its data, which stayed within the margin.

Review. OpenAI Codex reviewed the plan in three rounds, approving it before any multi-bubble cluster with logarithm insertions was computed, and inspected the code six times; the sixth inspection approved quoting the value with the disclosures above. After each of the last two rounds of fixes, every current record was rebuilt byte for byte with the revised checks.

What the calculation establishes and what it does not.

- It gives $C_8^{\mathrm{SR}} = 247.38118 \pm 0.13$ in a scheme that shares no input with the compendium or with the independent re-derivation, and that never uses A_9. It agrees with the record to 2.5×10^{-5}. Since $C_8^{\mathrm{SR}} - \frac{1}{4}J_9$ of the independent re-derivation agrees with the compendium (Sec. 2.6), the low independent value lies in its J_9, and within J_9 in $A_9[\varepsilon^0\ell^1]$, the one piece that deviates; whether that deviation is a fluctuation with an underestimated error or a systematic error is not known.
- With a different regulator (analytic continuation in v instead of a hard core) and different responses (the magnetization on the infinite line instead of finite-interval responses with an endpoint source), it reproduces the closed forms of C_3^{SR} to C_7^{SR} to 7×10^{-26} and the record of C_8^{SR} within its uncertainty. This is numerical support for the prescription of Sec. 2.1 and for the compendium's integrations, not a proof of either.
- The uncertainty is an empirical numerical estimate from three runs and measured roundoff: not a standard deviation and not a bound, and it cannot exclude an error common to all three runs. At 5×10^{-4} relative it can neither identify nor certify the rational coefficients of Eq. (2.15).
- It assumes, in its own coordinates, the same two-coupling description of the gas as the prescription (the analyticity relations test it order by order), the identification $\Delta_{\mathcal{E}} = \Delta_-$ of Eq. (2.6), the Bethe-ansatz equations of the anisotropic Kondo model and its identification with the gas at $s = 1$. It does not test whether the gas, or the dual, describes the critical chain.

I.8 Records

The records of the calculation are frozen and identified by their sha256 hashes; they are in `code/integ_d1/results/`, `make test` checks the hashes, and `make replay` rebuilds each current record byte for byte from the shipped engine outputs. The record of C_8^{SR} contains Table 8, the values of C_3^{SR} to C_7^{SR} and the amendment texts, and it is bound to the records of its inputs:

record	sha256
C_8^{SR} (`C8_record_v2.json`)	`0d463b29d9ec474cacf2ab96afff62ae56b0ec262d766684ebbf0b7b8b20ec0c`
two bubbles (`M2_flow_coefficients.json`)	`33cc2e1e9a6e9a57510b0b3c6a7ef419258317be892774c3054d04852161e833`
three bubbles, one logarithm (`M3_flow_coefficients.json`)	`4328cab931ea96e953e9e6ba3789dfbfc0859c94f26fc8d38c1b1161eb271bce`
f_{202} (`M4A_flow_coefficients.json`)	`584562ca72485e0f57c2f1e8ac6a05dcc5cc07b2acc73659a39587bd61641048`
f_{301} (`M4B_flow_coefficients_v2.json`)	`90542a62db08d875995f2944e84ec7d6ff8987acc1b50892cd459d7e632d9899`
confirmatory check (`M4B_cx_check_v2.json`)	`66287b2af2f4c5352a1c15dc07edf6b381c2215f54b5b979fe726f4febf0f1fb`

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Part III

Perturbative Hamiltonian truncation of the long-range Ising model

Finite-cutoff coefficients C_2 to C_8, window corrections, extrapolations in $d = 1, 2, 3$, and how to reproduce them

Companion note to the public repository, folder `HT_Pert/`
Draft

1 Summary

This note documents every perturbative Hamiltonian-truncation (HT) number quoted in the paper *The long-range Ising CFT: perturbative and nonperturbative approaches* and in its Supplemental Material, and how to reproduce each from the folder `HT_Pert/` of the public repository. The truncation at finite coupling is documented separately, in the note *Hamiltonian truncation of the long-range Ising model at finite coupling* (folder `HT/`).

The method expands the truncated Hamiltonian of the ϕ^4-deformed generalized free field in $\epsilon = 2s - d$ at fixed cutoff. Two exact relations fix the couplings, the gap $\Delta_\varepsilon = \sum_n C_n(\Lambda)\epsilon^n$ follows order by order, and the coefficients are extrapolated in the cutoff (Section 2). Two observations recur. First, in every dimension, truncating the total energy or level of a state leaves it a vacuum window that does not close as the cutoff grows. A window correction with no free parameter removes the resulting bias. In $d = 1$, with it, post hoc fits of the exact sums give C_4 and C_5 to 2×10^{-5} or better, and the solver data give C_4 to C_7 to 1×10^{-3} or better. In $d = 2$ it works through ϵ^4, and through ϵ^5 with two further rules that are not derived in full. In $d = 3$ it works through ϵ^4, with ϵ^5 provisional. It removes the ϵ^2 violation of the descendant spacings in all three dimensions (in $d = 3$ for the one state tested), and the ϵ^3 violation in the single-state sectors of $d = 1$; in $d = 2$ it removes at least 95% of the ϵ^3 violation of single-quantum descendants at $L = 14$ and 16. Second, convergence is slow: the corrections fall like $\Lambda^{-1/2}$ times powers of $\log \Lambda$ in $d = 1$ and like $1/L$ times powers of $\ln L$ in $d = 2$ and $d = 3$, so the tail model carries a large part of every limit and dominates the error. Section 2.2 gives the reference values (Table 1), the deviation $\delta = |C_n^{\mathrm{HT}}/C_n - 1|$ and the status words used below.

One dimension (Section 3). C_2 and C_3 come from exact sums over intermediate states to $\Lambda = 2000$, with a fit rule fixed in advance: $\delta = 4 \times 10^{-12}$ and 2×10^{-8}. The same sums extend C_4 to $\Lambda = 2000$ and C_5 to 500, and fits frozen before the largest runs predict them to 10^{-9}. The frozen C_4 fits land 1×10^{-6} from the exact value, with the basis and the median chosen post hoc. Fixing the $\log^3$ tail of C_5 by the exponentiated C_4 tail gives C_5 to 2×10^{-5}, post hoc. With the complete window, the solver data to $\Lambda = 118$ give C_6 to 8×10^{-4} and C_7 to 1×10^{-3}, with fit windows and tail models chosen after the data were seen. The same data give a C_4 range that lies below the exact value and excludes it (largest deviation 2×10^{-4}) and a C_5 range that contains it (largest deviation 3×10^{-4}). C_8 lies 7×10^{-4} from the reference, inside its uncertainty, when the tail models are calibrated on C_6 and C_7.

Two dimensions (Section 4). The complete window, built from the level left to each counterterm, removes the $\ln L$ divergence of C_4. C_2 to C_4 agree with the continuum values to better than 10^{-7}, and C_5 to 5×10^{-5}, if the C_5 data carry no residual $\ln L$ term. The data cannot exclude one; with the bound judged from the median of the fits it would move C_5 by up to about 0.1, and the spread of single fits would allow -0.7 to $+0.5$. C_6 to C_8 are not reproduced.

Three dimensions (Section 5). With the complete window of $d = 2$ carried over unchanged, exact sums over intermediate states (C_2 to $L = 8000$, C_3 and C_4 to $L = 600$) reproduce C_2 (15 digits), C_3 ($\delta = 1 \times 10^{-7}$) and C_4 ($\delta = 5 \times 10^{-6}$). C_5, from $L \le 40$, gives -6.0 ± 0.3 against -5.850 (provisional and calibrated on $C_4^{(3)}$; a term in $L^{-1/2}$ that $L \le 40$ cannot exclude would lower it by 0.5 to 1). The vacuum window alone diverges at order ϵ^4. C_6 to C_8 are not reproduced.

Section 8 maps every number to its code, records, command and tolerance.

2 Model, cutoffs and perturbation theory

2.1 Model and observable

The long-range Ising critical point in d dimensions is described by the generalized free field (GFF) ϕ with nonlocal kinetic term $\phi(-\partial^2)^{s/2}\phi$, deformed by a ϕ^4 interaction [1, 2, 3]. With $\epsilon = 2s - d$ the field dimension is

$$\Delta_\sigma = \frac{d-s}{2} = \frac{d-\epsilon}{4}, \tag{1}$$

exactly, since no local counterterm renormalizes the nonlocal kinetic term [3]. The nonlocal equation of motion gives the shadow relation $\Delta_{\phi^3} = d - \Delta_\sigma$ [3]. These two exact statements are the only physics inputs of the truncation: they fix the two couplings of the Hamiltonian order by order. The observable is the thermal operator $\varepsilon = [\phi^2]$,

$$\Delta_\varepsilon(\epsilon) = \sum_{n\geq 0} C_n \epsilon^n, \qquad C_0 = \frac{d}{2}, \qquad C_1 = -\frac{1}{6}. \tag{2}$$

We write $C_n(\Lambda)$ or $C_n(L)$ for a coefficient computed at finite cutoff, and C_n, or $C_n^{(d)}$ where the dimension must be named, for the continuum value.

2.2 Reference values and conventions

Table 1 lists the continuum coefficients that the HT results are compared with. They come from the ϵ expansion of the GFF with Feynman graphs. The note *Thermal dimensions in the long-range Ising model*, in the folder `GFF/` of the public repository, documents them, and the paper *The long-range Ising CFT: perturbative and nonperturbative approaches* summarizes them. In this note they are used only for comparison, except where a fit is explicitly labeled as calibrated on them.

	$d = 1$	$d = 2$	$d = 3$
C_2	$\frac{1}{9}(\pi + 4\log 2) = 0.657131263981$	$\frac{4}{9}\log 2 = 0.308065413582$	$\frac{1}{9}(2 + 4\log 2 - \pi) = 0.181221785405554$
C_3	-3.719856322902	-0.874850012653	-0.353581520433
C_4	45.42488010356	5.048868431	1.228063545
C_5	-731.13947662454	-38.833645896	-5.850351421
C_6	14331.68703275805	362.366 ± 0.004	33.5754 ± 0.0005
C_7	-322550 ± 41	-3876.1 ± 2.1	-220.35 ± 0.13
C_8	$(8.060 \pm 0.017) \times 10^6$	45990 ± 33	1602.6 ± 3.9

Table 1: Continuum coefficients $C_n^{(d)}$ used as references, from the GFF note (folder `GFF/`). C_2 to C_5 are closed forms, and $C_6^{(1)}$ is a finite representation in special-function values (an independent sum over all six-loop graphs gives 14331.6870328(5)). The other entries are Monte Carlo estimates; their $\pm$ is the empirical allowance of the GFF note, about four to five sampling standard errors, not a standard deviation.

Conventions. The deviation of an HT value from the reference is $\delta = |C_n^{\mathrm{HT}}/C_n - 1|$; for a range, δ is the largest deviation of its ends. Quoted HT uncertainties are spreads over fit models, fit windows or prescriptions, not standard errors; values are rounded with the PDG rule. Each result carries the status word of the analysis that produced it. The words are *final* (the analysis is complete; it may

still contain post hoc choices), *blind* (fit rules fixed before the data they predict existed), *post hoc* (a choice made after the data, or the continuum value, were seen), *calibrated* (tail models selected by their reproduction of known continuum coefficients), *provisional*, or *not reproduced*.

2.3 Radial quantization and the Hamiltonian

In radial quantization the GFF is a set of free oscillators. In $d = 1$ the "sphere" is the pair of points $\sigma = \pm 1$ and

$$\phi_\sigma = \sum_{n\geq 0} u_n\, \sigma^n \big(a_n + a_n^\dagger\big), \qquad u_n^2 = \frac{(2\Delta_\sigma)_n}{n!} = \frac{\big(\frac{1-\epsilon}{2}\big)_n}{n!}, \qquad E(a_n^\dagger) = \Delta_\sigma + n. \tag{3}$$

With the level $\ell = \sum_n nN_n$ and the particle number $P = \sum_n N_n$,

$$H = H_0 + \epsilon F_1 + r(\epsilon)\, M + g(\epsilon)\, V, \qquad H_0 = \ell + \tfrac{P}{4}, \qquad F_1 = -\tfrac{P}{4}, \tag{4}$$

where $M = \langle{:}\phi^2{:}\rangle$ and $V = \frac{1}{12}\langle{:}\phi^4{:}\rangle$ are averages over the sphere, normal ordered. Their matrix elements are used as they stand in the truncated space: operators are never commuted past each other, because in a truncated space a commutator identity no longer holds. The sphere average keeps only transitions that change ℓ by an even amount, so the parity of ℓ is conserved; the reference states below have $\ell = 0$, and odd-ℓ states such as $\partial\phi$ enter only the descendant tests. The field parity $(-1)^P$ is conserved as well. ϵ enters only through $u_n(\epsilon)$ and F_1; the Taylor coefficients of u_n are kept exactly to the working order. The $d = 2$ and $d = 3$ mode functions and weights are given in Sections 4 and 5.

2.4 Cutoffs

$d = 1$**: frozen energy cutoff.** The Fock space keeps the states whose energy at $\epsilon = 0$ satisfies

$$E^0 = \ell + \tfrac{P}{4} \leq \Lambda \qquad \Longleftrightarrow \qquad 4\ell + P \leq 4\Lambda, \tag{5}$$

and the thresholds are not moved with ϵ. Energies at $\epsilon = 0$ are multiples of $\frac{1}{4}$.

$d = 2, 3$**: level cutoff.** The zero mode (the mode with the lowest energy, Δ_σ) is kept untruncated; the other modes form the bath, whose total level is cut at L: $\ell \leq L$ in $d = 2$, $N_b \leq L$ in $d = 3$ (Sections 4, 5).

2.5 Tuning and the coefficients

The reference states are the zero-mode states $|p\rangle = (a_0^\dagger)^p|0\rangle/\sqrt{p!}$ with $\ell = 0$: $p = 0$ is the vacuum, $p = 1$ and $p = 3$ are ϕ and ϕ^3, $p = 2$ is ϕ^2, and $p = 6$ (ϕ^6) gives the gap channel used for the tail exponents. The couplings $r = \sum_{k\geq 1} r_k\epsilon^k$ and $g = \sum_{k\geq 1} g_k\epsilon^k$ are fixed order by order by

$$E_1 - E_0 = \Delta_\sigma, \qquad E_3 - E_0 = d - \Delta_\sigma, \qquad \Delta_\varepsilon = E_2 - E_0 = \sum_n C_n(\Lambda)\epsilon^n. \tag{6}$$

At $\epsilon = 0$, $\langle p|M|p\rangle - \langle 0|M|0\rangle = 2p$ and $\langle p|V|p\rangle - \langle 0|V|0\rangle = p(p-1)/2$, so the solve is triangular. At first order $r_1 = 0$ and $g_1 = \frac{1}{3}$, so $C_0 = \frac{d}{2}$ and $C_1 = -\frac{1}{6}$ at every cutoff. With $\mathcal{E}_p$ the order-ϵ^n energy of $|p\rangle$ without the r_n, g_n terms,

$$C_n = \mathcal{E}_2 - \mathcal{E}_1 - \tfrac{1}{3}\big(\mathcal{E}_3 - \mathcal{E}_0\big) = -\tfrac{1}{3}\,\Delta_p^3\mathcal{E}_p \qquad (n \geq 2)\,: \tag{7}$$

shifts linear or quadratic in p are absorbed by r and g.

Evaluation. The ϵ series are computed by formal power-series algebra at fixed cutoff, never by sampling ϵ. At low orders a Rayleigh–Schrödinger recursion on truncated power series in ϵ gives the wave functions. At high orders a trial state correct through ϵ^k (order k) gives the energy through ϵ^{2k+1} by its Rayleigh quotient, and the eighth order adds one scalar term (Section 3.1). At order k no intermediate state of $|p\rangle$ has more than $p + 4k$ particles, or more than $4k$ particles in modes $n \geq 1$, so these particle caps are exact at the order computed. Independently, the order-n Rayleigh–Schrödinger expressions can be summed exactly over the intermediate states of the truncated Hamiltonian, grouped by the levels of the intermediate states between consecutive vertices, without building a basis. This gives C_2 to C_4 to $\Lambda = 2000$ and C_5 to $\Lambda = 500$ in $d = 1$, C_4, C_5 to large L in $d = 2$, and C_2 to C_4 to $L = 600$ and C_5 to $L = 40$ in $d = 3$.

2.6 Large-cutoff tails

At second order the $d = 1$ truncation has a closed form: at every cutoff

$$C_2(\Lambda) = \sum_{\substack{n \text{ even} \\ 0 \leq n \leq \Lambda - 2}} \bar{u}_n^2 \Big[\frac{2}{81(n - \frac{1}{2})} + \frac{2}{9(n + \frac{1}{2})} + \frac{2}{81(n + 1)} + [n > 0] \frac{2}{9n} \Big], \qquad \bar{u}_n^2 = u_n^2 \big|_{\epsilon = 0} = \frac{(1/2)_n}{n!}, \tag{8}$$

which every solver run checks, and its tail is

$$C_2(\Lambda) - C_2 \simeq -\frac{40}{81\sqrt{\pi}} \, \Lambda^{-1/2} = -0.2786121 \, \Lambda^{-1/2}. \tag{9}$$

Cutoff effects scale as $\Lambda^{-(\Delta_{\text{irr}} - 1)}$ with Δ_{irr} the dimension of the leading irrelevant operator, ϕ^6. Its gap $\Delta_6 = E_6 - E_0$, computed in the same truncation, defines

$$\omega(\epsilon) = \Delta_6(\epsilon) - 1 = \tfrac{1}{2} + \tfrac{7}{2}\epsilon + \sum_{k \geq 2} \omega_k \epsilon^k, \tag{10}$$

with $\omega_1 = -\frac{6}{4} + 15 g_1 = \frac{7}{2}$ from the one-loop rule $[\epsilon]\Delta_{\phi^n} = -\frac{n}{4} + \binom{n}{2} g_1$, the same rule that gives $C_1 = -\frac{1}{6}$ for $n = 2$. The tail model of the extrapolations is

$$\sum_n \big[C_n(\Lambda) - C_n \big] \epsilon^n = \Lambda^{-1/2} \Big(\sum_{n \geq 2} a_n \epsilon^n \Big) \, e^{-[\omega(\epsilon) - \frac{1}{2}] \log \Lambda} + \rho(\epsilon, \Lambda), \tag{11}$$

with the remainder ρ built from the terms $\Lambda^{-p} \log^j \Lambda$ of one of six bases fixed in advance:

basis	terms (p, j) of ρ
A	$(1,0)$, $(\frac{3}{2},0)$, $(\frac{3}{2},1)$, $(2,0)$
B	A and $(2,1)$
C	$(1,0)$, $(1,1)$, $(\frac{3}{2},0)$, $(\frac{3}{2},1)$
D	C and $(2,0)$
E	D and $(2,1)$
F	$(1,0)$, $(\frac{3}{2},0)$, $(\frac{3}{2},1)$, $(\frac{3}{2},2)$, $(2,0)$

The model is fitted order by order, with C_2 and C_3 fixed to their continuum values and C_4 to C_8 fitted in sequence. Identifying the gap with the leading irrelevant operator is an assumption for this regulator.

Top logarithm. If the $\Lambda^{-1/2}$ tail comes from a single channel whose amplitude starts at ϵ^2, Eq. (11) fixes the coefficient of the highest power of $\log\Lambda$ at each order:

$$C_n(\Lambda) - C_n \ni c_n\,\Lambda^{-1/2}\log^{n-2}\Lambda, \qquad c_n = a_2\frac{(-\omega_1)^{n-2}}{(n-2)!}, \qquad a_2 = -\frac{40}{81\sqrt{\pi}}, \tag{12}$$

independently of $\omega_{k\geq 2}$, of $a_{k\geq 3}$ and of channels that start at ϵ^3 or later. The exact sums test it: c_3 agrees to 5×10^{-6} and c_4 to 0.3% (Section 3.4).

2.7 The vacuum window

Truncating the total energy of a state makes the cutoff effectively nonlocal and state dependent [5]. In ϕ^4 theory in $2+1$ dimensions this spoils the cancellation of disconnected vacuum bubbles in Hamiltonian perturbation theory and gives new UV divergences, which Ref. [6] addressed with perturbative state-dependent counterterms that add back the missing states. A basis state of energy E_i^0 is dressed only by vacuum bubbles with $E_b \leq \Lambda - E_i^0$, while the vacuum keeps all bubbles with $E_b \leq \Lambda$, so the state misses

$$E_b \in \big(\Lambda - E_i^0, \Lambda\big], \qquad N_{\mathrm{w},i} = \#\big\{\text{odd } k \in (\Lambda - E_i^0, \Lambda]\big\}. \tag{13}$$

At $\epsilon = 0$ every four-particle shell of odd energy k carries $-g^2/6$, so the deficit is proportional to the number N_{w} of missing shells, which is of order one at every cutoff and does not decay as $\Lambda\to\infty$. The descendant spacings $v_n(\mathcal{O}) = E(\partial^n\mathcal{O}) - E(\mathcal{O}) - n$, which vanish in a conformal theory, measure it: raw HT gives $v_n = \frac{N_{\mathrm{w}}}{54}\epsilon^2 + O(\epsilon^3)$.

Complete window, $d = 1$. We add to every basis state m the diagonal shift

$$D_m^{\mathrm{conn}} = E_c(\Lambda) - E_c\big(\lfloor\Lambda - E_m^0\rfloor_{1/2}\big), \tag{14}$$

with $\lfloor\cdot\rfloor_{1/2}$ the half-integer floor and $E_c(x)$ the vacuum energy at cutoff x of the truncated Hamiltonian plus the same correction, and retune r and g. Since $D_{s+b}^{\mathrm{conn}} = D_s^{\mathrm{conn}} + D_b^{[\Lambda - E_s]}$ for a disconnected bubble b, each window is restored exactly once; the literal vacuum-energy difference $E_{\mathrm{vac}}(\Lambda) - E_{\mathrm{vac}}(\Lambda - E_m^0)$ would count the bubble-bubble class twice. There is no free parameter. At second order in g, D^{conn} reduces to the isolated window g^2D of the second-order vacuum bubbles, the counterpart of Patch I of Ref. [6]: a state-dependent counterterm that adds back the same missing four-particle states, with the unperturbed energy of the level being computed in its energy denominator, where g^2D has the bubble energy alone. Its Patch II, called a temporary fix there, restores towers of iterated bubbles up to five bubbles with every bubble cut at the full cutoff; D^{conn} contains such towers through E_c, cut as in the truncated vacuum.

Level cutoff, $d = 2, 3$. With bath level $\ell \leq L$ the vacuum window is $D_L(\ell) = E_v(L) - E_v(L-\ell)$ (called the consistent window in the $d = 2$ production code). In $d = 2$ divergent sub-bubbles inside a state also see only the level left by the other quanta; the complete window W that cures this is defined in Section 4 and derived in Appendix A. The same complete window applies unchanged in $d = 3$ (Section 5).

2.8 Notation across dimensions

Table 2 collects the symbols that differ between the three dimensions.

	$d = 1$	$d = 2$	$d = 3$
cutoff	energy Λ, Eq. (5)	bath level L, $\ell \le L$	bath level L, $N_b \le L$
level of a state	ℓ	ℓ (operator $\hat{\ell}$)	N_b (operator $\hat{N}_b$); ℓ is the angular momentum
vacuum window	second-order part $g^2 D$ (isolated window)	consistent window $D_L(\ell)$	vacuum window $D_L(N_b)$
complete window	D^{conn}, Eq. (14)	$W = D^{\mathrm{zm}} + D^{M}$; at ϵ^5 the rules W_c, W_{cs}	W of $d = 2$; W_c, W_{cs} at ϵ^5
descendant spacing	$v_n(\mathcal{O})$	R_n (coefficient of ϵ^n)	as in $d = 2$
leading tail	$\Lambda^{-1/2} \log^k \Lambda$	$L^{-1} \ln^k L$	$L^{-1} \ln^k L$

Table 2: Notation that differs between dimensions. The "raw" truncation has no window in every dimension. D^{conn} has the self-consistent form of the $d = 2$ consistent window; "complete" names, in each dimension, the window that restores the tested descendant spacings and continuum coefficients there.

3 Results in one dimension

3.1 Computations and data

Table 3 lists the computations behind the $d = 1$ numbers. All of them use the Hamiltonian (4), the cutoff (5) and the tuning (6); they differ in the order reached, in how the ϵ series is evaluated, and in the window. The eighth order needs, besides third-order trial states, the scalar $\langle b_4, R_p b_4 \rangle$, with b_4 the fourth-order source vector of the reference state $|p\rangle$ and R_p its reduced resolvent. Production runs used SDSC Expanse and the symmetry cluster of the Perimeter Institute.

computation	reach	method, where
HT solver, thermal	C_0–C_7, every even $\Lambda \le 118$	Rayleigh quotient of third-order trial states in the Fock basis (up to 3.5×10^9 states at $\Lambda = 118$); Expanse, 111 new cutoffs; `code/lri_hpc_Hamitonian`
HT solver, ϕ^6 gap	Δ_6 to ϵ^5 for $\Lambda \le 160$, to ϵ^3 for $\Lambda \le 832$	same solver; exponents ω_k of Eq. (10)
eighth-order code	C_8, every even $\Lambda \le 92$	third-order trial states plus $\langle b_4, R_p b_4 \rangle$; raw and isolated $g^2 D$ window; symmetry; `code/lri_c8_hpc`
window engines	ΔC_4, ΔC_5 exact, $\Lambda \le 152$	two independent engines plus closed-form sums over the states at the top of the window; `code/lri_hpc_Hamitonian/analysis`, `code/lri_eh_hpc`
complete-window solver	C_0–C_8 with D^{conn}, every even $\Lambda \le 72$	numerical, eighth order; beyond 72 a continuation validated blind (Section 3.4); symmetry; `code/lri_d1_full_coefficient_flow`
exact sums	C_2–C_4 to $\Lambda = 2000$, C_5 to $\Lambda = 500$, raw and D^{conn}	Rayleigh–Schrödinger summed over intermediate states; symmetry; `code/lri_1d_c4_exact`, `code/lri_1d_c5_exact`

Table 3: The $d = 1$ computations. Where they overlap they agree to 2×10^{-11} or better (Section 6). The eighth-order code also reproduces the saved records of its original version to 2.1×10^{-8}.

3.2 Raw coefficients and slow convergence

The tails are long. At $\Lambda = 118$ the raw C_4 reaches only 67% of its limit and the raw C_7 only 44%, and even at $\Lambda = 500$ the exact $C_5^{\rm conn}(500) = -519.770442010804$ is still far from $C_5 = -731.14$. The tail model of Section 2.6 therefore carries a large part of every limit. Every extrapolation of the raw truncation, with any of the bases A–F, gives a C_4 about 0.19% below the exact value; the window explains this offset.

3.3 The window in one dimension

With $D^{\rm conn}$ added, the shift of the coefficients does not decay:

$$C_4^{\rm conn}(\Lambda) - C_4^{\rm raw}(\Lambda) \simeq 0.0803 + \frac{\log \Lambda}{17496}, \tag{15}$$

where $0.0803 = K + \eta_4$, with K the finite part of the isolated g^2D window and $\eta_4 = 0.0186910$ the limit of the remaining fourth-order shift. At $\Lambda = 118$, $\Delta C_4 = 0.080$ and $\Delta C_5 = 0.71$. The exact sums follow ΔC_5 to $\Lambda = 500$, where it is 0.717651334579: the two $\log \Lambda$ terms of its pieces nearly cancel, and ΔC_5 tends to a constant near 0.72. Beyond ϵ^5 the complete window is computed numerically to $\Lambda = 72$; there ΔC_6, ΔC_7, ΔC_8 are +1.1717361, -1.828185, -8.5641, respectively.

Descendant spacings. The spacings v_n test the window without any continuum input (Table 4). States are labeled by their particle number P and level ℓ; a sector is a pair (P, ℓ). The raw truncation gives $N_{\rm w}/54$ at ϵ^2 and about $0.146\, N_{\rm w}$ at ϵ^3. With $D^{\rm conn}$ the ϵ^2 violation vanishes, and the ϵ^3 one is within $0.003\, N_{\rm w}$ in the sectors with a single state; the degenerate block $P = 2$, $\ell = 2$ gives $+0.004$ to $+0.006\, N_{\rm w}$ over the fit models, and whether it vanishes is not settled. At ϵ^4, $D^{\rm conn}$ removes 96–100% of the violation left by g^2D alone. At ϵ^5 the violation is consistent with zero within about $\pm(0.1\text{–}0.2)\, N_{\rm w}$, against $+2.3$ to $+2.8\, N_{\rm w}$ raw and -1.3 to $-1.6\, N_{\rm w}$ with g^2D alone; the $\partial\phi$ residual is not converged.

order	raw	g^2D alone	with $D^{\rm conn}$
$[\epsilon^2]v_n$	$N_{\rm w}/54$	0	0
$[\epsilon^3]v_n$	$\approx 0.146\, N_{\rm w}$	$-N_{\rm w}/27$	$\lvert\cdot\rvert \le 0.003\, N_{\rm w}$ (single-state sectors)
$[\epsilon^4]v_n$		$\approx -0.46\, N_{\rm w}$	96–100% of the g^2D value removed
$[\epsilon^5]v_n$	$+2.3$ to $+2.8\, N_{\rm w}$	-1.3 to $-1.6\, N_{\rm w}$	0 within $\pm(0.1\text{–}0.2)\, N_{\rm w}$

Table 4: Violations of the descendant spacings in $d = 1$, per missing shell $N_{\rm w}$ (Eq. (13)). Values from ϵ^3 on are extrapolated in Λ; ranges are spreads over fit models. The raw value at ϵ^4 was not extrapolated. The g^2D column and the ϵ^5 raw value are from `analysis/DIAGNOSIS.md` of `lri_hpc_Hamitonian`.

Cutoff placement. If the cutoff sits at $\Lambda = 2m + \frac{1}{4}$ or $2m + \frac{1}{2}$, with m an integer, the window of ϕ is empty, and raw HT satisfies $E(\partial\phi) - E(\phi) = 1$ exactly through ϵ^3 (to 10^{-14}). At $\Lambda = 2m + \frac{1}{4}$ the ϵ^4 spacing of $\partial\phi$ measures the part η_4 of the C_4 window that g^2D leaves out, without extrapolating C_4: with g^2D alone it should equal

$$[\epsilon^4]\, v_1(\phi) = 3\eta_4 = 0.0561. \tag{16}$$

The fits give +0.0495 (median), 12% below this prediction, which lies just above the range of the fits. With D^{conn} a residual of about a tenth of $3\eta_4$, with the opposite sign, remains. The test thus confirms η_4 at the level of 12%.

3.4 Extrapolations of C_2 to C_8

Table 5 collects the $d = 1$ results.

C_2 and C_3. The exact $C_2(\Lambda)$, $C_3(\Lambda)$ at every even $\Lambda \le 2000$ are fitted by a rule fixed before any fit was run: free intercept; for C_2 the bases $\Lambda^{-j/2}$, $j \le 6$ and $j \le 8$; for C_3 the bases $\{\Lambda^{-1/2}, \Lambda^{-1/2}\log\Lambda, \Lambda^{-1}, \Lambda^{-1}\log\Lambda, \Lambda^{-3/2}, \Lambda^{-3/2}\log\Lambda, \Lambda^{-2}\}$ and the same with $\Lambda^{-2}\log\Lambda$, $\Lambda^{-5/2}$ added; lower cutoffs 100, 200, 400; the median of the six fits, with half their range. This gives $C_2 = 0.657131263984(10)$ ($\delta = 4\times10^{-12}$) and $C_3 = -3.7198563(10)$ ($\delta = 2\times10^{-8}$). The same rule on the solver range $\Lambda \le 118$ gives 0.657131272(35) and −3.71984(8). Earlier drafts of the paper quoted, instead, the closest of twenty solver fits to the exact values; that selection used the exact values and is no longer used.

C_4 from the solver. With the complete window, the bases A–F at lower cutoffs 40–70 give 45.415–45.419, all below the exact value. Two further extrapolations extend the range upward (`lri_1d_dconn_refit`). The first is an ensemble of joint fits of C_4 to C_8 that share their tail parameters, kept when their residuals reach the noise floor of the data. The second is a model-independent extrapolation that fits a grid of tail sets and selects among them with rules calibrated on held-out cutoffs. The union of the three is 45.415–45.420. It lies below the exact C_4 and excludes it; its largest deviation is 2×10^{-4}. The fit windows were chosen after the data were seen.

C_4 from the exact sums. With $C_4^{\mathrm{conn}}(\Lambda)$ to $\Lambda = 2000$, fits of the form $A + \sum c_{jk}\Lambda^{-j/2}\log^k\Lambda$ were frozen from the $\Lambda \le 1000$ result before the $\Lambda = 2000$ run existed, for several bases and lower cutoffs. In the basis L2 ($\Lambda^{-1/2}$ and Λ^{-1} with $\log^{0,1,2}$, $\Lambda^{-3/2}$ with $\log^{0,1}$, and Λ^{-2}) they predict $C_4^{\mathrm{conn}}(2000)$ to 4.0×10^{-10}, while fits without logarithms miss by up to 10^{-2}. Which frozen fits to quote was decided after their intercepts had been compared with the exact C_4: basis L2, lower cutoffs 100, 200, 300, median and half-range. This gives $C_4 = 45.4249(9)$, 1×10^{-6} from the exact value; the refit on all data gives 45.4249235266 at lower cutoff 200. The quoted uncertainty is the half-range of these three fits only; fits with larger lower cutoffs have condition numbers above 10^{12} and scatter by up to 3×10^{-3}.

C_5 from the solver. The bases A–F miss C_5 by −1.53 to −0.51, i.e. by up to 2×10^{-3} relative. Freeing the leading tail amplitudes (the coefficients of $\Lambda^{-1/2}$ and $\Lambda^{-1/2}\log\Lambda$ at orders 4 and 5) after the data were seen moves the misses to −0.22 to +0.06, and the ensemble of joint fits extends the range upward. The union is −731.36 to −730.97; it contains the exact C_5, and its largest deviation is 3×10^{-4}. The fits that reach the exact C_5 need an effective ω_3 of 380–430, not the gap value 556; Section 7 discusses this.

C_5 from the exact sums. The exact $C_5^{\mathrm{conn}}(\Lambda)$ to $\Lambda = 500$ carries $\Lambda^{-1/2}\log^k\Lambda$ up to $k = 3$. Frozen fits predict the unseen cutoffs to 3×10^{-9} (records frozen at $\Lambda \le 400$) and 1.5×10^{-8} ($\Lambda \le 320$). Their free intercepts, however, determine C_5 only to a few tenths: without the $\log^3$ term (basis L2) the intercept −730.8460638788 is biased; with it (basis M3, which adds $\Lambda^{-1/2}\log^3\Lambda$ to L2) the intercept is −731.0822822059 at the lowest cutoff and unstable above. With the intercept fixed to the exact C_5, only the bases with $\log^3$ fit at the noise floor, and the fitted $\log^3$ coefficient is $1.945 - -1.994$.

Eq. (12) predicts it from the C_4 data, $c_5 = -\frac{\omega_1}{3} c_4 = 1.9965$. The relation (12) holds at ϵ^3 to 5×10^{-6} and at ϵ^4 to 0.3% (Section 7); the step from c_4 to c_5 itself is not tested independently. Fixing c_5 at this value and fitting the rest with a free intercept gives $-731.1467, -731.1397, -731.1359$ for the three best-conditioned fits. A separately written fit of the same constrained model gives $C_5 = -731.13(10)$, the quoted determination. Its uncertainty is the spread over tail bases and fit windows, and the deviation from the exact value, 2×10^{-5}, lies well inside it. This is post hoc: the constraint was adopted after free-intercept fits had left C_5 open to about ± 0.3, with the exact value known, and the selection thresholds of the first constrained fit were set after a run had printed the exact C_5.

C_6, C_7 and C_8. Beyond $\Lambda = 72$ an ensemble of fits to the computed shifts continues ΔC_6, ΔC_7 to $\Lambda = 118$ and ΔC_8 to $\Lambda = 92$. The same rule, frozen on $\Lambda \leq 44$ before the data at $46 \leq \Lambda \leq 72$ existed, predicted those data within a tenth of its own uncertainty. The window changes the order-6 to order-8 extrapolations little; the spread of tail models dominates. The bases A–F, the ensemble of joint fits (16th to 84th percentiles) and the model-independent extrapolation give $C_6 \in 14\,320$–$14\,342$ (at most 8×10^{-4} from C_6) and $C_7 \in -322\,890$ to $-322\,230$ (at most 1×10^{-3}; the range includes the uncertainty of the continuation). For C_8 the eighth-order data to $\Lambda = 92$ with the continued window are used, and the tail models are selected by their reproduction of C_6 and C_7: the selected estimators span $(8.061$–$8.067) \times 10^6$, with central value 8.0652×10^6, 7×10^{-4} from the reference and inside its uncertainty (2×10^{-3}). Unselected tail models span $(8.045$–$8.073) \times 10^6$. This C_8 is calibrated on the diagrammatic C_6 and C_7 and is not an independent determination.

order	HT	δ	source, status
C_2	0.657131263984(10)	4×10^{-12}	exact sums, $\Lambda \leq 2000$; rule fixed in advance
C_3	−3.7198563(10)	2×10^{-8}	same
C_4	45.4249(9)	1×10^{-6}	exact sums to $\Lambda = 1000$, fits frozen before $\Lambda = 2000$; choice of fits post hoc
	45.415–45.420	2×10^{-4}	solver, $\Lambda \leq 118$, D^{conn}; fit windows post hoc; excludes C_4
C_5	−731.13(10)	2×10^{-5}	exact sums to $\Lambda = 500$, $\log^3$ tail fixed by Eq. (12); post hoc
	−731.36 to −730.97	3×10^{-4}	solver; leading tail freed post hoc (bases A–F alone: 2×10^{-3})
C_6	14 320–14 342	8×10^{-4}	solver plus complete window to 72; post hoc; spread of tail models
C_7	−322 890 to −322 230	1×10^{-3}	same
C_8	8.0652×10^6	7×10^{-4}	$\Lambda \leq 92$; calibrated on C_6, C_7

Table 5: $d = 1$ results; δ against Table 1. Ranges and uncertainties are spreads over tail models and fit windows, not standard errors.

3.5 A test protocol fixed before most of the data

A test protocol for the raw C_4 and C_5 was written after the first new thermal cutoff ($\Lambda = 84$) had been seen and before the rest. It set numeric thresholds for three criteria, evaluated in the bases A, B and C: stability under changes of the fit window, prediction of held-out cutoffs, and insensitivity to the tail model. Outcome: the held-out predictions pass for both orders; window stability fails for C_4 in one basis and for C_5 in two of the three; tail insensitivity passes for C_4 and fails for C_5.

The protocol's verdict for this mixed outcome is "unresolved", and no later analysis has revised it. Everything above beyond C_3, apart from the blind checks of the exact sums, is post hoc.

4 Results in two dimensions

In $d = 2$ the truncation is a level cutoff $\ell \le L$ and the coefficients are computed in three ways: a production solver to eighth order for $L \le 34$, exact-rational and Cauchy-integral solvers at small L, and sums over intermediate states that give $C_4(L)$ to $L = 2000$ and $C_5(L)$ to $L = 300$. Through ϵ^3 the truncation converges to the continuum. At ϵ^4 every window in use diverges like $\ln L$; a complete window, derived from the level budget of divergent sub-bubbles, removes the logarithm and gives C_4 to 5×10^{-8}. At ϵ^5 it needs two further pieces and then gives C_5 to 5×10^{-5}, if the data carry no residual $\ln L$ term (Sec. 4.5). C_6 to C_8 are not reproduced. Table 6 collects the results, and Appendices A to C give the derivations and the supporting data.

	HT, $L \to \infty$	continuum	$\lvert C_n^{\mathrm{HT}}/C_n - 1\rvert$	status
C_2	0.30806541358	$\frac{4}{9}\log 2$	35 digits	final
C_3	$-0.8748500(1)$	-0.8748500127	1×10^{-8}	final
C_4, complete window W	5.0488682(10)	5.0488684314	5×10^{-8}	final
C_5, scheme W_{cs} (Sec. 4.5)	$-38.835(13)$	-38.8336459	5×10^{-5}	final; no residual $\ln L$ assumed
C_6 to C_8	not reproduced	362.366(4), ...		not reproduced

Table 6: $d = 2$ perturbative results. "Final" refers to the quoted value in the sense of Sec. 2.2: the analysis is complete, with post hoc choices. The windows behind C_4 and C_5 were derived after their effects had been seen, and the rules W_c and W_{cs} behind C_5 are provisional (Table 31). Uncertainties are spreads over tail models and fit windows, not standard errors; that of C_4 is the half spread of nine fits, rounded up (Sec. 4.4). The C_5 uncertainty assumes that the data carry no residual $\ln L$ term. They cannot exclude one: with the bound judged from the median of the fits it would move C_5 by up to about 0.1, and the spread of single fits would allow -0.7 to $+0.5$ (Sec. 4.5). Continuum values from Table 1; in $d = 2$, $C_3 = \frac{4}{27}\log^2 2 - \frac{\Gamma(1/4)^8}{324\pi^4}$ and $C_4 = \frac{1}{648}\big[16P_4(2) - 64\log^3 2 - 32\log 2\, Q_2 - 359\zeta(3)\big]$, with $Q_2 = \frac{\Gamma(1/4)^8}{24\pi^4} = 12.77138119864696$ and the primitive four-loop wheel value $P_4(2) = 250.48731558186729$ (GFF note, section "Finite multivariable representations for the nonprimitive masters").

4.1 Model and level cutoff

On the unit circle the field has one oscillator A_{ab} for each $a, b \ge 0$, describing $\partial^a \bar\partial^b \phi$, with energy $\Delta_\sigma + a + b$, spin $a - b$ and weight

$$\phi(\theta) = \sum_{a,b} \sqrt{w_{ab}}\, \big(e^{\mathrm{i}(a-b)\theta} A_{ab}^\dagger + e^{-\mathrm{i}(a-b)\theta} A_{ab}\big), \qquad w_{ab} = \frac{(\Delta_\sigma)_a (\Delta_\sigma)_b}{a!\, b!}, \qquad \Delta_\sigma = \tfrac{1}{2} - \tfrac{\epsilon}{4}. \tag{17}$$

A Fock state has level $\ell = \sum (a + b)\, n_{ab}$ and particle number $P = \sum n_{ab}$, and its free energy is $\ell + P\Delta_\sigma = H_0 + \epsilon F_1$, with $H_0 = \ell + \frac{P}{2}$ and $F_1 = -\frac{P}{4}$ (the $d = 1$ Hamiltonian, Eq. (4), has $H_0 = \ell + \frac{P}{4}$). With $M = \langle{:}\phi^2{:}\rangle_\theta$ and $V = \frac{1}{12}\langle{:}\phi^4{:}\rangle_\theta$,

$$H(\epsilon) = H_0 + \epsilon F_1 + r(\epsilon) M + g(\epsilon) V + D, \tag{18}$$

where D is the window. We keep the states with $\ell \le L$ (even), spin zero and even under $a \leftrightarrow b$; the zero mode $(0,0)$ costs no level and is not truncated; the other modes form the bath. r and g are tuned at cutoff L, and $C_n(L)$ follows from the reference states $|p\rangle$, $p = 0, \dots, 3$, as in Sec. 2, with $C_0 = 1$, $C_1 = -\frac{1}{6}$, $r_1 = 0$, $g_1 = \frac{1}{3}$ at every L. Through ϵ^8 no intermediate state has more than $p + 16$ particles, so this cap is exact. We write $Z = A_{00} + A_{00}^\dagger$, $\chi = \phi - Z$ for the non-zero modes, $M' = M - :Z^2:$, and χ_{aa} for the one-quantum state of mode (a, a).

Windows. The raw truncation has $D = 0$. The consistent window is the vacuum window $D_L(\ell) = E_v(L) - E_v(L - \ell)$ with self-consistent vacua $E_v(\lambda)$ at sub-cutoffs $\lambda \le L$, each computed with the same window. The literal window is the same difference with the vacua $E_v(\lambda)$ of the raw truncation; it differs from the consistent window from ϵ^6 on. The zero-mode window replaces E_v by the self-consistent energy $E_z(\lambda)$ of the channel with the same number z of zero-mode quanta. All of them are diagonal.

Terms. The zero-mode channel $|z\rangle = (A_{00}^\dagger)^z|0\rangle/\sqrt{z!}$ has free energy $E_z^0 = \frac{z}{2}$; $|z, \chi\rangle$ is a state with z zero-mode quanta and bath part χ of level ℓ_χ, and $|k\rangle$ is an intermediate state of free energy E_k^0. In a time-ordered product of vertices, a *slice* is the intermediate state between two neighboring vertices. A sub-process (a bubble, sunset or mass insertion) absorbs or emits some χ quanta, its *legs*; the other quanta present are *spectators*, *hard* if their level is comparable with L and *soft* otherwise. A term of M' with one creator and one annihilator is a *hop*; one with two creators or two annihilators is a *pair* term. The *exchange sunset* of χ_{aa} is the ϵ^2 ordering in which the new χ_{aa} is emitted, together with a bubble, before the old one is absorbed, so that two (a, a) quanta are present at once. An *auxiliary channel* is the eigenvalue branch $E_z(\lambda)$, with right eigenvector $\psi_z(\lambda)$, of the Hamiltonian at a sub-cutoff $\lambda \le L$ through $|z\rangle$; the windows use it as the reference for zero-mode clusters inside larger states. A *ket-referenced (Bloch)* effective interaction uses the energy denominators of the state it acts on, so it is not symmetric in z and z'. Newton's forward-difference form in p, $E_p = E_0 + p\, e_1 + \binom{p}{2} e_2 + \dots$, defines the *Newton coefficients* $e_1 = E_1 - E_0$ and $e_2 = E_2 - 2E_1 + E_0$. A *tower* is a fit model: a constant plus terms $\ln^j L / L^k$ (Appendix A.3). A *tagged* line is an external quantum replaced by a copy that the cutoff and the windows see with an assigned level, its *charge* (Appendix C.2).

4.2 Solvers

The production solver `code/lri_d2_coeff_hpc/` computes C_0 to C_8, r_n, g_n and every reference energy for $L \le 34$ in the raw and consistent schemes, in float64 truncated power series in ϵ ($L = 16$ to 30 on the symmetry cluster, 32 and 34 on SDSC Expanse); the literal window is implemented and checked at small L. Two clean-room solvers, written from the model specification without seeing the production code or each other, check it: exact Rayleigh–Schrödinger perturbation theory in rational arithmetic ($L \le 18$) and Cauchy integrals of the non-perturbative truncated Hamiltonian at complex ϵ with 256-bit arithmetic ($L \le 8$). The production solver agrees with the exact one to 4×10^{-13} (largest scaled difference over all C_n, r_n, g_n, energies and vacua, $n \le 8$), and the Cauchy solver with the exact one to 5×10^{-46} (Sec. 6).

The large-L results come from separate codes in `code/lri_2d_coeff_independent_check/`. With V and M factorized into zero-mode and χ parts, a K-vertex chain is a sum over χ-line patterns of a Wick count, an exact zero-mode path sum and a χ level sum; at $\epsilon = 0$ the weights factorize holomorphically, $\sum_{ab} w_{ab} y^a \bar{y}^b = (1 - y)^{-1/2}(1 - \bar{y})^{-1/2}$, and the cutoff only bounds the largest slice level, so one run gives every even L. These codes reproduce the exact rationals and the production solver: $C_2(L)$ and $C_3(L)$ to 4×10^{-14} (absolute), $C_4(L)$ to 1.1×10^{-13} and $C_5(L)$ to 2.2×10^{-13} (relative). The C_5 engine was later optimized; per task it agrees with the original to 7.3×10^{-16}, and the assembled $C_5(L)$ reproduces the earlier data at $L \le 200$ to 7.6×10^{-13}.

The zero-mode scheme has its own exact solver (`zeromode_exact/`, $L \le 18$; Appendix A.3). Every other window scheme at $L \le 16$, and every descendant spacing, comes from two further exact solvers in the same package: `complete_exact/` (scheme W and its variants as explicit operators, exact through ϵ^8 for $L \le 16$) and its copy `residual_log/ce_wc/` with rule W_c and the external-line windows (Appendix B.2).

4.3 Second and third order

At ϵ^2 only $-\frac{P}{4} + g_1 V$ acts and

$$C_2(L) = \frac{1}{81}\Big[17 + \sum_{a=1}^{L/2} w_{aa}\Big(\frac{2}{2a-1} + \frac{18}{2a} + \frac{18}{2a+1} + \frac{2}{2a+2}\Big)\Big], \qquad C_2(L) - C_2 \simeq -\frac{40}{81\pi}\,\frac{1}{L}, \quad (19)$$

with w_{aa} at $\epsilon = 0$. Summed to $L = 10^5$ with its asymptotic tail it equals $\frac{4}{9}\log 2 = 0.30806541358\ldots$ to all 35 digits evaluated (numerical agreement, not a proof). $C_3(L)$ is exact to $L = 600$; its tail has a $\ln L/L$ term with coefficient $0.70734 \approx \frac{9}{2}\cdot\frac{40}{81\pi}$ (the factor $\frac{9}{2}$ identified numerically; Appendix A.2), and towers in $1/L^k \ln^j L$ give $C_3 = -0.8748500(1)$ against -0.8748500127. The schemes do not differ through ϵ^3.

4.4 Fourth order: the level budget and the complete window

Divergence of the raw, consistent and zero-mode windows. Exact $C_4(L)$ to $L = 2000$ in the raw, consistent and zero-mode schemes (Fig. 1a) has the large-L form $A + \beta \ln L$ (plus $-\kappa L$ in raw, $\kappa = 2.2073 \times 10^{-4}$):

scheme	A	fitted β	derived β
raw	5.0021	-0.026381	$-\frac{725}{8748\pi} = -0.0263803$
consistent	5.0026(5)	$-0.026563(42)$	$-\frac{365}{4374\pi} = -0.0265622$
zero-mode	5.0213(5)	$-0.010480(42)$	$-\frac{8}{243\pi} = -0.0104793$

For the consistent and zero-mode schemes, A and the fitted β are medians of twelve fits, with half their spread in brackets; the twelve include variants with the $\ln^2 L/L$ coefficient fixed to $-5/\pi$ (with T_{22} these fit 2.5 to 50 times worse, with $T_{22} + \ln^3 L/L$ about as well; `c4_exact/fits/fit_summary.txt`). The raw row follows from the consistent fit and the exact window shift. The consistent scheme thus diverges like $-0.027 \ln L$. Predictions for $L = 1200$ to 2000, recorded from fits to $L \le 1000$ before those data existed, miss by at most 8.3×10^{-8} with the logarithm; at $L = 2000$ the models without it miss by at least 4.1×10^{-7}.

Cause. The cutoff bounds the total level of an intermediate state, so a divergent sub-bubble inside a state with other quanta of level ℓ keeps only $L - \ell$, while r, g and the vacuum are tuned on level-0 states with the full L. In $d = 2$ the mass (sunset) sub-bubble diverges linearly, $r_2 \simeq \frac{2L}{27\pi}$ (exact slope 0.0235785 at $L = 2000$, $\frac{2}{27\pi} = 0.0235785$), so the deficit is proportional to ℓ and does not decay with L. It is an operator $\sim\{\hat{\ell}, M\}$; through the zero-mode amplitude it is cubic in p, so no choice of r and g absorbs it. Splitting the deficit by window class gives the derived coefficients above: the consistent window restores the vacuum class, the diagonal zero-mode window $\frac{3}{5}$ of the zero-mode sunset class, and the rest is the part of the sunset that changes the zero-mode number by ± 2 ($\frac{8}{243\pi}$; Appendix A.4).

Complete window. Each counterterm acting in a state is evaluated at the budget left by the quanta that do not take part. At the order that enters C_4, $D = D^{\rm zm} + D^M$ with

$$D^{\rm zm}|z,\chi\rangle = \sum_{z'} \left[h^{(2)}_{z'z}(L) - h^{(2)}_{z'z}(L-\ell_\chi)\right]|z',\chi\rangle, \qquad h^{(2)}_{z'z}(\lambda) = g_1^2 \sum_{k:\,\ell_k\le\lambda} \frac{\langle z'|V|k\rangle\langle k|V|z\rangle}{E^0_z - E^0_k}, \tag{20}$$

$$D^M = -\frac{g_1^2}{3\pi}\{\hat\ell, M'\} = -\frac{1}{2}\frac{dr_2}{dL}\{\hat\ell, M'\}, \tag{21}$$

where $\ell_\chi \ge 2$ is the level of the χ part and $h^{(2)}$ is the second-order effective interaction of the zero mode, a matrix in z in its ket-referenced (Bloch) form; its diagonal contains the consistent and zero-mode windows. The mass window D^M makes a mass insertion between states of levels ℓ_s, $\ell_{s'}$ use $r(L - \frac{1}{2}(\ell_s + \ell_{s'}))$. $D^{\rm zm}$ follows from the linked-cluster cancellation of two components that share only the zero mode; D^M is derived at leading order in $1/L$. Appendix B.1 gives the all-order form of D^M, with the Newton coefficient e_1 in place of r; at ϵ^2 the two forms agree up to $\mathcal{O}(1/L)$. The C_4 data to $L = 2000$ and the brute-force checks of Appendix A.5 use Eq. (21); the exact solvers `complete_exact` and `ce_wc` and the ϵ^5 data use the all-order form. Neither has a fitted parameter. The window codes agree with brute-force exact perturbation theory with D as an explicit operator at $L \le 12$ (Appendix A.5).

Result. With the complete window $C_4(L)$ has no $\ln L$ term (Fig. 1a). The full zero-mode window alone tends to 5.0386139; the mass window D^M adds the missing 0.0102543. Fits of $C_4^W(L)$ for $L \le 2000$ with tail towers in $\ln^j L/L^k$ give

$$C_4 = 5.0488682(10) \qquad \text{against} \qquad C_4^{(2)} = 5.0488684314, \tag{22}$$

with a half spread of 6×10^{-7} over the nine stable no-log fits (towers T_{22}, $T_{22} + \ln^3 L/L$ and T_{23}), rounded up to the quoted 10^{-6}. No-log fits outside this set move the limit by up to 2×10^{-5} (one over-fitted tower excepted). A free $\ln L$ term $\beta \ln L$ in the stable towers comes out at $|\beta| \le 3.5\times10^{-6}$ (T_{33} gives -8.1×10^{-6}) and moves the limit by up to 4.3×10^{-5} (Appendix A.5); the complete window has no $\ln L$ by construction (Table 12). C_2 to C_4 thus agree with the continuum to better than 10^{-7}.

4.5 Fifth order

All-order window and double counting. At all orders $D^{\rm zm}$ uses the Bloch matrix of the auxiliary channels $|z\rangle$ at sub-cutoff λ, $h(\lambda) = U\,\mathrm{diag}(E_z(\lambda))\,U^{-1}$ with $U_{z'z} = \langle z'|\psi_z(\lambda)\rangle$, and the mass window is $D^M = \frac{1}{4}\{e_1(L) - e_1(L-\hat\ell), M'\}$ with the Newton coefficient $e_1(\lambda) = E_1(\lambda) - E_0(\lambda)$; degenerate partners of $|z\rangle$ ($z \ge 5$) are projected out. If the auxiliary objects carry their own windows self-consistently, $C_5^W(L)$ diverges linearly, with coefficient -0.0019 ± 0.0004 (the uncertainty of the linear term of raw C_5; Table 16). The cause is a double counting: a sunset nested in an auxiliary channel is restored once by the outer D^M and once more inside $h(L-\ell)$. Rule W_c removes it: an auxiliary channel at sub-cutoff $\lambda < L$ represents a zero-mode cluster inside a state of the cutoff-L theory, so its windows are referred to the outer cutoff, $[h(L) - h(\lambda - \ell_\chi)]$ and $\frac{1}{4}\{e_1(L) - e_1(\lambda - \hat\ell), M'\}$. It leaves C_4 unchanged and removes the linear term (Appendices B.1 to B.5).

External-line window. In W_c the ϵ^3 spacing of χ_{aa} (Sec. 4.6) tends to a constant, $R_3 \approx 0.063\,a\,w_{aa}$ (0.062 to $0.063\,a\,w_{aa}$ at $L = 16$ for $a = 1, 2, 3$, changing slowly with L). An exact test with a tagged external line (Appendix C.2) locates it: if the external quantum is invisible to the cutoff and the windows (budget-free), $R_3 \to 0$; if it is charged its level, R_3 reproduces its value in W_c. D^M charges both legs of a mass sub-process with their average level in every slice, which is exact for the ϵ^2

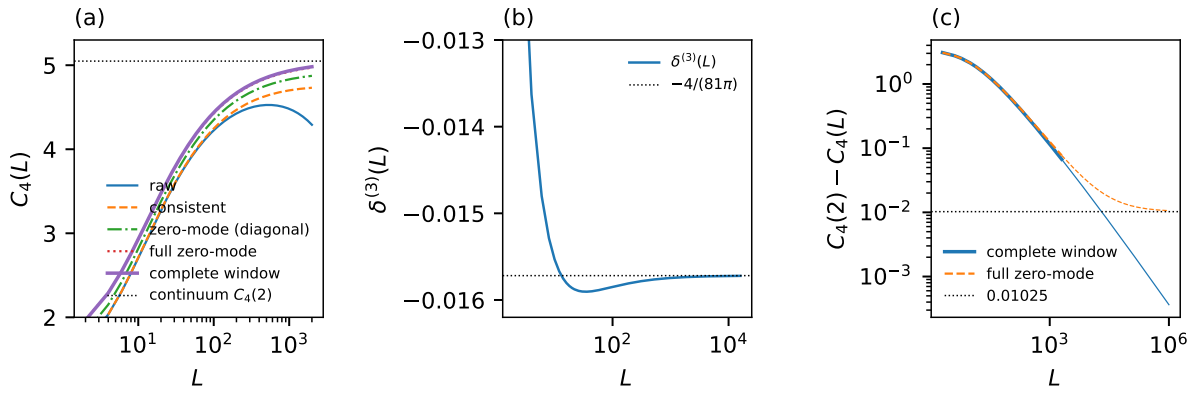


Figure 1: (a) Exact $C_4(L)$ in $d = 2$ for $L \le 2000$ in five windows: raw, consistent, diagonal zero-mode, full zero-mode (diagonal and off-diagonal) and complete; only the complete window converges to $C_4^{(2)}$ (dotted). (b) The ϵ^3 external-line coefficient $\delta^{(3)}(L)$ of Eq. (24) for $L \le 16000$, with its limit $-4/(81\pi)$ (dotted); it is not monotone. (c) Distance to the continuum value, $C_4^{(2)} - C_4(L)$, for the complete window (solid) and the full zero-mode window (dashed); thin lines continue each with its tail fit T_{22} ($300 \le L \le 2000$, Appendix A.5). The complete window falls like $\ln^2 L/L$; the full zero-mode window levels off at 0.01025 (dotted), the part that the mass window D^M supplies. Script `make_fig_d2.py`.

sunset but not for the mixed orderings at ϵ^3. Scheme W_{cs} adds at ϵ^3

$$D^{(3)} \mathrel{+}= \frac{\delta^{(3)}(L)}{2}\, X_0, \qquad \delta^{(3)}(L) = \tfrac{1}{4}\big[e_1^{(3)}(L-2) - e_1^{\mathrm{tag},(3)}(L;2,2)\big], \tag{23}$$

where X is M' with each term multiplied by the summed level of its χ legs (plus for hops, minus for pair creation and annihilation), X_0 its ϵ^0 part, and $e_1^{\mathrm{tag}}(\lambda;2,2)$ is $E - E_0$ of the ϕ channel $|1\rangle$ at cutoff λ whose external zero-mode quantum is tagged with charges $(2,2)$: it has level 0 in H_0, but the cutoff and the windows count it as level 2 before it is absorbed and after it is re-emitted (Appendix C.3). $\delta^{(3)}(L)$ is an exact rational at each L, one number per cutoff; nothing is fitted (Appendix C.3 derives the operator form). Chain sums give it for every even $L \le 16000$ (equal to the rationals at $L \le 16$ to 4×10^{-16}). It is not monotone, with a minimum -0.0159046 at $L = 34$ (Fig. 1b), and

$$\delta^{(3)}(L) = -\frac{4}{81\pi} - \frac{1}{54\pi}\,\frac{\ln L}{L} + \mathcal{O}(1/L), \tag{24}$$

where the limit $\delta_\infty = -\frac{4}{81\pi}$ (to 7×10^{-11} by free-limit fits) and the $\ln L/L$ coefficient are identified numerically, not derived. With this identification the shift $4a\, w_{aa}\delta_\infty = -\frac{16}{81\pi} a\, w_{aa}$ of $R_3(\chi_{aa})$ that the window produces is two thirds of the ϵ^2 exchange-sunset defect $\frac{8}{27\pi} a\, w_{aa}$ of R_2, and $\frac{16}{81\pi} = 0.0629$ is close to the W_c remainder above. The shift is therefore expected to cancel that remainder as $L \to \infty$; this is not established, since R_3 is known only to $L = 16$. W_{cs} leaves C_0 to C_4, $r_{\le 3}$ and $g_{\le 3}$ unchanged and shifts C_5 by $\delta^{(3)}(L)S(L)$, with $S(L)$ the C_5 response to $\frac{1}{2}X_0$; S has no $\ln L$ and $S_\infty \equiv S(\infty) = 0.367187$, so the shift tends to $\delta_\infty S_\infty = -0.005772$.

Data and tail (Appendix B.6). $C_5^{W_{cs}}(L)$ is raw C_5 to $L = 300$ plus the exact W_c shift and $\delta^{(3)}S$ (both known to $L = 1000$). The tails of C_2 to C_4 carry $\ln^{n-2} L/L$, and hold-out tests select $\ln^3 L/L$

for C_5. The fits use

$$C_5(L) = A + \sum_{j=0}^{3} a_j \frac{\ln^j L}{L} + \sum_{j=0}^{k_2} a'_j \frac{\ln^j L}{L^2}, \qquad k_2 \in \{0,1,2\}, \tag{25}$$

with no $\ln L$ term, on every even L from $L_{\min} \in \{30, 40, 60, 80, 100\}$ to 300. Fitted to $L \le 200$, these forms predict $L = 202$ to 300 to 4.3×10^{-6}, against 9.4×10^{-6} with a free $\ln L$; fitted to $L \le 240$, both predict $L = 242$ to 300 to about 6×10^{-7} (Appendix B.6). The median and half range of the 15 fits give

$$C_5 = -38.835(13) \qquad \text{against} \qquad C_5^{(2)} = -38.8336459, \tag{26}$$

with the unrounded median -38.83544, 0.0018 below the continuum value. Towers with one more power of $\ln L$ at $1/L$ ($k_1 = 4$), which the hold-out test does not exclude, move the median by only $+0.002$ but widen the half range of the fits from 0.013 to 0.050 (Appendix B.6); the quoted uncertainty does not include this. Without the external-line window, $C_5^{W_c} = -38.830(13)$ (median -38.8297), higher by about $-\delta_\infty S_\infty$ as expected. Both agree with the continuum value within the spread of the fits, so C_5 does not distinguish W_c from W_{cs}; the case for the external-line window rests on the spacing R_3 (Sec. 4.6).

Residual logarithm. A term $b \ln L$ that the data cannot see would move the limit. Over the 15 fits of Eq. (25), the median of the free coefficient drifts from -0.011 at $L_{\max} = 200$ to $+0.004$ at 300; a second set of twelve fits ($L_{\min} = 30$ to 80) drifts from -0.0134 to $+0.0054$ (Appendix B.6). On C_4^W, which has no $\ln L$, the 15-fit estimator returns $+0.0005$. On a synthetic series with a C_5-like tail and a known b, both sets read about 0.004 low; corrected for this bias, the median is about $+0.008$ (15 fits; $+0.009$ with 12). We take $|b| \lesssim 0.01$ as a judgement from this median, its drift and its bias; it is not a confidence bound. The individual fits of the 15-fit set range from -0.05 to $+0.07$, but the extreme ones have nearly as many parameters as points; taken as a bound they would allow shifts of C_5 between about -0.7 and $+0.5$. In the 15 fits, fixing $b = \mp 0.01$ moves the limit by about ± 0.1, and $b = +0.008$ would move it by about -0.08; bounding $|b|$ well below 0.01 would need $L \gtrsim 400$. The external-line class itself gives no $\ln L$: the $\ln L$ terms of S cancel within the hop and within the pair sector separately.

Status. The complete window has no fitted parameter, removes the $\ln L$ and is verified through ϵ^4. At ϵ^5 the rule W_c is needed to remove a linear divergence, and the external-line window to reduce R_3 (by at least 95% at $L = 14$ and 16). With both, C_5 agrees with the continuum. Both rules are provisional: they are not derived in full, and the window is not proven complete at this order. Like all windows here, they were derived after their effect had been seen and with the continuum values known.

4.6 Descendant spacings

A descendant lies an integer above its primary at every order. With

$$R_n(L) = \big[E(\chi_{aa}) - E(\phi) - 2a\big]_{\epsilon^n}, \tag{27}$$

the ϵ^n coefficient of the spacing of $\chi_{aa} = \partial^a \bar\partial^a \phi$ computed with the same r, g and window, and the same for the descendants $(\partial\bar\partial)^a \phi^2$ of ϕ^2, the exact small-L solvers `complete_exact` and `ce_wc` (Sec. 4.2) give every R_n, $n \le 8$, as rationals for $L \le 16$ in each scheme. These tests need no continuum input; Appendix C gives the data.

- ϵ^2: in the consistent and zero-mode windows the exchange sunset of χ_{aa} has only $L - 4a$, and $R_2 \to \frac{8a\, w_{aa}}{27\pi}$ (0.0243 for χ_{11} at $L = 16$). D^M shifts it by exactly $-\frac{8a\, w_{aa}}{27\pi}$, so in the complete window R_2 decays like $1/L$ ($0.0114/L$ for χ_{11}), also for the ϕ^2 descendants.

- ϵ^3, single-quantum descendants: $R_3(\chi_{11})$ is 0.0156 in W_c and -1.6×10^{-4} in W_{cs} at $L = 16$. At $L = 14$ and 16 the window leaves at most 3% of the W_c value for $a = 1$ to 3, so at least 95% of the defect is removed at $L = 14$ and 16. Free-limit fits of χ_{11} leave a few percent, and the window is expected to cancel the defect as $L \to \infty$ (above); R_3 itself is not computed beyond $L = 16$.
- ϵ^3, descendants of ϕ^2: at $L = 16$, $R_3 = -0.057$ (level 2) and -0.059 (level 4) in W_{cs}; the level-4 value still moves away from zero. Not settled.
- ϵ^4: in W_c and W_{cs}, $R_4(\chi_{11})$ grows, with a local slope of about 0.028 per unit of $\ln L$ at $L = 8$ to 16, while with budget-free legs it goes to zero: the same leg defect one order higher. A partial χ-line window ($W_{cs4\chi}$, calibrated on χ_{11}, with the ϵ^4 difference between the χ-line and zero-mode self-energies as a spectator term) closes 95% and 90% of the gaps of χ_{22} and χ_{33} to their budget-free lines at $L = 14$, but the χ_{22} gap is not shrinking, the ϕ^2 descendants still grow, and $R_5(\chi_{aa})$ grows. The window coefficient grows roughly like $\ln L$ at $L \le 16$, which suggests that C_6 in W_c carries about $+0.010 \ln L$ (a prediction, not a measurement). The ϵ^4 window is not yet known.

4.7 Sixth to eighth order

C_6 to C_8 are not reproduced. At $L = 16$, $C_6 \approx 165.4$ in W_c, W_{cs} and $W_{cs4\chi}$ (Table 30), 54% below the continuum value 362.366(4), and $C_6(L)$ still rises by about 7.5 per step of 2 in L. The external-line windows shift C_6 by about 10^{-4} of its value there, while W and W_c differ by about 1% (Appendix B.7). The production tables to $L = 34$ (raw and consistent) are shipped as finite-cutoff data, but those windows diverge from ϵ^4 on, and the external-line windows from ϵ^4 on are not known.

5 Results in three dimensions

Model and cutoff. In $d = 3$ the generalized free field is quantized on S^2 (Sec. 2, Ref. [4]). A mode (n, ℓ, m) has energy $\Delta_\sigma + 2n + \ell$, with ℓ the angular momentum; the zero mode $n = \ell = 0$ is the local oscillator, and all other modes form the bath. We keep the states with bath level

$$N_b = \sum_{(n,\ell,m)\neq(0,0,0)} (2n + \ell)\, n_{n\ell m} \le L \tag{28}$$

and do not truncate the zero mode. The perturbation is $H = H_0 + r\, M + g\, V$ with $M = \int d\Omega\, {:}\phi^2{:}/(4\pi)$ and $V = \int d\Omega\, {:}\phi^4{:}/(48\pi)$. r and g are tuned order by order to $E_1 - E_0 = \Delta_\sigma$ and $E_3 - E_0 = 3 - \Delta_\sigma$, and

$$\Delta_\varepsilon(\epsilon; L) = E_2 - E_0 = \tfrac{3}{2} - \tfrac{\epsilon}{6} + \sum_{n\ge 2} C_n(L)\, \epsilon^n, \qquad \Delta_\sigma = \frac{3 - \epsilon}{4}. \tag{29}$$

The continuum values $C_n^{(3)}$ are those of Table 1.

Two codes. The production package (`code/lri_3d_hpc/`) builds the truncated Hamiltonian explicitly in the SO(3)-scalar, inversion-even sector and solves the Rayleigh–Schrödinger recursion through ϵ^8 in x87 long double. It reached $L = 16$, 18 and 20 (196 585 scalar bath states at $L = 20$, from 182 727 113 before the projection; about 2550 service units on SDSC Expanse, charge units of about one per core-hour, with large-memory jobs charged more). It implements two prescriptions: raw, and the vacuum window $D_L(N_b) = E_v(L) - E_v(L - N_b)$.

The exact-sum package (`code/lri_3d_c4_exact/`) evaluates the same coefficients as sums over intermediate states, with exact partial-wave reductions on S^2 and exact zero-mode projections, at every even L:

- C_2 to $L = 8000$, C_3 and C_4 to $L = 600$, C_5 to $L = 40$;
- every window shift to $L = 8000$;
- every window scheme, including the complete window of $d = 2$ (Sec. 4).

At every even $L \leq 20$ the two codes agree on C_2 to C_5, r, g and the reference energies E_0 to E_3 to 3×10^{-13}. The window schemes that the production package lacks agree with a brute force, built as explicit operators on the package's own Hamiltonian at $L \leq 14$, to 4.3×10^{-13}. Table 7 gives the production values.

	consistent (vacuum window D_L)			raw		
	$L = 16$	$L = 18$	$L = 20$	$L = 16$	$L = 18$	$L = 20$
C_2	0.1799805130	0.1801641045	0.1803065210	0.1799805130	0.1801641045	0.1803065210
C_3	−0.3238079917	−0.3271370505	−0.3298433574	−0.3238079917	−0.3271370505	−0.3298433574
C_4	0.7154707349	0.7295143865	0.7400078407	0.7082381633	0.7206323697	0.7293066716
C_5	−3.157412985	−3.274129748	−3.376315158	−3.173233312	−3.292870751	−3.398184422
C_6	15.68169942	16.38726028	17.03798989	15.67552153	16.38079206	17.03198660
C_7	−84.42870757	−88.02482107	−91.14377531	−84.17501495	−87.66743530	−90.65688526
C_8	548.2872011	578.4398685	605.8501134	548.2146549	578.3577582	605.7652262

Table 7: Finite-cutoff coefficients $C_n(L)$ in $d = 3$ from the production package `lri_3d_hpc` (10 significant digits; the records hold 20), raw and with the vacuum window D_L. C_2 and C_3 do not depend on the window. The exact sums reproduce C_2 to C_5 of this table to 3×10^{-13}; from C_4 on, neither scheme converges (Fig. 2).

The vacuum window diverges at ϵ^4. $C_2(L)$ and $C_3(L)$ do not depend on the window and converge. From C_4 on, the vacuum window does not suffice (Fig. 2).

- As in $d = 2$, divergent sub-bubbles inside a state see only the level left by the other quanta, while r, g and the vacuum are tuned with the full L.
- In $d = 3$ the mass sub-bubble diverges like $L^{3/2}$ instead of L, so every term is one power worse.
- Uncorrected, $C_4(L)$ falls like $-2.135 \times 10^{-5}\, L^2$.
- With D_L alone it peaks at 0.759 near $L = 32$ and is -0.749 at $L = 600$. The terms are $-7.433 \times 10^{-5}\, L$ and $-\frac{10}{243\,\Gamma(7/2)}\sqrt{L}\ln L$, which the diagonal and zero-mode-number-changing parts of the zero-mode window remove.
- The mass window (χ-attached sunset term), which in $d = 2$ is a finite shift, here removes a further $3.614 \times 10^{-4}\sqrt{L}$.

Complete window. The complete window W of $d = 2$ is carried over unchanged: vacuum window, full zero-mode window, and mass window $-\frac{1}{2}r_2'(L)\{\hat{N}_b, M'\}$, with $\hat{N}_b$ the bath-level operator. With it, $C_4(L)$ converges like $1/L$. At $L = 600$ it is 1.21735. With the limit fixed at $C_4^{(3)}$, a free $\sqrt{L}$ term comes out below 7×10^{-8}. The mass window is used in its leading-order linear form; since r_2 is not linear in L in $d = 3$, the exact form may differ by a finite amount, which the agreement below suggests is small but which is not derived.

At ϵ^5 the nested windows are referred to the outer cutoff (rule W_c of $d = 2$); the external-line window (W_{cs}) changes $C_5(40)$ by -0.003.

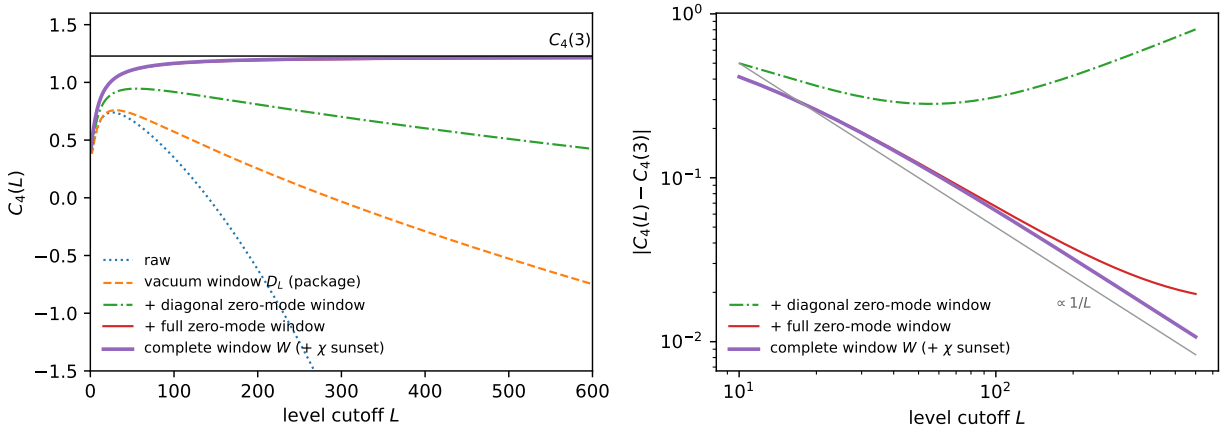


Figure 2: $C_4(L)$ in $d = 3$ from the exact sums, in each window scheme. Left: raw, the vacuum window D_L of the production package, and the zero-mode and mass windows added one by one (the full-zero-mode curve lies under W at this scale). Right: distance from $C_4^{(3)}$ on logarithmic axes. The complete window W converges like $1/L$; without its mass window the distance stops decreasing, has its minimum near $L \approx 1100$ and grows like $3.6 \times 10^{-4}\sqrt{L}$ beyond the plotted range.

Extrapolations. Fit forms (towers of $L^{-k}\ln^j L$ terms, from the $d = 2$ protocol) and windows were chosen after the data were seen, with the continuum values known; the quoted uncertainties are spreads over fits, not standard errors.

- $C_2 = 0.181221785405554$ (15 digits of the closed form; fit to $L \le 8000$).
- $C_3 = -0.353581479(20)$ ($\delta = 1 \times 10^{-7}$): mean and half-range of three fit windows.
- $C_4 = 1.228070(21)$ ($\delta = 5 \times 10^{-6}$): median of one tower over 12 fit windows; the uncertainty is the half-range, including a blind fit to $L \le 400$ that predicts $402 \le L \le 600$ to 1.6×10^{-8}. Towers with one more correction term spread to about 10^{-4}.
- C_5, provisional and calibrated on $C_4^{(3)}$: towers are kept if, on $C_4^W(L)$ at the same $L \le 40$, they reproduce $C_4^{(3)}$ to within 0.036 and pass a blind test. The median of their C_5 fits gives -6.0 ± 0.3 (median -6.005) against $C_5^{(3)} = -5.850351421$. The uncertainty is the worst C_4 miss relative to the remaining C_4 correction at $L = 40$ (0.036 of 0.147), times the remaining C_5 correction at $L = 40$ (1.28, which uses $C_5^{(3)}$ itself).
  - This is a weak test: $C_5^{W_c}(40) = -4.574$ is still 1.28 above the limit.

– A term in $L^{-1/2}$, which $L \le 40$ cannot exclude, would lower the limit by 0.5 to 1.

– A window without the mass window also fits at $L \le 40$, although it differs from W_c by a term growing like L^2.

– Going further needs a new representation of the angular channels: the cost grows like $L^{8.4}$ and the memory like $L^{4.5}$.

Descendant spacings. For the scalar descendant of ϕ (the $n = 1$, $\ell = 0$ quantum), the window removes the ϵ^2 violation of its spacing as $L \to \infty$. At ϵ^3 the external-line window acts in the right direction, but the violation is not shown to vanish at $L \le 16$.

6 Independent checks

One dimension.

- *Window engines.* The complete window through ϵ^5 was computed by two independent engines, one on the solver's kernels and one written from scratch with exact rationals, plus a third engine for a skeptic's check. Four algorithms within the second engine agree to 1.3×10^{-11}; the two engines agree on 3808 numbers to 9.6×10^{-12}; the third reproduces ΔC_5 at $\Lambda \le 28$ to 2.4×10^{-12} (quoted as 10^{-11} in the paper).

- *Eighth-order window.* The complete-window solver reproduces the exact ΔC_4, ΔC_5 at $\Lambda = 46$–72 to 2×10^{-11}.

- *Exact sums against the solver.* The exact sums reproduce the solver's raw C_4 and C_5 at $\Lambda \le 118$ to 1.4×10^{-12} and 10^{-11}, and the window pieces to 10^{-15}; brute-force Rayleigh–Schrödinger on explicit bases at small cutoffs agrees to 10^{-11}.

- *Eighth-order code.* The optimized eighth-order code reproduces the 38 saved records of its original version to 2.1×10^{-8} and the thermal solver's C_0–C_7 to 6×10^{-14} relative.

- *Blind protocol of the exact sums.* Fit records were frozen and hash-committed before each larger run existed: $\Lambda \le 1000$ before 2000 for C_4, $\Lambda \le 320$ before 400 and $\Lambda \le 400$ before 500 for C_5. Their predictions of the new cutoffs are quoted in Section 3.4.

- *Constrained C_5.* A separately written fit of the constrained extrapolation, made without reading the package's fit code, gives $-731.13(10)$; a separate derivation confirms $\omega_1 = \frac{7}{2}$ as the one-loop exponent of ϕ^6 and the closed form (12).

Two dimensions. Two solvers were written from a first-principles specification of the $d = 2$ model (Sec. 4.1) by agents that saw neither the production code nor each other's work. The first is plain Rayleigh–Schrödinger perturbation theory to ϵ^8 with full wavefunction corrections in exact rational arithmetic, in two codes (`fractions` over the full space, and a reflection-reduced code modulo many primes with rational reconstruction checked on held-out primes), for $L = 0, \ldots, 18$ in all three schemes. The second uses no perturbation theory: the truncated Hamiltonian is built at complex ϵ on a circle with 256-bit arithmetic, r and g are Newton-solved, the eigenvalues are followed from the references, and Cauchy integrals give the coefficients, for $L \le 8$. Over C_n, r_n, g_n, the reference energies and all sub-cutoff vacua, $n \le 8$, the production solver agrees with the exact one to 4×10^{-13} (scaled difference $\max |a - b| / \max(1, |b|)$; raw and consistent $L \le 18$, literal $L \le 14$), the Cauchy solver with the exact one to 5×10^{-46}, and the production C_n with the Cauchy

ones to 1.6×10^{-14}. The large-L sum codes of Sec. 4.2 reproduce these values at small L. The raw scheme was specified from first principles; for the consistent and literal schemes the definitions were taken from the production package, so there the check covers the implementation, not the choice of window. These checks verify that we compute the stated truncation; they do not test its continuum limit.

Three dimensions. These checks show that the codes compute the stated truncation; the windows and fit forms are the assumptions.

- *Two codes:* the exact sums and the production package agree to 3×10^{-13} at $L \leq 20$ (Sec. 5).
- *Original implementation:* the precomputed Hamiltonian of each production cutoff, solved at $L = 14$, agrees with the original implementation shipped in the package to 1.25×10^{-14}.
- *Stored Hamiltonians:* the one of $L = 20$ reproduces those of $L = 16$ and $L = 18$ to 9.8×10^{-15}.
- *Recursion:* every recursion residual is at most 5.1×10^{-19}.
- *Engine regression:* the final ϵ^5 engine reproduces the stored $L \leq 20$ run to 5×10^{-13} and all reduced $L = 40$ sub-sums bit for bit.

7 Caveats

One dimension.

- The windows were derived after their effects had been seen and with the continuum values known, though with no fitted parameter.
- The extrapolation dominates the error, and the exponents (10) are an assumption. Fits that reach the exact C_5 need an effective ω_3 of 380–430, not the gap value 556. A second channel with the leading power $\Lambda^{-1/2}$, from the phase operator of the cutoff kernel, enters at ϵ^4 (cited check `code/lri_d1_fullblock_tail_response`); it does not affect the top logarithm (12) but does affect the lower ones, so tying them to ω_2, ω_3 is not exact.
- The ϵ^4 top-log coefficient fitted from the exact $C_4(\Lambda)$ lies 0.3% from Eq. (12), depending on the basis; whether this is a real violation is undecided. Taking c_5 from the C_2 chain of Eq. (12) instead, a value 0.3% smaller, moves the constrained C_5 by 4×10^{-4} to 1.4×10^{-3} (absolute), far below its quoted uncertainty (`xchk/REPORT_blind.md` of `lri_1d_c5_exact`).
- A test protocol for the raw C_4 and C_5, fixed before most of the large-cutoff data had been seen, had mixed results: its held-out predictions passed; window stability failed for C_5 in two of three bases and for C_4 in one, and tail insensitivity failed for C_5. Every $d = 1$ fit beyond C_3 involves choices made after the data were seen, apart from the blind predictions of the exact sums. The constrained C_5 fit's selection thresholds were set after a run had printed the exact value; the separately written fit is the quoted determination.
- C_8, calibrated on the diagrammatic C_6 and C_7, is not an independent determination.
- We have not constructed an effective Hamiltonian beyond the windows.

Two dimensions. The windows of Secs. 4.4 and 4.5 were derived after their effects had been seen and with the continuum values known, though they contain no fitted parameter. The complete window is verified through ϵ^4; at ϵ^5 the rule W_c and the external-line window are needed as well, and the window is neither derived in full nor proven complete there. The limit $-4/(81\pi)$ of $\delta^{(3)}$ is identified numerically. The quoted limits are spreads over tail models and fit windows, not standard errors. The C_5 value assumes no residual $\ln L$ term; one with $|b| \lesssim 0.01$, a bound judged from the median of the fits rather than a confidence bound, is not excluded at $L \leq 300$ and would move C_5 by up to about 0.1. C_5 does not distinguish W_c from W_{cs}; the external-line window rests on the spacing R_3. The ϵ^3 spacings of the ϕ^2 descendants are not settled at $L \leq 16$, the ϵ^4 window is not known, and C_6 to C_8 are not reproduced. The production tables (raw and consistent schemes) diverge from ϵ^4 on, linearly in L (raw) and like $\ln L$ (consistent), so they are finite-cutoff data of those schemes, not continuum estimates.

Three dimensions. The fit forms and windows were chosen after the data were seen, with the continuum values known. The mass window is used in its leading-order linear form, which is not derived for $d = 3$. The C_4 uncertainty is the spread of one tower over 12 fit windows; towers with one more correction term spread to about 10^{-4}. C_5 is provisional: it is calibrated on $C_4^{(3)}$, the scale of its uncertainty uses $C_5^{(3)}$ itself, a term in $L^{-1/2}$ is not excluded at $L \leq 40$, and at $L \leq 40$ the test cannot separate W_c from the window without the mass window. The ϵ^3 violation of the descendant spacing (scalar descendant of ϕ) is not shown to vanish at $L \leq 16$. The last digit of C_3 is platform-sensitive: one fit window moves by about 10^{-9} between machines, and the replay checks it to a tenth of the stated uncertainty.

8 Reproduction

8.1 Layout

All paths are relative to the folder `HT_Pert/` of the public repository.

- `code/<package>/`: one package per solver or analysis, shipped at the commit its production ran. Each has a `public.mk` with the targets `test` and `replay` (some also `smoke`, `replay-full`, `fetch-data`) and a `public/` folder with the replay scripts.
- `data/HT_<d>d/`: the production records.
- `results/HT_<d>d/`: the expected value of every quoted number (`quoted_numbers.json`) and summaries.

`make test` and `make replay` in `HT_Pert/` run the targets of every package; in a package folder, `make -f public.mk test|replay`. A replay recomputes every quoted number of its package from the shipped records, without a production rerun, prints a PASS/FAIL table and exits non-zero on any failure.

8.2 Packages and records

package (`code/`)	d, role	content, records, cost
`lri_hpc_Hamitonian`	1: solver, analyses	HT solver to ϵ^7; joint tail fits, window engines, identities. Records `data/HT_1d/thermal_expanse` (111 cutoffs) and the analysis outputs
`lri_eh_hpc`	1: window	isolated g^2D window, closed form K
`lri_c8_hpc`	1: solver	eighth order, $\Lambda \leq 92$; records `data/HT_1d/c8`
`lri_d1_full_coefficient_flow`	1: solver	complete window through ϵ^8, $\Lambda \leq 72$; records `data/HT_1d/s10_production`
`lri_1d_dconn_refit`	1: analysis	extrapolations of C_4–C_8 with the complete window
`lri_1d_c4_exact`, `lri_1d_c5_exact`	1: exact sums	C_2–C_4 to $\Lambda = 2000$, C_5 to 500; records `data/HT_1d/exact_c4`, `exact_c5`; blind fit records
`lri_d1_fullblock_tail_response`	1: cited check	second $\Lambda^{-1/2}$ channel
`lri_d2_coeff_hpc`	2: solver	eighth-order production C_0–C_8, raw and consistent windows to $L = 34$ (literal window to $L = 14$); git ref `f847d38`; records `data/HT_2d/prod` (41 self-hashed records, tables, validation receipts); production 2.7 node-hours (symmetry $L \leq 30$: 0.6; Expanse $L = 32, 34$: 2.1, the largest point 48 min and 463 GiB); test 1.2–1.5 min (61 tests; native backend needs x86-64 Linux), smoke 10 s, replay 1 s
`lri_2d_coeff_independent_check`	2: checks, exact sums, windows	clean-room exact RS and Cauchy solvers; exact C_2, C_3 to $L = 600$, C_4 to $L = 2000$ in every scheme and with the complete window, raw C_5 to $L = 300$, $\delta^{(3)}$ to $L = 16000$; the all-order window solver (W_c, W_{cs}, $W_{cs4\chi}$, $L \leq 16$); dated snapshot (no git; derivations in Appendices A to C); records inside the package; production: C_5 to $L = 300$ about 55–60 node-hours with the optimized engine (measured: 130 node-hours for 40% of the tasks with the unmodified engine, 32 for the remaining 60% with the optimized one), C_4 to $L = 2000$ 1.3 node-hours (77 min on one 40-core node), $\delta^{(3)}$ to $L = 16000$ 11 min (18 GB), the exact solvers minutes to an hour per cutoff; test 1.6–1.8 min (with python-flint), replay 3 s, replay-full 15 s
`lri_3d_c4_exact`	3: exact sums	C_2 at every even $L \leq 8000$, C_3, C_4 to $L = 600$, C_5 to $L = 40$, window shifts to $L = 8000$, every window scheme; the paper's $C_2^{(3)}$ to $C_4^{(3)}$ and the C_5 footnote; git ref `d5afa4f`; results in the package (`results/`); test 1–3.5 min, replay about 3 s
`lri_3d_hpc`	3: solver	C_2–C_8 at $L = 16$, 18, 20, raw and vacuum window; the production check of the exact sums and the $L = 14$ gates against the original code; git ref `7c8a67e`; records `data/HT_3d/pert3d`, table `results/HT_3d/SUMMARY.md`; production about 2550 SU on SDSC Expanse; test 1.5–2.1 min, replay about 1 s

8.3 Rules that are part of the results

The fit families and windows, the selection and calibration rules, the rounding of quoted ranges and the status words are part of the results; the `REPRODUCE` file of each package lists them under "Decisions you must not make".

8.4 Result → code, record, command, tolerance

result	value	pkg	records	command	tolerance
$d = 1$ (packages: S = `lri_hpc_Hamitonian`, G = `lri_eh_hpc`, E = `lri_c8_hpc`, W = `lri_d1_full_coefficient_flow`, R = `lri_1d_dconn_refit`, X4 = `lri_1d_c4_exact`, X5 = `lri_1d_c5_exact`, F = `lri_d1_fullblock_tail_response`; records: T = `data/HT_1d/thermal_expanse`, A = the package's `analysis/`, C8 = `data/HT_1d/c8`, S10 = `data/HT_1d/s10_production`, R = the package's summary outputs, X4, X5 = `data/HT_1d/exact_c4`, `exact_c5`, F = the package's `results/`. Each package's integrity checks (code fingerprints, checksum lists, self-hashed records, gates) are merged into one row; `results/HT_1d/quoted_numbers.json` describes every row.)					
integrity of code and records	pass	S	T, A	replay `a1.code`, `a1.records`, `a1.xcheck`, `a1.gates`, `a1.reruns`	exact
largest cutoff of the thermal data	118	S	T, A	replay `a1.lam`	exact
basis size at $\Lambda = 118$	3.5×10^9	S	T, A	replay `a1.states`	as stated
raw $C_4(118)$, % of C_4	67	S	T, A	replay `a1.frac4`	as stated
raw $C_7(118)$, % of C_7	44	S	T, A	replay `a1.frac7`	as stated
tail of $C_2(\Lambda)$, Eq. (9)	$-40/(81\sqrt{\pi})\,\Lambda^{-1/2}$	S	T, A	replay `a1.T2`	bound
ΔC_4 at $\Lambda = 118$	0.080	S	T, A	replay `a1.win4`	as stated
ΔC_5 at $\Lambda = 118$	0.71	S	T, A	replay `a1.win5`	as stated
large-Λ constant of ΔC_4, Eq. (15)	0.0803	S	T, A	replay `a1.win4inf`	as stated
raw C_4 below C_4 (%)	0.19	S	T, A	replay `a1.offset4`	as stated
C_4 from the solver (union): check of its bases A–F part	45.415–45.420	S	T, A	replay `a1.c4AF`	as stated
largest relative C_5 miss, bases A–F	2×10^{-3}	S	T, A	replay `a1.c5pre`	as stated
C_5 from the solver (union): check of its freed-tail part	-731.36 to -730.97	S	T, A	replay `a1.c5freed`	as stated
ω_3 from the ϕ^6 gap	556	S	T, A	replay `a1.omega3`	as stated
effective ω_3 of the fits that reach C_5	380–430	S	T, A	replay `a1.omega3eff`	as stated
verdict of the test protocol	unresolved	S	T, A	replay `a1.protocol`	exact
agreement of the window engines through ϵ^5	10^{-11}	S	T, A	replay `a1.engine`	as stated
$[\epsilon^3]v_n$ with D^{conn}, per N_{w}	0.003	S	T, A	replay `a1.ident3`	reading
% of the g^2D violation at ϵ^4 removed	96–100	S	T, A	replay `a1.ident4`	reading
$[\epsilon^5]v_n$ with D^{conn}: zero within $\pm$, per N_{w}	0.1–0.2	S	T, A	replay `a1.ident5`	reading
cutoff placement: raw HT exact through ϵ^3	to 10^{-14}	S	T, A	replay `a1.phase`	bound
cutoff placement at ϵ^4, g^2D alone	$3\eta_4 = 0.0561$ required	S	T, A	replay `a1.phase.g2D`	info
raw $[\epsilon^2]v_n$	$N_{\mathrm{w}}/54$	S	T, A	replay `a1.W54`	bound

result	value	pkg	records	command	tolerance
C_4, bases A–F, lower cutoffs 40–70	45.415–45.419	S	T, A	replay `a1.c4AF.range`	exact
C_5 misses, bases A–F	-1.53 to -0.51	S	T, A	replay `a1.c5pre.miss`	exact
C_5 misses, leading tail freed	-0.22 to $+0.06$	S	T, A	replay `a1.c5freed.miss`	exact
$[\epsilon^4]v_1(\phi)$, g^2D alone (median)	$+0.0495$	S	T, A	replay `a1.phase.g2D.value`	exact
integrity of code and records	pass	G	A	replay `a1.eh.code`, `a1.eh.receipt`, `a1.eh.bench`	exact
finite part K of the g^2D shift of C_4	0.0616119308426799	G	A	replay `a1.eh.K`	as stated
exact g^2D shift $\Delta C_4(\Lambda)$	$\log\Lambda/17496 + K$	G	A	replay `a1.eh.exact`	bound
largest cutoff of the C_8 data	92	E	C8	replay `c8.reach`	exact
integrity of code and records	pass	E	C8	replay `c8.lanes`, `c8.envelopes`, `c8.exports`, `c8.source`, `c8.manifest`, `c8.gates`, `c8.validation`, `c8.datajson`	exact
$C_8(92)$, raw and g^2D	2883374.25, 2883433.21	E	C8	replay `c8.values`	exact
C_6, C_7, C_8 at every cutoff	table in the package	E	C8	replay `c8.table`	as stated
eighth-order code against its original version	2×10^{-8}	E	C8	replay `c8.original`	as stated
raw C_0–C_7 against the thermal solver	6.1×10^{-14}	E	C8	replay `c8.thermal`	as stated
integrity of code and records	pass	W	S10	replay `s10.manifest`, `s10.code`, `s10.records`, `s10.csv`	exact
largest cutoff of D^{conn} through ϵ^8	72	W	S10	replay `s10.winlam`	exact
ΔC_6, ΔC_7, ΔC_8 at $\Lambda = 72$	$+1.1717361$, -1.828185, -8.5641	W	S10	replay `s10.dC678`	exact
complete-window solver against the exact ΔC_4, ΔC_5	2×10^{-11}	W	S10	replay `s10.match`	exact
integrity of code and records	pass	R	R, A	replay `rf.inputs`	exact
blind check of the window continuation (per typical uncertainty)	0.099	R	R, A	replay `rf.blind`	exact

result	value	pkg	records	command	tolerance
C_4 from the solver (union)	45.415–45.420	R	R, A	replay `rf.C4`	exact
its largest deviation	2×10^{-4}	R	R, A	replay `rf.C4dev`	exact
C_5 from the solver (union)	-731.36 to -730.97	R	R, A	replay `rf.C5`	exact
its largest deviation	3×10^{-4}	R	R, A	replay `rf.C5dev`	exact
C_6	14 320–14 342	R	R, A	replay `rf.C6`	exact
its largest deviation	8×10^{-4}	R	R, A	replay `rf.C6dev`	exact
C_7	$-322\,890$ to $-322\,230$	R	R, A	replay `rf.C7`	exact
its largest deviation	1×10^{-3}	R	R, A	replay `rf.C7dev`	exact
sensitivity of C_7 to the weighting (not quoted)	not quoted	R	R, A	replay `rf.C7w`	info
C_8, calibrated	8.0652×10^{6}	R	R, A	replay `rf.C8`	exact
its deviation	7×10^{-4}	R	R, A	replay `rf.C8dev`	exact
calibrated range of C_8 ($\times 10^{6}$)	8.061–8.067	R	R, A	replay `rf.C8cal`	info
uncalibrated range of C_8 ($\times 10^{6}$)	8.045–8.073	R	R, A	replay `rf.C8uncal`	exact
window effect on C_6, C_7, C_8	C_6: -0.01 to -0.04; C_7: $+4$ to $+19$; C_8: $+17$ to $+32$, -240 to -490 (freed tails)	R	R, A	replay `rf.paired`	exact
noise of ΔC_6, ΔC_7, ΔC_8, $\Lambda = 46$–72	3.95×10^{-10}, 1.34×10^{-8}, 2.32×10^{-7}	R	R, A	replay `rf.noise`	exact
C_2 from the exact sums	0.657131263984(10)	X4	X4	replay `x4.c2`	exact
its δ	4×10^{-12}	X4	X4	replay `x4.c2`	exact
$\lvert C_2^{\mathrm{HT}} - C_2 \rvert$	3×10^{-12}	X4	X4	replay `x4.c2`	exact
C_3 from the exact sums	$-3.7198563(10)$	X4	X4	replay `x4.c3`	exact
its δ	2×10^{-8}	X4	X4	replay `x4.c3`	exact
$\lvert C_3^{\mathrm{HT}} - C_3 \rvert$	6×10^{-8}	X4	X4	replay `x4.c3`	exact
largest cutoff of the exact C_2–C_4 sums	2000	X4	X4	replay `x4.lexact`	exact
C_4 from the exact sums	45.4249(9)	X4	X4	replay `x4.c4`	exact
its δ	1×10^{-6}	X4	X4	replay `x4.c4`	exact
$C_4^{\mathrm{conn}}(1000)$	37.8351012026	X4	X4	replay `x4.values`	exact
$C_4^{\mathrm{conn}}(2000)$	39.4111422825	X4	X4	replay `x4.values`	exact
overlap of the runs to 1000 and 2000	3.3×10^{-13}	X4	X4	replay `x4.overlap`	bound
blind prediction of $1000 < \Lambda \le 2000$	4.0×10^{-10}	X4	X4	replay `x4.blind`	exact
frozen intercept, basis L2, lower cutoff 200	45.4249416926	X4	X4	replay `x4.intercept`	exact

result	value	pkg	records	command	tolerance
refit intercept, basis L2, lower cutoff 200	45.4249235266	X4	X4	replay `x4.refit`	as stated
C_2, same rule on $\Lambda \le 118$	0.657131272(35)	X4	X4	replay `x4.c23solver`	exact
C_3, same rule on $\Lambda \le 118$	$-3.71984(8)$	X4	X4	replay `x4.c23solver`	exact
C_5 from the exact sums	$-731.13(10)$	X5	X5	replay `x5.c5`	exact
its δ	2×10^{-5}	X5	X5	replay `x5.c5`	exact
largest cutoff of the exact C_5 sums	500	X5	X5	replay `x5.records`	exact
$C_5^{\mathrm{conn}}(400)$	-506.619063805112	X5	X5	replay `x5.values`	exact
$C_5^{\mathrm{conn}}(500)$	-519.770442010804	X5	X5	replay `x5.values`	exact
$\Delta C_5(500)$	0.717651334579	X5	X5	replay `x5.values`	exact
overlap of consecutive runs	6×10^{-12}	X5	X5	replay `x5.overlap`	bound
blind prediction, fits frozen at $\Lambda \le 400$	3×10^{-9}	X5	X5	replay `x5.blind`	exact
blind prediction, fits frozen at $\Lambda \le 320$	1.5×10^{-8}	X5	X5	replay `x5.blind`	exact
free intercept, basis L2	-730.8460638788	X5	X5	replay `x5.intercept`	exact
free intercept, basis M3	-731.0822822059	X5	X5	replay `x5.intercept`	exact
fitted $\log^3$ coefficient, intercept fixed to C_5	$1.945 - -1.994$	X5	X5	replay `x5.tail`	as stated
c_5 predicted from the C_4 tail	1.9965	X5	X5	replay `x5.constrained`	exact
constrained intercepts	-731.1467, -731.1397, -731.1359	X5	X5	replay `x5.constrained`	exact
separately written constrained fit	$-731.13(10)$	X5	X5	replay `x5.crosscheck`	as stated
top-log relation at ϵ^3	5×10^{-6}	X5	X5	replay `x5.toplog`	exact
top-log relation at ϵ^4	0.3%	X5	X5	replay `x5.toplog`	exact
integrity of code and records	pass	F	F	replay `fb.manifest`	exact
leading power of the second channel	$\Lambda^{-1/2}$ at $\epsilon = 0$	F	F	replay `fb.power`	exact
first order of the second channel	from ϵ^4	F	F	replay `fb.onset`	as stated
prefactor of the second channel (not quoted)	$2/(3\sqrt{\pi})$	F	F	replay `fb.amplitude`	as stated
log coefficient of the ϵ^4 phase response (not quoted)	0.4913887735	F	F	replay `fb.B`	as stated
finite-annulus test of the kernel	36 comparisons pass	F	F	replay `fb.annulus`	as stated

result	value	pkg	records	command	tolerance
what the paper uses from this package	structure only	F	F	replay `fb.caveat`	info
status of the second-channel coefficient	plausible only	F	F	replay `fb.unverified`	info
$d = 2$, *perturbative* (P = `code/lri_d2_coeff_hpc`, I = `code/lri_2d_coeff_independent_check`; records R = `data/HT_2d/prod`; inside I: RS = I/`rs_exact/results`, CY = I/`cauchy/results`, CT = I/`continuum/exact_C2_C3.csv`, C4 = I/`c4_exact/results/c4_L2000.npz`, WN = I/`residual_log/results/{offdiag_L20000,nm_L8000}.npz`, C5 = I/`residual_log/large_L/results/c5_L300.npz` with I/`residual_log/results/eps5/dC5_L1000_all.npz`, TG = I/`residual_log/large_L/results/tag3_L16000.npz`, S = I/`residual_log/results/legw/legw_L1000.npz`, CE = I/`residual_log/ce_wc/out_*`, CX = I/`complete_exact/results`, TF = I/`residual_log/results/tag`, WD = I/`c4_exact/results/windows_L20000.npz`, CW = I/`residual_log/C4W_by_L.csv`, BP = I/`c4_exact/fits`, RG = I/`residual_log/large_L/results/regression_opt.txt`; SRC = P/`src/lri_coeff`, REF = P/`reference/lri_d2_perturbative_C8/results`)					
record integrity	41 records, self-hashed	P	R	replay `p2.prod.records`	exact
campaign manifests	3, self-hashed	P	R	replay `p2.prod.campaigns`	exact
solver source digest in every record	550e9266…	P	R, SRC	replay `p2.prod.source`	exact
coefficient tables = records	20 rows	P	R	replay `p2.prod.tables`	exact
tables cited by HT	sha256 d86743c1…, ee72158e…	P	R, REF	replay `p2.prod.cited`	exact
production cutoff	$L \le 34$	P	R	replay `p2.prod.lprod`	exact
order	C_0–C_8	P	R	replay `p2.prod.degree`	exact
validation receipts (symmetry, Expanse)	passed	P	R	replay `p2.prod.validation`	receipts
exact RS vs package	4×10^{-13}	I	RS, R	replay `p2.impl_agree`	1 digit
exact RS cutoffs	$L \le 18$	I	RS	replay `p2.rs_lmax`	exact
Cauchy vs exact RS	5×10^{-46}	I	CY, RS	replay `p2.cauchy_vs_rs`	1 digit
package vs Cauchy	1.6×10^{-14}	I	CY, R	replay `p2.pkg_vs_cauchy`	2 digits
Cauchy cutoffs	$L \le 8$	I	CY	replay `p2.cauchy_lmax`	exact
exact C_2, C_3 code vs package	4×10^{-14}	I	CT, R	replay `p2.low_agree`	1 digit
large-L C_4 code vs exact RS, package	1.1×10^{-13}	I	C4, RS, R	replay `p2.c4_agree`	bound + record
large-L C_5 code vs package	2.2×10^{-13}	I	C5, R	replay `p2.c5_agree`	bound + record
optimized engine, per task	7.3×10^{-16}	I	RG	replay `p2.c5_opt_regress`	record
$L = 300$ assembly vs $L \le 200$ data	7.6×10^{-13}	I	C5	replay `p2.c5_assembly`	2 digits
$\delta^{(3)}$ chain sums vs rationals	4×10^{-16}	I	TG, TF	replay `p2.delta_rational`	1 digit

result	value	pkg	records	command	tolerance
$C_2(\infty)$	0.30806541358, 35 digits	I	closed form, CT	replay `p2.c2_digits`	≥ 35 digits
closed-form $C_2(L)$ = HT $C_2(L)$	$\leq 10^{-13}$	I	CT	replay `p2.c2_closed_form`	bound
$C_3(\infty)$	-0.8748500(1)	I	CT	replay `p2.c3_limit`	10^{-7}
C_3 deviation	1×10^{-8}	I	CT	replay `p2.c3_dev`	1 digit
$C_3(L)$ cutoffs	$L \leq 600$	I	CT	replay `p2.c3_lmax`	exact
C_3 tail, $\ln L/L$ coefficient	0.70734	I	CT	replay `p2.c3_tail`	5 decimals
r_2 slope ($r_2 \simeq 2L/(27\pi)$, Sec. 4.4)	0.0235785	I	C4	replay `p2.r2_slope`	7 decimals
C_2 to C_4	$\delta < 10^{-7}$	I	CT, C4, WN	replay `p2.low_dev`	bound
$C_4(L)$ cutoffs	$L \leq 2000$	I	C4	replay `p2.c4_lfour`	exact
κ (slope of consistent − raw)	2.2073×10^{-4}	I	WD	replay `p2.c4_kappa`	4 decimals
$\ln L$ coefficient, consistent	-0.026563(42) (-0.027)	I	C4	replay `p2.c4_log_consistent`	6 decimals
$\ln L$ coefficient, zero-mode	-0.010480(42)	I	C4	replay `p2.c4_log_zm`	6 decimals
$\ln L$ coefficient, raw	-0.026381	I	C4	replay `p2.c4_log_raw`	6 decimals
scheme limits A	5.0026(5), 5.0213(5), 5.0021	I	C4	replay `p2.c4_A_schemes`	4 decimals
blind test, $L = 1200$ to 2000	8.3×10^{-8}, 4.1×10^{-7}	I	C4, BP	replay `p2.c4_blind`	2 digits
C_4, complete window	5.0488682(10)	I	C4, WN	replay `p2.c4W_limit`	7 decimals
C_4 deviation	5×10^{-8}	I	C4, WN	replay `p2.c4_dev`	1 digit
C_4^W fit spread	6×10^{-7}	I	C4, WN	replay `p2.c4W_spread`	1 digit
C_4^W free $\ln L$	$\leq 3.5 \times 10^{-6}$	I	C4, WN	replay `p2.c4W_nolog`	2 digits
C_4^W table	= series	I	CW	replay `p2.c4W_table`	10^{-13}
zero-mode + off-diagonal window	5.0386139	I	C4, WN	replay `p2.c4_zmoff`	7 decimals
χ-line mass window	0.0102543	I	WN	replay `p2.c4_chi`	7 decimals
$C_5(L)$ cutoffs	$L \leq 300$	I	C5	replay `p2.c5_lfive`	exact
C_5, W_c + external-line window	-38.835(13) (median -38.83544)	I	C5, TG, S	replay `p2.c5_limit`	3 decimals
C_5 deviation	5×10^{-5}	I	C5, TG, S	replay `p2.c5_dev`	1 digit
residual $\ln L$: shift for $\lvert b\rvert = 0.01$	0.1	I	C5, TG, S	replay `p2.c5_logshift`	1 digit

result	value	pkg	records	command	tolerance
hold-out test, fit $\leq 200 \to 300$	4.3×10^{-6} vs 9.4×10^{-6}	I	C5, TG, S	replay `p2.c5_blind`	2 digits
free $\ln L$ coefficient by $L_{\max}$	-0.011 $\to$ +0.004	I	C5, TG, S	replay `p2.c5_bdrift`	3 decimals
free $\ln L$, single fits	-0.05 to +0.07	I	C5, TG, S	replay `p2.c5_bspread`	2 decimals
null test on C_4^W	+0.0005	I	C5	replay `p2.c5_null`	1 digit
estimator calibration	bias ≈ -0.004	I	C5, CW	replay `p2.c5_calibration`	4 decimals
tail power $k_1 = 4$ (15 fits)	shift +0.002, half range 0.050	I	C5, TG, S	replay `p2.c5_k1`	3 decimals
bias-corrected median b	+0.008 (15 fits)	I	C5, CW	replay `p2.c5_calibration15`	4 decimals
free $\ln L$, 12-fit medians	$-0.0134 \to +0.0054$	I	C5, TG, S	replay `p2.c5_bdrift_agent`	4 decimals
$C_5^{W_c}(\infty)$	-38.830(13) (median -38.8297)	I	C5	replay `p2.c5_wc`	3 decimals
$C_5^{W_{cs}} - C_5^{W_c}$ limits	≈ -0.005772	I	C5, TG, S	replay `p2.c5_wcs_minus_wc`	10^{-3}
C_5^W linear term	$-0.0019\,L$	I	C5	replay `p2.c5_W_linear`	6 decimals
$\delta^{(3)}(L)$ cutoffs	$L \leq 16000$	I	TG	replay `p2.delta_lmax`	exact
$\delta^{(3)}$ minimum	-0.0159046 at $L = 34$	I	TG	replay `p2.delta_min`	7 decimals
$\delta^{(3)} \to -4/(81\pi)$	within 7×10^{-11}	I	TG	replay `p2.delta_limit`	bound
$\delta^{(3)}$, $\ln L/L$ term	$-1/(54\pi)$	I	TG	replay `p2.delta_lnL`	5×10^{-7}
$S(\infty)$	0.367187	I	S	replay `p2.S_limit`	6 decimals
$\delta_\infty S_\infty$	-0.005772	I	S	replay `p2.delta_S`	6 decimals
window shifts to	$L = 1000$	I	S, C5	replay `p2.shift_lmax`	exact
R_2, window W	restored, $R_2 \approx 0.0114/L$	I	CX	replay `p2.ident2_W`	4 decimals
R_2, consistent window	0.0243	I	CX	replay `p2.ident2_cons`	4 decimals
R_3, single quantum	$\geq 95\%$ at $L = 14$, 16	I	CE	replay `p2.ident3_chi`	≥ 0.95
$[\epsilon^3]$ residual in W_c	$0.063\,a\,w_{aa}$	I	CE	replay `p2.ident3_wc`	10^{-3}
$R_3(\chi_{11})$ at $L = 16$	-1.6×10^{-4}; W_c: 0.0156	I	CE	replay `p2.ident3_L16`	2 digits
R_3, ϕ^2 descendants	-0.057, -0.059: not settled	I	CE	replay `p2.ident3_phi2`	3 decimals
$R_4(\chi_{11})$ local slope	0.028 per unit of $\ln L$	I	CE	replay `p2.eps4_R4slope`	3 decimals
ϵ^4 χ-line window	gaps 95%, 90%	I	CE, TF	replay `p2.eps4_chiwindow`	per cent

result	value	pkg	records	command	tolerance
C_6 shift identity (App. C.5)	$\delta^{(3)}S_1 = +0.0018$ to $+0.0025$	I	CE, S	replay `p2.c6_identity`	10^{-12}
$C_6(16)$ (not reproduced)	165.4 (+7.5 per step)	I	CE	replay `p2.c6_L16`	range
$d = 3$, *perturbative* (E = `code/lri_3d_c4_exact`, results E`/results`; P = `code/lri_3d_hpc`, records R = `data/HT_3d/pert3d`, table T = `results/HT_3d/SUMMARY.md`; ref = P`/results/reference_coefficients_L14.json`)					
$C_2^{(3)}$	0.181221785405554 (15 digits)	E	E`/results/windows_L8000.npz`	replay `C2-3D`	15 digits
$C_3^{(3)}$	-0.353581479(20), $\delta = 1 \times 10^{-7}$	E	E`/results/ c4_L600.npz`	replay `C3-3D`	PDG; central to 1/10 of the uncertainty
$C_4^{(3)}$, complete window	1.228070(21), $\delta = 5 \times 10^{-6}$	E	E`/results/` (c4, offdiag, dchi, windows)	replay `C4-3D`	PDG
blind fit, no $\sqrt{L}$ term	1.6×10^{-8}; $\lvert b\rvert < 7 \times 10^{-8}$	E	same	replay `C4-BLIND`, `C4-NODIV`	10^{-7}
$C_5^{(3)}$ (provisional)	-6.0 ± 0.3	E	E`/results/c5raw_all_L40.npz`, `dC5_L600.npz`	replay `C5-3D`, `C5-LHALF`	1 decimal
cutoffs	600, 40, 8000	E	E`/results/`	replay `CUTOFFS`	exact
fit summaries	regenerated	E	E`/results/`	replay `FITS-REGEN`, `FITS-QUOTED`	quoted lines
exact sums vs production, $L \le 20$	3×10^{-13}	E, P	ref, R`/results/`	replay `VAL-PKG-C4`, `VAL-PKG-C5`	10^{-12}
window schemes vs brute force, $L \le 14$	4.3×10^{-13}	E, P	E`/results/brute/`	replay `VAL-PKG-C5`	10^{-12}
sums recomputed at small L	11 stages	E, P	E`/results/`	test (all stages)	10^{-12} to 10^{-9}
production $C_n(L)$, $L = 16$, 18, 20 (Table 7, T)	6 rows	P	R`/results/L*/`	replay `TABLE`	stored strings
$C_4(20)/C_4^{(3)}$, read by `HT/`	60.26%	P	T	replay `C4-RATIO`	2 decimals
original implementation, $L = 14$	1.25×10^{-14}	P	R`/gates/`, ref	replay `GATE-L14`	5×10^{-13}
stored Hamiltonians, recursion	9.8×10^{-15}, 5.1×10^{-19}	P	R`/results/`	replay `CROSS-STORE`, `RESIDUAL`	5×10^{-13}, 10^{-16}
records and code identity	194 files	P	`data/HT_3d/MANIFEST.sha256`	replay `REC-MANIFEST`, `REC-SRC`, `REC-CHAIN`	sha256

A Two dimensions: operators, exact fourth order and the complete window

Appendices A to C give the derivations and the supporting data behind Section 4. This appendix covers the operators and schemes at low order, the solver checks, the exact $C_4(L)$, the cause of its $\ln L$ divergence class by class, and the complete window at ϵ^4. Appendix B treats order ϵ^5: the window at all orders, the double counting that rule W_c removes, and the tail of C_5. Appendix C treats the descendant spacings and the external-line windows W_{cs} and $W_{cs4\chi}$, and ends with the status of each window component. Paths are relative to the package `code/lri_2d_coeff_independent_check/` (package I of Section 8), and each subsection names its records. Numbers that Section 4 also quotes have reproduction rows in Section 8. The other numbers of these appendices are read from the records named in each subsection, or evaluated from them where the text says so; the reproduction scripts do not recompute them. "Exact" data are rationals, or float64 evaluations of exact finite sums with a measured rounding error of at most 1.4×10^{-11}. Limits $L \to \infty$ are fits; their uncertainties are spreads over tail models and fit windows, not standard deviations.

A.1 Operators and schemes at low order

Records: `SPEC.md`, `rs_exact/README.md`, `zeromode_exact/README.md`, `cauchy/README.md`.

Factorized vertices. With $Z = a_0 + a_0^\dagger$, $a_0 = A_{00}$, and $\chi = \phi - Z$, normal ordering factorizes:

$$V = \sum_{m=0}^{4} \frac{1}{12}\binom{4}{m} {:}Z^{4-m}{:} \otimes \langle {:}\chi^m{:}\rangle_\theta, \qquad M = \sum_{m=0}^{2} \binom{2}{m} {:}Z^{2-m}{:} \otimes \langle {:}\chi^m{:}\rangle_\theta. \tag{30}$$

At $\epsilon = 0$ the weights factorize holomorphically, $w_{ab} = v(a)v(b)$ with $v(a) = (\frac{1}{2})_a/a!$. The squared V-vertex coefficient times the Wick count of a bundle of m χ lines is $\gamma_m = m!\,\big[\binom{4}{m}/12\big]^2$: $\gamma_1 = \frac{1}{9}$, $\gamma_2 = \frac{1}{2}$, $\gamma_3 = \frac{2}{3}$, $\gamma_4 = \frac{1}{6}$.

Space and particle cap. The truncated space holds the occupation states with spin zero, even under $(a, b) \to (b, a)$, and $\ell \le L$; spin zero forces ℓ even, and the field parity $(-1)^P$ is conserved. Truncated operators are the projections of the full ones; nothing else is truncated. The level cutoff does not bound P, because zero-mode quanta cost no level. A path argument shows that a contribution to the order-ϵ^n energy of $|p\rangle$ reaches at most $P = p + 4\lfloor n/2 \rfloor$, which gives the cap $P \le p + 16$ through ϵ^8. It was also tested: caps $p + 18$ and $p + 20$ reproduce the $p + 16$ rationals identically, while $p + 14$ changes C_8 and nothing below. The Cauchy code confirms it with absolute caps: $P \le 21$ and $P \le 23$ reproduce $P \le 19$, and $P \le 17$, 18 change only C_8.

References and the projection Π. Each reference $|p\rangle$ is non-degenerate in its sector, so $E_p(\epsilon)$ is unique. Because $w_{00} = 1$ and every window has vanishing diagonal elements on level-0 states, $\langle p|M|p\rangle = 2p$ and $\langle p|V|p\rangle = \frac{1}{2}p(p-1)$ at every ϵ. With $\mathcal{E}_p^{(n)}$ the order-n energy without the r_n, g_n terms, Eq. (7) reads $C_n = \Pi[\mathcal{E}^{(n)}]$ with

$$\Pi f = f(2) - f(1) - \tfrac{1}{3}\big[f(3) - f(0)\big] = -\tfrac{1}{3}\Delta^3 f(0), \qquad \Pi[p^3] = -2, \qquad \Pi[p^4] = -12, \tag{31}$$

and Π annihilates every polynomial of degree at most two in p. Only the part of an energy shift that is at least cubic in p reaches C_n.

Where the schemes differ. The diagonal windows (consistent, literal, zero-mode) are of order ϵ^2 and vanish on $|p\rangle$, so they first enter through $\langle p|H_1 G_p D_2 G_p H_1|p\rangle$ at ϵ^4, with $H_1 = F_1 + g_1 V$ and the resolvent $G_p = Q_p/(\frac{p}{2} - H_0)$, Q_p the projection off $|p\rangle$. Raw and consistent therefore differ first at C_4. Raw and consistent vacua agree through ϵ^3, so the consistent and literal windows differ first at C_6; the exact data confirm both statements for every $L > 0$. At ϵ^4 only the ϵ^2 parts of the windows matter:

$$D_2^{\mathrm{cons}}(\ell) = E_v^{(2)}(L) - E_v^{(2)}(L-\ell), \qquad E_v^{(2)}(\lambda) = g_1^2 \langle 0|V G_0 V|0\rangle_\lambda = -\frac{(\lambda+2)(\lambda+4)}{864}, \tag{32}$$

where $\langle\cdot\rangle_\lambda$ restricts the intermediate states to level $\le \lambda$, and the zero-mode window is $D_2^{\mathrm{zm}}(z,\ell) = h_{zz}^{(2)}(L) - h_{zz}^{(2)}(L-\ell)$, the diagonal of Eq. (20). The r_2 and g_2 terms of the sub-cutoff energies cancel in these differences, and for $z = 0$ the two windows coincide.

Degenerate partners. For $z \ge 5$ the zero-mode channel $|z\rangle$ has degenerate partners with level $\ell = 2, 4, \cdots \le \frac{z-1}{2}$ and $P = z - 2\ell$. First order lifts the degeneracy (smallest separation $\frac{7}{3}$, at $z = 5$, $\ell = 2$), so the branch through $|z\rangle$ is unique; the references are never degenerate. The window objects project the partners out (Appendix B.1).

A.2 Solver checks in detail

Records: `compare_with_package.txt`, `compare_three_way.txt`, `rs_exact/README.md`, `cauchy/README.md`, the production records `../../data/HT_2d/prod/`, `continuum/exact_C2_C3.csv`, `continuum/asymptotics_C2_C3.txt`; code `continuum/exact_low_orders.py`.

The three computations. Section 4.2 and the independent-checks paragraph of Section 6 give their scope and agreement. The production solver propagates Taylor jets of the wave functions in float64, with scalar reductions in extended long double; its consistent run at $L = 34$ took 2893 s on 64 threads with a peak memory of 463 GiB. The exact solver works in a similarity frame in which every matrix element is rational; its modular code uses 72 primes with Chinese remaindering, and every rational reconstruction is verified on eight held-out primes. The Cauchy solver evaluates Δ_ε at $\epsilon_k = \rho\, e^{2\pi \mathrm{i} k/K}$ with $\rho = \frac{1}{64}$, $K = 128$ and 256-bit arithmetic, and $C_n = K^{-1} \sum_k \Delta_\varepsilon(\epsilon_k) \epsilon_k^{-n}$. It solves the consistent vacua exactly. On states with $\ell \ge 2$ the window is $E_v(\lambda) - E_v(\lambda - \ell)$, so the self-consistent condition $(H_\lambda + D_\lambda)\psi = E_v(\lambda)\psi$ becomes the linear generalized eigenproblem $A_\lambda \psi = E_v(\lambda)\, \mathcal{P}_0 \psi$, with $A_\lambda = H_\lambda - E_v(\lambda - \hat{\ell})(1 - \mathcal{P}_0)$, $\mathcal{P}_0$ the projection on level-0 states, and the sub-cutoff vacua $E_v(\lambda - \ell)$ known from smaller λ.

Internal checks of the exact codes.

- $C_0 = 1$, $C_1 = -\frac{1}{6}$, $r_1 = 0$, $g_1 = \frac{1}{3}$ for every L and scheme, and the tuning conditions hold exactly at every order. The fraction and modular codes agree identically for every number at $L \le 12$.
- A third, Hermitian construction of H was diagonalized at 110 digits at $\epsilon = 10^{-3}$, 10^{-5}, 10^{-8}, with the exact series for r, g and the vacua inserted ($L \le 6$, every scheme and channel). The residual divided by ϵ^9 tends to a constant, which confirms every coefficient through ϵ^8 to about 10^{-8}.
- The Cauchy results do not change with the circle radius, the number of points, the precision, the continuation path or the particle cap; the quoted errors (about 10^{-45} on C_8) are set by the most aliased variant. All 882 rational identifications have denominators of the form $2^a 3^b 5^c 7^d$.

C_2 and C_3. The closed form (19) gives $C_2(0) = \frac{17}{81}$, $C_2(2) = \frac{19}{72}$, $C_2(4) = \frac{179}{640}$ and $C_2(6) = \frac{833719}{2903040}$, equal to the exact rationals (evaluated against `rs_exact/results/raw_L{0,2,4,6}.json`). Its approach to the limit is $C_2(L) - C_2 = -\frac{40}{81\pi}L^{-1} + 0.235785\,L^{-2} + \dots$. An independent diagrammatic code, which contracts the factorized vertices (30) with the level cutoff on every intermediate state, gives $C_3(L)$ in float64 to $L = 600$, where $C_3(600) = -0.865794525554$. Its $\ln L/L$ coefficient is 0.70734, which is $4.4999 \times \frac{40}{81\pi}$; the factor $\frac{9}{2}$ is identified numerically from this fit. It is not the one-loop ϕ^6 value $\omega_1 = \frac{7}{2}$ of Eq. (10), and which operator sets the $d = 2$ tail is not derived. With the tower 1, $\ln L/L$, $1/L$, $\ln^k L/L^2$ ($k \le 2$), $\ln L/L^3$, $1/L^3$ the limit is -0.8748500372, -0.8748500106 and -0.8748500065 for $L_{\min} = 32$, 64 and 128, which gives $C_3 = -0.8748500(1)$.

A.3 Exact $C_4(L)$ to $L = 2000$

Records: `c4_exact/README.md`, `c4_exact/C4_by_L.csv`, `c4_exact/results/{c4_L1000,c4_L2000,windows_L20000}.npz`, `c4_exact/fits/fit_summary.txt`, `c4_exact/fits/blind_predictions_L1200_2000.json`, `c4_exact/validate_{rs,pkg,corrected}.txt`, `zeromode_exact/README.md`, `continuum/{fits_summary,zm_corrected_fits}.txt`; code `c4_exact/{c4lib,c4_compute,c4_run,windows,fitlib,fits_c4}.py`.

At $L \le 30$ the production data could not separate a slow power tail from a logarithm. Fits of the consistent C_4 with towers that contain $\ln L/L$ landed 2.4 to 5% below the continuum value, and the same fits with the zero-mode window added gave 4.90 to 5.04 (`continuum/zm_corrected_fits.txt`). The exact $C_4(L)$ settles the question.

Fourth-order perturbation theory. Write $H = H_0 + \sum_{k\ge1} \epsilon^k H_k$ with

$$H_1 = F_1 + g_1 V, \qquad H_2 = r_2 M + g_2 V + g_1 \dot{V} + D_2, \qquad H_3 = r_3 M + g_3 V + g_2 \dot{V} + \tfrac{1}{2} g_1 \ddot{V} + r_2 \dot{M} + D_3, \tag{33}$$

where $\dot{V} = \partial_\epsilon V$ and $\dot{M} = \partial_\epsilon M$ act through $w_{ab}(\epsilon)$. With $G = G_p$, $\langle\cdot\rangle = \langle p|\cdot|p\rangle$ and $E^{(1)} = \langle H_1\rangle$,

$$\begin{aligned} E^{(4)} = {} & \langle H_4\rangle + 2\langle H_1 G H_3\rangle + \langle H_2 G H_2\rangle + 2\langle H_1 G (H_1 - E^{(1)}) G H_2\rangle + \langle H_1 G (H_2 - \langle H_2\rangle) G H_1\rangle \\ & + \langle H_1 G (H_1 - E^{(1)}) G (H_1 - E^{(1)}) G H_1\rangle - \langle H_1 G H_1\rangle \langle H_1 G^2 H_1\rangle. \end{aligned} \tag{34}$$

A diagonal window enters C_4 only through $g_1^2 \langle V G D_2 G V\rangle$; the mass window D^M enters through more terms (Appendix A.5). The order-3 couplings r_3, g_3 need the complete $\mathcal{E}_p^{(3)}$ for $p = 0, 1, 3$, not only its part that survives Π.

Diagram evaluation. With Eq. (30), a K-vertex chain is a sum over χ-line patterns n_{ij} (lines between vertices $i < j$) of a Wick count, the vertex coefficients, an exact rational zero-mode path sum and a χ level sum. A time slice s with n_s χ lines and z_s zero-mode quanta has the energy denominator $\frac{1}{2}(z_s - p) + \frac{1}{2}n_s +$ (level of the slice). The χ sums use the holomorphic factorization: with $v_k(y) = (1-y)^{-k/2}$, $\sum_{ab} w_{ab} y^a \bar{y}^b = v_1(y) v_1(\bar{y})$, and a bundle of n χ lines has the generating function $[v_1(y)v_1(\bar{y}) - 1]^n = \sum_k \binom{n}{k} (-1)^{n-k} v_k(y) v_k(\bar{y})$. Angular averaging at every vertex forces spin zero in every slice, so the holomorphic and antiholomorphic levels of a slice are equal, j_s, and its level is $2j_s$. The density of a pattern is then a sum of squares of one-variable coefficients, with no angular-momentum sum left. ϵ-derivative vertices differentiate the line weights.

One run for all cutoffs. The densities do not depend on L; the cutoff only imposes $\max_s 2j_s \le L$. Accumulating by $\max_s j_s$ and summing cumulatively gives every even $L \le L_{\max}$ in one run. The window terms depend on L explicitly and are evaluated separately.

Pruning and symmetry. The zero-mode amplitude of a four-V chain with n χ lines is a polynomial of degree $8-n$ in p. Since Π kills degree ≤ 2, only $n \leq 5$ reaches C_4: 384 of the 641 four-vertex patterns. Time reversal maps them onto each other in 168 pairs with identical contributions, so 216 patterns are computed. The full set reproduces every individual $E_p^{(4)}$ of the exact solver.

Window shifts without cancellation. Evaluated naively, the window $h_{zz}^{(2)}(L) - h_{zz}^{(2)}(L-\ell)$ is built from terms of size $h_{zz}^{(2)} \sim L^2$, while the part that survives Π is of order $1/L$: a relative cancellation of about L^3. The script `windows.py` applies Π to the rational zero-mode polynomials first and only then multiplies by single-signed level sums. It runs to $L = 20000$ and agrees with the full computation to 6×10^{-13} for $L \leq 1000$.

Cost and precision. On one 40-core node the run to $L_{\max} = 600$ takes 51 s, to 1000 four minutes and to 2000 about 77 minutes (two Slurm jobs on the symmetry cluster, 59 minutes to the time limit and 17 minutes from the checkpoint; the package README's 80 to 90 minutes is the estimate made during the run). Everything is float64. Overlapping runs agree to 6.7×10^{-12} (600 against 1000, from the README) and 1.4×10^{-11} (1000 against 2000, evaluated from `c4_L1000.npz` and `c4_L2000.npz`), and the cancellation in the assembly is about 4×10^3 at $L = 1000$.

check	range	worst deviation
r_2, g_2, r_3, g_3, C_2, C_3 against the exact rationals	$L = 0$ to 18	3.4×10^{-15} relative (C_3: 1.3×10^{-14})
raw and consistent C_4 against the exact rationals	$L = 0$ to 18	1.1×10^{-13} relative
every $E_p^{(4)} - 2pr_4 - \frac{1}{2}p(p-1)g_4$, $p = 0$ to 3, raw, all 641 patterns	$L = 0$ to 18	2.1×10^{-14} relative
$E_v^{(2)}(\lambda)$ of Eq. (32)	$\lambda \leq 18$	2.2×10^{-16}; exact for $\lambda \leq 12$
C_2, C_3, raw and consistent C_4, r_3, g_3 against the production solver	$L = 2$ to 34	1.1×10^{-13} relative
zero-mode C_4 against brute-force fraction arithmetic	$L = 0$ to 12	1.1×10^{-13} relative
zero-mode C_4 against the exact zero-mode solver	$L = 0$ to 18	1.1×10^{-13} relative
pruned ($n \leq 5$) against the full pattern set	$L \leq 18$	10^{-13}
time-reversal pairs	168 pairs, $L = 40$	5.5×10^{-16}

Table 10: Validation of the large-L C_4 code (`c4_exact/README.md`, `c4_exact/validate_*.txt`).

Validation. Table 10 lists the checks. The zero-mode scheme itself was solved exactly through ϵ^8 for $L = 0$ to 18 (`zeromode_exact/`), by exact perturbation theory with the degenerate partners treated by a projected equation. Two implementations (modular and fraction) are identical for $L \leq 10$. A non-perturbative Cauchy check covers every coefficient and every sub-cutoff energy $E_z^{(k)}(\lambda)$ at $L \leq 6$, and closure tests show that this scheme needs only the zero-mode channels $z \leq 14$ through ϵ^8. The zero-mode C_4 minus the consistent C_4 equals an exact closed form in L (`zeromode_exact/closed_form_dC4.py`) at every L; for example $C_4^{\mathrm{zm}}(2) = \frac{6786263}{3732480}$.

Exact window shifts. Fits of the cancellation-free shifts to $L = 20000$ (residuals $\le 5 \times 10^{-14}$) give

$$\Delta C_4^{\rm cons}(L) \equiv C_4^{\rm cons} - C_4^{\rm raw} = \kappa L - \frac{5}{8748\pi} \ln L + 0.000446687 + o(1),$$
$$\Delta C_4^{\rm zm}(L) \equiv C_4^{\rm zm} - C_4^{\rm cons} = \Big(\frac{4}{81\pi} + \frac{5}{4374\pi}\Big) \ln L + 0.0187584 + o(1), \tag{35}$$

with $\kappa = 2.2073 \times 10^{-4}$; the analytic evaluation in `continuum/asymptotics_C2_C3.txt` gives $\kappa = 2.20729299645 \times 10^{-4}$. The $\ln L$ coefficients are those of Table 12. Split by the number of outer χ lines, $\Delta C_4^{\rm zm}$ has logarithms only for two lines ($\frac{4}{81\pi}$) and four lines (3.6387×10^{-4}). Evaluated from `C4_by_L.csv`, it has a minimum of 0.0982 near $L = 36$ and then grows logarithmically: 0.1028 at $L = 100$, 0.1313 at 1000 and 0.1418 at 2000.

Fits. The tail models (`fitlib.py`) are towers T_{jk}: a constant, $\ln^i L/L$ for $i \le j$ and $\ln^i L/L^2$ for $i \le k$; T_{jkm} adds $\ln^i L/L^3$ for $i \le m$. A model may add a free $\ln L$ and single terms such as $\ln^3 L/L$. The table of Section 4.4 uses T_{22} and $T_{22} + \ln^3 L/L$, each with a free $\ln L$, the $\ln^2 L/L$ coefficient free or fixed to $-5/\pi$ (end of Appendix A.5), and $L_{\rm min} \in \{200, 300, 400\}$, $L \le 2000$: twelve fits, with the median and half the full spread quoted. At $L = 2000$ the data are $C_4 = 4.29126$ (raw), 4.73178 (consistent) and 4.87359 (zero-mode). The raw fit uses $C_4^{\rm raw} = C_4^{\rm cons} - \Delta C_4^{\rm cons}$ with the exact shift. The difference of the two fitted β, 0.016082, equals the $\ln L$ coefficient of $\Delta C_4^{\rm zm}$. Subtracting the fitted $1/L$ and $1/L^2$ terms from $C_4(L)$ leaves straight lines in $\ln L$: $5.0026 - 0.02656 \ln L$ (consistent) and $5.0213 - 0.01048 \ln L$ (zero-mode). Without a $\ln L$ term every tower drifts with the fit window, and the limits, 4.881 to 4.896 (zero-mode) and 4.648 to 4.684 (consistent), are far from 5.0489 and from each other. The fitted β of the consistent scheme alone would not exclude the nearby simple value $-1/(12\pi) = -0.0265258$; the class decomposition of Appendix A.4 fixes it at $-\frac{365}{4374\pi}$.

Out-of-sample and blind tests. Fitted on $[200, 600]$ and used to predict $(600, 2000]$, the zero-mode model T_{22} with $\ln L$ misses by at most 9.0×10^{-8}, and models without $\ln L$ miss by 5.0×10^{-6} to 2.5×10^{-5}. Fitted on $[200, 1000]$, the models with $\ln L$ miss by 8.5×10^{-9} (T_{22}) and 5.5×10^{-8} (T_{232}), those without by 4.3×10^{-7} to 6.7×10^{-6}. Before the $L = 2000$ run finished, predictions for $L = 1200$ to 2000 were recorded from fits on $L \le 1000$. At $L = 2000$, prediction minus data is -8.6×10^{-9} (T_{22}) and -6.4×10^{-8} (T_{232}) for the models with $\ln L$ in the zero-mode and consistent schemes (-7.5×10^{-9} and -8.3×10^{-8} raw). For the models without $\ln L$ it is $+4.3 \times 10^{-7}$ to $+1.8 \times 10^{-6}$ (zero-mode), $+1.2 \times 10^{-6}$ to $+4.6 \times 10^{-6}$ (consistent) and -4.1×10^{-7} to -8.6×10^{-7} (raw). The data are accurate to about 10^{-11}, so the blind test favors the logarithm, by factors of 5 to 500 (the ratios of the deviations above).

A.4 Mechanism and the $\ln L$ coefficients by class

Records: `residual_log/results/log_coeffs.txt`; code `residual_log/{log_coeffs,offdiag_window, decompose}.py`.

The budget defect. The cutoff constrains the total level of each intermediate state. View a fourth-order process as an outer structure with a sub-bubble nested in one of its intermediate states. The sub-bubble may use only the level $L - \ell_{\rm other}$, where $\ell_{\rm other}$ is the level of everything else in that time slice, while the counterterm that cancels its divergence was tuned on level-0 references, where it has the full L. The mismatch is the density of the divergence at the cutoff times the level taken by the other quanta. Table 11 lists the three divergent sub-bubbles of the ϕ^4 vertex.

sub-bubble	divergence	deficit for spectators of level ℓ	restored by
vacuum (four χ lines)	$E_v^{(2)}(\lambda) = -\frac{(\lambda+2)(\lambda+4)}{864}$, quadratic	$\sim \ell L$	consistent window
sunset (three χ lines; mass)	$r_2(\lambda) = \frac{2\lambda}{27\pi} + \ldots$, linear	$\frac{2}{27\pi}\ell$ per mass insertion; does not decay	in part by the diagonal zero-mode window; fully by $D^{\rm zm}$ and D^M
coupling (two χ lines)	$g_2 \sim \ln \lambda$	$\ln \frac{L}{L-\ell} \sim \frac{\ell}{L}$	not needed for soft spectators; finite effects from hard ones

Table 11: Divergent sub-bubbles and their level-budget deficits in $d = 2$ ($\Delta_\sigma = \frac{1}{2}$ at $\epsilon = 0$).

The sunset slope is $dr_2/dL = \frac{2}{27\pi} = 0.0235785$ (the exact $[r_2(2000) - r_2(1998)]/2$ is 0.0235785). Because the divergence is linear, the deficit is proportional to the spectator level and independent of L, and it hits every insertion of the mass operator M, not only the diagonal of its zero-mode part.

Linked clusters with zero-mode transfer. Take two components that share only the zero mode: an outer bubble O with m χ lines and an inner bubble B with m'. Let E_O, E_B be their slice energies and $q = \frac{1}{2}\Delta z_B$ the zero-mode transfer, with Δz_B the change of z caused by B. The four vertices have six time orderings: two sequential ones ($OOBB$, $BBOO$), in which no slice holds lines of both components, and four overlapping ones ($OBBO$, $BOOB$, $OBOB$, $BOBO$), in which some slice holds both. Since $H_0 = \ell + \frac{P}{2}$ is additive, the six orderings and the unlinked terms of perturbation theory cancel in the continuum. At leading degree in p the zero-mode amplitude does not depend on the ordering, and the cancellation reduces to the identity

$$\frac{1}{E_O E_B (E_O + q)} + \frac{1}{E_O E_B (E_B - q)} - \frac{1}{q\, E_O (E_B - q)} + \frac{1}{q\, E_B (E_O + q)} = 0 \tag{36}$$

for every q, including the limit $q \to 0$; its first two terms come from the four overlapping orderings, the last two from the sequential orderings and the unlinked terms. With outer level $2a$ and inner level $2b$, the overlapping orderings need $a + b \le \frac{L}{2}$, and the others do not. The truncation therefore errs by

$$-\sum_{a,b \le L/2 < a+b} \Big[\frac{1}{E_O E_B (E_O + q)} + \frac{1}{E_O E_B (E_B - q)}\Big] \times (\text{zero-mode amplitude}). \tag{37}$$

This is the nested insertion, into every outer state of level ℓ, of the window part of the second-order zero-mode effective interaction,

$$D_\lambda(z \to z') = g_1^2 \sum_{k:\, \lambda - \ell < \ell_k \le \lambda} \frac{\langle z'|V|k\rangle\langle k|V|z\rangle}{E_z^0 - E_k^0} = h^{(2)}_{z'z}(\lambda) - h^{(2)}_{z'z}(\lambda - \ell), \tag{38}$$

in its ket-referenced (Bloch) form and including $z' \neq z$: the window $D^{\rm zm}$ of Eq. (20). The diagonal zero-mode scheme keeps only $q = 0$. At subleading degree in p the orderings differ by commutator terms. These are confined to the window region and are $\mathcal{O}(\ln L/L)$ for the numbers $(m, m') = (2, 2)$, $(1, 3)$ and $(3, 1)$ of outer and inner χ lines.

The ratio $\frac{2}{3}$. Take the outer two-χ bubble, with $:Z^2:$ at each outer vertex, and the inner zero-mode sunset, with $:Z:$ at each inner vertex. Per unit amplitude the window of the nested sunset at outer level $2a$ is $-\frac{2}{\pi}a$, and the outer weight (pattern density over the squared denominator) tends to $\frac{1}{4a^2}$. The window sum is therefore

$$\sum_{a\le L/2}\frac{1}{4a^2}\Big(-\frac{2a}{\pi}\Big)=-\frac{1}{2\pi}\ln L+\mathcal{O}(1). \tag{39}$$

At leading degree in p the zero-mode operators commute and act as $a_0\to\sqrt{p}\,e^{\mathrm{i}\theta}$, $a_0^\dagger\to\sqrt{p}\,e^{-\mathrm{i}\theta}$, so $:Z^2:\to p\,(e^{2\mathrm{i}\theta}+2+e^{-2\mathrm{i}\theta})$ and $:Z:\to\sqrt{p}\,(2\cos\theta)$. The closed-path amplitude is the angular average $p^3\langle(2\cos\theta)^6\rangle_\theta=20\,p^3$. Split it by the zero-mode transfer of the inner sunset:

- $\Delta z_B=0$ (inner vertices a_0, $a_0^\dagger$ in either order, two ways) times the phase-free part of the outer factor $(e^{2\mathrm{i}\theta}+2+e^{-2\mathrm{i}\theta})^2$, which is $1+4+1=6$: in total 12;
- $\Delta z_B=+2$ (inner $a_0^\dagger a_0^\dagger$) times the $e^{2\mathrm{i}\theta}$ coefficient of the outer factor, $2\cdot 2=4$, and likewise $\Delta z_B=-2$: in total 8.

So $20=12+8$. The coupling factor is $g_1^4\gamma_2\gamma_3=\frac{1}{81}\cdot\frac{1}{2}\cdot\frac{2}{3}=\frac{1}{243}$. With $\Pi[p^3]=-2$ the window logarithm restored in C_4 is

$$(-2)\cdot 20\cdot\Big(-\frac{1}{2\pi}\Big)\cdot\frac{1}{243}=\frac{20}{243\pi}=\underbrace{\frac{12}{243\pi}}_{\Delta z=0\ (=\frac{4}{81\pi})}+\underbrace{\frac{8}{243\pi}}_{\Delta z=\pm 2}. \tag{40}$$

The part that changes the zero-mode number is $\frac{8}{12}=\frac{2}{3}$ of the diagonal part, and a diagonal window captures $\frac{12}{20}=\frac{3}{5}$ of this class. The split $20=12+8$ and the γ_m follow from the algebra above; the factor $\frac{1}{243}$ is also in `log_coeffs.py`.

class (outer state × sub-bubble)	analytic	exact fit	raw	cons	zm	W
one χ line × vacuum bubble (also κL)	$-\frac{5}{8748\pi}=-1.81933\times10^{-4}$	-1.81933×10^{-4}	m	r	r	r
two χ lines × zero-mode sunset, $\Delta z=0$	$\frac{4}{81\pi}=0.0157190067$	0.015719006698	m	m	r	r
four χ lines × one-line zero-mode bubble	$\frac{5}{4374\pi}=3.63866\times10^{-4}$	3.63865887×10^{-4}	m	m	r	r
two χ lines × zero-mode sunset, $\Delta z=\pm2$	$\frac{8}{243\pi}=0.0104793378$	0.010479337749	m	m	m	r
all other zero-mode classes (diagonal and off-diagonal)	0	$\lvert\cdot\rvert<2\times10^{-10}$				
χ-attached sunsets (D^M)	0	-2.9×10^{-9}; -2.6×10^{-7}				

Table 12: $\ln L$ coefficients of the C_4 shift that each window class restores. "m": missing in the scheme; "r": restored. The fits use the exact shifts to $L=20000$ (the χ-attached class to $L=8000$) with $L_{\min}=2000$ and 4000 (`residual_log/results/log_coeffs.txt`).

The coefficients by class. Table 12 lists every class. The two other logarithmic classes follow the same pattern. For four χ lines times a one-line bubble, the outer weight tends to $\frac{1}{4}$, and when

the outer state has level $L-2j$ the inner one-line bubble keeps the budget $2j$; its window is then $-\frac{1}{2\pi}(\frac{1}{j}-\frac{2}{L})$, which sums to $-\frac{1}{8\pi}\ln L$; with $g_1^4\gamma_4\gamma_1=\frac{1}{4374}$ and the amplitude $20p^3$ this gives $+\frac{5}{4374\pi}$. For one χ line times the vacuum, Eq. (32) gives $E_v(L)-E_v(L-2a)=-[2a(2L+6)-4a^2]/864$: the term linear in L gives κL, and the $4a^2$ term with weight $1/(4\pi a^3)$ gives $(864\pi)^{-1}\ln L$, which the factor $-\frac{40}{81}$ of the one-line outer state (couplings, zero-mode amplitude and Π; `log_coeffs.py`) turns into $-\frac{40}{81\cdot 864\pi}=-\frac{5}{8748\pi}$ in C_4. The $\ln L$ coefficient of each scheme is minus the sum of the logarithms it misses:

$$\beta_{\rm raw}=-\Big(-\frac{5}{8748\pi}+\frac{20}{243\pi}+\frac{5}{4374\pi}\Big)=-\frac{725}{8748\pi},\qquad \beta_{\rm cons}=-\Big(\frac{20}{243\pi}+\frac{5}{4374\pi}\Big)=-\frac{365}{4374\pi},$$
$$\beta_{\rm zm}=-\frac{8}{243\pi},\qquad \beta_W=0, \tag{41}$$

in agreement with the fitted values of Section 4.4 to 10^{-6} or better. The χ-attached class has no net logarithm, but its two parts do: $+0.0369938\ln L$ directly at ϵ^4 and $-0.0369938\ln L$ through the retuned g_3. This is a local vertex renormalization, a sunset with one hard external χ line, which the tuning of g absorbs. The decomposition must be made by window class, not by diagram topology: single four-vertex topologies carry L, $\ln^2 L$ and $\ln L$ terms that cancel against counterterm chains (`decompose.py`), and the window classes are the combinations that are separately finite in the continuum.

Why the tuning cannot absorb the defect. The two tuning conditions fix two state-independent numbers, r_n and g_n. A shift of r or g changes the reference energies by $2p\,\delta r+\frac{1}{2}p(p-1)\,\delta g$, which Π removes. The budget defect is different in kind: it is an operator proportional to the level of the spectator quanta, $\sim\{\hat\ell, M\}$, and through the outer amplitude $20p^3$ it is cubic in p. No choice of tuning conditions can represent it.

Contrast with one dimension. In $d=1$ the cutoff is on the free energy $E^0=\ell+\frac{P}{4}$ (Eq. (5)). Every four-particle vacuum shell then carries the same energy, so the vacuum window is of order one per missing shell and does not decay, and $D^{\rm conn}$ repairs it (Section 2.7). The references have $E^0=\frac{p}{4}$, so their own windows hold vacuum shells as well, and how many depends on where the cutoff sits relative to them (the cutoff placement of Section 3.3). By power counting the sunset density falls like level$^{-1/2}$ ($r_2\sim\Lambda^{1/2}$), so its budget deficit $\sim\ell/\Lambda^{1/2}$ vanishes: this is why the vacuum correction suffices in $d=1$ and does not in $d=2$. The connected ("folded") ϵ^4 term that $D^{\rm conn}$ does not restore also decays; its contribution to the ϵ^4 spacing of $\partial\phi$ is -6.8×10^{-4} at $\Lambda=16.25$ and -3.8×10^{-4} at $\Lambda=64.25$ (`lri_hpc_Hamitonian`, record `analysis/identity_checks/challenge-theory/folded_ph0.25.json`). With the vacuum window alone the ϵ^2 descendant spacings vanish (Table 4). In $d=2$ every structure is one power worse: the vacuum window grows like L (the κL of the raw scheme), the sunset window is of order one, and the coupling window is $\sim\ell/L$. The references have level 0, so their own windows are empty and nothing depends on where the cutoff sits relative to them; the zero-mode dependence that $d=1$ carries through $E^0=\ell+\frac{P}{4}$ appears in $d=2$ as the separate zero-mode window.

A.5 The complete window at ϵ^4

Records: `residual_log/WINDOW_SPEC.md`, `residual_log/results/brute_vs_codes.txt`, `residual_log/C4W_by_L.csv`, `residual_log/results/{offdiag_L20000,nm_L8000}.npz`, `residual_log/fits/fits_final.txt`; code `residual_log/{nm_window,nm_window_exact,brute_full,brute_nm,nm_zz_check,offdiag_hermitian_check,fits_final}.py`.

Principle. Section 4.4 defines the window, $D = D^{\rm zm} + D^M$ with Eqs. (20) and (21), and states its principle. For a mass insertion between states of levels ℓ_s and $\ell_{s'}$ the budget is the average $L - \frac{1}{2}(\ell_s + \ell_{s'})$, which comes from averaging the internal orderings of the sunset: nested and exchange orderings for a χ hop $A^\dagger A$, and the two assignments for pair creation $A^\dagger A^\dagger$ and pair annihilation AA. $h^{(2)}$ contains all inner bubble types with one to four χ lines; the zero-line part and the r_2, g_2 terms do not depend on the level and cancel. Bra-referenced and Hermitian versions of $h^{(2)}$ give the same C_4 to 10^{-16}, because the product $\langle VG\,D^{\rm zm}GV\rangle$ through which it enters is symmetric.

How the window enters C_4. Since $D^{\rm zm}$ vanishes on level-0 states, its only contribution is $g_1^2\langle VG\,D^{\rm zm}GV\rangle$. D^M is symmetric, but it does not vanish on $|p\rangle$, because M' creates two quanta. It therefore enters already at ϵ^3, through $2g_1\langle VGD^M\rangle$, which changes r_3 and g_3 but not C_3. At ϵ^4

$$\begin{aligned}\delta\mathcal{E}_p^{(4)} &= g_1^2\langle VGD^MGV\rangle + 2g_1\big[g_1\langle VGVGD^M\rangle - g_1V_{pp}\langle VG^2D^M\rangle - \tfrac{1}{4}\langle VG(P-p)GD^M\rangle\big],\\ \delta C_4^M &= \Pi\big[\delta\mathcal{E}^{(4)}\big] + 2g_1\,\delta g_3\,\Pi\big[\langle VGV\rangle\big];\end{aligned} \tag{42}$$

r_3 drops out of C_4, and Π kills $\langle H_2GD^M\rangle$ and $\langle D^MGD^M\rangle$. The complete coefficient is

$$C_4^W(L) = C_4^{\rm zm}(L) + \delta C_4^{\rm off}(L) + \delta C_4^M(L), \tag{43}$$

where $\delta C_4^{\rm off}$ is the shift from the elements $z' \neq z$ of $D^{\rm zm}$. All three pieces are exact for every even $L \leq 2000$, and the shifts are known further, to $L = 20000$ (off-diagonal) and $L = 8000$ (D^M).

Derivation status. $D^{\rm zm}$ follows from the linked-cluster argument of Appendix A.4; it is exact at leading degree in p, and the differences at subleading degree vanish as $L \to \infty$. D^M is derived at leading order in $1/L$. Its slope can be replaced by the exact sum $t_M(j)$ of the sunset contributions that the cutoff removes for a spectator of level $2j$, with both zero-mode orderings averaged; $t_M(j) \to \frac{2}{\pi}j$ per unit amplitude (Appendix A.4). This changes δC_4^M by -4.5×10^{-4} at $L = 100$ and -2.4×10^{-4} at $L = 200$ (`nm_window_exact.py`), that is by $\mathcal{O}(1/L)$, with the same limit. The Newton-coefficient form of Appendix B.1, another refinement, changes C_4 much less (4.9×10^{-7} at $L = 100$), also with the same limit; why the two refinements differ this much at finite L was not analyzed.

check	L	agreement
`offdiag_window.py`, diagonal part, against $C_4^{\rm zm} - C_4^{\rm raw}$ of `c4_exact`	≤ 2000	4×10^{-11}
`nm_window.py` against `brute_nm.py` (exact perturbation theory with $\{\hat{\ell}, M'\}$ as an operator): δC_4 and δg_3	2, 4, 6	10^{-14}
`brute_full.py` (complete scheme as operators; off-diagonal window from auxiliary $\|z\rangle$ channels) against $C_4^{\rm zm}$, $C_4^{\rm zm} + \delta C_4^{\rm off}$ and C_4^W	2 to 12	$\leq 6.3\times10^{-14}$
`nm_window.py` on a workstation ($L \leq 400$) against the symmetry cluster ($L \leq 2000$, 8000)	overlap	6×10^{-14}

Table 13: Checks of the window codes against brute-force exact perturbation theory in fraction arithmetic, with D added as an explicit operator (`residual_log/results/brute_vs_codes.txt`).

Verification at finite L. Table 13 lists the checks. The brute-force values of the complete scheme are $C_4^W(2) = 1.966884466461$, $C_4^W(4) = 2.290042077448$, $C_4^W(6) = 2.546923269518$, $C_4^W(8) = 2.756245037529$, $C_4^W(10) = 2.928832263364$ and $C_4^W(12) = 3.073608283249$; the comparison is limited by the float64 precision of the large-L code, about 6×10^{-14}.

What does not work.

- Any diagonal window (consistent, zero-mode): it misses the transitions $\Delta z \neq 0$ and the χ parts by construction.
- Replacing $D^{\rm zm}$ by its leading form $-\frac{g_1^2}{3\pi}\{\hat{\ell}, :Z^2:\}$ (`nm_zz_check.py`). The difference tends to $0.042838 + \frac{5}{4374\pi}\ln L$: the inner coupling and one-line bubbles carry finite parts from hard outer states and the four-line × one-line logarithm, so the exact zero-mode window is required. Only the χ part may use the leading form.
- A modified tuning prescription (Appendix A.4).

scheme	$C_4(2000)$	large L	limit
raw	4.29126	$-\kappa L - \frac{725}{8748\pi}\ln L$	diverges
consistent (= literal at this order)	4.73178	$-\frac{365}{4374\pi}\ln L$	diverges
zero-mode (diagonal)	4.87359	$-\frac{8}{243\pi}\ln L$	diverges
+ off-diagonal zero-mode window	4.97161	no $\ln L$	5.0386139, 0.0102545 short
+ mass window D^M (= W)	4.98172	no $\ln L$	5.0488682
continuum			5.0488684314

Table 14: C_4 at $L = 2000$ and its large-L form, window by window (`residual_log/C4W_by_L.csv`, `residual_log/fits/fits_final.txt`).

Large L. Table 14 builds the complete window up piece by piece. The off-diagonal shift is $\delta C_4^{\rm off}(L) = \frac{8}{243\pi}\ln L + 0.0172932 + \mathcal{O}(\ln^2 L/L)$, from exact values to $L = 20000$. It removes the logarithm of the zero-mode scheme but leaves a finite miss of 0.0102545. The mass-window shift converges, $\delta C_4^M(\infty) = 0.0102543394$ with a spread of 1.5×10^{-8} over twelve fits of the exact values to $L = 8000$, and supplies the missing finite part: the separately extrapolated pieces give $5.0386139 + 0.0102543 = 5.0488682$. The approach is slow, like $\ln^2 L/L$: at $L = 2000$, $C_4^W - C_4^{(2)} = -0.0672$ (Fig. 1c).

Fits of C_4^W. The data are accurate to 10^{-10} or better, so the uncertainty is entirely the extrapolation.

- The towers T_{22}, $T_{22} + \ln^3 L/L$ and T_{23} without $\ln L$, with $L_{\rm min} \in \{200, 300, 400\}$ (nine fits), give 5.04886823 with a half spread of 5.6×10^{-7}, 2.0×10^{-7} below the continuum value; this is the value quoted in Section 4.4.
- No-log fits outside this set (towers $T_{22} + \ln^3 L/L$ and T_{23} at $L_{\rm min} = 600$, T_{33} at $L_{\rm min} = 400$) move the limit by up to 1.8×10^{-5} from the continuum value; T_{33} at $L_{\rm min} = 600$ is over-fitted ($\ln^2 L/L$ coefficient -25, limit 2.8×10^{-4} low).
- With a free $\ln L$ term the stable towers at $L_{\rm min} = 200$ to 300 give its coefficient between -1.8×10^{-6} and $+3.5 \times 10^{-6}$, and the limit moves by -4.3×10^{-5} to $+2.2 \times 10^{-5}$ (the tower T_{33}, outside this set, gives a coefficient of -8.1×10^{-6} at $L_{\rm min} = 200$). Richer towers at larger $L_{\rm min}$ become unstable, for example with a free $\ln L$ coefficient up to 5×10^{-4} and a $\ln^2 L/L$ coefficient of $+16$ to $+30$; these fits have more freedom than the data constrain.

Hence the quoted uncertainty: the half spread of the nine stable fits, rounded up to 10^{-6}. The fits outside the stable set show that the model dependence can reach 2×10^{-5}, and 4×10^{-5} with a free $\ln L$ term, which the complete window excludes by construction (Table 12). The fitted tail is about $-1.40 \ln^2 L/L$ and comes entirely from C_4^{zm}. Fixing it at $-5/\pi$, the value used as a prior in the fit scripts, makes the fits 500 times worse, so that prior is not supported. In the fits of the raw, consistent and zero-mode schemes (Section 4.4) the variants with the coefficient fixed only widen the quoted spread.

B Two dimensions: order ϵ^5

This appendix supports Section 4.5. It gives the window at all orders, shows that it diverges linearly at ϵ^5 and why, derives the rule W_c that removes the divergence, and gives the details of the C_5 extrapolation.

B.1 The window at all orders

Records: `residual_log/WINDOW_SPEC.md` (status table and §§2.2 to 2.10), `complete_exact/README.md`.

For every even $\lambda \le L$ the Hamiltonian at sub-cutoff λ is the truncation to $\ell \le \lambda$ of $H_0 + \epsilon F_1 + rM + gV$ plus $D_\lambda = D_\lambda^{\mathrm{zm}} + D_\lambda^M$, with one $r(\epsilon)$, $g(\epsilon)$ fixed at $\lambda = L$. Scheme W is this window with self-consistent auxiliary objects (last item below): through ϵ^4 it is the complete window of Section 4.4; at ϵ^5 it diverges (Appendix B.3), and rule W_c repairs it. The ingredients are the following.

- *Auxiliary channels.* $E_z(\lambda)$ and $\psi_z(\lambda)$ are the eigenvalue and the right eigenvector of this Hamiltonian on the branch through $|z\rangle$, normalized by $\langle z|\psi_z\rangle = 1$.

- *Bloch zero-mode window.* With $U_{z'z}(\lambda) = \langle z'|\psi_z(\lambda)\rangle$ and $h(\lambda) = U\,\mathrm{diag}(E_z(\lambda))\,U^{-1}$,

$$D^{\mathrm{zm}}|z,\chi\rangle = \sum_{z'} \big[h(\lambda) - h(\lambda - \ell_\chi)\big]_{z'z}|z',\chi\rangle \qquad (\ell_\chi \ge 2).$$

  Here h is the full Bloch matrix; at ϵ^2 its level-dependent part is $h^{(2)}$ of Eq. (20), so the window reduces to that equation.

- *Mass window.* $D^M = \frac{1}{4}\{e_1(\lambda) - e_1(\lambda - \hat{\ell}), M'\}$ with the Newton coefficient $e_1 = E_1 - E_0$, the combination that the mass counterterm renormalizes: $E_z = \sum_k e_k \binom{z}{k}$ and $\langle z|M|z\rangle = 2z$ (§2.4 of `WINDOW_SPEC.md`). At ϵ^2, $e_1^{(2)}(\lambda) - e_1^{(2)}(\lambda - \ell) = -\frac{4}{27\pi}\ell + \mathcal{O}(\ell/\lambda)$, so $D^{M,(2)}$ is Eq. (21) up to $\mathcal{O}(1/\lambda)$. This form changes C_4 relative to the slope form by 1.8×10^{-6}, 4.9×10^{-7}, 1.2×10^{-7} and 2.9×10^{-8} at $L = 50$, 100, 200 and 400, with the same limit, and it makes the scheme an exact rational at every L.

- *Coupling window* (scheme $W{+}V$ only). $D^V = \frac{1}{2}\{e_2(\lambda) - e_2(\lambda - \hat{\ell}), V'\}$ with $e_2 = E_2 - 2E_1 + E_0$ and $V' = V - \frac{1}{12}{:}Z^4{:}$.

- *Partners.* The degenerate partners of $|z\rangle$, $z \ge 5$ (Appendix A.1), are projected out of every window object. A reading that keeps them through a projected equation in the auxiliary channels differs by 2.8×10^{-5} in C_4 and -1.6×10^{-4} in C_5 at $L = 16$.

- *Self-consistency* (scheme W). $h(\lambda - \ell)$ and $e_1(\lambda - \ell)$ belong to the Hamiltonian at sub-cutoff $\lambda - \ell$ with its own window. Rule W_c (Appendix B.4) replaces this.

C_5 needs $D^{\mathrm{zm},(2)}$, $D^{\mathrm{zm},(3)}$, $D^{M,(2)}$ and $D^{M,(3)}$; $D^{M,(4)}$ does not enter. C_4 sees only the ϵ^2 windows, so C_4^W is the coefficient of Appendix A.5 up to the difference between the e_1 form and the slope form given above.

The coupling window is invalid. In the exact small-L solver $C_4(W{+}V) - C_4(W) = 0.484$, 0.491 and 0.492 at $L = 10$, 12 and 14. This is a finite shift, although power counting (§3 of `WINDOW_SPEC.md`) suggests that the effect of D^V vanishes. D^V also makes the ϵ^3 spacing grow (Table 22). It is not used below.

B.2 Exact ϵ^5 data

Records: `residual_log/results/eps5/{validation_c5,validation_vs_complete_exact}.txt`, `residual_log/results/eps5/{c5_L160,c5_L200,dC5_L1000_all}.npz`, `complete_exact/README.md`; code `residual_log/eps5/`, `residual_log/brute_W5.py`.

Two independent exact computations exist.

1. *Large-L diagram sums* (`residual_log/eps5/`). The C_4 machinery of Appendix A.3 is extended to five-vertex chains, ϵ-Taylor orders at every vertex, open zero-mode paths and window insertions; the perturbation-theory expression for $E^{(5)}$ is unrolled symbolically into 201 products of vertices, windows and resolvents. The code gives $C_n(L)$, $n \le 5$, for every even $L \le 160$ in the raw scheme and eleven window schemes (among them those of Table 16), raw C_5 to $L = 200$ (extended to $L = 300$ in Appendix B.6), and the scheme shifts $\delta C_n(L) = C_n(\text{scheme}) - C_n(\text{raw})$ for every even $L \le 1000$. The shifts need no five-vertex chain. The cost was about 25 node-hours on the symmetry cluster.

2. *Exact small-L solver* (`complete_exact/`). Scheme W and its variants as explicit operators in exact perturbation theory through ϵ^8, $L = 0$ to 16, modulo 208 primes ($L \le 12$) or 128 primes ($L = 14$, 16), every value verified on eight held-out primes, with descendant channels in every scheme. An independent fraction implementation agrees identically at $L = 2$ to 10; closure tests with random entries outside the claimed window sets change nothing; and the raw, consistent and zero-mode schemes reproduce the exact solvers of Appendix A.2 exactly to $L = 16$. Examples, with D^M in the e_1 form: $C_4^W(2) = \frac{11011547}{5598720}$ (the slope form of Appendix A.5 gives 1.966884466461) and $C_5^W(2) = -\frac{879593527}{80621568}$.

quantity	reference	worst deviation
raw, consistent C_2 to C_5	exact solver, $L \le 18$	1.0×10^{-13} relative
raw, consistent C_2 to C_5	production solver, $L \le 34$	2.2×10^{-13} relative
zero-mode C_2 to C_5	exact zero-mode solver, $L \le 18$	9×10^{-14} relative
C_5 of W and three diagnostic variants	explicit-operator fraction arithmetic (`brute_W5.py`), $L \le 8$	1.7×10^{-14} relative
C_3 to C_5 of W, W_2, raw, consistent, zero-mode	`complete_exact`, $L \le 16$	1.0×10^{-15} relative
full runs against raw plus shift	all schemes, $L \le 160$	1.7×10^{-11} absolute
raw C_5, run to $L = 200$ against run to $L = 160$	overlap	1.3×10^{-11} absolute

Table 15: Validation of the exact ϵ^5 data; scheme names as in Appendix B.3 (`residual_log/results/eps5/validation_c5.txt`, `validation_vs_complete_exact.txt`).

Table 15 lists the checks. Raw C_5 is -31.898 at $L = 160$ and -32.597 at $L = 200$, against the continuum value -38.834. The last row of the table is evaluated from `c5_L160.npz` and `c5_L200.npz`.

B.3 Scheme W diverges linearly

Records: `residual_log/results/eps5/classes_c5.txt`, `residual_log/fits/fits_c5.txt`, `residual_log/results/eps5/dC5_L1000_all.npz`, `residual_log/WINDOW_SPEC.md` (status after the exact ϵ^5 test, item 1); code `residual_log/eps5/{classes_c5,fits_c5}.py`.

The exact shifts have the form $aL + b_2 \ln^2 L + b_1 \ln L + c + \mathcal{O}(\ln^k L/L)$, with no $L \ln L$ term, and their linear coefficients are fixed to better than 2×10^{-6}. Raw C_5 to $L = 160$ fixes its own linear coefficient only roughly. A free fit gives $+0.0004 \pm 0.002$ (half range over tail towers and $L_{\min} = 30$, 40, 60; step 2 of `fits_c5.txt`). A profile over a fixed raw coefficient $a_{\rm raw}$ (fits of $C_5^{\rm raw}(L) - a_{\rm raw} L$ with logarithmic towers, $L \le 160$) is sharper: the intercept is most stable, and needs no logarithms, at $a_{\rm raw} = +0.0004(4)$; the (4) is the range of $a_{\rm raw}$ over which the intercept stays most stable, a judgement read off the profile (step 4 of `fits_c5.txt`), not a statistic. These fits were made with the data then available and were not repeated with the data to $L = 300$. Table 16 gives the coefficients. The scheme names are: subscript 2 for windows of order ϵ^2 only; consistent and zero-mode with their ϵ^3 windows; Bloch for the Bloch $D^{\rm zm}$ at ϵ^2 and ϵ^3 without D^M; W_{3e} for W with the eigenvalue-diagonal ϵ^3 zero-mode window $E_z^{(3)}(L) - E_z^{(3)}(L-\ell)$ in place of the Bloch one.

	consistent$_2$	consistent	zero-mode$_2$	zero-mode	Bloch$_2$	Bloch	W_2	W_{3e}	W
shift	-0.001429	-0.000392	$+0.003119$	-0.001381	$+0.000991$	-0.000364	-0.000974	-0.005474	-0.002329
total	-0.0010	$+0.0000$	$+0.0035$	-0.0010	$+0.0014$	$+0.0000$	-0.0006	-0.0051	-0.0019

Table 16: Linear coefficient of $C_5^S(L)$ by scheme S. "Shift": the coefficient of $\delta C_5^S(L) = C_5^S - C_5^{\rm raw}$, from the exact shifts to $L = 1000$ (`residual_log/results/eps5/classes_c5.txt`). "Total": shift plus the raw coefficient $+0.0004(4)$ from the profile fit of raw $C_5(L)$, $L \le 160$ (`residual_log/WINDOW_SPEC.md`); every total carries its uncertainty ± 0.0004.

So $C_5^W(L)$ carries about $-0.0019\,L$. The Bloch zero-mode windows alone (scheme Bloch) leave no linear term; the eigenvalue-diagonal ϵ^3 form leaves $-0.0051\,L$ and is excluded. The consistent scheme has no linear term but keeps about $+0.2 \ln L$.

Location. Differences of schemes isolate the window classes (Table 17). The whole linear defect of W is the Z–χ part $2\langle Z\chi\rangle_\theta$ of $D^{M,(2)}$, the only part of M' that connects a pure zero-mode state to a one-χ state. That part cannot simply be dropped: at ϵ^4 its $\ln L$ cancels that of the $\chi\chi$ parts, and together they give the finite $+0.0102$ that C_4 needs. A variant that evaluates the Z–χ window at the smaller of the two levels keeps the linear term of C_5 and adds $+0.00295 \ln L$ to C_4.

B.4 Double counting and the rule W_c

Records: `residual_log/WINDOW_SPEC.md` (§2.2a); code `residual_log/{brute_W5,brute_desc}.py`, `residual_log/ce_wc/`.

Diagnosis. A Z–χ sunset ($Z \to 3\chi \to \chi_{aa}$) followed by a vertex that absorbs χ_{aa} into the zero mode is a connected process from a pure zero-mode state to a pure zero-mode state, that is, a term of the Bloch matrix $h^{(3)}$. Inside an outer state with spectator level ℓ its sunset has only the budget

class	C_5: coefficient of L	C_4: finite part and $\ln L$
ϵ^2 vacuum window	-0.001429	the linear $+0.000221\,L$
ϵ^3 vacuum window	$+0.001037$	none
ϵ^2 diagonal z-dependent window	$+0.004548$	$+0.016083\ln L$
ϵ^2 off-diagonal Bloch window	-0.002129	$+0.010479\ln L$
ϵ^3 Bloch window	-0.001355	none
ϵ^3 eigenvalue-diagonal window (instead of Bloch)	-0.004500	none
$D^{M,(2)}$	-0.001965	$+0.010246$, no $\ln L$
$D^{M,(3)}$	0.000000	none
partner terms in $E_z^{(3)}$	0	0
$D^{M,(2)}$, Z–χ part	-0.001965	$+0.012293 - 0.002947\ln L$
$D^{M,(2)}$, χ hop (χ in, χ out)	0	$-0.000894 + 0.001474\ln L$
$D^{M,(2)}$, pair creation	0	$-0.000579 + 0.000737\ln L$
$D^{M,(2)}$, pair annihilation	0	$-0.000579 + 0.000737\ln L$

Table 17: Window classes at ϵ^5 and their ϵ^4 content (`residual_log/results/eps5/classes_c5.txt`). The lower block splits $D^{M,(2)}$ by the leg type of its M' vertex; the parts add up to $D^{M,(2)}$ within 8×10^{-11}. The C_4 column gives fits of the shifts to $L\le1000$: for the ϵ^2 vacuum window it lists only the linear term (its $\ln L$ is in Table 12), and the $D^{M,(2)}$ entry is a finite-L estimate of the limit 0.0102543 of Appendix A.5.

$L-\ell$. The outer $D^{M,(2)}$, acting on the transition $z\to(z\mp1,\chi_{aa})$, restores it once. The zero-mode window $D^{\mathrm{zm},(3)}=h^{(3)}(L)-h^{(3)}(L-\ell)$ restores it a second time: in scheme W the auxiliary channel at $L-\ell$ carries its own mass window, so $h(L-\ell)$ contains this sunset restored only to $L-\ell$, while $h(L)$ contains it restored to L, and their difference adds the restoration from $L-\ell$ to L once more. In general, with self-consistent auxiliary objects (h, e_1, E_0, …), every window nested inside them is counted twice. At ϵ^5 this affects only $D^{M,(2)}$ inside $h^{(3)}$ and $e_1^{(3)}$, which is why the defect is purely Z–χ and why C_4 is unaffected. From ϵ^6 on it affects D^{zm}, the hop and pair parts of D^M inside $h^{(4)}$, and the vacuum window inside $E_0^{(4)}$ of the consistent scheme.

The rule. An auxiliary channel at sub-cutoff $\lambda<L$ represents the zero-mode cluster embedded in a state of the cutoff-L theory whose spectators have level $L-\lambda$. The windows inside it are therefore the cutoff-L windows of that embedded state:

$$D^{\mathrm{zm}}_{\mathrm{aux}}|z,\chi\rangle=\sum_{z'}\big[h(L)-h(\lambda-\ell_\chi)\big]_{z'z}|z',\chi\rangle,\qquad D^{M}_{\mathrm{aux}}=\tfrac14\big\{e_1(L)-e_1(\lambda-\hat\ell),M'\big\}. \tag{44}$$

$D^{\mathrm{zm}}_{\mathrm{aux}}$ is absent on auxiliary level-0 states, where D^{M}_{aux} uses $e_1(L)-e_1(\lambda)$. The outer windows keep their form, with h and e_1 now computed under this rule. At the order that enters C_5,

$$\begin{aligned}h^{(3)}(\lambda;L)&=h^{(3)}(\lambda)+\tfrac12\big[e_1^{(2)}(L)-e_1^{(2)}(\lambda)\big]H_3^{M'}(\lambda),\\ e_1^{(3)}(\lambda;L)&=e_1^{(3)}(\lambda)+g_1\big[e_1^{(2)}(L)-e_1^{(2)}(\lambda)\big]\big[A_1(\lambda)-A_0(\lambda)\big],\qquad A_z(\lambda)=\langle z|M_0'G_zV_0|z\rangle_\lambda,\end{aligned} \tag{45}$$

where $H_3^{M'}$ is the response of $h^{(3)}$ to M_0' inserted with unit coefficient at ϵ^2, the subscript 0 marks the ϵ^0 part of an operator, and $G_z=Q_z/(\frac{z}{2}-H_0)$. The added terms complete the inner sunsets to the outer cutoff L, so that the outer difference no longer restores them a second time. The rule has no fitted parameter; its status is provisional (Appendix C.6). Three exact implementations agree: explicit-operator fraction arithmetic (`brute_W5.py`, `brute_desc.py`), the large-L diagram

code, and `ce_wc/`, a copy of the small-L solver with the auxiliary windows referred to L at all orders. They give identical C_5 at $L \le 8$, C_3 to C_5 within 10^{-15} at $L \le 16$, and the same ϵ^3 spacing of χ_{11} as rationals at $L = 8$ to 14.

B.5 Tests T1 to T3

Records: `residual_log/results/eps5/{classes_c5,ce_wc_tables}.txt`, `residual_log/fits/fits_c5.txt`, `residual_log/results/eps5/dC5_L1000_all.npz`.

We test each window in three ways: T1, C_4 stays unchanged and convergent; T2, C_5 converges to the continuum value; T3, the ϵ^3 spacing R_3 of the single-quantum descendants vanishes (Appendix C adds T4, the ϵ^4 spacing).

T1 passes. Windows of order ϵ^3 and higher do not enter C_4, and $|C_4^{W_c} - C_4^{W}| \le 6 \times 10^{-13}$ for $L \le 1000$. So $C_4^{W_c}$ has the limit 5.0488682(10) of Section 4.4.

T2 passes, with the caveat of Section 4.5. The W_c shift has the linear coefficient -0.0003922 (half range 2×10^{-6} over tail towers and fit windows; step 1 of `fits_c5.txt`), equal to the consistent scheme's -0.0003923, so with the raw $+0.0004(4)$ the scheme has no linear term. Its limit and the residual-logarithm question are those of Appendix B.6. Table 18 compares the schemes.

scheme	windows	linear	limit	limit − continuum
raw	none	+0.0004	needs L or $\ln L$ terms	
consistent	vacuum, ϵ^2 and ϵ^3	+0.0000	needs $\ln L$ ($+0.2 \ln L$)	
zero-mode	diagonal, ϵ^2 and ϵ^3	−0.0010	diverges	
Bloch	Bloch $D^{\rm zm}$, ϵ^2 and ϵ^3	+0.0000	≈ -38.5	$\approx +0.3$
W	$D^{\rm zm} + D^M$	−0.0019	diverges	
W_c	W with rule W_c	+0.0000	−38.830(13)	+0.004

Table 18: Large-L behavior and extrapolated C_5 by scheme. "Linear" is the total linear coefficient of $C_5^S(L)$ of Table 16, with uncertainty ±0.0004. The W_c limit uses the protocol of Appendix B.6 (towers with the $\ln^3 L/L$ term, $L \le 300$); for Bloch, whose tail also carries logarithms, the limit and the miss are only approximate. Raw: fits without L, $\ln^2 L$ and $\ln L$ terms miss the continuum value by 1.9 (step 3 of `fits_c5.txt`, $L \le 160$). Continuum value $C_5^{(2)} = -38.8336459$.

The scheme Bloch misses by about 0.3: the finite part of D^M is needed, and W_c keeps it.

T3 fails in W_c. W_c removes the slow growth of the ϵ^3 spacing that W has, but leaves a constant (Table 22). A repair of this defect shifts C_5 by less than the extrapolation spread: the external-line window of Appendix C, which identifies the cause and derives the repair, shifts it by -0.005772, against a spread of ±0.013. So T2 cannot decide it, and T3 is the decisive test.

B.6 The tail of C_5 and the residual logarithm

Records: `residual_log/large_L/fits/{c5_fits300,legw_logs}.txt`, `residual_log/large_L/results/{regression_opt,compare_c5_L300_vs_L200}.txt`, `residual_log/large_L/results/c5_L300.npz`, `residual_log/fits/parent_check_largeL/refit_c5_300.txt`; code `residual_log/large_L/{c5_fits300,c5_fits}.py`, `residual_log/large_L/lib_opt/`, `residual_log/fits/parent_check_largeL/refit_c5_300.py`.

C_5^{Wcs} (Section 4.5, Appendix C.3) differs from C_5^{Wc} by $\delta^{(3)}(L)S(L)$, which carries no $\ln L$. Towers without the $\ln^3 L/L$ term that C_5 needs bias the limit, because a free $\ln L$ then stands in for the missing tail.

Tail power. The tail of $C_n(L)$ carries $\ln^{n-2} L/L$: $C_2 \sim 1/L$, $C_3 \sim \ln L/L$, $C_4 \sim \ln^2 L/L$ and $C_5 \sim \ln^3 L/L$ (Appendices A.2 and A.5). Hold-out tests, made after the data to $L = 300$ existed, support this power (Table 19): fit $[L_{\min}, 240]$ and predict $L = 242$ to 300 with towers without $\ln L$ whose $1/L$ part has the powers $\ln^i L$, $i \le k_1$. The prediction error falls by one to two orders of magnitude for each added power up to $k_1 = n - 2$ ($k_1 = 2$ for C_4^W, 3 for C_5^{Wcs}) and changes by at most a factor of two beyond it. The test thus finds the smallest adequate tail, but it cannot exclude one more power: for C_4^W, $k_1 = 3$ is slightly better than 2.

	$k_1 = 0$	1	2	3	4
C_4^W: hold-out error	6.1×10^{-4}	9.7×10^{-6}	1.1×10^{-7}	$\mathbf{5.2 \times 10^{-8}}$	6.5×10^{-8}
C_5^{Wcs}: hold-out error	1.2×10^{-2}	3.5×10^{-4}	5.3×10^{-6}	$\mathbf{3.7 \times 10^{-7}}$	5.7×10^{-7}
C_5^{Wcs}: limit − continuum	+0.81	+0.16	+0.015	−0.003	+0.012

Table 19: Selection of the tail power k_1 (largest i in $\ln^i L/L$) by hold-out prediction of $L = 242$ to 300 from fits to $L \le 240$, towers without $\ln L$ ($1/L^2$ with $\ln^{0,1} L$ or $\ln^{0,1,2} L$; $L_{\min} = 40$, 60, 80; median of the largest errors), and the limit of the same towers fitted to $L \le 300$ (`residual_log/large_L/fits/c5_fits300.txt`, §1). Bold: smallest error.

With the 15 fits of Eq. (25) ($L_{\min} = 30$ to 100, $k_2 = 0$ to 2, $L \le 300$), $k_1 = 4$ gives a median of −38.8334, shifted by +0.002 from the $k_1 = 3$ value, with a half range of 0.050 instead of 0.013 (replay check `p2.c5_k1`); with the towers of Table 19 the limit moves by +0.015.

Raw C_5 to $L = 300$. The sum is split into 10012 independent tasks (groups of chain patterns and level ranges). The large-L engine as first run (`residual_log/large_L/lib/`, not the production solver) computed 3987 of them in about 130 node-hours. An optimized engine (`residual_log/large_L/lib_opt/engine.py`) finished the remaining 6025 tasks in 32 node-hours; only the density without ϵ-derivatives changes, through a memoized tree over the inclusion–exclusion terms, a parallel diagonal convolution and batched axis convolutions. Per task it agrees with the first engine to 7.3×10^{-16} (norm-relative; 1448 tasks, covering every kind of density kernel at $L = 300$ and all tasks with a bundle of six lines at $L = 200$). The $L = 300$ assembly, 40% old and 60% new tasks, reproduces the $L \le 200$ data to 7.6×10^{-13} (C_5, relative) and 5.1×10^{-13} (C_4). Raw $C_5(300) = -33.6509669991$.

Hold-out tests with $L \le 300$. Table 20 compares fits without and with a free $\ln L$ at fixed k_1.

With the misspecified $k_1 = 2$ a free logarithm helps. With $k_1 = 3$ and the fit ending at $L = 200$, the model without $\ln L$ predicts $L = 202$ to 300 better by a factor of two. With the fit ending at $L = 240$ the two are equal, so at that level the hold-out test no longer separates a logarithm from none; for C_4^W, which has no logarithm, a free one also predicts slightly better there.

Drift of the free $\ln L$ coefficient. With $k_1 = 3$ towers, $L_{\min} = 30$ to 80 (medians of twelve fits, §2 of `c5_fits300.txt`):

	fit $L \le 200$, predict 202 to 300		fit $L \le 240$, predict 242 to 300	
k_1	no $\ln L$	free $\ln L$	no $\ln L$	free $\ln L$
2	4.2×10^{-5}	7.1×10^{-6}	1.6×10^{-5}	2.0×10^{-6}
3	4.3×10^{-6}	9.4×10^{-6}	6.0×10^{-7}	5.7×10^{-7}
4	6.5×10^{-6}	1.4×10^{-5}	6.3×10^{-7}	7.7×10^{-7}

Table 20: $C_5^{W_{cs}}$: hold-out prediction errors, median over the 15 fits of Eq. (25) (`residual_log/fits/parent_check_largeL/refit_c5_300.txt`). The medians over twelve fits with $L_{\min} = 30$ to 80 (`c5_fits300.txt`) for the fit to $L \le 200$ with $k_1 = 3$ are 4.7×10^{-6} and 9.6×10^{-6}.

$L_{\max}$	200	220	240	260	280	300
free $\ln L$ coefficient b	-0.0134	-0.0080	-0.0005	$+0.0030$	$+0.0047$	$+0.0054$
calibration, injected 0	-0.0007	-0.0021	-0.0028	-0.0033	-0.0035	-0.0036
calibration, injected -0.03	-0.0307	-0.0321	-0.0328	-0.0333	-0.0335	-0.0336
$C_5^{W_{cs}}$: limit without $\ln L$ − continuum	-0.0049	-0.0051	-0.0043	-0.0032	-0.0024	-0.0017

The calibration applies the same estimator to the synthetic series $7.7\, C_4^W + 2 \ln^3 L/L + b \ln L$ with a known, injected b: C_4^W has no logarithm and a known limit, the factor $7.7 \approx |C_5/C_4|$ gives it the size of C_5, and the $\ln^3 L/L$ term gives it the top logarithm of C_5. The estimator settles by $L_{\max} \approx 250$ and then reads about 0.004 low. An independent refit (`refit_c5_300.py`), with the 15 fits of Eq. (25), gives the same drift (-0.011, -0.004, $+0.003$ and $+0.004$ at $L_{\max} = 200$, 240, 280 and 300), and on C_4^W, which has no logarithm, the same estimator returns $+0.0005$. Section 4.5 draws the conclusion.

The limit. The 15 fits of Eq. (25) to $L \le 300$ give Eq. (26); their range is -38.8430 to -38.8174. $C_5^{W_c}$ lies higher by about $-\delta_\infty S_\infty = +0.0058$, as the external-line window requires. As $L_{\max}$ grows, the W_c median moves away from the continuum value and the W_{cs} median toward it. The same protocol applied to C_4^W gives 5.04868 ± 0.00082 (half range of the 15 fits) against 5.0488684.

Classes excluded as sources of a logarithm. The ϵ^3 external-line class gives no $\ln L$ for any per-unit coefficients, because the $\ln L$ coefficients of $S(L)$ cancel within the hop sector and within the pair sector separately (6×10^{-7} and -1.7×10^{-6}; `legw_logs.txt`). The spectator window $D^{M,(3)}$ has no L, $\ln^2 L$ or $\ln L$ term within the fit uncertainties. The other classes that carry logarithms cannot be separated by fits of the exact shifts alone.

B.7 Consequences for C_6 to C_8

Records: `residual_log/results/eps5/ce_wc_tables.txt`, `complete_exact/README.md` (§9).

The double counting recurs, and grows, at every order with nested windows. Rule W_c removes it by construction and is implemented at all orders in `residual_log/ce_wc/`. At $L = 14$ the schemes W_c and W already differ by about 1%: $C_6 = 157.83$ against 159.46, $C_7 = -1439.9$ against -1458.3, and $C_8 = 14710$ against 14962. The ϵ^3 remainder of T3 is a constant defect attached to χ lines; it enters C_5 (through the shift $\delta^{(3)}S$, Appendix C.4) and every higher coefficient. Its repair (Appendix C) changes C_6 to C_8 by only about 0.01%, but the same defect recurs at ϵ^4 and ϵ^5. These differences between schemes are much smaller than the distance of C_6 to C_8 at $L \le 16$ from the continuum values, and C_6 to C_8 are not reproduced in any scheme (Section 4.7).

C Two dimensions: descendant spacings and the external-line windows

This appendix supports Section 4.6 and the external-line window of Section 4.5. The spacings R_n of Eq. (27) test a window at a fixed order without any continuum input; they exposed the defects that remain beyond the complete ϵ^4 window, and a tagged external line located the ϵ^3 defect exactly.

C.1 Descendant channels

Records: `complete_exact/README.md` (§§4, 8 and 9), `complete_exact/results/{descendant_summary.md,descendant_fits.txt}`, `complete_exact/results/descendant_<scheme>_L<L>.json`, `residual_log/results/descendant_e2.txt`, `residual_log/results/eps5/{descendant_eps3,ce_wc_tables}.txt`, `residual_log/fits/parent_check_wcs/verify.txt`; code `residual_log/descendant_e2.py`.

Two kinds of descendants are used: $\chi_{aa} = \partial^a\bar\partial^a\phi$ (one quantum (a,a), level $2a$, primary $|1\rangle$) and $(\partial\bar\partial)^a\phi^2$ (primary $|2\rangle$). The reference of the latter at $\epsilon = 0$ lies in a degenerate block and is fixed as the eigenvector of H_1 with eigenvalue $E_2^{(1)}$. The small-L solver `complete_exact/` and its W_c copy `residual_log/ce_wc/` compute these channels exactly in every scheme for $L \le 16$, and $R_0 = R_1 = 0$ exactly everywhere. A spacing counts as restored if it decays like $1/L$.

Order ϵ^2. Since $z = 0$ for χ_{aa}, the consistent and zero-mode windows coincide on it. $R_2(\chi_{aa})$ is a sum over two kinds of intermediate states. In the nested ones the external quantum is absorbed and re-emitted. In the exchange ones the first vertex emits the new χ_{aa} together with a bubble, and the second absorbs the old χ_{aa} with the bubble. The exchange states hold two (a,a) quanta, so the level cutoff leaves the bubble only $L - 4a$ instead of L. For the sunset this is a deficit of $\frac{2}{\pi}(2a) \times g_1^2 \cdot \frac{2}{3}w_{aa}$, and all other finite parts cancel, so

$$\lim_{L\to\infty} R_2(\chi_{aa}) = \frac{8a\,w_{aa}}{27\pi} \neq 0 \tag{46}$$

in the consistent, literal and zero-mode schemes; the raw truncation also lacks the vacuum window, which by Eq. (32) grows linearly in L. The mass window repairs it. On the one-quantum state $\langle\chi_{aa}|M'|\chi_{aa}\rangle = 2w_{aa}$ and $\ell = 2a$, while $\langle 1|D^M|1\rangle = 0$, so at first order D^M shifts R_2 by exactly $-\frac{1}{27\pi}\cdot 4a\cdot 2w_{aa} = -\frac{8a\,w_{aa}}{27\pi}$.

a	w_{aa}	$\frac{8a\,w_{aa}}{27\pi}$	$R_2(2000)$	$R_2(20000)$	with D^M, $L=2000$	with D^M, $L=20000$
1	0.25	0.023578510	0.023584405	0.023579100	5.90×10^{-6}	5.90×10^{-7}
2	0.140625	0.026525824	0.026532463	0.026526487	6.64×10^{-6}	6.63×10^{-7}
3	0.0976563	0.027631067	0.027637989	0.027631757	6.92×10^{-6}	6.91×10^{-7}
5	0.0605621	0.028559298	0.028566467	0.028560012	7.17×10^{-6}	7.14×10^{-7}
10	0.0310454	0.029280172	0.029287560	0.029280905	7.39×10^{-6}	7.33×10^{-7}

Table 21: The ϵ^2 spacing of χ_{aa} in the consistent window at large L, and the residual $R_2 - \frac{8a\,w_{aa}}{27\pi}$ left after adding D^M (`residual_log/results/descendant_e2.txt`).

Table 21 gives the numbers: with D^M the residual decays like $1/L$, with coefficient 1.2×10^{-2} ($a=1$) to 1.5×10^{-2} ($a=10$). The exact small-L solver confirms this for the full schemes at $L \le 16$. In the raw scheme R_2 grows (χ_{11}: 0.1077 at $L=16$). In the consistent and zero-mode schemes $R_2(\chi_{11})$ is constant (0.0243 at $L=16$), and the ϕ^2 descendant decreases only slowly (0.1459 at

$L = 16$, effective exponent -0.27). In W, and in W_c, which is identical at this order, R_2 is restored: $0.0114/L$ for χ_{11} (0.000714 at $L = 16$), $0.0146/L$ for χ_{22} and $0.634/L$ for the ϕ^2 descendant.

Order ϵ^3. None of the schemes W, W_c and their variants restores the ϵ^3 spacing. The external-line window W_{cs} (Appendix C.3) reduces it for χ_{aa} by at least 95% at $L = 14$ and 16, but a $1/L$ decay is not established. Table 22 collects $R_3(\chi_{11})$; the two sources agree where they overlap (scheme W to all printed digits at $L = 6$ to 14).

scheme	$L = 4$	$L = 8$	$L = 12$	$L = 14$	$L = 16$	behavior
raw	0.1617	0.2479	0.3355	0.3793	0.4232	grows linearly
consistent (= zero-mode)	0.04007	0.03941	0.04002	0.04037	0.04071	constant
ϵ^2 windows only	0.1874	0.2750	0.3643	0.4089		grows linearly
W without D^M	0.05598	0.05519	0.05577	0.05611		constant
W with $D^{M,(2)}$ only	0.05812	0.06237	0.06598	0.06749		grows slowly
W	0.02316	0.02686	0.02962	0.03074	0.03173	grows, $\sim 0.009 \ln L$
W+V	0.08723	0.1138	0.1362	0.1469	0.1573	grows
W_c	0.01785	0.01618	0.01579	0.01570	0.01565	constant
W_{cs}	0.004225	0.000896	0.000121	-0.000049	-0.000157	falls (Appendix C.4)

Table 22: R_3 of $\chi_{11} = \partial\bar{\partial}\phi$ by scheme, exact (`complete_exact/README.md`, `residual_log/results/eps5/{descendant_eps3,ce_wc_tables}.txt`, `residual_log/fits/parent_check_wcs/verify.txt`). "ϵ^2 windows only" is the window of Appendix A.5 with no ϵ^3 part; "without D^M" and "with $D^{M,(2)}$ only" are W without the mass window and with its ϵ^2 part only (their W_c versions are identical at this order).

For the other descendants at $L = 16$: χ_{22} has $R_3 = 0.04048$ (consistent), 0.03447 (W, growing) and 0.01755 (W_c, flat). The ϕ^2 descendant of level 2 has 0.09469 (consistent, growing), 0.03864 (W, growing after a minimum of 0.0074 near $L = 6$ to 8) and -0.0337 (W_c; -0.0306, -0.0413 and -0.0337 at $L = 6$, 10 and 16, not settled).

Structure of the ϵ^3 defect. For $a \ll L/2$ the ϵ^3 spacing of χ_{aa} is proportional to $a\, w_{aa} = \frac{1}{8}\langle\chi_{aa}|\{\hat{\ell}, M'\}|\chi_{aa}\rangle$ in every scheme. At $L = 12$, $R_3/(a\, w_{aa}) = 0.16$, 0.15, 0.15 for $a = 1$, 2, 3 in the consistent scheme, 0.22, 0.21, 0.21 in W without D^M, 0.119, 0.114, 0.110 in W, and 0.063 to 0.068 in W_c. This is the operator structure of a mass window attached to χ lines, as at ϵ^2. The diagnostics locate it:

- D^{zm} is not the problem at ϵ^3: W without D^M leaves a constant (0.056 for χ_{11}).
- In W, $D^{M,(2)}$ acting at third order produces the slow growth, and $D^{M,(3)}$ shifts the spacing by an almost L-independent -0.036. The growth is the double counting of Appendix B.4, which W_c removes.
- W_c leaves a near constant: at $L = 16$, $R_3/(a\, w_{aa}) = 0.06259$, 0.06242 and 0.06337 for $a = 1$, 2 and 3, changing slowly with L. Section 4.5 compares it with the first-order shift of the external-line window, $4a\, w_{aa}\, \delta^{(3)}(L) \to -\frac{16}{81\pi} a\, w_{aa}$, with $\frac{16}{81\pi} = \frac{2}{3} \cdot \frac{8}{27\pi} = 0.06288$.

The factor $\frac{2}{3}$ equals the $\Delta z = \pm 2$ share of the angular average of Appendix A.4, which suggested a missing channel of the ϵ^3 exchange sunset that changes the zero-mode number. The tagged-line test of Appendix C.2 shows instead that the defect is the level budget of the external legs, charged in each slice only where the legs are present. Whether the $\frac{2}{3}$ has a simple diagrammatic origin is not derived.

Order ϵ^4 and higher. The defect propagates. In W_c, $R_4(\chi_{11}) = 0.019$, 0.037, 0.049, 0.057, 0.063, 0.068, 0.073 and 0.076 for $L = 2, 4, \ldots, 16$; $R_4(\chi_{22})$ rises from 0.034 ($L = 6$) to 0.070 ($L = 16$), and R_4 of the ϕ^2 descendant from 0.14 to 0.40. In W, $R_4(\chi_{11}) = 0.097$ at $L = 16$, and orders 4 to 8 are violated by larger, growing amounts. Appendices C.4 and C.5 give the ϵ^4 results of the external-line windows.

C.2 The tagged-line test

Records: `residual_log/results/tag/tag_L<L>_a<a>_*.json`; code `residual_log/brute_desc_tag.py` (exact fraction arithmetic).

In the χ_{aa} channel the external line is tagged. The initial quantum is a separate copy $\tau_{\rm in}$ of the mode (a, a) that can only be absorbed, and the final quantum a copy $\tau_{\rm out}$ that can only be created, once. The three one-quantum states $|\tau_{\rm in}\rangle$, $|\tau_{\rm out}\rangle$ and the regular $|\chi_{aa}\rangle$ all stand for the real descendant. All three are projected out of every resolvent, as the reference state is in ordinary perturbation theory. The order-n energy is the sum of the components on $|\tau_{\rm in}\rangle$ and $|\tau_{\rm out}\rangle$ of the order-n right-hand side of the Rayleigh–Schrödinger recursion (the vector whose overlap with the reference gives $E^{(n)}$). Histories that end on the regular $|\chi_{aa}\rangle$ would count those ending on $|\tau_{\rm out}\rangle$ twice and are dropped. The tags count toward the level budget, and toward ℓ in every window, with charges $(\nu_{\rm in}, \nu_{\rm out})$.

- With charges $(2a, 2a)$ the construction is identical to the real channel; it reproduces W_c exactly ($L = 2$ to 14 for $a = 1$, $L = 6$ to 12 for $a = 2$).
- With charges $(0, 0)$ the external line is invisible to the cutoff and to the windows (budget-free). Its self-energy then has exactly the budget of the ϕ self-energy, on which r is tuned.

L	charged ($= W_c$): R_2	R_3	budget-free: R_2	R_3	in only: R_3	out only: R_3
2	0.004630	0.021917	0.017361	0.028412	0.024564	0.026879
6	0.001808	0.016678	0.008545	0.012127	0.012487	0.013542
10	0.001122	0.015932	0.005728	0.007330	0.009882	0.010568
14	0.000813	0.015702	0.004317	0.005055	0.008861	0.009368

Table 23: R_2 and R_3 of χ_{11} with a tagged external line, W_c windows (`residual_log/results/tag/`). "In only" and "out only" charge one leg by 2.

With the legs budget-free, $R_2 \approx 0.06/L$ and $R_3 \approx 0.07/L$ both go to zero (Table 23); charged, R_3 tends to 0.0156. For $a = 2$ the budget-free R_3 is 0.0313, 0.0227, 0.0177 and 0.0144 at $L = 6$ to 12, against 0.0221, 0.0195, 0.0185 and 0.0180 in W_c. So the whole ϵ^3 remainder of W_c is the level budget of the external legs; spectators, nested windows and the auxiliary objects are already treated correctly.

Why average-level charging is exact at ϵ^2 and not at ϵ^3. A mass-type sub-process on a χ line, one of the sub-bubbles that rM' cancels, has two external legs, the legs of the M' term. The truncation charges a leg only in the slices of the sub-process where it is present: an absorbed leg before its absorption, an emitted leg after its emission. D^M instead charges both legs with their average level $\frac{1}{2}(\ell_s + \ell_{s'})$ in every slice.

- At ϵ^2 the sub-process is the sunset. It has two orderings, normal (no leg present) and exchange (both legs present), whose weights are equal at large bubble level, so the average is exact and $R_2 \to 0$.

- At ϵ^3 there are also mixed orderings. In the vertex-corrected sunset (vertices v_1, v_2, v_3; lines v_1v_2, twice v_1v_3 and twice v_3v_2; the legs at v_1 and v_2) the six orderings have $(0,0)$, $(1,0)$, $(0,1)$, $(2,1)$, $(1,2)$ and $(2,2)$ external legs present in their two internal slices. Their weights differ, so the response to the level of a leg is not half the response to a spectator.
- The tadpole with a sunset on its loop has both legs at one vertex and charges exactly one leg in every slice, so it is unaffected.

C.3 The ϵ^3 external-line window W_{cs}

Records: `residual_log/WINDOW_SPEC.md` (§2.5a), `residual_log/large_L/fits/delta3_fits.txt`, `residual_log/large_L/results/{validation_small.txt,tag3_L16000.npz,tag3nu_L4000.npz}`, `residual_log/ce_wc/out_wcs/`; code `residual_log/{brute_desc_tag,delta3}.py`, `residual_log/large_L/{tagchan,delta3_fits}.py`, `residual_log/ce_wc/{ce_core,ce_solver,ce_run,tables_wcs}.py`.

Definition. Scheme W_{cs} is W_c with one more term at ϵ^3. The full ϵ^3 window is

$$D^{(3)} = \tfrac{1}{4}\big\{\Delta e_1^{(3)}(L,\hat{\ell}), M_0'\big\} + \tfrac{1}{4}\big\{\Delta e_1^{(2)}(L,\hat{\ell}), M_1'\big\} + \frac{\delta^{(3)}(L)}{2}\,X_0, \qquad \Delta e_1(L,\ell) = e_1(L) - e_1(L-\ell), \tag{47}$$

with $\delta^{(3)}$ of Eq. (23) and M_k' the ϵ^k part of M'.

- X is M' with every term multiplied by η times the sum of the levels of its χ legs. The crossing sign is $\eta = +1$ for terms with one creator and one annihilator (χ hops and $Z \leftrightarrow \chi$ hops) and $\eta = -1$ for pair creation and pair annihilation ($\chi\chi$ or $Z\chi$). Zero-mode legs have level 0.
- $e_1^{\rm tag}(\lambda;\nu_{\rm in},\nu_{\rm out})$ is $E - E_0$ of the ϕ channel $|1\rangle$ at sub-cutoff λ whose external zero-mode line is tagged as in Appendix C.2. The tags have level 0 in H_0 but add $\nu_{\rm in}$, $\nu_{\rm out}$ to the budget level; r and g are those of the references at L. With zero charges the tagged channel equals the ϕ channel exactly through ϵ^3. $e_1^{(3)}$ and $e_1^{\rm tag,(3)}$ are taken with r_3 removed.
- $\delta^{(3)}(L)$ is one number per outer cutoff, used in every channel, including the auxiliary ones (rule W_c). Everything else ($D^{(2)}$, $D^{\rm zm}$ at all orders, $D^{M,(k)}$ for $k \geq 4$, rule W_c) is unchanged.

Derivation of the operator form. For a hop $s \to s'$ with spectator level $\ell_{\rm spec}$ and χ legs of levels $\nu_{\rm in}$, $\nu_{\rm out}$, the slice-resolved window is $\frac{1}{2}[e_1(L) - e_1^{\rm tag}(L-\ell_{\rm spec};\nu_{\rm in},\nu_{\rm out})]M'_{s's}$. To first order in the charges this is $\frac{1}{2}[e_1'\ell_{\rm spec} + \mu(\nu_{\rm in}+\nu_{\rm out})]M'_{s's}$, with $e_1' = de_1/d\lambda$ and $\mu = \frac{1}{2}e_1' + \delta$ the response to one unit of leg level; δ is its departure from the average-level rule of D^M. As $L \to \infty$ in and out legs respond equally, by time reversal; at finite L they differ by $\mathcal{O}(1/L)$ (below). Crossing a leg exchanges the slices in which it is present, so a pair leg responds with $e_1' - \mu = \frac{1}{2}e_1' - \delta$. With $\frac{1}{2}(\ell_s + \ell_{s'}) = \ell_{\rm spec} + \frac{1}{2}(\nu_{\rm in}+\nu_{\rm out})$ this is D^M plus $\frac{\delta}{2}X$. δ is calibrated on the smallest charges, $(2,2)$: with $\delta_\nu^{(3)}(L) = \frac{1}{2\nu}\big[e_1^{(3)}(L-\nu) - e_1^{\rm tag,(3)}(L;\nu,\nu)\big]$ for symmetric charges (ν,ν), $\delta^{(3)} = \delta_2^{(3)}$. For χ_{11} the window is therefore exactly the slice-resolved one; for larger a it is its linearization, which becomes exact as $L \to \infty$ (below).

What is fixed and what is assumed. Nothing is fitted: $\delta^{(3)}(L)$ is an exact rational computed from the tagged ϕ channel at each L. The descendant tests do not probe the sign of the pair terms, because R_3 of any descendant involves only diagonal hop terms of X. The pair sign rests on the crossing argument and on the separate cancellation of $\ln L$ in the Z–χ and $\chi\chi$ parts of the C_5 response $S(L)$ (Appendix C.4).

$\delta^{(3)}(L)$ **at large** L**.** A chain-sum version of the tagged ϕ channel (`residual_log/large_L/tagchan.py`) evaluates $\delta^{(3)}(L)$ for every even $L \le 16000$ in one run of 11 minutes. It reproduces the exact rationals for $L = 2$ to 16 to 4×10^{-16} (relative 2.6×10^{-14}), including the values the W_{cs} runs used, and runs with different $L_{\max}$ agree to 10^{-11}. Table 24 gives selected values; Fig. 1b shows the curve.

L	$\delta^{(3)}(L)$	$\delta^{(3)}(L) + \frac{4}{81\pi}$	$\delta_4^{(3)}(L)$
6	-0.0148018	$+9.2 \times 10^{-4}$	-0.014450
10	-0.0155308	$+1.9 \times 10^{-4}$	-0.016733
16	-0.0158036	-8.5×10^{-5}	-0.016729
34 (minimum)	-0.0159046	-1.9×10^{-4}	
100	-0.0158494	-1.3×10^{-4}	-0.015993
400	-0.0157725	-5.3×10^{-5}	-0.015806
1000	-0.0157458	-2.7×10^{-5}	-0.015759
4000	-0.0157278	-8.8×10^{-6}	-0.015731
16000	-0.0157217	-2.7×10^{-6}	

Table 24: $\delta^{(3)}(L)$: exact rationals for $L \le 16$, chain sums beyond (`residual_log/large_L/fits/delta3_fits.txt`). Third column: distance from $-\frac{4}{81\pi} = -0.0157190$. Last column: the coefficient $\delta_4^{(3)}$ calibrated on charges $(4, 4)$ (the exact form for χ_{22}).

$\delta^{(3)}(L)$ falls to its minimum at $L = 34$ and then rises toward the limit of Eq. (24); the next term is $+0.013856/L$, followed by $\mathcal{O}(\ln L/L^2)$. Free-limit fits with a $\ln L/L$ term ($L_{\min} = 200$ to 2000, data to $L = 16000$) give $\delta_\infty + \frac{4}{81\pi}$ between -3×10^{-11} and $+7 \times 10^{-11}$; without $\ln L/L$ they miss by -5×10^{-7} to -1.5×10^{-6}. With the limit fixed, the $\ln L/L$ coefficient is -0.0058946, equal to $-\frac{1}{54\pi}$. A fit to $L \le 4000$ predicts $L = 16000$ to 4×10^{-10} with the $\ln L/L$ term (-2×10^{-6} without). Also $\delta^{(3)}(2) = -0.0093879$, $\delta^{(3)}(4) = -0.0136276$, and $\delta^{(3)}(16)$ is the rational $-17345700448126303825/1097581272385718321152$. Because $L \le 16$ lies on the falling branch before the minimum, extrapolations in $1/L^2$ from $L \le 16$, which do not see the $\ln L/L$ term, miss the limit: they give -0.0158 to -0.0165 (`residual_log/fits/parent_check_wcs/fits.txt`). With the limit identified numerically, $4|\delta_\infty|/\frac{8}{27\pi} = \frac{2}{3}$: the leg coefficient is two thirds of the ϵ^2 exchange-sunset defect.

The nonlinearities are finite-L **effects.** Define the single-leg coefficients d_{in} and d_{out} as $\delta_2^{(3)}$ with only one leg charged (by 2); with the $(2, 2)$ cross term they give $\delta_2^{(3)} = \frac{1}{2}(d_{\text{in}} + d_{\text{out}}) + \frac{1}{4}$(cross term). At $L \le 16$, d_{in} and d_{out} are smaller in magnitude than $\delta^{(3)}$, and the coefficients for larger charges are larger: at $L = 16$, $\delta_4^{(3)}$ and $\delta_6^{(3)}$ (the exact χ_{22} and χ_{33} forms) exceed $\delta_2^{(3)}$ in magnitude by 6% and 13% (evaluated from the $\nu = 4$ and $\nu = 6$ lines of `delta3_fits.txt`). At large L both effects disappear (`delta3_fits.txt`):

- d_{in} and d_{out} both tend to $-\frac{4}{81\pi}$. Their difference falls like $1/L$ (1.6×10^{-5} at $L = 1000$, 3.9×10^{-6} at 4000), so time reversal is restored, and the $(2, 2)$ cross term vanishes (-4.0×10^{-4} at $L = 1000$, -1.2×10^{-4} at 4000).
- $\delta_\nu^{(3)}$ for $\nu = 4$, 6, 8 and 16 tends to the same limit, with differences of about $-(\nu - 2) \times 0.0062/L$ from $\delta_2^{(3)}$ (last column of Table 24).

So the linearized window (47), calibrated on the charges $(2, 2)$, is asymptotically exact for every a.

C.4 Tests of W_{cs}

Records: `residual_log/fits/parent_check_wcs/{verify,fits,r3zero,blind16}.txt`, `residual_log/results/legw/legw_L1000.npz`, `residual_log/large_L/fits/legw_logs.txt`, `residual_log/ce_wc/{out_wcs,out_local,out_wcs4}/`, `residual_log/large_L/results/tag3_L16000.npz`; code `residual_log/{nm_legw,brute_legw}.py`.

T1 passes. C_0 to C_4, r_1 to r_3 and g_1 to g_3 of W_{cs} are identical to those of W_c as exact rationals for $L = 2$ to 16 (r_4 differs).

T2 passes, with the same caveat. $C_5^{W_{cs}}(L) - C_5^{W_c}(L) = \delta^{(3)}(L)\, S(L)$ holds to 1.2×10^{-16} or better for $L = 2$ to 16, with $S(L)$ the C_5 response to $\frac{1}{2}X$ inserted at ϵ^3 (`nm_legw.py`, checked by brute force at $L = 2$, 4, 6). $S(L)$ is known to $L = 1000$ and converges with no $\ln L$:

L	10	20	50	100	200	400	1000	∞
$S(L)$	0.286252	0.319024	0.344641	0.354908	0.360610	0.363699	0.365693	0.367187

A free $\ln L$ coefficient comes out at 7×10^{-7} or less (`fits.txt`, $L_{\min} = 100$ to 400; other tail towers give at most 4×10^{-6}, `legw_logs.txt`). By type of M' term the $\ln L$ coefficients of S are $+\frac{1}{16}$ (Z–χ hop), $-\frac{1}{16}$ (Z–χ pair), $-\frac{1}{16}$ (χ hop), $+\frac{1}{16} + \frac{1}{27}$ (pair creation) and $-\frac{1}{27}$ (pair annihilation). They cancel within the hop sector and within the pair sector separately, so a leg window with one coefficient per unit leg level for hops and another for pairs gives no $\ln L$ in C_5, whatever the two coefficients are; the large-L check confirms this to 10^{-6}. This holds for either sign of the pair terms. With the crossing sign the coefficients also cancel within the Z–χ sector and within the $\chi\chi$ sector; with an unsigned leg weight these two sectors would carry $\pm\frac{1}{8}$ each. Only the signed weight therefore keeps C_5 free of $\ln L$ if the Z–χ and $\chi\chi$ parts of the window have independent coefficients, and this is the support for the sign. With the exact $\delta^{(3)}(L)$, $C_5^{W_{cs}} - C_5^{W_c} = -0.005478$, -0.005625 and -0.005700 at $L = 50$, 100 and 200 (evaluated from `tag3_L16000.npz` and `legw_L1000.npz`), tending to $\delta_\infty S_\infty = -0.005772$ with no $\ln L$. The window shift is small compared with the effect of a residual $\ln L$ with $|b| \approx 0.01$, about ± 0.1 (Section 4.5).

T3 for χ_{aa}: passes at $L \le 16$, with a possible small remainder. Table 25 gives R_3 of χ_{11}, χ_{22} and χ_{33} in W_{cs}.

L	χ_{11}	χ_{22}	χ_{33}
6	+0.001876 [0.016678]	+0.005487 [0.022139]	+0.029714 [0.047060]
8	+0.000896 [0.016183]	+0.002279 [0.019477]	+0.012071 [0.029985]
10	+0.000401 [0.015932]	+0.000987 [0.018459]	+0.003591 [0.021791]
12	+0.000121 [0.015789]	+0.000345 [0.017971]	+0.001503 [0.019864]
14	−0.000049 [0.015702]	−0.000012 [0.017707]	+0.000554 [0.019012]
16	−0.000157 [0.015647]	−0.000225 [0.017554]	+0.000046 [0.018566]

Table 25: R_3 in W_{cs}, exact, with the W_c value in brackets (`residual_log/fits/parent_check_wcs/verify.txt`; $L = 16$ from `residual_log/ce_wc/out_wcs/wcs_L16.json`).

The table is the basis of the statement in Section 4.6 (at most 3% of the W_c value left at $L = 14$ and 16 for $a = 1$ to 3). In W_{cs}, $R_3(\chi_{11})$ is 0.0019 at $L = 6$ and below 2×10^{-4} from $L = 12$ on, against about 0.0157 in W_c; it crosses zero between $L = 12$ and 14. Whether it tends to zero

was tested by a hold-out test (`blind16.txt`, made when the $L = 16$ value already existed): fits to $L \leq 14$ predict $L = 16$.

- Free-limit models ($a + b/L^2$, $a + b/L^2 + c/L^3$, $a + b/L + c/L^2$; $L_{\min} = 6$, 8) give a χ_{11} limit of -0.0005 to -0.0007 and miss $L = 16$ by 5×10^{-6} to 2.8×10^{-5}.
- The zero-limit model $b/L + c/L^2 + d/L^3$ misses by 9×10^{-6} ($L_{\min} = 8$) and 2.1×10^{-5} ($L_{\min} = 6$), equally well.

So for χ_{11} a zero limit is not excluded, and a small negative remainder of 3 to 5% of the W_c value is slightly favored. For χ_{22} the same free-limit fits give -0.0014 to $+0.0001$, up to about 8% of its W_c value (0.0176); for χ_{33} they scatter between -0.008 and $+0.022$ and decide nothing. The exact statement behind the window is the budget-free test of Appendix C.2, where R_3 falls roughly like $1/L$ at $L \leq 14$ ($R_3 \approx 0.07/L$). The residual at $L \leq 16$ is consistent with finite-L effects (the linearization and the slow convergence of R_3 in W_c), but R_3 itself has not been computed beyond $L = 16$. The exact slice-resolved forms for χ_{22} and χ_{33} use $e_1^{\text{tag}}(L; 2a, 2a)$ instead of the linearization. They give -0.0004, -0.0009, -0.0012 (χ_{22}) and $+0.0019$, -0.0013, -0.0022 (χ_{33}) at $L = 10$, 12 and 14, drifting slowly negative. The linearized window is closer to zero at these L, and both forms share the same large-L limit.

T3 for the ϕ^2 descendants: not settled. The level-2 descendant is shifted by exactly $\frac{3}{2}\delta^{(3)}(L)$ (ratio 1.5000 at every L), the first-order expectation value of $\frac{\delta}{2}X$ in that state. W_{cs} is right for this state only if its W_c spacing tends to $-\frac{3}{2}\delta_\infty = \frac{6}{81\pi} = +0.0236$. Fits of the W_c spacing at $L \leq 16$ give $+0.005$ to $+0.039$ (`fits.txt`), so they neither confirm nor exclude it. At $L = 16$ the spacings are -0.057 (level 2; W_c: -0.0337) and -0.059 (level 4; W_c: -0.0257), and the level-4 one is still moving away from zero. Two-particle descendants have large finite-L terms: their R_2 is still 0.04 at $L = 16$, against 0.0007 for χ_{11}.

T4 (ϵ^4 spacing) is not restored by W_{cs}.

- W_{cs} leaves R_4 of every χ_{aa} exactly unchanged. The direct effect of the $\delta^{(3)}$ term on R_4 ($+0.002115$ at $L = 6$) is cancelled exactly by the change it makes in $e_1^{(4)}$ of the auxiliary channels, hence in $D^{M,(4)}$.
- R_4 still grows: $R_4(\chi_{11}) = 0.0489$, 0.0570, 0.0632, 0.0682, 0.0726 and 0.0765 at $L = 6$ to 16, a local slope of about 0.028 per unit of $\ln L$ (least-squares slope over $L = 8$ to 16).
- With budget-free legs (tagged, all windows of W_{cs} through ϵ^3), $R_4(\chi_{11}) = -0.0235$, -0.0164, -0.0118, -0.0086, -0.0062 and -0.0043 at $L = 4$ to 14, going to zero. So the growth is the same leg defect at ϵ^4.
- The direct transcription to ϵ^4 (diagnostic scheme W_{cs4}: $D^{(4)} \mathrel{+}= \frac{1}{2}[\delta^{(4)}X_0 + \delta^{(3)}X_1]$, with $\delta^{(4)}$ from the tagged ϕ channel) removes the growth but does not track the budget-free values: $R_4(\chi_{11}) = -0.0165$, -0.0133, -0.0115, -0.0108 and -0.0107 at $L = 6$ to 14, flattening at -0.0107. The reason is that the zero-mode proxy fails at ϵ^4 (Appendix C.5): in the tagged ϕ channel D^{zm} acts on the regular zero modes only, so the tag is a spectator of the zero-mode clusters, whereas the real external zero mode joins them. At zero charge the proxy already differs from $e_1^{(4)}$, and $\delta^{(4)}(L)$ grows roughly like $\ln L$ (-0.0802 at $L = 6$ to -0.0991 at $L = 14$; Table 28).

L		C_5	C_6	C_7	C_8
14	W_{cs}	-20.25163	157.8451	-1440.041	14711.12
14	W_c	-20.24684	157.8302	-1439.928	14710.24
14	relative change	-2.4×10^{-4}	$+9.4 \times 10^{-5}$	-7.8×10^{-5}	$+5.9 \times 10^{-5}$
16	W_{cs}	-21.06383	165.3880	-1521.286	15664.25
16	W_c	-21.05893	165.3722	-1521.165	15663.31
16	relative change	-2.3×10^{-4}	$+9.6 \times 10^{-5}$	-7.9×10^{-5}	$+6.0 \times 10^{-5}$

Table 26: C_5 to C_8 in W_{cs} and W_c at $L = 14$ and 16, exact rationals rounded (`residual_log/ce_wc/out_wcs/`, `residual_log/ce_wc/out_local/`).

Higher coefficients. W_{cs} changes C_6 to C_8 by about 0.01% at $L \le 16$ (Table 26), much less than the differences of about 1% between W_c and W. Every auxiliary object still follows rule W_c.

C.5 The ϵ^4 χ-line window $W_{cs4\chi}$

Records: `residual_log/WINDOW_SPEC.md` (§2.5b), `residual_log/fits/parent_check_p5/verify_p5.txt`, `residual_log/ce_wc/{out_wcs,out_wcs4,out_wcs4c,out_wcs4x,x4}/`, `residual_log/results/legw/legw_L1000.npz`, `residual_log/results/tag/{tag4bf_*,tag4_*,spec4_*}.json`; code `residual_log/{brute_desc_tag4,brute_desc_chi4,delta4chi,p5_prepare_x4}.py`, `residual_log/ce_wc/tables_p5.py`, `residual_log/fits/parent_check_p5/verify_p5.py`.

Why the zero-mode proxy fails at ϵ^4. Through ϵ^3 a χ line and the zero-mode line have the same self-energy function up to $\mathcal{O}(1/L)$. That is why D^M can use e_1 and why the leg window can be calibrated on the tagged ϕ channel. At ϵ^4 they differ: in the ϕ channel the external zero mode belongs to the zero-mode sector and joins the zero-mode clusters that D^{zm} windows, and a χ line cannot. Both parts of the χ mass window are then off.

- *Leg part.* The transcription W_{cs4} flattens at $R_4(\chi_{11}) = -0.0107$, while the budget-free χ_{11} line goes to zero.
- *Uniform (spectator) part.* Define the χ-line function $e_1^{\chi}(\lambda) = \Sigma_\chi(\lambda)/w_{11}(\epsilon)$, with $\Sigma_\chi(\lambda) = E(\chi_{11}\text{ line};\lambda) - E_0(\lambda)$ from the budget-free tagged χ_{11} channel at sub-cutoff λ under rule W_c, without a window on its own line, and the spectator function $Y_n(L,\ell) = \Delta e_1^{\chi,(n)}(L,\ell) - \Delta e_1^{(n)}(L,\ell)$ (Table 27). At fixed ℓ, Y_2 and Y_3 vanish like $1/L$, so D^M with e_1 is right for χ-line spectators through ϵ^3. $Y_4(L,2)$ does not vanish: it is 0.0829, 0.0824 and 0.0817 at $L = 12$, 14 and 16, about $+0.082$. $Y_4(L,4)$ levels off at 0.1426 at $L = 14$ and 16, $Y_4(L,6)$ is still growing there, and for hard spectators (ℓ near L) every Y_n is large.

Definition. Scheme $W_{cs4\chi}$ is W_{cs} plus, at ϵ^4,

$$D^{(4)} \mathrel{+}= \tfrac{1}{4}\{Y_4(L,\hat{\ell}), M'_{cc,0}\} + \tfrac{1}{2}\big[\delta_{4\chi}(L)\, X_0 + \delta^{(3)}(L)\, X_1\big], \tag{48}$$
$$\delta_{4\chi}(L) = R_4^{\mathrm{bf}}(L) - R_4^{W_{cs}}(\chi_{11};L) + \delta^{(3)}(L) - \tfrac{1}{4}Y_4(L,2).$$

- M'_{cc} is the $\chi\chi$ part of M' (hops and pair terms without a zero-mode leg). The anticommutator uses the same per-state rule as D^M, and in auxiliary channels Y_4 is evaluated at the outer reference (rule W_c).

L	$Y_2(L,2)$	$Y_3(L,2)$	$Y_4(L,2)$	$Y_4(L,4)$	$Y_4(L,6)$	$Y_4(L,L)$
6	−0.0114	−0.0599	+0.0624	−0.0567	−2.4772	−2.477
8	−0.0068	−0.0346	+0.0789	+0.0971	−0.1060	−2.925
10	−0.0045	−0.0225	+0.0826	+0.1312	+0.1085	−3.328
12	−0.0032	−0.0158	+0.0829	+0.1405	+0.1600	−3.706
14	−0.0024	−0.0117	+0.0824	+0.1426	+0.1756	−4.067
16	−0.0019	−0.0090	+0.0817	+0.1426	+0.1803	−4.421

Table 27: The spectator function $Y_n(L,\ell)$, exact rationals rounded (`residual_log/ce_wc/x4/`).

- $R_4^{\rm bf}(L)$ is the ϵ^4 spacing of the budget-free χ_{11} line at cutoff L, and X_1 is the ϵ^1 part of the leg operator X.
- The second term is a calibration on χ_{11}. On that state the window adds $w_0 Y_4(L,2)+4[w_0\delta_{4\chi}+w_1\delta^{(3)}]$, with $w_{11}=w_0+w_1\epsilon+\ldots$, $w_0=\frac{1}{4}$, $w_1=-\frac{1}{4}$; this is fixed so that the charged line has the ϵ^4 self-energy of the budget-free one. Hence $R_4(\chi_{11})=R_4^{\rm bf}(L)$ by construction, and χ_{11} is not a test.
- Nothing is fitted: $R_4^{\rm bf}$ and Y_4 are exact brute-force rationals at each $L\le 16$.

Two variants serve as comparisons. Variant "all" applies Y_4 also to the Z–χ terms of M'; whether a $Z\leftrightarrow\chi$ sub-process uses e_1, e_1^χ or a mixture is not derived, so the $\chi\chi$ form is the definition and "all" brackets the choice. The leg-only variant keeps the uniform part on e_1: $D^{(4)}\mathrel{+}=\frac{1}{2}[(\delta_{4\chi}+\frac{1}{4}Y_4(L,2))X_0+\delta^{(3)}X_1]$.

L	$\delta^{(3)}$	$\delta_{4\chi,\rm leg}$	$Y_4(L,2)$	$\delta_{4\chi}$	proxy $\delta^{(4)}$
6	−0.01480	−0.08012	+0.06243	−0.09572	−0.08023
8	−0.01529	−0.08409	+0.07888	−0.10381	−0.08554
10	−0.01553	−0.08729	+0.08258	−0.10794	−0.09023
12	−0.01567	−0.09007	+0.08293	−0.11080	−0.09469
14	−0.01575	−0.09264	+0.08240	−0.11324	−0.09908
16	−0.01580	−0.09509	+0.08172	−0.11552	

Table 28: Coefficients of the ϵ^4 windows (`residual_log/ce_wc/x4/`, `residual_log/results/tag/`). $\delta_{4\chi,\rm leg}=\delta_{4\chi}+\frac{1}{4}Y_4(L,2)$ is the coefficient of the leg-only variant; the proxy $\delta^{(4)}$ is that of W_{cs4}.

Exact invariances. For the leg-only variant ($L=2$ to 16) and for $W_{cs4\chi}$ and "all" ($L=4$ to 16), the following are identical to W_{cs} as exact rationals (`verify_p5.txt`): C_0 to C_5; r_1 to r_4 and g_1 to g_4; E_p through ϵ^4; R_2 and R_3 of every descendant. r_5, g_5 and C_6 change (at $L=16$, $r_5=10.4473$ against 10.2495 and $g_5=-2.2890$ against -2.2681). R_5 of χ_{aa} is exactly unchanged, as R_4 was under the ϵ^3 window, and it grows: $R_5(\chi_{11})=0.0737$, 0.0839, 0.0934, 0.1042, 0.1158, 0.1285 and 0.1420 at $L=4$ to 16. That is the ϵ^5 instance of the same leg defect.

T4: the ϵ^4 spacing. χ_{11} matches the budget-free line by construction (-0.0043 at $L=14$, -0.0028 at 16, going to zero). The tests are χ_{22} and χ_{33} against their own budget-free lines (Table 29).

	scheme	$L=6$	8	10	12	14	16
χ_{22}	W_{cs} $(=W_c)$	+0.0341	+0.0398	+0.0485	+0.0564	+0.0634	+0.0697
	proxy W_{cs4}	−0.0340	−0.0335	−0.0297	−0.0267	−0.0245	
	leg only	−0.0338	−0.0319	−0.0264	−0.0215	−0.0172	−0.0136
	$W_{cs4\chi}$	−0.0594	−0.0404	−0.0312	−0.0250	−0.0204	−0.0165
	budget-free	−0.0497	−0.0405	−0.0338	−0.0288	−0.0248	
χ_{33}	W_{cs} $(=W_c)$	+0.0894	+0.0715	+0.0448	+0.0474	+0.0544	+0.0619
	proxy W_{cs4}	+0.0220	−0.0013	−0.0331	−0.0354	−0.0334	
	leg only	+0.0221	+0.0004	−0.0296	−0.0300	−0.0259	−0.0212
	$W_{cs4\chi}$	−0.2381	−0.0331	−0.0432	−0.0387	−0.0329	−0.0275
	budget-free		−0.0723	−0.0569	−0.0490	−0.0429	

Table 29: R_4 of χ_{22} and χ_{33} by scheme, exact (`residual_log/fits/parent_check_p5/verify_p5.txt`). The $\chi\chi$ form ($W_{cs4\chi}$) and the variant "all" give identical R_4 for every descendant.

- χ_{22}: $W_{cs4\chi}$ closes 95% of the gap between W_{cs} and the budget-free line at $L=14$ (-0.0204 against -0.0248) and agrees to 10^{-5} at $L=8$ (-0.04044 against -0.04045). The remaining gap, +0.0026, +0.0037 and +0.0044 at $L=10$, 12 and 14, is not shrinking: L times the gap grows (0.026, 0.044, 0.062). A zero limit for $a\geq 2$ is therefore not established. The gap reflects the dependence of the ϵ^4 leg response on the charge, a few percent of the leg coefficient (about 3.5% at $L=14$), which the calibration on χ_{11}, with charges $(2,2)$, cannot see. $R_4(\chi_{22})$ itself falls ($LR_4=-0.31$, -0.30, -0.29, -0.26 at $L=10$ to 16); the budget-free χ_{22} line was not computed at $L=16$.
- χ_{33}: 90% of the gap is closed at $L=14$ (-0.0329 against -0.0429); the gap is +0.0137, +0.0103 and +0.0101 at $L=10$, 12 and 14.
- The leg-only variant misses χ_{22} by 0.0073 to 0.0076 and χ_{33} by 0.017 to 0.019 at $L=12$ and 14, so the uniform part Y_4 is needed. The proxy W_{cs4} happens to land near χ_{22} at $L=12$ and 14 but misses χ_{11} (-0.0064 at $L=14$) and χ_{33} (+0.0095).
- The ϕ^2 descendant of level 2 still grows in every scheme: $R_4=0.0404$, 0.1472, 0.2080, 0.2459, 0.2710 and 0.2885 in $W_{cs4\chi}$ at $L=6$ to 16 (W_{cs}: 0.1301 to 0.3949). It is not settled, as at ϵ^3.

L	W_c	W_{cs}	leg only	$W_{cs4\chi}$	"all"
12	149.30050	149.31434	149.29008	149.28512	149.28594
14	157.83017	157.84508	157.81947	157.81354	157.81431
16	165.37223	165.38804	165.36121	165.35460	165.35533

Table 30: C_6 by scheme, exact rationals rounded (`residual_log/fits/parent_check_p5/verify_p5.txt`); the continuum value is 362.366(4).

C_6. The leg-only variant and the proxy W_{cs4} add the same operator, $\frac{1}{2}[\delta X_0+\delta^{(3)}X_1]$, with different coefficients δ of X_0. Their C_6 values therefore fix the response of C_6 to $\frac{1}{2}X_0$ inserted at ϵ^4: it equals the C_5 response $S(L)$ of Appendix C.4 to 10^{-14} at $L=6$ to 14 (replay check `p2.c6_identity`). Relative to W_{cs}, the leg-only variant thus shifts C_6 by $\delta_{4\chi,\mathrm{leg}}(L)S(L)+\delta^{(3)}(L)S_1(L)$, where S_1 is

the C_6 response to $\frac{1}{2}X_1$; the second term is $+0.0018$ to $+0.0025$ at $L = 6$ to 14 ($S_1 = -0.12$ to -0.16). The shift is -0.0256 at $L = 14$ and -0.0268 at $L = 16$, and the uniform part of $W_{cs4\chi}$ adds -0.0035, -0.0050, -0.0059 and -0.0066 at $L = 10$ to 16. The total $W_{cs4\chi}$ shift relative to W_{cs} is -0.032 at $L = 14$ and -0.033 at $L = 16$ (Table 30).

The window coefficient grows with $\ln L$. $\delta_{4\chi,\mathrm{leg}} = -0.0801$, -0.0841, -0.0873, -0.0901, -0.0926 and -0.0951 at $L = 6$ to 16, a local slope of -0.014 to -0.018 per unit of $\ln L$ (-0.0183 between 14 and 16), and $\delta_{4\chi}$ runs from -0.0957 to -0.1155. This is forced: in W_{cs}, R_4 grows with a local slope of about 0.028 per unit of $\ln L$, while the budget-free line goes to zero, so the slope should approach -0.028 as the budget-free spacing flattens. Since the C_6 response to the ϵ^4 leg term is $S(L)$, the leg window's C_6 shift then tends to about $-0.028\, S_\infty \ln L \approx -0.010 \ln L$, plus a uniform part of order 0.01.

A structural prediction, not a measurement. If this window is complete at ϵ^4, C_6 in W_c and W_{cs} carries the opposite divergence, about $+0.010 \ln L$, from the missing χ-line budget, and $W_{cs4\chi}$ removes it. This parallels ϵ^3, where the missing constant leg window left $C_5^{W_c}$ about 0.006 above $C_5^{W_{cs}}$. It cannot be tested at $L \le 16$: there $C_6 \approx 165.4$ in every scheme of Table 30, against the continuum value 362.366(4), $C_6(L)$ still rises by about 7.5 per step of 2 in L, and the external-line windows shift C_6 by about 2×10^{-4} of its value. No statement about $C_6(\infty)$ is possible from $L \le 16$ in any scheme.

Cost. The χ-line objects are exact brute force. The budget-free line at $L = 16$ took 6.1 hours and 76 GB, the χ-line function 7.3 hours and 110 GB; $L = 18$ would need about 20 hours and more than 200 GB per process, more than one node holds. A large-L test needs an ϵ^6 version of the shift machinery and a diagrammatic tagged χ channel for $\delta_{4\chi}(L)$ and $Y_4(L, \ell)$; neither is built.

C.6 Status of the window components and outlook

Records: `residual_log/WINDOW_SPEC.md` (status table, §§2.5a and 2.5b), `zeromode_exact/README.md` (§§9 and 10), `complete_exact/README.md` (§§1 and 5), `continuum/fits_summary.txt`.

Structure at higher orders. Since $C_n = -\frac{1}{3}\Delta_p^3 \mathcal{E}_p^{(n)}$ at every order, only degree three or more in p survives, and at order n the number of surviving χ lines is at most $\frac{1}{2}(4n-6)$: five at ϵ^4, seven at ϵ^5, nine at ϵ^6. The following is conjectured from power counting.

- Nested χ sunsets with χ spectators start at ϵ^5. $\{\hat{\ell}, M'\}$ contains them, because $\hat{\ell}$ counts every quantum present.
- Coupling bubbles with hard χ spectators give windows of order one from ϵ^5 on. Their window behaves like $\ln \frac{L}{L-\ell}$, so the leading form is not enough; D^V as specified is not a correct form, and a correct one is open.
- The zero-mode effective Hamiltonian is needed at all orders as a matrix $h^{(k)}_{z'z}(\lambda)$, not only its eigenvalues; the slope of the mass window acquires a $\ln L$ at ϵ^3 ($r_3 \sim L \ln L$); and divergent three-vertex sub-bubbles with spectators appear.
- The double counting of Appendix B.4 recurs at every order with nested windows; rule W_c removes it by construction.

component	status
D^{zm} at ϵ^2, Eq. (20)	derived (linked clusters); exact agreement with brute force at $L \leq 12$; with D^M, the fitted limit of C_4^W agrees with the continuum (Section 4.4)
D^M at ϵ^2, Eq. (21), or in the e_1 form	derived at leading order in $1/L$; verified at ϵ^4
Bloch D^{zm} at ϵ^3	supported at ϵ^5: no linear term (scheme Bloch); the eigenvalue-diagonal form is excluded ($-0.0051\,L$)
self-consistent auxiliary windows (scheme W)	double counting; C_5^W diverges like $-0.0019\,L$
rule W_c (auxiliary windows referred to L)	provisional, not derived in full; T1 and T2 pass, T3 fails
coupling window D^V	invalid (finite C_4 shift of about 0.49; growing ϵ^3 spacing)
ϵ^3 external-line window (W_{cs}, Eq. (47))	provisional, not derived in full; $\delta_\infty = -\frac{4}{81\pi}$ identified numerically; linearization asymptotically exact; T1 and T2 pass; T3 for χ_{aa}: at least 95% removed at $L = 14$ and 16 ($a = 1$ to 3); extrapolated remainder 3 to 5% for χ_{11} (zero not excluded), up to about 8% for χ_{22}, undetermined for χ_{33}; cancellation as $L \to \infty$ expected but not established; ϕ^2 descendants not settled
C_5 residual logarithm	no $\ln L$ term preferred by the fits with the $\ln^3 L/L$ tail; $\lvert b\rvert \lesssim 0.01$ judged from the median of the fits to $L \leq 300$ (Section 4.5); bounding it well below 0.01 would need $L \gtrsim 400$
ϵ^4 χ-line window ($W_{cs4\chi}$, Eq. (48))	provisional; exact invariances through C_5 and R_3; χ_{11} by construction; χ_{22}, χ_{33} gaps 95%, 90% closed at $L = 14$ (χ_{22} gap not shrinking); ϕ^2 not settled; grows like $\ln L$
ϵ^5 leg window	open: $R_5(\chi_{aa})$ grows
coupling window with χ legs; higher-order windows; completeness	conjectured from power counting; not computed
C_6 to C_8 with a complete window	not available

Table 31: Status of the window components in $d = 2$. Every component was derived after its effect had been seen and with the continuum values known (Section 7); none has a fitted parameter.

Next steps.

1. The residual logarithm of C_5: bounding $|b|$ well below 0.01 needs exact $C_5(L)$ to $L \gtrsim 400$.
2. Large-L tests of the ϵ^4 window: the prediction $C_6^{W_c} \supset +0.010 \ln L$ and $C_6(\infty)$ need the ϵ^6 shift machinery and a diagrammatic tagged χ channel for $\delta_{4\chi}(L)$ and $Y_4(L, \ell)$, the analogue of `residual_log/large_L/tagchan.py` with an external χ line. The χ_{22} gap would then show whether the a-dependence of the leg response is a finite-L effect, as it was at ϵ^3.
3. The ϵ^5 leg window: $R_5(\chi_{aa})$ grows, and the same construction one order higher is needed before C_7.
4. The ϕ^2 descendants at ϵ^3 and ϵ^4 need larger L or a treatment of two-particle legs beyond the linearization.

Consequences for the production solver.

- The production solver computes its stated truncation correctly through ϵ^8 (Section 4.2); no implementation change is needed for what it claims to compute.
- The windows it implements (consistent, literal) are incomplete in $d = 2$ from ϵ^4 on. Its consistent C_4 follows $5.0026(5) - \frac{365}{4374\pi} \ln L + \ldots$, and its consistent C_5 needs about $+0.2 \ln L$

(`residual_log/WINDOW_SPEC.md`): fits without logarithms miss the continuum value by 1.6 to 1.7 (steps 3 and 4 of `residual_log/fits/fits_c5.txt`, $L \leq 160$). Fits of C_4 to C_8 from its data at $L \leq 30$ (the data available when they were made) fall short of the diagrammatic values in magnitude, by 6 to 13% for the medians over the tail towers (`continuum/fits_summary.txt`); they are no evidence against those values.

- The windows matter much more at high order. The diagonal zero-mode window alone changes C_4 by about +0.10 but C_8 by +67% at $L = 2$, +23% at $L = 14$ and +18.5% at $L = 18$ (§9 of `zeromode_exact/README.md`).
- A production implementation of the window must drop the assumption that D is diagonal in the occupation basis. It needs the Bloch matrices $h_{z'z}(\lambda)$ for every $\lambda \leq L$ (in scheme W, zero-mode channels $z \leq 20$ for the references through ϵ^8, $z \leq 21$ with descendant channels), the Newton coefficients $e_1(\lambda)$, rule W_c inside every auxiliary channel, and the partners projected out of the window objects. It needs the non-diagonal D^M, which does not vanish on the references and enters r_3 and g_3. The ϵ^3 external-line window needs the tagged ϕ channel that gives $\delta^{(3)}(L)$ and the reweighted operator X, and the ϵ^4 window the budget-free and spectator χ-line channels.
- The exact Fock basis grows fast (463 GiB at $L = 34$). A reliable extrapolation needs exact large-L sums of the kind used here, or at least the exact window shifts, to fix the $\ln L$ structure.

D Provenance and reviews

Each package folder carries `PROVENANCE` and `REVIEWS` files with commits, dates, authors and review rounds; the review logs and agent transcripts themselves are not published. Codex below is the Codex CLI with model gpt-6-astra at reasoning effort xhigh.

One dimension.

- *Solver* (`lri_hpc_Hamitonian`). Based on a supplied package written by OpenAI GPT-6 astra (via ChatGPT); the Expanse layer, fixes and the analyses (joint tail fits, window engines, identity checks) were written by Claude Code sessions in September and October 2026. Codex reviewed the plan (five rounds, approved) and inspected the code twice; the second round's fixes and all later analyses were not inspected by Codex. The analyses were checked by internal multi-agent workflows with independent engines and skeptic lenses.
- *Eighth-order code* (`lri_c8_hpc`). A GPT-6 astra rewrite of its own original eighth-order code; the port to the symmetry cluster by a Claude Code session; Codex inspection rounds 1–8, approved.
- *Complete-window solver* (`lri_d1_full_coefficient_flow`). Supplied by the author as a GPT-6 astra package; the extension on the symmetry cluster by Claude Code, Codex rounds 1–4, approved. The refits that use it (`lri_1d_dconn_refit`) were made by a Claude Code workflow with an independent verifier stage and had no Codex review.
- *Exact sums* (`lri_1d_c4_exact`, `lri_1d_c5_exact`). Written by the Claude Code session that also wrote this note, in October 2026, ported from the $d = 2$ exact-sum code. Codex approved the plans and inspected each package twice; the last fix round of each is not re-inspected. The constrained C_5 fit was checked by two separate Claude Code subagents (a blind re-implementation and a theory check).

- *Second channel* (`lri_d1_fullblock_tail_response`). A GPT-6 astra research package, shipped verbatim; reviewed by a Claude Code workflow, which rated the sign and normalization of its J term plausible but did not re-derive them.

Two dimensions, Appendices A to C. These appendices merge the method write-up of `lri_2d_coeff_independent_check`, which earlier shipped as a separate note in that package, in its public edition of 14 October 2026 (file paths made relative; an appendix on inconsistencies between internal sources and the values that later data superseded removed). The write-up and the computations behind it were made by Claude Code sessions (Claude Opus 5.5) between 8 and 14 October 2026: the coordinating session HT2dPert and the agents it launched. These were two clean-room solver agents that saw only the model specification, a continuum agent, the exact-C_4 agent, the zero-mode agent, the complete-window agent, the window-derivation agent, the large-L agent and a write-up agent. The parent checks were made by the coordinating session. HT2dPert merged the write-up into this note on 16 October 2026, leaving out material that the main text already states and values that later data superseded. The merged text was then read by a fresh agent with no project context and reviewed against the write-up by the reviewing session; their findings were checked against the records named in each subsection. None of this material had an external code review.

Two and three dimensions. See the provenance files of the packages named in Sections 4 and 5.

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Part IV

Hamiltonian truncation of the long-range Ising model at finite coupling

Method, window correction, extrapolation, results in $d = 1, 2, 3$, and how to reproduce them

Companion note to the public repository, folder `HT/`
Edition of 2026-10-18

Abstract

This note documents every finite-coupling Hamiltonian-truncation (HT) number quoted in the paper on the long-range Ising (LRI) model [4]. The perturbative HT coefficients C_2 to C_8 are documented separately, in `HT_Pert/`.

The LRI fixed point is the ϕ^4 deformation of the generalized free field (GFF) [5, 6, 7]. We diagonalize the deformed Hamiltonian in radial quantization, in a truncated Fock space [9, 1], at couplings tuned to the conformal values of two levels. The zero mode costs no level, so the cutoff does not cap its quanta. It is kept in a small basis of states, each a superposition of many of its quanta, and the full Hamiltonian, including the coupling of the zero mode to the bath, is diagonalized on the product of this basis with the bath of all other modes, truncated at a total level N.

In the truncated space an excited bath state can host fewer vacuum fluctuations than the vacuum. A correction for these missing fluctuations, the window correction, removes the dominant descendant-spacing violations of the raw truncation, which do not decay with N. A zero-mode basis selected from a reduced density matrix makes the strong-coupling cutoffs affordable.

In $d = 1$ we reach $N = 40$ at 13 values of Δ_σ and extrapolate in N. The results are final for $\Delta_\sigma \geq 0.1$, one-sided at 0.075, unstable at 0.05, and without a limit for $\Delta_\sigma \leq 0.025$, where the window-corrected model itself does not converge. In $d = 2$ ($N \leq 24$) and $d = 3$ ($N \leq 20$) we report finite-cutoff behaviour only.

Separately written solvers reproduce the finite-cutoff spectra in all three dimensions. In $d = 1$ they agree to 5.6×10^{-7} with the density-basis records behind the quoted values ($N = 12$–20) and to 5×10^{-9} with converged energy-basis records. In $d = 2$ they agree to 1.4×10^{-12} ($N = 12$, 16) and in $d = 3$ to 2×10^{-11} ($N = 8$–12). Section 9 maps each number to its code, records, command and tolerance.

1 Summary

What is computed. The long-range Ising (LRI) fixed point in d dimensions is reached from the generalized free field (GFF) ϕ of dimension Δ_σ by the relevant ϕ^4 deformation [5, 6, 7, 2]. In radial quantization the GFF becomes a set of free oscillators. We diagonalize the deformed Hamiltonian in a truncated Fock space and tune its two couplings, the mass r and the quartic g, so that two levels take their conformal values: $\sigma = \phi$ at Δ_σ and its shadow at $d - \Delta_\sigma$. The first excited $\mathbb{Z}_2$-even scalar level is then the prediction for the dimension Δ_ε of the energy operator ε (Section 2). The method is Hamiltonian truncation in radial quantization [9, 1], applied to a nonlocal GFF.

The three ingredients.

1. *Zero mode and bath* (Section 2.4). The zero mode costs no level, so its quanta are not capped by the level cutoff, and at strong coupling its fluctuations are large. The full Hamiltonian, including the couplings of the zero mode to the bath, is diagonalized on the product of n_z zero-mode states per parity, each a superposition of many zero-mode quanta, with the bath states of total level $\ell \leq N$. In $d = 1$ this zero-mode basis reproduces the Δ_ε of a converged zero mode to 10^{-6} (Section 2.5); only the bath cutoff N is extrapolated.

2. *Window correction* (Section 3). A bath state of level ℓ can host vacuum fluctuations only up to level $N - \ell$. The raw spectrum therefore violates the exact descendant spacings by amounts that do not decay with N. We add to every bath state the difference $D(\ell) = E_v(N) - E_v(N - \ell)$, where $E_v(M)$ is the vacuum energy at bath cutoff M of the corrected theory itself, zero mode included. With an energy cutoff, the same missing states were added back order by order with perturbative counterterms in ϕ^4 theory in $2 + 1$ dimensions [17].

3. *Zero-mode basis* (Section 2.5). If the n_z states are the lowest eigenstates of the zero-mode part H_z of H, the number needed at strong coupling grows with N, and the zero mode then sets the cost. We keep instead the leading eigenvectors of a reduced density matrix of the states the calculation needs, trained once per Δ_σ.

Results.

- In $d = 1$ (Section 4, Table 1) the cutoff reaches $N = 40$ at all 13 values $0.001 \le \Delta_\sigma \le 0.249$, and the series are extrapolated with a fixed family of fits.
  - At $\Delta_\sigma = 0.249$ the result differs from the ϵ expansion by -4×10^{-8}, well below the 10^{-6} to which we trust this check.
  - The values are *final* for $\Delta_\sigma \ge 0.1$ as limits of the window-corrected model. Their half-spreads are at most 1.2×10^{-4}; at $\Delta_\sigma \ge 0.15$ we judge the total error to be about 10^{-4}.
  - The quoted errors are half-spreads of the extrapolations in the cutoff, not standard deviations, and they do not include the dependence on the window prescription. That dependence is 7×10^{-4}–2.3×10^{-3} at $0.075 \le \Delta_\sigma \le 0.125$, as large as the gap there to the numerical renormalization group (NRG) of the sub-ohmic spin-boson model [11, 12] (Section 4.4).
  - At $\Delta_\sigma = 0.075$ the error is *one-sided*, and at 0.05 the extrapolation is *unstable*.
  - At $\Delta_\sigma \le 0.025$ the corrected sequence turns over and keeps falling, so it has *no limit*. This is a property of the window-corrected model, not of the numerics.
- In $d = 2$ (Section 5, $N \le 24$) and $d = 3$ (Section 6, $N \le 20$) the cutoffs are too few for a controlled extrapolation, and we report finite-N behaviour with status words.

Checks.

- Separately written solvers, built from the definition of the model without using the production code, reproduce the production spectra in each dimension (Section 7). The $d = 1$ and $d = 3$ solvers were written blind to the production code. The $d = 2$ solver is not blind: its author had earlier prepared the production code for the cluster.
- These checks verify that we compute the stated model. Whether its $N \to \infty$ limit is the CFT is a separate question, and the answer is negative in parts of the range (Section 8).

Status words. *Final*, *one-sided*, *unstable*, *no limit* and *finite-N* are part of the results (Section 4.2). A quoted error is a half-spread: half the difference between the largest and the smallest limit of the accepted fits. It is a model spread, not a standard deviation, and it covers the extrapolation only.

2 Model, truncation and tuning

2.1 Radial quantization of the GFF

Radial quantization turns the GFF of dimension Δ_σ into free oscillators a_k of energy $\Delta_\sigma + \kappa_k$:

$$\phi(\hat{n}) = \sum_k u_k \left[f_k(\hat{n})\, a_k + f_k(\hat{n})^* a_k^\dagger \right], \tag{1}$$

$$
\begin{aligned}
d=1:&\quad k=n, && \kappa=n, && f=(\pm1)^n, && u_n^2=\frac{(2\Delta_\sigma)_n}{n!},\\
d=2:&\quad k=(a,b), && \kappa=a+b, && f=e^{-\mathrm{i}(a-b)\vartheta}, && u_{ab}^2=\frac{(\Delta_\sigma)_a(\Delta_\sigma)_b}{a!\,b!},\\
d=3:&\quad k=(n,l,m), && \kappa=2n+l, && f=\sqrt{4\pi}\,Y_{lm}, && u_{nl}^2=\frac{(\Delta_\sigma)_{n+l}(\Delta_\sigma-\frac12)_n}{n!\,(\frac32)_{n+l}},
\end{aligned}
\tag{2}
$$

These reproduce the two-point function on $\mathbb{R}\times S^{d-1}$; κ_k is the level of mode k and ϑ the angle on the circle. In $d=1$ the "sphere" S^0 is the pair of points $x=\pm1$, and $f_n(\pm1)=(\pm1)^n$. A Fock state with occupation numbers n_k has:

- level $\ell=\sum_k \kappa_k n_k$;
- particle number $P=\sum_k n_k$;
- free energy $H_0=\ell+\Delta_\sigma P$.

The zero mode a_0 ($\kappa=0$, $u_0=1$) costs no level.

2.2 Hamiltonian and sectors

The tuning and Δ_ε use the sector of rotation scalars that are even under reflection (inversion in $d=3$). There we diagonalize

$$
H=H_0+r\,V_2+g\,V_4,\qquad V_2=\big\langle{:}\phi^2{:}\big\rangle,\qquad V_4=\tfrac{1}{12}\big\langle{:}\phi^4{:}\big\rangle,
\tag{3}
$$

with $\langle\cdot\rangle$ the average over S^{d-1}. Normal ordering is with respect to all oscillators, the zero mode included. In $d=1$:

- the reflection $x\to-x$ acts as $a_n\to(-1)^n a_n$, so the reflection-even sector contains the states of even total level;
- on that sector the sphere average equals the vertex at $x=+1$;
- the descendant diagnostic below also diagonalizes the reflection-odd sector (odd total level), which contains the level-1 descendants.

The Ising symmetry $\phi\to-\phi$ splits each sector by the parity of P.

2.3 Tuning conditions and the prediction

We denote by $E_v<E_\varepsilon$ the two lowest $\mathbb{Z}_2$-even scalar levels (the vacuum and the energy operator ε) and by $E_\sigma<E_{\phi^3}$ the two lowest odd ones. We impose

$$
E_\sigma-E_v=\Delta_\sigma,\qquad E_{\phi^3}-E_v=d-\Delta_\sigma,\qquad \Delta_\varepsilon=E_\varepsilon-E_v,
\tag{4}
$$

two conditions for (r,g) and one prediction. The shadow $d-\Delta_\sigma$ is ϕ^3 at weak coupling. The vacuum energy at cutoff N is $E_v=E_v(N)$ in the notation of Section 3. Conformality also requires the descendant spacings

$$
v_n(\mathcal{O})\equiv E(\partial^n\mathcal{O})-E(\mathcal{O})-n=0,
\tag{5}
$$

a test that needs no continuum input. In $d=1$ we evaluate v_1 and v_2 by identifying $\partial\mathcal{O}$ through its overlap with $L_{-1}\mathcal{O}$, where L_{-1} is the generator of the free GFF, with the zero-mode operator a_0 projected onto the zero-mode basis of Section 2.5. This is a diagnostic: L_{-1} is not the interacting generator, and ambiguous identifications are flagged.

At weak coupling, $\Delta_\sigma=(d-\epsilon)/4$ with $\epsilon\to0$:

- the solution has $g = \epsilon/3 + O(\epsilon^2)$ and $r = O(\epsilon^2)$;
- the prediction is $\Delta_\varepsilon = d/2 - \epsilon/6 + C_2(N)\epsilon^2 + \ldots$, with the cutoff-dependent coefficients $C_n(N)$ of the truncated model (Section 4.3). Their continuum values are the GFF coefficients C_n, written $C_n(d)$ where the dimension matters.

2.4 Zero mode and bath

We split the field into the zero mode and a bath of all other modes,

$$\phi = q + \chi, \qquad q = a_0 + a_0^\dagger, \qquad p = \mathrm{i}\big(a_0^\dagger - a_0\big), \qquad \chi = \sum_{k\neq 0} u_k \big[f_k\, a_k + f_k^* a_k^\dagger\big]. \tag{6}$$

Normal ordering the zero mode gives $:q^2: = q^2 - 1$, $:q^3: = q^3 - 3q$ and $:q^4: = q^4 - 6q^2 + 3$. Up to a constant,

$$H = H_z \otimes 1 + 1 \otimes H_b + \sum_{j=1}^{4} L_j \otimes W_j, \qquad W_j = \big\langle :\chi^j: \big\rangle, \tag{7}$$

$$H_z = \frac{\Delta_\sigma}{4} p^2 + A\, q^2 + \frac{g}{12} q^4, \qquad A = \frac{\Delta_\sigma}{4} + r - \frac{g}{2}, \tag{8}$$

$$L_1 = (2r - g)\, q + \frac{g}{3} q^3, \qquad L_2 = r - \frac{g}{2} + \frac{g}{2} q^2, \qquad L_3 = \frac{g}{3} q, \qquad L_4 = \frac{g}{12}. \tag{9}$$

- $H_b = \ell + \Delta_\sigma P_b$ is the free bath Hamiltonian.
- In H_z:
  - $\frac{\Delta_\sigma}{4}(p^2 + q^2)$ is the free part of the zero mode;
  - rq^2 comes from $r{:}\phi^2{:}$;
  - $-\frac{g}{2} q^2$ comes from normal ordering $:q^4:$.
- At the tuned couplings A can be negative, and H_z is then a double well. In $d = 1$, $A < 0$ at every cutoff for $\Delta_\sigma = 0.2$ and 0.175, and its sign changes with the cutoff at $\Delta_\sigma \leq 0.15$.
- L_j is the zero-mode operator that multiplies W_j in $r{:}\phi^2{:} + \frac{g}{12}{:}\phi^4{:}$. For example, L_1 collects $2rq\chi$ and $\frac{g}{3}{:}q^3{:}\chi$.

Truncation. We keep all bath states with level $\ell \leq N$, with no limit on the bath particle number, and n_z zero-mode states per parity (Section 2.5). The full H of Eq. (7), the couplings $L_j \otimes W_j$ included, is diagonalized on their product; the eigenstates of H_z alone serve at most to choose the zero-mode states. The normal-ordered W_j never pass through a level above both endpoints, so H at cutoff $M < N$ is exactly the $\ell \leq M$ projection of H at cutoff N. This property is used by the window correction. The same split is used in $d = 2, 3$, with the $J = 0$ projection of Section 6 in $d = 3$.

2.5 Zero-mode basis

H_z and the L_j are represented in an auxiliary harmonic-oscillator basis of N_{osc} states ($N_{\mathrm{osc}} = 384$ in production), with an adaptive scale t. Of this basis we keep n_z states per parity. The product space has $2n_z$ times as many states as the bath, so n_z sets the cost.

Energy basis. The original choice keeps the n_z lowest eigenstates of H_z in each parity. At strong coupling in $d = 1$ this basis is not converged at fixed n_z beyond $N \approx 24$, because the tuned g grows with N. At $\Delta_\sigma = 0.025$, $N = 28$, $n_z = 32$, 40 and 48 give $\Delta_\varepsilon = 0.710632, 0.710934, 0.710959$ (window). The energy-basis campaign `prod1--7` (Section 4.1) therefore needed n_z up to 56.

Density basis. Instead, we keep the zero-mode states that the low-lying spectrum actually uses: the leading eigenvectors, in each parity, of the reduced density matrix

$$\rho = \tfrac{1}{6}\big(\rho_{\mathrm{vac}} + \rho_\sigma + \rho_{\phi^3}\big) + \frac{1}{N_t} \sum_{M=0,2,\ldots}^{N_t-2} \rho_{\mathrm{vac}}(M). \tag{10}$$

- *What ρ averages.* It is obtained by tracing the bath out of the states the calculation needs. The first term covers the vacuum, σ and ϕ^3, which enter the tuning, at the training cutoff N_t. The sum covers the vacua at all lower even cutoffs, which enter the window correction. The ε state plays no role in the selection.

- *Training.* The basis is trained once per Δ_σ at $N_t = 24$, against a reference calculation in the energy basis with 48 states per parity (the *teacher*). The outer loop alternates density selection and retuning in the selected span. It stops when the couplings change by less than 2×10^{-8} and the teacher's tuning residuals are below 10^{-7}, after at least two sweeps.

- *Stored basis.* Each trained basis is stored in a file that keeps 24 states per parity: the isometry U and the scale t. Its first n_z columns give a nested smaller basis, and production uses $n_z = 16$.

- *Operators.* The powers q^j are formed in the full auxiliary basis before projection, $q_j = U^{\mathsf{T}}(t^j Q^j)U$; powers of the projected q are not used. For the descendant diagnostic the actual a_0 is projected.

- *What is frozen.* The span and t are fixed within a solve, while the window table is recomputed at every coupling.

In $d = 1$ the density basis reproduces the Δ_ε of a converged zero mode to 10^{-6} at a tenth of the cost. Two comparisons support this:

- against the $n_z = 48$ and 56 energy-basis records at $N = 32$: at most 6.4×10^{-7};

- against the blind solver's grid zero mode at $N \le 20$: at most 5.6×10^{-7}.

The difference between 24 and 16 kept states is at most 1.8×10^{-9} at $N = 36$, 40. That difference measures only the truncation within one span.

2.6 Tuning

The couplings are tuned by Newton's method on Eq. (4), with backtracking: the Newton step is halved until the norm of the two residuals decreases, at most 13 times. The Jacobian comes from Hellmann–Feynman derivatives of the levels; with the window it includes the window response, Eq. (13). In $d = 1$ the internal coordinates are $\theta = (\theta_0, \theta_1)$, with $g = 192e^{3\theta_1}/\Delta_\sigma^2$ and $A = 4e^{2\theta_1}\theta_0/\Delta_\sigma$ (as in $d = 2$, Section 5). Points are continued from weak coupling or from saved seeds; a seed is only a starting point.

Tolerances:

- tuning residuals $\le 5 \times 10^{-10}$;

- ARPACK tolerance 3×10^{-11}, eigenvector residuals $\leq 2 \times 10^{-7}$;
- window fixed points $\leq 10^{-12}$.

3 The window correction

The truncation artifact. In the CFT every descendant spacing $E(\partial^n \mathcal{O}) - E(\mathcal{O}) = n$ is exact. The raw truncation violates these spacings by amounts that do not decay with N. In $d = 1$ the level-1 violations are 0.06–0.25 at $\Delta_\sigma \leq 0.1$ over $N = 16$–28 (Table 3 shows $N = 28$). The mechanism is that a bath state of level ℓ can carry vacuum fluctuations of the bath only up to level $N - \ell$, whereas the vacuum keeps all fluctuations up to N. The same mechanism spoils the cancellation of disconnected vacuum diagrams in ϕ^4 theory in $2 + 1$ dimensions with an energy cutoff, where the effect does not decouple as the cutoff is removed; there it was addressed with perturbative state-dependent counterterms [17]. The missing vacuum window is the truncation artifact; in the perturbative truncation it is analysed in `HT_Pert/`.

The correction. We add to every bath state of level ℓ, in all sectors, the shift

$$D(\ell) = E_v(N) - E_v(N - \ell), \qquad D(0) = 0, \tag{11}$$

where $E_v(M)$ is the vacuum energy at bath cutoff M of the same Hamiltonian, with the same Δ_σ, couplings, zero mode and basis. The definition of E_v has three features:

- *Zero mode included.* Vacua of the bath alone miss the zero-mode tadpole $\frac{g}{2}\langle q^2 \rangle$, which cancels the $-\frac{g}{2}$ in A, and remove only part of the violation.
- *Consistent.* The production mode is *consistent*: $E_v(M)$ is the lowest eigenvalue of the *corrected* Hamiltonian at cutoff M,

  $$E_v(M) = \lambda_{\min}\big[H(M) + D_M\big], \qquad D_M(\ell) = E_v(M) - E_v(M - \ell), \tag{12}$$

  built level by level from the corrected lower cutoffs.
- *Odd cutoffs.* E_v of an odd M equals $E_v(M - 1)$, since only even levels exist in the sector.

The *literal* mode uses the uncorrected $\lambda_{\min} H(M)$ and serves as a diagnostic. Energies are measured from the lowest even eigenvalue of the zero mode alone, so $E_v(0) = 0$.

Precedent. $D(\ell)$ is the counterpart, with a level cutoff, of the state-dependent counterterms that Ref. [17] adds with an energy cutoff to restore the missing states of disconnected bubbles.

- At second order in the interaction, $D(\ell)$ corresponds to their single-bubble counterterm (Patch I, their Eq. (5.38)). Their denominators contain the unperturbed energy of the level being computed; ours contain the bubble energy alone.
- They add such terms order by order, including towers of repeated bubbles (Patch II, up to five bubbles), as a pragmatic fix. $D(\ell)$ is a nonperturbative analogue: it takes all disconnected bubbles at once from the vacuum energies of the same truncated Hamiltonian at lower cutoffs. The consistent definition, Eq. (12), also corrects the states inside those vacuum energies, which is the role of their Patch II.

Fixed point. At each M, Eq. (12) is a scalar fixed point $s = \lambda_{\min}[H(M) + D_M(s)]$, because $s = E_v(M)$ enters every $D_M(\ell > 0)$. We write $F(s) = \lambda_{\min} - s$. Then $F'(s) = -p_0$, with p_0 the weight of the vacuum vector on bath level 0. F is concave, so Newton's method converges monotonically after at most one step, and the fixed points are solved to 10^{-12}. At $M = N$, s is the ground energy of the main problem.

Derivatives. For the tuning Jacobian, Hellmann–Feynman on the stored vacuum vectors gives

$$\partial E_v(M) = \frac{\langle \partial H \rangle - \sum_{\ell>0} w_\ell \, \partial E_v(M-\ell)}{p_0}, \tag{13}$$

with w_ℓ the weight of the vacuum vector on bath level ℓ. The package tests compare it with finite differences.

What it removes and what it misses.

- In $d = 1$ the correction reduces the level-1 violation of σ by a factor 10–40 at $N = 28$, $\Delta_\sigma \leq 0.15$.
- At $\epsilon \geq 0.6$ ($\Delta_\sigma \leq 0.1$), non-decaying residuals remain:
  - about 0.01 for σ and ϕ^3;
  - up to 0.05 for ε at $\Delta_\sigma \leq 0.05$ (Table 3).
- The correction restores only disconnected vacuum fluctuations. At finite coupling it therefore models the truncation artifact rather than removing it completely, and the residuals show what it misses.
- Expanded in ϵ, the corrected $d = 1$ model changes Δ_ε only at $O(\epsilon^4)$: in the blind solver the difference to the raw Δ_ε scales as ϵ^4 (Section 7). C_2 and C_3 are therefore unaffected.
- The perturbative windows of `HT_Pert/` are built in the same spirit. In $d = 1$ the finite-coupling window has not been compared with them order by order, so whether its $O(\epsilon^4)$ term is the one the continuum requires is not known.
- In $d = 2, 3$ the consistent window is the same construction (Sections 5, 6). In $d = 2$ its ϵ expansion reproduces the consistent-window scheme of `HT_Pert/`, whose fourth-order coefficient grows like $\ln N$; the model is therefore correct only through third order. In $d = 3$ its coefficients $C_n(N)$ come from `HT_Pert/`: C_2 and C_3 approach their continuum values, while the fourth-order coefficient at $N = 20$ is 60% of $C_4(3)$ and still growing slowly, a finite-N statement.

We have not constructed an effective Hamiltonian beyond the window [15, 16].

4 One dimension

4.1 Campaigns

All $d = 1$ runs used the solver `HT/code/lri_dressed_phi4_hpc` on the symmetry cluster (Perimeter Institute). There were two campaigns.

`dens_v1` **(production, code 33998bb, 2026-10-07 to 10-09).**

- *Setup.* Consistent window, density basis with $n_z = 16$ (one stored basis per Δ_σ, trained at $N_t = 24$), auxiliary oscillator basis of $N_{\mathrm{osc}} = 384$ states.
- *Grid.* The 13 values $\Delta_\sigma = 0.249$, 0.24, 0.225, 0.2, 0.175, 0.15, 0.125, 0.1, 0.075, 0.05, 0.025, 0.01 and 0.001, that is $\epsilon = 1 - 4\Delta_\sigma$ from 0.004 to 0.996, at every even N from 12 to 40: 195 points.
- *Controls.* Six controls with 24 kept states at $N = 36$, 40. Two further controls, at $N = 40$ and $\Delta_\sigma = 0.025$, 0.01, stopped at their walltime and are not used.
- *Execution.* Items were packed on whole nodes (64-core AMD amdq; 40-core Intel preq), with 1 to 16 threads per item. Items resumed from Newton checkpoints after a cluster-wide node failure and after walltime stops.
- *Cost.* The longest items took 15 h ($N = 40$, $\Delta_\sigma = 0.025$, 16 threads). The total was about 10^4 core-hours of exclusive nodes. Every record carries its configuration, couplings, levels, window table, residuals and timing, and a sha256 over its content.

`prod1--7` **(code 4dfd833, 2026-10-02 to 10-05).**

- *Content.* Energy zero-mode basis with $n_z = 24$ to 56, in the raw, consistent and literal prescriptions, at $N = 12$ to 40: 680 records, about 1.06×10^4 core-hours.
- *Why it was superseded.* At $\Delta_\sigma \le 0.05$ its series needed an n_z that grows with N, and their extrapolations mixed values of different n_z. The production values now come from `dens_v1`.
- *What is still used.* Its records serve three roles: the inputs of the descendant-identity diagnostic (Section 4.5), the references of the basis check at $N = 32$, and the references of the blind-solver comparison.

4.2 Extrapolation

Fits. For each Δ_σ we fit

$$\Delta_\varepsilon(N) = \Delta_\infty + aN^{-\omega} \quad \big[+ bN^{-\omega-1} \text{ (F2) or } + bN^{-2\omega} \text{ (F3)} \big], \qquad N \ge N_{\min}, \tag{14}$$

with F1 the two-term form. The fit set is:

- $N_{\min} = 12$–24 in steps of 2, giving 21 fits per Δ_σ;
- each fit is a nonlinear least-squares fit from five starting exponents, keeping the best.

Acceptance. A fit is accepted if all three conditions hold:

- rms residual $< 3 \times 10^{-6}$;
- $0.2 < \omega < 4$;
- $|\Delta_\infty - \Delta_\varepsilon(N_{\max})| < 0.05$.

We quote the median of the accepted Δ_∞ and their half-spread, half the difference between the largest and the smallest accepted Δ_∞. It is a model spread, not a standard deviation.

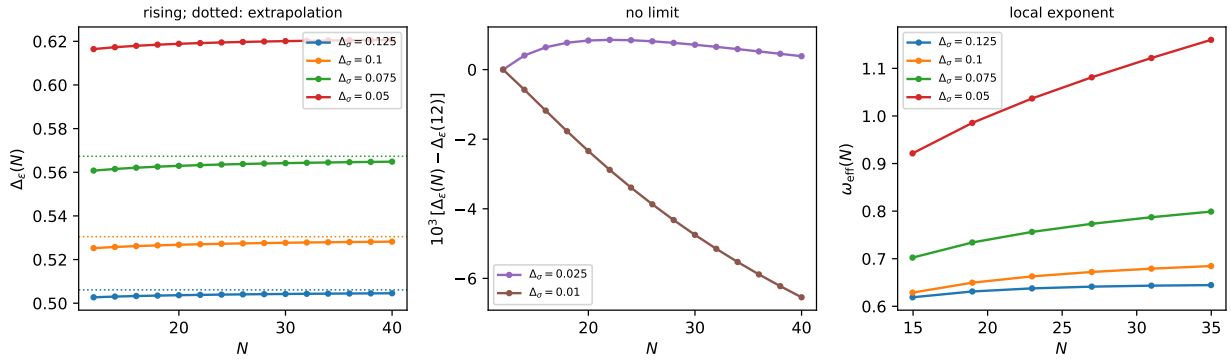


Figure 1: $d = 1$, `dens_v1`. *Left:* $\Delta_\varepsilon(N)$ at $0.05 \le \Delta_\sigma \le 0.125$; the dotted lines are the extrapolations of the stable series (0.075 to 0.125). *Middle:* change $\Delta_\varepsilon(N) - \Delta_\varepsilon(12)$ at $\Delta_\sigma = 0.025$, which peaks near $N = 22$ and then falls, and at 0.01, which falls at a nearly constant rate per unit of $\ln N$. *Right:* the local exponent (15), which levels off at 0.125 and 0.1 but rises at 0.075 and 0.05.

Stability. A result is *stable* if all three conditions hold:

- at least five fits are accepted;
- the median of their fitted ω is at least 0.45 (a condition on the fits, not on $\omega_{\rm eff}$ below);
- the half-spread is at most 2×10^{-3}.

The median exponents of the stable results lie at 0.6–0.9.

Status words. A stable result is *final* if the local exponent $\omega_{\rm eff}$ below levels off over the largest cutoffs, and *one-sided* if it still rises, which suggests that the fits overestimate the limit; this classification is a judgement. A series that turns over and keeps falling has *no limit*. Any other result that is not stable is *unstable*: we give the range of the accepted fits but no limit. In $d = 2, 3$, *finite-N* marks values of the truncated model at a given cutoff, with no extrapolation.

Local exponent. A fit-free check is

$$\omega_{\rm eff}(N) = -\frac{d \ln \left(d\Delta_\varepsilon / d \ln N \right)}{d \ln N}, \tag{15}$$

which is constant for a single power law. We evaluate it by finite differences. The slope on a pair of successive cutoffs, $s = [\Delta_\varepsilon(N'+2) - \Delta_\varepsilon(N')] / \ln \frac{N'+2}{N'}$, sits at $\bar{N} = \sqrt{N'(N'+2)}$. From the slopes on $[N', N'+2]$ and $[N'+4, N'+6]$ we take $\omega_{\rm eff} = -\ln(s_2/s_1)/\ln(\bar{N}_2/\bar{N}_1)$ and label it by the centre $N = N'+3$ of the four cutoffs, $N = 15, 19, \ldots, 35$.

Held-out test. Fits restricted to $N \le N_{\rm cut}$ predict the cutoffs above $N_{\rm cut}$ (Table 2).

4.3 Results

Table 1 gives Δ_ε at the largest cutoff and extrapolated. Figure 1 shows the strong-coupling series and their local exponents.

Δ_σ	ϵ	$\Delta_\varepsilon(N=40)$	$\Delta_\varepsilon(N\to\infty)$	status	NRG
0.249	0.004	0.4993430	0.49934358(9)	final	0.508030^{nc}
0.24	0.04	0.494186	0.4942198(26)	final	0.498922^{nc}
0.225	0.1	0.487982	0.488080(5)	final	0.489668^{nc}
0.2	0.2	0.482839	0.483024(5)	final	0.482980(708)
0.175	0.3	0.483541	0.483909(15)	final	0.483900(346)
0.15	0.4	0.490388	0.491209(33)	final	0.491106(115)
0.125	0.5	0.504561	0.50612(4)	final	0.505672(54)
0.1	0.6	0.528210	0.53048(12)	final	0.529608(33)
0.075	0.7	0.564851	0.56733(20)(−50)	one-sided	0.566260(30)
0.05	0.8	0.620671	0.6220–0.6224, or down to 0.618	unstable	0.622115(27)
0.025	0.9	0.710577		no limit	0.713422(29)
0.01	0.96	0.809111		no limit	0.809747(47)
0.001	0.996	0.969338		no limit	0.937424(111)

Table 1: Δ_ε in $d=1$ from finite-coupling HT, at the largest cutoff and extrapolated with Eq. (14). Errors in parentheses are half-spreads of the accepted fits (model spreads, not standard deviations), printed with the paper's rounding rule. *Weak coupling.* At $\Delta_\sigma \geq 0.15$ these spreads, at most 3.3×10^{-5}, are probably optimistic; we judge the total error there to be about 10^{-4}. At $\Delta_\sigma = 0.249$ this is conservative: the ϵ expansion agrees to -4×10^{-8} (Section 4.3). $\Delta_\sigma = 0.075$. The second error is one-sided: a further rise of ω_{eff} by 0.2 would lower the limit by about 5×10^{-4}. $\Delta_\sigma = 0.05$. The accepted F1 and F2 fits give the range shown, an upper estimate if ω_{eff} keeps rising. The F3 fits admit a later turnover and reach the lowest value shown. The data up to $N=40$ cannot decide, so we quote no limit there, only this range, and none at $\Delta_\sigma \leq 0.025$. The errors do not include the dependence on the window prescription (Section 4.4). *NRG.* Bosonic NRG of the sub-ohmic spin-boson model [11, 12] (NRG folder, final table), shown for comparison. Its error is the sum of three convergence diagnostics (chain depth, kept states, discretization), not a standard deviation; ^{nc}: not converged in chain depth near $\Delta_\sigma = \frac{1}{4}$.

Weak coupling and the ϵ expansion.

- At $\Delta_\sigma = 0.249$ the extrapolation differs from the ϵ series through ϵ^7 (GFF folder: $C_2 = (\pi + 4\ln 2)/9$, C_3 to C_7) by -4×10^{-8}. That is below the 10^{-6} at which we trust the check, because of the series and fit systematics.

- At $\Delta_\sigma = 0.24$ the value lies within the band 0.49419–0.49426 of the partial sums through ϵ^4 to ϵ^7.

- The blind solver of Section 7 computes the weak-coupling coefficient of the truncated model. With $\Delta_\varepsilon = \frac{1}{2} - \frac{\epsilon}{6} + C_2(N)\epsilon^2 + \ldots$, it reproduces

$$C_2(N) = \frac{34}{81} + \sum_{n \text{ even},\, 2\leq n\leq N} v_n\Big[\frac{2}{81(n-\frac{1}{2})} + \frac{2}{9n} + \frac{2}{9(n+\frac{1}{2})} + \frac{2}{81(n+1)}\Big], \qquad v_n = \frac{(\frac{1}{2})_n}{n!}, \tag{16}$$

  to at most 3×10^{-7} for $N=0$ to 16, raw and corrected. The limit of $C_2(N)$ is the GFF value; the approach is like $N^{-1/2}$.

Δ_σ	$\omega_{\text{eff}}(15)$	(19)	(23)	(27)	(31)	(35)
0.249	0.39	0.42	0.44	0.45	0.46	0.46
0.24	0.50	0.53	0.54	0.55	0.56	0.57
0.2	0.78	0.79	0.80	0.80	0.81	0.81
0.15	0.67	0.67	0.68	0.67	0.67	0.67
0.125	0.62	0.63	0.64	0.64	0.64	0.64
0.1	0.63	0.65	0.66	0.67	0.68	0.68
0.075	0.70	0.73	0.76	0.77	0.79	0.80
0.05	0.92	0.99	1.04	1.08	1.12	1.16
0.025	3.19	8.09	slope changes sign near $N = 22$			

Table 2: Local exponent (15) of the $d = 1$ series, labelled by the centre of its four cutoffs. Held-out test: fits on $N \leq 32$ predict the values at $N = 34$–40 to within 2.3×10^{-6} at every $\Delta_\sigma \geq 0.05$, inside the spread of the predictions. The test passes also at $\Delta_\sigma = 0.025$, where the series has no limit, so it checks the fits only within the range of the data, not their limit. All values: `results/HT_1d/holdout_and_local_exponent.md`.

Intermediate coupling.

- The extrapolations are final for $\Delta_\sigma \geq 0.1$.
- At $0.075 \leq \Delta_\sigma \leq 0.125$ they add 1.6×10^{-3}–2.5×10^{-3} to the $N = 40$ values.
- Over $15 \leq N \leq 35$ the local exponent levels off:
  - at 0.64 for $\Delta_\sigma = 0.125$;
  - at 0.68 for 0.1.
- It rises instead, over the same range of N:
  - 0.70–0.80 at 0.075;
  - 0.92–1.16 at 0.05.
- A rising ω_{eff} precedes the turnover seen at $\Delta_\sigma = 0.025$ and suggests that the fits overestimate the limit. This is why 0.075 carries a one-sided error. The classification final versus one-sided is a judgement based on these trends; they differ only modestly between 0.1 and 0.075.
- At weak coupling ω_{eff} also rises slowly (0.39–0.46 at $\Delta_\sigma = 0.249$), but the bias it implies is below 10^{-5}.
- Fits restricted to $N \leq 32$ predict $N = 34$–40 to 2.3×10^{-6} (Table 2).

Strong coupling.

- $\Delta_\sigma = 0.05$. The series still rises at $N = 40$. The accepted F1 and F2 fits give 0.6220–0.6224, a range that contains the NRG value of Table 1. The F3 fits reach down to 0.618. Their low branch moves up as cutoffs are added, which mildly argues against a turnover. The rising ω_{eff} argues the other way. No value is quoted.
- $\Delta_\sigma \leq 0.025$. The corrected sequence does not converge.

– At 0.025 it peaks near $N = 22$ and then falls ever faster, reaching -1.3×10^{-3} per unit of $\ln N$ at $N = 40$.

– At 0.01 it falls at a constant -6.2×10^{-3} per unit of $\ln N$.

– At 0.001 it falls slowly but lies 0.032 above the NRG value.

The zero-mode basis is converged and the solver is verified, so this is a property of the window model, not of the numerics. The uncorrected truncation converges, but to values that violate Eq. (5) (Table 3).

4.4 Model dependence and the comparison with NRG

The errors of Table 1 are half-spreads of one family of fits (F1–F3) on one prescription. They do not cover two further model dependences, which we measure on the shipped records (`public/model_dependence_ht1d.py`, table in `results/HT_1d/model_dependence.md`).

- *Tail model.* We tried two alternative descriptions of the approach $N \to \infty$ of the same window series:

  – F4, Eq. (14) with two free exponents;

  – a local exponent continued linearly in $\ln N$ from its last four values, with the slope at $N = 40$ integrated to infinity.

  Both move the limit down by at most 1.6×10^{-4} at $0.075 \le \Delta_\sigma \le 0.125$, of the order of the fit half-spread.

- *Prescription.* The same truncation without the window (raw, energy zero-mode basis) extrapolates, with the same fits, to limits $6.6 \times 10^{-4}, 2.3 \times 10^{-3}, 1.7 \times 10^{-3}$ below the window limits at $\Delta_\sigma = 0.125$, 0.1 and 0.075, respectively.

  – Neither prescription satisfies the descendant identities in the limit (Table 3), so the limit of the truncated model depends on the prescription at this level.

  – At $\Delta_\sigma \ge 0.15$ the two prescriptions agree to 1.3×10^{-4} or better.

The prescription dependence, 7×10^{-4}–2.3×10^{-3} at $0.075 \le \Delta_\sigma \le 0.125$, is thus as large as the gap to NRG below. We do not combine these shifts into an error. The truncation has no strict error bar, and an error built after the gap to NRG had been seen would make the comparison read as agreement within errors. The errors of Table 1 stay the fit spreads. The window limit is expected to be the better of the two, since its σ descendant violation is 10–40 times smaller.

Comparison with NRG. We compare with the final NRG values of Table 1 (NRG folder). Their errors are sums of three convergence diagnostics, not standard deviations.

- HT lies above NRG at all six points $0.075 \le \Delta_\sigma \le 0.2$, by up to 1.1×10^{-3}.
- The differences are 10^{-4} to 1.1×10^{-3} at $0.075 \le \Delta_\sigma \le 0.15$, and below 10^{-4} at 0.175 and 0.2.
- At $\Delta_\sigma = 0.125$, 0.1 and 0.075 the differences are $4.5 \times 10^{-4}, 8.7 \times 10^{-4}, 1.1 \times 10^{-3}$. They are at least five times both the fit half-spreads of Table 1 and the NRG errors, and of the size of the prescription dependence.

	raw			window		
Δ_σ	σ	ϕ^3	ε	σ	ϕ^3	ε
0.249	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000
0.2	+0.0030	+0.0067	+0.0048	+0.0007	+0.0044	+0.0024
0.15	+0.0207	+0.0297	+0.0238	+0.0007	+0.0096	+0.0040
0.125	+0.0388	+0.0492	+0.0393	−0.0011	+0.0088	−0.0003
0.1	+0.0644	+0.0743	+0.0580	−0.0044	+0.0046	−0.0100
0.075	+0.0990	+0.1056	+0.0797	−0.0088	−0.0024	−0.0248
0.05	+0.1481	+0.1439	+0.1040	−0.0123	−0.0110	−0.0423
0.025	$+0.2321^a$	+0.1869	+0.1273	-0.0063^a	−0.0186	−0.0548
0.01	$+0.3483^a$	+0.2062	+0.1320	$+0.0345^a$	−0.0201	-0.0519^a

Table 3: Level-1 descendant violations v_1 at $N = 28$, raw and with the consistent window, in the energy zero-mode basis of `prod1--7`: $n_z = 24$ at $\Delta_\sigma \geq 0.1$, 32 below. The ε columns use the first excited even state. ^{a}: ambiguous identification with the free L_{-1}. In the CFT all entries vanish.

- NRG lies between the raw and the window limits at these three points.
- NRG also lies inside the fit-free bracket $[\Delta_\varepsilon(40),\ \Delta_\varepsilon(40) + \frac{d\Delta_\varepsilon/d\ln N}{\omega_{\text{eff}}}]$ of the window series at every $0.075 \leq \Delta_\sigma \leq 0.2$. That bracket holds if $\Delta_\varepsilon(N)$ keeps rising and ω_{eff} does not decrease beyond $N = 40$.

We do not assume that either method is right. The Monte Carlo values that the MC folder labels *accepted* (MC column of the $d = 1$ results table of Ref. [4]) also lie above NRG at these three couplings, on the same side as HT, but their systematic errors exceed the HT–NRG differences, so they cannot decide between the two. The resummed expansions cannot decide at this coupling (`Resummation/` folder). The HT values at $0.075 \leq \Delta_\sigma \leq 0.125$ are limits of the window-corrected truncation at their fit errors. As estimates of the CFT they depend on the truncation prescription by the shifts above, which are as large as the gap to NRG.

4.5 Descendant identities

Table 3 gives the level-1 violations, Eq. (5), at $N = 28$, raw and with the window, from the identity diagnostic run on the `prod1--7` records. The other cutoffs are in `results/HT_1d/identity_violations.md`.

- The raw violations do not decay with N over $N = 16$–28 and grow with the coupling. At $\Delta_\sigma \leq 0.1$ they range over 0.06–0.25.
- The window reduces the σ violation by 10–40 times at $\Delta_\sigma \leq 0.15$.
- At $\epsilon \geq 0.6$ non-decaying residuals of about 0.01 remain for σ and ϕ^3, and up to 0.05 for ε at $\Delta_\sigma \leq 0.05$.

5 Two dimensions

5.1 Modes, sectors and couplings

In $d = 2$ the exponent s of the long-range interaction fixes $\Delta_\sigma = (d-s)/2 = 1 - s/2$, so $\epsilon = 2s - 2 = 2 - 4\Delta_\sigma$. The interacting window is $1 < s < s_*$, that is $\frac{1}{8} < \Delta_\sigma < \frac{1}{2}$. Its upper end is the crossover

to the short-range Ising CFT, at $s_* = 2 - \eta_{\rm SR} = \frac{7}{4}$ [6, 8, 2]. At $s = s_*$ the fixed point is the short-range Ising CFT decoupled from a generalized free field [8, 2], so there $\Delta_\varepsilon = 1$, the dimension of the energy operator of the two-dimensional Ising model [13]. The scan covers $\frac{1}{8} \le \Delta_\sigma < \frac{1}{2}$.

The modes are $k = (a, b)$, $a, b \ge 0$, of Eq. (2): the oscillator a_{ab} has level $\kappa = a+b$, spin $j = a-b$ and weight $u_{ab}^2 = (\Delta_\sigma)_a(\Delta_\sigma)_b/(a!\,b!)$. With $f_{ab} = e^{-{\rm i}(a-b)\vartheta}$, the two-point function of the field (1) is the cylinder covariance $z^{\Delta_\sigma}|1 - z\,e^{{\rm i}\vartheta}|^{-2\Delta_\sigma}$, with $z = e^{-\tau}$ the radial-time variable and ϑ the angle on the circle. The oscillator a_{00} is the zero mode of Eq. (6), and all others form the bath. A bath state with occupations n_{ab} has level, particle number and spin

$$\ell = \sum_{(a,b)\neq(0,0)} (a+b)\, n_{ab}, \qquad P = \sum n_{ab}, \qquad J = \sum (a-b)\, n_{ab}. \tag{17}$$

We keep every bath state with $\ell \le N$, with no limit on P (Section 2.4). The scalar spectrum is computed at $J = 0$, in the sector even under the reflection $a \leftrightarrow b$. Since $\ell - J = 2\sum b\, n_{ab}$ is even, every scalar bath state has even level. The truncations at $N = 2m$ and $2m+1$ therefore coincide, and we use the even cutoffs $N = 8, 10, \ldots, 24$.

The vertices $W_j = \langle{:}\chi^j{:}\rangle$ of Eq. (7) are built exactly in the retained scalar space. With $B = \sum_{(a,b)\neq(0,0)} u_{ab}\, a_{ab}$ the annihilation part of χ at $\vartheta = 0$, and S the embedding of the scalar space,

$$W_j = \sum_{i=0}^{j} \binom{j}{i} (B^i S)^\dagger (B^{j-i} S). \tag{18}$$

Between two scalar states the average over the circle equals the value at $\vartheta = 0$, so no further projection is needed; states of nonzero spin enter as intermediate states. Since B lowers the level, $B^{j-i}S$ stays inside the space $\ell \le N$, and the matrix elements are exact: no intermediate state is cut off.

The couplings are parametrized by $\theta = (\theta_0, \theta_1)$,

$$g = \frac{192\, e^{3\theta_1}}{\Delta_\sigma^2}, \qquad A = \frac{4\, e^{2\theta_1}\theta_0}{\Delta_\sigma}, \qquad r = A + \frac{g}{2} - \frac{\Delta_\sigma}{4}, \tag{19}$$

with A the coefficient of q^2 in H_z, Eq. (8). The two tuning conditions of Eq. (4), $E_\sigma - E_v = \Delta_\sigma$ and $E_{\phi^3} - E_v = 2 - \Delta_\sigma$, fix θ, and the prediction is $\Delta_\varepsilon = E_\varepsilon - E_v$.

5.2 Zero-mode basis and window

The zero-mode basis is the density basis of Section 2.5, Eq. (10), with the training cutoff $N_t = 6$. The teacher is an energy-basis calculation at $N = 6$ with 24 states per parity, taken from an auxiliary oscillator basis of $N_{\rm osc} = 192$ states. Each stored basis keeps the 20 leading eigenvectors of ρ per parity, which contain the nested bases of $n_z = 12$, 16 and 20 states. Production uses $n_z = 16$, and $n_z = 20$ and 12 are recomputed at every cutoff as checks. For ten values of Δ_σ the stored bases are those of the package; for $\Delta_\sigma = 0.125$, 0.135 and 0.175 new ones were trained with the same settings. No basis is transferred between values of Δ_σ. Over all chains and cutoffs,

$$|\Delta_\varepsilon(n_z{=}20) - \Delta_\varepsilon(n_z{=}16)| \le 9.4 \times 10^{-11}, \qquad |\Delta_\varepsilon(n_z{=}12) - \Delta_\varepsilon(n_z{=}16)| \le 5.2 \times 10^{-7},$$

the latter at $\Delta_\sigma = 0.125$, $N = 22$. Three reference points at $N = 14$ ($\Delta_\sigma = 0.126$, 0.25 and 0.4) use instead the energy basis with 32 states per parity, recomputed at every coupling from $N_{\rm osc} = 256$; they agree with $n_z = 16$ to 10^{-9}.

The window is the consistent correction of Section 3, Eqs. (11) and (12). Every bath state of level ℓ is shifted by $D(\ell) = E_v(N) - E_v(N-\ell)$. The vacua $E_v(M)$, $M = 0, 2, \ldots, N$, are computed with their own windows, at the same couplings and in the same zero-mode basis; they are not retuned. Their fixed points are solved as in Section 3 and accepted at 10^{-11}, and the derivatives (13) enter the tuning Jacobian. The other acceptance thresholds are a tuning residual $\leq 1.5 \times 10^{-10}$, an eigenvector residual $\leq 10^{-8}$ and an orthogonality defect $\leq 10^{-9}$. Every shipped record passes them (check `d2.records.gates`).

5.3 Campaign

Thirteen chains, $\Delta_\sigma \in \{0.125, 0.126, 0.135, 0.15, 0.175, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.475, 0.495\}$, ran from $N = 8$ to 24 in steps of two on Expanse (SDSC), one chain per 64-core, 64 GB allocation. Each cutoff was seeded by the previous one, and the first by the package's $N = 8$ reference solution or by the teacher. The 348 records (main runs, basis checks and reference points) took 1.1×10^4 core-hours; a point at $N = 24$ took 0.6–11 h and up to 24 GiB. The code is `lri_2d_window_hpc` at commit 889a968, and every record carries a hash of that source. The Python environment was frozen (Python 3.11.5, NumPy 2.3.5, SciPy 1.17.0, Numba 0.65.1). The lowest value, $\Delta_\sigma = 0.125$, is the crossover point $s = s_*$, where $\Delta_\varepsilon = 1$.

5.4 Finite-cutoff behaviour

Table 4 and Fig. 2 give $\Delta_\varepsilon(N)$. All of them are values of the truncated model at that cutoff (status: *finite-N*). Nine cutoffs up to $N = 24$ are too few, and the behaviour varies too much with Δ_σ, for a controlled extrapolation, so we quote no continuum value in $d = 2$. Steps are taken between successive cutoffs, $\delta(N) = \Delta_\varepsilon(N) - \Delta_\varepsilon(N-2)$.

- $\Delta_\sigma \geq 0.45$: the last step $\delta(24)$ is 2.54×10^{-6}, 1.28×10^{-6} and 1.57×10^{-7} at 0.45, 0.475 and 0.495, and the steps shrink by 18–22% per step.
- $0.25 \leq \Delta_\sigma \leq 0.4$: Δ_ε rises monotonically, with last steps 4.30×10^{-4}, 3.80×10^{-4}, 2.16×10^{-4} and 5.86×10^{-5}, which shrink by $1 - (\delta(24)/\delta(16))^{1/4} = 16$, 13, 11 and 8% per step. If the rise continues, $\Delta_\varepsilon(24)$ bounds the limit of the truncated model from below.
- $\Delta_\sigma = 0.2$: the steps change sign (+ up to $N = 22$, -4.2×10^{-5} at $N = 24$), so Δ_ε is not monotone.
- $0.125 \leq \Delta_\sigma \leq 0.175$: Δ_ε falls from $N = 10$ on, and the steps grow. The last step is -1.05×10^{-3} at 0.175 ($N = 24$) and -2.02×10^{-2} at 0.125 ($N = 22$), each larger than the previous one. At $\Delta_\sigma = 0.125$ the value at $N = 22$, 0.9400, lies 6% below the exact value 1 and moves away from it. This is the failure toward the short-range end discussed in Section 8.

5.5 No solution at $N = 24$ for $\Delta_\sigma = 0.125$, 0.126: a fold

At $\Delta_\sigma = 0.125$ and 0.126 the Newton solve at $N = 24$ did not converge. It was started from the $N = 22$ root and from the linear extrapolation $2\theta(22) - \theta(20)$, and repeated with a finite-difference Jacobian. Each time the residual stalled on a floor, at 2.9×10^{-3} and 2.0×10^{-3}. At the accepted roots of smaller N the condition number of the tuning Jacobian grows, from 84 at $N = 12$ to 888 at $N = 22$ ($\Delta_\sigma = 0.125$).

To decide whether a root exists, we scanned the valley of the residual. Write the conditions as

$$f_1(\theta) = E_\sigma - E_v - \Delta_\sigma, \qquad f_2(\theta) = E_{\phi^3} - E_v - (2 - \Delta_\sigma). \tag{20}$$

At six values $\theta_0 \in \{12.5, 13.25, \theta_0^{\rm best}, 14.3, 14.8, 15.5\}$ around the best point of the Newton solves, a secant iteration in θ_1 solves $f_2 = 0$ to $|f_2| \le 10^{-9}$, and we record f_1 there. Code, basis and solver settings are those of production, with $n_z = 16$. At $\theta_0^{\rm best}$ the first evaluation, before the secant, reproduces the stalled residual: $\max(|f_1|, |f_2|) = 2.93 \times 10^{-3}$ and 1.99×10^{-3}. A root would show as a sign change of f_1 along the curve $f_2 = 0$. There is none: $f_1 > 0$ at every point (smallest value 1.60×10^{-3}), with an interior minimum and a steep rise on both sides. A quartic fit in θ_0 puts the minimum at $f_1 = 2.686 \times 10^{-3}$ ($\theta_0 \approx 14.03$) for $\Delta_\sigma = 0.125$ and at 1.556×10^{-3} ($\theta_0 \approx 14.06$) for 0.126. Both are far above the effect of the zero-mode truncation ($\le 5 \times 10^{-7}$) and the eigensolver accuracy.

Hence, on the scanned range of θ_0, the two tuning conditions have no solution at $N = 24$ on the branch continued from smaller N. This is a statement about the truncated model at $N = 24$ and involves no extrapolation in N. The minimum falls by 1.13×10^{-3} per 0.001 in Δ_σ. Near a saddle-node (fold) it is linear in the parameter, which places the fold at

$$\Delta_\sigma^{\rm fold}(N = 24) \approx 0.1274, \tag{21}$$

an estimate from two values of Δ_σ. Roots exist above it ($\Delta_\sigma = 0.135$ converged at $N = 24$), and at $N = 22$ they exist down to 0.125. The fold has therefore moved past 0.125 between $N = 22$ and 24, and the growing condition number at the roots is its precursor. The $d = 2$ data at $\Delta_\sigma = 0.125$ and 0.126 end at $N = 22$.

5.6 Weak coupling and the ϵ expansion of the model

Expanded in ϵ, the truncated model agrees with the consistent-window scheme of the perturbative calculation (`HT_Pert/`). At $\Delta_\sigma = 0.495$ ($\epsilon = 0.02$) the [4/4] Padé approximant of that scheme's coefficients $C_0(N), \dots, C_8(N)$ reproduces $\Delta_\varepsilon(N)$ to 9.4×10^{-12} at every $N = 8, \dots, 24$, and at $\Delta_\sigma = 0.475$ ($\epsilon = 0.1$) to 7.6×10^{-8} (check `d2.pert_consistent`). A difference δC_4 in the fourth-order coefficient would show as $\delta C_4\, \epsilon^4$, so $|\delta C_4| \lesssim 6 \times 10^{-5}$ at every $N \le 24$. In that scheme the fourth-order coefficient grows with the cutoff, $C_4(N) \simeq 5.003 - \frac{365}{4374\pi} \ln N$, that is like $-0.027 \ln N$ (exact sums to $N = 2000$ in `HT_Pert/`), while the continuum value $C_4(2)$ is finite. The $d = 2$ window model is thus known to be correct only through third order in ϵ. In the perturbative calculation a larger correction, the complete window of `HT_Pert/`, removes this logarithm; it has no finite-coupling counterpart here.

6 Three dimensions

6.1 Modes, sectors and the truncated space

On $S^2 \times \mathbb{R}$ of unit radius we use the modes $k = (n, l, m)$ of Eq. (2), with $n, l \ge 0$ and $|m| \le l$, level $\kappa = 2n + l$ and energy $\Delta_\sigma + \kappa$. With $z = e^{-\tau}$ and μ the cosine of the angle between the two points, the two-point function of the generalized free field is

$$\frac{z^{\Delta_\sigma}}{(1 - 2z\mu + z^2)^{\Delta_\sigma}} = \sum_{\kappa \ge 0} z^{\Delta_\sigma + \kappa}\, G_\kappa(\mu), \qquad G_\kappa(\mu) = \sum_{2n+l=\kappa} (2l + 1)\, u_{nl}^2\, P_l(\mu). \tag{22}$$

Here $G_\kappa = C_\kappa^{(\Delta_\sigma)}$ is the Gegenbauer polynomial, written G to avoid a clash with the coefficients C_n, and P_l is the Legendre polynomial. Expanding G_κ in Legendre polynomials fixes the weights u_{nl}^2 of Eq. (2). The zero mode $k = (0, 0, 0)$ has $u = 1$; it is the oscillator a_0 of Eq. (6), with coordinate q. The bath field is $\chi = \sum_{k \ne (0,0,0)} u_{nl} \sqrt{4\pi}\, \big[Y_{lm}\, a_k + \text{h.c.}\big]$, and sphere averages use $d\Omega/4\pi$. A bath

Δ_σ	s	$N=16$	$N=20$	$N=22$	$N=24$	last step	behaviour
0.125	1.75	0.98916082	0.96025057	0.94002832	n.s.	-2.02×10^{-2}	falls; steps grow
0.126	1.748	0.99053540	0.96363343	0.94568294	n.s.	-1.80×10^{-2}	falls; steps grow
0.135	1.73	0.99948557	0.98399079	0.97537120	0.96202900	-1.33×10^{-2}	falls; steps grow
0.15	1.7	1.00548133	0.99836315	0.99471461	0.99022622	-4.49×10^{-3}	falls; steps grow
0.175	1.65	1.00330217	1.00162583	1.00076308	0.99971675	-1.05×10^{-3}	falls; steps grow
0.2	1.6	0.99490861	0.99526003	0.99534664	0.99530455	-4.21×10^{-5}	not monotone
0.25	1.5	0.97541470	0.97674862	0.97727730	0.97770747	$+4.30\times10^{-4}$	rises; shrink 16%/step
0.3	1.4	0.96158682	0.96267180	0.96311194	0.96349241	$+3.80\times10^{-4}$	rises; shrink 13%/step
0.35	1.3	0.95679429	0.95736832	0.95760910	0.95782471	$+2.16\times10^{-4}$	rises; shrink 11%/step
0.4	1.2	0.96152920	0.96167002	0.96173269	0.96179129	$+5.86\times10^{-5}$	rises; shrink 8%/step
0.45	1.1	0.97526784	0.97527566	0.97527858	0.97528111	$+2.54\times10^{-6}$	small steps
0.475	1.05	0.98580651	0.98581110	0.98581268	0.98581396	$+1.28\times10^{-6}$	small steps
0.495	1.01	0.99678104	0.99678155	0.99678174	0.99678190	$+1.57\times10^{-7}$	small steps

Table 4: $\Delta_\varepsilon(N)$ in $d=2$ at finite coupling ($n_z=16$, consistent window): finite-N values of the truncated model, not continuum values. n.s.: no solution at $N=24$ (Section 5.5). Last step: between the two largest cutoffs. Shrink: $1-(\delta(24)/\delta(16))^{1/4}$.

state with occupation numbers n_k has level $\ell=\sum_k \kappa_k n_k$. We keep all states with $\ell\le N$, with no limit on the particle number.

Sectors. Rotations, inversion $(-1)^\ell$ and field parity commute with H. The scalar operators live in the $J=0$, inversion-even sector, in both field parities. Inversion-even states have even ℓ, so an odd cutoff adds no scalar state, and we use even N only. The $J=0$ subspace is the kernel of J_+ within the $J_z=0$ subspace. It is built separately in every species-occupation block, a species being (n,l) with its $2l+1$ magnetic states, and it is computed as an SVD null space.

At $N\ge18$ LAPACK's default SVD driver (gesdd) fails to converge on a few blocks. The code then retries with the gesvd driver and full right singular vectors. It accepts the kernel Q only if its dimension is the nullity predicted by representation theory and if every matrix element of J_+Q and of $Q^{\mathsf T}Q-1$ is at most 10^{-10} in absolute value; a second failure is an error. The retry happened in 8 of 17 335 blocks at $N=18$ and in 62 of 40 033 at $N=20$, with $\max|J_+Q|=5.8\times10^{-15}$ (it does not depend on Δ_σ). The scalar bath space has 110, 371, 1285, 4486, 15 815, 55 859 and 196 585 states at $N=8,10,\dots,20$; with $n_z=16$ zero-mode states per parity, the scalar problem at $N=20$ has dimension 3.1×10^6 in each field parity.

Vertices. The couplings $W_j=\langle{:}\chi^j{:}\rangle$ of Eq. (7), restricted to $J=0$, equal the vertex at any single point of the sphere, by rotation invariance. The code evaluates them at the north pole. The $J=1$ and $J=2$ sectors are computed only as diagnostics; there the code uses the trace over the $2J+1$ magnetic components, and the original polar Gauss–Legendre quadrature (with $N+1$ nodes) is kept as an independent test. The vertices at lower cutoffs, which the window needs, are exact restrictions of those at N, because a normal-ordered W_j never passes through a level above both of its endpoints.

6.2 Window, zero-mode basis and tuning

The window of Eq. (11) is used in its self-consistent form. Only even sub-cutoffs carry distinct vacua, so a bath state of level ℓ is shifted by

$$D_N(\ell)=E_v(N)-E_v\big(2\lfloor(N-\ell)/2\rfloor\big),\qquad E_v(M)=\lambda_{\min}\big[H(M)+D_M\big],\quad E_v(0)=0,\tag{23}$$

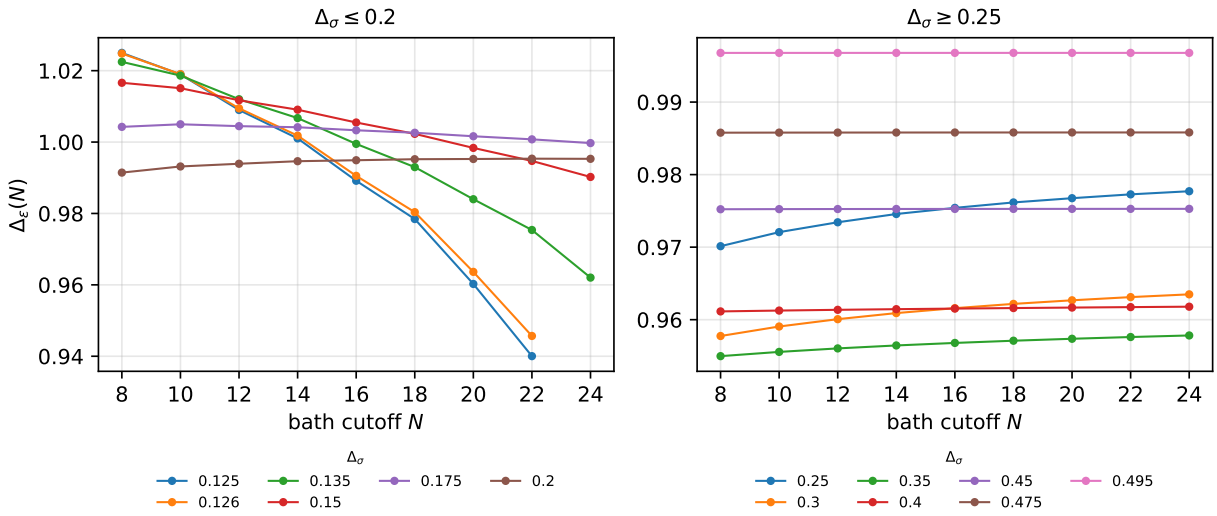


Figure 2: $\Delta_\varepsilon(N)$ in $d = 2$ against the bath cutoff N, for the 13 values of Δ_σ (made by `tools/make_d2_note_inputs.py` from `HT/results/HT_2d/prod1/points.csv`).

with $E_v(M)$ the lowest field-even scalar energy at sub-cutoff M, solved one sub-cutoff after the other. At each sub-cutoff the fixed point $v = E_v(M)$ solves $F(v) = \lambda_{\min}[H(M) + D(v)] - v = 0$. Newton's method converges on it because $F'(v) = -p_0$, with p_0 the bath-vacuum weight of the ground state. The iteration stops when $|F| \le \min\left(10^{-12}\max(1, |v|),\, 10^{-11}\right)$ and fails if $p_0 \le 10^{-12}$. The same D_N is applied, without retuning, in the $J = 1, 2$ sectors.

The zero-mode basis is the density basis of Section 2.5, trained once per Δ_σ. The teacher is a solve at $N_t = 6$ in the energy basis, with 24 states per parity taken from an auxiliary oscillator basis of $N_{\text{osc}} = 192$ states. The reduced density matrix of Eq. (10) is then $(\rho_{\text{vac}} + \rho_\sigma + \rho_{\phi^3})/6$ at N_t plus the vacua at $M = 0, 2, 4$, each with weight $1/6$. The stored basis keeps its 20 leading eigenvectors per parity, and every production solve at $N = 8, \ldots, 20$ uses the leading $n_z = 16$. As in $d = 1$, the ε state plays no role in the training.

The couplings (r, g) are tuned by Eq. (4), $E_\sigma - E_v = \Delta_\sigma$ and $E_{\phi^3} - E_v = 3 - \Delta_\sigma$, by safeguarded Newton steps with tolerance 3×10^{-11}. Each cutoff is started from the solution at $N - 2$, with the same zero-mode basis. A point is accepted only if

- both tuning conditions hold to 1.5×10^{-10} (five times the Newton tolerance);
- the eigenvector residuals are at most 10^{-8} and their orthogonality defects at most 10^{-9};
- every window fixed point holds to 10^{-11}.

The replay also checks that the two tuned gaps of every record equal their targets to 10^{-9}. The prediction is $\Delta_\varepsilon = E_\varepsilon - E_v$, the first excited field-even scalar gap.

6.3 Campaigns

We solve 15 values of Δ_σ: 0.7499, 0.745, 0.74, 0.725, 0.7, 0.675, 0.65, 0.625, 0.6, 0.575, 0.55, 0.535, 0.525, 0.52 and 0.518149. The last lies 1.9×10^{-7} above the Δ_σ of the 3d Ising model [3], which is not imposed. Each value is computed as one sequence of solves, the teacher at $N = 6$ followed

by $N = 8, 10, \ldots, 20$, each started from the previous one. Everything ran on SDSC Expanse (AMD EPYC 7742). Before production, each installed code release had to pass a validation job, the test suite and validators that `make test` runs. Every step then ran with the arguments fixed in advance in a campaign manifest (`code/lri_3d_window_hpc/public/PROVENANCE.md`).

The records are grouped in five campaigns, which name the folders of `data/HT_3d/campaigns`:

- `grid18` ($\Delta_\sigma \geq 0.52$, 14 values) and `grid18sri` ($\Delta_\sigma = 0.518149$, near the short-range Ising end): the teacher and $N = 8$ to 18. Each sequence used 16 cores. A point at $N = 18$, with $J = 1$ diagnostics, took 34–74 minutes at a peak of 38.5 GiB.

- `l20x06` ($\Delta_\sigma = 0.6$, with $J = 1$ diagnostics), `grid20` (the other 13 values $\Delta_\sigma \geq 0.52$) and `grid20sri` ($\Delta_\sigma = 0.518149$): $N = 20$, the last two in the scalar sector only. The bath graph and the scalar projector do not depend on Δ_σ, so they were built once into a cache that is never modified afterwards. On one large-memory node this took 9.0 h and 92 GiB, mostly in single-threaded LAPACK (from the Slurm log of a setup run, not a shipped record). The vertices depend on Δ_σ and are built in every point. A scalar-only point took 27–104 minutes at 84–87 GiB on 32 cores; the point at $\Delta_\sigma = 0.6$ with $J = 1$ diagnostics took 239 minutes at 328 GiB.

$N = 22$ was not attempted.

6.4 Results

Table 5 gives $\Delta_\varepsilon(N)$ at $N = 14$–20; all of $N = 8$–20 is in `results/HT_3d/finite_L_table.csv` (check `TABLE`). Figure 3 shows the steps $d_N = \Delta_\varepsilon(N) - \Delta_\varepsilon(N-2)$. We reach $N = 20$ at every Δ_σ and give finite-N values only. These cutoffs are too few for a controlled extrapolation, and the checks below show that small steps do not imply convergence.

Δ_σ	$N = 14$	$N = 16$	$N = 18$	$N = 20$	d_{20}	d_{20}/d_{18}
0.7499	1.49993336	1.49993336	1.49993336	1.49993336	2.3×10^{-11}	0.78
0.745	1.49673611	1.49673617	1.49673622	1.49673626	3.7×10^{-8}	0.75
0.74	1.49360198	1.49360214	1.49360225	1.49360232	7.3×10^{-8}	0.68
0.725	1.48486049	1.48485983	1.48485909	1.48485833	-7.6×10^{-7}	1.03
0.7	1.47194111	1.47193446	1.47192895	1.47192480	-4.2×10^{-6}	0.75
0.675	1.46046754	1.46048280	1.46050270	1.46052707	2.4×10^{-5}	1.23
0.65	1.45028710	1.45039728	1.45050697	1.45061574	1.1×10^{-4}	0.99
0.625	1.44146308	1.44172312	1.44196302	1.44218479	2.2×10^{-4}	0.92
0.6	1.43396549	1.43439430	1.43477460	1.43511242	3.4×10^{-4}	0.89
0.575	1.42767821	1.42827172	1.42878594	1.42922982	4.4×10^{-4}	0.86
0.55	1.42244491	1.42318494	1.42381704	1.42434921	5.3×10^{-4}	0.84
0.535	1.41973557	1.42054979	1.42124152	1.42181519	5.7×10^{-4}	0.83
0.525	1.41808151	1.41893773	1.41966339	1.42025885	6.0×10^{-4}	0.82
0.52	1.41729470	1.41816931	1.41890994	1.41951431	6.0×10^{-4}	0.82
0.518149	1.41700967	1.41789062	1.41863643	1.41924372	6.1×10^{-4}	0.81

Table 5: Δ_ε from the $d = 3$ window truncation at bath cutoffs $N = 14$–20, with the last step $d_{20} = \Delta_\varepsilon(20) - \Delta_\varepsilon(18)$ and its ratio to the previous one. Finite-cutoff values, not limits.

- *Weak coupling, $\Delta_\sigma \geq 0.74$: small steps.* The last step is below 10^{-7} (7.3×10^{-8} at most).

- $\Delta_\sigma = 0.7$*: not converged, and moving away.* The Borel–Padé resummation of the ϵ series through the GFF coefficient $C_7(3)$ [4] fixes the continuum value to 1.47214(3). This is the

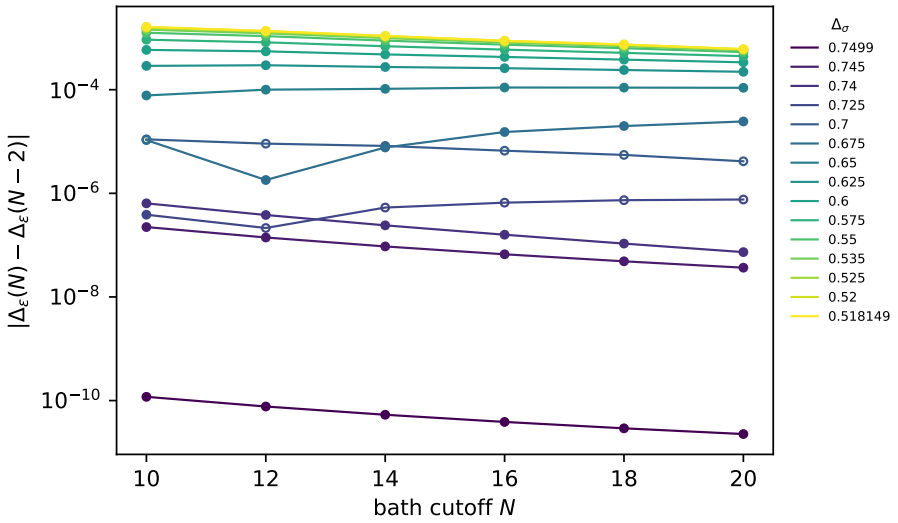


Figure 3: Size of the steps d_N against N, for every Δ_σ; filled symbols: $d_N > 0$, open: $d_N < 0$. Made by `public/note_d3.py` from `finite_L_table.csv`.

median and half-spread of 29 approximants, with Borel–Leroy $b = 0, \ldots, 4$ and Padé orders of total degree 4 and 5, rejecting those with a pole on the positive axis. The value at $N = 20$, 1.4719248, lies 2.2×10^{-4} below it, and the sequence $\Delta_\varepsilon(16) > \Delta_\varepsilon(18) > \Delta_\varepsilon(20)$ moves away. The continuum is reached, if at all, only after a turnover at larger N. An extrapolation in N that continues this trend lands below $\Delta_\varepsilon(20)$, further from the continuum, so none is quoted.

- $\Delta_\sigma = 0.65$ *and* 0.675*: no trend yet.* The steps do not yet shrink: $d_{20}/d_{18} = 0.99$ and 1.23. At 0.7 and 0.725 the values fall (last steps -4.2×10^{-6} and -7.6×10^{-7}), and from 0.675 down they rise, so the steps change sign in this range.

- $0.518 \leq \Delta_\sigma \leq 0.625$*: rising.* The values rise monotonically from $N = 8$ to 20. The last steps are $(2.2\text{–}6.1) \times 10^{-4}$, and they shrink by 8 to 19 percent per step. They shrink fastest near the short-range end.

- *Short-range end,* $\Delta_\sigma = 0.518149$*: fails near the endpoint.* The value at $N = 20$, 1.41924, lies above the 3d Ising value $\Delta_\varepsilon = 1.41263$ [3] and still rises. Near this end the long-range model is described by the short-range Ising model coupled, through its spin field, to an auxiliary generalized free field [8, 2]. That coupling is nearly marginal here, and the approach to the limit is expected to be logarithmic in N.

- *The window model at fourth order: a finite-N statement.* Expanded in ϵ at fixed N, the same window model has $C_2(N) \to C_2(3)$, and $C_3(N)$ is consistent with $C_3(3)$ (`HT_Pert/`). Its fourth-order coefficient at $N = 20$, 0.7400078, is 60% of $C_4(3) = 1.228063545$ [4] and still growing slowly with N; whether its $N \to \infty$ limit is wrong is not shown.

The $N = 20$ values are the $d = 3$ Hamiltonian-truncation results of Ref. [4]: rounded to five decimals in its $d = 3$ results table, and plotted as green squares in the right panel of its comparison figure.

7 Independent checks

In each dimension a second solver, written without using the production code, reproduces the production spectra at finite cutoff. These checks verify that we compute the stated model, not that its limit is the CFT.

One dimension.

- *Authorship.* A Claude Code subagent wrote the solver (`code/lri_dressed_phi4_independent_check`) from the definition of the model. It was not allowed to read the production code, results or analysis. Of the package documentation it read only the definition of the window (`docs/WINDOW.md`) and of the Hamiltonian (`docs/ALGORITHM.md`). It derived H itself from the GFF radial quantization, and its derivation agrees with Eqs. (8)–(9).

- *Own numerics.* The zero mode is a sinc discrete-variable representation, the bath Fock space is its own, and both the literal and consistent windows are implemented. Newton tuning continues from weak coupling. Every point is verified on 1.5 times finer and 1.5 times wider grids.

- *Comparison set.* 38 points: raw and consistent at $\Delta_\sigma = 0.249$, 0.2, 0.1, 0.05, 0.025 and 0.01 with $N = 12$, 16 and 20, plus literal at 0.2 and 0.1 with $N = 12$.

- *Agreement with the production records* (`dens_v1`, density basis, $n_z = 16$). At the 18 consistent-window points ($N = 12$, 16, 20) the records agree with the blind solver to 5.6×10^{-7} or better, the truncation error of the density basis.

- *Agreement with the energy-basis records* (`prod1--7`).

  - Δ_ε agrees to 1.3×10^{-9} where these records used $n_z = 48$ or $\Delta_\sigma \geq 0.2$.
  - It agrees to 4.8×10^{-9} where they used $n_z = 24$ or 32 at strong coupling, which is that basis's truncation error.
  - Over all 38 points this gives 5×10^{-9}. The other levels and the tuned (g, r) agree at the same level.

- *Weak coupling.* The solver reproduces $C_2(N)$, Eq. (16), to 3×10^{-7}. It also shows that the window changes Δ_ε only at order ϵ^4: the fitted exponent is 4.05 to 4.11.

Two dimensions. A second solver, `lri_2d_window_indep`, implements the $d = 2$ model of Section 5.1 from its written definition, without using the production code. Its numerics differ:

- bath vertices from explicit annihilation amplitudes, on a dictionary representation of the Fock states;

- reflection-even states as explicit symmetric combinations;

- the zero mode in the energy basis, the lowest eigenstates of H_z with K states per parity, instead of the density basis;

- its own Newton iteration for the window fixed points, and tuning by a finite-difference Newton method.

It is not blind: its author had earlier prepared the production code for the cluster. Its self-tests compare the vertices with a brute-force construction in the full Fock space at $N = 5$, and its Lanczos solver with dense diagonalization. At the production couplings it reproduces the production spectra (155 runs at 13 values of Δ_σ; each comparison is bound to the shipped record by its hash):

- With a converged zero-mode basis ($K \geq 32$) it reproduces Δ_ε to 1.4×10^{-12} at $N = 12$ and 16 for every Δ_σ, and the five lowest even and odd levels and the whole vacuum table $E_v(M)$ to 7.6×10^{-12}. At $N = 18$ ($\Delta_\sigma \leq 0.15$) Δ_ε agrees to 4.5×10^{-12}, and at $N = 20$ ($\Delta_\sigma \leq 0.175$) to 4.6×10^{-11}.

- At $N = 22$, $\Delta_\sigma = 0.125$ and 0.126, it agrees to 4.4×10^{-10} and 4.0×10^{-10} ($K = 32$). There the differences grow by a factor of about nine from one cutoff to the next and keep one sign. This is the signature of the truncation at $K = 32$, not of either code.

- With $K = 24$ the agreement degrades at strong coupling as K runs out (6×10^{-9} at $N = 16$ and 2.4×10^{-7} at $N = 20$, $\Delta_\sigma = 0.125$). This tests the check rather than production.

- Re-tuned independently from couplings shifted by $(0.05, -0.05)$ in θ, the solver returns to the production root: Δ_ε agrees to 9×10^{-11} at $N \leq 12$, and to 3.6×10^{-8} at $N = 16$ with $K = 24$.

The production zero-mode basis is itself converged: $n_z = 20$ and 16 agree to 9.4×10^{-11} over all chains and cutoffs. These checks verify that we compute the stated model; they do not test its continuum limit.

Three dimensions. A second solver, written from the definition of the model without reading or importing the production code (`code/lri_3d_window_indep`), reproduces the $d = 3$ spectra. It makes different choices at every step:

- a complex Y_{lm} Fock space with all magnetic sectors;

- the $J = 0$ space as the null space of the Casimir $J_- J_+$, from a dense eigensolver, rather than an SVD null space;

- the vertices at a generic point of the sphere, checked at a second point;

- the zero mode in a number-state basis truncated to its K lowest local eigenstates, not the trained density basis;

- window fixed points by bracketed root finding.

Its $J = 0$ state counts at $N = 2, 4, 6, 8$ (3, 10, 33, 110) equal those of the perturbative package in `HT_Pert/`. In 23 comparisons at $N = 8, 10, 12$, in the window and the raw scheme and over the whole range of Δ_σ, it reproduces the production Δ_ε:

- at the production couplings to at most 1.8×10^{-12}, with 16 or 24 local states;

- after tuning (r, g) independently from a start displaced by 2%, to at most 1.9×10^{-11}, quoted as 2×10^{-11}, with every tuning converged to 10^{-12}.

The production values it compares with are those of the shipped records. This verifies that the stated model is computed at $N \leq 12$, not that its limit is the CFT. The quoted results are at $N = 14$–20, beyond the reach of this check.

8 Caveats

The limit depends on the window model.

- The window restores only disconnected vacuum fluctuations.
- In $d = 1$ the corrected descendant spacings still miss by amounts that do not decrease with N (Table 3).
- In $d = 2$ the ϵ expansion of the window model is correct only through third order: its fourth-order coefficient grows like $\ln N$ (Section 5). In $d = 3$ the fourth-order coefficient at $N = 20$ is 60% of $C_4(3)$ and still growing slowly; whether its $N \to \infty$ limit is wrong is not shown (Section 6).

The method fails toward the short-range end in every dimension.

- The tuned coupling is largest there.
- In $d = 1$ it also grows with the cutoff, and toward the short-range end $\Delta_\sigma \to 0$ the σ gap that the tuning imposes, $E_\sigma - E_v = \Delta_\sigma$, closes.
- The failure sets in at $\Delta_\sigma \leq 0.05$ in $d = 1$, at $\Delta_\sigma \leq 0.175$ in $d = 2$, and near the endpoint in $d = 3$.

Errors.

- The quoted errors are half-spreads of the extrapolations in the cutoff, not standard deviations, and they do not include the dependence on the window prescription (Section 4.4).
- The exponents of Eq. (14) are fitted, not derived.
- At $\Delta_\sigma \geq 0.15$ in $d = 1$ the spreads are probably optimistic.
- The classification *final* versus *one-sided* is a judgement based on the local exponent.

Order of the analysis.

- The window was designed after its effect had been seen in the descendant spacings.
- The fit protocol of Section 4.2 was carried over unchanged from the earlier `prod1--7` analysis.
- The held-out test and the local exponent were added after the $N = 40$ data had been seen.
- The tail-model and prescription comparisons of Section 4.4 were made after the gap to NRG had been seen. For that reason they are given as shifts and are not turned into an error.

One dimension.

- *Strong coupling.* The $d = 1$ values at $\Delta_\sigma \leq 0.05$ are not limits. At 0.025 and below, the correction overshoots: the sequence turns over and falls without bound in $\ln N$.

- *Comparison with NRG.* Where the descriptions overlap ($0.075 \le \Delta_\sigma \le 0.2$), HT lies above NRG at all six points, by up to 1.1×10^{-3}. At $\Delta_\sigma = 0.1$ and 0.125 the gap exceeds the spread of our extrapolations. However, the uncorrected prescription extrapolates to limits 7×10^{-4}–2.3×10^{-3} lower, below NRG, so the dependence on the prescription is as large as the gap (Section 4.4).

- *Prescription dependence.* The quoted errors of Table 1 do not include the prescription dependence. The raw and window limits differ by $6.6 \times 10^{-4}, 2.3 \times 10^{-3}, 1.7 \times 10^{-3}$ at $\Delta_\sigma = 0.125$, 0.1 and 0.075.

Two dimensions. All $d = 2$ values are finite-N values of the truncated model. With cutoffs up to $N = 24$, and a behaviour that changes with Δ_σ, we quote no continuum limit and no extrapolation error. The window model is known to be correct only through third order in ϵ: expanded in ϵ, the model agrees with the consistent-window scheme, whose fourth-order coefficient grows like $-0.027 \ln N$ (Section 5.6). Even at weak coupling, the $N \to \infty$ limit of the truncated model is therefore not the continuum beyond order ϵ^3. Toward the short-range end the method fails: at $\Delta_\sigma \le 0.175$ the values fall with growing steps, at $\Delta_\sigma = 0.125$ the value at $N = 22$ lies 6% below the exact $\Delta_\varepsilon = 1$ and moves away, and at $N = 24$ the tuning conditions have no solution for $\Delta_\sigma \lesssim 0.1274$. The rise at $0.25 \le \Delta_\sigma \le 0.4$ bounds the truncated model's limit from below only if it continues.

The fold location is an estimate from two values of Δ_σ, and it assumes a saddle-node. The absence of a root at $N = 24$ is established on the scanned range $12.5 \le \theta_0 \le 15.5$, where f_1 has a single interior minimum and rises steeply at both ends. A root on another branch, far from the one continued from smaller N, is not excluded. Finally, the independent solver is not blind (Section 7).

Three dimensions.

- We reach $N = 20$ at every Δ_σ, too few cutoffs for a controlled extrapolation, and quote finite-N values only. Small steps do not mean convergence. At $\Delta_\sigma = 0.7$ the sequence moves away from the resummed ϵ series; it reaches the continuum, if at all, only after a turnover at larger N, so extrapolations of these data are biased low. At 0.65 and 0.675 the steps do not yet shrink.

- Expanded in ϵ, the window model has $C_2(N)$ and $C_3(N)$ approaching their continuum values (`HT_Pert/`). Its fourth-order coefficient at $N = 20$ is 60% of $C_4(3)$ and still growing slowly with N; whether its $N \to \infty$ limit is wrong is not shown.

- The method fails toward the short-range end, where the tuned couplings are largest. At $\Delta_\sigma = 0.518149$ the $N = 20$ value lies 6.6×10^{-3} above the 3d Ising value [3] and still rises; there an approach logarithmic in N is expected.

- The cost limits the cutoff. At $N = 20$ the scalar SO(3) projector takes about 9 h of single-threaded LAPACK, and an $N = 20$ point needs 84–87 GiB, or 328 GiB with the $J = 1$ diagnostics. $N = 22$ was not attempted.

- Four changes to the Expanse campaign tooling of `lri_3d_window_hpc` were not independently inspected: the fixes of the second Codex inspection round (630a217) with a test-fixture fix (e895443), and two campaign options (ccd055a, b330ee7). They change only the cluster tooling and its tests. The numerical source is unchanged, and every record carries its hash (check `REC-SRC`). The production results had no separate review; the independent solver and `HT_Pert/` cross-check them (`code/lri_3d_window_hpc/public/REVIEWS.md`).

9 Reproduction

9.1 Layout

All paths are relative to the folder `HT/` of the public repository.

Folders.

- `code/<package>/`: one package per solver or analysis, shipped at the commit its production ran. Each has a `public.mk` with the targets `test` and `replay` (optionally `smoke`), and a `public/` folder with the replay scripts and a README.
- `data/HT_<d>d/`: the production records.
- `results/HT_<d>d/`: the published tables and the expected values of every quoted number.

Commands.

- `make test` and `make replay` in `HT/` run the targets of every package.
- In a package folder: `make -f public.mk test|replay|smoke`.

A replay recomputes every quoted number of its package from the shipped records, without a production rerun. It prints a PASS/FAIL table and exits non-zero on any failure.

9.2 Packages and records

package (`code/`)	d, role	content, records, cost
`lri_dressed_phi4_hpc`	1: solver	commit 33998bb. Zero mode plus bath, window, density basis, launcher for the symmetry cluster. Records `data/HT_1d/dens_v1` (201) and `prod_window` (680). `test` about 3 min; `smoke` one item
`lri_dressed_phi4_hpc_analysis`	1: analysis	fits, held-out test, identity records, tables. `replay` about 1 min
`lri_dressed_phi4_independent_check`	1: blind check	DVR solver, its per-task outputs and comparison. `replay` seconds
`lri_2d_window_hpc`, `lri_2d_window_indep`	2	Section 5; records `data/HT_2d`
`lri_3d_window_hpc`, `lri_3d_window_indep`	3	Section 6; records `data/HT_3d`

9.3 Rules that are part of the results

- *Estimators.* The fit families, acceptance and stability rules, and estimators of Section 4.2 define the quoted numbers. So do the status words of Table 1.
- *Expected values.* `results/HT_1d/quoted_numbers.json` gives the estimator and comparison rule of every $d = 1$ number. The $d = 1$ numbers in this note are typeset from it.
- *Fit tolerance.* The cutoff fits of $d = 1$ are nonlinear least squares from several starting exponents, and other numpy/scipy versions may land in slightly different minima. The replay therefore checks each number twice:

– against the shipped analysis outputs, exactly;
– against a refit in the reader's environment, within a quarter of the quoted half-spread.

The largest shift measured on symmetry was 0.11 of the half-spread ($\Delta_\sigma = 0.175$).

9.4 Result → code, record, command, tolerance

result	value	pkg	records	command	tolerance
$d = 1$, *finite coupling* (A = `lri_dressed_phi4_hpc_analysis`, I = `lri_dressed_phi4_independent_check`; records D = `data/HT_1d/dens_v1`, R = `data/HT_1d/prod_window`, B = the stored bases in D, ID = `identities/` in A, S = the solver `lri_dressed_phi4_hpc`, NRG = `results/HT_1d/nrg_reference.json`; expected values `results/HT_1d/quoted_numbers.json`; tolerance "status: the stability flag behind the status word)					
smallest cutoff of the fitted series	12	A	D	replay `S1`	exact
largest cutoff, all 13 Δ_σ	40	A	D	replay `S2`	exact
$N_{\min}$ of the fits (smallest, largest)	12–24	A	D	replay `S3`	exact
fits per Δ_σ (F1, F2, F3 × 7 $N_{\min}$)	21	A	D	replay `S4`	exact
density-basis states per parity in production	16	A	D	replay `S5`	exact
training cutoff of the density basis	24	A	B	replay `S6`	exact
zero-mode states per parity of the teacher	48	A	B	replay `S7`	exact
states kept in each stored trained basis (dens16 = its first 16)	24	A	B	replay `S8`	exact
dens16 vs the largest-n_z reference (n_z 48/56) at N = 32: max |difference| ≤ bound	10^{-6}	A	D, R	replay `S9`	bound
stability rule: ≥ 5 fits, median $\omega \geq 0.45$, half-spread ≤ 2e-3 (dens_v1.py stable())	0.45, 2e-3	A	—	replay `S10`	judgement
median ω of the stable fits (min, max over Δ_σ)	0.6–0.9	A	D	replay `S11`	as quoted
raw level-1 violations, $\Delta_\sigma \leq 0.1$, N = 16..28, unambiguous (min, max)	0.06–0.25	A	ID	replay `V1`	as quoted
σ: |raw / window| level-1 violation at N = 28, $\Delta_\sigma \leq 0.15$ (min, max)	10–40	A	ID	replay `V2`	as quoted
window residual of σ and shadow, $\Delta_\sigma \leq 0.1$, N = 28: median $\|v_1\|$ ('about 0.01')	0.01	A	ID	replay `V3`	factor 2
window residual of ε, $\Delta_\sigma \leq 0.05$, N = 28: largest $\|v_1\|$ ('up to')	0.05	A	ID	replay `V4`	sig. figures
cutoffs of the raw identity records behind V1 ($\Delta_\sigma \leq 0.1$; min, max N)	16–28	A	ID	replay `V5`	as quoted
cutoff of the identity table (Table 3, copied in Ref. [4]) and of V2–V4: largest N of the identity records	28	A	ID	replay `V6`	as quoted

result	value	pkg	records	command	tolerance
rows of the identity table at N = 28, Δ_σ = 0.2, 0.15, 0.1, 0.075, 0.05, 0.025: raw and window level-1 violations of σ, ϕ^3 and E, as printed	6 table rows (Δ_σ = 0.2 to 0.025)	A	ID	replay `V7`	exact
ω_{eff} at N $\sim$ 35, Δ_σ = 0.125 (levels off)	0.64	A	D	replay `E1`	as quoted
ω_{eff} at N $\sim$ 35, Δ_σ = 0.1 (levels off)	0.68	A	D	replay `E2`	as quoted
ω_{eff} at N $\sim$ 15 and $\sim$ 35, Δ_σ = 0.075 (rising)	0.70–0.80	A	D	replay `E3`	as quoted
ω_{eff} at N $\sim$ 15 and $\sim$ 35, Δ_σ = 0.05 (rising)	0.92–1.16	A	D	replay `E4`	as quoted
fits on N $\leq$ 32 predict N = 34..40: max |actual - median prediction|, $\Delta_\sigma \geq 0.05$	2.3×10^{-6}	A	D	replay `E5`	sig. figures
Δ_σ = 0.075: slope at N = 40 $\times$ (1/w - 1/(w + 0.2)), w = ω_{eff}(35)	5×10^{-4}	A	D	replay `E6`	sig. figures
Δ_∞ - Δ_ε(N = 40) at Δ_σ = 0.075..0.125 (min, max)	1.6×10^{-3}–2.5×10^{-3}	A	D	replay `E7`	as quoted
Δ_σ = 0.025: cutoff of the largest Δ_ε(N)	22	A	D	replay `E8`	exact
Δ_σ = 0.025: d Δ_ε / d ln N between N = 38 and 40	-1.3×10^{-3}	A	D	replay `E9`	sig. figures
Δ_σ = 0.01: d Δ_ε / d ln N between N = 38 and 40 (constant)	-6.2×10^{-3}	A	D	replay `E10`	sig. figures
Δ_σ = 0.249: Δ_∞ - ϵ series through ϵ^7	-4×10^{-8}	A	D	replay `P1`	sig. figures
Δ_σ = 0.24: band of the partial sums through ϵ^4..ϵ^7	0.49419–0.49426	A	D	replay `P2`	as quoted
Δ_σ = 0.24: Δ_∞ lies inside that band	yes	A	D	replay `P3`	true
level at which the ϵ-series check is trusted	10^{-6}	A	—	replay `P4`	judgement
Δ_ε(N = 40), Δ_σ = 0.249	0.4993430	A	D	replay `T01`	as quoted
Δ_ε(N = 40), Δ_σ = 0.24	0.494186	A	D	replay `T02`	as quoted
Δ_ε(N = 40), Δ_σ = 0.225	0.487982	A	D	replay `T03`	as quoted
Δ_ε(N = 40), Δ_σ = 0.2	0.482839	A	D	replay `T04`	as quoted
Δ_ε(N = 40), Δ_σ = 0.175	0.483541	A	D	replay `T05`	as quoted
Δ_ε(N = 40), Δ_σ = 0.15	0.490388	A	D	replay `T06`	as quoted
Δ_ε(N = 40), Δ_σ = 0.125	0.504561	A	D	replay `T07`	as quoted
Δ_ε(N = 40), Δ_σ = 0.1	0.528210	A	D	replay `T08`	as quoted
Δ_ε(N = 40), Δ_σ = 0.075	0.564851	A	D	replay `T09`	as quoted
Δ_ε(N = 40), Δ_σ = 0.05	0.620671	A	D	replay `T10`	as quoted
Δ_ε(N = 40), Δ_σ = 0.025	0.710577	A	D	replay `T11`	as quoted
Δ_ε(N = 40), Δ_σ = 0.01	0.809111	A	D	replay `T12`	as quoted
Δ_ε(N = 40), Δ_σ = 0.001	0.969338	A	D	replay `T13`	as quoted
Δ_∞ (median of accepted fits, half-spread), Δ_σ = 0.249	0.49934358(9)	A	D	replay `D01`	as quoted
Δ_∞, Δ_σ = 0.24	0.4942198(26)	A	D	replay `D02`	as quoted
Δ_∞, Δ_σ = 0.225	0.488080(5)	A	D	replay `D03`	as quoted
Δ_∞, Δ_σ = 0.2	0.483024(5)	A	D	replay `D04`	as quoted

result	value	pkg	records	command	tolerance
Δ_∞, $\Delta_\sigma = 0.175$	0.483909(15)	A	D	replay `D05`	as quoted
Δ_∞, $\Delta_\sigma = 0.15$	0.491209(33)	A	D	replay `D06`	as quoted
Δ_∞, $\Delta_\sigma = 0.125$	0.50612(4)	A	D	replay `D07`	as quoted
Δ_∞, $\Delta_\sigma = 0.1$	0.53048(12)	A	D	replay `D08`	as quoted
Δ_∞, $\Delta_\sigma = 0.075$: half-spread, then the extra one-sided shift -5e-4 (E6) as a second bracket	0.56733(20)(−50)	A	D	replay `D09`	as quoted
$\Delta_\sigma = 0.05$: accepted F1 and F2 fits (min, max)	0.6220–0.6224	A	D	replay `D10`	as quoted
$\Delta_\sigma = 0.05$: lowest accepted F3 fit	0.618	A	D	replay `D11`	as quoted
total-error estimate at $\Delta_\sigma \geq 0.15$ (the fit spreads are optimistic)	10^{-4}	A	—	replay `D12`	judgement
largest fit half-spread at $\Delta_\sigma \geq 0.15$ ('at most' in the caption of the note's Table 1)	3.3×10^{-5}	A	D	replay `D13`	sig. figures
final vs one-sided ($\Delta_\sigma = 0.075$): judgement from the local exponent, still rising at 0.075 (E3: +0.10 over N ∼ 15..35) but levelling off at 0.125 and 0.1 (E1, E2); the computed rows U01–U13 check stable / unstable / no limit	one-sided at 0.075 only	A	—	replay `U00`	judgement
status, $\Delta_\sigma = 0.249$	final	A	D	replay `U01`	status
status, $\Delta_\sigma = 0.24$	final	A	D	replay `U02`	status
status, $\Delta_\sigma = 0.225$	final	A	D	replay `U03`	status
status, $\Delta_\sigma = 0.2$	final	A	D	replay `U04`	status
status, $\Delta_\sigma = 0.175$	final	A	D	replay `U05`	status
status, $\Delta_\sigma = 0.15$	final	A	D	replay `U06`	status
status, $\Delta_\sigma = 0.125$	final	A	D	replay `U07`	status
status, $\Delta_\sigma = 0.1$	final	A	D	replay `U08`	status
status, $\Delta_\sigma = 0.075$	one-sided	A	D	replay `U09`	status
status, $\Delta_\sigma = 0.05$	unstable	A	D	replay `U10`	status
status, $\Delta_\sigma = 0.025$	no limit	A	D	replay `U11`	status
status, $\Delta_\sigma = 0.01$	no limit	A	D	replay `U12`	status
status, $\Delta_\sigma = 0.001$	no limit	A	D	replay `U13`	status
max $\lvert\Delta_\infty$ - NRG$\rvert$ at $0.075 \leq \Delta_\sigma \leq 0.2$ (NRG: re-analysed final table, `results/HT_1d/nrg_reference.json`); at $\Delta_\sigma = 0.075$	1.1×10^{-3}	A	D, NRG	replay `C1`	sig. figures
ω_{eff} at N ∼ 15 and ∼ 35, $\Delta_\sigma = 0.249$ (slow rise at weak coupling)	0.39–0.46	A	D	replay `E11`	as quoted
$\Delta_\sigma = 0.249$: bias of a further rise of ω_{eff} by 0.2 (as E6) is below	10^{-5}	A	D	replay `E12`	bound
Δ_∞ - NRG at $\Delta_\sigma = 0.125$, 0.1, 0.075 (2 significant figures)	4.5×10^{-4}, 8.7×10^{-4}, 1.1×10^{-3}	A	D, NRG	replay `C2`	sig. figures

result	value	pkg	records	command	tolerance
$\Delta_\sigma = 0.001$: Δ_ε(N = 40) - NRG	0.032	A	D, NRG	replay `C3`	as quoted
A = $\Delta_\sigma/4$ + r - g/2 at the tuned couplings (dens16): largest Δ_σ at which its sign changes with the cutoff (A < 0 at every N for $\Delta_\sigma = 0.2$, 0.175; > 0 at $\Delta_\sigma \geq 0.225$)	0.15	A	D	replay `S12`	as quoted
energy zero-mode basis at $\Delta_\sigma = 0.025$, N = 28, consistent window: Δ_ε for n_z = 32, 40, 48 (prod records)	0.710632, 0.710934, 0.710959	A	R	replay `B1`	as quoted
largest fit half-spread among the final results ($\Delta_\sigma \geq 0.1$)	1.2×10^{-4}	A	D	replay `D14`	sig. figures
largest tail-model shift (F4 with two free exponents; drifting local exponent) of the window limit at $0.075 \leq \Delta_\sigma \leq 0.125$	1.6×10^{-4}	A	D	replay `M3`	sig. figures
NRG lies in the bracket [$\Delta_\varepsilon(40)$, $\Delta_\varepsilon(40)$ + slope(40)/$\omega_{\text{eff}}(35)$] of the window series at every $0.075 \leq \Delta_\sigma \leq 0.2$	yes	A	D, NRG	replay `M4`	true
the raw limit lies below NRG and the window limit above it at $0.075 \leq \Delta_\sigma \leq 0.125$	yes	A	D, R, NRG	replay `M5`	true
window minus raw limit at $\Delta_\sigma = 0.125$, 0.1, 0.075 (2 significant figures)	6.6×10^{-4}, 2.3×10^{-3}, 1.7×10^{-3}	A	D, R	replay `M6`	sig. figures
largest |window - raw| limit difference at $\Delta_\sigma = 0.2$, 0.175, 0.15	1.3×10^{-4}	A	D, R	replay `M7`	sig. figures
smallest Δ_∞ - NRG at $0.075 \leq \Delta_\sigma \leq 0.15$ (at 0.15)	10^{-4}	A	D, NRG	replay `C5`	sig. figures
Δ_∞ - NRG at $\Delta_\sigma = 0.175$, 0.2 is below 1e-4	10^{-4}	A	D, NRG	replay `C6`	bound
$\Delta_\infty >$ NRG at every $0.075 \leq \Delta_\sigma \leq 0.2$	yes	A	D, NRG	replay `C7`	true
Δ_∞ - NRG exceeds the fit half-spread at $\Delta_\sigma = 0.1$ and 0.125	yes	A	D, NRG	replay `C8`	true
max |Δ_∞ - NRG| / NRG at $0.075 \leq \Delta_\sigma \leq 0.2$ is at most 2e-3	2×10^{-3}	A	D	replay `C9`	bound
window minus raw limit at $0.075 \leq \Delta_\sigma \leq 0.125$ (min with 1, max with 2 significant figures)	7×10^{-4}–2.3×10^{-3}	A	D, R	replay `M10`	sig. figures
blind solver vs production, 38 points	5×10^{-9}	I	I `results/`, D, R	replay `B6` (B4, B5)	1 sig. fig.
blind solver vs dens16, $N \leq 20$	$\leq 10^{-6}$ (5.6×10^{-7})	I	same	replay `B7`	bound
weak coupling $C_2(N)$, $N \leq 16$	$\leq 3 \times 10^{-7}$	I	I `results/`	replay `B8`	bound

result	value	pkg	records	command	tolerance
window at $O(\epsilon^4)$ in the blind solver	exponent ≈ 4	I	I `results/`	replay `B9`	3.8–4.3
records intact and bound to the code	1106 records	A, S	D, R	replay `R1`–`R7`, S `F1`, `F2`	exact
rows, refits, tables	as shipped	A	D	replay `W1`, `A0`–`A4`, `T1`	exact; refit $\leq \frac{1}{4}$ half-spread

$d = 2$, *finite coupling.* Packages: P = `lri_2d_window_hpc`, I = `lri_2d_window_indep`, HP = `HT_Pert/code/lri_2d_coeff_independent_check`. Records: R = `data/HT_2d/prod1/chains`, V = `results/HT_2d/valley_N24`, O = `results/HT_2d/independent_check/out_sym`, C = `data/HT_2d/pert_crosscheck/consistent_C0_C8.csv`. Command: replay = `make -f public.mk replay` in the package, followed by the check name.

record integrity; source hash = code 889a968	348 records	P	R	replay `d2.records.*`	exact
largest cutoff (22 at $\Delta_\sigma = 0.125$, 0.126)	$N = 24$	P	R	replay `d2.Nmax`	exact
$\Delta_\varepsilon(N)$, $N = 8, \ldots, 24$, Table 4, Fig. 2	13 chains	P	R; `results/HT_2d/prod1/points.csv`	replay `d2.points`	exact
values at the largest N (Ref. [4], Fig. `fig:cmp2d3d`, Table `tab:num2`)	13 values	P	`results/HT_2d/prod1/largest_N.csv`	replay `d2.largest_N`; `d2.num2`	exact; 5 decimals
last step, $\Delta_\sigma \geq 0.45$ (largest at 0.45)	$\leq 2.6 \times 10^{-6}$ (2.54×10^{-6})	P	R	replay `d2.conv_step`	bound
shrink per step, $\Delta_\sigma \geq 0.45$	18–22%	P	R	replay `d2.conv_shrink`	integer %
rise, $0.25 \leq \Delta_\sigma \leq 0.4$	monotone	P	R	replay `d2.mono.rises`	sign
last steps, $0.25 \leq \Delta_\sigma \leq 0.4$	$(0.6\text{–}4.3) \times 10^{-4}$	P	R	replay `d2.mono_steps`	2 digits
shrink per step, $0.25 \leq \Delta_\sigma \leq 0.4$	8–16%	P	R	replay `d2.mono_shrink`	integer %
$\Delta_\sigma = 0.2$	not monotone	P	R	replay `d2.nonmono_0.2`	sign
fall, $0.125 \leq \Delta_\sigma \leq 0.175$	falls, steps grow	P	R	replay `d2.fall`	sign, ratio
last step, $\Delta_\sigma = 0.175$	-1.0×10^{-3}	P	R	replay `d2.fall_small`	2 digits
last step, $\Delta_\sigma = 0.125$, $N = 20 \to 22$	-2.0×10^{-2}	P	R	replay `d2.fall_large`	2 digits
n_z convergence: $n_z = 20$; 12 against 16	9.4×10^{-11}; 5.2×10^{-7}	P	R	replay `d2.nz.*`	10^{-10}; 6×10^{-7}
reference points ($N = 14$) against $n_z = 16$	$\leq 10^{-9}$	P	R (`ref*`)	replay `d2.refs`	10^{-9}
Jacobian condition at the roots, $\Delta_\sigma = 0.125$, $N = 12$; 22	84; 888	P	R	replay `d2.cond`	integer
stalled Newton residual, $N = 24$, $\Delta_\sigma = 0.125$; 0.126	2.93×10^{-3}; 1.99×10^{-3}	P	V	replay `d2.valley.best`	2 digits
no root at $N = 24$, $\Delta_\sigma = 0.125$, 0.126: $f_1 > 0$ on $f_2 = 0$	smallest f_1: 1.60×10^{-3}	P	V	replay `d2.no_root`	sign
valley minima, $\Delta_\sigma = 0.125$; 0.126	2.686×10^{-3}; 1.556×10^{-3}	P	V	replay `d2.valley.min`	10^{-6}
fold	$\Delta_\sigma \approx 0.1274$	P	V	replay `d2.fold`	5×10^{-5}

result	value	pkg	records	command	tolerance
weak coupling: model agrees with the consistent scheme, $\Delta_\sigma = 0.495$; 0.475	9.4×10^{-12}; 7.6×10^{-8}	P	R, C	replay `d2.pert_consistent.*`	10^{-11}; 10^{-7}
$\ln N$ coefficient of $C_4(N)$, $-365/(4374\pi)$	-0.027	HP	exact $C_4(N)$, $N \le 2000$	replay `p2.c4_log_consistent`	set in HP
independent solver, $N = 12$, 16	1.4×10^{-12}	I	O, R	replay `i2.dE.N12_16`	2 digits
independent solver, levels	7.6×10^{-12}	I	O, R	replay `i2.levels.N12_16`	10^{-11}
independent solver, $N = 18$; 20	4.5×10^{-12}; 4.6×10^{-11}	I	O, R	replay `i2.dE.N18`; `N20`	5×10^{-12}; 5×10^{-11}
independent solver, $N = 22$, $\Delta_\sigma = 0.125$; 0.126	4.4×10^{-10}; 4.0×10^{-10}	I	O, R	replay `i2.dE.N22`	2 digits
independent re-tuning, $N \le 12$; $N = 16$	9×10^{-11}; 3.6×10^{-8}	I	O, R	replay `i2.retune.*`	10^{-10}; 4×10^{-8}
$d = 3$, *finite coupling* (P = `lri_3d_window_hpc`, I = `lri_3d_window_indep`, C = `data/HT_3d/campaigns`)					
records intact (`REC-*`)	105 records, 1 source hash	P	every `window_L*.json` in C; `data/HT_3d/MANIFEST.sha256`	`make replay`	sha256, self-hash, seed and basis chain, acceptance gates: exact
$\Delta_\varepsilon(N)$, $N = 8$–20, 15 Δ_σ (Table 5; comparison figure of Ref. [4]) (`TABLE`)	`finite_L_table.csv`	P	the 105 window records	`make replay`	byte-identical rebuild
largest cutoff (`NMAX`)	$N = 20$	P	15 `window_L20.json`	`make replay`	exact
$N = 20$ values in the $d = 3$ results table of Ref. [4] (`NUM3`)	15 values, 1.41924 … 1.49993	P	15 `window_L20.json`	`make replay`	5 decimals
setup: teacher $N_t = 6$, 24 states, $N_{\mathrm{osc}} = 192$, 20 stored; $n_z = 16$ (`CONFIG`)	as stated	P	`teacher_L6.json`, window records	`make replay`	exact
scalar bath states, $N = 8$–20 (`SIZE`)	110 … 196 585	P	window records (dimension/$2n_z$)	`make replay`	exact
projector retries (`PROJECTOR`)	$N = 18$: 8/17 335; $N = 20$: 62/40 033; $\lvert J_+Q\rvert < 10^{-14}$	P	$N = 18$, 20 window records	`make replay`	exact counts
cost at $N = 18$ (`COST-L18`)	34–74 min, 38.5 GiB	P	$N = 18$ window records	`make replay`	integer minutes, 0.1 GiB
cost at $N = 20$ (`COST-L20`)	27–104 min, 84–87 GiB; 239 min, 328 GiB	P	$N = 20$ window records	`make replay`	integer minutes, GiB
last step at $\Delta_\sigma \ge 0.74$ (`CONV-STEP`)	$< 10^{-7}$ (7.3×10^{-8})	P	$N = 18$, 20	`make replay`	$< 10^{-7}$
last steps at $0.518 \le \Delta_\sigma \le 0.625$ (`MONO-STEPS`)	$(2.2\text{–}6.1) \times 10^{-4}$	P	$N = 18$, 20	`make replay`	2 significant digits
monotone rise there (`MONO-RISE`)	every step > 0, $N = 8$–20	P	$N = 8$–20	`make replay`	sign

result	value	pkg	records	command	tolerance
shrink per step there (`MONO-SHRINK`)	8 to 19%	P	$N = 16$–20	`make replay`	integer percent
no shrink at 0.65, 0.675 (`NO-SHRINK`)	$d_{20}/d_{18} = 0.99$, 1.23	P	$N = 16$–20	`make replay`	ratio > 0.99
short-range end (`SR-L20`)	1.41924, rising	P	`grid20sri`, $N = 20$	`make replay`	5 decimals
distance to the 3d Ising value (`SR-GAP`)	6.6×10^{-3}	P	same; Ising value 1.41262528 [3]	`make replay`	2 significant digits
resummed ϵ series at $\Delta_\sigma = 0.7$ (`BP-0.7`)	1.47214(3)	P	C_0–C_7(3) (GFF folder)	`make replay` (`public/bp_x07.py`)	median to 10^{-7}; half-spread 1 digit
$N = 20$ below it and moving away (`BP-DEV`)	2.2×10^{-4}	P	`grid20`, $N = 20$; `grid18`, $N = 16$, 18; at 0.7	`make replay`	2 significant digits; order
$C_4(N = 20)/C_4(3)$ (`C4-RATIO`)	60%	P	`HT_Pert/results/HT_3d/SUMMARY.md`, row 20 consistent	`make replay` (SKIP without `HT_Pert/`)	integer percent
independent solver, tuned (`INDEP-TUNED`)	2×10^{-11} ($N \le 12$)	I	`results/cmp_*.json` (23)	`make replay`	1 significant digit
independent solver, fixed couplings (`INDEP-FIXED`)	$\le 1.8 \times 10^{-12}$	I	same	`make replay`	$\le 3 \times 10^{-12}$
independent tunings converged (`INDEP-RESID`)	$\le 10^{-12}$	I	same	`make replay`	$\le 10^{-12}$
compared values are the records (`INDEP-PKG`)	exact	I	`cmp_*.json`, `indep_extract.json`, C	`make replay`	exact
$J = 0$ counts 3, 10, 33, 110 (`test`)	as stated	I	none	`make test`	exact

A Provenance and reviews

Who wrote what. Every package records its origin in `PROVENANCE.md`, and every review is summarized in `REVIEWS.md`.

$d = 1$ **solver.** `lri_dressed_phi4_hpc` went through three stages.

- The author supplied it as a matrix-free rewrite of an earlier dressed zero-mode solver (commit e637dfd), written by OpenAI GPT-6 astra.
- Claude Code sessions (Anthropic, model claude-opus-5-5) added the layer that runs it on the symmetry cluster (node packing, cross-node locks, validation records), the window correction and the descendant diagnostic (157a2c6 to e2ca07c). Codex (OpenAI, gpt-6-astra) reviewed the plan, which was approved in round 4, and inspected the code twice. All 20 findings were fixed; the last eight fixes were not re-inspected.
- The density basis was supplied by the author as a patch (33998bb). A fresh Codex inspection found two medium defects in scripts the production does not use; they were deferred.

Both supplied archives (e637dfd and the density-basis patch) were written by OpenAI GPT-6 astra via ChatGPT (model stated by the author).

$d = 1$ **analysis and records.**

- The analysis scripts, the campaigns and this note were written by Claude Code sessions.
- The blind $d = 1$ solver was written by a Claude Code subagent (Section 7).
- No Codex results review was run on the $d = 1$ finite-coupling numbers. The public-release replay found three paper values that the stated estimators did not reproduce: the $\Delta_\sigma = 0.05$ range, the bound on the weak-coupling spreads, and the basis error. It also found one value quoted beyond its reproducible precision, the ϵ-series difference at $\Delta_\sigma = 0.249$. All four were corrected on 2026-10-13.

$d = 2, 3$**.** See the provenance files of `lri_2d_window_hpc` and `lri_3d_window_hpc`.

This note. Written by the Claude Code session `phi4HT1d`. The $d = 2$ and $d = 3$ sections were written by the sessions `phi4HT2d` and `phi4HT3d` and edited here only for consistency.

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Part V

Thermal dimension of the one-dimensional long-range Ising model from bosonic NRG of the sub-ohmic spin–boson model

Method, data, checks, results, limitations and reproduction

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Status of 15 October 2026

Abstract

We compute the scaling dimension Δ_ε of the energy operator of the one-dimensional long-range Ising model, with couplings decaying as $1/r^{1+s}$, at thirteen values of the spin dimension $\Delta_\sigma = (1-s)/2$ between 0.249 and 0.001. The critical point is realized as the quantum critical point of the sub-ohmic spin–boson model, which we solve with a bosonic numerical renormalization group (NRG); the relevant eigenvalue $y_t = 1 - \Delta_\varepsilon$ is obtained from the exact derivative of the truncated NRG map with respect to the bath coupling. At ten points, $0.001 \le \Delta_\sigma \le 0.2$, we give final values, for example $\Delta_\varepsilon = 0.529608(33)$ at $s = 0.8$ and 0.4830(7) at $s = 0.6$. At eight of them, $0.001 \le \Delta_\sigma \le 0.15$, the local exponent settles on a plateau along the chain, and the error scales are (2.7–5.4) $\times 10^{-5}$ for $0.01 \le \Delta_\sigma \le 0.125$, 1.1×10^{-4} at $\Delta_\sigma = 0.15$ and 1.1×10^{-4} at $\Delta_\sigma = 0.001$. At $\Delta_\sigma = 0.175$ and 0.2 the local exponent is still drifting at the end of the usable chain and is extrapolated, with error scales 3.5×10^{-4} and 7.1×10^{-4}. Each error adds three parts linearly: the drift along the chain, the spread over kept states and oscillator levels, and a discretization error, the larger of the spread over the values of Λ that were run and the change of the $\Lambda = 2.5$ value when the number of kept states is doubled; the latter sets it at 8 of the 10 final points. The errors are not standard deviations. Near $s = 1/2$ the leading correction exponent $\omega \approx \varepsilon$ vanishes, and within the chain depth reachable in double precision the flow does not reach its fixed point: at $\Delta_\sigma = 0.249$, 0.24 and 0.225 the values where it comes closest lie $+8.7\times10^{-3}$, $+4.7\times10^{-3}$ and $+1.6\times10^{-3}$ above the ε expansion, and at $\Delta_\sigma = 0.249$ the inverse excess grows along the renormalization-group time at a rate of the size that one loop predicts. We label these three points not converged in chain depth, a label assigned after the comparison with the series, show their values, and use them in no comparison. Near $s = 1$ the results reproduce the strong–weak dual expansion of Δ_ε through its fifth coefficient. At the 5 points where the project's cluster Monte Carlo study accepts its own value, NRG differs from it by at most 2.9×10^{-3}. That is within the sum of the Monte Carlo statistical error, its fit-model half-range and the NRG error at 4 points, and 1.08 times that sum at the fifth; but it is up to 14 Monte Carlo statistical errors, with the same sign at neighboring points, so only the fit-model half-range reconciles the two. The Monte Carlo crossing extrapolations agree with NRG within 1.05 of their statistical errors at these points. A second solver, written by a separate Claude agent from a physics-only specification, reproduces the chain coefficients to better than 10^{-15} (rounding level), the critical couplings bitwise or to 7.2×10^{-14}, and Δ_E to 3.1×10^{-7} in every window its finite differences can reach. The code was written by GPT-6 astra (OpenAI) and ported to the cluster, run and analyzed by Claude Code (Anthropic); every number in this note is regenerated from the released records by `make replay`.

1 Overview

1.1 Scope

This note is written for a physicist outside the project. Section 2 states the problem and what is known analytically. Sections 3 and 4 describe the numerical method, the estimator and the error budget in enough detail to re-implement them. Section 5 describes the code and how it was validated, Sec. 6 the production runs, Sec. 7 the results, and Sec. 8 the checks against a second implementation, against two methods that do not use NRG and against published values. Section 9 collects the limitations and Sec. 10 explains how to recompute every number. The appendices give the data formats (App. A), the production on the cluster (App. B), the provenance of code, analysis and review (App. C), every production run (App. D), the estimator used before this revision and its values (App. E), and the terms and symbols (App. F).

Who did what. The NRG solver was written by GPT-6 astra (OpenAI, through ChatGPT) in two stages: a reference solver for an earlier, unreleased study of the project (the *original study*; shipped as `code/lri_spin_boson_hpc/reference/`) and an optimized package built on it. Claude Code (Anthropic) reviewed the package, ported it to a Slurm cluster, ran the production, wrote the analysis and drafted this note; OpenAI Codex reviewed the cluster port. Two rounds of read-only reviews of the analysis and of this note, three reviews each, were made by Claude agents on 13 October 2026, and an independent reader reviewed the text on 15 October. This version follows them; the changes made after the second round and after the reading were not reviewed again. There was no human referee. App. C gives the details, including which model ran where. In this note "the author" is Z. Zheng, who made the decisions marked as his, and "we" is the study as a whole: the author and the AI systems that worked for him.

Vocabulary. A *point* is one value of Δ_σ. An *item* is one complete NRG calculation at fixed $(\Delta_\sigma, \Lambda, K, n_b, N)$: the search for the critical coupling, the exact derivative of the critical flow, and the window audit (Sec. 3). Its *record* is the JSON file that holds the result. The *central item* of a point is the one at $\Lambda = 3$, $K = 128$, $n_b = 16$; all other items are convergence scans. A *campaign* is a set of items submitted together (Sec. 6), and *production* means the 197 items whose records are released. A *window* is a stretch of consecutive chain sites over which the growth of the derivative is fitted (Sec. 3.6). We write Δ_E for the NRG estimate of Δ_ε; despite the notation it is not an energy difference. App. F lists all terms and symbols.

1.2 Headline results

Table 1 is the main result, and Fig. 1 shows it with the references of Sec. 8. "Final" values are the estimates of Sec. 3.7 with the total error $\delta_{\rm tot}$ of Sec. 4. The three values in gray brackets are not converged in chain depth (Sec. 7.1): they are shown, carry no error, and enter no comparison and no agreement statement in this note. That label is not an outcome of the estimator; it was assigned after the comparison with the ε series (Sec. 3.7).

What the checks establish:

- **The code computes what it claims.** A second implementation written from a physics-only specification (Sec. 8.1) agrees with the production solver in eight configurations: chain coefficients to better than 10^{-15} (relative, rounding level), critical couplings bitwise in 5 of 8 configurations and to 7.2×10^{-14} in the others, and Δ_E window by window to 3.1×10^{-7} wherever its finite differences are reliable. This tests the numerical map, not the estimator.
- **Comparison with methods that do not use NRG.** Near $s = 1$ the final values reproduce the dual expansion of Δ_ε in $\sqrt{\Delta_\sigma}$ through its fifth coefficient (Sec. 7.5). At the 5 points where the Monte Carlo study accepts its own value they differ from it by at most 2.9×10^{-3}: within $\sigma_{\rm stat} + \sigma_{\rm sys} + \delta_{\rm tot}$ at 4 points and 1.08 times that sum at $\Delta_\sigma = 0.1$. The sum adds a standard deviation, a fit-model half-range and an error scale and is only a scale, not a test of statistical consistency. The differences are up to 14 Monte Carlo statistical errors and have the same sign at neighboring points, so only the Monte Carlo half-range $\sigma_{\rm sys}$ reconciles the two, and the comparison tests the Monte Carlo systematics more than NRG (Sec. 8.3). At $\Delta_\sigma = 0.2$ the resummed series has a spread of 9×10^{-3} and does not test NRG.

Table 1: Thermal dimension Δ_ε of the $d = 1$ long-range Ising model from bosonic NRG: $\Delta_E = 1 - y_t$ at $\Lambda = 3$, $K = 128$ kept states per parity sector and $n_b = 16$ oscillator levels, with the total error $\delta_{\rm tot}$ in parentheses (Sec. 4: chain depth plus K/n_b spread plus discretization error, the last mostly an allowance for the convergence in K at $\Lambda = 2.5$; PDG rounding). "Final" is the plateau value of Sec. 3.7; "final, extrapolated" is the extrapolation of the local exponent. Gray: not converged in chain depth; the settled value, no error, not used in any comparison, and its excess over the ε series in the last column. External test: the ε series (Sec. 8.2), whose spread at $\Delta_\sigma = 0.2$ is far larger than the NRG error; the label that the Monte Carlo study gives its own value (Sec. 8.3) and, where that value is accepted, NRG minus Monte Carlo with its ratio to $\sigma_{\rm stat} + \sigma_{\rm sys} + \delta_{\rm tot}$; at $\Delta_\sigma \le 0.01$ only the expansion about $s = 1$ of Sec. 7.5.

Δ_σ	s	ε	Δ_E	status	external test (Sec. 8)
0.249	0.502	0.004	[0.5080]	not converged	series 0.499344 (excess $+8.7 \times 10^{-3}$)
0.24	0.52	0.04	[0.4989]	not converged	series 0.49422 (excess $+4.7 \times 10^{-3}$)
0.225	0.55	0.1	[0.4897]	not converged	series 0.48808 (excess $+1.6 \times 10^{-3}$); MC label: conditional
0.2	0.6	0.2	0.4830(7)	final, extrapolated	series 0.48292 (spread 9×10^{-3}: no test)
0.175	0.65	0.3	0.48390(35)	final, extrapolated	MC label: cross-check failed (not used)
0.15	0.7	0.4	0.49111(11)	final	MC label: cross-check failed (not used)
0.125	0.75	0.5	0.50567(5)	final	MC label: accepted; NRG−MC -1.4×10^{-3} (ratio 0.61)
0.1	0.8	0.6	0.529608(33)	final	MC label: accepted; NRG−MC -2.9×10^{-3} (ratio 1.08)
0.075	0.85	0.7	0.566260(30)	final	MC label: accepted; NRG−MC -6.8×10^{-4} (ratio 0.38)
0.05	0.9	0.8	0.622115(27)	final	MC label: accepted; NRG−MC $+2.2 \times 10^{-3}$ (ratio 0.81)
0.025	0.95	0.9	0.713422(29)	final	MC label: accepted; NRG−MC $+2.3 \times 10^{-3}$ (ratio 0.22)
0.01	0.98	0.96	0.80975(5)	final	$s \to 1$ expansion only
0.001	0.998	0.996	0.93742(11)	final	$s \to 1$ expansion only

- **Near $s = 1/2$ the chain is too short.** The flow approaches its fixed point at a rate $\omega \approx \varepsilon \to 0$ in the RG time l. At $\Delta_\sigma = 0.249$ the inverse distance to the series value grows nearly linearly in l, with slopes 2.55 to 3.03, of the size of the one-loop prediction (3.14–3.17 to 3.44–3.47; Fig. 3c, Sec. 7.1). Convergence scans in K, n_b or Λ at fixed depth cannot reveal this, because once K and $n_b \ge 12$ are converged the effect depends only on l.

The errors are error scales built from measured spreads and a few stated judgements (Sec. 4); they are not standard deviations. In the text they are quoted with two significant digits; in the tables value(error) follows the PDG rounding (two digits if the leading digits are 10–35, else one). Bounds in the text ("at most", "agree to", "below") are rounded up, so they can exceed the corresponding table entry by one unit of the last digit. Monte Carlo values and the coefficients C_7, C_8 are quoted as their sources give them.

2 The problem and what is known

2.1 The long-range Ising chain and its thermal dimension

The ferromagnetic long-range Ising (LRI) chain has spins $S_i = \pm 1$ on the sites of a line ($d = 1$) and couplings $J_{ij} \propto 1/|i-j|^{1+s}$. It has a finite-temperature transition for $0 < s \le 1$; the transition is mean-field for $s < 1/2$, of Kosterlitz–Thouless type at $s = 1$ [6, 7], and for $1/2 < s < 1$ its critical point is neither mean-field nor short-range [4, 5]. There the spin field has the exactly known dimension

$$\Delta_\sigma = \frac{d-s}{2} = \frac{1-s}{2}, \qquad s = 1 - 2\Delta_\sigma, \qquad \varepsilon \equiv 2s - d = 2s - 1 = 1 - 4\Delta_\sigma, \tag{1}$$

because long-range interactions are not renormalized [5]. The quantity of interest is the dimension of the energy operator, the leading $\mathbb{Z}_2$-even scalar,

$$\Delta_\varepsilon = d - \frac{1}{\nu} = 1 - y_t, \tag{2}$$

where ν is the correlation-length exponent and $y_t = 1/\nu$ the renormalization-group (RG) eigenvalue of the temperature.

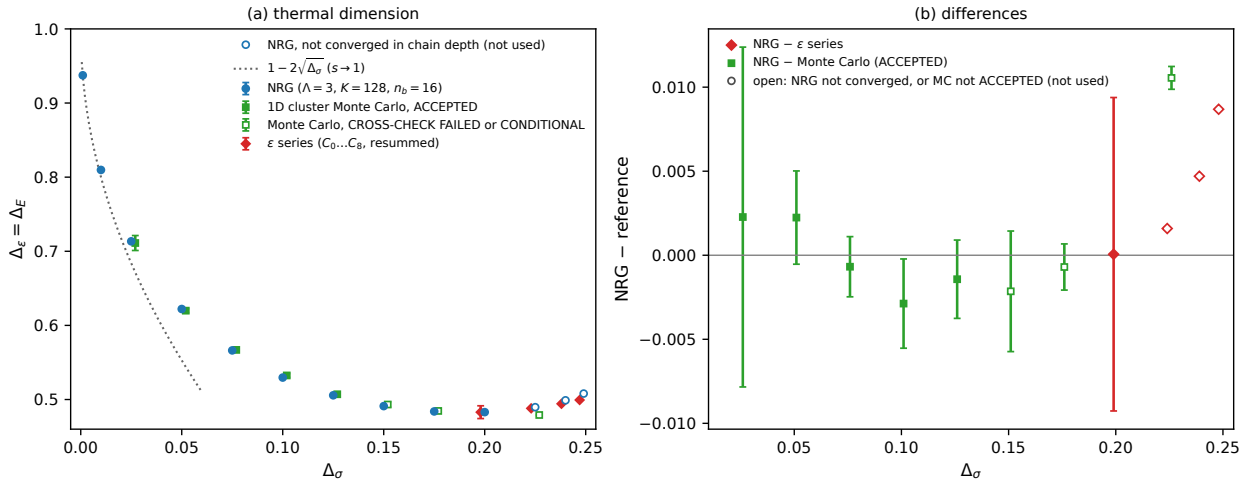


Figure 1: (a) Δ_E against Δ_σ: NRG (filled circles with $\delta_{\rm tot}$; open: not converged in chain depth, not used), the project's 1D cluster Monte Carlo study (filled squares: label ACCEPTED; open: CROSS-CHECK FAILED or CONDITIONAL, which that study does not certify as infinite-size values; bars $\sigma_{\rm stat}+\sigma_{\rm sys}$, added linearly), the resummed ε series (diamonds; bars: spread over Padé approximants, at $\Delta_\sigma=0.249$ the partial-sum error) and $1-2\sqrt{\Delta_\sigma}$ near $s=1$. (b) NRG minus reference; bars are $\delta_{\rm tot}$ plus the reference's error, added linearly. Open symbols are not used in any comparison.

Two limits are known analytically:

- $s\to1/2^+$ **($\Delta_\sigma\to1/4$, $\varepsilon\to0$).** The fixed point becomes Gaussian, $\Delta_\varepsilon\to1/2$, and Δ_ε has an expansion $\Delta_\varepsilon=\sum_n C_n\varepsilon^n$ with $C_0=1/2$, $C_1=-1/6$ and higher coefficients known through C_8 (Sec. 8.2, Ref. [1]). The series is asymptotic, but close to $s=1/2$ it is far more precise than any numerical method. The leading irrelevant coupling (φ^4) has the correction exponent $\omega=\varepsilon+O(\varepsilon^2)$, so corrections to scaling decay ever more slowly as $s\to1/2$.
- $s\to1^-$ **($\delta\equiv1-s=2\Delta_\sigma\to0$).** The two lowest $\mathbb{Z}_2$-even dimensions are $\Delta_\pm=1\pm\sqrt{2\delta}+\delta/4+O(\delta^{3/2})$ [8, 9]; the leading term is Kosterlitz's [6]. With $\Delta_-\sim\varphi^2$ the energy operator [8],

$$y_t=\sqrt{2\delta}-\frac{\delta}{4}+O(\delta^{3/2}),\qquad \omega=\Delta_+-1=\sqrt{2\delta}+\frac{\delta}{4}+O(\delta^{3/2}). \tag{3}$$

So $\omega\approx y_t$ near $s=1$: both are small, but they are comparable (Sec. 3.9). Beyond this order the dual description gives Δ_ε as a series in $\sqrt{\Delta_\sigma}$,

$$\Delta_\varepsilon=1+\sum_{n\ge1}C_n^{\rm SR}(1)\,\Delta_\sigma^{n/2},\qquad C_1^{\rm SR}(1)=-2,\quad C_2^{\rm SR}(1)=\tfrac12, \tag{4}$$

equivalent to Eq. (3) through its second term, with $C_3^{\rm SR}(1)$ to $C_8^{\rm SR}(1)$ computed in the accompanying paper [1]; $C_3^{\rm SR}(1)=\frac{39}{16}+2\zeta(2)$, $C_4^{\rm SR}(1)=4-4\zeta(2)-8\zeta(3)$, $C_5^{\rm SR}(1)=\frac{11249}{1024}-\frac{361}{16}\zeta(2)+24\zeta(3)+\frac{49}{2}\zeta(4)$ and $C_6^{\rm SR}(1)=26-38\zeta(2)+104\zeta(3)-80\zeta(4)-272\zeta(5)+112\zeta(2)\zeta(3)$.

2.2 The spin–boson model

The spin–boson model without a static bias field describes a two-level system coupled to a bath of harmonic oscillators [10]:

$$H=-\frac{\Omega}{2}\,\sigma_x+\sum_j\omega_j\,b_j^\dagger b_j+\frac{\sigma_z}{2}\sum_j\lambda_j\left(b_j+b_j^\dagger\right),\qquad \rho(\omega)\equiv\sum_j\lambda_j^2\,\delta(\omega-\omega_j)=2\alpha\,\omega^s\quad(0<\omega\le1). \tag{5}$$

Here $\sigma_{x,z}$ are Pauli matrices, Ω is the tunneling amplitude (often called Δ in the literature), b_j annihilates a bath oscillator of frequency ω_j, λ_j is its coupling, $\rho(\omega)$ is the bath spectral density (the convention of Ref. [12], equal to $J(\omega)/\pi$ in Ref. [10]), α is the dimensionless dissipation strength and $0<s<1$ ("sub-ohmic"). The bath cutoff frequency is the unit of energy, and $\Omega=0.1$ throughout; universality in Ω was not tested (Sec. 9).

For $0 < s < 1$ the model has a quantum phase transition at a critical coupling $\alpha_c(s, \Omega)$ [12, 16]. For $\alpha < \alpha_c$ (delocalized phase) the ground state is unique; for $\alpha > \alpha_c$ (localized phase) it is twofold degenerate, the spin being trapped in one of the eigenstates of σ_z. In the σ_z basis the imaginary-time path integral is a classical Ising chain along the imaginary-time axis, with lattice spacing of order the inverse cutoff [10, 16]: the tunneling Ω gives a short-range coupling between neighboring time slices, and integrating out the bath gives a long-range coupling $\propto \alpha/\tau^{1+s}$, so α plays the role of the inverse temperature. The quantum critical point is therefore in the universality class of the $d = 1$ LRI model with the same s: $\langle \sigma_z(\tau)\sigma_z(0)\rangle \propto \tau^{-2\Delta_\sigma}$ there, and $\alpha - \alpha_c$, which is $\mathbb{Z}_2$-even, couples to the energy operator, so its RG eigenvalue is the y_t of Eq. (2). The other even coupling, Ω, feeds the same single relevant even direction.

2.3 The thirteen points

The study covers the thirteen values of Δ_σ in Table 1, chosen for a point-by-point comparison with other methods of the project. The outer points $\Delta_\sigma = 0.249$ ($s = 0.502$) and 0.001 ($s = 0.998$) are the hardest for every method: near $s = 1/2$ the fixed point is almost Gaussian and the approach to it is slow, near $s = 1$ the exponent y_t is small.

3 Method: bosonic NRG with an exact α-derivative

NRG [11, 14], extended to bosonic baths in Refs. [12, 13], solves an impurity coupled to a bath by mapping the bath onto a chain whose sites carry exponentially decreasing energy scales, and then diagonalizing the chain one site at a time, keeping only the lowest states. We use the standard formulation; the non-standard elements are the way y_t is extracted (Secs. 3.5–3.7).

3.1 Logarithmic discretization

The bath interval $(0, 1]$ is divided into bins $I_m = (\Lambda^{-(m+1)}, \Lambda^{-m}]$, $m = 0, 1, 2, \ldots$, with a discretization ratio $\Lambda > 1$. Each bin is replaced by a single oscillator c_m that carries the bin's spectral weight at its mean frequency [12, 13]. With $q = 1/\Lambda$, this "star" representation has

$$\gamma_m^2 = \int_{I_m} \rho(\omega)\, d\omega = \frac{2\alpha\,(1 - q^{s+1})}{s+1}\, q^{m(s+1)}, \qquad \xi_m = \frac{\int_{I_m} \omega\rho(\omega)\, d\omega}{\gamma_m^2} = \xi_0\, q^m, \qquad \xi_0 = \frac{s+1}{s+2}\,\frac{1 - q^{s+2}}{1 - q^{s+1}}, \quad (6)$$

where γ_m is the coupling and ξ_m the frequency of oscillator m. Every decade of frequency gets the same number of modes, however low; dropping all other modes of a bin is the approximation of this step, which becomes exact as $\Lambda \to 1$.

3.2 Wilson chain

An orthogonal change of basis among the c_m (Lanczos tridiagonalization, started from the combination $\sum_m \gamma_m c_m$ that couples to the spin) maps the star onto a semi-infinite chain:

$$H = -\frac{\Omega}{2}\sigma_x + \frac{\lambda_{\mathrm{tot}}}{2}\,\sigma_z\,(b_0 + b_0^\dagger) + \sum_{n\ge 0} e_n\, b_n^\dagger b_n + \sum_{n\ge 1} t_{n-1}\,(b_{n-1}^\dagger b_n + b_n^\dagger b_{n-1}), \qquad \lambda_{\mathrm{tot}}^2 = \sum_m \gamma_m^2 = \frac{2\alpha}{s+1}. \quad (7)$$

The spin now couples only to site 0, and each site only to its neighbors; e_n are the on-site energies and t_n the hoppings. The star has weights $\gamma_m^2 \propto (aq)^m$ at the nodes $\xi_0 q^m$, with $a = q^s$: this is the orthogonality measure of the little q-Jacobi polynomials $p_n(x; a, 1|q)$ [23], so the recurrence coefficients are known in closed form. With

$$\mathcal{A}_n = \frac{(1 - aq^{n+1})^2}{(1 - aq^{2n+1})(1 - aq^{2n+2})}, \qquad \mathcal{B}_n = \frac{a\,(1 - q^n)^2}{(1 - aq^{2n})(1 - aq^{2n+1})},$$
$$\bar{e}_n \equiv \Lambda^n e_n = \xi_0\,(\mathcal{A}_n + \mathcal{B}_n), \qquad \bar{t}_n \equiv \Lambda^n t_{n-1} = \xi_0\,\sqrt{\mathcal{A}_{n-1}\mathcal{B}_n/q} \quad (n \ge 1). \quad (8)$$

(cf. also the orthogonal-polynomial chain mappings of Ref. [22]). The code evaluates these closed forms (`lri_spin.config.chain_coefficients`); the independent implementation of Sec. 8.1 obtains the same

numbers by a high-precision Lanczos recursion. Both rescaled coefficients tend to constants, so site n carries the energy scale Λ^{-n}. A plain Lanczos recursion on the star is unstable even in multiple-precision arithmetic: without reorthogonalization it loses the coefficients after 13 sites with 49 digits and after 33 sites with 268 digits (`independent_check/NOTES.md`). The closed form avoids the problem.

3.3 Iterative diagonalization and truncation

Let H_n be the chain (7) truncated to the sites $0, \dots, n$ and multiplied by Λ^n. Adding one site is the step

$$H_n = \Lambda\, H_{n-1} \otimes \mathbb{1} + \bar{e}_n\, \mathbb{1} \otimes b^\dagger b + \bar{t}_n \left(\bar{b}_{n-1}^\dagger \otimes b + \bar{b}_{n-1} \otimes b^\dagger\right), \qquad H_0 = -\frac{\Omega}{2}\sigma_x + \bar{e}_0\, b_0^\dagger b_0 + \frac{\lambda_{\mathrm{tot}}}{2}\sigma_z\,(b_0 + b_0^\dagger), \tag{9}$$

where $\mathbb{1}$ is the identity, $\bar{b}_{n-1}$ is the annihilation operator of site $n-1$ written in the basis kept at step $n-1$, and b acts on the new site. The factor Λ brings the previous levels to the new energy scale, so the levels of H_n stay of order one and their evolution with n is an RG flow; its plateaus are fixed points.

Three truncation parameters make each step finite:

- **Oscillator cutoff n_b.** Each new oscillator keeps its n_b lowest number states. This suffices near the delocalized and critical fixed points, where the chain sites hold few bosons [13].
- **Parity sectors.** The model conserves the parity $P = \sigma_x\,(-1)^{\sum_n b_n^\dagger b_n}$, which implements the Ising symmetry. H_n is diagonalized separately in the even and odd sectors, each of dimension at most Kn_b.
- **Kept states K.** In each sector the K lowest eigenstates are kept and the rest discarded. After each step all energies are shifted so that the ground state is at zero.

Together with Λ and the number N of chain sites (the last site sits at energy scale Λ^{-N}), these parameters define a finite numerical map from α to a sequence of spectra, which we call the *finite map*. Dense diagonalization costs of order $(Kn_b)^3$ per sector and site, which limits K and n_b.

3.4 Parity-gap flow and critical coupling

The observable is the rescaled parity gap

$$g_n(\alpha) = E_{0,n}^{\mathrm{odd}} - E_{0,n}^{\mathrm{even}}, \tag{10}$$

the difference between the lowest levels of H_n in the odd and even sectors. In the delocalized phase g_n flows to its value in the decoupled chain, where the spin is frozen by Ω and g_n is the lowest one-boson level; in the localized phase the ground state becomes degenerate and $g_n \to 0$; at α_c it stays on the value of the critical fixed point over an intermediate range of n.

For a chain of N sites, $n = 0, \dots, N-1$, the critical coupling of the finite map is located by bisection on the sign of

$$f(\alpha) = g_{N-1}(\alpha) - 0.3\, g_{N-1}(0), \tag{11}$$

the last gap measured against 30% of the last gap of the decoupled chain ($\alpha = 0$) with the same parameters; $f > 0$ on the delocalized side and $f < 0$ on the localized side. The rule is:

- **Bracket.** Start from $[0.01, 2]$; halve the lower end (up to 40 times) until $f > 0$; double the upper end until $f < 0$, never evaluating $\alpha > 100$. Without a sign change the item fails.
- **Bisection.** At most 48 steps, stopping when the bracket is narrower than $5 \times 10^{-14}\max(1, |\alpha|)$. A bisection that does not reach this width is rejected.
- **Result.** α_c is the midpoint of the final bracket. The *terminal residual* is $f(\alpha_c)$.

This follows the script of the original study, with two conservative differences: the original evaluated α up to 128 and accepted an exhausted bisection. The fraction 0.3 is a convention, like the level used to define the crossover scale in Refs. [12, 13]. The value of α_c is not universal: it depends on Λ, K and n_b (Fig. 6) and enters no result.

3.5 Exact α-derivative of the truncated map

A small detuning $\delta\alpha = \alpha - \alpha_c$ points along the relevant direction of the critical fixed point and grows by a factor Λ^{y_t} per site, so near the fixed point

$$\partial_\alpha g_n\big|_{\alpha_c} \propto \Lambda^{y_t n}. \tag{12}$$

The production code measures this growth without finite differences, by differentiating every step (9) with respect to α, including the change of the kept states. In each parity block write $H = U\,\mathrm{diag}(E)\,U^{\mathrm{T}}$, with kept eigenvectors U_k (energies E_k) and discarded eigenvectors U_d (energies E_d), and let $\dot H = \partial_\alpha H$. In the gauge without rotations among kept states,

$$\dot U_k = U_d\,Z, \qquad Z_{ij} = \frac{\left(U_d^{\mathrm{T}}\dot H\,U_k\right)_{ij}}{E_{k,j} - E_{d,i}}, \qquad T = U_k^{\mathrm{T}}\dot H\,U_k - \dot E_0\,\mathbb{1}, \tag{13}$$

where $\dot E_0$ is the derivative of the ground-state energy (the shift to zero is part of the step). T, the derivative of the kept Hamiltonian, and the derivative of the projected site operator, $\dot b = \dot U_k^{\mathrm{T}} B\,U_k + U_k^{\mathrm{T}} B\,\dot U_k$ with B the site-n annihilation operator in the unrotated product basis, are carried to the next step, where $\dot H_{n+1} = \Lambda\,T \otimes \mathbb{1} + \bar t_{n+1}(\dot b^\dagger \otimes b + \mathrm{h.c.})$. Only H_0 depends on α explicitly, through $\lambda_{\mathrm{tot}} \propto \alpha^{1/2}$. The derivative of the gap is $\partial_\alpha g_n = T_{00}^{\mathrm{odd}} - T_{00}^{\mathrm{even}}$. This is the exact derivative of the finite map wherever the kept and discarded levels are separated; if the gap between them falls below $5 \times 10^{-13}\max(1, |E|)$ the item fails and no record is written. A finite difference in α would instead turn nonlinear once $\delta\alpha\,\Lambda^{y_t n}$ is no longer small, and lose digits to rounding as $\delta\alpha \to 0$.

3.6 The window audit

Near the fixed point, Eq. (12) gives $\ln|\partial_\alpha g_n| = \kappa + y_t\,n\ln\Lambda$. The window audit, inherited from the analyzer of the original study, is:

1. **Windows.** Windows of $w = \max(12, \lfloor 0.13N \rfloor)$ consecutive sites, with first sites from $\max(8, \lfloor 0.13N \rfloor)$ up to (excluding) $\lfloor 0.76N \rfloor - w$. A window is skipped if any gap $g_n \le 0$ or any $\ln|\partial_\alpha g_n|$ is not finite.
2. **Fit.** In each window a least-squares line in n is fitted to $\ln|\partial_\alpha g_n|$; its slope divided by $\ln\Lambda$ is y_t, and $\Delta_E = 1 - y_t$. The *fit residual* is the largest absolute deviation from the line.
3. **Flatness.** The *relative variation* of the window is $(\max g_n - \min g_n)/|\overline{g_n}|$.
4. **Quality windows** have relative variation < 0.005 and fit residual < 0.003 (inherited conventions). Their number is n_q, and their Δ_E values span the *quality range*.

The audit uses no target value of any exponent.

3.7 The estimate: settled value, plateau and extrapolation

The window values of an item are not constant (Fig. 7). Along the chain the flow first approaches the fixed point, with irrelevant corrections decaying as $e^{-\omega l}$ in the *RG time* $l = n\ln\Lambda$; once the residual detuning has grown (Sec. 3.9) it leaves the fixed point again. The estimate is taken where the flow is closest to the fixed point, by a rule with three steps (`code/analysis/estimator.py`):

1. **Settled value.** The *settled window* is the window with the smallest fit residual plus relative variation; its Δ_E is the *settled value*. This is the window the original analyzer reports, stored in every record as `selected`.
2. **Plateau test.** Take the windows whose first site lies within $h = \lceil w/4 \rceil$ sites of that of the settled window. The largest $|\Delta_E - \text{settled value}|$ among them is the *plateau drift*. A point has a *plateau* if the plateau drift of its central item is at most $\tau_{\mathrm{p}} = 5 \times 10^{-5}$. Its value is then the settled value, and the plateau drift is its chain-depth error.
3. **Extrapolation**, where there is no plateau. The local exponent

$$\Delta_{\mathrm{loc}}(n) = 1 - \frac{\ln|\partial_\alpha g_{n+1}| - \ln|\partial_\alpha g_{n-1}|}{2\ln\Lambda} \tag{14}$$

is fitted with $\Delta_\infty + A\,e^{-\omega(l-l_0)}$ (free Δ_∞, A, ω; unweighted least squares) on $l_{\rm hi} - S \le l \le l_{\rm hi}$ for $S = 30, 25, 20$, where $l_{\rm hi}$ is the RG time of the last site of the settled window and l_0 that of the first fitted site. The value is the median of the three Δ_∞, and the chain-depth error is the larger of |extrapolated − settled| and the spread of the three fits. Details: a range with fewer than 8 sites is skipped; the start values are Δ_∞ = last point, A = first minus last, $\omega = 0.2$; the fit is Levenberg–Marquardt. The extrapolation counts as *determined* only if every fit converged with $\omega > 0.01$; otherwise it is reported as not determined (n.d.), and a point that needs it would stop the analysis.

Every item of a point is evaluated with that point's estimator, so the scans of Sec. 4 compare like with like. The rule was written on 13 October 2026 and its constants were not varied afterwards (Sec. 4, App. C). It replaces the midpoint of the quality range used before; App. E explains why and lists the old values.

At $\Delta_\sigma = 0.001$ the rule's own fit ranges lie entirely on the flat plateau, so its extrapolation is not determined (n.d.); the point has a plateau and is not extrapolated.

Classification of the points. The rule puts each point in one of two classes, *plateau* or *extrapolated*. The eight points with $\Delta_\sigma \le 0.15$ have a plateau: their plateau drift is at most 1.91×10^{-5} (at $\Delta_\sigma = 0.15$) and at most 9.5×10^{-6} for $\Delta_\sigma \le 0.125$. None of the five points with $\Delta_\sigma \ge 0.175$ has one: their smallest drift is 7.74×10^{-5} (at $\Delta_\sigma = 0.175$). Every threshold between these two numbers gives the same classification. The drift grows smoothly with Δ_σ, so the split also depends on h and on the window width w, which the rule fixes. At $\Delta_\sigma = 0.15$ the drift is 6.6×10^{-5} with $h = w/2$ and 9.3×10^{-5} with windows of 20 sites, above $\tau_{\rm p}$ in both cases, so the point would be extrapolated; every point with $\Delta_\sigma \le 0.125$ keeps its plateau (drifts below 2.2×10^{-5} and 1.6×10^{-5}). The values hardly depend on the class: extrapolating at $\Delta_\sigma = 0.15$ would move the value by -1.3×10^{-5}, windows of 20 sites move its settled value by $+1.8 \times 10^{-5}$, and taking the settled value at 0.175 would move it by $+8.6 \times 10^{-5}$, all well inside $\delta_{\rm tot}$. At the two final points without a plateau the extrapolation is the value (Table 4).

A third label, *not converged in chain depth*, is not an outcome of the rule. The author assigned it on 13 October 2026 to the three points closest to $s = 1/2$, $\Delta_\sigma = 0.249$, 0.24 and 0.225, when the comparison with the ε series was already known (App. C). The reasons: there $\omega\, l_{\rm max} \approx 61\varepsilon$ is small (Sec. 3.9); at $\Delta_\sigma = 0.249$ the excess over the series decays at a rate of the one-loop size (Sec. 7.1); and the excess of the settled values over the series falls smoothly with ε (Sec. 8.2). At $\Delta_\sigma = 0.225$ the rule alone would extrapolate: its three fits converge, with $\omega = 0.059$–0.092, and give 0.487529 with $\delta_{\rm depth} = 2.1 \times 10^{-3}$. The label there rests on the trend from the two other points and on the Borel–Padé sum of the series; optimal truncation of the series, 0.490(4), does not exclude the settled value. For the three labeled points the settled value is shown, the extrapolations are cross-checks only, and no error is quoted. That they "enter no comparison" means that no agreement or disagreement is claimed from their values; the label itself came from a comparison.

3.8 Cross-check fits (not adopted)

The rule's fits end at the settled window and contain no term for the departure from the fixed point, so on a plateau they return the plateau almost by construction. Two fits that include that departure test the rule from outside (`code/analysis/crosscheck.py`, written for the second review round, App. C). Neither is adopted, and no value in this note is taken from them. Both fit $\Delta_{\rm loc}(l)$ of the central item with an approach term plus a *detuning term* $A_{\rm d}\, e^{(1-\Delta_\infty)(l-l_{\rm end})}$, whose growth rate $1 - \Delta_\infty = y_t$ is that of the residual detuning (Sec. 3.9). Each fit range runs from a start $l_{\rm a}$ to $l_{\rm end} = l_N - k/y_t$, where $l_N = N \ln \Lambda$ is the end of the chain and y_t is taken from the settled value, so the last stretch of RG time, where the detuning dominates, is left out. The fits are Levenberg–Marquardt; the result is the median over the accepted fits, with their range and number $n_{\rm fit}$.

- **Late fit.** Approach term $A\, e^{-\omega(l-l_{\rm a})}$; $l_{\rm a} \in \{20, 25, 30, 35\}$ ($\{40, 50, 60, 70\}$ at $\Delta_\sigma = 0.001$), $k \in \{8, 10, 12\}$, at least 10 sites per range; a fit is accepted if its largest residual is below 2×10^{-6} (below 2×10^{-5} if no fit reaches 2×10^{-6}). It is computed at every point and shown for the final points (Table 2).
- **Crossover fit.** Approach term $A/(e^{\omega(l-l_0)} - 1)$, the one-loop crossover form of Eq. (17) with a free l_0, or $A\, e^{-\omega(l-l_{\rm a})}$; $l_{\rm a} \in \{13, 18, 22\}$, $k \in \{12, 15\}$, at least 8 sites per range; a fit is accepted if its largest residual is below 2×10^{-5} and $0 < \omega < 2$. It is computed at $\Delta_\sigma \ge 0.2$, where the approach is slowest

(Tables 4 and 10).

The late fit against the rule. It reproduces the plateau values to 3×10^{-6} at $0.001 \le \Delta_\sigma \le 0.125$, and differs from the final values by -1.7×10^{-5}, $+1.4 \times 10^{-5}$ and $+6.9 \times 10^{-5}$ at $\Delta_\sigma = 0.15$, 0.175 and 0.2. This is inside δ_{depth} except at $\Delta_\sigma = 0.001$, where it is $+2.9 \times 10^{-6}$ against $\delta_{\text{depth}} = 1.7 \times 10^{-6}$, far inside δ_{tot}. At $\Delta_\sigma = 0.01$ its fitted $\omega = 0.197$ agrees with $\sqrt{2\delta} + \delta/4 = 0.205$ from Eq. (3). At $\Delta_\sigma = 0.2$ it suggests that the rule's extrapolation, which ends where the detuning already bends Δ_{loc}, lies slightly low, by about a tenth of δ_{tot}. The crossover fit is discussed with the ε series in Sec. 8.2.

3.9 Chain depth

The bisection stops when its bracket is narrower than 5×10^{-14} (relative), so the critical coupling of the finite map is located to $\delta\alpha \approx 5 \times 10^{-14}$. This width is at the rounding level of the map: the same item run on 1 and on 2 threads gives values of α_c that differ by up to 2.6×10^{-14}, and the two implementations of Sec. 8.1 differ by up to 7.2×10^{-14} (relative), so a narrower bracket would not locate the critical coupling better in double precision. The residual detuning also grows as $\Lambda^{y_t n}$, and the flow therefore stays near the fixed point only up to

$$n \lesssim n_{\max} \simeq \frac{\ln(1/\delta\alpha)}{y_t \ln \Lambda} \approx \frac{30.6}{y_t \ln \Lambda}, \qquad l \lesssim l_{\max} = n_{\max} \ln \Lambda \approx \frac{30.6}{y_t}, \tag{15}$$

with $30.6 = \ln(2 \times 10^{13})$: the usable RG time is fixed whatever Λ. The chain lengths follow this estimate: at $\Lambda = 3$, $N = 65$ for $s \le 0.8$, then 70, 80, 90 and 105 for $s = 0.85$, 0.9, 0.95, 0.98, and $N = 250$ at $s = 0.998$; for other Λ, $N(\Lambda) = N(3) \ln 3 / \ln \Lambda$. Where $N \approx n_{\max}$ the terminal residual $|f(\alpha_c)|$ is of order 0.1–0.4, which is expected and not a failure. At $\Delta_\sigma = 0.025$, where $N = 90$ and $n_{\max} \approx 97$, the terminal residuals are 4×10^{-4} to 0.02; at $\Delta_\sigma \le 0.01$ the chains are clearly shorter than $n_{\max}$ (at $s = 0.98$, $N = 105$ against $n_{\max} \approx 146$) and they are at most 2×10^{-4} at $\Delta_\sigma = 0.01$ and 4×10^{-5} at 0.001 (App. D).

The flow reaches its fixed point within the usable depth only if $\omega\, l_{\max} \approx 30.6\, \omega/y_t$ is large. Near $s = 1$, Eq. (3) gives $\omega/y_t \to 1$, so $\omega l_{\max} \approx 30$ however small y_t becomes. Near $s = 1/2$, $\omega \approx \varepsilon$ and $y_t \approx 1/2$, so $\omega l_{\max} \approx 61\varepsilon$: about 0.25, 2.4 and 6 at $\Delta_\sigma = 0.249$, 0.24 and 0.225. This is why the plateau is lost as $s \to 1/2$ and not as $s \to 1$. Extended-precision arithmetic would lengthen $l_{\max}$ in proportion to the number of digits.

4 Error budget

For each final point the value is that of Sec. 3.7 for the central item. Its error has three components, each computed with the point's estimator:

- δ_{depth} (chain depth): the plateau drift of the central item; at an extrapolated point, the larger of |extrapolated − settled| and the spread of the three fits (Sec. 3.7).
- δ_K (K and n_b): the spread (maximum minus minimum) over all $\Lambda = 3$ items with $K \ge 64$ and $n_b \ge 10$; at $\Delta_\sigma \ge 0.2$, where $n_b = 10$ is not converged (Sec. 7.2), items with $n_b = 10$ are left out.
- δ_Λ (discretization): the larger of two numbers.

  (a) The spread over the items at different Λ, each at the largest K run there, with equal n_b (16 at $\Delta_\sigma \ge 0.2$, else 10): $\Lambda = 2.5$ and 3 at $K = 128$ at every point; for $\Delta_\sigma < 0.2$ also $\Lambda = 4$, run only at $K = 64$; and $\Lambda = 2$ at $K = 256$ at $\Delta_\sigma = 0.1$ and 0.05. The K convergence of $\Lambda = 4$ at $K = 64$ is assumed, not tested: larger Λ needs fewer kept states (Sec. 7.3), and at $\Lambda = 3$ the values are converged at $K = 48$ already (Sec. 7.2).

  (b) The shift of the $\Lambda = 2.5$ value under its one doubling of K, from 64 to 128, at $n_b = 10$, the only n_b run at $\Lambda = 2.5$, $K = 64$ (at $\Delta_\sigma \ge 0.2$, $n_b = 10$ is not converged).

  Part (b) allows for the untested K convergence at $\Lambda = 2.5$. It is the larger part at 8 of the ten final points, so at most points the quoted error is mainly this allowance, not a measured systematic of the central value.

The total is the linear sum

$$\delta_{\text{tot}} = \delta_{\text{depth}} + \delta_K + \delta_\Lambda, \tag{16}$$

listed in Table 2 and shown in Fig. 2. The points that are not converged in chain depth have no δ_{depth} and no δ_{tot}. At $\Delta_\sigma = 0.2$, δ_K mostly measures how the extrapolation itself scatters between items (the detuning fits of Sec. 3.7 bring those items within about 2×10^{-5} of each other), so it partly counts δ_{depth} twice; this is conservative.

What this error is, and is not. None of the components is a standard deviation: δ_{depth} is a resolution or an extrapolation length, δ_K and δ_Λ are ranges of convergence scans. They are therefore added linearly, not in quadrature. Five choices are judgements, not measurements:

- the plateau threshold τ_{p}, the neighborhood h and the window width w (the classification is the same for every τ_{p} between 1.91×10^{-5} and 7.74×10^{-5}; for h and w see Sec. 3.7);
- the fit form, the fit ranges S and the end point l_{hi} of the extrapolation;
- taking the full extrapolation length as δ_{depth} at an extrapolated point;
- the allowance (b) in δ_Λ for one K doubling at $\Lambda = 2.5$;
- no allowance below $\Lambda = 2$: over $2 \le \Lambda \le 4$ the values agree to 1.9×10^{-5} at the two points where $\Lambda = 2$ is K-converged ($\Delta_\sigma = 0.1$, 0.05), with a small monotone rise with Λ far inside the error, but the limit $\Lambda \to 1$ itself is not tested (Sec. 7.3).

The window rule of Sec. 3.6 (width, range, thresholds) is inherited and was not varied in production. Its quality thresholds do not set the value, because the settled window passes any reasonable threshold. The window width does enter the plateau test, and with it the class of $\Delta_\sigma = 0.15$, but not its value beyond 2×10^{-5} (Sec. 3.7).

When the choices were made. The rule and its constants were fixed on 13 October 2026, after the NRG data, the earlier comparisons with the ε series and the Monte Carlo study, and the reviews' exploratory comparisons had been seen; no protocol was registered in advance. App. C gives the timeline.

Table 2: Error budget. Columns: estimator (plateau, extrapolated, or settled for the points not converged in chain depth); Δ_E to six decimals; the chain-depth error δ_{depth}, the K/n_b spread δ_K, the discretization error δ_Λ and their linear sum δ_{tot}, in units of 10^{-4}; the number n_q of quality windows of the central item; the late fit of Sec. 3.8 minus Δ_E (cross-check, not adopted; shown for the final points). n.c.: not converged in chain depth.

Δ_σ	estimator	Δ_E	$10^4\delta_{\mathrm{depth}}$	$10^4\delta_K$	$10^4\delta_\Lambda$	$10^4\delta_{\mathrm{tot}}$	n_q	10^4 (late fit $-\Delta_E$)	status
0.249	settled	0.508030	n.c.	1.70	1.21	–	2	−−	not converged
0.24	settled	0.498922	n.c.	5.12	2.54	–	4	−−	not converged
0.225	settled	0.489668	n.c.	2.98	6.05	–	9	−−	not converged
0.2	extrapolated	0.482980	4.12	2.47	0.49	7.08	15	+0.69	final
0.175	extrapolated	0.483900	0.86	0.80	1.80	3.46	19	+0.14	final
0.15	plateau	0.491106	0.19	0.28	0.68	1.15	21	−0.17	final
0.125	plateau	0.505672	0.01	0.15	0.38	0.54	23	−0.00	final
0.1	plateau	0.529608	0.02	0.05	0.27	0.33	23	−0.01	final
0.075	plateau	0.566260	0.01	0.05	0.24	0.30	28	−0.00	final
0.05	plateau	0.622115	0.00	0.03	0.23	0.27	34	−0.00	final
0.025	plateau	0.713422	0.01	0.02	0.26	0.29	38	+0.00	final
0.01	plateau	0.809747	0.09	0.02	0.35	0.47	41	+0.01	final
0.001	plateau	0.937424	0.02	0.04	1.05	1.11	106	+0.03	final

5 Software and validation

5.1 Code

The solver is the Python package `lri_spin` (`code/lri_spin_boson_hpc/src/lri_spin`; NumPy, SciPy and threadpoolctl; float64 throughout).

- **Backend.** The `ordered` backend builds one parity block at a time in the basis order of the reference solver (`code/lri_spin_boson_hpc/reference/`) and reproduces its flows bitwise and its derivative to rounding (tested). The reordered `dense` and `banded` backends are experimental and were not used.
- **Workflows.** The critical search writes checkpoints every 8 sites and persists the bisection state, so an interrupted item resumes where it stopped.

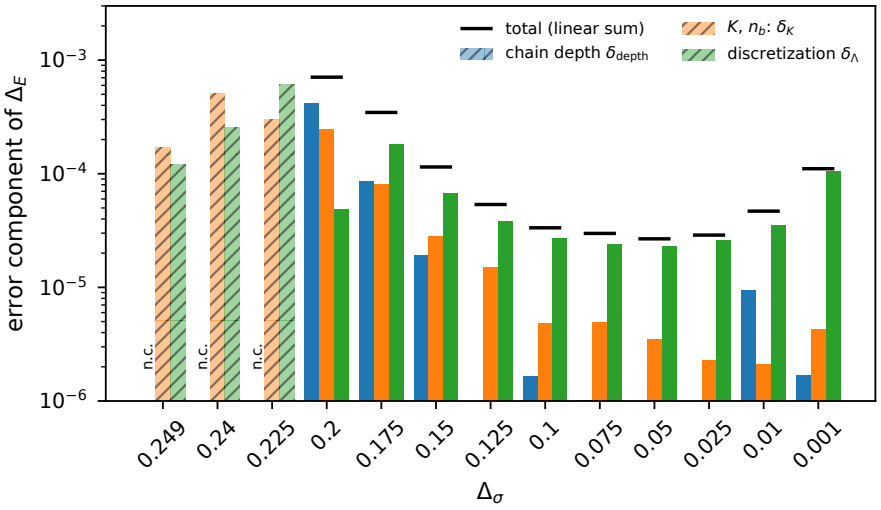


Figure 2: Components of the error at each Δ_σ and their linear sum δ_{tot}. Hatched: the three points not converged in chain depth, which have no δ_{depth} and no total. At the plateau points δ_Λ dominates, mostly through the K-doubling allowance at $\Lambda = 2.5$; at $\Delta_\sigma = 0.2$ the extrapolation length δ_{depth} dominates.

- **Records.** Each item ends in one result record (App. A) carrying a SHA-256 checksum and an *identity*: the configuration, the kind of run, bracket and tolerances, the SHA-256 of the solver source and the runtime (Python, NumPy, SciPy and OpenBLAS versions, BLAS kernel and threading layer). A record from another source or runtime is refused by the analysis.

Before production Claude Code read the package line by line against the reference solver (App. C); no numerical defect was found and no numerical code was changed.

5.2 Acceptance gates and tests

Every result record passed these gates before it was accepted: eigen-residual $\max|HU - UE| \le 10^{-8}$ and orthogonality error $\max|U^{\mathrm{T}}U - \mathbb{1}| \le 10^{-9}$ at sites $0, 1, 2$, every 16th site and the last; finite observables and sorted spectra; boson occupation within $[0, n_b - 1]$ and top-level probability (the weight of the highest kept level of the newest oscillator, p_{edge}) within $[0, 1]$; for derivatives, a positive gap between kept and discarded levels. Over all 197 production records the largest checked eigen-residual is 1.5×10^{-13} and the largest orthogonality error 5.0×10^{-13}.

The test suite (`make test`) has 85 tests. Physics: chain moments, small untruncated spectra against exact diagonalization, Kronecker products against block operators, reference trajectories, finite differences against the exact derivative. Software: workflows, record corruption, signals and locks, and the cluster tools of App. B (campaign ownership, generations, lanes, submission against a simulated Slurm, exports, stopping). Before production it passed on a compute node of the cluster for the final code, in the validation job that issued the receipt `inputs/validation_receipt/receipt.json`. The receipt binds the solver source to the node runtime: AMD EPYC 7543 nodes, Python 3.13.3, NumPy 2.1.2, SciPy 1.15.2, OpenBLAS 0.3.27/0.3.28 with the Haswell kernel.

5.3 Reproducibility across machines

With the same runtime and thread counts, results are bitwise identical across nodes: a repeated smoke run on a second node reproduced the first bitwise (App. B). Across runtimes, rounding differences are amplified along the relevant direction as $\Lambda^{y_t n}$: at a *fixed* α, late flows of two machines differ, and even two implementations with bitwise identical α_c (configuration c1 of Sec. 8.1) have flows that differ by more than 10^{-8} from site 30 on. Archived values of α_c are therefore not portable, and every item retunes α_c on its own runtime. The critical coupling itself and the exponent are robust: the production values of α_c reproduce the three couplings archived by the original study on another machine to 7.1×10^{-14} (relative); the three example items, run on 2 threads, reproduce the α_c of the same items run on 1 thread bitwise in 2 cases and to 2.6×10^{-14} in the

third; and the exponents of two implementations agree to 3.1×10^{-7} (Sec. 8.1).

6 Production campaigns

All items are critical searches followed by the exact derivative and the window audit, with $\Omega = 0.1$, bracket $[0.01, 2]$ and the `ordered` backend. Item IDs read `ds<`Δ_σ`>_L<`Λ`>_K<`K`>_B<`n_b`>_N<`N`>`; the example items use `s<`s`>_L<`Λ`>_K<`K`>_B<`n_b`>`. Items of the ds13 campaign and the Λ campaigns ran on 1 thread if $Kn_b \leq 1000$, 2 threads if $Kn_b < 1600$ and 4 otherwise; the example items ran on 2 threads, as in the package's example list.

Example (3 items). The package's example list: $s = 0.8$, 0.9, 0.98 at $K = 64$, $n_b = 10$, $\Lambda = 3$ (Table 3). They duplicate three items of the ds13 campaign, which makes them a test of thread-count independence (Sec. 5).

ds13 (146 items). All thirteen points at $\Lambda = 3$ with

- a scan in kept states, $K \in \{48, 64, 96, 128\}$ at $n_b = 10$;
- a scan in oscillator levels, $n_b \in \{8, 12, 16\}$ at $K = 64$;
- $\Lambda \in \{2, 4\}$ at $K = 64$, $n_b = 10$;
- the central item $K = 128$, $n_b = 16$;
- extras at the ends: $n_b \in \{20, 24\}$ for $\Delta_\sigma \geq 0.2$ and $K \in \{160, 192\}$ for $\Delta_\sigma \leq 0.05$.

Λ **pilot** (18 items). $\Delta_\sigma \in \{0.2, 0.1, 0.05\}$ at $\Lambda = 1.5$ ($K \in \{64, 128\}$ at $n_b = 10$, $K = 64$ at $n_b = 16$), $\Lambda = 2$ ($K = 128$) and $\Lambda = 2.5$ ($K \in \{64, 128\}$), $n_b = 10$ unless stated. ($\Lambda = 2$, $K = 64$ is part of the ds13 campaign.)

Λ **follow-up** (30 items). $\Lambda = 2.5$ at $K \in \{64, 128\}$ for the other ten points, plus $K = 128$, $n_b = 16$ for $\Delta_\sigma \geq 0.2$; and $\Lambda = 2$ at $K \in \{192, 256\}$ for $\Delta_\sigma \in \{0.2, 0.1, 0.05\}$.

The runs used 178 thread-hours (sum over items of wall time times threads) on whole cluster nodes. The allocations summed to 15 node-hours, i.e. about 970 core-hours as charged for 64-core nodes; the longest item took 2.7 h, and the production finished within two days including a cluster outage (Table 12 in App. B). The largest peak memory of an item was 495 MiB.

Table 3: The example items ($K = 64$, $n_b = 10$, $\Lambda = 3$, 2 threads): critical coupling, bisection steps, terminal residual $f(\alpha_c)$, number of quality windows, quality range and settled value. The terminal residuals near -0.16 reflect $N \approx n_{\max}$ (Sec. 3.9); at $s = 0.98$, $N < n_{\max}$.

item	α_c	bisection	terminal residual	quality windows	Δ_E range (settled)
`s0.80_L3_K64_B10`	0.43976978933704	46	-0.160	23	[0.52961, 0.53006] (0.52961)
`s0.90_L3_K64_B10`	0.68913573159160	46	-0.159	34	[0.62212, 0.62260] (0.62212)
`s0.98_L3_K64_B10`	1.08065976531605	46	$+0.000$	41	[0.80946, 0.81013] (0.80975)

7 Results

7.1 The approach to the fixed point

Figure 3 shows the local exponent (14) against the RG time, which is the quantity the estimator works on.

- **Plateau ($\Delta_\sigma = 0.1$, panel a).** After the transient, Δ_{loc} is flat over $30 \lesssim l \lesssim 55$ and the same at $\Lambda = 2$, 2.5, 3 and 4; the fitted correction exponent is $\omega = 0.339$. The settled window lies on the plateau.
- **No plateau ($\Delta_\sigma = 0.2$, panel b).** Δ_{loc} is still falling when the detuning sets in. The settled value, 0.483392, is the deepest usable value; the extrapolation (Table 4; $\omega = 0.142$) gives 0.482980, -4.1×10^{-4} from it, and the extrapolations at $\Lambda = 2.5$ and 3 agree to 2×10^{-5}. At $\Delta_\sigma = 0.175$ ($\omega = 0.199$) the shift is -8.6×10^{-5}.
- **Near $s = 1/2$ ($\Delta_\sigma = 0.249$, panel c).** At one loop, with the φ^4 coupling u normalized so that $du/dl = \varepsilon u - u^2$ (fixed point $u^* = \varepsilon$), the local dimension of the energy operator is $\frac{1}{2} - \frac{\varepsilon}{2} + \frac{u}{3}$; the

factor $\frac{1}{3} = (\mathcal{N}+2)/(\mathcal{N}+8)$ is that of the Ising case $\mathcal{N}=1$ of the O($\mathcal{N}$) model [4], and at u^* it gives $\Delta_\varepsilon = \frac{1}{2} - \frac{\varepsilon}{6}$. A strong bare coupling, $u \to \infty$ at some RG time l_0, then gives

$$\Delta_{\rm loc} - \Delta_\varepsilon = \frac{\varepsilon/3}{e^{\varepsilon(l-l_0)}-1}, \qquad \frac{1}{\Delta_{\rm loc}-\Delta_\varepsilon} = \frac{3}{\varepsilon}\big(e^{\varepsilon(l-l_0)}-1\big), \tag{17}$$

if $\Delta_{\rm loc}$ follows the running coupling. The inverse excess thus grows with local slope $3e^{\varepsilon(l-l_0)}$, which tends to 3 as $\varepsilon(l-l_0) \to 0$. With the series value $\Delta_\varepsilon = 0.499344$ the measured inverse excess grows nearly linearly in l at every K, n_b and Λ shown, with slope 2.55 for $15 \le l \le 25$, 2.87 over $20 \le l \le 45$ and 3.03 for $38 \le l \le 48$. Inferring l_0 from the data at $20 \le l \le 40$ gives $l_0 = 6$ to 9, for which Eq. (17) predicts local slopes 3.14–3.17 at $l = 20$ and 3.44–3.47 at $l = 43$. The measured slopes are 10–20% lower and rise with l as predicted. Terms of higher order in u, largest early in the flow where u is largest, could account for the difference, but we have not shown that. The test is semi-quantitative: it shows an approach of the one-loop kind, at a rate set by ε, not a precise coefficient. At $\Delta_\sigma = 0.249$, 0.24 and 0.225 the window Δ_E still changes by -2.5×10^{-4}, -2.0×10^{-4} and -1.9×10^{-4} per site at the settled window. Single-exponential fits give $\omega = 0.064$, 0.079 and 0.078, which approach ε (0.004, 0.04, 0.1) only at the largest ε, since an exponential cannot represent a $1/l$ approach, and the three fits scatter by up to 1×10^{-3} (Table 4). The data do not determine the fixed-point value at these points to the precision of the others: they are *not converged in chain depth.* A scan in K, n_b or Λ at fixed $l_{\max}$ cannot reveal this, because once K is converged and $n_b \ge 12$ the effect depends only on l ($n_b = 8$ and 10 are off by up to 9.5×10^{-3} and 8.9×10^{-4} near $s = 1/2$, Sec. 7.2).

- **Near $s = 1$ ($\Delta_\sigma = 0.001$, panel d).** $\Delta_{\rm loc}$ rises to a long plateau ($75 \lesssim l \lesssim 120$), as Eq. (15) allows. At the 10^{-4} level the plateau depends on the truncation away from $\Lambda = 3$: at $\Lambda = 2.5$ it moves by 1.05×10^{-4} between $K = 64$ and 128, at $\Lambda = 4$ ($K = 64$) it overshoots by about 10^{-4} before settling, and at $\Lambda = 2$, $K = 64$ is far from converged ($+5.6\times10^{-4}$). At the largest K run the values at $\Lambda = 2.5$, 3 and 4 agree to 2.3×10^{-5}, but at $\Lambda = 2.5$ that is a single doubling. The plateau value is therefore ambiguous at the 10^{-4} level; the budget carries this through the K-doubling allowance, which gives $\delta_\Lambda = 1.05\times10^{-4}$ at this point (Sec. 4).

Table 4: The points without a plateau (central items). Settled value; plateau drift; change of the window Δ_E per site at the settled window; the three fitted Δ_∞ and ω for $S = 30, 25, 20$; their median (the extrapolated value). At $\Delta_\sigma = 0.2$ and 0.175 the median is the final value. Last column: the crossover fit of Sec. 3.8 (median, number $n_{\rm fit}$ of accepted fits), computed at $\Delta_\sigma \ge 0.2$ only; a cross-check, not adopted. *Not converged in chain depth: cross-check only, not used.

				the rule (Sec. 3.7)			cross-check, not adopted
Δ_σ	settled	drift	per site	fits Δ_∞ ($S = 30, 25, 20$)	ω	median	crossover fit ($n_{\rm fit}$)
0.249*	0.508030	8×10^{-4}	-3×10^{-4}	0.50496, 0.50463, 0.50463	0.068, 0.064, 0.064	0.504628	0.50209 (6)
0.24*	0.498922	7×10^{-4}	-2×10^{-4}	0.49670, 0.49673, 0.49704	0.078, 0.079, 0.088	0.496734	0.49460 (6)
0.225*	0.489668	6×10^{-4}	-2×10^{-4}	0.48804, 0.48753, 0.48668	0.092, 0.078, 0.059	0.487529	0.48810 (7)
0.2	0.483392	2×10^{-4}	-7×10^{-5}	0.48305, 0.48298, 0.48291	0.151, 0.142, 0.130	0.482980	0.48305 (9)
0.175	0.483986	8×10^{-5}	-2×10^{-5}	0.48391, 0.48390, 0.48389	0.205, 0.199, 0.190	0.483900	–

7.2 Convergence in kept states and oscillator levels

Table 5 and Fig. 4 show the deviation of every $\Lambda = 3$ item from the central value.

- **Kept states.** The estimates are converged at $K = 48$ already: for $\Delta_\sigma \le 0.125$ the spread over K is at most 4.9×10^{-6}.
- **Oscillator levels.** n_b matters mainly near $s = 1/2$, where $n_b = 8$ is off by up to 9.5×10^{-3} and $n_b = 10$ by up to 8.9×10^{-4}, while at the final points all items with $n_b \ge 12$ (and $K \ge 64$) agree with the central value to 1.9×10^{-4}.
- **Top-level probability.** $p_{\rm edge}$ is the weight of the highest level, $n_b - 1$, of the newest oscillator in the ground state, evaluated at every site of the critical flow; App. D lists its maximum over the chain. That maximum is reached at the last sites, where the flow leaves the fixed point, and depends on the side it leaves to: for the items with $n_b \ge 12$ its median is 2×10^{-8} when the flow ends toward the localized

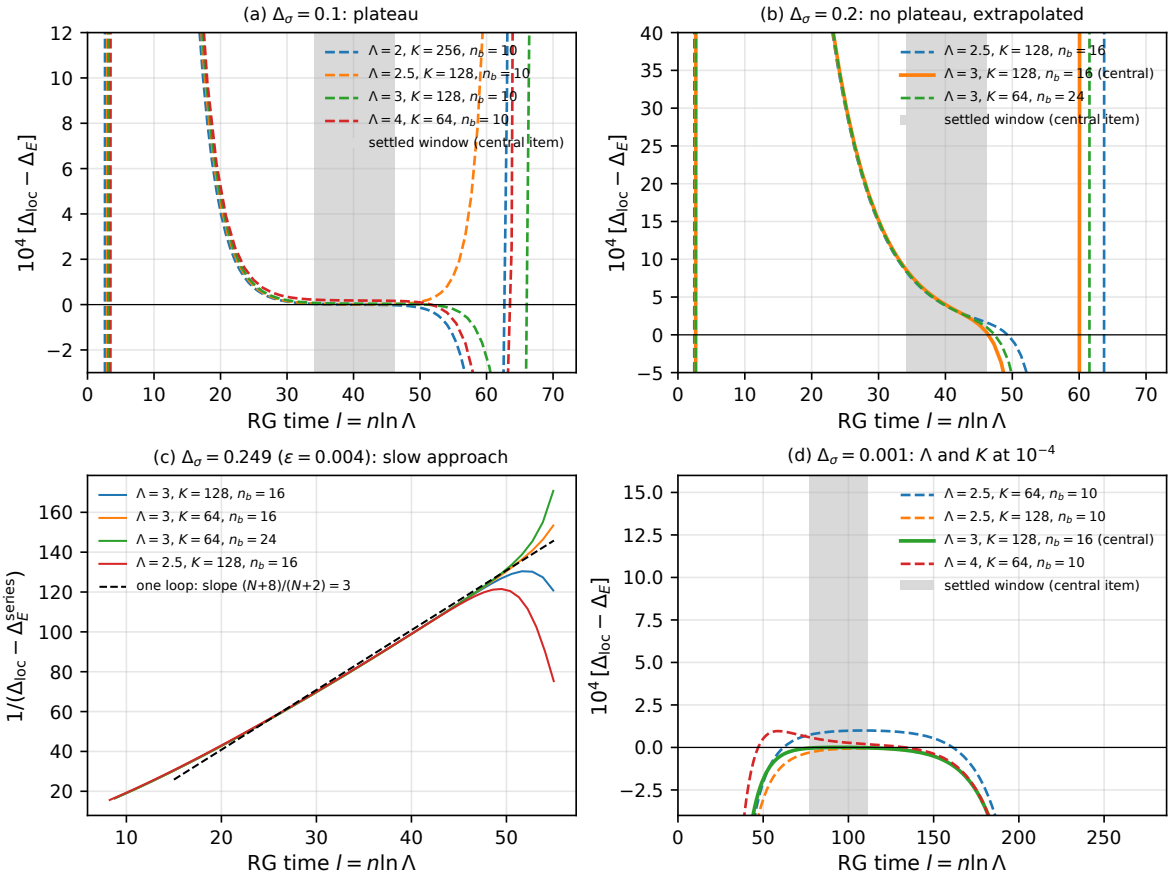


Figure 3: Local exponent $\Delta_{\rm loc}$, Eq. (14), against the RG time $l = n \ln \Lambda$. (a), (b), (d): $10^4\,(\Delta_{\rm loc} - \Delta_E)$ relative to the final value; gray band: the settled window of the central item. (a) $\Delta_\sigma = 0.1$, items at four values of Λ, each at the largest K run (at $\Lambda = 4$ only $K = 64$): a common plateau. (b) $\Delta_\sigma = 0.2$: no plateau; the zero line is the extrapolated value. (d) $\Delta_\sigma = 0.001$: Λ and K at the 10^{-4} level. (c) $\Delta_\sigma = 0.249$: inverse distance to the ε-series value against l, for four truncations; dashed: slope 3, the one-loop value for $\varepsilon(l - l_0) \to 0$ (Eq. (17)).

phase (negative terminal residual; the bosons are displaced there) and 3×10^{-19} when it ends toward the delocalized phase. This is why it changes by many decades between items that differ only in K. For $n_b \geq 12$ it is at most 4.9×10^{-4}, at most 2.1×10^{-4} for the central items, and below 5.4×10^{-6} at $\Delta_\sigma \leq 0.15$. It is largest near $s = 1/2$, where the critical fixed point holds the most bosons.

7.3 Dependence on the discretization Λ

The chain length is scaled as $N \propto 1/\ln \Lambda$, so all items of a point reach the same RG time. The Λ studies separate truncation from discretization (Tables 6 and 7, Fig. 5):

- **Smaller Λ needs many more kept states.** At $\Lambda = 1.5$, going from $K = 64$ to 128 moves Δ_E by 0.028–0.040, and $K = 128$ is still 0.008–0.024 away from the converged $\Lambda = 2$ value; raising n_b to 16 changes Δ_E by only 1.9×10^{-6}. At $\Lambda = 2$, $K = 192$ and 256 agree to 4.7×10^{-5}. At $\Lambda = 2.5$, $K = 64$ and 128 (one doubling) agree to 6.8×10^{-5} for $0.01 \leq \Delta_\sigma \leq 0.15$, to 1.05×10^{-4} at 0.001, and differ by up to 1.6×10^{-4} at the final points with $\Delta_\sigma \geq 0.175$ (up to 6.1×10^{-4} at the points not converged in chain depth). At $\Lambda = 2$, $K = 64$ is not converged: it lies $+2 \times 10^{-4}$ to $+6 \times 10^{-4}$ above $\Lambda = 3$ at $\Delta_\sigma \leq 0.05$.
- **A small Λ dependence, far inside the error.** At the largest K run at each Λ (at $\Lambda = 4$ only $K = 64$, whose convergence is assumed), the plateau values at $\Lambda = 2$, 2.5, 3 and 4 agree to 1.9×10^{-5} at $\Delta_\sigma = 0.1$ and 0.05, and rise monotonically with Λ: relative to $\Lambda = 3$, $\Lambda = 2$, 2.5 and 4 lie at -3×10^{-6}, -2×10^{-6} and $+2 \times 10^{-5}$ at $\Delta_\sigma = 0.1$, and at -1×10^{-6}, -1×10^{-6} and $+5 \times 10^{-6}$ at 0.05. Extended to $\Lambda \to 1$ this would put the $\Lambda = 3$ value a few 10^{-6} above the limit, far below $\delta_{\rm tot}$. Over $\Lambda = 2.5$–4

Table 5: Deviation $10^4\,[\Delta_E(K, n_b) - \Delta_E^{\text{central}}]$ at $\Lambda = 3$ (central: $K = 128$, $n_b = 16$), each item evaluated with its point's estimator. In the scan over K, the offsets at $\Delta_\sigma \geq 0.2$ come from $n_b = 10$, not from K. Empty cells were not run.

	K scan ($n_b = 10$)						n_b scan ($K = 64$)					
Δ_σ	K=48	K=64	K=96	K=128	K=160	K=192	n_b=8	n_b=10	n_b=12	n_b=16	n_b=20	n_b=24
0.249	+10.66	+8.88	+10.28	+10.92			+94.49	+8.88	+1.08	−0.30	+1.24	−0.45
0.24	+7.03	+6.80	+10.23	+7.21			+89.86	+6.80	+3.00	+5.12	+0.70	+3.32
0.225	+2.43	+2.12	+4.70	+4.34			+65.41	+2.12	+0.20	+0.28	+1.88	−1.10
0.2	+2.07	+2.68	+3.46	+3.15			+33.99	+2.68	+1.84	+1.48	−0.63	−0.15
0.175	+0.81	+0.42	+0.80	+0.65			+16.97	+0.42	+0.20	+0.22		
0.15	+0.21	+0.21	+0.15	+0.17			+9.45	+0.21	−0.07	−0.07		
0.125	+0.15	+0.15	+0.13	+0.10			+5.86	+0.15	+0.03	+0.03		
0.1	+0.08	+0.05	+0.05	+0.04			+3.90	+0.05	+0.01	+0.03		
0.075	+0.06	+0.05	+0.03	+0.03			+2.86	+0.05	+0.02	+0.02		
0.05	+0.05	+0.03	+0.02	+0.02	+0.01	+0.01	+2.33	+0.03	+0.01	+0.02		
0.025	+0.04	+0.02	+0.02	+0.01	+0.01	+0.01	+2.26	+0.02	+0.01	+0.01		
0.01	+0.03	+0.02	+0.02	+0.02	+0.01	+0.01	+2.83	+0.02	+0.01	+0.01		
0.001	+0.04	+0.04	+0.04	+0.04	+0.03	+0.01	+7.08	+0.04	+0.00	+0.00		

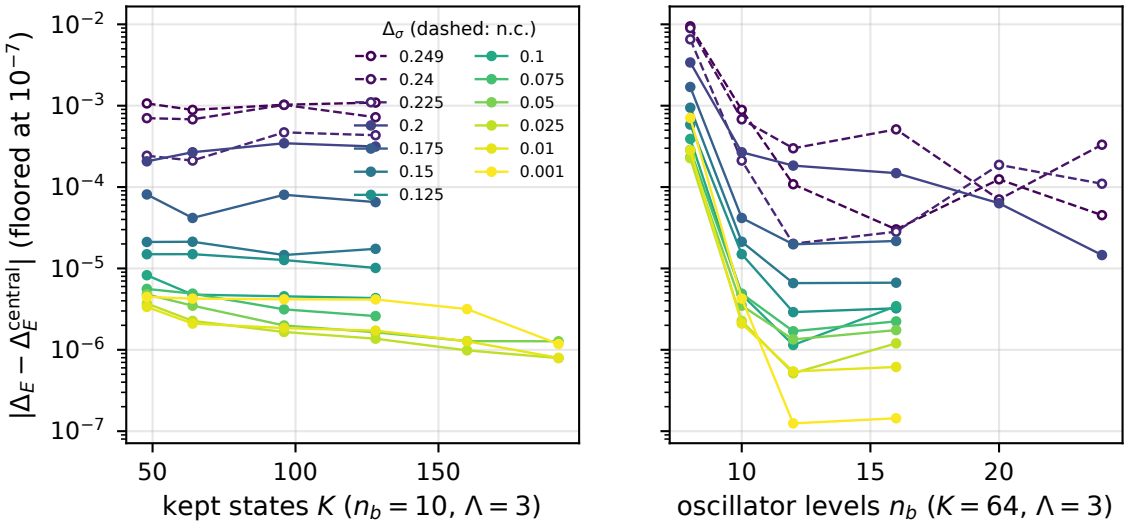


Figure 4: $|\Delta_E - \Delta_E^{\text{central}}|$ against K (left, $n_b = 10$) and n_b (right, $K = 64$) at $\Lambda = 3$; zero deviations are drawn at 10^{-7}. Flat curves mean converged. Dashed: the three points not converged in chain depth. In the left panel the level of the $\Delta_\sigma \geq 0.2$ curves is set by $n_b = 10$.

the largest spread at a plateau point is 6.8×10^{-5} (at $\Delta_\sigma = 0.15$, set by $\Lambda = 4$). How many kept states the limit $\Lambda \to 1$ needs was not determined, and the limit is not tested.

- **At** $\Delta_\sigma = 0.2$ only $\Lambda = 2.5$ and 3 were run with the converged $n_b = 16$; their extrapolations agree to 2×10^{-5}. At $n_b = 10$ the extrapolated values spread by 4×10^{-4} over $\Lambda = 2$–4 (Fig. 5). We attribute this to an interplay of Λ with the $n_b = 10$ truncation (the $\Lambda = 2.5$ minus $\Lambda = 3$ difference changes sign between $n_b = 10$ and 16), but that is inferred, not measured; it is within δ_{tot} only through the large δ_{depth}.

An earlier version of this note reported that Δ_E rises as Λ decreases and covered the continuum limit with twice the $\Lambda = 2.5$ shift. Both came from the midpoint estimator: the window rule counts sites while the chain length scales as $1/\ln\Lambda$, so smaller Λ admitted a larger share of transient windows (App. E).

7.4 The final table

Table 1 gives the final values and Table 2 their errors. Starting from the Gaussian value 1/2 at $s = 1/2$, Δ_ε first decreases (the ε series at $\Delta_\sigma \geq 0.2$), passes through a shallow minimum of about 0.4830 near $\Delta_\sigma = 0.2$, and rises toward 1 as $s \to 1$. The window audits of three central items are shown in Fig. 7. The number of quality windows falls from 106 at $\Delta_\sigma = 0.001$ to 2 at $\Delta_\sigma = 0.249$ (Table 2).

Table 6: The Λ study at three points ($n_b = 10$ unless stated): Δ_E with this note's estimator (plateau value at $\Delta_\sigma = 0.1$, 0.05; extrapolated at 0.2), the superseded midpoint with its half-width, and α_c. N is scaled with Λ so that the infrared scale Λ^{-N} is fixed. At $\Lambda = 1.5$ no item is converged in K.

Δ_σ	Λ	K	n_b	N	Δ_E (this note)	old midpoint (half-width)	α_c
0.2	1.5	64	10	176	0.536299	0.53659 (0.00036)	0.1626225790
0.2	1.5	64	16	176	0.536300	0.53660 (0.00036)	0.1626210299
0.2	1.5	128	10	176	0.507038	0.50754 (0.00102)	0.1603705735
0.2	2	64	10	103	0.484078	0.48534 (0.00177)	0.1676281822
0.2	2	128	10	103	0.483021	0.48471 (0.00165)	0.1676098637
0.2	2	192	10	103	0.483061	0.48448 (0.00193)	0.1676082707
0.2	2	256	10	103	0.483014	0.48449 (0.00188)	0.1676081346
0.2	2.5	64	10	78	0.483221	0.48426 (0.00131)	0.1761692489
0.2	2.5	128	10	78	0.483173	0.48440 (0.00101)	0.1761670918
0.2	3	64	10	65	0.483248	0.48404 (0.00063)	0.1847222031
0.2	3	128	10	65	0.483295	0.48406 (0.00061)	0.1847217711
0.2	4	64	10	52	0.483381	0.48401 (0.00034)	0.2008115424
0.1	1.5	64	10	176	0.567059	0.56714 (0.00019)	0.3786792319
0.1	1.5	64	16	176	0.567059	0.56714 (0.00019)	0.3786839607
0.1	1.5	128	10	176	0.539196	0.53968 (0.00049)	0.3733064368
0.1	2	64	10	103	0.529796	0.53042 (0.00063)	0.3936704020
0.1	2	128	10	103	0.529631	0.53025 (0.00063)	0.3936438035
0.1	2	192	10	103	0.529613	0.53024 (0.00063)	0.3936429089
0.1	2	256	10	103	0.529609	0.53023 (0.00064)	0.3936426886
0.1	2.5	64	10	78	0.529637	0.53010 (0.00046)	0.4167902745
0.1	2.5	128	10	78	0.529610	0.53007 (0.00046)	0.4167874486
0.1	3	64	10	65	0.529613	0.52984 (0.00022)	0.4397697893
0.1	3	128	10	65	0.529612	0.52984 (0.00023)	0.4397694449
0.1	4	64	10	52	0.529628	0.52981 (0.00018)	0.4825812310
0.05	1.5	64	10	217	0.670692	0.67093 (0.00026)	0.5909590331
0.05	1.5	64	16	217	0.670693	0.67092 (0.00026)	0.5909603112
0.05	1.5	128	10	217	0.630353	0.63061 (0.00026)	0.5780543351
0.05	2	64	10	127	0.622300	0.62262 (0.00032)	0.6120267877
0.05	2	128	10	127	0.622125	0.62241 (0.00036)	0.6119924970
0.05	2	192	10	127	0.622119	0.62244 (0.00033)	0.6119916485
0.05	2	256	10	127	0.622116	0.62242 (0.00034)	0.6119911289
0.05	2.5	64	10	96	0.622139	0.62240 (0.00026)	0.6507599139
0.05	2.5	128	10	96	0.622116	0.62235 (0.00029)	0.6507542509
0.05	3	64	10	80	0.622119	0.62236 (0.00024)	0.6891357316
0.05	3	128	10	80	0.622117	0.62236 (0.00024)	0.6891354544
0.05	3	192	10	80	0.622117	0.62236 (0.00024)	0.6891353149
0.05	4	64	10	63	0.622122	0.62223 (0.00013)	0.7603342269

7.5 The limit $s \to 1$

Table 8 compares $y_t = 1 - \Delta_E$ with the leading form $\sqrt{2\delta} = 2\sqrt{\Delta_\sigma}$ and the next-to-leading form of Eq. (3). The ratio to the leading form approaches 1 as $s \to 1$: $y_t = 0.06258$ against 0.06325 at $\Delta_\sigma = 0.001$ (1.1%) and 0.19025 against 0.20000 at 0.01 (4.9%). Against the next-to-leading form 0.06275 the difference at $\Delta_\sigma = 0.001$ is -1.7×10^{-4}. The remainder divided by $\delta^{3/2}$ changes smoothly over $0.001 \le \Delta_\sigma \le 0.075$ (-1.90 to -1.32): it is -1.68 at $\Delta_\sigma = 0.01$, where $\delta_{\rm tot}/\delta^{3/2} = 0.02$, and -1.90 at 0.001, where $\delta_{\rm tot}/\delta^{3/2} = 1.2$ is as large as the entry. The dual expansion (4) fixes the coefficient of $\delta^{3/2}$ in y_t, $-C_3^{\rm SR}(1)/2^{3/2} = -2.025$, which the remainder approaches as $s \to 1$.

The accompanying paper [1] tests the dual expansion directly at $\Delta_\sigma = 0.01$, where $\delta_{\rm tot} = 4.7 \times 10^{-5}$. The difference between Δ_E and the series (4) summed through $C_n^{\rm SR}(1)$ is $+4.7 \times 10^{-3}$, -9.8×10^{-4}, $+2.4 \times 10^{-4}$ and -5.3×10^{-5} for $n = 2$, 3, 4 and 5: it shrinks to the size of $\delta_{\rm tot}$ as $C_3^{\rm SR}(1)$, $C_4^{\rm SR}(1)$ and $C_5^{\rm SR}(1)$ are added. The term of $C_6^{\rm SR}(1)$, of magnitude 5.9×10^{-5}, is itself of the size of $\delta_{\rm tot}$ and brings the difference to $+5.8 \times 10^{-6}$; the higher terms lie below $\delta_{\rm tot}$. NRG thus reproduces the dual expansion through $C_5^{\rm SR}(1)$ and is consistent with $C_6^{\rm SR}(1)$.

8 Checks

A second implementation of the same numerical map tests the code (Sec. 8.1); two methods that do not use NRG test the method (Secs. 8.2 and 8.3); Sec. 8.4 lists published values. Figure 1 summarizes the comparisons.

8.1 A second implementation

Design. Claude Code, which knew the production code, wrote a specification of the physics and conventions only (`independent_check/SPEC.md`). It fixed the discretization, the chain, the truncation per parity sector,

Table 7: Λ and K shifts at all thirteen points, in units of 10^{-4}, relative to $\Lambda = 3$ at the same K and n_b ($\Lambda = 2.5$ uses $n_b = 16$ for $\Delta_\sigma \geq 0.2$, else 10). The $\Lambda = 2$, $K = 64$ column is not converged in K. Last columns: the K doubling at $\Lambda = 2.5$ ($n_b = 10$) and the number of quality windows of the $\Lambda = 2.5$, $K = 128$, $n_b = 10$ item.

	$10^4\times$ shift relative to $\Lambda = 3$			at $\Lambda = 2.5$	
Δ_σ	Λ=2, K=64	Λ=2.5, K=128	Λ=4, K=64	$10^4 \times$ (K=64 minus K=128)	n_q
0.249	+41.37	+1.21	+17.22	+0.88	13
0.24	+34.03	+0.71	+16.43	+2.54	13
0.225	+19.17	+2.29	+10.71	+6.05	17
0.2	+8.30	+0.18	+1.33	+0.49	22
0.175	+3.71	−0.53	+1.50	+1.55	25
0.15	+2.38	−0.10	+0.54	+0.67	28
0.125	+1.90	−0.07	+0.23	+0.38	31
0.1	+1.84	−0.02	+0.15	+0.27	32
0.075	+1.75	−0.02	+0.06	+0.24	36
0.05	+1.81	−0.01	+0.03	+0.23	43
0.025	+1.62	−0.01	+0.03	+0.26	47
0.01	+2.02	−0.00	+0.06	+0.35	49
0.001	+5.63	−0.11	+0.12	+1.05	126

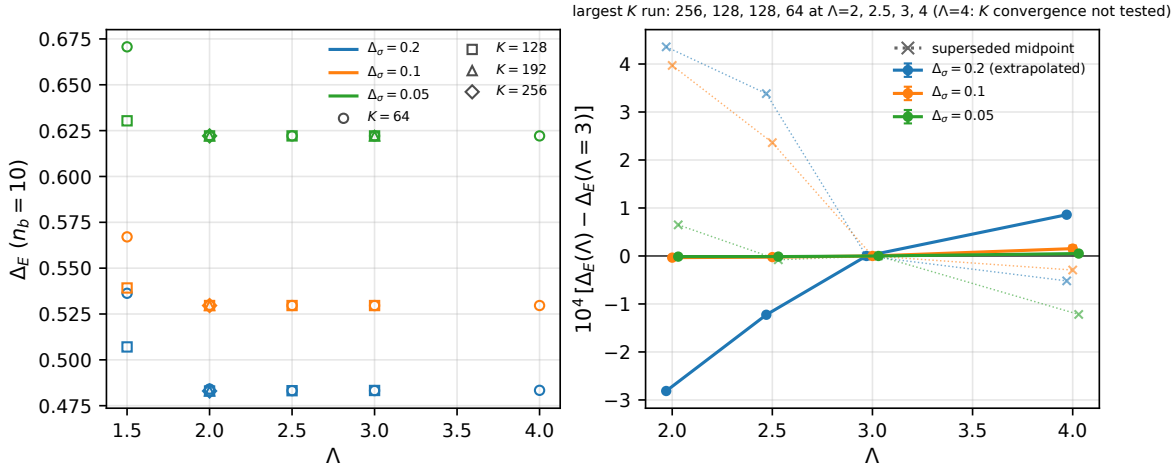


Figure 5: Left: Δ_E against Λ for all K at $n_b = 10$ (open symbols by K, color by Δ_σ); at $\Lambda = 1.5$ no item is converged in K. Right: K-converged items relative to $\Lambda = 3$ with this note's estimator (bars: plateau drift) and, dotted, the superseded midpoints, whose apparent Λ dependence is an artifact of the window rule.

the critical-search rule and the window rule, and forbade access to the production code, data and this note. From it a separate Claude agent wrote a second solver (`independent_check/indep_nrg.py`); that it did not open the forbidden files rests on the instruction and on the agent's own statement in `NOTES.md`, not on an access log. Every algorithmic choice differs from production:

	production (`lri_spin`)	second implementation (`indep_nrg.py`)
chain coefficients	closed form (8)	Lanczos with full reorthogonalization on the star, in mp-math
parity blocks	basis order of the reference solver	NumPy Kronecker order, generic operators
eigensolver	partial spectrum (flows), full (derivative)	full spectrum everywhere
y_t	exact derivative (13)	central finite differences in α, relative steps $\eta = 10^{-9}, 10^{-10}, 10^{-11}$
window audit	package code	written from the specification

The Lanczos recursion runs with $40 + 1.3N \log_{10}\Lambda$ decimal digits and is repeated with 40 more digits and 40 more star modes to check convergence. The second implementation has its own 38 self-tests (for example, the chain reproduces the star to 3×10^{-50}, and the NRG reproduces exact diagonalization of untruncated systems to 5×10^{-15}). Eight configurations c1–c8 with $0.502 \leq s \leq 0.998$, $\Lambda = 2$, 2.5 and 3, $K = 48$ or 64 and $n_b = 10$–16 ran on one cluster node, one single-core process each, the longest in 28 min; each is compared with the production record of the same item.

Result. Table 9 lists the comparison.

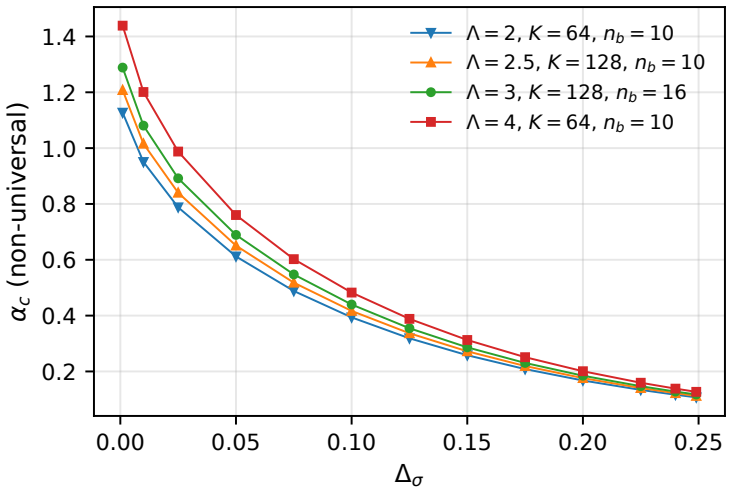


Figure 6: The critical coupling is not universal: it depends on Λ and weakly on K and n_b. It enters no result.

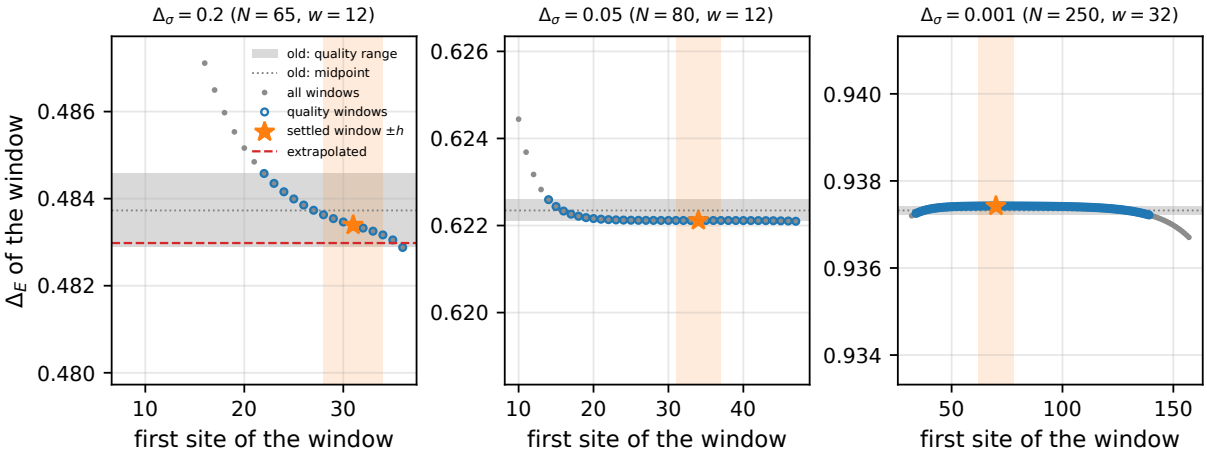


Figure 7: Window audit of three central items: Δ_E of every window (gray dots) and of the quality windows (open circles) against the first site of the window; star and orange band: the settled window and its neighborhood $\pm h$; gray band and dotted line: the quality range and its midpoint, the superseded estimate; dashed (at $\Delta_\sigma = 0.2$): the extrapolated value. The vertical scales differ between panels. At $\Delta_\sigma = 0.05$ and 0.001 the quality windows end on a flat plateau, at 0.2 they drift downward to the end.

- The chain coefficients agree to better than 10^{-15} (relative): to rounding, whose last bits depend on the platform. The closed form is confirmed.
- The critical couplings are bitwise identical in 5 configurations and agree to 7.2×10^{-14} in the others. The critical flows agree to 5.1×10^{-13} over the first eleven sites.
- A window's finite-difference Δ_E is *trusted* when two neighboring steps η agree to 10^{-6}. In all 316 trusted windows (17 to 126 per configuration) the two implementations agree to 3.1×10^{-7}, with median 4×10^{-9}, far below the quoted errors. In the 3 configurations whose settled window is trusted, they agree there to 8.2×10^{-8}.
- The latest windows of a chain lie beyond the reach of finite differences, because $\eta\, \alpha\, \Lambda^{y_t n}$ is no longer small there. Only 228 of the 296 production quality windows are within reach. All quality windows of a configuration are within reach only in $c3$ and $c6$; at $s = 0.502$ (c5) none is. This is a limit of the check, not a disagreement.

The check verifies the numerical map at $K \leq 64$. It does not test the central items at $K = 128$, $n_b = 16$ directly (they differ only in K and n_b), the estimator (it shares the window rule by construction), or the limit $\Lambda \to 1$. Both implementations come from the same model family, and the specification was written by the

Table 8: $y_t = 1 - \Delta_E$ near $s = 1$ against the leading form $\sqrt{2\delta}$ ($\delta = 1 - s = 2\Delta_\sigma$) and the next-to-leading form $\sqrt{2\delta} - \delta/4$ of Refs. [8, 9]; the remainder in units of $\delta^{3/2}$, and $\delta_{\rm tot}$ in the same units (the uncertainty of the remainder).

Δ_σ	$\delta = 1 - s$	$y_t = 1 - \Delta_E$	$\sqrt{2\delta} = 2\sqrt{\Delta_\sigma}$	ratio	$\sqrt{2\delta} - \delta/4$	$(y_t - \sqrt{2\delta} + \delta/4)/\delta^{3/2}$	$\delta_{\rm tot}/\delta^{3/2}$
0.075	0.15	0.43374	0.54772	0.792	0.51022	-1.32	0.00051
0.05	0.1	0.37788	0.44721	0.845	0.42221	-1.40	0.00085
0.025	0.05	0.28658	0.31623	0.906	0.30373	-1.53	0.0026
0.01	0.02	0.19025	0.20000	0.951	0.19500	-1.68	0.017
0.001	0.002	0.06258	0.06325	0.989	0.06275	-1.90	1.2

agent that knew the production code: the check tests the code, not shared conceptual choices.

Table 9: Second implementation against production. Columns: configuration (s, Λ, K, n_b, N); largest relative difference of the rescaled chain coefficients (a rounding-level bound); relative difference of α_c; number of trusted windows; largest window-by-window difference $|\delta\Delta_E|$ over them; production quality windows within finite-difference reach (covered/all); production quality range.

id	(s, Λ, K, n_b, N)	chain	α_c	trusted	$\max\|\delta\Delta_E\|$	covered	quality range (production)
c1	$(0.8, 3, 64, 10, 65)$	$< 10^{-15}$	bitwise	19	2×10^{-7}	13/23	[0.529612, 0.530062]
c2	$(0.9, 3, 64, 10, 80)$	$< 10^{-15}$	4×10^{-14}	27	9×10^{-8}	23/34	[0.622119, 0.622597]
c3	$(0.98, 3, 64, 10, 105)$	$< 10^{-15}$	bitwise	53	3×10^{-7}	41/41	[0.809460, 0.810127]
c4	$(0.6, 3, 64, 12, 65)$	$< 10^{-15}$	bitwise	17	2×10^{-7}	3/15	[0.483316, 0.484567]
c5	$(0.502, 3, 64, 16, 65)$	$< 10^{-15}$	bitwise	18	2×10^{-7}	0/2	[0.508000, 0.508268]
c6	$(0.998, 3, 48, 10, 250)$	$< 10^{-15}$	4×10^{-14}	126	2×10^{-7}	106/106	[0.937223, 0.937429]
c7	$(0.8, 2, 64, 10, 103)$	$< 10^{-15}$	7×10^{-14}	34	2×10^{-7}	28/46	[0.529796, 0.531050]
c8	$(0.7, 2.5, 64, 10, 78)$	$< 10^{-15}$	bitwise	22	2×10^{-7}	14/29	[0.491178, 0.492328]

8.2 The ε expansion near $s = 1/2$

Reference. The accompanying paper gives $\Delta_\varepsilon = \sum_{n=0}^{8} C_n \varepsilon^n$ for $d = 1$ [1], with $C_0 = 1/2$, $C_1 = -1/6$, $C_2 = 0.657131263981$, $C_3 = -3.719856322902$, $C_4 = 45.42488010356$, $C_5 = -731.13947662454$ and $C_6 = 14331.6870328$ (exact to the digits shown), $C_7 = -322550 \pm 40$ and $C_8 = (8.060 \pm 0.017) \times 10^6$. The uncertainties of C_7 and C_8 are the paper's allowances, mostly four sampling standard errors. They are not part of the Borel–Padé spread below; shifting C_7 or C_8 by about its allowance moves the [4/4] approximant by at most 1.1×10^{-5} at $\Delta_\sigma = 0.225$ and 4.6×10^{-4} at 0.2, less than the spread at both points. The coefficients grow factorially, so the series is asymptotic. We evaluate it in two ways (`method_comparison/compare_methods.py`):

- **optimal truncation**: the partial sum up to, but excluding, the smallest term, with that term as its error;
- **Borel–Padé**: Padé approximants $[L/M]$ of the Borel transform $\sum_n C_n u^n/n!$ with $L + M = 6, 7, 8$, keeping those without a pole on the positive axis and with a value between 0.3 and 1.2, Laplace-transformed back. The central value is the upper of the two middle values of the $L + M = 8$ approximants (there are four at $0.2 \le \Delta_\sigma \le 0.24$), not their mean; the error is the half-range of all approximants with $L + M \ge 7$. The $L + M = 6$ approximants are computed but enter neither. The choice of the upper middle value is a convention of the comparison script of 8 October 2026, unchanged since and not chosen with the present NRG values in view; it has no further justification. At $\Delta_\sigma \ge 0.225$, where the approximants agree to within the quoted spread, it hardly matters; at $\Delta_\sigma = 0.2$ it does (below).

The reference is the partial sum at $\Delta_\sigma = 0.249$ ($\varepsilon = 0.004$), where all terms decrease and the error is 5×10^{-13} (Borel–Padé gives 0.499344 too), and the Borel–Padé sum at larger ε. At $\Delta_\sigma = 0.24$ the two evaluations agree to 3×10^{-5}. At 0.225 the Borel–Padé sum, 0.48808, has a small spread, but optimal truncation gives 0.490(4), so the reference there depends on the resummation. At $\Delta_\sigma = 0.2$ the spread, 9×10^{-3}, is too large to test NRG at the 10^{-3} level.

Result. Table 10. At $\Delta_\sigma = 0.2$ the series does not test NRG: the extrapolated NRG value lies $+6 \times 10^{-5}$ from the central approximant, a coincidence of the convention above, and $+2.1 \times 10^{-3}$ from the median 0.4809

of the 4 approximants with $L + M = 8$, which range from 0.468 to 0.483; both differences are well inside the spread, and neither is evidence for or against NRG. At the three points not converged in chain depth the settled values lie above the series by $+8.7 \times 10^{-3}$, $+4.7 \times 10^{-3}$ and $+1.6 \times 10^{-3}$, decreasing with ε, as the slow approach of Sec. 7.1 predicts. Two extrapolations serve as cross-checks; neither is adopted:

- **The rule's own extrapolation** (Sec. 3.7) is not stable there. Over the three fit ranges and $\Lambda = 2.5$, 3 it gives 0.4934–0.4970 (-8×10^{-4} to $+3 \times 10^{-3}$ from the series) at $\Delta_\sigma = 0.24$, and 0.4867–0.4888 (-1×10^{-3} to $+7 \times 10^{-4}$) at 0.225. At $\Delta_\sigma = 0.249$, where $\varepsilon l_{\max} \approx 0.25$, a single exponential cannot represent the $1/l$ approach, and the extrapolation (0.5046–0.5057) removes less than half of the excess.
- **The crossover fit** of Sec. 3.8 is stable and moves the values toward the series, but not onto it. At $\Delta_\sigma = 0.225$ it gives 0.48810 (7 accepted fits, 0.48808 to 0.48854, i.e. $+4 \times 10^{-6}$ to $+5 \times 10^{-4}$ from the series value 0.48808). At 0.24 it gives 0.49460 (0.49442 to 0.49501): every accepted fit lies above the series value 0.49422, by $+2 \times 10^{-4}$ to $+8 \times 10^{-4}$, against $+4.7 \times 10^{-3}$ for the settled value. At 0.249 it gives 0.50209, $+3 \times 10^{-3}$ from the series: there the fit is ill-conditioned, and the one-loop test of Sec. 7.1 is the relevant check.

Interpretation. The excess is the expected crossover from the strong bare coupling of the spin–boson model to the interacting fixed point, whose correction exponent $\omega \approx \varepsilon$ vanishes at $s = 1/2$; within $l_{\max}$ the flow has not arrived. It depends only on the RG time, not on K, n_b or Λ at fixed depth (Sec. 7.1), and at $\Delta_\sigma = 0.249$ it decays at a rate of the one-loop size. Logarithmic corrections of this kind were already seen in NRG at $s = 1/2$ [15]. An earlier version of this note attributed the excess to the known truncation errors of bosonic NRG near Gaussian fixed points (Hilbert-space truncation and mass flow [15, 17, 18]). The data do not support that: those errors affect order-parameter exponents below $s = 1/2$, and for $s > 1/2$ the mass-flow correction changes the flows only quantitatively (at $s = 0.6$, "no qualitative difference between the flows without and with mass-flow correction" [17]); we have no run that isolates either effect. The same mechanism acts at $\Delta_\sigma = 0.2$ and 0.175 at a smaller level (shifts -4.1×10^{-4} and -8.6×10^{-5} from settled to extrapolated), where the extrapolation is stable and its full length is in δ_{depth}.

Decision. The three values at $\Delta_\sigma = 0.249$, 0.24 and 0.225 are kept as computed (the settled values), marked "not converged in chain depth", shown in the tables and figures, and used in no comparison and no agreement statement. This is the author's decision. The label is not an outcome of the estimator, and it was assigned when this comparison was already known (Sec. 3.7); App. C gives its history, including an earlier recommendation that the author overruled.

Table 10: NRG against the ε series near $s = 1/2$. Optimal truncation with its smallest term; Borel–Padé central value with the spread over approximants; NRG minus the reference (the partial sum at $\Delta_\sigma = 0.249$, the Borel–Padé sum otherwise; at $\Delta_\sigma = 0.2$ this difference is not meaningful, see text); the range of the rule's extrapolations and the crossover fit of Sec. 3.8 (cross-checks, not adopted). Bracketed: settled values not converged in chain depth, not used.

Δ_σ	ε	NRG Δ_E	optimal truncation	Borel–Padé (spread)	NRG − reference	rule's extrapolation	crossover fit
0.249	0.004	[0.508030]	0.499344 ($\pm 5 \times 10^{-13}$)	0.49934 (6×10^{-15})	$+8.7 \times 10^{-3}$	0.5046–0.5057	0.50209
0.24	0.04	[0.498922]	0.494194 ($\pm 5 \times 10^{-5}$)	0.49422 (2×10^{-7})	$+4.7 \times 10^{-3}$	0.4934–0.4970	0.49460
0.225	0.1	[0.489668]	0.489905 ($\pm 4 \times 10^{-3}$)	0.48808 (1×10^{-4})	$+1.6 \times 10^{-3}$	0.4867–0.4888	0.48810
0.2	0.2	0.4830(7)	0.466667 (± 0.03)	0.48292 (9×10^{-3})	$+5.9 \times 10^{-5}$	adopted	0.48305

8.3 The 1D cluster Monte Carlo study

Reference. The Monte Carlo study of the same model in this repository (`MC/` folder [2]; Fukui–Todo cluster algorithm [24, 25] with finite-size scaling) is the project's own, run by the same AI systems; it is independent of NRG in method, not in authorship. We use only its quoted values and labels; their definitions are in its note [2]. In brief, its value of Δ_ε comes from a finite-size-scaling fit of all lattice sizes, with a statistical error σ_{stat} and a systematic scale σ_{sys}, the half-range of the values over a family of fit models (a sensitivity scale, not a standard deviation). As a cross-check it also extrapolates in the lattice size the crossings, between pairs of sizes, of a finite-size-scaling observable; this *crossing extrapolation* is quoted as $\Delta_\varepsilon^\infty(\sigma_\infty)(\Delta_{\omega'})$, with error σ_∞ and sensitivity $\Delta_{\omega'}$ to the assumed correction exponent ω' ($\Delta_{\omega'}$ is a sensitivity, not a dimension).

Each value carries a label: **accepted**, the fit value and the crossing extrapolation agree within twice the error of their difference ("σ_{sys}-dominated" marks values whose σ_{sys} is much larger than σ_{stat}); **cross-check failed**, they differ by more, and the study does not certify the value as an infinite-size value; **cross-check unavailable**, fewer than two size series are available for the crossing extrapolation; **conditional**, σ_{sys} could not be estimated.

Result. Table 11.

- **Accepted ($\Delta_\sigma = 0.125, 0.1, 0.075, 0.05, 0.025$).** NRG and Monte Carlo differ by at most 2.9×10^{-3} (at $\Delta_\sigma = 0.1$). The differences exceed the Monte Carlo statistical errors, by up to $14\,\sigma_{\mathrm{stat}}$, and have the same sign at neighboring points (below). They lie within σ_{sys} at 3 of the 5 points and within $\sigma_{\mathrm{stat}} + \sigma_{\mathrm{sys}} + \delta_{\mathrm{tot}}$ at 4, the largest ratio being 1.08 at $\Delta_\sigma = 0.1$. This linear sum adds a standard deviation, a fit-model half-range and an NRG error scale; we use it as a scale for the size of the differences, not as a test of statistical consistency. At $\Delta_\sigma = 0.1$ the Monte Carlo crossing extrapolation agrees with NRG to $0.10\,\sigma_\infty$; at all 5 accepted points it agrees within $1.05\,\sigma_\infty$, and within 0.19 of $\sigma_\infty + \Delta_{\omega'} + \delta_{\mathrm{tot}}$. The NRG errors are smaller than σ_{sys} by factors 36–333, so the comparison tests the Monte Carlo systematics more than it tests NRG; only σ_{sys} reconciles the two.
- **Cross-check failed ($\Delta_\sigma = 0.175$ and 0.15).** These are not infinite-size values, and we use them only to say where NRG lies: at 0.175 between the fit and the crossing extrapolation (-7.0×10^{-4} from the fit, $+8.6 \times 10^{-4}$ from the crossing), at 0.15 below both (-2.1×10^{-3} and -7.2×10^{-4}). Including these two points, the crossing extrapolations agree with NRG within $1.06\,\sigma_\infty$ at all 7 common points.
- **Correlated signs.** NRG lies below the Monte Carlo fit at 5 consecutive points of the 7 (0.175 to 0.075) and above it at 0.05 and 0.025. The differences are therefore not independent fluctuations; they look like a smooth systematic of one of the two methods, which the Monte Carlo σ_{sys} is meant to cover.
- $\Delta_\sigma = 0.225$ is conditional in the Monte Carlo study and not converged in NRG; the Monte Carlo value lies -9.0×10^{-3} from the series. It enters no comparison.

Table 11: NRG against the 1D cluster Monte Carlo study at the common points. MC value with σ_{stat} and σ_{sys} in units of the last digit, and its label (Sec. 8.3); NRG minus MC; "ratio" is $|\mathrm{NRG} - \mathrm{MC}|$ divided by $\sigma_{\mathrm{stat}} + \sigma_{\mathrm{sys}} + \delta_{\mathrm{tot}}$ (shown also for the cross-check-failed points, which are not infinite-size values); MC crossing extrapolation with σ_∞ and $\Delta_{\omega'}$. MC values are quoted as that study gives them. *Not converged in NRG, not used.

Δ_σ	NRG Δ_E	MC Δ_ε $(\sigma_{\mathrm{stat}})(\sigma_{\mathrm{sys}})$	MC label	NRG − MC	ratio	MC crossing $(\sigma_\infty)(\Delta_{\omega'})$
0.225*	[0.4897]	0.47912(68)(–)	conditional (no σ_{sys})	$+0.011$	–	–
0.175	0.48390(35)	0.48460(35)(67)	cross-check failed	-7.0×10^{-4}	0.51	0.48304(81)(1)
0.15	0.49111(11)	0.49325(41)(306)	cross-check failed	-2.1×10^{-3}	0.60	0.49183(71)(70)
0.125	0.50567(5)	0.50710(35)(193)	accepted	-1.4×10^{-3}	0.61	0.50640(172)(392)
0.1	0.529608(33)	0.53248(20)(242)	accepted (σ_{sys}-dom.)	-2.9×10^{-3}	1.08	0.52975(135)(248)
0.075	0.566260(30)	0.56694(37)(139)	accepted	-6.8×10^{-4}	0.38	0.56843(207)(1044)
0.05	0.622115(27)	0.61987(55)(220)	accepted	$+2.2 \times 10^{-3}$	0.81	0.62393(348)(773)
0.025	0.713422(29)	0.71115(49)(959)	accepted (σ_{sys}-dom.)	$+2.3 \times 10^{-3}$	0.22	0.70906(902)(1509)

8.4 Published values

- **1D LRI Monte Carlo.** Uzelac, Glumac and Aničić [26] give $1/\nu = 0.50(2)$, 0.51(2), 0.47(2) and 0.40(1) at $s = 0.6$, 0.7, 0.8 and 0.9, i.e. $\Delta_\varepsilon = 0.50(2)$, 0.49(2), 0.53(2) and 0.60(1) at $\Delta_\sigma = 0.2$, 0.15, 0.1 and 0.05. NRG differs from them by -0.85, $+0.055$, -0.020 and $+2.2$ of their errors; at $\Delta_\sigma = 0.05$ the project's Monte Carlo value, 0.61987(55)(220), also lies above theirs. Angelini, Parisi and Ricci-Tersenghi [27] give $1/\nu = 0.4124(13)$ at $s = 0.875$ ($\Delta_\varepsilon = 0.5876(13)$ at $\Delta_\sigma = 0.0625$) and 0.506(14) at $s = 0.65453$; neither point is on our grid. Four interpolations of our final values (a cubic spline over $0.01 \le \Delta_\sigma \le 0.15$ and polynomials through three or four neighboring points) give 0.5898–0.5918 at $\Delta_\sigma = 0.0625$, 1.7 to 3.2 of their errors above 0.5876; the project's Monte Carlo study gives 0.59003(35)(150) there (label: cross-check unavailable).
- **The spin–boson model.** Bosonic NRG [12, 15] and continuous-time quantum Monte Carlo [20] give $1/\nu$ (or νz) for $s > 1/2$ only in figures. Read off the figures by eye, the NRG values of

Ref. [15] are about 0.50, 0.50, 0.46, 0.37 and 0.28 at $s = 0.6$, 0.7, 0.8, 0.9 and 0.95, against our $y_t = 0.517, 0.509, 0.470, 0.378$ and 0.287, and the quantum Monte Carlo values agree with both within their error bars; this is agreement at the resolution of the figures, a few hundredths, not a quantitative test. Other methods for this model [21, 19] report order-parameter exponents, not ν.

- **Functional RG and the hierarchical model** [28] give $1/\nu$ for $1/2 < s < 1$ in figures; near $s = 1$ they quote the form of Eq. (3).

9 Limitations and open problems

1. **Chain depth near $s = 1/2$.** At $\Delta_\sigma = 0.249$, 0.24 and 0.225 the flow does not reach its fixed point within the RG time that double precision allows (Sec. 7.1). The crossover fit of Sec. 3.8 (cross-check, not adopted) is stable there and gives 0.48810 and 0.49460 at $\Delta_\sigma = 0.225$ and 0.24, closer to the series than the settled values but above it at 0.24; it is ill-conditioned at 0.249 (Sec. 8.2). Extended precision (longer $l_{\max}$) would remove the need to extrapolate. At $\Delta_\sigma = 0.2$ and 0.175 the values are extrapolations; they depend on the fit form, and δ_{depth} covers the full extrapolation length.
2. **Continuum limit.** Over $2 \le \Lambda \le 4$ the plateau values change by at most a few 10^{-5}, monotonically, but $\Lambda < 2$ is not reached, K convergence at $\Lambda = 2.5$ rests on one doubling (covered by an allowance) and at $\Lambda = 4$ is assumed. At $\Delta_\sigma = 0.001$ the plateau depends on K at $\Lambda \neq 3$ at the 10^{-4} level. Runs at $\Lambda \le 1.5$ with more than 128 kept states, or at $\Lambda = 2$ with $K \ge 192$ at all points, would improve this. At $K = 512$ and $n_b = 10$ a parity block has dimension about 5000, which dense diagonalization handles, but it must be repeated at every site of every bisection step.
3. **Universality in Ω.** All runs use $\Omega = 0.1$. A second Ω at one or two points would test universality and the reading of the approach as a correction to scaling (the l dependence should move, the plateau value not).
4. **Error model.** The errors are scales, not standard deviations; the linear sum, the plateau threshold, the fit form and ranges, the full extrapolation length and the K-doubling allowance are judgements (Sec. 4). The estimator was chosen after the NRG data and after an earlier comparison had been seen, and the label "not converged in chain depth" was assigned after the comparison with the ε series.
5. **Coverage of the second implementation.** It ran at $K \le 64$, could not reach the latest windows, including both quality windows at $s = 0.502$, and shares the window rule (Sec. 8.1).
6. **Review.** The analysis, the estimator and this note were reviewed only by AI agents: two rounds on 13 October 2026 by agents of the same family as the AI that wrote them, and a reading on 15 October (App. C). The changes made after the second round and after the reading have not been reviewed again.
7. **Not needed.** Hamiltonian-truncation calculations of the same model [1, 3] need a correction for a fixed energy cutoff on a free Fock space. NRG truncates iteratively in the interacting basis, keeps the complete new site for every kept state and is non-perturbative in α, so no such correction applies, and none was added.

10 Reproduction

10.1 The released folder

The `NRG/` folder of the repository contains:

- `code/lri_spin_boson_hpc/`: the solver, its tests, the reference solver of the original study and the cluster tools (App. C lists what was removed from the original tree);
- `code/campaigns/`: the generators of the item lists in `inputs/items/`;
- `code/analysis/`: the analysis stages (`summarize.py`, `estimator.py`, `build_tables.py`, `make_figures.py`, `note_numbers.py`, `check_figures.py`) and the collector of cluster records, `collect_records.py`;

- `inputs/`: the item lists, the frozen manifests that were submitted, the validation receipt and the environment record;
- `data/`: the 197 result records, the per-job lane logs and the job accounting;
- `results/`: the final table (`ds13_final.json`, `ds13_final_table.md`) and the Λ study;
- `independent_check/`, `independent_check_compare/`, `method_comparison/`: the checks of Sec. 8;
- `verify/`: the verifier, the expected values and the smoke-test reference; `slurm/`: job templates; `note/`: this note.

`README.md`, `REPRODUCE.md`, `PROVENANCE.md`, `REVIEWS.md` and `data/DATA.md` in that folder give the details.

10.2 How to run

From `NRG/`, with Python ≥ 3.10, NumPy, SciPy, threadpoolctl, mpmath, matplotlib and pytest:

- `make test` (about 75 s on one core of a recent laptop): the 85 solver tests, the checksum list of the code, and the 38 self-tests of the second implementation.
- `make replay` (20–30 s): checks the solver source hash, verifies every record (checksum, identity against its manifest and the receipt, numerical gates), recomputes all results, tables, in-text numbers and figures of this note in a hidden workspace `.work/`, and compares them with the released files and with `verify/expected.json`. It also needs the Monte Carlo table of the repository, `../MC/data/1d/scan_data/FIT_V3_SCAN.json` (`make replay MCSCAN=<path>` to point elsewhere). The extrapolations are least-squares fits whose last bits depend on the NumPy/SciPy build and the CPU, so the comparison uses tolerances (about 10^{-9} for values, one unit of the last printed digit for tables), not byte identity (`REPRODUCE.md`).
- `make smoke` (about 15 s): one small critical search with both solvers, compared with each other and with the reference values in `verify/smoke_expected.json`.
- `make note`: rebuilds this note in place from the released tables and figures.

Rerunning production needs a Slurm cluster: 178 thread-hours, about 15 node-hours of whole-node allocations (`REPRODUCE.md`). On another runtime the exponents are expected to agree within the quoted errors, while α_c moves at the 10^{-13} level and late flows at fixed α differ (Sec. 5); no full rerun on a second runtime was made.

References

[1] The accompanying paper on the long-range Ising model (in preparation): ε-expansion coefficients $C_0, \ldots, C_8$ for $d = 1$, with their derivations in the `GFF/` folder of this repository.

[2] *Monte Carlo determination of the thermal dimension $\Delta_\varepsilon(\Delta_\sigma)$ of the long-range Ising model in $d = 1, 2, 3$*, note in the `MC/` folder of this repository (2026).

[3] Notes on Hamiltonian truncation of the long-range Ising model, in the `HT/` folder of this repository (2026).

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A Data formats

Result records (`data/records/<campaign>/<item>.json`; campaign folders `example_K64_B10`, `ds13` (the ds13 campaign), `lambda_pilot`, `lambda_followup`). JSON with three keys:

- `sha256`: SHA-256 of the canonical JSON (sorted keys, no spaces) of the identity and the payload;
- `identity`: configuration (`s`, `Lambda` $= \Lambda$, `nb` $= n_b$, `keep` $= K$, `nsites` $= N$, `tunneling` $= \Omega$, `nlevels`, threads), kind, bracket, fraction (0.3), tolerance, iteration limit, whether the derivative was computed, the solver source hash (`source_sha256`, `3a40deb333e1...`) and the runtime;
- `payload`: `critical_flow` (the flow at α_c with per-site records and the bisection summary), `tuning` (bisection history and final bracket), `tangent` (the derivative flow, with per-site $\partial_\alpha g_n$ and $\ln|\partial_\alpha g_n|$), `analysis` (the window audit: every window with its Δ_E, fit residual and variation, the quality range and the settled window under the key `selected`) and `scope` (a statement of what the audit means).

Per-site records hold the parity gap `odd_gap` $= g_n$, the lowest levels of each sector, the boson occupation and top-level probability of the newest site, and correlations of the spin with the bath. `INDEX.json` in each campaign folder lists the generation (App. B) of every record.

Manifests and receipt (`inputs/manifests`, `inputs/validation_receipt`). A manifest is the frozen list of items submitted to the cluster together with the solver source hash; the receipt is itself a record, issued by the validation job, that binds the source hash to the compute-node runtime. `summarize.py` refuses any record whose identity differs from its manifest item or whose runtime differs from the receipt.

Summaries and results. `make replay` writes per-item summaries (critical coupling, bisection data, quality range, settled window, largest top-level probability and occupation, gate maxima) and per-item details (all windows, per-site gap and $\ln|\partial_\alpha g_n|$) to `.work/data/summaries`. `results/ds13_final.json` holds, per Δ_σ, the estimator, Δ_E, all error components, the settled and extrapolated values with every fit, the Λ values used in δ_Λ, the number of quality windows, α_c, the limiting forms, the superseded midpoint budget, and the late and crossover fits of `crosscheck.py` under `crosschecks`. An extrapolation that is not determined is written as `null`.

B Production on the cluster

All production ran on the AMD nodes of the *symmetry* cluster of the Perimeter Institute (whole nodes with 64 cores and 240 GB, 24 h limit, Slurm batch scheduler). One node with AVX-512 was excluded, since its runtime would not match the receipt. Code was used from a read-only frozen copy with its commit recorded in `GIT_SHA`.

Lanes. One Slurm job (a *lane*) runs many items concurrently on one node, admitting an item while the total threads stay ≤ 64 and the total memory reservation ≤ 232 GiB. Each lane verifies the SHA-256 of its manifest and receipt and logs the start, end, exit code, threads and peak memory of every item (`data/lane_logs`). The memory planner of the code overestimates the need by factors of 1.7 to 3.6 (median 2.8) against the peaks in the lane logs.

Ownership. The shared file system of the cluster does not support file locks across nodes, so items are owned explicitly. Only the submitter, running on the login host under a campaign-wide lock, creates an item's owner file (a random token, a generation number, the job, and the hash of the item definition). A writer must present the current token, and generation g writes only inside its own subdirectory, seeded from the previous generation's checkpoints, so a surviving old process cannot touch new files. An item is reclaimed only when its job is gone from the queue and terminal in the accounting; jobs are never requeued silently.

Validation, smoke runs and drills. The test suite passed on a compute node for each code state, and the last run issued the receipt. A smoke lane of five small items ran packed on one node; a repeat on a second node of the same runtime reproduced it bitwise. At a warning signal 300 s before the wall limit, or

by hand, a lane asks every item to checkpoint and exit: an interrupted item resumed from its checkpoint in a second generation and completed; writes without the token were refused from the login node and from another compute node; and two simultaneous reclaim calls produced exactly one winner.

Outage. Two lanes of the Λ follow-up ended with `NODE_FAIL` when every compute node of the cluster stopped responding (3 October 2026, for about 22 h). Their 21 unfinished items were reclaimed as generation 2 and resumed from their checkpoints.

Table 12: Every job of the study (node names anonymized). Thread-h is the sum over items of wall time times threads, from the lane logs; each job held a whole node for its elapsed time.

job	purpose	state	start	elapsed	node	items	thread-h
569091	validation (build)	COMPLETED	2026-10-02 23:34:20	00:01:08	amd-5		
569093	smoke lane	COMPLETED	2026-10-02 23:35:55	00:00:18	amd-5		
569096	USR1 drill, gen. 1	FAILED	2026-10-02 23:36:27	00:00:28	amd-5		
569097	cross-node ownership drill	COMPLETED	2026-10-02 23:36:32	00:00:02	amd-3		
569098	USR1 drill, gen. 2 (resumed)	COMPLETED	2026-10-02 23:37:06	00:03:23	amd-5		
569134	re-validation (fix round 1)	COMPLETED	2026-10-02 23:59:29	00:01:10	amd-6		
569135	second smoke	COMPLETED	2026-10-03 00:01:04	00:00:16	amd-6		
569161	validation of final code	COMPLETED	2026-10-03 00:15:53	00:01:13	amd-6		
569565	example manifest	COMPLETED	2026-10-03 08:48:32	00:08:32	amd-4	3/3	0.6
569572	ds13 lane 0	COMPLETED	2026-10-03 08:54:47	02:39:45	amd-7	27/27	20.4
569573	ds13 lane 1	COMPLETED	2026-10-03 08:54:47	02:22:00	amd-8	29/29	21.7
569574	ds13 lane 2	COMPLETED	2026-10-03 08:54:47	01:34:52	amd-1	30/30	23.1
569575	ds13 lane 3	COMPLETED	2026-10-03 08:54:47	01:09:04	amd-2	30/30	22.7
569576	ds13 lane 4	COMPLETED	2026-10-03 08:54:49	00:58:52	amd-3	30/30	21.8
569602	Λ pilot	COMPLETED	2026-10-03 11:55:38	00:52:39	amd-7	18/18	12.9
569626	Λ follow-up lane 0	NODE_FAIL	2026-10-03 12:52:49	00:15:29	amd-7	3/14	0.2
569627	Λ follow-up lane 1	NODE_FAIL	2026-10-03 12:52:49	00:15:29	amd-8	6/16	0.5
569640	follow-up lane 0, gen. 2	COMPLETED	2026-10-04 11:36:31	02:41:50	amd-2	11/11	27.1
569641	follow-up lane 1, gen. 2	COMPLETED	2026-10-04 11:36:31	02:03:51	amd-3	10/10	26.7

C Provenance and review

Origin of the code. The reference solver and analyzer (`code/lri_spin_boson_hpc/reference/`) and the three archived critical couplings (`code/lri_spin_boson_hpc/inputs/`) come from the original study, an earlier, unreleased study of the project (archive `lri_extended_comparison_study.zip`, SHA-256 `04bb8d30...`; `REFERENCE_PROVENANCE.json`). The optimized package `lri_spin_boson_hpc` was built on it and supplied by the author as the starting point of this study. Both were written by GPT-6 astra (OpenAI), used through ChatGPT; the package itself carries no authorship information.

Review and port to the cluster. Claude Code read the package line by line against the reference solver (no numerical defect; operational defects only) and adapted it to the cluster in a plan–review–build–inspect loop with OpenAI Codex as reviewer. Codex (requested model gpt-6-astra at extra-high reasoning effort; the command-line tool did not report which model ran) approved the plan after four rounds and inspected the built code twice in fresh sessions (10 and 4 findings, all fixed and tested). The last fix round (commit `af9b1dc`: scripts and benchmarks only, solver source unchanged) was not re-inspected. Codex reviewed the cluster port: plan and code. It did not review the numerics, the analysis, the estimator or the table. `REVIEWS.md` summarizes all reviews.

Analysis, checks and this note. Claude Code wrote the analysis scripts (3–8 October), the specification of the second implementation and the comparison scripts (8 October), and drafted this note (13 October; this revision with Claude Opus 5.5). The models of the earlier Claude Code sessions are not recorded in the package. The second implementation was written by a separate Claude agent from the specification only (Sec. 8.1).

The review of 13 October 2026 and this revision. Three read-only reviews of the previous version of this note (physics, statistics, readability) were made by Claude Opus 5.5 agents, launched one per review from a written brief by a separate Claude Code session that coordinated the public release; they did not see the session that drafted this note or each other's reports. They are independent of each other, not of the

model family. They found that the quality-range midpoint sits on the transient before the plateau, that the Λ dependence and the factor 2 were artifacts of it, and that the excess near $s = 1/2$ is slow approach to the fixed point, not NRG truncation. Claude Code verified these findings from the records with its own scripts before changing anything, and the author decided (13 October): (i) the settled late-window value with a chain-depth error, extrapolated where there is no plateau, with the rule fixed before the comparisons were recomputed; (ii) the label "not converged in chain depth" for the three points near $s = 1/2$, which stay shown and unused, the extrapolation at $\Delta_\sigma = 0.225$ being a cross-check only; (iii) the measured Λ spread instead of the factor 2; (iv) the old values in App. E. The use of the midpoint was this study's choice of 4 October, not part of the inherited analyzer, whose own output is the settled window; the package's analyzer calls the quality range "not a physical uncertainty interval".

Second review round (13 October 2026). Three further read-only reviews by the same kind of agents (numbers against the records, note against code on the assembled release, and a recheck of the first round with a scrutiny of the new rule) reproduced every value and error component of Table 1 independently from the raw records, and found the rule's classification robust and the errors not too small. They found that `make replay` failed with the oldest library versions that `requirements.txt` allows (the fits differ in their last bits), several wrong literal numbers and wording issues, and they supplied the fits that are now the late and crossover fits of Sec. 3.8. All were addressed in this version; these last fixes have not been reviewed again.

Reading of 15 October 2026. At the author's request an independent reader without project context, coordinated by a separate Claude Code session, reviewed this note for precision and readability. This version addresses its findings: the status of the label "not converged in chain depth", the definition of the cross-check fits, the one-loop test, the statements about the Monte Carlo comparison and the error budget, the uncertainties near $s = 1$, and notation. These changes were not reviewed again.

Timeline of the choices. The production ran on 3–4 October 2026, after validation on 2 October. On 4 October, after the NRG data had been seen but before any external comparison, Claude Code fixed the first estimate: the quality-range midpoint, the central item, the K/n_b set, the $\Lambda = 2.5$ follow-up, a factor 2 on the Λ shift and a quadrature sum. The comparisons with the ε series and the Monte Carlo study followed on 8 October. The reviews of 13 October showed that the midpoint sits on the transient before the plateau (App. E). The present rule was written the same day, before the comparisons were recomputed, and none of its constants was changed afterwards. It was not written blind, however. The earlier comparisons were known, and the reviews had already compared candidate estimators with the references: the last quality window at $\Delta_\sigma = 0.2$ with the series, an extrapolation at 0.225 with the series, and the plateau value at 0.1 with the Monte Carlo crossing value. The exploration that fixed the rule's constants read only the NRG records. No protocol was registered in advance. The label "not converged in chain depth" was assigned by the author on 13 October, with the series comparison known (Sec. 3.7).

Earlier decision on the points near $s = 1/2$. After the comparisons of 8 October a Claude Code analysis (not released) recommended excluding $\Delta_\sigma = 0.249$ and 0.24 or replacing them by series values, and enlarging the error at 0.225. The author overruled this and kept the three values as computed, with their excess stated, and unused (Sec. 8.2).

The released code tree. `code/lri_spin_boson_hpc` is commit `af9b1dc` (`GIT_SHA`) minus files not needed to test, replay or rerun the study: `adapters/`, the benchmark results and all benchmark scripts except the four that the tests run, `PERFORMANCE.md`, `VALIDATION.md`, `validation/`, `slurm/` (replaced by `NRG/slurm/`), `review/` (summarized in `REVIEWS.md`), `docs/ALGORITHM.md` (Sec. 3 of this note), `cluster/symmetry/freeze.sh` and a duplicate of `inputs/items/example_items.json`. The package's own `README.md` and `docs/SYMMETRY.md` still point to some of these; use `NRG/slurm/` and this note instead. The checksum list `MANIFEST.sha256.json` was regenerated for the remaining files. The solver `src/lri_spin/` is byte-identical to `af9b1dc`, so its SHA-256 still binds every record. Outside the code tree the analysis was revised as described above (`estimator.py` is new; `build_tables.py`, `make_figures.py`, `note_numbers.py` and `check_figures.py` changed), the comparison scripts take their paths as arguments and no longer

compute a quadrature "pull", and C_8 in `compare_methods.py` was updated to the current value, which changes the series references by at most 3×10^{-6} (at $\Delta_\sigma = 0.2$; 2×10^{-7} at 0.225; at $\Delta_\sigma = 0.24$ optimal truncation now stops one order later). `crosscheck.py` (the late and crossover fits) is new and `run_all.sh` now only calls `make replay`.

D All production items

The 194 records of the three campaigns D (ds13, folder `ds13`), P (Λ pilot) and F (Λ follow-up); the three example items are in Table 3. Columns: critical coupling; number of quality windows n_q; quality range; settled value; plateau drift; largest top-level probability p_{edge} over the sites of the critical flow (Sec. 7.2).

campaign	item	α_c	n_q	Δ_E quality range	settled	plateau drift	max p_{edge}
D	ds0.249_L2_K64_B10_N103	0.1071795160	31	[0.51298, 0.51732]	0.513056	3e-04	1e-15
F	ds0.249_L2.5_K64_B10_N78	0.1122397288	12	[0.50803, 0.51056]	0.508166	5e-04	2e-13
F	ds0.249_L2.5_K128_B10_N78	0.1122368788	13	[0.50808, 0.51105]	0.508078	6e-04	2e-13
F	ds0.249_L2.5_K128_B16_N78	0.1122377084	12	[0.50789, 0.51046]	0.508151	6e-04	2e-21
D	ds0.249_L3_K64_B8_N65	0.1173295097	13	[0.51730, 0.51890]	0.517479	3e-04	2e-03
D	ds0.249_L3_K48_B10_N65	0.1173263397	5	[0.50910, 0.50996]	0.509096	6e-04	5e-12
D	ds0.249_L3_K64_B10_N65	0.1173260725	4	[0.50892, 0.50966]	0.508918	7e-04	9e-04
D	ds0.249_L3_K96_B10_N65	0.1173255595	5	[0.50906, 0.50995]	0.509058	6e-04	5e-12
D	ds0.249_L3_K128_B10_N65	0.1173254942	4	[0.50886, 0.50965]	0.509122	8e-04	9e-04
D	ds0.249_L3_K64_B12_N65	0.1173260502	3	[0.50814, 0.50862]	0.508138	8e-04	1e-04
D	ds0.249_L3_K64_B16_N65	0.1173261829	2	[0.50800, 0.50827]	0.508000	8e-04	2e-04
D	ds0.249_L3_K128_B16_N65	0.1173255478	2	[0.50803, 0.50828]	0.508030	8e-04	4e-19
D	ds0.249_L3_K64_B20_N65	0.1173261902	3	[0.50815, 0.50860]	0.508155	7e-04	6e-24
D	ds0.249_L3_K64_B24_N65	0.1173261918	2	[0.50798, 0.50826]	0.507985	9e-04	1e-28
D	ds0.249_L4_K64_B10_N52	0.1269362680	0	–	0.510640	1e-03	5e-03
D	ds0.24_L2_K64_B10_N103	0.1166435376	15	[0.50223, 0.50699]	0.503006	8e-04	7e-06
F	ds0.24_L2.5_K64_B10_N78	0.1222308934	13	[0.49911, 0.50160]	0.499379	6e-04	2e-13
F	ds0.24_L2.5_K128_B10_N78	0.1222282882	13	[0.49902, 0.50166]	0.499125	5e-04	2e-13
F	ds0.24_L2.5_K128_B16_N78	0.1222289164	13	[0.49819, 0.50145]	0.498993	8e-04	2e-05
D	ds0.24_L3_K64_B8_N65	0.1278472683	13	[0.50788, 0.50929]	0.507908	2e-04	1e-09
D	ds0.24_L3_K48_B10_N65	0.1278438327	5	[0.49962, 0.50057]	0.499625	7e-04	6e-12
D	ds0.24_L3_K64_B10_N65	0.1278435265	5	[0.49960, 0.50056]	0.499602	7e-04	6e-12
D	ds0.24_L3_K96_B10_N65	0.1278430054	5	[0.49982, 0.50058]	0.499944	6e-04	6e-12
D	ds0.24_L3_K128_B10_N65	0.1278429538	5	[0.49964, 0.50057]	0.499643	7e-04	6e-12
D	ds0.24_L3_K64_B12_N65	0.1278435098	4	[0.49906, 0.49967]	0.499222	7e-04	2e-14
D	ds0.24_L3_K64_B16_N65	0.1278436618	4	[0.49913, 0.49967]	0.499434	8e-04	3e-19
D	ds0.24_L3_K128_B16_N65	0.1278429945	4	[0.49892, 0.49962]	0.498922	7e-04	3e-19
D	ds0.24_L3_K64_B20_N65	0.1278436705	2	[0.49899, 0.49929]	0.498992	9e-04	9e-05
D	ds0.24_L3_K64_B24_N65	0.1278436707	0	–	0.499254	9e-04	6e-05
D	ds0.24_L4_K64_B10_N52	0.1384395657	0	–	0.501245	9e-04	5e-04
D	ds0.225_L2_K64_B10_N103	0.1339530997	30	[0.49078, 0.49557]	0.491797	4e-04	7e-06
F	ds0.225_L2.5_K64_B10_N78	0.1405232485	17	[0.49033, 0.49270]	0.490435	3e-04	2e-13
F	ds0.225_L2.5_K128_B10_N78	0.1405208812	17	[0.48974, 0.49231]	0.489830	3e-04	2e-13
F	ds0.225_L2.5_K128_B16_N78	0.1405214369	17	[0.48974, 0.49220]	0.489897	4e-04	7e-22
D	ds0.225_L3_K64_B8_N65	0.1471196556	14	[0.49618, 0.49761]	0.496210	2e-04	1e-09
D	ds0.225_L3_K48_B10_N65	0.1471158185	9	[0.48991, 0.49132]	0.489911	4e-04	8e-04
D	ds0.225_L3_K64_B10_N65	0.1471154781	9	[0.48988, 0.49131]	0.489880	4e-04	8e-04
D	ds0.225_L3_K96_B10_N65	0.1471149602	9	[0.48979, 0.49134]	0.490139	5e-04	8e-04
D	ds0.225_L3_K128_B10_N65	0.1471149168	9	[0.48973, 0.49132]	0.490102	5e-04	8e-04
D	ds0.225_L3_K64_B12_N65	0.1471154761	9	[0.48927, 0.49099]	0.489689	6e-04	5e-04
D	ds0.225_L3_K64_B16_N65	0.1471156481	9	[0.48931, 0.49098]	0.489697	6e-04	2e-04
D	ds0.225_L3_K128_B16_N65	0.1471149209	9	[0.48924, 0.49098]	0.489668	6e-04	2e-04
D	ds0.225_L3_K64_B20_N65	0.1471156611	8	[0.48944, 0.49098]	0.489856	7e-04	1e-04
D	ds0.225_L3_K64_B24_N65	0.1471156769	9	[0.48956, 0.49099]	0.489559	4e-04	7e-30
D	ds0.225_L4_K64_B10_N52	0.1595443361	3	[0.49095, 0.49129]	0.490952	6e-04	3e-10
P	ds0.2_L1.5_K64_B10_N176	0.1626225790	89	[0.53623, 0.53696]	0.536303	1e-07	2e-14
P	ds0.2_L1.5_K128_B10_N176	0.1603705735	76	[0.50652, 0.50855]	0.507049	1e-05	3e-10
P	ds0.2_L1.5_K64_B16_N176	0.1626210299	89	[0.53623, 0.53696]	0.536303	1e-07	3e-26
D	ds0.2_L2_K64_B10_N103	0.1676281822	34	[0.48357, 0.48710]	0.484345	2e-04	5e-06
P	ds0.2_L2_K128_B10_N103	0.1676098637	36	[0.48306, 0.48637]	0.483192	9e-05	3e-14
F	ds0.2_L2_K192_B10_N103	0.1676082707	34	[0.48256, 0.48641]	0.483363	2e-04	5e-06
F	ds0.2_L2_K256_B10_N103	0.1676081346	35	[0.48261, 0.48637]	0.483290	2e-04	5e-06
P	ds0.2_L2.5_K64_B10_N78	0.1761692489	23	[0.48295, 0.48558]	0.483530	2e-04	1e-04
P	ds0.2_L2.5_K128_B10_N78	0.1761670918	22	[0.48338, 0.48541]	0.483420	1e-04	2e-13
F	ds0.2_L2.5_K128_B16_N78	0.1761675436	23	[0.48307, 0.48538]	0.483233	1e-04	2e-22
D	ds0.2_L3_K64_B8_N65	0.1847272377	17	[0.48643, 0.48778]	0.486523	1e-04	1e-03
D	ds0.2_L3_K48_B10_N65	0.1847227575	15	[0.48333, 0.48467]	0.483334	1e-04	6e-04
D	ds0.2_L3_K64_B10_N65	0.1847222031	15	[0.48341, 0.48467]	0.483442	1e-04	5e-12
D	ds0.2_L3_K96_B10_N65	0.1847218040	15	[0.48350, 0.48468]	0.483517	2e-04	5e-12
D	ds0.2_L3_K128_B10_N65	0.1847217711	15	[0.48344, 0.48467]	0.483457	1e-04	5e-12
D	ds0.2_L3_K64_B12_N65	0.1847223271	15	[0.48332, 0.48457]	0.483330	1e-04	1e-14
D	ds0.2_L3_K64_B16_N65	0.1847224093	15	[0.48330, 0.48457]	0.483321	1e-04	7e-20
D	ds0.2_L3_K128_B16_N65	0.1847217485	15	[0.48288, 0.48458]	0.483392	2e-04	1e-04
D	ds0.2_L3_K64_B20_N65	0.1847225105	15	[0.48292, 0.48457]	0.483316	2e-04	7e-05
D	ds0.2_L3_K64_B24_N65	0.1847225110	15	[0.48302, 0.48457]	0.483326	2e-04	3e-08
D	ds0.2_L4_K64_B10_N52	0.2008115424	8	[0.48366, 0.48435]	0.483664	2e-04	3e-03
D	ds0.175_L2_K64_B10_N103	0.2084469699	40	[0.48393, 0.48650]	0.484362	4e-05	3e-06
F	ds0.175_L2.5_K64_B10_N78	0.2194635288	27	[0.48411, 0.48571]	0.484116	4e-05	2e-13
F	ds0.175_L2.5_K128_B10_N78	0.2194614482	25	[0.48362, 0.48532]	0.483990	6e-05	9e-05
D	ds0.175_L3_K64_B8_N65	0.2304807533	19	[0.48562, 0.48647]	0.485623	3e-05	6e-05

campaign	item	α_c	n_q	Δ_E quality range	settled	plateau drift	max p_{edge}
D	ds0.175_L3_K48_B10_N65	0.2304755377	19	[0.48403, 0.48505]	0.484042	6e-05	5e-12
D	ds0.175_L3_K64_B10_N65	0.2304749169	19	[0.48395, 0.48505]	0.483987	4e-05	5e-04
D	ds0.175_L3_K96_B10_N65	0.2304744981	19	[0.48403, 0.48506]	0.484044	7e-05	5e-12
D	ds0.175_L3_K128_B10_N65	0.2304744735	19	[0.48400, 0.48505]	0.484007	4e-05	5e-12
D	ds0.175_L3_K64_B12_N65	0.2304749974	19	[0.48395, 0.48501]	0.483951	3e-05	1e-14
D	ds0.175_L3_K64_B16_N65	0.2304751345	19	[0.48396, 0.48501]	0.483974	5e-05	2e-20
D	ds0.175_L3_K128_B16_N65	0.2304744393	19	[0.48377, 0.48502]	0.483986	8e-05	1e-04
D	ds0.175_L4_K64_B10_N52	0.2511400536	11	[0.48414, 0.48461]	0.484150	9e-05	2e-10
D	ds0.15_L2_K64_B10_N103	0.2580502776	42	[0.49114, 0.49310]	0.491365	2e-05	2e-06
F	ds0.15_L2.5_K64_B10_N78	0.2721823788	29	[0.49118, 0.49233]	0.491181	8e-06	8e-13
F	ds0.15_L2.5_K128_B10_N78	0.2721802333	28	[0.49111, 0.49226]	0.491113	2e-05	8e-13
D	ds0.15_L3_K64_B8_N65	0.2862878455	21	[0.49191, 0.49271]	0.492051	3e-05	7e-04
D	ds0.15_L3_K48_B10_N65	0.2862812698	21	[0.49102, 0.49187]	0.491127	3e-05	2e-04
D	ds0.15_L3_K64_B10_N65	0.2862807211	21	[0.49112, 0.49187]	0.491127	2e-05	4e-12
D	ds0.15_L3_K96_B10_N65	0.2862803155	21	[0.49105, 0.49187]	0.491120	2e-05	3e-04
D	ds0.15_L3_K128_B10_N65	0.2862802913	21	[0.49099, 0.49187]	0.491123	3e-05	3e-04
D	ds0.15_L3_K64_B12_N65	0.2862808093	21	[0.49110, 0.49185]	0.491099	1e-05	7e-15
D	ds0.15_L3_K64_B16_N65	0.2862809076	21	[0.49105, 0.49185]	0.491099	2e-05	2e-08
D	ds0.15_L3_K128_B16_N65	0.2862802558	21	[0.49110, 0.49185]	0.491106	2e-05	5e-06
D	ds0.15_L4_K64_B10_N52	0.3126750056	14	[0.49118, 0.49169]	0.491181	4e-05	2e-10
D	ds0.125_L2_K64_B10_N103	0.3186931853	45	[0.50588, 0.50735]	0.505878	4e-06	2e-12
F	ds0.125_L2.5_K64_B10_N78	0.3367673600	31	[0.50570, 0.50675]	0.505714	6e-06	1e-09
F	ds0.125_L2.5_K128_B10_N78	0.3367650662	31	[0.50567, 0.50672]	0.505676	3e-06	3e-08
D	ds0.125_L3_K64_B8_N65	0.3547738461	23	[0.50623, 0.50693]	0.506258	1e-05	2e-04
D	ds0.125_L3_K48_B10_N65	0.3547649005	23	[0.50569, 0.50640]	0.505687	8e-06	6e-12
D	ds0.125_L3_K64_B10_N65	0.3547642364	23	[0.50568, 0.50640]	0.505687	1e-05	6e-12
D	ds0.125_L3_K96_B10_N65	0.3547638772	23	[0.50568, 0.50640]	0.505685	6e-06	6e-12
D	ds0.125_L3_K128_B10_N65	0.3547638514	23	[0.50568, 0.50640]	0.505683	5e-06	6e-12
D	ds0.125_L3_K64_B12_N65	0.3547643489	23	[0.50567, 0.50639]	0.505675	6e-06	5e-15
D	ds0.125_L3_K64_B16_N65	0.3547644468	23	[0.50567, 0.50639]	0.505676	6e-06	1e-11
D	ds0.125_L3_K128_B16_N65	0.3547638114	23	[0.50567, 0.50639]	0.505672	9e-07	5e-21
D	ds0.125_L4_K64_B10_N52	0.3883742704	15	[0.50571, 0.50609]	0.505710	1e-05	2e-10
P	ds0.1_L1.5_K64_B10_N176	0.3786792319	87	[0.56695, 0.56733]	0.567059	6e-08	6e-12
P	ds0.1_L1.5_K128_B10_N176	0.3733064368	83	[0.53919, 0.54017]	0.539196	4e-07	6e-12
P	ds0.1_L1.5_K64_B16_N176	0.3786839607	87	[0.56695, 0.56733]	0.567059	7e-08	5e-22
D	ds0.1_L2_K64_B10_N103	0.3936704020	46	[0.52980, 0.53105]	0.529796	2e-06	9e-12
P	ds0.1_L2_K128_B10_N103	0.3936438035	46	[0.52962, 0.53089]	0.529631	2e-06	9e-12
F	ds0.1_L2_K192_B10_N103	0.3936429089	46	[0.52961, 0.53087]	0.529613	3e-06	4e-11
F	ds0.1_L2_K256_B10_N103	0.3936426886	46	[0.52960, 0.53087]	0.529609	2e-06	9e-12
P	ds0.1_L2.5_K64_B10_N78	0.4167902745	32	[0.52964, 0.53056]	0.529637	1e-06	7e-11
P	ds0.1_L2.5_K128_B10_N78	0.4167874486	32	[0.52961, 0.53054]	0.529610	2e-06	4e-11
D	ds0.1_L3_K64_B8_N65	0.4397848811	24	[0.53000, 0.53063]	0.529998	2e-06	2e-06
D	ds0.1_L3_K48_B10_N65	0.4397704546	23	[0.52962, 0.53006]	0.529616	3e-06	2e-11
D	ds0.1_L3_K64_B10_N65	0.4397697893	23	[0.52961, 0.53006]	0.529613	1e-06	2e-08
D	ds0.1_L3_K96_B10_N65	0.4397694732	23	[0.52961, 0.53006]	0.529613	3e-07	2e-11
D	ds0.1_L3_K128_B10_N65	0.4397694449	23	[0.52961, 0.53006]	0.529612	2e-06	3e-09
D	ds0.1_L3_K64_B12_N65	0.4397699011	23	[0.52961, 0.53006]	0.529609	2e-06	6e-11
D	ds0.1_L3_K64_B16_N65	0.4397699761	23	[0.52961, 0.53006]	0.529611	5e-06	2e-14
D	ds0.1_L3_K128_B16_N65	0.4397693853	23	[0.52961, 0.53006]	0.529608	2e-06	4e-21
D	ds0.1_L4_K64_B10_N52	0.4825812310	16	[0.52962, 0.52999]	0.529628	6e-06	2e-10
D	ds0.075_L2_K64_B10_N111	0.4882361312	51	[0.56644, 0.56743]	0.566440	6e-07	4e-11
F	ds0.075_L2.5_K64_B10_N84	0.5179599027	36	[0.56628, 0.56707]	0.566284	5e-07	7e-11
F	ds0.075_L2.5_K128_B10_N84	0.5179561214	36	[0.56625, 0.56705]	0.566260	7e-07	7e-11
D	ds0.075_L3_K64_B8_N70	0.5474812521	28	[0.56655, 0.56708]	0.566546	4e-07	6e-08
D	ds0.075_L3_K48_B10_N70	0.5474526311	28	[0.56627, 0.56681]	0.566265	6e-07	1e-10
D	ds0.075_L3_K64_B10_N70	0.5474520316	28	[0.56624, 0.56681]	0.566265	2e-06	1e-10
D	ds0.075_L3_K96_B10_N70	0.5474517839	28	[0.56626, 0.56681]	0.566263	9e-07	1e-10
D	ds0.075_L3_K128_B10_N70	0.5474517479	28	[0.56625, 0.56681]	0.566262	1e-06	1e-10
D	ds0.075_L3_K64_B12_N70	0.5474520799	28	[0.56626, 0.56681]	0.566261	1e-06	1e-13
D	ds0.075_L3_K64_B16_N70	0.5474521815	28	[0.56626, 0.56681]	0.566262	1e-06	4e-17
D	ds0.075_L3_K128_B16_N70	0.5474516143	28	[0.56625, 0.56681]	0.566260	1e-06	6e-20
D	ds0.075_L4_K64_B10_N55	0.6022718795	18	[0.56627, 0.56657]	0.566270	6e-07	7e-08
P	ds0.05_L1.5_K64_B10_N217	0.5909590331	100	[0.67067, 0.67118]	0.670692	3e-07	1e-10
P	ds0.05_L1.5_K128_B10_N217	0.5780543351	102	[0.63035, 0.63087]	0.630353	1e-07	1e-10
P	ds0.05_L1.5_K64_B16_N217	0.5909603112	100	[0.67066, 0.67119]	0.670693	3e-07	9e-20
D	ds0.05_L2_K64_B10_N127	0.6120267877	59	[0.62230, 0.62295]	0.622300	5e-08	2e-10
P	ds0.05_L2_K128_B10_N127	0.6119924970	59	[0.62205, 0.62277]	0.622125	4e-07	2e-10
F	ds0.05_L2_K192_B10_N127	0.6119916485	59	[0.62211, 0.62277]	0.622119	8e-08	2e-10
F	ds0.05_L2_K256_B10_N127	0.6119911289	59	[0.62208, 0.62277]	0.622116	3e-07	2e-10
P	ds0.05_L2.5_K64_B10_N96	0.6507599139	43	[0.62214, 0.62266]	0.622139	1e-07	3e-10
P	ds0.05_L2.5_K128_B10_N96	0.6507542509	43	[0.62206, 0.62264]	0.622116	4e-07	3e-10
D	ds0.05_L3_K64_B8_N80	0.6892086918	34	[0.62234, 0.62282]	0.622349	2e-07	2e-07
D	ds0.05_L3_K48_B10_N80	0.6891363592	34	[0.62211, 0.62260]	0.622120	3e-07	6e-10
D	ds0.05_L3_K64_B10_N80	0.6891357316	34	[0.62212, 0.62260]	0.622119	3e-07	6e-10
D	ds0.05_L3_K96_B10_N80	0.6891355001	34	[0.62208, 0.62260]	0.622117	7e-07	6e-10
D	ds0.05_L3_K128_B10_N80	0.6891354544	34	[0.62212, 0.62260]	0.622117	2e-07	6e-10
D	ds0.05_L3_K160_B10_N80	0.6891354015	34	[0.62210, 0.62260]	0.622117	5e-07	6e-10
D	ds0.05_L3_K192_B10_N80	0.6891353149	34	[0.62212, 0.62260]	0.622117	2e-07	1e-09
D	ds0.05_L3_K64_B12_N80	0.6891353868	34	[0.62212, 0.62260]	0.622117	3e-07	4e-12
D	ds0.05_L3_K64_B16_N80	0.6891355208	34	[0.62212, 0.62260]	0.622117	4e-07	9e-19
D	ds0.05_L3_K128_B16_N80	0.6891349793	34	[0.62210, 0.62259]	0.622115	5e-07	9e-19
D	ds0.05_L4_K64_B10_N63	0.7603342269	23	[0.62211, 0.62236]	0.622122	1e-06	1e-09
D	ds0.025_L2_K64_B10_N143	0.7879305686	63	[0.71355, 0.71403]	0.713586	6e-07	1e-09
F	ds0.025_L2.5_K64_B10_N108	0.8401456498	47	[0.71340, 0.71387]	0.713448	7e-07	2e-09
F	ds0.025_L2.5_K128_B10_N108	0.8401346083	47	[0.71338, 0.71384]	0.713422	7e-07	2e-09
D	ds0.025_L3_K64_B8_N90	0.8921068317	38	[0.71361, 0.71402]	0.713648	7e-07	9e-07
D	ds0.025_L3_K48_B10_N90	0.8918504547	38	[0.71339, 0.71380]	0.713425	7e-07	3e-09
D	ds0.025_L3_K64_B10_N90	0.8918498061	38	[0.71339, 0.71380]	0.713424	7e-07	3e-09

campaign	item	α_c	n_q	Δ_E quality range	settled	plateau drift	max p_{edge}
D	ds0.025_L3_K96_B10_N90	0.8918496024	38	[0.71339, 0.71380]	0.713423	7e-07	3e-09
D	ds0.025_L3_K128_B10_N90	0.8918495416	38	[0.71339, 0.71380]	0.713423	7e-07	3e-09
D	ds0.025_L3_K160_B10_N90	0.8918494283	38	[0.71339, 0.71380]	0.713423	7e-07	3e-09
D	ds0.025_L3_K192_B10_N90	0.8918492517	38	[0.71338, 0.71380]	0.713423	7e-07	3e-09
D	ds0.025_L3_K64_B12_N90	0.8918472563	38	[0.71339, 0.71380]	0.713422	7e-07	9e-12
D	ds0.025_L3_K64_B16_N90	0.8918474343	38	[0.71339, 0.71380]	0.713423	7e-07	2e-17
D	ds0.025_L3_K128_B16_N90	0.8918469295	38	[0.71339, 0.71380]	0.713422	7e-07	2e-17
D	ds0.025_L4_K64_B10_N71	0.9876957393	26	[0.71341, 0.71366]	0.713427	1e-06	8e-09
D	ds0.01_L2_K64_B10_N166	0.9503649771	66	[0.80960, 0.81027]	0.809951	8e-06	5e-09
F	ds0.01_L2.5_K64_B10_N126	1.0158010470	49	[0.80949, 0.81011]	0.809784	7e-06	8e-09
F	ds0.01_L2.5_K128_B10_N126	1.0157777118	49	[0.80945, 0.81008]	0.809749	7e-06	8e-09
D	ds0.01_L3_K64_B8_N105	1.0814250819	41	[0.80974, 0.81041]	0.810031	1e-05	3e-06
D	ds0.01_L3_K48_B10_N105	1.0806604206	41	[0.80946, 0.81013]	0.809751	9e-06	1e-08
D	ds0.01_L3_K64_B10_N105	1.0806597653	41	[0.80946, 0.81013]	0.809749	9e-06	1e-08
D	ds0.01_L3_K96_B10_N105	1.0806595638	41	[0.80946, 0.81013]	0.809749	9e-06	1e-08
D	ds0.01_L3_K128_B10_N105	1.0806594892	41	[0.80946, 0.81013]	0.809749	9e-06	1e-08
D	ds0.01_L3_K160_B10_N105	1.0806592593	41	[0.80946, 0.81013]	0.809749	9e-06	1e-08
D	ds0.01_L3_K192_B10_N105	1.0806588400	41	[0.80946, 0.81013]	0.809748	9e-06	1e-08
D	ds0.01_L3_K64_B12_N105	1.0806490638	41	[0.80946, 0.81013]	0.809748	9e-06	5e-11
D	ds0.01_L3_K64_B16_N105	1.0806491182	41	[0.80946, 0.81013]	0.809748	9e-06	2e-16
D	ds0.01_L3_K128_B16_N105	1.0806486418	41	[0.80946, 0.81012]	0.809747	9e-06	2e-16
D	ds0.01_L4_K64_B10_N83	1.2010212236	30	[0.80947, 0.81002]	0.809756	1e-05	3e-08
D	ds0.001_L2_K64_B10_N396	1.1270646401	150	[0.93780, 0.93799]	0.937991	4e-06	2e-08
F	ds0.001_L2.5_K64_B10_N300	1.2079818456	125	[0.93729, 0.93752]	0.937523	3e-06	3e-08
F	ds0.001_L2.5_K128_B10_N300	1.2078925099	126	[0.93719, 0.93742]	0.937418	3e-06	3e-08
D	ds0.001_L3_K64_B8_N250	1.2911723978	105	[0.93793, 0.93813]	0.938132	2e-06	7e-06
D	ds0.001_L3_K48_B10_N250	1.2885532639	106	[0.93722, 0.93743]	0.937429	2e-06	5e-08
D	ds0.001_L3_K64_B10_N250	1.2885527136	106	[0.93722, 0.93743]	0.937428	2e-06	5e-08
D	ds0.001_L3_K96_B10_N250	1.2885525078	106	[0.93722, 0.93743]	0.937428	2e-06	5e-08
D	ds0.001_L3_K128_B10_N250	1.2885524382	106	[0.93722, 0.93743]	0.937428	2e-06	5e-08
D	ds0.001_L3_K160_B10_N250	1.2885515952	106	[0.93722, 0.93743]	0.937427	2e-06	5e-08
D	ds0.001_L3_K192_B10_N250	1.2885497372	106	[0.93722, 0.93743]	0.937425	2e-06	5e-08
D	ds0.001_L3_K64_B12_N250	1.2885056066	106	[0.93722, 0.93742]	0.937424	2e-06	3e-10
D	ds0.001_L3_K64_B16_N250	1.2885051164	106	[0.93722, 0.93742]	0.937424	2e-06	2e-15
D	ds0.001_L3_K128_B16_N250	1.2885046794	106	[0.93722, 0.93742]	0.937424	2e-06	2e-15
D	ds0.001_L4_K64_B10_N198	1.4387239566	78	[0.93724, 0.93751]	0.937441	8e-06	1e-07

E The superseded midpoint estimator

Until 13 October 2026 the value of a point was the midpoint of the quality range of its central item, with the half-width δ_{w} as one error component and the total $\delta_{\text{tot}}^{\text{old}} = [\delta_{\text{w}}^2 + \delta_K^2 + (2\delta_\Lambda^{\text{old}})^2]^{1/2}$, where $\delta_\Lambda^{\text{old}}$ was the midpoint shift from $\Lambda = 3$ to 2.5. Claude Code chose it when the final table was first built (4 October); the inherited analyzer reports the settled window. Table 14 lists the old values.

Why it was replaced. Along the quality windows Δ_E falls monotonically at 11 of 13 points (Fig. 7): the early quality windows are the transient toward the fixed point. The midpoint therefore lies 1.7 to 3.7×10^{-4} above the plateau at $0.025 \le \Delta_\sigma \le 0.15$ (and -1.0×10^{-4} from it at $\Delta_\sigma = 0.001$, where the windows rise), and δ_{w} measures the length of the transient, not a resolution of the plateau; the earlier claim of a precision floor of about 10^{-4} was wrong. Because the window rule counts sites while the chain length scales as $1/\ln\Lambda$, a smaller Λ admits a larger share of transient windows, which produced an apparent rise of Δ_E as Λ decreases (the midpoint at $\Lambda = 2$ lay $+4.0 \times 10^{-4}$ and $+6.5 \times 10^{-5}$ above $\Lambda = 3$ at $\Delta_\sigma = 0.1$ and 0.05, against -3×10^{-6} and -1×10^{-6} on plateau values). The factor 2 on $\delta_\Lambda^{\text{old}}$ multiplied this artifact. At $0.01 \le \Delta_\sigma \le 0.15$ the new errors are smaller than the old ones by factors 6–16. At 13 of the 13 points the old value lies within its old $\delta_{\text{tot}}^{\text{old}}$ of the new one, but only because δ_{w} equalled the offset: the old interval was in effect one-sided.

F Terms and symbols

Abbreviations. LRI: long-range Ising; NRG: numerical renormalization group; RG: renormalization group; MC: Monte Carlo; BLAS: basic linear algebra subprograms; SHA-256: a cryptographic checksum.

term	meaning
point	one value of Δ_σ
item, record	one NRG calculation at fixed $(\Delta_\sigma, \Lambda, K, n_b, N)$; the JSON file holding its result
central item	the item at $\Lambda = 3$, $K = 128$, $n_b = 16$
campaign, production	a set of items submitted together; the released items
finite map	the truncated NRG at fixed Λ, K, n_b, N as a function of α

term	meaning
window, quality window	w consecutive sites over which $\ln\lvert\partial_\alpha g_n\rvert$ is fitted; a window with relative variation < 0.005 and fit residual < 0.003
quality range	the range of Δ_E over the quality windows of an item
settled window, settled value	the window with the smallest fit residual plus relative variation; its Δ_E
plateau, plateau drift	a point whose central item has plateau drift $\le \tau_\mathrm{p}$; the largest change of Δ_E within $\pm h$ sites of the settled window
extrapolated	a point without a plateau; its value is the extrapolation of Δ_loc (Sec. 3.7)
not converged in chain depth	label assigned by the author to $\Delta_\sigma = 0.249$, 0.24, 0.225, not an outcome of the estimator: the flow does not reach its fixed point within l_max; no error quoted, not used
trusted window	a window whose finite-difference Δ_E in the second implementation is stable under a change of the step η
late fit, crossover fit	cross-check fits of Δ_loc with a detuning term (Sec. 3.8); not adopted
n.d.	not determined: an extrapolation whose fitted ω is below 0.01
fit residual, terminal residual, eigen-residual	largest deviation from the window fit; $f(\alpha_c)$; $\max\lvert HU - UE\rvert$
top-level probability p_edge	weight of the highest level $n_b - 1$ of the newest oscillator in the ground state, at one site of the critical flow
receipt, manifest, identity, runtime	see App. A
lane, generation, drill	see App. B

symbol	meaning
d, s	dimension (1); exponent of the coupling decay $1/r^{1+s}$ and of the bath spectrum $\rho \propto \omega^s$
Δ_σ, ε, δ	spin dimension $(1-s)/2$; $2s - 1 = 1 - 4\Delta_\sigma$; $1 - s = 2\Delta_\sigma$
Δ_ε, y_t, ν, ω	dimension of the energy operator; thermal RG eigenvalue $1/\nu$; $\Delta_\varepsilon = 1 - y_t$; leading correction exponent (also, in $\rho(\omega)$ and ω_j, a frequency)
$C_n^\mathrm{SR}(1)$	coefficients of the dual expansion (4) of Δ_ε in $\sqrt{\Delta_\sigma}$
u, l_0, $\mathcal{N}$	φ^4 coupling of the one-loop flow; RG time where it is infinite; number of components of the O($\mathcal{N}$) model
Δ_E, $\Delta_E^\mathrm{central}$	NRG estimate of Δ_ε; that of the central item
Δ_loc, l, l_max	local exponent, Eq. (14); RG time $n \ln\Lambda$; its usable maximum, Eq. (15)
Δ_∞, A, l_0, l_hi, S	extrapolated value, amplitude, first and last RG time and span of an extrapolation fit
A_d, l_a, l_end, k, n_fit	detuning amplitude, first and last RG time, end offset k/y_t and number of accepted fits of a cross-check fit
τ_p, h	plateau threshold; half-width of the plateau neighborhood, $\lceil w/4 \rceil$ sites
τ	imaginary time (Sec. 2)
C_n	coefficients of the ε expansion of Δ_ε for $d = 1$
Ω, α, α_c, $\delta\alpha$	tunneling amplitude (0.1); dissipation strength; its critical value for the finite map; residual detuning
$\rho(\omega)$, $J(\omega)$, ω_j, λ_j, b_j	bath spectral density; $\pi\rho$; oscillator frequencies, couplings and annihilation operators
Λ, q, I_m	discretization ratio; $1/\Lambda$; frequency bins
c_m, γ_m, ξ_m, ξ_0	star oscillators, their couplings and frequencies $\xi_m = \xi_0 q^m$
e_n, t_n, $\bar{e}_n$, $\bar{t}_n$	chain on-site energies and hoppings; rescaled by Λ^n
$\mathcal{A}_n$, $\mathcal{B}_n$, a	little q-Jacobi recurrence quantities of Eq. (8); $a = q^s$
λ_tot	total coupling of the spin to the bath, $\sqrt{2\alpha/(s+1)}$
H_n, $\bar{b}_n$, $\mathbb{1}$	rescaled Hamiltonian of sites $0, \ldots, n$; site-n operator in the kept basis; identity
K, n_b, N, $N(\Lambda)$	kept states per parity sector; oscillator levels; chain sites; chain sites at Λ
P	parity $\sigma_x(-1)^{\sum b^\dagger b}$
g_n, $E_{0,n}^\mathrm{odd/even}$	parity gap; lowest level of H_n in each sector
$f(\alpha)$	bisection function, Eq. (11)
U_k, U_d, E_k, E_d	kept and discarded eigenvectors and energies

term	meaning
$\dot{H}$, Z, T, $\dot{E}_0$, B	$\partial_\alpha H$; rotation of the kept states; derivative of the kept Hamiltonian; of the ground energy; site operator
w, κ, n_q	window width; intercept of the window fit; number of quality windows
δ_{depth}, δ_K, δ_Λ, δ_{tot}	chain-depth error; K/n_b spread; discretization error; their sum
δ_{w}, $\delta_\Lambda^{\mathrm{old}}$, $\delta_{\mathrm{tot}}^{\mathrm{old}}$	quantities of the superseded estimator (App. E)
$n_{\max}$	estimated last site near the fixed point in double precision
σ_{stat}, σ_{sys}, σ_∞, $\Delta_{\omega'}$	statistical error and systematic scale of the Monte Carlo fit; error and sensitivity of its crossing extrapolation (not a dimension)
η	relative step of the finite differences of the second implementation (`h` in its code)
L, M, u	Padé orders and Borel variable
c1–c8; D, P, F	configurations of the second implementation; campaign tags of App. D

Table 14: The superseded estimator. Old value: quality-range midpoint with $\delta_{\mathrm{tot}}^{\mathrm{old}}$ (two significant digits, as given in the earlier version of this note); the half-width δ_{w} and $2|\delta_{\Lambda}^{\mathrm{old}}|$ in units of 10^{-4}; the value of this note; the difference; the old status labels: (a) far above the ε series compared with the old error, attributed to a truncation bias of bosonic NRG, not used; (b) above the Borel–Padé sum by about the old error, same attribution, not used; (c) at $\Delta_\sigma = 0.2$, the series cannot test the point at the 10^{-3} level. The attribution to a truncation bias has since been withdrawn (Sec. 8.2).

Δ_σ	old: midpoint ($\delta_{\mathrm{tot}}^{\mathrm{old}}$)	$10^4\delta_{\mathrm{w}}$	$10^4 \cdot 2\delta_{\Lambda}^{\mathrm{old}}$	this note	$10^4\times$(old − this note)	old status
0.249	0.5082(21)	1.3	20.5	[0.508030]	+1.3	biased (a)
0.24	0.4993(12)	3.5	10.9	[0.498922]	+3.5	biased (a)
0.225	0.4901(19)	8.7	17.2	[0.489668]	+4.4	biased (b)
0.2	0.4837(13)	8.5	9.9	0.4830(7)	+7.5	final (c)
0.175	0.48440(65)	6.2	1.1	0.48390(35)	+4.9	final
0.15	0.49148(64)	3.8	5.1	0.49111(11)	+3.7	final
0.125	0.50603(48)	3.6	3.2	0.50567(5)	+3.6	final
0.1	0.52983(52)	2.3	4.7	0.529608(33)	+2.2	final
0.075	0.56653(36)	2.8	2.4	0.566260(30)	+2.7	final
0.05	0.62235(25)	2.5	0.2	0.622115(27)	+2.3	final
0.025	0.71359(21)	2.1	0.3	0.713422(29)	+1.7	final
0.01	0.80979(34)	3.3	0.6	0.80975(5)	+0.4	final
0.001	0.93732(11)	1.0	0.4	0.93742(11)	-1.0	final

Part VI

Monte Carlo determination of the thermal dimension $\Delta_\varepsilon(\Delta_\sigma)$ of the long-range Ising model in $d = 1, 2, 3$

Methods, data, results, limitations and reproduction

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Abstract

We determine, by cluster Monte Carlo simulation, the scaling dimension Δ_ε of the leading $\mathbb{Z}_2$-even scalar of the ferromagnetic long-range Ising model with couplings $J(r) \propto r^{-(d+s)}$, as a function of the exactly known spin dimension $\Delta_\sigma = (d-s)/2$, in $d = 1, 2, 3$. Δ_ε is extracted from finite-size scaling of the scaled susceptibility and the Binder cumulants and their exact temperature derivatives. The analysis rules were reviewed by a separate AI system (OpenAI Codex with the model gpt-6-astra, which also drafted the simulation engines). In $d = 2, 3$ they were fixed before the production data were analyzed (one later amendment is noted below); in $d = 1$ they were revised after 8 000 of the 10 240 chains had been analyzed, and two gates were replaced after couplings had failed them (section 5). In $d = 1$ ($N \le 2^{19}$, 20 couplings) nine values pass the protocol, with statistical errors (2–5.5) $\times 10^{-4}$; they differ from the eight-loop resummed ε expansion by -0.0024 to $+0.0032$, up to 13 times their statistical errors. Their systematic error is the half-range of a family of fit models, a sensitivity scale rather than a standard deviation, and an audit by the same AI system finds the values at three couplings sensitive to the size cut or the correction model. At the three couplings closest to the Gaussian end ($\varepsilon \le 0.15$) the Monte Carlo values lie 0.0065–0.0104 below the expansion. In $d = 2$ ($L \le 1024$) the finite-size effective dimensions are precise ($\sigma \approx 0.002$) and agree with the resummed expansion to 0.0015 for $s = 1.2$–1.4, about as well as the expansion itself is determined there; no infinite-size value passes the protocol. In $d = 3$ ($L \le 128$) one value passes, $\Delta_\varepsilon(s{=}1.6) = 1.47314 \pm 0.00118\,(\text{stat}) \pm 0.00230\,(\text{sys})$, only under an amendment adopted after the data had been evaluated; it agrees with the expansion (1.4721). Near the short-range end the 3D extrapolations lie 0.022–0.033 above the expansion. We describe the model, algorithms, data, rules, results and limitations, and how to reproduce every number from the released records.

1 Overview

1.1 Scope

This note is written for a physicist outside the project. It explains the problem and what is known exactly (section 2), the Monte Carlo and analysis methods in enough detail to re-implement them (sections 3–4), how and when the analysis rules were fixed (section 5), all results including negative ones (sections 6–9), the limitations and open problems (sections 10–11), and how to recompute every number from the released code and records (section 12). The work was carried out by AI agents under the direction of the author. The simulation engines (section 3) were drafted by GPT-6 astra (OpenAI, via ChatGPT) in discussions with the author and supplied as three separate packages: the initial 1D package (reference engine, campaign tools and analysis core), the specialized 1D engine, and the 2D/3D engine. The fit pipelines, protocols, evaluators and analyses were written by Claude (Anthropic) agents. Protocols and code were reviewed by a second AI system, OpenAI Codex with the model gpt-6-astra, the same model that drafted the engines, in sessions separate from the ones that wrote the protocols and fits. There was no human referee. Numbers in tables and figures are generated by script from the production records; nothing here is a new measurement except the diagnostics explicitly marked as such.

1.2 Headline results

Table 1: What the Monte Carlo establishes. "Passes the protocol" means the label ACCEPTED or ACCEPTED (σ_{sys}-DOMINATED) of section 4; it is a statement about the protocol (in $d = 3$ as amended by P1, section 7.2), not a guarantee of correctness. The theory comparison uses the eight-loop resummed ε expansion of the accompanying article [1] (CB8, section 6.5).

d	result	limitations
1	Nine of 20 couplings pass the protocol: Δ_ε with $\sigma_{\text{stat}} = (2.0\text{–}5.5) \times 10^{-4}$ and $\sigma_{\text{sys}} = (0.8\text{–}9.6) \times 10^{-3}$ (table 5); they differ from the resummed ε expansion by -0.0024 to $+0.0032$ (table 12).	σ_{sys} is a model-family half-range; the values at $\Delta_\sigma = 0.025$, 0.175, 0.225 are sensitive to the size cut or correction model; two labels (0.1625, 0.1727335) depend on the platform; quoted fits at 0.1 and 0.125 have $P_{\chi^2} \approx 5 \times 10^{-5}$; for $\Delta_\sigma \geq 0.0875$ the fit lies above the crossing extrapolation by a common $+0.0010(3)$ (section 6.4). At $\Delta_\sigma = 0.2125$–0.2375 the Monte Carlo values are 0.0065–0.0104 below the expansion.
2	Finite-size effective dimensions $\Delta_{\text{eff}}(U_4, 512 \to 1024) = 0.958$–$0.977$ with $\sigma \approx 0.002$ at all eight couplings; they agree with the resummed expansion to 0.0015 for $s = 1.2$–1.4, where the expansion itself depends on how its end point is imposed by up to 0.005 (table 18).	No infinite-size value passes the protocol. Two single-series extrapolations ($s = 1.05, 1.1$) lie 0.05–0.065 below the expansion; near both ends of the window the finite-size values drift away from it.
3	One value passes, $\Delta_\varepsilon(s{=}1.6) = 1.47314 \pm 0.00118 \pm 0.00230$, consistent with the expansion (1.4721) and with the finite-size $\Delta_{\text{eff}}(U_4, 64 \to 128) = 1.4715(31)$.	Accepted only under amendment P1, adopted after the data were evaluated, whose last code fixes are uninspected. Near the short-range end the extrapolations are 0.022–0.033 above the expansion, and the Y and U_4 estimates split.

2 The model and what is known exactly

2.1 Hamiltonian and conventions

Ising spins $S_i = \pm 1$ on a periodic hypercubic lattice of linear size L ($N = L^d$ sites) interact through the fully periodized power law

$$H = -\sum_{i<j} J_L(i-j)\, S_i S_j, \qquad J_L(\mathbf{r}) = \sum_{\mathbf{n}\in\mathbb{Z}^d} |\mathbf{r} + L\mathbf{n}|^{-d-s} \quad (\mathbf{r} \not\equiv 0 \bmod L), \tag{1}$$

with no Kac normalization (the nearest-neighbor coupling of the infinite lattice is one) and no extra short-range term. Inverse temperatures β, including critical ones, are quoted in this normalization. Below, a *coupling*

always means one value of s (equivalently of Δ_σ) on the simulation grid. In $d = 1$ the lattice size is denoted N, and $J_N(r) = N^{-1-s}[\zeta(1+s, r/N) + \zeta(1+s, 1-r/N)]$ with the Hurwitz zeta function.

2.2 Long-range fixed points and their correction exponents

For $d/2 < s < s_*$ the critical point is the long-range fixed point [4, 5, 6]. Because the kinetic term is non-local, the spin field is not renormalized and

$$\Delta_\sigma = \frac{d-s}{2} \qquad \text{exactly,} \tag{2}$$

i.e. $\eta = 2 - s$. The thermal dimension Δ_ε (dimension of the leading $\mathbb{Z}_2$-even scalar, $1/\nu = d - \Delta_\varepsilon$) is the unknown. The ends of the window and the leading corrections to scaling are:

- *Gaussian end* $s \to d/2$: mean field, $\Delta_\varepsilon \to d - s = d/2$. With $\varepsilon = 2s - d$, $\Delta_\varepsilon = d - s + \varepsilon/3 + O(\varepsilon^2)$; the ϕ^4 coupling is nearly marginal and the leading correction exponent is $\omega = \varepsilon + O(\varepsilon^2)$, so corrections are slow ($\omega \approx \varepsilon$, which is 0.05 at the $d = 1$ coupling closest to this end and 0.1 in $d = 2, 3$).
- *Short-range end* $s \to s_* = 2 - \eta_{\mathrm{SR}}$ ($d = 2, 3$; Sak's criterion [5]): $s_* = 7/4$ in $d = 2$, $s_* \simeq 1.963702$ in $d = 3$, where $\Delta_\varepsilon \to \Delta_\varepsilon^{\mathrm{SR}} = 1$ ($d = 2$) and 1.412625 ($d = 3$; the 3D values from the conformal bootstrap [2]). Near s_* the fixed point is the short-range Ising fixed point weakly coupled to a generalized free field χ through $g \int \sigma\chi$ [7, 8]: with $\delta = (s_* - s)/2$, $dg/d\ell = \delta g - \beta_3 g^3$, where ℓ is the logarithm of the length scale, and $\gamma_\varepsilon \equiv \Delta_\varepsilon - \Delta_\varepsilon^{\mathrm{SR}} = c_\varepsilon g^2$, with $\beta_3 = 12.26(3)$ and $c_\varepsilon = 3.3(5)$ in $d = 3$ [8, eqs. (2.24), (3.41)] (and $\gamma_\varepsilon = O(\delta^2)$ in $d = 2$). The $\sigma\chi$ coupling approaches its fixed point as $\ell^{-(s_*-s)}$, a slow correction with exponent $s_* - s = 2\delta$ (0.05 at 2D $s = 1.7$, 0.014 at 3D $s = 1.95$). A lattice model with a pure power law starts at strong bare coupling, so the approach is a crossover rather than a small correction. It may contribute to the Y–U_4 splits seen near s_*, but in $d = 3$ the split grows with L, whereas at leading order this flow makes it shrink (sections 8, 9).
- $d = 1$, $s \to 1$ ($\Delta_\sigma \to 0$): the chain approaches the inverse-square point [9, 10], where the thermal perturbation becomes marginal ($\Delta_\varepsilon \to 1$); the leading correction exponent tends to zero together with the thermal exponent $y_t = 1 - \Delta_\varepsilon$ (at leading order of the Kosterlitz flow they are equal). Our fits near this end indeed give $\omega \approx y_t$ (table 7).

In $d = 2$ both ends have $\Delta_\varepsilon = 1$, so $\Delta_\varepsilon(s)$ dips below 1 in the interior; in $d = 3$ it interpolates between 1.5 and 1.4126.

2.3 Observables and the effective dimension

With $m = N^{-1}\sum_i S_i$ define the scaled second moment, the Binder cumulant [11] and higher moment ratios

$$Y = L^{2\Delta_\sigma}\langle m^2\rangle, \quad U_4 = 1 - \frac{\langle m^4\rangle}{3\langle m^2\rangle^2}, \quad U_6 = \frac{\langle m^6\rangle}{15\langle m^2\rangle^3}, \quad U_8 = \frac{\langle m^8\rangle}{105\langle m^2\rangle^4}. \tag{3}$$

All are dimensionless at criticality up to corrections and are functions of the scaling variable $(\beta - \beta_c)L^{d-\Delta_\varepsilon}$. For a size-doubling pair $(L, 2L)$ let $\beta_\times$ be the crossing $U_4^{(L)}(\beta_\times) = U_4^{(2L)}(\beta_\times)$. At this same temperature

$$\Delta_{\mathrm{eff}}^{Q}(L) = d - \log_2 \frac{Q'_{2L}(\beta_\times)}{Q'_L(\beta_\times)}, \qquad Q \in \{U_4, U_6, U_8, Y\}, \quad Q' = \partial_\beta Q \tag{4}$$

(the quotient method [12]). Every Q gives Δ_ε as $L \to \infty$; their differences at finite L measure corrections to scaling. In $d = 1$ the Binder cumulant is written U and sizes are N.

3 Monte Carlo method

3.1 Cluster algorithm

All simulations use the extended-Poisson Swendsen–Wang [13] construction of Fukui and Todo [14], whose cost per sweep is $O(N)$ for long-range couplings (see also ref. [16]; for an application to the long-range Ising model, ref. [15]). Per sweep a Poisson number of candidate bond events is drawn with mean $\beta N \sum_{\mathbf{r}} J(\mathbf{r})$; each event

picks a uniform source site and a displacement $\mathbf{r}$ with probability $\propto J(\mathbf{r})$, so each unordered pair receives events at rate $2\beta J_{ij}$. An event is kept only if the two spins are aligned; kept bonds are joined by union–find and every cluster receives an independent random sign. The number of kept events, K (counted with multiplicity, including repeated and cycle-closing events), makes the weight of a bond configuration proportional to β^K; cumulants of K therefore give exact β-derivatives and exact multi-temperature reweighting.

3.2 Engines

$d = 1$**, reference engine** (`code/1d/src/lri_mc.cpp`, C++17): Walker alias table over all $N - 1$ displacements, Mersenne twister, checkpoint/restart with atomic writes, one completion record per chain. Used for the pilot, the two-point campaign and stages 0, A and B (the $d = 1$ campaigns of table 4).

$d = 1$**, specialized engine** (`code/1d/src_1d/`): alias table for $r < 256$, geometric envelope with rejection for the tail, xoshiro256++ generator; 1.5–2.1× faster single core and 2.5–3.3× faster when 30 chains share a node. It agrees with the reference engine within statistics on 24 observable comparisons. Used for the N2 campaign (table 4).

$d = 2, 3$ (`code/2d3d_engine/code/src/`): exact periodic-image sampling without an L^d coupling table. Displacements in the core cube $[-16, 15]^d$ use exact weights via an alias table; the rest is a union of shells $C_{k+1} \setminus C_k$, $C_k = [-16 \cdot 2^k, 16 \cdot 2^k - 1]^d$, sampled from the envelope $(16 \cdot 2^k)^{-d-s}$ with rejection and folded modulo L; the 8 (2D) or 26 (3D) nearest offsets are separate Poisson processes. Records per sweep: K, improved estimators of m^2, m^4, m^6, m^8 from the cluster fractions c_j ($S_{2k} = \sum_j c_j^{2k}$: $m_2 = S_2$, $m_4 = 3S_2^2 - 2S_4$, $m_6 = 15S_2^3 - 30S_2S_4 + 16S_6$, $m_8 = 105S_2^4 - 420S_2^2S_4 + 140S_4^2 + 448S_2S_6 - 272S_8$), the sampled m^2, the largest cluster and the number of clusters. The engine was checked against exact enumeration of 4×4 and 2^3 systems and against the reference engine, and restarts reproduce bitwise.

3.3 Derivatives and reweighting

For a graph observable A, $\partial_\beta \langle A \rangle = \mathrm{Cov}(A, K)/\beta$ and $\partial_\beta^2 \langle A \rangle = [\langle A(K - \bar{K})^2 \rangle - \langle A \rangle \mathrm{Var} K - \mathrm{Cov}(A, K)]/\beta^2$ (the last term arises because the weight is β^K). In $d = 1$ the analysis keeps, per (N, β), the six quantities (Y, Y', Y'', U, U', U'') with a 6×6 jackknife covariance from blocks of 2048, 4096 (primary) and 8192 sweeps. In $d = 2, 3$ each chain is packed into histograms in K at blocks of 1000, 2000 (primary) and 4000 sweeps, and the five simulated temperatures of a size are combined by multi-temperature reweighting (MBAR [17], the multiple-histogram method [18]) with the exact factor $(\beta/\beta_0)^K$.

3.4 Crossings and bootstrap in $d = 2, 3$

For each pre-registered pair, the crossing $\beta_\times$ is the unique root of $U_4^{(2L)} - U_4^{(L)}$ inside the shared temperature bracket (exactly one root required, reweighting effective sample size ≥ 0.2). Errors come from 500 joint bootstrap draws (seed 571239): in each draw the blocks of every chain are resampled once (multinomial weights), the same resample serves both pairs that contain that size, and crossings and slopes are recomputed; the draws give the full covariance across pairs and observables.

3.5 Equilibration and provenance

Every (L, β) group has hot and cold replicas with 5×10^4 warm-up sweeps. In $d = 2, 3$ the equilibration rule of a coupling requires (i) every chain's block mean of m^2 within $|z| \leq 4$ of its group, and at most 2 % of the coupling's chains beyond $|z| = 3$; and (ii) no group in which half or more of the chains fail the drift test (first against second half of the chain, $|z| \leq 4$), the condition block length $\geq 20\,\tau_{\mathrm{int}}$ at every block size used, or finiteness. In $d = 3$ the rule is also applied to the reblocking records. In $d = 1$ (protocol v3) every group must have its hot–cold difference and every replica's first–second-half difference within $|z| \leq 4$ and τ_{int} below block/20; across the coupling at most 2 % of the groups may exceed $|z| = 3$ in the hot–cold test and at most 2 % of the replicas in the half-chain test. A failure labels the coupling EQUILIBRATION FAILED; no chain is removed. In the preflight runs (short runs at the production sizes, before each production, that check rates, autocorrelation times and crossing brackets) τ_{int} was at most 17 sweeps ($d = 1$), 29 ($d = 2$, $L = 1024$) and

34 ($d = 3$, $L = 128$). Each chain writes a completion record; snapshot manifests list the SHA-256 of every summary, block archive, registry and code file; output directories are created afresh and never overwritten.

4 How a label is assigned

Each coupling receives exactly one label from a fixed decision table, applied in the order of table 2; the first condition that applies gives the label. Only the two ACCEPTED labels denote values that passed the protocol (in $d = 2, 3$ the frozen rule or, where it is calibrated, amendment P1; section 7.2). A *gate* is a pass/fail condition on a fit or an extrapolation series; a $d = 1$ fit that passes its gates is *quotable* (section 6.2), and a $d = 2, 3$ series that passes its gates is *eligible* (section 7).

The two errors. σ_{stat} is a bootstrap error of the quoted value. σ_{sys} is half the range of the quoted values over a family of admissible fit models (in $d = 1$) or extrapolation series (in $d = 2, 3$): a *model-family sensitivity scale*, not a standard deviation and not a bound. It covers only the variation inside that family, it is symmetric about the quoted value although the family's excursions need not be, and there is no justified rule for adding it to σ_{stat} in quadrature. Values are written $v(\sigma_{\mathrm{stat}})(\sigma_{\mathrm{sys}})$ in units of the last digits.

Table 2: The decision labels and what triggers them. "Cross-check": two extrapolations of the same data (in $d = 1$ the global fit and the crossing series, in $d = 2, 3$ the U_4 and Y series) must agree within $2\sigma_D$, σ_D being the error of their paired difference.

label	trigger in $d = 1$	trigger in $d = 2, 3$
EQUILIBRATION FAILED	a group fails the equilibration rule	same
COVERAGE-LIMITED	a pre-registered size or temperature missing	a pre-registered pair missing
MODEL-LIMITED	no fit passes the quality gates (finite-size values reported)	no extrapolation series is eligible
CONDITIONAL	a quoted fit exists but σ_{sys} is not available: fewer than four quotable fits, or fewer than two minimum sizes or two variants among them	exactly one eligible series, so no σ_{sys}
REBLOCKING FAILED/UNAVAILABLE	σ_{stat} at blocks 2048 or 8192 outside $\pm 30\,\%$ of the primary	same at blocks 1000 or 4000
CROSS-CHECK UNAVAILABLE	fewer than two eligible crossing series	U_4 and Y not both eligible
CROSS-CHECK FAILED	global fit and median crossing extrapolation differ by more than $2\sigma_D$	extrapolated U_4 and Y differ by more than $2\sigma_D$
ACCEPTED	none of the above	none of the above
ACCEPTED (σ_{sys}-DOMINATED)	accepted and $\sigma_{\mathrm{sys}} > 3\times$ the median σ_{stat} of the quotable fits	–

Suffix "[σ]" ($d = 2, 3$): $\sigma(\Delta_{\mathrm{eff}}^{U_4})$ of the largest pair exceeds the target 0.003. A label is prefixed NON-PRODUCTION when the provenance checks of its inputs do not pass. Tables abbreviate cross-check to x-check, unavailable to unavail., dominated to dom., reblocking to reblock. and equilibration to equil.

5 How the rules were fixed

The aim was to write each analysis protocol, and have it reviewed, before the data it governs were analyzed, with selection rules blind to Δ_ε. Table 3 records how far this held. Reviews were done by the second AI system named in section 1: plan reviews of the protocol texts, each ending with the verdict "approved" (no unresolved material defect in the reviewed revision) or "revise", and inspections of the code that implements them. The protocol texts, their hashes and their review status are released with the code (appendix D); `REVIEWS.md` summarizes every review round and verdict.

$d = 1$: rules changed after failures. Two gates of v2 were replaced in v3 after couplings had failed them on these data. The absolute cap on σ_{sys} (0.0015 for $\Delta_\sigma \le 1/16$) failed at all four couplings $\Delta_\sigma \le 0.05$; v3 has no absolute cap and labels a large σ_{sys} instead (ACCEPTED (σ_{sys}-DOMINATED)). The per-group equilibration

Table 3: When the rules were fixed, relative to the data they govern.

rules	review status	data seen when the rules were fixed
$d = 1$ (FIT_MODEL_V3)	approved at revision 14 (2026-09-19). The first evaluation of all 20 couplings with v3 (called N1, 2026-09-18) ran under revision 8, which had not been reviewed (the review of revision 7 had asked for changes). The last fixes to the implementation are uninspected.	The previous protocol (v2, approved 2026-09-16) had been applied to stages A and B (8 000 of the 10 240 chains of the final fits). v3 was written after those results; its temperature windows (the frozen $(\beta_c^{\mathrm{pre}}, \Delta_\varepsilon^{\mathrm{pre}})$) come from fits of those data, and two v2 gates were replaced after couplings had failed them (below). The rule of the N0 diagnostic (table 4) was amended twice after its data had been seen (below). Only the N2 data (2 240 chains) were taken after approval.
$d = 2$ (FIT_MODEL_2D)	reviewed, never approved; the last fixes applied without a further round	Frozen 2026-09-20, before the production; stage-0 and preflight data had been seen.
$d = 3$ (FIT_MODEL_3D)	revision 5 approved; production under revision 7, whose review returned one finding, fixed without a further round	Revision 7 changed one preflight condition (a τ_{int} cap of 30 to the protocol's own block/$\tau \geq 20$ rule) after the preflight had run; its record also showed preliminary Δ_{eff} values. No production chain had run.
$d = 2, 3$ amendment P1	approved; code inspected twice, the second round's fixes uninspected	Drafted, reviewed and approved after both production evaluations were known, with its effect on every label already computed (section 7.2).

rule failed at 0.0875 and 0.125, each on one group of 50 ($|z| = 3.04$ and 3.05); v3 applies a population rule (no group above 4σ, at most 2 % of groups above 3σ). Both changes have a statistical reason (a cap fixed on two couplings; a per-group rule that a 3σ excursion fails at about 13 % of couplings by chance), but four of the nine ACCEPTED labels depend on them: 0.025 and 0.05 on the cap, 0.0875 and 0.125 on the equilibration rule.

$d = 1$: the N0 diagnostic. N0 tested, at $\Delta_\sigma = 0.175$, whether the planned N2 window could be described by the fit polynomials, and so decided whether N2 ran and with which window. It was evaluated under three rules, the second and third written after its data had been seen. The pre-registered rule gave INCONCLUSIVE; its prescribed remedy, a retest at half the temperature spacing, was not run. The second rule gave FAIL, which would have stopped N2 and returned the plan to review; that failure came from selecting too low a polynomial degree. The third rule, a joint polynomial test of values and derivatives, gave PASS and released N2 with the reduced window $|x| \leq 0.35$.

6 $d = 1$

6.1 Grid and campaigns

Twenty couplings $\Delta_\sigma = k/80$ ($k = 1 \ldots 19$) and $\Delta_\sigma = 0.1727335$ ($s = 0.654533$, the coupling studied in ref. [20]). Every chain of the final fits (stages A, B and N2) has 5×10^4 warm-up and 2×10^6 measured sweeps, every (N, β) group has 8 hot/cold replicas, and each size has five temperatures; the other campaigns of table 4 differ.

6.2 Fit model and selection

Model. For $Q \in \{Y, U\}$, $r = N/1024$ and the scaling variable $x = 100\,\tau(\beta)\, r^{1-\Delta_\varepsilon}$, where $\tau = (\beta - \beta_c)[1 + a_1(\beta - \beta_c)]$ for field order $k_\tau = 2$ ($a_1 = 0$ for $k_\tau = 1$) and the factor 100 only makes x of order one,

$$Q_N(\beta) = F_Q(x) + r^{-\omega} G_Q(x)\ [+\, r^{-2\omega} H_Q(x)]\ [+\, r^{-s}(b_0 + b_1(\beta - \beta_c)) \text{ on } Y]\ [\times(1 + c_Q/\ln(N/256)) \text{ on } F_Q]. \quad (5)$$

F_Q is a polynomial of degree $p = 3 \ldots 7$ and G_Q, H_Q are quadratic. The four *variants* are: `leading` (no bracketed term), `second` (the $r^{-2\omega}$ term), `background` (the analytic r^{-s} background of Y) and `log` (the logarithmic factor; only for $\Delta_\sigma \leq 0.05$). Each (N, β) contributes six rows to the fit: Y, Y', Y'', U, U', U'',

Table 4: $d = 1$ campaigns. Only stages A, B and N2 enter the final fits: 10 240 chains, 2.05×10^{10} measured sweeps, 0.655 TB of raw trajectories.

campaign	design	chains	in final fits?
pilot	bracketing at $\Delta_\sigma = 0.1, 1/16$, $N \le 32\,768$	–	no
two-point	$\Delta_\sigma = 0.1, 1/16$, $N = 2^{10} \dots 2^{19}$, half-width $h(N) = 0.005(N/1024)^{-0.35}$	800	no
stage 0	bracket finder for the temperature centers	–	no
stage A	20 points, $N = 2^{10} \dots 2^{16}$	5 600	yes
stage B	20 points, $N = 2^{17} \dots 2^{19}$	2 400	yes
N0	diagnostic at $\Delta_\sigma = 0.175$, $N = 2^{16}, 2^{18}$, 9 temperatures	72	no
N2	14 points ($\Delta_\sigma = 1/16$ and ≥ 0.1), $N = 2^{16} \dots 2^{19}$, temperatures matched in the scaling variable, specialized engine	2 240	yes

the derivatives following from the model by the chain rule, with the full 6×6 covariance. The nonlinear parameters $(\beta_c, \Delta_\varepsilon, \omega, a_1[, c_Q])$ are fitted by profiling over all linear amplitudes.

Configurations. Data are selected by windows $|x| \le x_{\max} \in \{0.35, 0.45, 0.6\}$, computed with frozen $(\beta_c^{\mathrm{pre}}, \Delta_\varepsilon^{\mathrm{pre}})$ (appendix A), and minimum sizes $N_{\min} \in \{4096, 8192, 16384\}$; at least four sizes with three temperatures each must remain. Windows $\times$ variants $\times$ $N_{\min} \times k_\tau$ give 54 (72 with `log`) configurations per coupling; at $\Delta_\sigma = 0.075$ and 0.0875 the window 0.35 leaves too few temperatures, so these couplings have 36, and the 20 couplings have 1 116 in all.

Degree ladder. For each configuration the degree starts at $p = 3$ and is raised while an F-test prefers $p + 1$ (F-test probability below 0.01); the ladder stops at the first missing or flagged fit (a fit is flagged if a nonlinear parameter sits on its search bound, a matrix is rank deficient or the optimizer fails). A fit at $p + 1$ must agree with p to $3\sigma_{\mathrm{stat}}$. We write P_{χ^2} for the χ^2 probability of a fit and keep p for the degree.

Quotable fits. A configuration's fit is quotable if $\chi^2/\mathrm{dof} \le 1.5$, it is unflagged, the mean |pull| of each observable is ≤ 1.2, at least 270 of 300 bootstrap refits are valid, and a test of residual structure passes (runs of pull signs and lag-one correlation of pulls along β, with thresholds calibrated on synthetic data drawn from the fit itself).

The quoted value is the quotable fit with, in order of preference, the smallest $N_{\min}$, the fewest parameters, the window 0.45 before 0.6 before 0.35, the variant `leading` before `background` before `second` before `log`, the lower k_τ, the lower degree. σ_{stat} is half the 16–84 % range of its 300 bootstrap refits, in which the blocks of every chain are resampled and all nonlinear parameters are refitted. σ_{sys} is half the range over all quotable fits and requires at least four fits spanning two $N_{\min}$ and two variants.

Crossing cross-check. Independently of the global fit, local polynomials in β through the in-window temperatures of each size give the U-crossings of the four largest usable doubling pairs (a size is usable if its local fit is unflagged with $\chi^2/\mathrm{dof} \le 2$, a pair if both sizes are usable and the crossing lies inside their common bracket). There are up to four *crossing series*, one per crossing window $|x| \le x_c \in \{0.45, 0.6\}$ and local-degree rule (the degree p^* chosen by an F-test, or $p^* + 1$), named `xc0.45_star`, `xc0.45_plus`, `xc0.6_star` and `xc0.6_plus`. Each is extrapolated as $\Delta_{\mathrm{eff}}^Q(N) = \Delta_\varepsilon^\infty + c_Q N^{-\omega'}$ with $\omega' \in [0.2, 2]$ profiled on a grid of step 0.01 in every bootstrap draw; a series is eligible if $\chi^2/\mathrm{dof} \le 2$ and ω' is off the grid bounds. With at least two eligible series, the test is $|\Delta_\varepsilon^{\mathrm{fit}} - \mathrm{median}_E\, \Delta_\varepsilon^\infty| \le 2\sigma_D$, where, as pre-registered in the protocol, σ_D is the half 16–84 % width of the paired bootstrap difference *enlarged in quadrature* by half the spread of the central values of the eligible series. This makes σ_D itself a mixed statistical and sensitivity quantity; table 7 gives z with and without the spread term. The crossing value is quoted as $\Delta_\varepsilon^\infty(\sigma_\infty)(\Delta_{\omega'})$: σ_∞ is the bootstrap error of $\Delta_\varepsilon^\infty$ with ω' re-profiled in every draw, and $\Delta_{\omega'}$ is a spread of $\Delta_\varepsilon^\infty$ (not of ω'): the between-series half-spread plus the half-range of $\Delta_\varepsilon^\infty$ over the union of the series' $\Delta\chi^2 \le 1$ intervals of ω'. It counts the between-series spread twice and is an enlarged sensitivity measure.

Worked examples. At $\Delta_\sigma = 0.1727335$, 4 of 54 configurations are quotable; the preferred one (window 0.35, variant `second`, $N_{\min} = 4096$, $k_\tau = 2$, $p = 5$) gives 0.48368(52); the four quotable values span $\pm 0.00078 = \sigma_{\text{sys}}$; four crossing series are eligible, their median lies 0.00013 below the fit with $\sigma_D = 0.00078$, so the cross-check passes and the label is ACCEPTED. At $\Delta_\sigma = 0.15$, 11 configurations are quotable, the quoted value is 0.49325(41) with $\sigma_{\text{sys}} = 0.00306$, and the crossing median lies 0.00143 below with $\sigma_D = 0.00068$: CROSS-CHECK FAILED.

6.3 Results

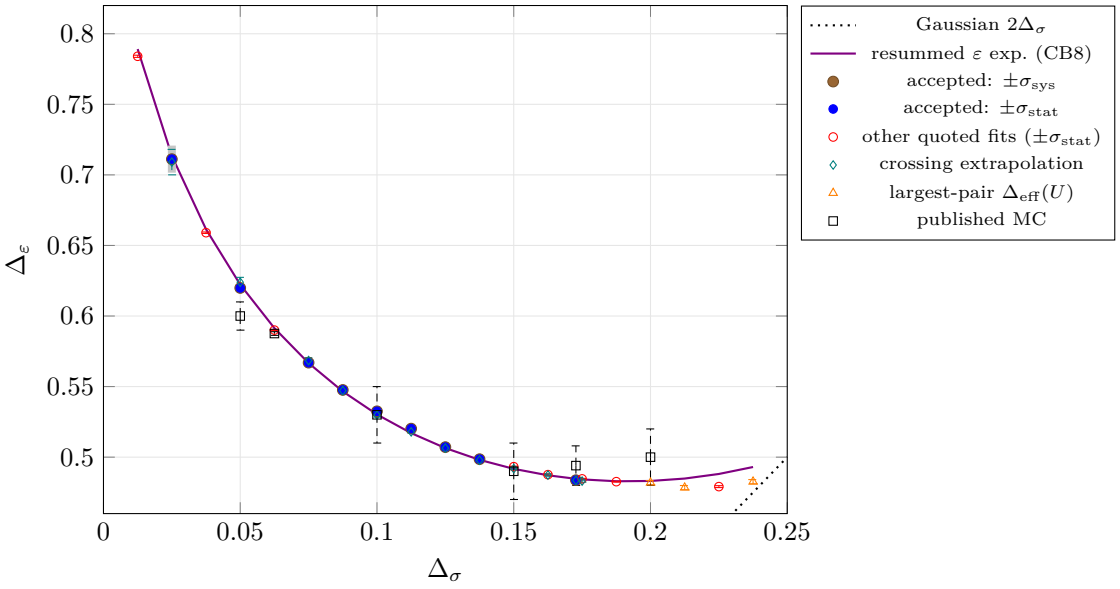


Figure 1: $d = 1$: $\Delta_\varepsilon(\Delta_\sigma)$. Values that passed the protocol with σ_{stat} (blue bars) and, separately, the model-family half-range σ_{sys} (gray bands), which is not a standard deviation; other quoted fits; crossing extrapolations with σ_∞; largest-pair effective dimensions where no fit is quotable; published Monte Carlo values; and the resummed ε expansion. Table 12 gives the differences.

Reading the results.

Near $s \to 1$ ($\Delta_\sigma \leq 0.05$). 68–72 configurations are quotable, but $\sigma_{\text{sys}} = 4$–$20 \times \sigma_{\text{stat}}$: the fitted correction exponents ($\omega \approx 0.26$–0.40) and the variants do not settle the correction sector, as expected near a marginal point.

Middle ($0.0625 \leq \Delta_\sigma \leq 0.1375$). Six of seven couplings pass, with $\sigma_{\text{stat}} = (2\text{–}4) \times 10^{-4}$ and $\sigma_{\text{sys}} = (1.4\text{–}2.4) \times 10^{-3}$. At $\Delta_\sigma = 1/16$ the value is stable across campaigns: 0.59003 (final fit, with N2), 0.59046 (stages A and B alone, under protocol v2) and 0.59071 (the separate two-point campaign); but the crossing series fail their own goodness-of-fit gate, hence CROSS-CHECK UNAVAILABLE.

Towards the Gaussian end ($\Delta_\sigma \geq 0.15$). $\Delta_\sigma = 0.1727335$ passes. At 0.15 and 0.175 the cross-check fails by 2.1 and $2.0\sigma_D$; at 0.1625, 0.1875 and 0.225 only fits of variant `second` with $k_\tau = 2$ are quotable (CONDITIONAL); at 0.2, 0.2125 and 0.2375 no fit is quotable, because the correction exponent $\omega \approx \varepsilon \leq 0.2$ is too small to separate from the scaling function over $N \leq 2^{19}$. There the finite-size values of table 6 are reported; they are not infinite-size estimates.

6.4 Robustness of the $d = 1$ results

Table 7 collects, per coupling, the quantities that bear on how much a value or label can be trusted; the paragraphs below explain them.

Goodness of fit. The quality gate $\chi^2/\text{dof} \leq 1.5$ admits, at about 200 degrees of freedom, fits with $P_{\chi^2} \sim 10^{-5}$. The quoted fits at $\Delta_\sigma = 0.1$ and 0.125 have $P_{\chi^2} \approx 5 \times 10^{-5}$, at 0.025 $P_{\chi^2} = 3 \times 10^{-4}$, and at 0.1125 $P_{\chi^2} = 1.5 \times 10^{-3}$. The preference for the smallest $N_{\min}$ selects the member of the family with

Table 5: $d = 1$ results. Δ_ε is value(σ_{stat})(σ_{sys}) in units of 10^{-5}; †: σ_{sys} not available. "quot.": number of quotable fits; quoted configuration: window, variant, $N_{\min}$, field order and polynomial degree (e.g. `x0.45_leading_Nmin4096_K1 d4`: $|x| \le 0.45$, variant `leading`, $N_{\min} = 4096$, $k_\tau = 1$, $p = 4$); crossing value $\Delta_\varepsilon^\infty(\sigma_\infty)(\Delta_{\omega'})$ (section 6.2). Published values: UGA [19], APR [20].

Δ_σ	s	decision	Δ_ε (stat)(sys)	quot.	quoted configuration	crossing value	published MC
0.0125	0.975	CROSS-CHECK UNAVAILABLE	0.78404(77)(1003)	72	x0.45_leading_Nmin4096_K1 d3	–	
0.025	0.95	ACCEPTED (σ_{sys}-DOMINATED)	0.71115(49)(959)	68	x0.45_leading_Nmin4096_K1 d4	0.70906(902)(1509)	
0.0375	0.925	CROSS-CHECK UNAVAILABLE	0.65904(51)(425)	71	x0.45_leading_Nmin4096_K1 d4	–	
0.05	0.9	ACCEPTED	0.61987(55)(220)	72	x0.45_leading_Nmin4096_K1 d4	0.62393(348)(773)	0.60(1) UGA
0.0625	0.875	CROSS-CHECK UNAVAILABLE	0.59003(35)(150)	50	x0.45_leading_Nmin4096_K1 d5	–	0.5876(13) APR
0.075	0.85	ACCEPTED	0.56694(37)(139)	30	x0.45_leading_Nmin4096_K2 d4	0.56843(207)(1044)	
0.0875	0.825	ACCEPTED	0.54761(32)(194)	34	x0.45_leading_Nmin4096_K1 d5	0.54711(113)(328)	
0.1	0.8	ACCEPTED (σ_{sys}-DOMINATED)	0.53248(20)(242)	48	x0.45_leading_Nmin4096_K1 d5	0.52975(135)(248)	0.53(2) UGA
0.1125	0.775	ACCEPTED (σ_{sys}-DOMINATED)	0.52020(25)(217)	41	x0.45_leading_Nmin4096_K2 d5	0.51790(86)(944)	
0.125	0.75	ACCEPTED	0.50710(35)(193)	9	x0.35_background_Nmin4096_K2 d5	0.50640(172)(392)	
0.1375	0.725	ACCEPTED (σ_{sys}-DOMINATED)	0.49856(30)(180)	17	x0.45_background_Nmin4096_K2 d4	0.49783(94)(85)	
0.15	0.7	CROSS-CHECK FAILED	0.49325(41)(306)	11	x0.35_background_Nmin4096_K2 d5	0.49183(71)(70)	0.49(2) UGA
0.1625	0.675	CONDITIONAL	0.48761(58)(†)	5	x0.35_second_Nmin4096_K2 d5	0.48732(75)(297)	
0.1727335	0.6545	ACCEPTED	0.48368(52)(78)	4	x0.35_second_Nmin4096_K2 d5	0.48355(62)(84)	0.494(14) APR
0.175	0.65	CROSS-CHECK FAILED	0.48460(35)(67)	6	x0.45_second_Nmin4096_K2 d6	0.48304(81)(1)	
0.1875	0.625	CONDITIONAL	0.48263(30)(†)	4	x0.45_second_Nmin4096_K2 d5	–	
0.2	0.6	MODEL-LIMITED	–	0	–	–	0.50(2) UGA
0.2125	0.575	MODEL-LIMITED	–	0	–	–	
0.225	0.55	CONDITIONAL	0.47912(68)(†)	1	x0.35_second_Nmin16384_K2 d7	–	
0.2375	0.525	MODEL-LIMITED	–	0	–	–	

Table 6: $d = 1$: effective dimensions $\Delta_{\mathrm{eff}}(U)/\Delta_{\mathrm{eff}}(Y)$ at the U-crossing of the two largest usable doubling pairs (errors in units of 10^{-4}, 300 draws). For the MODEL-LIMITED couplings these finite-size values are the only result.

Δ_σ	pair 1	pair 2
0.0125	128k–256k: 0.7606(57) / 0.7583(26)	256k–512k: 0.7747(61) / 0.7671(28)
0.025	128k–256k: 0.7034(32) / 0.7015(17)	256k–512k: 0.7094(37) / 0.7044(20)
0.0375	128k–256k: 0.6617(31) / 0.6551(19)	256k–512k: 0.6545(34) / 0.6528(20)
0.05	128k–256k: 0.6211(26) / 0.6174(17)	256k–512k: 0.6185(26) / 0.6173(17)
0.0625	128k–256k: 0.5889(13) / 0.5850(9)	256k–512k: 0.5921(14) / 0.5893(10)
0.075	128k–256k: 0.5670(18) / 0.5628(14)	256k–512k: 0.5676(20) / 0.5659(16)
0.0875	128k–256k: 0.5452(17) / 0.5418(13)	256k–512k: 0.5472(19) / 0.5440(15)
0.1	128k–256k: 0.5302(12) / 0.5251(10)	256k–512k: 0.5296(13) / 0.5279(11)
0.1125	128k–256k: 0.5194(12) / 0.5144(10)	256k–512k: 0.5185(13) / 0.5140(11)
0.125	128k–256k: 0.5058(12) / 0.5003(10)	256k–512k: 0.5084(13) / 0.5020(12)
0.1375	128k–256k: 0.5005(12) / 0.4921(11)	256k–512k: 0.4989(13) / 0.4918(12)
0.15	128k–256k: 0.4942(13) / 0.4819(12)	256k–512k: 0.4930(14) / 0.4837(13)
0.1625	128k–256k: 0.4875(12) / 0.4746(11)	256k–512k: 0.4889(14) / 0.4771(13)
0.1727335	128k–256k: 0.4835(13) / 0.4673(14)	256k–512k: 0.4852(14) / 0.4715(13)
0.175	128k–256k: 0.4839(14) / 0.4668(13)	256k–512k: 0.4851(14) / 0.4700(13)
0.1875	128k–256k: 0.4846(14) / 0.4645(13)	256k–512k: 0.4803(14) / 0.4625(14)
0.2	128k–256k: 0.4866(14) / 0.4587(13)	256k–512k: 0.4819(15) / 0.4570(15)
0.2125	128k–256k: 0.4855(15) / 0.4499(15)	256k–512k: 0.4784(15) / 0.4518(15)
0.225	128k–256k: 0.4814(15) / 0.4405(16)	256k–512k: 0.4844(16) / 0.4464(16)
0.2375	16k–32k: 0.4825(23) / 0.4150(19)	128k–256k: 0.4826(15) / 0.4307(16)

Table 7: $d = 1$ robustness. n_q: quotable fits; P_{χ^2}: χ^2 probability of the quoted fit; ω: its correction exponent; fit − crossing: quoted value minus the median crossing extrapolation; z: that difference over σ_D as pre-registered (with the spread term) and over the bootstrap part alone; P: label platform-sensitive (section 6.4.1).

Δ_σ	decision	n_q	P_{χ^2}	ω	fit − crossing	z	z (bootstrap)	
0.0125	X-CHECK UNAVAIL.	72	0.01	0.262	–	–	–	
0.025	ACCEPTED (σ_{sys}-DOM.)	68	3.1×10^{-4}	0.309	+0.00209	+0.23	+0.24	
0.0375	X-CHECK UNAVAIL.	71	0.21	0.348	–	–	–	
0.05	ACCEPTED	72	0.36	0.397	-0.00406	-0.92	-1.16	
0.0625	X-CHECK UNAVAIL.	50	0.29	0.427	–	–	–	
0.075	ACCEPTED	30	0.51	0.418	-0.00148	-0.33	-0.70	
0.0875	ACCEPTED	34	0.41	0.385	+0.00050	+0.35	+0.50	
0.1	ACCEPTED (σ_{sys}-DOM.)	48	5.2×10^{-5}	0.367	+0.00273	+1.73	+2.13	
0.1125	ACCEPTED (σ_{sys}-DOM.)	41	1.5×10^{-3}	0.368	+0.00230	+0.61	+2.73	
0.125	ACCEPTED	9	4.6×10^{-5}	0.368	+0.00070	+0.31	+0.42	
0.1375	ACCEPTED (σ_{sys}-DOM.)	17	0.03	0.365	+0.00073	+0.77	+0.79	
0.15	X-CHECK FAILED	11	2.1×10^{-3}	0.320	+0.00143	+2.10	+2.24	
0.1625	CONDITIONAL	5	0.05	0.194	+0.00029	+0.22	+0.38	P
0.1727335	ACCEPTED	4	0.08	0.195	+0.00013	+0.17	+0.20	P
0.175	X-CHECK FAILED	6	0.03	0.192	+0.00155	+2.00	+2.00	
0.1875	CONDITIONAL	4	1.5×10^{-4}	0.155	–	–	–	
0.2	MODEL-LIMITED	0	–	–	–	–	–	
0.2125	MODEL-LIMITED	0	–	–	–	–	–	
0.225	CONDITIONAL	1	7.4×10^{-4}	0.083	–	–	–	
0.2375	MODEL-LIMITED	0	–	–	–	–	–	

the most data, which is also often the one with the largest χ^2. For these couplings σ_{stat} understates the point-to-point scatter, and the values should be read with σ_{sys}.

The cross-check and its σ_D. Without the spread term in σ_D the test would fail at $\Delta_\sigma = 0.1$ ($z = 2.13$) and 0.1125 ($z = 2.73$), both labeled ACCEPTED. Moreover, for all nine couplings with $\Delta_\sigma \geq 0.0875$ and a cross-check the fit lies above the crossing extrapolation: the inverse-variance mean difference is $+0.00105 \pm 0.00028$ with $\chi^2 = 8.1$ for 8 degrees of freedom about it. (Inverse-variance weights from the bootstrap part.) The two failures at 0.15 and 0.175 are among the points with the smallest σ_D, not isolated anomalies. The global fit and the crossings use different correction exponents (for example $\omega = 0.32$ and $\omega' = 0.31$–0.32 at 0.15, $\omega = 0.19$ and $\omega' = 0.27$ at 0.175) and the same data, so their comparison is a consistency check of two extrapolation models, not an independent measurement.

The failures and the ω' grid. The failures at 0.15 and 0.175 are not an artifact of conditional errors: the crossing bootstrap refits ω' in every draw, and fixing it at its central value would make σ_D smaller (table 8). They are, however, numerically fragile: refining only the ω' grid from step 0.01 to 0.001 and 0.0001, with everything else fixed, moves the ratio $|\Delta_\varepsilon^{\mathrm{fit}} - \Delta_\varepsilon^{\infty}|/\sigma_D$ from 2.101 to 1.884 and 1.907 at 0.15, and from 2.003 to 1.897 and 1.931 at 0.175, below the threshold. The labels stand under the frozen protocol (step 0.01); whether a fit–crossing difference of this size is real is unresolved, and the common offset above suggests a systematic difference between the two extrapolation models.

Table 8: The cross-check at $\Delta_\sigma = 0.15, 0.175$ with ω' refitted per draw (as recorded) and fixed at its central value. Last column: central ω' (16–84 % of the per-draw values) of each eligible crossing series.

Δ_σ	fit − crossing	σ_D (per draw)	z	σ_D (fixed ω')	z	series
0.15	+0.00143	0.00068	+2.10	0.00062	+2.30	xc0.45_star: 0.32 (0.31–0.32), xc0.45_plus: 0.31 (0.30–0.32)
0.175	+0.00155	0.00078	+2.00	0.00067	+2.30	xc0.6_star: 0.27 (0.26–0.27), xc0.6_plus: 0.27 (0.26–0.27)

Audit. On 2026-09-22 the second AI system audited the finished analysis, with diagnostic refits (no new data) on the same records and bootstrap draws. It found no defect in the sampler, estimators or implementation, and these sensitivities:

- The quoted values at $\Delta_\sigma = 0.1$, 0.1375 and 0.1727335 are stable under stronger size cuts ($N \geq 32\,768$, $N \geq 65\,536$); at 0.1727335 the cuts give 0.48421 and 0.48386 against 0.48368(52).
- At 0.025 a model outside the family (analytic background plus second power together) gives 0.69348, below the family's lower end 0.69686, a shift of 7.0 paired-bootstrap scales (half the 16–84 % width of the matched difference). At 0.175, $N \geq 32\,768$ shifts the value by -0.00237 (2.2 scales); at 0.225, $N \geq 65\,536$ shifts it by $+0.00483$ (3.7 scales).
- The family is asymmetric: at 0.1375 it extends 0.00020 below and 0.00340 above the quoted value.
- A regular magnetic background gives an N^{-s} correction to U as well as to Y, but the `background` variant puts it only on Y. In the six cases the audit reports (0.025, 0.1, 0.1375, 0.1727335, 0.175, 0.225; variant with background and second power), adding it moves the central value by at most 0.0021 (at 0.025). At 0.05 such a model gives 0.61602, 0.0039 below the quoted value and below the family range [0.61845, 0.62286] (central fit only, not bootstrapped).

The degree ladder. In 160 of the 1 116 configurations, at 15 of the 20 couplings, the ladder cannot take its first step ($p = 3$ against $p = 4$) because one of the two fits is missing or flagged. In 140 of them it is the $p = 3$ fit, although that fit is typically a gross misfit that the F-test would reject by many orders of magnitude; in the other 20 it is the $p = 4$ fit. n_q and σ_{sys} are computed on the families trimmed in this way. At two couplings this decides the label (section 6.4.1). A sensitivity study with a ladder that does not stop on a flagged lower-degree fit is reported in section 6.4.2.

6.4.1 Reproducibility of the fitting stage across platforms

The fit inputs of every coupling (merged summaries and block archives) are released, so the whole fitting stage can be rerun (`make refit1d`). It was rerun for all 20 couplings on 7 October 2026 on a workstation

(Intel i9-13900K; Python 3.12.3, NumPy 1.26.4, SciPy 1.11.4, OpenBLAS) with the production arguments and compared with the production run (AMD EPYC 7742; Python 3.8.5, NumPy 1.19.2, SciPy 1.5.2); table 9. Eighteen labels and all quoted configurations are reproduced, and so is σ_{sys} wherever both runs have it; quoted values agree to $\leq 10^{-8}$ and σ_{stat} to $\leq 1.2 \times 10^{-7}$ absolute, because nonlinear optima agree only to $\sim 10^{-7}$. Two labels change, in opposite directions, for the same reason: the $p = 3$ start of the ladder of configuration `x0.45_background_Nmin8192_K2`, a gross misfit ($\chi^2 \approx 6$–8×10^4 for 178 degrees of freedom, rejected by the F-test at a probability below 10^{-100}), has two minima, one interior and a lower one with a_1 on its bound -50, and which one the multistart optimizer finds depends on floating-point details (table 10). A bound minimum is flagged and stops the ladder, so the configuration never reaches its good $p \geq 5$ fits.

- At $\Delta_\sigma = 0.1727335$ production found the interior minimum: four quotable fits, $\sigma_{\text{sys}} = 0.00078$, ACCEPTED. The workstation (with its own libraries, and with the production versions in OpenBLAS and MKL builds) and an AMD EPYC 7543 node with the production versions find the bound minimum: three fits remain, σ_{sys} needs four, CONDITIONAL. The fourth fit, which exists only on the production platform, has $\chi^2/\text{dof} = 1.47$ against the cap 1.5.
- At $\Delta_\sigma = 0.1625$ it is the reverse: production found the bound minimum (five quotable fits, all of variant `second`, so σ_{sys} is not available: CONDITIONAL); the workstation finds the interior minimum, a sixth fit of variant `background` becomes quotable, $\sigma_{\text{sys}} = 0.00024$ and the label is ACCEPTED.

The values and σ_{stat} are robust; these two labels should be read as platform-sensitive.

Table 9: $d = 1$ fitting stage rerun from the released snapshots (workstation, 2026-10-07) against production. n_q: quotable fits; $|\delta\Delta_\varepsilon|$: difference of the quoted values. Records: `results/1d/refit_workstation_20261007/`.

	production			refit (workstation)				
Δ_σ	decision	n_q	σ_{sys}	decision	n_q	σ_{sys}	$\lvert\delta\Delta_\varepsilon\rvert$	
0.0125	X-CHECK UNAVAIL.	72	0.01003	X-CHECK UNAVAIL.	72	0.01003	3e-09	
0.025	ACCEPTED (σ_{sys}-DOM.)	68	0.00959	ACCEPTED (σ_{sys}-DOM.)	68	0.00959	1e-09	
0.0375	X-CHECK UNAVAIL.	71	0.00425	X-CHECK UNAVAIL.	71	0.00425	2e-09	
0.05	ACCEPTED	72	0.00220	ACCEPTED	72	0.00220	5e-09	
0.0625	X-CHECK UNAVAIL.	50	0.00150	X-CHECK UNAVAIL.	50	0.00150	4e-09	
0.075	ACCEPTED	30	0.00139	ACCEPTED	30	0.00139	2e-09	
0.0875	ACCEPTED	34	0.00194	ACCEPTED	34	0.00194	2e-09	
0.1	ACCEPTED (σ_{sys}-DOM.)	48	0.00242	ACCEPTED (σ_{sys}-DOM.)	48	0.00242	8e-10	
0.1125	ACCEPTED (σ_{sys}-DOM.)	41	0.00217	ACCEPTED (σ_{sys}-DOM.)	41	0.00217	2e-09	
0.125	ACCEPTED	9	0.00193	ACCEPTED	9	0.00193	3e-09	
0.1375	ACCEPTED (σ_{sys}-DOM.)	17	0.00180	ACCEPTED (σ_{sys}-DOM.)	17	0.00180	2e-09	
0.15	X-CHECK FAILED	11	0.00306	X-CHECK FAILED	11	0.00306	2e-10	
0.1625	CONDITIONAL	5	–	ACCEPTED	6	0.00024	2e-09	changed
0.1727335	ACCEPTED	4	0.00078	CONDITIONAL	3	–	3e-09	changed
0.175	X-CHECK FAILED	6	0.00067	X-CHECK FAILED	6	0.00067	3e-09	
0.1875	CONDITIONAL	4	–	CONDITIONAL	4	–	6e-09	
0.2	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	–	
0.2125	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	–	
0.225	CONDITIONAL	1	–	CONDITIONAL	1	–	7e-09	
0.2375	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	–	

Table 10: The $p = 3$ start of the ladder of `x0.45_background_Nmin8192_K2` on each platform: χ^2 (178 dof), the field coefficient a_1 (bounds ± 50), and the consequence for the ladder.

Δ_σ	platform	χ^2	a_1	minimum	ladder
0.1625	production (AMD EPYC 7742)	60656.3	-50.00	bound	stops at $p = 3$
0.1625	Intel i9-13900K, Py 3.12/NumPy 1.26/SciPy 1.11, OpenBLAS	67496.2	-38.29	interior	continues
0.1727335	production (AMD EPYC 7742)	75400.9	-45.49	interior	continues
0.1727335	Intel i9-13900K, Py 3.12/NumPy 1.26/SciPy 1.11, OpenBLAS	68429.4	-50.00	bound	stops at $p = 3$
0.1727335	Intel i9-13900K, Py 3.8.5/NumPy 1.19.2/SciPy 1.5.2, OpenBLAS	68429.4	-50.00	bound	stops at $p = 3$
0.1727335	Intel i9-13900K, same versions, MKL	68429.4	-50.00	bound	stops at $p = 3$
0.1727335	AMD EPYC 7543, same versions, MKL	68429.4	-50.00	bound	stops at $p = 3$

6.4.2 Sensitivity to the ladder rule

To measure how much the trimmed families matter, all 20 couplings were refitted twice in one job on one node (AMD EPYC 7543; Python 3.8.5, NumPy 1.19.2, SciPy 1.5.2, MKL): once with the frozen fitter, and once with a copy whose only change is that a flagged lower-degree fit no longer stops the ladder (its χ^2 still enters the F-test). This is a diagnostic, not the protocol, and table 5 is not modified; records and the modified fitter

are in `results/1d/robust_ladder_20261013/`. Table 11 gives the result. On this node the frozen ladder reproduces the workstation refit of section 6.4.1, including its two label changes. With the robust ladder:

- every quoted value is unchanged to 2×10^{-5};
- the number of quotable fits grows at eleven couplings, and σ_{sys} grows at 0.125 (0.00193 to 0.00221), 0.1375 (0.00180 to 0.00192), 0.1625 (0.00024 to 0.00071) and 0.175 (0.00067 to 0.00115);
- both platform-sensitive couplings pass: 0.1625 with $\sigma_{\text{sys}} = 0.00071$, and 0.1727335 with nine quotable fits and the production value $\sigma_{\text{sys}} = 0.00078$;
- 0.125 passes as σ_{sys}-dominated, 0.1875 becomes CROSS-CHECK UNAVAILABLE with $\sigma_{\text{sys}} = 0.00063$, and 0.2 becomes CONDITIONAL with two quotable fits.

The trimming therefore mostly shrinks the model families; it decides a label only where few fits are quotable. Under the robust rule ten couplings would pass, and the two platform-sensitive labels would no longer depend on which minimum the degree-3 fit finds.

Table 11: $d = 1$ ladder sensitivity: production, and the frozen and the robust ladder refitted on one node. n_q: quotable fits; $|\delta\Delta_\varepsilon|$: quoted value, robust minus production; last column: what the robust ladder changes relative to the frozen ladder on the same node.

	production			frozen ladder			robust ladder				
Δ_σ	decision	n_q	σ_{sys}	decision	n_q	σ_{sys}	decision	n_q	σ_{sys}	$\lvert\delta\Delta_\varepsilon\rvert$	
0.0125	X-CHECK UNAVAIL.	72	0.01003	X-CHECK UNAVAIL.	72	0.01003	X-CHECK UNAVAIL.	72	0.01003	1e-09	
0.025	ACCEPTED (σ_{sys}-DOM.)	68	0.00959	ACCEPTED (σ_{sys}-DOM.)	68	0.00959	ACCEPTED (σ_{sys}-DOM.)	68	0.00959	2e-10	
0.0375	X-CHECK UNAVAIL.	71	0.00425	X-CHECK UNAVAIL.	71	0.00425	X-CHECK UNAVAIL.	71	0.00425	9e-10	
0.05	ACCEPTED	72	0.00220	ACCEPTED	72	0.00220	ACCEPTED	72	0.00220	6e-09	
0.0625	X-CHECK UNAVAIL.	50	0.00150	X-CHECK UNAVAIL.	50	0.00150	X-CHECK UNAVAIL.	50	0.00150	3e-09	
0.075	ACCEPTED	30	0.00139	ACCEPTED	30	0.00139	ACCEPTED	30	0.00139	6e-09	
0.0875	ACCEPTED	34	0.00194	ACCEPTED	34	0.00194	ACCEPTED	35	0.00194	2e-09	n_q
0.1	ACCEPTED (σ_{sys}-DOM.)	48	0.00242	ACCEPTED (σ_{sys}-DOM.)	48	0.00242	ACCEPTED (σ_{sys}-DOM.)	49	0.00242	1e-10	n_q
0.1125	ACCEPTED (σ_{sys}-DOM.)	41	0.00217	ACCEPTED (σ_{sys}-DOM.)	41	0.00217	ACCEPTED (σ_{sys}-DOM.)	44	0.00217	4e-09	n_q
0.125	ACCEPTED	9	0.00193	ACCEPTED	9	0.00193	ACCEPTED (σ_{sys}-DOM.)	18	0.00221	2e-05	label
0.1375	ACCEPTED (σ_{sys}-DOM.)	17	0.00180	ACCEPTED (σ_{sys}-DOM.)	17	0.00180	ACCEPTED (σ_{sys}-DOM.)	24	0.00192	1e-09	n_q
0.15	X-CHECK FAILED	11	0.00306	X-CHECK FAILED	11	0.00306	X-CHECK FAILED	19	0.00306	1e-09	n_q
0.1625	CONDITIONAL	5	–	ACCEPTED	6	0.00024	ACCEPTED	15	0.00071	4e-09	n_q
0.1727335	ACCEPTED	4	0.00078	CONDITIONAL	3	–	ACCEPTED	9	0.00078	2e-09	label
0.175	X-CHECK FAILED	6	0.00067	X-CHECK FAILED	6	0.00067	X-CHECK FAILED	13	0.00115	5e-11	n_q
0.1875	CONDITIONAL	4	–	CONDITIONAL	4	–	X-CHECK UNAVAIL.	10	0.00063	2e-09	label
0.2	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	CONDITIONAL	2	–	–	label
0.2125	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	–	
0.225	CONDITIONAL	1	–	CONDITIONAL	1	–	CONDITIONAL	1	–	5e-09	
0.2375	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	–	

6.5 Comparison with theory and earlier simulations

Resummed ε expansion. Table 12 compares with the eight-loop ε expansion of the accompanying article [1], resummed by conformal Borel resummation [3]. In $d = 1$ it is resummed in the variable $v = 1 - 2\sqrt{\Delta_\sigma}$, with the short-range series near $s = 1$ imposed; we call this curve CB8 (values and band from the article's resummation package, `data/reference/theory_1d_CB8.csv`). The band is a sensitivity range over resummation prescriptions, not a confidence interval. The nine values that passed the protocol differ from CB8 by -0.0024 to $+0.0032$, that is 1 to 13 times their σ_{stat}; four of them (0.025, 0.05, 0.1, 0.1125) lie outside the band. As a scale only, σ_{sys} exceeds the difference except at 0.05, 0.1125 and 0.1727335 (1.1, 1.5 and 1.2 times σ_{sys}). Near the Gaussian end the series needs little resummation (at $\Delta_\sigma = 0.2375$, $\varepsilon = 0.05$, its partial sums are stable to 3×10^{-4}). There the largest-pair effective dimensions (0.2, 0.2125, 0.2375) and the single quoted fit at 0.225 lie 0.0012, 0.0065, 0.0104 and 0.0090 below it. A slow approach with exponent $\omega \approx \varepsilon$, which the fit windows and the crossing grid ($\omega' \geq 0.2$) cannot resolve, would give a deficit of this sign; we have not estimated its size. These values are finite-size results.

Earlier simulations. Table 13 lists the published Monte Carlo values at our couplings (earlier estimates from exact finite-range scaling [24] are not simulations and are not listed). Published values exist at six of our couplings; at three of them (0.05, 0.1, 0.1727335) our value passed the protocol, and at $\Delta_\sigma = 0.2$ the comparison is with a finite-size value. The equilibrium values (UGA, APR, Luijten's thesis) agree with ours to within two standard deviations of the difference, computed from the published error and our σ_{stat}; σ_{sys} is not

Table 12: $d = 1$ against the resummed ε expansion (CB8, with band). MC: the quoted global fit (σ_{stat} only), or the largest-pair $\Delta_{\mathrm{eff}}(U)$ where no fit is quotable.

Δ_σ	ε	CB8	band	decision	MC kind	MC	MC − CB8
0.0125	0.950	0.7892	[0.7891, 0.7892]	X-CHECK UNAVAIL.	fit	0.78404(77)	-0.0051
0.025	0.900	0.7134	[0.7133, 0.7136]	ACCEPTED (σ_{sys}-DOM.)	fit	0.71115(49)	-0.0023
0.0375	0.850	0.6615	[0.6611, 0.6619]	X-CHECK UNAVAIL.	fit	0.65904(51)	-0.0024
0.05	0.800	0.6222	[0.6215, 0.6230]	ACCEPTED	fit	0.61987(55)	-0.0024
0.0625	0.750	0.5914	[0.5903, 0.5924]	X-CHECK UNAVAIL.	fit	0.59003(35)	-0.0014
0.075	0.700	0.5666	[0.5650, 0.5679]	ACCEPTED	fit	0.56694(37)	+0.0003
0.0875	0.650	0.5466	[0.5446, 0.5479]	ACCEPTED	fit	0.54761(32)	+0.0011
0.1	0.600	0.5302	[0.5280, 0.5316]	ACCEPTED (σ_{sys}-DOM.)	fit	0.53248(20)	+0.0022
0.1125	0.550	0.5170	[0.5146, 0.5188]	ACCEPTED (σ_{sys}-DOM.)	fit	0.52020(25)	+0.0032
0.125	0.500	0.5064	[0.5040, 0.5086]	ACCEPTED	fit	0.50710(35)	+0.0007
0.1375	0.450	0.4981	[0.4958, 0.5005]	ACCEPTED (σ_{sys}-DOM.)	fit	0.49856(30)	+0.0005
0.15	0.400	0.4917	[0.4898, 0.4942]	X-CHECK FAILED	fit	0.49325(41)	+0.0015
0.1625	0.350	0.4871	[0.4857, 0.4895]	CONDITIONAL	fit	0.48761(58)	+0.0005
0.1727335	0.309	0.4846	[0.4835, 0.4867]	ACCEPTED	fit	0.48368(52)	-0.0010
0.175	0.300	0.4842	[0.4832, 0.4862]	X-CHECK FAILED	fit	0.48460(35)	+0.0004
0.1875	0.250	0.4829	[0.4822, 0.4844]	CONDITIONAL	fit	0.48263(30)	-0.0003
0.2	0.200	0.4831	[0.4828, 0.4840]	MODEL-LIMITED	largest pair	0.4819(15)	-0.0012
0.2125	0.150	0.4848	[0.4847, 0.4852]	MODEL-LIMITED	largest pair	0.4784(15)	-0.0065
0.225	0.100	0.4881	[0.4881, 0.4882]	CONDITIONAL	fit	0.47912(68)	-0.0090
0.2375	0.050	0.4930	[0.4930, 0.4930]	MODEL-LIMITED	largest pair	0.4826(15)	-0.0104

added, because it is not a standard deviation. The largest ratio, 1.98, is at $\Delta_\sigma = 0.05$, where the difference is nine times our σ_{sys}. The nonequilibrium-relaxation values of Tomita lie below; his own estimate of η misses the exact $\eta = 2 - s$ by 2.5–3.5 quoted errors at $s = 0.6$–0.8, so his quoted errors do not cover the systematic errors of that method, and the disagreement at $\Delta_\sigma = 0.1$ is with an estimate that fails an exact check. APR's correction exponent $\omega = 0.201(11)$ at $\Delta_\sigma = 0.1727335$ agrees with our fitted $\omega = 0.195$ (table 7).

Table 13: Published $d = 1$ values ($\Delta_\varepsilon = 1 - 1/\nu$ where ν is quoted) against ours. UGA: Uzelac–Glumac–Aničić; APR: Angelini–Parisi–Ricci-Tersenghi; NER: nonequilibrium relaxation. Ours: value(σ_{stat})(σ_{sys}), or a finite-size value.

Δ_σ	source	method	published	ours	our label	published − ours
0.05	Tomita 2009 [23]	MC, NER	0.5633(362)	0.61987(55)(220)	ACCEPTED	-0.0566
0.05	UGA 2001 [19]	MC, largest cluster	0.6000(100)	0.61987(55)(220)	ACCEPTED	-0.0199
0.0625	APR 2014 [20]	MC, FSS	0.5876(13)	0.59003(35)(150)	X-CHECK UNAVAIL.	-0.0024
0.1	Luijten 1997 (thesis) [21]	MC, FSS	0.5392(106)	0.53248(20)(242)	ACCEPTED (σ_{sys}-DOM.)	+0.0067
0.1	Tomita 2009 [23]	MC, NER	0.4709(140)	0.53248(20)(242)	ACCEPTED (σ_{sys}-DOM.)	-0.0616
0.1	UGA 2001 [19]	MC, largest cluster	0.5300(200)	0.53248(20)(242)	ACCEPTED (σ_{sys}-DOM.)	-0.0025
0.15	Tomita 2009 [23]	MC, NER	0.4536(508)	0.49325(41)(306)	X-CHECK FAILED	-0.0397
0.15	UGA 2001 [19]	MC, largest cluster	0.4900(200)	0.49325(41)(306)	X-CHECK FAILED	-0.0033
0.1727335	APR 2014 [20]	MC, FSS	0.4940(140)	0.48368(52)(78)	ACCEPTED	+0.0103
0.2	Tomita 2009 [23]	MC, NER	0.4444(648)	0.4819(15) [finite size]	MODEL-LIMITED	-0.0375
0.2	UGA 2001 [19]	MC, largest cluster	0.5000(200)	0.4819(15) [finite size]	MODEL-LIMITED	+0.0181

6.6 Status of the design goals

The campaign had three design goals: ACCEPTED at every $\Delta_\sigma \in [0.0625, 0.1875]$, which was not met (7 of 12 couplings); a definite label at every outer coupling, which was met; and exact reproduction of the two-point campaign's values in its original windows, which was met. An extension to $N = 2^{20}$ (called N3) was not run.

7 $d = 2$ and $d = 3$: common protocol

7.1 Design and analysis

For each coupling, five sizes form four size-doubling pairs, the minimum for an extrapolation with a free exponent. Temperatures are centered on crossings measured in a short stage-0 campaign (frozen tables `data/2d/production_centers.csv`, `data/3d/production_centers_3d.csv`), five per size. Every (L, β) group has 12 hot/cold replicas with 10^6 measured sweeps; each dimension has 2 400 chains.

1. For each pair and $Q \in \{U_4, U_6, U_8, Y\}$ the effective dimension (4) is computed at the U_4 crossing; the bootstrap gives the covariance over pairs and observables.
2. Each series $\Delta_{\mathrm{eff}}^{Q}(L)$ is extrapolated as $\Delta_\varepsilon + c_Q L^{-\omega'}$ by generalized least squares, with ω' profiled on $[0.2, 2]$ (step 0.01); a *joint* series fits U_4 and Y with common Δ_ε and ω' and separate amplitudes (written

U4+Y).

3. A series is eligible if $\chi^2/\mathrm{dof} \le 2$, its $\Delta\chi^2 \le 1$ interval of ω' lies inside $[0.22, 1.98]$, and its σ_∞ is at most 5 times the error of the largest pair.
4. The quoted value is that of the joint series if eligible, else U_4, else the first eligible one of U_6, U_8, Y; σ_{sys} is half the range of Δ_ε over the eligible series plus half the range of the quoted series' Δ_ε over its $\Delta\chi^2 \le 1$ interval of ω'.
5. Cross-check: the extrapolated U_4 and Y must agree within $2\sigma_D$; reblocking at blocks 1000 and 4000 against 2000.

The grid floor $\omega' \ge 0.2$ excludes the correction exponent expected near the Gaussian end ($\omega \approx \varepsilon = 0.1$ at 2D $s = 1.05$ and 3D $s = 1.55$), and near the short-range end the relevant exponent $s_* - s$ is also below 0.2; there the labels partly reflect this design choice.

7.2 The statistical error of an extrapolation: amendment P1

The frozen protocols compute σ_{stat}, σ_D and the reblocking comparison by refitting each bootstrap draw at the *central* ω', i.e. conditionally on a fitted exponent. A review of the finished evaluations identified this. Amendment P1 re-profiles ω' in every draw (same grid and covariance; draws on a grid edge are kept). It leaves the eligibility gates, σ_{sys}, the choice of series and the decision order unchanged. The gate $\sigma_\infty \le 5\times$ (error of the largest pair) is therefore still evaluated with the fixed-ω' σ_∞. With the profiled one, three series would fail it: Y at 2D $s = 1.05$ and U_4 at 3D $s = 1.55$ and 1.7. 2D $s = 1.05$ would then have no eligible series, and the 3D cross-check would become unavailable at 1.55 and 1.7. The series of the accepted value at 3D $s = 1.6$ pass the gate either way.

P1 was calibrated before adoption on synthetic data whose truths are the recorded fits: for every coupling, single-series ensembles for each of U_4, U_6, U_8, Y and a joint ensemble, 2 000 synthetic data sets each with 500 synthetic draws, run through the full fit, gates and both error rules. A coupling is evaluated under P1 only if every series it uses has profiled 1σ coverage ≥ 0.60 and 2σ coverage ≥ 0.90 (with at least 200 eligible fits) and, when both U_4 and Y are eligible, the profiled cross-check's false-failure rate is ≤ 0.10; otherwise the frozen rule applies (appendix C).

Results:

- The fixed-ω' rule under-covers mainly for single series: in the strata that pass the adoption rules its 1σ coverage is 0.00–0.38 for U_4 and 0.12–0.29 for Y, against 0.59–0.67 for the joint series (pooled over all strata, the 1σ coverage is 0.287 for 45 440 eligible synthetic fits). In the same strata the profiled rule covers 0.81–1.00 for the Binder series, 0.69–0.83 for Y and 0.67–0.70 for the joint series.
- The false-failure rate of the cross-check at 3D $s = 1.6$ is 0.7 % profiled against 61 % fixed.

The calibration is *conditional on the single-power model*. We tested one misspecified family, which adds an $L^{-2\omega'}$ term that reproduces the recorded $U_4 - Y$ difference. In it the joint series at 3D $s = 1.6$ is biased by -0.0062 with profiled coverage 0, and the profiled cross-check does not detect the bias. But U_4 is then ineligible in every synthetic set, so that coupling could not have been accepted under this misspecification. The calibration therefore does not test model errors; the one misspecification studied would have blocked the label rather than biased it. Exactly one label changes under P1: 3D $s = 1.6$, from CROSS-CHECK FAILED to ACCEPTED. This was predicted before implementation, but after the frozen $z = -2.87$ was known.

8 $d = 2$

Couplings $s \in \{1.05, \dots, 1.7\}$ ($\Delta_\sigma = 0.475 \dots 0.15$), sizes $L = 64 \dots 1024$, temperature half-width $w(L) = 0.0015(L/64)^{-0.75}$. Stage 0 used $L \le 512$; the preflight confirmed every $512 \to 1024$ crossing inside its bracket. The production ran on 2026-09-20 to 22 ($L = 1024$ at 88 ms per sweep, 25–31 h per chain).

Reading. No 2D coupling has a value that passed the protocol. The robust statement is finite-size: $\Delta_{\mathrm{eff}}(U_4, 512 \to 1024) = 0.958$–$0.977$ at all eight couplings ($\sigma \approx 0.002$), moving by ≤ 0.01 over the four pairs and always below the endpoint value 1. Against the resummed expansion (table 18) the largest-pair U_4 agrees

Table 14: $d = 2$ finite-size data (block 2000): for each doubling pair the crossing inverse temperature $\beta_\times$ (error in units of 10^{-7}) and $\Delta_{\text{eff}}(U_4)$, $\Delta_{\text{eff}}(Y)$ (errors in units of 10^{-4}, 500 bootstrap draws).

		$64 \to 128$			$128 \to 256$			$256 \to 512$			$512 \to 1024$		
s	Δ_σ	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y
1.05	0.475	0.1300733(16)	0.9633(13)	0.8390(14)	0.1300298(10)	0.9661(17)	0.8658(18)	0.1300145(6)	0.9708(20)	0.8863(21)	0.1300084(4)	0.9702(25)	0.8964(23)
1.1	0.45	0.1355249(16)	0.9609(14)	0.8608(15)	0.1354801(10)	0.9639(17)	0.8912(18)	0.1354691(6)	0.9664(20)	0.9045(19)	0.1354627(4)	0.9659(23)	0.9172(23)
1.2	0.4	0.1463564(17)	0.9610(12)	0.9142(13)	0.1463343(10)	0.9576(16)	0.9281(16)	0.1463269(5)	0.9616(19)	0.9391(19)	0.1463241(3)	0.9631(22)	0.9478(22)
1.3	0.35	0.1571230(17)	0.9581(12)	0.9561(12)	0.1571157(9)	0.9586(15)	0.9683(15)	0.1571170(6)	0.9605(17)	0.9709(18)	0.1571177(3)	0.9584(20)	0.9698(20)
1.4	0.3	0.1677986(17)	0.9599(12)	1.0024(12)	0.1678175(9)	0.9590(14)	1.0032(14)	0.1678263(5)	0.9582(17)	0.9986(16)	0.1678310(3)	0.9627(20)	0.9977(19)
1.5	0.25	0.1783705(17)	0.9608(12)	1.0484(11)	0.1784129(10)	0.9615(14)	1.0423(13)	0.1784331(5)	0.9640(16)	1.0339(15)	0.1784417(3)	0.9661(18)	1.0245(17)
1.6	0.2	0.1888074(18)	0.9653(12)	1.0977(11)	0.1888717(10)	0.9652(13)	1.0885(12)	0.1889039(5)	0.9691(15)	1.0729(13)	0.1889149(3)	0.9698(18)	1.0620(15)
1.7	0.15	0.1990663(19)	0.9671(11)	1.1516(10)	0.1991553(10)	0.9699(13)	1.1381(11)	0.1991952(5)	0.9751(16)	1.1254(14)	0.1992121(3)	0.9773(17)	1.1088(14)

Table 15: $d = 2$ labels under the frozen rule and under P1 (with the rule actually applied), the quoted value, $\sigma_{\rm stat}$ under both rules, $\sigma_{\rm sys}$, and the $U_4 - Y$ cross-check (z with the fixed and the profiled σ_D). The CONDITIONAL values at $s = 1.05$ and 1.1 are single-series (Y) extrapolations 0.065 and 0.047 below the resummed expansion (table 18); they are not infinite-size estimates.

s	Δ_σ	frozen label	P1 label	rule	Δ_ε	$\sigma_{\rm stat}$ frozen	$\sigma_{\rm stat}$ P1	$\sigma_{\rm sys}$	$U_4 - Y$; z fixed / profiled
1.05	0.475	CONDITIONAL	CONDITIONAL	profiled	0.92078	0.00181	0.01773	–	–
1.1	0.45	CONDITIONAL	CONDITIONAL	profiled	0.92846	0.00145	0.01012	–	–
1.2	0.4	CONDITIONAL	CONDITIONAL	profiled	0.96202	0.00162	0.00182	–	–
1.3	0.35	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.4	0.3	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.5	0.25	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.6	0.2	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.7	0.15	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–

within 0.0015 for $s = 1.2$–1.4 and falls short by 0.005–0.02 towards both ends. At $s = 1.2$, 1.3 and 1.4 the other ways of imposing the end point move the expansion by up to 0.0005, 0.0023 and 0.0047, so there the agreement is as good as the expansion is determined. Y approaches U_4 as L doubles at every coupling, slowly and from below at $s \le 1.2$ and from above at $s \ge 1.4$; near s_* this is consistent with the slow crossover of section 2. A single power $L^{-\omega'}$ with $\omega' \in [0.2, 2]$ is either unresolved or inconsistent over $L = 64$–1024 (appendix B). The three CONDITIONAL values rest on one series without $\sigma_{\rm sys}$; at $s = 1.05$ and 1.1 that series is Y, and its extrapolations (0.921, 0.928) lie 0.065 and 0.047 below the expansion, 3.7 and 4.6 times their $\sigma_{\rm stat}$. **These two values should not be used as infinite-size estimates.** The crossing inverse temperatures $\beta_\times$ still move by up to 1.7×10^{-5} between the last two pairs; they are finite-size values.

9 $d = 3$

Couplings $s \in \{1.55, \ldots, 1.95\}$ ($\Delta_\sigma = 0.725 \ldots 0.525$), sizes $L = 8 \ldots 128$, half-width $w(L) = 10^{-3}(L/16)^{-1}$. An $L = 128$ chain runs at 105 ms per sweep alone but at about 250 ms when four chains share a 4-core CCX (a group of cores sharing a 16 MB L3 cache); the production therefore pinned one $L = 128$ chain per CCX (32 per 128-core node, ≈ 30 h per chain). The preflight gate and its amendment are described in section 5. The production ran on 2026-09-21 to 23.

The value at $s = 1.6$. At $s = 1.6$ ($\Delta_\sigma = 0.70$) all five series are eligible: U_4 1.4726, U_6 1.4720, U_8 1.4709, Y 1.4750, U4+Y 1.4731. The quoted value comes from the joint series: $\Delta_\varepsilon = 1.47314 \pm 0.00118\,(\text{stat}) \pm 0.00230\,(\text{sys})$ ($\omega' = 1.24$, $\chi^2 = 2.38/4$); the range over the U_4 and Y series alone is 1.4726–1.4750. It passed the protocol only under amendment P1, and the precision suffix remains ($\sigma = 0.0031 > 0.003$ at $64 \to 128$). Three facts limit what the label shows. The cross-check has little power here: the extrapolated $Y - U_4$ changes sign between $s = 1.55$ and 1.7 (-0.033, $+0.002$, $+0.068$), so their agreement at 1.6 is partly a crossing of two biased estimates, and the profiled $\sigma_D \approx 0.0055$ cannot test a $\sigma_{\rm stat}$ of 0.0012. The fitted $\omega' = 1.24$ is far above the leading exponents expected at $s = 1.6$ ($\omega \approx \varepsilon \approx 0.2$; $s_* - s \approx 0.36$), so the single power is an effective description. And the label changed under an amendment adopted after the frozen result was known. The value is *consistent* with the resummed expansion (1.4721) and with the finite-size $\Delta_{\rm eff}(U_4, 64 \to 128) = 1.4715(31)$, which is the main reason to trust it.

Reading. The finite-size U_4 exponent at $64 \to 128$ decreases from 1.473 ($s = 1.55$) to 1.449 ($s = 1.95$), between the Gaussian value 1.5 and the short-range value 1.4126. Y does not converge towards U_4 at $s \ge 1.7$: it lies above U_4 from $16 \to 32$ on and keeps rising (1.53–1.68 at $64 \to 128$). Against the resummed expansion (table 18), the extrapolated values at $s = 1.7$–1.9 lie 0.022–0.033 above. At 1.7 and 1.85 the label is CROSS-CHECK FAILED, and the differences ($+0.022$, $+0.030$) are smaller than $\sigma_{\rm sys}$ (0.052, 0.091), which is large because the series disagree; at 1.9 (CROSS-CHECK UNAVAILABLE, $\sigma_{\rm sys} = 0.0045$) the difference is 4.6 times $\sigma_{\rm stat}$. Near s_* the U_4 series therefore sits high, and the Y–U_4 split matches in size the slow crossover of section 2 at the largest pair, although its growth with L up to 128 is not reproduced by that leading-order picture.

Table 16: $d = 3$ finite-size data (block 2000), as table 14.

		$8 \to 16$			$16 \to 32$			$32 \to 64$			$64 \to 128$		
s	Δ_σ	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y
1.55	0.725	0.0894786(14)	1.4427(11)	1.2535(12)	0.0893412(7)	1.4583(15)	1.3545(16)	0.0893146(4)	1.4677(22)	1.4028(23)	0.0893087(2)	1.4729(30)	1.4269(30)
1.6	0.7	0.0916831(15)	1.4398(11)	1.2914(13)	0.0915467(8)	1.4596(15)	1.3970(17)	0.0915234(4)	1.4661(21)	1.4382(23)	0.0915178(2)	1.4715(31)	1.4628(30)
1.7	0.65	0.0959967(15)	1.4411(11)	1.3761(12)	0.0958738(8)	1.4528(16)	1.4673(17)	0.0958528(3)	1.4612(22)	1.5085(21)	0.0958497(2)	1.4650(29)	1.5263(29)
1.75	0.625	0.0981147(16)	1.4377(11)	1.4115(12)	0.0979933(8)	1.4535(15)	1.5071(16)	0.0979751(3)	1.4536(21)	1.5379(21)	0.0979722(2)	1.4615(27)	1.5596(27)
1.8	0.6	0.1002020(16)	1.4372(11)	1.4513(12)	0.1000841(8)	1.4505(14)	1.5417(15)	0.1000677(3)	1.4563(20)	1.5795(20)	0.1000665(2)	1.4515(27)	1.5818(27)
1.85	0.575	0.1022550(15)	1.4364(11)	1.4929(12)	0.1021468(8)	1.4490(15)	1.5776(16)	0.1021325(3)	1.4538(20)	1.6113(20)	0.1021317(2)	1.4569(27)	1.6221(27)
1.9	0.55	0.1042865(17)	1.4370(10)	1.5325(12)	0.1041795(8)	1.4458(14)	1.6148(15)	0.1041684(4)	1.4526(19)	1.6464(20)	0.1041687(2)	1.4511(26)	1.6488(25)
1.95	0.525	0.1062845(16)	1.4320(11)	1.5694(11)	0.1061853(7)	1.4466(15)	1.6539(15)	0.1061755(4)	1.4481(20)	1.6771(21)	0.1061771(2)	1.4493(28)	1.6790(27)

Table 17: $d = 3$ labels, as table 15. At $s = 1.7$ and 1.85 P1 is not calibrated (rule "fixed", appendix C): the profiled z is shown for information only, and the label uses the fixed z. The extrapolated values at $s = 1.7$–1.9 lie 0.022–0.033 above the resummed expansion (table 18).

s	Δ_σ	frozen label	P1 label	rule	Δ_ε	σ_{stat} frozen	σ_{stat} P1	σ_{sys}	$U_4 - Y$; z fixed / profiled
1.55	0.725	X-CHECK FAILED [σ]	X-CHECK FAILED [σ]	profiled	1.48118	0.00165	0.01632	0.02716	+0.0326; +29.48 / +2.49
1.6	0.7	X-CHECK FAILED [σ]	ACCEPTED [σ]	profiled	1.47314	0.00112	0.00118	0.00230	-0.0025; -2.87 / -0.45
1.7	0.65	X-CHECK FAILED	X-CHECK FAILED	fixed	1.47375	0.00174	0.00174	0.05237	-0.0676; -56.57 / -4.90
1.75	0.625	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.8	0.6	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.85	0.575	X-CHECK FAILED	X-CHECK FAILED	fixed	1.45830	0.00103	0.00103	0.09064	-0.1723; -232.64 / -28.05
1.9	0.55	X-CHECK UNAVAIL.	X-CHECK UNAVAIL.	profiled	1.45449	0.00113	0.00720	0.00452	–
1.95	0.525	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–

9.1 $d = 2, 3$ against theory and earlier work

Table 18 compares with CB8 and figure 2 shows the effective dimensions. Table 19 lists published values inside the window. In $d = 2$ the Monte Carlo values (APR, Luijten's thesis, Picco) have errors 0.02–0.034 and differ from our largest-pair $\Delta_{\text{eff}}(U_4)$ by at most two of their errors; they do not distinguish our finite-size values from CB8. In $d = 3$, at $s = 1.6$, Luijten's $L \leq 64$ values (1.477(16), 1.500(20)) and the functional-RG value (1.475) lie 0.002–0.027 above our accepted value, within 1.3 of their errors. Nearer s_* the bootstrap values agree with CB8 to about 0.01 or lie slightly below it, while the functional-RG values are close to our largest-pair $\Delta_{\text{eff}}(U_4)$, up to 0.03 above CB8 (both read off figures). Published values at the ends of the window, $s = d/2$ in $d = 3$ [22] and $s = s_*$ in $d = 2$ [26], are in the reference files but outside our grid.

Table 18: $d = 2$ (top) and $d = 3$ (bottom) against the resummed ε expansion of the accompanying article (CB8, with band; `data/reference/theory_2d3d_CB8.csv`). CB8 imposes the short-range end: in $d = 2$ the value 1, zero slope and the curvature, in $d = 3$ the value and the slope 0.27315, itself an estimate (range 0.269–0.276). The band is a sensitivity range over prescriptions; it excludes the way the end point is imposed, which in $d = 2$ moves the curve by up to 0.0005, 0.0023, 0.0047, 0.0057, 0.0070 and 0.0055 at $s = 1.2, \ldots, 1.7$, and the 3D slope range. Last two columns: quoted value minus CB8, and largest-pair $\Delta_{\text{eff}}(U_4)$ minus CB8.

s	P1 label	Δ_ε	σ_{stat}	σ_{sys}	$\Delta_{\text{eff}}(U_4)$ largest pair	CB8	band	Δ_ε−CB8	U_4−CB8
1.05	CONDITIONAL	0.92078	0.01773	–	0.9702(25)	0.9858	[0.9858, 0.9858]	-0.0651	-0.0157
1.1	CONDITIONAL	0.92846	0.01012	–	0.9659(23)	0.9754	[0.9753, 0.9754]	-0.0470	-0.0095
1.2	CONDITIONAL	0.96202	0.00182	–	0.9631(22)	0.9626	[0.9609, 0.9628]	-0.0005	+0.0006
1.3	MODEL-LIMITED	–	–	–	0.9584(20)	0.9581	[0.9546, 0.9589]	–	+0.0003
1.4	MODEL-LIMITED	–	–	–	0.9627(20)	0.9612	[0.9572, 0.9624]	–	+0.0015
1.5	MODEL-LIMITED	–	–	–	0.9661(18)	0.9711	[0.9684, 0.9722]	–	-0.0050
1.6	MODEL-LIMITED	–	–	–	0.9698(18)	0.9854	[0.9845, 0.9858]	–	-0.0157
1.7	MODEL-LIMITED	–	–	–	0.9773(17)	0.9978	[0.9977, 0.9978]	–	-0.0205
1.55	X-CHECK FAILED [σ]	1.48118	0.01632	0.02716	1.4729(30)	1.4849	[1.4849, 1.4849]	-0.0037	-0.0119
1.6	ACCEPTED [σ]	1.47314	0.00118	0.00230	1.4715(31)	1.4721	[1.4721, 1.4721]	+0.0010	-0.0006
1.7	X-CHECK FAILED	1.47375	0.00174	0.05237	1.4650(29)	1.4515	[1.4515, 1.4516]	+0.0222	+0.0134
1.75	MODEL-LIMITED	–	–	–	1.4615(27)	1.4430	[1.4429, 1.4431]	–	+0.0185
1.8	MODEL-LIMITED	–	–	–	1.4515(27)	1.4352	[1.4352, 1.4353]	–	+0.0163
1.85	X-CHECK FAILED	1.45830	0.00103	0.09064	1.4569(27)	1.4281	[1.4280, 1.4281]	+0.0302	+0.0288
1.9	X-CHECK UNAVAIL.	1.45449	0.00720	0.00452	1.4511(26)	1.4212	[1.4212, 1.4213]	+0.0333	+0.0299
1.95	MODEL-LIMITED	–	–	–	1.4493(28)	1.4145	[1.4145, 1.4145]	–	+0.0348

Table 19: Published values inside the long-range window, $d = 2$ (top) and $d = 3$ (bottom), with our largest-pair $\Delta_{\mathrm{eff}}(U_4)$ and CB8 where the coupling is on our grid. Bootstrap and functional-RG values are read off figures.

s	source	method	$L_{\max}$	Δ_ε	our $\Delta_{\mathrm{eff}}(U_4)$	CB8
1.2	APR 2014 [20]	Monte Carlo	–	0.976(34)	0.9631(22)	0.9626
1.2	Luijten thesis 1997 [21], Tab. 5.7	Monte Carlo	400	1.000(20)	0.9631(22)	0.9626
1.2	Luijten thesis 1997 [21], Tab. 5.8	Monte Carlo	400	0.980(30)	0.9631(22)	0.9626
1.30906	APR 2014 [20]	Monte Carlo	–	0.986(33)	–	–
1.4	Luijten thesis 1997 [21], Tab. 5.7	Monte Carlo	400	1.000(20)	0.9627(20)	0.9612
1.4	Luijten thesis 1997 [21], Tab. 5.8	Monte Carlo	400	0.960(30)	0.9627(20)	0.9612
1.6	APR 2014 [20]	Monte Carlo	–	1.004(33)	0.9698(18)	0.9854
1.6	Luijten thesis 1997 [21], Tab. 5.7	Monte Carlo	400	1.020(30)	0.9698(18)	0.9854
1.6	Luijten thesis 1997 [21], Tab. 5.8	Monte Carlo	400	0.970(20)	0.9698(18)	0.9854
1.6	Picco 2012 [25]	Monte Carlo	5120	0.958(22)	0.9698(18)	0.9854
1.57	Behan 2019 [27]	bootstrap (figure-read)	–	1.484(10)	–	–
1.59274	Behan et al. 2024 [28]	bootstrap (figure-read)	–	1.470(5)	–	–
1.6	Defenu et al. 2015 [29]	functional RG (figure-read)	–	1.475(5)	1.4715(31)	1.4721
1.6	Luijten thesis 1997 [21], Tab. 5.12	Monte Carlo	64	1.477(16)	1.4715(31)	1.4721
1.6	Luijten thesis 1997 [21], Sec. 5.5.2	Monte Carlo	64	1.500(20)	1.4715(31)	1.4721
1.615	Behan 2019 [27]	bootstrap (figure-read)	–	1.468(10)	–	–
1.66	Behan 2019 [27]	bootstrap (figure-read)	–	1.461(10)	–	–
1.68548	Behan et al. 2024 [28]	bootstrap (figure-read)	–	1.450(5)	–	–
1.7	Defenu et al. 2015 [29]	functional RG (figure-read)	–	1.459(5)	1.4650(29)	1.4515
1.704	Behan 2019 [27]	bootstrap (figure-read)	–	1.451(10)	–	–
1.77822	Behan et al. 2024 [28]	bootstrap (figure-read)	–	1.432(5)	–	–
1.8	Defenu et al. 2015 [29]	functional RG (figure-read)	–	1.452(5)	1.4515(27)	1.4352
1.8	Luijten thesis 1997 [21], Tab. 5.12	Monte Carlo	64	1.460(40)	1.4515(27)	1.4352
1.87096	Behan et al. 2024 [28]	bootstrap (figure-read)	–	1.415(5)	–	–
1.9	Defenu et al. 2015 [29]	functional RG (figure-read)	–	1.453(5)	1.4511(26)	1.4212

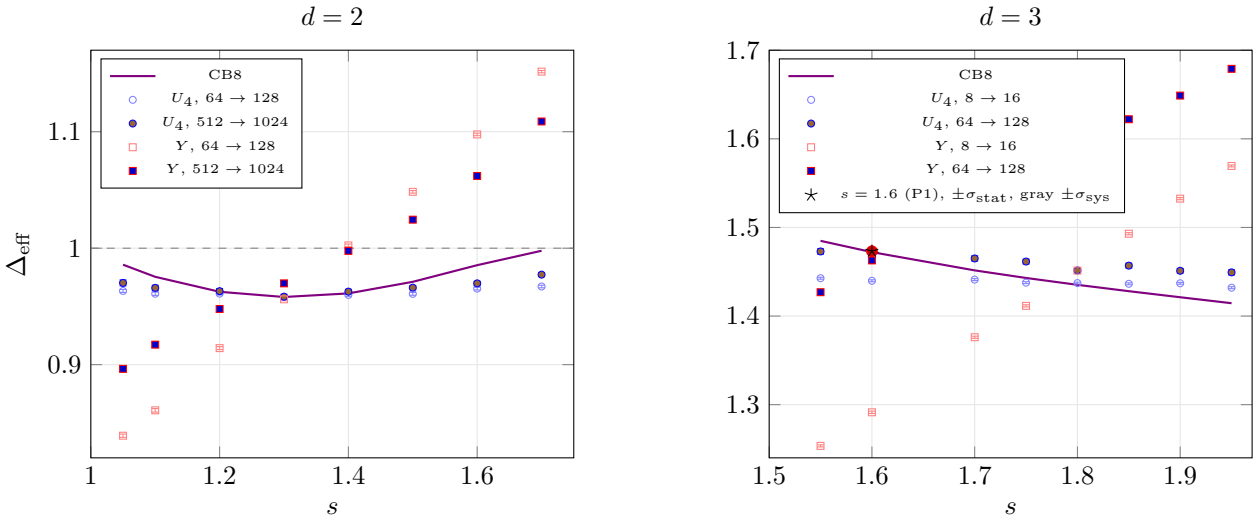


Figure 2: Effective dimensions at the smallest and the largest doubling pair, with the resummed ε expansion (CB8). Left, $d = 2$ (dashed: $\Delta_\varepsilon = 1$ at both ends of the window). Right, $d = 3$; the star is the value at $s = 1.6$ that passed the protocol under P1, with σ_{stat} (bar) and σ_{sys} (gray band) shown separately. At the largest pair the Y points (red) and the U_4 points (blue) agree in the middle of the window and differ near its ends. In $d = 2$ the difference shrinks as L doubles at every coupling; in $d = 3$ it shrinks at $s = 1.55$ and 1.6 and grows at $s \geq 1.7$.

10 Synthesis

1. In $d = 1$ nine couplings pass the protocol, with $\sigma_{\text{stat}} = (2\text{–}5.5) \times 10^{-4}$; they differ from the resummed ε expansion by -0.0024 to $+0.0032$. σ_{sys} is a sensitivity scale. The audit supports the values at 0.1, 0.1375 and 0.1727335, which are stable under stronger size cuts. It flags 0.025 (a model outside the family) and 0.175 and 0.225 (stronger size cuts). The labels at 0.1625 and 0.1727335 are platform-sensitive; two quoted fits have $P_{\chi^2} \approx 5 \times 10^{-5}$; for $\Delta_\sigma \geq 0.0875$ the global fits lie above the crossing extrapolations by a common 0.0010(3).
2. In $d = 2, 3$ the finite-size data are precise ($\sigma(\Delta_{\text{eff}}^{U_4}) \approx 0.0017$–$0.0031$ at the largest pair) and the bottleneck is the extrapolation: four doubling pairs over a factor 16 in L do not determine the correction sector, especially near the ends of the window, where the expected correction exponents are below the grid floor. More statistics at the same sizes would sharpen, not resolve, this.
3. One 3D value passes, at $s = 1.6$, under the post hoc amendment P1, and agrees with the expansion.
4. With a fitted correction exponent, a bootstrap at fixed exponent under-covers badly for single series; re-profiling the exponent in every draw restores coverage, conditionally on the extrapolation model.
5. The published equilibrium Monte Carlo values agree with ours within about two of their errors (in $d = 1$ computed with σ_{stat}, not σ_{sys}); quantitative statements should use σ_{stat} and σ_{sys} separately.

11 Open issues and next steps

1. **1D uncertainties:** a pre-declared broader model family (a U background, two correction powers, stronger size cuts), asymmetric model ranges, a converged ω' minimization and a ladder rule that does not stop on a flagged lower-degree fit, followed by a full rerun of the selection. Priority couplings: 0.025, 0.175, 0.225. A label-stability study (seeds, ladder rules, grid steps) belongs here.
2. **1D goodness of fit:** a p-value gate or a scale factor instead of $\chi^2/\text{dof} \leq 1.5$.
3. **Near the Gaussian end** in all dimensions: a crossover form $D(L) = 1 + CL^{-\varepsilon}$ with free amplitudes, or ω fixed at its perturbative value, instead of a grid with floor 0.2.
4. **Near the short-range end** ($d = 2, 3$): the crossover flow of section 2 as a model for the Y–U_4 split; measuring the effective η at the crossings would test it.
5. **Larger lattices**, only where they answer a question: 2D $L = 2048$ at $s = 1.2$ and 1.5; 3D $L = 256$ at $s = 1.9$ (about 1.5×10^4 core-hours), to tell an ordinary correction power from the slow short-range crossover by whether the Y–U_4 split grows or shrinks at $128 \to 256$.
6. **Critical inverse temperatures:** extrapolate $\beta_\times(L)$ with errors from the existing bootstrap draws.
7. **Uninspected code:** the last fixes to the P1 implementation and to the 1D tooling that records which campaigns make up each coupling (the registry).

12 Reproduction

12.1 The released folder

This note is released with the code and the records in the `MC/` folder of the public repository (code under the MIT license, everything else under CC BY 4.0, except the delivered 2D/3D engine, which keeps its own MIT license); paths below are relative to `MC/`. `README.md` states what is computed and its status; `REPRODUCE.md` gives the steps in order (environment, tests, replay, smoke run, production, merge, checks) and the decisions that a reproduction must not take on its own.

`code/1d/` 1D engines (`src/`, `src_1d/`), campaign tools, analysis (`analyze.py`, `merge_summaries.py`, `equilibration.py`, `union_check.py`), fitting (`select_model_v3.py`, `select_model.py`, `fit_fss.py`, `quotients.py`), the 20-point report (`scan_v3_report.py`) and tests.

`code/2d3d_engine/` the $d = 2, 3$ sampler and its analysis (`pack_histograms.py`, `multihist.py`, `precision_analysis.py`) and tests.

`code/2d/, code/3d/` the evaluators (frozen rule and P1) and planners.

`code/reanalysis_p1/` `calibrate_p1.py` and the tests of amendment P1.

`code/package_tools/` workspace assembly, the output comparison and the generator of this note's tables.

`protocols/` the analysis protocols, the amendment and their history (`PROTOCOL_HISTORY.md`; appendix D).

`data/` the merged production records (`DATA.md`): 1D per-coupling selected fit runs, snapshot manifests, completion evidence and registries; 2D/3D analysis records at three block sizes, campaign manifests, histogram indices and the pinned production evaluations; reference values (`data/reference/`).

`results/` expected outputs: the P1 evaluations, the P1 calibration (an input of the P1 evaluation), the 1D tables and figures, and the cited diagnostics (cross-platform refits, the robust-ladder study with its modified fitter, the cross-check study).

`slurm/` submission templates with placeholders for account, partition and paths.

The evaluators bind themselves, the calibration script and the protocols by SHA-256 and locate their inputs relative to the layout they ran in; the make targets therefore assemble that layout in a hidden `.work/` directory (code copied byte for byte, data linked), so every check passes without any edit. The production records are released byte-identical, with one exception: 25 records (from the $d = 1$ couplings $\Delta_\sigma = 0.225$ and 0.2375, and the P1 calibration) named a personal machine and were scrubbed after the results; only host labels, a home directory and file names were replaced, and the hashes inside them re-pinned. `RE-PINNED.json` lists their original and published SHA-256; the comparison maps one to the other. The other records name the cluster paths, user and compute nodes on which they were written.

12.2 How to run

Requirements: Python with NumPy and SciPy (production: Python 3.8.5, NumPy 1.19.2, SciPy 1.5.2 for the 1D fits; Python 3.8.8, NumPy 1.20.1 for the P1 calibration and evaluations; also tested with 3.12.3, 1.26.4, 1.11.4), pytest, matplotlib, a C++17 compiler and GNU make for the engines, and pdflatex for this note.

target	runs	reproduces
`test`	builds the engines; 1D protocol tests; the tests of amendment P1	the verification claims of section 7.2
`replay`	everything below that needs only the records in git, then the comparison with `results/`	every table and figure of this note except those needing `fetch-data`
`smoke`	engine validation: a tiny 1D campaign with bitwise resume; exact enumeration and bitwise restart in $d = 2, 3$	that the samplers are exact on small systems
`eval-fixed`, `eval-p1`	the 2D/3D evaluators under the frozen rule and under P1	tables 15, 17; appendix B
`calibration`	`calibrate_p1.py` (1 780 tasks, ≈ 7 min on 16 cores)	appendix C and the coverage numbers
`scan1d`, `tables1d`	`scan_v3_report.py`, `make_tables_and_figure.py`	tables 5, 6, appendix A
`refit1d DS=...`	`select_model_v3.py` on the released snapshot of one coupling (6–156 s on 20 cores); needs `fetch-data`	the fits, crossing series and label of that coupling; section 6.4.1
`note-tables`	`make_note_tables.py -check`	every generated line of this note
`fetch-data`, `build`, `note`	downloads the data not in git; the engines; this note (three passes)	–

12.3 Verified reproduction

On 7 October 2026 the folder, in its layout before the public release, was assembled afresh on the workstation and every target was run through Slurm. The engines built and all tests passed. The 2D/3D evaluations reproduced the pinned production evaluations in all per-point fields (maximum relative difference 3×10^{-11}), and so did the P1 evaluations and the calibration regenerated from scratch (2×10^{-11}). The 1D cross-check study, the 20-point 1D table and the 1D report tables were reproduced, and the 1D fitting stage reproduced 18 of 20 labels and all values (section 6.4.1). The comparison reports those two labels as known to be platform-sensitive.

On 13 October 2026 the folder in its public layout was exported afresh and tested through Slurm on two platforms: a personal desktop (Python 3.12.3, NumPy 1.26.4, SciPy 1.11.4) and an Expanse debug node (Python 3.8.8, NumPy 1.20.1, SciPy 1.6.2, GCC 10.2.0). On both, the engines build, all tests pass (3.1 and 7.5 min), and `make replay` (14 and 43 s) reports 9 checks PASS, none FAIL and one NOT RUN, the regenerated calibration, which needs `make calibration`; `note-tables` finds all 355 generated lines in this note. The cited 1D refits need `fetch-data` and were not part of this test.

12.4 What is not in git

`make fetch-data` downloads two archives from the project's Zenodo data record, doi:10.5281/zenodo.23466766, and checks their SHA-256 (`data/HOSTED.json`). `refit1d` needs the first: the 1D block archives of the selected snapshots (1.43 GB) and their merged summaries (25 MB). The second holds the per-degree fits of the selected runs (140 MB), for inspection; no target needs them. Regenerating the records from scratch needs the raw data: the 1D trajectories (0.655 TB) and the 2D/3D trees of chains, packed histograms and analyses (257.5 and 237.2 GiB). They are not hosted; they are available from the author on request. The records were produced from these by `analyze.py` and `merge_summaries.py` ($d = 1$) and by `pack_histograms.py` and `precision_analysis.py` ($d = 2, 3$).

A $d = 1$ frozen window parameters

The window parameters $(\beta_c^{\mathrm{pre}}, \Delta_\varepsilon^{\mathrm{pre}})$ were frozen per coupling from earlier fits of the stage A/B data (section 5); h' is the N2 temperature half-width.

Table 20: Frozen $(\beta_c^{\mathrm{pre}}, \Delta_\varepsilon^{\mathrm{pre}})$ per coupling and N2 half-widths h' (units 10^{-4}) at $N = 2^{16}$ and 2^{19}.

Δ_σ	s	β_c^{pre}	$\Delta_\varepsilon^{\mathrm{pre}}$	$h'(2^{16})$	$h'(2^{19})$	N2
0.0125	0.975	0.596534	0.7910	14.67	9.50	no
0.025	0.95	0.558588	0.7179	10.83	6.02	no
0.0375	0.925	0.527150	0.6575	8.42	4.13	no
0.05	0.9	0.499660	0.6234	7.31	3.34	no
0.0625	0.875	0.474858	0.5898	6.35	2.71	yes
0.075	0.85	0.452099	0.5674	5.79	2.35	no
0.0875	0.825	0.430939	0.5462	5.30	2.06	no
0.1	0.8	0.411095	0.5292	4.94	1.86	yes
0.1125	0.775	0.392343	0.5191	4.74	1.74	yes
0.125	0.75	0.374529	0.5074	4.51	1.62	yes
0.1375	0.725	0.357528	0.4991	4.36	1.54	yes
0.15	0.7	0.341247	0.4930	4.25	1.48	yes
0.1625	0.675	0.325592	0.4863	4.13	1.42	yes
0.1727335	0.654533	0.313200	0.4845	4.10	1.40	yes
0.175	0.65	0.310504	0.4843	4.10	1.40	yes
0.1875	0.625	0.295925	0.4814	4.05	1.38	yes
0.2	0.6	0.281805	0.4808	4.04	1.37	yes
0.2125	0.575	0.268101	0.4775	3.98	1.34	yes
0.225	0.55	0.254785	0.4802	4.03	1.37	yes
0.2375	0.525	0.241813	0.4809	4.04	1.37	yes

B $d = 2, 3$ extrapolation series

For every coupling and series: Δ_ε of the extrapolation, ω' and its $\Delta\chi^2 \leq 1$ interval, χ^2/dof, σ with ω' fixed and profiled per draw (P1), eligibility.

Table 21: $d = 2$ series (P1 evaluation; central values identical to the frozen evaluation).

s	series	Δ_ε	ω' [interval]	χ^2/dof	σ fixed	σ profiled	eligible
1.05	U4	0.9750	0.56 [0.20, 2.00]	0.51/1	0.0020	0.0083	no: ω' edge
	U6	0.9719	0.84 [0.20, 2.00]	0.29/1	0.0014	0.0080	no: ω' edge
	U8	0.9700	0.96 [0.20, 2.00]	0.14/1	0.0011	0.0087	no: ω' edge
	Y	0.9208	0.60 [0.43, 0.76]	0.44/1	0.0018	0.0177	yes
	U4+Y	0.9840	0.26 [0.26, 0.27]	8.33/4	0.0037	0.0042	no: χ^2
1.1	U4	0.9670	1.17 [0.20, 2.00]	0.11/1	0.0011	0.0060	no: ω' edge
	U6	0.9663	1.33 [0.20, 2.00]	0.10/1	0.0010	0.0060	no: ω' edge
	U8	0.9651	1.41 [0.39, 2.00]	0.03/1	0.0009	0.0067	no: ω' edge
	Y	0.9285	0.80 [0.61, 1.00]	1.10/1	0.0015	0.0101	yes
	U4+Y	0.9720	0.35 [0.35, 0.36]	15.28/4	0.0028	0.0031	no: χ^2
1.2	U4	0.9654	0.20 [0.20, 2.00]	3.57/1	0.0045	0.0043	no: χ^2
	U6	0.9657	0.20 [0.20, 1.44]	3.66/1	0.0044	0.0042	no: χ^2
	U8	0.9686	0.20 [0.20, 0.78]	2.73/1	0.0040	0.0044	no: χ^2
	Y	0.9806	0.34 [0.20, 0.60]	0.00/1	0.0027	0.0237	no: ω' edge
	U4+Y	0.9620	0.55 [0.53, 0.57]	5.85/4	0.0016	0.0018	yes
1.3	U4	0.9595	2.00 [0.20, 2.00]	0.49/1	0.0007	0.0030	no: ω' edge
	U6	0.9595	2.00 [0.20, 2.00]	0.56/1	0.0007	0.0029	no: ω' edge
	U8	0.9595	2.00 [0.20, 2.00]	0.70/1	0.0007	0.0036	no: ω' edge
	Y	0.9714	2.00 [1.58, 2.00]	0.72/1	0.0007	0.0012	no: ω' edge
	U4+Y	0.9832	0.20 [0.20, 0.20]	201.09/4	0.0042	0.0042	no: χ^2
1.4	U4	0.9624	0.20 [0.20, 2.00]	2.38/1	0.0042	0.0035	no: χ^2
	U6	0.9643	0.20 [0.20, 2.00]	2.30/1	0.0041	0.0038	no: χ^2
	U8	0.9689	0.20 [0.20, 1.32]	1.83/1	0.0039	0.0044	no: ω' edge
	Y	0.9868	0.20 [0.20, 0.76]	1.85/1	0.0042	0.0051	no: ω' edge
	U4+Y	0.9730	0.20 [0.20, 0.20]	93.97/4	0.0040	0.0040	no: χ^2
1.5	U4	0.9759	0.20 [0.20, 0.98]	0.48/1	0.0042	0.0055	no: ω' edge
	U6	0.9777	0.20 [0.20, 0.92]	0.46/1	0.0041	0.0056	no: ω' edge
	U8	0.9819	0.20 [0.20, 0.91]	0.29/1	0.0039	0.0064	no: ω' edge
	Y	0.9810	0.20 [0.20, 0.27]	3.15/1	0.0038	0.0055	no: χ^2
	U4+Y	0.9792	0.20 [0.20, 0.20]	18.48/4	0.0038	0.0038	no: χ^2
1.6	U4	0.9799	0.20 [0.20, 0.97]	1.09/1	0.0041	0.0053	no: ω' edge
	U6	0.9822	0.20 [0.20, 1.10]	0.59/1	0.0041	0.0057	no: ω' edge
	U8	0.9839	0.25 [0.20, 1.47]	0.18/1	0.0031	0.0063	no: ω' edge
	Y	0.9919	0.20 [0.20, 0.23]	6.34/1	0.0035	0.0050	no: χ^2
	U4+Y	0.9915	0.20 [0.20, 0.20]	56.29/4	0.0035	0.0035	no: χ^2
1.7	U4	0.9984	0.20 [0.20, 0.66]	0.51/1	0.0039	0.0087	no: ω' edge
	U6	1.0012	0.20 [0.20, 0.75]	0.50/1	0.0038	0.0093	no: ω' edge
	U8	0.9935	0.36 [0.20, 0.97]	0.57/1	0.0021	0.0097	no: ω' edge
	Y	1.0311	0.20 [0.20, 0.23]	5.82/1	0.0034	0.0038	no: χ^2
	U4+Y	1.0260	0.20 [0.20, 0.20]	225.45/4	0.0034	0.0034	no: χ^2

Table 22: $d = 3$ series (P1 evaluation).

s	series	Δ_ε	ω' [interval]	χ^2/dof	σ fixed	σ profiled	eligible
1.55	U4	1.4812	0.75 [0.44, 1.11]	0.00/1	0.0017	0.0163	yes
	U6	1.4809	0.79 [0.51, 1.10]	0.02/1	0.0016	0.0139	yes
	U8	1.4790	0.92 [0.71, 1.15]	0.02/1	0.0013	0.0075	yes
	Y	1.4486	1.05 [0.99, 1.11]	0.02/1	0.0014	0.0058	yes
	U4+Y	1.4789	0.78 [0.78, 0.79]	44.76/4	0.0015	0.0017	no: χ^2
1.6	U4	1.4726	1.29 [0.93, 1.74]	0.28/1	0.0011	0.0072	yes
	U6	1.4720	1.35 [1.02, 1.75]	0.29/1	0.0011	0.0062	yes
	U8	1.4709	1.39 [1.14, 1.67]	0.37/1	0.0010	0.0040	yes
	Y	1.4750	1.21 [1.14, 1.28]	1.51/1	0.0013	0.0048	yes
	U4+Y	1.4731	1.24 [1.23, 1.25]	2.38/4	0.0011	0.0012	yes
1.7	U4	1.4737	0.66 [0.28, 1.09]	0.07/1	0.0017	0.0168	yes
	U6	1.4724	0.74 [0.40, 1.14]	0.03/1	0.0015	0.0157	yes
	U8	1.4700	0.92 [0.67, 1.20]	0.02/1	0.0012	0.0082	yes
	Y	1.5414	1.16 [1.08, 1.24]	0.02/1	0.0012	0.0050	yes
	U4+Y	1.4862	2.00 [2.00, 2.00]	7052.05/4	0.0008	0.0008	no: χ^2
1.75	U4	1.4587	1.69 [1.03, 2.00]	2.89/1	0.0009	0.0062	no: χ^2
	U6	1.4583	1.75 [1.17, 2.00]	3.47/1	0.0009	0.0047	no: χ^2
	U8	1.4581	1.65 [1.28, 2.00]	3.84/1	0.0009	0.0033	no: χ^2
	Y	1.5654	1.35 [1.27, 1.44]	4.12/1	0.0011	0.0041	no: χ^2
	U4+Y	1.6546	0.20 [0.20, 0.20]	10550.66/4	0.0050	0.0050	no: χ^2
1.8	U4	1.4553	2.00 [1.53, 2.00]	2.02/1	0.0009	0.0016	no: χ^2
	U6	1.4551	2.00 [1.59, 2.00]	2.37/1	0.0009	0.0015	no: χ^2
	U8	1.4542	2.00 [1.62, 2.00]	2.76/1	0.0008	0.0016	no: χ^2
	Y	1.5918	1.55 [1.46, 1.64]	5.46/1	0.0010	0.0030	no: χ^2
	U4+Y	1.6897	0.20 [0.20, 0.20]	11986.04/4	0.0050	0.0050	no: χ^2
1.85	U4	1.4583	1.20 [0.71, 1.85]	0.04/1	0.0010	0.0075	yes
	U6	1.4589	1.17 [0.76, 1.68]	0.01/1	0.0010	0.0062	yes
	U8	1.4592	1.19 [0.92, 1.50]	0.00/1	0.0009	0.0045	yes

$d = 3$ *series, continued*

s	series	Δ_ε	ω' [interval]	χ^2/dof	σ fixed	σ profiled	eligible
	Y	1.6306	1.39 [1.30, 1.48]	0.20/1	0.0010	0.0034	yes
	U4+Y	1.6685	0.20 [0.20, 0.20]	14600.83/4	0.0045	0.0045	no: χ^2
1.9	U4	1.4545	1.15 [0.63, 1.87]	0.89/1	0.0011	0.0072	yes
	U6	1.4544	1.23 [0.78, 1.81]	0.83/1	0.0011	0.0057	yes
	U8	1.4546	1.29 [1.00, 1.64]	0.83/1	0.0009	0.0034	yes
	Y	1.6563	1.63 [1.53, 1.74]	2.91/1	0.0009	0.0026	no: χ^2
	U4+Y	1.6896	0.20 [0.20, 0.20]	13764.62/4	0.0046	0.0046	no: χ^2
1.95	U4	1.4500	2.00 [1.56, 2.00]	0.49/1	0.0008	0.0014	no: ω' edge
	U6	1.4498	2.00 [1.61, 2.00]	0.39/1	0.0008	0.0013	no: ω' edge
	U8	1.4490	2.00 [1.64, 2.00]	0.05/1	0.0007	0.0014	no: ω' edge
	Y	1.6827	2.00 [1.88, 2.00]	0.70/1	0.0008	0.0014	no: ω' edge
	U4+Y	1.6808	0.20 [0.20, 0.20]	19130.26/4	0.0048	0.0048	no: χ^2

C P1 calibration strata

Each cell: eligible synthetic fits: profiled $1\sigma/2\sigma$ coverage (fixed-ω' 1σ coverage), status (P pass, F fail, nc not calibrated, i.e. fewer than 200 eligible fits). Last column: synthetic sets with U_4 and Y both eligible: false-failure rate of the profiled cross-check (fixed), status.

D Protocol identities and history

SHA-256 prefixes of the files whose identity matters (full hashes in `protocols/` and in the records): FIT_MODEL_V3 rev. 14 `f05b835b`; FIT_MODEL_2D rev. 5 `4d2efc63`; FIT_MODEL_3D rev. 5 (approved) `b4ccf436` and rev. 7 (production) `682858f0`; amendment P1 rev. 4 `2a18c0ae`; frozen evaluators `evaluate_2d.py` rev. 5 `0fe221e8` and `evaluate_3d.py` rev. 1 `6bd35d7a`; P1 evaluators rev. 6 `3c3e35fc` and rev. 2 `d635c917`; `calibrate_p1.py` `d94b59f2`; calibration `CALIBRATION.json` `897142c1`; production evaluations `fbfc0ef9` (2D) and `ae9cf237` (3D); P1 evaluations `f9272978` (2D) and `b3fd6364` (3D). Records that named a personal machine were scrubbed after the results and re-pinned; `RE-PINNED.json` lists the original and published hashes (for example `CALIBRATION.json`: original `897142c1`, published `1afae79a`, which differ only in a host label). The prefixes above are the original ones.

The file `FIT_MODEL_3D.md.sha256` records the hash at approval (revision 5), while the evaluators pin revision 7. The approved revision 5, revision 6 and the difference to revision 7 are released (`protocols/FIT_MODEL_3D_rev5_approved.md`, ...); the two differ only in the preflight condition and the gate records described in section 5. `FIT_MODEL_2D.md` has no approval hash because it was never approved. The status lines inside the protocol files (for example "UNREVIEWED") predate their reviews; `PROTOCOL_HISTORY.md` gives the review status of each revision.

E The pre-campaign ε-expansion reference

Before the campaign, six-loop reference values were computed and frozen with the records (`data/1d/reference/published_and_epsilon_reference.csv`); they played no role in any selection. "GFF only" (generalized free field, i.e. the Gaussian end) is the ordinary Borel–Padé [4/2] of the six-loop series about the Gaussian end; "endpoint assisted" is the Borel–Padé [3/3] of the remainder R in $\Delta_\varepsilon = 1 - \sqrt{1-e} + (1-e)/8 + (1-e)^{3/2} R(e)$, $e = \varepsilon$, which builds in the $s \to 1$ endpoint behavior. They used an older value of the six-loop coefficient of the $d = 1$ series, $C_6(1) = 14331.8 \pm 0.5$; the final value, 14331.687, is given in the accompanying article [1]. They are superseded by the eight-loop values of section 6.5. The same file contains columns from an exploratory pre-campaign analysis (`new_MC_conditional_fit` and related), which are also superseded.

F Glossary

Δ_σ, Δ_ε dimensions of the spin and of the leading $\mathbb{Z}_2$-even scalar; $1/\nu = d - \Delta_\varepsilon$.

Δ^Q_{eff} finite-size effective dimension from a doubling pair at the U_4 crossing, eq. (4).

σ_{stat}, σ_{sys}, σ_D bootstrap error of the quoted value; half-range over the model family (a sensitivity scale, not a standard deviation); error of a paired difference.

Table 23: Calibration strata of amendment P1 (top $d = 2$, bottom $d = 3$).

$d = 2$, s	U_4	U_6	U_8	Y	U4+Y	cross-check
1.05	0 nc	0 nc	19: 1.00/1.00 (0.16) nc	1493: 0.83/0.95 (0.12) P	1812: 0.68/0.95 (0.59) P	0 nc
1.1	0 nc	1: 1.00/1.00 (0.00) nc	8: 1.00/1.00 (0.00) nc	1672: 0.83/0.95 (0.15) P	1799: 0.69/0.96 (0.63) P	0 nc
1.2	0 nc	0 nc	0 nc	490: 0.33/0.82 (0.00) F	1805: 0.67/0.95 (0.62) P	0 nc
1.3	0 nc	0 nc	0 nc	196: 1.00/1.00 (0.01) nc	194: 0.10/0.44 (0.01) nc	0 nc
1.4	0 nc	0 nc	0 nc	0 nc	15: 0.87/1.00 (0.87) nc	0 nc
1.5	1: 0.00/1.00 (0.00) nc	0 nc	3: 0.00/1.00 (0.00) nc	234: 0.00/0.02 (0.00) F	0 nc	0 nc
1.6	0 nc	0 nc	2: 1.00/1.00 (0.00) nc	222: 0.00/0.03 (0.00) F	0 nc	0 nc
1.7	62: 0.00/0.13 (0.00) nc	69: 0.00/0.10 (0.00) nc	144: 0.40/1.00 (0.00) nc	214: 0.00/0.04 (0.00) F	0 nc	0 nc

$d = 3$, s	U_4	U_6	U_8	Y	U4+Y	cross-check
1.55	1225: 0.86/0.98 (0.19) P	1396: 0.87/0.97 (0.17) P	1666: 0.83/0.96 (0.17) P	1663: 0.69/0.95 (0.20) P	1771: 0.68/0.95 (0.63) P	1062: 0.007 (0.69) P
1.6	1122: 0.99/1.00 (0.31) P	1165: 0.98/1.00 (0.31) P	1378: 0.88/1.00 (0.29) P	1668: 0.69/0.95 (0.21) P	1786: 0.70/0.95 (0.67) P	1076: 0.007 (0.61) P
1.7	825: 0.92/1.00 (0.20) P	1107: 0.90/1.00 (0.19) P	1603: 0.89/0.97 (0.16) P	1671: 0.69/0.96 (0.18) P	218: 0.29/0.89 (0.06) F	108: 0.000 (0.43) nc
1.75	428: 1.00/1.00 (0.01) P	420: 0.98/1.00 (0.02) P	684: 0.81/1.00 (0.15) P	1710: 0.69/0.95 (0.23) P	47: 0.00/0.09 (0.00) nc	108: 0.000 (0.76) nc
1.8	228: 0.99/1.00 (0.00) P	246: 0.95/1.00 (0.01) P	229: 0.41/1.00 (0.02) F	1645: 0.70/0.96 (0.26) P	1: 0.00/0.00 (0.00) nc	79: 0.000 (0.87) nc
1.85	717: 1.00/1.00 (0.32) P	996: 1.00/1.00 (0.27) P	1483: 0.93/1.00 (0.23) P	1695: 0.70/0.96 (0.22) P	0 nc	93: 0.108 (1.00) nc
1.9	531: 1.00/1.00 (0.38) P	764: 1.00/1.00 (0.32) P	1289: 0.98/1.00 (0.32) P	1663: 0.71/0.97 (0.29) P	0 nc	92: 0.326 (1.00) nc
1.95	216: 1.00/1.00 (0.00) P	237: 0.99/1.00 (0.01) P	254: 0.47/1.00 (0.00) F	182: 0.15/0.84 (0.01) nc	0 nc	39: 0.641 (1.00) nc

Table 24: $d = 1$: the frozen six-loop reference values against the Monte Carlo result.

Δ_σ	$\varepsilon = 1 - 4\Delta_\sigma$	six-loop, GFF only	six-loop, endpoint assisted	MC (this work)
0.025	0.900	0.5703	0.7109	0.71115(49)(959)
0.05	0.800	0.5459	0.6182	0.61987(55)(220)
0.0625	0.750	0.5351	0.5871	0.59003(35)(150)
0.1	0.600	0.5085	0.5262	0.53248(20)(242)
0.15	0.400	0.4871	0.4897	0.49325(41)(306)
0.1727335	0.309	0.4829	0.4836	0.48368(52)(78)
0.2	0.200	0.4828	0.4829	largest pair

p, k_τ polynomial degree of the scaling function; order of the thermal field ($d = 1$ fits).

P_{χ^2} χ^2 probability of a fit.

coupling one value of s (or Δ_σ) of the simulation grid; inverse temperatures are β.

window, variant, configuration in $d = 1$: the range $|x| \le x_{\max}$ of the scaling variable kept in a fit; the form of the correction terms (`leading`, `second`, `background`, `log`); one choice of window, variant, $N_{\min}$ and k_τ (section 6.2).

gate a pass/fail condition on a fit or series.

quotable, eligible a $d = 1$ fit that passes its gates; a $d = 2, 3$ extrapolation series that passes its gates.

series $\Delta^{Q}_{\text{eff}}(L)$ over the doubling pairs for one Q ($d = 2, 3$) or one crossing window and degree rule ($d = 1$), extrapolated to $L \to \infty$.

U4+Y joint extrapolation of U_4 and Y with common Δ_ε and ω'.

x-check cross-check (table 2).

$[\sigma]$ label suffix: $\sigma(\Delta^{U_4}_{\text{eff}})$ of the largest pair above the target 0.003.

frozen fixed before the data in question were analyzed (a protocol, the $d = 1$ window parameters); "the frozen rule" is the $d = 2, 3$ protocol before amendment P1.

fixed-ω' / profiled rule bootstrap at the central ω' (the frozen rule) / ω' re-profiled in every draw (amendment P1).

preflight short runs at the production sizes, before a production, that check rates, autocorrelation times and crossing brackets.

stage 0, A, B; N0, N2, N3 1D campaigns (table 4); N3, an extension to $N = 2^{20}$, was not run.

N1 the first evaluation of all 20 $d = 1$ couplings with protocol v3 (section 5).

snapshot an immutable, hashed set of merged summaries from which one fit run is made.

CB8 the eight-loop ε expansion of the accompanying article, resummed by conformal Borel resummation with the short-range end imposed.

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Part VII

Borel resummation of the long-range Ising ϵ expansion in one, two and three dimensions

Eighth-order GFF coefficients, short-range endpoint data and the critical instanton

LRI project note

15 October 2026

Abstract

We reconstruct the thermal scaling dimension $\Delta_{\mathcal{E}}$ of the critical long-range Ising model across its whole interacting window in $d = 1, 2, 3$, by Borel resummation of the generalized-free-field (GFF) ϵ expansion through eighth order combined with the expansion about the short-range end. In $d = 1$, with the short-range series in $\sqrt{\Delta_\sigma}$ imposed through eighth order, the conformal Borel curve gives $\Delta_{\mathcal{E}} = 0.5302$ at $\Delta_\sigma = 0.1$, with a sensitivity range $[0.5280, 0.5316]$ that is nowhere wider than 4.7×10^{-3}. This range contains every final Hamiltonian-truncation (HT) value and every converged numerical-renormalization-group (NRG) value within its quoted error, and every accepted Monte Carlo (MC) value reaches it within the sum of its statistical error and its fit-model half-range. In $d = 2$ and 3 the corresponding ranges are at most 5.2×10^{-3} and 1.5×10^{-4} wide; in $d = 2$ the way the short-range endpoint is imposed matters more, up to 7.5×10^{-3}. The critical fractional instanton supplies the candidate leading Borel singularity in each dimension; in $d = 1$ its one-loop amplitude $K_I = 6.25 \times 10^{-3}$ has the sign of every computed coefficient, which at eighth order reach 16% of its leading asymptotic form. We explain why the GFF series alone cannot reach the short-range end in $d = 1$, where $\Delta_{\mathcal{E}}$ has a square-root branch point at $\Delta_\sigma = 0$, while it nearly does in $d = 2, 3$. No HT, NRG or MC value enters a curve or is used to choose between resummation prescriptions; in $d = 1$, however, the number of imposed short-range coefficients was fixed after these data had been seen. Every quoted range is a sensitivity range, not a confidence interval.

1 Introduction and conventions

1.1 The model and its window

We consider the ferromagnetic Ising model in d dimensions with interaction $J(r) \propto r^{-(d+s)}$. In a window $d/2 < s < s_*(d)$ the critical point is a nontrivial long-range fixed point whose spin dimension is protected,

$$x := \Delta_\sigma = \frac{d-s}{2}, \qquad \epsilon := 2s - d = d - 4x, \qquad F(\epsilon) := \Delta_{\mathcal{E}} = \sum_{n\geq 0} C_n(d)\,\epsilon^n. \tag{1.1}$$

Here $\Delta_{\mathcal{E}}$ denotes the scaling dimension of the leading $\mathbb{Z}_2$-even (thermal) operator, as in all figures, and $\nu^{-1} = d - \Delta_{\mathcal{E}}$. This operator is often called ε; we write $\mathcal{E}$ to keep it apart from the expansion parameter ϵ. At $s = d/2$ ($\epsilon = 0$) the theory is a generalized free field (GFF) and $\Delta_{\mathcal{E}} = d/2$. At the upper end s_* the long-range fixed point joins the short-range (SR) Ising universality class. Table 1 collects the windows.

d	window in s	window in $x = \Delta_\sigma$	ϵ at SR end	GFF end $(x, \Delta_{\mathcal{E}})$	SR-end data used
1	$\frac{1}{2} < s < 1$	$0 < x < \frac{1}{4}$	1	$(\frac{1}{4}, \frac{1}{2})$	$\Delta_{\mathcal{E}} = 1 + \sum_{n\leq 8} C_n^{\mathrm{SR}} x^{n/2}$
2	$1 < s < \frac{7}{4}$	$\frac{1}{8} < x < \frac{1}{2}$	$\frac{3}{2}$	$(\frac{1}{2}, 1)$	$\Delta_{\mathcal{E}} = 1 + c_2(x - \frac{1}{8})^2 + \ldots$
3	$\frac{3}{2} < s < s_*$	$S < x < \frac{3}{4}$	$3 - 4S \simeq 0.9274$	$(\frac{3}{4}, \frac{3}{2})$	$\Delta_{\mathcal{E}} = E + c_1(x - S) + \ldots$

Table 1: Interacting windows and endpoint information. In $d = 3$, $S = 0.518148806$ and $E = 1.41262528$ are the short-range Ising spin and energy dimensions and $s_* = 3 - 2S = 1.963702388$. In $d = 1$ the values at $x = 0$ are limiting values; the short-range data are a series in $\tau = \sqrt{x}$.

1.2 What resummation is for

The GFF coefficients are now known through $n = 8$ in all three dimensions [2]. They alternate in sign and grow factorially; in $d = 1$, $C_8 \simeq 8.06 \times 10^6$. The physical window, however, reaches $\epsilon = 1$, $3/2$ and 0.93 at the short-range end in $d = 1, 2, 3$. A truncated power series is useless there. Borel resummation removes the factorial growth, analytic information about the Borel transform (the location of its leading singularity, supplied by the critical instanton) extends the transform beyond its radius of convergence, and the known expansion about the short-range end pins the far side of the interval. The aim of this note is a curve $\Delta_{\mathcal{E}}(\Delta_\sigma)$ in each dimension with an honest account of how sensitive it is to the choices made, together with an understanding of which pieces of information actually control it.

The coefficients, the short-range data and the comparison values come from the literature and from the research notes of the release of Ref. [1]: the GFF, SRI, HT, NRG and MC notes [2, 3, 4, 5, 6]. This note is the resummation note of that release, and the paper's section on resummation summarizes it.

1.3 Terms used below

Approximants. CBm and PBm are the conformal Borel and Padé–Borel approximants of order m, built from $C_0, \ldots, C_m$ and any imposed endpoint data (Secs. 4.1 and 4.2); CB8 is the eighth-order conformal Borel approximant. PB$[L/M]$ is a Padé–Borel approximant with numerator degree L and denominator degree M. The Borel–Leroy parameter b_{L} (Eq. (3.1)) and the exponent α of the conformal Borel ansatz at the boundary of the map are free parameters; a scan varies them.

Endpoint constructions (Sec. 4.3) impose the short-range end. In $d = 1$: endpoint-v, resummed in $v = 1 - 2\sqrt{\Delta_\sigma}$ with the short-range series imposed through C_K^{SR}, K being the number of imposed short-range coefficients; endpoint-ϵ, the same resummed in ϵ; the archived construction

is endpoint-v with $K = 4$, used in earlier versions of this note. In $d = 2$: curvature (value, zero slope and curvature imposed), tangent (value and zero slope) and value (value only). In $d = 3$: slope (value and slope) and value. GFF-only curves resum the ϵ series directly, without short-range input.

Baseline: CB8 of endpoint-v with $K = 8$ ($d = 1$), of the curvature construction ($d = 2$) and of the slope construction ($d = 3$), at the central b_{L} and $\alpha = 0$ (Table 7).

Core band: the pointwise minimum and maximum, over orders $6, 7, 8$, three values of b_{L} and $\alpha = -1, 0, 1$, of one construction (27 approximants).

Sensitivity envelope ($d = 1$): the union of the core bands of endpoint-v for $K = 5, \ldots, 8$. Core bands and the envelope measure sensitivity, not error (Sec. 1.5).

Instanton: the critical bubble, a real stationary field of the action at negative coupling; its action A_g and the Borel-plane scale $A_\epsilon = 3A_g$ (Sec. 3).

1.4 Main results

1. **One dimension.** The baseline (CB8 of endpoint-v, with the short-range series through C_8^{SR}) gives, for example, $\Delta_{\mathcal{E}}(0.05) = 0.62223$, $\Delta_{\mathcal{E}}(0.1) = 0.53025$, $\Delta_{\mathcal{E}}(0.15) = 0.49171$, $\Delta_{\mathcal{E}}(0.2) = 0.48312$, with a shallow minimum 0.48281 at $\Delta_\sigma = 0.192$. The sensitivity envelope is 1.45×10^{-3}, 3.64×10^{-3} and 4.40×10^{-3} wide at $\Delta_\sigma = 0.05, 0.1, 0.15$ (Sec. 5.2).

2. **Two and three dimensions.** The core bands of the baselines (curvature construction in $d = 2$, slope construction in $d = 3$) are at most 5.24×10^{-3} wide (near $\Delta_\sigma = 0.308$) and 1.53×10^{-4} (near $\Delta_\sigma = 0.595$). In $d = 2$ the choice of endpoint construction moves the curve by up to 7.5×10^{-3}; for $\Delta_\sigma < 0.29$ this shift exceeds the width of the core band (Sec. 5.3).

3. **Comparisons.** In $d = 1$ the sensitivity envelope contains every final HT value and every converged NRG value within its quoted error, and every accepted MC value reaches it within the sum of its statistical error and fit-model half-range. The NRG values lie up to 7.6×10^{-4} below the baseline. At $\Delta_\sigma = 0.1$ and 0.125 they lie below the final HT values by 5.8 and 4.6 times the sum of the two quoted errors. The gap is of the size of the dependence of HT on how its truncation artifact is corrected, which the quoted HT error does not include. No comparison data favor the $K = 8$ baseline over the archived $K = 4$ construction (Table 13); $K = 8$ is motivated a priori only, and it was fixed after these data had been seen. In $d = 2$ finite-cutoff HT lies above every endpoint construction at $s = 1.6$–1.65, by about 3×10^{-3} beyond the highest of their core bands; its cutoff dependence there does not show that the excess survives at infinite cutoff. Published MC values (quoted errors 0.006 to 0.034) cannot separate the curves. In $d = 3$ the project's MC point at $s = 1.6$ lies 1.0×10^{-3} above the curve, and bootstrap kinks read off a figure lie 0.004 to 0.010 below it at $s = 1.59$–1.87, the gap growing toward the short-range end (Sec. 6).

4. **Instanton.** The critical real bubble gives the candidate Borel singularity $-A_\epsilon$ with $A_\epsilon = 9\pi^2/\Gamma(\frac{1}{4})^4$, $\frac{9}{8}$ and $\frac{9}{4}[\Gamma(\frac{9}{4})/\Gamma(\frac{3}{4})]^2$ for $d = 1, 2, 3$. In $d = 1$ the regular one-instanton sector predicts $C_n \sim K_I(-1/A_\epsilon)^n\Gamma(n + 4)$ with $K_I = 6.2466 \times 10^{-3}$; the coefficients have the predicted sign, but at $n = 8$ they are only 0.158 of this leading form and still rising (Sec. 3).

5. **Analytic structure.** In $d = 1$, $\Delta_{\mathcal{E}}$ has a square-root branch point at $\Delta_\sigma = 0$; without short-range input the GFF series cannot reproduce it and the GFF-only curve ends at 0.567 instead of 1. In $d = 2, 3$ the short-range end is approached analytically and the GFF-only curve reaches 1.0049 (exact 1) and 1.4161 (exact 1.4126) (Sec. 4.4).

1.5 Status of the statements

Table 2 separates what is established from what is conditional and from what is our interpretation. We use the following words throughout: a *band*, *envelope* or *range* is the pointwise minimum and maximum over a stated set of resummation prescriptions. It measures sensitivity to those choices; it is not a confidence interval, it has no probabilistic coverage, and it does not bound the error of the approximant. Printed digits identify an approximant reproducibly and are not claimed digits of the physical dimension. No HT, NRG or MC value enters any curve or is used to choose between prescriptions. In $d = 1$, however, the number $K = 8$ of imposed short-range coefficients was fixed after these data had been seen; it is motivated a priori only (Secs. 5.2 and 6.2). Uncertainties of different kinds are never added in quadrature: a difference is compared with each quoted error separately or with their linear sum, as stated each time (Sec. 6).

Statement	Status
Critical bubble and its action A_g	Explicit classical solution; least action among nonzero real finite-energy stationary fields (sharp fractional Sobolev inequality).
$A_\epsilon = 3A_g$ as the nearest singularity of the full Borel transform	Conditional. It is the exponential scale of the one-instanton sector; dominance is not proved.
$b_I = (d + 5)/2$	$d = 1$: formal renormalized one-instanton prediction, with the leading mass insertion checked. $d = 2, 3$: power counting only.
$K_I = 6.25 \times 10^{-3}$ $(d = 1)$	One-loop prediction of the stated prescription, which reproduces the ϕ^4 amplitude in $d = 4$; the treatment of the regulator dependence of the bubble action (Sec. 3.4) is argued, not derived in full. No $d = 2, 3$ amplitude.
Square-root branch point at $\Delta_\sigma = 0$ $(d = 1)$	Established within the short-range construction: odd powers of $\sqrt{\Delta_\sigma}$ come from an eigenvalue splitting.
Fixed-point merger of Kosterlitz–Thouless type at $s = 1$	Our interpretation of that structure.
Radius $\simeq \frac{1}{2}$ of the short-range series in $\sqrt{\Delta_\sigma}$	Suggestive only (eight coefficients).
$K = 8$ in $d = 1$	Motivated a priori (all available short-range coefficients); fixed after the comparison data had been seen.
Eighth-order curves	Reproducible approximants; not fitted to comparison data.
Bands and envelopes	Sensitivity ranges, not confidence intervals.
C_9	Not computed; all C_9 values are predictions or scenarios.

Table 2: Logical status of the main statements.

Earlier versions. This note merges and supersedes three earlier project notes, for $d = 1$ (7 October 2026) and for $d = 2$ and $d = 3$ (12 October). Four things have changed since then. The short-range series in $d = 1$ is imposed through C_8^{SR} ($K = 8$) instead of C_4^{SR} (the archived $K = 4$ construction, kept as a control). $C_8(1)$ is the final value of Sec. 2 instead of a four-batch value. The $d = 1$ HT and NRG values are the final ones of the HT and NRG notes [4, 5], which replace earlier extrapolations. And the statement that $K = 8$ removes an offset of $K = 4$ against HT and NRG is withdrawn (Sec. 6.2).

2 Inputs

2.1 GFF coefficients

Table 3 lists the coefficients used. In every dimension $C_0 = d/2$ and $C_1 = -1/6$. The coefficients C_0 to C_3 are published [7]; C_4 to C_8 are from the GFF note [2]. Through C_5 they are exact expressions in an explicit master basis; the lowest ones read

$$C_2(d) = \frac{\mathcal{A}_d}{9}, \qquad C_3(d) = \frac{\mathcal{A}_d^2 + \mathcal{R}_d - 8Q_d}{108}, \qquad \mathcal{A}_d = \psi(\tfrac{d}{2}) + \psi(1) - 2\psi(\tfrac{d}{4}), \quad \mathcal{R}_d = \psi_1(\tfrac{d}{2}) - \psi_1(1), \tag{2.1}$$

with Q_d the three-loop primitive period (one third of the tetrahedron period, as corrected in the erratum of Ref. [7]). For $d = 1$ this gives $\mathcal{A}_1 = \pi + 4\log 2$; for $d = 2$, $C_2 = \frac{4}{9}\log 2$. The values of C_5 are numerical evaluations of the exact expression ($d = 3$: a supplied evaluation of the master basis). The coefficients C_6, C_7, C_8 are numerical estimators with empirical allowances.

n	$C_n(1)$	$C_n(2)$	$C_n(3)$
0	1/2	1	3/2
1	−1/6	−1/6	−1/6
2	0.6571312639810638	0.3080654135821979	0.181221785405554
3	−3.719856322901833	−0.8748500126526337	−0.353581520433173
4	45.42488010356	5.048868431354847	1.22806354532762
5	−731.13947662454	−38.83364589622536	−5.85035142075133
6	$14331.68703276 \pm 4.9 \times 10^{-7}$	362.36647 ± 0.004	33.57543 ± 0.0005
7	-322549.889 ± 41	-3876.126 ± 2.1	-220.3545 ± 0.13
8	8059766.7 ± 17000	45990.0 ± 33	1602.605 ± 3.9

Table 3: GFF coefficients used [2]. The ± entries are empirical reporting allowances, not standard deviations: for the Monte Carlo estimates ($C_6(2)$, $C_6(3)$, C_7, C_8) four sampling standard errors, rounded up; for the deterministic graph sum $C_6(1)$ a convergence-and-rounding allowance. Sampling standard errors: $C_6(2)$ 8.0×10^{-4}; $C_6(3)$ 1.19×10^{-4}; $C_7(1)$ 10.09; $C_7(2)$ 0.524; $C_7(3)$ 0.0318; $C_8(1)$ 4210; $C_8(2)$ 8.22; $C_8(3)$ 0.952. Full stored digits are used in the calculations.

Three remarks on Table 3. First, $C_6(1)$ is a deterministic sum over all six-loop graphs with a convergence-and-rounding allowance; a closed form in nine special-function values gives 14331.68703275805, which differs by -1.03×10^{-8} and is not used. Second, the final value $C_8(1) = 8059766.723129955$ is the mean of fourteen production batches (sampling standard error 4209.6, pre-registered allowance ±17000, quoted $(8.060 \pm 0.017) \times 10^6$). It replaces the four-batch value 8066723.343 ± 33000 used in earlier versions of this note; every one-dimensional result below uses the final value. Third, the eighth-order Monte Carlo campaigns, whose C_8 includes the power-containing graph sector of the GFF note, also estimate the lower coefficients; those estimates serve as cross-checks only, and the more precise dedicated values of $C_0, \ldots, C_7$ in the table are used. No joint covariance of C_6, C_7, C_8 is available, so coefficient sensitivities below are sums of absolute responses to the allowances.

2.2 Short-range data

One dimension. Near $s = 1$ the thermal dimension is a series in $\tau = \sqrt{x}$ [3],

$$\Delta_{\mathcal{E}} = 1 + \sum_{n \geq 1} C_n^{\mathrm{SR}}\, \tau^n, \qquad \begin{aligned} &C_1^{\mathrm{SR}} = -2, \quad C_2^{\mathrm{SR}} = \tfrac{1}{2}, \quad C_3^{\mathrm{SR}} = \tfrac{39}{16} + 2\zeta(2), \\ &C_4^{\mathrm{SR}} = 4 - 4\zeta(2) - 8\zeta(3), \end{aligned} \tag{2.2}$$

with closed forms in multiple zeta values through C_7^{SR}. Numerically $C_{3,\ldots,8}^{\mathrm{SR}} = 5.72736813$, -12.19619149, 29.23781158, -58.66569747, 51.06898861 and 247.38120480. The leading orders

reproduce the published dual description [8, 9], $\Delta_\pm = 1 \pm \sqrt{2(1-s)} + (1-s)/4 + O((1-s)^{3/2})$, with $1 - s = 2x$. The higher coefficients are results of the project's construction and are conditional on its regulator, matching and two-coupling renormalization assumptions, which the SRI note states in its Appendix A and examines in its Appendix D, "The two-coupling closure and the prescription" [3]. C_8^{SR} is a completed numerical assembly; a short zeta-value form agrees with it to a recorded residual of about 10^{-86} but is unproved. An independent re-derivation gives $C_8^{\mathrm{SR}} = 242.8 \pm 2.4$ (one statistical standard error), 1.9 standard errors lower [3]; using its central value instead moves the baseline curve by at most 3×10^{-5}, so this tension does not matter here.

The magnitudes $|C_n^{\mathrm{SR}}|^{1/n}$ are 1.96, 1.97, 1.75 and 1.99 for $n = 5, 6, 7, 8$. This suggests a radius of convergence close to $\frac{1}{2}$ in τ, that is, singularities at $\tau = \pm\frac{1}{2}$: $\tau = \frac{1}{2}$ is the GFF point $\Delta_\sigma = \frac{1}{4}$, and $\tau = -\frac{1}{2}$ is its image on the second sheet (Sec. 4.4). With eight coefficients this is suggestive, not a determination.

Two dimensions. With $\delta = x - \frac{1}{8}$ [10, 3],

$$\Delta_{\mathcal{E}} = 1 + c_2\,\delta^2 + O(\delta^3), \qquad c_2 = -4.066270148132521 55274\ldots, \tag{2.3}$$

with zero linear term, so $F(\frac{3}{2}) = 1$, $F'(\frac{3}{2}) = 0$ and $F''(\frac{3}{2}) = c_2/8$ in ϵ derivatives. The vanishing linear term is published [10]; the value of c_2, from the SRI note [3], is a formal result in a critically tuned analytic minimal-subtraction prescription. A Monte Carlo evaluation of the regulated integral gives $c_2 = -4.075 \pm 0.019$ (one statistical standard error) [3]. A uniform proof of the nested subtractions is not available.

Three dimensions. With $S = 0.518148806(24)$ and $E = 1.41262528(29)$ from the conformal bootstrap [11] and $\delta = x - S$,

$$\Delta_{\mathcal{E}} = E + c_1\,\delta + O(\delta^2), \qquad c_1 = \frac{a_0}{b_0} = 0.27315242382601645, \tag{2.4}$$

where $a_0 = 3.3474594916917857$ and $b_0 = 12.254914105481044$ are the crossover coefficients of $\gamma_{\mathcal{E}} = a_0 g_{\mathrm{SR}}^2 + \ldots$ and $\beta_{\mathrm{SR}} = -\delta g_{\mathrm{SR}} + b_0 g_{\mathrm{SR}}^3 + \ldots$ in the scaling theory of Ref. [10], computed in the SRI note [3] from the 3d Ising spectrum of Refs. [12, 11]. That note quotes a working range $c_1 \in [0.2690, 0.2760]$, conditional on its spectral and truncation assumptions (60 retained primaries through dimension 24, gaps for omitted operators, inputs within twice their errors); it is not a statistical interval.

3 Large orders and the critical instanton

3.1 Borel transform and the dispersion relation

A large-order behavior $C_n \sim \kappa(-1/A)^n \Gamma(n + b + 1)$ corresponds to a singularity of the Borel transform at $t = -A$. With the Borel–Leroy transform

$$\mathcal{B}_{b_{\mathrm{L}}}(t) = \sum_{n \geq 0} \frac{C_n\, t^n}{\Gamma(n + b_{\mathrm{L}} + 1)}, \qquad \mathcal{R}[F](\epsilon) = \int_0^\infty e^{-u} u^{b_{\mathrm{L}}}\, \mathcal{B}_{b_{\mathrm{L}}}(\epsilon u)\, \mathrm{d}u, \qquad b_{\mathrm{L}} > -1, \tag{3.1}$$

termwise integration returns $C_n \epsilon^n$. If the discontinuity of F across the negative axis has a sector $\mathrm{Disc}\, F(-u) \sim D\, u^{-p} e^{-A/u}$, a dispersion relation gives the contribution

$$C_n^{(I)} \sim (-1)^{n+1}\, \frac{D}{2\pi i}\, A^{-n-p}\, \Gamma(n + p), \tag{3.2}$$

so $b = p - 1$. The instanton supplies the candidate location $-A$ and, after a fluctuation and renormalization analysis, a candidate strength. It constrains the tail of the series; it does not supply the missing coefficients. Eight coefficients cannot determine the singularity by themselves: adding $\eta[(1 + t/t_0)^{-4} - T_8(1 + t/t_0)^{-4}]$, with T_8 the Taylor truncation, to the Borel transform reproduces all known terms and places a singularity at $t = -t_0$ for any $t_0 > 0$.

3.2 The critical bubble in d dimensions

At the expansion point $s = d/2$ normalize the free covariance to $G(x) = |x|^{-d/2}$, so that the kinetic operator is $\mathcal{K} = |D|^{d/2}/(2\pi)^{d/2}$, where $|D|^\rho$ has symbol $|p|^\rho$. At negative quartic coupling the action $\mathcal{S} = \frac{1}{2}\int \phi\mathcal{K}\phi - \frac{h_{\rm phys}}{4!}\int \phi^4$ has the stationary family

$$\phi_I(y) = \pm\frac{B_d}{\sqrt{h_{\rm phys}}}\left[\frac{R}{R^2 + |y - y_0|^2}\right]^{d/4}, \qquad B_d^2 = \frac{6\,\Gamma(3d/4)}{\pi^{d/2}\,\Gamma(d/4)}, \tag{3.3}$$

labeled by position, size and sign. In the project coupling g, normalized so that $\beta_g = -\epsilon g + 3g^2 + \ldots$ and $g_* = \epsilon/3 + \ldots$, the action at $g = -h$ is $A_g(d)/h$ with

$$A_g(d) = \frac{3}{2}\,\frac{\Gamma(3d/4)^2}{\Gamma(d/4)^2\,\Gamma(d)}, \qquad A_\epsilon = 3A_g. \tag{3.4}$$

The candidate Borel singularity of the physical ϵ series is at $t = -A_\epsilon$:

$$\begin{aligned} A_\epsilon(1) &= \frac{9\pi^2}{\Gamma(\frac{1}{4})^4} = 0.51406490367502158, \qquad A_\epsilon(2) = \frac{9}{8}, \\ A_\epsilon(3) &= \frac{9}{4}\left[\frac{\Gamma(\frac{9}{4})}{\Gamma(\frac{3}{4})}\right]^2 = 1.92343325557014548, \end{aligned} \tag{3.5}$$

and $A_\epsilon(4) = 3$ in the same normalization. In $d = 1$, $A_g = 3\pi^2/\Gamma(\frac{1}{4})^4 = 0.171354967891674$; the normalization was checked independently by rescaling the field and the coupling.

Two properties fix the meaning of this saddle. The sharp fractional Sobolev (Hardy–Littlewood–Sobolev) inequality $T \geq \kappa_{\rm S}\sqrt{V}$, with $T = \int \phi\mathcal{K}\phi$ and $V = \int \phi^4$, combined with the virial identity $T = hV/6$ of any nonzero real stationary field, gives $\mathcal{S} \geq A_g/h$ with equality on the bubble [13, 14]. The bubble is therefore the least-action nonzero real stationary field in the finite-energy class. This does not classify complex saddles, integration cycles or the singularities of the renormalized theory. Second, dilations give $(2s - d)T = 0$ for a finite-action massless solution, so the closed form exists only at $s = d/2$. Using it for the large-order behavior of the expansion *about* $\epsilon = 0$ is consistent, but continuing the saddle to physical s requires a regulated (constrained) treatment.

3.3 Fluctuations, power counting and the exponent b_I

Mapped conformally to the sphere, the Hessian relative to the free kinetic form has eigenvalues $1 - 3p_0/p_\ell$ with $p_\ell = \Gamma(\ell + 3d/4)/\Gamma(\ell + d/4)$: one negative mode ($\ell = 0$), $d + 1$ zero modes ($\ell = 1$: d translations and the dilation) and positive modes for $\ell \geq 2$. Bubble nondegeneracy excludes further zero modes in its class [15]. The $d + 1$ collective coordinates, normalized with the kinetic metric (the flat L^2 norm of the dilation mode diverges), contribute $h^{-(d+1)/2}$; the four amputated legs contribute h^{-2}. After the size integral and the minimal-subtraction residue, $\mathrm{Disc}\,\beta(-h) \propto h^{-(d+5)/2}e^{-A_g/h}$. The fixed-point shift divides by $\beta'(g_*) = O(\epsilon)$ and adds one inverse power, so

$$\mathrm{Disc}\,F(-u) \propto u^{-(d+7)/2}e^{-A_\epsilon/u}, \qquad C_n^{(I)} \sim K_I\,(-1/A_\epsilon)^n\,\Gamma(n + b_I + 1), \qquad b_I = \frac{d+5}{2}. \tag{3.6}$$

This gives $b_I = 3, \frac{7}{2}, 4$ for $d = 1, 2, 3$ and reproduces the established $\frac{9}{2}$ of the short-range ϵ expansion in $d = 4$ [16, 18].

In $d = 1$ the analysis goes beyond counting. It includes the kinetic-metric measure $G_{ij} = \frac{3A_g}{8hR^2}\delta_{ij}$, the cancellation of the determinant pole $3A_g/z$ against the one-loop coupling counterterm (an independent check of the one-loop coefficient 3), and the leading direct mass insertion, whose discontinuity is down by one power of u and therefore cannot cancel the leading term within the same regular semiclassical expansion. Three different statements must be kept apart:

1. $b_{\mathrm{L}} = 3$ is a *choice* of the freely adjustable Borel–Leroy parameter;
2. $b_I = 3$ is the exact integer predicted by the regular one-instanton sector, not a fit;
3. that this sector controls the complete coefficient sequence has not been proved. It requires a uniform renormalized saddle expansion and global dominance with nonzero net amplitude; competing complex saddles, multi-instanton sectors and non-instanton mechanisms have not been classified [19].

In $d = 2, 3$ only the action and the power counting are available; no renormalized determinant, Stokes amplitude or competing-singularity analysis has been done.

3.4 The one-loop amplitude in one dimension

Following the minimal-subtraction procedure of Refs. [16, 17] with the project's analytic regulator $G_z(y) = |y|^{-(1-z)/2}$, the leading one-instanton sector gives

$$\begin{aligned} C_n &\sim K_I\,(-1/A_\epsilon)^n\,\Gamma(n+4), \qquad K_I = 6.2466 \times 10^{-3}, \\ \beta_n &\sim (-1)^{n+1} C_\beta\, A_g^{-(n+1)} \Gamma(n+4), \qquad C_\beta = 0.024373, \end{aligned} \tag{3.7}$$

the second relation for the coefficients β_n of the minimal-subtraction beta function $\beta_0(g) = \sum_n \beta_n g^{n+1}$. The assembly is

$$K_I = \frac{\mathcal{N}\, e^{-2\mathcal{A}_1 A_g}}{3\pi A_g}, \qquad C_\beta = \frac{\mathcal{N}}{\pi}, \qquad \mathcal{N} = \frac{9}{2\sqrt{2}\,\pi^2 A_g} \exp\Big(-\tfrac{1}{2}L_{\mathrm{sub}} + 9A_g \log 2 - \tfrac{21}{4} + \delta_{\mathrm{p}}\Big) = 0.076571, \tag{3.8}$$

with the following ingredients:

- $L_{\mathrm{sub}} = -1.769169774220360678$, the convergent subtracted spectral sum over the positive modes; its terms fall like $n^{-3/2}$, so it is summed with an exact asymptotic tail (a default automatic summation gets the third decimal wrong);
- the finite part $9A_g \log 2 = 1.06897$ of the regulated bubble trace, after its pole $3A_g/z$ cancels against the coupling counterterm, and the subtraction $\Sigma_{\mathrm{spec}} = \frac{21}{2}$ of the modes $n = 0, \pm 1$, which enters the exponent as $-\Sigma_{\mathrm{spec}}/2 = -\frac{21}{4}$;
- the prefactor $9/(2\sqrt{2}\pi^2 A_g) = 2 \cdot \frac{1}{2\sqrt{2}} \cdot \frac{3}{16\pi} \cdot \frac{24}{\pi} \cdot \frac{1}{A_g}$: the negative mode, $1/(2\sqrt{2})$ in magnitude per saddle sign, with both signs contributing equally; the collective-coordinate measure $\sqrt{\det G}/(2\pi) = 3A_g/(16\pi h R^2)$; and the four amputated legs, $(24/\pi A_g)(A_g/h)^2 R$. The remaining factor $(A_g/h)^3$ enters the size integral, whose simple pole leaves $(A_g/h)^2 e^{-A_g/h}$;
- the constrained-profile shift $\delta_{\mathrm{p}} = \frac{3}{2}A_g(\pi - 4\log 2) = 0.0948460$;
- the fixed-point factor $e^{-2\mathcal{A}_1 A_g} = 0.13175005$, with $\mathcal{A}_1 = \pi + 4\log 2$.

The positive K_I fixes the sign of C_n to that of $(-1)^n$, in agreement with all computed coefficients.

The same programs, with only dimension-specific inputs changed, reproduce three known results: the zero-dimensional quartic integral; McKane's general ϕ^6 formula in $d = 3$ to fifteen digits; and the minimal-subtraction beta-function amplitude of ϕ^4 theory in $d = 4 - \epsilon$ to twelve digits, once the corresponding constrained-profile shift ($-\frac{9}{2}$ there) is included [16, 17]. The remaining qualification concerns the regulator dependence of the classical action, $A(z) = A_g + z\vartheta A_g$ with $\vartheta = \frac{\pi}{2} - 2\log 2$. The adopted prescription treats it as a rescaling of the size coordinate, which drops out of the simple-pole residue; this cancellation is argued and supported by the $d = 4$ benchmark, but not yet derived while tracking the integration limits, determinant, measure, legs and counterterms explicitly. The variants $K_I = 5.681 \times 10^{-3}$ (without δ_{p}) and 5.167×10^{-3} (with $-\delta_{\mathrm{p}}$ in place of δ_{p}) are disfavored by the adopted derivation, not excluded; their difference is not an error bar. No $d = 2$ or $d = 3$ amplitude has been computed, and the one-dimensional value is not borrowed.

3.5 Finite-order mismatch

Table 4 compares the one-dimensional coefficients with the leading form. Every coefficient has the predicted sign. In magnitude they reach 10% to 16% of $K_I(-1/A_\epsilon)^n\Gamma(n+4)$, rising from $n = 4$ without a plateau. The column with the alternative normalization $n!\,n^3$ is not an independent test: the two denominators differ by $(n+1)(n+2)(n+3)/n^3$, which is 1.93359375 at $n = 8$. For the minimal-subtraction beta function of ϕ^4 theory in $d = 4$ the corresponding fractions at one to six loops are 43%, 12.5%, 9.6%, 9.4%, 10.4% and 12.1% [18]. The similarity makes a slow approach to the asymptotic regime plausible; it does not show that both theories approach it in the same way.

n	$\dfrac{C_n}{K_I(-1/A_\epsilon)^n\Gamma(n+4)}$	$\dfrac{C_n}{K_I(-1/A_\epsilon)^n\,n!\,n^3}$	$r_n = -\dfrac{C_n}{nC_{n-1}}$	$\widehat{a}_n = -\dfrac{C_n}{(n+3)C_{n-1}}$
3	0.112	0.499	1.887	0.943
4	0.101	0.331	3.053	1.744
5	0.104	0.280	3.219	2.012
6	0.117	0.272	3.267	2.178
7	0.135	0.283	3.215	2.251
8	0.158	0.305	3.123	2.272

Table 4: One-dimensional coefficients against the leading instanton form, with the final $C_8(1)$. If Eq. (3.6) already dominated, $\widehat{a}_n$ would equal $1/A_\epsilon = 1.9453$.

The power-corrected ratio $\widehat{a}_8$ lies 16.8% above $1/A_\epsilon$ and is still increasing. Write the next correction as $C_n \sim K_I(-1/A_\epsilon)^n\Gamma(n+4)[1 + d_1A_\epsilon/(n+3) + \dots]$. It changes $\widehat{a}_nA_\epsilon$ by $-d_1A_\epsilon/[(n+2)(n+3)]$, so if it alone explained the drift, $n^2(\widehat{a}_nA_\epsilon - 1)$ would approach the constant $-d_1A_\epsilon$; it takes the values 4.31, 7.69 and 10.74 at $n = 6, 7, 8$. The coefficient d_1 is only partly known: its known pieces (fixed-point composition and the residue operation) sum to -6.40, a -30% correction at $n = 8$; the two-loop fluctuations, the subleading size integral and the direct mass insertion are not computed. Fits of the first column of Table 4 with corrections in powers of $1/(n+3)$ are unstable, depending on which orders are fitted. One correction term gives an asymptotic value 0.24 to 0.34 (and $d_1 \simeq -8.5$ to -11.6); two terms give 0.58 to 0.73 (and $d_1 \simeq -22$ to -25), values of d_1 that would give corrections beyond -100% at $n = 8$. The eighth order is therefore not yet a sharp test of K_I.

The same diagnostic in $d = 2$ and 3, with $\widehat{a}_n = -C_n/[(n+b_I)C_{n-1}]$ (in $d = 1$, $b_I = 3$) and the candidate b_I, gives Table 5. At $n = 8$ the ratio exceeds the candidate limit by 16.1% in $d = 2$ and 16.6% in $d = 3$, close to the 16.8% of $d = 1$, and is still rising in all three dimensions. Eighth order does not establish asymptotic dominance in any dimension, and no amplitude is inferred from these ratios.

n	2	3	4	5	6	7	8
$d = 2$, $\widehat{a}_n$ (limit 0.8889)	0.3361	0.4369	0.7695	0.9049	0.9822	1.0187	1.0317
$d = 3$, $\widehat{a}_n$ (limit 0.5199)	0.1812	0.2787	0.4342	0.5293	0.5739	0.5966	0.6061

Table 5: Corrected ratios $\widehat{a}_n = -C_n/[(n+b_I)C_{n-1}]$ in $d = 2$ ($b_I = \frac{7}{2}$) and $d = 3$ ($b_I = 4$), against the candidate limit $1/A_\epsilon$.

4 Resummation methods

4.1 Padé–Borel and admissibility

For the series $f(z) = \sum_{n \le m} f_n z^n$ actually being resummed, the Padé–Borel (PB) approximant replaces $\mathcal{B}_{b_L}$ of Eq. (3.1) by a rational function $[L/M]$ with $L + M = m$. It receives no instanton information. Candidates have denominator degree $M = \lfloor m/2 \rfloor - 1, \ldots, \lfloor m/2 \rfloor + 1$ (one more for odd m), restricted to $1 \le M \le m$, ordered by $(|m - 2M|, M)$; the first candidate without a positive-real denominator root, defined by $\Re t > 0$ and $|\Im t| < 10^{-8} \max(1, |t|)$, is selected. No principal value is taken. If no candidate survives, the configuration is reported as unavailable, never replaced silently. All poles and rejected candidates are archived.

PBm denotes the approximant this rule selects at order m. The rule fixes the PB controls of the curves in $d = 2, 3$ (PB8 is $[5/3]$ in $d = 2$ at $b_L = 3.5$; PB7 is $[3/4]$ in $d = 3$ at $b_L = 4$) and the shapes of all held-out predictions (Sec. 7). In $d = 1$ the curve controls have fixed shapes: those the rule selects for the archived $K = 4$ construction at $b_L = 3$, $[4/2]$, $[5/2]$ and $[5/3]$ at orders 6, 7 and 8, held fixed for every K so that a matched comparison changes only the Borel approximation. At order 8 the rule itself selects $[5/3]$ also for $K = 5$–7; at $K = 8$ only $[3/5]$ survives.

4.2 Conformal Borel

Assuming that the Borel transform is analytic in the plane cut along $(-\infty, -A]$, the map

$$w(t) = \frac{\sqrt{1 + t/A} - 1}{\sqrt{1 + t/A} + 1}, \qquad t = \frac{4Aw}{(1 - w)^2}, \tag{4.1}$$

sends the cut to the unit circle and the positive integration axis into the disk. The conformal Borel (CB) approximant is the Taylor-matched polynomial $\widehat{\mathcal{B}}(t) = (1 - w)^{-\alpha} \sum_{k \le m} p_k w^k$ (Appendix A), integrated as in Eq. (3.1). It uses the candidate action A, but the amplitude K_I is not imposed. The baseline has $\alpha = 0$; α parametrizes the behavior at $w = 1$ and is not a derived exponent. A CB versus PB comparison compares two approximation families, not the effect of one isolated input. The map can fail if the assumed nearest singularity is wrong, if there are other off-axis singularities, or if subleading terms are large.

4.3 Endpoint constructions

All constructions below preserve every supplied GFF coefficient through order m: the series of the remainder is formed by exact series operations, truncated after each step. Regularity of the resummed remainder at the short-range end then enforces the endpoint constraints. Imposing endpoint information is supplied input; agreement at that endpoint is not a prediction.

One dimension. With $F_K^{\mathrm{SR}}(x) = \sum_{n \le K} C_n^{\mathrm{SR}}\, x^{n/2}$ (and $C_0^{\mathrm{SR}} = 1$),

$$\begin{gathered} H = \frac{F - F_K^{\mathrm{SR}}}{(4x)^{(K+1)/2}}, \qquad v = 1 - \sqrt{1 - \epsilon} = 1 - 2\sqrt{x}, \quad \epsilon = 2v - v^2, \\ F = F_K^{\mathrm{SR}} + (4x)^{(K+1)/2}\, \mathcal{R}_m\big[H(2v - v^2)\big](v). \end{gathered} \tag{4.2}$$

The series of H in ϵ is composed with $\epsilon = 2v - v^2$ and resummed in v. The action transforms with the expansion variable: since $A_\epsilon/\epsilon(v) = A_\epsilon/(2v) + A_\epsilon/4 + O(v)$, the CB map in v uses $A_v = A_\epsilon/2 = 0.2570324518375$. This change of the physical expansion variable is distinct from the conformal map of the Borel plane. The archived construction of earlier versions of this note is $K = 4$, with $F_4^{\mathrm{SR}} = 1 - \sqrt{q} + \frac{q}{8} + (\frac{39}{128} + \frac{\pi^2}{24}) q^{3/2} + (\frac{1}{4} - \frac{\pi^2}{24} - \frac{\zeta(3)}{2}) q^2$, $q = 4x$. The baseline of this note is $K = 8$, which uses all available short-range coefficients. A control resums H in ϵ instead of v (endpoint-ϵ), and the GFF-only control resums F directly in ϵ.

Two dimensions. With $q = 1 - \epsilon/\frac{3}{2} = (x - \frac{1}{8})/\frac{3}{8}$ and $F^{\rm SR} = 1 + \frac{9c_2}{64}q^2$, the baseline (*curvature*) construction is $F = F^{\rm SR} + q^3\mathcal{R}_m[(F - F^{\rm SR})/q^3]$; it imposes value, zero slope and curvature. The *tangent* construction $F = 1+q^2\mathcal{R}_m[(F-1)/q^2]$ imposes value and zero slope; the *value* construction $F = 1 + q\,\mathcal{R}_m[(F-1)/q]$ imposes only the value; the GFF-only curve resums F itself. All resum in ϵ with $A_\epsilon = \frac{9}{8}$.

Three dimensions. With $X = \frac{3}{4}-S$, $q = (x-S)/X = 1-\epsilon/\epsilon_*$ ($\epsilon_* = 3-4S$) and $F^{\rm SR} = E+c_1Xq$, the baseline (*slope*) construction is $F = F^{\rm SR} + q^2\mathcal{R}_m[(F - F^{\rm SR})/q^2]$. The *value* construction is $F = E + q\,\mathcal{R}_m[(F - E)/q]$, and the GFF-only curve resums F. All resum in ϵ with $A_\epsilon(3)$. In $d = 2, 3$ the one-dimensional square-root variable is not used, for the reason explained next.

4.4 Analytic structure at the short-range end

What is established in $d = 1$. In the project's construction of the short-range end, the two lowest even operators are governed by a two-coupling flow whose stability matrix at the fixed point has trace $T = (1-s)/2+O((1-s)^2)$ and determinant $D = -2(1-s)+O((1-s)^2)$, with $1-s = 2x$. The fixed-point fugacity is of order $\sqrt{1-s}$ even though the other couplings are of order $1-s$. The two dimensions are

$$\Delta_\pm = 1 + \frac{T}{2} \pm \sqrt{\frac{T^2}{4} - D}, \qquad \Delta_\mathcal{E} = \Delta_-, \tag{4.3}$$

and the square root is $\sqrt{2(1-s)}$ times a power series in $1-s$ [9]. Odd powers of $\tau = \sqrt{x}$ in Eq. (2.2) come from this eigenvalue splitting. In the GFF variable, $1-\epsilon = 4x = 4\tau^2$, so $F(\epsilon)$ has a square-root branch point at $\epsilon = 1$, where $\Delta_\mathcal{E} - 1 \simeq -2\sqrt{\Delta_\sigma}$ and the slope $\mathrm{d}\Delta_\mathcal{E}/\mathrm{d}\Delta_\sigma$ diverges. Continuing around the branch point ($\tau \to -\tau$) exchanges Δ_- and Δ_+. The variable $v = 1 - 2\tau$ removes this branch point: the GFF point $\tau = \frac{1}{2}$ is $v = 0$, the short-range end is $v = 1$, and the image of the GFF point on the second sheet, $\tau = -\frac{1}{2}$, is $v = 2$.

Interpretation. We read Eq. (4.3) as the structure of a fixed-point merger at $s = 1$: the coupling vanishes as $\sqrt{\Delta_\sigma}$, two operators become degenerate at dimension 1, and their mixing is linear in the coupling, so the dimensions split as $\pm\sqrt{\Delta_\sigma}$. This is the pattern of a Kosterlitz–Thouless-type endpoint, consistent with the inverse-square Ising chain at $s = 1$ [20]. In $d = 2, 3$ the short-range end is instead an analytic crossover [10]: a single coupling $g_{\rm SR}$ with $\beta = -\delta g_{\rm SR} + b_0 g_{\rm SR}^3$ has $g^2_{{\rm SR}*} = \delta/b_0$, and since $\gamma_\mathcal{E} = a_0 g^2_{\rm SR} + \ldots$ is even in the coupling, $\Delta_\mathcal{E} = E + (a_0/b_0)\delta + O(\delta^2)$ is analytic in $\delta = x - S$; in $d = 2$ the linear term vanishes and Eq. (2.3) holds. The analytic forms (2.3)–(2.4) are inputs; the merger reading of $d = 1$ is ours.

Consequence for GFF-only resummation. Each finite-order Borel approximant of the ϵ series, even with the Borel-plane singularity built in, is analytic at $\epsilon = 1$ and cannot develop the infinite slope. The GFF-only CB8 curve in $d = 1$ reaches 0.567 at $\Delta_\sigma = 0$ instead of 1. In $d = 2, 3$, where the endpoint is analytic, the same GFF-only prescription reaches 1.0049 at $\Delta_\sigma = \frac{1}{8}$ (exact value 1, core band $[0.944, 1.071]$) and 1.4161 at $\Delta_\sigma = S$ (exact 1.4126, core band $[1.4124, 1.4203]$).

How much analytic structure can the GFF series carry? The test of Table 6 resums the GFF coefficients alone, without short-range input, in increasingly informed variables: ϵ; v, which removes the branch point (with $A_v = A_\epsilon/2$); $\lambda = 2v/(2-v)$, which also sends the mirror point $v = 2$ to infinity; and v with only the exact value $\Delta_\mathcal{E}(0) = 1$ imposed, $F = 1 + (1-v)\mathcal{R}[\cdot]$. Each step moves the center toward the $K = 8$ baseline: at $\Delta_\sigma = 0.1$ the deficit shrinks from -25.6×10^{-3} to -15.8×10^{-3}, -6.0×10^{-3} and $+0.6\times10^{-3}$. Their core bands, however, are 20 to 65 times wider than that of the $K = 8$ baseline at $\Delta_\sigma = 0.1$ (2.0×10^{-3}), and in v and λ they grow rapidly toward $\Delta_\sigma = 0$. Analytic structure helps the center; only the short-range coefficients control the spread.

Δ_σ	$K = 8$ baseline	GFF only, ϵ	GFF only, v	GFF only, λ	v with $\Delta_{\mathcal{E}}(0) = 1$
0.001	0.937424	−371.3 [188.0]	−237.9 [776.2]	−53.3 [3018]	−2.1 [45.7]
0.025	0.713432	−164.5 [144.9]	−104.6 [367.0]	−33.1 [1036]	−0.6 [108.0]
0.05	0.622235	−89.9 [106.0]	−56.4 [212.0]	−19.4 [510.2]	+0.9 [88.2]
0.1	0.530246	−25.6 [47.3]	−15.8 [71.7]	−6.0 [127.6]	+0.6 [42.8]
0.15	0.491712	−5.0 [13.6]	−3.2 [16.8]	−1.5 [21.6]	−0.2 [13.6]
0.2	0.483116	−0.3 [1.2]	−0.2 [1.4]	−0.1 [1.5]	−0.1 [1.3]

Table 6: GFF-only CB8 in different variables. The second column is the $K = 8$ baseline; the other columns give each GFF-only curve minus the baseline, in units of 10^{-3}, with the width of its core band (orders 6–8, $b_{\mathrm{L}} = 2, 3, 4$, $\alpha = -1, 0, 1$) in brackets, in the same units. At $\Delta_\sigma = 0.001$ the four GFF-only values are 0.566, 0.700, 0.884 and 0.935; the exact limit at $\Delta_\sigma = 0$ is 1.

4.5 Scans and what the bands mean

Table 7 lists the prescriptions. The *core band* is the pointwise minimum and maximum over orders $6, 7, 8$, three Borel–Leroy parameters and $\alpha = -1, 0, 1$ (27 approximants). In $d = 1$ the reported *sensitivity envelope* is the union of the core bands for $K = 5, 6, 7, 8$, so it also covers the number of imposed short-range coefficients. It starts at $K = 5$, one coefficient beyond the archived construction. Including $K = 4$ would widen it from at most 4.67×10^{-3} to at most 6.12×10^{-3} (near $\Delta_\sigma = 0.0975$). For $\Delta_\sigma \leq 0.04$ the $K = 4$ curve lies below the $K = 5$–8 envelope (by 5.8×10^{-5} at $\Delta_\sigma = 0.025$, Table 9). Every final HT and converged NRG value lies inside either envelope within its quoted error (Sec. 6.2). Wider scans (b_{L} from 0 to 6 or 7, α from -2 to 2) and action variations ($1/A$ scaled by 0.8, 0.9, 1.1, 1.2) are archived as diagnostics. No HT, NRG or MC value was used to choose these settings. In $d = 1$, however, the number $K = 8$ of imposed short-range coefficients, and with it the range $K = 5$–8 of the envelope, was fixed after these data had been seen (Sec. 5.2).

d	Baseline	Core b_{L}	PB control shown	Action in ϵ
1	CB8, endpoint-v, $K = 8$, $b_{\mathrm{L}} = 3$, $\alpha = 0$	$2, 3, 4$	PB[5/3], $b_{\mathrm{L}} = 3$ ($K \leq 7$)	0.5140649
2	CB8, curvature, $b_{\mathrm{L}} = 3.5$, $\alpha = 0$	$2.5, 3.5, 4.5$	PB8 [5/3], $b_{\mathrm{L}} = 3.5$	9/8
3	CB8, slope, $b_{\mathrm{L}} = 4$, $\alpha = 0$	$3, 4, 5$	PB7 [3/4], $b_{\mathrm{L}} = 4$	1.9234333

Table 7: Baseline prescriptions and core scans. The core scan always uses orders $6, 7, 8$ and $\alpha = -1, 0, 1$. In $d = 1$ PB[5/3] is admissible only for $K \leq 7$; in $d = 3$ every eighth-order PB candidate at $b_{\mathrm{L}} = 4$ is rejected, so the seventh-order control is shown.

5 Results

5.1 From three loops to eight loops and the endpoint

Table 8 shows, in each dimension, how the curve changes as information is added: CB3 uses only $C_0, \ldots, C_3$, the coefficients known before this work [7], in the GFF-only prescription; CB8 adds $C_4, \ldots, C_8$; the baseline adds the short-range end. In $d = 1$ the GFF series alone is far from the answer over most of the window. In $d = 2$ the eighth-order GFF-only curve is already within 5.2×10^{-3} of the baseline everywhere (5.24×10^{-3} at $\Delta_\sigma = 0.193$), and in $d = 3$ within 3.4×10^{-3} (at the endpoint; 1.6×10^{-3} at $\Delta_\sigma = 0.55$), while CB3 is off by up to 0.087 in $d = 2$.

d	Δ_σ	CB3, GFF only	CB8, GFF only	CB8 + short-range end (baseline)
1	0.05	0.458989	0.532338	0.622235
1	0.1	0.463610	0.504636	0.530246
1	0.15	0.470013	0.486725	0.491712
1	0.2	0.479926	0.482857	0.483116
2	0.125	0.913472	1.004921	1 (imposed)
2	0.2	0.921916	0.980233	0.985419
2	0.3	0.936589	0.960577	0.961182
2	0.4	0.958418	0.962702	0.962567
3	S	1.410634	1.416062	1.412625 (imposed)
3	0.55	1.418920	1.422866	1.421232
3	0.6	1.433555	1.435599	1.435241
3	0.65	1.450835	1.451576	1.451541

Table 8: Progression of the resummed $\Delta_{\mathcal{E}}$ with the information used. GFF-only curves resum F directly in ϵ with $\alpha = 0$ and the baseline b_{L} of each dimension.

5.2 One dimension

Figure 1 shows the baseline over the full interval and its residuals with respect to the comparison data, and Table 9 gives representative values. The curve falls from the imposed limit 1 at $\Delta_\sigma = 0$, reaches a shallow minimum 0.482813 at $\Delta_\sigma = 0.192$ and rises to the GFF value $\frac{1}{2}$. The sensitivity envelope is 1.45×10^{-3}, 3.64×10^{-3} and 4.40×10^{-3} wide at $\Delta_\sigma = 0.05$, 0.1, 0.15 and at most 4.67×10^{-3} wide (at $\Delta_\sigma = 0.134$); it is not drawn in the figures. The coefficient-allowance budget is at most 6.0×10^{-5} (at $\Delta_\sigma = 0.141$), far below it.

Δ_σ	Baseline CB8	$K = 5$–8 envelope	$K = 4$ (archived)	endpoint-ϵ, $K = 8$	GFF only
0.025	0.713432	[0.713301, 0.713615]	0.713243	0.713437	0.548904
0.05	0.622235	[0.621524, 0.622977]	0.621623	0.622296	0.532338
0.075	0.566627	[0.565044, 0.567868]	0.565583	0.566838	0.517472
0.1	0.530246	[0.527952, 0.531597]	0.528948	0.530651	0.504636
0.125	0.506428	[0.503992, 0.508580]	0.505179	0.506948	0.494227
0.15	0.491712	[0.489809, 0.494205]	0.490810	0.492177	0.486725
0.175	0.484240	[0.483202, 0.486233]	0.483803	0.484506	0.482703
0.2	0.483116	[0.482766, 0.484000]	0.483012	0.483189	0.482857
0.225	0.488085	[0.488058, 0.488177]	0.488081	0.488088	0.488077

Table 9: One-dimensional curves: CB8 with $b_{\mathrm{L}} = 3$, $\alpha = 0$ and the final $C_8(1)$. The envelope column is the union of the core bands for $K = 5$–8; the endpoint columns impose $C_1^{\mathrm{SR}}, \ldots, C_K^{\mathrm{SR}}$; the last column is the GFF-only curve.

Dependence on the number of short-range coefficients. The short-range input is the most important ingredient after the GFF series itself. The core band narrows from $K = 4$ to $K = 7$ and widens slightly at $K = 8$: at $\Delta_\sigma = 0.1$ its width is 6.11×10^{-3}, 3.64×10^{-3}, 2.33×10^{-3}, 1.52×10^{-3} and 2.00×10^{-3} for $K = 4, \ldots, 8$, and at $\Delta_\sigma = 0.05$ it falls from 3.4×10^{-3} ($K = 4$) to 2.3×10^{-4} ($K = 8$). The central value moves with K, non-monotonically, by up to 1.08×10^{-3} among $K = 5, \ldots, 8$ (near $\Delta_\sigma = 0.121$) and by up to 1.32×10^{-3} from $K = 4$ to $K = 8$. This K dependence is part of the envelope. Replacing C_8^{SR} by the independent value 242.83 moves the curve by at most 3×10^{-5}.

Endpoint variable and Padé controls. At $K = 8$ the endpoint-ϵ control differs from the v baseline by at most 5.2×10^{-4} (at $\Delta_\sigma = 0.13$) and stays inside the envelope. The matched PB[5/3]

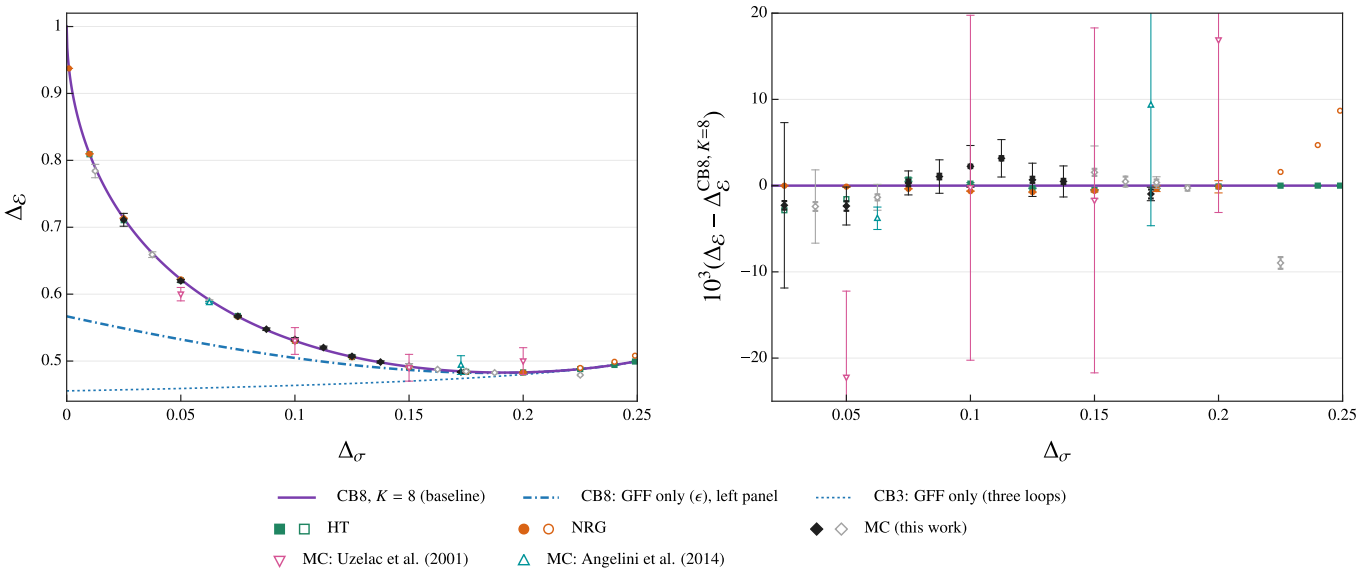


Figure 1: One dimension. Left: $\Delta_\mathcal{E}$ over the full interval. Purple: baseline (CB8, endpoint-v, $K = 8$). Blue dash-dot: GFF-only CB8. Blue dotted: CB3, the GFF-only curve from $C_0, \ldots, C_3$. Right: residuals $10^3(\Delta_\mathcal{E} - \Delta_\mathcal{E}^{\mathrm{baseline}})$ for $0.02 \le \Delta_\sigma \le 0.25$. Green squares: HT [4] (filled: final; at $\Delta_\sigma = 0.075$ one-sided, with a dashed extra bar of -5×10^{-4}; open: unstable or no limit, value at cutoff $N = 40$). Orange circles: NRG [5] (open: not converged in chain depth, no error, not compared). Diamonds, labeled "MC (this work)" as in Ref. [1]: the project's MC [6] (filled black: accepted; open light gray: other labels), with separate statistical and systematic bars. Triangles: published MC with quoted errors, one color per reference (rose, inverted: Uzelac et al. [21]; teal: Angelini et al. [22]); the right panel extends to -25 and $+20$ for them, and three of their bars are cut at the frame (full bars in Fig. 3). Labels and errors as in Table 12. The envelope and the PB controls are not drawn.

control ($b_\mathrm{L} = 3$, same F_K^{SR} and variable) is admissible only for $K = 4, \ldots, 7$; at $K = 8$ and order 8 only [3/5] survives the pole screen, and no $K = 8$ PB curve is substituted. The matched CB−PB difference changes sign and size with K; its signed extreme is -5.5×10^{-4}, $+3.4 \times 10^{-4}$, -4.6×10^{-5} and -1.5×10^{-4} for $K = 4, 5, 6, 7$ (Fig. 4, top). The CB/PB comparison is therefore a comparison of families, not evidence for either.

The choice $K = 8$. We impose $K = 8$ because these are all the short-range coefficients we have. This choice, and the range $K = 5$–8 of the envelope, were not fixed before the comparison data had been seen. Choosing the K with the narrowest core band (with an admissible PB control) would give $K = 7$, and the comparison data do not favor $K = 8$ over $K = 4$ (Sec. 6.2). Whether the envelope should also include the choice of endpoint variable is open; at present the endpoint-ϵ curve lies inside it.

5.3 Two and three dimensions

Figure 2 shows the baselines with the GFF-only curves and the comparison data; Tables 10 and 11 give values.

Two dimensions. The curvature baseline has a minimum $\Delta_\mathcal{E} = 0.958072$ at $\Delta_\sigma = 0.346$. Adding C_8 changed the seventh-order curve by at most 8.29×10^{-4}, and the matched PB8 control ([5/3], $b_\mathrm{L} = 3.5$) differs from CB8 by at most 7.88×10^{-4}; at $b_\mathrm{L} = 0$ every eighth-order PB candidate has a positive-real pole. The propagated coefficient allowance is at most 7.75×10^{-5}. The largest uncertainty is the endpoint construction (Table 10). The tangent and value constructions differ from the baseline by up to 5.7×10^{-3} and 7.5×10^{-3}. The tangent construction lies outside the baseline's core band for all $\Delta_\sigma < 0.476$, the value construction for $\Delta_\sigma < 0.26$; for $\Delta_\sigma < 0.29$ the

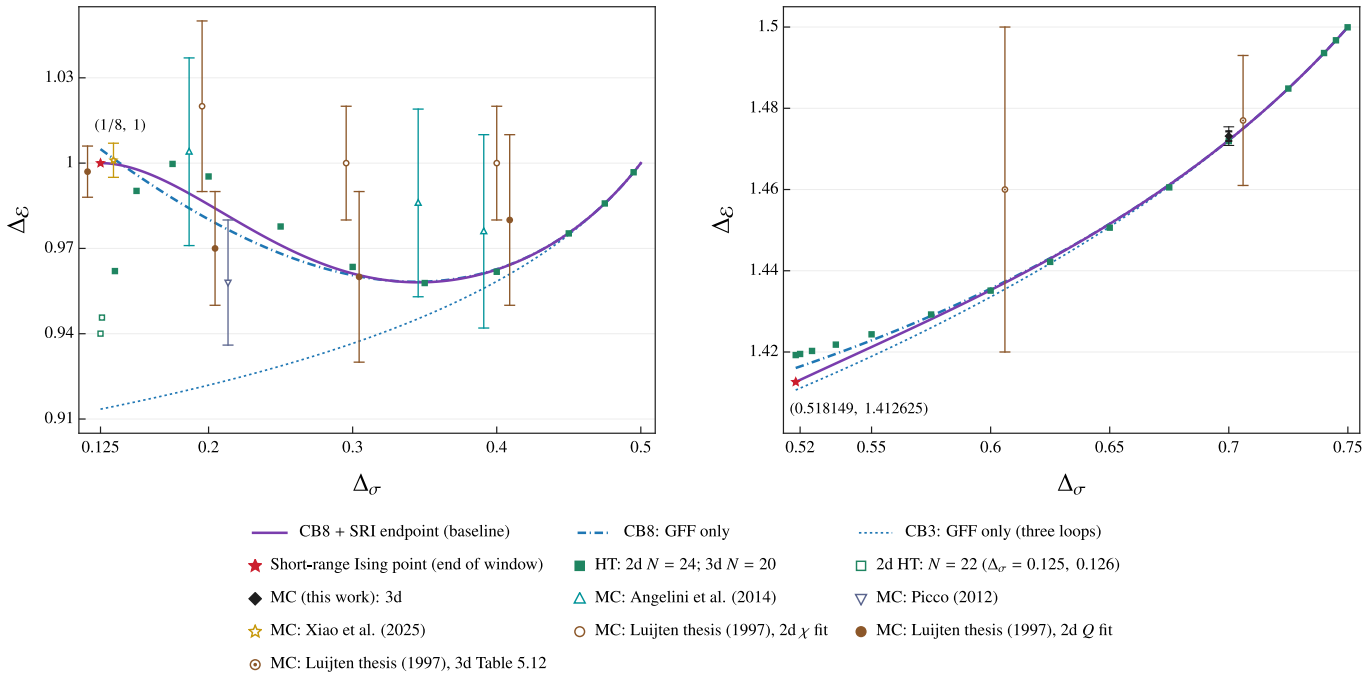


Figure 2: Two dimensions (left) and three dimensions (right). Purple: baseline. Blue dash-dot: GFF-only CB8. Blue dotted: CB3. Green squares: finite-cutoff HT [4] (2d $N = 24$; open: $N = 22$ at $\Delta_\sigma = 0.125, 0.126$, where the tuning has no solution at $N = 24$; 3d $N = 20$), without error bars. Black diamond: the project's MC value in 3d [6], accepted only under amendment P1 (Table 12). Red star: the short-range Ising point, where the baseline is pinned: (1/8, 1) in 2d and (0.518149, 1.412625) in 3d [11]. Other markers: published MC with quoted errors, one color and shape per reference (2d: Angelini et al. [22], teal triangles; Picco [24], slate inverted triangle; Xiao et al. [25], ochre star; the χ (open) and Q (filled) estimators of Luijten's thesis [23], brown circles; 3d: Luijten's thesis, brown circles with a central dot), shifted horizontally by at most 0.0135 (2d) and 0.006 (3d) where they share a Δ_σ. Not drawn: the core bands of the baselines (within -4.07×10^{-3} and $+1.23 \times 10^{-3}$ of the 2d curve, -7.24×10^{-5} and $+8.09 \times 10^{-5}$ of the 3d curve); the 2d value and tangent constructions; the value of Angelini et al. at $s = \frac{7}{4}$ (1.02 ± 0.10).

Δ_σ	curvature CB8	core band	tangent CB8	value CB8	GFF-only CB8	PB8 [5/3]
0.15	0.997797	[0.997747, 0.997823]	0.998560	0.992255	0.995913	0.997816
0.2	0.985419	[0.984485, 0.985834]	0.989418	0.978404	0.980233	0.985708
0.25	0.971143	[0.968443, 0.972158]	0.976830	0.967441	0.968198	0.971821
0.3	0.961182	[0.957192, 0.962396]	0.965863	0.960349	0.960577	0.961953
0.35	0.958099	[0.954550, 0.958909]	0.960426	0.958296	0.958332	0.958584
0.4	0.962567	[0.960930, 0.962810]	0.963108	0.962704	0.962702	0.962702
0.45	0.975429	[0.975260, 0.975440]	0.975448	0.975436	0.975436	0.975434

Table 10: Two dimensions: the three endpoint constructions and the GFF-only curve (CB8, $b_{\mathrm{L}} = 3.5$, $\alpha = 0$), the core band of the baseline, and the PB control PB8 [5/3] of the curvature construction.

larger of the two deviations exceeds the width of the core band. Varying $1/A_\epsilon$ by $\pm 20\%$ moves the baseline by at most 1.7×10^{-3}.

Δ_σ	slope CB8	core band	value CB8	GFF-only CB8	PB7 [3/4]	PB8 [5/3]
0.55	1.421232	[1.421196, 1.421266]	1.421119	1.422866	1.421226	1.421225
0.575	1.428068	[1.428002, 1.428137]	1.427959	1.428886	1.428059	1.428057
0.6	1.435241	[1.435170, 1.435321]	1.435165	1.435599	1.435231	1.435229
0.65	1.451541	[1.451512, 1.451582]	1.451526	1.451576	1.451537	1.451537
0.7	1.472143	[1.472141, 1.472146]	1.472142	1.472143	1.472143	1.472143

Table 11: Three dimensions: the slope (baseline) and value constructions and the GFF-only curve (CB8, $b_{\mathrm{L}} = 4$, $\alpha = 0$), the core band of the baseline, and PB controls of the slope construction: PB7 at $b_{\mathrm{L}} = 4$ and PB8 at $b_{\mathrm{L}} = 0$, since at order 8 and $b_{\mathrm{L}} = 4$ every PB candidate is rejected.

Three dimensions. The slope baseline increases monotonically across the window. Adding C_8 changed the seventh-order curve by at most 1.13×10^{-5}; the admissible PB8 at $b_{\mathrm{L}} = 0$ ([5/3]) differs by at most 1.21×10^{-5}, and the propagated coefficient allowance is at most 1.66×10^{-5}. The estimated range of the short-range slope, $c_1 \in [0.269, 0.276]$ (status conditional), moves the curve by at most 6.97×10^{-5}, comparable to the width of the core band; the value construction stays within 1.2×10^{-4} of the baseline. In $d = 3$ the resummation is therefore tightly constrained; its uncertainty is dominated by the short-range inputs and by assumptions shared by all prescriptions.

6 Comparison with HT, NRG and Monte Carlo

6.1 The comparison data and their status

The comparison values come from the HT [4], NRG [5] and MC [6] notes of the release of Ref. [1] and from the literature; frozen copies ship with this note (Appendix B). None of them enters a curve. Their uncertainties have different meanings, summarized in Table 12, and their labels (final, converged, accepted and so on) were assigned in those notes by their own criteria, not by comparison with the curves of this note. The errors must not be combined as independent Gaussian standard deviations: below, a difference is compared with each quoted error separately or with the linear sum of the errors involved, never with a quadrature sum, and the ratios are descriptive, not χ^2 statistics.

6.2 One dimension

Figure 3 shows all 48 displayed comparison points relative to the baseline (the unconverged HT point at $\Delta_\sigma = 0.001$ is omitted). With $K = 8$:

- The final HT values lie within -5.0×10^{-4} to $+2.3 \times 10^{-4}$ of the baseline for $0.1 \le \Delta_\sigma \le 0.249$; the one-sided point at $\Delta_\sigma = 0.075$ is $+7.0 \times 10^{-4}$ above.
- The final NRG values ($0.001 \le \Delta_\sigma \le 0.2$) lie 1×10^{-7} to 7.6×10^{-4} below the baseline, the largest offset at $\Delta_\sigma = 0.125$.
- The nine accepted MC values lie within -2.4×10^{-3} to $+3.2 \times 10^{-3}$.

The $K = 5$–8 envelope contains every final and one-sided HT value and every final NRG value within its quoted error; all but two lie inside it, and HT at $\Delta_\sigma = 0.249$ and NRG at 0.001 lie 4×10^{-8} and 9×10^{-8} outside. Every accepted MC value reaches the envelope within the sum of its statistical error and σ_{sys}; the largest distance is 0.60 of that sum (at $\Delta_\sigma = 0.05$). All of this also holds for the envelope extended to $K = 4$. Among the points with other labels, the conditional MC

Data	Meaning and status
HT, $d = 1$	Hamiltonian truncation corrected for the truncation artifact, with the zero mode in a density-matrix basis, extrapolated to infinite cutoff from $N \leq 40$ [4]. *Final* ($0.1 \leq \Delta_\sigma \leq 0.249$): a stable extrapolation; error = half the spread of the accepted fits, a model spread. *One-sided* ($\Delta_\sigma = 0.075$): the local exponent still rises, so the fits may overestimate the limit; extra allowance -5×10^{-4}. *Unstable* ($\Delta_\sigma = 0.05$): the fit families disagree; no value quoted. *No limit* ($\Delta_\sigma \leq 0.025$): the corrected sequence turns over and keeps falling; the value at $N = 40$ is shown. The error does not include the dependence on how the artifact is corrected (Sec. 6.2).
HT, $d = 2, 3$	Finite-cutoff values only [4] (2d: $N = 24$, $N = 22$ near $\Delta_\sigma = \frac{1}{8}$; 3d: $N = 20$). Solver residuals and last cutoff steps are not continuum error bars.
NRG, $d = 1$	Bosonic NRG of the sub-ohmic spin–boson model [5]: the settled value of the flow, a plateau or (at $\Delta_\sigma = 0.175$, 0.2) an extrapolation in chain depth. Error: chain depth, spread over the numbers of kept states and oscillator levels, and discretization, added linearly; not a standard deviation. *Final* (converged; $0.001 \leq \Delta_\sigma \leq 0.2$); *not converged in chain depth* ($\Delta_\sigma = 0.225$, 0.24, 0.249, near $s = \frac{1}{2}$): settled value shown without error, not compared.
Project MC	Cluster MC of the lattice model [6]; σ_{stat} is a bootstrap error, σ_{sys} half the range of the values over a family of admissible fits or extrapolations, a sensitivity scale, not a standard deviation. Labels from a fixed decision table: *accepted*, the value passed every check of the fit protocol (σ_{sys}*-dominated*: with σ_{sys} more than three times the median σ_{stat}); *conditional*, a single extrapolation, so no σ_{sys}; *cross-check unavailable*, fewer than two independent extrapolations pass their gates; *cross-check failed*, two independent extrapolations differ by more than twice the error of their difference; *model-limited*, no fit passes. $d = 1$: 17 values, 9 accepted. $d = 3$: one value, at $\Delta_\sigma = 0.7$, accepted only under amendment P1, which re-profiles the correction exponent in every bootstrap draw of σ_{stat} and was adopted after the protocol had been applied to the production data. Where no infinite-size value exists, the finite-size exponent $\Delta_{\text{eff}}(U)$ of the two largest lattices, from the slopes of the Binder cumulant U at its crossing for sizes L and $2L$, is shown.
Published MC	Quoted errors. $d = 1$: Uzelac et al. [21] (4 points), Angelini et al. [22] (2). $d = 2$: 13 values from Refs. [22, 23, 24, 25]. $d = 3$: Luijten's thesis [23] (2).
Bootstrap, FRG	$d = 3$ only, read off published figures [26, 27]; we assign ± 0.005 to the bootstrap kinks; not plotted.

Table 12: Comparison data, their uncertainty conventions and labels.

value at $\Delta_\sigma = 0.225$ lies 9.0×10^{-3} below, and the NRG points not converged in chain depth rise to 8.7×10^{-3} above at $\Delta_\sigma = 0.249$. Of the published MC values, four lie within one quoted error of the envelope, while Uzelac et al. at $\Delta_\sigma = 0.05$ and Angelini et al. at $\Delta_\sigma = 0.0625$ lie 2.2 and 2.0 quoted errors below it.

Agreement among the methods. Away from the GFF end, HT, NRG and MC agree at the sub-percent level. The largest relative difference among accepted MC, final HT and converged NRG values at a common coupling is 0.54%, between MC and NRG at $\Delta_\sigma = 0.1$, where HT and MC differ by 0.38%. HT and NRG agree to within two parts in a thousand wherever HT has a continuum limit ($0.075 \le \Delta_\sigma \le 0.2$; the largest relative difference, 1.9×10^{-3}, is at the one-sided point $\Delta_\sigma = 0.075$). Near the GFF end the MC values at finite size lie 0.006 to 0.010 below the resummed curve: the exponents $\Delta_{\mathrm{eff}}(U)$ of the largest lattice pair at $\Delta_\sigma = 0.2125$ and 0.2375 (Table 18) and the conditional fit at 0.225. There the ϵ series is accurate and these MC values are not infinite-size estimates.

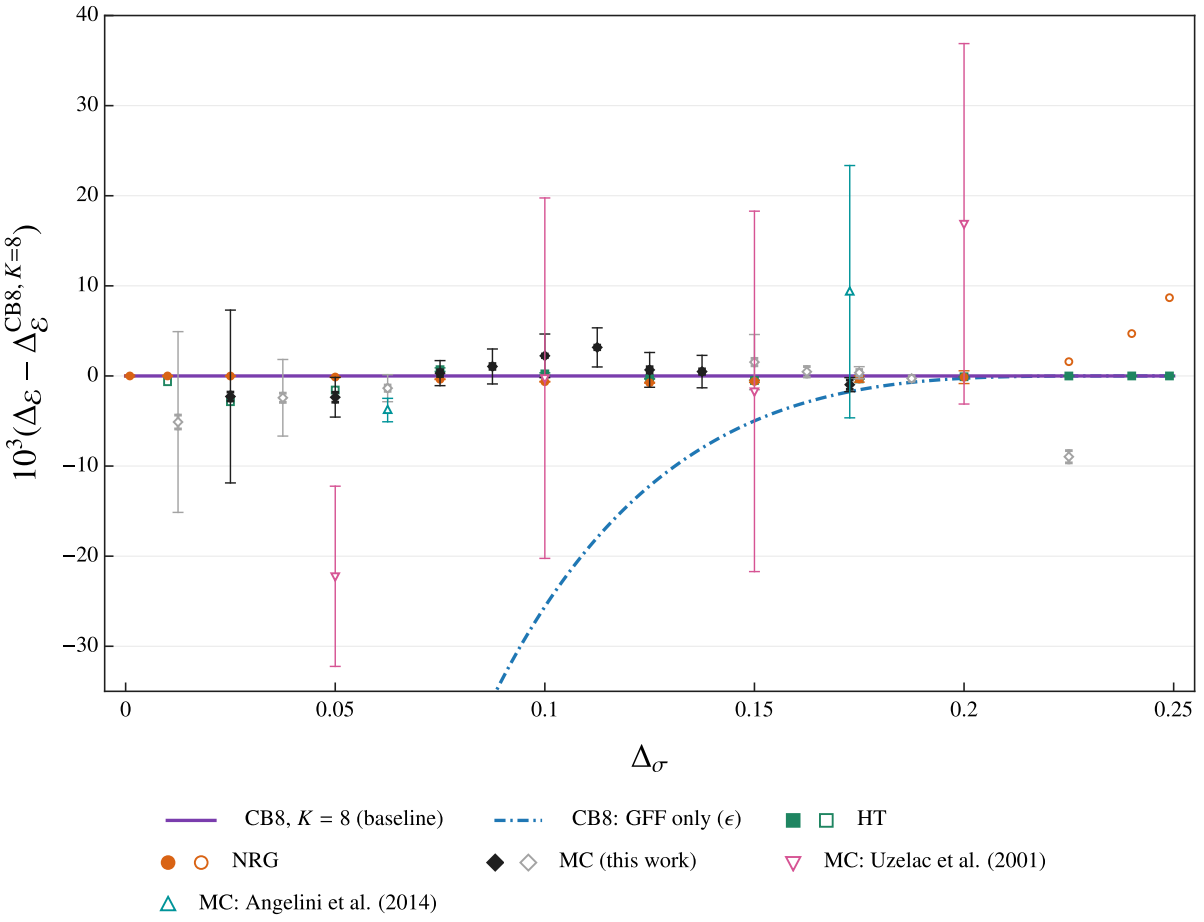


Figure 3: One dimension: deviations $10^3(\Delta_{\mathcal{E}} - \Delta_{\mathcal{E}}^{\mathrm{baseline}})$ of all displayed reference points from the $K = 8$ baseline, with full error bars, including the published MC values (rose inverted triangles: Uzelac et al. 2001; teal triangles: Angelini et al. 2014). Markers as in Fig. 1. Blue dash-dot: GFF-only CB8, below the frame for $\Delta_\sigma < 0.088$ (it reaches -433 at $\Delta_\sigma = 0$). The $K = 5$–8 envelope is not drawn; its width is at most 4.67×10^{-3}, that is 4.67 in these units.

HT and NRG. At $\Delta_\sigma = 0.15$–0.2 the final HT and NRG values agree (differences at most 1.0×10^{-4}, below the sum of their errors). At $\Delta_\sigma = 0.1$ and 0.125 the NRG values lie 8.7×10^{-4} and 4.5×10^{-4} below the HT values, 5.8 and 4.6 times the sum of the HT fit half-spread and the NRG error. The gap is of the size of the dependence of HT on the correction of its truncation artifact: without the correction the truncation extrapolates 2.3×10^{-3} and 6.6×10^{-4} lower at these

points (1.7×10^{-3} at 0.075), and NRG lies between the two limits. The HT note does not combine this dependence into its error, because the truncation has no strict error bar and an error built after the gap to NRG had been seen would make the two methods appear to agree [4]. The HT values therefore keep their fit half-spreads, and the difference between the corrected and uncorrected limits is shown separately.

Do the data select $K = 8$ over $K = 4$? The archived $K = 4$ and the baseline $K = 8$ curves differ by at most 1.32×10^{-3} (at $\Delta_\sigma = 0.109$). That is less than the width of the $K = 8$ core band at the same points (2.0×10^{-3} at $\Delta_\sigma = 0.1$, 3.0×10^{-3} at 0.125), so a preference between them is a choice inside the prescription dependence. Table 13 compares the separation with the quoted errors and lists the dependence of HT on the artifact correction separately.

- *MC* does not resolve the two curves anywhere: the separation is at most 0.59 of the sum of σ_{stat} and σ_{sys} of an accepted value.
- *HT.* At $\Delta_\sigma = 0.075$ and 0.1 HT lies above both curves, nearer $K = 8$, but its dependence on the artifact correction (1.7×10^{-3}, 2.3×10^{-3}) exceeds the separation of the curves (1.0×10^{-3}, 1.3×10^{-3}), so HT does not decide there. At 0.125 it lies between the curves, nearer $K = 8$. At 0.15–0.2, where the prescription dependence is small ($\leq 1.3 \times 10^{-4}$), it lies between the curves and nearer $K = 4$: 1×10^{-5} to 4×10^{-4} above $K = 4$ (2.6–12 errors) and 9×10^{-5} to 5×10^{-4} below $K = 8$ (15–21 errors).
- *NRG* selects $K = 8$ only at $\Delta_\sigma = 0.025$, where the separation is 1.9×10^{-4} and $K = 4$ is 6.2 errors away. At 0.05–0.15 it lies between the curves, more than 2.5 errors from each: nearer $K = 8$ at 0.05, 0.075 and, marginally, 0.1 (19.1 against 19.7 errors), nearer $K = 4$ at 0.125 and 0.15. At 0.175, 0.2 and $\Delta_\sigma \leq 0.01$ the separation is below two errors.

So the precise data resolve the $K = 4$–$K = 8$ difference at several points, but no method favors $K = 8$ over its range. Where both curves are resolved, HT and NRG mostly lie between them, and at 0.15–0.2, where HT is most precise and depends least on the artifact correction, they lie nearer $K = 4$. An earlier version of this note justified $K = 8$ by its agreement with HT and NRG. That argument relied on the HT fit half-spread alone at $\Delta_\sigma = 0.1$–0.125 and on earlier NRG values; it does not survive the dependence of HT on the artifact correction, which exceeds the separation at $\Delta_\sigma = 0.075$ and 0.1, and it is withdrawn. The choice $K = 8$ rests only on its a priori motivation: these are all the short-range coefficients we have, and at fixed K the prescriptions then spread much less than with $K = 4$ (core band at $\Delta_\sigma = 0.05$: 2.3×10^{-4} against 3.4×10^{-3}) and depend less on b_{L} (Secs. 5.2 and 7). It was fixed after the comparison data had been seen, and taken together these data do not favor it over $K = 4$.

6.3 Two dimensions

The finite-cutoff HT values [4] at the largest cutoff ($N = 24$ at eleven couplings, $N = 22$ at $\Delta_\sigma = 0.125$ and 0.126) compare with the curvature baseline as follows (Fig. 2, left):

- At weak coupling, $\Delta_\sigma \geq 0.35$ ($s \leq 1.3$), HT agrees with CB8 to at most 8×10^{-4}, inside the core band.
- In mid-window HT lies above the baseline: $+2.3 \times 10^{-3}$ at $\Delta_\sigma = 0.3$, $+6.6 \times 10^{-3}$ at 0.25, $+9.9 \times 10^{-3}$ at 0.2 and $+7.3 \times 10^{-3}$ at 0.175. At $s = 1.4$–1.5 ($\Delta_\sigma = 0.3$–0.25) HT lies within the spread of the endpoint constructions, inside the core band of the tangent construction. At $s = 1.6$–1.65 ($\Delta_\sigma = 0.2$–0.175) HT is above every construction, by about 3×10^{-3} above the upper edge of the tangent core band. The cutoff dependence does not show whether this excess survives at infinite cutoff. At $s = 1.4$ and 1.5 HT rises monotonically with N, so if the rise continues the $N = 24$ values are lower bounds. At $s = 1.6$, however, it is flat from $N = 18$ on and falls by 5×10^{-5} in the last step, and at $s = 1.65$ it falls steadily from $N = 10$

		HT (corrected)					NRG			
Δ_σ	$10^4(F_4-F_8)$	$10^4(y-F_8)$	err	z_8, z_4	corr.−unc.		$10^4(y-F_8)$	err	z_8, z_4	
0.025	−1.89	·	·	·	·		−0.10	2.9×10^{-5}	−0.4, +6.2	near 8
0.05	−6.12	·	·	·	·		−1.19	2.7×10^{-5}	−4.5, +18.4	b(8)
0.075	−10.43	+6.99	2.0×10^{-4}	+3.5, +8.8	1.7×10^{-3}	o(8)†	−3.67	3.0×10^{-5}	−12.3, +22.7	b(8)
0.1	−12.99	+2.31	1.2×10^{-4}	+2.0, +13.2	2.3×10^{-3}	o(8)†	−6.39	3.3×10^{-5}	−19.1, +19.7	b(8)
0.125	−12.48	−3.09	4.3×10^{-5}	−7.2, +21.8	6.6×10^{-4}	b(8)	−7.55	5.4×10^{-5}	−14.1, +9.2	b(4)
0.15	−9.01	−5.03	3.3×10^{-5}	−15.4, +12.2	1.3×10^{-4}	b(4)	−6.06	1.1×10^{-4}	−5.3, +2.6	b(4)
0.175	−4.37	−3.32	1.5×10^{-5}	−21.4, +6.8	3.3×10^{-5}	b(4)	−3.40	3.5×10^{-4}	−1.0, +0.3	n.r.
0.2	−1.04	−0.92	4.7×10^{-6}	−19.4, +2.6	4.5×10^{-6}	b(4)	−1.36	7.1×10^{-4}	−0.2, −0.0	n.r.

Table 13: Can the comparison data tell the $K = 4$ and $K = 8$ curves apart? F_K: CB8 with the short-range series through C_K^{SR}; y: the HT or NRG value; $z_K = (y - F_K)/\mathrm{err}$. HT err: half-spread of the accepted fits (quoted); at $\Delta_\sigma = 0.075$ its one-sided allowance -5×10^{-4} is not included and would lower z_8 to 1.0. corr.−unc.: the HT limit corrected for the truncation artifact minus the uncorrected limit, shown separately and not combined with err (Sec. 6.2). NRG err: chain-depth, kept-state and discretization errors added linearly. Labels: near 8, within one error of F_8 and more than two from F_4; b(K), between the curves, and o(K), outside both, more than one error from each and nearer F_K; n.r., separation below two errors. †: the dependence on the artifact correction exceeds the separation. MC resolves neither curve anywhere. The errors are not standard deviations; the ratios are descriptive.

on (by 5.3×10^{-3} up to $N = 24$, 1.0×10^{-3} in the last step), so larger cutoffs could reduce the excess there.

- Near the short-range end ($\Delta_\sigma \leq 0.15$) HT drifts downward, away from the exact endpoint value 1 (0.940 at $\Delta_\sigma = \frac{1}{8}$, $N = 22$), and is not usable.

The discrepancy is not assigned to either method: the cutoff dependence and the artifact correction of HT, the uncomputed orders and the endpoint assumptions all remain. The thirteen published MC values have quoted errors from 0.006 to 0.034 (and 0.10 for Angelini et al. at $s = \frac{7}{4}$). Away from $s = \frac{7}{4}$ they lie within two quoted errors of both CB8 and HT, so they cannot separate them; at $s = \frac{7}{4}$ they agree with the exact value 1, which HT misses. Luijten's two estimators on the same data differ by up to 0.05. The project's own 2d MC values are conditional or model-limited [6], so only finite-size exponents are available (Table 14). The exponents $\Delta_{\mathrm{eff}}(U_4)$ of the largest lattice pair ($L = 512 \to 1024$) agree with the curvature baseline to 1.5×10^{-3} at $s = 1.2$–1.4. There the endpoint constructions differ from it by up to 4.7×10^{-3}, so the exponents do not select one. Toward both ends ($s = 1.05$, 1.1 and 1.5–1.7) they fall short of the baseline by 0.005 to 0.02.

s	Δ_σ	label	$\Delta_{\mathrm{eff}}(U_4)$	curvature CB8	10^3(MC−CB8)	10^3 max |constr. − CB8|
1.05	0.475	conditional	0.9702(25)	0.985836	−15.65	0.00
1.1	0.45	conditional	0.9659(23)	0.975429	−9.51	0.02
1.2	0.4	conditional	0.9631(22)	0.962567	+0.56	0.54
1.3	0.35	model-limited	0.9584(20)	0.958099	+0.28	2.33
1.4	0.3	model-limited	0.9627(20)	0.961182	+1.54	4.68
1.5	0.25	model-limited	0.9661(18)	0.971143	−5.03	5.69
1.6	0.2	model-limited	0.9698(18)	0.985419	−15.66	7.02
1.7	0.15	model-limited	0.9773(17)	0.997797	−20.49	5.54

Table 14: Two dimensions: finite-size exponents $\Delta_{\mathrm{eff}}(U_4)$ of the project's MC for the largest lattice pair, $L = 512 \to 1024$, with their statistical errors [6], against the curvature baseline. Last column: the largest deviation of the tangent and value constructions and the GFF-only curve from the baseline. These are not infinite-size values.

6.4 Three dimensions

Against the slope baseline (Fig. 2, right):

- HT at $N = 20$ is within 1×10^{-3} of CB8 for $\Delta_\sigma \geq 0.6$ ($s \leq 1.8$), for example -2.2×10^{-4} at $\Delta_\sigma = 0.7$ and -9.3×10^{-4} at $\Delta_\sigma = 0.65$. Toward the short-range end it lies above, by 1.2×10^{-3} at $\Delta_\sigma = 0.575$ rising to 6.6×10^{-3} at $\Delta_\sigma = S$, where HT gives 1.41924 against the exact $E = 1.41262528$; as in $d = 2$, HT fails near the short-range end, here lying above the curve rather than below it.
- The project's MC value at $s = 1.6$ [6], $\Delta_\mathcal{E} = 1.47314 \pm 0.00118\,(\text{stat}) \pm 0.00230\,(\text{sys})$, lies 1.0×10^{-3} above CB8 (1.47214). It is accepted only under amendment P1 (Table 12), adopted after the protocol had been applied to the production data; σ_{sys} is the half-range of the values over the admissible extrapolation series.
- The project's extrapolations that are not accepted, at $s = 1.7$, 1.85 and 1.9 (cross-check failed or unavailable), lie 0.022 to 0.033 above CB8.
- Luijten's thesis gives 1.477(16) at $s = 1.6$ and 1.46(4) at $s = 1.8$, +0.3 and +0.6 quoted errors from CB8; Luijten calls these results rather poor (lattices up to $L = 64$, correction exponent fixed by hand).
- Bootstrap kinks read off Fig. 11 of Ref. [26] (conjectured identification) give 1.470, 1.450, 1.432, 1.415 at $s = 1.593$, 1.685, 1.778, 1.871 (read-off error about ± 0.005), below CB8 by 0.004, 0.004, 0.007, 0.010. The gap grows toward the short-range end, where CB8 imposes $c_1 = 0.273$; the kink at $s = 1.871$ implies a smaller slope. A functional-RG truncation [27] departs by up to 0.03 near the short-range end, where its thermal exponent, about 1.543 as read off its Fig. 2, stays below the 3d Ising value 1.5874.

7 Robustness

7.1 Held-out coefficients

A finite-order test that uses no comparison data is to withhold the last known coefficient, construct the approximant from the lower ones and re-expand it to predict the withheld one. This tests finite-order predictive power, not asymptotic dominance. Tables 15 and 16 collect the results; Fig. 4 shows the one-dimensional ones together with the matched CB−PB differences.

Construction ($b_\text{L} = 3$)	C_6	C_7	C_8
GFF only, CB	11.03	1.58	1.55
GFF only, PB ($[3/2], [4/2], [5/2]$)	5.70	0.44	0.33
Endpoint-v, $K = 8$, CB	10.95	5.93	0.94
Endpoint-v, $K = 8$, PB ($[4/1], [4/2], [3/4]$)	3.86	0.17	0.50
Endpoint-v, $K = 4$, CB	7.23	0.37	0.81
Endpoint-v, $K = 4$, PB ($[3/2], [4/2], [5/2]$)	2.53	0.16	0.44

Table 15: One dimension: absolute relative errors (%) of held-out coefficients. The targets C_6, C_7 are unchanged; for C_8 the final value is the target, and the predictions themselves do not depend on it.

Different orders, dimensions and constructions favor different families. For the newly measured C_8, the GFF-only PB prediction is closer in $d = 1$ and $d = 2$, CB is closer in $d = 3$. In $d = 1$ the $K = 8$ CB prediction of C_8 is 0.94% off, which is 4.4 times the allowance of $C_8(1)$ (0.21%). An accurate PB prediction of C_8 also does not guarantee an admissible eighth-order PB curve of the matched shape once C_8 is supplied: at $K = 8$ in $d = 1$ only [3/5] survives at eighth order, and in $d = 3$ no eighth-order candidate survives at $b_\text{L} = 4$. These tests support keeping both families and

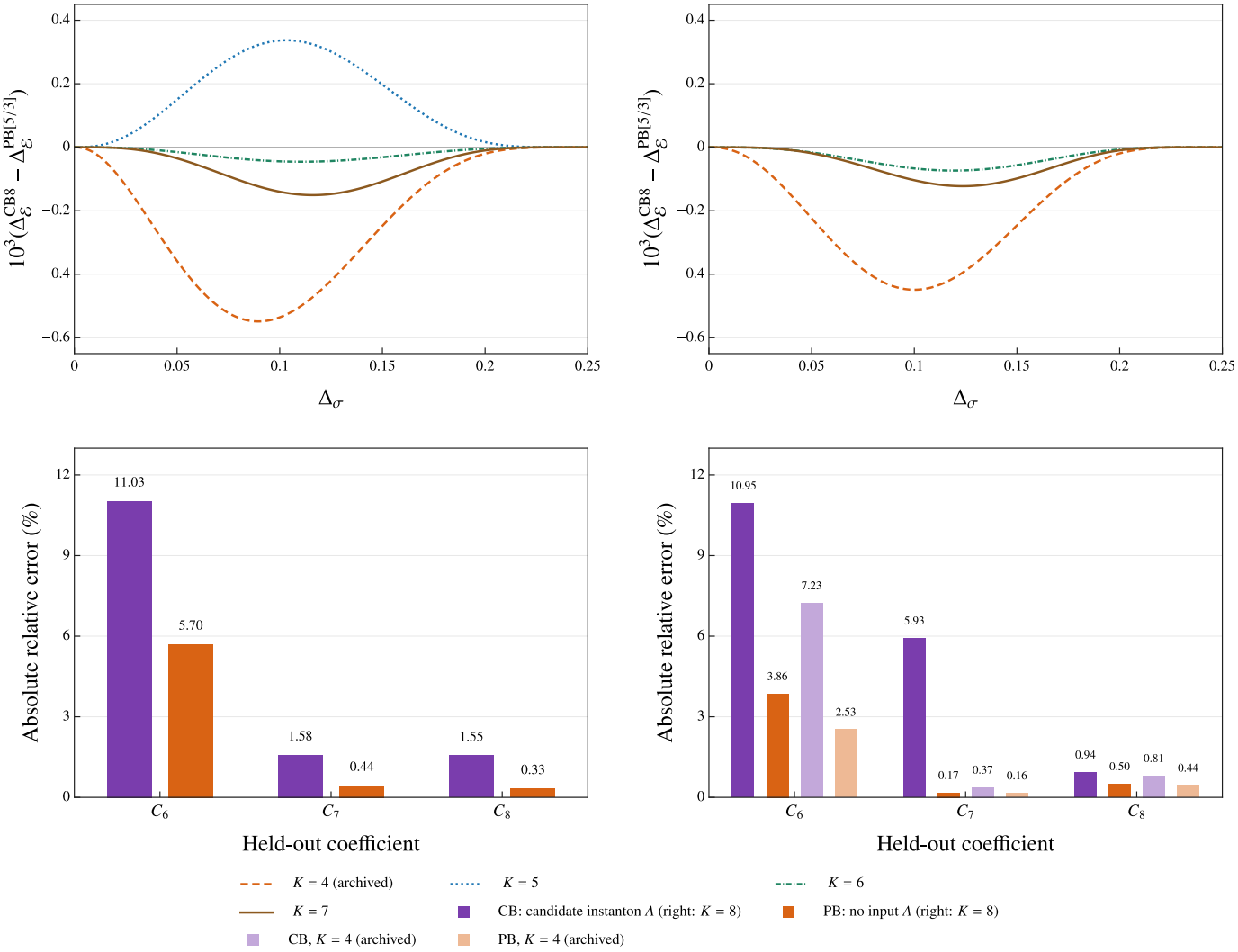


Figure 4: One dimension: conformal Borel against Padé–Borel. Top: matched differences $10^3(\Delta_{\mathcal{E}}^{\mathrm{CB8}} - \Delta_{\mathcal{E}}^{\mathrm{PB}[5/3]})$ with the same coefficients, endpoint input F_K^{SR}, order, variable and $b_{\mathrm{L}} = 3$; left in v for $K = 4$ (archived), 5, 6, 7; right in ϵ for $K = 4, 6, 7$ (where PB[5/3] is admissible). Bottom: absolute relative errors of held-out C_6, C_7, C_8 predicted from the preceding coefficients; left for the GFF-only series (independent of K), right for the endpoint-v construction with $K = 8$ (dark) and $K = 4$ (light). PB shapes as selected by the rule of Sec. 4.1 ($K = 8$: [4/1], [4/2], [3/4]; $K = 4$ and GFF-only: [3/2], [4/2], [5/2]). The C_8 bars use the final $C_8(1)$ (allowance about 0.21%).

	$d = 2$				$d = 3$			
	GFF only		with SR input		GFF only		with SR input	
Target	CB	PB	CB	PB	CB	PB	CB	PB
C_5	7.75	37.99	44.88	2.26	1.67	1.56	9.72	7.00
C_6	9.25	5.16	22.31	4.57	4.32	2.42	0.80	18.25
C_7	0.81	0.18	5.82	6.93	1.64	1.16	1.99	0.10
C_8	1.24	0.22	2.99	0.31	0.37	0.52	0.33	0.16

Table 16: Two and three dimensions: absolute relative errors (%) of held-out coefficients at the baseline b_{L}. GFF only: the series resummed directly; with SR input: the curvature ($d = 2$) and slope ($d = 3$) constructions. PB shapes as selected by the rule of Sec. 4.1.

the explicit admissibility diagnostics. They do not validate an instanton amplitude or establish an advantage of supplying the action.

7.2 Order and Borel–Leroy stability

Figure 5 shows the order and b_{L} dependence in $d = 1$. At $K = 8$ the CB curves at orders 6 and 7 differ from CB8 by at most 8.8×10^{-4} and 2.2×10^{-4}; changing b_{L} to 2 or 4 moves the curve by at most 1.4×10^{-4}, while $b_{\mathrm{L}} = 0$ moves it by up to 2.9×10^{-3} (at $K = 4$: 1.8×10^{-2}). The short-range input thus also stabilizes the Borel–Leroy dependence. Varying $1/A_\epsilon$ by $\pm 20\%$ moves the $K = 8$ baseline by at most 1.0×10^{-3} (scale 0.8), and by at most 1.6×10^{-4} for the other variations. In $d = 2$ and 3, adding C_8 changed the curves by at most 8.3×10^{-4} and 1.1×10^{-5} (Sec. 5.3). Small order-to-order changes are stability observations, not an estimate of the total error.

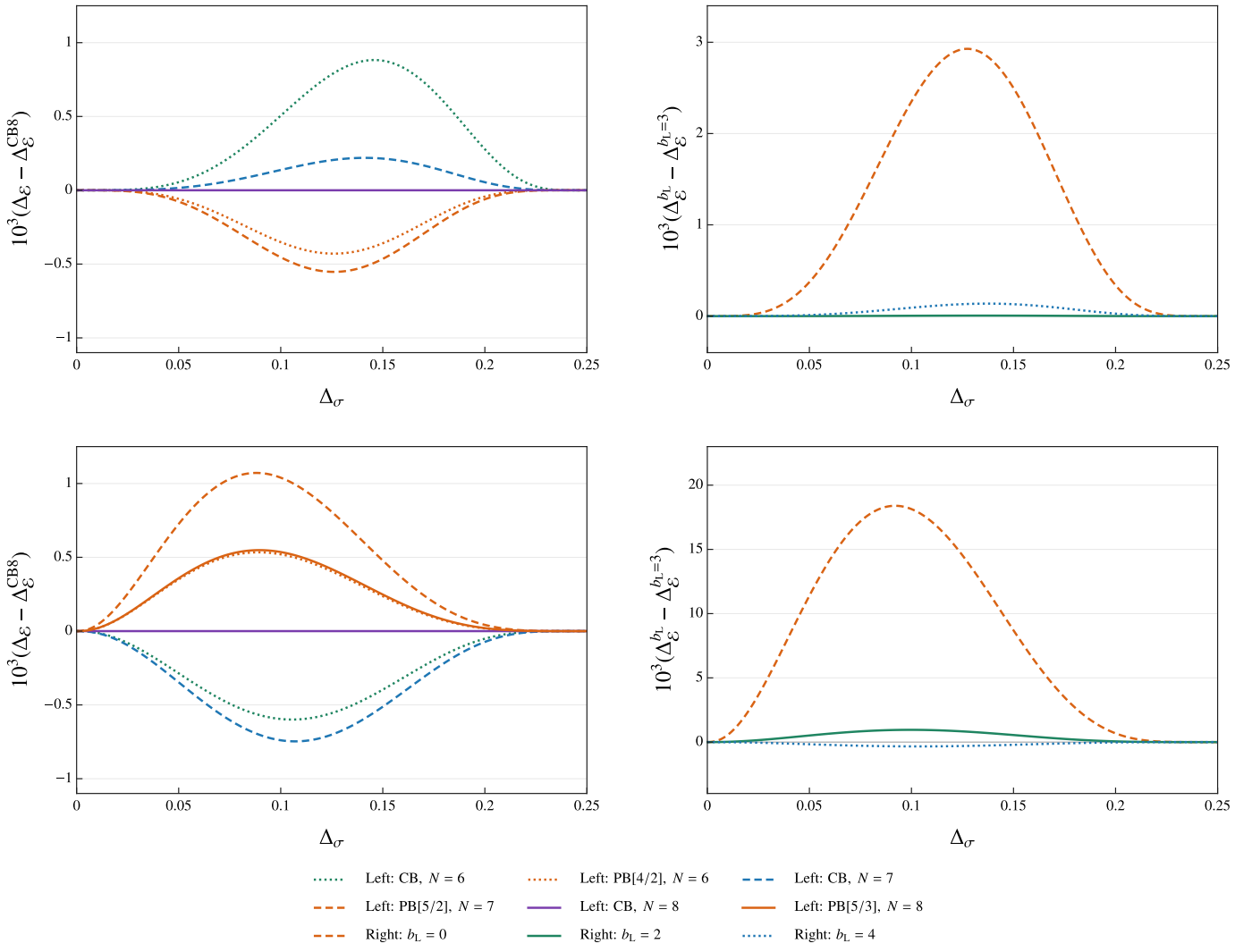


Figure 5: One dimension: finite-order stability. Top row: $K = 8$ baseline; bottom row: the archived $K = 4$ construction (with the final $C_8(1)$). Left: deviations $10^3(\Delta_{\mathcal{E}} - \Delta_{\mathcal{E}}^{\mathrm{CB8}})$ of CB at orders 6, 7 and of the PB controls [4/2], [5/2], [5/3] ($b_{\mathrm{L}} = 3$, drawn only where admissible; at $K = 8$ and order 8 no PB is drawn). Right: deviations of CB8 at $b_{\mathrm{L}} = 0, 2, 4$ from $b_{\mathrm{L}} = 3$. Vertical ranges differ between the right panels.

7.3 Coefficient precision, the final $C_8(1)$ and a ninth coefficient

For fixed CB parameters the reconstruction is affine in the input coefficients, so coefficient sensitivity can be computed exactly from unit responses $J_n = \partial F/\partial C_n$. We report $U = \sum_{n=6}^{8} |J_n|\, \delta C_n$, the sum of absolute responses to the allowances. This would be a worst-case bound only if the allowances were rigorous simultaneous bounds, which they are not; with a covariance matrix Σ the variance would be $J^{\mathsf{T}}\Sigma J$. Four sources of uncertainty must be kept separate: numerical integration (negligible), input coefficients (U), prescription and endpoint construction (the bands), and the theoretical assumptions about the endpoint and the instanton.

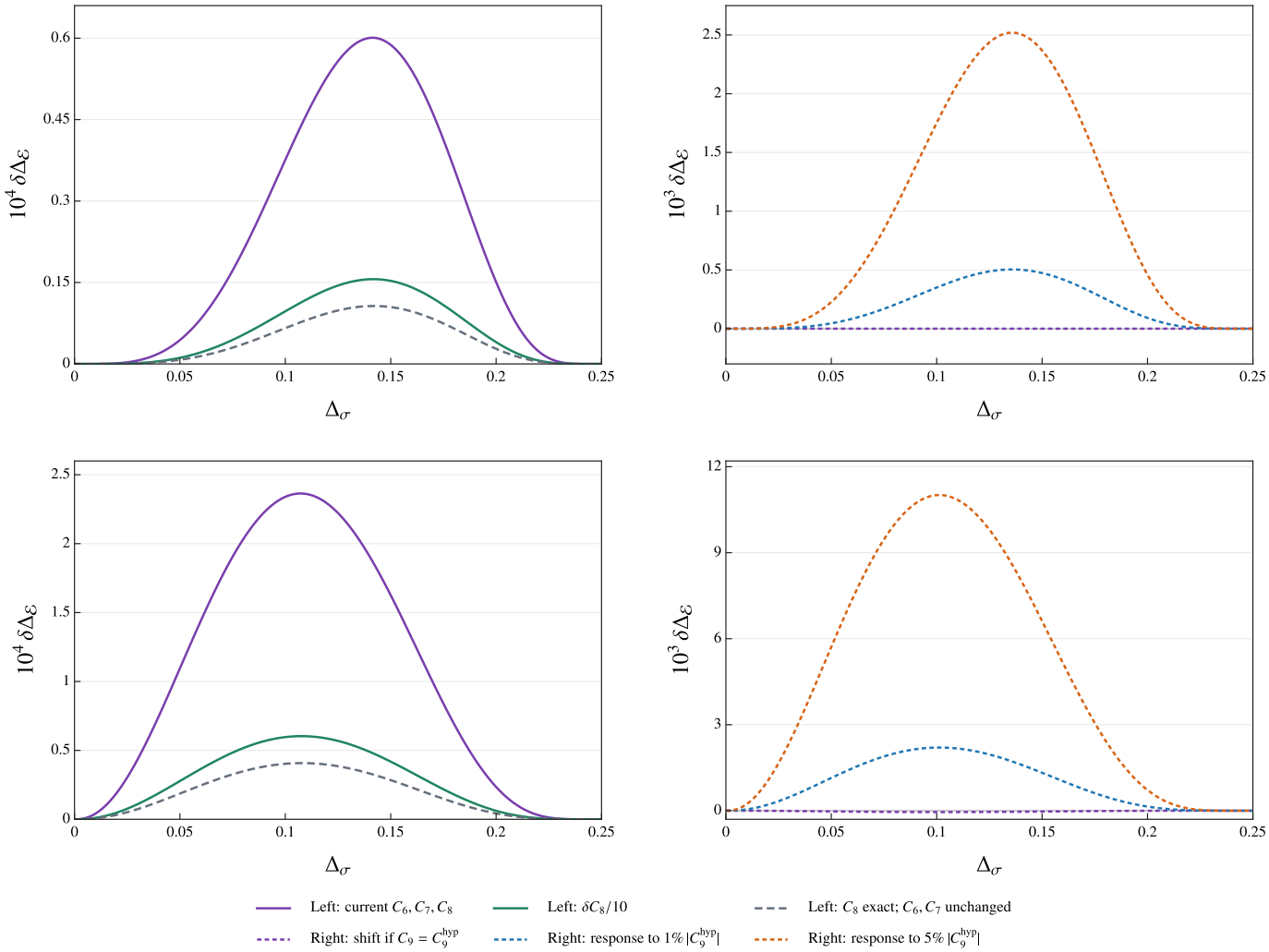


Figure 6: One dimension: coefficient precision. Top row $K = 8$, bottom row $K = 4$ (archived construction, current inputs). Left: $10^4\,U$ for the current allowances ($\delta C_8 = 17000$, $\delta C_7 = 41$, $\delta C_6 = 4.9 \times 10^{-7}$), for $\delta C_8/10$, and with C_8 exact. Right: $10^3\,\delta\Delta_{\mathcal{E}}$ for a hypothetical C_9: the shift if $C_9 = C_9^{\rm hyp}$ and the responses to 1% and 5% of $|C_9^{\rm hyp}|$. Top: $C_9^{\rm hyp} = -2.178401719 \times 10^8$, the CB prediction of the $K = 8$ baseline, so its shift vanishes identically. Bottom: $C_9^{\rm hyp} = -2.190506633 \times 10^8$, the archived $K = 4$ PB[5/3] prediction made with the previous $C_8(1)$, kept as a fixed scenario value. These are sensitivity scenarios, not confidence intervals; C_9 has not been calculated.

At $\Delta_\sigma = 0.1$ and $K = 8$, $U = 3.76 \times 10^{-5}$ with the final allowance; dividing δC_8 by ten gives 9.69×10^{-6}, and the floor with C_8 exact (set by δC_7) is 6.59×10^{-6}. The corresponding values at $K = 4$ are 2.34×10^{-4}, 5.96×10^{-5} and 4.03×10^{-5}. A tenfold better C_8 would thus improve the input budget by a factor 3.9 but leave the prescription envelope, about a hundred times larger, unchanged.

What the final $C_8(1)$ changed. Replacing 8066723.343 ± 33000 by 8059766.723 ± 17000 (a shift of -6957) moves the $K = 8$ baseline down by at most 2.0×10^{-5} (at $\Delta_\sigma = 0.141$; -1.27×10^{-5} at $\Delta_\sigma = 0.1$). It lowers the upper edge of the envelope by up to 1.9×10^{-4}: the width at $\Delta_\sigma = 0.1$ goes from 3.84×10^{-3} to 3.64×10^{-3}, and the maximum width 4.67×10^{-3} is unchanged. It roughly halves the C_8 term of U ($6.02 \times 10^{-5} \to 3.10 \times 10^{-5}$ at $\Delta_\sigma = 0.1$, so that U falls from 6.68×10^{-5} to 3.76×10^{-5}), and it changes the mismatch ratio at $n = 8$ from 0.1578 to 0.1576. With the previous value every program reproduces the archived outputs to 8.4×10^{-15}.

A ninth coefficient. C_9 has not been computed. At fixed CB parameters the ninth-order curve obeys the exact identity

$$F_9(x) - F_8(x) = J_9(x)\,[C_9 - C_9^{\mathrm{CB8}}], \qquad J_9 = \frac{\partial F_9}{\partial C_9}, \tag{7.1}$$

where C_9^{CB8} is the coefficient implied by the eighth-order approximant; supplying that value adds no information. The $K = 8$ baseline implies $C_9^{\mathrm{CB8}} = -2.178401719 \times 10^8$; the only admissible eighth-order PB at $K = 8$, [3/5], predicts -2.181155×10^8, and the $K = 4$ CB8 implies -2.19002×10^8. These are correlated predictions, not a calculation. At $\Delta_\sigma = 0.1$ a change of 1% of $|C_9|$ would move the $K = 8$ curve by 3.5×10^{-4} (at $K = 4$: 2.2×10^{-3}). Adding the ninth order also amplifies the inherited sensitivity to C_8: at $\Delta_\sigma = 0.1$ the C_8 term of U grows from 3.1×10^{-5} to 2.9×10^{-4} at $K = 8$ (from 1.9×10^{-4} to 1.8×10^{-3} at $K = 4$). Simply appending C_9 is therefore not guaranteed to improve the curve: a useful ninth order needs an independently computed C_9, sufficient precision on C_8, and renewed prescription checks.

8 Limitations and open problems

1. **Reporting prescription.** The number $K = 8$ of imposed short-range coefficients, and the range $K = 5$–8 of the envelope, were fixed after the comparison data had been seen. $K = 8$ is motivated a priori only (all short-range coefficients); choosing the narrowest core band would give $K = 7$. Whether the endpoint variable (v or ϵ) belongs in the envelope is open.

2. **Endpoint assumptions.** The constructions assume that the resummed remainder is regular at the short-range end. Logarithms beyond the imposed terms are not excluded. The two-dimensional c_2 carries a formal analytic-MS qualification, the three-dimensional c_1 a conditional spectral range; the one-dimensional C_n^{SR} beyond the published orders are project results with their own assumptions, and the short form of C_8^{SR} is unproved. In $d = 2$ the endpoint construction is the largest uncertainty in mid-window.

3. **The amplitude.** The cancellation of the regulator dependence of the bubble action in the one-loop amplitude (Sec. 3.4) should be derived explicitly. The derivation must keep the integration limits, determinant, measure, legs and counterterms through the simple-pole residue, for the size coordinate and for its rescaled form.

4. **Next order and approach to large order.** The coefficient d_1 needs the two-loop fluctuations, the subleading size integral and the direct mass insertion, and possibly logarithms. Its known part already gives -30% at $n = 8$; further corrections may be needed before eighth order can test K_I.

5. **Dominance.** The Sobolev bound identifies the least-action real saddle; it does not settle complex saddles, integration cycles, multi-instanton sectors, non-instanton factorial mechanisms or the uniformity of the renormalized expansion.

6. **Two and three dimensions.** Only the actions and the power counting are available. A renormalized determinant and amplitude would be needed for amplitude-based tests.

7. **Coefficients.** A covariance model for C_6, C_7, C_8 is missing. A ninth coefficient would be a useful test of the predictions only with adequate precision on C_8 (Sec. 7).
8. **Comparisons.** In $d = 1$ the final HT and NRG values differ at $\Delta_\sigma = 0.1$ and 0.125 by 5.8 and 4.6 times the sum of their quoted errors. The gap is of the size of the dependence of HT on the artifact correction, which the HT error does not include (Sec. 6.2), and its origin is open. HT and NRG resolve the $K = 4$–$K = 8$ difference at several points but lie between the two curves there (Table 13), so they cannot justify the choice of K. In $d = 2$ and 3 HT is available only at finite cutoff; the $d = 2$ mid-window excess of HT over every endpoint construction (about 3×10^{-3} beyond their core bands at $s = 1.6$–1.65) is unresolved, and the cutoff dependence there does not show that it survives at infinite cutoff. Published MC is too imprecise to discriminate, and in $d = 3$ there is a single accepted project MC point. Continuum HT extrapolations and MC with controlled systematics would be the most informative additions.
9. **Interpretation of the $d = 1$ endpoint.** The fixed-point-merger reading of the square-root branch point, and the radius $\simeq \frac{1}{2}$ suggested by the short-range series, should be checked against more coefficients or an independent argument.

A Formulas and parameters

Conformal Borel coefficients. For the resummed series $f(z) = \sum_{j \le m} f_j z^j$ with action A,

$$p_k = \sum_{j=0}^{k} \frac{f_j\,(4A)^j}{\Gamma(j + b_{\mathrm{L}} + 1)}\,(-1)^{k-j} \binom{\alpha - 2j}{k - j}, \qquad \mathcal{R}_m^{\mathrm{CB}}[f](z) = \int_0^\infty e^{-u} u^{b_{\mathrm{L}}} \frac{\sum_{k=0}^m p_k\, w(zu)^k}{[1 - w(zu)]^\alpha}\,\mathrm{d}u, \tag{A.1}$$

which reproduces the Taylor polynomial through order m; for integer $\alpha - 2j$ the generalized binomial coefficient is the polynomial one. The inverse map $w(t) = \sum_{k \ge 1} (-1)^{k-1} \frac{1}{k+1} \binom{2k}{k} (t/4A)^k$ gives an independent round-trip check.

Padé–Borel. With $d_k = f_k / \Gamma(k + b_{\mathrm{L}} + 1)$, the denominator of $[L/M]$ solves $\sum_{j=1}^{M} q_j d_{k-j} = -d_k$ for $k = L + 1, \ldots, m$, and $p_k = \sum_{j=0}^{\min(k,M)} q_j d_{k-j}$ with $q_0 = 1$. The Padé denominator depends nonlinearly on the coefficients, so PB curves are not affine in the inputs.

Series operations and quadrature. Products, powers $(1 - z)^p$ and compositions are formed with truncation after each operation, in 65-digit arithmetic. The Laplace integral uses 240-point generalized Gauss–Laguerre quadrature for weight $u^{b_{\mathrm{L}}} e^{-u}$ (Jacobi matrix with diagonal $2k + b_{\mathrm{L}} + 1$ and off-diagonal $\sqrt{k(k + b_{\mathrm{L}})}$), checked against 120 nodes, adaptive integration and selected 65-digit integrals. For the one-dimensional CB curves the 240-node and adaptive results agree to 1.1×10^{-15}, the 65-digit checks to 5.2×10^{-16}, and the displayed PB curves to 4.0×10^{-15}. Formal re-expansion reproduces the input coefficients. These checks validate the evaluation of each approximant, not its distance from the exact theory.

Grids. The one-dimensional dense grid has 1004 points (1001 equally spaced on $[0, \frac{1}{4}]$ plus the special comparison abscissae); the matched CB/PB comparison uses 502 points. The two- and three-dimensional curves have 754 and 763 stored samples.

Scan parameters. Table 17 collects all prescription parameters.

	$d = 1$	$d = 2$	$d = 3$
Resummed variable	$v = 1 - 2\sqrt{\Delta_\sigma}$	ϵ	ϵ
CB action	$A_v = A_\epsilon/2 = 0.25703245$	$A_\epsilon = 9/8$	$A_\epsilon = 1.92343326$
Endpoint input	$C_1^{\mathrm{SR}}, \ldots, C_8^{\mathrm{SR}}$ $(K = 8)$	1, 0, c_2	E, c_1
Endpoint controls	$K = 4$–7, endpoint-ϵ	tangent, value	value
Without SR input	GFF-only	GFF-only	GFF-only
Orders	$6, 7, 8$	$6, 7, 8$	$6, 7, 8$
Core b_{L}	$2, 3, 4$	$2.5, 3.5, 4.5$	$3, 4, 5$
Wide b_{L}	$0, 1, \ldots, 6$	$0, 1.5, 2.5, \ldots, 6.5$	$0, 2, 3, \ldots, 7$
α (core; wide)	$-1, 0, 1$; $-2, \ldots, 2$	$-1, 0, 1$; $-2, \ldots, 2$	$-1, 0, 1$; $-2, \ldots, 2$
Action variations	$1/A \times 0.8, 0.9, 1.1, 1.2$	same	same
PB control	[5/3] $(K \leq 7)$	PB8 [5/3]	PB7 [3/4]; PB8 [5/3] at $b_{\mathrm{L}} = 0$
Stored grid	1004 points	754 points	763 points

Table 17: Resummation parameters. The baseline uses order 8, the middle core b_{L} and $\alpha = 0$; the PB controls use the baseline b_{L} unless stated.

One-dimensional comparison data. Table 18 lists every one-dimensional value compared in this note, with its residual from the baseline.

Table 18: One-dimensional comparison data [4, 5, 6, 21, 22], labels as in Table 12. "err": for HT the half-spread of the accepted fits, for NRG the chain-depth, kept-state and discretization errors added linearly, for MC σ_{sys}, for published MC the quoted error; "stat": the MC statistical error. A dot means none supplied. MC Δ_{eff}: finite-size exponents of the largest lattice pair (sizes in parentheses), not infinite-size values. The last column is $10^3(\Delta_{\mathcal{E}} - \Delta_{\mathcal{E}}^{\mathrm{baseline}})$.

Method	Δ_σ	$\Delta_{\mathcal{E}}$	err	stat	status	resid.
HT	0.249	0.4993436	9.3×10^{-8}	·	final	−0.00
HT	0.24	0.4942198	2.6×10^{-6}	·	final	+0.00
HT	0.225	0.4880804	5.2×10^{-6}	·	final	−0.00
HT	0.2	0.4830244	4.7×10^{-6}	·	final	−0.09
HT	0.175	0.4839086	1.5×10^{-5}	·	final	−0.33
HT	0.15	0.4912086	3.3×10^{-5}	·	final	−0.50
HT	0.125	0.5061186	4.3×10^{-5}	·	final	−0.31
HT	0.1	0.5304771	1.2×10^{-4}	·	final	+0.23
HT	0.075	0.5673260	2.0×10^{-4}	·	one-sided	+0.70
HT	0.05	0.6206712	·	·	unstable $(N = 40)$	−1.56
HT	0.025	0.7105771	·	·	no limit $(N = 40)$	−2.86
HT	0.01	0.8091106	·	·	no limit $(N = 40)$	−0.64
HT	0.001	0.9693379	·	·	no limit (not shown)	+31.91
NRG	0.249	0.508030	·	·	not converged	+8.69
NRG	0.24	0.498922	·	·	not converged	+4.70
NRG	0.225	0.489668	·	·	not converged	+1.58
NRG	0.2	0.482980	7.1×10^{-4}	·	final	−0.14
NRG	0.175	0.483900	3.5×10^{-4}	·	final	−0.34
NRG	0.15	0.491106	1.1×10^{-4}	·	final	−0.61
NRG	0.125	0.505672	5.4×10^{-5}	·	final	−0.76
NRG	0.1	0.529608	3.3×10^{-5}	·	final	−0.64
NRG	0.075	0.566260	3.0×10^{-5}	·	final	−0.37
NRG	0.05	0.622115	2.7×10^{-5}	·	final	−0.12
NRG	0.025	0.713422	2.9×10^{-5}	·	final	−0.01
NRG	0.01	0.809747	4.7×10^{-5}	·	final	−0.00
NRG	0.001	0.937424	1.1×10^{-4}	·	final	−0.00
MC	0.0125	0.78404	0.01003	0.00077	cross-check unavailable	−5.11
MC	0.025	0.71115	0.00959	0.00049	accepted (σ_{sys}-dominated)	−2.28
MC	0.0375	0.65904	0.00425	0.00051	cross-check unavailable	−2.42

continued on the next page

Method	Δ_σ	$\Delta_\mathcal{E}$	err	stat	status	resid.
MC	0.05	0.61987	0.00220	0.00055	accepted	−2.36
MC	0.0625	0.59003	0.00150	0.00035	cross-check unavailable	−1.36
MC	0.075	0.56694	0.00139	0.00037	accepted	+0.31
MC	0.0875	0.54761	0.00194	0.00032	accepted	+1.05
MC	0.1	0.53248	0.00242	0.00020	accepted (σ_{sys}-dominated)	+2.23
MC	0.1125	0.52020	0.00217	0.00025	accepted (σ_{sys}-dominated)	+3.17
MC	0.125	0.50710	0.00193	0.00035	accepted	+0.67
MC	0.1375	0.49856	0.00180	0.00030	accepted (σ_{sys}-dominated)	+0.48
MC	0.15	0.49325	0.00306	0.00041	cross-check failed	+1.54
MC	0.1625	0.48761	·	0.00058	conditional	+0.47
MC	0.1727335	0.48368	0.00078	0.00052	accepted	−0.97
MC	0.175	0.48460	0.00067	0.00035	cross-check failed	+0.36
MC	0.1875	0.48263	·	0.00030	conditional	−0.28
MC	0.225	0.47912	·	0.00068	conditional	−8.96
MC Δ_{eff}	0.2125	0.4784	·	0.0015	model-limited (256k→ 512k)	−6.43
MC Δ_{eff}	0.2375	0.4826	·	0.0015	model-limited (128k→ 256k)	−10.41
Uzelac 2001	0.05	0.600	0.010	·	published	−22.23
Angelini 2014	0.0625	0.5876	0.0013	·	published	−3.79
Uzelac 2001	0.1	0.53	0.02	·	published	−0.25
Uzelac 2001	0.15	0.49	0.02	·	published	−1.71
Angelini 2014	0.1727335	0.494	0.014	·	published	+9.35
Uzelac 2001	0.2	0.50	0.02	·	published	+16.88

B Reproduction

Everything in this note is reproduced from the folder `Resummation/` of the public repository. Paths below are relative to it. The programs regenerate the resummations, the sensitivity analyses and the comparisons from the coefficients and the comparison data; they do not regenerate graph integrals, HT spectra, NRG runs or MC samples, which come from the sibling folders `GFF/`, `HT/`, `NRG/` and `MC/` (frozen copies in `data/inputs/`, listed with their sources in `data/inputs/INPUTS.json`).

Packages. `code/resum_d1/`: the one-dimensional resummation of Secs. 4–7 (endpoint construction with the short-range series through C_K^{SR}, the $C_8(1)$ switch, curves, envelopes, matched CB/PB comparisons, held-out tests, C_9 scenarios, allowance budgets, reference residuals, the instanton-mismatch table and the $K = 4/K = 8$ table). `code/resum_d23/`: the two- and three-dimensional resummations, the GFF-only CB3 curves and the table of published MC values. `code/instanton/`: the critical bubble, the amplitude K_I and its benchmarks (Sec. 3). `code/comparison/`: the comparison numbers of Sec. 6. The production files ship unchanged with their production layout and a dated sha256 list (`public/SNAPSHOT.sha256`); the tools added for the public replay are in each package's `public/` and `public.mk`.

Commands. From `Resummation/`:

```
make test           # fast tests of every package (1 min)
make replay         # every quoted number, shipped records (1 min)
make replay-full    # rerun the resummations, compare outputs (3 min)
make check-inputs   # frozen inputs against the sibling folders
```

`make replay` recomputes every number of the sections on resummation and on the comparison of the paper [1] and of its paragraph on higher orders, and the entries of its results tables that come from this folder: the resummed values in $d = 1, 2, 3$, the HT values at the largest cutoff in

$d = 2, 3$ and the MC Binder exponents in $d = 2$ (`results/quoted_numbers.json`). It compares each value, rounded half-up to the printed digits, with the printed text (the MC exponents with their errors by the PDG rule). The map from each number to its record, command and tolerance is in `results_map.md` (per package in `code/<package>/public/results_map.md`). It also regenerates, byte for byte, the tables derived from the frozen inputs (the table of one-dimensional comparison data, the $K = 4/K = 8$ table of Sec. 6, the published-MC table and the comparison tables in `results/comparison/`), and checks the data behind the two figures of the paper (`results/figures/*.csv`). `make replay-full` reruns the one-dimensional chain (`code/resum_d1/update_2026-10-13/C8_1_ext/run_chain.sh`, about 2 min on one core) and the ten 2d/3d scripts; every output must agree with the shipped one to $10^{-12} + 10^{-12}|x|$ (fits of the coefficient-model analysis to 10^{-6} relative), and the quoted numbers are recomputed from the rerun. Measured on the symmetry cluster with Python 3.12, numpy 2.3, scipy 1.17: all differences below 5×10^{-15} except those fits (2×10^{-6}).

Numbers of this note that the paper does not quote. They are read from the records of the packages: the one-dimensional curves, bands and key points from `code/resum_d1/update_2026-10-13/C8_1_ext/pipeline_K8/results/` (`curve_K.csv`, `key_points_K.csv`, `ablation_K.csv`, `held_out_K.csv`, `c9_*`, `precision_budget_K.csv`, `comparison_points_K.csv`) and `.../instanton_mismatch/`; the two- and three-dimensional ones from `code/resum_d23/\{2d,3d\}/results/`; the amplitude from `code/instanton/results.json`. `make replay-full` regenerates all of these records.

Figures. The six figures are rendered with Wolfram Language and MaTeX (`vendor/MaTeX/`) by `code/resum_d1/plot_lri_comparisons.wl` (Figs. 1, 4, 5 and 6), `code/resum_d1/plot_lri_dispersion.wl` (Fig. 3) and `code/resum_d23/plot_lri_2d3d.wl` (Fig. 2), through `run_plot.wls`; `make figures` rebuilds them where `wolframscript` and TeX are available. The replay checks the plotted data, not the pixels.

Superseded constructions. The archived $K = 4$ construction (short-range series through C_4^{SR}) and the previous $C_8(1) = (8.067 \pm 0.033) \times 10^6$ appear only where this note compares with them; the previous-$C_8(1)$ outputs are not shipped (they are regenerated by `LRI_C8_1=previous`).

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